Theory of Topoi and Étale Cohomology of Schemes. Volumes 1 to 3

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Contents

EXPOSÉ I

Presheaves

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In numbers 0 through 5 of this exposé, we present the elementary, and most often well-known, properties of categories of presheaves[i]. These properties are used constantly throughout the rest of the seminar, and knowledge of them is essential for understanding the subsequent exposés. The proofs are immediate; most often, they are omitted. In numbers 6 through 9, we take up a few themes used on various occasions later. The reader in a hurry may omit them on a first reading. Number 10 fixes the terminology used. Appendix 11 is due to N. Bourbaki. 1

0. Universes

A *universe* is a nonempty set[1] $\mathscr{U}$ having the following properties:

($\mathscr{U}$ 1) If $x \in \mathscr{U}$ and $y \in x$, then $y \in \mathscr{U}$.
($\mathscr{U}$ 2) If $x, y \in \mathscr{U}$, then $\{x, y\} \in \mathscr{U}$.
($\mathscr{U}$ 3) If $x \in \mathscr{U}$, then $\mathscr{P}(x) \in \mathscr{U}$.
($\mathscr{U}$ 4) If $(x_i, i \in I)$ is a family of elements of $\mathscr{U}$ and $I \in \mathscr{U}$, then $\cup_{i \in I} x_i \in \mathscr{U}$.

From the preceding axioms one readily deduces the following properties: 2

- If $x \in \mathscr{U}$, the set $\{x\}$ belongs to $\mathscr{U}$.
- If x is a subset of $y \in \mathscr{U}$, then $x \in \mathscr{U}$.
- If $x, y \in \mathscr{U}$, the ordered pair $(x, y) = \{\{x, y\}, x\}$ (Kuratowski's definition) is an element of $\mathscr{U}$.
- If $x, y \in \mathscr{U}$, the union $x \cup y$ and the product $x \times y$ are elements of $\mathscr{U}$.
- If $(x_i, i \in I \in \mathscr{U})$ is a family of elements of $\mathscr{U}$, the product $\prod_{i \in I} x_i$ is an element of $\mathscr{U}$.
- If $x \in \mathscr{U}$, then $\mathrm{card}(x) < \mathrm{card}(\mathscr{U})$ (strictly). In particular, the relation $\mathscr{U} \in \mathscr{U}$ does not hold.

We can therefore carry out all the usual operations of set theory starting from the elements of a universe without thereby causing the final result to cease to be an element of the universe.

The first advantage of the notion of a universe is that it provides a definition of the usual categories: the category of sets belonging to the universe $\mathscr{U}$ ($\mathscr{U}$-Ens), the category of topological spaces belonging to the universe $\mathscr{U}$, the category of commutative groups belonging to the universe $\mathscr{U}$ ($\mathscr{U}$-Ab), the category of categories belonging to the universe $\mathscr{U}$....

However, the only known universe is the set of symbols of the form $\{\{\varnothing\}, \{\{\varnothing\}, \varnothing\}\}$ etc.. (all the elements of this universe are finite sets, and this universe is countable). In particular, no universe is known that contains an element of infinite cardinality. We are

[i]The reader may also consult SGA 3 I § § 1 to 3.

[1]N.D.E.: This definition contradicts Bourbaki's appendix *below*. However, the usefulness of empty universes seems debatable...

therefore led to add to the axioms of set theory the axiom: ($\mathcal{U}A$) *For every set x there exists a universe $\mathcal{U}$ such that $x \in \mathcal{U}$.*

3 Since the intersection of a family of universes is a universe, it follows immediately that every set is an element of a smallest universe. One can show that axiom ($\mathcal{U}A$) is independent of the axioms of set theory.

We shall also add the axiom:

($\mathcal{U}B$) *Let $R\{x\}$ be a relation and let $\mathcal{U}$ be a universe. If there exists an element $y \in \mathcal{U}$ such that $R\{y\}$, then $\tau_x R\{x\} \in \mathcal{U}$.*

The consistency of axioms ($\mathcal{U}A$) and ($\mathcal{U}B$) relative to the other axioms of set theory has not been proved and, it seems, cannot be proved.

Let $\mathcal{U}$ be a universe and let $c(\mathcal{U})$ be the supremum of the cardinalities of the elements of $\mathcal{U}$ ($c(\mathcal{U}) \le \mathrm{card}(\mathcal{U})$). The cardinal $c(\mathcal{U})$ has the following properties:

(FI) If $a < c(\mathcal{U})$, then $2^a < c(\mathcal{U})$.

(FII) If $(a_i, i \in I)$ is a family of cardinals strictly less than $c(\mathcal{U})$ and if $\mathrm{card}(I)$ is strictly less than $c(\mathcal{U})$, then $\Sigma_{i\in I} a_i < c(\mathcal{U})$.

The cardinals that have properties (FI) and (FII) are called *strongly inaccessible cardinals.*

The cardinal 0 and the countably infinite cardinal are strongly inaccessible.

Axiom ($\mathcal{U}A$) implies:

($\mathcal{U}A'$) *Every cardinal is strictly less than a strongly inaccessible cardinal.*

Conversely, one can show (11) that the consistency of (UA') implies the consistency of (UA), and that the consistency of these axioms entails that of axiom ($\mathcal{U}B$).

4 Call an *Artinian set* any set E for which there is no infinite family $(x_n, n \in N)$ such that $x_\circ \in E$, $x_{n+1} \in x_n$. One can then show [*loc. cit.*] that there is a one-to-one correspondence between the strongly inaccessible cardinals and the Artinian universes, defined as follows: to every strongly inaccessible cardinal c we associate the unique Artinian universe U_c such that

$$\mathrm{card}(\mathcal{U}_c) = c.$$

Let $c < c'$ be two strongly inaccessible cardinals. We have:

$$\mathcal{U}_c \in \mathcal{U}'_{c'}.$$

In particular, the Artinian universes of cardinality less than a given cardinal form a well-ordered set (with respect to the membership relation). Axiom ($\mathcal{U}A'$) is equivalent to the axiom:

($\mathcal{U}A''$). *Every Artinian set is an element of an Artinian universe.*

Observe that all the usual sets ($\mathbf{Z}, \mathbf{Q}, \dots$) are Artinian sets.

1. $\mathcal{U}$-categories. Presheaves of sets

1.0. *Throughout the remainder of the seminar and unless expressly stated otherwise, the universes under consideration will contain an element of infinite cardinality.* Let $\mathcal{U}$ be a universe. A set is said to be *$\mathcal{U}$-small* (or, when no confusion can result, *small*) if it is *isomorphic* to an element of $\mathcal{U}$. We shall also use the terminology: small group, small ring, small category[2] ...We shall often assume, without explicit mention, that the
5 schemes, topological spaces, indexing sets...with which we work are $\in \mathcal{U}$, or at least have cardinality $\in \mathcal{U}$; nevertheless, many of the categories with which we shall work will not be $\in \mathcal{U}$.

[2]N.D.E.: A category C is regarded as a set of arrows (equipped with the subset of objects, regarded as the set of identity arrows, and with the "source, target" maps). Thus, the expressions "C is an element of $\mathcal{U}$" or "C is $\mathcal{U}$-small" make sense.

Definition 1.1. *Let $\mathscr{U}$ be a universe and $\boldsymbol{C}$ a category. We say that $\boldsymbol{C}$ is a $\mathscr{U}$-category if, for every pair (x, y) of objects of $\boldsymbol{C}$, the set $\mathrm{Hom}_{\boldsymbol{C}}(x, y)$ is $\mathscr{U}$-small.*

1.1.1. Let $\boldsymbol{C}$ and $\boldsymbol{D}$ be two categories and let $\mathrm{Fonct}(\boldsymbol{C}, \boldsymbol{D})$ be the category of functors from $\boldsymbol{C}$ to $\boldsymbol{D}$. The following assertions are immediate:

a) If $\boldsymbol{C}$ and $\boldsymbol{D}$ are elements of a universe $\mathscr{U}$ (resp. are $\mathscr{U}$-small), the category $\mathrm{Fonct}(\boldsymbol{C}, \boldsymbol{D})$ is an element of $\mathscr{U}$ (resp. is $\mathscr{U}$-small).
b) If $\boldsymbol{C}$ is $\mathscr{U}$-small and $\boldsymbol{D}$ is a $\mathscr{U}$-category, $\mathrm{Fonct}(\boldsymbol{C}, \boldsymbol{D})$ is a $\mathscr{U}$-category.

Remark 1.1.2. Let $\boldsymbol{D}$ be a category having the following properties:

(C1) The set $\mathrm{ob}(\boldsymbol{D})$ is contained in the universe $\mathscr{U}$.
(C2) For every pair (x, y) of objects of $\boldsymbol{D}$, the set $\mathrm{Hom}_{\boldsymbol{D}}(x, y)$ is an element of $\mathscr{U}$.

(The usual categories constructed from a universe $\mathscr{U}$ have these two properties: $\mathscr{U}$-Ens, $\mathscr{U}$-Ab,...). Let $\boldsymbol{C}$ be a category belonging to $\mathscr{U}$. Then the category $\mathrm{Fonct}(\boldsymbol{C}, \boldsymbol{D})$ does not in general have properties (C1) and (C2). For example, the category $\mathrm{Fonct}(\boldsymbol{C}, \mathscr{U}\text{-Ens})$ has neither property (C1) nor property (C2). This justifies the adopted definition of $\mathscr{U}$-category, in preference to the more restrictive notion given by conditions (C1) and (C2) above.

Definition 1.2. *Let $\boldsymbol{C}$ be a category. The category of presheaves of sets on $\boldsymbol{C}$ relative* 6
to the universe $\mathscr{U}$ (or, when no confusion can result, the category of presheaves on $\boldsymbol{C}$) is the category of contravariant functors on $\boldsymbol{C}$ with values in the category of $\mathscr{U}$-sets.

We denote by $\widehat{\boldsymbol{C}}_{\mathscr{U}}$ (or more simply, when no confusion can result, by $\widehat{\boldsymbol{C}}$) the category of presheaves of sets on $\boldsymbol{C}$ relative to the universe $\mathscr{U}$. The objects of $\widehat{\boldsymbol{C}}_{\mathscr{U}}$ are called $\mathscr{U}$-presheaves (or simply presheaves) on $\boldsymbol{C}$. When $\boldsymbol{C}$ is $\mathscr{U}$-small, the category $\widehat{\boldsymbol{C}}_{\mathscr{U}}$ is a $\mathscr{U}$-category. When $\boldsymbol{C}$ is a $\mathscr{U}$-category, $\widehat{\boldsymbol{C}}_{\mathscr{U}}$ is not necessarily a $\mathscr{U}$-category.

Construction-Definition 1.3. Let x be an object of a $\mathscr{U}$-category $\boldsymbol{C}$. The *$\mathscr{U}$-functor represented by x* is the functor $h_{\mathscr{U}}(x) : \boldsymbol{C}^{\circ} \to \mathscr{U}\text{-Ens}$ whose construction is as follows [2]. Let y be an object of $\boldsymbol{C}$.

a) If $\mathrm{Hom}_{\boldsymbol{C}}(y, x)$ is an element of $\mathscr{U}$, then:
$$h_{\mathscr{U}}(x)(y) = \mathrm{Hom}_{\boldsymbol{C}}(y, x)$$
b) Suppose that $\mathrm{Hom}_{\boldsymbol{C}}(y, x)$ is not an element of $\mathscr{U}$ and let $\boldsymbol{R}(Z, x, y)$ be the relation: "The set Z is the codomain of an isomorphism $\mathrm{Hom}_{\boldsymbol{C}}(y, x) \xrightarrow{\sim} Z$". We then set:
$$h_{\mathscr{U}}(x)(y) = \tau_Z \boldsymbol{R}(Z).$$

(By axiom ($\mathscr{U}\boldsymbol{B}$), $h_{\mathscr{U}}(x)(y)$ is an element of $\mathscr{U}$). Let $\boldsymbol{R}'(u, x, y)$ be the relation: « u is a bijection
$$u : \mathrm{Hom}_{\boldsymbol{C}}(y, x) \xrightarrow{\sim} h_{\mathscr{U}}(x)(y) \text{ ».}$$
We then set: 7
$$\varphi(y, x) = \tau_u \boldsymbol{R}'(u).$$

It should be noted that, in cases (a) and (b), we have *a* canonical isomorphism:
$$\varphi(y, x) : \mathrm{Hom}_{\boldsymbol{C}}(y, x) \xrightarrow{\sim} h_{\mathscr{U}}(x)(y).$$

[2] $\boldsymbol{C}^{\circ}$ will always denote the category opposite to $\boldsymbol{C}$.

(In case a) $\varphi(y, x)$ is the identity). Now let $u : y \to y'$ be an arrow of C. By composition of morphisms, the morphism u defines a map:

$$\operatorname{Hom}_C(u, x) : \operatorname{Hom}_C(y', x) \longrightarrow \operatorname{Hom}_C(y, x).$$

We then set

$$h_{\mathscr{U}}(x)(u) = \varphi(y, x) \operatorname{Hom}_C(x, u) \varphi(y, x)^{-1}.$$

It is immediate that $h_{\mathscr{U}}(x)$, so defined, is a functor $\widehat{C} \to \mathscr{U}$-Ens.

1.3.1. Similarly, using the isomorphisms φ, for every morphism $y : x \to x'$ we define a morphism of functors:

$$h_{\mathscr{U}}(y) : h_{\mathscr{U}}(x) \longrightarrow h_{\mathscr{U}}(x')$$

and it is immediate that this defines a functor

$$h_{\mathscr{U}} : C \longrightarrow \widehat{C}_{\mathscr{U}}.$$

1.3.2. Now let $\mathscr{V}$ be a universe such that $\mathscr{U} \subset \mathscr{V}$. There is then a canonical fully faithful functor:

$$\mathscr{U}\text{-Ens} \hookrightarrow \mathscr{V}\text{-Ens}$$

whence a fully faithful functor:

$$\widehat{C}_{\mathscr{U}} \hookrightarrow \widehat{C}_{\mathscr{V}}$$

8 and the diagram:

$$\begin{tikzcd} C \arrow[r, "h_{\underline{U}}"] \arrow[dr, "h_{\underline{V}}"'] & C^{\wedge}_{\underline{U}} \arrow[d, hook] \\ & C^{\wedge}_{\underline{V}} \end{tikzcd}$$

commutes up to canonical isomorphism. When C is an element of $\mathscr{U}$, this canonical isomorphism is the identity.

1.3.3. *In practice, the universe $\mathscr{U}$ is fixed once and for all and is not mentioned.* We then use the notations $\widehat{C}$ (for the category of $\mathscr{U}$-presheaves of sets) and

$$h : C \longrightarrow \widehat{C}.$$

For every object x of C, the presheaf $h(x)$ is called the *presheaf represented by x*, and we shall always identify the value at $y \in \operatorname{ob}(C)$ of the presheaf $h(x)$ with $\operatorname{Hom}_C(y, x)$.

Proposition 1.4. *Let C be a $\mathscr{U}$-category, F a presheaf on C, and X an object of C. There is an isomorphism functorial in X and in F:*

$$i : \operatorname{Hom}_{\widehat{C}}(h(X), F) \xleftarrow{\sim} F(X),$$

where the map i, when F is of the form $h(Y)$, is none other than the map:

$$h : \operatorname{Hom}(h(X), h(Y)) \longleftarrow \operatorname{Hom}(X, Y).$$

In particular, the functor h is fully faithful.

1.4.1. This proposition justifies the customary abuse of language identifying an ob-
9 ject of C with the corresponding contravariant functor. A presheaf isomorphic to an object in the image of h (or, using the abuse of language noted above, isomorphic to an object of C) is called a *representable presheaf.*

1.4.2. Let $\mathscr{V}$ be a universe containing the universe $\mathscr{U}$. The category of $\mathscr{U}$-presheaves is a *full* subcategory of the category of $\mathscr{V}$-presheaves, and consequently a $\mathscr{U}$-presheaf is representable if and only if its image in the category of $\mathscr{V}$-presheaves is a representable $\mathscr{V}$-presheaf.

2. Projective and inductive limits

Let C be a $\mathscr{U}$-category and I a small category. Denote by $\mathrm{Fonct}(I, C)$ the category of functors from I to C. The category $\mathrm{Fonct}(I, C)$ is a $\mathscr{U}$-category. To every object X of C, associate the subcategory $\mathscr{X}$ of C whose only object is X and whose only arrow is the identity of X. Let $i_X : \mathscr{X} \to C$ denote the inclusion functor. There is a unique functor $e_X : I \to \mathscr{X}$, and we denote by $k_X : I \to C$ the functor $i_X \circ e_X$. (We say that k_X is the constant functor with value X.) The correspondence $X \mapsto k_X$ is clearly functorial in X, which allows us to define, for every functor $G : I \to C$, the presheaf $X \mapsto \mathrm{Hom}_{\mathrm{Fonct}(I,C)}(k_X, G)$.

Definition 2.1. *The projective limit of G, denoted by $\varprojlim_I G$, is the presheaf:*

$$X \longmapsto \mathrm{Hom}_{\mathrm{Fonct}(I,C)}(k_X, G).$$

When the presheaf $\varprojlim_I G$ is representable, we also denote by $\varprojlim_I G$ an object of C rep- 10
resenting it. Thus the object $\varprojlim_I G$ is defined only up to isomorphism. When no confusion can result, we use the abbreviated notation $\varprojlim G$.

As G varies, $\varprojlim G$ is a functor from the category $\mathrm{Fonct}(I, C)$ to $\widehat{C}$.

2.1.1. By symmetry (reversing the direction of the arrows in C), one similarly defines the inductive limit of a functor: it is a covariant functor on C with values in the category of $\mathscr{U}$-sets. We shall use the notations $\varinjlim_I G$ or $\varinjlim G$.

Observe that products, fiber products, and kernels are projective limits. Similarly, sums, amalgamated sums, and cokernels are inductive limits.

Definition 2.2. *Let I and C be two categories and $G : I \to C$ a functor. The projective limit of G is said to be representable if there exists a universe $\mathscr{U}$ such that:*

1) *The category I is $\mathscr{U}$-small.*
2) *The category C is a $\mathscr{U}$-category.*
3) *The presheaf $\varprojlim G$, with values in the category of $\mathscr{U}$-sets, is representable.*

It follows from n° 1 that the object $\varprojlim G$ representing the presheaf $\varprojlim G$ is independent, up to isomorphism, of the universe $\mathscr{U}$. Note also that there exists a smallest universe $\mathscr{U}'$ having properties (1) and (2), and that the presheaf $\varprojlim G$ necessarily takes values in the category of $\mathscr{U}'$-sets.

2.2.1. Let I and C be two categories. We say that *the I-projective limits in C are* 11
representable if, for every functor $G : I \to C$, the projective limit of G is representable. Finally, let C be a category and $\mathscr{U}$ a universe. We say that the *$\mathscr{U}$-projective limits in C are representable* if, for every category I that is $\mathscr{U}$-small and every functor $G : I \to C$, the projective limit of G is representable.

Proposition 2.3. *Let C be a category and $\mathscr{U}$ a universe. The following assertions are equivalent:*

i) *The $\mathcal{U}$-projective limits in* $\mathbf{C}$ *are representable.*
ii) *Products indexed by a small set are representable, and kernels of pairs of arrows are representable.*
iii) *Products indexed by a small set are representable, and fiber products are representable.*

Proof. It suffices to observe that there is an isomorphism, functorial in G,

$$\varprojlim G = \operatorname{Ker}\Big(\prod_{i\in \operatorname{ob}(I)} G(i) \rightrightarrows \prod_{u\in \mathrm{F}\ell(\mathrm{I})} G(\operatorname{but}(u))\Big),$$

where the pair of arrows is defined by the morphisms

$$\prod_{i\in \operatorname{Ob}(I)} G(i) \xrightarrow{\mathrm{pr}_{\operatorname{but}(u)}} G(\operatorname{but}(u))$$

$$\prod_{i\in \operatorname{Ob}(I)} G(i) \xrightarrow{\mathrm{pr}_{\operatorname{source}(u)}} G(\operatorname{source}\,(u)) \xrightarrow{G(u)} G(\operatorname{but}(u)).$$

Moreover, it is clear that $\operatorname{Ker}(X \overset{u}{\underset{v}{\rightrightarrows}} Y) = X \prod_{X\times Y} X$, where the two morphisms from X to $X \times Y$ are $\mathrm{id}_X \times u$ and $\mathrm{id}_X \times v$.

2.3.1. There are of course analogous definitions and assertions for inductive limits,
12 which we shall not spell out. The same applies to finite projective and inductive limits (i.e. those relative to a finite category I).

Corollary 2.3.2. *Denote by* $\mathcal{U}$-Ens *the category of* $\mathcal{U}$*-sets and by* $\mathcal{U}$-Ab *the category of abelian group objects in* $\mathcal{U}$-Ens. *The* $\mathcal{U}$*-projective and inductive limits in* $\mathcal{U}$-Ens *and in* $\mathcal{U}$-Ab *are representable.*

Proposition 2.3.3. *Let I be a small category, $F : I \to \mathcal{U}$*-Ens *a functor, and $G \in$* ob *I a small set of objects of I such that, for every object X of I, there exists an object $Y \in G$ and a morphism $Y \to X$ (resp. $X \to Y$). Then $\varprojlim_I F$ (resp. $\varinjlim_I F$) is representable, and* $\operatorname{card}(\varprojlim_I F) \leq \Pi_{Y\in G} \operatorname{card}(F(Y))$ *(resp.* $\operatorname{card}(\varinjlim_I F) \leq \Sigma_{Y\in G} \operatorname{card}(F(Y))$*).*

Proof. One immediately verifies that $\varprojlim_I F'$ (resp. $\varinjlim_I F'$) is isomorphic to a subobject (resp. a quotient) of $\Pi_{Y\in G} F(Y)$ (resp. $\coprod_{Y\in G} F(Y)$); hence the proposition.

Definition 2.4.1. *Let $\mathbf{C}$ be a category in which finite projective (resp. inductive) limits are representable, and let $u : \mathbf{C} \to \mathbf{C}'$ be a functor. We say that u is left exact (resp. right exact) if it "commutes" with finite projective (resp. inductive) limits. A functor that is both left and right exact is called an exact functor.*

2.4.2. It follows from 2.3 that, for a functor to be exact on the left, it is necessary and sufficient that it take the terminal object (= empty product) to the terminal object, the product of two objects to the product of their images, and the kernel of a pair of arrows to the kernel of the image pair; equivalently, that it take the terminal object to the terminal object and fiber products to fiber products (we assume that in $\mathbf{C}$ finite $\varprojlim$ are representable).

13 2.5.0. Let I, J, and C be three categories, and let $G : I \times J \to C$ be a functor (i.e. a functor from I with values in the category of functors from J to C). Suppose that the projective limits of the functors

$$\begin{aligned} G_i : J &\longrightarrow C \qquad i \in \mathrm{ob}(I) \\ j &\longmapsto G(i \times j) \end{aligned}$$

are representable, and that the functor

$$i \longmapsto \varprojlim G_i$$

admits a representable projective limit. It is then clear that the functor G admits a representable projective limit and that there is a canonical isomorphism

$$\varprojlim_{I\times J} G \xrightarrow{\sim} \varprojlim_I \varprojlim_J G_i,$$

and consequently a canonical isomorphism

$$\varprojlim_I \varprojlim_J G_i \xrightarrow{\sim} \varprojlim_J \varprojlim_I G_j.$$

Hereafter we shall say more briefly that projective limits commute with projective limits. Similarly, inductive limits commute with inductive limits. But in general it is not true that inductive limits commute with projective limits.

Definition 2.5. *Let C be a category with representable fiber products, I a category, $G : I \to C$ a functor, $g : G \to X$ a morphism from G to an object of C (i.e. a morphism from G to the constant functor associated with X), and $m : Y \to X$ a morphism in C. Let $G \times_X Y : I \to C$ be the functor* 14

$$i \longmapsto G(i) \times_X Y.$$

We say that the inductive limit of G is universal if, for every object X, every morphism $g : G \to X$, and every morphism $m : Y \to X$,

a) *the inductive limit of the functor $G \times_X Y$ is representable,*
b) *the canonical morphism $\varinjlim(G \times_X Y) \to (\varinjlim G) \times_X Y$ is an isomorphism.*

Proposition 2.6. *Let $\mathcal{U}$ be a universe. The inductive limits in $\mathcal{U}$-*Ens *that are representable (in particular, the $\mathcal{U}$-inductive limits (2.2.1) in $\mathcal{U}$-*Ens*) are universal.*

We shall also use another commutation result for projective and inductive limits, which we now present.

Definition 2.7. *A category I is pseudo-filtered if it has the following properties:*

PS 1) *Every diagram of the form*

$$\begin{tikzcd} & j \\ i \arrow[ur] \arrow[dr] & \\ & j' \end{tikzcd}$$

can be embedded in a commutative diagram

$$\begin{tikzcd} & j \arrow[dr] & \\ i \arrow[ur] \arrow[dr] & & k \\ & j' \arrow[ur] & \end{tikzcd}$$

PS 2) *Every diagram of the form*

$$\begin{tikzcd} i \arrow[r, shift left, "u"] \arrow[r, shift right, "v"'] & j \end{tikzcd}$$

15 *can be embedded in a diagram*

$$\begin{tikzcd} i \arrow[r, shift left, "u"] \arrow[r, shift right, "v"'] & j \arrow[r, "w"] & k \end{tikzcd}$$

such that

$$w \circ u = w \circ v.$$

A category I is said to be filtered if it is pseudo-filtered, nonempty, and connected, i.e. if any two objects of I can be joined by a sequence of arrows (with no condition imposed on the directions of the arrows); in the presence of PS 2, *this also means that $I \neq \varnothing$ and that, for any two objects a, b of I, there is always an object c of I and arrows $a \to c$ and $b \to c$. We also say that a category I is cofiltered if I° is filtered.*

Example 2.7.1. *If, in I, amalgamated sums (resp. sums of two objects) and cokernels of pairs of parallel arrows are representable, then I is pseudo-filtered (resp. filtered).*

Proposition 2.8. *Let $\mathcal{U}$ be a universe. Filtered $\mathcal{U}$-inductive limits in $\mathcal{U}$-*Ens *commute with finite projective limits.*

We are immediately reduced to proving that filtered $\mathcal{U}$-inductive limits commute with fiber products. The proof is left to the reader. One may use the description of the limit given by the following

Lemma 2.8.1. *Let I be a small filtered category and $i \mapsto X_i$ a functor from I to $\mathcal{U}$-*Ens. *On the disjoint union $\coprod_{i\in \mathrm{ob}\, I} X_i$, let R be the relation:*

(R) *Two elements $x_i \in X_i$ and $x_j \in X_j$ are related if there exists an object $k \in \mathrm{ob}\, I$ and two morphisms $u : i \to k$ and $v : j \to k$ such that the images in X_k of x_i and x_j under the transition maps u and v, respectively, are equal.*

16 *Then:*

1) *R is an equivalence relation.*
2) *The quotient $\coprod_{i\in \mathrm{ob}\, I} X_i/R$ is canonically isomorphic to $\varinjlim_I X_i$.*
3) *Two elements $x_i \in X_i$ and $x_j \in X_j$ are respectively equivalent under R to two elements of the same X_k.*
4) *For every $i \in \mathrm{ob}\, I$, two elements α and β of X_i are equivalent under R if and only if there is a morphism $u : i \to j$ such that $u(\alpha) = u(\beta)$.*

Assertion 1) follows from (PS 1) (2.7). To prove 2), one checks that $\coprod_{i\in \mathrm{ob}\, I} X_i/R$ has the universal property of the inductive limit (only (PS 1) is used). Assertion 3) follows from the fact that I is connected. Assertion 4) follows from (PS 2).

Corollary 2.9. *Let γ be a kind of algebraic structure "defined by finite projective limits." (The reader is asked to give a mathematical meaning to the preceding phrase. We merely note that the structures of groups, abelian groups, rings, modules, etc. are structures of this kind.) Denote by $\mathcal{U}$-γ the category of γ-objects in $\mathcal{U}$-*Ens. *The functor that associates with each object of $\mathcal{U}$-γ its underlying set commutes with filtered $\mathcal{U}$-inductive limits. Consequently, filtered $\mathcal{U}$-inductive limits in $\mathcal{U}$-γ commute with finite projective limits.*

Corollary 2.10. *Pseudo-filtered $\mathcal{U}$-inductive limits in $\mathcal{U}$-*Ab *commute with finite projective limits.*

(One reduces to filtered limits by decomposing the indexing category into connected components.)

We now record a result that we shall use constantly: let C and C' be two $\mathcal{U}$- 17
categories, and let u and v be two functors $C \underset{u}{\overset{v}{\rightleftarrows}} C'$, with *$u$ left adjoint* to v. Recall that this means that there is an isomorphism between the bifunctors with values in $\mathcal{U}$-Ens:

$$\mathrm{Hom}_{C'}(u(X), X') \xrightarrow{\sim} \mathrm{Hom}_C(X, v(X')).$$

Proposition 2.11. *The functor u commutes with representable inductive limits; the functor v commutes with representable projective limits.*

This assertion means that, for every category I and every functor $G : I \to C'$ such that the projective limit of G is representable, the functor $v \circ G$ has a representable projective limit and there is a canonical isomorphism

$$v(\varprojlim G) \longrightarrow \varprojlim(v \circ G).$$

The reader may spell out the assertion concerning the functor u.

Finally, we record a computation of projective limits in the case where the indexing category has products.

Proposition 2.12. *Let I and C be two categories, and $\mathcal{V}$ a universe. Assume that the $\mathcal{V}$-projective limits in C are representable and that there is in I a family of objects $(i_\alpha)_{\alpha \in A}$, where A is an element of $\mathcal{V}$, such that:*

1) *the products $i_\alpha \times i_\beta$ are representable for every pair $(\alpha, \beta) \in A \times A$,*
2) *every object of I admits a morphism to at least one of the i_α.*

Then, for every contravariant functor F from I to C,

$$F : I^\circ \longrightarrow C,$$

the projective limit of F exists, and there is an isomorphism functorial in F:

$$\varprojlim F \xrightarrow{\sim} \mathrm{Ker}\left(\prod_{\alpha \in A} F(i_\alpha) \rightrightarrows \prod_{(\alpha,\beta) \in A \times A} F(i_\alpha \times i_\beta)\right),$$

where the two arrows are defined by the projections of the products $i_\alpha \times i_\beta$ onto their factors. 18

3. Exactness properties of the category of presheaves

Let C be a $\mathcal{U}$-category and let $\widehat{C}$ be the category of presheaves on C. All the exactness properties of $\widehat{C}$ follow from the following proposition:

Proposition 3.1. *The $\mathcal{U}$-projective and inductive limits in $\widehat{C}$ are representable. For every object X of C, the functor on $\widehat{C}$*

$$F \longmapsto F(X) \qquad F \in \widehat{C}$$

commutes with inductive and projective limits.

In other words, "inductive and projective limits in $\widehat{C}$ are computed argument by argument." We mention a few corollaries.

Corollary 3.2. *Let γ be an algebraic structure defined by finite projective limits. The category of contravariant functors with values in $\mathcal{U}$-γ is equivalent to the category of γ-objects of $\widehat{C}$.*

Corollary 3.3. *A morphism of $\widehat{C}$ that is both a monomorphism and an epimorphism is an isomorphism. A morphism of $\widehat{C}$ factors uniquely as an epimorphism followed by a monomorphism. The representable inductive limits in $\widehat{C}$ are universal. The filtered $\mathcal{U}$-inductive limits in $\widehat{C}$ commute with finite projective limits. The canonical functor $C \to \widehat{C}$ commutes with representable projective limits.*

19 More generally, one may say that the category $\widehat{C}$ inherits all the properties of the category of $\mathcal{U}$-sets that involve inductive and projective limits.

3.4.0. Let F be an object of $\widehat{C}$. We denote by C/F the following category. The objects of C/F are pairs consisting of an object X of C and a morphism u from X to F. Let (X, u) and (Y, v) be two objects. A morphism from (X, u) to (Y, v) is a morphism g from X to Y such that the following diagram is commutative:

$$\begin{tikzcd} X \arrow[rr, "g"] \arrow[dr, "u"'] & & Y \arrow[dl, "v"] \\ & F & \end{tikzcd} .$$

Proposition 3.4. *With the notation of (3.4.0), the source functor $C/F \to \widehat{C}$ admits a representable inductive limit in $\widehat{C}$. The canonical morphism*

$$\varinjlim_{C/F} \text{source}\,(.) \longrightarrow F$$

is an isomorphism.

Corollary 3.5. *Let F and H be two objects of $\widehat{C}$. There is a canonical isomorphism*

$$\mathrm{Hom}(F, H) \xrightarrow{\sim} \varprojlim_{(X,u)\in \mathrm{ob}(C/F)} H(X).$$

Proof. The corollary follows immediately from 3.4 and 1.2.

20 Proposition 3.6. *Let C be a $\mathcal{U}$-category, let $\mathcal{V}$ be a universe containing $\mathcal{U}$, and let $i : \widehat{C}_{\mathcal{U}} \to \widehat{C}_{\mathcal{V}}$ be the natural inclusion functor of the corresponding categories of presheaves (1.3). The functor i commutes with inductive and projective limits. Moreover, for every object F of $\widehat{C}_{\mathcal{U}}$, every subobject of $i(F)$ is isomorphic to the image under i of a unique subobject of F (thus giving a bijection from the set of subobjects of F to the set of subobjects of $i(F)$).*

4. Sieves

Definition 4.1. *Let C be a category. A* sieve *in the category C is a full subcategory D of C with the following property: every object of C from which there exists a morphism to an object of D belongs to D. If X is an object of C, the sieves of X are (by an abuse of language) the sieves in the category C/X.*

Let $\mathscr{U}$ be a universe such that C is a $\mathscr{U}$-category. Let $\widehat{C}$ be the corresponding category of presheaves. To every sieve of X one associates a subobject of X in $\widehat{C}$ as follows: to every object Y of C, one assigns the set of morphisms $f : Y \to X$ such that the object (Y, f) belongs to the sieve.

Proposition 4.2. *The map defined above establishes a bijection between the set of sieves of X and the set of subobjects of X in $\widehat{C}$.*

Proof. We shall only describe what is the inverse map. To every subfunctor R of X we associate the category C/R of objects of C over R (3.4). One immediately checks that C/R is a sieve of X.

Remark 4.2.1. One sees likewise that the sieves of C are in canonical bijective cor- 21
respondence with the set of subfunctors of the "terminal functor" on C (the "terminal" object of $\widehat{C}$).

4.3. Let C be a $\mathscr{U}$-category. By an abuse of language, we shall also call the subobjects of X in the category $\widehat{C}$ the sieves of X. This abuse of language allows us, for every presheaf F and every sieve R of X, to define $\operatorname{Hom}_{\widehat{C}}(R, F)$ as the set of morphisms from the functor R to F. There is moreover a canonical isomorphism, functorial in F (3.5):

$$\operatorname{Hom}_{\widehat{C}}(R, F) \xrightarrow{\sim} \varprojlim_{C/R} F(.),$$

which also gives a direct definition[3]. Likewise, Proposition 4.2 allows us to transfer the usual operations on functors to sieves. We mention the following:

4.3.1. *Base change.* Let R be a sieve of X and let $f : Y \to X$ be a morphism of objects of C. The fiber product $R \times_X Y$ is a sieve of Y, called the *sieve deduced from R by base change.* The corresponding subcategory of C/Y is the inverse image of the subcategory of C/X defined by R under the canonical functor $C/Y \to C/X$ defined by f.

4.3.2. *Order relation, intersection, union.* The inclusion relation on the subfunctors of X is an order relation. The union and intersection of a family of sieves indexed by an arbitrary set may be defined as the least upper bound and the greatest lower bound of the family of corresponding subpresheaves.

4.3.3. *Image, generated sieve.* Let $(F_\alpha)_{\alpha \in A}$ be a family of presheaves and, for every 22
$\alpha \in A$, let $f_\alpha : F_\alpha \to X$ be a morphism, where X is an object of C. The *image* of this family of morphisms is the union of the images of the f_α. The image of this family is therefore a sieve of X. In particular, if all the F_α are objects of C, the image sieve will be called the sieve generated by the morphisms f_α. The category C/R is the full subcategory of C/X formed by the objects $X' \to X$ over X for which there exists an X-morphism from X' to one of the F_α.

[3] N.D.E.: More simply, C/R is identified with R, so that one has the formula $\operatorname{Hom}_{\widehat{C}}(R, F) \xrightarrow{\sim} \varprojlim_R F$, where, by an abuse of notation, F denotes the composite of F and the tautological "source" functor $R \to C/X \to C$.

The reader may, as an exercise, translate the relations and operations defined here on subfunctors into the language of the categories C/R. One will then observe that these relations and operations do not depend on the universe $\mathscr{U}$ for which C is a $\mathscr{U}$-category, and that they are therefore defined for every category, without having to specify the universe to which the morphism sets belong; this could moreover have been foreseen *a priori* from 3.6.

5. Functoriality of categories of presheaves

5.0. Let C, C', and D be three categories, and let $u : C \to C'$ be a functor. We denote by u^* the functor

$$u^* : \left\{ \begin{array}{ccc} \mathbf{Hom}(C'^{\circ}, D) & \longrightarrow & \mathbf{Hom}(C^{\circ}, D) \\ G & \longmapsto & G \circ u \end{array} \right.$$

obtained by composition with the functor u. The functor u^* commutes with inductive and projective limits.

23 Proposition 5.1. *Suppose that C is small and that the $\mathscr{U}$-inductive (resp. projective) limits in D are representable. The functor u^* has a left adjoint $u_!$ (resp. a right adjoint u_*). Thus there is an isomorphism*

$$\mathrm{Hom}_{\mathbf{Hom}(C^{\circ},D)}(F, u^*G) \xrightarrow{\sim} \mathrm{Hom}_{\mathbf{Hom}(C'^{\circ},D)}(u_!F, G)$$
$$(\textit{resp. } \mathrm{Hom}_{\mathbf{Hom}(C^{\circ},D)}(u^*G, F) \simeq \mathrm{Hom}_{\mathbf{Hom}(C'^{\circ},D)}(G, u_*F)).$$

Proof. We shall give only the proof of the existence of the left adjoint functor. The resp. part of the proposition will then follow formally from the isomorphisms

$$\mathbf{Hom}(C^{\circ}, D) \xrightarrow{\sim} \mathbf{Hom}(C, D^{\circ})^{\circ} \xrightarrow{\sim} \mathbf{Hom}((C^{\circ})^{\circ}, D^{\circ})^{\circ}.$$

Let Y be an object of C'. Denote by I_u^Y the following category. The objects of I_u^Y are pairs (X, m), where X is an object of C and m is a morphism $Y \to u(X)$. Let (X, m) and (X', m') be two objects of I_u^Y. A morphism from (X, m) to (X', m') is a morphism $\xi : X \to X'$ such that $m' = u(\xi)m$. Composition of morphisms is defined in the evident way.

Let $f : Y \to Y'$ be a morphism of C'. By composition, the morphism f defines a functor $I_u^f : I_u^Y \to I_u^{Y'}$.

There is moreover a functor $\mathrm{pr}_Y : I_u^Y \to C$ that assigns the object X to the object (X, m). Notice that the diagram

$$(*) \qquad \begin{tikzcd} I_u^Y \arrow[r] \arrow[dr, "\mathrm{pr}_Y"'] & I_u^f \arrow[r] & I_u^{Y'} \arrow[dl, "\mathrm{pr}_{Y'}"] \\ & C & \end{tikzcd}$$

is then commutative.

24 Now let F be a presheaf on C, and set

$$u_!F(Y) = \varinjlim_{I_u^Y} F \circ \mathrm{pr}_Y(.). \tag{5.1.1}$$

The commutativity of diagram (*) and the functoriality of the inductive limit make $u_! F$ a presheaf on C'. Let us show that the functor $u_!$ is a left adjoint of the functor u^*. For this, we show that for every presheaf G on C', there is a functorial isomorphism

$$\mathrm{Hom}(u_! F, G) \xrightarrow{\sim} \mathrm{Hom}(F, u^* G).$$

Let $\xi \in \mathrm{Hom}(u_! F, G)$. Thus, for every object X of C, there is a morphism

$$\xi_X : u_!(u(X)) \longrightarrow G(u(X)).$$

But $(u(X), \mathrm{id}_{u(X)})$ is an object of $I_u^{u(X)}$, and by the definition of the inductive limit there is a canonical morphism

$$F(X) \longrightarrow u_! F(u(X)).$$

For every object X of C, we therefore obtain a morphism

$$\eta_X : F(X) \longrightarrow G(u(X))$$

which is visibly functorial in X. Hence a morphism

$$\eta : F \longrightarrow u^* G.$$

Conversely, let $\eta \in \mathrm{Hom}(F, u^* G)$. For every object Y of C, this gives a morphism of functors

$$\eta_Y : F \circ \mathrm{pr}_Y \longrightarrow (u^* G)\, \mathrm{pr}_Y\,,$$

and hence, by composition with the evident morphism from the functor $(u^* G)\, \mathrm{pr}_Y$ to the constant functor $G(Y)$, a morphism

$$\xi_Y : u_! F(Y) \longrightarrow G(Y)$$

which is functorial in Y. Hence a morphism

$$\xi : u_! F \longrightarrow G.$$

The reader will check that the two maps thus defined are inverse to one another, 25
thereby completing the proof.

Proposition 5.2. *Suppose that in D the $\mathcal{U}$-inductive limits are representable, finite projective limits are representable, and filtered $\mathcal{U}$-inductive limits commute with finite projective limits. Suppose also that finite projective limits are representable in C and that the functor u is left exact (2.3.2). Then finite projective limits are representable in $\mathbf{Hom}(C^\circ, D)$ and in $\mathbf{Hom}(C'^\circ, D)$, and the functor $u_!$ is left exact.*

Proof. The first assertion is trivial. Let us prove the second. By the proof of 5.1, for every presheaf F on C and every object Y of C', one has

$$u_! F(Y) \xrightarrow{\sim} \varinjlim_{I_u^Y} F \circ \mathrm{pr}_Y\,.$$

It therefore suffices to show that the category $(I_u^Y)^\circ$ satisfies axioms (PS 1) and (PS 2) (2.7), and that this category is connected. The verification is left to the reader.

5.3. Specializing these results to the case where D is the category of $\mathcal{U}$-sets, one obtains a sequence of three functors:

$$u_!, u^*, u_*,$$

which is a "sequence of adjoint functors" in the sense that for any two consecutive functors in the sequence, the one on the right is right adjoint to the other. Their essential 26
properties are summarized in the following proposition:

Proposition 5.4. *Let C be a small category, C' a $\mathcal{U}$-category, and $u : C \to C'$ a functor.*

1) *The functor $u^* : \widehat{C'} \to \widehat{C}$ commutes with inductive and projective limits.*
2) *The functor $u_* : \widehat{C} \to \widehat{C'}$ commutes with projective limits. For every presheaf F on C and every object Y of C', one has:*

$$u_*F(Y) \xrightarrow{\sim} \mathrm{Hom}_{\widehat{C}}(u^*(Y), F).$$

3) *The functor $u_! : \widehat{C} \to \widehat{C'}$ commutes with inductive limits. The functor $u_!$ is defined only up to isomorphism, but it can always be chosen so that the diagram*

$$\begin{tikzcd} C \arrow[r, "u"] \arrow[d, "h"'] & C' \arrow[d, "h'"] \\ C^\wedge \arrow[r, "u_!"] & C'^\wedge \end{tikzcd}$$

(h and h' are the canonical inclusion functors) is commutative.
For every presheaf F on C, one has:

$$u_!F \xrightarrow{\sim} \varinjlim_{C/F} h' \circ u.$$

4) *If finite projective limits are representable in C and if u is left exact (2.3.2), the functor $u_!$ is left exact.*

Proof. Assertion (1) is trivial. Assertion (2) follows from the fact that u_* is a right adjoint functor, by (2.11) and (1.4). The same is true of assertion (3), but one also applies (3.4). Finally, assertion (4) is simply (5.2), which can be applied thanks to (2.7).

27 Proposition 5.5. *Let C and C' be two small categories and let $C \underset{u}{\overset{v}{\rightleftarrows}} C'$ be a pair of functors, where v is left adjoint to u. There are then isomorphisms, compatible with the adjunction isomorphisms:*

$$v^* \xrightarrow{\sim} u_!$$

$$v_* \xrightarrow{\sim} u^* \quad .$$

Proof. It suffices to exhibit an isomorphism $v^* \xrightarrow{\sim} u_!$; the other isomorphism follows from it by adjunction. Let F be a presheaf on C and let Y be an object of C'. Then:

$$v^*F(Y) \xrightarrow{\sim} \mathrm{Hom}(v(Y), F).$$

Next, using (3.4):

$$\mathrm{Hom}(v(Y), F) \xrightarrow{\sim} \varinjlim_{C/F} \mathrm{Hom}(v(Y), .).$$

But v is left adjoint to u, and hence:

$$\varinjlim_{C/F} \mathrm{Hom}(v(Y), .) \xrightarrow{\sim} \varinjlim_{C/F} \mathrm{Hom}(Y, u(.)).$$

Now using (5.4.3)), one obtains:

$$\varinjlim_{C/F} \mathrm{Hom}(Y, u(.)) \xrightarrow{\sim} \mathrm{Hom}(Y, u_! F) \xrightarrow{\sim} u_! F(Y).$$

Thus, for every object Y of C', we have determined an isomorphism $v^* F(Y) \xrightarrow{\sim} u_! F(Y)$ that is clearly functorial in Y and in F, C.Q.F.D.

Corollary 5.5.1. *Let $u : C \to C'$ be a functor that admits a left adjoint. The functor $u_! : \widehat{C} \to \widehat{C'}$ commutes with projective limits (recall that it commutes with inductive limits by (1.4.3)).*

Remark 5.5.2. One thus obtains a "sequence of *four* adjoint functors" (cf. 5.3): 28

$$v_!, v^* = u_!, v_* = u^*, u_*,$$

of which the first three (resp. the last three) therefore commute with $\varinjlim$ (resp. $\varprojlim$).

Proposition 5.6. *The hypotheses are those of 5.4. The following conditions are equivalent:*

i) *The functor u is fully faithful.*
ii) *The functor $u_!$ is fully faithful.*
iii) *The adjunction morphism $\mathrm{id}_{\widehat{C}} \to u^* u_!$ is an isomorphism.*
iv) *The functor u_* is fully faithful.*
v) *The adjunction morphism $u^* u_* \to \mathrm{id}_{\widehat{C}}$ is an isomorphism.*

Proof. It is clear that ii) ⇔ iii) and iv) ⇔ v) (general properties of adjoint functors), and that ii) ⇒ i) (5.4.3)). Let us show that i) ⇒ iii). The functors $\mathrm{id}_{\widehat{C}}$, u^*, and $u_!$ commute with inductive limits. By (3.4), it therefore suffices to prove that $H \to u^* u_! H$ is an isomorphism when H is representable, which is evident. Let us show that iii) is equivalent to v). For every object H (resp. K) of $\widehat{C}$, denote by $\Phi(H) : H \to u^* u_! H$ (resp. by $\Psi(K) : u^* u_* \to K$) the adjunction morphism. We then have a commutative diagram:

$$\begin{tikzcd}
\mathrm{Hom}_{\widehat{C}}(H, u^* u_* K) \arrow[rr, "\mathrm{Hom}(H,\Psi(K))"] \arrow[d, dash, "\wr"'] & & \mathrm{Hom}_{\widehat{C}}(H, K) \\
\mathrm{Hom}_{\widehat{C}}(u_! H, u_* K) \arrow[d, dash, "\wr"'] & & \\
\mathrm{Hom}_{\widehat{C}}(u^* u_! H, K) \arrow[uurr, "\mathrm{Hom}(\Phi(H),K)"'] & &
\end{tikzcd}$$

.

It follows that $\Phi(H)$ is an isomorphism for every H if and only if $\Psi(\mathscr{K})$ is an isomorphism for every K, C.Q.F.D. 29

Remark 5.7. a) The equivalences ii) ⇔ iii) ⇔ iv) ⇔ v) are general results on adjoint functors.

b) The explicit form of $u_!$ resp. u_* given in the proof of 1.4 immediately shows that, under the general hypotheses of 1.1, if u is fully faithful, then the adjunction morphism $\mathrm{id} \to u^* u_!$ (resp. $u^* u_* \to \mathrm{id}$) is an isomorphism, i.e. that $u_!$ (resp. u_*) is fully faithful.

5.8.0. Let γ be a type of algebraic structure defined by finite projective limits, let $\mathcal{U}$-γ-Ens be the category of γ-objects of $\mathcal{U}$-Ens, and let esj: $\mathcal{U}$-γ-Ens $\to$ $\mathcal{U}$-Ens be the underlying-set functor (for simplicity, we suppose that the type of structure under consideration has only one underlying set). Let C be a category. Composition with esj gives a functor denoted by

$$\mathrm{esj}^{\wedge} : \mathbf{Hom}(C^{\circ}, \mathcal{U}\text{-}\gamma\text{-}\mathrm{Ens}) \longrightarrow \widehat{C}.$$

Since projective limits in $\widehat{C}$ are computed argument by argument, the functor $\mathrm{esj}^{\wedge}$ factors as an equivalence

$$\mathbf{Hom}(C^{\circ}, \mathcal{U}\text{-}\gamma\text{-}\mathrm{Ens}) \xrightarrow{\approx} \widehat{C}_{\gamma}, \tag{5.8.1}$$

where $\widehat{C}_{\gamma}$ is the category of γ-objects of $\widehat{C}$, followed by a functor, still denoted by

$$\mathrm{esj}^{\wedge} : \widehat{C}_{\gamma} \longrightarrow \widehat{C},$$

and called the "*underlying presheaf of sets*" functor.

30 5.8.2. Suppose that the functor esj : $\mathcal{U}$-γ-Ens $\to$ $\mathcal{U}$-Ens has a left adjoint Lib : $\mathcal{U}$-Ens $\to$ $\mathcal{U}$-γ-Ens [3] (in fact, one can show that this condition is always satisfied). Composition with Lib gives a functor

$$\mathrm{Lib}^{\wedge} : \widehat{C} \longrightarrow \mathbf{Hom}(C^{\circ}, \mathcal{U}\text{-}\gamma\text{-}\mathrm{Ens})$$

and, after composition with the equivalence (5.8.1), a functor still denoted by

$$\mathrm{Lib}^{\wedge} : \widehat{C} \longrightarrow \widehat{C}_{\gamma}$$

and called the "freely generated presheaf of γ-objects" functor. The functor $\mathrm{Lib}^{\wedge}$ is left adjoint to the functor $\mathrm{esj}^{\wedge}$.

Proposition 5.8.3. *Let γ be a type of algebraic structure defined by finite projective limits such that, in the category of γ-objects of $\mathcal{U}$-*Ens*, the $\mathcal{U}$-inductive limits are representable* [3]*; let C be a category belonging to $\mathcal{U}$, let C' be a $\mathcal{U}$-category, and let $u : C \to C'$ be a functor. Denote by $\widehat{C}_{\gamma}$ (resp. $\widehat{C'}_{\gamma}$) the category of γ-objects of $\widehat{C}$ (resp. $\widehat{C'}$), and by $u^{*\gamma}$ the functor on γ-objects induced by the functor u^*. It follows from 5.1 and the equivalence 5.8.1 that the functor $u^{*\gamma}$ has a left (resp. right) adjoint. This functor is denoted by $u_{!_{\gamma}}$ (resp. $u_{*_{\gamma}}$).*

(1) *The functor $u^{*\gamma}$ commutes with inductive and projective limits. The diagram*

$$(*) \qquad \begin{tikzcd} \widehat{C'}_{\gamma} \arrow[r, "u^{*\gamma}"] \arrow[d, "\mathrm{esj}'^{\wedge}"'] & \widehat{C}_{\gamma} \arrow[d, "\mathrm{esj}^{\wedge}"] \\ \widehat{C'} \arrow[r, "u_*"] & \widehat{C} \end{tikzcd}$$

31 *is commutative* ($\mathrm{esj}'^{\wedge}$ *and* $\mathrm{esj}^{\wedge}$ *denote the "underlying-set" functors).*

[3] The functor Lib is the "free generated γ-object" functor. Examples: free group, free abelian group, free A-module, etc.

[3] One can show that this condition is always satisfied.

(2) *The functor $u_{*\gamma}$ commutes with projective limits. The diagram*

(**)
$$\begin{tikzcd}
\widehat{C}_\gamma \arrow[r, "u_{*\gamma}"] \arrow[d, "\mathrm{esj}^\wedge"'] & \widehat{C'}_\gamma \arrow[d, "\mathrm{esj}'^\wedge"'] \\
\widehat{C} \arrow[r, "u_*"] & \widehat{C'}
\end{tikzcd}$$

commutes up to isomorphism.

(3) *The functor $u_{!\gamma}$ commutes with inductive limits. Suppose that* $\mathrm{esj}^\wedge$ *(resp.* $\mathrm{esj}'^\wedge$*) has a left adjoint* $\mathrm{Lib}^\wedge$ *(resp.* $\mathrm{Lib}'^\wedge$*) (5.8.2). The diagram*

(***)
$$\begin{tikzcd}
\widehat{C} \arrow[r, "u_!"] \arrow[d, "\mathrm{Lib}^\wedge"'] & \widehat{C'} \arrow[d, "\mathrm{Lib}'^\wedge"'] \\
\widehat{C}_\gamma \arrow[r, "u_{!\gamma}"] & \widehat{C'}_\gamma
\end{tikzcd}$$

commutes up to isomorphism.

Suppose that $u_!$ commutes with finite projective limits (5.4) and (5.6). Then the diagram

(****)
$$\begin{tikzcd}
\widehat{C}_\gamma \arrow[r, "u_{!\gamma}"] \arrow[d, "\mathrm{esj}^\wedge"'] & \widehat{C'}_\gamma \arrow[d, "\mathrm{esj}'^\wedge"'] \\
\widehat{C} \arrow[r, "u_!"] & \widehat{C'}
\end{tikzcd}$$

commutes up to isomorphism, and $u_{!\gamma}$ commutes with finite projective limits.

Proof. Assertion (1) is evident. So is assertion (2), since u_* is a right adjoint and therefore (2.11) commutes with projective limits and, in particular, with finite projective limits; this gives the commutativity of diagram (**). The commutativity of diagram (***) follows from the uniqueness, up to isomorphism, of the left adjoint functor, and 32
finally the commutativity of diagram (****) follows immediately from the fact that $u_!$ commutes with finite projective limits.

Notation 5.9. By an abuse of notation, the functors $u^{*\gamma}$ and $u_{*\gamma}$ will often be denoted by u^* and u_*. In view of the commutativity of diagrams (*) and (**), this cannot cause confusion. On the other hand, when $u_!$ does not commute with finite projective limits, diagram (****) does not commute up to isomorphism, and the notations $u_!$ and $u_{!\gamma}$ must be used to avoid confusion.

5.10. Let C be a small category. For every object X of $\widehat{C}$, let C/X denote the category of arrows with target X and source an object of C. The source functor defines a functor $j_X : C/X \to C$. Let $m : Y \to X$ be a morphism of $\widehat{C}$. Composition of morphisms defines a functor $j_m : C/Y \to C/X$. The diagram

$$\begin{tikzcd} C/Y \arrow[rr, "j_m"] \arrow[dr, "j_Y"'] & & C/X \arrow[dl, "j_X"] \\ & C & \end{tikzcd}$$

is commutative. It follows from 5.1 that, for every object F of $(C/X)\hat{}$ and every object Y of C, one has

$$j_{X!}F(Y) = \coprod_{u \in \operatorname{Hom}_{\widehat{C}}(Y,X)} F(u). \tag{5.10.1}$$

Formula (5.10.1) defines $j_{X!}$ when C is a $\mathcal{U}$-category, and one checks that the functor
33 $j_{X!}$ thus defined is always right adjoint to the functor $j_X^* : \widehat{C} \to (C/X)\hat{}$ (5.0).

Proposition 5.11. *Let C be a $\mathcal{U}$-category and let X be a presheaf on C.*

1) *The functor*

$$j_{X!} : (C/X)\hat{} \longrightarrow \widehat{C}$$

factors through the category $\widehat{C}/X$:

$$(C/X)\hat{} \xrightarrow{e_X} \widehat{C}/X \longrightarrow \widehat{C}.$$

The functor e_X is an equivalence of categories.

2) *The functor $e_X \circ j_X^* : \widehat{C} \to \widehat{C}/X$ is canonically isomorphic to the functor $H \mapsto (H \times X \xrightarrow{\mathrm{pr}_2} X)$.*

Proof. 1) Let f be the terminal object of $(C/X)\hat{}$. There is a canonical isomorphism $f \xrightarrow{\sim} \varinjlim_{Y \in \operatorname{ob} C/X} Y$, and hence $j_{X!}(f) = \varinjlim_{Y \in \operatorname{ob} C/X} j_X(Y)$ (5.4). But $j_{X!}(f) \simeq X$, which gives the factorization. To show that e_X is an equivalence, it is enough to exhibit a quasi-inverse functor. To every object $H \to X$ of $\widehat{C}/X$, associate the presheaf on C/X

$$(Y \to X) \longmapsto \operatorname{Hom}_{\widehat{C}/X}((Y \to X), (H \to X)).$$

2) The functor $e_X \circ j_X^*$ is right adjoint to the forgetful functor, and therefore the functor $e_X \circ j^*$ is canonically isomorphic to the functor $H \mapsto (H \times H \xrightarrow{\mathrm{pr}_2} X)$.

5.12. Let $m : Y \to X$ be a morphism of $\widehat{C}$. By (5.11), the morphism m is canonically isomorphic to the image under e_X of an object of $(C/X)\hat{}$, which we denote by $[m]$. By restriction to the subcategories, the functor e_X defines an equivalence

$$(C/X)/[m] \xrightarrow{e_m} C/Y.$$

34 The diagram

$$\begin{tikzcd} (C/X)/[m] \arrow[rr, "e_m"] \arrow[dr, "j_{[m]}"'] & & C/Y \arrow[dl, "j_m"] \\ & C/X & \end{tikzcd}$$

commutes up to canonical isomorphism.

5.13. We record a result that will be useful in Exp. VI. Let $u : C \to C'$ be a functor between small categories. For every object H of $\widehat{C}$, denote by $u/H : C/H \to C'/u_!H$ the functor that assigns to every morphism $m : X \to H$ the morphism $u_!X \xrightarrow{u_!m} u_!H$ (we know (5.4) that we may always put $u_!X = uX$). The following diagram is commutative:

(5.13.1)
$$\begin{tikzcd} C/H \arrow[r, "u/H"] \arrow[d, "J_H"'] & C'/u_!H \arrow[d, "j_{u_!H}"] \\ C \arrow[r, "u"] & C' \end{tikzcd} .$$

We therefore have a diagram commutative up to isomorphism: and since $(u/H)_!$ takes the terminal object of $(C/H)\hat{}$ to the terminal object of $(C'/u_!H)\hat{}$, the diagram

(5.13.3)
$$\begin{tikzcd} (C/H)\hat{} \arrow[r, "(u/H)_!"] \arrow[d, "e_H"'] & (C'/u_!H)\hat{} \arrow[d, "e_{u_!H}"] \\ C\hat{}/H \arrow[r, "u_!/H"] & C'\hat{}/u_!H \end{tikzcd}$$

commutes up to isomorphism (6.1). 35

Proposition 5.14. *Let $u : C \to C'$ be a functor between small categories. Suppose that u has the following property:*

(PPF) *For every object X of C, the functor $u/X : C/X \to C'/uX$ is fully faithful.*

Then:

1) *Let f be the terminal object of $\widehat{C}$. The functor u factors as*
$$C = C/f \xrightarrow{u/f} C'/u_!f \xrightarrow{j_{u_!f}} C'.$$
The functor u/f is fully faithful.

2) *The functor $u_! : \widehat{C} \to \widehat{C'}$ factors as*
$$\widehat{C} = (C/f)\hat{} \xrightarrow{(u/f)_!} (C'/u_!f)\hat{} \xrightarrow{e_{u_!f}} \widehat{C'}/u_!f \longrightarrow \widehat{C'},$$
where the functor $(u/f)_!$ is fully faithful, the functor $e_{u_!f}$ is an equivalence, and the functor $\widehat{C'}/u_!f \to \widehat{C'}$ is the forgetful functor.

3) *In particular, the functor $u_!$ is faithful, and therefore the unit morphism $\mathrm{id} \xrightarrow{\Phi} u^*u_!$ is a monomorphism. Moreover, for every morphism $\alpha : H \to K$ of $\widehat{C}$, the diagram*
$$\begin{tikzcd} H \arrow[r, hook, "\Phi(H)"] \arrow[d, "\alpha"'] & u^*u_!H \arrow[d, "u.u^*(\alpha)"] \\ K \arrow[r, hook, "\Phi(K)"] & u^*u_!K \end{tikzcd}$$
is cartesian.

Proof. 1) The factorization comes from diagram (5.13.1). The functor u is 36
faithful. Hence u/f is faithful. Let us show that it is fully faithful. Let X and Y be two objects of C, let $\mathrm{can}_X : uX \to u_! f$ (resp. $\mathrm{can}_Y : uY \to u_! f$) be the canonical morphisms, and let

$$\begin{tikzcd} uX \arrow[rr, "m"] \arrow[dr, "\mathrm{can}_X"'] & & uY \arrow[dl, "\mathrm{can}_Y"] \\ & u^{\cdot} f & \end{tikzcd}$$

be a morphism of $C'/u_! f$. We have $u_! f = \varinjlim_{Z \in \mathrm{ob}\, C} uZ$, and therefore (3.1)

$$\mathrm{Hom}_{\widehat{C'}}(uX, u_! f) = \varinjlim_{Z \in \mathrm{ob}\, C} \mathrm{Hom}_{C'}(uX, uZ).$$

By the definition of the inductive limit, to say that $\mathrm{can}_Y\, m = \mathrm{can}_X$ is equivalent to saying that there exist

a) a finite sequence of objects X_i of C, $i \in [0, n]$, with $X_\circ = X$ and $X_n = Y$,

b) for every i, a morphism $m_i : uX \to uX_i$ ($m_\circ = \mathrm{id}_X, m_n = m$),

c) for every pair $(i, i+1)$, a morphism $f_i : X_i \to X_{i+1}$ or else $f_i : X_{i+1} \to X_i$,

such that the diagrams

$$\begin{tikzcd} & uX \arrow[dl, "m_i"'] \arrow[dr, "m_{i+1}"] & \\ uX_i \arrow[rr, "u(f_i)"] & & uX_{i+1} \end{tikzcd} \quad \text{or else} \quad \begin{tikzcd} & uX \arrow[dl, "m_i"'] \arrow[dr, "m_{i+1}"] & \\ uX_i & & uX_{i+1} \arrow[ll, "u(f_i)"'] \end{tikzcd},$$

are commutative.

It then follows immediately, by induction on i and using property (PPF), that m_i is of the form $u(p_i)$. In particular $m = u(p)$, and hence u/f is fully faithful.

37 2) The factorization is immediate. The functor $(u/f)_!$ is fully faithful by 5.6. The other assertions follow from 5.11.

3) The functor $u_!$ is the composite of the faithful forgetful functor and fully faithful functors. It is therefore faithful. It follows from the general properties of adjoint functors that the unit morphism $\mathrm{id} \to u^* u_!$ is a monomorphism. By 2), the functor $u_!$ appears as the composite of a fully faithful functor $v : \widehat{C} \to \widehat{C'}/u_! f$ and the forgetful functor. The functor u^*, right adjoint to $u_!$, is therefore the composite of the "product with $u_! f$" functor, right adjoint to the forgetful functor, and a functor w right adjoint to v. Moreover, since v is fully faithful, the unit morphism $\mathrm{id} \to wv$ is an isomorphism. The last assertion follows readily.

38 6. Faithful functors and conservative functors

Definition 6.1. *Let E be a category and let $(\varphi_i : E \to F_i)_{i \in I}$ be a family of functors*

$$\varphi_i : E \longrightarrow F_i.$$

The family of functors (φ_i) is said to be faithful if, for every pair of objects X, Y of E and every pair of arrows u, $v : X \rightrightarrows Y$, the relation $\varphi_i(u) = \varphi_i(v)$ for every $i \in I$ implies $u = v$ (in other words, if the map $\mathrm{Hom}(X, Y) \to \Pi_i \mathrm{Hom}(\varphi_i(X), \varphi_i(Y))$ *defined by (φ_i) is injective). The family of functors (φ_i) is said to be conservative if every arrow u of E for which $\varphi_i(u)$ is an isomorphism for every $i \in I$ is an isomorphism. We say that*

(φ_i) is conservative for monomorphisms (resp. for epimorphisms, resp. ...) if the preceding condition holds whenever u is a monomorphism (resp. an epimorphism, resp. ...).

6.1.1. If we introduce the unique functor

$$\varphi : E \longrightarrow F = \prod_{i \in I} F_i$$

defined by the family of functors (φ_i), it is clear that this family is faithful (resp. conservative, resp. conservative for monomorphisms, resp. ...) if and only if the functor φ is faithful (resp. conservative, resp. ...) (by which we mean that the family consisting only of φ is faithful, resp. conservative, resp. ...). Thus, without major inconvenience, we could restrict ourselves below to a family consisting of a single functor. Nevertheless, for ease of future reference, we shall state the following results for families.

The notions of 6.1 are especially useful when the φ_i satisfy suitable exactness prop- 39
erties, and in that case they tend to coincide:

Proposition 6.2. *The notation is that of 5.1.*

(i) *If the kernels of double arrows, or the cokernels of double arrows, are representable in E, and if the φ_i commute with them, then one has the implication*

$$(\varphi_i) \text{ conservative } \Longrightarrow (\varphi_i) \text{ faithful.}$$

(ii) *Suppose that fiber products (resp. amalgamated sums) are representable in E, and that the φ_i commute with them. Suppose that (φ_i) is faithful or conservative; then for every arrow u of E, u is a monomorphism (resp. an epimorphism) if and only if the same holds for $\varphi_i(u)$ for every $i \in I$.*

(iii) *Suppose that fiber products and amalgamated sums are representable in E and that the φ_i commute with them, and that every arrow in E which is a bimorphism (i.e. both a monomorphism and an epimorphism) is an isomorphism. Then one has the implication*

$$(\varphi_i) \text{ faithful } \Longrightarrow (\varphi_i) \text{ conservative.}$$

(iv) *Suppose that fiber products (resp. amalgamated sums) are representable in E, and that the φ_i commute with them. Then, if (φ_i) is conservative for monomorphisms (resp. for epimorphisms), it is in fact conservative.*

(v) *Let D be a type of diagram, let $F : d \mapsto F(d)$ be a diagram of type D in E, let X be an object of E, and let $u = (u_d)_{d \in D}$ be a family of arrows*

$$X \longrightarrow F(d) \quad (\text{resp. } F(d) \longrightarrow X).$$

Suppose that (φ_i) is conservative, that projective (resp. inductive) limits of type 40
D are representable in E, and that the φ_i commute with them. Then u makes X a projective (resp. inductive) limit of F in E if and only if, for every $i \in I$, $\varphi_i(u)$ makes $\varphi_i(X)$ a projective (resp. inductive) limit of $\varphi_i(D)$ in E_i.

Proof.

(i) For the non-resp. statement, given a double arrow $u, v : X \rightrightarrows Y$, it suffices to express the equality $u = v$ by the condition that the inclusion $\mathrm{Ker}(u, v) \to X$ is an isomorphism. Here and below, we omit repetition of the dual argument for the dual statement.

(ii) If (φ_i) is faithful, the condition that $u : X \to Y$ be a monomorphism is expressed by the equality $\mathrm{pr}_1 = \mathrm{pr}_2$ for the fiber product $X \times_Y X$. If (φ_i) is conservative, it is expressed by the condition that the diagonal morphism $\delta : X \to X \times_Y X$ be an isomorphism.

(iv) Since in the latter argument the morphism δ is a monomorphism, we see that it would in fact have been enough to suppose that (φ_i) is conservative for monomorphisms. But this then implies that (φ_i) is conservative without qualification. Indeed, if $u \in F\ell\, E$ is such that the $\varphi_i(u)$ are isomorphisms, the preceding argument shows that these are monomorphisms, and hence they are isomorphisms by the hypothesis on (φ_i).

(iii) This is a trivial consequence of (ii).

(v) This is a trivial consequence of the definitions.

We record the following consequence of (i), (ii), and (iv):

41 Corollary 6.3. *Suppose that fiber products and amalgamated sums are representable in E and that the φ_i commute with them, and that kernels of double arrows or cokernels of double arrows are representable and that the φ_i commute with them. (It suffices, for example, that finite projective limits and finite inductive limits be representable in E, and that the φ_i be exact functors.) Then the following conditions are equivalent:*

a) (φ_i) *is faithful.*
b) (φ_i) *is conservative.*
c) (φ_i) *is conservative for monomorphisms.*
c$'$) (φ_i) *is conservative for epimorphisms.*

We also record, for reference:

Proposition 6.4. *Let $\varphi : E \to F$ be a functor with a right adjoint ψ (thus* $\mathrm{Hom}(\varphi(X), Y) \simeq \mathrm{Hom}(X, \psi(Y))$*). The functor φ (resp. ψ) is faithful if and only if, for every element X of E (resp. every element Y of F), the unit morphism $X \to \psi\varphi(X)$ is a monomorphism. The functor φ (resp. ψ) is fully faithful if and only if the preceding unit morphism is an isomorphism.*

Indeed, if X', X are two objects of E, the map

$$\mathrm{Hom}(X', X) \longrightarrow \mathrm{Hom}(\varphi(X'), \varphi(X)) \tag{$*$}$$

is identified with the map obtained from

$$X \longrightarrow \psi\varphi(X) \tag{$**$}$$

42 by applying the functor $\mathrm{Hom}(X', -)$. Thus, for $(*)$ to be a monomorphism (resp. an isomorphism) for every X', with X fixed, it is necessary and sufficient that the unit morphism $(**)$ be a monomorphism (resp. an isomorphism).

Proposition 6.5. *Let $p : E' \to E$ be a functor. The following conditions are equivalent:*

(i) *p is faithful, conservative, and fibered* (SGA 1 VI 6.1).

(ii) *p is a fibered functor whose fibers are discrete categories.*

(iii) *For every $X' \in \mathrm{ob}\, E'$, the functor $E'_{/X'} \to E_{/p(X)}$ induced by p is an equivalence of categories that is surjective on objects.*

(iv) *(When E is a $\mathcal{U}$-category.) There exists an element $F \in \mathrm{ob}\, \widehat{C}$ and an equivalence of categories over E* (SGA 1 VI 4.3) *$E' \xrightarrow{\approx} E_{/F}$ (where $E_{/F}$ is the full subcategory of $\widehat{E}_{/F}$ formed by the arrows $X \to F$ whose source belongs to E).*

6.5.1. Recall that a category C is said to be *discrete* if it is a *groupoid* (i.e. every arrow in it is invertible) and if it is *rigid* (i.e. the automorphism group of every object is reduced to the trivial group); equivalently, the category is equivalent to the category C' defined by a set I (with $\mathrm{ob}C' = I$, and with the identity arrows as its only arrows). When C is already assumed to be a groupoid, saying that C is discrete amounts to saying that for two objects X, Y of C, there exists at most one arrow from X to Y, i.e. that C is isomorphic to the category defined by a preordered set.

The equivalence of conditions (i) and (ii) of 6.5 is an immediate consequence of the 43
preceding observations and of the following

Lemma 6.5.2. *Let $p : E' \to E$ be a fibered functor. Then:*

(i) *For p to be conservative, it is necessary and sufficient that its fiber categories be groupoids.*
(ii) *For p to be faithful, it is necessary and sufficient that its fiber categories be ordered categories.*

Proof of 6.5.2.

(i) Suppose p is conservative. For every arrow u in a fiber E'_X, $p(u) = \mathrm{id}_X$ is an isomorphism, so u is an isomorphism in E', hence also in E'_X (since an inverse of u in E' will clearly be an inverse in E'_X). Thus E'_X is a groupoid. Conversely, suppose that the E'_X are groupoids, and let u' be an arrow of E' such that $p(u')$ is an isomorphism; let us prove that u is an isomorphism. To this end, note that, since p is fibered, we can factor $u' : X' \to Y'$ as a composite $X' \to u^*(Y') \to Y'$, where the first arrow is an X-morphism (N.B. $X = p(X')$, $u = p(u')$), and the second is a cartesian morphism over u. The first arrow is an isomorphism since E'_X is a groupoid, and the second is as well, since a *cartesian* morphism of a fibered category is clearly an isomorphism whenever its projection is an isomorphism.
(ii) Suppose p is faithful, and let X', Y' be two objects of a fiber category E_X. Then two arrows from X' to Y' lie over the same arrow id_X of E, and hence are identical, so E_X is ordered. Conversely, suppose that the fiber categories are 44
ordered, and let us prove that p is faithful. Thus let $u', v' : X' \rightrightarrows Y'$ be arrows of E' lying over the same arrow $u : X \to Y$ of E. They then factor as $X' \rightrightarrows u^*(Y) \to Y'$, where the two arrows $X' \rightrightarrows u^*(Y')$ are arrows of E'_X with the same source and target; they are therefore equal, and hence $u' = v'$, C.Q.F.D.

Let us return to the proof of 6.5. We have proved (i) $\Leftrightarrow$ (ii). On the other hand, (ii) $\Leftrightarrow$ (iv) is fairly clear: indeed, on the one hand, the category $E_{/F}$ is fibered over E, with fiber categories the discrete categories defined by the sets $F(X)$, as follows immediately from the definitions; on the other hand, if p is as in (ii), then by the sorites of SGA 1 VI 8 the fibered category E' over E is E-equivalent to the split category over E defined by the functor $E \to (\mathrm{Cat})$ determined by the functor $F : E \to (\mathrm{Ens})$, which assigns to every $X \in \mathrm{ob}\,E$ the set of isomorphism classes of objects of E'_X. Now this split category is E-isomorphic to the category $E_{/F}$. Since (iv) $\Rightarrow$ (iii) is clear, it remains to prove (iii) $\Rightarrow$ (i). Now it is clear that for p to be faithful (resp. conservative), it is necessary and sufficient that the induced functors $E'_{/X'} \to E_{/p(X)}$ be so; *a fortiori*, it suffices that these be fully faithful; therefore it remains only to prove that (iii) implies that p is fibered. But one again sees that a functor p is fibered if and only if the induced functors $E'_{/X'} \to E_{/p(X')}$ are fibered. This is in particular the case if they are equivalences of categories that are surjective on objects.

7. Generating and cogenerating subcategories

45

Definition 7.1. *Let E be a category and C a full subcategory of E. We say that C is a subcategory of E* generating by strict epimorphisms (resp. by epimorphisms) *if, for every object X of E, the family of arrows of E with target X and source $X' \in \operatorname{ob} C$ is epimorphic (resp. strictly epimorphic) (1.3). We say that C is a subcategory of E* generating (resp. generating for monomorphisms, resp. generating for strict monomorphisms) *if, for every arrow $u : Y \to X$ (resp. every monomorphism of E, resp. every strict monomorphism of E (10.5)) such that, for every $X' \in \operatorname{ob} C$, the corresponding map* $\operatorname{Hom}(X', Y) \to \operatorname{Hom}(X', X)$ *is bijective, u is an isomorphism. Finally, we say that a family (X_i) of objects of E is generating by strict epimorphisms (resp. ...) if the full subcategory C of E generated by this family is generating by strict epimorphisms (resp. ...).*

7.1.1. Observe that, in terms of the family $(h_{X'})_{X' \in \operatorname{ob} C}$ of functors

$$h_{X'} : E \longrightarrow (\mathrm{Ens}) \ (X' \in \operatorname{ob} C)$$

represented by the $X' \in \operatorname{ob} C$, the condition that C be generating (resp. generating for monomorphisms, resp. generating for strict monomorphisms) can be expressed by saying that the family $(h_{X'})$ is *conservative* (resp. conservative for monomorphisms, resp. conservative for strict monomorphisms) (6.1). It also follows immediately from the
46 definitions that C is generating by epimorphisms if and only if the family $(h_{X'})_{X' \in \operatorname{ob} C}$ is *faithful* (6.1). We shall also give below (7.2 (i)) an analogous interpretation of the condition that C be generating by strict epimorphisms.

7.1.2. As with the notions introduced in 6.1, the notions of 7.1 are especially useful when E has suitable exactness properties, in which case the various notions introduced have a marked tendency to be all equivalent (7.3). Thus the question of which of these notions in 7.1 should be regarded as the most important scarcely arises; in the most important cases, these notions coincide, and the term "generating subcategory" may therefore be interpreted indifferently as referring to any one of the properties considered in 7.1 (for example the first, which is the strongest of all, as we shall see (7.2 (ii))).

7.1.3. Suppose that E is a $\mathscr{U}$-category, and consider the canonical functor

$$\varphi : E \longrightarrow \widehat{C} = \mathbf{Hom}(C, \mathscr{U}\text{-}\mathrm{Ens}) \tag{7.1.3.1}$$

which is the composite of the functors $E \to \widehat{E} \to \widehat{C}$, where the first functor is the canonical functor (1.3.3), and the second is the restriction functor to C. Observe that it is plainly equivalent to say that the preceding functor φ is conservative (resp. faithful), or to say that the family of functors $h_{X'} : X \to \operatorname{Hom}(X', X) = \varphi(X)(X')$, for variable $X' \in \operatorname{ob} C$, is a conservative (resp. faithful) family, that is, also (7.1.2), that C is gener-
47 ating (resp. generating by epimorphisms). Similarly, φ is conservative for monomorphisms (resp. for strict monomorphisms) if and only if the family of the $h_{X'} (X' \in \operatorname{ob} C)$ is conservative for monomorphisms (resp. for strict monomorphisms), i.e. if and only if the subcategory C of E is generating for monomorphisms (resp. for strict monomorphisms).

Proposition 7.2. *Let E be a $\mathscr{U}$-category and C a full subcategory.*

(i) *The following conditions are equivalent:*
   a) *C is a subcategory generating by strict epimorphisms.*
   b) *For every $X \in \operatorname{ob} E$, let $C_{/X}$ denote the full subcategory of $E_{/X}$ consisting of arrows $X' \to X$ with source $X' \in \operatorname{ob} C$. The natural arrow from the inclusion functor $j : C_{/X} \to E$ to the constant functor on $C_{/X}$ with value X makes X*

an inductive limit of j:

$$X \xleftarrow{\sim} \varinjlim_{C_{/X}} X'.$$

c) *The canonical functor φ of (7.1.3.1) is fully faithful.*

(ii) *Among the notions of 7.1 there are the following implications :*

1) *C generating by strict epimorphisms* (φ fully faithful)

$\Downarrow$

2) *C generating by epimorphisms* (φ faithful)

3) *C generating* (φ conservative)

$\Downarrow$

4) *C generating for monomorphisms* (φ conservative for mon.)

$\Downarrow$

5) *C generating for strict monomorphisms* (φ conservative for strict mon.).

(iii) *There are the following conditional implications:* 48

a) *If in E epimorphic families of arrows are strictly epimorphic, then* 2) $\Rightarrow$ 1). *If in E monomorphisms are strict, then* 5) $\Rightarrow$ 4).

b) *If in E the kernels of pairs of arrows (resp. fiber products) are representable, then* 3) $\Rightarrow$ 2) *(resp.* 4) $\Rightarrow$ 3)*).*

c) *If in E every family of morphisms $X_i \to X$ with common target X factors as a strictly epimorphic (resp. epimorphic) family $X_i \to Y$ followed by a monomorphism (resp. by a strict monomorphism) $Y \to X$, then* 4) $\Rightarrow$ 1) *(resp.* 5) $\Rightarrow$ 2)*).*

Let us immediately record the following.

Corollary 7.3. *All the notions considered in 6.1 (and repeated in the implication diagram in* (ii) *above) are equivalent in each of the following two cases:*

(i) *In E, the kernels of pairs of arrows and the fiber products are representable, the monomorphisms are strict, and the epimorphic families of arrows are strictly epimorphic.*

(ii) *In E, every family $(X_i \to X)_{i\in I}$ of arrows with common target X factors as an epimorphic family $(X_i \to Y)$ followed by a monomorphism $Y \to X$, every monomorphism of E is strict, and every epimorphic family of arrows of E is strict.*

Indeed, in case (i) we have 4) $\Rightarrow$ 3) and 3) $\Rightarrow$ 2) by b), and 5) $\Rightarrow$ 4) and 2) $\Rightarrow$ 1) by a). 49
In case (ii), by c), we have the implications 4) $\Rightarrow$ 1) and 5) $\Rightarrow$ 2). We therefore conclude by the implication diagram 6.2 (ii).

Proof of 7.2.

(i) The implication b) $\Rightarrow$ a) follows at once from the definitions. Let us prove a) $\Rightarrow$ b). Thus, under hypothesis a), we must prove that, for all $X, Y \in \mathrm{ob}\, E$, every system of arrows

$$u_{X'} : X' \longrightarrow Y$$

indexed by the $X' \in \operatorname{ob} C_{/X}$, such that $u_{X'} f = u_{X''}$ for every arrow $f : X'' \to X'$ in $C_{/X}$, factors through an arrow (necessarily unique by hypothesis a)) $X \to Y$. By hypothesis a), it suffices to check that, for every object Z of $E_{/X}$ and every pair of morphisms $v' : Z \to X'$, $v'' : Z \to X''$ in $E_{/X}$, with X' and X'' in $C_{/X}$, we have $u_{X'} v' = u_{X''} v''$. But, by hypothesis a), the family of arrows $w : X''' \to Z$, with $X''' \in \operatorname{ob} C$, is epimorphic, and it therefore suffices to check that, for every such w, we have $(u_{X'} v')w = (u_{X''} v'')w$, which can also be written $u_{X'}(v'w) = u_{X''}(v''w)$ and follows at once from the hypothesis made on the family of the u.

Let us now prove the equivalence of conditions b) and c. For this, observe that, for every $X \in \operatorname{ob} E$, the object $\varphi(X)$ of $\widehat{C}$ is the inductive limit in $\widehat{C}$ of the canonical functor $\widehat{C}_{/\varphi(X)} \to \widehat{C}$ (3.4); but $\widehat{C}_{/\varphi(X)}$ is canonically isomorphic to $C_{/X}$, so that in $\widehat{C}$ we have

$$\varphi(X) = \varinjlim_{C_{/X}} X',$$

50 where the object X' of C is identified with the functor $\in \widehat{C}$ that it represents. Consequently, for a second object Y of E, we have a canonical isomorphism

$$(*)\qquad \begin{aligned} \operatorname{Hom}(\varphi(X), \varphi(Y)) \simeq \varprojlim_{C_{/X}} \operatorname{Hom}(X', \varphi(Y)) &\simeq \varprojlim_{C_{/X}} \varphi(Y)(X')\text{(1.4)} \\ &\simeq \varprojlim_{C_{/C}} \operatorname{Hom}(X', Y). \end{aligned}$$

This being so, the map

$$\operatorname{Hom}(X, Y) \longrightarrow \operatorname{Hom}(\varphi(X), \varphi(Y))$$

defined by φ is none other, via the isomorphism between the extreme terms of $(*)$, than the map

$$\operatorname{Hom}(X, Y) \longrightarrow \varprojlim_{C_{/X}} \operatorname{Hom}(X', Y)$$

deduced from the inductive system of arrows $X' \to X$ indexed by $C_{/X}$ considered in 7.2 (i) b). Thus the first map is bijective for every Y (with X fixed) if and only if X is an inductive limit of the inclusion functor $j : C_{/X} \to E$, which proves the equivalence of b) and c).

(ii) The implications 1) $\Rightarrow$ 2) and 3) $\Rightarrow$ 4) $\Rightarrow$ 5) are trivial by virtue of the definitions. The implication 1) $\Rightarrow$ 3) is obtained by interpreting property 1) as the full faithfulness of φ by (i), and by observing that a fully faithful functor is conservative. But we have already observed (7.1.1) that 3) means that φ is conservative.

51 (iii) Assertion a) is a tautology. Assertion b) follows from 6.2 (i) (resp. 6.2 (iv)) applied to the functor φ, taking into account that the latter is left exact. Finally, let us prove c). For $X \in \operatorname{ob} E$, consider the family of all the morphisms $X_i \to X$, with $X_i \in \operatorname{ob} C$; by hypothesis on E it factors as a family $X_i \to Y$ that is effective epimorphic (resp. epimorphic), followed by a monomorphism (resp. by a strict monomorphism) $Y \to X$. Then hypothesis 4) (resp. 5)) implies that $Y \to X$ is an isomorphism, so the family considered is strictly epimorphic (resp. epimorphic), C.Q.F.D.

Proposition 7.4. *Let E be a category, C a full subcategory of E, and X an object of E. Suppose that C is generating in E (resp. C is generating for monomorphisms, and that the fiber product of two subobjects of X over X is representable in E). Then a strict subobject (10.11) (resp. a subobject) X' of X is determined when, for every $T \in \operatorname{ob} C$, one knows the subset of $\operatorname{Hom}(T, X)$ which is the image of $\operatorname{Hom}(T, X')$. Consequently, the cardinality of the set of strict subobjects (resp. of the set of subobjects) of X is bounded above by $\prod_{T \in \operatorname{ob} C} 2^{\operatorname{card} \operatorname{Hom}(T,X)}$, and if $X \in \operatorname{ob} C$, it is bounded above by $2^{\operatorname{card} F\ell\, C}$.*

Let us first prove the parenthetical assertion. Let X' and X'' be two subobjects of X such that, for every $T \in \operatorname{ob} C$, the images of $\operatorname{Hom}(T, X')$ and $\operatorname{Hom}(T, X'')$ in $\operatorname{Hom}(T, X)$ are equal. They are therefore also equal to the image of $\operatorname{Hom}(T, X''')$, where X''' is the fiber product of X' and X'' over X. Since C is generating for monomorphisms, it follows that the monomorphisms $X''' \to X'$ and $X''' \to X''$ are isomorphisms, so X' and X'' 52
are equal, each being equal to the subobject X''' of X.

Let us prove the nonparenthetical assertion. By definition of the notion of strict subobject, it suffices to check that knowledge of the subset $\operatorname{Hom}(T, X')$ of $\operatorname{Hom}(T, X)$ for every $T \in \operatorname{ob} C$ implies knowledge of those pairs of arrows $X \overset{u,v}{\rightrightarrows} T$ such that $ui = vi$, where $i : X' \to X$ is the canonical injection. But since C is generating, the relation $ui = vi$ is equivalent to the relation $(ui)f = (vi)f$ for every $f \in \operatorname{Hom}(T, X')$, i.e. to $ug = vg$ for every $g \in \operatorname{Hom}(T, X)$ coming from $\operatorname{Hom}(T, X')$ (i.e. of the form if, with $f \in \operatorname{Hom}(T, X')$), which proves our assertion.

Corollary 7.5. *Let E be a category, C a full generating subcategory, and X an object of C. Then a strict quotient (10.8) X' of X is determined when, for every $T \in \operatorname{ob} C$, one knows the subset of $\operatorname{Hom}(T, X)^2$ consisting of pairs (u, v) such that $qu = qv$, where $q : X \to X'$ is the canonical morphism. Thus the cardinality of the set of strict quotients of X is bounded above by $\prod_{T \in \operatorname{ob} C} 2^{\operatorname{card} \operatorname{Hom}(T,X)^2}$.*

Indeed, by definition, a strict quotient X' of X is determined when one knows, for every object Y of E, the subset of $\operatorname{Hom}(Y, X) \times \operatorname{Hom}(Y, X)$ consisting of pairs (u, v) such that $qu = qv$, where $q : X \to X'$ is the canonical morphism. But the relation $qu = qv$ is equivalent to the relation $(qu)f = (qv)f$ for every morphism $f : T \to Y$ with source T in C, since C is generating. This relation may also be written $q(uf) = q(vf)$, which proves the first assertion of 7.5. The second follows at once.

7.5.1. We can generalize 7.5 by introducing, for a family of objects $(X_i)_{i \in I}$ of E, 53
the notion of a *strict quotient in E of the family*, by which we mean a strictly epimorphic family $(p_i : X_i \to X')_{i \in I}$ of morphisms of E, — with the understanding, as for ordinary quotients, that we identify two such families $(p_i : X_i \to X')$ and $(q_i : X_i \to X'')$ if one can find an isomorphism $v : X' \to X''$ (necessarily unique) such that $f p_i = q_i$ for every $i \in I$. With this terminology, the proof of 7.5 applies *ne varietur* to give the

Variant 7.5.2. Let E, C be as in 7.5, and let $(X_i)_{i \in I}$ be a family of objects of E. Then a strict quotient X' of $(X_i)_{i \in I}$ in E (7.5.1) is determined by the following data: for every $T \in \operatorname{ob} C$, and every pair $(i, j) \in I \times I$, the subset of $\operatorname{Hom}(T, X_i) \times \operatorname{Hom}(T, X_j)$ consisting of the pairs (u, v) such that $p_i u = p_j v$, where for every $i \in I$, $p_i : X_i \to X'$ denotes the canonical morphism. Consequently, the cardinality of the set of strict quotients of $(X_i)_{i \in I}$ in E is bounded above by $\prod_{T \in \operatorname{ob} C} \prod_{i,j \in I} 2^{\operatorname{card} \operatorname{Hom}(T,X_i) \times \operatorname{card} \operatorname{Hom}(T,X_j)}$.

7.5.3. It is immediately clear that the analogous conclusion holds if one assumes only that C is generating for strict monomorphisms, provided that the products $X_i \times X_j$ are representable and that one restricts to the *effective* quotients of the family $(X_i)_{i \in I}$, i.e. to the strict quotients X' such that the fiber products $X_i \times_{X'} X_j$ are representable in E (which is no restriction if E is stable under fiber products).

54 PROPOSITION 7.6. *Let E be a category, C a full subcategory of E generating by epimorphisms (7.1), $\Pi_\circ$ = an infinite cardinal $\geq \operatorname{card} F\ell C$, $\Pi \geq \Pi_\circ$ a cardinal, and $(u_i : X_i \to X)_{i \in I}$ a universally epimorphic family (10.3) in E consisting of pullbackable morphisms (10.7) with sources $X_i \in \operatorname{ob} C$, and such that $\operatorname{card} I \leq \Pi$. Then $\operatorname{card} F\ell C_{/X} \leq \Pi^{\Pi_\circ}$.*

Let $T \in \operatorname{ob} C$; it will suffice to prove that

$$\operatorname{card} \operatorname{Hom}(T, X) \leq \Pi^{\Pi_\circ}, \tag{7.6.1}$$

for one then obtains successively

$$\operatorname{card} \operatorname{ob} C_{/X} \leq (\Pi^{\Pi_\circ}) \times \operatorname{card} \operatorname{ob} C_{/X} \leq (\Pi^{\Pi_\circ}) \times \Pi_\circ = \Pi^{\Pi_\circ},$$

and finally $\operatorname{card} F\ell C_{/X} \leq ((\Pi^{\Pi_\circ})^2) \times \Pi_\circ = \Pi^{\Pi_\circ}$, since for two objects S, T of $C_{/X}$, one has $\operatorname{card} \operatorname{Hom}_{C_{/X}}(S, T) \leq \operatorname{car} \operatorname{Hom}_C(S, T) \leq \Pi_\circ$. To prove 7.5.3, first note the

LEMMA 7.6.2. *Let J be a set such that $\operatorname{card} J = \Pi_\circ$. For every object T of C, and every morphism $f : T \to X$, there exists an epimorphic family $(v_j : S_j \to T)_{j \in J}$, with sources $S_j \in \operatorname{ob} C$, and, for every $j \in J$, an $i(j) \in I$ and a $g_j : S_j \to X_{i(j)}$ such that $u_{i(j)} g_j = f v_j$.*

Indeed, the family of $X_i \times_X T \to$ is epimorphic by hypothesis; on the other hand, since C is generating by strict epimorphisms, for every i the family of arrows $S \to X_i \times_X T$ with source $S \in \operatorname{ob} C$ is epimorphic, and hence, by transitivity, the family of arrows $v : S \to T$ with source in $\operatorname{ob} C$ which factor through one of the $X_i \times_X T$ is epimorphic. These are precisely the arrows $v : S \to T$ with source in C for which there
55 exist an $i \in I$ and a $g : S \to X_i$ such that $u_i g = f v$. Since the set of these arrows $v : S \to T$ is contained in $F\ell\, C$, and therefore has cardinality $\leq \Pi_\circ$, it can be indexed by the index set J, whence the lemma.

Now note that a morphism $f : T \to X$ is determined once the $f v_j$ are known, and these are determined once the g_j are known. Thus $\operatorname{card} \operatorname{Hom}(T, X)$ is bounded above by the cardinality of the set of families $(i(j), S_j, v_j, g_j)_{j \in J}$; since the cardinality of the set of maps $j \to I$ is $\leq \Pi^{\Pi_\circ}$, and since, for a fixed such map $j \mapsto i(j)$, the cardinality of the set of corresponding families S_j, f_j, v_j is bounded above by that of the set of maps from J to $F\ell C \times F\ell C$, which is $\leq \Pi_\circ^{\Pi_\circ}$ because $\operatorname{card}(F\ell C \times F\ell C) = \Pi_\circ^2 = \Pi_\circ$, we find that the left-hand side of (7.6.1) is bounded above by $\Pi^{\Pi_\circ} \times \Pi_\circ^{\Pi_\circ} = \Pi^{\Pi_\circ}$, which completes the proof of 7.6.

PROPOSITION 7.7. *Let E be a category in which fiber products are representable, $(X_\alpha)_{\alpha \in A}$ a generating family of objects of E, and $(\varphi_i)_{i \in I}$ a family of functors $\varphi_i : E \to E_i$ commuting with fiber products. The family (φ_i) is conservative (5.1) if and only if, for every $\alpha \in A$ and every subobject X' of X_α distinct from X_α, there exists an $i \in I$ such that $\varphi_i(X') \to \varphi_i(X_\alpha)$ is not an isomorphism. In this case, if E is a $\mathscr{U}$-category and if A is $\mathscr{U}$-small, there exists a $\mathscr{U}$-small subset J of I such that $(\varphi_j)_{j \in J}$ is already a conservative family of functors.*

The necessity of the stated conservativity condition being obvious, let us prove its sufficiency. By 6.1 (iv), it suffices to prove that (φ_i) is conservative for monomorphisms.
56 Let $u : Y' \to Y$ be a monomorphism in E which is not an isomorphism; we must prove that there exists $i \in I$ such that $\varphi_i(u)$ is not an isomorphism. By the hypothesis on (X_α), there exist an α and a morphism $X_\alpha \to Y$ which does not factor through Y', in

other words, such that the inverse image X' of Y' is a subobject of X_α distinct from X_α. By hypothesis, there exists an $i \in I$ such that $\varphi_i(X') \to \varphi(X)$ is not an isomorphism. Since $\varphi_i(X') \simeq \varphi(X_\alpha) \times_{\varphi_{i(Y)}} \varphi_i(Y')$, it indeed follows that $\varphi_i(Y') \to \varphi_i(Y)$ is not an isomorphism.

The second assertion of 7.7 follows immediately from the first, in view of 7.4.

Corollary 7.7.1. *Let E be a $\mathscr{U}$-category in which fiber products are representable, and which admits a $\mathscr{U}$-small generating family of objects. Then every generating family $(Y_i)_{i\in I}$ of E contains a $\mathscr{U}$-small generating subfamily $(Y_j)_{j\in J}$* (card $J \in \mathscr{U}$).

Indeed, it suffices to apply 7.7 to a $\mathscr{U}$-small generating family $(X_\alpha)_{\alpha\in A}$ of E and to the family of functors φ_i $(i \in I)$ represented by the Y_i.

Proposition 7.8. *Let E be a category, C a full subcategory generating by strict epimorphisms, D a category, and $\mathbf{Hom}'(E, D)$ the full subcategory of $\mathbf{Hom}(E, D)$ consisting of the functors which commute with inductive limits of type $C_{/X}$, where X is an arbitrary object of E. Then the functor $F \to F|C$ induces a fully faithful functor*

$$\mathbf{Hom}'(E, D) \longrightarrow \mathbf{Hom}(C, D).$$

Consequently, if $\mathbf{Hom}(C, D)$ is a $\mathscr{U}$-category (1.1.), *for example* (1.1.1. b)) *if C is $\mathscr{U}$-small and D is a $\mathscr{U}$-category, then $\mathbf{Hom}'(E, D)$ is also a $\mathscr{U}$-category.* 57

Let $F, G : E \to D$ be two functors in the first member, and let $u : F \to G$ be a natural transformation. If $X = \varinjlim X_i$ in E and if F and G commute with the inductive limit under consideration, then $u(X) : F(X) \to G(X)$ is identified with the inductive limit of the morphisms $u(X_i) : F(X_i) \to G(X_i)$, and is therefore determined once the $u(X_i)$ are known. This shows that the functor considered in 7.8 is faithful, in view of the implication a) ⇒ b) in 7.2 (i). Conversely, let $v : F|C \to G|C$ be a natural transformation; let us prove that it comes from a natural transformation $u : F \to G$. For every $X \in \operatorname{ob} E$, define

$$u(X) : F(X) = \varinjlim_{C_{/X}} F(X_i) \longrightarrow G(X) = \varinjlim_{C_{/X}} G(X_i)$$

as the inductive limit of the $v(X_i)$. It is immediate that one thus obtains maps functorial in X, hence a natural transformation $u : F \to G$, and finally that the morphism induced by u from $F|C$ to $G|C$ is v, which completes the proof.

7.9. **Cogenerating families and subcategories.** Let E be a category and C a full subcategory of E. We say that C is cogenerating by strict monomorphisms (resp. cogenerating by monomorphisms, resp. cogenerating, resp. cogenerating for epimorphisms, resp. cogenerating for strict epimorphisms) if the full subcategory C° of E° is generating by strict epimorphisms (resp. etc.). The terminology for families is analogous. All the results of this section concerning the notion of a generating subcategory and its variants (7.1) therefore yield corresponding results for the dual notions, which we leave to the reader to formulate for their personal satisfaction. 58

Proposition 7.10. *Let E be a category, C a full generating subcategory* (7.1), *and D a full subcategory of E. The subcategory D is cogenerating* (7.9) *if and only if, for every pair of parallel arrows $T \overset{u,v}{\rightrightarrows} X$ in E with source $T \in \operatorname{ob} C$ and $u \neq v$, there exists an arrow $w : X \to I$ with target $I \in \operatorname{ob} D$ such that $wu \neq wv$.*

By definition, to say that D is cogenerating means that for every pair of parallel arrows $Y \overset{u,v}{\rightrightarrows} X$ in E such that $u \neq v$, there exists an arrow $w : X \to I$, with $I \in \mathrm{ob}\, D$, such that $wu \neq wv$. Thus 7.10 simply means that it suffices to test this property when $Y \in \mathrm{ob}\, C$. Now, since C is generating, the hypothesis $u \neq v$ implies that there exists $f : T \to Y$ such that $uf \neq vf$, whence, by hypothesis, there exists a $w : X \to I$ with target $I \in \mathrm{ob}\, D$ such that $w(uf) \neq w(vf)$, and hence $wu = wv$, C.Q.F.D.

Corollary 7.11. *With the notation of 7.10, suppose that the objects I of D are injective objects of E, i.e. such that for every monomorphism $X \to Y$ in E, the corresponding map* $\mathrm{Hom}(Y, I) \to \mathrm{Hom}(X, I)$ *is surjective. Suppose moreover that every pair of parallel arrows $T \overset{u,v}{\rightrightarrows} X$ in E factors as an epimorphic (resp. effective epimorphic) pair of parallel arrows $Y \overset{u',v'}{\rightrightarrows} X$ followed by a monomorphism $i : X' \to X$. Then, in the criterion of 7.10 for D to be cogenerating, one may restrict to the pairs (u, v) which are epimorphic (resp. effective epimorphic).*

59 We conclude:

Corollary 7.12. *Let E be a $\mathscr{U}$-category admitting a small generating subcategory C, and such that every object of E is the source of a monomorphism into an injective object of E. Suppose moreover that, for every $T \in \mathrm{ob}\, C$, the coproduct $T \amalg T$ in E is representable, and that every morphism with source $T \amalg T$ factors as an effective epimorphism followed by a monomorphism. Then E admits a small full cogenerating subcategory D. More precisely, one may take D such that* $\mathrm{card}\,\mathrm{ob}\, D \leq \prod_{T,T' \in \mathrm{ob}\, C} 2^{\mathrm{card}(\mathrm{Hom}(T', T \amalg T))^2}$.

Indeed, by 7.11, it suffices, for every $T \in \mathrm{ob}\, C$ and every effective quotient X of $T \amalg T$, to choose an embedding of X into an injective object I of E, and to take for D the full subcategory of E spanned by these I. The conclusion then follows from 7.5.

Examples 7.13. To construct small cogenerating subcategories in terms of small generating subcategories, we are thus led to seek conditions under which a $\mathscr{U}$-category E has “enough injectives”, i.e. every object embeds into an injective object (by a monomorphism). It is well known [Tohoku] that this condition holds in a abelian $\mathscr{U}$-category with exact (small) filtered inductive limits (axiom $AB\,5$ of *loc. cit.*) which admits a small generating family. Moreover, the construction of *loc. cit.* is not essentially tied to abelian categories, and also works in the category of sheaves of $\mathscr{U}$-sets on a topological space $X \in \mathscr{U}$. We shall not state here the exactness properties which make the construction
60 in question work, and shall merely point out that, in the special case of the category of sheaves of sets on X, one immediately reduces to the case of *loc. cit.* as follows. Note that if $\mathscr{O}_X$ is a ring on X, then every injective object I of the category of $\mathscr{O}_X$-modules is also injective as an object of the category of sheaves of sets. Indeed, if F is a sheaf of sets, and $\mathscr{O}_X[F]$ the “free $\mathscr{O}_X$-module generated by F”, there is by definition a homomorphism of sheaves of sets

$$(*) \qquad F \longrightarrow \mathscr{O}_X[F]$$

giving rise to an isomorphism, functorial in F,

$$\mathrm{Hom}_{\mathscr{O}_X}(\mathscr{O}_X[F], I) \xrightarrow{\sim} \mathrm{Hom}(F, I).$$

Since the functor $F \mapsto \mathscr{O}_X[F]$ manifestly takes monomorphisms to monomorphisms, it follows immediately that if I is an injective $\mathscr{O}_X$-module, it is also an injective sheaf of sets. It follows that if the morphism $(*)$ is a monomorphism (as is the case if one takes for $\mathscr{O}_X$ a constant ring with value a ring $A \neq 0$), then an embedding of $\mathscr{O}_X[F^*]$ into an injective $\mathscr{O}_X$-module I gives an embedding of F into the injective sheaf of sets underlying I.

The same argument applies, without change, to the category of sheaves of $\mathscr{U}$-sets on a $\mathscr{U}$-site, which will be introduced in the following exposé.

The chief interest of the existence of a small cogenerating subcategory lies in the representability criterion 8.12.7 below.

8. Ind-objects and pro-objects

8.1. Cofinal functors and cofinal subcategories. 61

Definition 8.1.1. *Let*

$$\varphi : I \longrightarrow I' \tag{8.1.1.1}$$

be a functor. The functor φ is said to be cofinal if, for every category C and every functor $u : I' \to C$, regarding $\varinjlim u$ and $\varinjlim u \circ \varphi$ as objects of $\mathbf{Hom}(G(\mathscr{V}\text{-Ens}))^\circ$ *(2.1, 2.3.1) (where $\mathscr{V}$ is a universe such that $I, I' \in \mathscr{V}$ and C is a $\mathscr{V}$-category), the canonical morphism*

$$\varinjlim u \circ \varphi \longrightarrow \varinjlim u \tag{8.1.1.2}$$

is an isomorphism. When φ is the inclusion functor of a subcategory of I', the subcategory I is said to be cofinal if φ is cofinal.

8.1.2. Notice that for given φ and u, the bijectivity of (8.1.1.2) does not depend on the choice of the universe $\mathscr{V}$.

Returning to the definition of the terms occurring in (8.1.1.2) (2.1, 2.3.1), we see that to say that φ is cofinal also means that, for every universe $\mathscr{V}$ such that I and I' are $\mathscr{V}$-small, and every functor

$$v : I'^\circ \longrightarrow (\mathscr{V}\text{-Ens}),$$

the canonical homomorphism

$$\varprojlim v \longrightarrow \varprojlim v \circ \varphi \tag{8.1.2.1}$$

is an isomorphism, i.e. a bijection. To see that this condition is indeed necessary, observe, on putting $v = v^\circ$, that it also says that the condition of Definition 8.1.1 is satisfied when $C = (\mathscr{V}\text{-Ens})^\circ$ (which implies that the inductive limits considered in (8.1.1.2) are 62
representable in C).

. It follows immediately from the definition that *the composite of two cofinal functors is cofinal.*

Proposition 8.1.3. *Let $\varphi : I \to I'$ be a functor.*

a) *For φ to be cofinal, it is necessary that φ satisfy the condition:*
   F 1) *For every object i' of I', there exists an object i of I such that $\varphi(i)$ is an upper bound of i' (i.e. such that* $\mathrm{Hom}(i', \varphi(i)) \neq \phi$*).*

b) *Suppose that I is filtered. For φ to be cofinal, it is necessary and sufficient that φ satisfy condition* (F 1) *and the following condition:*
   F 2) *For every object i of I and every double arrow $i' \overset{f',g'}{\rightrightarrows} \varphi(i)$ in I' with target $\varphi(i)$, there exists an arrow $h : i \to j$ in I such that $\varphi(h)f' = \varphi(h)g'$.*

   Moreover, if φ is cofinal, I' is filtered.

c) *Suppose that I' is filtered and φ is fully faithful. For φ to be cofinal, it is necessary and sufficient that it satisfy condition* F 1) *of* a)*. This implies that I is filtered.*

Proof. For necessity in a) and b), use Definition 8.1.1 only in the case where u is the canonical inclusion functor (1.3.3)

$$u : I' \hookrightarrow \widehat{I'}$$

(having chosen a universe $\mathscr{U}$ such that I and I' are $\mathscr{U}$-small). We may then interpret
63 (8.1.1.2) as an arrow of $\widehat{I'}$ (3.1) whose target is the *terminal* functor on I' (3.4). It is immediately clear that condition F 1) says that this arrow is an epimorphism of $\widehat{I'}$, i.e. that it is surjective at each argument (taking into account that inductive limits in $\widehat{I'}$ are computed "argument by argument"). This proves a). Now suppose that I is filtered and φ is cofinal. Then I' is filtered: indeed, the condition that two objects of I' be bounded above by a third follows at once from the same condition on I and from F 1), and it remains only to prove condition PS 2) of 2.7, i.e. that for every double arrow

$$f', g' : i' \rightrightarrows j'$$

in I', there exists an arrow $h' : j' \to k'$ of I' such that $h'f' = h'g'$. By F 1), we may suppose that k' is of the form $\varphi(i)$. But since φ is cofinal, for every object i' of I' the set $\varinjlim_i \mathrm{Hom}(i', \varphi(i))$ consists of a single element. The standard description of *filtered* inductive limits in (Ens) (2.8.1) therefore implies that there exists an arrow $h : i \to j$ in I such that $\varphi(h)f' = \varphi(h)g'$. This completes the proof that I' is filtered and at the same time proves condition F 2).

To prove that the conditions in b) are sufficient for φ to be cofinal, use the form 8.1.2 of the definition. One sees at once that condition F 1) implies that (8.1.2.1) is a monomorphism, i.e. is injective at each argument, while condition F 2) (together with F 1)) ensures that it is bijective (taking into account the argument-by-argument computation of projective limits in $\widehat{I'^{\circ}}$). This proves b).

64 Finally, if I' is filtered and φ is fully faithful, it is immediate that condition F 1) implies that I is filtered and implies condition F 2). Thus c) follows from a) and b).

8.1.4. When I' is a filtered category and I is a full subcategory of I', we see from 8.1.3 c) that the condition that I be cofinal in I' depends only on the subset Ob I of the preordered set $OB\, I'$ (for the preorder relation "$x \leq y$" $\Leftrightarrow$ "$\mathrm{Hom}(x, y) \neq \varnothing$"). If J' is a preordered set and J a subset of J', we shall sometimes say that J is a *cofinal subset of* J' when every element of J' is bounded above by an element of J. When J' is filtered, this therefore means that the inclusion functor $\mathscr{J} \hookrightarrow \mathscr{J}'$ for the associated categories is cofinal.

8.1.5. In what follows, we shall use the notion of cofinal functor only when the categories I and I' are filtered. Classically, one even restricted oneself to categories associated with preordered sets (i.e. categories in which there is at most one arrow with given source and target). It turns out, however, that this restriction is inconvenient in applications, since the "natural" filtered categories arising in many applications are *not* ordered categories. The following result, due to P. Deligne, shows nevertheless that there is no essential difference between the two points of view:

65 Proposition 8.1.6. *Let I be a small filtered category. Then there exist a small ordered set E and a cofinal functor $\varphi : \mathscr{E} \to I$, where $\mathscr{E}$ denotes the category associated with E.*

First suppose that the preordered set Ob I has no greatest element. A subdiagram of I is a pair $D = (\mathscr{O}, F)$ consisting of a subset F of $F\ell(I)$ and a subset $\mathscr{O}$ of Ob I, such that for every $f \in F$, the source and target of f belong to $\mathscr{O}$. An element e of $\mathscr{O}$ is called a *terminal object* of the subdiagram D if, for every $x \in D'$, the set $\mathrm{Hom}(x, e) \cap F$ of arrows of D with source x and target y has exactly one element f_x, if $f_x = f_y \circ g$ for every arrow

$g : x \to y$ of D, and if $f_e = \mathrm{id}_e$. Let E be the set of *finite* subdiagrams D of I having a *unique* terminal element $\varphi(D)$, and order E by inclusion. If D, D' are two elements of E and $D \subset D'$, there is a unique arrow $\varphi(D', D)$ of D' with source $\varphi(D)$ and target $\varphi(D')$. If $D \subset D' \subset D''$ are inclusions in E, one clearly has $\varphi(D'', D) = \varphi(D'', D')\varphi(D', D)$; one also has $\varphi(D, D) = \mathrm{id}_{\varphi(D)}$. Thus we obtain a functor $\varphi : \mathscr{E} \to I$, and it remains to prove that E is filtered and φ is cofinal. In view of 8.1.3 b), three verifications are required:

1) Condition F 1) of 8.1.3: for every $i \in I$, take for D the subdiagram of I consisting only of the object i and its identity arrow; then $i \leq \varphi(D) = i$. This condition F 1) also shows that $E \neq \varnothing$.
2) Condition F 2): For every $i \in \mathrm{Ob}\, I$, every $D \in E$, and every double arrow $i \rightrightarrows \varphi(D)$, find a diagram $D' \in E$ containing D such that $\varphi(D', D) : \varphi(D) \to \varphi(D')$ *equalizes* the double arrow. Since I is filtered, there is an object j of I and an 66
arrow $f : \varphi(D) \to j$ that equalizes the double arrow. Since I has no greatest element, we may suppose that j is a *strict* upper bound of $\varphi(D)$, which implies that it is distinct from every other object of D. Let D' be the subdiagram of I whose set of objects is $\mathscr{O} \cup \{j\}$ and whose set of arrows consists of the arrows of D, together with the composites $x \xrightarrow{f_x} \varphi(D) \xrightarrow{f} j$, where x is an object of D and f_x is the unique arrow of D with source x and target $\varphi(D)$, and id_j. It is then clear that D' is a finite subdiagram of I, that it has j as its unique terminal object, hence that $D' \in E$, and that D' satisfies the required condition.
3° Two subdiagrams $D, D' \subset E$ are contained in the same $D'' \in E$. Indeed, one may find a strict upper bound j of $\varphi(D)$ and $\varphi(D')$,

$$\begin{tikzcd}
\varphi(D) \arrow[dr, "f"] & \\
& j \\
\varphi(D') \arrow[ur, "f'"'] &
\end{tikzcd},$$

and take for D'' the subdiagram whose set of objects is the union of the sets of objects of D and D' with $\{j\}$, and whose set of arrows is the union of the sets of arrows of D and D', the set of composites $f \circ f_x$ (x an object of D) and $f' \circ f'_{x'}$ (x' an object of D'), and $\{\mathrm{id}_j\}$. This indeed gives a finite subdiagram of E. Let us show that, after replacing j, f, f' by j', gf, gf' for a suitable $g : j \to j'$, we may arrange that j is a terminal object of D'. Equivalently, for every x that is an object of both D and D', one must have $ff_x = f'f'_{x'}$. This amounts to a *finite* set of double arrows to be equalized by a $g : j \to j'$, which is possible 67
since I is filtered.

This completes the proof in the case considered. The general case is reduced to it by introducing the filtered category $\mathbf{N}$ associated with the ordered set $\mathbf{N}$ of natural integers, and observing that the category $\mathbf{N} \times I$ is filtered, has no greatest element, and that the projection $\mathbf{N} \times I \to I$ is a cofinal functor.

Corollary 8.1.7. *Let I be a $\mathscr{U}$-category. There exist a small filtered ordered set E and a cofinal functor $\mathscr{E} \to I$ if and only if I is filtered and* Ob I *has a small cofinal subset I' (8.1.4).*

Necessity follows from 8.1.3 a), and sufficiency follows from 8.1.6 applied to the full subcategory of I defined by I'.

Definition 8.1.8. *Let I be a filtered category. We say that I is essentially small if I is a $\mathscr{U}$-category and satisfies the equivalent conditions of 8.1.7.*

8.2. Ind-objects and ind-representable functors.

8.2.1. In what follows, we fix a $\mathscr{U}$-category C, which we always regard as embedded in the category $\widehat{C}$ of $\mathscr{U}$-presheaves by means of the canonical functor (1.3.1)

$$h : C \hookrightarrow \widehat{C}. \tag{8.2.1.1}$$

An *ind-object of C* (or an *inductive system* of C) is any functor

$$\varphi : I \longrightarrow C, \tag{8.2.1.2}$$

68 where I is a *filtered* category (2.7), called the *index category* of the ind-object. (Apart from this, I is arbitrary; in particular, we do not require I to be a $\mathscr{U}$-category.) It will often be convenient to denote an ind-object φ by the indexed notation $(X_i)_{i\in \mathrm{ob}\, I}$, or even (by another abuse of notation) $(X_i)_{i\in I}$, or $(X_i)_i$, or (X_i), where $X_i = \varphi(i)$ for $i \in \mathrm{ob}\, I$. This notation is no more and no less abusive than the notation classically used for inductive systems indexed by filtered ordered sets (where the transition morphisms are likewise not mentioned). The X_i are called the *component objects* of the ind-object $\mathscr{X}$. Note that, in the general case considered here, the "*transition morphisms*" $\varphi(f)$ of the inductive system are indexed by the set $F\ell I$ of arrows of I, which in general is not identified with a subset of $\mathrm{Ob}\, I \times \mathrm{Ob}\, I$; in other words, for given i and j, with j an upper bound of i, there may exist more than one transition morphism from X_i to X_j.

8.2.2. The most useful ind-objects $\mathscr{X} = (X_i)_{i\in I}$ of C are those for which the index category I is essentially small (8.1.8); such an ind-object is called *essentially small.* If $(X_i)_{i\in I}$ is such an object, then, using the fact that small inductive limits are representable in $\widehat{C}$ (3.1), we may consider

$$L(\mathscr{X}) \overset{\text{def}}{=} \varinjlim h \cdot \mathscr{X} = \varinjlim_i X_i \in \mathrm{ob}\, \widehat{C}, \tag{8.2.2.1}$$

where the last limit is taken *in* $\widehat{C}$. Thus $L(\mathscr{X})$ is the presheaf

$$L(\mathscr{X}) : Y \longmapsto \varinjlim_i \mathrm{Hom}(Y, X_i) \tag{8.2.2.2}$$

69 on C. This presheaf is called the *presheaf ind-represented by the ind-object $\mathscr{X} = (X_i)_{i\in I}$ of C.* A presheaf F on C is called an *ind-representable presheaf* if it is isomorphic to a presheaf $L(\mathscr{X})$ ind-represented by an essentially small ind-object. It follows moreover from 8.1.6 that, in this definition, we may suppose that I is a small category, and even that I is associated with an ordered set $\in \mathscr{U}$.

8.2.3. Consider an ind-object $\mathscr{X} = (X_i)_{i\in I}$ given by a functor $\varphi : I \to C$, and let

$$u : I' \longrightarrow I$$

be a functor, where I' is a second filtered category. Then the functor $\varphi\circ u$ is an ind-object $\mathscr{X}'$ of C with index category I'; in indexed notation it is $(X_{u(i')})_{i'\in I'}$. It is called the ind-object of C *deduced from $(X_i)_{i\in I}$ by a change of index categories* using the functor u. If I and I' (i.e. $\mathscr{X}$ and $\mathscr{X}'$) are essentially small, one obtains a canonical morphism

$$L(\mathscr{X}') = \varinjlim_{i'} X_{u(i')} \longrightarrow L(\mathscr{X}) = \varinjlim_i X_i \tag{8.2.3.1}$$

between the presheaves ind-represented by $\mathscr{X}'$ and $\mathscr{X}$, respectively. When u is a cofinal functor (8.1.1), $\mathscr{X}$ is essentially small if and only if $\mathscr{X}'$ is (8.1.7), and the preceding homomorphism (8.2.3.1) is an isomorphism

$$L(\mathscr{X}') \simeq L(\mathscr{X}). \tag{8.2.3.2}$$

8.2.4. Let 70

$$\mathscr{X} = (X_i)_{i\in I}, \mathscr{Y} = (Y_j)_{j\in I} \tag{8.2.4.1}$$

be two essentially small ind-objects of $\boldsymbol{C}$, indexed by the (filtered and essentially small) categories $\boldsymbol{I}$ and $\boldsymbol{J}$. A *morphism* from the ind-object $\mathscr{X}$ to the ind-object $\mathscr{Y}$ is any morphism

$$L(\mathscr{X}) \longrightarrow L(\mathscr{Y}) \tag{8.2.4.2}$$

between the presheaves that they ind-represent. We put

$$\mathrm{Hom}_{\mathrm{ind-ob}}(\mathscr{X}, \mathscr{Y}) = \mathrm{Hom}_{\widehat{C}}(L(\mathscr{X}), L(\mathscr{Y})). \tag{8.2.4.3}$$

(The subscript ind-ob on Hom is omitted when there is no risk of confusion.) Composition of morphisms of ind-objects of $\mathscr{C}$ is defined as composition of the morphisms of the presheaves that they ind-represent. If $\mathscr{V} \supset \mathscr{U}$ is a universe containing $\mathscr{U}$, and if we restrict to the essentially small ind-objects that belong to $\mathscr{V}$, these form the set of objects of a category whose set of arrows consists of triples $(\mathscr{X}, \mathscr{Y}, u)$, where $\mathscr{X}$ and $\mathscr{Y}$ are essentially small ind-objects $\in \mathscr{V}$ and $u : \mathscr{X} \to \mathscr{Y}$ is a morphism of ind-objects, i.e. a morphism $L(\mathscr{X}) \to L(\mathscr{Y})$. The category thus obtained is denoted by

$$\mathrm{Ind}_{\mathscr{V}}(\boldsymbol{C}, \mathscr{U}). \tag{8.2.4.4}$$

When $\mathscr{V} = \mathscr{U}$, it is denoted simply by

$$\mathrm{Ind}_{\mathscr{U}}(\boldsymbol{C}) \text{ or } \mathrm{Ind}(\boldsymbol{C}), \tag{8.2.4.5}$$

and is called the *category of ind-objects of* $\boldsymbol{C}$ relative to $\mathscr{U}$, with the mention of $\mathscr{U}$ omitted when no confusion is likely.

Clearly, if $\mathscr{V} \subset \mathscr{V}'$ are two universes containing $\mathscr{U}$, then $\mathrm{Ind}_{\mathscr{V}}(\boldsymbol{C}, \mathscr{U})$ is a full sub- 71
category of $\mathrm{Ind}_{\mathscr{V}'}(, \mathscr{U})$; it follows from 8.2.3 that the inclusion functor is an *equivalence of categories*:

$$\mathrm{Ind}_{\mathscr{V}}(\boldsymbol{C}, \mathscr{U}) \xrightarrow{\approx} \mathrm{Ind}_{\mathscr{V}'}(\boldsymbol{C}, \mathscr{U}). \tag{8.2.4.6}$$

This shows in particular that *every essentially small ind-object of* $\boldsymbol{C}$ *defines an element of* $\mathrm{Ind}(\boldsymbol{C})$*, up to unique isomorphism.* This observation justifies the fairly common abuse of language of identifying every essentially small ind-object of $\boldsymbol{C}$ with an object of $\mathrm{Ind}(\boldsymbol{C})$.

It follows at once from the definitions that, for every universe $\mathscr{V}$, there is a canonical functor

$$L : \mathrm{Ind}_{\mathscr{V}}(\boldsymbol{C}, \mathscr{U}) \longrightarrow \widehat{\boldsymbol{C}} \tag{8.2.4.7}$$

which is fully faithful. By (8.2.4.6), these functors are clearly determined up to unique isomorphism once one of them is known, and in particular once the canonical functor

$$L : \mathrm{Ind}(\boldsymbol{C}) \longrightarrow \widehat{\boldsymbol{C}}. \tag{8.2.4.8}$$

is known. Since this functor is fully faithful, it is often used to identify an object of the first member with the functor that it ind-represents, or even to identify $\mathrm{Ind}(\boldsymbol{C})$ with its essential image in $\widehat{\boldsymbol{C}}$, consisting of the ind-representable presheaves. Note, however, that this identification has distinctly more disadvantages than the analogous identification of C with a full subcategory of $\widehat{C}$ by the functor h (8.2.1.1), since, unlike the latter, the functor (8.2.4.8) is not in general injective on objects.

72 8.2.5. Let us express Definition (8.2.4.3) explicitly in terms of the indexed expressions (8.2.4.1) of the ind-objects under consideration. Taking Definition (8.2.2.1) into account, one obtains

$$\mathrm{Hom}(\mathscr{X},\mathscr{Y}) \simeq \varprojlim_i \mathrm{Hom}(X_i,\mathscr{Y}) \overset{\mathrm{dfn}}{=} \varprojlim_i \mathrm{Hom}(X_i, L(\mathscr{Y})) = \varprojlim_i \varinjlim_j \mathrm{Hom}_C(X_i, Y_j),$$

where the last equality follows from (8.2.2.2) applied to $\mathscr{Y}$. Thus one obtains a canonical bijection

$$\mathrm{Hom}((X_i)_{i\in I},(Y_j)_{j\in J}) \simeq \varprojlim_{i\in I} \varinjlim_{j\in J} \mathrm{Hom}_C(X_i, Y_j). \tag{8.2.5.1}$$

We leave to the reader the task of expressing composition of morphisms of ind-objects explicitly in terms of this formula. Notice that it follows at once from this formula that the set $\mathrm{Hom}(\mathscr{X},\mathscr{Y})$ of homomorphisms of ind-objects is *$\mathscr{U}$-small*, i.e. that the categories (8.2.4.4) are $\mathscr{U}$-categories. Equivalently (by (8.2.4.6)), the *category* $\mathrm{Ind}(C)$ *is a* $\mathscr{U}$*-category*, i.e. the second member of (8.2.5.1) is small when $I, J \in \mathscr{U}$. This follows at once from the fact that C is a $\mathscr{U}$-category, so the sets $\mathrm{Hom}_C(X_i, Y_j)$ are small.

8.2.6. Let I be an essentially small filtered category. It follows from (8.2.5.1), or from the functoriality of the inductive limit of a functor $I \to \widehat{C}$ with respect to that functor, that there is a canonical functor

$$\mathbf{Hom}(I, C) \longrightarrow \mathrm{Ind}(C). \tag{8.2.6.1}$$

Note that this functor is in general *not* fully faithful, nor even faithful.

73 REMARK 8.2.7. The definitions of 8.2.4 make clear the notion of *isomorphism* of two (essentially small) ind-objects, which also means the isomorphism of the presheaves that they ind-represent. Note that this is a very weak notion of isomorphism when compared with the notion of isomorphism in categories of the form $\mathbf{Hom}(I, C)$. Thus, many fairly natural conditions that may be imposed on an ind-object (such as having its components in a given strictly full subcategory, or being strict (8.12 below)) *are not* invariant under isomorphism. Hence, if one wishes to work with notions invariant under isomorphism of ind-objects, the notions in question must be "saturated" by passing to the strictly full subcategory of $\mathrm{Ind}(C)$ generated by the subcategory of $\mathrm{Ind}(C)$ formed by the objects satisfying the condition under consideration. See, for example, the notion of essentially constant ind-object introduced below (8.4).

EXERCISE 8.2.8. Let $\mathscr{U} \subset \mathscr{V}$ be two universes and let C be a $\mathscr{U}$-category belonging to $\mathscr{V}$. Denote by $\mathrm{SysInd}(C)$ the following category:

a) The objects of $\mathrm{SysInd}(C)$ are the functors $\varphi : I \to C$ where I is an essentially $\mathscr{U}$-small filtered category belonging to $\mathscr{V}$.
b) Let (I,φ) and (J,ψ) be two objects of $\mathrm{SysInd}(C)$. A morphism in $\mathrm{SysInd}(C)$ from (I,φ) to (J,ψ) is a pair (m,u), where $m : I \to J$ is a functor and $u : \varphi \to \psi \circ m$ is a morphism of functors. Composition of morphisms in $\mathrm{SysInd}(C)$ is defined in the evident way.

74 Let S be the set of morphisms $(m,u) : (I,\varphi) \to (J,\Psi)$ of $\mathrm{SysInd}(C)$ such that m is cofinal and u is an isomorphism. Let (I,φ) be an object of $\mathrm{SysInd}(C)$. The category $F\ell(I)$ of arrows of I maps to I by two natural functors: source and target. These two functors are moreover linked by the canonical morphism of functors v : source $\to$ target. Hence

there are two morphisms in SysInd(C) from $(F\ell(I), \varphi \circ \text{ source})$ to (I, φ): $p_1(I,\varphi) = (\text{source, id})$ and $p_2(I,\varphi) = (\text{target}, \varphi^* v : \varphi\circ\text{source} \to \varphi\circ \text{ target})$. Let $p : \mathrm{SysInd}(C) \to \mathrm{Ind}(C)$ be the evident functor. Let B be a category. Show that the functor $F \mapsto F \circ p$

$$\mathbf{Hom}(\mathrm{Ind}(C), B) \longrightarrow \mathbf{Hom}(\mathrm{SysInd}(C), B)$$

is fully faithful, and that a functor $G : \mathrm{SysInd}(C) \to B$ belongs to its essential image if and only if it has the following two properties:

1) For every $s \in S$, $F(s)$ is an isomorphism of B.
2) For every object (I, φ) of SysInd(C), $F(p_1(I,\varphi)) = F(p_2(I,\varphi))$.

8.3. Characterization of ind-representable functors.

Proposition 8.3.1. *Let F be an ind-representable presheaf on C. Then F is left exact, i.e. (2.3.2) for every finite category J and every functor $\varphi : J \to C$ such that $\varinjlim \varphi$ is representable (i.e. $\varprojlim \varphi^\circ$ is representable), the canonical map*

$$F(\varinjlim \varphi) \longrightarrow \varprojlim F \circ \varphi$$

is bijective.

Indeed, since representable presheaves are clearly left exact, it follows at once from 75
2.8 that the same is true of every filtered inductive limit of such functors, C.Q.F.D.

8.3.2. Given a presheaf F on C, we shall often have occasion to work with the category

$$C_{/F} \hookrightarrow \widehat{C}_{/F}, \tag{8.3.2.1}$$

which denotes the full subcategory of the latter category formed by arrows $X \to F$ *whose source lies in C*. It follows from 1.4 that the objects of this category may be identified with pairs (X, u), where X is an object of C and $u \in F(X)$. A morphism from (X,u) to (X', u') is then interpreted as an arrow $f : X \to X'$ in C such that $F(f)(u') = u$. The functor from $\widehat{C}_{/F}$ to $\widehat{C}$ that "forgets the target" induces a functor (called *canonical*).

$$C_{/F} \longrightarrow C, \tag{8.3.2.2}$$

already considered in 3.4, where it is proved that the inductive limit of this functor exists in $\widehat{C}$ (without any smallness condition on $C_{/F}$!) and is canonically isomorphic to F.

. Note that if finite inductive limits are representable in C, and if F is left exact (i.e. takes them to finite projective limits in (Ens)), then the same is true in $C_{/F}$, and a fortiori (2.7.1) $C_{/F}$ is filtered. If, more generally, sums of two objects and cokernels of pairs of parallel arrows are representable in C, and if F takes them to products resp. kernels, then $C_{/F}$ is closed under the same types of finite inductive limits, hence it is filtered if 76
and only if it is nonempty, i.e. if and only if the functor F is not the constant functor with value $\varnothing$.

Theorem 8.3.3. *Let C be a $\mathcal{U}$-category, and let $F \in \mathrm{ob}\,\widehat{C}$, $F : C^\circ \to (\mathcal{U}\text{-}\mathrm{Ens})$ be a $\mathcal{U}$-presheaf on C. The following conditions are equivalent:*

(i) *F is ind-representable (8.2.2).*
(ii) *The category $C_{/F}$ (8.3.2) is filtered and essentially small.*
(iii) *(if finite inductive limits are representable in C.) The functor F is left exact, and $\mathrm{ob}\,C_{/F}$ has a small cofinal subset (8.1.4).*

(iii bis) *(If sums of two objects and cokernels of pairs of parallel arrows are representable in C.) The functor F takes sums of two objects of C to products, and cokernels of pairs of parallel arrows of C to kernels; the category $C_{/F}$ is nonempty, i.e. the functor F is not the constant functor with value $\varnothing$; finally, there exists a small family of objects of $C_{/F}$ such that every object X of $C_{/F}$ is dominated by an object X_i, i.e. admits an F-morphism $X_i \to X$.*

(iv) *(if the category C is equivalent to a small category.) The category $C_{/F}$ is filtered.*

(v) *(If the category C is equivalent to a small category, and if filtered inductive limits are representable in it.) The functor F is left exact.*

77 *Proof.* (i) $\Rightarrow$ (ii). Suppose that F is ind-represented by $(X_i)_{i\in I}$, with I small. Let us prove that $C_{/F}$ is filtered. Let X, X' be two objects of $C_{/F}$, i.e. objects of C equipped with morphisms $X \to F$, $X' \to F$; let us prove that they have a common upper bound in $C_{/F}$. The morphisms $X \to F$, $X' \to F$ come from morphisms $X \to X_i$, $X' \to X'_i$, and, after replacing i, i' by a common upper bound in Ob I, we may assume that $i = i'$, and take as a common upper bound of X, X' the object X_i equipped with the canonical morphism $X_i \to F$. Now let $X \overset{f,g}{\rightrightarrows} X' \xrightarrow{h} F$ be a pair of parallel arrows in $C_{/F}$; let us prove that it is equalized by an arrow $X' \to X''$ in $C_{/F}$. Now $h : X' \to F$ is given by a morphism $h_i : X' \to X_i$, and the condition $hf = hg$ means that there exists $\alpha : i \to j$ in I such that $\varphi(\alpha)(h_i f) = \varphi(\alpha)(h_i g)$, i.e. after replacing h_i by $\varphi(\alpha)h_i$, this morphism equalizes f and g. This proves that $C_{/F}$ is filtered. It is then immediate that it is essentially small, since the objects $(X_i \to F)_{\in \mathrm{Ob}\, I}$ form a small family in Ob $C_{/F}$ that is cofinal.

(ii) $\Rightarrow$ (i). Put $I = C_{/F}$, and consider the canonical functor (8.3.2.1)

$$\varphi : I = C_{/F} \longrightarrow F.$$

We know that the presheaf ind-represented by this ind-object of C is F (3.4), so F is ind-representable by definition (8.2.2).

(ii) $\Leftrightarrow$ (iii). Indeed, (ii) $\Rightarrow$ (iii) because an ind-representable functor is left exact (8.3.1),
78 and (iii) $\Rightarrow$ (ii) because we noted (8.3.1) that if F is left exact, then $C_{/F}$ is filtered, and we apply Definition 8.1.8 to conclude that this category is essentially small.

(ii) $\Leftrightarrow$ (iii bis) is proved in the same way as (ii) $\Leftrightarrow$ (iii).

The equivalences (ii) $\Leftrightarrow$ (iv) and (iii) $\Leftrightarrow$ (v) are immediate, since if C is equivalent to a small category, then $C_{/F}$ is clearly likewise equivalent to a small category and *a fortiori* it is automatically essentially small as soon as it is filtered. This completes the proof of 8.3.3.

Remark 8.3.4. Let $F : C' \to C''$ be a functor between $\mathscr{U}$-categories. It appears that the notion of left exactness (2.3.2) is of little practical interest unless finite projective limits are representable in C'. In the case where $C'' = (\mathscr{U}\text{-Ens})$, and where C' is $\mathscr{U}$-small, writing C' in the form C° (thus $C = C'^\circ$), one should regard the "right notion" that replaces, in the general case, that of left exactness as the ind-representability of F (regarded as a presheaf on C°; i.e. the pro-representability of F, regarded as a functor on C', cf. 8.10). This notion does indeed coincide with that of left exactness when finite projective limits are representable in C' *i.e.* finite inductive limits in C are representable (8.3.3). When C' is no longer assumed $\mathscr{U}$-small, or at least equivalent to a $\mathscr{U}$-small category, the two notions for a functor F, left exactness and pro-representability, no longer necessarily coincide, even if finite projective limits in C' are representable (cf. 8.12.9). In
79 fact, it seems that in this case, left exact functors that are not ind-representable should

be regarded as pathological, while the "good" objects remain the ind-representable functors; compare 8.13.3. For the case of functors $F : C' \to C''$, with C' and C'' once again arbitrary $\mathscr{U}$-categories, we shall develop below (8.11.5), more generally, a notion that "improves" upon left exactness (namely, that of a functor admitting a pro-adjoint).

8.4. Constant and essentially constant ind-objects. Choose a terminal category e, i.e. one such that $\mathrm{Ob}\, e$ and $F\ell\, e$ each consist of a single element (for example, one may take for e the full subcategory of (Ens) on the empty set, if a canonical choice is desired). This is obviously a small filtered category. If X is an object of C, we associate with it the constant ind-object indexed by e, with value X, denoted by $c(X)$. It is clear that, as X varies, this gives a fully faithful functor

$$c : C \longrightarrow \mathrm{Ind}(C), \tag{8.4.1}$$

which is moreover injective on objects, and by which we shall identify C with a full subcategory of $\mathrm{Ind}(C)$. Note that the composite functor

$$C \xrightarrow{c} \mathrm{Ind}(C) \xrightarrow{L} \widehat{C} \tag{8.4.2}$$

is the canonical functor h (8.2.1.1).

8.4.3. More generally, let I be a filtered category. The constant functor on I whose 80
value is an object X of C is also called *the constant ind-object with value X indexed by I*. It is clear that, if I is essentially small, this ind-object ind-represents the functor $h(X)$ represented by X, hence this ind-object is isomorphic to $c(X)$.

8.4.4. An ind-object $\mathscr{X}$ of C is said to be *essentially constant* if it is isomorphic (in a category $\mathrm{Ind}_{\mathscr{V}}(C, \mathscr{V}')$, for suitable $\mathscr{V}$, $\mathscr{V}'$) to an ind-object of the form $c(X)$, $X \in \mathrm{ob}\, C$. The object X, which is then determined up to canonical isomorphism, is called the *value* of the essentially constant ind-object under consideration. Thus, by definition, the functor c of (8.4.1) induces an equivalence of C with the full subcategory of $\mathrm{Ind}(C)$ consisting of the ind-objects that are essentially constant, this subcategory being the essential image of the functor c.

Clearly a constant ind-object is essentially constant, whereas the converse holds only in the trivial case in which C is empty or a one-point category.

8.5. Filtered inductive limits in $\mathrm{Ind}(C)$.

Proposition 8.5.1. *Let C be a $\mathscr{U}$-category. In* $\mathrm{Ind}(C)$, *small filtered inductive limits are representable, and the canonical functor* (8.2.4.8)

$$L : \mathrm{Ind}(C) \longrightarrow \widehat{C}$$

commutes with them.

Since L is fully faithful, the assertions of the proposition are equivalent to the fol- 81
lowing:

Corollary 8.5.2. *Every small filtered inductive limit (in $\widehat{C}$) of ind-representable presheaves is ind-representable.*

This follows easily from criterion 8.3.3 (ii). The details of the verification are left to the reader.

8.5.3. The “tautological” case of filtered inductive limits in $\mathrm{Ind}(C)$ is that in which one starts with an essentially small ind-object $\mathscr{X} = (X_i)_{i\in I}$ of C. Then, in $\widehat{C}$ and hence also in $\mathrm{Ind}(C)$ (or in $\widehat{C}$), one has:

$$\mathscr{X} = \varinjlim_I X_i. \tag{8.5.3.1}$$

When writing this formula, care must be taken that it is not an inductive limit in C, and that even when the latter exists, it is not isomorphic in $\mathrm{Ind}(C)$ to $\mathscr{X}$: indeed, the canonical functor (8.4.1) $c : C \to \mathrm{Ind}(C)$ *does not in general commute with filtered inductive limits.* (Cf. 8.5.5 below.)

To avoid this possible confusion in writing (8.5.3.1), “some authors” (= P. Deligne) prefer to write it as

$$\mathscr{X} = «\varinjlim_I» X_i, \tag{8.5.3.2}$$

the quotation marks indicating that the inductive limit is taken in a category of ind-objects $\mathrm{Ind}(C)$. By extension, one should then denote by «lim» every inductive-limit operation in $\mathrm{Ind}(C)$ (without the components of the inductive system under consideration in $\mathrm{Ind}(C)$ necessarily lying in the image, or the essential image, of $c : C \to \mathrm{Ind}(C)$).

82 8.5.4. To compute the inductive or projective limit of a functor

$$\varphi : J \longrightarrow \mathrm{Ind}(C), \tag{8.5.4.1}$$

where J is now an arbitrary category, it is often convenient to make φ explicit by means of a functor

$$\Phi : J \times I \longrightarrow C, \tag{8.5.4.2}$$

where I is an essentially small filtered category, or preferably a small one. Such a Φ indeed defines a functor

$$J \longmapsto (i \longmapsto \Phi(j,i)) : J \longmapsto \mathbf{Hom}(I, C),$$

whence, by composing with the canonical arrow (8.2.6.1) $\mathbf{Hom}(I, C) \to \mathrm{Ind}(C)$, a functor

$$J \longmapsto (i \longmapsto \Phi(j,i)) : J \longrightarrow \mathrm{Ind}(C). \tag{8.5.4.3}$$

We shall say that Φ is an *indexed expression of φ, indexed by the (filtered) category I*, if the corresponding functor (8.5.4.3) is isomorphic to φ. We shall study below (8.8) general conditions for the existence of an indexed expression for a given functor φ, and here we shall restrict ourselves to starting from a functor φ given in indexed form (8.5.4.3), in the case where J is a small filtered category, in order to indicate in this case how to compute $«\varinjlim» \varphi = «\varinjlim_j» \varphi(j)$. Formula (8.5.3.2), together with the associativity formula for inductive limits (2.5.0), then immediately gives the “*tautological computation*” of $«\varinjlim» \varphi$:

$$«\varinjlim_J» \varphi = «\varinjlim_{J\times I}» \Phi, \tag{8.5.4.4}$$

83 i.e.

$$«\varinjlim_j» (i \longmapsto \Phi(j,i)) = «\varinjlim_{j,i}» \Phi(j,i), \tag{8.5.4.5}$$

Thus, the sought-for inductive system is none other than Φ itself (the indexing category being $J \times I$, which is indeed filtered since J and I are).

Exercise 8.5.5. Let C be a $\mathscr{U}$-category

a) Prove that the following conditions are equivalent:

(i) In C, small inductive limits are representable, and the functor $c : C \to \mathrm{Ind}(C)$ commutes with them.
(iii) The functor c is an equivalence of categories.
(ii) In C, small inductive limits are representable, and for every $X \in \mathrm{Ob}\, C$, the functor $Y \mapsto \mathrm{Hom}(X, Y)$ commutes with these limits.
(iv) Small filtered inductive limits are representable in C, and the functor $\lim c$: $\mathrm{Ind}\, C \to C$ (cf. 8.7.1.5) is fully faithful (or equivalently, an equivalence).

b) If C is a finite category, prove that the preceding conditions are equivalent to the following: for every projector in C (10.6), its image is representable in C.

8.6. Extension of a functor to ind-objects.

8.6.1. Let

$$f : C \longrightarrow C' \tag{8.6.1.1}$$

be a functor between $\mathscr{U}$-categories. For every ind-object

$$\varphi : I \longrightarrow C$$

of C (I a filtered category), the composite functor $f \circ \varphi$ is an ind-object of C', also 84
denoted by $\mathrm{Ind}(f)(\varphi)$. In indexed notation, if φ is written as $\mathscr{X} = (X_i)_{i \in I}$, one obtains

$$\mathrm{Ind}(f)(X_i)_{i \in I} = (f(X_i))_{i \in I}. \tag{8.6.1.2}$$

Let $\mathscr{Y} = (Y_j)_{j \in J}$ be a second inductive system of C. One then obtains an evident map, also denoted by $u \to \mathrm{Ind}(f)(u)$:

$$\begin{aligned} \mathrm{Hom}(\mathscr{X}, \mathscr{Y}) = \varprojlim_i \varinjlim_j \mathrm{Hom}(X_i, Y_j) \to \\ \mathrm{Hom}(\mathrm{Ind}(f)(\mathscr{X}), \mathrm{Ind}(f)(\mathscr{Y})) \simeq \varprojlim_i \varinjlim_j \mathrm{Hom}(f(X_i), f(Y_j)). \end{aligned} \tag{8.6.1.3}$$

Referring to the unwritten formula for composition of morphisms of ind-objects (8.2.5), one immediately sees that these maps are compatible with composition of morphisms of ind-objects. Thus, for every universe $\mathscr{V}$, we have obtained a functor

$$\mathrm{Ind}(f) : \mathrm{Ind}_{\mathscr{V}}(C, \mathscr{U}) \longrightarrow \mathrm{Ind}_{\mathscr{V}}(C', \mathscr{U}) \tag{8.6.1.4}$$

(notation of (8.2.4.5)), and in particular a functor

$$\mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow \mathrm{Ind}(C'). \tag{8.6.1.5}$$

If there is a second functor

$$g : C' \longrightarrow C'',$$

then one clearly has an identity of functors between Ind-categories (or $\mathrm{Ind}_{\mathscr{V}}$-categories, as desired):

$$\mathrm{Ind}(gf) = \mathrm{Ind}(g) \circ \mathrm{Ind}(f), \tag{8.6.1.6}$$

and, when $C' = C$, the perennial formula 85

$$\mathrm{Ind}(\mathrm{id}_C) = \mathrm{id}_{\mathrm{Ind}(C)}. \tag{8.6.1.7}$$

One may therefore say, heuristically, that the category $\mathrm{Ind}(C)$ *depends functorially on* C (covariantly) [3]. We leave to the reader the task of making explicit even a 2-*functorial* dependence by defining, for every pair of $\mathscr{U}$-categories C, C', a canonical functor

$$\mathbf{Ind}(f) : \mathbf{Hom}(C, C') \longrightarrow \mathbf{Hom}(\mathrm{Ind}(C), \mathrm{Ind}(C')), \tag{8.6.1.8}$$

[3] For a contravariant dependence, under certain conditions, cf. 8.11.

and specifying the functorial nature of the identical isomorphisms (8.6.1.6) and (8.6.1.7).

8.6.2. Return to the functor (8.6.1.1), and consider the corresponding functor (5.1)

$$f_! : \widehat{C} \longrightarrow \widehat{C'}, \tag{8.6.2.1}$$

and the diagram of functors

$$\begin{tikzcd} \mathrm{Ind}(C) \arrow[r, "\mathrm{ind}(f)"] \arrow[d, "L_C"'] & \mathrm{Ind}(C') \arrow[d, "L_{C'}"] \\ \widehat{C} \arrow[r, "f_!"] & \widehat{C}' \end{tikzcd} \tag{8.6.2.2}$$

where the vertical arrows denote the canonical functors L of (8.2.4.8). Since the functor $f_!$ "extends f" and commutes with inductive limits (5.4.3), and the functors L_C and $L_{C'}$ commute with small filtered inductive limits (8.5.1), it follows from (8.5.3.2) that, for every ind-object $(X_i)_{i\in I}$ of C, there is a canonical isomorphism in $\widehat{C'}$

$$f_!(L_C((X_i)_{i\in I})) \simeq \varinjlim_i f_! L_C(X_i) \simeq \varinjlim_i L_{C'} f(X_i) = L_{C'}(f(X_i)_{i\in I})$$

86 i.e. a canonical isomorphism

$$f_! L_C(\mathscr{X}) \simeq L_{C'}(\mathrm{Ind}(f)(\mathscr{X})).$$

We leave to the reader the verification that the latter is functorial, i.e. that it corresponds to a canonical isomorphism

$$f_! L_C \simeq L_{C'} \circ \mathrm{Ind}(f). \tag{8.6.2.3}$$

In other words, the *square (8.6.2.2) commutes up to canonical isomorphism.* Taking into account the fact that $f_!$ commutes with small inductive limits, and using 8.5.1, we conclude:

Proposition 8.6.3. *Let $f : C \to C'$ be a functor between $\mathscr{U}$-categories. Then the functor*

$$\mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow \mathrm{Ind}(C')$$

commutes with filtered inductive limits. Moreover, it makes the following diagram commutative:

$$\begin{tikzcd} C \arrow[r, "f"] \arrow[d, "c_C"'] & C' \arrow[d, "c_{C'}"] \\ \mathrm{Ind}(C) \arrow[r, "\mathrm{Ind}(f)"] & \mathrm{Ind}(C') \end{tikzcd} ,$$

where the vertical arrows c_C, $c_{C'}$ are the canonical functors (8.4.1).

The last assertion is trivial from the definitions and has been included for ease of reference. Notice moreover that the properties stated in 8.6.3 *characterize the functor*

87 $\mathrm{Ind}(f)$ up to unique isomorphism, as the functor induced by $f_!$ (8.6.2.1), as follows from the proof of (8.6.2.3) just given.

Proposition 8.6.4. *The notation is that of 8.6.3.*

a) *The functor $\mathrm{Ind}(f)$ is faithful (resp. fully faithful) if and only if f is.*

b) *The functor* $\mathrm{Ind}(f)$ *is an equivalence of categories if and only if* f *is fully faithful and every object of* C' *is isomorphic to a direct factor (10.6) of an object in the image of* f.

Proof.

a) Necessity clearly follows from the fact that f is induced by $\mathrm{Ind}(f)$. For sufficiency, it is enough to use the form (8.6.1.3) of $\mathrm{Ind}(f)$ on the sets $\mathrm{Hom}(\mathscr{X}, \mathscr{Y})$, recalling that filtered inductive limits of sets and arbitrary projective limits take monomorphisms to monomorphisms and isomorphisms to isomorphisms.

b) We may already suppose that f, hence $\mathrm{Ind}(f)$, is fully faithful. Since every object of $\mathrm{Ind}(C')$ is a small filtered inductive limit of objects of C', the full faithfulness of $\mathrm{Ind}(f)$ implies that, for this functor to be essentially surjective, it is equivalent that every object X' of C' belong to the essential image. But if there is an isomorphism

$$X' \xrightarrow{\sim} «\varinjlim» f(X_i),$$

this isomorphism factors through one of the $f(X_i)$, showing that X' is a direct factor of $f(X_i)$ and proving necessity in b). For sufficiency, using the full faithfulness of $\mathrm{Ind}(f)$, the assertion follows immediately.

Corollary 8.6.5. *In* $\mathrm{Ind}(E)$ *the images of projectors (10.6) are representable, and the functor* $\mathrm{Ind}(f) : \mathrm{Ind}(C) \to \mathrm{Ind}(C')$ *commutes with the formation of these images.* 88

Indeed, this follows from the fact that small filtered inductive limits are representable in $\mathrm{Ind}(C)$ (8.5.1) and that $\mathrm{Ind}(f)$ commutes with them (8.6.3), taking into account that the image of a projector $p : X \to X$ may be interpreted as the inductive limit of a filtered inductive system indexed by $\mathbf{N}$

$$X \xrightarrow{p} X \xrightarrow{p} X \longrightarrow \dots$$

or, equivalently, as the inductive limit of the evident functor on the filtered category P having one object and one nonidentity arrow Π such that $\Pi^2 = \Pi$.

8.7. *The functor* $\varinjlim_C : \mathrm{Ind}(C) \to C$. *Universal characterizations of the category* $\mathrm{Ind}(C)$.

8.7.1. Let us return to a $\mathscr{U}$-category C and the canonical functor

$$c : C \longrightarrow \mathrm{Ind}(C). \tag{8.7.1.1}$$

Let $\mathscr{X} = (X_i)_{i\in I}$ be an object of $\mathrm{Ind}(C)$. It follows immediately from the definition of representable inductive limits (2.1, 2.1.1) that $\varinjlim_i X_i$ *is representable in* C *if and only if the left adjoint of* c *is defined at* $\mathscr{X}$, and that in this case $\varinjlim_i X_i$ (computed in C) is precisely the value at $\mathscr{X}$ of that left adjoint:

$$\mathrm{Hom}_C(\varinjlim_i X_i, Y) \simeq \mathrm{Hom}_{\mathrm{Ind}(C)}((X_i)_{i\in I}, c(Y)). \tag{8.7.1.2}$$

If we denote by 89

$$\mathrm{Ind}(C)' \subset \mathrm{Ind}(C) \tag{8.7.1.3}$$

the full subcategory of $\mathrm{Ind}(C)$ formed by the ind-objects of C that admit an inductive limit *in* C, the preceding observation shows that this subcategory is *strictly full*, and that there is a canonical functor

$$\varinjlim_C : \mathrm{Ind}(C)' \longrightarrow C, \tag{8.7.1.4}$$

whose value at each object $(X_i)_{i\in I}$ of $\mathrm{Ind}(C)'$ is its inductive limit in C. In the special case where small filtered inductive limits are representable in C, one therefore obtains a natural functor

$$\varinjlim_C : \mathrm{Ind}(C) \longrightarrow C. \tag{8.7.1.5}$$

. Of course, the preceding functors may also be defined on categories of the form $\mathrm{Ind}_{\mathscr{V}}(C,\mathscr{U})'$ and $\mathrm{Ind}_{\mathscr{V}}(C,\mathscr{U})$, but in view of (8.2.4.6) they are already determined (up to unique isomorphism) by the preceding functors corresponding to the case $\mathscr{V} = \mathscr{U}$.

. The preceding construction of $\varinjlim_C$ as a left adjoint functor immediately implies that this functor *commutes with arbitrary inductive limits*, and in particular with small filtered inductive limits (the latter being representable in $\mathrm{Ind}(C)$ (8.5.1)). Notice also that $\mathrm{Ind}(C)'$ always contains the essential image of c (formed by the essentially constant ind-objects), and that there is a canonical isomorphism, functorial in $X \in \mathrm{ob}\,\mathrm{Ind}(C)'$:

$$\varinjlim_C c(X) \simeq X, \tag{8.7.1.8}$$

90 i.e. , in the favorable case where C is stable under small inductive limits, there is a canonical isomorphism

$$\varinjlim_C \circ\, c \simeq \mathrm{id}_C\,. \tag{8.7.1.9}$$

8.7.2. Now consider a functor

$$f : C \longrightarrow E, \tag{8.7.2.1}$$

where E is a $\mathscr{U}$-category in which small inductive limits are representable, so that there is a functor

$$\varinjlim_E : \mathrm{Ind}(E) \longrightarrow E.$$

Since we have also defined (8.6)

$$\mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow \mathrm{Ind}(E),$$

we may consider the composite

$$\overline{f} = \varinjlim_E \circ \mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow E, \tag{8.7.2.2}$$

sometimes called the *canonical extension of f to ind-objects* (but not to be confused with $\mathrm{Ind}(f)$!). As a composite of two functors that commute with small filtered inductive limits (8.6.3, 8.7.1.7), this functor itself commutes with small filtered inductive limits. Moreover, the isomorphism (8.7.1.9) applied to E, together with (8.6.2.3), shows that $\overline{f}$ *extends* f up to isomorphism, i.e. there is a canonical isomorphism

$$\overline{f} \circ c_C \simeq f\,. \tag{8.7.2.3}$$

91 It is moreover fairly clear, in view of the fact that every object of $\mathrm{Ind}(C)$ is a small filtered inductive limit of objects of C, that the two preceding properties still *characterize* $\overline{f}$ up to unique isomorphism. This can be sharpened to obtain at the same time a universal characterization, up to equivalence, of $\mathrm{Ind}(C)$ among the $\mathscr{U}$-categories E in which small filtered inductive limits are representable. If E and F are two such $\mathscr{U}$-categories, denote by

$$\mathbf{Hom}(E,F)' \subset \mathbf{Hom}(E,F) \tag{8.7.2.4}$$

the full subcategory of $\mathbf{Hom}(E,F)$ formed by the functors that commute with small filtered inductive limits. Then:

PROPOSITION 8.7.3. *Let C be a $\mathscr{U}$-category, and use the notation (8.7.2.4) above. Then the canonical functor*

$$c_C : C \longrightarrow \operatorname{Ind}(C)$$

is 2-universal among the functors with source C and target a $\mathscr{U}$-category having representable small filtered inductive limits. In other words, $\operatorname{Ind}(C)$ *is such a category (8.2.5, 8.5.1), and if E is such a category, the functor*

$$g \longmapsto g \circ c_C : \mathbf{Hom}(\operatorname{Ind}(C), E)' \longrightarrow \mathbf{Hom}(C, E) \tag{8.7.3.1}$$

is an equivalence of categories.

To prove the latter point, it is enough to verify that the map $f \mapsto \overline{f}$ defined by (8.7.2.2) can be refined to a *functor*

$$f \longmapsto \overline{f} = \varinjlim_E \cdot \operatorname{Ind}(f) : \mathbf{Hom}(C, E) \longrightarrow \mathbf{Hom}(\operatorname{Ind}(C), E)', \tag{8.7.3.2}$$

and that the latter is a quasi-inverse of (8.7.3.1). The details of the essentially trivial 92
verification are left to the reader. Alternatively, one may invoke 7.8 to conclude first that (8.7.3.1) is fully faithful, and then (8.7.2.3) to conclude that it is essentially surjective.

8.7.4. Return to a functor between $\mathscr{U}$-categories

$$f : C \longrightarrow E, \tag{8.7.4.1}$$

where E is a $\mathscr{U}$-category in which small filtered inductive limits are representable, giving a functor

$$\overline{f} : (X_i)_{i \in I} \longmapsto \lim_i f(X_i) : \operatorname{Ind}(C) \longrightarrow E. \tag{8.7.4.2}$$

We propose to study conditions on f ensuring that $\overline{f}$ is fully faithful, resp. an equivalence of categories. Since f is isomorphic to the composite $\overline{f} \circ c_C$, and $c_C : C \to \operatorname{Ind}(C)$ is fully faithful, we see that for $\overline{f}$ to be fully faithful, it is *necessary* that f be so. Replacing C by its essential image in E, we therefore lose essentially no generality by supposing that f is the inclusion functor of a subcategory C of E, as we shall do below to simplify the notation.

PROPOSITION 8.7.5. *The notation is that of 8.7.4.*

a) *The functor $\overline{f}$ is fully faithful if and only if:*
   (i) *Every object X of C satisfies the following condition:*
   (PF) *For every small filtered inductive system $(Y_i)_{i \in I}$ in C whose inductive limit $\varinjlim_i Y_i$ exists in E, the canonical map* 93

$$\varinjlim_i \operatorname{Hom}(X, Y_i) \longrightarrow \operatorname{Hom}(X, \varinjlim_i Y_i) \tag{8.7.5.1}$$

   is bijective.

b) *The functor $\overline{f}$ is an equivalence of categories if and only if C satisfies condition* (i) *and the following two conditions:*
   (ii) *C is a generating subcategory of E by strict epimorphisms (7.1).*
   (iii) *For every object X of E, the full subcategory $C_{/X}$ of $E_{/X}$ formed by the objects X' over X whose source is in* $\operatorname{Ob} C$ *is filtered and essentially small.*

Condition (iii) *is satisfied in particular if C is equivalent to a small category, the finite inductive limits in C are representable, and the inclusion functor $f : C \to E$ commutes with them (i.e. for every finite category J and every functor $\varphi : J \to C$, the inductive limit of $f\varphi$ is representable and is isomorphic in E to an object of C).*

Proof.

a) With the notation of condition (i), let $\mathscr{Y}$ denote the ind-object (Y_i) and let $\mathscr{X}$ denote the constant ind-object defined by X. Then (8.7.5.1) is precisely the canonical map

$$\mathrm{Hom}(\mathscr{X},\mathscr{Y}) \longrightarrow \mathrm{Hom}(\overline{f}(\mathscr{X}),\overline{f}(\mathscr{Y})),$$

so if $\overline{f}$ is fully faithful, this map is indeed bijective. Conversely, if $\mathscr{X}=(X_j)_{j\in J}$ and $\mathscr{Y}=(Y_i)_{i\in I}$ are arbitrary ind-objects of E, then the map $u : \mathrm{Hom}(\mathscr{X},\mathscr{Y}) \to$
94 $\mathrm{Hom}(\overline{f}(\mathscr{X}),\overline{f}(\mathscr{Y}))$ is the projective limit over j of the maps $u_j : \mathrm{Hom}(\mathscr{X}_j,\mathscr{Y}) \to \mathrm{Hom}(\overline{f}(\mathscr{X}_j),\mathscr{Y})$, where $\mathscr{X}_j$ is the constant ind-object with value X_j. Thus u is bijective if the u_j are, and (i) therefore implies that $\overline{f}$ is fully faithful.

b) Suppose that (i), (ii), and (iii) hold, and let us prove that $\overline{f}$ is an equivalence. It remains to prove that it is essentially surjective, hence that every $X\in \mathrm{ob}\,E$ belongs to the essential image. But hypothesis (ii) means that $\varinjlim_{C_{/X}} Z \to X$ is an isomorphism, while (iii) says that the category $C_{/X}$ is filtered and essentially small. Thus X is the image of the ind-object of E defined by the natural functor $C_{/X}\to E$. Conversely, suppose that $\overline{f}$ is an equivalence. In view of a), it remains to verify (ii) and (iii). We may suppose that $E=\mathrm{Ind}(C)$, with C identified with the subcategory $c(C)$ of E. But we know (8.3.3 (ii)) that for every $X\in \mathrm{Ind}(C)$, identified if desired with the functor F on C° that it ind-represents, $C_{/X}=C_{/F}$ is an essentially small filtered category (which proves (iii)), and that the inductive limit in $\widehat{C}$ of the functor $C_{/F}\to\widehat{C}$ is F. It is therefore also so in the full subcategory E of $\widehat{C}$, since $F=X$ belongs to E, which proves (ii). It remains to prove the last assertion of b). But the hypothesis on C clearly implies that finite inductive limits are representable in the category $C_{/X}$; *a fortiori*, the category $C_{/X}$ is filtered.

95 Remark 8.7.5.2. The preceding argument shows more generally that, when condition (i) is satisfied, the essential image of the fully faithful functor $\overline{f}$ (8.7.4.2) consists of the $X\in\mathrm{ob}\,E$ such that the category $C_{/X}$ is filtered and essentially small (a condition automatically satisfied if C satisfies the conditions stated at the end of b)), *and* such that the inductive limit of the canonical functor $C_{/X}\to E$ is X.

Corollary 8.7.6. *Let E_{PF} be the full subcategory of E formed by the objects satisfying condition (PF) of 8.7.5 a). Then, for every functor $J\to E_{PF}$, where J is a finite category, that admits an inductive limit X in E, one has $X\in\mathrm{ob}\,E_{PF}$. The canonical functor $(X_i)_{i\in I}\to \varinjlim_{i}{}_E X_i$ deduced from the inclusion $g : E_{PF}\hookrightarrow E$*

$$\overline{g} : \mathrm{Ind}(E_{PF}) \longrightarrow E \tag{8.7.6.1}$$

is fully faithful. If finite inductive limits are representable in E and E_{PF} is equivalent to a small category, then the essential image of the functor $\overline{f}$ consists of the objects X of E such that the canonical morphism $\varinjlim_{(E_{PF})_{/X}} Y\to X$ is an isomorphism. If, moreover, the subcategory E_{PF} of E is generating by strict epimorphisms (7.1), then the functor (8.7.6.1) is an equivalence of categories.

All the assertions are evident, in the order in which they are stated, in view of 8.7.5, using 2.8 for the first assertion.

96 Corollary 8.7.7. *Suppose that finite inductive limits are representable in E. Let C be a full subcategory of E, equivalent to a small category, and consider the functor $(X_i)_{i\in I} \to \varinjlim_{i} {}_E X_i$:*

$$\overline{f} : \mathrm{Ind}(C) \to E. \tag{8.7.7.1}$$

Let E_{PF} be as in 8.7.6. The following conditions are equivalent:

(i) *The functor $\overline{f} : \mathrm{Ind}(C) \to E$ is an equivalence.*

(ii) *The category C is generating in E by strict epimorphisms, is contained in E_{PF}, and every object of E_{PF} is isomorphic in E (or in E_{PF}, which is equivalent (8.7.6)) to a direct factor of an object of C.*

When the subcategory C of E is stable under direct factors (10.6), the preceding conditions are also equivalent to the following:

(ii bis) *$C = E_{PF}$, and C is generating in E by strict epimorphisms.*

(iii) *The subcategory C of E is generating by strict epimorphisms, is contained in E_{PF}, and finite inductive limits are representable in it.*

When, moreover, finite inductive limits are representable in E, these conditions are also equivalent to

(iii bis) *The subcategory C of E is generating by strict epimorphisms, is contained in E_{PF}, and is stable in E under finite inductive limits (or, equivalently, under finite sums and cokernels of double arrows).*

We already know (8.7.5) that condition (i) implies $C \subset E_{PF}$ and that C is generating 97
by strict epimorphisms. Let us also prove that every object X of E_{PF} is then isomorphic to a direct factor of an object of C. Indeed, we know that X is a filtered inductive limit $\varinjlim_i X_i$ in E of objects of C, and by the definition of E_{PF}, for a suitable i the canonical homomorphism $X_i \to X$ has a left inverse, so X is indeed a direct factor of the object X_i of C. This proves (i) ⇒ (ii) (without any hypothesis on C or E, moreover). Let us prove (ii) ⇒ (i). Since C is equivalent to a small category and every object of E_{PF} is a direct factor of an object of C, we see that E_{PF} is also equivalent to a small category. Thus, by 8.7.6, the functor (8.7.6.1) is an equivalence of categories, and we conclude by 8.6.4 b) applied to the inclusion $E \hookrightarrow E_{PF}$.

When every direct factor in E of an object of C belongs to C, it is clear that (ii) ⇔ (ii bis). On the other hand, (ii bis) implies (iii) resp. (iii bis) by 8.7.6. Finally, (iii) (and *a fortiori* (iii bis)) implies (i) by 8.7.5. This completes the proof.

Exercise 8.7.8. (*Karoubi envelopes*). Let C be a $\mathcal{U}$-category, let $E = \mathrm{Ind}(C)$, and let $E_{PF} \supset C$ be the full subcategory of E defined in 8.7.6.

a) Prove that images of projectors are representable in E_{PF} and that every object of E_{PF} is a direct factor of an object of C.

b) Call a category F *Karoubian* if images of projectors are representable in it. Let

$$\varphi : C \longrightarrow K$$

be a fully faithful functor such that K is Karoubian and every object of K is a 98
direct factor of an object in the image of C. Prove that φ is 2-*universal* among functors from C with values in Karoubian categories, more precisely: for every Karoubian category F, the functor

$$g \longmapsto g \circ \varphi : \mathbf{Hom}(K, F) \longrightarrow \mathbf{Hom}(K, F)$$

is an equivalence of categories. (Use the fact that every functor commutes with images of projectors.) The category K equipped with φ, determined up to equivalence (itself determined up to unique isomorphism) by the preceding properties, is called the *Karoubi envelope* $\widetilde{C}$ of C. (Compare also IV 7.5.)

c) Deduce from a) and b) that $\mathrm{Ind}(C)_{PF}$, equipped with the functor $C \to \mathrm{Ind}(C)_{PF}$ induced by $c : C \to \mathrm{Ind}(C)$, is a Karoubi envelope of C.

d) Show that every functor $f : C \to C'$ extends in an essentially unique way to a functor $\widetilde{f} : \widetilde{C} \to \widetilde{C'}$ of Karoubi envelopes, and that if these Karoubi envelopes are taken as in c), $\widetilde{f}$ is the functor induced by $\mathrm{Ind}(f) : \mathrm{Ind}(C) \to \mathrm{Ind}(C')$. Show that the conditions of 8.5.4 b) are also equivalent to the following: $\widetilde{f}$ is an equivalence of categories.

Exercise 8.7.9. Let E be a $\mathcal{U}$-category.

a) Show that the following conditions are equivalent:
   (i) E is equivalent to a category of the form $\mathrm{Ind}(C)$, where C is a $\mathcal{U}$-category.
99 (i bis) As in (i), with C moreover Karoubian (8.7.8 b)).
   (ii) The subcategory E_{PF} of E (8.7.6) is generating by strict epimorphisms, and for every object X of E, $(E_{PF})_{/X}$ is an essentially small filtered category.

   Show that if C is a *Karoubian* $\mathcal{U}$-category such that E is equivalent to $\mathrm{Ind}(C)$, then C is equivalent to E_{PF} (by an equivalence determined up to unique isomorphism). Establish a 2-equivalence between the 2-category formed by the categories E satisfying the preceding conditions and the functors between them that *commute with small filtered inductive limits*, and the 2-category formed by *Karoubian* $\mathcal{U}$-categories and arbitrary functors between them.

b) Show that the following conditions are equivalent:
   (i) E is equivalent to a category of the form $\mathrm{Ind}(C)$, where C is a $\mathcal{U}$-category in which finite inductive limits are representable.
   (i bis) As in (i), but with C moreover Karoubian.
   (ii) Finite inductive limits in E are representable, the subcategory E_{PF} of E is generating by strict epimorphisms, and for every object X of E, there exists a small subset I of $\mathrm{Ob}\, E_{PF/X}$ such that every object of $E_{PF/X}$ is bounded above by an object of I. (Use 8.9.5 b).)

100

8.8. Indexed representation of a functor $J \to \mathrm{Ind}(C)$.

8.8.1. Let

$$\varphi : J \longrightarrow \mathrm{Ind}(C) \text{ i.e. } \varphi \in \mathrm{Ob}\,\mathbf{Hom}(J, \mathrm{Ind}(C))$$

be a functor, where C is a $\mathcal{U}$-category. Recall (8.5.4) that an *indexed representation of* φ is, by definition, an ind-object of $\mathbf{Hom}(J, C)$

$$\psi : I \longrightarrow \mathbf{Hom}(J, C),$$

with I an essentially small filtered category, together with a specified isomorphism from its inductive limit in $\mathbf{Hom}(J, \mathrm{Ind}(C))$ to φ. Thus φ admits an indexed representation if and only if it is isomorphic to an essentially small filtered inductive limit in $\mathbf{Hom}(J, \mathrm{Ind}(C))$ of objects of the full subcategory $\mathbf{Hom}(J, C)$. We shall say that the category J is *admissible for* C if every functor $\varphi : J \to \mathrm{Ind}(C)$ admits an indexed representation.

Since in $\mathrm{Ind}(C)$ small filtered inductive limits are representable (8.5.1), the same is true in $\mathbf{Hom}(J, \mathrm{Ind}(C))$, and they are computed "argument by argument". Consequently, the inclusion functor

$$\mathbf{Hom}(J, C) \hookrightarrow \mathbf{Hom}(J, \mathrm{Ind}(C)) \tag{8.8.1.1}$$

extends canonically to a functor

$$\mathrm{Ind}(\mathbf{Hom}(J, C)) \longrightarrow \mathbf{Hom}(J, \mathrm{Ind}(C)) \tag{8.8.1.2}$$

(8.7.2). The objects of the essential image of this functor are then precisely the φ that admit an indexed representation; thus, to say that J is admissible for C also means that the functor (8.8.1.2) is essentially surjective. In this connection, we note:

Proposition 8.8.2. *If the category J is equivalent to a finite category, the canonical functor (8.8.1.2) is fully faithful; thus J is admissible for C if and only if this functor is an equivalence of categories.* 101

By 8.7.5 a), it suffices to prove that for every small filtered inductive system $(\varphi_i)_{i\in I}$ in $\mathbf{Hom}(J, C)$ and every object φ of $\mathbf{Hom}(C, J)$, the canonical map

$$\varinjlim_i \mathrm{Hom}(\varphi, \varphi_i) \longrightarrow \mathrm{Hom}(\varphi, \varinjlim_i \varphi_i) \tag{8.8.2.1}$$

is bijective, where $\varinjlim_i \varphi_i$ denotes the inductive limit taken in $\mathrm{Hom}(J, \mathrm{Ind}(C))$. Now, if φ, ψ are two functors $J \to C$, there is an exact diagram of sets, functorial in φ and ψ:

$$\mathrm{Hom}(\varphi, \psi) \longrightarrow \prod_{X\in \mathrm{Ob}\, J} \mathrm{Hom}(\varphi(X), \psi(X)) \rightrightarrows \prod_{\substack{f\in F\ell J\\ f:X\to Y}} \mathrm{Hom}(\varphi(X), \psi(Y)).$$

Since filtered inductive limits commute with kernels of double arrows and with finite products, the bijectivity of (8.8.2.1) follows when J is finite. The case in which J is equivalent to a finite category reduces immediately to the preceding case.

Proposition 8.8.3. *Let $\varphi : J \to \mathrm{Ind}(C)$ be a functor, with J equivalent to a small category. For φ to admit an indexed representation, the following two conditions are sufficient, and they are also necessary when J is equivalent to a finite category:*

a) *The category $\mathbf{Hom}(J, C)_{/\varphi}$ (formed by the arrows in $\mathbf{Hom}(J, \mathrm{Ind}(C))$ with target φ and source in $\mathrm{Hom}(J, C)$) is filtered.*
b) *For every object j of J, the functor* 102

$$\psi \longmapsto \psi(j) : \mathbf{Hom}(J, C)_{/\varphi} \longrightarrow C_{/\varphi(j)} \tag{8.8.3.1}$$

is cofinal, i.e. satisfies conditions F 1), F 2) *of* 8.1.3.

Proof. Suppose a) and b) hold; let us prove that φ admits an indexed representation. We may clearly suppose that J is small. Let

$$I = \mathbf{Hom}(J, C)_{/\varphi}, \tag{8.8.3.2}$$

which is a filtered category by hypothesis, and first suppose that I is essentially small, so that the "inclusion" of I in $\mathbf{Hom}(J, C)$ defines an ind-object of $\mathbf{Hom}(J, C)$, $i \mapsto \psi_i$. Moreover, there is a canonical homomorphism

$$\varinjlim_i \psi_i \longrightarrow \varphi, \tag{8.8.3.3}$$

given argument by argument by the homomorphism

$$\varinjlim_I \psi_i(j) \longrightarrow \varphi(j) = \varinjlim_{C_{/\varphi(j)}} X$$

induced by the functor (8.8.3.1). Since the latter is cofinal, we conclude that (8.8.3.3) is an isomorphism, which proves the assertion. In the case where we do not suppose I essentially small, it is enough to construct an essentially small full subcategory I'

such that (8.8.3.3) remains an isomorphism when $\varinjlim_{I'}$ is used in place of $\varinjlim_{I}$. For this purpose, it is enough that the composite functors

$$I' \longrightarrow I \longrightarrow C_{/\varphi(j)}$$

103 induced by the functors (8.8.3.1) all be cofinal. We then conclude by the following lemma, whose proof is immediate from criterion (8.1.3 b)) and is left to the reader:

Lemma 8.8.3.4. *Let I be a filtered $\mathscr{U}$-category, and let*

$$f_j : I \longrightarrow I_j \quad (j \in J)$$

be a small family of cofinal functors from I to essentially small filtered categories. Then there exists a full subcategory I' of I which is filtered and essentially small, and such that the functors induced by the f_j are cofinal.

Finally, let us prove necessity in 8.8.3 when J is assumed equivalent to a finite category. An indexed representation of φ by means of an essentially small filtered indexing category I defines a functor ψ

$$\begin{tikzcd} I \arrow[r, "\psi"] \arrow[dr, dashed, "\psi_j"'] & \mathbf{Hom}(J, C)_{/\varphi} \arrow[d, dashed] \\ & C_{/\varphi(j)} \end{tikzcd} \tag{8.8.3.5}$$

and it is enough to prove that ψ and each of the functors ψ_j induced by it are cofinal: by 8.1.3 b), it will then indeed follow that $\mathbf{Hom}(J, C)_{/\varphi}$ is filtered, and by definition 8.1.1 that the vertical arrow in (8.8.3.5) is also cofinal. By 8.8.2, we may identify φ with an object of $\mathrm{Ind}(E)$, where $E = \mathbf{Hom}(J, C)$, and the fact that the functor

$$\psi : I \longrightarrow E_{/\varphi},$$

104 arising from the ind-object φ of E indexed by I is cofinal is a general fact, which follows immediately from criteria (F 1) and (F 2) of 8.1.3. The same argument (with E, φ replaced by C, $\varphi(j)$) shows that ψ_j is cofinal, which completes the proof.

Remark 8.8.4. a) In the case where J is equivalent to a finite category, if φ admits an indexed representation, we have seen that (8.8.3.5) is a cofinal functor, which implies that the category $\mathbf{Hom}(J, C)_{/\varphi}$ is itself essentially small.

b) Suppose that in C finite inductive limits are representable. Then the same is true in $E = \mathbf{Hom}(J, C)$, and these limits are computed argument by argument and are also inductive limits in $\mathbf{Hom}(J, \widehat{C})$ and *a fortiori* in $\mathbf{Hom}(J, \mathrm{Ind}(C))$. It follows immediately that condition a) of 8.8.3 is then automatically satisfied, and it remains only to consider condition b). I do not know whether it is automatically satisfied when J is in addition assumed small.

We now come to the main result of this section:

Proposition 8.8.5. *Let J be a category equivalent to a finite category, and suppose in addition that J is rigid, i.e. that for every $j \in \mathrm{Ob}\, J$, every endomorphism of j is the identity. Then J is admissible for C (for any $\mathscr{U}$-category C), i.e. every functor $J \to \mathrm{Ind}(C)$ admits an indexed representation.*

105 Upon replacing J by an equivalent category if necessary, we may suppose that J is *skeletal*, i.e. that two isomorphic objects of J are equal. Then J is in fact finite. We proceed by induction on $\operatorname{card} \operatorname{Ob} J$; the case where this number is ≤ 0 is trivial, so we shall suppose it is ≥ 1. Let $j_\circ$ be a *maximal* object of J, i.e. such that for every arrow $j_\circ \to j$, there exists an arrow $j \to j_\circ$; in view of the fact that J is rigid, this implies that $j_\circ \to j$ is an isomorphism, and since J is skeletal, it implies that $j_\circ = j$. Let J' therefore be the full subcategory obtained from J by removing the object $j_\circ$, and let J'' be the full subcategory consisting of the object $j_\circ$. The proposition then follows from the

Lemma 8.8.5.1. *Let J be a finite category, and let J' and J'' be two full subcategories such that* $\operatorname{Ob} J = \operatorname{Ob} J' \cup \operatorname{Ob} J''$, *and such that for every $j' \in \operatorname{Ob} J'$ and $J'' \in \operatorname{Ob} J''$, one has* $\operatorname{Hom}(j'', j') = \varnothing$. *If J' and J'' are admissible for C, then so is J.*

It is enough to prove criteria a) and b) of 8.8.3, which amounts to carrying out six elementary verifications, namely the last two filteredness conditions for $\mathbf{Hom}(J, C)_{/\varphi}$, and conditions F 1) and F 2) for the functors (8.8.3.1), successively in the cases $j \in \operatorname{Ob} J'$ and $j \in \operatorname{Ob} J''$; the latter conditions moreover imply that $\mathbf{Hom}(J, C)_{/\varphi}$ is nonempty. The rather tedious verification presents no difficulty and is left to the reader. (The editor sought in vain for an elegant proof that would circumvent these unpleasant verifications.)

Exercise 8.8.6. a) Let J, J' be two categories admissible for C, one of them being finite. Prove that $J \times J'$ is admissible for C. (Use 8.8.2.) 106

b) Prove that a small discrete category is admissible for every category C.

c) Let J be a category satisfying the following conditions: (i) J is rigid, (ii) $\operatorname{Ob} J$ is countable, (iii) For any two objects j, j' of J, $\operatorname{Hom}(j, j')$ is finite, (iv) For every object j of J, the set of objects of J that are bounded above by j, i.e. that are sources of an arrow with target j, is finite. Show that J is admissible for every $\mathscr{U}$-category C. (First show that J can be written as a filtered union of finite full subcategories J_n, such that every object of J bounded above by an object of J_n belongs to J_n. Then verify criteria a), b) of 8.8.3.)

Exercise 8.8.7. In our notation we identify an ordered set with the category it defines.

a) Let C be an ordered set. Let $C^{\sim}$ be the set of subsets A of C that are filtered and such that $x \in A$ and $y \leq x$ imply $y \in A$. Let $L_\circ : C \to C^{\sim}$ be the map that associates to every $x \in C$ the set $L_\circ(x)$ consisting of those $y \in A$ such that $y \leq x$. Show that L is injective and that the order on C is induced by that on $C^{\sim}$. For every $A \in C^{\sim}$, consider the inclusion $f(A) : A \to C$, which is an ind-object of C; for every ind-object $\varphi : I \to C$ of C, let $g(\varphi) \in C^{\sim}$ be the subset of C consisting of the elements of C bounded above by an element of the form $\varphi(i)$. Show that one thus obtains two mutually quasi-inverse equivalences 107

$$\operatorname{Ind}(C) \xleftrightarrow[\approx]{f,\,g} C^{\sim},$$

transforming the functor $L_\circ$ into the canonical functor $L : C \to \operatorname{Ind}(C)$.

b) Suppose that for every $x \in C$, the set of elements bounding x from above (resp. from below) is finite. Show that every ind-object (resp. pro-object) of C is essentially constant, and indeed that for every such $\varphi : I \to C$ (resp. $\varphi : I^\circ \to C$) there exists an $i_\circ \in \operatorname{Ob} I$ such that the functor induced on $I_{/i_\circ}$ is constant (i.e. takes every arrow to an isomorphism).

c) Let $C \subset \mathbf{N}^{\circ} \times \mathbf{N}$ be formed by the pairs of natural numbers (j, i) such that $i \geq j$. Show that $\mathbf{N}^{\sim}$ is isomorphic to the ordered set obtained from $\mathbf{N}$ (identified with an ordered subset of $\mathbf{N}^{\sim}$ as in a)) by adjoining a greatest element ∞. Show that $C^{\sim}$ is isomorphic to $\mathbf{N}^{\circ} \times \mathbf{N}^{\sim}$. Consider the functor

$$\varphi : \mathbf{N}^{\circ} \longrightarrow C^{\sim} \quad , \quad \varphi(j) = (j, \infty).$$

Show that:

1) the projective limit of φ is not representable in $C^{\sim}$, although filtered projective limits in C are representable (and are essentially constant, by b)).
2) For every functor $\psi : \mathbf{N}^{\circ} \to C$, one has $\mathrm{Hom}(\psi, \varphi) = \varnothing$, and *a fortiori* the functor φ admits no indexed representation.

108 Exercise 8.8.8. Let $X = \mathbf{N} \amalg \mathbf{N}$ be the disjoint union of two copies of $\mathbf{N}$, let s be the symmetry of X, and for every $i \in \mathbf{N}$, let X_i be the subset $\Delta_{i+1} \amalg \Delta_i$ of $\mathbf{N}$ (where Δ_i denotes the subset of $\mathbf{N}$ formed by the j such that $j \leq i$). Let C be the subcategory of the category of subsets of X whose objects are the X_i and whose arrows are the maps between the X_i induced by the identity of X or by its symmetry s, and let $C^{\sim}$ be the category defined analogously, but with the additional object X.

a) Show that $\mathrm{Ind}(C)$ is equivalent to $C^{\sim}$, with the canonical functor $C \to \mathrm{Ind}(C)$ corresponding to the inclusion $C \to C^{\sim}$. Show that there exists no pair (Y, f) consisting of an object Y of C and an endomorphism f of Y, together with a morphism of objects with endomorphism in $\mathrm{Ind}(C)$ from (Y, f) to (X, s).
b) Deduce that if J is the category having a single object and, in addition to the identity arrow, a single arrow of order 2, then the functor $J \to \mathrm{Ind}(C)$ defined by (X, s) admits no indexed representation.

8.9. Exactness properties of $\mathrm{Ind}(C)$.

Proposition 8.9.1. *Let C be a $\mathcal{U}$-category.*

a) *The canonical functors L (8.2.4.8) and c (8.4.1)*

$$C \xrightarrow{c} \mathrm{Ind}(C) \xrightarrow{L} \widehat{C}$$

commute with projective limits.

b) *Suppose that finite inductive limits are representable in C and that C is equivalent to a small category. Then small projective limits are representable in* $\mathrm{Ind}(C)$.

109 c) *If small projective limits (resp. finite projective limits) are representable in C, then the same is true in* $\mathrm{Ind}(C)$. *Let J be a category that is finite and rigid, or discrete; if projective limits of type J are representable in C, projective limits of the same type are representable in* $\mathrm{Ind}(C)$.

d) *Small filtered inductive limits in* $\mathrm{Ind}(C)$ *(which are representable by virtue of 8.5.1) are left exact, i.e. commute with finite projective limits.*

Proof.

a) The fact that L commutes with projective limits follows from the fact that it is fully faithful and from the computation of projective limits in $\widehat{C}$ "argument by argument," which implies that, to verify that a projective system of morphisms $F \to F_i$ $(i \in I)$ in $\widehat{C}$ exhibits F as a projective limit of the F_i, it suffices to verify that the analogous assertion holds for the projective systems of maps of sets $\mathrm{Hom}(X, F) \to \mathrm{Hom}(X, F_i)$, for every $X \in \mathrm{Ob}\, C$. But $C \subset \mathrm{Ind}(C)$.

The fact that c commutes with projective limits follows formally from the fact that L does so, as does the composite $L \circ c : C \to \widehat{C}$.

b) In view of a), the assertion amounts to saying that every projective limit of ind-representable presheaves is ind-representable, which follows at once from the criterion 8.3.3 (v), since a projective limit of left exact functors is left exact (which follows from the fact that "projective limits commute with projective limits" (2.5.0)).

c) This follows formally from the analogous property of $\widehat{C}$ (3.3), and from the fact 110
that L is conservative (being fully faithful) and commutes with limits of the type under consideration (a) and 8.5.1).

d) It is well known (cf. 2.3) that the representability of finite projective limits is equivalent to that of projective limits of the types

$$\text{empty}\,, \dots, \qquad \begin{tikzcd} \cdot \arrow[dr] & & \cdot \arrow[dl] \\ & \cdot & \end{tikzcd}$$

(corresponding to certain finite ordered sets), and the representability of small projective limits amounts to that of products and finite projective limits. Thus the second assertion made in (c) implies the first.

Suppose first that I is finite. Using the result 8.8.5 on the representability of functors $\varphi : J \to \mathrm{Ind}(C)$ in indexed form (8.5.4)

$$(8.9.1.1) \qquad \phi : J \times I \longrightarrow C,$$

the existence of $\varprojlim \varphi$ is therefore a special case of the following more precise and more general result:

Corollary 8.9.2. *Consider a functor $\varphi : J \to \mathrm{Ind}(C)$ given in indexed form Φ (8.9.1.1), where J is a finite category. If for every $i \in \mathrm{Ob}\, I$, the partial functor $j \to \Phi(j,i)$ has a projective (resp. inductive) limit representable in C, then φ has a projective (resp. inductive) limit representable in* $\mathrm{Ind}(C)$, *and there is a canonical isomorphism*

$$(8.9.2.1) \qquad \varprojlim \varphi \simeq \text{«}\varinjlim_i\text{»} \varprojlim_j \Phi(j,i)$$

(resp. 111

$$(8.9.2.2) \qquad \varinjlim \varphi \simeq \text{«}\varinjlim_i\text{»} \varprojlim_j \Phi(j,i)),$$

where $\varprojlim_j$ (resp. . $\varinjlim_j$) is computed in C.

For the first formula, one simply uses the fact that in $\mathrm{Ind}(C)$ filtered inductive limits commute with $\varprojlim_j$, and that c commutes with $\varprojlim_J$ (8.9.1 d) and a)):

$$\varprojlim \varphi = \varprojlim_j \text{«}\varinjlim_i\text{»} \Phi(j,i) \xleftarrow{\sim} \text{«}\varinjlim_i\text{»} \varprojlim_j{}^{\mathrm{Ind}(C)} \Phi(j,i) \simeq \text{«}\varinjlim_i\text{»} \varprojlim_j{}^{C} \Phi(j,i).$$

The second is proved in the same way, using the commutation of inductive limits of type I with inductive limits of type J (2.5.0) and the fact that c commutes with inductive limits of type J, proved in 8.9.4 a) below.

The case where J is discrete. This case is contained in the following more precise assertion:

Corollary 8.9.3. *Let the I_α be essentially small filtered categories indexed by a small set A, and for every $\alpha \in A$, let $X(\alpha) = (X(\alpha)_{i_\alpha})_{i_\alpha \in I_\alpha}$ be an ind-object of C indexed by I_α. Let I be the product category of the I_α, and suppose that for every $\imath = (i_\alpha)_{\alpha\in A} \in I$, the product $\prod_{\alpha\in A} X(\alpha)_{i_\alpha}$ is representable in C. Then the product $\prod_{\alpha\in A} X(\alpha)$ is representable in* Ind(C)*, and it is canonically isomorphic to the ind-object indexed by I given by the formula*

$$\prod_{\alpha\in A} \text{«}\varinjlim_{i_\alpha \in I_\alpha}\text{»}\, X(\alpha)_{i_\alpha} \simeq \text{«}\varinjlim_{\imath=(i_\alpha)_{\alpha\in A}\in I}\text{»} \left(\prod_{\alpha\in A} A(\alpha)_{i_\alpha}\right), \tag{8.9.3.1}$$

112 *where the product on the right-hand side is the product computed in C. (Note that I is filtered and essentially small, since the I_α are.)*

Indeed, it is well known (and immediate by reduction to the case in which one works in the category of sets) that the formula in question holds when the limits are computed in $\widehat{C}$. The conclusion then follows from the fact that L commutes with the limits in question (8.9.1 a) and 8.5.1).

Remarks 8.9.4. The proof given for 8.9.2 and 8.9.3 shows, more generally, that if for every $i \in \operatorname{Ob} I$, $\varprojlim_j \Phi(j,i)$ (resp. $\varinjlim_j \Phi(j,i)$) *computed in* Ind(C) is representable, then the same is true of $\varprojlim_i \varphi$ (resp. of $\varinjlim_j \varphi$). This, together with the argument of c), shows that for a given category J arising from a finite or discrete ordered set (or more generally, one that is C-admissible (8.3.1)), projective limits (resp. inductive limits) of type J are representable in Ind(C) if (and only if) for every functor $\varphi : J \to C$, the projective (resp. inductive) limit of φ *computed in* Ind(C) is representable. In the nonparenthetical case, this also means, by virtue of (a), that every projective limit of type J of representable presheaves on C is ind-representable.

Proposition 8.9.5. *Let C be a $\mathcal{U}$-category.*

a) *The canonical functor*
$$c : C \longrightarrow \operatorname{Ind}(C)$$
is right exact (hence exact, in view of 8.9.1 a)*).*

b) *If finite inductive limits (resp. finite sums) are representable in C, then small inductive limits (resp. small sums) are representable in* Ind(C)*. Let J be a preordered*
113 *set that is finite or discrete; if inductive limits of type J are representable in C, the same is true in* Ind(C)*.*

Proof.

a) Suppose that in C one has $X \simeq \varinjlim_J X_j$, where J is a finite category and X and the X_j belong to C. Then, for every ind-object $\mathcal{Y} = (Y_i)_{i\in I}$ of C, one has:
$$\operatorname{Hom}(X,\mathcal{Y}) \simeq \varinjlim_i \operatorname{Hom}(X, Y_i) \simeq \varinjlim_i \varprojlim_j \operatorname{Hom}(X_j, Y_i)$$
$$\overset{2.8}{\simeq} \varprojlim_j \varinjlim_i \operatorname{Hom}(X_j, Y_i) \simeq \varprojlim_j \operatorname{Hom}(X_j, \mathcal{Y}),$$
and the desired conclusion follows by comparing the outer terms.

b) We already noted in 8.8.4 that the assertions made follow from a) and 8.9.2, at least for finite inductive limits. To prove the stated conclusions in the infinite cases, we are reduced to the case of sums, which can be interpreted as a filtered inductive limit of finite sums, and we therefore conclude by 8.5.1.

Remark 8.9.6. We have already observed that the functor c does not in general commute with filtered inductive limits, and hence does not commute with infinite sums either, in contrast to what happens for $L : \mathrm{Ind}(C) \to \widehat{C}$. On the other hand, apart from its commutation with *filtered* inductive limits (8.5.1), L has practically no commutation properties with any other type of inductive limit (initial object, sum of two objects, cokernels of double arrows). Thus, if C admits an initial object $\varnothing_C$, which is therefore an initial object of $\mathrm{Ind}(C)$ by a), $\varnothing_C$ is never an initial object of $\widehat{C}$ (i.e. identical to the constant presheaf with value $\varnothing$), since $\mathrm{Hom}(\varnothing_C, \varnothing_C) \neq \varnothing$!

Proposition 8.9.7. *Let* 114

$$f : C \longrightarrow C'$$

be a functor between $\mathscr{U}$-categories, and let J be a finite rigid category (8.8.5). If inductive (resp. projective) limits of type J are representable in C and f commutes with them, then $\mathrm{Ind}(f) : \mathrm{Ind}(C) \to \mathrm{Ind}(C')$ *also commutes with limits of this type.*

This follows at once from 8.8.5 and from the calculation 8.9.2 of finite limits in a category of ind-objects, for a functor represented in indexed form.

Corollary 8.9.8. *If finite inductive (resp. projective) limits are representable in C, and if f is right exact (resp. left exact), then the same is true of* $\mathrm{Ind}(f)$*; in the unparenthesized case,* $\mathrm{Ind}(f)$ *even commutes with all small inductive limits.*

The first assertion follows from 8.9.7. The second follows from the first, taking into account that $\mathrm{Ind}(f)$ commutes with filtered inductive limits (8.6.3).

Exercise 8.9.9. Let C be a $\mathscr{U}$-category.

a) Suppose that finite sums are representable in C. Show that if *finite* sums in C are disjoint (resp. universal) (cf. II 4.5), then in $\mathrm{Ind}(C)$ *small* sums are disjoint (resp. universal).
b) Suppose that every morphism in C factors as an epimorphism followed by a monomorphism (resp. as an effective epimorphism followed by a monomor- 115
phism, resp. as an epimorphism followed by an effective monomorphism, resp. as an effective epimorphism followed by an effective monomorphism). Show that $\mathrm{Ind}(C)$ satisfies the same property. In the last case, it follows that every epimorphism of $\mathrm{Ind}(C)$ is effective, every monomorphism of $\mathrm{Ind}(C)$ is effective, and every bimorphism of $\mathrm{Ind}(C)$ is an isomorphism; show that in this case every epimorphism (resp. monomorphism) of $\mathrm{Ind}(C)$ can be represented by an inductive system of epimorphisms (resp. monomorphisms) of C. If, in addition, every epimorphism (resp. every monomorphism) in C is assumed to be *universal*, the same property holds in $\mathrm{Ind}(C)$.
c) Suppose that C is additive (resp. abelian); then so is $\mathrm{Ind}(C)$.
d) Let $X = « \varinjlim_I » X_i$ be an ind-object of C, and let $R \rightrightarrows X$ be an equivalence relation on X. For every $i \in \mathrm{Ob}\, I$, let $R_i \rightrightarrows X_i$ be the equivalence relation on X_i (regarded as an object of $\mathrm{Ind}(C)$) induced by R via $X_i \to X$. (We assume that the fiber product $R_i = R \times_{X \times X} X_i \times X_i$ in $(\mathrm{Ind}(C))$ is representable by an object of $\mathrm{Ind}(C)$, as is the case if finite projective limits are representable in C.) Show that, for R to be effective, it suffices that the R_i be effective.
e) Let X be an object of C, and let R be an equivalence relation on $c(X)$, i.e. on X regarded as an object of $\mathrm{Ind}(C)$. Suppose that fiber products are representable in C. For R to be effective, it is necessary that R be of the form $« \varinjlim_i » R_i$, where

the R_i are equivalence relations on X (regarded as an object of C), and this
116 condition is sufficient if equivalence relations in C are assumed to be effective.

f) Suppose that every morphism in C factors as an effective epimorphism followed by an effective monomorphism, and that finite projective limits are representable. Let X be an object of C, and let R be a subobject of $X \times X$, regarded as an object of $\mathrm{Ind}(C)$. Write R in the form « $\varinjlim$ ” Z_i, where $(Z_i)_{i\in I}$ is a filtered increasing family of subobjects of $X \times X$ *in* C (this is possible by b)). Show that, for R to be an equivalence relation on X, it is necessary and sufficient that, for every $i \in \mathrm{Ob}\, I$, there exists an index $j \in \mathrm{Ob}\, I$ whose corresponding subobject contains $s(Z_i)$ and $Z_i \circ Z_i$, where s is the symmetry of $X \times X$.

g) Suppose that in C finite projective limits and finite sums are representable, that every equivalence relation there is effective, and that every morphism there factors as an effective epimorphism followed by an effective monomorphism. Show that equivalence relations in $\mathrm{Ind}(C)$ are universal if and only if C satisfies the following condition:

ST) For every object X of C and every subobject Z of $X \times X$ in C, if the sequence of subobjects Z_n $(n \geq 0)$ of $X \times X$ in C is defined recursively by $Z_\circ = Z$, $Z_n = \mathrm{Sup}(Z_n, s(Z_n), Z_n \circ Z_n)$ (where Sup is taken in the set of subobjects of $X \times X$, which exists by the hypotheses made on C), then the sequence $(Z_n)_{n\geq 0}$ is *stationary.*

k) Show that equivalence relations in Ind((Ens)) are not necessarily effective.

* NB. For criteria ensuring that $\mathrm{Ind}(C)$ is a topos, cf. VI 8.9.9.$_*$

117 Exercise 8.9.10. Let C be a $\mathscr{U}$-category. Put $\mathrm{Pro}(C) = (\mathrm{Ind}(C^\circ))^\circ$ (cf. 8.11).

a) Suppose that finite sums (resp. small sums) are representable in C. Show that if they are disjoint (cf. II 4.5), then $\mathrm{Pro}(C)$ satisfies the same condition.

b) Let J be a small set such that sums indexed by J are representable in C, and hence also in $\mathrm{Pro}(C)$. Let $(X(\alpha))_{\alpha\in J}$ be a family of objects of $\mathrm{Pro}(C)$, with $X(\alpha) =$ « $\varprojlim_{I_\alpha}$ » X_{i_α}. Show that, for the sum of the $X(\alpha)$ in $\mathrm{Ind}(C)$ to be universal, it suffices that the same hold for each of the families $\varprojlim_J X_{i_\alpha}$, where $(i_\alpha)_{a\in I} \in \prod_{\alpha\in J} I_\alpha$. Deduce that, if in C sums of type J are universal, then for the same to hold in $\mathrm{Pro}(C)$ it is necessary and sufficient that, for every family $(X(\alpha))_{\alpha\in J}$ as above, with $I_\alpha = I$ for every $\alpha \in J$, the canonical homomorphism in $\mathrm{Pro}(C)$

$$\text{«} \varprojlim_{I} \text{»} \coprod_\alpha X(\alpha)_i \longleftarrow \text{«} \varprojlim_{I^J} \text{»} \coprod_{\alpha\in J} X(\alpha)_{i_\alpha}$$

be an isomorphism. Deduce that in Pro((Ens)), sums of type J are universal if and only if J is finite.

Exercise 8.9.11. Let C be a $\mathscr{U}$-category.

a) Let $(X_i \to X)_{i\in I}$ be a family of morphisms in C. Show that, for it to be epimorphic in $\mathrm{Ind}(C)$, it suffices that there exists a *finite* subfamily that is epimorphic in C. Prove that this condition is also necessary when it is assumed that in C every finite family of morphisms with target X_i factors as an epimorphic family,
118 followed by an effective monomorphism (10.5). (For this last assertion, let P be the set of finite subsets of I, ordered by inclusion, and let $X' = (X'_J)_{J\in P}$ be the ind-object formed by the images of the finite subfamilies of the given family;

finally, let $X'' = X \amalg_{X'} X = « \lim » X \amalg_{X'_J} X$. Show that the two canonical morphisms $X \rightrightarrows X''$ are distinct, but coincide on the X'_J.)

b) Suppose that every finite family of morphisms with the same target in C factors as an epimorphic family followed by an effective monomorphism. Prove that $\mathrm{Ind}(C)$ admits a small subcategory generating by epimorphisms (7.1) if and only if C admits a small subcategory C' such that every object of C is the target of a *finite* epimorphic family with source in C'. If C is assumed to admit a small generating subcategory (7.1), then $\mathrm{Ind}(C)$ admits a small subcategory generating by epimorphisms if and only if C is equivalent to a small category. (For this last statement, use 7.5.2).

c) $\mathrm{Ind}((\mathrm{Ens}))$ does not admit a small generating subcategory.

Exercise 8.9.12. Let C be a $\mathscr{U}$-category, and consider $\mathrm{Pro}(C) = \mathrm{Ind}(C^\circ)^\circ$.

a) Show that if $\mathrm{Pro}(C)$ admits a small subcategory P' generating by epimorphisms (7.1), then there exists a small subcategory C' of C that is generating by epimorphisms in $\mathrm{Pro}(C)$. (Take the category of components of the objects of P'.)

b) Show that the category $\mathrm{Pro}((\mathrm{Ens}))$ has no small subcategory generating by epi- 119
morphisms. (If C' is as in a), choose a set X whose cardinality strictly exceeds the cardinalities of the objects of C', and consider the pro-set $\mathscr{X}$ formed by the complements in X of subsets of cardinality $< \mathrm{card}(X)$. Show that, for every nonempty set Y of cardinality $< \mathrm{card}(X)$, one has $\mathrm{Hom}(Y, \mathscr{X}) = \varnothing$, but that $\mathscr{X}$ is not isomorphic to the constant pro-set $\varnothing$.)

8.10. Dual notions: pro-objects, pro-representable functors. Let C be a $\mathscr{U}$-category. A *pro-object* of C is a functor

$$\varphi : I^\circ \longrightarrow C, \tag{8.10.1}$$

where I is an essentially small filtered category (called the index category), and, as usual, I° denotes the opposite category. Let $\mathscr{V}$ be a universe such that $\mathscr{V} \supset \mathscr{U}$. One should note that the pro-objects of C indexed by $I \in \mathscr{V}$ are in one-to-one correspondence with the ind-objects of C° indexed by $I \in \mathscr{V}$: to every such ind-object $\psi : I \to C^\circ$ one associates the pro-object $\varphi = \psi^\circ : I^\circ \to C$. Following this correspondence, one defines "by transport of structure and reversal of arrows" the notion of a morphism between pro-objects of C from the analogous notion (8.2.4.2), (8.2.5.1) for ind-objects, and consequently defines the category of pro-objects of C indexed by $I \in \mathscr{V}$, together with a canonical isomorphism

$$\mathrm{Pro}_{\mathscr{V}}(C, \mathscr{U}) \simeq \mathrm{Ind}_{\mathscr{V}}(C^\circ, \mathscr{U})^\circ; \tag{8.10.2}$$

for $\mathscr{V} = \mathscr{U}$ one simply speaks of the *category of pro-objects of* C, denoted $\mathrm{Pro}_{\mathscr{U}}(C)$ or 120
simply $\mathrm{Pro}(C)$, which is thus defined by

$$\mathrm{Pro}(C) = \mathrm{Pro}_{\mathscr{U}}(C, \mathscr{U}) \simeq \mathrm{Ind}(C^\circ)^\circ. \tag{8.10.3}$$

If two pro-objects are given in *indexed form*

$$\mathscr{X} = (X_i)_{i \in I} \quad , \quad \mathscr{Y} = (Y_j)_{j \in J}, \tag{8.10.4}$$

formula (8.2.5.1) takes here the form

$$\mathrm{Hom}(\mathscr{X}, \mathscr{Y}) \simeq \varprojlim_j \varinjlim_i \mathrm{Hom}(X_i, Y_j). \tag{8.10.5}$$

The functor (8.4.1), applied to C°, gives, upon passing to opposite categories, a canonical functor (denoted by the same letter c when there is no risk of confusion) that identifies C with a full subcategory of $\mathrm{Pro}(C)$:

$$c : C \longrightarrow \mathrm{Pro}(C); \tag{8.10.6}$$

it is in order to have such a functor, rather than $C \to \mathrm{Pro}(C)^\circ$, that we "reversed the arrows" in the definition of morphisms of pro-objects starting from the analogous definition for ind-objects. Pro-objects in the essential image of (8.10.6) are again called *essentially constant pro-objects.*

Put

$$\check{C} = (C^\circ)\hat{} = \mathbf{Hom}(C, (\mathrm{Ens})), \tag{8.10.7}$$

then the functor (8.2.4.7) may be regarded as a canonical fully faithful functor

$$L : \mathrm{Pro}_{\mathscr{V}}(C, \mathscr{U}) \longrightarrow \check{C}^\circ, \tag{8.10.8}$$

121 and in particular one obtains

$$L : \mathrm{Pro}(C) \hookrightarrow \check{C}^\circ. \tag{8.10.9}$$

We leave it to the reader to translate, as the need arises, the results of the preceding sections and of those that follow from the language of ind-objects into that of pro-objects. Depending on the mathematical context, one language or the other is more useful, not to mention the cases in which both notions arise simultaneously, for example when one must consider complex categories such as $\mathrm{Pro}(\mathrm{Ind}(C))$ or $\mathrm{Ind}(\mathrm{Pro}(C))$. We confine ourselves to giving a few additional indications, in order to fix the terminology and notation and to make the "yoga" precise.

8.10.10. A *covariant functor* $F : C \to (\mathrm{Ens})$ is called a *pro-representable functor* if it belongs to the essential image of (8.10.9) (or of 8.10.8, which amounts to the same thing); the full subcategory of $\check{C}$ formed by these functors is therefore equivalent to $\mathrm{Pro}(C)^\circ$. In general one will prefer to work in the opposite category, the essential image *as a subcategory of* (8.10.9), which is therefore equivalent to $\mathrm{Pro}(C)$ itself. In accordance with this usage, it is often preferable to regard covariant functors

$$F : C \longrightarrow (\mathrm{Ens})$$

as objects of $\mathbf{Hom}(C, (\mathrm{Ens}))^\circ = \check{C}^\circ$, and consequently to write the category of morphisms

$$X \longrightarrow F \text{ in } \check{C}^\circ,$$

122 i.e. of pairs (X, u) consisting of an object X of C and an element $u \in F(X)$, as

$${}_F\backslash C\ . \tag{8.10.11}$$

With this convention, one thus obtains a *covariant* functor (source functor)

$${}_F\backslash C \to C, \tag{8.10.12}$$

and 3.4 is rewritten in the form

$$F \xrightarrow{\sim} \text{“}\varprojlim_{({}_F\backslash C)^\circ}\text{”}\, X \tag{8.10.13}$$

8.10.14. Criterion 8.3.3, applied to C°, becomes a pro-representability criterion for F: it is necessary and sufficient that $({}_F\backslash C)^\circ$ be filtered and essentially small, the latter condition being superfluous if C is equivalent to a small category; if, moreover, finite projective limits are representable in C, this is equivalent to saying that F commutes with them, i.e. is *left exact.*

8.10.15. In $\mathrm{Pro}(C)$ small filtered projective limits are representable and the functor (8.10.9) commutes with them; in other words, this functor carries projective limits in $\mathrm{Pro}(C)$ to *inductive* limits in $\check{C} = \mathbf{Hom}(C, (\mathrm{Ens}))$, just like its composite $C \to \check{C}^\circ$ with (8.10.6), which shares with it the unfortunate property of being contravariant when regarded as taking values in $\check{C}$. One should note that pro-representable functors from C 123
to (Ens) are the functors that are small filtered *inductive* limits of representable functors (and not projective limits, as the terminology might perhaps suggest).

8.11. Ind-adjoints and pro-adjoints.

8.11.1. Let

$$f : C \longrightarrow C' \tag{8.11.1.1}$$

be a functor between $\mathscr{U}$-categories, whence a functor $F \mapsto F \circ f$

$$f^* : \widehat{C}' \longrightarrow \widehat{C}. \tag{8.11.1.2}$$

We say that f *admits an ind-adjoint* if the preceding functor carries ind-representable functors to ind-representable functors. Since f^* commutes with inductive limits, and since the full subcategory of $\widehat{C}'$ formed by the ind-representable functors is stable under small filtered inductive limits, it is equivalent to say that f admits an ind-adjoint, or that f^* carries objects of C' (identified with a full subcategory of $\widehat{C}'$) to ind-representable functors on C, i.e.

$$X \longmapsto \mathrm{Hom}(f(X), Y') \text{ is ind-representable for every } Y' \in \mathrm{Ob}\, C'. \tag{8.11.1.3}$$

Another way to express the condition that f admit an ind-adjoint is to say that there exists a functor

$$g : \mathrm{Ind}(C') \longrightarrow \mathrm{Ind}(C) \tag{8.11.1.4}$$

essentially induced by f^*, i.e. such that there is an isomorphism of bifunctors

$$\mathrm{Hom}_{\mathrm{Ind}(C)}(X, g(\mathscr{Y}')) \simeq \mathrm{Hom}_{\mathrm{Ind}(C')}(f(X), \mathscr{Y}'), \tag{8.11.1.5}$$

where $X \in \mathrm{Ob}\, C$ and $\mathscr{Y}' \in \mathrm{Ob}\, \mathrm{Ind}(C')$. In fact, it even suffices to have a functor 124

$$g_\circ : C' \longrightarrow \mathrm{Ind}(C) \tag{8.11.1.6}$$

and an isomorphism of bifunctors

$$\mathrm{Hom}_{\mathrm{Ind}(C)}(X, g_\circ(Y')) \simeq \mathrm{Hom}_{C'}(f(X), Y') \tag{8.11.1.7}$$

in $X \in \mathrm{Ob}\, C$, $Y' \in \mathrm{Ob}\, C'$. The functor g (respectively $g_\circ$) is evidently determined up to unique isomorphism by f, and conversely f can be reconstructed up to canonical isomorphism from the knowledge of this g (respectively $g_\circ$). It is clear from (8.11.1.5) that the functor g commutes with inductive limits and that it "extends" the functor $g_\circ$; this therefore determines it up to unique isomorphism in terms of $g_\circ$ (8.7.2). The functor g, and sometimes also the functor $g_\circ$ that it extends, is called the *ind-adjoint functor of* f (there is no need here to specify: on the right, since the other one, if it exists, will be called the pro-adjoint; cf. 8.11.5 below).

Of course, when f admits a right adjoint

$$f^{\mathrm{ad}} : C' \longrightarrow C,$$

it admits an ind-adjoint g, and the latter is canonically isomorphic to the canonical extension of f^{ad} to ind-objects:

$$g \simeq \mathrm{Ind}(f^{\mathrm{ad}}). \tag{8.11.1.8}$$

The notion of an ind-adjoint is thus a natural generalization of the notion of a right adjoint.

125 Now consider the canonical extension

$$\mathrm{Ind}(f) : \mathrm{Ind}(C) \longrightarrow \mathrm{Ind}(C'). \tag{8.11.1.9}$$

We then have:

Proposition 8.11.2. *For the functor $f : C \to C'$ to admit an ind-adjoint, it is necessary and sufficient that the functor* $\mathrm{Ind}(f)$ (8.11.1.9) *admit a right adjoint. The latter is canonically isomorphic to the ind-adjoint* (8.11.1.4).

Sufficiency and the last assertion are immediate from the ind-adjunction formula (8.11.1.5). For necessity, note that by passing to the projective limit of this formula over the X_i, one obtains an adjunction isomorphism for the ind-objects $\mathscr{X} = (X_i)_{i\in I}$ and $\mathscr{Y}'$:

$$\mathrm{Hom}_{\mathrm{Ind}(C)}(\mathscr{X}, g(\mathscr{Y}')) \simeq \mathrm{Hom}_{\mathrm{Ind}(C')}(\mathrm{Ind}(f)(\mathscr{X}), \mathscr{Y}'), \tag{8.11.2.1}$$

C.Q.F.D.

Corollary 8.11.3. *If f admits an ind-adjoint, then* $\mathrm{Ind}(f)$ *commutes with inductive limits, and the ind-adjoint g commutes with projective limits.*

Proposition 8.11.4. *For the functor $f : C \to C'$ to admit an ind-adjoint, it is necessary that f be right exact, and this condition is sufficient when C is equivalent to a small category.*

This follows from criterion (8.11.1.3), and from 8.3.1 and 8.3.3 (iv).

For another criterion in terms of the notion of an accessible functor, cf. 8.13.3.

126 8.11.5. Now consider the functor $F \mapsto F \circ f$

$$f^{\circ *} : \check{C}' \longrightarrow \check{C} \tag{8.11.5.1}$$

induced by f. We say that f *admits a pro-adjoint* if the preceding functor carries pro-representable functors to pro-representable functors, i.e. if there exists a functor (called the *pro-adjoint functor of f*)

$$g : \mathrm{Pro}(C') \longrightarrow \mathrm{Pro}(C), \tag{8.11.5.2}$$

and an isomorphism of bifunctors

$$\mathrm{Hom}_{\mathrm{Pro}(C)}(g(\mathscr{Y}'), \mathscr{X}) \simeq \mathrm{Hom}_{\mathrm{Pro}(C')}(\mathscr{Y}', f(\mathscr{X})). \tag{8.11.5.3}$$

Of course, to say that f admits a pro-adjoint means that f° admits an ind-adjoint, so that the notions and results for ind-adjoints translate immediately into terms of pro-adjoints. We merely point out that f admits a pro-adjoint if and only if $\mathrm{Pro}(f) : \mathrm{Pro}(C) \to \mathrm{Pro}(C')$ admits a *left* adjoint, and that in this case the preceding functor g is such a left adjoint of $\mathrm{Pro}(f)$; and that for this it is necessary that f be *left* exact, this condition also being sufficient when C is equivalent to a small category. In this case, f is therefore exact if and only if it admits both an ind-adjoint and a pro-adjoint.

Example 8.11.6. Consider the case of a functor

$$f : C \longrightarrow (\mathrm{Ens}).$$

If this functor admits a pro-adjoint, it is pro-representable, and the converse is true if and only if the full subcategory of $\check{C}$ formed by the pro-representable functors is stable
127 under small products; this is the case in particular if C is equivalent to a small category (8.10.14) or if small products are representable in C (8.9.5 b), applied to C°). Up to minor

qualifications, one may thus say that for a functor $f : C \to C'$, the existence of a pro-adjoint is the *natural generalization of the notion of pro-representability of* f, which is defined when $C' = (\mathrm{Ens})$.

8.12. Strict ind-objects and pro-objects. Application to a representability criterion.

8.12.1. Let

$$\mathscr{X} = (X_i)_{i \in I}$$

be an ind-object of the $\mathscr{U}$-category C, and let F be the presheaf that it ind-represents. It is then immediately clear that it is equivalent to say that the canonical morphisms

$$X_i \longrightarrow F = \text{“}\varinjlim_i\text{”}\, X_i$$

are monomorphisms in $\mathrm{Ind}(C)$ (or, equivalently, in $\widehat{C}$, i.e. monomorphisms of functors “argument by argument”), or to say that for every arrow $i \to j$ of I, the corresponding transition arrow

$$X_i \longrightarrow X_j$$

is a monomorphism. When these conditions are satisfied, and if moreover I is an ordered category, we say that $\mathscr{X}$ is a *strict ind-object.* One should note that this condition 128
is not invariant under isomorphism of ind-objects; an ind-object will be called *essentially strict* if it is isomorphic to a strict ind-object. A presheaf will be called *strictly ind-representable* if it is ind-representable by a strict ind-object; thus $F = \text{“}\varinjlim_i\text{”}\, X_i$ is strictly ind-representable if and only if $\mathscr{X} = (X_i)_{i \in I}$ is essentially strict.

. Let F be a presheaf on C, and consider the full subcategory of $C_{/F}$ formed by the arrows $X \to F$ with source in C that are monomorphisms. We shall call it the *category of representable subfunctors of* F; it is the category associated with the ordered set of representable subfunctors of F, ordered by the order induced from that of the set of subobjects of F. The following then follows immediately from the definitions:

Proposition 8.12.2. *For the presheaf F on C to be strictly ind-representable, it is necessary and sufficient that the (ordered) category I of representable subfunctors of F be filtered and essentially small, and that one have*

$$\varinjlim_I X_i \xrightarrow{\sim} F. \tag{8.12.2.1}$$

When, for every object X of C, the set of subobjects of X in C is small (for example, if C admits a small generating subcategory (7.4)), this implies that the category of representable subfunctors is itself small.

. If F is strictly ind-representable, there is thus a preferred way of ind-representing 129
it by an ind-object, and the latter is even a strict ind-object: one takes the representation (8.12.2.1). A strict ind-object $\mathscr{Y}$ is called *saturated* if it is isomorphic to an ind-object of the form (X_i) occurring in (8.12.2.1), for a suitable F; thus, up to unique isomorphism, there exists a single saturated strict ind-object isomorphic to the given strict ind-object, namely the one considered in 8.12.2, taking $F = \varinjlim \mathscr{Y}$ (the limit in $\widehat{C}$).

. Suppose that it is already known that one can find a small cofinal subset in the set $\mathrm{Ob}\, C_{/F}$, as is the case in particular if F is ind-representable; it then follows that the same condition is satisfied in the full subcategory I considered in 8.12.2, so in criterion 8.12.2 one may omit the condition that I be essentially small.

8.12.3. Let F be a presheaf on C, and consider an object

$$u : X \longrightarrow F$$

of $C_{/F}$. We say that u (or the pair (X, u)) is *minimal* if, for every factorization of u as

$$X \xrightarrow{p} X' \xrightarrow{u'} F, \tag{8.12.3.1}$$

with p a strict epimorphism (10.2), p is an isomorphism. Regarding u as an element of
130 $F(X)$ (1.4), to say that u is minimal thus means that every strict epimorphism $p : X \to X'$ such that $u \in \operatorname{Im}(F(p) : F(X') \to F(X))$ is an isomorphism. This notion is clarified by part a) of the following lemma:

Lemma 8.12.4. *Let F be a presheaf on C.*

a) *Suppose that F carries cokernels to kernels, and consider a morphism*

$$u : X \longrightarrow F,$$

with $X \in \operatorname{Ob} C$. For u to be a monomorphism, it is sufficient, when cokernels of pairs of arrows are representable in C, that u be minimal; this condition is also necessary if fiber products are representable in C.

b) *Suppose that finite $\varinjlim$ are representable in C. For the subcategory of representable subfunctors of F (8.12.1.1) to be filtered (not necessarily small) and to have F as its inductive limit, it is sufficient that F be left exact and that every morphism $u : X \to F$, with $X \in \operatorname{Ob} C$, factor as*

$$X \to X' \xrightarrow{u'} F,$$

with u' minimal; this condition is also necessary if fiber products are representable in C.

c) *Suppose that C admits a small generating subcategory (7.1), and that I, the category of representable subfunctors of F, is filtered; then I is small.*

131 *Proof.*

a) Suppose u is minimal; let us prove that u is a monomorphism, i.e. that for every pair of arrows $v, v' : Y \rightrightarrows X$ such that $uv = uv'$, one has $v = v'$. Indeed, if $X' = \operatorname{Coker}(v, v')$, then u factors as $X \xrightarrow{p} X' \xrightarrow{u'} F$ (since F carries cokernels to kernels), and since p is a strict epimorphism by construction, it follows that p is an isomorphism, i.e. $v = v'$. Suppose that u is a monomorphism; let us prove that it is minimal. Consider a factorization (8.12.3.1); since p is a strict epimorphism, it is the cokernel of the canonical pair of arrows $v, v' : X \times_{X'} X \rightrightarrows X$, and since $(u'p)v = (u'p)v'$ and $u'p = u$ is a monomorphism, one has $v = v'$, hence p is an isomorphism.

b) Sufficiency: Since F is left exact, $C_{/F}$ is filtered. Since we know that $F = \varinjlim_{C_{/F}} X$, by 8.1.3 c) it remains to prove that the full subcategory I of $C_{/F}$ consisting of the subobjects of F is cofinal in $C_{/F}$; by the "sufficient" part of a), this is precisely what is ensured by the hypothesis that every object of $C_{/F}$ is dominated by a "minimal" object. Necessity: Since every filtered inductive limit of right exact functors is again right exact, the first condition is trivially necessary. The second then follows from the "necessary" part of a).

c) A representable subfunctor $X \hookrightarrow F$ of F is known once one knows the full subcategory $C'_{/X}$ of $C'_{/F}$, where C' is a fixed small generating subcategory of C. (Use the hypothesis that I is filtered.) Since $C'_{/F}$ is small, the set of its full subcategories is small, whence the conclusion.

Proposition 8.12.5. *Let C be a $\mathcal{U}$-category in which finite inductive limits are representable, and which admits a small generating subcategory (7.1). Let F be a presheaf on C. For F to be strictly ind-representable, it is sufficient that it satisfy the following two conditions, and these conditions are also necessary if fiber products are representable in C:* 132

a) *F is left exact.*
b) *Every pair (X, u), with $X \in \operatorname{Ob} C$ and $u \in F(X)$, is dominated in $C_{/F}$ by a minimal pair (8.12.3), i.e. there exists a minimal pair (X', u') $(X' \in \operatorname{Ob} C, u' \in F(X'))$ and a morphism $f : X \to X'$ such that $u = F(f)(u')$.*

Sufficiency follows from 8.12.2 and 8.12.4 b), c), and necessity from 8.3.1 and 8.12.4 a).

Remark 8.12.6. When F carries amalgamated sums in C to fiber products, it is immediately clear that for a given pair (X, u) as in b), the set of strict quotients X' of X such that $u \in \operatorname{Im}(F(X') \to F(X))$ is filtered decreasingly; this implies that if the set of strict quotients of X is Artinian, then condition b) of 8.12.5 is automatically satisfied. Thus, if the preceding condition on X is satisfied for every object X of C (one then also says, sometimes, that the objects of C° are *Artinian*), it follows from 8.12.5 that F is strictly ind-representable if and only if F is left exact; in this case, F is therefore strictly ind-representable as soon as it is ind-representable (8.3.1).

Corollary 8.12.7. *Let C be a $\mathcal{U}$-category in which small projective limits are representable, and which admits a small cogenerating subcategory (7.9, 7.13). Then a functor $F : C \to (\mathrm{Ens})$ is representable if and only if F commutes with small projective limits.* 133

Necessity is clear. For sufficiency, apply 8.12.5 to the opposite category C°. To prove first that F is (strictly) pro-representable, it remains to show that every pair (X, u), $X \in \operatorname{Ob} C$ and $u \in F(X)$, is dominated by a "minimal" pair. Now the family $(X_i)_{i \in I}$ of *strict* subobjects of X such that $u \in \operatorname{Im}(F(X_i) \to F(X))$ is small (7.5 in dual form). Since F is left exact, this family is cofiltered (8.12.6), and since F commutes with small projective limits, one sees that $X' = \varprojlim_i X_i$ is a least object of this family. (Use the fact that, being left exact, F carries monomorphisms to monomorphisms.) If $u' \in F(X')$ is the unique element whose image in $F(X)$ is u, one then sees that (X', u') is a minimal pair dominating (X, u). This proves that F is pro-representable, and the conclusion then follows from the

Lemma 8.12.8.1. *Let $F : C \to (\mathrm{Ens})$ be a functor, where C is a $\mathcal{U}$-category in which small projective limits are representable. For F to be representable, it is necessary and sufficient that it be pro-representable and commute with small projective limits.*

Necessity is clear; let us prove sufficiency. To say that F is representable evidently means that ${}_F\backslash C$ admits an initial object (in fact, the initial objects of ${}_F\backslash C$ are precisely the isomorphisms $F \to X$, i.e. the representation data for F). Now, since F is pro-representable, $({}_F\backslash C)^\circ$ is filtered and equivalent to a small category, so $X = \varprojlim_{({}_F\backslash C)^\circ} X$ is representable in C, and since F commutes with the limit under consideration, it follows that X is an initial object of ${}_F\backslash C$, C.Q.F.D. 134

COROLLARY 8.12.8. *Under the hypotheses on C of 8.12.7, let $f : C \to C'$ be a functor from C to a $\mathscr{U}$-category C'. For f to admit a left adjoint, it is necessary and sufficient that f commute with small projective limits.*

Indeed, this reduces immediately to 8.12.7, by applying that statement to composite functors of the form $X \mapsto \operatorname{Hom}(Y', f(X))$.

EXAMPLES 8.12.9. As was pointed out in 7.13, the hypotheses on C in 8.12.7 and 8.12.8 are satisfied if C is the category of $\mathscr{U}$-sheaves of sets on a topological space $X \in \mathscr{U}$ (or, more generally, on a $\mathscr{U}$-site (II (3.0.2)). We give an instructive example[3] showing that the hypothesis of the existence of a small cogenerating subcategory D of C is not superfluous in 8.12.7. Take for C the category of groups that are elements of $\mathscr{U}$. Let J be the set of isomorphism classes of simple groups in C; for every $j \in J$ choose a simple group G_j in the class j, and let I be the filtered ordered set of $\mathscr{U}$-small subsets of J, and, for $i \in I$,
135 let $X_i = \amalg_{j\in i} G_j$. The X_i then form a projective system $(X_i)_{i\in I}$ in C, whose transition morphisms are epimorphisms, and the corresponding functor

$$(*) \qquad F(X) = \varinjlim_{i} \operatorname{Hom}(X_i, X)$$

takes its values in ($\mathscr{U}$-Ens) (although the index set I clearly does not have cardinality in $\mathscr{U}$); more precisely, let us show that for every small full subcategory $C_\circ$ of C, there exists an $i_\circ \in I$ such that the restriction of F to $C_\circ$ is represented by $X_{i_\circ}$, which will prove both that F takes values in $\mathscr{U}$-Ens and that it commutes with small projective limits. To prove our assertion, it suffices to note that for every $X \in \operatorname{Ob} C_\circ$, the cardinality of the set $J(X)$ of $j \in J$ for which there exists a nontrivial homomorphism from G_j to X is necessarily small, since such a morphism is necessarily a monomorphism (G_j being simple); consequently, if $i_\circ$ is the subset of J given by the union of the $J(X)$ for $X \in \operatorname{ob} C_\circ$, then $i_\circ$ is small, i.e. $i_\circ \in I$, and it has the required property. On the other hand, it is clear that F is not representable, since $\operatorname{card} I \notin \mathscr{U}$. From this and 8.12.7 one therefore concludes that *the category C of groups in $\mathscr{U}$ admits no small full cogenerating subcategory.* Since, on the other hand, the object $\mathbf{Z}$ of C is a generator, it then follows from the proof of 7.12 that there exists a two-generator group G that cannot be embedded in an injective object of the category C of groups in $\mathscr{U}$. It moreover seems plausible that C has no injective objects other than the trivial groups.

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8.13. Pro-representable functors and accessible functors.

8.13.1. In this section, we use some notions and results from the following section, in particular 9.11 and 9.13, to obtain a pro-representability criterion that we shall use (incidentally) in IV 9.16. In what follows, C denotes a $\mathscr{U}$-category satisfying condition L of 9.1 b), this condition being satisfied, for example, if small filtered inductive limits are representable in C.

PROPOSITION 8.13.2. *Let C be as above, and let*

$$f : C \longrightarrow (\mathrm{Ens})$$

be a functor.

a) *Suppose that every object of C is accessible (9.3). If f is pro-representable, then f is accessible (9.2) and left exact.*
b) *Suppose that finite projective limits are representable in C, and that C admits a cardinal filtration (9.12). If f is accessible and left exact, then f is pro-representable.*

[3](due to H. BASS).

Proof.

a) The hypothesis on C means that the covariant representable functors from C to (Ens) are accessible. The same is therefore true of every small inductive limit of such functors (9.6 (i)), and hence also of every pro-representable functor.
b) By 8.3.3 (iii), it remains to prove that $\mathrm{Ob}({}_F\backslash C)^{\circ}$ contains a small cofinal subcategory. By hypothesis, there exists a cardinal Π such that f is Π-accessible. Let (X,u), $u \in F(X)$, be an object of ${}_F\backslash C$. With the notation of 9.12 c), we then have 137
$X = \varinjlim_i X_i$, with I filtered and large relative to Π and the X_i in $C' = \mathrm{Filt}^{\Pi}(C)$, whence $F(X) \xleftarrow{\sim} \varinjlim_i F(X_i)$. This shows that the small subcategory $({}_F\backslash C')^{\circ}$ is cofinal in $({}_F\backslash C)^{\circ}$, and completes the proof.

Corollary 8.13.3. *Let C be a $\mathscr{U}$-category satisfying the following conditions:*

a) *Finite projective limits are representable in C.*
b) *Small filtered inductive limits are representable in C.*
c) *The functor* Ker *on the category of double arrows of C is accessible (for example, it commutes with small filtered $\varinjlim$).*
d) *Every strict epimorphism of C is a universal strict epimorphism (10.2).*
e) *There exists a small subcategory of C generating by strict epimorphisms (7.1).*

Under these conditions, a functor $f : C \to$ (Ens) is pro-representable if and only if it is left exact and accessible (9.2).

Indeed, the conditions of 8.13.2 a) and b) on C are satisfied, by 9.11 and 9.13 respectively.

Corollary 8.13.4. *Let $f : C \to C'$ be a functor between $\mathscr{U}$-categories satisfying conditions* a) *through* e) *of 8.13.3. For f to admit a pro-adjoint, it is necessary and sufficient that f be left exact and accessible.*

Since the representable functors $h_{Y'} : X' \to \mathrm{Hom}(Y', X') : C' \to$ (Ens) are 138
left exact and accessible (9.11), if f has the same properties, so do its composites with the preceding functors; these are therefore pro-representable by 8.13.3, i.e. f admits a pro-adjoint. Conversely, suppose that f admits a pro-adjoint, and let C'_1 be a small generating subcategory of C, and Π a cardinal such that the $Y' \in \mathrm{Ob}\, C'$ are Π-accessible; to prove that f is Π-accessible, it therefore suffices to prove the same of its composites with the $h_{Y'}$, $Y' \in \mathrm{Ob}\, C'$; by hypothesis, these composites are pro-representable, hence accessible (8.13.3), and hence Π-accessible provided Π is taken sufficiently large, C.Q.F.D.

9. Accessible functors, cardinal filtrations, and the construction of small generating subcategories

The present section, which is more technical in nature than the other sections of this exposé, will be used in this seminar only in IV 9 and VI 4, which are not used elsewhere in the Seminar. The reader should therefore omit the present section, at least on a first reading!

9.0. All categories considered in this section are assumed to be $\mathscr{U}$-categories. Except for the small index categories $I, J \dots$ that we shall have occasion to use, the developments that follow will apply primarily to "large" categories $E, F \dots$ that are stable under small filtered inductive limits. Most often, however, it will suffice that a slightly weaker condition be satisfied (condition L in 9.1 below). All cardinals considered in the 139 present section are assumed to belong to $\mathscr{U}$.

Following a suggestion of P. Deligne, we shall study, for a functor $f : E \to F$ between large categories, a condition that f commute with certain types of filtered inductive limits. This condition is remarkably stable, and is satisfied by the most important functors occurring in nature. The applications we have in mind for our seminar are 9.13.3, 9.13.4 (used in VI 4), and especially 9.25, which gives, in a nontrivial case, the existence of a small generating family in a category of sections of a fibered category; this result will be used in IV 9.16.

Definition 9.1. a) *Let I be a preordered set and pi a cardinal. We say that I is large relative to pi if I is filtered and every subset of I of cardinality $\leq \pi$ has an upper bound in I.*

b) *Let E be a category. If pi is a cardinal, we say that E satisfies condition L_π if, for every small ordered set I large relative to π, E is stable under inductive limits of type I. We say that E satisfies condition L if there exists a cardinal $\pi \in \mathscr{U}$ such that E satisfies condition L_π.*

9.1.1. When, in 9.1 a), one has $\pi \geq 2$, the second stated condition already implies that I is filtered, and if π is finite, for I to be large relative to π simply means that I is filtered. In what follows we shall be concerned almost exclusively with the case in which π is infinite. Note that if π, π' are two cardinals such that $\pi' \geq \pi$, then I being large relative to π' evidently implies that it is large relative to π.

140 9.1.2. As announced in 9.0, conditions L_π and L should be regarded as technical variants of the stronger condition of stability under small filtered inductive limits. It is clear that if π, π' are cardinals such that $\pi' \geq \pi$, then condition L_π implies condition $L_{\pi'}$.

Definition 9.2. *Let $f : E \to F$ be a functor. If Π is a cardinal, we say that f is Π-preaccessible (respectively Π-accessible) if E satisfies L_Π (9.1) and if, for every ordered set $I \in \mathscr{U}$ large relative to Π, and every inductive system $(X_i)_{i\in I}$ in E of type I, the canonical morphism*

$$\varinjlim f(X_i) \longrightarrow f(\varinjlim X_i)$$

is a monomorphism (respectively an isomorphism). We say that f is preaccessible (respectively accessible) (relative to the universe $\mathscr{U}$) if there exists a cardinal $\Pi \in \mathscr{U}$ such that f is Π-preaccessible (respectively Π-accessible).

The category of Π-accessible (respectively accessible) functors from E to F will be denoted $\mathbf{Hom}(E, F)_\Pi$, respectively $\mathbf{Hom}(E, F)_{\mathrm{acc}}$.

9.2.1. Evidently, a functor commuting with small filtered $\varinjlim$ (for example, a functor admitting a right adjoint) is Π-accessible for every cardinal $\Pi \geq 2$.

Definition 9.3. *Let E be a category, X an object of E,*

$$h^\circ_X : E \longrightarrow (\mathscr{U}\text{-Ens})$$

the covariant functor it represents, and $\Pi \in \mathscr{U}$ a cardinal. We say that X is a Π-
141 preaccessible (respectively Π-accessible) object of E *if the functor h°_X is Π-preaccessible (respectively Π-accessible); we say that X is* preaccessible (respectively accessible) *if there exists a cardinal $\Pi \in \mathscr{U}$ such that X is a Π-preaccessible (respectively Π-accessible) object of E.*

9.3.1. We denote by E_Π the full subcategory of E formed by the Π-accessible objects of E.

When Definition 9.3 is applied to a category of the form $\mathbf{Hom}(C, F)$, the terminology just introduced presents *a priori* an ambiguity with the analogous terminology introduced in 9.2, when the objects of $\mathbf{Hom}(C, F)$ are interpreted as functors; there does not, however, seem to be any serious risk of confusion.

Definition 9.4. *Let E be a category and let $\pi \in \mathcal{U}$ be a cardinal. We say that E is a π-preaccessible (respectively π-accessible) category if there exists in E a small full subcategory C that is generating (7.1) and whose objects are π-preaccessible (respectively π-accessible) (9.3). We say that E is a preaccessible (respectively accessible) category if there exists a cardinal $\pi \in \mathcal{U}$ such that E is π-preaccessible (respectively π-accessible).*

For important examples, cf. 9.11.3 below.

Proposition 9.5. *Let $f : E \to F$ be a functor between categories such that F is accessible and every object of E is accessible. Then, if f admits a left adjoint, f is accessible.*

Indeed, by hypothesis, F admits a small conservative family of representable func- 142
tors $F \to (\mathcal{U}\text{-}\mathrm{Ens})$ that are accessible, so we are immediately reduced to showing that the composite of f with each of the preceding functors is accessible. But since f admits a left adjoint, these composites are functors $E \to (\mathcal{U}\text{-}\mathrm{Ens})$ that are *representable,* and hence accessible by hypothesis on E.

Proposition 9.6. *Let E and F be two categories, let $\pi \in \mathcal{U}$ be a cardinal, and consider the full subcategory $\mathbf{Hom}(E, F)_\Pi$ of $\mathbf{Hom}(E, F)$ formed by the Π-accessible functors (9.2) from E to F.*

- (i) *This subcategory is stable under every type of $\varinjlim$ that is representable in F.*
- (ii) *Suppose that F is π-accessible (9.4), and let J be a category such that* $\operatorname{card} F\ell\, J \leq \pi$ *and that the $\varprojlim$ of type J are representable in F, so the $\varprojlim$ of type J are representable in $\mathbf{Hom}(E, F)$. Then the subcategory $\mathbf{Hom}(E, F)_\Pi$ is stable under the $\varprojlim$ of type J.*

Corollary 9.7. *Let E and F be two categories, with F accessible (9.4). Then the full subcategory $\mathbf{Hom}(E, F)_{\mathrm{acc}}$ of $\mathbf{Hom}(E, F)$ formed by the accessible functors is stable under every type of inductive or projective limit, indexed by a small category J, that is representable in F (and hence in $\mathbf{Hom}(E, F)$).*

Proof of 9.6. Assertion (i) follows trivially from the commutation of the functor $\varinjlim$ with arbitrary inductive limits. For (ii), let $(f_j)_{j \in J}$ be a projective system of Π-accessible functors $E \to F$, and let $f = \varprojlim f_j$ be its projective limit, computed "argument by 143
argument." We prove that f is Π-accessible, i.e. that for every ordered set $I \in \mathcal{U}$ large relative to Π, and every inductive system $(X_i)_{i \in I}$ in E, the canonical morphism

$$\varinjlim_i((\varprojlim_j f_j)(X_i)) \longrightarrow (\varprojlim_j f_j)(\varinjlim_i X_i)$$

is an isomorphism. Now the "argument-by-argument" computation of the functor $\varprojlim_j f_j$ allows us to identify the preceding canonical morphism with the canonical morphism

$$\varinjlim_i \varprojlim_j f_j(X_i) \longrightarrow \varprojlim_j \varinjlim_i f_j(X_i)$$

associated with the bifunctor

$$(i, j) \longmapsto f_j(X_i) : I \times J \longrightarrow F.$$

Thus 9.6 is a consequence of the following more general assertion:

Corollary 9.8. *Let F be a Π-accessible category, I an ordered set large relative to Π, J a small category such that* card $F\ell \le \Pi$ *and such that projective limits of type J are representable in F, and $h : I \times J \to F$ a functor; then the canonical morphism*

$$\varinjlim_i \varprojlim_j h(i,j) \longrightarrow \varprojlim_j \varinjlim_i h(i,j) \tag{9.8.1}$$

is an isomorphism. In other words, the functor

$$\varinjlim_i : \mathbf{Hom}(I,F) \longrightarrow F \tag{9.8.2}$$

144 *commutes with projective limits of type J (for every small category J such that* card $F\ell(J) \le \Pi$ *and such that projective limits of type J are representable in F). Equivalently, for every J as above, the functor*

$$\varinjlim_j : \mathbf{Hom}(J,F) \longrightarrow F \tag{9.8.3}$$

is Π-accessible.

By hypothesis, F admits a conservative family of covariant representable Π-accessible functors $g : F \to (\mathscr{U}\text{-Ens})$, so it is immediately clear that it suffices to prove 9.8 in the case $F = \mathscr{U}\text{-Ens}$. We shall then prove the assertion in the form that (9.8.2) commutes with projective limits of type J, where card $F\ell\, J \le \Pi$. Since I is filtered, we know that (9.8.2) commutes with finite projective limits (2.8). By a standard argument (cf. 2.3), it therefore remains to prove that it commutes with products indexed by a set J such that card $J \le \Pi$. This reduces us to proving the bijectivity of (9.8.1) when J is discrete. Let us prove injectivity: consider two elements a, b of the first member; they therefore come from $\prod_j h(i_\circ, j)$ for a suitable $i_\circ \in I$, say $\prod_j a(i_\circ, j)$ and $\prod_j b(i_\circ, j)$; suppose that the elements $\prod_j h(\infty, j)$ of the second member that they define are equal, i.e. that for every j, there exists $i(j) \ge i_\circ$ such that $a(i_\circ, j)$ and $b(i_\circ, j)$ have the same image in $h(i,j)$. Since I is large relative to card J, it follows that there exists a common upper bound $i_1 \in I$ of all the $i(j)$, which implies that the two elements under consideration of $\prod_j h(i_\circ, j)$ have the same image in $\prod_j h(i_1, j)$ and therefore define the same element of the first member
145 of (9.8.1), establishing injectivity. For surjectivity, let $\prod_j a(\infty, j)$ be an element of the second member; thus, for every $j \in J$, $a(\infty, j)$ comes from an element of $h(i(j), j)$ for a suitable $i(j) \in I$. As before, one can find a common upper bound $i \in I$ of the $i(j)$, so $\prod_j a(\infty, j)$ comes from an element $\prod_j a(i,j)$ of $\prod_j h(i,j)$, and hence lies in the image of (9.8.1). This completes the proof.

Corollary 9.9. *Let F be a Π-accessible (respectively accessible) category. Then, for every category J such that* card $F\ell\, J \le \Pi$ *(respectively every small category J), and for every functor $(X_j)_{j \in J} : J \to F$ whose projective limit in F is representable, if the X_j are Π-accessible (respectively accessible), then so is $\varprojlim X_j$.*

Indeed, the functor $F \to (\mathscr{U}\text{-Ens})$ represented by $\varprojlim X_j$ is the projective limit of the functors represented by the X_j, and one applies 9.6 (ii) to the projective system formed by these functors.

Remark 9.10. In the statements 9.6, 9.7, 9.8, and 9.9, one may everywhere replace the words "Π-accessible", "accessible" by "Π-preaccessible", "preaccessible". Indeed, the proof given also proves this variant of the preceding statements.

Proposition 9.11. *Let E be a category satisfying condition L (9.1), and let C be a small full generating subcategory (7.1). Suppose that the following condition is satisfied:*

a) *E is stable under kernels of pairs of arrows, and the functor* Ker *on the category of pairs of arrows in E is accessible (for example, it commutes with small filtered* 146
inductive limits).

Then every object of E is preaccessible (9.3), a fortiori E is preaccessible. Suppose that C is even generating by strict epimorphisms (7.1), and suppose that E satisfies the following conditions:

b) *E is stable under fiber products.*

c) *Every small strictly epimorphic family in E is universally strictly epimorphic (10.3), or small direct sums are representable in E and every strict epimorphism of E is a universal strict epimorphism.*

Then every object of E is accessible, a fortiori E is accessible.

The fact that, under (a), every object of E is preaccessible follows from the fact that for every object X of E, the set of strict subobjects of X is small (7.4), and from the

Lemma 9.11.1. *Under the conditions of (9.11a), let $X \in \operatorname{ob} E$ and let Π be a cardinal such that E satisfies L_Π (9.1), the functor* Ker *in (9.11a) is Π-accessible, and the set of strict subobjects of X has cardinality bounded by Π. Then X is Π-preaccessible.*

Indeed, let I be an ordered set large relative to Π, let $(Y_i)_{i\in I}$ be an inductive system in E with inductive limit Y, and let $(u_i, v_i : X \rightrightarrows Y_i)$ be a pair of arrows such that the composite pair $X \overset{u,v}{\rightrightarrows} Y$ satisfies $u = v$; let us prove that there exists $j \geq i$ in I such that $u_j = v_j$. To this end, consider the inductive system of pairs of arrows $(u_j, v_j :$ 147
$X \rightrightarrows Y_j)_{j\geq i}$, whose inductive limit is (u, v). By hypothesis one has $X = \operatorname{Ker}(u, v) = \varinjlim_j \operatorname{Ker}(u_j, v_j) = \varinjlim_j X_j$, where $X_j = \operatorname{Ker}(u_j, v_j)$. Now the X_j are strict subobjects of X, so there exists a subset I' of I, formed by indices $j \geq i$, such that $\operatorname{card} I' \leq \Pi$ and every X_j is equal to one of the $X_{i'}$ $(i' \in I')$. Since I is large relative to Π, there exists an upper bound j of I' in I. Then X_j contains all the $X_{j'}$ for $j' \geq i$, so $X_j \to X$ is an epimorphism (since the family of the $X_{j'} \to X$ is epimorphic), and hence an isomorphism because it is a strict monomorphism. Thus $u_j = v_j$, which proves 9.11.1.

The second assertion of 9.11 follows from the first and from the

Lemma 9.11.2. *Under the conditions of (9.11a), b), c), let X be an object of E, and let Π be an infinite cardinal such that E satisfies L_Π (9.1), that* $\operatorname{card}\operatorname{ob} C_{/X} \leq \Pi$*, that for two objects $X' \to X$ and $X'' \to X$ of $C_{/X}$, $X' \times_X X''$ is Π-preaccessible, and finally that X is Π-preaccessible. Then X is Π-accessible.*

With the notation of the proof of 9.11.1, it suffices to prove that every morphism $u : X \to Y$ comes from a morphism $u_i : X \to Y_i$. Consider the family of morphisms $Y_i \to Y$, which is strictly epimorphic; by b) and c), the family of morphisms $Y_i \times_Y X \to X$ is also strictly epimorphic; on the other hand, for every i, since C is generating by strict epimorphisms, the family of arrows

$$T \longrightarrow Y_i \times_Y X$$

with source in C is strictly epimorphic. In the second alternative considered in c), one 148
can find such a strictly epimorphic arrow whose source is a (small) sum of objects of C. By the transitivity of the notion of a universally strictly epimorphic family (II 2.5), it then follows that the family of arrows $T_\alpha \xrightarrow{f_\alpha} X$ with source in C that factor through one of

the $Y_i \times_Y X$ is strictly epimorphic. Let J be the index set of this family; its cardinality is bounded by $\operatorname{card} \operatorname{ob} C_{/X} \leq \Pi$. For every $\alpha \in J$, choose an $i = i(\alpha) \in I$ and an X-morphism $T_\alpha \to Y_i \times_Y X$, or, equivalently, a $v_\alpha : T_\alpha \to Y_i$ such that $p_i v_\alpha = u f_\alpha$, where $p_i : Y_i \to Y$ is the canonical morphism. Since I is large relative to Π, one can choose $i(\alpha)$ independently of α, say equal to i. For every pair of indices $\alpha, \beta \in J$, consider the composites

$$\mathrm{pr}_1 v_\alpha, \mathrm{pr}_2 v_\beta : T_\alpha \times_X T_\beta \rightrightarrows Y_i.$$

Their composites with $Y_i \to Y$ are equal, so, since $T_\alpha \times_X T_\beta$ is Π-preaccessible by hypothesis, there exists an index $i' = i(\alpha, \beta) \geq i$ such that the composites of the arrows under consideration with $Y_i \to Y_{i'}$ are equal. Since the set of pairs α, β has cardinality $\leq \Pi^2 = \Pi$ (Π being infinite), it follows once again that i' can be chosen independently of α, β. We may evidently suppose that $i' = i$. But then, since the family $(f_\alpha : T_\alpha \to X)$ is strictly epimorphic, one can find a morphism $u_i : X \to Y_i$ such that $u_i f_\alpha = v_\alpha$. One then has $p_i u_i = u$, since for every α one has $(p_i u_i) f_\alpha = p_i(u_i f_\alpha) = p_i v_\alpha = u f_\alpha$, and the family of the f_α is epimorphic. This completes the proof of 9.11.2.

149 Remark 9.11.3. As a special case of 9.11, one sees that every object in the category E is accessible in each of the following two cases: 1) E is a $\mathscr{U}$-abelian category with exact filtered inductive limits and admitting a small generating subcategory (here it is the *second* alternative of c) that applies). 2) E is the category of sheaves of sets on a topological space $X \in \mathscr{U}$. More generally, it suffices that E be the category of sheaves of sets on a $\mathscr{U}$-site (II 2.1), or that E be a $\mathscr{U}$-topos (IV 1.1).

Definition 9.12. *Let E be a category. A cardinal filtration of E is an increasing filtration $(\mathrm{Filt}^\Pi(E))_{\Pi \geq \Pi_\circ}$ of E by strictly full subcategories $\mathrm{Filt}^\Pi(E)$, indexed by the cardinals $\Pi \in \mathscr{U}$ such that $\Pi \geq \Pi_\circ$ (where $\Pi_\circ$ is a fixed infinite cardinal, depending on the cardinal filtration under consideration), and satisfying the following conditions:*

a) *For every $\Pi \geq \Pi_\circ$, $\mathrm{Filt}^\Pi(E)$ is equivalent to a small category.*
b) *E satisfies $L_{\Pi_\circ}$ (9.3), and for every $\Pi \geq \Pi_\circ$, $\mathrm{Filt}^\Pi(E)$ is stable in E under filtered inductive limits indexed by ordered sets I large relative to $\Pi_\circ$ such that $\operatorname{card} I \leq \Pi$.*
c) *For every $\Pi \geq \Pi_\circ$, and every $X \in \operatorname{ob} E$, one can find an isomorphism*

$$X \simeq \varinjlim_I X_i,$$

150 *where $(X_i)_{i \in I}$ is an inductive system in E indexed by an ordered set I large relative to Π (9.1), and where the X_i belong to $\mathrm{Filt}^\Pi(E)$. Moreover, if $X \in \operatorname{ob} \mathrm{Filt}^\Pi(E)$, $\Pi' \geq \Pi$, one may choose I such that $\operatorname{card} I \leq \Pi'^\Pi$.*

9.12.1. Note that it follows from b) and c) that every $X \in \operatorname{ob} E$ belongs to some $\mathrm{Filt}^\Pi(E)$, for Π sufficiently large.

Example 9.12.2. Take $E = (\mathscr{U}\text{-Ens})$, $\Phi_\circ = \aleph_\circ$, $\mathrm{Filt}^\Pi(E) =$ the full subcategory of E consisting of the sets such that $\operatorname{card} X \leq \Pi$. More generally:

Proposition 9.13. *Let E be a $\mathscr{U}$-category. Suppose that E is stable under small filtered $\varinjlim$, under sums of two objects, and under cokernels of pairs of arrows, that E is stable under fiber products, and that the epimorphisms in E are universal strict epimorphisms. Let C be a small full subcategory of E that is generating by strict epimorphisms. Let $\Pi_\circ$ be an infinite cardinal $\geq \operatorname{card} F\ell\, C$. For every cardinal $\Pi \geq \Pi_\circ$, let $\mathrm{Filt}^\Pi(E)$ be the strictly full subcategory of E consisting of the objects X of E for which there exists a strictly epimorphic* *family* $(X_i \to X)_{i\in I}$ *with target* X*, such that* $\operatorname{card} I \leq \Pi$ *and* $X_i \in \operatorname{ob} C$ *for every* $i \in I$*. Then* $(\operatorname{Filt}^{\Pi}(E))_{\Pi\geq\Pi_\circ}$ *is a cardinal filtration of* E*. Moreover, for every* $X \in \operatorname{ob}\operatorname{Filt}^{\Pi}(E)$*, the cardinality of the set of arrows of* $C_{/X}$ *(and a fortiori, of the set of objects of* $C_{/X}$*) is bounded by* $\Pi^{\Pi_\circ}$*.*

This last assertion is precisely 7.6.

To show that this is a cardinal filtration, one must prove conditions a), b), and c) of 9.12. Condition a) follows at once from 7.5.2. For b), suppose that $X = \varinjlim_I X_i$, with $\operatorname{card} I \leq \Pi$ and the $X_i \in \operatorname{ob}\operatorname{Filt}^{\Pi}(E)$. Thus the family of canonical morphisms $X_i \to X$ 151
is strictly epimorphic, and by the hypothesis on the X_i, for every $i \in I$ there exists a strictly epimorphic family $(X_{ij} \to X_i)_{j\in J_i}$, with $\operatorname{card} J_i \leq \Pi$. Taking the sums $\amalg_i X_i$ and $\amalg_{i,j} X_{ij}$, one sees by the transitivity of universal strict epimorphisms (II 2.5) that the family of composites $X_{ij} \to X_i \to X$ (indexed by the disjoint union of the J_i, for $i \in I$) is strictly epimorphic. But $\operatorname{card} J \leq \Pi^2 = \Pi$, whence the conclusion. (NB. we have used only the fact that $\varinjlim_I X_i$ exists, and not the fact that I is large relative to $\Pi_\circ$, or even that it is filtered.) Finally, let us prove c). Note that for every $X \in \operatorname{ob} E$, the family of arrows $X_i \to X$ with source in $\operatorname{ob} C$ is strictly epimorphic, since C is generating by strict epimorphisms, and is plainly small, so there exists a cardinal $\Pi \geq \Pi_\circ$ such that $X \in \operatorname{ob}\operatorname{Filt}^{\Pi}(E)$. It remains to prove that if $\Pi' \geq \Pi \geq \Pi_\circ$ are cardinals, and if $X \in \operatorname{ob}\operatorname{Filt}^{\Pi'}(E)$, then there is an isomorphism

$$X \simeq \varinjlim_{I} X_i,$$

with I large relative to Π, $\operatorname{card} I \leq \Pi'^{\Pi}$, and the X_i in $\operatorname{ob}\operatorname{Filt}^{\Pi}(E)$. Now, since C is generating by strict epimorphisms, one has

$$(*) \qquad X \simeq \varinjlim_{C_{/X}} T.$$

Let us also note that by successively adjoining to C sums and cokernels of pairs of arrows in C, and doing the same for the category thus obtained, and so on, one reduces to the case where C is stable under sums of two objects in E and under cokernels of pairs of arrows in E, without losing the hypothesis $\Pi_\circ \geq \operatorname{card} F\ell\, C$. We may suppose moreover that, if E contains a chosen initial object $\varnothing_E$, then $\varnothing_E \in \operatorname{ob} C$. This implies that the 152
category $C_{/X}$ is stable under sums of two objects and cokernels of pairs of arrows. It is then filtered, since it is nonempty: if it were empty, the relation $(*)$ would show that X is an initial object, and hence $C_{/X}$ would contain the arrow $\varnothing_E \to X$, a contradiction. Let I be the set, ordered by inclusion, of full subcategories i of $C_{/X}$ that are filtered and such that $\operatorname{card}\operatorname{ob} i \leq \Pi$. For every $i \in I$, let $X_i \in \operatorname{ob} E$ be the inductive limit of the composite functor $i \to C_{/X} \to E$, which exists by hypothesis. Then plainly

$$\varinjlim_{I} X_i \simeq \varinjlim_{C/X} T \simeq X.$$

On the other hand, we have already noted that $\operatorname{card}\operatorname{ob} C_{/X} \leq \Pi'^{\Pi_\circ}$. It follows that I is large relative to Π, and that $\operatorname{card} I \leq (\Pi'^{\Pi_\circ})^{\Pi} = \Pi'^{\Pi\Pi_\circ} = \Pi'^{\Pi}$ by the following lemma, whose proof is left to the reader (where we take $I = C_{/X}$, $c = \Pi'^{\Pi_\circ}$):

Lemma 9.13.1. *Let* J *be a filtered category, and let* c *and* Π *be two cardinals such that* $c \geq \operatorname{Sup}(\operatorname{card} F\ell\, J, \Pi)$*, and suppose that* $\operatorname{card}\operatorname{Hom}(j, J') \leq \Pi_\circ$ *for every pair of objects* j, j' *of* J*. Let* I *be the set of full filtered subcategories* i *of* J *such that* $\operatorname{card}\operatorname{ob} i \leq \Pi$*. Then, ordered by inclusion,* I *is large relative to* Π*, and one has* $\operatorname{card} I \leq c^{\Pi}$*.*

This completes the proof of 9.13.

153 REMARK 9.13.2. A slight additional effort should make it possible to replace in 9.13 the hypothesis that E is stable under small filtered inductive limits by the hypothesis that E satisfies L (9.1), provided E is stable under small sums, or that in E every epimorphic *family* is universally epimorphic. In the proof of c), one must then choose $\Pi_\circ$ such that E satisfies $L_{\Pi_\circ}$, and restrict oneself to those full subcategories i of $C_{/X}$ that are not only filtered, but such that the preordered set ob i is large relative to $\Pi_\circ$. Using Section 8 and the hypothesis that E satisfies $L_{\Pi_\circ}$, one then finds that the X_i exist, and one should be able to conclude by a suitable variant of 9.13.1, which the editor has not verified.

PROPOSITION 9.14. *Let $f : E \to F$ be an accessible functor (9.2) between two categories equipped with cardinal filtrations $(\mathrm{Filt}^{\Pi}(E))_{\Pi \geq \Pi_\circ}$ and $(\mathrm{Filt}^{\Pi}(F))_{\Pi \geq \Pi'_\circ}$. Then there exists a cardinal $\Pi_1 \geq \mathrm{Sup}(\Pi_\circ, \Pi'_\circ)$ such that for every cardinal $\Pi \geq \Pi_1$, one has*

$$f(\mathrm{Filt}^{\Pi}(E)) \subset \mathrm{Filt}^{\Pi^{\Pi_1}}(F). \tag{9.14.1}$$

In particular, if $\Pi = 2^c$ with $c \geq \Pi_1$, whence $\Pi^{\Pi_1} = 2^{c\Pi_1} = 2^c = \Pi$, one has

$$f(\mathrm{Filt}^{\Pi}(E)) \subset \mathrm{Filt}^{\Pi}(F). \tag{9.14.2}$$

We immediately point out the

154 COROLLARY 9.15. *Let E be a category equipped with two cardinal filtrations $(\mathrm{Filt}^{\Pi}(E))_{\Pi \geq \Pi_\circ}$ and $(\mathrm{Filt'}^{\Pi}(E))_{\Pi \geq \Pi'_\circ}$. Then there exists a cardinal $\Pi_1 \geq \mathrm{Sup}(\Pi_\circ, \Pi'_\circ)$ such that, for every cardinal $c \geq \Pi_1$, putting $\Pi = 2^c$, one has $\mathrm{Filt}^{\Pi}(E) = \mathrm{Filt'}^{\Pi}(E)$.*

Proof of 9.14. Let c be a cardinal such that $c \geq \mathrm{Sup}(\Pi_\circ, \Pi'_\circ)$ and such that f is c-accessible. Let, moreover, $\Pi_1 \geq c$ be such that

$$f(\mathrm{Filt}^{c}(E)) \subset \mathrm{Filt}^{\Pi_1}(F). \tag{$*$}$$

Such a Π_1 exists because $\mathrm{Filt}^c(E)$ is equivalent to a small category (9.12 a)), and hence so is $f(\mathrm{Filt}^c(E))$, so that one can apply 9.12.1 to the objects of the latter to find a $\mathrm{Filt}^{\Pi}(F)$ containing them all (taking into account that the $\mathrm{Filt}^{\Pi}(F)$ are full subcategories). Let then $\Pi \geq \Pi_1$ and $X \in \mathrm{ob}\,\mathrm{Filt}^{\Pi}(E)$; let us prove that $f(X) \in \mathrm{Filt}^{\Pi^{\Pi_1}}(F)$. Indeed, write

$$X = \varinjlim_{I} X_i,$$

where the $X_i \in \mathrm{ob}\,\mathrm{Filt}^c(E)$, I is large relative to c, and $\mathrm{card}\, I \leq \Pi^c \leq \Pi^{\Pi_1}$ (9.12c)). Since f is c-accessible and I is large relative to c, one has

$$f(X) \simeq \varinjlim_{I} f(X_i),$$

and since $\mathrm{card}\, I \leq \Pi^{\Pi_1}$ and $f(X_i) \in \mathrm{ob}\,\mathrm{Filt}^{\Pi_1}(F) \subset \mathrm{ob}\,\mathrm{Filt}^{\Pi^{\Pi_1}}(F)$ by $(*)$, one has $f(X) \in \mathrm{Filt}^{\Pi^{\Pi_1}}(F)$ by 9.12 b), C.Q.F.D.

9.15.1. The notion of cardinal filtration 9.12 is of interest mainly when the objects of E are accessible. We point out that it follows from 9.11 that this condition is satisfied
155 when, in addition to the hypotheses of 9.13, one supposes that the functor Ker on the category of pairs of arrows in E is accessible. We also point out:

PROPOSITION 9.16. *Let E be a category equipped with a cardinal filtration $(\mathrm{Filt}^{\Pi}(E))_{\Pi \geq \Pi_\circ}$. Suppose that the objects of $\mathrm{Filt}^{\Pi_\circ}(E)$ are accessible objects of E; then there exists a cardinal*

$\Pi_1 \geq \Pi_\circ$ in $\mathcal{U}$ such that the objects of $\mathrm{Filt}^{\Pi_\circ}(E)$ are Π_1-accessible; if Π_1 is chosen in this way, then for every cardinal $\Pi \geq \Pi_1$, we have (with the notation of 9.3.1):

$$E_\Pi \subset \mathrm{Filt}^{\Pi}(E) \subset E_{(\Pi^{\Pi_\circ})}. \tag{9.16.1}$$

The existence of Π_1 follows immediately from the fact that $\mathrm{Filt}^{\Pi_\circ}(E)$ is equivalent to a small category (9.12a)). Now let $\Pi \geq \Pi_1$. If X belongs to $\mathrm{Filt}^{\Pi}(E)$, write $X \simeq \varinjlim_I X_i$ with $\mathrm{card}\, I \leq \Pi^{\Pi_\circ}$, $X_i \in \mathrm{ob}\,\mathrm{Filt}^{\Pi_\circ}(E)$ for every i (9.12c)). It then follows from 9.9 that X is $\Pi^{\Pi_\circ}$-accessible, whence the second inclusion (9.16.1). Suppose that X is Π-accessible, and write $X \simeq \varinjlim_I X_i$, with I large relative to Π and the $X_i \in \mathrm{ob}\,\mathrm{Filt}^{\Pi}(E)$ (9.12c)). By the hypothesis on X, the given isomorphism $X \xrightarrow{\sim} \varinjlim_I X_i$ factors through one of the X_i, so X is isomorphic to a direct factor of that X_i. We conclude that $X \in \mathrm{ob}\,\mathrm{Filt}^{\Pi}(E)$, and hence obtain the first inclusion (9.16.1), by the following lemma.

Lemma 9.16.2. *Every object of E that is a direct factor of an object of $\mathrm{Filt}^{\Pi}(E)$ belongs to $\mathrm{Filt}^{\Pi}(E)$.*

Indeed, if X is the image of a projector p on an object Y of E (i.e. an endomorphism p such that $p^2 = p$), and if I is an ordered filtered set, X is the inductive limit of the 156
filtered inductive system $(Y_i)_{i\in I}$ defined by $Y_i = Y$ for every $i \in I$, $p : Y_i \to Y_j$ if $i < j$. Taking I large relative to $\Pi_\circ$ and $\mathrm{card}\, I \leq \Pi$, we thus see that if $Y \in \mathrm{Filt}^{\Pi}(E)$, then the same holds for X, by 9.12 b).

Corollary 9.17. *Under the conditions of 9.16, for every cardinal $c \geq \Pi_1$, setting $\Pi = 2^c$, $\mathrm{Filt}^{\Pi}(E)$ coincides with the strictly full subcategory E_Π of E consisting of the Π-accessible objects.*

Corollary 9.18. *Let E be a category satisfying condition L_Π (9.1), where Π is a cardinal, and let C be a full subcategory of E. Denote by $\mathrm{Ind}(C)_\Pi$ the full subcategory of the category $\mathrm{Ind}(C)$ of ind-objects of C (8.2) consisting of ind-objects of the form $(X_i)_{i\in I}$, where I is an ordered set large relative to Π. Consider the canonical functor*

$$(X_i)_{i\in I} \longmapsto \varinjlim_I X_i \,:\, \mathrm{Ind}(C)_\Pi \longrightarrow E. \tag{9.18.1}$$

a) *This functor is fully faithful if and only if every object X of C is a Π-accessible object (9.3) of E.*
b) *Assume the conditions of 9.17, in particular $\Pi = 2^c$, and take $C = \mathrm{Filt}^{\Pi}(E)$. Then the functor (9.18.1) is an equivalence of categories.*

Assertion a) is an immediate generalization of 8.7.5 a), and is proved in the same way. Part b) then follows from 9.17 and condition 9.12 c) on cardinal filtrations, which implies that the functor under consideration is essentially surjective.

Corollary 9.19. *Under the conditions of 9.17, let $C = \mathrm{Filt}^{\Pi}(E)$, let F be a $\mathcal{U}$-category, and consider the functor* 157

$$\mathbf{Hom}(E, F)_\Pi \longrightarrow \mathbf{Hom}(C, F) \tag{9.19.1}$$

induced by the "restriction to C" functor $f \mapsto f|C$, where the source of (9.19.1) is the category of Π-accessible functors from E to F (9.2). The preceding functor is fully faithful. If F satisfies condition L_Π (9.1), then the functor (9.19.1) is an equivalence of categories.

The first assertion is proved as in 7.8, using 9.18 b). The second is obtained by constructing a quasi-inverse functor to (9.19.1), associating to every functor $f : C \to F$ the functor $(X_i)_{i \in I} \to \varinjlim_i f(X_i)$ from $\mathrm{Ind}(E_\Pi)$ to F, and using 9.18 b) to deduce from it a functor $\overline{f} : E \to F$. It remains only to show that thelatter is Π-accessible. But this is proved in the same way as the analogous assertion 8.7.3.

Corollary 9.20. *Let E be a category admitting a cardinal filtration and such that every object of E is accessible (cf. 9.16), and let F be a category. Then:*

a) *The category $\mathbf{Hom}(E, F)_{acc}$ of accessible functors from E to F (9.2) is a $\mathscr{U}$-category. (*NB *we recall (9.0) that the given categories E, F are assumed to be $\mathscr{U}$-categories.)*
b) *Suppose that F is stable under small filtered inductive limits. For every full subcategory C of E equivalent to a small category, with inclusion functor $i : C \to E$, consider the corresponding functor*

$$i_! : \mathbf{Hom}(C, F) \longrightarrow \mathbf{Hom}(E, F)$$

158 *(5.1). A functor $f : E \to F$ is accessible if and only if there exists a small subcategory C of E such that f belongs to the essential image of the preceding functor $i_!$.*

Proof.

a) It suffices to prove that for every cardinal Π such that E satisfies L_Π, the full subcategory $\mathbf{Hom}(E, F)_\Pi$ of $\mathbf{Hom}(E, F)_{acc}$ is a $\mathscr{U}$-category. It is clearly enough to verify this for cardinals of the form 2^c, with c sufficiently large. But then this follows from 9.19, since ($C = \mathrm{Filt}^\Pi(E)$ being essentially small) $\mathbf{Hom}(C, F)$ is obviously a $\mathscr{U}$-category.
b) By transitivity of the formation of the functors $i_!$, one may restrict in the statement to subcategories C of the form $\mathrm{Filt}^\Pi(E)$, where Π is as in 9.17. One then sees readily that the composite functor $\mathbf{Hom}(\mathrm{Filt}^\Pi(E), F) \to \mathbf{Hom}(E, F)_\Pi \to \mathbf{Hom}(E, F)$, where the first arrow is quasi-inverse to (9.19.1), and the second is the inclusion, is none other than the functor $i_!$, up to isomorphism. Thus assertion b) follows from 9.19.

*Exercise 9.20.1. (The present exercise uses the notions of site and topos, developed in Exposés II and IV.) Let E be a $\mathscr{U}$-topos, let $\mathscr{V}$ be a universe such that $\mathscr{U} \in \mathscr{V}$, let C be a small full generating subcategory of the $\mathscr{U}$-topos E, and let $\Pi_\circ \in \mathscr{U}$ be an infinite cardinal such that $\Pi_\circ \in \mathrm{card}\, F\ell\, C$. For every cardinal $\Pi \geq \Pi_\circ$, $\Pi \in \mathscr{U}$, let $\mathrm{Filt}^\Pi(E)$ be the strictly full subcategory of E consisting of the objects X such that there exists
159 a strictly epimorphic family $(X_i \to X)_{i \in I}$ with target X, such that $\mathrm{card}\, I \leq \Pi$ and $X_i \in \mathrm{Ob}\, C$ for every $i \in I$.

a) Show that $(\mathrm{Filt}^\Pi(E))_{\Pi \geq \Pi_\circ}$ is a cardinal filtration of the $\mathscr{U}$-category E, and that one can choose $\Pi_\circ$ such that for every $\Pi \geq \Pi_\circ$, one has the inclusions

$$E_\Pi \subset \mathrm{Filt}^\Pi(E) \subset E_{\Pi^{\Pi_\circ}},$$

where, for every cardinal c, E_c denotes the strictly full subcategory of E consisting of the c-accessible objects.
b) Choosing a cardinal $\Pi \geq \Pi_\circ$ such that $\Pi = \Pi^{\Pi_\circ}$ (for example, Π of the form 2^c, with $c > \Pi_\circ$), and putting $C = \mathrm{Filt}^\Pi(E)$, show that there is an equivalence of categories

$$\mathrm{Ind}(C)_\Pi \xrightarrow{\approx} E,$$

(notation $\mathrm{Ind}(C)_{\Pi}$ of 9.18).

c) Retain the notation of b), and let $\mathscr{V}$ be a universe such that $\mathscr{U} \subset \mathscr{V}$, let $E^{\sim}_{\underline{V}}$ be the $\mathscr{V}$-topos of $\mathscr{V}$-sheaves on the site E (endowed with its canonical topology). Denote by $\mathrm{Ind}(C, \mathscr{V})_{\Pi}$ the category of $\mathscr{V}$-Ind-objects of C indexed bypreordered index sets *large relative to* Π. Show that there is an equivalence of categories

$$\mathrm{Ind}(C, \mathscr{V})_{\Pi} \xrightarrow{\approx} E^{\sim}_{\mathscr{V}}.$$

d) Let $c \in \mathscr{V}$ be a cardinal, and let $\mathrm{Ind}(E, \mathscr{V})'_c$ be the full subcategory of $\mathrm{Ind}(E, \mathscr{V})$ consisting of the $\mathscr{V}$-ind-objects of E indexed by a preordered set that is large relative to every cardinal $< c$. Taking $c = \operatorname{card} \mathscr{U}$, show that there is an equiv- 160
alence of categories

$$\mathrm{Ind}(E, \mathscr{V})'_c \xrightarrow{\approx} E^{\sim}_{\mathscr{V}}.$$ *

9.21. The present section 9.21 develops technical preliminaries for the proof of Theorem 9.22 below, which is the main result of the present section 9. Let

$$p : E \to B \tag{9.21.1}$$

be a fibered functor, where B is a small category, and where the inverse-image functors

$$f^* : E_\beta \longrightarrow E_\alpha$$

associated with the arrows $f : \alpha \to \beta$ of B are accessible (9.2). In particular, the fiber categories E_α ($\alpha \in \operatorname{ob} B$) satisfy condition L (9.1), and therefore, since B is small, there exists a cardinal $\Pi'_\circ \in \mathscr{U}$ such that all the categories E_α satisfy condition $L_{\Pi'_\circ}$. Let $\Pi_\circ \in \mathscr{U}$ be an infinite cardinal $> \Pi'_\circ$, so that one has

$$\Pi_\circ > \Pi'_\circ,\ E_\alpha \text{ satisfies } L_{\Pi'_\circ} \text{ for every } \alpha \in \operatorname{ob} B. \tag{9.21.2}$$

Moreover, the hypotheses made immediately imply the existence of a cardinal $c \in \mathscr{U}$ satisfying the following conditions:

$$\left\{\begin{array}{ll} \text{a)} & c \text{ is infinite,} \\ \text{b)} & c \geq \operatorname{card} F\ell\, B, \\ \text{c)} & \text{for every arrow } f : \alpha \to \beta \text{ in } B, \\ & \text{the functor } f^* : E_\beta \to E_\alpha \text{ is } c\text{-} \\ & \text{accessible.} \end{array}\right. \tag{9.21.3}$$

Suppose moreover that, for every $\alpha \in \operatorname{ob} B$, one can find a full subcategory C_α of E_α 161
satisfying the following conditions:

$$\left\{\begin{array}{ll} \text{a)} & \text{For every } X \in \operatorname{ob} C_\alpha,\ X \text{ is } c\text{-} \\ & \text{accessible in } E_\alpha \text{ (9.3).} \\ \text{b)} & \text{For every } X \in \operatorname{ob} E_\alpha, \text{ one can find} \\ & \text{an isomorphism } X \simeq \varinjlim_I X_i \text{ in} \\ & E_\alpha, \text{ where the } X_i \text{ belong to } C_\alpha \text{ and} \\ & I \text{ is an ordered set large relative to} \\ & c. \end{array}\right. \tag{9.21.4}$$

Let Π be a cardinal such that

$$\Pi \geq \Pi_\circ \quad , \qquad \text{hence } \Pi > \Pi'_\circ, \tag{9.21.5}$$

and, for every $\alpha \in \operatorname{ob} B$, let

$$C^{\Pi}_\alpha \subset E_\alpha \tag{9.21.6}$$

be formed by the objects that can be represented in the form $\varinjlim_I X_i$, where I is an ordered set large relative to $\Pi'_\circ$, such that card $I \leq \Pi$, and where the X_i belong to C_α. It is then clear, by 7.5.2, that C_α^Π is essentially small, i.e. is equivalent to a small category.

Let

$$F = \mathbf{Hom}_B(B, E) \tag{9.21.7}$$

be the category of sections of E over B, and let

$$F^\Pi \subset F \tag{9.21.8}$$

be the strictly full subcategory of F formed by the sections $X : \alpha \mapsto X(\alpha)$ such that for every $\alpha \in \mathrm{ob}\, B$ one has

$$X(\alpha) \in C_\alpha^\Pi .$$

162 Since the C_α^Π are essentially small, it is clear that the same is true of the category F^Π. We shall show that this category is generating, and more precisely:

Lemma 9.21.9. *Under the preceding conditions and with the preceding notation, the following statements hold:*

(i) *Every object of F is accessible (9.3). If d is a cardinal $\geq \Pi'_\circ$, and if $\alpha \mapsto X(\alpha)$ is an element of F such that, for every α, $X(\alpha)$ is d-accessible in E_α, then X is d-accessible in F.*

(ii) *Suppose that $\Pi \leq c$, or that $\Pi'_\circ$ is finite (i.e. the E_α are stable under small filtered $\varinjlim$). Then every object X of F is isomorphic to an object of the form $\varinjlim_I X_i$, where the X_i belong to F^Π and I is large relative to Π.*

To prove (i), note that by (9.21.4) and 9.9, every object of E_α is accessible. On the other hand, F satisfies condition $L_{\Pi'_\circ}$ by the

Lemma 9.21.10. *Let $p : E \to B$ be a fibered functor, let $F = \mathbf{Hom}_B(B, E)$, let I be a category, and let $i \mapsto X_i$ be a functor from I to F. For $\varinjlim_I X_i$ to be representable in F, it suffices that, for every $\alpha \in \mathrm{ob}\, B$, $\varinjlim_I X_i(\alpha)$ be representable in the fiber category E_α; when this is so, $\varinjlim_I X_i$ is "computed argument by argument". In particular, if the fiber categories satisfy condition $L_{\Pi'_\circ}$ (where $\Pi'_\circ$ is a given cardinal), then so does F.*

163 We leave the details of the (easy) proof of 9.21.10 to the reader, merely noting that it is convenient to use the following result, whose proof is immediate:

Corollary 9.21.10.1. *With the hypotheses and notation of 9.21.10 for $p : E \to B$ and for I, for every $\alpha \in \mathrm{ob}\, B$, the inclusion functor $E_\alpha \to E$ commutes with inductive limits of type I.*

Returning now to the general conditions of 9.21.9, let X be an object of F. Since, for every $\alpha \in \mathrm{ob}\, B$, $X(\alpha)$ is accessible in E_α, there exists a cardinal $d \in \mathcal{U}$ such that, for every α, $X(\alpha)$ is d-accessible in E_α. We may choose $d \geq \Pi'_\circ$, so that F satisfies L_d by 9.21.10. Using 9.21.10 once more to compute the inductive limits $\varinjlim_I Y_i$ in F, with I large relative to d, one immediately sees that X is d-accessible. This proves (i).

We shall now prove 9.21.9 (ii) in several steps ((9.21.11) to (9.21.16)).

Let then

$$X : \alpha \longmapsto X(\alpha)$$

be an object of F. By (9.21.4 b)), for every $\alpha \in \mathrm{ob}\, B$ one can find an ordered set I_α large relative to c, and an isomorphism $X(\alpha) = \varinjlim_{i \in I_\alpha} X(\alpha)_i$, where $i \mapsto X(\alpha)_i$ is an inductive system of type I_α in E_α, with the $X(\alpha)_i$ in C_α. Consider the product ordered set $I = \Pi_\alpha I_\alpha$; it is clear that it is large relative to c, and that the preceding inductive systems give rise to inductive systems in the E_α, 164

$$i \longmapsto X(\alpha)_i \in \operatorname{ob} C_\alpha, i \in \operatorname{ob} I, \tag{9.21.11}$$

indexed by the same ordered set I, and to isomorphisms

$$X(\alpha) \xrightarrow{\sim} \varinjlim_I X(\alpha)_i \text{ in } E_\alpha, \tag{9.21.12}$$

for every $\alpha \in \operatorname{ob} B$. In what follows we suppose that data (9.21.11) and (9.21.12) have been fixed.

Lemma 9.21.13. *Under the preceding conditions, one can find a map $\varphi : I \to I$ such that $\varphi(i) \geq i$ for every $i \in I$, and a map $(i, f) \mapsto \lambda(i, f)$ from $I \times \mathrm{Fl}\, B$ to $\mathrm{Fl}\, E$, satisfying the following conditions:*

a) *For $i \in I$ and $(f : \alpha \to \beta) \in F\ell B$, $\lambda(i, f)$ is an f-morphism from $X(\alpha)_i$ to $X(\beta)_{\varphi(i)}$, making the diagram*

$$\begin{tikzcd}
X(\alpha) \arrow[rr, "X(f)"] & & X(\beta) \\
& & X(\beta)_{\varphi(i)} \arrow[u] \\
X(\alpha)_i \arrow[uu] \arrow[urr, "\lambda(i,f)"] & &
\end{tikzcd}$$

$$\alpha \xrightarrow{\quad f \quad} \beta \quad ,$$

commutative, where the vertical arrows are the canonical morphisms deduced from (9.21.12).

b) *For every $i \in I$, and every $(f : \alpha \to \beta) \in \mathrm{Fl}\, B$, the diagram* 165

$$\begin{tikzcd}
& & X(\beta)_{\varphi^2(i)} \\
X(\alpha)_{\varphi}(i) \arrow[urr, "\lambda(\varphi(i),f)"] & & X(\beta)_{\varphi(i)} \arrow[u] \\
X(\alpha)_i \arrow[u] \arrow[urr, "\lambda(i,f)"'] & &
\end{tikzcd}$$

$$\alpha \xrightarrow{\quad f \quad} \beta \quad .$$

is commutative.

c) *For every pair of consecutive arrows $\alpha \xrightarrow{f} \beta \xrightarrow{g} \gamma$ of $\boldsymbol{B}$, the following diagram is commutative:*

$$\begin{tikzcd}
X(\alpha) \arrow[r, "X(f)"] & X(\beta) \arrow[r, "X(g)"] & X(\gamma) \\
& & X(\gamma)_{\varphi^2(i)} \arrow[u] \\
& X(\beta)_{\varphi(i)} \arrow[uu] \arrow[ur, "\lambda(\varphi(i),g)"] & X(\gamma)_{\varphi(i)} \arrow[u] \\
X(\alpha)_i \arrow[uuu] \arrow[ur, "\lambda(i,f)"] \arrow[urr, "\lambda(i,gf)"'] & & \\
\alpha \arrow[r, "f"] & \beta \arrow[r, "g"] & \gamma
\end{tikzcd} \quad ,$$

where the vertical arrows are those deduced from (9.21.12).

166 One should moreover note that the commutativity of the two upper trapezoids in c) is already contained in a), so c) in fact asserts the commutativity of the lower triangle of the diagram under consideration.

Let us prove 9.21.13. Let $i \in \boldsymbol{I}$, and let $f : \alpha \to \beta$ be an arrow in $\boldsymbol{B}$. Consider the composite $X(\alpha)_i \to X(\alpha) \xrightarrow{X(f)} X(\beta) \simeq \varinjlim_j X(\beta)_j$; it defines a morphism in $\boldsymbol{E}$:

$$X(\alpha)_i \to f^*(X(\beta)) \simeq f^*(\varinjlim_j X(\beta)_j) \simeq \varinjlim_j f^*(X(\beta)_j),$$

where the last equality follows from the fact that f^* is c-accessible (9.21.3 c)and that $\boldsymbol{I}$ is large relative to c. By (9.21.4 a)), since $X(\alpha)_i \in \operatorname{ob} \boldsymbol{C}_\alpha$, the morphism under consideration factors through some $f^*(X(\beta)_j)$, where j *a priori* depends on i and on $f \in \mathrm{Fl}\,(\boldsymbol{B})$. But by (9.21.3 b)), and since $\boldsymbol{I}$ is large relative to c, one may choose j independently of f, say $j = \varphi(i)$. Since $\boldsymbol{I}$ is filtered, one may suppose $\varphi(i) \geq i$. One thus obtains, for $i \in \boldsymbol{I}$ and $f \in \mathrm{Fl}\,\boldsymbol{B}$, a morphism

$$X(\alpha)_i \longrightarrow f^*(X(\beta)_{\varphi(i)}),$$

or, equivalently, an f-morphism

$$\lambda(i, f) : X(\alpha)_i \longrightarrow X(\beta)_{\varphi(i)},$$

which by construction makes the diagram

$$D(f) \qquad \begin{tikzcd}
X(\alpha) \arrow[r, "X(f)"] & X(\beta) \\
& X(\beta)_{\varphi(i)} \arrow[u] \\
X(\alpha)_i \arrow[uu] \arrow[ur, "\lambda(i,f)"] & \\
\alpha \arrow[r, "f"] & \beta
\end{tikzcd} \quad .$$

167 commutative. Now , for given i, f, g, consider the lower triangle of the diagram in 9.21.13

c); it is not clear that it is commutative, but the two composites $X(\alpha)_i \rightrightarrows X(\gamma)_{\varphi^2(i)}$ become equal after composition with $X(\gamma)_{\varphi^2(i)} \to X(\gamma)$, as follows from the commutativity of the two upper trapezoids and of the outer trapezoid, which are respectively the diagrams $D(f)$, $D(g)$, and $D(gf)$. Thus, since $X(\alpha)_i$ is c-accessible (9.21.4 a)) and $X(\gamma) \simeq \varinjlim_{j\in I} X(\gamma)_j$, with I large relative to c, it follows that one can find an element $j \geq \varphi^2(i)$ of I such that the two arrows under consideration become equal after composition with $X(\gamma)_{\varphi^2(i)} \to X(\gamma)_j$. A priori, j depends on i, f, g. But, for fixed i, the set of possible pairs f, g has cardinality $\leq c^2 = c$, by (9.21.3 a) and b)), and hence, since I is large relative to c, one may choose j independently of f and g, say $j = \varphi'(i) \geq \varphi^2(i) \geq \varphi(i)$. For every $i \in I$ and $f : \alpha \to \beta$, let then

$$\lambda'(i, f) : X(\alpha)_i \longrightarrow X(\beta)_{\varphi'(i)}$$

be the composite f-morphism $X(\alpha)_i \to X(\beta)_{\varphi(i)} \to X(\beta)_{\varphi'(i)}$, where the second arrow is the transition morphism. It is then immediate from the construction that (φ', λ') satisfies conditions 9.21.13 a) and c) for (φ, λ). Proceeding in the same way for condition b), one sees that φ' may be chosen so that this condition is also satisfied for (φ', λ'). This completes the proof of (9.21.13).

9.21.14. Now let **168**

$$J \subset I$$

be a subset of I satisfying the following conditions:

a) J is filtered,
b) for every $j \in J$, one has $\varphi(j) \in J$,
c) For every $\alpha \in \mathrm{ob}\, B$, $X_J(\alpha) = \varinjlim_{i\in J} X(\alpha)_i$ is representable in E_α.

Conditions a) and c) are satisfied in particular if J is large relative to $\Pi'_\circ$ (9.21.2). It then follows from 9.21.10.1 that, for every $\alpha \in \mathrm{ob}\, B$, $X_J(\alpha)$ is the inductive limit $\varinjlim_{i\in J} X(\alpha)_i$ in E. Now, for an arrow $f : \alpha \to \beta$ of B, the arrows $\lambda(i, f)$, with i varying in J, define, by (9.21.13 b)), a morphism of ind-objects from $(X(\alpha)_i)_{i\in J}$ to $(X(\beta)_i)_{i\in J}$, whence a morphism

$$X_J(f) : X_J(\alpha) \to X_J(\beta)$$

on the inductive limits, which is manifestly an f-morphism. Using (9.21.13 (c)), one obtains the transitivity relations

$$X_J(gf) = X_J(g)X_J(f),$$

so that a section $X_J \in \mathrm{ob}\, F$ of E over B has been defined. Finally, (9.21.13 a)) shows us that the canonical morphisms

$$X_J(\alpha) \longrightarrow X(\alpha) \quad , \quad \alpha \in \mathrm{ob}\, B,$$

are functorial in α, so there is a canonical morphism

$$u_J : X_J \longrightarrow X.$$

On the other hand, if **169**

$$J' \supset J$$

is another subset of I satisfying conditions a), b), c) above, one obtains a canonical morphism

$$u_{J',J} : X_J \longrightarrow X_{J'},$$

so that the X_J form an inductive system in F, parametrized by the set (ordered by inclusion) of subsets J of I satisfying the conditions under consideration. Finally, the morphisms u_J above define a morphism from this inductive system to (the constant inductive system defined by) X.

9.21.15. Now let K be a set of subsets J of I satisfying conditions a), b), c) considered in 9.21.14, and suppose that K is filtered and has union I. It is then clear that one has

$$\varinjlim_{J \in K} X_J \xrightarrow{\sim} X,$$

using 9.21.10.1, which reduces us to checking that there is an isomorphism argument by argument.

Take, for example, for K the set of all subsets J of I that are large relative to $\Pi'_\circ$, stable under φ, and such that $\operatorname{card} J \leq \Pi$. Then, by the definition (9.21.6) of C^{Π}_{α}, for every $\alpha \in \operatorname{ob} B$ one has $X_J(\alpha) \in \operatorname{ob} C^{\Pi}_{\alpha}$, hence $X_J \in \operatorname{ob} F^{\Pi}$. Consequently, 9.21.9 (ii) will be proved if we establish that K is large relative to Π (and hence filtered) and has union I. For this, it will evidently suffice to prove that every subset S of I such that
170 $\operatorname{card} S \leq \Pi$ is contained in a $J \in K$. This will indeed follow from the preliminary hypothesis stated in 9.21.9 (ii), and from the

Lemma 9.21.16. *Let I be an ordered set, let Π be an infinite cardinal, and let $\varphi : I \to I$ be a map from I to itself.*

a) *Suppose I is filtered. Then, for every subset S of I such that $\operatorname{card} S \leq \Pi$, there exists a filtered subset $J \supset S$ of I such that $\varphi(J) \subset J$ and $\operatorname{card}(J) \leq \Pi$.*
b) *Let $\Pi'_\circ$ be a cardinal $< \Pi$, and suppose that I is large relative to Π. Then, for every subset S of I such that $\operatorname{card} S \leq \Pi$, there exists a subset $J \supset S$ of I large relative to $\Pi'_\circ$, such that $\varphi(J) \subset J$ and $\operatorname{card} J \leq \Pi$.*

Let us prove b), for example (the proof of a) being similar and simpler). Let P be the set of subsets of I of cardinality $\leq \Pi$, and let $T \mapsto i_T : P \to I$ be a map such that, for $T \in P$, i_T is an upper bound of T in I; the existence of this map merely expresses the hypothesis that I is large relative to Π. Let A be a well-ordered set such that $\operatorname{card} A = \Pi$, and such that, for every $a \in A$, the set of $a' \leq a$ has cardinality $< \Pi$. It follows in particular, since $\Pi'_\circ < \Pi$, that A *is large relative to* $\Pi'_\circ$. Define, by transfinite recursion, maps

$$S \longmapsto S_a : P \longrightarrow P \quad (a \in A),$$

by the following formulas:

$S_\circ = S$, $S_{a+1} = S_a \cup \varphi(S_a) \cup \{i_{S_a}\}$, $S_a = \bigcup_{a'<a} S_{a'}$ if a is a limit ordinal.
171 Then put, for $S \in P$,

$$J(S) = \bigcup_{a \in A} S_a.$$

It is then clear that $J(S) \supset S$, that $J(S)$ is stable under φ, that $\operatorname{card} J(S) \leq \Pi$ (since $\operatorname{card} A = \Pi$), and finally that $J(S)$ is large relative to $\Pi'_\circ$ (using the fact that A is large relative to $\Pi'_\circ$).

This proves 9.21.16 and completes the proof of 9.21.9.

We also point out a variant of 9.21.9:

Lemma 9.21.17. *The notation is that of 9.21.9. Let F' denote the full subcategory $\mathbf{Homcart}_B(B, E)$ of $F = \mathbf{Hom}_B(B, E)$ formed by the cartesian sections of E over B. Then:*

(i) *Every object of F' is accessible. More precisely, let $\alpha \mapsto X(\alpha)$ be an element of F', and let d be a cardinal such that $d \geq \Pi'_\circ$, $d \geq c$, and such that, for every $\alpha \in \operatorname{ob} B$, $X(\alpha)$ is a d-accessible object of E_α. Then X is a d-accessible object of F'.*

(ii) *Suppose $\Pi \leq c$ and that the functors $f^* : E_\beta \to E_\alpha$ are $\Pi'_\circ$-accessible, or that the categories E_α are stable under small filtered inductive limits and the functors $f^* : E_\beta \to E_\alpha$ commute with these inductive limits. Then every object X of F' is isomorphic to an object of the form $\varinjlim_I X_i$, where, for every $i \in I$, X_i is an object of the full subcategory $F'^{\Pi} = F' \cap F^{\Pi}$ of F', and where I is large relative to Π.*

The proof being entirely analogous to that of 9.21.9, we shall merely indicate the 172
points at which the latter must be modified. First note:

Lemma 9.21.18. *The statement of 9.21.10 remains valid when $F = \mathbf{Hom}_B(B, E)$ is replaced there by $F' = \mathbf{Homcart}_B(B, E)$, provided that the inverse image functors $f^* : E_\beta \to E_\alpha$ commute with inductive limits of type I.*

Thus, under the conditions of 9.21.17, if d is a cardinal such that $d \geq c$ and $d \geq \Pi'_\circ$, it follows from the above and from hypothesis (9.21.3 c)) that, for every ordered set I large relative to d, inductive limits of type I are representable in F' and are computed argument by argument, whence the conclusion 9.21.17 (i) follows by an immediate argument.

To prove (ii), one must supplement 9.21.13:

Lemma 9.21.19. *Under the conditions of 9.21.13, suppose that $X \in \operatorname{ob} F'$ i.e. that X is a cartesian section of E over B. Then the conclusion can be strengthened by asserting that there exists a function μ which assigns to every $i \in I$ and every arrow $f : \alpha \to \beta$ of B a morphism $\mu(i, f) : f^*X(\beta)_i \to X(\alpha)_{\varphi(i)}$ in E_α, such that:*

d) *For every $i \in I$ and every arrow $f : \alpha \to \beta$ of B, the following two diagrams are commutative:*

$$\begin{tikzcd}
 & f^*X(\beta)_{\varphi^2(i)} \\
X(\alpha)_{\varphi(i)} \arrow[ur, "\lambda(\varphi(i),f)"] & \\
 & f^*X(\beta)_i \arrow[ul, "\mu(i,f)"] \arrow[uu]
\end{tikzcd}
\qquad
\begin{tikzcd}
X(\alpha)_{\varphi^2(i)} & \\
 & f^*X(\alpha)_{\varphi(i)} \arrow[ul, "\mu(\varphi(i),f)"'] \\
x(\alpha)_i \arrow[uu] \arrow[ur, "\lambda(i,j)"'] &
\end{tikzcd} .$$

To prove this, one proceeds as in 9.21.13, by considering the composite morphism 173
$f^*(X(\beta)_i) \to f^*(X(\beta)) \xrightarrow{X(f)^{-1}} X(\alpha)$, where (as already in the formulation of condition d)) we identify notationally an f-morphism $R \to S$ of E with the corresponding morphism $R \to f^*(S)$ of E_α. Since $X(\alpha) = \varinjlim_j X(\alpha)_j$, and I is large relative to c, one can factor the preceding morphism through one of the $X(\alpha)_j$, where j *a priori* depends on i *and* on f, but can be chosen independently of f, say $\psi(i)$. After enlarging the function φ of 9.21.13, one may suppose that $\psi = \varphi$. Proceeding as for conditions b) and c) of 9.21.13, one sees that, after further enlarging the map φ, one may suppose that the two diagrams of 9.21.19 d) are commutative.

9.21.20. With the map φ chosen as in 9.21.19, let us repeat the argument of 9.21.14, except that we must assume that J satisfies, in addition to the stated conditions a), b), c), the following condition:

d) For every arrow $f : \alpha \to \beta$ in B, the functor $f^* : E_\beta \to E_\alpha$ commutes with inductive limits of type J.

174 We claim that, under this additional condition, the section X_J of E over B is cartesian, i.e. for every arrow $f : \alpha \to \beta$ of B, the morphism

$$(*) \qquad X_J(\alpha) \longrightarrow f^* X_J(\beta)$$

is an isomorphism. Indeed, by condition d) above, the target of the morphism under consideration is identified with $\varinjlim_{i\in J} f^*(X(\beta)_i)$, and the morphism is obtained by taking $\varinjlim_J$ of the morphism of ind-objects in E_α

$$(X(\alpha)_i)_{i\in J} \longrightarrow (f^*)(X(\beta)_i)_{i\in J},$$

induced by the morphisms $\lambda(i, f) : X(\alpha)_i \to f^*(X(\beta)_{\varphi(i)})$ for $i \in J$. Now the conditions set out in 9.21.19 ensure that the preceding homomorphism of ind-objects is in fact an isomorphism, an inverse being obtained from the homomorphism induced by the system of $\mu(i, f)$, for $i \in J$. This proves our assertion that $(*)$ is an isomorphism, and hence that $X \in \operatorname{ob} F'$.

9.21.21. We shall now assume that the functors $f^* : E_\beta \to E_\alpha$ are *$\Pi'_\circ$-accessible.* Then the conditions on J considered in 9.21.20 are satisfied if J is large relative to $\Pi'_\circ$, stable under φ, and such that card $J \leq \Pi$, and we complete the proof of 9.21.17 just as in 9.21.15.

175 Theorem 9.22. *Let $p : E \to B$ be a fibered functor, where B is a category essentially equivalent to a small category, and where for every arrow $f : \alpha \to \beta$ in B the inverse image functor $f^* : E_\beta \to E_\alpha$ is accessible (9.2). Suppose moreover that for every $\alpha \in \operatorname{ob} E$, the fiber category E_α admits a cardinal filtration (9.12) and that every object of E_α is accessible (9.3). Then each of the categories $F = \mathbf{Hom}_B(B, E)$ and $F' = \mathbf{Homcart}_B(B, E)$ has a small full subcategory generating by strict epimorphisms (7.1), and every object of either category is accessible.*

It is immediate that the statement is not essentially changed when B is replaced by a full subcategory $B_\circ$ such that the inclusion functor $B_\circ \to B$ is an equivalence, and E by $E_\circ = E \times_B B_\circ$. This allows us to assume that B is a small category.

For every $\alpha \in \operatorname{ob} B$, let $(\mathrm{Filt}^\Pi(E))_{\Pi \geq \Pi_\alpha}$ be a cardinal filtration of E_α. Let $c_\circ \in \mathscr{U}$ be a cardinal satisfying the following conditions:

$$(9.22.1) \qquad \begin{cases} c_\circ \geq \operatorname*{Sup}_{\alpha\in obB} \Pi_\alpha \quad , \quad c_\circ \geq \operatorname{card} \mathrm{F}\ell\, B; \\ \text{for every arrow } f : \alpha \to \beta \text{ of } B,\ f^* : E_\beta \to E_\alpha \text{ is } c_\circ\text{-accessible;} \\ \text{for every } \alpha \in \operatorname{ob} B, \text{ the objects of } \mathrm{Filt}^{\Pi_\alpha}(E_\alpha) \text{ are } c_\circ\text{-accessible.} \end{cases}$$

Set

$$c = 2^{c_\circ},$$

so that $c > c_\circ$, and, for every $\alpha \in \operatorname{ob} B$, let

$$C_\alpha = \mathrm{Filt}^c(E_\alpha).$$

176 We claim that the preliminary conditions of 9.21.9 and 9.21.17 are satisfied on taking $\Pi = \Pi_\circ = c$. This is clear for (9.21.2) and (9.21.3). For (9.21.4 a)), this follows from 9.17, and (9.21.4 b)) follows from condition 9.12 c) for a cardinal filtration. Let us also note that condition 9.12 b) on cardinal filtrations implies that for every $\alpha \in \operatorname{ob} B$, we have $C^\Pi_\alpha = C_\alpha = \mathrm{Filt}^C(E_\alpha)$, where C^Π_α is defined in (9.21.6). Consequently, 9.22 follows from 9.21.9 and 9.21.17, and more precisely, we have proved the

Corollary 9.23. *Under the conditions of 9.22, if $c_\circ \in \mathscr{U}$ is a cardinal satisfying conditions (9.22.1), and if $c = 2^{c_\circ}$, then the subcategory F^c of F (respectively F'^c of F') formed by the X such that $X(\alpha) \in \operatorname{ob}\operatorname{Filt}^c(E_\alpha)$ for every $\alpha \in \operatorname{ob} B$ is essentially small and generating by strict epimorphisms; more precisely, every object X of F (respectively F') is isomorphic to an object of the form $\varinjlim_I X_i$, where the X_i belong to F^c (respectively F'^c), and where I is an ordered set large relative to c.*

Under slightly stronger hypotheses in 9.22, one can moreover make 9.22 and 9.23 considerably more precise:

Corollary 9.24. *Under the conditions of 9.22, suppose that each of the categories E_α ($\alpha \in \operatorname{ob} B$) is stable under small filtered inductive limits, and that, for every $\Pi \geq \Pi_\alpha$, $\operatorname{Filt}^\Pi(E_\alpha)$ is stable under filtered inductive limits indexed by filtered ordered sets I such that $\operatorname{card} I \leq \Pi$ (which slightly strengthens condition 9.12 b) for cardinal filtrations). In* 177
the case where F' is being considered, suppose moreover that, for every arrow $f : \alpha \to \beta$ of B, the functor $f^* : E_\beta \to E_\alpha$ commutes with small filtered inductive limits. Finally, let $c_\circ \in \mathscr{U}$ be a cardinal satisfying conditions (9.22.1), and, for every cardinal $\Pi \geq c = 2^{c_\circ}$, consider the strictly full subcategory F^Π (respectively F'^Π) of F (respectively F') formed by the X such that $X(\alpha) \in \operatorname{Filt}^\Pi(E_\alpha)$ for every $\alpha \in \operatorname{ob} B$. Then the subcategories under consideration define a cardinal filtration (9.12) of F (respectively F').

Conditions a), b), c) of 9.12 must be checked. Conditions a) and b) follow from the analogous conditions for the given cardinal filtrations of the E_α, and from 9.21.10 (respectively 9.21.18, taking account of the second of conditions (9.22.1)). It remains to prove c), whose first part follows immediately from 9.21.9 (respectively 9.21.17), by taking $\Pi'_\circ = 0$ and $\Pi_\circ = c$. It remains to prove that if $\Pi' \geq \Pi \geq c$ are two cardinals, then for every X in $F^{\Pi'}$ (respectively $F'^{\Pi'}$) one has $X \simeq \varinjlim_{i \in K} X_i$, with the X_i in F^Π (respectively F'^Π), K large relative to Π, and $\operatorname{card} K \leq \Pi'^\Pi$. For this, return to the proofs of 9.21.9 (ii) and 9.21.17 (ii). By condition 9.12 c) on the cardinal filtration of E_α, one may suppose that each I_α has cardinality $\leq \Pi'$, and therefore one will have

$$\operatorname{card} I \leq (\Pi'^\Pi)^{\operatorname{card}\operatorname{ob} B} \leq (\Pi'^\Pi)^c = \Pi'^{\Pi c} = \Pi'^\Pi.$$

The index set K used in 9.21.15 (respectively 9.21.21) is the set of subsets J of I that are filtered (i.e. large relative to $\Pi'_\circ = 0$), stable under φ, and such that $\operatorname{card} J \leq \Pi$. It is then immediate that

$$\operatorname{card} K \leq (\operatorname{card} I)^\Pi \leq (\Pi'^\Pi)^\Pi = \Pi'^{\Pi^2} = \Pi'^\Pi,$$

which completes the proof of 9.24. 178

To conclude, it is appropriate to give a statement freed from the somewhat technical hypotheses of 9.22, replacing them by the conjunction, for the E_α, of the hypotheses that occur in 9.11 (which ensure the accessibility of the objects of the E_α) and in 9.13 (which ensure the existence of a cardinal filtration in E_α):

Corollary 9.25. *Let $p : E \to B$ be a fibered functor, where B is a category equivalent to a small category, and where, for every arrow $f : \alpha \to \beta$ of B, the functor $f^* : E_\beta \to E_\alpha$ is accessible (9.2). Suppose moreover that, for every $\alpha \in \operatorname{ob} B$, the fiber category E_α satisfies the following conditions:*

a) *E_α admits a small full subcategory generating by strict epimorphisms (7.1).*

b) *E_α is stable under fiber products, kernels of pairs of arrows, sums of two objects, and cokernels of pairs of arrows, and finally under small filtered inductive limits*[3].

c) *The functor* $\mathrm{Ker}(u, v)$ *on the category of pairs of arrows of* E_α *is accessible (for example, it commutes with small filtered inductive limits).*

d) *Every strict epimorphism of* E_α *(10.2) is a universal strict epimorphism.*

Under these conditions, each of the categories $F = \mathbf{Hom}_B(B, E)$ *and* $F' = \mathbf{Homcart}_B(B, E)$ *admits a small subcategory generating by strict epimorphisms, and all its objects are accessible (9.3).*

179 Moreover, 9.24 gives us:

Corollary 9.26. *Under the conditions of 9.25, and in the case where one considers* $F' = \mathbf{Homcart}_B(B, E)$*, suppose that the functors* $f^* : E_\beta \to E_\alpha$ *commute with small filtered inductive limits. Then* F *(respectively* F'*) admits a cardinal filtration, which can be made explicit by the procedure of 9.24.*

10. Glossary

For the reader's convenience, we collect here the definitions of some terms used in the preceding sections. We denote a category by C.

10.1. Cartesian, cocartesian. A diagram in C

$$\begin{tikzcd} A \arrow[r] \arrow[d] & B \arrow[d] \\ C \arrow[r] & D \end{tikzcd}$$

is called *cartesian* if it is commutative and the canonical morphism from A to the fiber product $C \times_D B$ is an isomorphism. It is called *cocartesian* if the corresponding diagram in the category opposite to C is cartesian.

180 **10.2. Epimorphism, strict epimorphism.** etc. Cf. 10.3.

10.3. Epimorphic family, strictly epimorphic family. etc.: A family

$$f_i : A_i \longrightarrow B, i \in I,$$

of arrows with the same target in C is called an *epimorphic family* if, for every object C of C, the map induced by the f_i,

$$\mathrm{Hom}_C(B, C) \longrightarrow \prod_i \mathrm{Hom}_C(A_i, C), \tag{10.3.1}$$

is injective.

The family under consideration is called *strictly epimorphic* if the image of the map (10.3.1) consists of the families (g_i) such that, for every object R of C, every pair of indices $i, j \in I$, and every pair of arrows $u : R \to A_i$, $v : R \to A_j$ with $f_i u = f_j v$, one also has $g_i u = g_j v$. When the fiber products $A_i \times_B A_j$ are representable for $i, j \in I$, this is equivalent to saying that the image of (10.3.1) consists of the (g_i) such that, for every pair of indices $i, j \in I$, one has $g_i \,\mathrm{pr}_1 = g_j \,\mathrm{pr}_2$, where $\mathrm{pr}_1, \mathrm{pr}_2$ are the two projections of $A_i \times_B A_j$. The family $(f_i)_{i \in I}$ is called an *effective epimorphic family* if the preceding

[3] This last condition is probably unnecessary; cf. 9.13.2.

condition is satisfied, i.e. if it is strictly epimorphic and the fiber products $A_i \times_B A_j$ are representable.

The family $(f_i)_{i\in I}$ is called a *universally epimorphic* (respectively *universally effective epimorphic*) family if the morphisms f_i are pullbackable (10.3), and if, for every arrow $B' \to B$, the family of arrows $f'_i : A'_i = A_i \times_B B' \to B'$, $i \in I$, deduced from the family $(f_i)_{i\in I}$ by the base change $B' \to B$, is epimorphic (respectively effective epimorphic).

The notion of a *universally strictly epimorphic family* will be defined in (II 2.5, 2.6). **181**
Note that when the morphisms f_i are pullbackable (for example, if fiber products are representable in C), this notion coincides with that of a universally effective epimorphic family (use II 2.4). In the general case, the six variants of the notion of epimorphicity under consideration are related by the logical implications:

$$\begin{tikzcd}
& & \text{universally effective epimorphic} \arrow[dll, Rightarrow] \arrow[dr, Rightarrow] & & \\
\text{effective epimorphic} \arrow[ddr, Rightarrow] & & & \text{universally strictly epimorphic} \arrow[dl, Rightarrow] \arrow[dr, Rightarrow] & \\
& & \text{strictly epimorphic} \arrow[dl, Rightarrow] & & \text{universally epimorphic} \arrow[dlll, Rightarrow] \\
& \text{epimorphic} & & &
\end{tikzcd}$$

A morphism $f : A \to B$ of C is called an *epimorphism (respectively a strict epimorphism, respectively an effective epimorphism, respectively a universal epimorphism, respectively a universal effective epimorphism, respectively a universal strict epimorphism)* if the family of morphisms consisting of the single element f is epimorphic (respectively ...).

10.4. Monomorphic family, strictly monomorphic family. etc. A family of arrows $f_i : A_i \to B$ of C is called *monomorphic (respectively strictly monomorphic,...)* if, regarded as a family of arrows of the opposite category C°, it is epimorphic (respectively strictly epimorphic,...). An arrow $f : A \to B$ of C is called a *monomorphism (respectively a strict monomorphism,...)* if the family consisting only of f is monomorphic (respectively strictly monomorphic,...), i.e. if f, regarded as an arrow of the opposite **182**
category C°, is an epimorphism (respectively a strict epimorphism,...).

10.5. Monomorphism, strict monomorphism. etc.Cf. 10.4.

10.6. Projector, image of a projector, direct factor of an object. Let $f : X \to X$ be an endomorphism of an object of C. The endomorphism f is called a *projector* if $f^2 = f$. The pair (f, id_X) then has a representable cokernel X' if and only if it has a representable kernel X''; when this is the case, there exists a unique isomorphism $u : X' \to X''$ in C such that $iup = f$, where $p : X \to X'$ and $i : X'' \to X$ are the canonical morphisms.[4] One then generally identifies X' with X'' and calls it the *image of the projector* f; the projector f is said to *admit an image* if $\mathrm{Ker}(f, \mathrm{id}_X)$ is representable, i.e. if $\mathrm{Coker}(f, \mathrm{id}_X)$

[4]N.D.E.: Representability of the kernel, or of the cokernel, is also equivalent to the existence of morphisms i, p, and an isomorphism u such that $iup = f$ and $pi = u^{-1}$. In particular, this property is preserved by every functor.

is representable. A subobject (respectively a quotient) Y of X is called a *direct factor* of X if there exists a projector f on X that factors through Y and has Y as its image as a subobject (respectively as a quotient) of X. Sometimes, by abuse of language, an object of C is called a *direct factor of* X if it is isomorphic to the image of a projector on X.

10.7. Pullbackable. An arrow $f : A \to B$ of C is called *pullbackable* if, for every arrow $B' \to B$ of C, the fiber product $A \times_B B'$ is representable in C.

183 **10.8. Quotient, strict quotient.** etc.
Let A be an object of C. Two epimorphisms $f : A \to B$ and $f' : A \to B'$ with source A are called *equivalent* if there exists an isomorphism $u : B \xrightarrow{\sim} B'$ such that $uf = f'$. This defines an equivalence relation on the set of epimorphisms with source A, whose classes are called the *quotients, or quotient objects,* of A. For each quotient of A, one generally chooses an element of this class, say $f : A \to B$, and one often speaks (by abuse of language) of the quotient B of A (or of the quotient $f : A \to B$ of A). The quotient B is called a *strict quotient* (respectively an *effective quotient, respectively a universal quotient,...*) if the morphism $f : A \to B$ is a strict epimorphism (respectively an effective epimorphism, respectively a universal epimorphism,...) (10.3).

10.9. Equivalence relations. Let F and G be two presheaves of sets on C. A diagram $p_1, p_2 : F \rightrightarrows G$ is called an *equivalence relation on* G if, for every object A of C, the map

$$(p_1(A), p_2(A)) : F(A) \longrightarrow G(A) \times G(A)$$

induces a bijection from $F(A)$ onto the graph of an equivalence relation on the set $G(A)$.

A diagram $p_1, p_2 : B \rightrightarrows C$ in C is called an equivalence relation on C if the corresponding diagram of presheaves is an equivalence relation. When finite products and fiber products are representable in C, a functor $\Phi : C \to C'$ commuting with finite products and fiber products carries equivalence relations on C to equivalence relations on $\Phi(C)$.

184 Let $\pi : C \to D$ be a morphism of C such that the fiber product $C \times_D C$ is representable. Then the diagram $\mathrm{pr}_1, \mathrm{pr}_2 : C \times_D C \rightrightarrows C$ is an equivalence relation on C.

10.10. Effective equivalence relation, universally effective equivalence relation. An equivalence relation $p_1, p_2 : R \rightrightarrows A$ on an object A of C is called an *effective equivalence relation* if there exists a morphism $\pi : A \to B$ such that the square

$$\begin{tikzcd} R \arrow[r, "p_2"] \arrow[d, "p_1"'] & A \arrow[d, "\pi"] \\ A \arrow[r] & B \end{tikzcd}$$

is cartesian and cocartesian. (The morphism π is the cokernel of the pair (p_1, p_2) and is therefore determined up to unique isomorphism.) The morphism π is then an effective epimorphism; if it is a universal effective epimorphism (10.3), the *equivalence relation* is said to be *universally effective.*

10.11. Subobject, strict subobject. etc. These are the notions dual to those of quotient and strict quotient etc. (10.8).

Bibliography

[1] B. Mitchell, "Theory of Categories," Academic Press (1965).

II. Appendix: Universes (by N. Bourbaki(*))

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1. Definition and first properties of universes

Definition 1. *A set U is called a universe if it satisfies the following conditions:*

(U.I) *if $x \in U$ and $y \in x$, then $y \in U$;*
(U.II) *if $x, y \in U$, then $\{x, y\} \in U$;*
(U.III) *if $x \in U$, then $\mathscr{P}(x) \in U$;*
(U.IV) *if $(x_\alpha)_{\alpha\in I}$ is a family of elements of U, and if $I \in U$, then the union $\bigcup_{\alpha\in I} x_\alpha$ belongs to U;*
(U.?) *if $x, y \in U$, then the ordered pair $(x, y) \in U$.*

N.B. Since it has, I believe, been decided that in future editions the ordered pair will be defined à la Kuratowski by $(x, y) = \{\{x, y\}, \{x\}\}$, condition (U.?) is unnecessary, since it follows from (U.II).

Examples. 1) The *empty* set is a universe, denoted by $U_\circ$.

2) Consider the nonempty finite words formed with the four symbols "{"," }"," ," and "∅" (cf. Alg. I). Define, by induction on the length n of such a word, the
186 notion of a *meaningful* word:

(a) For $n = 1$, only the word ∅ is meaningful;
(b) for a word A of length n to be meaningful, it is necessary and sufficient that there exist p distinct meaningful words $A_1, \dots, A_p$ $(p \geq 1)$ of lengths $< n$ such that

$$A = \{A_1, A_2, \dots, A_p\}.$$

For example, $\varnothing$, $\{\varnothing\}$, $\{\{\varnothing\},\{\{\varnothing\}\},\{\varnothing,\{\varnothing\}\}\}$ are meaningful words. It is clear, by induction on the length, that every meaningful word denotes a term of Set Theory; for example, if $A = \{A_1, A_2, \dots, A_p\}$, A denotes the set whose elements are $A_1, A_2, \dots,$ and A_p; these terms are plainly *finite sets.* Let U_1 be the set of sets thus obtained. It is readily checked that U_1 satisfies conditions (U.I), (U.II), (U.III), and (U.IV) of Definition 1 (but not that idiotic condition (U.?), which is no problem if one is willing to decanulate the ordered pair). Thus U_1 is a *universe.* Note that the elements of U_1 are finite, and that U_1 is countable.

In the statements that follow, U *denotes a universe.*

Proposition 1. *If $X \in U$ and if $y \subset x$, then $y \in U$.*

Indeed, $\mathscr{P}(x) \in U$ by (U.III), whence $y \in \mathscr{P}(x)$ and $y \in U$ by (U.I).

Corollary. *If $x \in U$, every quotient set y of x is an element of U.*

Indeed, y is a subset of $\mathscr{P}(x)$. Hence $y \in U$ by (U.III) and Proposition 1.

187 Proposition 2. *If $x \in U$, then $\{x\} \in U$.*

This follows from (U.II), applied with $y = x$.

Proposition 3. *Every ordered pair, ordered triple, and ordered quadruple of elements of U is an element of U.*

This is true for ordered pairs by the N.B. or (U.?). The case of ordered triples follows, since $(x, y, z) = ((x, y), z)$, and then that of ordered quadruples, since $(x, y, z, t) = ((x, y, z), t)$.

[4]We reproduce here, with his permission, secret papers of N. Bourbaki. The references in this text are to his learned work.

Proposition 4. *If $X, Y \in U$, then $X \times Y \in U$.*

Indeed, for $x \in X$, $\{x\} \times Y$ is the union of the family $\{(x, y)\}_{y \in Y}$; since $\{(x, y)\} \in U$ by (U.I), (U.II), and Proposition 3, we have $\{x\} \times Y \in U$ by (U.IV). Finally, $X \times Y$ is the union of the family $(\{x\} \times Y)_{x \in X}$; it is therefore an element of U, again by (U.IV).

Corollary 1. *If $X, Y, Z, \dots$ are elements of U, all the sets of the scale constructed on $X, Y, Z, \dots$ are elements of U* (cf. Chapter IV, §).

This indeed follows by successive applications of Proposition 4 and of (U.III).

Corollary 2. *If $(x_\alpha)_{\alpha \in I}$ is a family of elements of U and if $I \in U$, the sum set $\Sigma_{\alpha \in I} x_\alpha$ is an element of U.*

Indeed, this sum set is a subset of the product $(\bigcup_{\alpha \in I} x_\alpha) \times I$ (Chapter II), a product both of whose factors are elements of U (by (U.IV) and the hypothesis). We then apply Propositions 4 and 1.

Proposition 5. *If X and Y are elements of U, every correspondence between X and* **188**
Y (in particular, every map from X to Y) is an element of U.

Indeed, such a correspondence C is an ordered triple (X, Y, Γ), where Γ is a subset of $X \times Y$ (the graph of C) (Chapter II, §). We have $\Gamma \in U$ by Propositions 4 and 1. Hence $C \in U$ by Proposition 3.

Proposition 6. *If $X, Y \in U$, every set Z of correspondences between X and Y (in particular, of maps from X to Y) is an element of U.*

Indeed, Z is a subset of $\{X\} \times \{Y\} \times \mathscr{P}(X \times Y)$. Now this product is an element of U by Proposition 2 and the corollary to Proposition 4. Therefore $Z \in U$ by Proposition 1.

Corollary. *If $(x_\alpha)_{\alpha \in I}$ is a family of elements of U, and if $I \in U$, then $\Pi_{\alpha \in I} x_\alpha \in U$.*

Indeed, this product is a set of maps from I to $\bigcup_{\alpha \in I} x_\alpha$, and this union is an element of U by (U.IV).

Proposition 7. *If X is a subset of U whose cardinality is at most that of an element of U, then X is an element of U.*

Let I be an element of U such that $\mathrm{card}(X) \leq \mathrm{card}(I)$. There is a surjection $i \mapsto x_i$ from a subset I' of I onto X. Then X is the union of the family $(\{x_i\})_{i \in I'}$; now this union is an element of U by Proposition 1, Proposition 2, and (U.IV).

Corollary. *If U is nonempty, every finite subset of U is an element of U, and U has* **189**
elements of arbitrarily large finite cardinality.

On the other hand, if $U = \varnothing$, then $\varnothing$ is a finite subset of $\varnothing$ but not an element of $\varnothing$.

Indeed, if U is nonempty, Proposition 2 shows that a one-element set $x_\circ$ belongs to U. By induction on n, put $x_{n+1} = \mathscr{P}(x_n)$. We have $x_n \in U$ by (U.III), and $\mathrm{card}(x_{n+1}) = 2^{\mathrm{card}(x_n)}$, so that U has elements of arbitrarily large finite cardinality.

Remark. It follows from Proposition 2 and the corollary to Proposition 7 that every nonempty universe U contains the universe U_1 of Example 2; indeed, $\varnothing \in U$ by Proposition 1. Thus U_1 is the intersection of all nonempty universes. More generally:

Proposition 8. *If $(U_\lambda)_{\lambda \in L}$ is a nonempty family of universes, then $U = \bigcap_{\lambda \in L} U_\lambda$ is a universe.*

This follows at once from Definition 1.

2. Universes and species of structures

Let $\mathscr{E}$ be a species of structure; suppose, to fix ideas and simplify the exposition, that every structure of species $\mathscr{E}$ is defined on an underlying set. Let (X, S) be a structure of species $\mathscr{E}$ (where X is the underlying set and S the structure), and let U be a *universe.* If $X \in U$, then all the constituent objects of the structure S on X are elements of U (by Cor. 1 of Prop. 4, n° 1, and by (U.I)), so the structure (X, S) is an element of U.

190 Suppose now that $\mathscr{E}$ is a species of structure with *morphisms.* If X and X' are elements of U endowed with structures of species $\mathscr{E}$, then the set of morphisms from X to X' is again an element of U (n° 1, Prop. 6).

Now consider the *category* $(U\text{-}\mathscr{E})$ defined in § 1, n° 2, Ex. c). Since the objects and arrows of $(U\text{-}\mathscr{E})$ are elements of U, $(U\text{-}\mathscr{E})$ is a *pair of subsets* of U, endowed with a quadruple of maps; in general, it is not an element of U. The notation $X \in (U\text{-}\mathscr{E})$ will mean that X is an element of U endowed with a structure of species $\mathscr{E}$.

The category $(U\text{-}\mathscr{E})$ is *stable* under many operations:

a) If $X \in (U\text{-}\mathscr{E})$, and if X' is a subset of X that admits an *induced structure*, then $X' \in (U\text{-}\mathscr{E})$. This follows from Prop. 1, n° 1.
b) If $X \in (U\text{-}\mathscr{E})$, and if X'' is a quotient set of X that admits a *quotient structure*, then $X'' \in (U\text{-}\mathscr{E})$ (Cor. of Prop. 1, n° 1).
c) If $X, Y \in (U\text{-}\mathscr{E})$, and if $X \times Y$ admits a *product structure*, then $X \times Y \in (U\text{-}\mathscr{E})$ (Prop. 4, n° 1). More generally, if $(X_i)_{i \in I}$ is a family of elements of $(U\text{-}\mathscr{E})$, *if* $I \in U$, and if $\Pi_{i \in I} X_i$ admits a *product structure*, then $\Pi_{i \in I} X_i \in (U\text{-}\mathscr{E})$ (Cor. of Prop. 6). The analogous assertions hold for *sum structures* (Cor. 2 of Prop. 4).
d) Let I be a preordered set, and let $(X_i, f_{ij})_{i,j \in I}$ be a *projective system* (resp. *inductive system*) of sets endowed with structures of species $\mathscr{E}$ and morphisms; let L be the limit of this system, in the sense of Chap. III. If $X_i \in U$ for every i, if $I \in U$, and if L admits a projective-limit structure (resp. an inductive-limit
191 structure), then $L \in (U\text{-}\mathscr{E})$: indeed, L is a subset of $\Pi_{i \in I} X_i$ (resp. a quotient set of $\Sigma_{i \in I} X_i$), and is therefore an element of U by n° 1.

Here are a few more particular examples:

* 1) Let $X, Y \in (U\text{-}\mathrm{Top})$ be two *locally compact* spaces. Then the set $\mathscr{C}(X, Y)$ of continuous maps from X to Y, endowed with the topology of compact convergence (General Topology X), is an element of $(U\text{-}\mathrm{Top})$. Indeed, $\mathscr{C}(X, Y) \in U$ by Prop. 6 of n° 1.⁎
* 2) Let $X, Y \in (U\text{-}\mathrm{Ab})$. Then the group $\mathrm{Hom}_{\mathbf{Z}}(X, Y)$ is an element of $(U\text{-}\mathrm{Ab})$ by Prop. 6 of n° 1. If, in addition, $\mathbf{Z} \in U$, the tensor product $X \otimes_{\mathbf{Z}} Y$ is an element of $(U\text{-}\mathrm{Ab})$; indeed, this tensor product is a quotient of $\mathbf{Z}^{X \times Y}$, which is a subset of $\mathbf{Z}^{X \times Y}$, which in turn is an element of U by Props. 4 and 6 of n° 1.⁎
* 3) Let $X \in (U\text{-}\mathrm{Unif})$ be a uniform space. Then its *completion* $\widehat{X}$ is an element of $(U\text{-}\mathrm{Unif})$. Indeed, $\widehat{X}$ is a set of equivalence classes of Cauchy filters on X; now a filter on X is an element of $\mathscr{P}(\mathscr{P}(X))$, so an equivalence class of filters is an element of $\mathscr{P}(\mathscr{P}(\mathscr{P}(X)))$, whence $\widehat{X}$ is an element of $\mathscr{P}(\mathscr{P}(\mathscr{P}(\mathscr{P}(X))))$, and hence an element of U by (U.III) and (U.I).⁎

N.B. Moral: Bourbaki will have to take care to "canonicalize" its constructions properly. For example, in constructions in which an element ∞ is adjoined (Alexandroff
192 compactification, projective field), it will be advantageous to take $\infty = \varnothing$ (since $\varnothing$ is an element of every nonempty universe) and to form the disjoint union of $\{\varnothing\}$ and the

given set. Of course, one should also "canonicalize" the two-element set used to construct disjoint unions; thus $\{\varnothing, \{\varnothing\}\}$ seems to me a good candidate, since it belongs to every nonempty universe.

3. Universes and categories

Let U be a universe and C a category. We write $C \in U$ if $\mathrm{Ob}(C) \in U$ and $\mathrm{F}l\,(C) \in U$. This notation is justified by the fact that C is the sextuple consisting of $\mathrm{Ob}(C)$, $\mathrm{F}l\,(C)$, and the four structure maps; thus, if $\mathrm{Ob}(C) \in U$ and $\mathrm{F}l\,(C) \in U$, these four maps are elements of U (n° 1, Corollary 1 of Proposition 4), and hence so is the sextuple C (n° 1, Proposition 3, *cum grano salis*). This is moreover a special case of n° 2, if one considers the species of structure "cat" of a category. With the notation of n° 2, the relation $C \in U$ is also written $C \in (U\text{-}\mathrm{cat})$.

One should note that a category such as $(U\text{-}\mathrm{Ens})$, $(U\text{-}\mathrm{Ord})$, $(U\text{-}\mathrm{Top})$, or $(U\text{-}\mathrm{Gr})$ is *not*, in general, an element of U.

Let C, D be two categories and U a universe such that $C, D \in U$. If C' is a *subcategory* of C, then $C' \in U$; the *product category* $C \times D$, the sum category of C and D, and the opposite categories C° and D° are also elements of U (cf. n° 2). The *functor category* $E = \mathbf{Hom}(C, D)$ is likewise an element of U: indeed, $\mathrm{Ob}(E)$ is a set of pairs of maps $\mathrm{Ob}(C) \to \mathrm{Ob}(D)$, $F\ell\,(C) \to F\ell\,(D)$, and hence $\mathrm{Ob}(E) \in U$ by n° 1; as for $F\ell\,(E)$, it is 193
a set of natural transformations, that is, of maps $\mathrm{Ob}(C) \to F\ell\,(D)$, whence $F\ell\,(E) \in U$ by Proposition 6 of n° 1.

PROPOSITION 9. *Let $\mathscr{E}$ be a species of structure with morphisms, and let U, U' be two universes such that $U' \subset U$. Then $(U'\text{-}\mathscr{E})$ is a full subcategory of $(U\text{-}\mathscr{E})$.*

Indeed, if x and y are objects of $(U'\text{-}\mathscr{E})$, they are by definition objects of $(U\text{-}\mathscr{E})$. As for $\mathrm{Hom}(x, y)$, it is the same set in both categories (cf. § 1, n° 2, Example c)).

N.B. One should note that the hypothesis that U and U' are universes is unnecessary, so that Proposition 9 would be more appropriately placed in § 1, n° 4. But the Tribe asked that it be put here.

REMARK. It may happen that the axioms of the species of structure $\mathscr{E}$ imply that the *cardinals* of sets endowed with structures of species $\mathscr{E}$ are *bounded* by a fixed cardinal, say c (* for example, the species of structure of finite group, finitely generated group, finitely generated module over a fixed ring A, or finitely generated algebra over A *). Suppose then that there exists an element z of U' such that $c \leq \mathrm{card}(z)$. Then, for every set x endowed with a structure of species $\mathscr{E}$, there exists an element x' of U' equipotent to x, for example a subset of z; endow x' with the structure deduced from that of x by transport of structure; this gives an element of $(U'\text{-}\mathscr{E})$. It follows that *the inclusion functor from $(U'\text{-}\mathscr{E})$ to $(U\text{-}\mathscr{E})$ is then essentially surjective* (§ 4, n° 2, Definition 2) *and is therefore an equivalence of categories* (§ 4, n° 3, Theorem 1).

4. The universe axiom

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The very pleasant stability results of n° 2 are of interest only if they can be applied to something other than the two small universes $U_\circ$ and U_1 described in n° 1. We shall therefore add the following axiom to the axioms of Set Theory:

(A.6) *For every set x, there exists a universe U such that $x \in U$.*

This axiom implies that, if $(x_\alpha)_{\alpha\in I}$ is a *family* of sets, there exists a universe U such that $x_\alpha \in U$ for every $\alpha \in I$: indeed, it suffices to apply (A.6) to $x = \bigcup_{\alpha\in I} x_\alpha$ and then apply Proposition 1 of n° 1.

In particular, given a *category* C, there exists a universe U such that $C \in U$ in the sense of n° 3: apply the preceding result to the family $(\mathrm{Ob}(C), F\ell(C))$. This applies to categories of the form $(V\text{-}\mathscr{E})$, where $\mathscr{E}$ is a species of structure with morphisms and V is a set, for example a universe; in general one has $V \neq U$.

For example, if V is a universe, then $\mathrm{Ob}(V\text{-}\mathrm{Ens}) = V$ and $F\ell(V\text{-}\mathrm{Ens}) \subset V$ (n° 1, Proposition 5). The universes U such that $(V\text{-}\mathrm{Ens}) \in U$ are therefore those such that $V \in U$. But the relation $V \in V$ is impossible for a universe: indeed, for every subset A of V, one would have $A \in V$ (n° 1, Proposition 1), whence $\mathrm{card}(\mathscr{P}(V)) \leq \mathrm{card}(V)$, which is impossible.

5. Universes and strongly inaccessible cardinals

195 Let U be a universe. By condition (U.I) of Definition 1, every element x of U is a subset of U; hence $\mathrm{card}(x) \leq \mathrm{card}(U)$. Since the cardinals of the subsets of U form a well-ordered set, the cardinal

$$\mathscr{c}(U) = \sup_{x \in U} \mathrm{card}(x) \tag{1}$$

exists. Note that, for every cardinal $\mathscr{c} < \mathscr{c}(U)$, there exists an element x of U such that $\mathrm{card}(x) = \mathscr{c}$; indeed, by definition there exists $y \in U$ such that $\mathscr{c} \leq \mathrm{card}(y) \leq \mathscr{c}(U)$, and one takes for x a suitable subset of y. Conversely, if $x \in U$, then $\mathrm{card}(x) < \mathscr{c}(U)$; indeed, $\mathscr{P}(x) \in U$ by (U.III), whence $2^{\mathrm{card}(x)} \leq \mathscr{c}(U)$. Thus the cardinal $\mathscr{c}(U)$ has the following properties:

1) If $\mathscr{c}$ is a cardinal $< \mathscr{c}(U)$, then $2^{\mathscr{c}} < \mathscr{c}(U)$; indeed, if x is an element of U of cardinality $\mathscr{c}$, then $\mathscr{P}(x) \in U$ by (U.III), whence $2^{\mathscr{c}} < \mathscr{c}(U)$.
2) If $(\mathscr{c}_\lambda)_{\lambda \in I}$ is a family of cardinals such that $\mathscr{c}_\lambda < \mathscr{c}(U)$ for every $\lambda \in I$ and $\mathrm{card}(I) < \mathscr{c}(U)$, then the sum cardinal $\Sigma_{\lambda \in I} \mathscr{c}_\lambda$ is $< \mathscr{c}(U)$. Indeed, let $x_\lambda \in U$ be such that $\mathrm{card}(x_\lambda) = \mathscr{c}_\lambda$; after replacing I by an equipotent index set if necessary, one may suppose that $I \in U$; then the sum set of the x_λ is an element of U (Corollary 2 of Proposition 4 of n° 1), which proves our assertion.

We make the following definition:

196 DEFINITION 2. *A cardinal $\mathscr{d}$ is called strongly inaccessible if:*

(FI.1) *If $\mathscr{c}$ is a cardinal such that $\mathscr{c} < \mathscr{d}$, then $2^{\mathscr{c}} < \mathscr{d}$.*
(FI.2) *If $(\mathscr{c}_\lambda)_{\lambda \in I}$ is a family of cardinals such that $\mathscr{c}_\lambda < \mathscr{d}$ for every $\lambda \in I$, and if* $\mathrm{card}(I) < \mathscr{d}$, *then $\Sigma_{\lambda \in I} \mathscr{c}_\lambda < \mathscr{d}$.*

Examples. The cardinal 0 and the countably infinite cardinal are strongly inaccessible. No nonzero finite cardinal is strongly inaccessible. Thus we have just proved that the universe axiom (A.6) implies the statement:

(A′.6) *Every cardinal is strictly bounded above by a strongly inaccessible cardinal.*

Conversely:

THEOREM 1. *Statement (A′.6) implies the universe axiom (A.6).*

Indeed, let A be a set. We must construct a universe U of which A is an element. Define recursively a sequence $(A_n)_{n \geq 0}$ of sets by:

(2) $A_0 = A$, A_{n+1} = the union of the elements of A_n = the set of elements of the elements of A_n.

Put $B = \bigcup_{n=0}^{\infty} A_n$. By (A′.6), let $\mathscr{c}$ be a strongly inaccessible cardinal such that $\mathrm{card}(B) < \mathscr{c}$.

There exists a *well-ordered* set I such that $\operatorname{card}(I) = \mathscr{c}$. After replacing I by its smallest segment of cardinality $\mathscr{c}$ if necessary, we may suppose that *every segment of I distinct from I has cardinality $< \mathscr{c}$*. We denote by $\mathscr{E}$ the *least element of I*. It follows 197
from the hypothesis on segments that I has no greatest element (otherwise we would remove it), and hence that every element of I has a successor; for $\alpha \in I$, we denote the *successor* of α by $s(\alpha)$.

Now define, by *transfinite recursion*, a family $(B_\alpha)_{\alpha\in I}$ of sets by:

$$(3)\qquad \begin{cases} B_{\mathscr{E}} = B \\ B_{s(\alpha)} = B_\alpha \cup \mathscr{P}(B_\alpha) \\ B_\alpha = \bigcup\limits_{\beta<\alpha} B_\beta \text{ if } \alpha \text{ has no predecessor.} \end{cases}$$

Put $U = \bigcup_{\alpha\in I} B_\alpha$. We shall show that U is the desired universe.

First, $A \in U$, since $A = A_\circ$ is a subset of $B = B_{\mathscr{E}}$, and therefore an element of $\mathscr{P}(B_{\mathscr{E}}) \subset B_{s(\mathscr{E})}$.

Next note that, for every $\alpha \in I$, one has:

$$(4)\qquad \operatorname{card}(B_\alpha) < \mathscr{c}.$$

Indeed, proceed by transfinite recursion on α. This is true for $\alpha = \mathscr{E}$ by hypothesis. From (4) one obtains $\operatorname{card}(B_{s(\alpha)}) \leq \operatorname{card}(B_\alpha) + 2^{\operatorname{card}(B_\alpha)} < \mathscr{c} + \mathscr{c}$ (by (FI.1)) $= \mathscr{c}$ (since $\mathscr{c}$ is infinite). Finally, if α has no predecessor, the set I' of $\beta < \alpha$ has cardinality $< \mathscr{c}$ by construction; hence (if $\operatorname{card}(B_\beta) < \mathscr{c}$ for every $\beta < \alpha$), $\operatorname{card}(B_\alpha) = \operatorname{card}(\bigcup_{\beta\in I'} B_\beta) \leq \Sigma_{\beta\in I'} \operatorname{card}(B_\beta) < \mathscr{c}$ (by (FI.2)). This proves (4).

Let us now show that 198

$$(5)\qquad \operatorname{card}(x) < \mathscr{c} \text{ for every } x \in U.$$

Indeed, it suffices to show that, for every $\alpha \in I$, one has "$\operatorname{card}(x) < \mathscr{c}$ for every $x \in B_\alpha$". Proceed again by transfinite recursion on α. This is true for $\alpha = \mathscr{E}$, since, if $x \in B_{\mathscr{E}} = B$, there exists $n \geq 0$ such that $x \in A_n$; hence $x \subset A_{n+1}$ by (2), $x \subset B$, and $\operatorname{card}(x) \leq \operatorname{card}(B) < \mathscr{c}$. The case where α has no predecessor is evident. Finally, if $x \in B_\alpha$ and $\alpha = s(\beta)$, either $x \in B_\beta$, in which case the assertion "$\operatorname{card}(x) < \mathscr{c}$" is true by recursion, or $x \in \mathscr{P}(B_\beta)$, whence $x \subset B_\beta$ and the assertion "$\operatorname{card}(x) < \mathscr{c}$" is true by (4).

We are now in a position to prove that U is indeed a *universe*:

(U.I) Let $x \in U$ and $y \in x$. We must show that $y \in U$. In other words, we must show that, for every $\alpha \in I$, one has:

$$\text{“}x \in B_\alpha \text{ and } y \in x \Longrightarrow y \in B_\alpha\text{”}.$$

Proceed again by transfinite recursion on α. This is true for $\alpha = \mathscr{E}$, since $x \in B_{\mathscr{E}} = B$ implies $x \in A_n$ for some n; hence, if $y \in x$, then $y \in A_{n+1}$, whence $y \in B = B_{\mathscr{E}}$. Passage to an element α with no predecessor is evident. Finally, pass from α to $s(\alpha)$: if $x \in B_{s(\alpha)}$ and $y \in x$, either $x \in B_\alpha$, whence $y \in B_\alpha \subset B_{s(\alpha)}$ by recursion, or $x \in \mathscr{P}(B_\alpha)$, whence again $y \in B_\alpha \subset B_{s(\alpha)}$.

(U.II) Let $x, y \in U$. Since the family $(B_\alpha)_{\alpha\in I}$ is increasing by (3), there exists $\alpha \in I$ such that $x, y \in B_\alpha$. Then $\{x, y\} \in \mathscr{P}(B_\alpha) \subset B_{s(\alpha)} \subset U$.

(U.III) Let $x \in U$. Then there exists $\alpha \in I$ such that $x \in B_\alpha$. It therefore suffices to 199
show that, for every $\alpha \in I$, one has:

$$\text{“}x \in B_\alpha \Longrightarrow \mathscr{P}(x) \in B_{s(s(\alpha))}\text{”}.$$

Proceed again by transfinite recursion on α. If $\alpha = \mathscr{E}$, then $x \in A_n$ for some n, whence $y \subset A_{n+1} \subset B = B_{\mathscr{E}}$ for every subset y of x; thus $y \in \mathscr{P}(B_{\mathscr{E}}) \subset B_{s(\mathscr{E})}$ for every $y \in \mathscr{P}(x)$, whence $\mathscr{P}(x) \subset B_{s(\mathscr{E})}$ and $\mathscr{P}(x) \in \mathscr{P}(B_{s(\mathscr{E})}) \subset B_{s(s(\mathscr{E}))}$, so our assertion is true for $\alpha = \mathscr{E}$. Passage to an element α with no predecessor is immediate, since $\beta < \alpha$ then implies $s(s(\beta)) < \alpha$. Finally, pass from α to $s(\alpha)$; let $x \in B_{s(\alpha)}$; if $x \in B_\alpha$, then $\mathscr{P}(x) \in B_{s(s(\alpha))} \subset B_{s(s(s(\alpha)))}$ by recursion; if $x \in \mathscr{P}(B_\alpha)$, then $x \subset B_\alpha$, whence $\mathscr{P}(x) \subset \mathscr{P}(B_\alpha)$ and $\mathscr{P}(x) \in \mathscr{P}(\mathscr{P}(B_\alpha)) \in B_{s(s(s(\alpha)))}$.

(U.IV) Let $(x_\lambda)_{\lambda \in K}$ be a family of elements of U such that $K \in U$. We must show that the union $x = \bigcup_{\lambda \in K} x_\lambda$ is an element of U. For every $\lambda \in K$, choose an $\alpha(\lambda) \in I$ such that $x_\lambda \in B_{\alpha(\lambda)}$. Let us show that the set $\alpha(K)$ of the $\alpha(\lambda)$ is *bounded above* in I: indeed, if it were not, one would have

$$I = \bigcup_{\lambda \in K} [\mathscr{E}, \alpha(\lambda)],$$

which would contradict (FI.2), since $\operatorname{card}(K) < \mathfrak{c}$ and $\operatorname{card}([\mathscr{E}, \alpha(\lambda)]) < \mathfrak{c}$ for every $\lambda \in K$. Let then β be an upper bound of $\alpha(K)$; one has $x_\lambda \in B_\beta$ for every $\lambda \in K$, since the family $(B_\alpha)_{\alpha \in I}$ is increasing. Every element of x_λ then belongs to B_β by what was proved in (U.I), whence $x = \bigcup_{\lambda \in K} x_\lambda \subset B_\beta$. By (3), it follows that $x \in B_{s(\beta)}$, whence $x \in U$. C.Q.F.D.

200 DEFINITION 3. *Let x, y be two sets and let n be an integer ≥ 0. We say that y is a component of order n of x if there exists a sequence $(x_j)_{j=0,\dots,n}$ such that $x_0 = x$, $x_n = y$, and $x_{j+1} \in x_j$ for $j = 0, \dots, n-1$.*

Thus x is the only component of order 0 of x. The components of order 1 (resp. 2) of x are the elements of x (resp. the elements of the elements of x). We say that y is a *component* of x if there exists $n \geq 0$ such that y is a component of order n of x. The relation "y is a component of x" is a *preorder relation*. By the selection–union scheme (Chapter II, ...), the relation "y is a component of x" is collectivizing with respect to y, so that the components of x form a *set*.

DEFINITION 4. *Let $\mathfrak{c}$ be a cardinal. A set x is said to be of type $\mathfrak{c}$ (resp. of strict type $\mathfrak{c}$, of finite type) if all the components of x have cardinalities $\leq \mathfrak{c}$ (resp. $< \mathfrak{c}$, finite cardinalities).*

Examples. The elements of the universe U_1 (n° 1, Example 2) are all of finite type. If $\mathfrak{c}$ is a cardinal and x is of type $\mathfrak{c}$ (resp. of strict type $\mathfrak{c}$, of finite type), then every component of x and every subset of x is of type $\mathfrak{c}$ (resp. of strict type $\mathfrak{c}$, of finite type); likewise $\mathscr{P}(x)$ is of type $2^{\mathfrak{c}}$ (resp. of strict type $2^{\mathfrak{c}}$, of finite type). If x is of type $\mathfrak{c}$ and if $\mathfrak{c} \leq \mathfrak{c}'$, then x is of type $\mathfrak{c}'$.

201 LEMMA. *Let $\mathfrak{c}$ be an uncountable strongly inaccessible cardinal, and let X be a set of strict type $\mathfrak{c}$. Then there exists a cardinal $\mathfrak{d} < \mathfrak{c}$ such that X is of type $\mathfrak{d}$.*

Indeed, for every integer n, let $\mathfrak{d}_n$ denote the cardinality of the set X_n of components of order n of X. We have $\mathfrak{d}_0 = 1$, $\mathfrak{d}_1 = \operatorname{card}(X) < \mathfrak{c}$, and, since $X_n = \bigcup_{y \in X_{n-1}} y$, induction on n using (FI.2) gives $\mathfrak{d}_n < \mathfrak{c}$ for every n. The $\mathfrak{d}_n$ are therefore bounded above by the cardinal $\mathfrak{d} = \Sigma_{j \geq 0} \mathfrak{d}_j$, which is $< \mathfrak{c}$ by (FI.2) and the hypothesis that $\mathfrak{c}$ is uncountable. If Y is a component of order n of X, then $\operatorname{card}(Y) \leq \operatorname{card}(X_{n+1}) \leq \mathfrak{d}$. C.Q.F.D.

PROPOSITION 10. *Let U be a universe and let $\mathfrak{c}$ be a strongly inaccessible cardinal. Then the set U' of those $x \in U$ that are of strict type $\mathfrak{c}$ is a universe.*

Indeed, if $x, y \in U'$ and if $z \in x$, then plainly $z \in U'$ and $\{x, y\} \in U'$. If $x \in U'$, then $\mathrm{card}(x) < c$, whence $\mathrm{card}(\mathscr{P}(x)) < c$ by (FI.1); since a component of $\mathscr{P}(x)$ is either $\mathscr{P}(x)$ itself, a subset of x, or a component of x, we have $\mathscr{P}(x) \in U'$. Finally, if $(x_\alpha)_{\alpha\in I}$ is a family of elements of U' such that $I \in U'$, then $\mathrm{card}(\bigcup_\alpha x_\alpha) < c$ by (FI.2); since every component of order > 0 of $\bigcup_\alpha x_\alpha$ is a component of some x_α, we indeed have $\bigcup_\alpha x_\alpha \in U'$. C.Q.F.D.

Remark on the cardinal $c(U)$. Let U be a universe and let $c(U)$ be the cardinal defined by (1), i.e.

$$c(U) = \sup_{x\in U} \mathrm{card}(x).$$

We plainly have $c(U) \leq \mathrm{card}(U)$, since, as we recall, $x \in U$ implies $x \subset U$. But 202
the equality $c(U) = \mathrm{card}(U)$ does not always hold. For example, let $(x_\alpha)_{\alpha\in I}$ be an *uncountable* family of symbols subject to the axioms:

$$«x_\alpha = \{x_\alpha\} \text{ for every } \alpha \in I» \quad , \quad «x_\alpha \neq x_\beta \text{ if } \alpha \neq \beta»$$

(in other words, x_α is a one-element set, namely itself). As in Example 2 of n° 1, form the "significant words" constructed from the symbols $\varnothing$, x_α, $\{$, $\}$ and so forth. Each denotes a *finite* set. Let U be the set of these words. One readily verifies that the conditions of Definition 1 are satisfied, so that U is a *universe*. Since the significant words are finite sequences of elements of a set of cardinality $\mathrm{card}(I) + 4 = \mathrm{card}(I)$, we have $\mathrm{card}(U) = \mathrm{card}(I)$ (Chapter III; in fact, $\mathrm{card}(U) \geq \mathrm{card}(I)$ suffices, and this is obvious). On the other hand, $c(U)$ is the countably infinite cardinal, and hence $c(U) < \mathrm{card}(U)$.

The strict inequality $c(U) < \mathrm{card}(U)$ is due to our having introduced sets x_α such that $x_\alpha \in x_\alpha$. If one rules out horrors of this kind, one obtains $\mathrm{card}(U) = c(U)$ for every universe U, as well as other very pretty results "which cannot possibly be useful." This is what we shall do in the next section.

6. Artinian sets and universes

DEFINITION 5. *A set x is said to be Artinian if there is no infinite sequence $(x_n)_{n\geq 0}$ such that $x_\circ = x$ and $x_{n+1} \in x_n$ for every $n \geq 0$.*

EXAMPLES. The sets $\varnothing$, $\{\varnothing\}$, $\{\varnothing, \{\varnothing\}\}$, and, more generally, the elements of the universe U_1 of n° 1 are Artinian.

In pictorial terms, if x is Artinian, and if one takes an element x_1 of x, then an 203
element x_2 of x_1, etc., the process must stop, and one arrives at a component x_n of x that is *empty*. In other words, Artinian sets are "built from $\varnothing$"; this will be made more precise later (cf. (*) in the proof of Theorem 2).

Artinian sets plainly have the following properties:

(AR.I) If x is Artinian, every subset of x and every component of x is Artinian. For x to be Artinian, it is necessary and sufficient that every element of x be Artinian.
(AR.II) If x and y are Artinian, then $\{x, y\}$ is Artinian.
(AR.III) If x is Artinian, $\mathscr{P}(x)$ is Artinian.
(AR.IV) Every union of Artinian sets is an Artinian set.

These properties immediately yield the following result:

PROPOSITION 11. *If U is a universe, the set of Artinian elements of U is a universe, necessarily Artinian.*

COROLLARY. *If x is an Artinian set, there exists an Artinian universe U such that $x \in U$.*

Indeed, by axiom (A.6), x is an element of a universe V; take for U the set of Artinian elements of V.

The following proposition is even less useful than the rest of n° 6:

204 PROPOSITION 12. *Let A be an Artinian set. Then:*

a) *For every $x \in A$, one has $x \notin x$;*
b) *If $x, y \in A$, one cannot have both $x \in y$ and $y \in x$;*
c) *For every $x \in A$, the relation "y is a component of z" on the components y, z of x is an order relation;*
d) *For every nonempty element x of A, there exists $y \in x$ such that $x \cap y = \varnothing$.*

Indeed, the negation of a) (resp. of b)) entails the existence of an infinite sequence $(x_n)_{n \geq 0}$ contradicting Definition 5, namely $(x, x, x, \dots)$ (resp. $(x, y, x, y, x, y, \dots)$). The negation of c) means that there exist $x \in A$, distinct components y, z of x, and membership chains:

$$y \in y_1 \in \cdots \in y_q \in z \quad , \quad z \in z_1 \in \cdots \in z_r \in y;$$

whence, as in b), an infinite sequence $(x_n)_{n \geq 0}$ contradicting Definition 5. Finally, if d) is false, there exists a nonempty element x of A such that $x \cap y \neq \varnothing$ for every $y \in x$; set $x_\circ = x$, and take for x_1 an element of x; since $x \cap x_1 \neq \varnothing$, take for x_2 an element of $x \cap x_1$, and so on; more formally, define by induction an infinite sequence $(x_n)_{n \geq 1}$ of elements of x by means of $x_1 \in x$, $x_{n+1} \in x_n \cap x$ for $n \geq 1$; then the sequence $(x_n)_{n \geq 0}$ contradicts Definition 5.

Remark. Let B be a set. Then B is Artinian if and only if every set A whose elements are sets of components of B satisfies condition d) of Proposition 12. Indeed, necessity
205 follows from (AR.I) and Proposition 12. Conversely, if B is not Artinian, there exists an infinite sequence $(x_n)_{n \geq 0}$ with $x_\circ = B$ and $x_{n+1} \in x_n$ for every $n \geq 0$; then take for A the set whose sole element is the set X of the x_n; thus A contains a nonempty element X such that, for every element y of X, one has $y \cap X \neq \varnothing$ (indeed, y is of the form x_n, and one has $x_{n+1} \in y \cap X$).

THEOREM 2. *Let $\mathscr{c}$ be an infinite cardinal. Then:*

a) *The relation "x is an Artinian set of type $\mathscr{c}$ (resp. of strict type $\mathscr{c}$)" is collectivizing with respect to x; the set $A_{\mathscr{c}}$ of Artinian sets of type $\mathscr{c}$ has cardinality $2^{\mathscr{c}}$.*
b) *If $\mathscr{c}$ is strongly inaccessible, the set $U_{\mathscr{c}}$ of Artinian sets of strict type $\mathscr{c}$ is a universe of cardinality $\mathscr{c}$; the cardinal $\mathscr{c}(U_{\mathscr{c}}) = \sup_{x \in U_{\mathscr{c}}} \operatorname{card}(x)$ is $\mathscr{c}$.*
c) *If a universe U contains an element of cardinality $\mathscr{c}$, every Artinian set of type $\mathscr{c}$ belongs to U (in other words, $A_{\mathscr{c}} \subset U$).*
c′) *If a universe U is nonempty, every Artinian set of finite type is an element of U.*

Before proving Theorem 2, let us deduce from it a few illuminating corollaries:

COROLLARY 1. *If a universe U is Artinian, then* $\operatorname{card}(U)$ *is strongly inaccessible, and U is the set of Artinian sets of strict type* $\operatorname{card}(U)$.

206 Indeed, set $\mathscr{c}(U) = \sup_{x \in U} \operatorname{card}(x)$. This is a strongly inaccessible cardinal (beginning of n° 5). First suppose that it is uncountable; then, for every infinite cardinal $\mathscr{c} < \mathscr{c}(U)$, every Artinian set of type $\mathscr{c}$ is an element of U by c); hence every Artinian set of strict type $\mathscr{c}(U)$ is an element of U by the lemma of n° 5. This last assertion remains valid if $\mathscr{c}(U)$ is countable, by c′). Conversely, by (U.I), every set in U is of strict type $\mathscr{c}(U)$. Thus U is the universe $U_{\mathscr{c}(U)}$ of b), whence $\mathscr{c}(U) = \operatorname{card}(U)$ by b).

It follows from Corollary 1 that an Artinian universe is uniquely *determined* by its cardinality (which is moreover strongly inaccessible). Thus there is a "one-to-one correspondence" between Artinian universes and strongly inaccessible cardinals. In particular:

COROLLARY 2. *The inclusion relation $U \subset U'$ on Artinian universes is a well-ordering relation.*

Indeed, the relation $\mathcal{c} \leq \mathcal{c}'$ on cardinals is a well-ordering relation (Chapter III), and, with the notation of b) of Theorem 2, the relations $\mathcal{c} \leq \mathcal{c}'$ and $U_{\mathcal{c}} \subset U_{\mathcal{c}'}$ are equivalent.

Observe that b) of Theorem 2 gives a second proof of Theorem 1 (n° 5).

We now turn to the proof of Theorem 2. Given a set A, we shall call a *chain* of A any finite sequence $(x_j)_{j=0,\ldots,n}$ such that $x_0 = A$ and $x_{j+1} \in x_j$ for $j = 0, \ldots, n-1$. The chains of A form a *set*, by the selection–union scheme; denote it by $G(A)$. Given a 207
chain $X = (x_j)_{j=0,\ldots,n}$, the chains of the form $(x_i)_{i=0,\ldots,q}$ with $q \leq n$ will be called less than X; this endows $G(A)$ with the structure of an *ordered set*. For A to be Artinian, it is necessary and sufficient that $G(A)$ be a "*Noetherian*" ordered set (i.e. one satisfying the equivalent conditions of Chapter III, § 6, n° 5).

We shall prove that:

(*) *If A is Artinian, it is uniquely determined by the isomorphism class of the ordered set $G(A)$.*

Indeed, given an ordered set G and an element $g \in G$, let $S(g)$ denote the set of all $g' \in G$ such that $g < g'$ and such that $g \leq h \leq g'$ implies $h = g$ or $h = g'$ (in other words, the set of "immediate successors" of g). Consider the map θ that assigns to every chain $X = (x_0, \ldots, x_n)$ in $G(A)$ the set x_n; then, for every $X \in G(A)$, we have:

$$\theta(X) = \{\theta(X') | X' \in S(X)\}.$$

Since A is the image under θ of the least element $X_0 = (A)$ of $G(A)$, it will suffice to show that θ is uniquely determined by the isomorphism class of the ordered set $G(A)$. This follows from the following lemma:

LEMMA 1. *Let G be a Noetherian ordered set and let φ be a function with domain G* 208
such that, for every $g \in G$, one has:

$$\varphi(g) = \{\varphi(g') | g' \in S(g)\}. \tag{1}$$

Then φ is uniquely determined. Moreover, if U is a universe containing an element equipotent to G, then φ takes its values in U.

The hypothesis implies that $\varphi(g) = \varnothing$ if g is a maximal element of G.

Indeed, let φ and φ' be two functions satisfying (1). If $\varphi \neq \varphi'$, the set of all $g \in G$ such that $\varphi(g) \neq \varphi'(g)$ is nonempty and hence, since G is Noetherian, has a *maximal* element h. We then have $\varphi(g) = \varphi'(g)$ for every $g > h$, in particular for every immediate successor $g \in S(h)$; hence $\varphi(h) = \varphi'(h)$ by (1), which is a contradiction. Thus indeed $\varphi = \varphi'$. Now let U be a universe containing an element x equipotent to G; let us show that φ takes its values in U. Otherwise, let h be a maximal element among those $g \in G$ such that $\varphi(g) \notin U$. Then $\varphi(g') \in U$ for every $g' \in S(h)$, so that

$$\varphi(h) = \{\varphi(g') | g' \in S(h)\}$$

is a subset of U. Since its cardinality does not exceed card(G), and hence does not exceed the cardinality of an element of U, we have $\varphi(h) \in U$ (n° 1, Proposition 7). This contradiction shows that φ takes its values in U.

With this established, let us prove a) of Theorem 2. We may confine ourselves to the 209 assertion without the resp. clause, since the other follows at once. Let $\mathscr{c}$ be an *infinite* cardinal. If A is a set of type $\mathscr{c}$, then:

$$\operatorname{card}(G(A)) \leq \mathscr{c}\ . \tag{2}$$

Indeed, if A_n denotes the set of components of order n of A, then $\operatorname{card}(A_1) \leq \mathscr{c}$ and $\operatorname{card}(A_{n+1}) \leq \mathscr{c} \cdot \operatorname{card}(A_n)$, whence $\operatorname{card}(A_n) \leq \mathscr{c}^n$ by induction. But $\mathscr{c}^n = \mathscr{c}$, since $\mathscr{c}$ is infinite (Chapter III); hence $\operatorname{card}(\bigcup_{n=0}^{\infty} A_n) \leq \operatorname{card}(\mathrm{N}) \cdot \mathscr{c} = \mathscr{c}$ (Chapter III). Now $G(A)$ is a set of *finite* sequences of elements of $\bigcup_n A_n$, so inequality (2) indeed holds (Chapter III). Now, if E is a set of cardinality $\mathscr{c}$, specifying an order structure on a subset E' of E is equivalent to specifying the subset of $E \times E$ consisting of the (x, y) such that $x \in E'$, $y \in E'$, and $x \leq y$. Consequently, by (2), the isomorphism classes of the ordered sets $G(A)$ (where A is of type $\mathscr{c}$) form a set $\mathfrak{F}_{\mathscr{c}}$, and $\operatorname{card}(\mathfrak{F}_{\mathscr{c}}) \leq \operatorname{card}(\mathscr{P}(E \times E)) = 2^{\mathscr{c}}$. Let $\mathfrak{F}'_{\mathscr{c}}$ be the subset of $\mathfrak{F}_{\mathscr{c}}$ consisting of the classes of Noetherian ordered sets having a least element. If, to every $G \in \mathfrak{F}'_{\mathscr{c}}$, one assigns the value of the function φ of Lemma 1 at the least element of G, one obtains a map θ on $\mathfrak{F}'_{\mathscr{c}}$ whose image contains *all* Artinian sets of type $\mathscr{c}$. The latter therefore do indeed form a set $A_{\mathscr{c}}$, and we have $\operatorname{card}(A_{\mathscr{c}}) \leq \operatorname{card}(\mathfrak{F}'_{\mathscr{c}}) \leq \operatorname{card}(\mathfrak{F}_{\mathscr{c}}) \leq 2^{\mathscr{c}}$.

210 It remains to show that $\operatorname{card}(A_{\mathscr{c}}) = 2^{\mathscr{c}}$. For this it suffices to show that there exists an Artinian set B of type $\mathscr{c}$ and cardinality $\mathscr{c}$, since the subsets of B will then be elements of $A_{\mathscr{c}}$. This existence follows from the following lemma:

LEMMA 2. *For every cardinal $\mathscr{c}$, there exists an Artinian set B of type $\mathscr{c}$ and cardinality $\mathscr{c}$.*

Proceed by transfinite induction on $\mathscr{c}$. For $\mathscr{c} = 0$ there is no choice, and one takes $B = \varnothing$. If $\mathscr{c}$ has a predecessor $\mathscr{c}'$, let B' be an Artinian set of type $\mathscr{c}'$ and cardinality $\mathscr{c}'$; one then has $\mathscr{c} \leq 2^{\mathscr{c}'}$, so there exists a subset B of $\mathscr{P}(B')$ of cardinality $\mathscr{c}$; the elements of B are subsets of B' and therefore have cardinalities $\leq \mathscr{c}' \leq \mathscr{c}$; the higher-order components of B are components of B', and hence also have cardinalities $\leq \mathscr{c}' \leq \mathscr{c}$. Finally, if $\mathscr{c}$ has no predecessor, choose, for every cardinal $\mathscr{c}_\lambda < \mathscr{c}$, an Artinian set B_λ of type $\mathscr{c}_\lambda$ and cardinality $\mathscr{c}_\lambda$; then $B = \bigcup_\lambda B_\lambda$ has the required property. This proves Lemma 2 and completes the proof of part a).

We turn to b). Let $\mathscr{c}$ be a strongly inaccessible cardinal. By a), we already know that the Artinian sets of strict type $\mathscr{c}$ form a set $U_{\mathscr{c}}$. The fact that $U_{\mathscr{c}}$ is a universe follows immediately from properties (AR.I) to (AR.IV) of Artinian sets (beginning of n°),
211 and from cardinal estimates analogous to those of Proposition 10 of n° 5. The relation $\sup_{x \in U_{\mathscr{c}}} \operatorname{card}(x) = \mathscr{c}$ followsfrom Lemma 2, applied to cardinals $< \mathscr{c}$. Finally, to show that $\operatorname{card}(U_{\mathscr{c}}) = \mathscr{c}$, first suppose that $\mathscr{c}$ is *uncountable*; bythe lemma of n° 5, $U_{\mathscr{c}}$ is the union $\bigcup_{\mathscr{d} < \mathscr{c}} A_{\mathscr{d}}$, where $A_{\mathscr{d}}$ denotes the set of Artinian sets of type $\mathscr{d}$; now $\operatorname{card}(A_{\mathscr{d}}) = 2^{\mathscr{d}} < \mathscr{c}$ (by a)); on the other hand, the set of cardinals $\mathscr{d} < \mathscr{c}$ has cardinality $\leq \mathscr{c}$ (Chapter III); whence $\operatorname{card}(U_{\mathscr{c}}) \leq \mathscr{c} \cdot \mathscr{c} = \mathscr{c}$, and also $\operatorname{card}(U_{\mathscr{c}}) \geq \mathscr{c}$, since $\operatorname{card}(U_{\mathscr{c}}) \geq \operatorname{card}(A_{\mathscr{d}}) = 2^{\mathscr{d}}$ for every $\mathscr{d} < \mathscr{c}$.

The case $\mathscr{c} = 0$ being trivial, there remains the case where $\mathscr{c}$ is the *countably infinite cardinal.* In this case $U_{\mathscr{c}}$ is the set of Artinian sets of finite type (i.e. finite together with all their components), and one uses a pleasant result of a combinatorial nature.

LEMMA 3. (D. K"onig?). *Consider two infinite sequences $(E_n)_{n\geq 1}$, $(f_n)_{n\geq 1}$ of finite sets E_n and maps $f_n : E_{n+1} \to E_n$. If there exists no infinite sequence $(x_n)_{n\geq 1}$ such that $x_n \in E_n$ and $f_n(x_{n+1}) = x_n$ for every n, then E_n is empty for all sufficiently large n.*

In other words, if all sequences (x_n) such that $f_n(x_{n+1}) = x_n$ are *finite*, their lengths are *bounded*. This can be expressed in terms of projective limits: a projective limit of nonempty finite sets is nonempty (cf. General Topology, Chapter I, 2nd ed., § 9, n° 6, Proposition 8, 2°)).

Indeed, argue by contradiction. If there exist nonempty E_n for arbitrarily large n, no E_n is empty (since $E_n = \varnothing$ implies $E_{n+1} = \varnothing$ in view of the existence of $f_n : E_{n+1} \to E_n$). Call a finite sequence $(x_j)_{1 \le j \le n}$ *coherent* if $f_j(x_{j+1}) = x_j$ for $j = 1, \dots, n-1$. We prove by induction on n the existence of a coherent sequence $(a_1, \dots, a_n)$ $(a_i \in E_i)$ that, for every $q \ge n$, can be extended to a coherent sequence $(a_1, \dots, a_n, x_{n+1}, \dots, x_q)$ of length q. This is evident for $n = 0$, since no E_q is empty. Pass from n to $n+1$. If, for every $x \in f_n^{-1}(\{a_n\}) \subset E_{n+1}$, all coherent sequences extending $(a_1, \dots, a_n, x)$ had lengths bounded by an integer $q(x)$, then all coherent sequences extending $(a_1, \dots, a_n)$ would have bounded lengths (by $\sup_x q(x)$), since E_{n+1} is finite; hence there exists $a_{n+1} \in f_n^{-1}(\{a_n\})$ such that the coherent sequence $(a_1, \dots, a_n, a_{n+1})$ admits extensions of arbitrary length. We then obtain an infinite sequence $(a_n)_{n \ge 1}$ contradicting the hypothesis. 212

It follows from Lemma 3 that if A is an Artinian set of finite type, then the ordered set $G(A)$ of its chains is *finite*: indeed, take for E_n the set of chains $(x_n \in x_{n-1} \in \cdots \in x_1 \in A)$ with $n+1$ terms (which is finite because the components of A of order $\le n+1$ are finite in number and are all finite), and for f_n the map $(x_{n+1} \in x_n \in x_{n-1} \in \dots) \mapsto (x_n \in x_{n-1} \in \dots)$. Now the set of isomorphism classes of finite ordered sets is *countable*: indeed, giving an order structure on a finite subset of $\mathbf{N}$ is equivalent to giving its graph, which is a finite subset of $\mathbf{N} \times \mathbf{N}$; on the other hand, the set of finite subsets of a countable set is countable (Chapter III). It follows from (*) and Lemma 1 that the set $\mathfrak{a}$ of Artinian sets of finite type is countable. It is infinite by Lemma 2 (or, more simply, by using the sequence $(z_n)_{n \ge 0}$ defined by $z_0 = \varnothing$, $z_{n+1} = \{z_n\}$). This completes the proof of b). 213

Remark. We have just proved that if A is an Artinian set of finite type, it has only finitely many components. There therefore exists an integer n such that A is of type n.

We turn to the proof of c). Let $\mathscr{c}$ be an infinite cardinal, let U be a universe admitting an element of cardinality $\mathscr{c}$, and let A be an Artinian set of type $\mathscr{c}$. Then the ordered set $G(A)$ has cardinality $\le \mathscr{c}$ (formula (2) above). The second assertion of Lemma 1, applied to $G(A)$, shows that the map $(x_0, \dots, x_n) \rightsquigarrow x_n$ from $G(A)$ takes its values in U. In other words, all the components of A are elements of U. This proves c). The proof of c′) is analogous: if A is an Artinian set of finite type, we have just seen that $G(A)$ is finite; since U is nonempty, it contains an element equipotent to $G(A)$ by the corollary to Proposition 7 (n° 1). C.Q.F.D.

7. Vague metamathematical remarks

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a) The axiom "*every set is Artinian*" is *harmless*. Indeed, if one has a model M of Set Theory, the set M' of the Artinian elements of M is also a model (cf. Proposition 11).
b) The universe axiom (A.6) (and the equivalent axiom $(A'.6)$ on strongly inaccessible cardinals) is *independent* of the rest of Set Theory. Indeed, let $\mathscr{c}$ be the first uncountable strongly inaccessible cardinal. The universe $U_{\mathscr{c}}$ of Artinian sets of strict type $\mathscr{c}$ (Theorem 2, b)) is a model of Set Theory without (A.6): the elements of $U_{\mathscr{c}}$ are called "sets," the "membership relation" is the restriction to $U_{\mathscr{c}}$ of the usual one, etc. The "universes" of the model are therefore the ordinary universes that are elements of $U_{\mathscr{c}}$. But we have seen that the only universes

which are elements of U_c are the two pequeños $U_0 = \varnothing$ and U_1. Thus U_1 is a "set" which is not an element of any "universe." We therefore have a model of Set Theory in which (A.6) is false.

c) Bourbaki was too cautious in merely "presuming" that the *axiom of infinity* (A.5) is *independent* of the preceding axioms and schemes. It is indeed independent, since the countable universe U_1 of Artinian sets of finite type is a model in which (A.5) is false and the preceding axioms and schemes are true.

d) It would be very interesting to prove that the universe axiom (A.6) is harmless. That looks difficult—and in fact it is unprovable, says Paul Cohen.

215 The adjective "vague" in the title means that, when constructing models, we did not take the trouble to fiddle with the symbol τ so that it could not take us outside the model. The trick, if one is considering a model M, is to replace the ordinary $\tau_x(\mathscr{R}(x))$ by:

$$\tau_x(\mathscr{R}(x) \text{ and } x \in M).$$

One still has to check that this really turns the ordinary quantifiers into the quantifiers formerly called "typical":

$$(\forall x \in M) \quad , \quad (\exists x \in M).$$

Exercises. 1) Let n be an integer ≥ 1. Show that the set of Artinian sets of type n is infinite (the sets $z_0 = \varnothing$, $z_1 = \{\varnothing\}$, $z_{q+1} = \{z_q\}$ are of type 1).

2) Define the *height* of a set A to be the supremum (finite or infinite) of the integers n for which there exists a sequence $x_n \in x_{n-1} \in \cdots \in x_0 = A$. Show that the sets of height $\leq n$ are finite and form a finite set, whose cardinality p_n can be calculated from $p_0 = 1$, $p_{n+1} = 2^{p_n}$ (proceed by induction on n, noting that the elements of a set A of height $\leq n$ are sets of height $\leq n-1$).

3) Let G be a Noetherian ordered set having a least element g_0, and suppose that, for every $h \in G$, the set of $g \in G$ such that $g \leq h$ is finite and totally ordered.

216 a) Show that every element $h \neq g_0$ of G has a unique predecessor.

b) Let φ be the map on G defined in Lemma 1 (i.e. $\varphi(g)$ is the set of all $\varphi(g')$ as g' ranges over the successors of g). Put $A = \varphi(g_0)$. Show that A is an Artinian set. For $g \in G$, let $g_0 < g_1 < \cdots < g_n = g$ be the sequence of elements $\leq g$; put $f(g) = (\varphi(g_0), \varphi(g_1), \ldots, \varphi(g_n))$; show that f is an increasing surjection from G onto the ordered set $G(A)$ of the text. Show that, if f is *injective*, it is an *isomorphism* from G onto $G(A)$.

c) For $g \in G$, let M_g be the set of upper bounds of g. Suppose that, for every pair of distinct elements g, g' having the same predecessor p, the ordered sets M_g and $M_{g'}$ are not isomorphic. Show that the map f of b) is then an *isomorphism* from G onto $G(A)$. [If $f(g) = f(g')$ with $g \neq g'$, and if $g_0 < g_1 < \cdots < g_n = g$ and $g'_0 < g'_1 < \cdots < g'_{n'} = g'$ are the sequences of elements $\leq g$ and $\leq g'$, respectively, show that $n = n'$, and that one may assume that g and g' have the same predecessor p; then consider the set of $p \in G$ for which there exist two distinct successors g, g' of p such that $f(g) = f(g')$, a maximal element q of this set, and two distinct successors h, h' of q such that $f(h) = f(h')$; note that the restrictions of f to M_h and $M_{h'}$ are injective and hence (by b)) are isomorphisms from M_h onto $G(\varphi(h))$ and from $M_{h'}$ onto $G(\varphi(h'))$; deduce from the hypothesis that M_h and $M_{h'}$ are not isomorphic that $\varphi(h) \neq \varphi(h')$, which contradicts $f(h) = f(h')$].

217 N.B. Exercise 3) gives very precise information on how Artinian sets are "made."

We already knew that such a set A is determined by the isomorphism class of the ordered set $G(A)$ (Lemma 1). We now know how to *characterize* the ordered sets G isomorphic to some $G(A)$:

α) G is Noetherian and has a least element;

β) For every $g \in G$, the set of $h \in G$ such that $h \leq g$ is totally ordered and finite (whence the existence and uniqueness of the predecessor of every nonleast g);

γ) If g, g' are distinct elements having the same predecessor, the set M_g of upper bounds of g and the set $M_{g'}$ of upper bounds of g' are not isomorphic.

4) Let $(A_n)_{n\geq 0}$ be the sequence of finite *ordinals* defined by $A_0 = \varnothing$, $A_{n+1} = A_n \cup \{A_n\}$. Show that the ordered set $G(A_n)$ has 2^n elements, and that the number of its elements of height q (in the sense of Exercise 2) is $\binom{n}{q}$.

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EXPOSÉ II

Topologies and sheaves

J.-L. Verdier

After defining topologies and pretopologies (Section 1) and sheaves of sets (Section 2), in Section 3 we turn to the central theorem of this exposé (3.4): the existence of the sheaf associated with a presheaf. In order to cover all the cases encountered in practice, the existence of this functor is proved for presheaves on a $\mathscr{U}$-site (3.0.2). The consequences of this theorem for the behavior of inductive and projective limits in categories of sheaves are drawn in Sections 4 and 5. In Section 6 we define and study sheaves with values in arbitrary categories, immediately focusing attention on sheaves of rings, groups, modules, etc. 218

1. Topologies, covering families, pretopologies

Definition 1.1. *A topology on a category C consists, for every object X of C, of a set $J(X)$ of sieves on X, subject to the following axioms:*

- T 1) (Stability under base change). *For every object X of C, every sieve $R \in J(X)$, and every morphism $f : Y \to X$ ($Y \in \mathrm{ob}(C)$), the sieve $R \times_X Y$ on Y belongs to $J(Y)$.*
- T 2) (Local character). *If R and R' are two sieves on X, if $R \in J(X)$, and if, for every $Y \in \mathrm{ob}(C)$ and every morphism $Y \to R$, the sieve $R' \times_X Y$ belongs to $J(Y)$, then R' belongs to $J(X)$.* 219
- T 3) *For every object X of C, X belongs to $J(X)$.*

1.1.1. The sieves belonging to $J(X)$ will be called the *sieves covering* X, or the *refinements of* X. It follows immediately from axioms (T1), (T2), and (T3) that the set of sieves covering X is stable under finite intersections, and that every sieve containing a covering sieve is itself a covering sieve. Thus the set $J(X)$, ordered by inclusion, is cofiltered (I 8).

1.1.2. Let C be a category, and let T and T' be two topologies on C. The topology T is said to be *finer than the topology* T' if, for every object X of C, every refinement of X for the topology T' is a refinement of X for the topology T. This defines an order structure on the set of topologies.

1.1.3. Let $(T_i)_{i\in I}$ be a family of topologies on C. For every object X of C, the set of sieves on X that are covering for all the topologies T_i satisfies axioms (T 1), (T 2), and (T 3), and hence defines a topology: the *intersection topology* of the T_i, i.e. the infimum of the T_i. It is the finest topology that is coarser than all the topologies T_i. Consequently, the family $(T_i)_{i\in I}$ also admits a *supremum*: the intersection topology of the topologies finer than each of the T_i.

1.1.4. In particular, taking $J(X)$ to be the set of all sieves on X gives a topology finer than every topology on C; we call it the *discrete topology*. 220

There is a topology coarser than every topology on C: the topology for which $J(X) = \{X\}$ for every X of C. This topology is called the *coarse* or *chaotic* topology.

1.1.5. A category C equipped with a topology is called a *site*. The category C is called the *category underlying the site.*

Definition 1.2. *Let C be a site and X an object of C. A family of morphisms $(f_\alpha : X_\alpha \to X)$, $\alpha \in A$, is called a covering family if the sieve generated by the family f_α* (I 4.3.3) *is a sieve covering X.*

1.1.6. Let C be a category. Suppose that, for each object X of C, we are given a set of families of morphisms with target X. There is then a coarsest topology T for which the given families are covering, namely the intersection (1.1.3) of all the topologies in question. This topology is called the *topology generated by the given sets of families of morphisms.* In general, it is difficult to describe all the covering sieves of this topology. The situation is more favorable, however, in the following case:

Definition 1.3. *Let C be a category. A pretopology on C consists, for each object X of C, of a set* $\mathrm{Cov}(X)$ *of families of morphisms with target X, subject to the following axioms:*

221 PT 0) *For every object X of C, the morphisms in the families belonging to* $\mathrm{Cov}(X)$ *are pullbackable. (Recall that a morphism $Y \to X$ is called pullbackable if, for every morphism $Z \to X$, the fiber product $Z \times_X Y$ is representable.)*

PT 1) *For every object X of C, every family $(X_\alpha \to X)_{\alpha \in A}$ belonging to* $\mathrm{Cov}(X)$, *and every morphism $Y \to X$ ($Y \in \mathrm{ob}(C)$), the family $(X_\alpha \times_X Y \to Y)_{\alpha\in A}$ belongs to* $\mathrm{Cov}(Y)$. *(Stability under base change.)*

PT 2) *If $(X_\alpha \to X)_{\alpha \in A}$ belongs to* $\mathrm{Cov}(X)$ *and if, for every $\alpha \in A$, $(X_{\beta_\alpha} \to X_\alpha)_{\beta_\alpha \in B_\alpha}$ belongs to* $\mathrm{Cov}(X_\alpha)$, *then the family $(X_\gamma \to X)_{\gamma \in \coprod_{\alpha\in A} B_\alpha}$ (where, for every $\gamma = (\alpha, \beta_\alpha)$, $\alpha \in A$, $\beta_\alpha \in B_\alpha$, the morphism $X_\gamma \to X$ is the composite morphism $X_\gamma = X_{\beta_\alpha} \to X_\alpha \to X$) belongs to* $\mathrm{Cov}(X)$. *(Stability under composition.)*

PT 3) *The family $X \xrightarrow{\mathrm{id}_X} X$ belongs to* $\mathrm{Cov}(X)$.

1.3.1. The definition in 1.1.6 allows us to consider, for a given *pretopology* on C, the *topology* on C generated by that pretopology. Note that if C is a category in which fiber products are representable, then every topology T on C can be defined by a pretopology, namely the one for which $\mathrm{Cov}(X)$ consists of *all* the families covering X for the topology T.

Proposition 1.4. *Let C be a category, E a pretopology on C, T the topology defined by the pretopology E* (1.1.6), *and X an object of C. Denote by $J_E(X)$ the set of sieves on X generated by families of the pretopology, and by $J_T(X)$ the set of refinements of X for*
222 *the topology T. Then $J_E(X)$ is cofinal in $J_T(X)$. In other words, a sieve R on X belongs to $J_T(X)$ if and only if there exists a sieve $R' \in J_E(X)$ such that $R' \subset R$.*

Proof. For every object X of C, let $J'(X)$ be the set of sieves on X that contain a sieve belonging to $J_E(X)$. Clearly, $J'(X) \subset J_T(X)$. To prove that $J'(X) = J_T(X)$, it is enough to show that the data $J'(X)$ define a topology on C. Now the $J'(X)$ clearly satisfy axioms (T 1) and (T 3). It therefore remains to verify (T 2). For this, it is enough to prove that if R' is a subsieve of $R \in J_E(X)$ such that, for every $Y \to R$, the sieve $R' \times_X Y$ on Y belongs to $J'(Y)$, then the sieve R' belongs to $J'(X)$. Now the sieve R is generated by a family $(X_\alpha \to X)$ belonging to $\mathrm{Cov}(X)$, and for every α the sieve $R' \times_X X_\alpha$ contains a sieve generated by a family $(X_{\beta_\alpha} \to X_\alpha)$ belonging to $\mathrm{Cov}(X_\alpha)$. It follows that the sieve R' contains a sieve generated by the family $(X_{\beta_\alpha} \to X)$. Thus, by axiom (PT 2), R' contains a sieve belonging to $J_E(X)$ and consequently belongs to $J'(X)$, C.Q.F.D.

2. Sheaves of sets

Definition 2.1. *Let C be a site whose underlying category is a $\mathscr{U}$-category. A presheaf F with values in $\mathscr{U}-$*Ens *is said to be separated (resp. to be a sheaf) if, for every sieve R covering an object X of C, the map*

$$\mathrm{Hom}_{C^\wedge}(X, F) \longrightarrow \mathrm{Hom}_{C^\wedge}(R, F)$$

is injective (resp. bijective). The full subcategory of $C^\wedge$ whose objects are the sheaves is 223
called the category of sheaves of sets on C, and is most often denoted by $\widetilde{C}$ [4]*. When no ambiguity can result, we shall simply say the category of sheaves on C; on the other hand, we shall sometimes be more precise and say the category of sheaves with values in $\mathscr{U}-$*Ens*, or the category of $\mathscr{U}$-sheaves.*

Proposition 2.2. *Let C be a $\mathscr{U}$-category, and let $\mathscr{F} = (F_i)_{i\in I}$ be a family of $\mathscr{U}$-presheaves on C. For each object X of C, denote by $J_{\mathscr{F}}(X)$ the set of sieves $R \to X$ such that, for every morphism $Y \to X$ in C with target X, the sieve $R \times_X Y$ has the following property: the map*

$$\mathrm{Hom}_{C^\wedge}(Y, F_i) \longrightarrow \mathrm{Hom}_{C^\wedge}(R \times_X Y, F_i)$$

is bijective (resp. injective) for every $i \in I$. Then the sets $J_{\mathscr{F}}(X)$ define a topology on C, which is the finest topology for which each F_i is a sheaf (resp. separated presheaf).

Proof. The $J_{\mathscr{F}}(X)$ clearly satisfy axioms (T 1) and (T 3). It remains to show that they satisfy (T 2). For this it suffices to show the following:

1) If $R' \hookrightarrow R \hookrightarrow X$ are two sieves on X such that $R \in J_{\mathscr{F}}(X)$ and such that, for every $Y \to R$ (where Y is an object of C), the sieve $R' \times_X Y$ belongs to $J_{\mathscr{F}}(Y)$, then the sieve R' belongs to $J_{\mathscr{F}}(X)$.
2) If $R' \hookrightarrow R \hookrightarrow X$ are two sieves on X such that $R' \in J_{\mathscr{F}}(X)$, then the sieve R belongs to $J_{\mathscr{F}}(X)$.

Note that, in case 2), for every $Y \to R$ (where Y is an object of C), the sieve $R' \times_X Y$ belongs to $J_{\mathscr{F}}(Y)$. Now, in cases 1) and 2), R is the inductive limit of the objects Y of C over R (I 3.4). The inductive limits in $C^\wedge$ are universal (I 3.3). Consequently, in cases 1) 224
and 2), R' is the inductive limit of the $R' \times_X Y$. Now, in cases 1) and 2), $R' \times_X Y$ belongs to $J_{\mathscr{F}}(Y)$. Passing to the inductive limit over the objects Y of C over R, we deduce that the map

$$\mathrm{Hom}_{C^\wedge}(R, F_i) \longrightarrow \mathrm{Hom}_{C^\wedge}(R', F_i)$$

is a bijection (resp. an injection) for every $i \in I$. It follows that the maps

$$\mathrm{Hom}_{C^\wedge}(X, F_i) \longrightarrow \mathrm{Hom}_{C^\wedge}(R', F_i) \quad , \quad \mathrm{Hom}_{C^\wedge}(X, F_i) \longrightarrow \mathrm{Hom}_{C^\wedge}(R, F_i)$$

are, in cases 1) and 2), bijections (resp. injections). Moreover, hypotheses 1) and 2) are plainly stable under arbitrary base change $Y \to X$ (Y an object of C). It then follows that, for every object Y of C over X and every $i \in I$, the maps

$$\mathrm{Hom}_{C^\wedge}(Y, F_i) \longrightarrow \mathrm{Hom}_{C^\wedge}(R \times_X Y, F_i)$$
$$\mathrm{Hom}_{C^\wedge}(Y, F_i) \longrightarrow \mathrm{Hom}_{C^\wedge}(R' \times_X Y, F_i)$$

are in both cases bijections (resp. injections); this shows that R' and R belong to $J_{\mathscr{F}}(X)$,
C.Q.F.D.

[4] or $C^\sim$, depending on the mood of the typewriter.

Corollary 2.3. *Let C be a $\mathscr{U}$-category, and for every X of C let $K(X)$ be a set of sieves on X. Suppose that the $K(X)$ satisfy axiom* (T 1) *of* (1.1). *A presheaf F is a sheaf (resp. a separated presheaf) for the topology generated (1.1.6) by the $K(X)$ if and only if, for every object X of C and every sieve $R \in K(X)$, the map*

$$\mathrm{Hom}_{C^\wedge}(X, F) \longrightarrow \mathrm{Hom}_{C^\wedge}(R, F)$$

is a bijection (resp. an injection).

225 Corollary 2.4. *In particular, let C be a $\mathscr{U}$-category equipped with a pretopology. A presheaf F is a sheaf (resp. a separated presheaf) if and only if, for every object X of C and every covering family $(X_\alpha \to X)$ belonging to* $\mathrm{Cov}(X)$, *the diagram of sets*

$$F(X) \to \prod_{\alpha \in A} F(X_\alpha) \rightrightarrows \prod_{(\alpha,\beta) \in A \times A} F(X_\alpha \times_X X_\beta)$$

is exact (resp. the map

$$F(X) \longrightarrow \prod_{\alpha \in A} F(X_\alpha)$$

is injective).

Proof. Apply 2.3, then (I 3.5) and (I 2.12).

Corollary 2.4 recovers the definition given in [1].

Definition 2.5. *Let C be a $\mathscr{U}$-category. The canonical topology of C is the finest topology for which the representable functors are sheaves (2.2). A sieve covering X for the canonical topology will be called a universally strictly epimorphic sieve. A covering family for the canonical topology will be called a universally strictly epimorphic family. If, in addition, the morphisms of the covering family are pullbackable, the family will be called universally effective epimorphic* [2].

Remark 2.5.1. For almost all the sites that have been used up to now, the topology is *coarser* than the canonical topology; in other words, the representable functors on C
226 are sheaves, i.e. the covering families of C are universally strictly epimorphic. The only exception to this rule is the site $\widehat{C}$ studied in n° 5 (whose topology is *finer*, and most often strictly finer, than the canonical topology).

Proposition 2.6. *A sieve $R \hookrightarrow X$ is universally strictly epimorphic if and only if, for every object $Y \to X$ of C over X, and for every object Z of C, the map*

$$\mathrm{Hom}_C(Y, Z) \longrightarrow \varprojlim_{C/(Y \times_X R)} \mathrm{Hom}_C(., Z)$$

is a bijection.

Proof. Immediate upon applying 2.2 and then (I 5.3).

Remarks 2.7.

1) Proposition 2.6 gives a characterization of the universally strictly epimorphic sieves of a category C that is independent of the universe in which the presheaves take their values, provided only that the morphism sets $\mathrm{Hom}(X, Y)$ of C belong to that universe. It thus makes it possible to define the canonical topology for any category C, without having to specify the universes.
2) Let C be a site whose underlying category is a $\mathscr{U}$-category, let F be a $\mathscr{U}$-presheaf, and let $\mathscr{V}$ be a universe containing $\mathscr{U}$. The category underlying C is a $\mathscr{V}$-category, and F may be regarded as a $\mathscr{V}$-presheaf. The $\mathscr{U}$-presheaf F is a sheaf (resp. a separated presheaf) if and only if the $\mathscr{V}$-presheaf F is a sheaf

(resp. a separated presheaf). In other words, the condition for a presheaf F to be a sheaf (resp. a separated presheaf) does not depend (in the sense just specified) on the universe in which the presheaf F takes its values. In particular, let C be a site and let F be a $\mathscr{U}$-presheaf; we say that F is a sheaf (resp. a separated presheaf) if there exists a universe $\mathscr{V}$ containing $\mathscr{U}$ such that the category C is a $\mathscr{V}$-category and such that F is a $\mathscr{V}$-sheaf (resp. a separated $\mathscr{V}$-presheaf). This property does not depend on the universe $\mathscr{V}$. 227

3. The sheaf associated with a presheaf

Definition 3.0.1. *Let C be a site. A topologically generating family (or, when no confusion can result, a generating family) of C is a set G of objects of C such that every object of C is the target of a covering family of morphisms of C (1.2) whose sources are elements of G.*

Definition 3.0.2. *Let $\mathscr{U}$ be a universe. A $\mathscr{U}$-site is a site C whose underlying category is a $\mathscr{U}$-category* (I 1.1)*, and which possesses a small topologically generating family. Let C be a $\mathscr{U}$-category; a $\mathscr{U}$-topology on C is a topology on C making C a $\mathscr{U}$-site. A site C is called $\mathscr{U}$-small, or, by abuse of language, small, if its underlying category is small* (I 1.0)*.*

3.0.3. It follows immediately from the definitions that every topology finer than a $\mathscr{U}$-topology is a $\mathscr{U}$-topology, and that a small site is a $\mathscr{U}$-site.

Proposition 3.0.4. *Let C be a $\mathscr{U}$-site and G a small topologically generating family of C. For every $X \in \operatorname{ob} C$, denote by $J_G(X)$ the set of sieves covering X generated by a family of morphisms* 228

$$(Y_\alpha \xrightarrow{u_\alpha} X), \alpha \in A, \text{ where } Y_\alpha \in G.$$

1) *The set $J_G(X)$ is small.*
2) *The set $J_G(X)$ is cofinal in the set $J(X)$ of all sieves covering X, ordered by inclusion.*
3) *For every $R \in J_G(X)$, there exists a small epimorphic family* (I 10.2) *$(u_\alpha : Y \to R)$, $\alpha \in A$, with $Y \in G$.*

Proof. 1) Put $A(X) = \coprod_{Y \in G} \operatorname{hom}(Y, X)$. The set $A(X)$ is small (I 0) and $\operatorname{card}(J_G(X)) \leq 2^{\operatorname{card}(A(X))}$. Consequently $J_G(X)$ is small.

2) Let $R \in J(X)$. Put $A(R) = \coprod_{Y \in G} \operatorname{Hom}(Y, R)$, and let R' be the sieve of X generated by the family $(Y \xrightarrow{u} R \hookrightarrow X)$, $u \in A(R)$. One has $R' \subset R$, and it suffices to show that R' is covering. By axiom (T 2) of 1.1, it suffices to show that for every morphism $Z \to R$, $Z \in \operatorname{ob} C$, the sieve $R' \times_X Z$ of Z is covering. Now the sieve $R' \times_X Z$ contains the sieve generated by the family of all morphisms $Y \to Z$ with $Y \in G$, which is covering by hypothesis. Therefore the sieve $R' \times_X Z$ of Z is covering (axiom T2 of 1.1).

3) Let $R \in J_G(X)$. The family $(Y \xrightarrow{u} R)$, $u \in A(R) = \coprod_{Y \in G} \operatorname{Hom}(Y, R)$, is epimorphic by hypothesis. Now, for every $Y \in \operatorname{ob} C$, $\operatorname{Hom}(Y, R)$ is contained in $\operatorname{Hom}(Y, X)$, which is small. Consequently $A(R)$ is small.

3.0.5. Let C be a $\mathscr{U}$-site, $\mathscr{V}$ a universe such that $C \in \mathscr{V}$ and $\mathscr{U} \subset \mathscr{V}$, and G a $\mathscr{U}$-small topologically generating family of C. The category $\hat{C}$ of presheaves of $\mathscr{U}$-sets on C is a $\mathscr{V}$-category (I 1.1.1). Let X be an object of C. The set $J(X)$ of sieves covering X is $\mathscr{V}$-small and, ordered by inclusion, is cofiltered (1.1.1). For every $\mathscr{U}$-presheaf F, the inductive limit $\varinjlim_{J(X)} \operatorname{Hom}_{\hat{C}}(R, F)$ is therefore representable by an element of $\mathscr{V}$ (I 229

2.4.1). Moreover, it follows from (3.0.4 3)) that, for every $R \in J_G(X)$, $\mathrm{Hom}_{C^\wedge}(R, F)$ is $\mathscr{U}$-small, and since $J_G(X)$ is a $\mathscr{U}$-small set cofinal in $J(X)$ (*loc. cit.*), it follows from (I 2.4.2) that $\varinjlim_{J(X)} \mathrm{Hom}_{C^\wedge}(R, F)$ is $\mathscr{U}$-small. Choose, for every F and every X, an element of $\mathscr{U}$ representing this inductive limit, and put

$$LF(X) = \varinjlim_{J(X)} \mathrm{Hom}_{C^\wedge}(., F).$$

Let $g : Y \to X$ be a morphism of C. The base-change functor $g^* : J(X) \to J(Y)$ defines a map

$$LF(g) : LF(X) \longrightarrow LF(Y),$$

which makes $X \mapsto LF(X)$ a presheaf on C.

Since the morphism $\mathrm{id}_X : X \to X$ is an element of $J(X)$, for every object X of C one has a map

$$\ell(F)(X) : F(X) \longrightarrow LF(X),$$

which visibly defines a morphism of functors

$$\ell(F) : F \longrightarrow LF.$$

It is moreover clear that $F \mapsto LF$ is a functor in F and that the morphisms $\ell(F)$ define a morphism

$$\ell : \mathrm{Id} \longrightarrow L.$$

Finally, let $R \hookrightarrow X$ be a covering sieve of X. The definition of $LF(X)$ and (I 1.4) provide a map

$$Z_R : \mathrm{Hom}_{C^\wedge}(R, F) \longrightarrow \mathrm{Hom}_{C^\wedge}(X, LF),$$

230 and, for every morphism $Y \xrightarrow{g} X$ of C, the definition of the functor LF shows that the following diagram is commutative:

$$(*)\qquad \begin{tikzcd}
Z_R : & \mathrm{Hom}_{C^\wedge}(R, F) \arrow[r] \arrow[d] & \mathrm{Hom}_{C^\wedge}(X, LF) \arrow[d] \\
Z_{R\times_X Y} : & \mathrm{Hom}_{C^\wedge}(R \times_X Y, F) \arrow[r] & \mathrm{Hom}_{C^\wedge}(Y, LF)
\end{tikzcd}$$

(the vertical arrows are the evident arrows).

Let us now copy [2].

Lemma 3.1. 1) *For every covering sieve $i_R : R \hookrightarrow X$ and every $u : R \to F$, the diagram*

$$(**)\qquad \begin{tikzcd}
F \arrow[r, "\ell(F)"] & LF \\
R \arrow[u, "u"] \arrow[r, hook, "i_R"'] & X \arrow[u, "Z_R(u)"']
\end{tikzcd}$$

is commutative.

2) *For every morphism $v : X \to LF$, there exists a covering sieve R of X and a morphism $u : R \to F$ such that $Z_R(u) = v$.*

3) *Let Y be an object of C, and let $u, v : Y \rightrightarrows F$ be two morphisms such that $\ell(F) \circ u = \ell(F) \circ v$. Then the kernel of the pair (u, v) is a covering sieve of Y.*

4) *Let R, R' be two covering sieves of X, and let $u : R \to F$, $u' : R' \to F$ be two morphisms. For $Z_R(u) = Z_{R'}(u')$, it is necessary and sufficient that u and u' coincide on a covering sieve $R'' \hookrightarrow R \times_X R'$.*

Proof. The only nontrivial assertion is assertion 1). We must show that $Z_R(u) \circ i_R = \ell(F) \circ u$. For this it suffices to show (I 3.4) that the composites of these morphisms with every morphism $Y \xrightarrow{g} R$ (Y an object of C) are equal. Consider $f = i_R \circ g$ and the fiber product $R \times_X Y$: 231

$$\begin{tikzcd}
F \arrow[r, "\ell(F)"] & LF \\
R \arrow[u, "u"] \arrow[r, hook, "i_R"'] & X \arrow[u, "Z_R(u)"'] \\
R \times_X Y \arrow[u, "g'"] \arrow[r, "\sim", "i'"'] & Y. \arrow[u, "f"'] \arrow[ul, "g"]
\end{tikzcd}$$

In the diagram above, i' is a monomorphism admitting a section. Consequently (1.3 3) i' is an isomorphism. By the definition of the morphism $\ell(F)$, the morphism $Z_{R\times_X Y}(u \circ g')$ is equal to $\ell(F) \circ u \circ g$. On the other hand, the commutativity of diagram (*) gives the equality $Z_{R\times_X Y}(u \circ g') = Z_R(u) \circ f$, and hence one indeed has $\ell(F) \circ u \circ g = Z_R(u) \circ i_R \circ g$.

Proposition 3.2. 1) *The functor L is left exact* (I 2.3.2).

2) *For every presheaf F, LF is a separated presheaf.*
3) *The presheaf F is separated if and only if the morphism $\ell(F) : F \to LF$ is a monomorphism. The presheaf LF is then a sheaf.*
4) *The following properties are equivalent:*
   (i) *$\ell(F) : F \to LF$ is an isomorphism.*
   (ii) *F is a sheaf.*

Proof. 1) It suffices to show (I 3.1) that, for every object X of C, the functor $F \mapsto LF(X)$ commutes with finite projective limits. By the definition of a projective limit, the functor

$$F \longmapsto \mathrm{Hom}_{\hat{C}}(R, F) \quad R \hookrightarrow X \in \mathrm{ob}(J(X))$$

commutes with projective limits, and the inductive limit $\varinjlim_{J(X)}$ commutes with finite projective limits because $J(X)$ is a cofiltered set (I 2.7). 232

2) Let X be an object of C, and let $f, g : X \rightrightarrows LF$ be two morphisms that coincide on a covering sieve $R \hookrightarrow X$ of X. By 3.1 2), there exists a covering sieve $R' \hookrightarrow X$, which may always be supposed contained in R, and two morphisms $u, v : R' \rightrightarrows F$ such that $Z_{R'}(u) = f$ and $Z_{R'}(v) = g$. By 3.1 1), one then has $\ell(F) \circ u = \ell(F) \circ v$. Consequently (3.1 4)) u and v coincide on a covering sieve $R'' \hookrightarrow R'$. Let w be the restriction of u to R''. One has $Z_{R''}(w) = Z_{R'}(u) = Z_{R'}(v)$, and consequently $f = g$. Thus the presheaf LF is separated.
3) If F is separated, the morphism $\ell(F)$ is a monomorphism because a filtered inductive limit of monomorphisms is a monomorphism. If $\ell(F)$ is a monomorphism, the presheaf F is a subpresheaf of a separated presheaf, and therefore is separated. Let us show that LF is then a sheaf. Let $i : R \hookrightarrow X$ be a sieve

covering an object X of C, and let $u : R \to LF$ be a morphism. It suffices to show that u factors through X. Put $R' = F \times_{LF} R$ and $v = Z_{R'}(\mathrm{pr}_1)$:

$$\begin{tikzcd}
F \arrow[r, "\ell(F)"] & LF & \\
R' \arrow[u, "\mathrm{pr}_1"] \arrow[r, "\mathrm{pr}_2"] & R \arrow[u, "u"'] \arrow[r, "i"] & X \arrow[ul, dash, "v"'] \\
R'' \arrow[u, "p_1"] \arrow[r, "p_2"] & Y \arrow[u, "m"'] & .
\end{tikzcd}$$

233 To show that $u = v \circ i$, it suffices to show (I 3.4) that for every morphism $m : Y \to R$ (Y an object of C), $v \circ i \circ m = u \circ m$. Put $R'' = Y \times_R R'$, and let p_1, p_2 be the projections. The projection $R'' \overset{p_2}{\hookrightarrow} Y$ is a monomorphism and makes R'' a covering sieve of Y (3.1 2)). One has $v \circ i \circ \mathrm{pr}_2 = \ell(F) \circ \mathrm{pr}_1$ (3.1 1)), and hence $v \circ i \circ \mathrm{pr}_2 = u \circ m \circ p_2$. It follows that $v \circ i \circ \mathrm{pr}_2 \circ p_1 = u \circ \mathrm{pr}_2 \circ p_1$, i.e. $v \circ i \circ m \circ p_2 = u \circ m \circ p_2$. Since the presheaf LF is separated, one has $v \circ i \circ m = u \circ m$, C.Q.F.D.

4) Clear.

Remark 3.3. Let $J'(X)$ be a cofinal subset of $J(X)$. One has

$$LF(X) = \varinjlim_{J'(X)} \mathrm{Hom}_{C^{\wedge}}(., F).$$

In particular, if the topology of C is defined by a pretopology $X \mapsto \mathrm{Cov}(X)$ (1.1.3), the functor L can be described using the covering families of $\mathrm{Cov}(X)$ (I 2.12 and I 3.5). Writing the formulas explicitly, one recovers the construction of [

Theorem 3.4. *Let C be a $\mathcal{U}$-site. The inclusion functor $i : C^{\sim} \hookrightarrow C^{\wedge}$ from sheaves into presheaves admits a left adjoint $\underline{a}$, which is left exact (I 2.3.2):*

$$\begin{tikzcd}
C^{\sim} \arrow[r, hook, shift right, "i"'] & C^{\wedge} \arrow[l, shift right, "\underline{a}"']
\end{tikzcd} .$$

The functor $i \circ \underline{a}$ is canonically isomorphic to the functor $L \circ L$ (cf. 3.0.5). For every presheaf F, the adjunction morphism $F \to i \circ \underline{a}(F)$ is deduced, via the preceding isomorphism, from the morphism $\ell(LF)\ell(F) : F \to \overline{L} \circ L(F)$.

234 Definition 3.5. *The sheaf $\underline{a}F$ is called the sheaf associated with the presheaf F.*

Theorem 3.4 follows immediately from Proposition 3.2.

Proposition 3.6. *Let C be a $\mathcal{U}$-site and let $\mathcal{V} \supset \mathcal{U}$ be a universe. Denote by $C^{\wedge}_{\mathcal{U}}$ and $C^{\sim}_{\mathcal{U}}$ (respectively $C^{\wedge}_{\mathcal{V}}$ and $C^{\sim}_{\mathcal{V}}$) the categories of $\mathcal{U}$-presheaves and $\mathcal{U}$-sheaves (respectively $\mathcal{V}$-presheaves and $\mathcal{V}$-sheaves), and $\underline{a}_{\mathcal{U}} : C^{\wedge}_{\mathcal{U}} \to C^{\sim}_{\mathcal{U}}$ (respectively $\underline{a}_{\mathcal{V}} : C^{\wedge}_{\mathcal{V}} \to C^{\sim}_{\mathcal{V}}$) the* *corresponding "associated sheaf" functors. The diagram*

$$\begin{tikzcd}
\hat{C}_U \arrow[r, "\underline{a}_U"] \arrow[d, hook] & \widetilde{C}_{\underline{U}} \arrow[d, hook] \\
\hat{C}_{\underline{V}} \arrow[r, "\underline{a}_V"] & \widetilde{C}_{\underline{V}},
\end{tikzcd}$$

where the vertical functors are the canonical inclusions, is commutative up to canonical isomorphism.

Proof. This follows from the construction of the functors $\underline{a}_U$ and $\underline{a}_V$ (3.4 and 3.0.5).

4. Exactness properties of the category of sheaves

The exactness properties of the category of sheaves follow from the exactness properties of the category of presheaves via Theorem 3.4. This section makes this philosophy explicit through some of the most useful standard statements.

Theorem 4.1. *Let C be a $\mathcal{U}$-site, let $\widetilde{C}$ be the category of sheaves, let $\underline{a} : \hat{C} \to \widetilde{C}$ be the associated-sheaf functor, and let $i : \widetilde{C} \to \hat{C}$ be the inclusion functor.*

1) *The functor $\underline{a}$ commutes with inductive limits and is exact.*
2) *The $\mathcal{U}$-inductive limits in $\widetilde{C}$ are representable. For every small category I and* 235
every functor $E : I \to \widetilde{C}$, the canonical morphism

$$\varinjlim_I E \longrightarrow \underline{a}(\varinjlim_I i \circ E)$$

is an isomorphism.
3) *The $\mathcal{U}$-projective limits in $\widetilde{C}$ are representable. For every object X of C, the functor on $\widetilde{C} : F \mapsto F(X)$ commutes with projective limits, i.e. the inclusion functor $i : \widetilde{C} \to \widehat{C}$ commutes with projective limits.*

Proof. These properties follow essentially from Theorem 3.4 and from (I 2.11).

Thus, in the category of sheaves, products indexed by an element of $\mathcal{U}$, fiber products, sums indexed by an element of $\mathcal{U}$, amalgamated sums, kernels, cokernels, images, and coimages are representable.

Corollary 4.1.1. *Let C be a $\mathcal{U}$-site and let F be a sheaf of sets on C. The canonical homomorphism*

$$\varinjlim_{C/F} \underline{a}X \longrightarrow F$$

is an isomorphism.

Proof. This follows from (I 3.4) and the fact that $\underline{a}$ commutes with inductive limits.

Proposition 4.2. *Every morphism of $\widetilde{C}$ that is both an epimorphism and a monomorphism is an isomorphism.*

Proof. Let $u : G \to H$ be a morphism of $\widetilde{C}$ that is both an epimorphism and a monomorphism. First observe that the morphism u is a monomorphism of presheaves 236
(4.1 1)). Construct the amalgamated sum $H \amalg_G H = K$ and the two canonical morphisms

$i_1, i_2 : H \rightrightarrows K$ in the category of presheaves. Since u is a monomorphism of presheaves, one immediately verifies that the following diagram is cartesian and cocartesian (I 10.1):

$$\begin{tikzcd} G \arrow[r, "u"] \arrow[d, "u"'] & H \arrow[d, "i_2"] \\ H \arrow[r, "i_1"] & K. \end{tikzcd}$$

Applying the "associated sheaf" functor, one therefore obtains a cartesian and cocartesian diagram in the category of sheaves (4.1 1)):

(*)
$$\begin{tikzcd} G \arrow[r, "u"] \arrow[d, "u"'] & H \arrow[d, "\underline{a}i_2"] \\ H \arrow[r, "\underline{a}i_1"] & \underline{a}K. \end{tikzcd}$$

Since u is an epimorphism of sheaves, the morphism $\underline{a}i_1$ is an isomorphism, and since diagram (*) is cartesian, the morphism u is an isomorphism.

PROPOSITION 4.3. 1) *The inductive limits in $C^{\sim}$ that are representable are universal* (I 2.5).

2) *Every epimorphic family* (I 10.2) *of morphisms is universally effective epimorphic (2.6).*
3) *Every equivalence relation is universally effective* (I 10.6).
4) *Filtered $\mathscr{U}$-inductive limits commute with finite projective limits* (I 2.6).

237 PROOF. Assertions 1) through 4) hold in the category of sets, hence in the category of presheaves (I 3.1). Assertions 1) and 4) then follow immediately from the corresponding assertions for presheaves and from 4.1. Let us prove 3). Let $p_1, p_2 : R \rightrightarrows X$ be an equivalence relation in $C^{\sim}$. The diagram $R \overset{p_1}{\underset{p_2}{\rightrightarrows}} X$ is then an equivalence relation of presheaves (4.1.1). There is therefore a morphism of presheaves $u : X \to Y$ such that the diagram

$$\begin{tikzcd} R \arrow[r, "p_2"] \arrow[d, "p_1"'] & X \arrow[d, "u"] \\ X \arrow[r, "u"] & Y \end{tikzcd}$$

is cartesian and cocartesian in the category of presheaves. Applying the "associated sheaf" functor, one obtains a cartesian and cocartesian diagram in the category of sheaves (4.1 1)). It follows that $R \overset{p_1}{\underset{p_2}{\rightrightarrows}} X$ is an effective equivalence relation. Since every equivalence relation is effective by what precedes, every equivalence relation is universally effective. Let us prove 2). Let $(u_i : X_i \to X)$, $i \in I$, be an epimorphic family in $C^{\sim}$. By (II 2.6) and (I 2.12), it suffices to prove the following assertions:

a) For every morphism of sheaves $v : Y \to X$, the family of morphisms $\mathrm{pr}_{2,i} : X_i \times_X Y \to Y$, $i \in I$, is an epimorphic family in $C^{\sim}$.

b) For every sheaf Z, the following diagram of sets is exact:

$$\mathrm{Hom}(X,Z)\to\prod_{i\in I}\mathrm{Hom}(X_i,Z)\rightrightarrows\prod_{(i,k)\in I\times I}\mathrm{Hom}(X_i\times_X X_k,Z)$$

Let X' be the union of the images of the morphisms u_i, formed in the category of presheaves, and let $j : X' \hookrightarrow X$ be the canonical injection. For every $i \in I$, the morphism $u_i : X_i \to X$ factors as a morphism $u'_i : X_i \to X'$ followed by j, and the family $u'_i : X_i \to X'$, $i \in I$, is an epimorphic family in $C\hat{}$. Since j is a monomorphism, the morphism $\underline{a}j : \underline{a}X' \to X$ is a monomorphism of sheaves (4.1). Since, for every $i \in I$, one has $u_i = \underline{a}j\underline{a}u'_i$, $\underline{a}j$ is an epimorphism of sheaves. It follows that $\underline{a}j$ is an isomorphism (4.2). Let $v : Y \to X$ be a morphism of sheaves. Base change then yields a morphism $j_v : X' \times_X Y \to Y$ and morphisms $u'_{i,v} : X_i \times_X Y \to X' \times_X Y$. Since $\underline{a}$ commutes with fiber products, $\underline{a}j_v$ is an isomorphism. Since epimorphic families in $C\hat{}$ remain epimorphic under base change, the family $u'_{i,v}$, $i \in I$, is epimorphic in $C\hat{}$. Since $\underline{a}$ commutes with inductive limits, the family $\underline{a}u'_{i,v} : X_i \times_X Y \to \underline{a}(X' \times_X Y)$, $i \in I$, is epimorphic in $C\tilde{}$, proving a). Let us prove b). Let Z be a sheaf. Since $\underline{a}j$ is an isomorphism, the map 238

$$\mathrm{Hom}(X,Z)\longrightarrow\mathrm{Hom}(X',Z)$$

is a bijection. Since the u'_i, $i \in I$, form an epimorphic family in $C\hat{}$, the following diagram of sets is exact:

$$\mathrm{Hom}(X',Z)\to\prod_{i\in I}\mathrm{Hom}(X_i,Z)\rightrightarrows\prod_{(i,k)\in I\times I}\mathrm{Hom}(X_i\times_{X'} X_k,Z)$$

Finally, since X' is a subpresheaf of X, the presheaf $X_i \times_{X'} X_k$, $(i,k) \in I \times I$, is canonically isomorphic to $X_i \times_X X_k$, proving b).

Remarks 4.3.1. 1) Proposition 4.3 2) formally gives us a second proof of 4.2: it is clear that, in any category, a morphism that is both a universal effective epimorphism and a monomorphism is in fact an isomorphism. 239

2) It follows from the preceding propositions that every morphism in the category of sheaves factors uniquely as an effective epimorphism followed by an effective monomorphism. This property naturally generalizes, to nonadditive categories, axiom (AB2) for abelian categories (coim $\simeq$ im) [Tohoku].

4.4.0. Let C be a $\mathcal{U}$-site. Composing the functor $h : C \to C\hat{}$ (I 1.3.1) with the associated sheaf functor yields a functor

$$\epsilon_C : C \longrightarrow C\tilde{},$$

called the *canonical functor* from C to $C\tilde{}$, which will be used constantly in what follows. The functor ϵ_C commutes with finite projective limits. When the topology on C is coarser than the canonical topology, ϵ_C is fully faithful and commutes with projective limits; moreover, it is then defined even when C is not assumed to possess a small topologically generating family. We shall not study in detail the behavior of ϵ_C with respect to inductive limits (cf. [SGA 3 IV]). We shall, however, need the results below.

Theorem 4.4. *Let C be a $\mathcal{U}$-site and let $(u_i : X_i \to X)$, $i \in I$, be a family of morphisms of C with target X. The following conditions are equivalent* [4]*:*

i) *The family $(\epsilon_C u_i : \epsilon_C X_i \to \epsilon_C X)$, $i \in I$, is an epimorphic family in $C\tilde{}$* (I 10.2). 240

[4] When the site C does not necessarily possess a small topologically generating family and when the topology on C is coarser than the canonical topology, one has ii) $\Rightarrow$ i) (cf. the proof of 4.4).

ii) *The family* $(u_i : X_i \to X)$, $i \in I$, *is a covering family of* C *(1.2).*

Proof. ii) $\Rightarrow$ i). Let $R \hookrightarrow X$ be the sieve generated by the u_i, $i \in I$ (I 4.3.3), and let $u'_i : X_i \to R$ be the morphisms induced by the u_i. The morphisms u'_i, $i \in I$, form an epimorphic family in $C^{\wedge}$, and R is a covering sieve of X. Consequently, for every sheaf F, the map

$$\mathrm{Hom}(X, F) \longrightarrow \mathrm{Hom}(R, F)$$

is bijective, and the map

$$\mathrm{Hom}(R, F) \longrightarrow \prod_i \mathrm{Hom}(X_i, F)$$

is injective. Thus the map

$$\mathrm{Hom}(X, F) \longrightarrow \prod_i \mathrm{Hom}(X_i, F)$$

is injective, and consequently the map

$$\mathrm{Hom}(\underline{a}X, F) \longrightarrow \prod_i \mathrm{Hom}(\underline{a}X_i, F)$$

is injective, which proves i).
i) $\Rightarrow$ ii). With the notation introduced above, let $J_R : R \hookrightarrow X$ be the canonical inclusion into X of the sieve generated by the u_i, $i \in I$. It follows from i) that the morphism of sheaves $\underline{a}J_R : \underline{a}R \to \underline{a}X$ is both an epimorphism and a monomorphism in the category of sheaves. It is therefore an isomorphism of sheaves (4.2). With the notation of n° 3, we have a commutative diagram:

$$\begin{tikzcd}
X \arrow[r, "\ell(X)"] & LX \arrow[r, "\ell(LX)"] & \underline{a}X \\
R \arrow[u, hook, "J_R"] \arrow[r, "\ell(R)"] & LR \arrow[u, hook, "LJ_R"] \arrow[r, "\ell(LR)"] & \underline{a}R. \arrow[u, "\underline{a}J_R"', "\wr"]
\end{tikzcd}$$

241 First of all, by 3.1 2), there exist a covering sieve $J_{S_1} : S_1 \to X$ and a morphism $u_1 : S_1 \to LR$ such that

$$\ell(LX) \circ \ell(X) \circ J_{S_1} = \underline{a}J_R \circ \ell(LR) \circ u_1 = \ell(LX) \circ LJ_R \circ u_1. \tag{4.6.1}$$

Put $m = \ell(X) \circ J_{S_1}$ and $n = LJ_R \circ u_1$. The morphisms m and n are morphisms from S_1 to LX. Let $S_2 \hookrightarrow S_1$ be the kernel of the pair of arrows (m, n). For every object Y of C and every morphism $\alpha : Y \to S_1$, one has, by 4.6.1, $\ell(LX) \circ m \circ \alpha = \ell(LX) \circ n \circ \alpha$. It then follows from (3.1 3)) that the kernel $S_2 \times_{S_1} Y$ of the pair of arrows $(m \circ \alpha, n \circ \alpha)$ is a covering sieve of Y. Consequently, by axiom (T 2) for topologies, $J_{S_2} : S_2 \hookrightarrow X$ is a covering sieve of X. Thus there is a morphism $u_2 : S_2 \to LR$ and a commutative diagram:

$$
(4.6.2)\qquad
\begin{tikzcd}
S_2 \arrow[rr, hook, "J_{S_2}"] \arrow[dd, "u_2"'] & & X \arrow[dd, "\ell(X)"] \\
& R \arrow[ur, hook, "J_R"'] \arrow[dl, "\ell(R)"] & \\
LR \arrow[rr, "LJ_R"] & & LX
\end{tikzcd}
.
$$

Let Y be an object of C and let $\beta : Y \to S_2$ be a morphism. By 3.1 2) and 1), there exist a covering sieve $J_{Q_1} : Q_1 \hookrightarrow Y$ and a morphism $v_1 : Q_1 \to R$ such that $u_2 \circ \beta \circ J_{Q_1} = \ell(R) \circ v_1$. Put $f = J_{S_2} \circ \beta \circ J_{Q_1}$ and $g = J_R \circ v_1$. Let Q_2 be the kernel of the pair $(f, g) : Q_1 \rightrightarrows X$, and let $J_{Q_2} : Q_2 \hookrightarrow Y$ be the canonical inclusion. For every object Z of C and every morphism $\gamma : Z \to Q_1$, the commutativity of (4.6.2) gives:

$$
\begin{aligned}
\ell(X) \circ f \circ \gamma = \ell(X) \circ J_{S_2} \circ \beta \circ J_{Q_1} \circ \gamma = LJ_R \circ u_2 \circ \beta \circ J_{Q_1} \circ \gamma = \\
= LJ_R \circ \ell(R) \circ v_1 \circ \gamma = \ell(X) \circ J_R \circ v_1 \circ \gamma = \ell(X) \circ g \circ \gamma.
\end{aligned}
$$

It then follows from 3.1 3) that the kernel $Q_2 \times_{Q_1} Z$ of the pair $(f \circ \gamma, g \circ \gamma)$ is a covering 242
sieve of Z. By axiom (T 2) for topologies, $J_{Q_2} : Q_2 \hookrightarrow Y$ is a covering sieve of Y. Thus, for every morphism $\beta : Y \to S_2$, there exist a covering sieve $J_{Q_2} : Q_2 \hookrightarrow Y$ and a morphism $v_2 : Q_2 \to R$ such that the following diagram is commutative:

$$
\begin{tikzcd}
Y \arrow[r, "\beta"] & S_2 \arrow[r, "J_{S_2}"] & X \\
Q_2 \arrow[u, hook, "J_{Q_2}"] \arrow[r, "v_2"] & R \arrow[ur, hook, "J_R"'] &
\end{tikzcd}
$$

Denote by $S_2 \cap R$ the fiber product of S_2 and R over X. The sieve (of Y) $(S_2 \cap R) \times_{S_2} Y$ contains the sieve Q_2, and hence $(S_2 \cap R) \times_{S_2} Y$ is a covering sieve of Y. It then follows from axiom (T 2) for topologies that $S_2 \cap R$ is a covering sieve of X, and consequently the sieve R, which contains $S_2 \cap R$, is a covering sieve of X, C.Q.F.D.

COROLLARY 4.4.4. *Let T be the topology of the $\mathcal{U}$-site C, and let T' (resp. T'') be the finest topology on C among those for which the objects of $\widetilde{C}$ are sheaves* (resp. separated presheaves) *(2.2). Then $T = T' = T''$.*

Indeed, one trivially has $T \leq T' \leq T''$, and 4.4 and 2.2 imply that $T'' \leq T$, C.Q.F.D.

DEFINITION 4.5. *An initial object of a category A is an object ϕ_A that represents the empty inductive limit, i.e. such that, for every $X \in$ ob A, there exists a unique arrow $\phi_A \to X$. An object ϕ_A is called* strict initial *if it is initial and every morphism with target ϕ_A is an isomorphism. Let (S_i), $i \in I$, be a family of objects of a category A. Suppose that the sum $S = \coprod_{i \in I} S_i$ is representable. The sum S is called disjoint if the structure morphisms $S_i \to S$ are pullbackable, if they are monomorphisms, and if, for every pair i, j with $i \neq j$, the fiber product $S_i \times_S S_j$ is an initial object of A. The sum S is called disjoint and universal if it is disjoint and remains a disjoint sum after every base change $T \to S$; it follows that,* 243
for every pair i, j with $i \neq j$, the objects $S_i \times_S S_j$ are strict initial objects of A.

Example 4.5.1. In the category of sets, sums are disjoint and universal. The same is therefore true in every category of presheaves of sets (I 3.1), and hence in every category of sheaves of sets on a $\mathscr{U}$-site (4.1); in particular, the initial object of $\tilde{C}$ is strict.

Proposition 4.6. *Let C be a $\mathscr{U}$-category and let $(s_i : X_i \to X)$, $i \in I$, be a family of morphisms of C. For every $\mathscr{U}$-topology $\mathscr{T}$ on C (3.0.2), denote by $\tilde{C}_{\mathscr{T}}$ the corresponding category of sheaves and by $\epsilon_{\mathscr{T}} : C \to \tilde{C}_{\mathscr{T}}$ the corresponding functor.*

1) *Let $\mathscr{T}$ be a $\mathscr{U}$-topology such that*

$$\coprod_{i\in I} \epsilon_{\mathscr{T}}(X_i) \xrightarrow{(\epsilon_{\mathscr{T}}(s_i))} \epsilon_{\mathscr{T}}(X)$$

is an isomorphism. Then, for every topology $\mathscr{T}'$ finer than $\mathscr{T}$, the morphism

$$\coprod_{i\in I} \epsilon_{\mathscr{T}'}(X_i) \xrightarrow{(\epsilon_{\mathscr{T}'}(s_i))} \epsilon_{\mathscr{T}'}(X)$$

is an isomorphism.

2) *Let $\mathscr{T}$ be a $\mathscr{U}$-topology on C. The following properties* (i) *and* (ii) *are equivalent:*

i) a) *The family $(s_i : X_i \to X)$, $i \in I$, is covering for $\mathscr{T}$.*

b) *For every $i \in I$, the diagonal morphism of presheaves $\Delta_i : X_i \hookrightarrow X_i \times_X X_i$ is sent by the associated sheaf functor (for $\mathscr{T}$) to an isomorphism (as is the case if the s_i are monomorphisms).*

244 c) *For every pair (i, j) of elements of I with $i \neq j$, the presheaf $X_i \times_X X_j$ is sent by the associated sheaf functor (for $\mathscr{T}$) to the initial object of $\tilde{C}_{\mathscr{T}}$.*

ii) *$\epsilon_{\mathscr{T}}(X)$ is the sum of the $\epsilon_{\mathscr{T}}(X_i)$.*

Proof. 1) A sheaf for $\mathscr{T}'$ is a sheaf for $\mathscr{T}$. Consequently, for every sheaf F for $\mathscr{T}'$, the morphism

$$\mathrm{Hom}_{C^\wedge}(X, F) \longrightarrow \prod_{i\in I} \mathrm{Hom}_{C^\wedge}(X_i, F)$$

is an isomorphism, which proves the assertion.

2) By definition, property ii) holds if and only if the morphism of presheaves

$$\Phi = (h(s_i), i \in I) : \coprod_i h(X_i) \longrightarrow h(X)$$

is sent by the associated sheaf functor to an isomorphism, i.e. (4.2) if and only if $\underline{a}(\Phi)$ is both an epimorphism and a monomorphism of sheaves. By (4.4), the morphism $\underline{a}(\Phi)$ is an epimorphism if and only if property a) holds. The diagonal morphism

$$\Delta : \coprod_i h(X_i) \longrightarrow (\coprod_i h(X_i)) \times_{h(X)} (\coprod_i h(X_i))$$

is the direct sum of a family $\Delta_{i,j}$, $(i, j) \in I \times I$, of morphisms defined as follows:

β) When $i = j$, $\Delta_{i,i}$ is the diagonal morphism $h(X_i) \to h(X_i) \times_{h(X)} h(X_i)$.

γ) When $i \neq j$, $\Delta_{i,j}$ is the morphism $\phi_{C^\wedge} \to h(X_i) \times_{h(X)} h(X_j)$ ($\phi_{C^\wedge}$ denotes the initial object of $\hat{C}$).

245 The morphism $\underline{a}(\Phi)$ is a monomorphism if and only if $\underline{a}(\Delta)$ is an isomorphism, and by (4.1) $\underline{a}(\Delta)$ is an isomorphism if and only if the $\underline{a}(\Delta_{i,j})$ are isomorphisms, i.e. if and only if properties b) and c) hold.

Corollary 4.6.1. *Let C be a $\mathscr{U}$-category and X an object of C.*

1) *Let $\mathscr{T}$ be a $\mathscr{U}$-topology on C. The object X of C is sent by the associated sheaf functor (for the topology $\mathscr{T}$) to the initial object of $C^{\sim}_{\mathscr{T}}$ if and only if the empty sieve covers X.*
2) *When X is a strict initial object of C, the sheaf associated with X, for every $\mathscr{U}$-topology finer than the canonical topology, is an initial object.*

Proof. 1) Take the indexing set I in 4.6 2) to be the empty set.

2) By 1) and 4.6 1), it is enough to show that the empty sieve covers X for the canonical topology; i.e. by (2.6), it is enough to show that, for every object $Y \to X$ over X, Y is an initial object of C, which follows from the definition (4.5).

Corollary 4.6.2. *Let C be a $\mathscr{U}$-site and let $(s_i : X_i \to X)$, $i \in I$, be a family of pullbackable morphisms of C with the same target and having the following properties:*

α) *The family of the s_i, $i \in I$, is covering.*
β) *For every $i \in I$, $s_i : X_i \to X$ is a monomorphism.*
γ) *For every pair (i, j) of elements of I with $i \neq j$, $X_i \times_X X_j$ is covered by the empty sieve, i.e. for every sheaf F on C, $F(X_i \times_X X_j)$ is a singleton set.*

Then $\epsilon_C(X)$ is the sum of the $\epsilon_C(X_i)$, $i \in I$.

Proof. This follows immediately from 4.6 2) and 4.6.1 1). 246

Corollary 4.6.3. *Let C be a $\mathscr{U}$-category and let $(s_i : X_i \to X)$, $i \in I$, be a family of pullbackable morphisms with the same target. Suppose that the canonical topology of C is a $\mathscr{U}$-topology. The following conditions are equivalent:*

i) *There exists a $\mathscr{U}$-topology $\mathscr{T}$ on C, coarser than the canonical topology, such that $\epsilon_{\mathscr{T}}(X)$ is the sum of the $\epsilon_{\mathscr{T}}(X_i)$, $i \in I$.*
ii) *The object X is the disjoint and universal sum (4.5) (in C) of the X_i.*

Proof. ii) $\Rightarrow$ i). Since X is the universal sum of the X_i, $i \in I$, the family of the s_i, $i \in I$, is covering for the canonical topology of C (2.6). Condition α) of 4.6.2 is therefore satisfied when C is endowed with the canonical topology. Condition β) is evidently satisfied, and property γ) follows from 4.6 1) 2), whence i).
i) $\Rightarrow$ ii). Let $\mathscr{T}$ be a topology on C coarser than the canonical topology such that $\epsilon_{\mathscr{T}}(X)$ is the sum of the $\epsilon_{\mathscr{T}}(X_i)$. For every sheaf F for $\mathscr{T}$, the canonical map $\mathrm{Hom}(X, F) \to \prod_i \mathrm{Hom}(X_i, F)$ is a bijection. Moreover, every representable presheaf is a sheaf. It follows immediately that X is the sum in C of the X_i, $i \in I$. Applying this to the case where the set I is empty, one sees that if an object of C is carried to the initial object of $C^{\sim}$, that object is an initial object of C. The functor $\epsilon_{\mathscr{T}} : C \to C^{\sim}$ is fully faithful and commutes with finite projective limits. Consequently, condition b) of 4.6.2 implies that the s_i, $i \in I$, are monomorphisms of C, and condition c) implies, by what precedes, that the $X_i \times_X X_j$, $i \neq j$, are initial objects of C. Consequently X is the disjoint sum of the X_i. Since $\epsilon_{\mathscr{T}}$ commutes with fiber products, this last property is stable under every base change $Y \to X$, and therefore X is the disjoint and universal sum of the X_i, $i \in I$. 247

Corollary 4.6.4. *Let C be a $\mathscr{U}$-category and X an object of C. Suppose that the canonical topology on C is a $\mathscr{U}$-topology. The following properties are equivalent:*

i) *There exists on C a topology coarser than the canonical topology such that $\epsilon_{\mathscr{T}}(X)$ is an initial object of $C^{\sim}_{\mathscr{T}}$.*
ii) *X is a strict initial object of C.*

Proof. Take the empty set for the set I in 4.6.3.

Proposition 4.7. *Let C be a $\mathcal{U}$-category, let $C^{\sim}$ be the category of sheaves on C for the canonical topology, let $\epsilon_C : C \to C^{\sim}$ be the canonical functor (4.4.0), and let $R \rightrightarrows X$ be an equivalence relation in C admitting a cokernel Y in C. Let $\Pi : X \to Y$ be the canonical morphism and suppose it is pullbackable. Consider the following properties:*

i) *$\epsilon_C(Y)$ is the quotient of the equivalence relation $\epsilon_C(R) \rightrightarrows \epsilon_C(X)$.*
ii) *The equivalence relation $R \rightrightarrows X$ is universally effective* (cf. n° 7).

One has ii) $\Rightarrow$ i). *When C possesses a $\mathcal{U}$-topology coarser than the canonical topology, one has* i) $\Rightarrow$ ii).

Proof. ii) $\Rightarrow$ i). The canonical morphism $R \to X \times_Y X$ is an isomorphism. Let $S \to Y$ be the sieve generated by $\pi : X \to Y$. For every presheaf F, there is an exact diagram (I 2.12):

$$\mathrm{Hom}(S, F) \to \mathrm{Hom}(X, F) \rightrightarrows \mathrm{Hom}(R, F).$$

248 It therefore suffices to show that, for every sheaf F for the canonical topology, the map

$$\mathrm{Hom}(X, F) \longrightarrow \mathrm{Hom}(S, F)$$

is a bijection, i.e. it suffices to show that S is covering for the canonical topology, or equivalently that the sieve generated by $\pi : X \to Y$ is covering, which follows from Definition 2.5.

i) $\Rightarrow$ ii). The functor $\epsilon_C : C \to C^{\sim}$ commutes with finite projective limits and is fully faithful. The canonical morphism $\epsilon_C(R) \to J_C(X) \times_{J_C(Y)} J_C(X)$ is an isomorphism (4.3). Consequently the canonical morphism $R \to X \times_Y X$ is an isomorphism. The morphism $\epsilon_C(X) \to \epsilon_C(Y)$ is an epimorphism, which implies (4.4) that $\pi : X \to Y$ is a covering morphism of C for the canonical topology, C.Q.F.D.

Proposition 4.8. *Let C be a $\mathcal{U}$-site. The category of sheaves on C has the following properties:*

a) *Finite projective limits are representable.*
b) *Direct sums indexed by an element of $\mathcal{U}$ are representable. They are disjoint and universal (4.5).*
c) *Equivalence relations are universally effective.*

This was proved in 4.1, 2) and 3), 4.3, 3), and 4.5.1.

These properties have been singled out because they will later allow us to characterize the $\mathcal{U}$-categories equivalent to categories of sheaves on categories belonging to $\mathcal{U}$ (IV 1.2).

Remark. Recall that property a) is equivalent to the property:
There exists a terminal object, and fiber products are representable (I 2.3.1).

249 4.9. Let C be a site whose underlying category is a $\mathcal{U}$-category. Then the category $\widetilde{C}$ of $\mathcal{U}$-sheaves on C satisfies the conditions considered in I 7.3 ((i) or (ii), as desired), and therefore, by *loc. cit.*, the various variants of the notion of a *generating family* in $\widetilde{C}$ considered in I 7.1 coincide. This being understood:

Proposition 4.10. *With the preceding notation, consider the canonical functor $\epsilon : C \to \widetilde{C}$ (4.4.0), and let $G \subset \mathrm{ob}\, C$ be a topologically generating family in C (3.0.1). Then the family $(\epsilon(X))_{X \in G}$ of objects of $\widetilde{C}$ is a generating family. In particular, the family $(\epsilon(X))_{X \in \mathrm{ob}\, C}$ of objects of $\widetilde{C}$ is generating.*

We must prove that every morphism $u : F \to F'$ in $\widetilde{C}$ such that

$$\mathrm{Hom}(\epsilon(X), F) \longrightarrow \mathrm{Hom}(\epsilon(X), F') \tag{4.10.1}$$

is a bijection for every $X \in G$, is an isomorphism. The preceding map identifies with the map

$$u(X) : F(X) \longrightarrow F'(X), \tag{4.10.2}$$

and we must show that if the latter is bijective for every $X \in G$, then it is so for every $X \in \mathrm{ob}\, C$. Let $(X_i \to X)_{i\in I}$ be a covering family of X by objects of G, and, for every pair of indices (i, j) in I, consider the set of all morphisms $X_{ijk} \to X_i \times_X X_j$ in $\widehat{C}$, with $X_{ijk} \in G$ (k varying in an index set $I_{i,j}$). One then obtains a morphism of exact diagrams of sets

$$\begin{tikzcd}
F(X) \arrow[r] \arrow[d] & \prod_i F(X_i) \arrow[r, shift left] \arrow[r, shift right] \arrow[d, "\wr"] & \prod_{ijk} F(X_{ijk}) \arrow[d, "\wr"] \\
F'(X) \arrow[r] & \prod_i F'(X_i) \arrow[r, shift left] \arrow[r, shift right] & \prod_{ijk} F'(X_{ijk})
\end{tikzcd} ,$$

where , by hypothesis, the last two vertical arrows are bijections. The same is therefore true of the first vertical arrow, C.Q.F.D. 250

COROLLARY 4.11. *Suppose that C is a $\mathscr{U}$-site (3.0.2). Then the category $\widetilde{C}$ is a $\mathscr{U}$-category and admits a $\mathscr{U}$-small generating family.*

Indeed, by hypothesis one may take for G in 4.10 a small generating family, which proves the existence of a small generating family. On the other hand, if $F, F' \in \mathrm{ob}\, \widetilde{C}$, a morphism from F to F' is known once one knows the morphism (4.10.1), i.e. (4.10.2), for every $X \in \mathrm{ob}\, G$ (I 7.1.1), whence the map

$$\mathrm{Hom}(F, F') \longrightarrow \prod_{X\in G} \mathrm{Hom}(F(X), F'(X))$$

is injective. Since the second member is $\mathscr{U}$-small, so is the first, which proves that $\widetilde{C}$ is a $\mathscr{U}$-category.

COROLLARY 4.12. *Let C be a $\mathscr{U}$-site. Then, for every object X of $\widetilde{C}$, the set of subobjects of X and the set of quotient objects of X are $\mathscr{U}$-small.*

This follows from I 7.4, respectively I 7.5, which apply by 4.11.

5. Extension of a topology from C to $C\hat{}$

PROPOSITION 5.1. *Let C be a $\mathscr{U}$-site and let $f : H \to K$ be a morphism of $C\hat{}$. The following conditions are equivalent:*

i) *For every $X \to K$, with $X \in \mathrm{ob}\, C$, the image of the corresponding morphism $H \times_K X \to X$ is a covering sieve of X.*

ii) *The morphism $\underline{a}(f) : \underline{a}H \to \underline{a}K$ between the associated sheaves is an epimorphism in $C\tilde{}$.* 251

ii bis) *For every sheaf F on C, the map $\mathrm{Hom}(K, F) \to \mathrm{Hom}(H, F)$ induced by f is injective.*

PROOF. It is clear that ii) is equivalent to ii bis).

Let us prove that (ii) ⇒ (i). By 4.4, (i) means that $u : \underline{a}(H \times_K X) \to \underline{a}(X)$ is an epimorphism. Now, since $\underline{a}(H \times_K X) \simeq \underline{a}(H) \times_{\underline{a}(K)} \underline{a}(X)$, because $\underline{a}$ commutes with finite $\varprojlim$ (4.1), the morphism in question is obtained from $\underline{a}(f) : \underline{a}(H) \to \underline{a}(K)$ by base change. Since the epimorphisms of $\widetilde{C}$ are universal, this proves (ii) ⇒ (i). Conversely, since the family of all $X_i \to K$, with $X_i \in \operatorname{ob} C$, is epimorphic, the same is true of the family of $\underline{a}(X_i) \to \underline{a}(K)$ in $\widetilde{C}$ (4.1). Hence, to verify that $\underline{a}(f) : \underline{a}(H) \to \underline{a}(K)$ is epimorphic, it suffices to do so after every base change of the preceding kind $\underline{a}(X_i) \to \underline{a}(K)$, which proves i) ⇒ ii), C.Q.F.D.

Definition 5.2. 1) *A morphism $f : H \to K$ satisfying the three equivalent conditions of 5.1 is called a* covering morphism. *A family of morphisms $f_i : H_i \to K$, $i \in I$, with the same target is called* covering *if the corresponding morphism $f : \amalg_i H_i \to K$ is covering.*

2) *A morphism $f : H \to K$ is called* bicovering *if it is covering and if the diagonal morphism $H \to H \times_K H$ is covering. A family $f_i : H_i \to K$, $i \in I$, with the same target is called bicovering if the corresponding morphism $f : \amalg_i H_i \to K$ is bicovering.*

By condition i) of 5.1, to say that a family $(f_i : H_i \to K)$, $i \in I$, is covering means that, for every morphism $X \to K$, with $X \in \operatorname{ob} C$, the image of the family $H_i \times_K X \to X$,
252 $i \in I$, is a covering sieve of X; or equivalently, by ii), that the family of morphisms $\underline{a}(f_i) : \underline{a}H_i \to \underline{a}K$ in $C^{\tilde{}}$ is epimorphic (taking into account that the functor $\underline{a}$ commutes with direct sums (4.1)).[4]

Proposition 5.3. *Let C be a $\mathcal{U}$-site and let $f : H \to K$ be a morphism of $C^{\hat{}}$. The following conditions are equivalent:*

i) *The morphism f is bicovering (5.2).*

i bis) *The morphism f is covering and, for every object X of C and every pair of morphisms $u, v : X \rightrightarrows H$ such that $fu = fv$, the kernel of (u, v) is a covering sieve of X.*

ii) *The morphism $\underline{a}(f) : \underline{a}H \to \underline{a}K$ is an isomorphism in $C^{\tilde{}}$.*

ii bis) *For every sheaf F on C, the map $\operatorname{Hom}(K, F) \to \operatorname{Hom}(H, F)$ is a bijection.*

Proof. The equivalence i) ⇔ i bis) follows from condition i) of 5.1, applied to the diagonal morphism $H \to H \times_K H$; the equivalence ii) ⇔ ii bis) is trivial.
i) ⇒ ii). The morphism $\underline{a}(f) : \underline{a}H \to \underline{a}K$ is an epimorphism (5.1). Since the functor $\underline{a}$ commutes with the formation of fiber products (4.1), the diagonal morphism $\underline{a}H \to \underline{a}H \times_{\underline{a}K} \underline{a}H$ is an epimorphism (5.1). Since a diagonal morphism is always a monomorphism, it is an isomorphism (4.2). It follows that the morphism $\underline{a}(f)$ is a monomorphism. It is therefore an isomorphism (4.2).
ii) ⇒ i). Since the functor $\underline{a}$ commutes with the formation of fiber products, the morphism f and the diagonal morphism $H \to H \times_K H$ are carried by $\underline{a}$ to isomorphisms. In particular, they are carried by $\underline{a}$ to epimorphisms. They are therefore covering, C.Q.F.D.

253 5.3.1. It follows from 5.3, and from the fact that $\underline{a}$ commutes with direct sums, that a family $(f_i : H_i \to K)$, $i \in I$, of morphisms of $C^{\hat{}}$ is bicovering if and only if the f_i induce an isomorphism from the direct sum $\amalg_i \underline{a}H_i$ onto $\underline{a}K$ on the associated sheaves,

[4] Note that the property of a morphism or a family of morphisms of $C^{\hat{}}$ being covering (resp. bicovering) is stable under base change.

or equivalently if and only if, for every sheaf F, the map

$$\mathrm{Hom}(K, F) \longrightarrow \prod_i \mathrm{Hom}(H_i, F)$$

is bijective.

Proposition 5.4. *Let C be a $\mathcal{U}$-site. There exists a (necessarily unique) topology on $C^{\wedge}$ such that a family $H_i \to K$ of arrows of $C^{\wedge}$ with the same target is covering for this topology if and only if it is covering in the sense of 5.2. It is also the coarsest topology on $C^{\wedge}$ among the topologies T having the following properties:*

a) *T is finer than the canonical topology of $C^{\wedge}$ (i.e. every epimorphic family in $C^{\wedge}$ is covering for T).*
b) *Every covering family in C is covering in $C^{\wedge}$.*

Proof. We shall confine ourselves to giving indications. We leave it to the reader to show that the families which are covering in the sense of 5.2 are the covering families of a topology T_C on $C^{\wedge}$ (one uses 4.1). The covering families of the canonical topology on $C^{\wedge}$ are the epimorphic families in $C^{\wedge}$ (2.6 and I 3.1). Since $\underline{a}$ commutes with inductive limits (4.1), the topology T_C is finer than the canonical topology of $C^{\wedge}$. Moreover, the covering families of C are covering families for T_C (5.1). Thus, if T' denotes the coarsest topology on $C^{\wedge}$ having properties a) and b), T_C is finer than T'. Let $(f_i : H_i \to K)$, $i \in I$, be a covering family for T_C. We show that $(f_i, i \in I)$ is a covering family for T'. 254
Let $s_i : H_i \to H = \amalg_i H_i$ be the canonical monomorphisms, and let $f = (f_i, i \in I) : H = \amalg_i H_i \to K$ be the morphism defined by the f_i. The family of the s_i is covering for T'. Thus, by axiom (T 2) for topologies, to show that $(f_i, i \in I)$ is a covering family for T', it suffices to show that the morphism $f : H \to K$ is covering for T'. There exists an epimorphic family in $C^{\wedge}$, $u_\lambda : X_\lambda \to K$, $\lambda \in \Lambda$, with $X_\lambda \in \mathrm{ob}\, C$ (I 3.4). Therefore, again by axiom (T 2) for topologies, to show that $f : H \to K$ is covering for T', it suffices to show that, for every $\lambda \in \Lambda$, the morphism $\mathrm{pr}_2 : H \times_K X_\lambda \to X_\lambda$ is covering for T'. Let $v_j : Y_j \to H \times_K X_\lambda$, $j \in J$, with $Y_j \in \mathrm{ob}\, C$, be an epimorphic family in $C^{\wedge}$. The family $(\mathrm{pr}_2 \circ v_j, j \in J)$ is covering for T_C. It is therefore a covering family in C (5.1), and hence a covering family for T'. Consequently, the sieve generated by $\mathrm{pr}_2 : H \times_K X_\lambda \to X_\lambda$ contains a covering sieve for T'. It is therefore covering for T', and hence $\mathrm{pr}_2 : H \times_K X_\lambda \to X_\lambda$ is covering, C.Q.F.D.

Remark 5.4.1. The proof of 5.4 in fact shows that every topology T' on $C^{\wedge}$ which is finer than the canonical topology of $C^{\wedge}$ is the coarsest of the topologies T on $C^{\wedge}$ having the following properties:

a) T is finer than the canonical topology of $C^{\wedge}$.
b) Every family covering for T' of the form $u_i : X_i \to X$, $i \in I$, where X and the X_i are objects of C, is covering for T.

In particular, every topology T' on $C^{\wedge}$ which is finer than the canonical topology is uniquely determined by the families $u_i : X_i \to X$, $i \in I$, with X_i and X objects of C, which are covering for T'.

Remark 5.4.2. It is easy to show that, for every topology T' on $C^{\wedge}$ which is finer 255
than the canonical topology, the set of families of morphisms $(X_i \to X)$, $i \in I$, with the same target in C, which are covering for T' is the set of covering families of a topology on C. Thus 5.4 and 5.4.1 establish a one-to-one correspondence between the topologies on C and the topologies on $C^{\wedge}$ finer than the canonical topology.

5.5.0. Let C be a small category. Denote by Caf the set of strictly full subcategories (every object isomorphic to an object of the subcategory is an object of the subcategory) of $\widehat{C}$ whose inclusion functor admits a left adjoint which commutes with finite projective limits. Also denote by $\mathscr{T}$ the set of topologies on C. Theorem 3.4 defines a map

$$\Phi : \mathscr{T} \longrightarrow \mathrm{Caf}\,\mathit{quoi}.$$

We shall define a map in the opposite direction. For this purpose, to every element $e = i' : C' \underset{\longrightarrow}{\overset{\underline{a}'}{\longleftarrow}} \widehat{C}$ of Caf, with $\underline{a}'$ left adjoint to i', we must associate a topology T_e on C. For every object X of C, define $J_e(X)$ to be the set of subobjects of X whose inclusion morphism is carried by $\underline{a}'$ to an isomorphism. Using the hypotheses made on $\underline{a}'$, one verifies immediately that this defines a topology T_e on C. We have therefore defined a map

$$\Psi : \mathrm{Caf} \longrightarrow \mathscr{T}\,\mathit{quoi}.$$

We then have the following result (due to J. GIRAUD):

256 THEOREM 5.5. *The map Φ is a bijection, and Ψ is its inverse.*

PROOF. The map $\Psi \circ \Phi$ is the identity. Indeed, this follows immediately from 5.1. The map $\Phi \circ \Psi$ is the identity. Indeed, let $i' : C' \underset{\longrightarrow}{\overset{\underline{a}'}{\longleftarrow}} \widehat{C}$ be an element of Caf, let T_e be the topology corresponding to it under Ψ, and let C_e be the category of sheaves for T_e. Using the definition of the topology T_e and the definition of bicovering morphisms (5.2), one proves that the following assertions are equivalent:

i) The morphism u of $\widehat{C}$ is bicovering for the topology T_e.
ii) The morphism u of $\widehat{C}$ is carried by $\underline{a}'$ to an isomorphism.

It is clear that C' is a full subcategory of C_e. It therefore suffices to show that every sheaf F for the topology T_e is an object of C'. But, by the equivalence above, the morphism $F \to i' \circ a'(F)$ is bicovering, and since its source and target are sheaves, it follows from 5.3 (ii) that $F \to i' \circ a'(F)$ is an isomorphism, C.Q.F.D.

6. Sheaves with values in a category

6.0. Let C and D be two categories. A contravariant functor from C to D, $F : C^\circ \to D$, is called a *presheaf on C with values in D*.

DEFINITION 6.1. *Let C be a site and D a category. A presheaf $F : C^\circ \to D$ is called a* sheaf on C with values in D *(or, more briefly, a sheaf with values in D) if, for every object S of D, the presheaf of sets*

$$X \longmapsto \mathrm{Hom}_D(S, F(X)) \quad , \quad X \in \mathrm{ob}(C)$$

is a sheaf. The full subcategory of $\mathbf{Hom}(C^\circ, D)$ *formed by the sheaves on C with values in D will be denoted by* $\mathbf{Hom}^\sim(C^\circ, D)$.

257 REMARKS 6.2. 1) The condition of 6.1 means that for every object X of C, every covering sieve R of X, and every object S of D, one has an isomorphism (I 3.5 and 2.1):

$$\mathrm{Hom}_D(S, F(X)) \xrightarrow{\sim} \varprojlim_{C/R} F(.).$$

Thus, when D is the category of $\mathscr{U}$-sets, one recovers the definition of sheaves of sets (*loc. cit.*).

2) Let D' be a $\mathscr{U}$-category and let $G : D \to D'$ be a functor commuting with projective limits. By composition, the functor G carries sheaves with values in D to sheaves with values in D'.

6.3.0. Let γ be a type of algebraic structure defined by finite projective limits (I 2.9), and let $\mathscr{U} - \gamma$ be the category of γ-sets belonging to $\mathscr{U}$. Denote by $\mathrm{esj} : \mathscr{U} - \gamma \to \mathscr{U} - \mathrm{Ens}$ the "underlying set" functor. The functor esj commutes with projective limits and therefore, by the preceding remark, defines by composition the functors

$$\mathrm{esj}^{\wedge} : \mathbf{Hom}(C^{\circ}, \mathscr{U} - \gamma) \longrightarrow C^{\wedge}$$
$$\mathrm{esj}^{\sim} : \mathbf{Hom}^{\sim}(C^{\circ}, \mathscr{U} - \gamma) \longrightarrow C^{\sim}.$$

The functor $\mathrm{esj}^{\sim}$ is called the "underlying sheaf of sets" functor. In fact, it factors canonically through the category $\gamma - \widetilde{C}$ of γ-objects of $C^{\sim}$. Denoting again by $\widetilde{\mathrm{esj}}$ the resulting functor

$$\mathbf{Hom}^{\sim}(C^{\circ}, \mathscr{U} - \gamma) \longrightarrow \gamma - \widetilde{C},$$

we may state:

Proposition 6.3.1. *The functor* $\mathrm{esj}^{\sim}$ *establishes an equivalence among the category of sheaves on* C *with values in* $\mathscr{U} - \gamma$*, the category of* γ*-objects in the category of sheaves of sets, and the category of* γ*-objects in the category of presheaves of sets whose underlying presheaf is a sheaf.*

Proof. The proof essentially uses (I 3.2) and the fact that a finite projective limit of 258
sheaves in the category of presheaves is a sheaf (4.1).

6.3.2. The preceding proposition justifies the abuse of language whereby a sheaf with values in $\mathscr{U} - \gamma$ is identified with the corresponding γ-object of $C^{\sim}$. We shall henceforth make this abuse of language systematically.

We shall use the classical terminology: *sheaves of groups, sheaves of commutative groups* (which we shall most often call *abelian sheaves*), *sheaves of rings, sheaves of* A*-modules* etc.

6.3.3. We denote by

$$C^{\sim}_{\gamma} (\textit{resp. } C^{\wedge}_{\gamma})$$

the category of γ-objects of $C^{\sim}$ (resp. of $C^{\wedge}$). When γ is the type of structure "A-module", we also write $C^{\sim}_{A}$, or simply $\widetilde{C}_{\mathrm{ab}}$ when $A = \mathbf{Z}$.

Suppose that C is a $\mathscr{U}$-site. The "associated sheaf" functor is left exact (4.1), and therefore:

Proposition 6.4. *The inclusion functor* $i_{\gamma} : C^{\sim}_{\gamma} \to C^{\wedge}_{\gamma}$ *admits a left adjoint* a_{γ} *which is left exact; that is, one has an isomorphism*

$$\mathrm{Hom}_{C^{\sim}_{\gamma}}(a_{\gamma}, \ ) \xrightarrow{\sim} \mathrm{Hom}_{C^{\wedge}_{\gamma}}(\ , i_{\gamma}).$$

Let X *be an object of* $C^{\wedge}_{\gamma}$*. The underlying sheaf of sets of* $a_{\gamma}(X)$ *is canonically isomorphic to the sheaf of sets associated with the underlying presheaf of sets of* X*. The morphism of underlying presheaves of sets of the adjunction morphism* $\mathrm{id} \to i_{\gamma} a_{\gamma}$ *is identified with the adjunction morphism applied to the underlying presheaf of sets.*

Proposition 6.5. *Suppose that the functor* $\mathrm{esj} : \gamma - \mathscr{U} \to \mathscr{U} - \mathrm{Ens}$ *admits a left* 259 *adjoint*[4]*:*

$$\mathrm{Hom}_{\mathcal{U}-\mathrm{Ens}}(.,\mathrm{esj}(.)) \xrightarrow{\sim} \mathrm{Hom}_{\gamma-\mathcal{U}}(\mathrm{Lib}(.),.).$$

The functor $\mathrm{esj}^{\sim} : C^{\sim}_{\gamma} \to C^{\sim}$ *(resp.* $\mathrm{esj}^{\wedge} : C^{\wedge}_{\gamma} \to C^{\wedge}$*) admits a left adjoint* $\mathrm{Lib}^{\sim}$ *(resp.* $\mathrm{Lib}^{\wedge}$*), and one has a canonical isomorphism*

$$\mathrm{Lib}^{\sim} = a_{\gamma}\,\mathrm{Lib}^{\wedge}\, i.$$

Proof. The proof is formal and is left to the reader.

Corollary 6.6. *Let* (X_i)*,* $i \in I$*, be a generating family of* $C^{\sim}$ *(4.9). Then the family* $\mathrm{Lib}^{\sim}(X_i)$ *is a generating family of* $C^{\sim}_{\gamma}$*.*

Proof. The proof is formal once one observes that the functor $\mathrm{esj}^{\sim}$ commutes with projective limits and is conservative ($\mathrm{esj}^{\sim}(u)$ is an isomorphism $\Leftrightarrow$ u is an isomorphism).

Proposition 6.7. *Let* A *be a* $\mathcal{U}$*-sheaf of rings, or a small ring, and let* $C^{\sim}_{A}$ *(resp.* $C^{\wedge}_{A}$*) be the category of sheaves (resp. of presheaves) of unital* A*-modules (6.3.3) on a* $\mathcal{U}$*-site* C*. Then* $C^{\sim}_{A}$ *is an abelian* $\mathcal{U}$*-category satisfying the axioms* (AB 5) *("existence of left exact filtered inductive limits") and* (AB 3)* *("existence of infinite products") of* [Tohoku]*. It has a family of generators indexed by an element of* $\mathcal{U}$*.*

Proof. It is clear that the category $C^{\wedge}_{A}$ is an abelian category satisfying the axioms
260 (AB 3), (AB 4), and (AB 5) (I 2.8 and I 3.3). It then follows from 6.4 that the category $C^{\sim}_{A}$ is an additive category in which kernels and cokernels are representable. More precisely, let i_A and a_A be the inclusion and associated-sheaf functors, respectively, and let $u : X \to Y$ be a morphism of $C^{\sim}_{A}$. The functor a_A is left exact and commutes with inductive limits. The functor i_A commutes with projective limits. Consequently the functor $i_A a_A : C^{\wedge}_{A} \to C^{\wedge}_{A}$ is an additive left exact functor. One deduces canonical isomorphisms

$$i_A(\mathrm{Ker}(u)) \xrightarrow{\sim} \mathrm{Ker}(i_A(u)), \tag{$*$}$$

$$\mathrm{coker}(u) \xrightarrow{\sim} a_A\,\mathrm{coker}(i_A(u)). \tag{$**$}$$

Consequently there is a canonical isomorphism

$$\mathrm{coim}(u) \xrightarrow{\sim} a_A\,\mathrm{coim}(i_A(u)). \tag{$***$}$$

Let us show that the canonical morphism $\mathrm{coim}(u) \to \mathrm{im}(u)$ is an isomorphism. For this, by (*), it suffices to show that the sequence

$$0 \longrightarrow i_A(\mathrm{coim}(u)) \longrightarrow i_A(Y) \longrightarrow i_A(\mathrm{coker}(u))$$

is exact. Using the isomorphisms ($**$) and ($***$), and observing that $a_A i_A(Y)$ is isomorphic to Y, one sees that this sequence is isomorphic to the sequence

$$0 \longrightarrow i_A a_A(\mathrm{coim}(i_A(u))) \longrightarrow i_A a_A i_A(Y) \longrightarrow i_A a_A(\mathrm{coker}(i_A(u))),$$

which is precisely the image under the functor $i_A a_A$ of the sequence

$$0 \longrightarrow \mathrm{coim}(i_A(u)) \longrightarrow i_A(Y) \longrightarrow \mathrm{coker}(i_A(u)).$$

This sequence is exact, and the functor $i_A a_A$ is left exact. Thus the category $C^{\sim}_{A}$ is an
261 abelian category. The functor $a_A : C^{\wedge}_{A} \to C^{\sim}_{A}$ is left exact and commutes with arbitrary

[4] One proves that such an adjoint always exists [C.F.]: free group, free abelian group, free A-module etc.

inductive limits. Consequently it is additive, exact, and commutes with inductive limits. Since the category $\hat{C}_A$ satisfies axiom (AB 5), the same is true of the category $\tilde{C}_A$. Finally, it is clear that in the category $\tilde{C}_A$, products indexed by an element of $\mathscr{U}$ are representable, and hence axiom (AB 3)* is satisfied. This completes the proof of the first assertion.

For every ring A which is an element of $\mathscr{U}$, the functor

$$\mathrm{esj} : \mathscr{U} - A - \text{module} \longrightarrow \mathscr{U} - \text{Ens}$$

admits a left adjoint $X \longmapsto X_A$ (the free A-module generated by X). It therefore suffices to apply 4.10 and 6.6 to complete the proof.

Notation 6.8. Let H be a sheaf of sets. The sheaf of A-modules associated with the presheaf (cf. notation 6.5)

$$X \longmapsto \mathrm{Lib}_A\, H(X) \qquad X \in \mathrm{ob}\, C$$

is denoted by A_H. When $H = \underline{a}(X)$ (the sheaf associated with the presheaf represented by X), one sometimes simply writes A_X (by abuse of notation). The proof of 6.7 shows that the family of sheaves of A-modules A_X, $(X \in \mathrm{ob}(C))$, is a generating family, indexed by an element of $\mathscr{U}$, of the category of sheaves of A-modules.

Remark 6.9. According to [Tôhoku], 6.7 shows that the category $\tilde{C}_A$, when A is an element of $\mathscr{U}$, has enough injectives. It is also known that infinite products are not necessarily exact in $\tilde{C}_A$, and therefore that, in general, the category $\tilde{C}_A$ does not have enough projectives [Roos].

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EXPOSÉ III

Functoriality of categories of sheaves

J.-L. Verdier

In I 5, we studied the behavior of categories of presheaves with respect to functors 263
between the categories of arguments. In this exposé, we extend this study to the case of sites and categories of sheaves (n° 1 and 2). After introducing the induced topology (n° 3), we turn in n° 4 to the comparison lemma, which will play an important role in Exp. IV. In number 5, for the patient reader, we study certain commutative diagrams related to localized categories (of the form C/X). In Theorem I 5.1, we impose smallness hypotheses on the categories under consideration. In general, these hypotheses are not satisfied by the sites encountered in practice ($\mathcal{U}$-sites). This necessitates a few contortions (4.2, 4.3, 4.4).

1. Continuous functors

Definition 1.1. *Let C and C' be two $\mathcal{U}$-sites and let $u : C \to C'$ be a functor between their underlying categories. We say that u is continuous if, for every sheaf of sets F on C', the presheaf $X \mapsto F \circ u(X)$ on C is a sheaf.*

This notion of continuity depends a priori on the universe $\mathcal{U}$ with respect to which the two sites are $\mathcal{U}$-sites. Proposition 1.5 shows that it is independent of it.

1.11. Denote by $i : C^{\sim} \to C^{\wedge}$ (resp. by $i' : C'^{\sim} \to C'^{\wedge}$) the canonical inclusion 264
functor from sheaves into presheaves. By Definition 1.1, the functor u is continuous if and only if there exists a functor $u_s : C'^{\sim} \to C^{\sim}$ such that $iu_s = u^* i'$ (where $u^* F = F \circ u$ (I 5.0)).

Proposition 1.2. *Let C be a small site, C' a $\mathcal{U}$-site, and $u : C \to C'$ a functor between their underlying categories. The following properties are equivalent:*

i) *The functor u is continuous.*
ii) *For every object X of C and every sieve $R \hookrightarrow X$ covering X, the morphism $u_! R \hookrightarrow u(X)$ is bicovering in $C'^{\wedge}$* (II 5.2 and I 5.1). [4]
iii) *For every bicovering family $H_i \to K$, $i \in I$, in $C^{\wedge}$, the family of morphisms $u_! H_i \to u_! K$, $i \in I$, is bicovering in $C'^{\wedge}$.*
iv) *There exists a functor $u^s : C^{\sim} \to C'^{\sim}$, commuting with inductive limits and "extending u", i.e. such that the diagram of canonical functors*

$$\begin{tikzcd} C \arrow[r, "u"] \arrow[d, "\epsilon_C"'] & C' \arrow[d, "\epsilon_{C'}"] \\ C^{\sim} \arrow[r, "u^s"] & C'^{\sim}, \end{tikzcd}$$

commutes up to isomorphism.

[4] "Covering" in place of "bicovering" did not necessarily suffice; cf. Example 1.9.3 below.

Moreover, when C is no longer necessarily a small site but is a $\mathscr{U}$-site, one still has i) $\Leftrightarrow$ iv)*, and the functor u^s of* iv) *is necessarily a left adjoint of the functor $u_s : \widetilde{C}' \to \widetilde{C}$, and is consequently determined up to a unique isomorphism.*

265 Proof. The proof of i) $\Leftrightarrow$ iv) when C is a $\mathscr{U}$-site will be given in 4.2. Assume that C is small. It is clear that iii) $\Rightarrow$ ii).

i) $\Rightarrow$ iii): Let $H_i \to K$, $i \in I$, be a bicovering family in $C^\wedge$. For every sheaf F on C', the presheaf $u^* F$ (I 5.0) is a sheaf on C. Hence (II 5.3) the canonical map

$$\operatorname{Hom}(K, u^* F) \longrightarrow \prod_i \operatorname{Hom}(H_i, u^* F),$$

is bijective. By adjunction (I 5.1), it follows that the canonical map

$$\operatorname{Hom}(u_! K, F) \longrightarrow \prod_i \operatorname{Hom}(u_! H_i, F)$$

is bijective. Consequently (II 5.3) the family $u_! H_i \to u_! K$, $i \in I$, is bicovering.

ii) $\Rightarrow$ i): Let X be an object of C, let $R \hookrightarrow X$ be a covering sieve, and let F be a sheaf on C'. By (II 5.3), the map

$$\operatorname{Hom}(u_! X, F) \longrightarrow \operatorname{Hom}(u_! R, F)$$

is bijective. By adjunction (I 5.1), it follows that the map

$$\operatorname{Hom}(X, u^* F) \longrightarrow \operatorname{Hom}(R, u^* F)$$

is bijective. Consequently $u^* F$ is a sheaf.

i) $\Rightarrow$ iv): We use the customary notation $\underline{a}$ and i (resp. $\underline{a}'$ and i') for the "associated sheaf" and "inclusion into presheaves" functors. For every sheaf G on C, set $u^s G = \underline{a}' u_! i G$. For every sheaf F on C', we have the following sequence of natural isomorphisms:

$$\begin{aligned} \operatorname{Hom}(G, u_s F) &\simeq \operatorname{Hom}(iG, iu_s F) \simeq \operatorname{Hom}(iG, u^* i' F) \\ &\simeq \operatorname{Hom}(u_! iG, i' F) \simeq \operatorname{Hom}(\underline{a}' u_! iG, F). \end{aligned}$$

266 (The first isomorphism follows from the full faithfulness of i; the second follows from the definition of u_s; the third and fourth are obtained by adjunction.) Consequently u^s is left adjoint to u_s. In particular, u^s commutes with inductive limits. For every presheaf K on C and every sheaf F on C', we have the following sequence of natural isomorphisms: $\operatorname{Hom}(u^s \underline{a} K, F) \simeq \operatorname{Hom}(\underline{a} K, u_s F) \simeq \operatorname{Hom}(K, iu_s F) \simeq \operatorname{Hom}(K, u^* i' F) \simeq \operatorname{Hom}(u_! K, i' F) \simeq \operatorname{Hom}(\underline{a}' u_! K, F)$ (the third isomorphism follows from the definition of u_s; the others are obtained by adjunction); whence a functorial isomorphism $u^s \underline{a} K \simeq \underline{a}' u_! K$. In particular, using this isomorphism when K is representable, and taking I 5.4 3) into account, we obtain a diagram commutative up to isomorphism:

$$\begin{tikzcd} C \arrow[r, "u"] \arrow[d, "\epsilon_C"'] & C' \arrow[d, "\epsilon_{C'}"] \\ \widetilde{C} \arrow[r, "u^S"] & \widetilde{C'} \end{tikzcd}$$

iv) $\Rightarrow$ ii): Let $h : C \to C\hat{}$ (resp. $h' : C' \to C'\hat{}$) be the functor that assigns to an object the presheaf it represents. Consider the diagram:

$$\begin{tikzcd}
C \arrow[r, "u"] \arrow[d, "h"'] & C' \arrow[d, "h'"] \\
C\hat{} \arrow[r, "u_!"] \arrow[d, "\underline{a}"'] & C'\hat{} \arrow[d, "\underline{a}'"] \\
C^{\sim} \arrow[r, "u^S"] & C'^{\sim}.
\end{tikzcd}$$

The upper square is commutative (1.4 3)), and we have $\underline{a}h = \epsilon_C$, $\underline{a}'h' = \epsilon_{C'}$. 267
For every object K of $C\hat{}$, we have a canonical isomorphism (I 3.4):

$$\varinjlim_{(h(X)\to K)\in \mathrm{ob}(C/K)} h(X) \xrightarrow{\sim} K.$$

Since the functors $u_!$, u^s, $\underline{a}$, and $\underline{a}'$ commute with inductive limits, we obtain isomorphisms:

$$\varinjlim_{(h(X)\to K)\in \mathrm{ob}(C/K)} u^s\underline{a}h(X) \xrightarrow{\sim} u^s\underline{a}K,$$

$$\varinjlim_{(h(X)\to K)\in \mathrm{ob}(C/K)} \underline{a}'u_!h(X) \xrightarrow{\sim} \underline{a}'u_!K.$$

We have $u_!h = h'u$ and, by hypothesis, an isomorphism $\underline{a}'h'u \simeq u^s\underline{a}h$, whence an isomorphism

$$u^s\underline{a}K \simeq \underline{a}'u_!K,$$

which is immediately seen to be functorial in K. The diagram

$$\begin{tikzcd}
C\hat{} \arrow[r, "u_!"] \arrow[d, "\underline{a}"'] & C'\hat{} \arrow[d, "\underline{a}'"] \\
C^{\sim} \arrow[r, "u^S"] & C',
\end{tikzcd}$$

is therefore commutative up to isomorphism. Since $\underline{ai}$ is isomorphic to the identity functor, we have an isomorphism $u^s \simeq \underline{a}'u_!i$, whence the uniqueness of u^s. Let $v : H \to K$ be a bicovering morphism in $C\hat{}$. Then $\underline{a}(v)$ is an isomorphism (II 5.3), and consequently $\underline{a}'u_!(v)$ (isomorphic to $u^s\underline{a}(v)$) is an isomorphism. Thus $u_!(v)$ is carried by $\underline{a}'$ to an isomorphism, and consequently $u_!(v)$ is bicovering (II 5.3), whence ii).

The additional assertion, in the case where C is small, was proved in the course of 268
the proof, C.Q.F.D.

1.2.1. The functor $u^s : C^{\sim} \to C'^{\sim}$ of 2.2 iv) may often be interpreted as an "inverse image" functor for a "morphism of topoi" $C'^{\sim} \to C^{\sim}$, cf. IV 4.9. Its properties are summarized in the following proposition:

Proposition 1.3. *Let $u : C \to C'$ be a continuous functor from a small site C to a U-site C'.*

1) *The functor u^s is left adjoint to the functor u_s.*

2) *There is a canonical isomorphism* $u^s \simeq \underline{a}' u_! i$.
3) *There is a canonical isomorphism* $u^s \underline{a} \simeq \underline{a}' u_!$.
4) *The functor* u^s *commutes with inductive limits.*
5) *When the functor* $u_!$ *is left exact (cf.* I 5.4 *4)), the functor* u^s *is also left exact. More generally, the functor* u^s *commutes with those types of finite projective limits with which the functor* $u_!$ *commutes.*

Proof. Assertions 1), 2), 3), and 4) were proved in the course of the proof of 1.2. Assertion 5) follows from the isomorphism in 2), since i commutes with projective limits and $\underline{a}'$ commutes with finite projective limits (II 4.1).

Remark 1.4. Combining I 5.4 3) and 1.3 3), we obtain a diagram:

$$\begin{tikzcd}
C \arrow[r, "u"] \arrow[d, "h"'] & C' \arrow[d, "h'"] \\
C^\wedge \arrow[r, "u_!"] \arrow[d, "\underline{a}"'] & C'^\wedge \arrow[d, "\underline{a}'"] \\
C^\sim \arrow[r] & C'^\sim,
\end{tikzcd}$$

269 where the upper square commutes and the lower square commutes up to isomorphism. But the functors $\underline{a}$, $\underline{a}'$, $u_!$, and u^s are defined only up to isomorphism. The reader may verify that the functors $\underline{a}$, $\underline{a}'$, and u^s can be chosen so that:

1) The composite functors $\underline{a}h$ and $\underline{a}'h'$ are injective on the sets of objects;
2) the diagram

$$\begin{tikzcd}
C \arrow[r, "u"] \arrow[d, "\underline{a}h"'] & C' \arrow[d, "\underline{a}'h'"] \\
C^\sim \arrow[r, "u^s"] & C'^\sim
\end{tikzcd}$$

commutes.

More precisely, the functors $\underline{a}$ and $\underline{a}'$ can be chosen so as to satisfy condition 1). Once the functors $\underline{a}$ and $\underline{a}'$ have been chosen, the functor u^s can be chosen so as to satisfy condition 2).

Proposition 1.5. *Let* $\mathscr{U} \subset \mathscr{V}$ *be two universes, let* C *and* C' *be two* $\mathscr{U}$*-sites, and let* $u : C \to C'$ *be a functor between the underlying categories. Then* u *is continuous relative to* $\mathscr{U}$ *if and only if it is continuous relative to* $\mathscr{V}$*. Moreover, when* u *is continuous, denote by* $u^s_{\mathscr{U}}$ *(resp.* $u^s_{\mathscr{V}}$*) the functor between the categories of* $\mathscr{U}$*-sheaves (resp.* $\mathscr{V}$*-sheaves) introduced in* 1.2 iv)*. Then the diagram*

(1.5.1)
$$\begin{tikzcd}
\widetilde{C}_{\mathscr{U}} \arrow[r, "u^s_{\underline{U}}"] \arrow[d] & \widetilde{C'}_{\underline{U}} \arrow[d] \\
C^\sim_{\underline{V}} \arrow[r, "u^s_{\underline{V}}"] & C'^\sim_{\underline{V}},
\end{tikzcd}$$

270 *where the vertical functors are the canonical inclusion functors, commutes up to canonical isomorphism.*

Proof. We give the proof only in the case where C is $\mathscr{U}$-small. The general case will be proved in 4.3. It is clear that if the functor u^* carries every $\mathscr{V}$-sheaf on C' to a $\mathscr{V}$-sheaf on C, then it carries every $\mathscr{U}$-sheaf on C' to a $\mathscr{U}$-sheaf on C. Suppose that u^* carries every $\mathscr{U}$-sheaf on C' to a $\mathscr{U}$-sheaf on C. Denote by $\hat{C}_{\mathscr{U}} \hookrightarrow \hat{C}_{\mathscr{V}}$ (resp. $\hat{C'}_{\mathscr{U}} \hookrightarrow \hat{C'}_{\mathscr{V}}$) the categories of $\mathscr{U}$-presheaves and $\mathscr{V}$-presheaves, and by $u_{\mathscr{U}!}$ and $u_{\mathscr{V}!}$ the "inverse image" functors for $\mathscr{U}$-presheaves and $\mathscr{V}$-presheaves, respectively. We have a commutative diagram (cf. the explicit construction of $u_!$ in I 5.1):

$$(*)\qquad \begin{tikzcd} C\hat{}_{\underline{U}} \arrow[r, "u_{\underline{U}!}"] \arrow[d, hook] & C'\hat{}_{\underline{V}} \arrow[d, hook] \\ C\hat{}_{\underline{V}} \arrow[r, "u_{\underline{V}!}"] & C'\hat{}_{\underline{V}}. \end{tikzcd}$$

The inclusion functors from $\mathscr{U}$-presheaves into $\mathscr{V}$-presheaves are fully faithful and commute with projective limits. It follows from (II 5.1 i)) that a bicovering morphism of $\mathscr{U}$-presheaves is a bicovering morphism of $\mathscr{V}$-presheaves. Now let $R \hookrightarrow X$ be a sieve covering an object X of C. By 1.2 ii), the morphism $u_{\mathscr{U}!}R \to u_{\mathscr{U}!}X$ is bicovering. Using the commutativity of $(*)$ and what precedes, we deduce that the morphism $u_{\mathscr{V}!}R \to u_{\mathscr{V}!}X$ is bicovering. Consequently (1.2), the functor u^* carries $\mathscr{V}$-sheaves on C' to $\mathscr{V}$-sheaves on C. The commutativity of (1.5.1) then follows from the commutativity of $(*)$, from II 3.6, and from 1.3 2).

Proposition 1.6. *Let C and C' be two $\mathscr{U}$-sites, and let $u : C \to C'$ be a functor* 271
between the underlying categories. Consider the following conditions:

i) *u is continuous* (cf. . 1.1 and 1.2).
ii) *For every covering family $(X_i \to X)$, $i \in I$, in C, the family $(uX_i \to uX)$, $i \in I$, in C' is covering.*

We have the implication i) $\Rightarrow$ ii).

Suppose that the topology of C can be defined by a pretopology T (II 1.3) *and that the functor u commutes with fiber products.*[4] *Then conditions* i) *and* ii) *are equivalent and are also equivalent to the following condition:*

iii) *The functor u carries the covering families of T to covering families in C'.*

In particular, suppose that fiber products in C are representable and that the functor u commutes with fiber products. Then conditions i) *and* ii) *are equivalent.*

Proof. Let us prove that i) $\Rightarrow$ ii). Let $X_i \to X$, $i \in I$, be a covering family in C. For every sheaf F on C', u^*F is a sheaf. Consequently, the map

$$\mathrm{Hom}(X, u^*F) \longrightarrow \prod_i \mathrm{Hom}(X_i, u^*F)$$

is injective (II 5.1). Hence there is an injective map

$$\mathrm{Hom}(u_!X, F) \longrightarrow \prod_i \mathrm{Hom}(u_!X_i, F).$$

Therefore the family $u_!X_i \to u_!X$, $i \in I$, is covering in C' (II 5.1).

Every covering family of the pretopology T is a covering family in C, whence ii) $\Rightarrow$ iii). Let us prove that iii) $\Rightarrow$ i). Let X be an object of C and let $X_i \to X$, $i \in I$, be a 272

[4] In fact, it suffices that u commute with the fiber products occurring in base changes of morphisms belonging to covering families for T.

covering family in $\mathrm{Cov}(X)$ (II 1.3). The morphisms $X_i \to X$ are pullbackable, and the functor u commutes with fiber products. Consequently, the fiber products $u(X_i) \times_{u(X)} u(X_j)$ are representable and canonically isomorphic to $u(X_i \times_X X_j)$. Let F be a sheaf on C'. Then the diagram of sets

$$F(u(X)) \to \prod_i F(u(X_i)) \rightrightarrows \prod_{i,j} F(u(X_i \times_X X_j))$$

is exact (II 2.1, I 3.5, and I 2.12). Consequently, the diagram of sets

$$u^*F(X) \to \prod_i u^*F(X_i) \rightrightarrows \prod_{i,j} u^*F(X_i \times_X X_j)$$

is exact. Therefore u^*F is a sheaf (II 2.4), whence i). The final assertion of 1.6 follows from what precedes and from the fact that, when fiber products are representable in C, all covering families in C[4] define a pretopology on C whose associated topology is the topology of C.

Proposition 1.7. *Let C be a small site, C' a $\mathscr{U}$-site, and $u : C \to C'$ a continuous functor. Let γ be an algebraic structure defined by finite projective limits, such that in the category of γ-objects of $\mathscr{U} -$* Ens *the $\mathscr{U}$-indexed inductive limits are representable* [4]. *We use the notation of Proposition* II 6.4. *The functor u_s commutes with projective limits and consequently defines a functor $u_s^\gamma : C_\gamma^\sim \to C'^\sim_\gamma$. The functor u_s^γ has a left adjoint u_γ^s with the following properties:*

273 1) *There is a canonical isomorphism $u_\gamma^s \xrightarrow{\sim} \underline{a}'_\gamma u_{\gamma!} i_\gamma$, and u_γ^s commutes with every finite projective limit with which $u_{\gamma!}$ commutes.*

2) *There is a canonical isomorphism $u_\gamma^s \underline{a}_\gamma \xrightarrow{\sim} \underline{a}'_\gamma u'_{\gamma!}$.*

3) *The functor u_γ^s commutes with inductive limits.*

4) *If u^s is left exact, the diagram* (in the notation of II 6.3.0)

$$\begin{tikzcd} C^\sim{}_\gamma \arrow[r, "u_\gamma^s"] \arrow[d, "\mathrm{esj}^\sim"'] & C'^\sim{}_\gamma \arrow[d, "\mathrm{esj}'^\sim"] \\ C^\sim \arrow[r, "u^s"] & C'^\sim \end{tikzcd}$$

is commutative up to isomorphism, and the functor u_γ^s is exact.

5) *Suppose that the functor* esj$^\sim$ *(resp.* esj$^\sim$*) has a left adjoint* Lib$^\sim$ *(resp.* Lib$'^\sim$*)* (II 6.5). *There is a canonical isomorphism $u_\gamma^s\,\mathrm{Lib}^\sim \xrightarrow{\sim} \mathrm{Lib}'^\sim u^s$.*

Moreover, when C is not necessarily a small site but is a $\mathscr{U}$-site, the functor u_s^γ has a left adjoint u_γ^s. Finally, if V is a universe containing $\mathscr{U}$ and if $u^s_{\gamma\mathscr{U}}$ (resp. $u^s_{\gamma\mathscr{V}}$) denotes the left adjoint of u_s relative to the universe $\mathscr{U}$ (resp. $\mathscr{V}$), the following diagram, analogous to diagram (1.5.1):

[4] indexed by sets belonging to a suitable universe.

[4] In fact, one can show that this condition is always satisfied.

(1.7.1)
$$\begin{tikzcd} \boldsymbol{C}^{\sim}_{\gamma\underline{U}} \arrow[r, "u^s_{\gamma\underline{U}}"] \arrow[d, hook] & \widetilde{\boldsymbol{C}}'_{\gamma\underline{U}} \arrow[d, hook] \\ \boldsymbol{C}^{\sim}_{\gamma\underline{V}} \arrow[r, "u^s_{r\underline{V}}"] & \widetilde{\boldsymbol{C}}'_{\gamma\underline{V}} \end{tikzcd}$$

is commutative up to canonical isomorphism.

Proof. The proof, in the case where $\boldsymbol{C}$ is a small site, is left to the reader. It will be completed in 4.3 in the case where $\boldsymbol{C}$ is a $\mathscr{U}$-site.

Notation 1.8. Henceforth we shall use the notation u_s for the functor u^γ_s. This notation is unlikely to cause confusion, because the functor u_s (defined on sheaves of sets) commutes with finite projective limits. 274

Likewise, when u^s is left exact, we shall use the notation u^s for the functor u^s_γ. This abuse of notation is justified by 1.7 4).

Example 1.9.1. Let X be a small topological space. Denote by $\mathrm{Ouv}(X)$ the category of open subsets of X (the objects of $\mathrm{Ouv}(X)$ are the open subsets of X, and the morphisms of $\mathrm{Ouv}(X)$ are inclusions), equipped with the canonical topology. A family $(U_i \subset U)$, $i \in I$, is covering if and only if the open subsets U_i cover U (II 2.6). The sheaves of sets on $\mathrm{Ouv}(X)$ are therefore the sheaves of sets in the sense of [TF]. Let $f : X \to Y$ be a continuous map. It induces a functor $f^{-1} : \mathrm{Ouv}(Y) \to \mathrm{Ouv}(X)$: for every open subset U of Y, $f^{-1}(U)$ is the inverse image of U under the map f. The functor f^{-1} is continuous (1.6). The functor $f^{-1}_s : \mathrm{Ouv}(X)^\sim \to \mathrm{Ouv}(Y)^\sim$ is the direct-image functor for sheaves, in the sense of [TF]. The functor $(f^{-1})^s : \mathrm{Ouv}(Y)^\sim \to \mathrm{Ouv}(X)^\sim$ is the inverse-image functor for sheaves in the sense of [TF]. The functor $(f^{-1})^s$ commutes with finite projective limits, by 1.3 5).

Example 1.9.2. Let X be a topological space and let $Y \hookrightarrow X$ be an open subset of X. Every open subset of Y is an open subset of X, and thus we obtain a functor $\boldsymbol{\Phi} : \mathrm{Ouv}(Y) \to \mathrm{Ouv}(X)$. The functor $\boldsymbol{\Phi}$ is continuous (1.6). The functor $\boldsymbol{\Phi}_s : \mathrm{Ouv}(X)^\sim \to \mathrm{Ouv}(Y)^\sim$ is the "restriction to Y" functor. The functor $\boldsymbol{\Phi}^s_{\mathrm{ab}} : \mathrm{Ouv}(Y)^\sim_{\mathrm{ab}} \to \mathrm{Ouv}(X)^\sim_{\mathrm{ab}}$ (II 6.3.3, I 1.7) is the "extension by zero outside Y" functor introduced in [TF]. The functor $\boldsymbol{\Phi}^s : \mathrm{Ouv}(Y)^\sim \to \mathrm{Ouv}(X)^\sim$ (1.3) is analogous to the "extension by zero" functor: for every sheaf H on Y and every point $y \in Y$, the stalk of $\boldsymbol{\Phi}^s H$ at y is canonically isomorphic to the stalk of H at y; for every point $x \in X$, $x \notin Y$, the stalk of $\boldsymbol{\Phi}^s H$ at x is empty. The functor $\boldsymbol{\Phi}^s$ is called the "extension by the empty set outside Y" functor. The functor $\boldsymbol{\Phi}^s$ commutes with fiber products and with finite products over a nonempty index set, but it does not carry the terminal object of $\mathrm{Ouv}(Y)^\sim$ to the terminal object of $\mathrm{Ouv}(X)^\sim$. 275

Example 1.9.3. Let $\boldsymbol{C}$ and $\boldsymbol{C}'$ be two small sites, and let $u : \boldsymbol{C} \to \boldsymbol{C}'$ be a functor between the underlying categories that carries every covering family of $\boldsymbol{C}$ to a covering family of $\boldsymbol{C}'$. The functor u is not continuous in general (cf. nevertheless 1.6). Here is a counterexample. Let S be a set of infinite cardinality in the universe $\mathscr{U}$, and let $\boldsymbol{C}$ be the full subcategory of the category of sets whose objects are the finite, nonempty subsets of S. Equip $\boldsymbol{C}$ with the canonical topology. Observe that the covering families of $\boldsymbol{C}$ are the surjective families of maps and that covering families are nonempty. Observe also that, up to isomorphism, there are only two constant sheaves on $\boldsymbol{C}$: the sheaf with value the

empty set and the sheaf with value a one-element set. Set $C = C'$ and let $u : C \to C'$ be a constant functor. The functor u carries the covering families of C to covering families of C'. The functor u is not continuous. Indeed, since the representable presheaves of
276 C' are sheaves, for every object X of C' there exists a sheaf F on C' such that the cardinality of $F(X)$ is ≥ 2, and consequently the presheaf $u^* F$ is not a sheaf.

2. Cocontinuous functors

Definition 2.1. *Let C and C' be two $\mathcal{U}$-sites, and let $u : C \to C'$ be a functor between the underlying categories. We say that u is cocontinuous if it has the following property:*

(COC) For every object Y of C and every covering sieve $R \hookrightarrow u(Y)$, the sieve on Y generated by the arrows $Z \to Y$ such that $u(Z) \to u(Y)$ factors through R covers Y.

Proposition 2.2. *Let C be a small site, let C' be a $\mathcal{U}$-site, and let $u : C \to C'$ be a functor between the underlying categories. Denote by $\hat{u}_* : C^\wedge \to C'^\wedge$ the functor right adjoint to the functor $F \mapsto F \circ u = \hat{u}^* F$ (I 5.1). The functor u is cocontinuous if and only if, for every sheaf G on C, $\hat{u}_* G$ is a sheaf on C'.*

Proof. *The condition is sufficient.* Suppose that for every sheaf G on C, the presheaf $\hat{u}_* G$ is a sheaf. Let Y be an object of C, and let $R \hookrightarrow u(Y)$ be a covering sieve. There is an isomorphism $\operatorname{Hom}(u(Y), u_* G) \xrightarrow{\sim} \operatorname{Hom}(R, u_* G)$, hence, by adjunction (I 5.1), an isomorphism $\operatorname{Hom}(\hat{u}^* \circ u(Y), G) \xrightarrow{\sim} \operatorname{Hom}(\hat{u}^* R, G)$. It follows (II 5.3) that the morphism $\hat{u}^* R \to \hat{u}^* u(Y)$ is bicovering. But the functor $\hat{u}^*$ commutes with projective limits, and therefore the morphism $\hat{u}^* R \to \hat{u}^* u(Y)$ is a monomorphism. It is thus a covering monomorphism. On the other hand, there is a canonical morphism $Y \xrightarrow{p} \hat{u}^* u(Y)$ deduced
277 from the adjunction morphism $\mathrm{id}_C \to \hat{u}^* u_!$ (I 5.4 3)). We then make the base change $Y \xrightarrow{p} \hat{u}^* u(Y)$. This gives a covering sieve on Y:

$$\hat{u}^* R \times_{\hat{u}^* u(Y)} Y \longrightarrow Y,$$

and, on determining the arrows $Z \to Y$ that factor through this sieve, one obtains the sieve described by property (COC).

The condition is necessary. Let S be an object of C' and let $R \hookrightarrow S$ be a covering sieve on S. We must prove that, for every sheaf G on C, there is an isomorphism

$$\operatorname{Hom}(S, u_* G) \xrightarrow{\sim} \operatorname{Hom}(R, u_* G).$$

Using Proposition II 5.3 and the adjunction isomorphisms, we see that it suffices to prove that the monomorphism $\hat{u}^* R \to \hat{u}^* S$ is covering. For this, it suffices to prove (II 5.1) that for every base change $Y \to \hat{u}^* S$, the sieve $\hat{u}^* R \times_{\hat{u}^* S} Y$ is covering, where Y is an object of C. But, by the properties of adjoint functors and I 5.4 3), the morphism $Y \to \hat{u}^* S$ factors through $Y \xrightarrow{p} \hat{u}^* u(Y) \to \hat{u}^* S$, and consequently the sieve $\hat{u}^* R \times_{\hat{u}^* S} Y \hookrightarrow Y$ is obtained, by the base change $Y \xrightarrow{p} \hat{u}^* u(Y)$, from the monomorphism $\hat{u}^* R \times_{\hat{u}^* S} \hat{u}^* u(Y) \xhookrightarrow{q} \hat{u}^* u(Y)$. Now the functor $\hat{u}^*$ commutes with projective limits, and consequently the monomorphism q is the image under $\hat{u}^*$ of the sieve $R \times_S u(Y) \hookrightarrow u(Y)$, which is covering. Hypothesis (COC) then allows us to conclude that the sieve $\hat{u}^* R \times_{\hat{u}^* S} Y \hookrightarrow Y$ is covering, C.Q.F.D.

Proposition 2.3. *Let C and C' be two $\mathcal{U}$-sites, and let $u : C \to C'$ be a cocontinuous functor. Denote by $u^* : \widehat{C}' \to \widehat{C}$ the functor $u^* = \underline{a}\hat{u}^* i'$, where $\underline{a}$ denotes the "associated*
278 *sheaf" functor for $C^\wedge$, $\hat{u}^*$ the functor $F \mapsto F \circ u$, and i' the inclusion functor $\widetilde{C}' \to C'^\wedge$.*

1) *The functor* u^* *commutes with inductive limits and is exact.*
2) *There is a canonical isomorphism* $u^*\underline{a}' \simeq \underline{a}\hat{u}^*$, *where* $\underline{a}'$ *denotes the "associated sheaf" functor for* $C'^{\wedge}$.
3) *The functor* u^* *has a right adjoint* $u_* : \widetilde{C} \to \widetilde{C}'$. *When* C *is small, the diagram*

(2.3.1)
$$\begin{tikzcd} C^{\sim} \arrow[r, "u_*"] \arrow[d, hook, "i"'] & \widetilde{C}' \arrow[d, hook, "i'"] \\ C^{\wedge} \arrow[r, "\hat{u}_*"] & C'^{\wedge}, \end{tikzcd}$$

in which the lower functor is right adjoint to the functor $\hat{u}^*$ *(I 5.1), commutes up to canonical isomorphism.*
4) *Let* $\mathscr{V} \supset \mathscr{U}$ *be a universe. Denote by* $u_{*\mathscr{U}} : C^{\sim}_{\mathscr{U}} \to \widetilde{C}'_{\mathscr{U}}$ *(resp.* $u_{*\mathscr{V}} : C^{\sim}_{\mathscr{V}} \to \widetilde{C}'_{\mathscr{V}}$*) the right adjoint relative to the universe* $\mathscr{U}$ *(resp.* $\mathscr{V}$*) whose existence is asserted in 3). The diagram*

(2.3.2)
$$\begin{tikzcd} C^{\sim}{}_{\underline{U}} \arrow[r, "u_{*\underline{U}}"] \arrow[d, hook] & \widetilde{C}'_{\underline{U}} \arrow[d, hook] \\ C^{\sim}{}_{\underline{V}} \arrow[r, "u_{*\underline{V}}"] & \widetilde{C}'_{\underline{V}} \end{tikzcd}$$

commutes up to isomorphism.

Proof. Assertions 1) and 2) follow immediately from II 3.4. Assertion 3), when C is small, follows from 2.2. When C is a $\mathscr{U}$-site, it will be proved in 4.4. Assertion 4), when C is small, follows from the analogous commutativity assertion when the topologies on C and C' are chaotic and from I 3.5. When the topologies on C and C' are chaotic, the 279
commutativity of (2.3.2) follows immediately from the explicit description of u_* (I 5.1).

2.4. We shall not develop here the considerations concerning γ-objects of categories of sheaves. It will suffice to note that the functors u^* and u_* introduced in this section always commute with finite projective limits (unlike what happened in the preceding section for the functor u^s). Consequently, they extend naturally to functors defined on γ-objects; these functors are adjoint to one another and commute with the "underlying sheaf of sets" functors. We shall make the notational abuses mentioned in 1.8, using u^* and u_* also for the extensions to γ-objects.

Proposition 2.5. *Let* C *and* C' *be two* $\mathscr{U}$*-sites, and let* $C \underset{u}{\overset{v}{\rightleftarrows}} C'$ *be a pair of functors, where* v *is left adjoint to* u. *The following properties are equivalent:*

i) *The functor* u *is continuous.*
ii) *The functor* v *is cocontinuous.*

Moreover, under these equivalent conditions, there are canonical isomorphisms $v_* \simeq u_s$ *and* $v^* \simeq u^s$. *In particular, the functor* u^s *commutes with finite projective limits.*

Proof. The property of a functor being continuous or cocontinuous does not depend on the universe $\mathscr{U}$ for which C and C' are $\mathscr{U}$-sites (this is immediate for a cocontinuous functor, and follows from 1.5 for a continuous functor). We may therefore, after enlarging the universe, suppose that C and C' are small. The proposition then follows from 2.2 and I 5.6.

280 Proposition 2.6. *Let $u : C \to C'$ be a continuous and cocontinuous functor between two $\mathscr{U}$-sites. Then the functor $u^* : C'^\sim \to C^\sim$ (2.3) commutes with $\mathscr{U}$-inductive and projective limits. Denote by $u_! : C^\sim \to C'^\sim$ and $u_* : C^\sim \to C'^\sim$ the functors left and right adjoint to u^*, respectively. The functor $u_!$ is fully faithful if and only if the functor u_* is fully faithful. When u is fully faithful, $u_!$ is fully faithful, and the converse holds when the topologies of C and C' are coarser than the canonical topology.*

The functor u^* has a right adjoint u_* (2.3) and a left adjoint $u_!$ (1.2). The functor u^* therefore commutes with inductive and projective limits (I 2.11). The fact that $u_!$ is fully faithful if and only if u_* is fully faithful is a general property of adjoint functors (I 5.7 1). When u is fully faithful, the functor $\hat{u}_* : C^\wedge_{\mathscr{V}} \to C'^\wedge_{\mathscr{V}}$ on $\mathscr{V}$-presheaves, for a sufficiently large universe $\mathscr{V}$, is fully faithful (I 5.7). Consequently, by the commutativity of (2.3.1) and (2.3.2), the functor $u_* : C^\sim \to \widetilde{C}'$ is fully faithful, and hence so is $u_!$ by what precedes. The converse follows from the existence of the commutative diagram in 1.2 iv), taking into account the fact that the functors ϵ_C and $\epsilon_{C'}$ are fully faithful when the topologies are coarser than the canonical topology, C.Q.F.D.

Example 2.7. With the notation of 1.9.2, the functor

$$\boldsymbol{\Phi} : \mathrm{Ouv}(Y) \longrightarrow \mathrm{Ouv}(X)$$

is continuous (1.9.2) and cocontinuous (2.2). Moreover, let $i : Y \to X$ be the canonical
281 inclusion and let $i^{-1} : \mathrm{Ouv}(X) \to \mathrm{Ouv}(Y)$ be the functor that it defines (1.9.1). The functor i^{-1} is right adjoint to the functor $\boldsymbol{\Phi}$. There is therefore a sequence of three adjoint functors:

$$\begin{array}{ccccc} \boldsymbol{\Phi}^s & , & \boldsymbol{\Phi}_s & & \\ & & \| & & \\ & & \boldsymbol{\Phi}^* & , & \boldsymbol{\Phi}_*, \end{array}$$

and canonical isomorphisms $\boldsymbol{\Phi}^* \simeq (i^{-1})^s$ and $\boldsymbol{\Phi}_* \simeq i_s^{-1}$.

3. Induced topology

3.1. Let C' be a site, C a category, and $u : C \to C'$ a functor. For every universe $\mathscr{U}$ such that C' is a $\mathscr{U}$-site and C is a $\mathscr{U}$-small category, denote by $\mathscr{T}_{\mathscr{U}}$ the finest among the topologies T on C that make u continuous (1.1). (Such a topology exists by II 2.2.) The topology $\mathscr{T}_{\mathscr{U}}$ does not depend on the universe $\mathscr{U}$. Indeed, if $\mathscr{V} \supset \mathscr{U}$ is a universe, then $\mathscr{T}_{\mathscr{U}} = \mathscr{T}_{\mathscr{V}}$ (1.1 and 1.5). The topology $\mathscr{T}_{\mathscr{U}}$ is called *the topology induced on C by the topology of C' by means of the functor u* [4].

Proposition 3.2. *Let C be a small category, C' a $\mathscr{U}$-site, $u : C \to C'$ a functor, and $\mathscr{T}$ the topology on C induced by u. Let X be an object of C, and let $R \to X$ be a sieve on X. The following properties are equivalent:*

i) *The sieve $R \hookrightarrow X$ is covering for $\mathscr{T}$.*
ii) *For every base change $Y \to X$, where Y is an object of C, the morphism $u_!(R \times_X Y) \to u(Y)$ is bicovering in $C'^\wedge$* (II 5.2).

[4] When no confusion can result, this topology is called the *topology induced on C by the topology of C'*.

Proof. i) $\Rightarrow$ ii): This follows from axiom (T 1) for topologies and from 1.2.
ii) $\Rightarrow$ i): For every sheaf F on C', the map

$$\mathrm{Hom}(Y, u^* F) \longrightarrow \mathrm{Hom}(R \times_X Y, u^* F)$$

is isomorphic, by adjunction (I 5.1), to the map 282

$$\mathrm{Hom}(u(Y), F) \longrightarrow \mathrm{Hom}(u_!(R \times_X Y), F),$$

which is bijective (II 5.3). Consequently (II 2.2), the sieve $R \hookrightarrow X$ is covering for $\mathcal{T}$.

Corollary 3.3. *Let C' be a site, C a category, $u : C \to C'$ a functor, and $\mathcal{T}$ the topology on C induced by the topology of C'. Let $(X_i \to X)$, $i \in I$, be a family of pullbackable morphisms in C, and suppose that u commutes with fiber products* [4]. *The following properties are equivalent:*

i) *The family $X_i \to X$, $i \in I$, is covering for $\mathcal{T}$.*
ii) *The family $(u(X_i) \to u(X))$, $i \in I$, is covering.*

Proof. i) $\Rightarrow$ ii): This follows from 1.6.
ii) $\Rightarrow$ i): Let $\mathcal{U}$ be a universe such that C is $\mathcal{U}$-small and C' is a $\mathcal{U}$-site. Let $R \hookrightarrow X$ be the sieve generated by the $X_i \to X$. The presheaf R is the coequalizer of the pair of arrows (I 2.12 and I 3.5)

$$\coprod_{i,j} X_i \times_X X_j \rightrightarrows \coprod_i X_i$$

(the coproduct is taken here in $C^\wedge$). Since the functor $u_!$ commutes with inductive limits (I 5.4), the presheaf $u_! R$ is the coequalizer of the pair of arrows

$$\coprod_{i,j} u(X_i \times_X X_j) \rightrightarrows \coprod_i u(X_i).$$

Since the functor u commutes with fiber products, the presheaf $u_! R$ is the coequalizer of the pair of arrows

$$\coprod_{i,j} u(X_i) \times_{u(X)} u(X_j) \rightrightarrows \coprod_i u(X_i),$$

and consequently $u_! R \to u(X)$ is the sieve on $u(X)$ generated by the $u(X_i) \to u(X)$, $i \in I$. Using once again the fact that u commutes with fiber products, one shows that for every base change $Y \to X$, where Y is an object of C, $u_!(R \times_X Y) \to u(Y)$ is the sieve on $u(Y)$ generated by the $u(X_i) \times_{u(X)} u(Y) \to u(Y)$, $i \in I$. Since the family $u(X_i) \to u(X)$, $i \in I$, is covering, the family $u(X_i) \times_{u(X)} u(Y) \to u(Y)$, $i \in I$, is covering. Thus $u_!(R \times_X Y) \to u(Y)$ is a covering sieve, and consequently (3.2) $R \to X$ is covering for $\mathcal{T}$, C.Q.F.D. 283

Corollary 3.4. *Let C' be a $\mathcal{U}$-site, let C be a full subcategory of C', and let $u : C \to C'$ be the inclusion functor. Suppose that fiber products are representable in C and that u commutes with fiber products. The following conditions are equivalent:*

i) a) *For every object X of C, every covering family $(Y_j \to X)$, $j \in J$, in C' is refined by a covering family $X_i \to X$, $i \in I$, where the X_i are objects of C.*
b) *There exists a small set G of objects of C such that every object of C is the target of a covering family in C' of morphisms whose sources belong to G.*

[4] In fact, it suffices that u commute with fiber products of the form $X_i \times_X X'$.

ii) *The topology induced by the topology of C' is a $\mathscr{U}$-topology* (I 3.0.2), *and the functor u is continuous and cocontinuous for this topology.*

Proof. This follows immediately from 3.3 and 2.1.

284 Proposition 3.5. *Let C be a $\mathscr{U}$-site and let $\epsilon_C : C \to C^{\sim}$ be the canonical functor* (II 4.4.0). *Equip $C^{\sim}$ with the canonical topology. Then the topology of the site C is the topology induced by the topology of $C^{\sim}$.*

Proof. The covering families of $C^{\sim}$ for the canonical topology are the universally effective epimorphic families (II 2.5), that is (II 4.3), the epimorphic families. Let T be the topology of the site C, and let $\mathscr{T}_{\mathscr{U}}$ be the finest among the topologies T' on C such that, for every sheaf F on $C^{\sim}$, $F \circ \epsilon_C$ is a sheaf for T'. It suffices to show that $T = \mathscr{T}_{\mathscr{U}}$, for then $\mathscr{T}_{\mathscr{U}}$ is a $\mathscr{U}$-topology and consequently $\mathscr{T}_{\mathscr{U}}$ is the induced topology.

A) *The topology T is finer than $\mathscr{T}_{\mathscr{U}}$*: Let $(X_i \to X)$, $i \in I$, be a covering family for $\mathscr{T}_{\mathscr{U}}$. Then, for every sheaf F on $C^{\sim}$, the map $F(\epsilon_C(X)) \to \prod_i F(\epsilon_C(X_i))$ is injective, and consequently (II 5.2) $(\epsilon_C X_i \to \epsilon_C X)$, $i \in I$, is covering for the canonical topology of $C^{\sim}$, hence (II 5.2) $(X_i \to X)$, $i \in I$, is covering for T.
B) *The topology T is coarser than $\mathscr{T}_{\mathscr{U}}$*: It suffices to show that $\epsilon_C : C \to C^{\sim}$ is continuous (1.1). Now we shall prove in IV n° 1 that every sheaf on $C^{\sim}$ is representable, and consequently, for every sheaf F on $C^{\sim}$, $F \circ \epsilon_C$ is a sheaf on C. (We shall not use 3.5 before IV 1.)

We record two results that will be used in VI 7.

Proposition 3.6. *Let $(C_i)_{i\in I}$ be a family of sites, let C be a category, and, for every $i \in I$, let $u_i : C_i \to C$ be a functor. Let $\mathscr{U}$ be a universe such that the categories C_i and C are $\mathscr{U}$-small. There exists a topology $\mathscr{T}_{\mathscr{U}}$ on C that is the coarsest among the topologies for which the u_i are continuous. The topology $\mathscr{T}_{\mathscr{U}}$ does not depend on the universe $\mathscr{U}$ relative to which the categories under consideration are small.*

The last assertion of 3.6 follows from 1.5.

Let T be a topology on C. The functors u_i are continuous if and only if, for every
285 $i \in I$ and every covering sieve $R \hookrightarrow X$ on an object X of C_i, the morphism

$$u_{i!}(R) \longrightarrow u_i(X)$$

in $\widehat{C}$ is bicovering (1.2 (ii)). Thus 3.6 is a consequence of the following lemma.

Lemma 3.6.1. *Let C be a small category and let $(u_i : F_i \to G_i)_{i\in I}$ be a family of arrows in $\widehat{C}$. There exists on C a topology that is coarsest among those making the morphisms u_i covering (resp. bicovering)* (II 5.2).

To say that $u : F \to G$ is covering for a given topology T means that, for every arrow $X \to G$, with $X \in \operatorname{Ob} C$, the corresponding arrow $F \times_G X \to X$ is covering; equivalently, the family of arrows $X' \to X$ in C that factor through the preceding arrow is covering. The existence of a coarsest topology among those for which the $u_i : F_i \to G_i$ are covering thus follows from I 1.1.6, proving the assertion of 3.6.1 without the parenthetical alternative. The parenthetical assertion follows by recalling that a morphism $u : F \to G$ is bicovering if and only if the morphisms $u : F \to G$ and $\operatorname{diag}_u : F \to F \times_G F$ are covering.

Proposition 3.7. *Let $(C_i)_{i\in I}$ be a family of sites, let C be a category, for every $i \in I$ let $u_i : C_i \to C$ be a functor, and let $\mathscr{U}$ be a universe such that the categories C_i and C are $\mathscr{U}$-small. There exists a topology $\mathfrak{T}_{\mathscr{U}}$ that is the finest for which the u_i are cocontinuous.*

The topology $\mathfrak{T}_{\mathscr{U}}$ does not depend on the universe $\mathscr{U}$ relative to which the categories under consideration are small.

Let $\mathscr{U}$ be a universe relative to which the categories under consideration are small. The functors u_i are cocontinuous for a topology $\mathfrak{T}$ on C if and only if, for every $i \in I$ and every sheaf F on C_i, the presheaf $\hat{u}_{i*}F$ is a sheaf for $\mathfrak{T}$ (2.2). It follows that the topology $\mathfrak{T}_{\mathscr{U}}$ is the finest topology for which the presheaves $\hat{u}_{i*}F$, $i \in I$, $F \in C_i^{\sim}$, are sheaves (II 2.2). The last assertion follows from 2.2.

4. Comparison lemma

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Theorem 4.1. (comparison lemma). *Let C be a small category, let C' be a site whose underlying category is a $\mathscr{U}$-category, and let $u : C \to C'$ be a fully faithful functor. Endow C with the topology induced by u (3.1). Consider the following properties:*

i) *Every object of C' can be covered by objects coming from C.*
ii) *The functor $F \mapsto F \circ u$ induces an equivalence from the category of sheaves on C' to the category of sheaves on C.*

We always have i) $\Rightarrow$ ii). *When C' is a $\mathscr{U}$-site and the topology of C' is coarser than the canonical topology, we have* ii) $\Rightarrow$ i).

4.1.1. Let us first prove i) $\Rightarrow$ ii). The proof proceeds in two steps.

First step. For every presheaf H on C', the adjunction morphism $\varphi : u_! u^* H \to H$ (I 5.1) is bicovering (II 5.3), and the functor $u_s : \widetilde{C'} \to \widetilde{C}$ (1.1.1) is fully faithful.

Let C/H be the small category whose objects are the objects Y of C equipped with a morphism $u(Y) \to H$, and whose morphisms are the commutative diagrams

$$\begin{tikzcd} u(Y) \arrow[rr, "u(m)"] \arrow[dr] & & u(Y') \arrow[dl] \\ & H & \end{tikzcd}.$$

We have $u^* H = \varinjlim_{Y \in \mathrm{ob}\, C/H} Y$ (I 3.4), and hence (I 5.4) $u_! u^* H = \varinjlim_{Y \in \mathrm{ob}\, C/H} u(Y)$. The adjunction morphism is the evident morphism arising from the description of $u_! u^* H$ as an inductive limit. The morphism φ has the following property: $(*)$ For every object Y 287
of C, every morphism $m : u(Y) \to H$ factors uniquely as the canonical morphism $\alpha(m) : u(Y) \to u_! u^* H$ followed by φ. It follows immediately from i) that φ is covering (II 5.1); let us show that it is bicovering. Let $p, q : Z \rightrightarrows u_! u^* H$ be two morphisms from an object Z of C' such that $\varphi p = \varphi q$. For every object Y of C and every morphism $n : u(Y) \to Z$, we have $\varphi p n = \varphi q n$. Property $(*)$ then implies that $pn = qn$, and since the $u(Y)$ cover Z, the kernel of (p, q) is a covering sieve of Z. Thus φ is bicovering (II 5.3). For every sheaf H on C', $u^* H$ is a sheaf on C, denoted by $u_s H$ (1.1.1). Moreover, $u^s u_s H = \underline{a}' u_! u_s H$ (1.3), and the adjunction morphism $u^s u_s H \to H$ is obtained by applying the "associated sheaf" functor to $\varphi : u_! u^* H \to H$. Consequently (II 5.3), the adjunction morphism $u^s u_s H \to H$ is an isomorphism. Hence $u_s : \widetilde{C}' \to \widetilde{C}$ is fully faithful.

Second step. The functor u is cocontinuous and the functor $u^s : C^{\sim} \to C'^{\sim}$ (1.2) is fully faithful. Consequently $u_s : C'^{\sim} \to C^{\sim}$ is an equivalence. Let Y be an object of C and let $i : R \hookrightarrow u(Y)$ be a covering sieve. Since the functor u^* commutes with projective limits (I 5.5), the morphism $u^*(i) : u^*(R) \hookrightarrow u^* u(Y)$ is a monomorphism. Since u is fully faithful, we have $u^* u(Y) \simeq Y$, and thus obtain a sieve of Y, $u^*(i) : u^*(R) \to Y$. To show that u is cocontinuous, it suffices to show that $u^*(i) : u^*(R) \hookrightarrow Y$ is a covering sieve for the

induced topology on C (2.1). For this, it suffices to show (3.2) that for every base change $m : X \to Y$, the morphism $u_!(u^*(R) \times_Y X) \to u(X)$ is bicovering. But since u^* commutes
288 with projective limits, we have $u^*(R) \times_Y X = u^*(R \times_{u(Y)} u(X))$, and $R \times_{u(Y)} u(X)$ is a covering sieve of $u(X)$. It therefore suffices to show that for every object Y of C and every covering sieve $i : R \hookrightarrow u(Y)$, the morphism in $C'^{\wedge}$, $u_!u^*(R) \xrightarrow{u_!u^*(i)} u(Y)$, is bicovering. Now this morphism factors as the adjunction morphism $u_!u^*(R) \to R$, which is bicovering (first step), followed by the covering monomorphism $R \hookrightarrow u(Y)$. It is therefore bicovering (II 5.3). This shows that u is cocontinuous. Since u is continuous and fully faithful, it follows from 2.6 (used in the case where C is small) that u^s (denoted by $u_!$ in 2.6) is fully faithful. Since u^s and u_s are adjoint to one another and are fully faithful, they are quasi-inverse functors, and hence u_s is an equivalence.

4.1.2. Let us now prove ii) ⇒ i). The objects Y of C form a generating family of $C^\sim$ (II 4.10), and consequently, for every object X of C', there exists an epimorphic family (in $C^\sim$) $v_i : Y_i \to u_s X$, $i \in I$. We have $u_s u(Y_i) = Y_i$, and, since u_s is an equivalence of categories, $v_i = u_s(w_i)$, where the $w_i : u(Y_i) \to X$ form an epimorphic family in $C'^\sim$. It then follows from II 5.1 that the family $w_i : u(Y_i) \to X$, $i \in I$, is covering.

4.2. **Completion of the proof of 1.2.** Let G be a full subcategory of C whose objects form a small topologically generating family of C (II 3.0.1). Endow G with the topology induced by the inclusion functor $i : G \to C$ (3.1). The functor $(u \circ i)_s = i_s \circ u_s$ (1.1.1) has
289 a left adjoint by the first part of the proof of 1.2. Since i_s is an equivalence of categories (4.1), the functor u_s has a left adjoint u^s, and there is a functorial isomorphism $u^s \simeq (u \circ i)^s \circ i_s$. Let us show that the diagram

(4.2.1)
$$\begin{tikzcd} C \arrow[r, "u"] \arrow[d, "\epsilon_C"'] & C' \arrow[d, "\epsilon_{C'}"] \\ C^\sim \arrow[r, "u^s"] & \widetilde{C'} \end{tikzcd}$$

commutes up to isomorphism. For every sheaf H on C', we have $\mathrm{Hom}_{\widetilde{C}'}(\epsilon_{C'} uX, H) \simeq u_s H(X)$ for every $X \in \operatorname{ob} C$. Moreover, $\mathrm{Hom}_{\widetilde{C}'}(u^s \epsilon_C X, H) \simeq \mathrm{Hom}_{\widetilde{C}}(\epsilon_C X, u_s H)$ by adjunction, and then $\mathrm{Hom}_{\widetilde{C}}(\epsilon_C X, u_s H) \simeq u_s H(X)$ by the definition of ϵ_C. Hence there is an isomorphism $\epsilon_{C'} uX \simeq u^s \epsilon_C X$ for every $X \in \operatorname{ob} C$. This proves i) ⇒ iv). Let us prove iv) ⇒ i). We have a commutative diagram

(4.2.2)
$$\begin{tikzcd} G \arrow[r, "i"] \arrow[d, "\epsilon_G"'] & C \arrow[r, "u"] \arrow[d, "\epsilon_C"'] & C' \arrow[d, "\epsilon_{C'}"'] \\ \widetilde{G} \arrow[r, "i^s"] & \widetilde{C} \arrow[r, "u^s"] & \widetilde{C'}, \end{tikzcd}$$

and consequently, by the first part of the proof, the functor $u \circ i : G \to C'$ is continuous. It follows at once from 4.1 and 1.1 that u is continuous, whence iv) ⇒ i). This also gives the uniqueness of u^s, since the first part of the proof gives uniqueness when C is small.

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4.3. **Completion of the proofs of 1.5 and 1.7.** Let $u : C \to C'$ be a functor between two $\mathcal{U}$-sites, and let G be a full subcategory of C whose set of objects is a small topologically generating family of C (II 3.0.1). Endow G with the topology induced by the inclusion functor $i : G \to C$. It follows immediately from 4.1 and 1.1 that u is continuous if and only if $u \circ i$ is continuous. This observation and the first part of the proof of 1.5 give the general case. The same observation also reduces the proof of 1.7 to the case where the category C is small.

4.4. **Completion of the proof of 2.3.** Let $u : C \to C'$ be a functor between two $\mathscr{U}$-sites, and let G be a full subcategory of C whose set of objects is a small topologically generating family of C (II 3.0.1), endowed with the topology induced by the inclusion functor $i : G \to C$. It follows from the proof of 4.1 (4.1.1, second step) that i is cocontinuous. Consequently, when u is cocontinuous, the functor $u \circ i : G \to C'$ is cocontinuous. Moreover, $i^* : \widetilde{C} \to \widetilde{G}$ is an equivalence of categories (4.1). Therefore the functor $u^* : \widetilde{C}' \to \widetilde{C}$ has a right adjoint. This proves 3). To prove 4), it suffices to observe that $(u \circ i)_* \simeq u_* \circ i_*$ and that i_* is an equivalence (4.1). The commutativity of (2.3.2) then follows from the commutativity of these diagrams when C is small.

5. Localization 291

5.1. Now let C be a $\mathscr{U}$-site and X an object of $C^{\wedge}$. Unless expressly stated otherwise, the category C/X will be endowed with the topology $\mathscr{T}$ induced by the functor $j_X : C/X \to C$ (3.1). The notation C/X will denote the category C/X endowed with the topology $\mathscr{T}$. Proposition I 5.11 shows that the functor $j_{X!}$ commutes with fiber products and consequently sends every monomorphism to a monomorphism. In particular, for every object $(Y \to X)$ of C/X, the functor $j_{X!}$ establishes a one-to-one correspondence between the sieves, in the category C/X, on the object $(Y \to X)$ and the sieves, in C, on the object Y.

Proposition 5.2. *Let C be a $\mathscr{U}$-site, X an object of $C^{\wedge}$, and $j_X : C/X \to C$ the corresponding continuous functor.*

1) *A sieve R on an object $(Z \to X)$ is covering in C/X if and only if the sieve $j_{X!}(R) \hookrightarrow Z$ is covering in C.*
2) *The functor $j_X : C/X \to C$ is cocontinuous and continuous.*
3) *Let $m : Y \to X$ be a morphism of $C^{\wedge}$. We then have the commutative diagram*

$$\begin{tikzcd} C/Y \arrow[rr, "j_m"] \arrow[dr, "j_Y"'] & & C/X \arrow[dl, "j_X"] \\ & C & \end{tikzcd} .$$

The topology induced by the functor j_Y on C/Y is equal to the topology induced by j_m on C/Y.

4) *The topology induced by $j_X : C/X \to C$ is a $\mathscr{U}$-topology (II 3.0.2).*

Proof. 1) If the sieve R on $(Z \to X)$ is covering, the sieve $j_{X!}(R) \hookrightarrow Z$ is 292
covering (1.6).

Conversely, if the sieve $j_{X!}(R) \to Z$ is covering in C, one sees that the same is true for every sieve obtained by a base change in C/X. The sieve R is therefore covering (3.2).

2) This follows immediately from 1) by applying 2.1.
3) This follows immediately from the description of covering sieves given in 1).

4) Let $(G_i)_{i\in I}$ be a small topologically generating family of C. One verifies immediately that the small family $(u : G_i \to X)$, $u \in \coprod_{i\in I} \mathrm{Hom}_{C^\wedge}(G_i, X)$, is topologically generating in C/X.

Terminology and notation 5.3. By the preceding proposition, the functor j_X is both continuous and cocontinuous. It therefore defines a sequence of three adjoint functors (4.3.2) between the categories of sheaves of sets (1.3 and 2.3):

$$\begin{aligned} j_X^s &: (C/X)^\sim \longrightarrow C^\sim \\ j_X^* = j_{X,s} &: C^\sim \longrightarrow (C/X)^\sim \\ j_{X*} &: (C/X)^\sim \longrightarrow C^\sim . \end{aligned}$$

In the particular situation of Proposition 5.2, we shall use the following terminology and notation:

1) The functor j_{X*} will be called the *direct image functor.*
2) The functor $j_{X,s} = j_X^*$ will be denoted by j_X^* and will be called the *restriction functor to* C/X.
293 3) The functor j_X^s on sheaves of sets will be called the “extension by the empty set to C” functor and will be denoted by $j_{X!}$ (cf. 2.9.2).

Thus we have a sequence of three adjoint functors between $(C/X)^\sim$ and $C^\sim$:

$$j_{X!},\ j_X^*,\ j_{X*}.$$

Proposition 5.4. *The functor* $j_{X!} : (C/X)^\sim \to C^\sim$ *factors through the category* $C^\sim/\underline{a}X$ *($\underline{a}$ is the associated sheaf functor):*

$(C/X)^\sim \xrightarrow{e_X^\sim} C^\sim/\underline{a}X \to C^\sim$. *The functor*

$$e_X^\sim : (C/X)^\sim \longrightarrow \widetilde{C}/\underline{a}X$$

is an equivalence of categories. The restriction functor to C/X*, followed by the equivalence* $e_X^\sim$*, is isomorphic to the “base change by* $\underline{a}X \to \epsilon$*” functor ($\epsilon$ being the terminal object of* $C^\sim$*):* $F \mapsto (F \times \underline{a}X \xrightarrow{\mathrm{pr}_2} \underline{a}X)$.

Proof. The image under $j_{X!}$ of the terminal object of $(C/X)^\sim$ is the object $\underline{a}X$, which gives the factorization. To show that the functor $e_X^\sim$ is an equivalence, we shall confine ourselves to a few indications. By I 5.11, a presheaf on C/X is defined by a presheaf F on C equipped with a morphism $F \to X$. One then proves that the following two properties are equivalent:

i) The presheaf on C/X defined by $F \to X$ is a sheaf.
ii) The following diagram is cartesian (where i and $\underline{a}$ denote the inclusion in the category of presheaves and the associated sheaf functor, respectively):

$$\begin{tikzcd} F \arrow[r] \arrow[d] & i\underline{a}F \arrow[d] \\ X \arrow[r] & i\underline{a}X \end{tikzcd} .$$

294 It follows immediately that $e_X^\sim$ is an equivalence. The last assertion is trivial.

Proposition 5.5. 1) *Let C be a $\mathscr{U}$-site and X an object of $C^{\wedge}$. The diagram below of categories and functors commutes up to canonical isomorphisms:*

$$\begin{tikzcd}
C/X \arrow[r, "h_X"] \arrow[d] & (C/X)^{\wedge} \arrow[r, "\underline{a}_X"] \arrow[d, "e^{\wedge}_X"] & (C/X)^{\sim} \arrow[rr, "i_X"] \arrow[d, "e^{\sim}_X"] & & (C/X)^{\wedge} \arrow[d, "e^{\wedge}/X"] \\
C/X \arrow[r, "h/X"] & C^{\wedge}/X \arrow[r, "\underline{a}/X"] & C^{\sim}/\underline{a}X \arrow[r, "i/\underline{a}X"] & C^{\wedge}/i\underline{a}X \arrow[r, "\pi"] & C^{\wedge}/X .
\end{tikzcd}$$

The notations $\underline{a}_X$ and i_X denote the "associated sheaf" and inclusion into presheaves functors for the site C/X. The notation $\underline{a}/X$ denotes the natural extension of the functor $\underline{a}$ (the associated sheaf functor for the site C) to the category of arrows with target X; similarly for the notation $i/\underline{a}X$. Finally, the notation π denotes the base-change functor along the canonical arrow $X \to i\underline{a}X$.

2) *Let, in addition, $m : Y \to X$ be a morphism of $C^{\wedge}$. The following diagram commutes up to canonical isomorphism:*

$$\begin{tikzcd}
C^{\sim}/\underline{a}X/\underline{a}Y \arrow[dd, equal] & (C/X)^{\sim}/\underline{a}_X Y \arrow[l, "f"'] & \\
& & (C/X/Y)^{\sim} \arrow[ul, "g"'] \arrow[dl, "e^{\sim}_m"] \\
C^{\sim}/\underline{a}Y & (C/Y)^{\sim} \arrow[l, "e^{\sim}_Y"'] & .
\end{tikzcd}$$

The arrow f is the natural extension of the equivalence $e^{\sim}/X$ to the category 295
of arrows with target $\underline{a}_X Y$, and is consequently an equivalence of categories. The arrow g is precisely the equivalence of 5.4 applied to the situation $(C/X)/[m]$.

3) *Denote by $j_{\underline{a}X!} : C^{\sim}/\underline{a}X \to C^{\sim}$ the "forget $\underline{a}X$" functor, and by $j^*_{\underline{a}X} : C^{\sim} \to C^{\sim}/\underline{a}X$ the "product with $\underline{a}X$" functor. The following diagrams commute up to canonical isomorphism:*

$$\begin{tikzcd}
(C/X)^{\sim} \arrow[r, "j_{X!}"] \arrow[d] & C^{\sim} \arrow[d, equal] \\
C^{\sim}/\underline{a}X \arrow[r, "j_{\underline{a}X!}"] & C^{\sim}
\end{tikzcd} \quad , \quad
\begin{tikzcd}
C^{\sim} \arrow[r, "j^*_X"] \arrow[d, equal] & (C/X)^{\sim} \arrow[d, "e^{\sim}/X"] \\
C^{\sim} \arrow[r, "j^*_{\underline{a}X}"] & C^{\sim}/\underline{a}X .
\end{tikzcd}$$

Proof. The proof of these assertions follows immediately from the definition of the equivalences $e^{\sim}_X$ (5.4) and $e^{\wedge}_X$ (I 5.11).

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EXPOSÉ IV

Topoi

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0. Introduction

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0.1. We saw in II various exactness properties of categories of the form $\widetilde{\mathscr{C}}$ = the category of sheaves of sets on $\mathscr{C}$, where $\mathscr{C}$ is a small site. These properties can be expressed by saying that in many respects such categories (which we shall call *topoi*) inherit the familiar properties of the category (Ens) of (small) sets. On the other hand, experience has shown that various situations in Mathematics should be regarded *above all as a technical means of constructing the corresponding categories of sheaves* (of sets), i.e. *the corresponding "topoi."* It appears that all the genuinely important notions attached to a site (for example its cohomological invariants, studied in V; various other "topological" invariants, such as its homotopy invariants recently studied by M. Artin and B. Mazur [1]; and the notions studied in J. Giraud's book on noncommutative cohomology) are in fact expressed directly in terms of the associated topos. From this point of view, two sites should be regarded as essentially equivalent when their associated topoi are equivalent categories. The datum of a site (at least in the case, especially important in practice, where its topology is coarser than its canonical topology) should be regarded as amounting to the datum of a topos $\mathscr{E}$ (namely the associated topos formed by the sheaves of sets on the site) together with a generating family of objects of $\mathscr{E}$ (cf. II 4.9, and 1.2.1 300
below). This viewpoint is analogous to associating a group with a system of generators and a system of relations among those generators, and attaching interest rather to the structure of that group than to the system of generators and relations used to generate it (which are regarded as auxiliary data of the situation). Moreover, the "comparison lemma" III 4.1 provides numerous examples of pairs of sites $\mathscr{C}$, $\mathscr{C}'$ that are not isomorphic, and are not even equivalent as categories, but give rise to equivalent topoi; thus $\mathscr{C}$ and $\mathscr{C}'$ should be regarded as essentially equivalent.

0.2. In the present exposé we give a characterization of topoi by simple exactness properties (due to J. Giraud); we study the natural notion of a morphism of topoi, inspired by the notion of a continuous map from one topological space to another; and we develop in the setting of topoi certain constructions familiar from ordinary sheaf theory (sheaves **Hom**, tensor-product sheaves, supports). Finally, following M. Artin, we show how a topos can be reconstructed from an "open" part, its complementary "closed" part, and a certain left exact functor connecting them, which may in fact be chosen almost arbitrarily.

0.3. This gives a procedure for gluing topoi which, when applied to topoi arising from ordinary topological spaces, will in general produce a topos no longer of the same type. This is a first indication of the remarkable stability of the notion of a topos under various natural constructions, a stability lacking for the notion of a topological space (from which the notion of a topos is inspired). As a second remarkable example, let 301 us also mention the relative classifying topos of a group in a topos (cf. [1] or 5.9 below), inspired by the classical notion of the classifying space of a topological group, and the notion of a "moduli topos" associated with various "moduli problems" in Algebraic Geometry or Analytic Geometry [10] [13].

Other topoi, such as the *étale topos* of a scheme (systematically studied in the present Seminar beginning with Exp. VII) or the *crystalline topos* of a relative scheme [6], arise naturally when one wishes to develop usable cohomology theories for abstract algebraic varieties (and, more generally, schemes), replacing the classical Betti cohomology of algebraic varieties over the field of complex numbers.

0.4. Thus one may say that the notion of a topos, a natural outgrowth of the *sheaf-theoretic viewpoint* in Topology, in turn constitutes a substantial enlargement of the notion of a topological space[4], embracing a large number of situations which formerly were not regarded as belonging to topological intuition. The characteristic feature of such situations is the availability of a notion of "localization," a notion formalized precisely by that of a site and, ultimately, by that of a topos (through the topos associated with the site). As the term "topos" itself is intended precisely to suggest, it seems reasonable and legitimate to the authors of the present Seminar to regard the subject of Topology as the study of *topoi* (and not merely of topological spaces).

0.5. It seemed useful to include in this general exposé on topoi a fairly large number
302 of examples, many of which are only remotely related to the original aim of this seminar (namely, the study of étale cohomology). The reader in a hurry, interested exclusively in étale cohomology, may of course omit these examples without inconvenience, as well as most of the present exposé, consulting it only when the need arises.

1. Definition and characterization of topoi

Definition 1.1. *A category E is called a $\mathcal{U}$-topos, or simply a topos when no confusion is likely, if there exists a site $C \in \mathcal{U}$ such that E is equivalent to the category $C^{\sim}$ of $\mathcal{U}$-sheaves of sets on C.*

1.1.1. Let E be a $\mathcal{U}$-topos. We always regard E as endowed with its canonical topology (II 2.5), which therefore makes it a *site*, and even, by 1.1.2 d) below, a $\mathcal{U}$-site (II 3.0.2). Unless expressly stated otherwise, we shall consider no topology on E other than the one just specified.

1.1.2. We saw in II 4.8, II 4.11 that a $\mathcal{U}$-topos E is a $\mathcal{U}$-category (I 1.1) satisfying the following conditions:

a) Finite projective limits are representable in E.
b) Direct sums indexed by an element of $\mathcal{U}$ are representable in E. They are disjoint and universal (II 4.5).
c) Equivalence relations in E are universally effective (I 10.10).
d) E admits a generating family (II 4.9) indexed by an element of $\mathcal{U}$.

303 In fact, we shall see that these intrinsic properties *characterize* $\mathcal{U}$-topoi:

Theorem 1.2. (J. Giraud). *Let E be a $\mathcal{U}$-category. The following properties are equivalent:*

i) *E is a $\mathcal{U}$-topos* (1.1).
ii) *E satisfies conditions* a), b), c), *and* d) of 1.1.2.

[4] Cf. [9], or 4.1 and 4.2 below, for the precise relations between the notion of a topos and that of a topological space.

iii) *The $\mathscr{U}$-sheaves on E for the canonical topology are representable, and E has a small generating family (condition* 1.1.2 d)*).*

iv) *There exist a category $C \in \mathscr{U}$ and a fully faithful functor $i : E \to \widehat{C}$ (where $\widehat{C}$ denotes the category of $\mathscr{U}$-presheaves on C) having a left adjoint a which is left exact.*

i′) *There exists a site $C \in \mathscr{U}$ such that projective limits are representable in C and the topology of C is coarser than its canonical topology* (II 2.5)*, and such that E is equivalent to the category $C^{\sim}$ of $\mathscr{U}$-sheaves of sets on C.*

Moreover:

Corollary 1.2.1. *Let E be a $\mathscr{U}$-topos, and let C be a full subcategory of E, endowed with the topology induced* (III 3.1) *by that of E (1.1.1). Consider the functor*

$$E \longrightarrow C^{\sim}$$

which associates to every $X \in \operatorname{ob} E$ the restriction to C of the functor represented by X. This functor is an equivalence of categories if and only if ob C *is a generating family of E.*

The equivalence i) ⇔ iv) follows immediately from II 5.5, and we have already re- **304**
called above that i) ⇒ ii). Since i′) ⇒ i) trivially, it remains to prove ii) ⇒ iii) and iii) ⇒ i′), which will be done in 1.2.4 and 1.2.3 below. The corollary is then obtained by observing that the functor in question factors as $E \to E^{\sim} \to C^{\sim}$; since $E \to E^{\sim}$ is an equivalence by virtue of (iii), the question reduces to determining when $E^{\sim} \to C^{\sim}$ is an equivalence. The conclusion then follows from the "comparison lemma" III 4.1; cf. the proof in 1.2.3 below.

1.2.3. *Proof of* iii) ⇒ i′). Let $\mathfrak{X} = (X_i)_{i \in I}$, $I \in \mathscr{U}$, be a small generating family of E. Since E is a $\mathscr{U}$-category, the set of isomorphism classes of finite diagrams in E whose objects are members X_i is $\mathscr{U}$-small. Consequently, the smallest set $\mathfrak{X}'$ of objects of E containing the finite projective limits of objects of $\mathfrak{X}$ is a countable union of small sets and is therefore small [4]. Setting $\mathfrak{X}^{(n+1)} = (\mathfrak{X}^{(n)})'$, and $\mathfrak{X} = \bigcup_n \mathfrak{X}^{(n)}$, we see that, after enlarging the generating family, we may suppose that $\mathfrak{X}$ is stable under finite projective limits. Let $\mathscr{V}$ be a universe containing $\mathscr{U}$ such that E is $\mathscr{V}$-small. For every object H of E, denote by $I(H)$ the set $\coprod_{i \in I} \operatorname{Hom}(X_i, H)$. Let H be an object of E and let $H' \hookrightarrow H$ be the subsheaf of H, for the canonical topology, which is the "union" of the images of the morphisms $u : X_i \to H$, $(u, i) \in I(H)$ (II 4.1). Since H' is a subsheaf of a $\mathscr{U}$-sheaf, H' is a $\mathscr{U}$-sheaf. It is therefore representable. Since the family $\mathfrak{X}$ is generating and, for every $i \in I$, the map $\operatorname{Hom}(X_i, H') \to \operatorname{Hom}(X_i, H)$ is bijective, the morphism $H' \hookrightarrow H$ is an isomorphism (II 4.9). Consequently (II 5.2), the family $(u : X_i \to H)$, $(u, i) \in I(H)$, is covering for the canonical topology of E. Thus every object of E can be covered, **305**
for the canonical topology of E, by objects of $\mathfrak{X}$. Let C be the full subcategory of E defined by the objects of $\mathfrak{X}$, and let $u : C \hookrightarrow E$ be the inclusion functor. The topology $\mathscr{C}$ induced on C by the canonical topology of E is coarser than the canonical topology of C, and when $\mathscr{U}$ contains an element of infinite cardinality, finite projective limits are representable in C. It follows from III 4.1 that the functor $F \mapsto F \circ u$ is an equivalence from E to the category of $\mathscr{U}$-sheaves on C for the topology $\mathscr{C}$, C.Q.F.D.

1.2.4. *Proof of* ii) ⇒ iii). The proof consists of four steps.

Let $\mathscr{V}$ be a universe containing $\mathscr{U}$, such that E is an element of $\mathscr{V}$. Let $\widetilde{E}$ be the category of $\mathscr{V}$-sheaves on E for the canonical topology, and let $J_E : E \to \widetilde{E}$ be the canonical functor.

[4] Here we reason with classes of objects of E up to isomorphism.

. Let $(g_i : G_i \to H)$, $i \in I \in \mathscr{U}$, be an epimorphic family in $\widetilde{E}$. If the G_i and the fiber products $G_i \times_H G_j$ are representable, then the sheaf H is representable.

Indeed, the direct sum $\coprod_{i\in I} G_i$ is represented in $\widetilde{E}$ (II 4.1) by a representable sheaf G (property b) and II 4.6). The same holds for the direct sum $K = \coprod_{(i,j)\in I\times I} G_i \times_H G_j$. Moreover, the diagram $K \rightrightarrows G \to H$ is exact and K is the fiber square of G over H. (This latter property holds in the category of presheaves, and hence in the category of sheaves (II 4.1).) It follows from c) and II 4.7 that the sheaf H is representable.

. Let X_α, $\alpha \in A \in \mathscr{U}$, be a generating family of E. For every sheaf H, denote by $I(H)$
306 the set $\coprod_{\alpha\in A} \mathrm{Hom}_E(X_\alpha, H)$. The family $(u : X_\alpha \to H, (u,\alpha) \in I(H))$ is epimorphic in $\widetilde{E}$. When H is a $\mathscr{U}$-sheaf, $I(H)$ is an element of $\mathscr{U}$.

Indeed, since every sheaf H is the target of an epimorphic family of morphisms whose sources are representable sheaves (I 3.4 and II 4.1), it is enough to prove the first assertion when the sheaf H is representable. Let G then be the image of the family $(u : X_\alpha \to H, (u,\alpha) \in I(H))$. The morphism $G \to H$ is a monomorphism. We are therefore in the situation of the first step, since $X_\alpha \times_G X_\beta$ is isomorphic to $X_\alpha \times_H X_\beta$, which is representable, and moreover $I(H)$ is an element of $\mathscr{U}$. It follows that G is representable. But since X_α is a generating family, $G \to H$ is an isomorphism. The last assertion is immediate from the definition of $\mathscr{U}$-sheaves.

. Every subsheaf of a representable sheaf is representable.

Indeed, let $G \to H$ be a subsheaf of a representable sheaf. Then G is a $\mathscr{U}$-sheaf. The family $(u : X_\alpha \to G, (u,\alpha) \in I(G))$ is therefore epimorphic in $\widetilde{E}$ and indexed by an element of $\mathscr{U}$. Moreover, the fiber products $X_\alpha \times_G X_\beta$ are isomorphic to the fiber products $X_\alpha \times_H X_\beta$, which are representable. Thus we are in the situation of the first step and G is representable.

. Every $\mathscr{U}$-sheaf is representable.

Indeed, by the first and second steps, it is enough to show that the fiber products $X_\alpha \times_H X_\beta$ are representable. But these fiber products are subobjects of the products $X_\alpha \times X_\beta$. The conclusion therefore follows from the third step. This completes the proof of Theorem 1.2.

307 Remark 1.3. Of course, for a given $\mathscr{U}$-topos E, there is in general no preferred way of representing it, up to equivalence, in the form $C^\sim$, with C a *small* site; or, what amounts essentially to the same thing by virtue of 1.2.1 when one restricts to C whose topology is coarser than the canonical topology, there is no preferred *small* generating family in E. When one ceases to impose on C the condition that C be small, there is, on the other hand (by virtue of 1.2 iii)), an entirely canonical choice of a $\mathscr{U}$-site C such that E is equivalent to $C^\sim$, namely E itself! This is one of the technical reasons why in practice it is inconvenient to work only with small sites: in fact, the most important sites of all, namely $\mathscr{U}$-topoi, are not small! Moreover, the sites generating topoi that arise in many questions in algebraic geometry (and even in topology, cf. 2.5) are not small either; examples include the étale site of a scheme (VII.1).

Proposition 1.4. *Let E be a $\mathscr{U}$-topos, let E' be a category (not necessarily a $\mathscr{U}$-category), and let F be a presheaf on E with values in E'. In order that F be a sheaf with values in E' (II 6.1), it is necessary and sufficient that F take $\mathscr{U}$-inductive limits in E to projective limits in E'.*

Let $\mathscr{V}$ be a universe containing $\mathscr{U}$ and such that E' is a $\mathscr{V}$-category. By composing F with the functors $\mathrm{Hom}(X', -) : E' \to \mathscr{V}\text{-}\mathrm{Ens}$ defined by the objects X' of E', we are

reduced to the case where $E' = \mathscr{V}\text{-}\mathrm{Ens}$, with $\mathscr{V}$ a universe. Suppose that F takes $\mathscr{U}$-inductive limits to projective limits. It then follows at once from the definitions that F is a sheaf, since a covering family $X_i \to X$ in $\mathscr{E}$, by the definition of the canonical topology 308
of $\mathscr{E}$, allows one to regard X as an inductive limit of the diagram $\left(X_i \times_X X_j \begin{array}{l} \nearrow X_i \\ \searrow X_j \end{array} \right)$.
Suppose that F is a sheaf, and let us prove that it takes $\mathscr{U}$-inductive limits to projective limits. If $\mathscr{V} \subset \mathscr{U}$, we may suppose that $\mathscr{V} = \mathscr{U}$, and it suffices to apply criterion 1.2 iii). To treat the general case, a little more work is needed. Let C be a full subcategory of E generated by a generating family indexed by an element of $\mathscr{U}$, and let $u : C \to E$ be the inclusion functor. Endow C with the topology induced by the canonical topology of E (III 3.5). Denote by $C^{\sim}_{\mathscr{U}}$ and $C^{\sim}_{\mathscr{V}}$ the categories of sheaves on C, by $E^{\sim}_{\mathscr{V}}$ the category of V-sheaves on E for the canonical topology, by $J_E : E \to E^{\sim}_{\mathscr{V}}$ the canonical functor, and by $i_{\mathscr{U},\mathscr{V}} : C^{\sim}_{\mathscr{U}} \to C^{\sim}_{\mathscr{V}}$ the inclusion functor. The diagram

(∗)
$$\begin{tikzcd}
C^{\sim}_{\underline{U}} \arrow[r, "i_{\underline{U},\underline{V}}"] & C^{\sim}_{\underline{V}} \\
E \arrow[u] \arrow[r, "J_E"] & E^{\sim}_{\underline{V}} \arrow[u]
\end{tikzcd} ,$$

where the vertical arrows are induced by the functor $F \mapsto F \circ u$, is commutative. It follows from the explicit construction of the associated sheaf functor (II 3) and from II 4.1 that the functor $i_{\mathscr{U},\mathscr{V}}$ commutes with $\mathscr{U}$-inductive limits. Moreover, it follows from III 5.1 and 1.5 that the vertical arrows in (∗) are equivalences of categories. Consequently, $J_E : E \to E^{\sim}_{\mathscr{V}}$ commutes with $\mathscr{U}$-inductive limits. For every object X of E, we have

$$F(X) \simeq \mathrm{Hom}_{E^{\sim}_{\mathscr{V}}}(J_E(X), F).$$

Thus F takes $\mathscr{U}$-inductive limits in E to projective limits. 309

Corollary 1.5. *Let E be a $\mathscr{U}$-topos, let E' be a $\mathscr{U}$-category, and let $f : E \to E'$ be a functor. In order that f admit a right adjoint, it is necessary and sufficient that f commute with $\mathscr{U}$-inductive limits.*

This is an immediate consequence of 1.4 in the case $E' = \mathscr{U}\text{-}\mathrm{Ens}$.

Corollary 1.6. *Let E and E' be two $\mathscr{U}$-topoi, and let $f : E \to E'$ be a functor. The following conditions are equivalent:*

i) *f commutes with $\mathscr{U}$-inductive limits.*
ii) *f admits a right adjoint.*
iii) *f is continuous* (III 1.1).

The equivalence i) ⇔ ii) was proved in 1.5. To prove i) ⇔ iii), apply the definition of continuous functors, choosing a universe $\mathscr{V}$ such that $\mathscr{U} \in \mathscr{V}$ (so E and E' are $\mathscr{V}$-small sites). We must express that for every $\mathscr{V}$-sheaf F on E', the composite $F \circ f$ is a sheaf on E, that is (1.4), that it takes $\mathscr{U}$-inductive limits to projective limits. For this it is sufficient that i) hold, since by 1.4 F itself takes $\mathscr{U}$-inductive limits to projective limits; it is also necessary, as one sees by taking F to be a representable functor.

Corollary 1.7. *With the notation of 1.6, in order that f be the inverse-image functor u^* for a morphism of topoi* (3.1) *$u : E' \to E$, it is necessary and sufficient that f be left exact and take epimorphic families to epimorphic families.*

310 The necessity is trivial by definition (NB every right exact functor takes epimorphisms to epimorphisms). The sufficiency follows from III 1.6, which implies that f is continuous under the stated conditions, and hence commutes with $\mathscr{U}$-inductive limits by 1.6.

Remark 1.8. With the notation of 1.5, one can show that f admits a left adjoint if and only if it commutes with $\mathscr{U}$-projective limits. In other words, a *covariant* functor $E \to (\mathscr{U}\text{-}\mathrm{Ens})$ is representable if and only if it commutes with $\mathscr{U}$-projective limits [4]. We indicate only the principle of the proof, which proceeds in three steps:

a) Standard arguments [5, n° 195, § 3] show that if F commutes with $\varprojlim$, it is prorepresentable by a strict projective system $(T_i)_{i \in I}$, where I is a filtered ordered set, *not necessarily small.* We may suppose that if $i > j$, then $T_i \to T_j$ is not an isomorphism, and under this hypothesis F is representable if and only if I is small (which in fact implies that the projective system is essentially constant).
b) To prove that I is small, knowing that for every object X of E the set $F(X) = \varinjlim_i \mathrm{Hom}(T_i, X)$ is small, it suffices to have a *small cogenerating family* $(X_j)_{j \in J}$ (i.e. one that is generating for the opposite category E°). Now one shows that every $\mathscr{U}$-topos E has a small cogenerating family.
c) To prove this last point, one observes by standard arguments [Toh] that every object X of E admits a monomorphism into an "injective" object; and then that
311 for every generating family (L_α) of E, if each L_α is embedded in this way into an injective object I_α, the family (I_α) is cogenerating.

2. Examples of topoi

2.0. We have brought together in this section a fairly large number of typical examples of topoi, which the reader will already have had occasion to encounter elsewhere, and which are intended to facilitate his access to the "yoga" of topoi. For other examples (drawn from algebraic geometry) of topologies on sites, giving rise to as many topoi, he may consult SGA 3 IV 6, and (for the étale topos) Exposé VII of the present seminar. Since later on we shall hardly refer to the present section except for questions of notation or terminology, we leave it to the reader, as an exercise, to verify the statements with which we have supplemented these examples for his general instruction. All the examples of the present section will be made more precise in 4, where their functorial dependence on the data will be examined, and in the following sections by way of illustration of the general notions concerning topoi.

2.1. Topos associated with a topological space. Let X be a small topological space, and let $\mathrm{Ouv}(X)$ be the category of open subsets of X, endowed with the canonical topology (III 1.9.1). We shall denote by $\mathrm{Top}(X)$ the topos of $\mathscr{U}$-sheaves on $\mathrm{Ouv}(X)$. This topos is equivalent to the category of étale topological spaces over X, by associating with every such space X' over X the sheaf $U \mapsto \Gamma(X'/U) = \mathrm{Hom}_X(U, X')$ on $\mathrm{Ouv}(X)$ [TF][5]
312 By an abuse of language, we shall sometimes be unable to refrain from denoting by the same letter X the topos $\mathrm{Top}(X)$ defined by the topological space X.

It is of course the preceding example that served chiefly as a guide and as an intuitive basis for the development of the theory of topoi. One should note, however, that the topoi obtained from topological spaces are of a very special nature, owing in particular

[4] A more general statement is found in I 8.12.8, I 8.12.9; the outline of the proof following a) through c) corresponds to the proof given in *loc. cit.*

[5] N.D.E.: Here [TF] denotes reference [8].

to the fact that they have been described by sites $C = \mathrm{Ouv}(X)$ in which *all morphisms are monomorphisms* (and hence whose underlying category reduces to a preordered set). It follows in particular that the sheaves represented by the objects of C are subsheaves of the terminal sheaf, and consequently that *the subsheaves of the terminal sheaf form a generating family of the topos under consideration.* This property is not shared by most topoi that arise naturally in algebraic geometry or in algebra, cf. the examples below. It is, up to minor qualifications, characteristic of topoi of the form $\mathrm{Top}(X)$ (cf. 7.1.9 below).

It is easily verified that the map $\mathrm{Ouv}(X) \to \mathrm{Top}(X)$, which associates with every open subset of X the sheaf it represents, is a *bijection* from $\mathrm{Ouv}(X)$ onto the set of subobjects of the terminal object of $\mathrm{Top}(X)$; this bijection is even an isomorphism for the natural order structures, i.e. it induces an isomorphism of the corresponding categories. This suggests that it should be possible to reconstruct the topological space X, up to homeomorphism, from $\mathrm{Top}(X)$ known up to equivalence. We shall see below (4.2) that this is indeed the case, subject to a slight restriction on X.

2.2. Point topos or terminal topos, and empty topos or initial topos. When X is a 313
topological space consisting of a single point, the functor

$$F \longrightarrow F(X) : \mathrm{Top}(X) \longrightarrow \mathscr{U}\text{-}\mathrm{Ens}$$

is an equivalence of categories. This shows in particular that the *category* $\mathscr{U}$-Ens is a $\mathscr{U}$*-topos.* We saw by examples in Exp. II that this $\mathscr{U}$-topos is typical from the point of view of exactness properties, in that the verification of many properties (especially exactness properties) of general topoi reduces to this particular topos. The interpretation given here in terms of the one-point topological space justifies the abuse of language by which a topos equivalent to the category $\mathscr{U}$-Ens is called a *point topos* (although, as a category, it is not at all equivalent to the one-point category!). This is the terminology that corresponds to the correct geometric intuition of the role played by these topoi. By another abuse of language, a point topos is sometimes called a *terminal topos*, cf. 4.3; we shall say "*the* terminal topos" for the topos ($\mathscr{U}$-Ens).

When X is the empty topological space, so that $\mathrm{Ouv}(X)$ is the one-point category, a presheaf F on $\mathrm{Ouv}(X)$ is a sheaf if and only if its value at the unique object ϕ of $\mathrm{Ouv}(X)$ is a singleton set. It follows that $\mathrm{Top}(\varnothing)$ is isomorphic to the category of singleton $\mathscr{U}$-sets, a category equivalent to the one-point category. From this one concludes in particular that the one-point category (as well as any $\mathscr{U}$-category equivalent to it) is a $\mathscr{U}$-topos. By an abuse of language, it is sometimes called the *empty topos* or the *initial topos* (cf. 4.4); 314
one should take care that it is not equivalent to the empty category.

2.3. Topos associated with a space with operators. Let X be a topological space and G a discrete group acting on X by homeomorphisms. In [Toh][6] 5.1 the category of G*-sheaves on* X, or, as we shall also say, sheaves on (X, G), was defined; these are sheaves (of sets) on X *equipped* with actions of G compatible with those of G on X. One sees at once, using Giraud's criterion 1.2 iii), that this category is a topos (NB $\mathscr{U}$ is understood throughout), which we shall simply denote by $\mathrm{Top}(X, G)$. When G is the trivial group, we recover the example of 2.1; when X is the one-point space, we obtain the topos of sets on which G acts on the left (or G*-sets*), also called the *classifying topos of the discrete group* G, and denoted by B_G. It is easily verified that the only subobjects of the terminal object e of the topos B_G are e and the empty sheaf ϕ; in particular, if G is not the trivial group, the subobjects of the terminal object of the classifying topos B_G

[6] N.D.E.: Here [Toh] denotes reference [4].

do not form a generating family of B_G. Thus B_G is then not equivalent to a topos of the type $\mathrm{Top}(X)$ considered in 2.1.

The notion of a G-sheaf was introduced in *loc. cit.* in order to develop the cohomological theory of abelian G-sheaves. Interpreting the latter as the abelian sheaves of the topos (X, G), that theory becomes part of the theory of Exp. V, developed in the setting of general topoi.

315 More generally, one may seek to associate an appropriate topos with a topological space X equipped with a topological group G (not necessarily discrete) of automorphisms, which would give rise to an adequate cohomological theory. The same may be done in the context of differentiable manifolds, real or complex analytic manifolds, or schemes. This is indeed possible, cf. 2.5 below.

2.4. Classifying topos of a group object. Let E be a topos, and let G be a group object of E. Let (E, G) be the category of objects of E on which G acts. It follows at once from Giraud's criterion that this is a topos. It is called the *classifying topos* of the group object G, and is denoted by B_G. When E is the point topos (2.2), i.e. when G is an ordinary group, we recover the classifying topos of 2.3.

The terminology adopted here is justified by the fact that the topos B_G plays a universal role in the classification of "torsors" (or homogeneous principal bundles) under G, or more generally under $G_{E'} = f^*(G)$, where E' is a topos "over E", i.e. endowed with a morphism $f : E' \to E$ (cf. 3.1 below). This role, made explicit in [3 Chap V] or in 5.9 below, shows that B_G plays, in the context of topoi, the same role as the classical classifying spaces of topological groups in the homotopy theory of topological spaces. The latter may be regarded (cf. 2.5) as a weakened version of the former, obtained by retaining from the classifying topos only the "homotopy type" of that topos, in a suitable sense which need not be specified here.

2.5. "Big site" and "big topos" of a topological space. Classifying topos of a topo-
316 **logical group.** [6]

Let $U-\mathrm{Esp}$, or simply (Esp), be the category of topological spaces $\in \mathscr{U}$. It is known that finite projective limits are representable in (Esp). Consider on (Esp) the pretopology (II 1.3) for which $\mathrm{Cov}(X)$ is the set of surjective families of open embeddings $u_i : X_i \to X$. We regard (Esp) as a site by means of the topology generated by the preceding pretopology. For every object X of (Esp), regard the category

$$(\mathrm{Esp})_{/X}$$

of objects of (Esp) over X, i.e. topological spaces over X, as a site, using the topology induced by that of (Esp) via the forgetful functor $(\mathrm{Esp})_{/X} \to (\mathrm{Esp})$ (III 5.2 4)). This site is called the *big site* associated with X. One should note that it is not $\in \mathscr{U}$; nor is it a $\mathscr{U}$-site in the sense of II 3.0.2, so precautions are necessary when applying the usual results to it. To remedy this inconvenience, one may choose a universe $\mathscr{V}$ such that $\mathscr{U} \in \mathscr{V}$, so that $(\mathrm{Esp})_{/X}$ becomes a $\mathscr{V}$-site, and work with the associated $\mathscr{V}$-topos $(\mathrm{Esp})_{/X}$, which may be denoted by $\mathrm{TOP}(X)$ and will be called the *big topos of* X. If one is reluctant to enlarge $\mathscr{U}$, one may choose a cardinal c bounding the cardinalities of X and of all the topological spaces one intends to use in the arguments (most often, $\mathrm{Sup}(\mathrm{card}\, X, \mathrm{card}\, \mathbb{R})$ will suffice!), and replace $(\mathrm{Esp})_{/X}$ by the subcategory $(\mathrm{Esp})'_{/X}$ consisting of those X' over X such that $\mathrm{card}\, X' \leq c$, endowed with the induced topology; one then denotes by
317 $\mathrm{TOP}(X)$ the topos of sheaves on this site. To fix ideas, suppose that the first definition

[6] The introduction of these sites and topoi is due to M. Giraud, who also brought out their advantages over the traditional "small" site.

has been adopted.

The advantage of the big topos of X over the small one is that the site defining it contains $(\mathrm{Esp})_{/X}$ as a full subcategory; since the topology of this site is manifestly coarser than the canonical topology, the canonical functor from $(\mathrm{Esp})_{/X}$ to $\mathrm{TOP}(X)$, associating with every space X' over X the sheaf it represents, is seen to be *fully faithful.* Consequently, *a space X' over X is determined up to X-isomorphism by the sheaf* ($\in \mathrm{Top}(X)$) *it defines*; thus the notion of a sheaf on (the big site of) X may be regarded as a *generalization* of the notion of a topological space over X, by means of which all constructions of sheaf theory acquire a meaning for topological spaces over X.

Thus, when G is a group object of the category $(\mathrm{Esp})_{/X}$ of topological spaces over X, one may associate with it the classifying topos B_G (2.4), hence classifying cohomology groups, classifying homotopy groups, etc. (defined as the corresponding invariants of the $\mathscr{V}$-topos B_G). In particular, when X is a one-point space, G is identified with an ordinary topological group. Subject to the usual local conditions ensuring that the singular cohomology of the Cartesian products G^n agrees with cohomology in the sense of sheaves (for constant coefficients, say), for example if G is locally contractible, one may verify that the cohomology of the classifying topos of G is canonically isomorphic to that of the classifying space of G in the sense of topologists.

The introduction of classifying topoi (via "big sites") has over classifying spaces **318**
the advantage of providing a richer theory, since in particular they provide useful cohomological invariants for coefficients more general than constant or locally constant coefficients. Moreover, the definition considered here adapts in an evident way to the other customary contexts: differentiable manifolds, real or complex analytic manifolds or spaces, and schemes. This point of view makes it possible in particular to connect the study of characteristic classes from the traditional and the "arithmetic" points of view, by regarding the "classical groups" as arising from schemes defined over the ring of integers; cf. [7] for indications in this direction. Likewise, the general results of J. Giraud [3] on the classification of extensions of group objects, developed in the very general and flexible setting of topoi, may, by means of "big topoi", be specialized to results on the classification of extensions of topological groups, or of real or complex Lie groups, which seem to have been scarcely known to topologists except in the case of extensions with abelian kernel [11].

2.6. Topoi of the form $\widehat{C}$. Let C be a small category. Then the category $\widehat{C}$ of U-presheaves on C is plainly a $\mathscr{U}$-topos, since it is of the form $C^{\sim}$, where C is endowed with the chaotic topology. We shall give below some details on the relations between C and $\widehat{C}$. Let us note here only that a topos E equivalent to a topos of the form $\widehat{C}$ is of a rather special nature, because *it has a small generating family (X_i) consisting of connected projective objects*, i.e. objects X such that the functor $Y \mapsto \mathrm{Hom}(X, Y)$ takes **319**
epimorphisms to epimorphisms and coproducts to coproducts: indeed, it suffices to take in $\widehat{C}$ the generating family consisting of the functors represented by the $X \in \mathrm{ob}\, C$. Moreover, note that if in a topos E one has a covering family $X_i \to X$, where the X_i are projective and connected, then every other covering family of X is refined by the former (I 4.3.2, I 4.3.3). Consequently, in a topos E of the form $\widehat{C}$ every object X admits a covering family refining all the others. A topos of the form $\mathrm{Top}(X)$ (2.1), where X is a topological space whose points are closed, has the preceding property only if X is discrete.

When the category C has a single object, C is identified with a monoid G. A presheaf on C is then identified with a set on which G acts *on the right* (since it is a functor

$G^\circ \to (\mathrm{Ens})$), and $\widehat{C}$ is the topos of sets with a monoid of right operators, which may also be denoted by B_{G° in view of 2.3: it is the *topos of sets with monoid of operators G° (the monoid opposite to G)*. When G is a group, using the isomorphism $g \mapsto g^{-1}$ from G onto G°, one recovers the classifying topos B_G of 2.3.

2.7. Classifying topos of a pro-group.

2.7.1. Let $\mathscr{G} = (G_i)_{i \in I}$ be a projective system of groups, with G_i, $I \in \mathscr{U}$. Assume that the projective system is *strict*, i.e. that the transition morphisms $G_j \to G_i$ are surjective. If E is a set, by a (left, say) *action of $\mathscr{G}$ on E* we mean the following structure: a) a family $(E_i)_{i \in I}$ of subsets of E, whose union is E; b) for every $i \in I$, an action of
320 the group G_i on the set E_i; these data are furthermore required to satisfy the following condition: for $j \geq i$, E_i is the subset of E_j consisting of the elements fixed under the kernel of $G_j \to G_i$. We also say that $\mathscr{G}$ *acts on E* (on the left) if a (left) action of $\mathscr{G}$ on E has been given. The sets $\in \mathscr{U}$ endowed with an action of $\mathscr{G}$ form a category in the evident way. By Giraud's criterion, one sees at once that this category is a $\mathscr{U}$-topos. It is denoted by $B_{\mathscr{G}}$ and called the *classifying topos of $\mathscr{G}$*. When I has an initial object $i_\circ$, on putting $G = G_{i_\circ}$ one recovers the classifying topos of 2.3.

2.7.2. Another important example is that in which the groups G_i are finite, so that

$$G = \varprojlim G_i$$

is a compact totally disconnected topological group, or *profinite group.* An action of $\mathscr{G}$ on E then amounts to a continuous action of G on E, or, equivalently, to one for which the stabilizer of every point of E is an open subgroup of G. The classifying topos $B_{\mathscr{G}}$ will also be denoted by B_G, where, of course, G is to be regarded as endowed with its profinite topology.

2.7.3. Using the remarks of 2.6, it is easy to verify that the topos $B_{\mathscr{G}}$ defined by a strict projective system $\mathscr{G} = (G_i)_{i \in I}$ of groups is equivalent to a topos of the form $\widehat{C}$ only if this projective system is essentially constant; in the case of a profinite group, this means that the group is in fact finite.

2.7.4. The following geometric interpretation of the classifying topos B_G of a discrete group G is useful for giving a correct geometric intuition for these topoi. (Cf. also, in the same vein, 4.5, 5.8, 5.9, and 7.2 below.) Let X be a connected, locally connected,
321 and locally simply connected topological space, let x be a point of X, and let G be its fundamental group at x. (NB It is known that, up to isomorphism, every discrete group G may be obtained in this way.) Then the Galois theory of coverings of X gives an equivalence between the category B_G of G-sets and the category of *étale coverings* of X, i.e. spaces X' over X that are locally X-isomorphic to X-spaces of the form $X \times I$, where I is a discrete space. (Compare SGA 1 V 4,5.) The latter can moreover be interpreted as the category of locally constant sheaves on X, i.e. the category of locally constant objects (IX 2.0) of the topos $\mathrm{Top}(X)$.

When G is a profinite group, there is an analogous geometric interpretation of B_G as the category of X-schemes that are coproducts of finite étale coverings of a connected scheme X, equipped with a geometric point x and an isomorphism $G \simeq \pi_1(X, x)$. It is again known that every profinite group may be obtained as the fundamental group of a suitable connected scheme (the spectrum of a field, if desired). Finally, pro-groups (not necessarily profinite or essentially constant) also occur in the classification of coverings of connected and locally connected spaces that are not locally simply connected, or in the classification of étale coverings, not necessarily finite or ind-finite, of connected nonnormal schemes. For the latter case, cf. SGA 3 X 6.

EXERCISE 2.7.5. For a topos E, define the notions of connectedness, local connectedness [6], simple connectedness, and local simple connectedness. Define the notion of a constant and locally constant object of E (cf. IX 2.0). Let $f : E' \to E$ be a morphism of 322
topoi (3.1), with E' simply connected (for example, E' the point topos (2.2)), and E connected and locally connected. Define a strict pro-group $\pi_1(E, f) = (G_i)_{i\in I} = \mathscr{G}$ (called the *fundamental pro-group of E at f*) and an equivalence of categories between $B_{\mathscr{G}}$ and the category of locally constant objects of E.[6] Show that, when E is locally simply connected, $\pi_1(E, f)$ is essentially constant and hence is identified with an ordinary discrete group $\pi_1(E, f)$, called the *fundamental group of E at f*. Show that every strict pro-group $\mathscr{G}$ is isomorphic (as a pro-group) to the fundamental pro-group of a suitable connected and locally connected topos at a suitable f, with E' the point topos (take $E = B_{\mathscr{G}}$, and $f : E' \to E$ defined by the forgetful functor $f^* : E \to E' = (\mathrm{Ens})$). When $\mathscr{G}$ is essentially constant, i.e. isomorphic (as a pro-group) to an ordinary discrete group, prove that one may take E above to be locally simply connected (again take $E = B_G$).

2.8. **Example of a false topos**. Let $\mathscr{G} = (G_i)_{i\in I}$ be a strict pro-group, where I is a filtered ordered set and where $i > j$ implies that $G_i \to G_j$ is not an isomorphism. Suppose that $\mathrm{card}(I) \notin \mathscr{U}$. Consider the category of sets $E \in \mathscr{U}$ on which $\mathscr{G}$ acts on the left (2.7.1). It is a $\mathscr{U}$-category, and one sees as in 2.7.1 that this category satisfies conditions a), b), and c) of 1.1.2. Nevertheless, it is not a $\mathscr{U}$-topos, since it has no generating family that is $\mathscr{U}$-small. Similarly, it is not a $\mathscr{V}$-topos for any universe $\mathscr{V}$.

3. Morphisms of topoi

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DEFINITION 3.1. *Let E and E' be two $\mathscr{U}$-topoi. A morphism*[7] *from E to E', or sometimes (by abuse of language) a continuous map from E to E', is a triple $u = (u_*, u^*, \varphi)$, consisting of functors*

$$u_* : E \longrightarrow E' \quad , \quad u^* : E' \longrightarrow E$$

and an "adjunction" isomorphism φ of bifunctors in $X' \in \mathrm{Ob}\, E'$, $Y \in \mathrm{Ob}\, E$:

$$\varphi : \mathrm{Hom}_E(u^*(X'), Y) \xrightarrow{\sim} \mathrm{Hom}_{E'}(X', u_*(Y)),$$

where the functor u^* is moreover required to be left exact, i.e. to commute with finite projective limits. The functor u_* is called the direct-image functor for the morphism of topoi u, the functor u^* is called the inverse-image functor for the morphism of topoi u, and the isomorphism φ is called the adjunction isomorphism for u.

3.1.1. Unless expressly stated otherwise, for a morphism of topoi $u : E \to E'$, we shall henceforth denote the corresponding direct- and inverse-image functors by u_* and u^* [7]. Note that, since u_* is right adjoint to u^* and u^* is left adjoint to u_* by the adjunction isomorphism φ, each of the two functors u_*, u^* determines the other up to unique isomorphism, by the well-known general facts about adjoint functors [14]. In practice, depending on the circumstances, it may be more convenient to define a morphism of topoi $u : E \to E'$ either by specifying $u^* : E' \to E$, or by specifying $u_* : E \to E'$; in the first case, one need only verify that the given functor u^* has a 324
right adjoint and is left exact. In the second case, one must verify that the given functor u_* has a left adjoint which is left exact. In either case, from the partial data, by *choosing*

[6] cf. 8.7 l).

[6] One may take inspiration from SGA 3 X 6.

[7] N.D.E.: Some authors speak of a *geometric morphism*.

[7] It is sometimes appropriate also to write u^{-1} instead of u^*, cf. 13.2.3.

an adjoint functor and an adjunction isomorphism, one obtains a morphism of topoi $u : E \to E'$, and the latter will be "unique up to unique isomorphism" in terms of the datum u^* resp. u_*, in a sufficiently clear sense which will moreover be made completely explicit below (3.2.1).

3.1.2. If $u : E \to E'$ is a morphism of topoi, it follows from the properties of adjoint functors (I 2.11) that the direct-image functor $u_* : E \to E'$ commutes with projective limits, and the functor $u^* : E' \to E$ commutes with inductive limits [7]. Since the latter is assumed in addition to be left exact, i.e. to commute with finite projective limits, we see in particular that u^* *is exact.* We may therefore say that *it is the inverse-image functor* u^*, in the pair (u_*, u^*), *which has the most remarkable exactness properties.* These properties ensure that, for every type of algebraic structure Σ whose data can be described in terms of arrows between the underlying sets and sets deduced from them by repeated application of operations of finite projective limits and arbitrary inductive limits, and for every "object of $\mathscr{E}'$ endowed with a structure of type Σ" (a notion which makes sense by virtue of the internal exactness properties of the topos $\mathscr{E}'$ (II 4.1)), its image under u^* is endowed with the same structures. Rather than undertake the unrewarding task
325 of giving a precise meaning to this statement and formally justifying it, we advise the reader to spell it out and convince himself of its validity for types of structure such as groups, rings, modules over a ring, comodules over a coring, bialgebras over a ring, and torsors under a group. (In these examples, the first three types of structure are defined exclusively in terms of finite projective limits, while the other notions implicitly involve constructions which also use inductive limits.) Furthermore, the functor u^* "*commutes*" with all the usual functorial operations performed in terms of such structures, more precisely with all operations which can be expressed in terms of $\varinjlim$ and finite $\varprojlim$: constructions of free objects (free groups or modules, for example) generated by an object, tensor products (cf. § 12 below), etc.

As for the direct-image functor $u_* : E \to E'$, since it commutes with projective limits, it consequently "preserves" every algebraic structure on an object (or a family of objects) of E which can be defined exclusively in terms of projective limits (such as the structures of group, ring, or module over a ring among the preceding examples). On the other hand, the functor u_* is not in general right exact, i.e. it does not in general commute with finite inductive limits, and it does not even in general take epimorphisms to epimorphisms (it is, moreover, this failure of exactness of u_* which is the source of its cohomological properties; these will be studied, from the viewpoint of commutative homological algebra, in the following exposé). Consequently it does not, in general,
326 extend to a functor on objects of the type comodule, bialgebra, or torsor under a group, and does not in general commute with operations such as "free module generated by", tensor product of modules, etc.

3.1.3. In practice, when one is given a functor $f : E \to F$ from one $\mathscr{U}$-topos to another, one should make all of its exactness properties explicit, including the possible existence of left or right adjoint functors (cf. the footnote on page 162), in order to understand the "geometric nature" of f, an understanding which will generally be an indispensable guide to a correct geometric intuition of the situation. Thus, if it turns out that f commutes with arbitrary inductive limits and finite projective limits, one should write f in the form

$$f = u^*,$$

[7] Moreover (1.5 and 1.8), for a given functor $u_* : E \to E'$ resp. $u^* : E' \to E$, this functor has a left adjoint (resp. a right adjoint) if and only if it commutes with $\mathscr{U}$-projective limits (resp. $\mathscr{U}$-inductive limits).

where

$$u : F \longrightarrow E$$

is a morphism of topoi, i.e. interpret f as an "inverse-image" functor for a "continuous map" of topoi. When f commutes with arbitrary projective limits, so that it has a left adjoint, and if the latter (which a priori commutes with arbitrary inductive limits) *also* commutes with finite projective limits, one should write f in the form

$$f = v_*,$$

where

$$v : E \longrightarrow F$$

is a morphism of topoi. In certain cases it may happen that f satisfies both of the prop- 327
erties under consideration (cf. 4.10 for an example). In that case one should introduce simultaneously the morphisms of topoi

$$u : F \longrightarrow E \quad , \quad v : E \longrightarrow F,$$

which give rise to a sequence of three adjoint functors (I 5.3):

$$v^* \quad , \quad v_* = u^* = f \quad , \quad u_*.$$

One must then distinguish carefully between the morphisms of topoi u and v, lest one lose the geometric intuition of the situation.

In this connection, note that when between two topoi E, F one has a sequence of three adjoint functors

$$e, f, g \ (e, g : F \rightrightarrows E, f : E \to F),$$

then f commutes with arbitrary inductive and projective limits, and therefore can always be written in the form u^*, where $u : F \to E$ is a morphism of topoi. Thus g can be written $g = u_*$. On the other hand, of course, e can be written in the form v^* (and then f in the form v_*) only if it also commutes with finite projective limits. This will evidently be the case if e is itself the right adjoint of a fourth functor d. Without a condition of this kind, one will often write

$$e = u_!$$

for the left adjoint of an inverse-image functor $f = u^*$ when such a left adjoint exists, this notation being suggested by the example of an open immersion $u : X \to Y$ of topological spaces. Similarly, if e is of the form v^*, i.e. if f is of the form v_*, one sometimes writes $g = v^!$ for the right adjoint of a direct-image functor v_*, when this adjoint exists [7].

3.2. Let E, E' be two $\mathcal{U}$-topoi, and let 328

$$u = (u_*, u^*, \varphi) \quad , \quad v = (v_*, v^*, \Psi) : E \rightrightarrows E'$$

be two morphisms of topoi from E to E'. A *morphism from u to v* is any morphism from u_* to v_* (in the sense of the category $\mathbf{Hom}(E, E')$ of functors from E to E'). Morphisms of morphisms of topoi compose in the evident way, thereby defining a category, which is in fact a *$\mathcal{U}$-category* (I 7.8), denoted

$$\mathbf{Homtop}(E, E'),$$

and called the *category of morphisms* (or *continuous maps*) *from E to E'*. One then has an evident functor

$$u \longmapsto u_* : \mathbf{Homtop}(E, E') \longrightarrow \mathbf{Hom}(E, E'),$$

[7] Cf. below 14.4, in the case of abelian sheaves.

which is fully faithful by the definition of arrows in the first member (but is not injective on objects).

3.2.1. Care must be taken that, if u and v are given as above, the theory of adjoint functors supplies a canonical bijection

$$\mathrm{Hom}(u_*, v_*) \xrightarrow{\sim} \mathrm{Hom}(v^*, u^*);$$

in particular, one obtains a *contravariant functor* on $\mathbf{Homtop}(E, E')$:

$$u \longmapsto u^* : \mathbf{Homtop}(E, E')^\circ \longrightarrow \mathbf{Hom}(E, E').$$

In accordance with geometric intuition, according to which the direct-image functor u_* "goes in the same direction" as the continuous map which gives rise to it, one should therefore define the direction of arrows for morphisms between morphisms of topoi *in terms of the direct-image functors*, and not in terms of the inverse-image functors (although the latter, as we have seen, possess the exactness properties characteristic of the notion of morphism of topoi).

329 3.2.2. Having defined the category $\mathbf{Homtop}(E, E')$ of morphisms from the topos E to the topos E', the notion of an *isomorphism* between two morphisms u, v from E to E' is also defined. In practice there is no need to distinguish essentially between two isomorphic morphisms of topoi, at least when one has a *canonical* isomorphism between the two (just as there is often no need to distinguish between two objects of a category when a canonical isomorphism between them is given). Let us point out in this connection that, most often, when one deals with diagrams of morphisms of topoi and questions of commutativity of such diagrams (a notion which makes sense by (3.3)), one means only commutativity up to ("canonical") isomorphism; by abuse of language, one then treats these diagrams as actually commutative diagrams.

One might try to justify this abuse of language by introducing the set $\mathrm{Homtop}(E, E')/\mathrm{Isom}$ of morphisms from E to E' up to isomorphism, and calling such an isomorphism class a morphism, instead of following Definition 3.2. But this runs into the very serious difficulties which arise whenever one tries to identify two isomorphic objects of a category without having a canonical isomorphism between them. Experience shows that such a viewpoint is impracticable, and that one must retain the "fine" notion 3.1 of a morphism between morphisms of topoi, even if one is sometimes obliged to struggle with compatibilities between canonical isomorphisms [7].

330 3.3.1. Let E, E', E'' be three $\mathscr{U}$-topoi, and consider morphisms of topoi

$$u : E \longrightarrow E' \quad , \quad v : E' \longrightarrow E''.$$

The theory of adjoint functors then gives us an adjunction isomorphism between the composite functors $v_* u_*$ and $u^* v^*$, in terms of the adjunction isomorphisms for the pairs (u_*, u^*) and (v_*, v^*). On the other hand, the functor $u^* v^*$ is left exact, as the composite of two left exact functors. Consequently one obtains a morphism from E to E'', called the *composite of the morphisms u and v*, and denoted vu:

$$vu : E \longrightarrow E''.$$

One verifies at once that composition of morphisms is associative, and that for every $\mathscr{U}$-topos E there is a morphism from E to itself which is a two-sided identity for composition: it is the morphism $(\mathrm{id}_E, \mathrm{id}_E, \varphi)$, where φ is the evident adjunction isomorphism

[7] For examples of such struggles (apparently successful), we refer the reader to Mme M. Hakim's book on relative schemes [9].

of id_E with itself. Let $\mathscr{V}$ be a universe such that $\mathscr{U} \in \mathscr{V}$. One defines a category

$$(\mathscr{V}\text{-}\mathscr{U}\text{-}\mathrm{Top}),$$

whose objects are the $\mathscr{U}$-topoi which belong to $\mathscr{V}$, whose arrows are morphisms between such $\mathscr{U}$-topoi, and whose composition of arrows is the one just described.

3.3.2. In fact, the composition map

$$\mathrm{Homtop}(E, E') \times \mathrm{Homtop}(E', E'') \longrightarrow \mathrm{Homtop}(E, E'')$$

is the map induced on objects by a "functor of composition of morphisms":

$$\mathbf{Homtop}(E, E') \times \mathbf{Homtop}(E', E'') \longrightarrow \mathbf{Homtop}(E, E''),$$

whose effect on arrows is the usual "convolution product" operation for morphisms **331**
between functors (here, direct-image functors). These composition functors satisfy a (strict) associativity property, for four topoi E, E', E'', E''', which refines the associativity of composition of morphisms of topoi. One may also say, in language which is beginning to become familiar [2], [9], that the $\mathscr{U}$-topoi are the objects (or 0-arrows) of a 2-*category*, whose 1-arrows are the morphisms of topoi and whose 2-arrows are morphisms of morphisms of topoi.

It is the fact that the $\mathscr{U}$-topoi (elements of a universe $\mathscr{V}$) form a 2-category, rather than merely an ordinary category like ordinary topological spaces, which from the technical viewpoint constitutes the most important difference between the theory of topoi and that of topological spaces. This fact is the source of certain technical complications already alluded to, but also of facts essentially new in comparison with traditional topology.

3.4. The fact that the $\mathscr{U}$-topoi (elements of a universe $\mathscr{V}$) form a 2-category (3.3.2) makes it possible in particular to define the notion of *equivalence of two $\mathscr{U}$-topoi E, E'*: one says that E and E' are *equivalent* if there exist morphisms of topoi $u : E \to E'$ and $v : E' \to E$ such that the composites uv and vu are isomorphic respectively to the identity morphisms of E and E'; the morphisms u and v are then said to be *quasi-inverse* **332**
equivalences of one another.

One sees at once that for the morphism of topoi $u : E \to E'$ to be an equivalence, it is necessary and sufficient that u_* be an equivalence, or, what amounts to the same thing, that u^* be an equivalence. (Use the fact that a functor $f : E \to E'$ between two topoi which is an equivalence is both of the form u_* and of the form v^*, with u and v morphisms of topoi); and for E and E' to be equivalent in the sense of the preceding paragraph, it is necessary and sufficient that they be equivalent as categories (i.e. as objects of the 2-category $(\mathscr{V}\text{-}\mathrm{cat})$). As it should, this shows that the notions of equivalence introduced do not depend on the choice of the universe $\mathscr{V}$ in 3.3.1.

3.4.1. In practice, there is most often no reason to distinguish essentially between equivalent $\mathscr{U}$-topoi, just as there is often no reason to distinguish essentially between two equivalent categories—provided, however, that one has an explicit equivalence from one to the other, or at least an equivalence defined up to unique isomorphism. Here it is the notion of equivalence of topoi which replaces the traditional notion of homeomorphism between two topological spaces. See the example in 4.2 below for the precise relations between these two notions.

4. Examples of morphisms of topoi

Here we return to the examples of 2, using the notion of a morphism of topoi. The comments of 2.0 apply equally to the present section. The universe $\mathscr{U}$ will generally be understood.

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4.1. The topos $\mathrm{Top}(X)$ for a varying topological space X.

4.1.1. Let

$$f : X \longrightarrow Y$$

be a continuous map of topological spaces. We shall associate with it canonically a morphism of topoi

$$\mathrm{Top}(f) \text{ or } f : \mathrm{Top}(X) \to \mathrm{Top}(Y),$$

with the notation of 2.1. When $\mathrm{Top}(X)$ is defined as $\mathrm{Ouv}(X)^{\sim}$, the most convenient description of $\mathrm{Top}(f)$ is by means of the direct-image functor on sheaves

$$f_* : \mathrm{Top}(X) \longrightarrow \mathrm{Top}(Y),$$

defined by the formula

$$f_*(F) = F \circ f^{-1},$$

where

$$f^{-1} : \mathrm{Ouv}(Y) \longrightarrow \mathrm{Ouv}(X)$$

is the evident functor $U \to f^{-1}(U)$. We have already noted that this functor is continuous and left exact, and therefore indeed defines the functor f_* above, which admits a left adjoint f^* that is left exact (III 1.9.1). Of course, strictly speaking, the morphism of topoi $\mathrm{Top}(f)$ depends on the choice of the left adjoint f^* of f_* and hence is defined only up to canonical isomorphism. In what follows we shall refrain from explicitly pointing out phenomena of this kind.

When one adopts the "étale spaces" point of view in defining $\mathrm{Top}(X)$, it is the inverse-image functor

$$f^* : \mathrm{Top}(Y) \longrightarrow \mathrm{Top}(X)$$

334 that is the most convenient for defining the morphism of topoi $\mathrm{Top}(f)$, by simply setting

$$f^*(Y') = X \times_Y Y'$$

for every étale space Y' over Y; it is clear that the fiber product is indeed an étale space over X, and that the functor f^* so obtained is left exact and commutes with arbitrary $\varinjlim$, and therefore defines a morphism of topoi $\mathrm{Top}(f)$. For the compatibility of the two definitions obtained, we refer to [TF].

For two composable continuous maps

$$X \xrightarrow{f} Y \xrightarrow{g} Z,$$

there is a canonical isomorphism $\mathrm{Top}(gf) \simeq \mathrm{Top}(g)\,\mathrm{Top}(f)$ of morphisms of topoi. These *transitivity isomorphisms*, for three composable continuous maps f, g, h, satisfy a compatibility relation which we shall refrain from writing here, and which is precisely the one considered in SGA 1 VI 7.4 B) (for $\epsilon = (\mathrm{Esp})^\circ$). This may be expressed by saying that, as X varies in the category (Esp),

$$X \longmapsto \mathrm{Top}(X)$$

is a "pseudofunctor"

$$(\mathscr{U}\text{-}\mathrm{esp}) \longrightarrow (\mathscr{V}\text{-}\mathscr{U}\text{-}\mathsf{T}), \tag{4.1.1.1}$$

or, in the terminology of 2-categories, that one has a *non-strict* functor of 2-categories [9]. In practice, we shall most often permit ourselves the abuse of language consisting in identifying $\mathrm{Top}(gf)$ and $\mathrm{Top}(g)\,\mathrm{Top}(f)$, that is, reasoning as if (4.1.1.1) were an actual functor between ordinary categories. We shall permit ourselves analogous abuses of language in the other examples treated below. 335

4.1.2. The preceding considerations extend immediately to topoi associated with topological spaces equipped with operator groups (2.3). If $f = (f^{\mathrm{es}}, f^{\mathrm{gr}})$,

$$f : (X, G) \longrightarrow (Y, H)$$

is a morphism of spaces with operators (where

$$f^{\mathrm{es}} : X \longrightarrow Y \quad , \quad f^{\mathrm{gr}} : G \longrightarrow H$$

are respectively continuous maps and group homomorphisms, compatible in the evident sense), one associates with it a morphism of topoi

$$\mathrm{Top}(f) \text{ or } f : \mathrm{Top}(X, G) \longrightarrow \mathrm{Top}(Y, H),$$

whose definition is left to the reader. When the groups G and H are trivial, one recovers the definition of 4.1.1; when, on the other hand, the spaces X and Y are each reduced to a point, the inverse-image functor f^* is the "restriction of the operator group" functor

$$f^* : B_H \longrightarrow B_G,$$

which associates with every H-set the G-set that it defines by means of $f : G \to H$. We shall encounter this example again in other forms in 4.5 and 4.6.1.

4.1.3. Given a continuous map $f : X \to Y$ of topological spaces, one also associates with it a morphism between the corresponding "large topoi" (2.5)

$$\mathrm{TOP}(f) \text{ or } f : \mathrm{TOP}(X) \longrightarrow \mathrm{TOP}(Y),$$

most conveniently defined by the inverse-image functor 336

$$f^* : \mathrm{TOP}(Y) \longrightarrow \mathrm{TOP}(X),$$

which is simply the *restriction functor*. This morphism $\mathrm{TOP}(f)$ is a special case of the morphism called an "inclusion" for an induced topos, which will be studied in 5. Thus, subject to the same qualifications as in 4.1.1, the topos $\mathrm{TOP}(X)$ may be regarded as a functor of X, as X varies in (Esp).

4.2. Fidelity properties of $X \mapsto \mathrm{Top}(X)$. We propose to specify to what extent a topological space X can be reconstructed in terms of the topos $\mathrm{Top}(X)$. For this purpose it is appropriate, for two spaces X and Y, to describe the category of morphisms from $\mathrm{Top}(X)$ to $\mathrm{Top}(Y)$ (3.2), so as to determine the fidelity properties of the "functor" $X \mapsto \mathrm{Top}(X)$. We shall merely state the results obtained, referring the reader to [9] for details. The reader who wishes to carry out the verification himself may consult Exercise 7.8.

4.2.1. Recall that a topological space X is called *sober* if every irreducible closed subset of X has exactly one generic point. Almost all spaces used in practice are sober; this is true in particular of a Hausdorff space, more generally of a space all of whose points are closed, or of the underlying space of a scheme. With a topological space X one associates (*loc. cit.* or EGA 0_I, new edition) a sober topological space X_{sob} and a continuous map 337

$$\varphi : X \longrightarrow X_{\mathrm{sob}} \tag{4.2.1.1}$$

which is *universal* for continuous maps from X to sober spaces; in other words, one constructs a left adjoint $X \mapsto X_{\mathrm{sob}}$ to the inclusion functor (Espsob) $\to$ (Esp) from the category of sober spaces to that of "arbitrary" topological spaces (the quotation marks serving as a reminder that a universe is involved!). The explicit construction takes as the points of X_{sob} the irreducible closed subsets of X, and as its open subsets the sets of the form U', where U is an open subset of X and $U' \subset X_{\mathrm{sob}}$ denotes the set of irreducible closed subsets of X that meet U. The map (4.2.1.1) is obtained by associating with each $x \in X$ the closure of $\{x\}$. The space X is sober if and only if the preceding map is bijective, and hence a homeomorphism.

One checks that the functor

$$\varphi^{-1} : \mathrm{Ouv}(X_{\mathrm{sob}}) \longrightarrow \mathrm{Ouv}(X)$$

induced by φ is an isomorphism, which implies that the morphism of topoi

$$\mathrm{Top}(\varphi) : \mathrm{Top}(X) \longrightarrow \mathrm{Top}(X_{\mathrm{sob}})$$

defined by φ is also an isomorphism. This explains a priori why X_{sob} must necessarily enter into the question of reconstructing X from $\mathrm{Top}(X)$: since the latter depends only on X_{sob} up to isomorphism, the question can have an affirmative answer only if X is
338 sober. We shall specify below (7.1) how X_{sob} can in fact be reconstructed in terms of $\mathrm{Top}(X)$, by interpreting its points as "points" of the topos $\mathrm{Top}(X)$ (or, equivalently, as fiber functors).

4.2.2. On every topological space one should introduce the order relation $\leq$ for which

$$x \leq y \Longleftrightarrow \overline{\{x\}} \subset \overline{\{y\}} \text{ i.e. } x \in \overline{\{y\}}$$

(one also expresses this by saying that x is a *specialization* of y, or that y is a *generization* of x). For a space of the form X_{sob}, this is simply the inclusion relation among the irreducible closed subsets of X.

With this understood, on the set of maps from a space X to another space Y one should introduce the order relation called "specialization", deduced from that of Y, namely

$$f \leq g \Longleftrightarrow f(x) \leq g(x) \text{ for every } x \in X.$$

With these conventions, one has the following result:

4.2.3. *Let X, Y be two topological spaces, with Y sober, and let f and g be two continuous maps from X to Y. Then there is at most one morphism from* $\mathrm{Top}(f)$ *to* $\mathrm{Top}(g)$, *and such a morphism exists if and only if g is a specialization of f. Finally, every morphism of topoi* $\mathrm{Top}(X) \to \mathrm{Top}(Y)$ *is isomorphic to a morphism of the form* $\mathrm{Top}(f)$, *where $f : X \to Y$ is a continuous map (uniquely determined by the first assertion).*

339 If , for an ordered set I, one defines the associated category $\mathrm{cat}(I)$ by declaring that, for $i \geq j$, there is exactly one arrow from i to j, the preceding result may be summarized by saying that there is a canonical equivalence of categories

$$\mathrm{cat}(\mathrm{Hom}_{(\mathrm{esp})}(X, Y)) \xrightarrow{\approx} \mathbf{Homtop}(\mathrm{Top}(X), \mathrm{Top}(Y))$$

(where the second member is defined in 3.2).

The following is a formal consequence of these results:

Corollary 4.2.4. a) *Let $f : X \to Y$ be a continuous map. Then* $\mathrm{Top}(f) : \mathrm{Top}(X) \to \mathrm{Top}(Y)$ *is an equivalence of topoi if and only if $f_{\mathrm{sob}} : X_{\mathrm{sob}} \to Y_{\mathrm{sob}}$ is a homeomorphism (and hence, when X and Y are sober, if and only if f is a homeomorphism).*

b) *Let X and Y be topological spaces. The topoi* $\mathrm{Top}(X)$ *and* $\mathrm{Top}(Y)$ *are equivalent if and only if X_{sob} and Y_{sob} are homeomorphic (and hence, if X and Y are sober, if and only if X and Y are homeomorphic).*

4.3. **Morphisms into the terminal topos: constant objects of a topos, section functors**. Let P (the initial letter of "point") denote the standard terminal topos, i.e. $P =$ (Ens) (2.2). Let E be any topos. We shall see that *up to unique isomorphism, there exists a unique morphism of topoi*

$$f : E \to P$$

or, more precisely, that *the category* $\mathbf{Homtop}(E, P)$ *is equivalent to the one-point category*: between any two objects of this category there is therefore a unique arrow, and it is an isomorphism. (This justifies to some extent the name "terminal topos".) For this, recall (3.2.1) that $\mathbf{Homtop}(E, P)$ is equivalent to the full subcategory of $\mathbf{Hom}(P, E)^{\circ}$ formed by the functors 340

$$f^* : P = (\mathrm{Ens}) \longrightarrow E$$

that commute with $\varinjlim$ and are left exact. Let e be a singleton set. Every set X can then be written canonically as the "coproduct of X copies of e", from which it follows that the category of functors $g : (\mathrm{Ens}) \to E$ that commute with $\varinjlim$ is equivalent to the category E, by associating with each g the object $g(e)$ of E. In terms of $T = g(e)$, the functor g is reconstructed, up to unique isomorphism, by $g(I) = T \times I$, where one sets $T \times I = \amalg_{i \in I} T_i$, with $T_i = T$ for every $i \in I$. The fact that the functor of I so defined by T does indeed commute with $\varinjlim$ follows from the fact that it is manifestly left adjoint to the functor $X \mapsto \mathrm{Hom}(T, X)$ from E to (Ens). This said, for g to be left exact, it is plainly necessary that $g(e) = T$ be a terminal object of E (since e is a terminal object of (Ens)), and it follows easily from the fact that in E "coproducts are universal" (1.1.2 b)) that this condition is also sufficient. Thus the category of inverse-image functors f^* is equivalent to the category of terminal objects of E, which is itself plainly equivalent to the terminal category.

It follows from the foregoing that the choice of a morphism (4.3.1) is essentially equivalent to the choice of a terminal object of E, say e_E. In terms of this object, one then has canonical isomorphisms of functors

$$f^*(I) \simeq e_E \times I = \text{ coproduct of } I \text{ copies of } e_E \qquad \text{for } I \in \mathrm{ob}(\mathrm{Ens}), \tag{4.3.2}$$

and 341

$$f_*(X) = \mathrm{Hom}(e_E, X) \qquad \text{for } X \in \mathrm{Ob}\, E. \tag{4.3.3}$$

4.3.4. The preceding two functors will play an important role below. For a set I, the object $f^*(I) = e_E \times I$ of (4.3.2) is called the *constant object with value* I in E (or, when E is presented as a category $\tilde{C}$ in terms of a site C, the *constant sheaf with value* I *on* C). It will also often be denoted by I_E, or by I_C when E is defined by the site C. The fact that $I \mapsto I_E$ is the inverse-image functor of a morphism of topoi specifies its exactness properties, which imply in particular that this functor respects all the usual kinds of algebraic structure, transforming a group into a group object of E, and so on (3.1.2). If, for example, one has a group G of E, one says that it is a *constant group* (or, where appropriate, a *constant sheaf of groups*) if it is isomorphic to a group of the form $G_{\circ E}$, where $G_{\circ}$ is an ordinary group. The same terminology is used for every other kind of "algebraic" structure, in the sense specified (more or less) in 3.1.2.

4.3.5. Notice that the functor $I \mapsto I_E$ is not necessarily fully faithful (or even faithful: take E to be the "empty topos" (2.2)); thus a constant object of E does not in general determine, up to unique isomorphism, the set I from which it arises. We say that E is 0-*acyclic*, or *connected and nonempty*, if the functor $I \mapsto I_E$ is fully faithful. By the general properties of adjoint functors, this amounts to saying that the adjunction morphism

$$I \longrightarrow f_*(f^*(I)) = f_*(I_E) = \mathrm{Hom}(e_E, I_E)$$

342 is an isomorphism (so that the functor (4.3.3) recovers the "value" of a constant object of E). It is easily checked that this holds if and only if e_E is not the initial object ϕ_E of E, i.e. if E is not an "empty topos" (which expresses the *faithfulness* of the functor $I \mapsto I_E$[7]), *and* if e_E is a *connected* object of E, i.e. is not the sum of two objects of E which are not "empty" (i.e. which are not initial objects of E).

4.3.6. The functor (4.3.3) is also often called the *sections functor* and is denoted by Γ_E, $\Gamma(E, -)$, or simply Γ:

(4.3.6.1) $$\mathrm{Hom}(e_E, X) = \Gamma_E(X) = \Gamma(E, X) = \Gamma(X).$$

It commutes with arbitrary projective limits, but is not in general right exact; its derived functors (on abelian group objects) will be studied in the next exposé.

4.4. Morphisms from the "empty topos". Let ϕ_{top} be an empty topos, corresponding to a category of sheaves Φ equivalent to the terminal category (2.2). Let E be a topos. The category of functors from E to Φ is clearly equivalent to the one-point category, and every such functor commutes with inductive and projective limits (without any merit in this), and hence may be regarded as an inverse-image functor for a morphism of topoi $\phi_{\mathrm{top}} \to E$. It follows that the *category* $\mathbf{Homtop}(\phi_{top}, E)$ *is equivalent to the one-point cat-*
343 *egory*, and in particular that *there exists, up to unique isomorphism, exactly one morphism of topoi*

(4.4.1) $$\phi_{\mathrm{top}} \longrightarrow E.$$

This justifies to some extent the terminology "initial topos" introduced in 2.2.

One may also determine the morphisms of topoi

(4.4.2) $$E \longrightarrow \phi_{\mathrm{top}}\ ;$$

one checks at once that such a morphism exists if and only if the initial object of E is also a terminal object, i.e. if and only if E itself is an "empty topos," and that in this case the category $\mathbf{Homtop}(E, \phi_{\mathrm{top}})$ is again equivalent to the one-point category. The unique morphism (4.4.2) (up to isomorphism) is then an equivalence of topoi.

4.5. The classifying topos B_G for a varying group G.

4.5.1. Let E be a topos, and let

$$f : G \longrightarrow H$$

be a morphism of group objects in E. It induces a "restriction of the group of operators" functor

$$f^* : B_H \longrightarrow B_G,$$

where the notation is that of 2.4. It is immediate that this functor commutes with inductive and projective limits; a fortiori, it may be interpreted as an inverse-image functor associated with a morphism of topoi

$$B_f \text{ or } f : B_G \longrightarrow B_H.$$

344 The corresponding direct-image functor

$$f_* : B_G \longrightarrow B_H$$

[7] or equivalently the fact that this functor is conservative (I 6.3).

is readily made explicit by the formula

$$f_*(X) = \mathbf{Hom}_G(H_s, X),$$

where X is an object of E with a left action of G, where H_s is H regarded as endowed with the left actions by G induced by f, and where $\mathbf{Hom}_G$ denotes the evident subobject of the object $\mathbf{Hom}$ defined below (10.2); the left action of H on $\mathbf{Hom}_G(H_s, X)$ is induced by the right actions of H on H_s by *right* translations.

Since the inverse-image functor f^* commutes with arbitrary $\varprojlim$ (and not only with finite $\varprojlim$), it is itself the right adjoint of a functor

$$f_! : B_G \longrightarrow B_H,$$

so that there is a sequence of three adjoint functors as in 3.1.3:

$$f_!, f^*, f_*.$$

The functor $f_!$ is readily made explicit by the formula

$$f_!(X) = H \times^G X,$$

where the right-hand side denotes the "contracted product" induced by the actions of G on X (on the left) and on H (on the right, via right translations and f), defined as the quotient of $H \times X$ by the action of G given by

$$g \circ (h, x) = (hg^{-1}, gx).$$

The functor $f_!$, being a left adjoint, clearly commutes with inductive limits, but is not in general left exact (i.e. it cannot in turn be regarded as an inverse-image functor 345
for a morphism of topoi $B_H \to B_G$). In fact, one checks easily that it can be left exact only if $f : G \to H$ is an isomorphism. Likewise, the functor f_*, which commutes with projective limits, is not in general right exact and, a fortiori, does not in general admit a right adjoint. At least when E is the punctual topos, i.e. when G and H are ordinary groups, f_* is right exact only if f is an isomorphism.

Given a second morphism of groups $g : H \to K$, one obtains, as in 4.1.1, transitivity up to canonical isomorphism, so that, subject to the usual reservation, the classifying topos B_G may be regarded as depending *functorially* on the group G. We leave it to the reader to generalize this functorial behavior to the case where both G *and* the topos E vary.

4.5.2. *The topos $B_{\mathscr{G}}$ for a varying pro-group $\mathscr{G}$.* We leave it to the reader to make precise the covariant character of the topos $B_{\mathscr{G}}$ (2.7) with respect to $\mathscr{G}$, by following the account given in 4.5.1. One should note, however, that in the case of a morphism $f : \mathscr{G} \to \mathscr{H}$ of pro-groups which are not essentially constant, the corresponding morphism of topoi $B_{\mathscr{G}} \to B_{\mathscr{H}}$ does not in general permit the definition of a functor $f_!$ (whose right adjoint would be the functor f^* restricting the pro-group of operators). Suppose, to simplify the statement, that $\mathscr{G}$ and $\mathscr{H}$ are defined by *profinite* groups G and H. It is then easily seen that $f_!$ exists if and only if the image of the morphism in question $f : G \to H$ has finite index in H, and in that case $f_!$ is given by the same formula as 346
in 4.5.

4.6. The topos $\widehat{C}$ for a varying category C.

4.6.1. Let

$$f : C \longrightarrow C'$$

be a functor from a category $C \in \mathscr{U}$ to another category C', and hence let

$$f^* : \widehat{C}' \longrightarrow \widehat{C} \quad , \quad f^*(F) = F \circ f .$$

It is immediate that this functor commutes with projective and inductive limits; a fortiori, it may be regarded as the inverse-image functor for a morphism of topoi

$$\widehat{f} \text{ or } f : \widehat{C} \to \widehat{C}'.$$

The corresponding direct-image functor

$$f_* : \widehat{C} \longrightarrow \widehat{C}'$$

is none other than the functor likewise denoted f_* in I 5.1. Moreover (as was to be expected from the fact that f^* also commutes with arbitrary $\varprojlim$), f^* also admits a left adjoint

$$f_! : \widehat{C} \longrightarrow \widehat{C}',$$

(which was likewise denoted $f_!$ in I 5.1). Thus one obtains a sequence of three adjoint functors

$$f_!, f^*, f_*,$$

the first of which is moreover an extension of $f : C \to C'$ to the categories of presheaves (for the usual embeddings of C and C' in the categories $\widehat{C}$ and $\widehat{C}'$). One should note that $f_!$ is not in general left exact, nor is f_* in general right exact; this removes any ambiguity
347 about the direction of variance of the topos $\widehat{C}$ associated with the varying category C: this topos is a *covariant* functor of C, subject to the usual reservation arising from the transitivity isomorphisms (cf. 4.1.1). When the sets of objects of C and C' each consist of a single element, so that C and C' identify with monoids G and G', the corresponding topoi are the classifying topoi B_{G° and $B_{G'^\circ}$, and one recovers their variance with respect to the varying G, already encountered from other viewpoints in 4.1.2 and 4.5 (where the restriction to groups rather than monoids was inessential).

4.6.2. The 2-functorial dependence of the topos $\widehat{C}$ on C can be made more precise by introducing, for two categories $C, C' \in \mathscr{U}$, a canonical functor

$$\mathbf{Hom}(C, C') \longrightarrow \mathbf{Homtop}(\widehat{C}, \widehat{C}'). \tag{4.6.2.1}$$

It remains to define this functor on arrows. For this, note that $f \mapsto f^*$ identifies (up to equivalence of categories) the right-hand side with a full subcategory of $\mathbf{Hom}(\widehat{C}', \widehat{C})^\circ$ (3.2.1). If $f, g : C \to C'$ are two functors, every morphism of functors $u : f \to g$ defines, for $F \in \operatorname{ob} \widehat{C}'$, a morphism of functors $F \circ g \to F \circ f$ (since F is a *contra*variant functor), hence a morphism $g^* \to f^*$ and therefore a morphism $\widehat{f} \to \hat{g}$, as asserted.

When C is the one-point category, $\widehat{C}$ is the punctual topos (2.2), denoted P, and (4.6.2.1) is interpreted as a natural functor

$$C' \longrightarrow \mathbf{Homtop}(P, \widehat{C}') \overset{\text{dfn}}{=} \mathbf{Points}(\widehat{C}') \tag{4.6.2.2}$$

348 from C' to the "*category of points*" of $\widehat{C}'$, which will be studied in 6. This functor is not necessarily an equivalence of categories (7.3.3); a fortiori, (4.6.2.1) is not necessarily an equivalence of categories.

On the other hand, the functor (4.6.2.1) is always *fully faithful*. To see this, note that if $f, g : E \to E'$ are two morphisms of topoi for which $f_!$ and $g_!$ are defined (3.1.3), then the theory of adjoint functors gives canonical isomorphisms

$$\mathrm{Hom}(f_!, g_!) \simeq \mathrm{Hom}(g^*, f^*) \simeq \mathrm{Hom}(f_*, g_*) \overset{\text{dfn}}{=} \mathrm{Hom}(f, g).$$

Applying this to functors of the form $\widehat{f}$ and $\hat{g}$ associated with $f, g : C \to C'$, one obtains the asserted result, taking into account that the natural extension map $\mathrm{Hom}(f, g) \to \mathrm{Hom}(f_!, g_!)$ is bijective (I 7.8).

4.6.3. One may ask when the functor (4.6.2.1) is an equivalence of categories, i.e. when it is essentially surjective. This is a special case of the problem of determining all morphisms of topoi $\widehat{C} \to \widehat{C}'$. More generally, if C is a category in $\mathscr{U}$ and E is a topos, one may seek to determine the morphisms of topoi

$$f : \widehat{C} \longrightarrow E.$$

The category of these morphisms is equivalent to the opposite of the category of functors $f^* : E \to \widehat{C}$ commuting with arbitrary inductive limits and finite projective limits. A functor $E \to \widehat{C}$ may be interpreted as a functor $E \times C^\circ \to (\mathscr{U}\text{-Ens}) = (\mathrm{Ens})$, or equivalently as a functor $F : C^\circ \to \mathbf{Hom}(E, (\mathrm{Ens}))$. The exactness property in question is then expressed by requiring that, for every object X of C, the functor $F(X) : E \to (\mathrm{Ens})$ commute with arbitrary inductive limits and finite projective limits (or, as we shall say in 6, that $F(X)$ is a "*fiber functor*" on E). This amounts to saying that $F(X)$ is the **349**
inverse-image functor for a morphism of topoi $P \to E$, and one therefore obtains finally a canonical equivalence of categories

$$\mathbf{Homtop}(\widehat{C}, E) \xrightarrow{\approx} \mathbf{Hom}(C, \mathbf{Points}(E)), \tag{4.6.3.1}$$

where, for brevity, we write

$$\mathbf{Points}(E) = \mathbf{Homtop}(P, E)$$

for the category of points of the topos E.

When E is of the form $\widehat{C}'$, it follows immediately from the definitions that the composite of (4.6.2.1) and the preceding equivalence (4.6.3.1) is the functor

$$\mathbf{Hom}(C, C') \longrightarrow \mathbf{Hom}(C, \mathbf{Points}(\widehat{C}')) \tag{4.6.3.2}$$

defined by $F \mapsto i \circ F$, where $i : C' \to \mathbf{Points}(\widehat{C}')$ is the canonical embedding (4.6.2.2). It follows at once that *for* (4.6.2.1) *to be an equivalence, it is necessary and sufficient that C be empty or that* (4.6.2.2) *be essentially surjective* (and hence an equivalence of categories). This last condition on C' is satisfied in certain interesting cases, notably when C' is the one-object category defined by a group G (7.2.5).

Remark 4.6.4. [8] The association of a topos $\widehat{C}$ with an arbitrary category C suggests that a category C has invariants of a topological nature (cohomology groups, homotopy groups, etc.), just as a topos does. The cohomology groups of $\widehat{C}$ with coefficients in an abelian group object F (in the general sense studied in V) are precisely the values of the derived functors $\varprojlim^{(n)}$ of the functor $\varprojlim$, already familiar to topologists. J. L. **350**
Verdier and (independently) D. G. Quillen verified that, when one restricts to constant, or more generally locally constant, coefficients, these cohomology groups identify with the cohomology groups of the semi-simplicial set Nerf(C) canonically associated with C

[8]N.D.E.: For a recent perspective on the notion of the homotopy type of a topos, see [15] § 7.1.6.

[5, n° 212, prop. 4.1], and moreover that, up to isomorphism in the "homotopy category" of [2], every semi-simplicial set can be obtained from a category C. With a suitable notion of homotopy type for topoi, which we do not specify here, one may say that the semi-simplicial homotopy types of topologists are precisely the homotopy types of topoi of the special form $\widehat{C}$, and more generally of topoi E in which every object admits a covering that refines every other one (2.6) [8]. In the absence of this condition on E, its homotopy type can at best be expressed by a suitable *projective system* of semi-simplicial sets [1].

4.7. **The topos $C^{\sim}$ for a variable site C (cocontinuous functors).** Let

$$f : C \longrightarrow C'$$

be a cocontinuous functor (III 2.1) between sites in $\mathscr{U}$, i.e. a functor such that the functor $\widehat{f}_*$ from $\widehat{C}$ to $\widehat{C}'$ (4.6.1) maps $C^{\sim}$ into $C'^{\sim}$, that is, induces a functor

$$\widetilde{f}_* : \widetilde{C} \longrightarrow C'^{\sim} \tag{4.7.1}$$

making the following diagram of functors commutative:

$$\begin{tikzcd} C^{\sim} \arrow[r, "\widetilde{f}_*"] \arrow[d, "i_*"'] & C'^{\sim} \arrow[d, "i'_*"] \\ \widehat{C} \arrow[r, "\widehat{f}_*"] & \widehat{C}' \end{tikzcd} \tag{4.7.2}$$

351 where i_* and i'_* are the inclusion functors. We saw (III 2.3) that the functor $\widetilde{f}_*$ then has a left adjoint $\widetilde{f}^*$, and that the latter is left exact. In other words, $\widetilde{f}_*$ is the direct-image functor associated with a morphism of topoi

$$\widetilde{f} \text{ or } f : C^{\sim} \longrightarrow C'^{\sim},$$

the corresponding inverse-image functor being, of course, $\widetilde{f}^*$. Taking left adjoints of the functors involved, the commutative diagram (4.7.2) also gives a diagram commutative up to isomorphism:

$$\begin{tikzcd} C^{\sim} & C'^{\sim} \arrow[l, "\widetilde{f}^*"'] \\ \widehat{C} \arrow[u, "\underline{a}"] & \widehat{C}' \arrow[l, "\widehat{f}^*"'] \arrow[u, "\underline{a}'"'] \end{tikzcd} \tag{4.7.3}$$

where $\underline{a}$ and $\underline{a}'$ are the "associated sheaf" functors. This diagram immediately recovers the formula (III 2.3)

$$\widetilde{f}^* = \underline{a}\widehat{f}^* i'.$$

The transitivity property for the morphisms of topoi $\widehat{f} : \widehat{C} \to \widehat{C}'$ implies the same property for the morphisms of topoi $\widetilde{f} : C^{\sim} \to C'^{\sim}$ associated with cocontinuous functors. Thus one may say that *the topos $C^{\sim}$ varies functorially with C* in a *covariant* manner, when cocontinuous functors are taken as the "morphisms" of sites.

352 In all the cases encountered thus far, the cocontinuous functor f used is also con-

[8] and, more generally still, of topoi that are "locally ∞-connected" in an evident sense which we leave it to the reader to make precise.

tinuous, that is (III 1.1), it "extends" to a functor

$$\widetilde{f}_! : C^\sim \longrightarrow C'^\sim$$

commuting with inductive limits, which is left adjoint to $\widetilde{f}^*$, so that there is a sequence of three adjoint functors

$$\widetilde{f}_!, \widetilde{f}^*, \widetilde{f}_*.$$

One should note that the functor $\widetilde{f}_!$ is not in general left exact, nor is $\widetilde{f}_*$ right exact. This removes any ambiguity concerning the direction of the variance of the topos $C^\sim$, when C varies through functors assumed only to be cocontinuous, or even continuous and cocontinuous.

Remark 4.7.4. Given a morphism of topoi

$$F : E \longrightarrow E',$$

for there to exist a functor $F_!$ left adjoint to F^*, it is necessary and sufficient that one be able to "realize" (up to equivalence) E and E' in the form $C^\sim$ and $C'^\sim$, for two sites $C, C' \in \mathscr{U}$, and that one be able to find a *continuous and cocontinuous* functor $f : C \to C'$ such that F identifies with $\widetilde{f}$. This is clearly sufficient; for necessity, it is enough to take for C and C' small full generating subcategories of E and E', respectively, such that

$$F_!(\operatorname{ob} C) \subset \operatorname{ob} C',$$

endowed with the topologies induced by those of E and E', and to take for f the functor induced by $F_!$ (cf. 1.2.1). Recall (1.8) that the existence of $F_!$ also means that F^* commutes with projective limits, or, equivalently here since this functor is left exact, that it 353
commutes with products.

4.7.5. If one asks which morphisms of topoi $F : E \to E'$ can be realized by a cocontinuous (not necessarily continuous) functor of sites, one likewise finds the following answer: the set of objects X of E such that the functor $X' \mapsto \operatorname{Hom}(X, F^*(X'))$ from E' to (Ens) commutes with projective limits (or, equivalently, with products) must be a *generating family of E*.[9]

4.8. The morphism of topoi $\widetilde{C} \to \widehat{C}$ for a site C. Let C be a small site, with which the two topoi $C^\sim$ and $\widehat{C}$ are therefore associated (the latter does not depend on the topology placed on C). In II 3.4 we defined the "associated sheaf" functor

$$\underline{a} : \widehat{C} \longrightarrow C^\sim,$$

and proved that it is left exact and commutes with inductive limits. It is therefore the inverse-image functor associated with a morphism of topoi

$$p : \widetilde{C} \longrightarrow \widehat{C}, \ p^* = \underline{a},$$

the corresponding direct-image functor being the canonical inclusion

$$i = p_* : C^\sim \longrightarrow \widehat{C}.$$

[9]N.D.E.: This statement does not appear to be correct: the condition expressed is equivalent to the assertion that F^* commutes with arbitrary products, or equivalently that F^* has a left adjoint $F_!$. For every site C, the identity functor $F = id : C \longrightarrow C$ is cocontinuous when the term on the right is endowed with the trivial topology. The functor $F^* : \widehat{C} \longrightarrow \widetilde{C}$ is then the associated sheaf functor, and does not in general commute with arbitrary products.

It is well known that the latter functor is not in general right exact (its derived functors on abelian objects give rise to the cohomology presheaves $\mathscr{X}^n(F)$ of V 2), and that $\underline{a}$ does not in general commute with arbitrary $\varprojlim$. This removes any ambiguity concerning the direction of the "natural" morphism of topoi between $\widehat{C}$ and $C^{\sim}$.

Given a cocontinuous functor

$$f : C \longrightarrow C'$$

354 of sites, one obtains a diagram of morphisms of topoi

$$\begin{tikzcd} C^{\sim} \arrow[r, "\widetilde{f}"] \arrow[d, "p_*"'] & C'^{\sim} \arrow[d, "p'_*"] \\ \widehat{C} \arrow[r, "\widehat{f}"] & \widehat{C}' \end{tikzcd}$$

which is commutative up to canonical isomorphism: this is precisely what the commutativity of the diagram of functors (4.7.2) expresses. Thus one may say that the morphism of topoi $p : C^{\sim} \to \widehat{C}$ is functorial in C, when C is varied through *cocontinuous* functors between sites.

4.9. Effect of a continuous functor of sites. Morphisms of sites.

4.9.1. Let

$$f : C \longrightarrow C'$$

be a continuous functor from C to C', i.e. such that the functor $\widehat{f}^* : \widehat{C}' \to \widehat{C}$ maps $C'^{\sim}$ into $C^{\sim}$, and hence induces a functor

$$f_s : C'^{\sim} \longrightarrow C^{\sim}.$$

We saw (III 1.2) that the latter admits a left adjoint

$$f^s : C^{\sim} \longrightarrow C'^{\sim},$$

which moreover "extends" f in the evident sense. One should note that in general f^s is not left exact (even if f is also cocontinuous), nor does f_s commute with inductive limits, so that the datum of f does not, without further hypotheses, describe a morphism of topoi in either direction between $C^{\sim}$ and $C'^{\sim}$. The case where f is cocontinu-
355 ous, i.e. where f_s commutes with inductive limits and may therefore be regarded as an inverse-image functor for a morphism of topoi $\widetilde{f} : C^{\sim} \to C'^{\sim}$, was examined in 4.7. We shall examine the case where the functor f^s is left exact, and may therefore be regarded as an inverse-image functor for a morphism of topoi in the *opposite direction*:

$$\mathrm{Top}(f) = g : C'^{\sim} \longrightarrow C^{\sim}. \tag{4.9.1.1}$$

Notice that *we have deliberately not denoted this morphism by the letter $\widetilde{f}$ or f, so as to avoid confusion with the situation of 4.7, in accordance with the general recommendations of 3.1.3.* A continuous functor $f : C \to C'$ is sometimes called a *morphism of sites from C' to C* (note: *not from C to C'*) if the functor f^s is left exact; in other words, if there exists a morphism of topoi (4.9.1.1) such that the corresponding inverse-image functor

$$g^* : C^{\sim} \longrightarrow C'^{\sim}$$

extends the functor f, i.e. makes the following diagram of functors commute up to isomorphism:

$$\begin{tikzcd} C \arrow[r, "f"] \arrow[d, "\epsilon"'] & C' \arrow[d, "\epsilon'"] \\ C^{\sim} \arrow[r, "g^*"] & C'^{\sim} \end{tikzcd},$$

where ϵ and ϵ' are the canonical functors of II 4.4.0.

4.9.2. In practice, one recognizes that a continuous functor $f : C \to C'$ is a morphism of sites from C' to C from the fact that finite projective limits are representable in C and that f commutes with them (III 1.3 5)). Subject to the indicated condition 356
on C (which is almost always satisfied in practice), and assuming in addition that the topology of C' is coarser than the canonical topology (also almost always satisfied), the preceding sufficient condition (f left exact) for f to be a morphism of sites from C' to C is also necessary.

4.9.3. In the spirit of the preceding discussion, if C and C' are two $\mathcal{U}$-sites, the category of morphisms of sites $\mathbf{Morsite}(C', C)$ from C to C' should be defined as the full subcategory of the *opposite category* $\mathbf{Hom}(C, C')^\circ$ of the category of functors from C to C', formed by those functors which are morphisms of sites (from C' to C). In this way one obtains a canonical functor (defined up to unique isomorphism)

$$\mathbf{Morsite}(C', C) \longrightarrow \mathbf{Homtop}(C'^{\sim}, C^{\sim}).$$

Proposition 4.9.4. *Let E be a $\mathcal{U}$-topos and C a $\mathcal{U}$-site. The functor $f \mapsto f^*|C = f^* \circ \epsilon_C$, associating with every morphism of topoi $f : E \to C^{\sim}$ the "restriction" to C of the associated inverse-image functor $f^* : C^{\sim} \to E$, induces an equivalence of categories*

$$\mathbf{Homtop}(E, C^{\sim}) \xrightarrow{\approx} \mathbf{Morsite}(E, C)(\hookrightarrow \mathbf{Hom}(C, E)^\circ).$$

When finite $\varprojlim$ are representable in C, the corresponding fully faithful functor

$$\mathbf{Homtop}(E, C^{\sim}) \longrightarrow \mathbf{Hom}(C, E)^\circ$$

has as its essential image the functors $g : C \to E$ which are left exact and continuous, or equivalently those which are left exact and carry covering families to covering families. 357

The last assertion follows from the first by 4.9.2. On the other hand, it follows from III 1.2 iv) and IV 1.2 iii) that the functor $G \to G \circ \epsilon$ induces an equivalence between the category of *continuous* functors from $C^{\sim}$ to E and the category of *continuous* functors from C to E, a quasi-inverse being given by $g \to g^s$ in the notation of *loc. cit.* On the other hand, by definition, g is a morphism of sites if and only if g^s is the inverse-image functor associated with a morphism of topoi $E \to C^{\sim}$, whence the conclusion by 3.2.1; the "or equivalently" follows from III 1.6.

The main point to retain from 4.9.4 is that (when finite $\varprojlim$ are representable in C) "it amounts to the same thing" to give a morphism of topoi $f : E \to C^{\sim}$ or to give a functor $g : C \to E$ which is left exact and carries covering families to covering families.

4.9.5. Using the developments of 4.9.1 and 4.9.3, one sees as usual that, for a varying $\mathcal{U}$-site C, via the notions of a morphism of sites and a morphism between morphisms of sites just made explicit, the topos $C^{\sim}$ depends functorially (or more exactly, 2-functorially) on the site C. Note that, with the terminology just introduced, $C^{\sim}$ depends *covariantly* on the site C.

4.9.6. It is immediate that (unlike what happens for the notion of a cocontinuous functor, cf. 4.7 4) *every* morphism of topoi $F : E \to E'$ can be realized by means of a morphism of sites $f : C \to C'$ (i.e. a continuous functor $f : C' \to C$ such that
358 ...): it suffices to choose small full generating subcategories C and C' of E and E', respectively, endowed with the topologies induced by those of E and E', such that

$$F^*(\operatorname{ob} C') \subset \operatorname{ob} C,$$

and to take for f the functor induced by F^*. This explains why most morphisms of topoi encountered in practice are indeed described by morphisms of sites (rather than by cocontinuous functors as in 4.7).

4.10. Relations between the small and large topoi associated with a topological space X. We retain the notation of 2.5. In particular, $\mathscr{V}$ is a universe such that $\mathscr{U} \in \mathscr{V}$. We depart from the convention of 2.1 by using $\mathrm{Top}(X)$ to denote the topos of $\mathscr{V}$-sheaves (rather than $\mathscr{U}$-sheaves) on X. Thus, we shall work with $\mathscr{V}$-topoi and not $\mathscr{U}$-topoi. (NB it would be possible to retain $\mathscr{U}$-topoi by adopting the appropriate convention for the definition of $\mathrm{TOP}(X)$, so that the latter is a $\mathscr{U}$-topos, cf. 2.5). With this understood, we shall define TWO morphisms of topoi

(4.10.1)
$$\begin{cases} f : \mathrm{Top}(X) \longrightarrow \mathrm{TOP}(X) \\ g : \mathrm{TOP}(X) \longrightarrow \mathrm{Top}(X) \quad , \quad gf \simeq \mathrm{id}_{\mathrm{Top}(X)}\,, \end{cases}$$

giving rise to a sequence of three adjoint functors

(4.10.1.1)
$$g^* = f_! \quad , \quad g_* = f^* \quad , \quad g^! = f_*,$$

359 with the notation of 3.1.3. We shall successively define the first two functors in this sequence, the third then being defined as the right adjoint of the second.

4.10.2. The functor

$$g_* = f^* : \mathrm{TOP}(X) \longrightarrow \mathrm{Top}(X)$$

is defined simply as the *restriction functor* from $(\mathrm{Esp})_{/X}$ to the site $\mathrm{Ouv}(X)$ of open subsets of X; it does indeed take sheaves to sheaves, as follows immediately from the definition. Using underlined letters for sheaves on the large site of X, for such a sheaf $\underline{F}$ we denote by $\underline{F}_X$ its restriction to the site $\mathrm{Ouv}(X)$, thereby obtaining a functor

(4.10.2.1)
$$\mathrm{Restr} : \underline{F} \longmapsto \underline{F}_X : \mathrm{TOP}(X) \longrightarrow \mathrm{Top}(X).$$

It is clear that this functor commutes with $\mathscr{V}$-projective limits, since these are computed pointwise (one could also invoke the existence of the left adjoint constructed in 4.10.4 below). I claim that it also commutes with $\mathscr{V}$-inductive limits. To see this, we shall give a very convenient interpretation of "large" sheaves on X, i.e. of objects of $\mathrm{TOP}(X)$, in terms of ordinary sheaves (NB these are $\mathscr{V}$-sheaves) on topological spaces X' over X.

4.10.3. For a large sheaf $\underline{F}$ on X, and for every space X' over X (understood that $X' \in \mathscr{U}$), one defines in the obvious way, as in 4.10.2, the "small" sheaf $\underline{F}_{X'}$, the restriction of $\underline{F}$ to X'. If

$$u : X'' \longrightarrow X'$$

360 is a morphism in $(\mathrm{Esp})_{/X}$, one then defines in the obvious way a morphism $u_*(\underline{F}_{X''}) \to \underline{F}_{X'}$, or, what amounts to the same thing, a "*transition morphism*"

(4.10.3.1)
$$\varphi_u : u^*(\underline{F}_{X'}) \longrightarrow \underline{F}_{X''}.$$

As u varies, these morphisms satisfy an evident transitivity condition for a composite

$$X''' \xrightarrow{u} X'' \xrightarrow{v} X'$$

of morphisms in $(\mathrm{Esp})_{/X}$, whose explicit formulation is left to the reader. In this way one obtains a natural functor from the category $\mathrm{TOP}(X)$ of large sheaves on X to the category of systems

$$(\underline{F}_{X'}) \quad (X' \in \mathrm{ob}(\mathrm{Esp})_{/X}), (\varphi_u) \quad (u \in Fl(\mathrm{Esp})_{/X}),$$

satisfying the transitivity condition just described and, in addition, such that for every morphism $u : X'' \to X'$ which is an *open immersion* (or, more generally, a local homeomorphism), the transition morphism φ_u is an isomorphism. Since the functors $u^* : \mathrm{Top}(X') \to \mathrm{Top}(X'')$ used to describe the φ_u commute with $\mathscr{V}$-inductive limits, it follows at once that, in the preceding description of the objects of $\mathrm{TOP}(X)$ in terms of "small" sheaves $F_{X'}$, $\mathscr{V}$-inductive limits are computed pointwise, i.e. the functors of the form $\underline{F} \to \underline{F}_{X'}$ commute with $\mathscr{V}$-inductive limits.

In particular, this holds for the functor $\underline{F} \mapsto \underline{F}_X$ considered in 4.10.2. It therefore does have a right adjoint (1.5). It remains to construct a left adjoint (whose existence also follows a priori from 1.8) and to verify that the latter is left exact. This will complete 361
the definition of the morphisms of topoi announced in (4.10.1) in terms of the functors (4.10.1.1).

4.10.4. The functor

$$g^* = f_! : \mathrm{Top}(X) \longrightarrow \mathrm{TOP}(X)$$

is obtained by associating with every small sheaf F on X the system of its inverse images $(F_{X'})$, $X' \in \mathrm{ob}(\mathrm{Esp})_{/X}$, under the structural morphisms $X' \to X$, the transition morphism φ_u for $u : X'' \to X'$ being the transitivity isomorphism for inverse images of sheaves (4.1.1). One sees at once that this gives a functor

$$\mathrm{Prol} : \mathrm{Top}(X) \longrightarrow \mathrm{TOP}(X),$$

which is *fully faithful*, and whose essential image consists of the large sheaves $\mathscr{F}$ on X for which all the transition morphisms φ_u of (4.10.3.1) are isomorphisms. We leave to the reader the task of defining an adjunction isomorphism between this canonical prolongation functor and the restriction functor of 4.10.2, proving that the latter is right adjoint to the former. This is immediate from the description in 4.10.3 of the category $\mathrm{TOP}(X)$.

Remarks 4.10.5. a) The construction in 4.10.4 also shows that $g^* = f_!$ is fully faithful (or, what amounts to the same thing, that $g_* = f^*$ is a localization functor, or again that $g^! = f_*$ is fully faithful). The large sheaves on X belonging to the essential image of the functor $g^* = f_! = \mathrm{Prol}$ deserve the name of large *étale sheaves* on X, since they form a category equivalent to the category of ordinary sheaves on X, or again to that of étale spaces over X. It is moreover 362
easy to see that if X' is a topological space over X, then the large sheaf on X represented by it is étale in the preceding sense if and only if X' is an étale space over X.

b) The full faithfulness of f_* may also be expressed by saying that the adjunction morphism $f^* f_* \to \mathrm{id}_{\mathrm{Top}(X)}$ is an isomorphism, i.e. that $g_* f_* \simeq \mathrm{id}$, i.e. that $gf \simeq \mathrm{id}$. Thus g makes $\mathrm{TOP}(X)$ into a *topos over* $\mathrm{Top}(X)$, admitting a "section" f over $\mathrm{Top}(X)$.

c) The fact that the functor $g^* = f_!$ is fully faithful, exact, and commutes with $\mathscr{V}$-inductive limits partially justifies the very convenient point of view (due to J. Giraud) according to which, in practically all questions in the theory of sheaves on X, it is harmless to replace the usual sheaves, or "small" sheaves, by the associated "large" sheaves. This applies in particular to cohomological questions, since g_* is exact, the functors $R^i g_*$ (V 5) vanish for $i > 0$, and hence (V 5.4) for every *large* sheaf $\mathscr{F}$ on X there are canonical isomorphisms

$$H^i(\mathrm{TOP}(X), \mathscr{F}) \simeq H^i(\mathrm{Top}(X), \mathscr{F}_X) = H^i(X, F).$$

Applying this to a sheaf of the form $\mathrm{Prol}(F)$, where F is a small sheaf on X, we obtain (since $\mathrm{Prol}(F)_X \simeq F$ canonically) a canonical isomorphism

$$H^i(X, F) = H^i(\mathrm{TOP}(X), \mathrm{Prol}(F)).$$

Thus, *the cohomological invariants of X, computed by means of the small or the large topos of X, are essentially identical.* The same result is moreover valid in noncommutative cohomology.

363 Exercise 4.10.6. Let S be a $\mathscr{U}$-site whose topology is coarser than the canonical topology, and let $\underline{M}$ be a subset of $Fl\ S$ satisfying the following conditions:

a) The morphisms of $\mathscr{M}$ are pullbackable (I 10.7), and $\underline{M}$ is stable under base change.
b) $\underline{M}$ contains the identity arrows and is stable under composition.
c) An arrow $u : X \to Y$ such that there exists a covering family $Y_i \to Y$, with the arrows $X \times_Y Y_i \to Y_i$ in $\underline{M}$, itself belongs to $\underline{M}$.
d) For every $X \in \mathrm{ob}\, S$, every covering family of X is refined by a covering family $(f_i : X_i \to X)$, with $f_i \in \underline{M}$.

For every $X \in \mathrm{ob}\, S$, consider the site $S(X)$ (the "small site of X"), whose underlying category is the full subcategory of $S_{/X}$ consisting of the objects whose structural morphism belongs to $\underline{M}$, endowed with the topology induced (III 3.1) by that of S.

1°) For every arrow $u : X \to Y$ of S, show that base change by u from $S(Y)$ to $S(X)$ is a continuous functor, hence gives a functor commuting with inductive limits (III 1.2 iv))

$$S(u)^x : S(Y)^\sim \longrightarrow S(X)^\sim.$$

2°) Define an equivalence between the topos $S^\sim$ and the category of systems

$$(F_X) \quad (X \in \mathrm{ob}\, S) \quad , \quad (\varphi_u) \quad (u \in Fl S)$$

364 consisting of objects $F_X \in \mathrm{ob}\, S(X)^\sim$ and, for every arrow $u : X \to Y$ in S, of a morphism $\varphi_u : S(u)^x(F_Y) \to F_X$, these systems being subject to a transitivity condition for a composite $v \circ u$ of arrows of S, and to the condition that $u \in \underline{M}$ imply that φ_u is an isomorphism.

3°) Define "restriction" and "prolongation" functors

$$\mathrm{Res}_X : (S_{/X})^\sim \longrightarrow S(X)^\sim, \mathrm{Prol}_X : S(X)^\sim \longrightarrow (S_{/X})^\sim$$

Show that Res_X commutes with small inductive and projective limits and that Prol_X is *fully faithful*, its essential image consisting of the sheaves F such that φ_u is an isomorphism for every arrow u of $S_{/X}$.

4°) Define an adjunction morphism making Res_X the right adjoint of Prol_X. Deduce that there exists a morphism of topoi

$$f : S(X)^\sim \to (S_{/X})^\sim$$

making $S(X)^\sim$ a subtopos of $(S_{/X})^\sim$ and such that

$$f_! = \mathrm{Prol}_X$$
$$f^x = \mathrm{Res}_X\,.$$

Show that Prol_X takes abelian sheaves to abelian sheaves.

5°) Show that if $S(X)$ admits fiber products, then Prol_X is exact. Deduce that there then exists a morphism of topoi $g : (S_{/X})^\sim \to S(X)^\sim$ which is a left retraction of f, i.e. such that

$$g^x = \mathrm{Prol}_X$$
$$g_x = \mathrm{Res}_X\,.$$

(For an example in which $S(X)$ does not admit fiber products, take for S the category of schemes endowed with the étale topology, for M the smooth morphisms, and for X a Noetherian scheme of dimension > 0.) **365**

6°) Show that for every abelian sheaf F of $(S_{/X})^\sim$ there is an isomorphism

$$H^q(X, F) \simeq H^q(X, \mathrm{Res}_X F) \qquad \forall q.$$

Show, using hypercoverings for example, that for every abelian sheaf G of $S(X)^\sim$ there is an isomorphism

$$H^q(X, G) \simeq H^q(X, \mathrm{Prol}_X G) \qquad \forall q.$$

7°) Buy a chocolate medal for the editor.

5. Induced topos

5.1. Let E be a topos and X an object of E. Then the category $E_{/X}$ of objects of E over X is a topos, as follows, for example, immediately from Giraud's criterion 1.2 ii). Alternatively, by 1.2.1, one may realize E as a category of sheaves $C^\sim$, where C is a generating subcategory of E that may be chosen so that $X \in \mathrm{ob}\, C$; one then knows **366**
that $E_{/X} = C^\sim_{/X}$ is equivalent to $(C_{/X})^\sim$ (III 5.4), and hence is a topos.
The topos $E_{/X}$ is called the *topos induced* on the object X of E.

5.2. We shall define a canonical morphism of topoi

$$j_X : E_{/X} \longrightarrow E, \tag{5.2.1}$$

which is called the *inclusion morphism* of the induced topos $E_{/X}$ into the ambient topos E, or, better (cf. 5.7), the *localization morphism of E at X*. This morphism corresponds to a sequence of three adjoint functors (cf. 3.1.3)

$$j_{X!} \quad , \quad j_X^* \quad , \quad j_{X*}, \tag{5.2.2}$$

which may be described explicitly as follows:

a) The functor

$$j_{X!} : E_{/X} \longrightarrow E$$

is the functor that *forgets the structural arrow* from $E_{/X}$ to E.

b) The functor

$$j_X^* : E \longrightarrow E_{/X}$$

is defined by

$$j_X^*(Z) = (X \times Z, \mathrm{pr}_1),$$

where $\mathrm{pr}_1 : X \times Z \to X$ is the first projection. It may also be interpreted as the *base-change functor* relative to the morphism

$$X \longrightarrow e_E,$$

where e_E is the terminal object of E. By the definition of the base-change functor, it is immediate that j_X^* is indeed right adjoint to $j_{X!}$. In particular, it com-
367 mutes with projective limits. It also commutes with inductive limits, by virtue of the special exactness properties of topoi (II 4).

c) It follows from this last fact (1.6) that j_X^* has a right adjoint

$$j_{X*} : E_{/X} \longrightarrow E.$$

For an object X' over X, the object $j_{X*}(X')$ is also sometimes denoted by one of the symbols $\prod_{X/e_E}(X'/X)$, $\mathbf{Hom}_{X/e_E}(X, X')$, or $\mathrm{Res}_{X/e_E}(X'/X)$ ("Weil restriction"), one or another of which is doubtless already familiar to the learned reader of our modest work.

5.3. One should note that the functor $j_{X!}$ commutes with fiber products and takes monomorphisms to monomorphisms, but it is not in general left exact for all that. It is left exact only if X is a terminal object of E (since $X = j_{X!}(X, \mathrm{id}_X)$ and (X, id_X) is a terminal object of $E_{/X}$), that is, if and only if j_X is in fact an equivalence of topoi (and hence all the functors in (5.2.2) are equivalences). [9]

Likewise, the functor j_{X*} is not in general right exact and does not necessarily take epimorphisms to epimorphisms.

It follows from these observations that the direction of the morphism of topoi relating $E_{/X}$ to E, for an object X of the topos E, is determined without any possible ambiguity.

5.4. The functor j_X^* is often called the *localization functor* or the *restriction functor.* The latter terminology is justified by identifying the objects of E resp. of $E_{/X}$ with the sheaves on E resp. on $E_{/X}$ (1.2 iii)), and observing that, under this identification, the adjunction formula between $j_{X!}$ (the forgetful functor) and j_X^* may be interpreted as
368 saying that $j_X^*(F)$ *is the restriction to* $E_{/X}$ *of the sheaf* F *on* E (where "restriction" is understood in the generalized sense: composition with the "inclusion" functor $j_{X!} : E_{/X} \to E$). In accordance with notation familiar from other contexts, we shall also often write, interchangeably:

(5.4.1) $$j_X^*(F) = F|X = F_X = \text{ restriction of } F \text{ to } X.$$

Likewise, when E is of the form $C^\sim$, where C is a (small) site, and X is of the form $\epsilon(S)$ with $S \in \mathrm{ob}\, C$, so that one has the equivalence of categories recalled in 5.1

$$E_{/X} = C^\sim_{/\epsilon(S)} \xrightarrow{\approx} (C_{/S})^\sim$$

(where $C_{/S}$ is endowed with the topology induced by that of C), the functor j_X^* is identified simply with the functor that "restricts" a variable sheaf F on C to the category $C_{/S}$.

These considerations also make it possible to foresee the important role that induced topoi and the localization morphisms (5.2.1) will play in all questions in which one is led to argue by "localization on E" (cf. 8), that is, in practically all questions involving sites or topoi.

[9]For a commutation property of $j_{X!}$ under change of topos, cf. XVII 5.1.2.

5.5. Let

$$f : X \longrightarrow Y$$

be an arrow of E, which thus allows us to regard X (or, more correctly, (X, f)) as an object of $E_{/Y}$. It is clear that there is a canonical isomorphism

$$(E_{/Y})_{/X} \xrightarrow{\sim} E_{/X} \tag{5.5.1}$$

(*transitivity of induced topoi*). Applying the construction of 5.1 to $E_{/Y}$ in place of E, one therefore obtains a canonical morphism of topoi

$$\mathrm{loc}(f) \text{ or } f : E_{/X} \longrightarrow E_{/Y}, \tag{5.5.2}$$

also called the *localization morphism (associated with f).* When Y is the terminal object of E, one essentially recovers (5.2.1). The localization morphism for f is associated with a sequence of three adjoint functors 369

$$f_! \quad , \quad f^* \quad , \quad f_*, \tag{5.5.3}$$

which may be interpreted respectively as a forgetful functor, a restriction functor, and a functor denoted, according to preference (with X' as argument), by $\prod_{X/Y}(X'/X)$, $\mathbf{Hom}_{X/Y}(X, X')$, or $\mathrm{Res}_{X/Y}(X'/X)$.

5.6. The transitivity of the forgetful functors implies that, for two composable morphisms

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

of E, there is a canonical isomorphism of morphisms of topoi

$$\mathrm{loc}(gf) \simeq \mathrm{loc}(g)\,\mathrm{loc}(f) : E_{/X} \longrightarrow E_{/Y} \longrightarrow E_{/Z}.$$

Moreover, for three composable morphisms, the preceding transitivity isomorphisms satisfy the usual compatibility relation. Thus, subject to the customary abuse of language, this allows one to regard

$$X \longmapsto E_{/X}$$

as a covariant functor from E to the category ($\mathscr{V}$-$\mathscr{U}$-top) (3.3.1). More precisely, one obtains a 2-functor (not strict in general) from E to the 2-category ($\mathscr{V}$-$\mathscr{U}$-top).

Before continuing with the general theory of induced topoi (5.10), let us give a few instructive examples.

5.7. Let X be a topological space, and hence let $\mathrm{Top}(X)$ be a topos (2.1). Let X' 370
be an object of $\mathrm{Top}(X)$, which it will be convenient to interpret as an étale space over X, $p : X' \to X$. The category $\mathrm{Top}(X)_{/X'}$ is then identified with the category of étale spaces X'' over X, equipped with an X-morphism $X'' \to X'$. It is known that such a morphism makes X'' an étale space over X', and in this way one obtains a canonical equivalence of topoi

$$\mathrm{Top}(X)_{/X'} \xrightarrow{\approx} \mathrm{Top}(X').$$

The localization morphism is therefore identified with a morphism of topoi $\mathrm{Top}(X') \to \mathrm{Top}(X)$, and one sees immediately that the latter is none other than the morphism

$$\mathrm{Top}(p) : \mathrm{Top}(X') \longrightarrow \mathrm{Top}(X)$$

associated with the structural continuous map $p : X' \to X$. Intuitively, the localization morphism is thus simply the translation, into the language of topoi, of the étale map $p : X' \to X$. This both explains the appropriateness of the term “localization morphism” and counsels caution in using the term “inclusion morphism”, which seems chiefly appropriate when p is an open immersion. It will consequently be prudent in 5.1 to reserve the term “inclusion morphism” for the case where the morphism $X \to e_E$ is a monomorphism, i.e. where X is identified with a subobject of the terminal object of E.

5.8. Let E be a topos, G a group in E, and H a subgroup of G, and let

$$X = G/H$$

be the quotient homogeneous space, regarded as an object of E with a left action by G,
371 i.e. as an object of the classifying topos B_G (2.4). We shall determine the induced topos

$$(B_G)_{/X} \quad (X = G/H),$$

by defining an equivalence of topoi

(5.8.1) $$c : B_H \xrightarrow{\approx} (B_G)_{/X},$$

such that the following diagram commutes up to canonical isomorphism:

(5.8.2) $$\begin{tikzcd} B_H \arrow[r, "c"] \arrow[rr, hook, bend right, "B_i"'] & (B_G)_{/X} \arrow[r, "j_X"] & B_G \end{tikzcd}$$

where j_X is the localization morphism in B_G, and where

$$B_i : B_H \longrightarrow B_G$$

is the morphism deduced from the inclusion $i : H \to G$ (4.5).

To define (5.8.1), we shall simply give the descriptions of the inverse-image functor c^* and the direct-image functor c_*, leaving it to the reader to verify that these are indeed quasi-inverse equivalences and give rise to the commutative diagram (5.8.2) of morphisms of topoi. Let e be the subobject of E underlying X, the image of the section of X (over a chosen terminal object e_E of E) deduced from the unit section of G. If X' is an object of B_G over X, then $c^*(X')$ denotes the inverse image of e in X',

$$c^*(X') = X' \times_X e,$$

which is stable under the left action of H on X', obtained by restricting the given action of G on X'. This indeed defines a functor

$$c^* : (B_G)_{/X} \longrightarrow B_H.$$

372 On the other hand, if X' is an object of B_H, then the morphism from X' to the terminal object e_H of B_H (i.e. the terminal object e_E of E with the trivial action of H) gives, upon application of the functor $i_!$ (4.5), a morphism in B_G

$$i_!(X') \longrightarrow i_!(e_H) \simeq X,$$

which allows $i_!(X')$ to be regarded as an object of $(B_G)_{/X}$, and hence gives the desired functor

$$c_* : X' \longmapsto (i_!(X') \longrightarrow X) : B_H \longrightarrow B_G.$$

5.8.3. Thus, by (5.8.2), one sees that if G is a group in a topos and H a subgroup, *the functor "restriction of the operators from G to H" may be interpreted as a localization functor.* In particular, taking H to be the unit subgroup, one sees that the functor that *forgets the actions of G* may be interpreted as a localization functor. More generally, it follows that if (X, G) is an object of E with an action of G (i.e. an object of the classifying topos B_G), then the functor that "forgets the actions of G",

$$E' = (B_G)_{/(X,G)} \longrightarrow E_{/X}$$

may be interpreted as a localization functor with respect to a suitable object $E_{G,(X,G)}$ of E' covering the terminal object. It suffices to take $E_{G,(X,G)} = E_G \times (X, G)$, giving the cartesian diagram (where $E_G = G_S = G$ with the action of G by left translation)

(5.8.3.1)
$$\begin{tikzcd} E_G \arrow[d] & E_{G,(X,G)} = Z \arrow[l] \arrow[d] \\ e_G & (X,G) \arrow[l] \end{tikzcd} ,$$

and to observe that, by transitivity of induced topoi, the induced topos $E' = ((B_G)_{X/G})_{/Z}$ 373
is equivalent to the topos $((B_G)_{/E_G})_{/Z}$; since $(B_G)_{/E_G}$ is equivalent to E by the functor c^* of (5.8.1) (with $H = e$), it suffices to verify that this equivalence takes Z to the object X of E (which is immediate) in order to deduce the desired equivalence of categories

$$E'_{/Z} \xrightarrow{\approx} E_{/X}.$$

We leave it to the reader to verify that the composite of this equivalence with the localization functor

$$E' = (B_G)_{/(X,G)} \longrightarrow E'_{/Z}$$

is the functor that "forgets the actions of G". Thus one sees that, upon passing to induced topoi, the cartesian diagram above gives a diagram of morphisms of topoi, commutative up to canonical isomorphism (5.6):

(5.8.3.2)
$$\begin{tikzcd} (B_G)_{/E_G} \approx E \arrow[d, "j"'] & (B_G)_{/Z} \approx E'_{/Z} \approx E_{/X} \arrow[l, "i"'] \arrow[d, "j'"] \\ B_G & (B_G)_{/(X,G)} = E' \arrow[l, "f"'] \end{tikzcd} ,$$

where the inverse-image functors associated with the vertical arrows j, j' are the functors that "forget the actions of G", and where the inverse-image functor i^* is identified with the localization functor $E \to E_{/X}$. This diagram is, moreover, "2-cartesian" in the 374
sense of 5.11 below.

5.8.4. We leave it to the reader to extend the preceding considerations to the case of a pro-group $\mathscr{G} = (G_i)$, in the particular case where H is defined as the "kernel" of $\mathscr{G} \to G_{i_\circ}$. (In the case of profinite groups, this means restricting attention to *open* subgroups H of G, i.e. subgroups that are closed and of *finite index*.)

Exercise 5.9. Let E be a topos, G a group of E, B_G the classifying topos of G (2.4), and

$$\pi : B_G \longrightarrow E$$

the morphism of topoi deduced from the homomorphism from G to the unit group e of E (taking into account that $B_e \approx E$).

a) Let

$$E_G = G_s$$

be the object of B_G whose underlying E-object is G, with the actions of G defined by left translations. Regard $\pi^*(G)$ as a group of B_G (it is the group G of E with the trivial action of G on it). Show that the morphism of E

$$G_s \times G \longrightarrow G_s$$

defined by the right translations of G on G_s is compatible with the actions of G on G_s and on $G = \pi^*(G)$, and defines a morphism of B_G

$$E_G \times \pi^*(G) \longrightarrow E_G.$$

Show that this morphism equips E_G with a right action of the B_G-group $\pi^*(G)$,
375 and that this action makes E_G a *torsor* under $\pi^*(G)$ (that is, the terminal object of B_G can be covered by objects X_i such that, for every i, the restriction of E_G to X_i is isomorphic to the trivial bundle with operators $(\pi^*(G)_{X_i})_d$).

b) Let

$$q : E' \longrightarrow E$$

be a topos over E. For every topos B over E, define the category $\mathbf{Homtop}_E(E', B)$ of morphisms of topoi from E' to B compatible, up to a specified isomorphism, with the structural morphisms $E' \to E$ and $B \to E$. Taking $B = B_G$, define, by the formula $f \mapsto f^*(E_G)$, an equivalence of categories

$$\mathbf{Homtop}_E(E', B_G) \longrightarrow \mathbf{Tors}(E, q^*(G)),$$

where the right-hand side denotes the category of (right) $q^*(G)$-torsors on E.

c) Let H be a subgroup of G, so that $\pi^*(H)$ is a subgroup of $\pi^*(G)$ and hence acts on E_G by restriction of the operator group, giving a quotient object

$$X = E_G/\pi^*(H),$$

which is simply the object G/H endowed with the usual left action of G. Let e_G be the terminal object of B_G (that is, the terminal object e_E of E with the necessarily trivial action of G), and consider in B_G the diagram of morphisms

$$\begin{tikzcd}
E_G \arrow[dd] \arrow[dr] & \\
& X = E_G/\pi^*(H) \arrow[dl] \\
e_G &
\end{tikzcd}$$

376 which, upon passing to the topoi induced by B_G on these objects, gives a diagram of morphisms of topoi, commutative up to canonical isomorphism (5.6). Show that this diagram is equivalent to the diagram

$$\begin{tikzcd}
E = B_e \arrow[dd] \arrow[dr] & \\
& B_H \arrow[dl] \\
B_G &
\end{tikzcd} ,$$

whose three arrows are the arrows associated by 4.5 with the group morphisms $e \to H \to G$ in E. (Thus, *the classifying topos B_H may intuitively be interpreted as a homogeneous space over B_G, with group $\pi^*(G)$, associated with the universal torsor* (= homogeneous principal bundle) *E_G over B_G*.)

d) Reconsider the example of 5.7 in the case where X' is an *étale covering* (locally trivial, cf. 2.7.4) of X, interpreting it in terms of the considerations of the present exercise.

5.10. Inverse images of induced topoi. Let

$$f : E' \to E$$

be a morphism of $\mathscr{U}$-topoi, let X be an object of E, and let $X' = f^*(X)$ be its inverse image in E'. We shall define a diagram of morphisms of topoi

(5.10.1)
$$\begin{tikzcd} E_{/X} \arrow[d, "j_X"'] & E'_{/X'} \arrow[l, "f_{/X}"'] \arrow[d, "j_{X'}"] \\ E & E' \arrow[l, "f"'] \end{tikzcd} ,$$

where j_X and $j_{X'}$ are the localization morphisms (5.2), and where $f_{/X}$ is defined below; 377
this diagram will be commutative up to canonical isomorphism. We shall define $f_{/X}$ here by means of its inverse image functor

(5.10.2)
$$(f_{/X})^* : E_{/X} \longrightarrow E'_{/X'},$$

which we simply take to be the functor "induced" by f^*, in the evident sense. Since the inclusion functors $j_{X!} : E_{/X} \to E$ and $j_{X'!} : E'_{/X'} \to E'$ commute with $\mathscr{U}$-inductive limits and fiber products, and are conservative, it follows at once that the functor $(f_{/X})^*$ commutes with $\mathscr{U}$-inductive limits and fiber products, just as does the functor f^* which induces it. Moreover, since $(f_{/X})^*$ plainly takes the terminal object X to the terminal object X', it is left exact and hence is associated with a morphism of topoi

(5.10.3)
$$f_{/X} : E'_{/X'} \longrightarrow E_{/X},$$

defined up to unique isomorphism (3.1.1). On the other hand, the diagram deduced from (5.10.1) by passing to the inverse-image functors is commutative (up to canonical isomorphism) by construction and by the fact that f^* commutes with the products $X \times Y$. Thus (5.10.1) itself is commutative, up to a unique isomorphism inducing the canonical isomorphism on the inverse-image functors of the two composites $f \circ j_{X'}$ and $j_X \circ f_{/X}$ (3.2.1).

5.10.4. When $f : E' \to E$ is associated with a continuous map $Y' \to Y$ of topological spaces (4.1), so that X is identified with an étale space over Y, and X' with the fiber product $X \times_Y Y'$, then, identifying the induced topoi $\mathrm{Top}(Y)_{/X}$ and $\mathrm{Top}(Y')_{/X'}$ with $\mathrm{Top}(X)$ and $\mathrm{Top}(X')$ respectively (5.7), one sees that the morphism $f_{/X}$ just constructed is (up to unique isomorphism) the morphism $\mathrm{Top}(X') \to \mathrm{Top}(X)$ deduced from the 378
canonical continuous map $X' \to X$. In the light of this example, one may therefore say that the topos $E'_{/X'}$ plays the role of a fiber product of $E_{/X}$ and E' over E. This intuition is made precise by the following result:

Proposition 5.11. *With the notation of 5.10, the diagram (5.10.1), equipped with the compatibility isomorphism $\alpha : f \circ j_{X'} \to j_X \circ f_{/X}$, is "2-cartesian." More precisely, for every $\mathscr{U}$-topos F, if to each morphism of topoi $g : F \to E'_{/X'}$ one associates the triple*

$(f_{/X} \circ g, j_{X'} \circ g, \alpha * g) = (g_1, g_2, \beta)$, where $g_1 : F \to E_{/X}$ and $g_2 : F \to E'$ are morphisms of topoi and $\beta : j_X \circ g_1 \to f \circ g_2$ is an isomorphism of morphisms of topoi from F to E, then one obtains an equivalence between the category $\mathbf{Homtop}(F, E'_{/X'})$ *and the category* $\mathbf{Homtop}(F, E_{/X}) \times^2_{\mathbf{Homtop}(F,E)} \mathbf{Homtop}(F, E')$ *of all triples (g_1, g_2, β) as above.*

(NB. A 2 has been placed above the cartesian-product sign to recall that this is not an ordinary fiber product of categories, but a "2-fiber product," in the sense made explicit in the statement.)

To prove 5.11, we shall describe morphisms of topoi explicitly by means of their associated inverse-image functors. For this we shall need the auxiliary results given in 5.11.1 through 5.12 below.

5.11.1. Consider, in general, the situation in which two categories E and E' are given, together with an object X of E which is assumed to be pullbackable, that is, such that the functor

$$j : A \longmapsto A \times X : E \longrightarrow E_{/X}$$

379 is defined, and finally a left exact functor

$$(*) \qquad \varphi : E_{/X} \longrightarrow E'.$$

Since $E_{/X}$ has a terminal object, namely $(X, \mathrm{id}_X) = X$, the same is therefore true of E', which has the terminal object

$$e' \simeq \varphi(X).$$

Having chosen the terminal object e' of E', we shall associate with every φ as above a pair

$$(**) \qquad (\Psi, u), \text{ with } \Psi = \varphi \circ j : E \to E', \text{ and } u \in \mathrm{Hom}(e', \Psi(X)).$$

To define u, note that

$$j(X) = X \times X$$

has a canonical section δ_X over the terminal object X of $E_{/X}$, namely the diagonal section, and define

$$u = \varphi(\delta_X) : \varphi(X) = e' \longrightarrow \varphi(X \times X) = \Psi(X).$$

I claim that knowledge of the pair $(**)$ makes it possible to reconstruct the functor φ of $(*)$ up to unique isomorphism. For this, for every object

$$p : X' \longrightarrow X$$

of $E_{/X}$, consider the following cartesian diagram in $E_{/X}$:

$$\begin{tikzcd}
X' \arrow[r] \arrow[d] & j(X') = X' \times X \arrow[d, "j(p)"] \\
X \arrow[r, "\delta_X"] & j(X) = X \times X
\end{tikzcd} ,$$

where the first horizontal arrow is the graph morphism $\Gamma_p = (\mathrm{id}_X, p)$. Applying the left exact functor φ to this diagram, and using the definition of Ψ as $\varphi \circ j$, gives a cartesian diagram

$$\begin{tikzcd} \varphi(X', p) \arrow[r] \arrow[d] & \Psi(X') \arrow[d, "\Psi(p)"] \\ e' \arrow[r] & \Psi(X) \end{tikzcd} \quad ,$$

or, in other words, a canonical isomorphism 380

(5.11.1.1) $$\varphi(X', p) \simeq \Psi(X') \times_{\Psi(X)} e'$$

It is immediately seen to be functorial in the object (X', p) of $E_{/X}$, which describes explicitly how the functor $\varphi : E_{/X} \to E'$ is reconstructed, up to isomorphism, from the pair (Ψ, u) of $(**)$. Notice also that the functor Ψ is left exact. More generally, Ψ commutes with every type of projective limit with which φ commutes, because j commutes with projective limits.

Conversely, suppose now that finite projective limits are representable in E and E', and start with a pair

$$(\Psi, u), \Psi : E \longrightarrow E', u \in \operatorname{Hom}(e', \Psi(X)),$$

where the functor Ψ is left exact. Formula (5.11.1.1) then defines a functor $\varphi : E_{/X} \to E'$. One checks at once that this functor is left exact and, more generally, that it commutes with every type of projective limit representable in E with which Ψ commutes. This follows immediately from the cartesian diagram

$$\begin{tikzcd} \overset{E}{\varprojlim_I} X'_i \arrow[d] & \overset{E_{/X}}{\varprojlim_I} X'_i \arrow[l] \arrow[d] \\ \overset{E}{\varprojlim_I} X & X \arrow[l, "\delta"'] \end{tikzcd}$$

relating the projective limits computed in E and in $E_{/X}$ (where δ denotes the diagonal 381
morphism into the projective limit of the constant functor with value X): apply the functor Ψ to it, obtaining a cartesian diagram in E', and compose the latter with the cartesian square deduced from $\Psi(X) \leftarrow e'$. The resulting composite cartesian square expresses the compatibility of φ with the projective limit under consideration.

The pair (Ψ, u) is reconstructed up to isomorphism from φ by observing that there is a canonical isomorphism

(5.11.1.2) $$\Psi(X') = \varphi(j(X')),$$

deduced from (5.11.1.1) by replacing (X', p) there with $j(X') = (X' \times X, \mathrm{pr}_2)$ and observing that $\Psi(X' \times X) \simeq \Psi(X') \times \Psi(X)$. The preceding isomorphism is manifestly functorial in X', that is, it gives an isomorphism

$$\Psi \simeq \varphi \circ j,$$

and one likewise checks that, under this isomorphism, $u : e' \to \Psi(X)$ is identified with $\varphi(\delta_X)$. We have thus obtained the essential part of the following result.

Lemma 5.11.2. *Let E and E' be two categories in which finite projective limits are representable, let X be an object of E, let* $\mathbf{Sex}(E, E')$ *be the category of left exact functors Ψ from E to E', and let* $\mathbf{Sex}(E_{/X}, E')$ *be the analogous category of left exact functors φ from $E_{/X}$ to E'. There is then an equivalence between the category* $\mathbf{Sex}(E_{/X}, E')$ *and the category* $\mathbf{Sex}(E, E')_{/\Gamma}$ *of pairs (Ψ, u), where $\Psi \in \operatorname{ob}\mathbf{Sex}(E, E')$ and $u : e' \to \Psi(X)$ (with e' the terminal object of E'). This equivalence, as well as a quasi-inverse functor, is described explicitly in 5.11.1. If φ and (Ψ, u) correspond, then φ commutes with a given type*
382 *of projective limit if and only if Ψ does. If the base-change functors in E and E' commute with a given type of inductive limit, then φ commutes with inductive limits of that type if and only if Ψ does.*

We leave it to the reader to make explicit the category structure on $\mathbf{Sex}(E, E')_{/\Gamma}$ corresponding to the pairs (Ψ, u). If the functor $\Gamma : \mathbf{Sex}(E, E') \to (\mathrm{Ens})$ is defined by $\Gamma(\Psi) = \operatorname{Hom}(e', \Psi(X))$, then u in the pair (Ψ, u) may be interpreted as an element of $\Gamma(\Psi)$, that is, as a homomorphism $\Psi \to \Gamma$ in the category of presheaves on $\mathbf{Sex}(E, E')^{\circ}$. This justifies the notation $_{/\Gamma}$ used in the statement of 5.11.1. To finish the proof of 5.11.2, it remains to make explicit in 5.11.1 the functorial nature of the constructions considered when φ, respectively (Ψ, u), varies. This is essentially trivial and is left to the reader. It also remains to verify the assertion concerning commutation with inductive limits; this too follows immediately from the explicit formulas (5.11.1.1) and (5.11.1.2) relating these functors.

In particular, when E and E' are $\mathscr{U}$-topoi, and morphisms of topoi are interpreted by means of their associated inverse-image functors, one obtains:

Proposition 5.12. *Let E and E' be two $\mathscr{U}$-topoi, let X be an object of E, and consider on* $\mathbf{Homtop}(E', E)$ *the contravariant functor*

$$\gamma : f \longmapsto \Gamma(E', f^*(X)) = \operatorname{Hom}(e', f^*(X)) : \mathbf{Homtop}(E', E)^{\circ} \longrightarrow (\mathrm{Ens}).$$

There is then an equivalence of categories

$$\mathbf{Homtop}(E', E_{/X}) \xrightarrow{\approx} \mathbf{Homtop}(E', E)_{/\gamma}, \tag{5.12.1}$$

383 *where the right-hand side is the full subcategory of* $\mathbf{Homtop}(E', E)\hat{}_{/\gamma}$ *consisting of pairs (f, u) with $f \in \operatorname{Homtop}(E', E)$ and $u \in \gamma(f)$, that is, $u \in \Gamma(E', f^*(X))$. This functor is obtained by associating with every morphism of topoi $h : E' \to E_{/X}$ the pair (f, u) consisting of the composite*

$$f = j_X \circ h : E' \to E_{/X} \to E,$$

and the morphism

$$u : e' = h^*(X, \mathrm{id}_X) \longrightarrow f^*(X) = h^*(j_X^*(X)) = h^*(X \times X)$$

deduced from the diagonal morphism $\delta_X : X \to X \times X$ by applying h^* to it.

5.12.2. Using 5.12, the proof of 5.11 becomes almost evident, and essentially reduces to the following (which will serve in place of a formal proof that would be no more instructive): to give a morphism of topoi $g : F \to E'_{/X'}$ "amounts to the same thing" as to give a morphism of topoi $g_2 : F \to E'$ and a section u of $g_2^*(X')$. But $g_2^*(X') = g_2^*(f^*(X)) = (f g_2)^*(X)$, so giving u is also equivalent to giving a lifting of the morphism of topoi $f g_2 : F \to E$ to a morphism of topoi $g_1 : F \to E_{/X}$. This expresses precisely that giving a morphism of topoi $g : F \to E'_{/X'}$ is essentially equivalent to giving a triple (g_1, g_2, α) as in 5.11.

REMARK 5.13. We shall see (§ 15[10]) that for every diagram

$$\begin{tikzcd} & E_1 \arrow[d] \\ & E & E_2 \arrow[l] \end{tikzcd}$$

of topoi, there exists a 2-fiber product in the sense of the 2-category ($\mathscr{V}$-$\mathscr{U}$-Top) of $\mathscr{U}$-topoi belonging to $\mathscr{V}$ (independent, up to equivalence, of the choice of a universe $\mathscr{V}$ such that $\mathscr{U} \in \mathscr{V}$). For natural examples of "fiber products of topoi" other than 5.11, see the final chapter of GIRAUD's book [3]. 384

EXERCISE 5.14. a) Let F and G be two topoi over a topos E. Define the category $\mathbf{Homtop}_E(F, G)$ of pairs (f, α), where $f : F \to G$ is a morphism of topoi and $\alpha : p \to q \circ f$ is an isomorphism of morphisms of topoi from F to E (where $p : F \to E$ and $q : G \to E$ are the structural morphisms of topoi). Define pairing functors $\mathbf{Homtop}_E(F, G) \times \mathbf{Homtop}_E(G, H) \to \mathbf{Homtop}_E(F, H)$, and associativity isomorphisms.

b) Let X and Y be two objects of a topos E. Define a natural functor from the discrete category defined by the set $\mathrm{Hom}(X, Y)$ to the category $\mathbf{Homtop}_E(E_{/X}, E_{/Y})$. Establish compatibility with the composition of morphisms in E and with the pairings considered in a).

c) Prove that the functor considered in b) is an equivalence of categories. In particular, an object X of a topos E can be reconstructed up to unique isomorphism when one knows, "up to E-equivalence," the induced topos $E_{/X}$ as a topos *over* E. [10]

6. Points of a topos and fiber functors

6.0. Let P be the standard $\mathscr{U}$*-punctual topos* (2.2)

$$P = (\mathscr{U}\text{-}\mathrm{Ens}).$$

In view of the rather different intuitions attached on the one hand to the symbol P, representing a geometric object of the nature of a point, and on the other hand to ($\mathscr{U}$-Ens), representing a "large category" to be interpreted as "the category of sheaves on P," one should use one or the other symbol exclusively according to context, although from a strictly logical viewpoint they denote one and the same object. 385

DEFINITION 6.1. *Let E be a $\mathscr{U}$-topos. A point of E is a morphism of topoi $P \to E$ from the punctual topos P (6.0) to E. The category* $\mathbf{Homtop}(P, E)$ *(3.2) is called the category of points of E and is denoted by* $\mathbf{Points}(E)$ *or* $\mathrm{Pt}(E)$*. If $p : P \to E$ is a point of E, with associated inverse-image functor $p^* : E \to (\mathscr{U}\text{-}\mathrm{Ens})$, and if F is an object of E, the set $p^*(F)$ is called the fiber of F at p and is denoted by F_p.*

6.1.1. Of course, when F is a group object (resp. ring object, resp. ...) of E, its fiber F_p at p is a group (resp. a ring, resp. ...) (3.1.2).

[10] N.D.E.: This section does not exist. See, however, [16], Proposition 4.47, for a general existence statement for fiber products of topoi (noting that all geometric morphisms between Grothendieck topoi are bounded).

[10] This result is due to P. DELIGNE. The case where E is the punctual topos had previously been treated by Mme HAKIM.

6.1.2. By 3.2.1, the category of points of E is equivalent, via the functor $p \mapsto p^*$, to the *opposite* of the category of functors

$$\varphi : E \longrightarrow (\mathscr{U}\text{-Ens})$$

which commute with $\mathscr{U}$-inductive limits and are left exact. Thus it amounts essentially to the same thing to give a point of E or to give such a functor φ.

Definition 6.2. *A fiber functor of the $\mathscr{U}$-topos E is a functor $\varphi : E \to (\mathscr{U}\text{-Ens})$ which commutes with $\mathscr{U}$-inductive limits and is left exact. The category of fiber functors*
386 *on E, denoted by* $\mathbf{Fib}(E)$*, is the full subcategory of* $\mathbf{Hom}(E, (\mathscr{U}\text{-Ens}))$ *formed by the fiber functors on E.*

6.2.3. By 6.1.2, the category $\mathbf{Fib}(E)$ of fiber functors on E is therefore equivalent to the *opposite* of the category $\mathbf{Point}(E)$ of points of E, via the functor $p \mapsto p^*$ from the latter to the former:

$$\mathbf{Point}(E) \xrightarrow{\approx} \mathbf{Fib}(E)^\circ. \tag{6.2.1.1}$$

A functor

$$\varphi : E \longrightarrow (\mathscr{U}\text{-Ens})$$

is a fiber functor if and only if it is the fiber functor $F \mapsto F_p$ associated with a point p of E (6.1). In other words, the functor (6.2.1.1) is surjective. Let us also note that a functor φ is a fiber functor if and only if it is left exact and carries covering families to surjective families (1.7). This last condition may also be expressed by saying that φ commutes with $\mathscr{U}$-sums and carries epimorphisms to epimorphisms.

6.3. Let $C \in \mathscr{U}$ be a site. A *point of the site C* is a point of the topos $C^\sim$, and the *category of points of the site C* is the category

$$\mathbf{Point}(C) = \mathbf{Point}(C^\sim).$$

Consider, on the other hand, the functor

$$\varphi \longrightarrow \varphi|C = \varphi \circ \epsilon_C : \mathbf{Fib}(C^\sim) \longrightarrow \mathbf{Hom}(C, (\mathscr{U}\text{-Ens})), \tag{6.3.1}$$

which is fully faithful and whose essential image is the full subcategory $\mathbf{Morsite}((\mathscr{U}\text{-Ens}), C)^\circ$ (4.9.4), which we shall also denote by $\mathbf{Fib}(C)$. A functor

$$\Psi : C \longrightarrow (\mathscr{U}\text{-Ens})$$

387 is called a *fiber functor on the site C* if it belongs to the essential image of (6.3.1), just described. Such a functor is therefore *continuous* (a fortiori (III 1.6), it carries covering families to covering families, i.e. to surjective families), and it is left exact. When finite projective limits are representable in C (a condition satisfied in almost every case encountered in practice), the preceding properties *characterize* the fiber functors on C: they are then precisely the functors $\Psi : C \to (\mathscr{U}\text{-Ens})$ which are left exact and carry covering families to surjective families (4.9.4).

One should retain that the functor $\varphi \to \varphi|C$ is an equivalence of categories

$$\mathbf{Fib}(C^\sim) \xrightarrow{\approx} \mathbf{Fib}(C),$$

so that *it amounts essentially to the same thing to give a fiber functor on the topos $C^\sim$* (or equivalently a point of that topos) *or to give a fiber functor on the site C*. Given a fiber functor Ψ on C, arising up to isomorphism from a fiber functor φ on $C^{\sim}$, the latter is reconstructed from Ψ, up to canonical isomorphism, by the formula

$$\varphi(F) \simeq \varinjlim_{C_{/F}} \Psi(X), \tag{6.3.2}$$

where $C_{/F}$ is the category of objects X of C endowed with a morphism $X \to F$ (in $\widehat{C}$), i.e. an element of $F(X)$. Formula (6.3.2) is an immediate consequence of the fact that φ commutes with inductive limits and of the canonical isomorphism, functorial in F (II 4.1.1):

$$F \simeq \varinjlim_{C_{/F}} \epsilon_C(X).$$

More generally, when G is a presheaf on C, there is a canonical isomorphism, functorial in G: **388**

$$\varphi(\underline{a}G) \simeq \varinjlim_{C_{/G}} \Psi(X), \tag{6.3.3}$$

where $\underline{a}G$ is the sheaf associated with G (II 4.1.1). Indeed, one has the formula

$$\underline{a}G \simeq \varinjlim_{C_{/G}} \epsilon_C(X).$$

6.4.0. We refer to I 6.1 for the notion of a *conservative family of functors*

$$\varphi_i : E \longrightarrow E_i \qquad i \in I.$$

Note that when E and the E_i are $\mathscr{U}$-topoi and the functors φ_i are inverse-image functors f_i^* associated with morphisms of topoi

$$f_i : E_i \longrightarrow E,$$

all the exactness properties assumed in I 6.2, I 6.3, and I 6.4 are satisfied, provided that in I 6.2 (v) D is assumed finite. The following conditions are then equivalent: $(f_i^*)_{i\in I}$ is conservative; it is conservative for *monomorphisms* (I 6.1); it is conservative for *epimorphisms*; and it is *faithful*.

When the family of functors f_i^* is conservative, one also sometimes says that *the family of morphisms of topoi* $(f_i)_{i\in I}$ *is conservative*. In particular: **389**

Definition 6.4.1. *Let E be a $\mathscr{U}$-topos and let $(p_i : P \to E)_{i\in I}$ be a family of points of E. This family of points is called conservative if the family of associated fiber functors $F \mapsto F_{p_i}$ from E to $(\mathscr{U}\text{-}\mathrm{Ens})$ is conservative (6.4.0). The topos E is said to have enough points if it admits a conservative family of points (i.e. if the family of all points of the topos is conservative).*

6.4.2. Let us note that all $\mathscr{U}$-topoi used thus far have enough points. It is nevertheless possible, "on purpose," to construct topoi which do not have enough points (7.2.6 e) and 7.4); such a topos is necessarily not "empty" in the geometric sense of 2.2. A topos having enough points admits a conservative family of points indexed by some $I \in \mathscr{U}$ (6.5). Note, however, that if E is a $\mathscr{U}$-topos, the set of isomorphism classes of points of E is not necessarily $\mathscr{U}$-small (7.3), although it is so in many cases encountered in practice. Finally, for an interesting existence theorem (due to P. Deligne) providing enough points and covering all cases arising in algebraic geometry, see VI 9.

6.4.3. When $(p_i)_{i\in I}$ is a conservative family of points of the topos E, the remarks of I 6.2 apply. In particular, they imply that two arrows $u, v : F \rightrightarrows G$ in E are equal if and only if, for every $i \in I$, the induced arrows on fibers $u_{p_i}, v_{p_i} : F_{p_i} \rightrightarrows G_{p_i}$ are
390 equal; that an arrow $u : F \to G$ in E is a monomorphism (resp. an epimorphism) if and only if, for every $i \in I$, the map $u_{p_i} : F_{p_i} \to G_{p_i}$ is so; that an object F of E is initial (resp. terminal) if and only if, for every $i \in I$, F_{p_i} is empty (resp. consists of one point); and that an object F is a subobject of the terminal object e_E if and only if, for every $i \in I$, F_{p_i} has at most one point.

Proposition 6.5. a) *Let C be a $\mathcal{U}$-site and let $(\varphi_i)_{i\in I}$ be a family of fiber functors on C (6.3). The family $(p_i)_{i\in I}$ of corresponding points of the topos $E = C^\sim$ is conservative if and only if every family $(X_j \to X)_{j\in J}$ in C for which, for every $i \in I$, the corresponding family $(\varphi_i(X_j) \to \varphi_i(X))_{j\in J}$ is surjective is a covering family.*

b) *Let E be a $\mathcal{U}$-topos. If E has enough points (6.4.1), then E admits a $\mathcal{U}$-small conservative family of points.*

Assertion b) is a special case of I 7.7. To prove a), apply I 7.7 to the generating family in E formed by the $\epsilon(X)$, $X \in \operatorname{ob} C$. Recall (II 4.4) that the family $X_j \to X$ is covering if and only if the family $\epsilon(X_j) \to \epsilon(X)$ is epimorphic. If the family of points $(p_i)_{i\in I}$ is conservative, this amounts (6.4.3) to saying that, for every $i \in I$, the family $(X_j)_{p_i} \to (X)_{p_i}$ is surjective, i.e. that the family $\varphi_i(X_j) \to \varphi_i(X)$ is surjective. This proves the "only if" part of a). For the "if" part, apply the criterion I 7.7, which reduces
391 us to verifying that every monomorphism $F \to \epsilon(X)$ for which $F_{p_i} \to \epsilon(X)_{p_i}$ is an isomorphism for every $i \in I$ is an isomorphism, or equivalently an epimorphism. Now one can find a covering family of morphisms $\epsilon(X_j) \to F$, and, after refining further, one may suppose that the morphisms $\epsilon(X_j) \to \epsilon(X)$ are induced by morphisms $X_j \to X$. By hypothesis, for every $i \in I$, the composite morphisms $(X_j)_{p_i} \to F_{p_i} \to (X)_{p_i}$ form a surjective family (as the composite of two surjective families), i.e. the family $\varphi_i(X_j) \to \varphi_i(X)$ is surjective. It follows from the hypothesis that the family $X_j \to X$ is covering, hence that the family $\epsilon(X_j) \to \epsilon(X)$ is epimorphic, and a fortiori that $F \to \epsilon(X)$ is an epimorphism.

Corollary 6.5.1. *Let C be a category. A topology on C making C a $\mathcal{U}$-site with enough fiber functors (i.e. such that the associated topos has enough points) is completely determined by the full subcategory $\boldsymbol{\Phi}$ of* $\mathbf{Hom}(C, \mathrm{Ens})$ *formed by the fiber functors on C.*

Indeed, by 6.5 a), one can then describe the covering families in terms of the family of fiber functors.

In fact, we shall see below (6.8.5) that the subcategory $\boldsymbol{\Phi}$ is contained in the category of pro-representable functors on C, so that $\boldsymbol{\Phi}^\circ \simeq \mathbf{Point}(C^\sim)$ identifies, up to equivalence, with a full subcategory of Pro $-C$.

Compare 6.5.1 with Giraud's theorem II 5.5, which establishes a one-to-one correspondence between the set of topologies on C and a certain set of full subcategories of $\mathbf{Hom}(C^\circ, \mathrm{Ens})$.

392 Exercise 6.5.2. Let C be a category equivalent to a category in $\mathcal{U}$, and let P be a subcategory of Pro(C). For every $p \in P$, denote by $X \mapsto X_p$ the functor pro-represented by p. A family $(X_i \to X)_{i\in I}$ in C is called *P-covering* if, for every $p \in P$, the family $(X_{ip} \to X_p)_{i\in I}$ is surjective. Show that there is a topology T_p on C whose covering families are precisely the P-covering families, and that T_p is the finest topology on C

for which the functors $X \mapsto X_p$, $p \in P$, are fiber functors. Show that the topology T_p makes C a site with *enough* fiber functors. Show that, for every topology T on C, among the topologies T' finer than T which have enough fiber functors there is a coarsest one: it is the topology T_p, where P is the strictly full subcategory of $\mathrm{Pro}(C)$, equivalent to $\mathbf{Point}(C)$, described in (6.8.5) below.

Problem 6.5.3. Characterize the subcategories (strictly full, stable under $\mathscr{U}$-filtered projective limits, ...) P of $\mathrm{Pro}(C)$ which can arise from a topology on C as the essential image of $\mathbf{Point}(C^{\sim})$ (6.8.5). [10]

6.6. Let

$$f : E \longrightarrow E'$$

be a morphism of $\mathscr{U}$-topoi. It induces a canonical functor

$$\mathbf{Point}(f) = (p \longmapsto f \circ p) : \mathbf{Point}(E) \longrightarrow \mathrm{Point}(E').$$

Given two composable morphisms of topoi

$$E \xrightarrow{f} E' \xrightarrow{g} E'',$$

one checks immediately that 393

$$\mathbf{Point}(gf) = \mathbf{Point}(g) \circ \mathbf{Point}(f) : \mathbf{Point}(E) \longrightarrow \mathbf{Point}(E') \longrightarrow \mathbf{Point}(E'').$$

This makes precise the functorial dependence (without any abuse of language for once) of $\mathbf{Point}(E)$ on the topos E: if $\mathscr{V}$ is a universe such that $\mathscr{U} \in \mathscr{V}$, one obtains a (genuine!) functor

$$\mathbf{Point} : (\mathscr{V}\text{-}\mathscr{U}\text{-top}) \longrightarrow (\mathscr{V}\text{-cat})$$

from the category of $\mathscr{U}$-topoi which are elements of $\mathscr{V}$ to the category of categories which are elements of $\mathscr{V}$.

6.7. Let E be a $\mathscr{U}$-topos. Every object F of E then defines a contravariant functor

$$p \longmapsto F_p = p^*(F) : \mathbf{Point}(E)^{\circ} \longrightarrow (\mathscr{U}\text{-Ens}),$$

and hence, for varying F, a canonical functor

$$E \longrightarrow \mathbf{Point}(E)^{\hat{}} = \mathbf{Hom}(\mathbf{Point}(E)^{\circ}, (\mathscr{U}\text{-Ens})), \tag{6.7.1}$$

which, by definition (6.5), is faithful if and only if E has enough points. (The functor (6.7.1) bears a certain formal analogy to a Fourier transform...)

Let X be an object of E, which therefore defines a presheaf $\widehat{X}$ on

$$C = \mathbf{Point}(E).$$

We may therefore define the category

$$C_{/\widehat{X}} = \mathbf{Point}(E)_{/\widehat{X}}$$

of morphisms $p \to \widehat{X}$ in the category $\widehat{C}$ of presheaves on C, i.e. the category of pairs (p, ξ) where p is a point of E and ξ is an element of $\widehat{X}(p) = X_p$, the fiber of X at p. With this understood, I claim that *there is a canonical equivalence of categories*

$$C' = \mathbf{Point}(E_{/X}) \xrightarrow{\approx} C_{/\widehat{X}} \quad , \qquad \text{where} \quad C = \mathbf{Point}(E), \tag{6.7.2}$$

whose composite with the inclusion functor $C_{/\widehat{X}} \to C$ is none other than the functor 394

[10] cf. 9.1.8 a) and e) for necessary conditions, and 9.1.12 b) for necessary and sufficient conditions in the cases of $\mathrm{Top}(X)$, with X a sober locally Noetherian space.

$$\mathbf{Point}(j_X) : C' = \mathbf{Point}(E_{/X}) \longrightarrow C = \mathbf{Point}(E)$$

induced functorially (6.6) by the localization morphism (5.2.1)

$$j_X : E_{/X} \longrightarrow E.$$

This is simply the special case of 5.12 obtained by taking $E' = P = (\text{Ens})$.

Corollary 6.7.3. *Let E be a topos. If E has enough points, then so does every induced topos $E_{/X}$, $X \in \operatorname{ob} E$. Conversely, if $(X_i)_{i\in I}$ is a family of objects of E covering the terminal object and, for every $i \in I$, $E_{/X_i}$ has enough points, then E has enough points.*

For the first assertion, let $f : X' \to X''$ be a morphism in $E_{/X}$ which is not an isomorphism. We prove that there exists a point q of $E_{/X}$ such that f_q is not an isomorphism. Since E has enough points, there is a point p of E such that $f_p : X'_p \to X''_p$ is not an isomorphism. This implies that there exists $q \in X_p$ such that the map induced by f_p on the fibers X'_q and X''_q of X'_p and X''_p over X_p at q is not an isomorphism. But by the equivalence (6.7.2), q may be identified with a point of $E_{/X}$, and the map $X'_q \to X''_q$ just considered is none other than the map on fibers at q induced by $f : X' \to X''$, which proves the assertion.

Conversely, suppose that the X_i cover the terminal object of E and that the $E_{/X_i}$
395 have enough points. We prove the same for E. Let $f : X' \to X''$ be a morphism in E which is not an isomorphism. There is then some $i \in I$ such that $f_i : X'_i \to X''_i$ is not an isomorphism, where the subscript i denotes restriction to X_i. Hence there is a point p_i of $E_{/X_i}$ such that $f_{ip_i} : X'_{ip_i} \to X''_{ip_i}$ is not an isomorphism. If p denotes the image of p_i in E under the localization morphism $E_{/X_i} \to E$, this says that $X'_p \to X''_p$ is not an isomorphism, which proves that E has enough points.

Remark 6.7.4. The preceding argument shows that if (p_α) is a conservative family of points of E, then the family of points of $E_{/X}$ lying over one of the p_α (a family indexed by the disjoint union of the X_{p_α}) is conservative. Likewise, if the X_i cover the terminal object of E and, for every $i \in I$, P_i is a conservative set of points of $E_{/X_i}$, then the set of points of E which are the *images* of the $p \in P_i$ under the localization morphisms $E_{/X_i} \to E$ (with $i \in I$ and p varying) is conservative.

6.8. Let E be a topos and p a point of E. A *neighborhood of the point p of the topos E* is a pair (X, u), where $X \in \operatorname{ob} E$ and $u \in X_p$. By (6.7.2), one may also interpret u as a lift of p to a point of the induced topos $E_{/X}$. Writing φ_p for the fiber functor associated with the point p, one may also interpret a neighborhood of the point p as an object of the category $E_{/\varphi_p}$ (the full subcategory of $\widehat{(E^\circ)}_{/\varphi_p}$ formed by the objects whose source lies in $E \subset \widehat{E}$). This latter category, i.e. the category of pairs (X, u) as above, will naturally be called the *category of neighborhoods* of the point p of the topos E; it is also denoted by
396 $\mathbf{Vois}(p)$. Since φ_p is left exact and finite projective limits are representable in E, finite projective limits are representable in the category $\mathbf{Vois}(p)$, and the canonical functor $\mathbf{Vois}(p) \to E$ commutes with them (I 8.10.10). A fortiori, the opposite category $\mathbf{Vois}(p)^\circ$ is filtered. Moreover, it is clear (I 3.4) that there is a canonical isomorphism, functorial in $F \in \operatorname{ob} E$,

$$\varphi_p(F) = F_p \simeq \varinjlim_{\mathbf{Vois}(p)^\circ} F(X). \tag{6.8.1}$$

In other words, *the functor φ_p, "fiber at the point p," is isomorphic to the filtered inductive limit of the functors represented in the topos E by the neighborhoods of the point p of E.* In fact, *the fiber functor is pro-representable* (I 8), which also means that it is represented by a pro-object $(X_i)_{i\in I}$ of E, where I is a filtered ordered set such that $I \in \mathcal{U}$. Indeed, by *loc. cit.*, this is equivalent to saying that the category $\mathbf{Vois}(p)$ admits a small full cofinal subcategory. Now let C be a small full generating subcategory of E; I claim that the full subcategory $C_{/\varphi_p}$ of $\mathbf{Vois}(p) = E_{/\varphi_p}$, formed by the neighborhoods (X,u) of p such that $X \in \operatorname{ob} C$ (a category that is obviously small), is cofinal in $\mathbf{Vois}(p)$. Indeed, let (X,u) be a neighborhood of p; one must find a neighborhood (X',u') of p lying over (X,u), with $X' \in \operatorname{ob} C$. There is a covering family $X'_i \to X$, with the X'_i in C, from which it follows that the X'_{ip} cover X_p; hence there exist an $X' = X'_{i_\circ}$ and a $u' \in X'_p$ such that (X',u') lies over (X,u), as claimed.

6.8.2. Let C be a $\mathcal{U}$-site and p a point of C, i.e. a point of the topos $C^\sim$. By an abuse 397
of notation, we again denote by φ_p the "restriction" of the fiber functor φ_p to C, i.e. the composite $\varphi_p \circ \epsilon$; for an object X of C, we shall also often write

$$X_p = \varphi_p(X) = \epsilon(X)_p.$$

A *neighborhood of the point* p *in the site* C is a pair (X,u), where $X \in \operatorname{ob} C$ and $u \in X_p = \varphi_p(X)$. These neighborhoods again form a category $C_{/\varphi_p}$, which may be denoted by $\mathbf{Vois}_C(p)$. Let us show that *the category* $\mathbf{Vois}_C(p)^\circ$ *is again filtered, that it admits a* $\mathcal{U}$*-small full cofinal subcategory, and that there is an isomorphism, functorial in the sheaf* F *on* C*:*

$$F_p \simeq \varinjlim_{\mathbf{Vois}_C(p)^\circ} F(X). \tag{6.8.3}$$

First note, by repeating the argument of (6.8.1), that if C' is a full subcategory of C that is topologically generating (II 3.0.1), then $\mathbf{Vois}_{C'}(p)^\circ$ is cofinal in $\mathbf{Vois}_C(p)^\circ$. Taking $C' \in \mathcal{U}$ to be small, we are thus reduced, in proving the assertion, to the case where $C \in \mathcal{U}$. In this case $\widehat{C}$ is a $\mathcal{U}$-topos, and there is an associated-sheaf functor $\underline{a} : \widehat{C} \to C^\sim$, which allows us to construct on $\widehat{C}$ the composite fiber functor $\varphi_p\underline{a}$. Applying (6.8.1) to the topos $\widehat{C}$ and to its full generating subcategory C, one finds that the category $C^\circ_{/\varphi_p\underline{a}}$, which is manifestly isomorphic to $\mathbf{Vois}_C(p)^\circ$, is filtered, and that there is an isomorphism functorial in the presheaf P on C:

$$\varphi_p\underline{a}(P) \simeq \varinjlim_X P(X).$$

If F is a sheaf on C, then $F = \underline{a}F$, and applying the preceding isomorphism to F 398
regarded as a presheaf gives (6.8.3). At the same time, this argument establishes the more general formula

$$(\underline{a}P)_p \simeq \varinjlim_{\mathbf{Vois}_C(p)^\circ} P(X), \tag{6.8.4}$$

an isomorphism functorial in an arbitrary presheaf P on C, at least when C is small. The general case follows by introducing a universe $\mathcal{V}$ such that $C \in \mathcal{V}$, by regarding φ_p on C as a fiber functor with values in ($\mathcal{V}$-Ens), and using the compatibility of the formation of the associated sheaf with enlargement of universes (II 3.6).

6.8.5. To every point p of the topos $C^\sim$ we have associated a pro-object of the $\mathcal{U}$-site C, defined by the canonical functor $\mathbf{Vois}_C(p) \to C$, taking into account the fact that the category of neighborhoods of p in C is cofiltered and admits a small full cofinal subcategory. One immediately verifies that, as p varies, this gives a functor

$$\mathbf{Point}(C^\sim) \longrightarrow \operatorname{Pro}(C). \tag{6.8.5.1}$$

This functor is *fully faithful.* Via the associated fiber functors, it may be interpreted as a functor $\mathrm{Fib}(C) \to \mathrm{Fib}(C^\sim) \to \mathrm{Pro}(C)^\circ$. The first functor is an equivalence by 4.9.4 (recalled for fiber functors in 6.3), and the second is fully faithful by the general formalism (I 8) of pro-objects.

6.8.6. Consider the special case where C is $\mathscr{U}$-small and endowed with the chaotic topology, so that $C^\sim = \widehat{C}$. I claim that in this case the preceding functor is even an equivalence of categories

$$\mathbf{Point}(\widehat{C}) \xrightarrow{\approx} \mathrm{Pro}(C). \tag{6.8.6.1}$$

399 Indeed, it remains to prove that this functor is essentially surjective, which follows from the fact that the functors of the form $P \mapsto P(X)$ on $\widehat{C}$ are fiber functors, and from the evident fact that every filtered inductive limit of fiber functors on a topos is again a fiber functor.

Identifying C with a full subcategory of $\mathrm{Pro}(C)$ (I 8), the functor

$$C \longrightarrow \mathbf{Point}(\widehat{C})$$

induced by a quasi-inverse to (6.8.6.1) is manifestly isomorphic to the canonical functor already considered in (4.6.2.2); the quasi-inverse in question may be regarded as its canonical extension (I 8), taking into account the fact that $\mathbf{Point}(\widehat{C})$ is stable under small projective limits.

6.8.7. More generally, let $(X_i)_{i\in I}$ be a pro-object of the $\mathscr{U}$-site C, and hence a functor on $C^\sim$

$$\varphi(F) = \varinjlim_{I} F(X_i), \tag{6.8.7.1}$$

and, in view of (6.8.5), it is natural to ask when this functor is a fiber functor. The following conditions are equivalent:

(i) The preceding functor φ is a fiber functor on $C^\sim$.

(ii) The "restriction" of φ to C is a fiber functor on the site C (6.3).

(iii) For every covering family $Y_\alpha \to Y$ of an object Y of C, every $i_\circ \in I$, and every morphism $X_{i_\circ} \to Y$, there exist an $i \geq i_\circ$, an α, and a morphism $X_i \to Y_\alpha$ making the diagram

$$\begin{tikzcd} X_i \arrow[r] \arrow[d] & Y_\alpha \arrow[d] \\ X_{i_o} \arrow[r] & Y \end{tikzcd} .$$

commutative.

400 Indeed, condition (iii) simply expresses the fact that $\varphi|C$ carries covering families to covering families. The implications (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) are trivial, and it remains to prove (iii) $\Rightarrow$ (i). Since the functor φ is manifestly left exact, it remains to prove (4.9.4) that it carries covering families $F_\alpha \to F$ in $C^\sim$ to covering families, i.e. that for every $i_\circ$ and every $x_\circ \in F(X_{i_\circ})$, there exist an $i \geq i_\circ$, an α, and a $y \in F_\alpha(X_i)$ such that $x_\circ$ and y have the same image in $F(X_i)$. Since the family $F_\alpha \to F$ is covering, there is a covering family $Z_\gamma \to X_{i_\circ}$ of $X_{i_\circ}$ such that, for every γ, the inverse image of $x_\circ$ in $F(Z_\gamma)$ lifts to an element of some $F_\alpha(Z_\gamma)$. By hypothesis (iii), there exist an $i \geq i_\circ$ and an $X_{i_\circ}$-morphism $X_i \to Z_\gamma$. The image of $x_\circ$ in $F(X_i)$ therefore lifts to an element y of some $F_\alpha(X_i)$, C.Q.F.D.

Exercise 6.9. Let $C \in \mathscr{U}$ be a site, and let φ be a fiber functor on C, pro-represented by an object $(X_i)_{i \in I}$ of $\mathrm{Pro}(C)$. For every index $i \in I$, every object Y of C, every morphism $u : X_i \to Y$, and every covering sieve R on Y, choose an index $j \geq i$ such that the composite $X_j \to X_i \xrightarrow{u} Y$ factors through R. For fixed i, let $J(i)$ be the set formed by i and by the j obtained as Y, u, and R vary; for a subset I' of I, likewise let $J(I')$ be the union of the $J(i)$ for $i \in I'$. On the other hand, if I' is a finite subset of I, let $m(I') \in I$ be an upper bound of I', and for an arbitrary subset I' of I, let $M(I')$ be the union of the $m(I'')$, where I'' ranges over the set of finite subsets of I'. We assume 401
that the function $I' \mapsto m(I')$ is chosen so that, when I' consists of a single element i, one has $m(I') = i$; this implies that, for every subset I' of I, one has $I' \subset M(I')$. Consider the iterates $(MJ)^n$ of the map MJ from the set of subsets of I to itself, and, for every subset I' of I, let $P(I')$ be the union of the $(MJ)^n(I')$.

a) Show that, for every subset I' of I, $P(I')$ is a filtered subset of I for the induced order, and that I is the increasing filtered union of the $P(I')$ as I' ranges over the set of *finite* subsets of I.
b) Show that there is a cardinal $c \in \mathscr{U}$, depending only on the cardinal $\mathrm{card}\,\mathrm{fl}(C) = a$, such that, for every finite subset I' of I, one has $\mathrm{card}(P(I')) \leq c$. (If a is infinite, one may take $c = 2^a$.)
c) Show, using 6.8.7, that for every subset I' of I the functor $\varphi_{I'}$ on C, pro-represented by the pro-object $(X_i)_{i \in P(I')}$, is a fiber functor on C, and that the fiber functor φ is the filtered inductive limit of the functors $\varphi_{I'}$ as I' ranges over the set of finite subsets of I.
d) Deduce that there is a small full subcategory Φ of $\mathbf{Fib}(C) \approxeq \mathbf{Fib}(C)$ such that every object of $\mathbf{Fib}(C)$ is a filtered inductive limit of objects of Φ. (If a is infinite, one may take Φ to have cardinal $\leq 2^{2^a}$; if a is finite, one may of course take $\Phi = C$.)

Exercise 6.10. Let C be a $\mathscr{U}$-site, and let D be a $\mathscr{U}$-category in which small filtered $\varinjlim$ are representable; let $\mathrm{Fais}(C, D)$ be the category of sheaves on C with values in D (II 6.1). By extending (6.8.3), define a "fiber functor"

$$F \longrightarrow F_p : \mathrm{Fais}(C, D) \longrightarrow D. \tag{6.10.1}$$

If finite $\varprojlim$ are representable in D and commute with filtered $\varinjlim$, then the preceding functor is left exact.

7. Examples of fiber functors and points of topoi

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7.1. **The case of** $\mathrm{Top}(X)$ **for a topological space** X. Let x be a point of X. Regard the singleton $\{x\}$ as a topological space, and consider the inclusion

$$i_x : \{x\} \longrightarrow X.$$

By 4.1.1, it defines a morphism of topoi (defined up to unique isomorphism)

$$\mathrm{Top}(i_x) : \mathrm{Top}(\{x\}) \longrightarrow \mathrm{Top}(X). \tag{7.1.1}$$

On the other hand, one may identify P with $\mathrm{Top}(\{x\})$ (2.2), and one thereby obtains a point p_x of $\mathrm{Top}(X)$:

$$p_x : P \longrightarrow \mathrm{Top}(X), \tag{7.1.2}$$

defined by x up to unique isomorphism. The associated fiber functor is given by the familiar formula [TF], which moreover follows immediately from I 5.1:

$$p_x^*(F) = F_x = F_{p_x} = \varinjlim_{U \ni x} F(U), \tag{7.1.3}$$

the limit being taken over the open neighborhoods U of x in X.

Since $\mathrm{Top}(X)$ is known to be isomorphic to $\mathrm{Top}(X_{\mathrm{sob}})$ (4.2.1), one sees more generally that every point of X_{sob}, i.e. every irreducible closed subset Z of X, defines a point of $\mathrm{Top}(X)$, or equivalently a fiber functor on $\mathrm{Top}(X)$, which we shall still denote by $F \mapsto F_Z$. Returning to its definition by formula (7.1.3) on X_{sob}, one obtains

$$F_Z = \varinjlim_{U \cap Z \neq \phi} F(U), \tag{7.1.4}$$

403 the inductive limit being taken over the open subsets U of X which meet Z.

7.1.5. It is well known that the fiber functors $F \mapsto F_x$ ($x \in X$) already form a conservative family. This is especially clear from the interpretation of sheaves on X in terms of étale spaces, since the fiber of F at x is then simply its fiber in the sense of spaces fibered over X. Note that the index set of the conservative family of fiber functors under consideration is X, and hence is $\mathscr{U}$-small.

7.1.6. By 4.2.3, the category $\mathbf{Point}(\mathrm{Top}(X))$ is equivalent to the category associated with the set X_{sob} ordered by the specialization relation. In other words, every fiber functor on $\mathrm{Top}(X)$ is isomorphic to a fiber functor $F \mapsto F_Z$, where Z is a uniquely determined element of X_{sob}, i.e. a uniquely determined irreducible closed subset of X; moreover, if $Z, Z' \in X_{\mathrm{sob}}$, then the set $\mathrm{Hom}(F_Z, F_{Z'})$ is either empty or consists of a single point, the latter case occurring if and only if Z is a specialization of Z' in X_{sob}, i.e. if and only if $Z \subset Z'$ as a subset of X. When X itself is sober, one may replace X_{sob} by X itself in these statements.

Note that the automorphism group of a fiber functor of $\mathrm{Top}(X)$ (opposite to the automorphism group of the corresponding point of $\mathrm{Top}(X)$) is always the trivial group (more generally, 4.2.3 tells us the same thing about the automorphism group of every morphism of topoi associated with topological spaces). This is a phenomenon very special to the
404 case under consideration: see the example in 7.2, as well as VIII 7. In the case of topoi associated with "moduli problems" (cf. for example [10]), the automorphism groups of fiber functors moreover have a remarkable interpretation as the automorphism groups of the algebraic structures (over algebraically closed fields) which one proposes to classify.

7.1.7. We have just seen how one can reconstruct a sober space X (or the sober space X_{sob} associated with an arbitrary topological space), at least as a set ordered by the specialization relation, from the topos $\mathrm{Top}(X)$ (which it even suffices to know up to equivalence), as the set of isomorphism classes of "points" of $\mathrm{Top}(X)$. From what was said in 2.1, one also reconstructs the topology of X, i.e. the family of its open subsets, as follows: for every subobject U of the terminal object e_E of E, let $\mathrm{Pt}(U)$ be the set of $x \in X$ such that $U_x \neq \phi$ (which moreover identifies, by (6.7.2), with the set of isomorphism classes of points of the induced topos $E_{/U}$). Then $U \mapsto \mathrm{Pt}(U)$ is an isomorphism of ordered sets from the set of subobjects of e_E onto the set of open subsets of X.

7.1.8. The determination in 7.1.6 of the category of points of the topos $\mathrm{Top}(X)$ defined by a topological space X leads to the following terminology for points p, p' of an arbitrary topos E: one says that p is a *specialization* of p', or that p' is a *generization* of p, when there exists a morphism from p' to p (in the sense of the category $\mathbf{Point}(E)$ of 6.1). These relations are still transitive, but, using 6.8.6, one easily finds examples in
405 which p and p' are each a specialization of the other, without p and p' being isomorphic

(much less equal). The category of generizations of a point p of the topos E, by which we mean the category $\mathbf{Point}(E)_{/p}$, plays in some respects a role analogous to that of passing to the localization of a scheme at a point. This role will be made precise in VI 8.4 by the construction of the topos obtained by localizing E at the point p, whose category of points is canonically equivalent to the category of generizations of p.

Exercise 7.1.9. Let E be a $\mathscr{U}$-topos. Prove that E is equivalent to a topos of the form $\mathrm{Top}(X)$, where X is a topological space $\in \mathscr{U}$, if and only if it satisfies the following two conditions:

a) The family of subobjects of the terminal object e_E is generating for the topos E.
b) E has enough points (6.5). (This condition is not superfluous: cf. 7.4.)

Compare with 7.8 c).

Exercise 7.1.10. Let X be a topological space endowed with an operator group G ($X, G \in \mathscr{U}$), and let $E = \mathrm{Top}(X, G)$ (2.3).

a) Show that, for every object (X', G) of E, the induced topos $E_{/(X',G)}$ is canonically equivalent to the topos $\mathrm{Top}(X', G)$ (compare 5.7).
b) When G acts properly and freely on X' (Bourbaki, Gen. Top. Chap. III 4), show that the morphism of spaces with operators $(X', G) \to (X'/G, e)$ induces (4.1.2) an equivalence of topoi

$$\mathrm{Top}(X', G) \longrightarrow \mathrm{Top}(X'/G).$$

c) Deduce from b) that there exists an object Z of E such that the induced topos $E_{/Z}$ is equivalent to $\mathrm{Top}(X)$, the localization functor $E \to E_{/Z}$ being isomorphic to the functor that "forgets the actions of G". (Imitate the argument of 5.8.3.) 406
d) Deduce from c) that every fiber functor on E is induced, via the functor that "forgets the actions of G", by a fiber functor on $\mathrm{Top}(X)$, and can therefore be defined by a point of X_{sob}.
e) Determine the structure of the category $\mathbf{Point}(E)$ (up to equivalence) in terms of the space with operators (X_{sob}, G). Conclude in particular that two points of X_{sob} define isomorphic fiber functors on E if and only if they are conjugate under the action of G.

Exercise 7.1.11. Let $X \in \mathscr{U}$ be a topological space. Prove that the $\mathscr{V}$-topos $\mathrm{TOP}(X)$ (2.5) has enough points. More precisely, for every object X' of $\mathrm{TOP}(X)$ and every $x' \in X'$, define a fiber functor $F \mapsto F_{X',x'}$ on $\mathrm{TOP}(X)$ which depends only on the "germ" of the space (X', x') over X, and prove that this family of fiber functors is conservative (and indexed by a $\mathscr{V}$-small index set). Show, using a suitable filtered projective system $(X'_i, x'_i)_{i \in I}$, with I not $\mathscr{U}$-small, that one does not obtain in this way all the fiber functors of the $\mathscr{V}$-topos $\mathrm{TOP}(X)$. Show that the set of isomorphism classes of such fiber functors does not have cardinality $\in \mathscr{V}$.

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7.2. **Points of a classifying topos B_G.** Let E be a $\mathscr{U}$-topos and G a group of E, whence a classifying topos B_G (2.4). If e is the unit group, the morphisms $e \to G \to e$ define (4.5) morphisms of topoi (taking into account that $B_e \approx E$)

$$E \longrightarrow B_G \longrightarrow E, \tag{7.2.1}$$

whose composite is isomorphic to id_E; passage to the categories of points therefore gives functors

$$\mathbf{Point}(E) \longrightarrow \mathbf{Point}(B_G) \longrightarrow \mathbf{Point}(E), \tag{7.2.2}$$

whose composite is isomorphic to the identity. Use the fact that the first morphism of topoi in (7.2.1) identifies with a localization morphism relative to the object $E_G = G$ of B_G (5.8); by (6.7.2), it follows that the first functor in (7.2.2) identifies with the natural "inclusion" functor

$$\mathbf{Point}(B_G)_{/\widehat{E}_G} \longrightarrow \mathbf{Point}(B_G), \tag{7.2.3}$$

where $\widehat{E}_G$ denotes the presheaf $p' \mapsto (E_G)_{p'}$ on $\mathbf{Point}(B_G)$. Since E_G plainly covers the terminal object of B_G, its fibers $(E_G)_p$ are nonempty, from which it follows that (7.2.3) is essentially surjective; this means that *every fiber functor on B_G is induced (up to isomorphism) by a fiber functor on E, via the functor that "forgets the actions of G" $B_G \to E$*. One concludes in particular that *if E has enough points (resp. a small conservative family of points),* then the same is true of B_G.

Remark 7.2.4. One can go further and determine (up to equivalence) the structure of the category $C' = \mathbf{Point}(B_G)$ in terms of the category $C = \mathbf{Point}(E)$ and the presheaf of groups

$$\widehat{G} : p \longmapsto G_p : C^\circ \longrightarrow (\text{Groups})$$

408 on it. Indeed, define a new category C_1, having the same objects as C, and such that for two objects p, q of C, one has

$$\mathrm{Hom}_{C_1}(p, q) = \mathrm{Hom}_C(p, q) \times \widehat{G}(p),$$

composition of arrows in C_1 being induced by that in C and by the group law of the presheaf $\widehat{G}$. The datum of a functor $\varphi_1 : C_1 \to D$ into an arbitrary category D is then equivalent to the datum of a functor $\varphi : C \to D$ endowed with an action of the presheaf $\widehat{G}$ on the latter (i.e. the datum, for every $p \in \mathrm{ob}\, C$, of an action of $\widehat{G}(p)$ on $\varphi(p)$, satisfying an evident functoriality condition as p varies). Using this observation, one defines a canonical functor

$$\varphi_1 : C_1 \longrightarrow C', \tag{7.2.4.1}$$

corresponding to the functor $\varphi : C \to C'$ of (7.2.2), which associates with every point p of E the point $\varphi(p)$ induced on B_G, together with the natural actions of $\widehat{G}(p) = G_p$ on the latter, arising from the actions of G_p on the X_p as X ranges over B_G. We leave to the reader the proof that (7.2.4.1) is an equivalence of categories, using the structure (7.2.3) of the first functor in (7.2.2), and noting that the natural inclusion functor $C \to C_1$ has an analogous structure, relative to the inverse image P_1 of the presheaf $P' = \widehat{E}_G$ on C'.

409 7.2.5. One concludes in particular that the *isomorphism classes of points of B_G correspond exactly to the isomorphism classes of points of E: every point of the classifying topos B_G is induced, up to a* non-unique *isomorphism, by a point of E*. In particular, when E is the punctual topos P, and hence G is an ordinary group, the category $\mathbf{Point}(B_G)$ is a connected groupoid with fundamental group G: every fiber functor on B_G is isomorphic (noncanonically) to the functor ω that "forgets the actions of G", and the monoid of endomorphisms of the latter functor is the group G (hence every endomorphism of ω is an automorphism).

Exercise 7.2.6. a) Let (Tors) be the category of pairs $(\Gamma, P) \in \mathscr{U}$, where Γ is a group and P is a right torsor under Γ. The functor $(\Gamma, P) \mapsto \Gamma$

$$(\text{Tors}) \longrightarrow (\text{Groupes}) \tag{7.2.6.1}$$

is a cofibered functor. With the notation of 7.2.4, consider the functor

$$\widehat{G} : C^{\circ} \longrightarrow \text{(Groupes)},$$

and let D' be the cofibered category over C° obtained by inverse image from the cofibered category (7.2.6.1), and let D be the corresponding category fibered over C, having the same fiber categories as D', so that for $p \in \operatorname{ob} C$ one has

$$D_p = \text{ category of right torsors under } G_p.$$

Construct canonical functors

$$D \longrightarrow C' = \mathbf{Point}(B_G) \quad , \quad C' \longrightarrow D, \tag{7.2.6.2}$$

quasi-inverse to one another. In particular, conclude that the category of points of B_G above a given point p of E is canonically equivalent to the category of right torsors under the group G_p.

b) Let $\mathcal{G} = (G_i)_{i \in I}$ be a strict projective system of groups of the $\mathcal{U}$-topos E ($I \in \mathcal{U}$ **410**
a filtered preordered set), whence a classifying topos $B_{\mathcal{G}}$, by imitating the definition of 2.7.1. Determine the category of points of $B_{\mathcal{G}}$ by imitating a). (NB. Replace the categories (Tors) and (Groupes) by the categories of projective systems indexed by I in these categories.) Conclude that if I admits a countable cofinal subset, then every fiber functor on $B_{\mathcal{G}}$ is isomorphic to a functor induced by a fiber functor on E (via the functor that "forgets the actions of G"), the latter being determined up to non-unique isomorphism.

c) Taking E to be the punctual topos, so that $\mathcal{G}$ is an ordinary pro-group, show that the conclusion of b) is likewise valid if I is arbitrary but $\mathcal{G}$ is profinite (i.e. the G_i are finite): every fiber functor is isomorphic to the functor that "forgets the actions of G".

d) Still taking the case where E is the punctual topos, give an example of a fiber functor on $B_{\mathcal{G}}$, for a suitable projective system $\mathcal{G} = (G_i)_{i \in I}$, which is not isomorphic to the functor that "forgets the actions of G".

e) Regard $\mathcal{G} = (G_i)$ as a group of the topos $\widehat{I}$, and let $\mathcal{T}$ be a *gerbe* on I with *band* G [3]. Show how one can "twist" the classifying topos B_G by means of the gerbe T, to obtain a topos $B^T_{\mathcal{G}}$, an inductive limit of subcategories $B^T(i)$ equivalent (noncanonically) to the classifying topoi B_{G_i}. Show that the topos $B^T_{\mathcal{G}}$ admits a fiber functor if and only if the gerbe T is "neutral", by establishing
an equivalence of categories between the category of fiber functors on $B^T_{\mathcal{G}}$ and **411**
the category of sections of T over I. Conclude, if the G_i are commutative, so that gerbes on I with band $\mathcal{G}$ are classified by $\varprojlim_I^{(2)} G_i = H^2(I, \mathcal{G})$ (*loc. cit.*), that there is an example of a topos (not "empty" (2.2)) of the form $B^T_{\mathcal{G}}$ which has no points. (Take an example in which $\varprojlim_I^{(2)} G_i \neq 0$.)

7.3. **Points of presheaf topoi $\widehat{C}$; examples of $\mathcal{U}$-presheaf topoi $\widehat{C}$ whose category of points is not equivalent to a small category.** Let E be a $\mathcal{U}$-topos, and let $(\varphi_i)_{i \in I}$ be a filtered family of fiber functors on E. It follows from the exactness properties of the functors $\varinjlim$ (I 2.8) that the functor $\varphi = \varinjlim \varphi_i$ is also a fiber functor: *every filtered inductive limit of fiber functors is a fiber functor.* Consequently, the category $\mathbf{Fib}(E)$ admits $\mathcal{U}$-filtered inductive limits (and the inclusion functor $\mathbf{Fib}(E) \to \mathbf{Hom}(E, (\mathcal{U}\text{-Ens}))$

commutes with them); in other words, the category $\mathbf{Point}(E)$ admits $\mathscr{U}$-filtered projective limits. This fact was already used in Exercise 7.1.11 to give an example of "large" categories of fiber functors.

Considerably simpler examples can be constructed, with presheaf topoi of the form $E = \widehat{C}$, $C \in \mathscr{U}$, by using (6.8.6.1). This immediately gives examples in which $\mathbf{Fib}(\widehat{C})$ is not equivalent to a category $\in \mathscr{U}$, i.e. in which the cardinal of the set of isomorphism classes of objects of this $\mathscr{U}$-category is not $\in \mathscr{U}$. This means that the category of pro-objects $\operatorname{Pro}-(C)$ is not equivalent to a category $\in \mathscr{U}$. For example, it suffices to take
412 $D = C^{\circ}$ to be the category of finite sets of the form $[0, n]$, where $n \geq 0$ is an integer; in this case $\operatorname{Ind}(D) \simeq \operatorname{Pro}(C)^{\circ}$ is equivalent to the category of nonempty sets. The topos E is then the familiar category of *cosimplicial sets*. One could also take C to be the category of nonempty finite sets; one then finds that the category of points of the presheaf topos $\widehat{C}$ of simplicial sets is equivalent to the category of profinite sets, or equivalently to the category of totally disconnected compact spaces.

7.4. **Nonempty topoi without points**. The following example is due to P. Deligne. (For another example, cf. 7.2.6 e).) Let K be a compact space equipped with a measure μ, and let U be the ordered set of measurable subsets of K, modulo sets of measure zero. Make U into a category $\underline{U}$ with $\operatorname{ob}\underline{U} = U$, the morphisms of $\underline{U}$ being the "inclusion morphisms" between elements of U. Make $\underline{U}$ into a site by taking the pretopology for which $\operatorname{Cov}(E)$ (for $E \in U$) consists of the *countable* families of elements E_i of U bounded above by E, such that E is the union of the E_i modulo a set of measure zero. This gives a topos $\operatorname{Top}(\mu) = \underline{U}^{\sim}$ whose set of subobjects of the terminal object is a generating family (a topos which appears to have escaped the attention of probabilists). This topos is an "empty topos" (2.2) if and only if $\mu = 0$. On the other hand, the category of points of this topos is equivalent to the discrete category defined by the set of points $x \in K$ such that $\mu(\{x\}) \neq 0$ (proof left to the reader). It is therefore empty if K has no such points, for example if K is the unit interval in the real line, with the measure induced by Lebesgue measure.

413 Exercise 7.5. (*karoubian categories and morphisms of topoi*)[10]

a) Let C be a category, let X be an object of C, and let p be an endomorphism of X. We say that p is a *projector* if $p^2 = p$. Prove that, if p is a projector, $\operatorname{Ker}(\mathrm{id}_X, p)$ is representable if and only if $\operatorname{Coker}(\mathrm{id}_X, p)$ is representable, and that the two objects of C so obtained are canonically isomorphic. We then say that the projector p *admits an image*, and $\operatorname{Ker}(\mathrm{id}_X, p) \simeq \operatorname{Coker}(\mathrm{id}_X, p)$ is called the *image of the projector* p; according to the context, it is identified with either a subobject or a quotient object of C. An object isomorphic to $\operatorname{Im} p$, for a suitable projector p in X, is called a *direct factor of the object* X *of* C. We say that C is a *category with direct factors*, or a *karoubian category*, if every projector in an object of C admits an image. If $F : C \to C'$ is a functor that commutes with kernels or cokernels, then F sends a projector admitting an image to a projector admitting an image[11].
b) Show that, for every category C, one can find a functor $\varphi : C \to \operatorname{kar}(C)$ from C to a Karoubian category (determined up to equivalence) such that, for every Karoubian category C', the functor

$$f \mapsto f \circ \varphi : \mathbf{Hom}(\operatorname{kar}(C), C') \longrightarrow \mathbf{Hom}(C, C')$$

[10] Compare I 8.7.8.

[11] N.D.E.: No condition on F is needed here: every functor preserves projectors admitting images.

is an equivalence of categories; $\mathrm{kar}(C)$ is called the *Karoubi envelope* of the category C. (Hint: take $\mathrm{ob}\,\mathrm{kar}(C)$ to be the set of pairs (X, p), with $X \in \mathrm{ob}\, C$ and p a projector in X, and take $\mathrm{Hom}((X, p), (Y, q))$ to be the subset of $\mathrm{Hom}(X, Y)$ consisting of those $f : X \to Y$ such that $f = qfp$.) Show that φ is fully faithful.

c) Let $F : E \to E'$ be a functor from one category to another. Show that if F commutes with a given type of inductive or projective limits, then the same is 414
true of every direct factor of F. In particular, if E and E' are $\mathscr{U}$-topoi and F is an inverse-image functor for a morphism of topoi $E' \to E$, then the same is true of every direct factor of F; consequently, the category $\mathbf{Homtop}(E', E)$ is Karoubian, and in particular the category $\mathbf{Point}(E)$ is Karoubian.

d) Show that, for every $\mathscr{U}$-category C, the category $\mathrm{Ind}(C)$ is Karoubian (where $\mathrm{Ind}(C)$ is formed using inductive systems in C indexed by a filtered preordered set $I \in \mathscr{U}$). (If $C \in \mathscr{U}$, use, for example, the fact (7.3) that $\mathrm{Ind}(C)$ is equivalent to the category $\mathbf{Fib}(\widehat{(C^\circ)}) = \mathbf{Point}(\widehat{(C^\circ)}{}^\circ)$.) Deduce another construction of $\mathrm{kar}(C)$ as the full subcategory of $\mathrm{Ind}(C)$ formed by the images in $\mathrm{Ind}(C)$ of the projectors of objects X of C.

Exercise 7.6. (*Essential morphisms of topoi, essential points*).

a) Let $f : E \to F$ be a morphism of topoi. Show that the following conditions are equivalent:
   i) $f_!$ exists, i.e. f^* admits a left adjoint.
   ii) f^* commutes with $\mathscr{U}$-projective limits.
   ii$'$) f^* commutes with $\mathscr{U}$-products.

   We then say that f is an *essential morphism* from the topos E to the topos F.

   Show that if f satisfies these conditions, then the same is true of every direct factor of f. (Use 1.8 and 7.5 c).)

b) Let E be a topos. An *essential point of the topos* E is a point $p : P \to E$ of E such that $p_!$ exists. Show that if $E = \mathrm{Top}(X)$, where X is a topological space, and if p is the point of E defined by an $x \in X$, then p is essential if and only if x has 415
a smallest open neighborhood U (which means, if the points of X are closed, that x is an *isolated point* of X). For a morphism of topoi $f : E \to F$ to be essential (a)), it is necessary that f send essential points to essential points, and this condition is also sufficient when E has enough essential points (i.e. when the family of these points is conservative) (cf. d)).

c) A point p of the topos E is essential if and only if the associated fiber functor is representable by an object X of E. For the covariant functor $\varphi : E \to (\mathrm{Ens})$ represented by a given object X of E to be a fiber functor (i.e. to define a point, necessarily an essential point, of E), it is necessary and sufficient that X be nonempty and connected (4.3.5), and projective (i.e. that the functor φ send epimorphisms to epimorphisms). (Hint: first show that φ commutes with coproducts if and only if X is nonempty and connected, and then use the criterion 4.9.4.) Deduce an equivalence between the category $\mathbf{Pointess}(E)$ and the full subcategory P of E formed by the nonempty connected projective objects.

d) Show that the topology on P induced by that of E is the chaotic topology. Deduce the equivalence of the following conditions on E (due to J. E. Roos [12 c), prop. 1]): (i) The family of essential points of E is conservative; (ii) The full subcategory P of E formed by the nonempty connected projective objects is generating; (iii) E is equivalent to a presheaf topos of the form $\widehat{C}$, where C is a category equivalent to a category $\in \mathscr{U}$ (or, if one prefers, $C \in \mathscr{U}$). (Use c) and

the fact that, if P is generating, the natural functor $E \to P^{\sim}$ is an equivalence
416 of categories.)

e) The category of essential points of a topos E is Karoubian (cf. 7.5 c)). Let C be a category equivalent to a category $\in \mathscr{U}$. Prove that the canonical fully faithful functor (4.6.2)
$$C \longrightarrow \mathbf{Point}(\widehat{C})$$
factors through a functor
$$C \longrightarrow \mathbf{Pointess}(\widehat{C}),$$
which makes $\mathbf{Pointess}(\widehat{C})$ a *Karoubi envelope* (7.5 b)) *of* C. In particular, this functor is an equivalence of categories if and only if the category C is Karoubian. (Hint: using *c*), prove that every essential point of $\widehat{C}$ is isomorphic to a direct factor of some $X \in \operatorname{ob} C$.)

f) Let C and C' be two categories equivalent to categories $\in \mathscr{U}$. Prove that the canonical fully faithful functor (4.6.2)
$$\mathbf{Hom}(C, C') \longrightarrow \mathbf{Homtop}(\widehat{C}, \widehat{C}')$$
takes its values in the full subcategory $\mathbf{Homtopess}(\widehat{C}, \widehat{C}')$ of essential morphisms *of topoi* (a)), and that the induced functor
$$\mathbf{Hom}(C, C') \longrightarrow \mathbf{Homtopess}(\widehat{C}, \widehat{C}')$$
is an equivalence of categories if and only if C is empty or C' is Karoubian. (Use *g*) below.)

g) Let C be a category equivalent to a category $\in \mathscr{U}$, and let E be a topos. Define an equivalence between the category $\mathbf{Homtopess}(\widehat{C}, E)$ of essential morphisms
417 of topoi $\widehat{C} \to E$, and the category $\mathbf{Hom}(C, \mathbf{Pointess}(E))$. (Use *b*) and the fact that $\widehat{C}$ has enough essential points.)

h) Let C be a finite category. Prove that every point of the presheaf topos $\widehat{C}$ is essential, and that $\mathbf{Point}(\widehat{C})$ is equivalent to a finite category. (Note that the Karoubi envelope of C is finite, and, using (6.8.6.1), reduce to proving that if C is a finite Karoubian category, then the canonical functor $C \to \operatorname{Pro}(C)$ is an equivalence of categories.) Using b), deduce that *every* morphism of topoi $\widehat{C}' \to \widehat{C}$ is essential, and that $\mathbf{Hom}(C, C') \to \operatorname{Hom}(\widehat{C}, \widehat{C}')$ is an equivalence if and only if C is empty or C' is Karoubian.

i) Let X be a topological space, and let $f : \operatorname{Top}(X) \to P$ be the morphism of topoi induced by the continuous map from X to the one-point space. Show that f is essential if and only if the space X is *locally connected*, i.e. if it satisfies the following condition: for every $x \in X$ and every open neighborhood U of x, the connected component of x in U is a neighborhood of x; or equivalently: for every open subset U of X, the connected components of U are open. (cf. 8.7 l) for a generalization).

Exercise 7.7. (*Unusual points of a classifying topos*)[11].

a) Let G be a monoid. The monoid G contains an element g such that $g \neq e$ and $g^2 = g$ if and only if the classifying topos B_G admits an essential point that is not isomorphic to the trivial point (corresponding to the fiber functor "forget the G-action"). (Use 7.6 e)).

[11] cf. also 7.2.6 d).

b) Let G be the additive monoid of the integers ≥ 0. Show that every essential point of the classifying topos B_G is isomorphic to the trivial point. Construct a point of B_G that is not isomorphic to the trivial point. Show that the category $\mathbf{Point}(B_G)$ is not equivalent to a category $\in \mathscr{U}$.

Exercise 7.8. (*Topology on* Point(E)*, and topoi associated with ordered sets*).

a) Let E be a topos. Denote by Point(E) the set of isomorphism classes of points of E. For every open U of E, let U' be the set of classes of points p of E such 418
that $U_p \neq \phi$. Show that $U \mapsto U'$ is an order-preserving map from the ordered set Ouv(E) of opens of E to the set of subsets of $X = \mathrm{Point}(E)$, and that this map commutes with finite Inf and arbitrary Sup. Deduce that the subsets of X of the form U' define a topology on X (called the *canonical topology on the set of classes of points of the topos* E). Show that if E has enough points, the map $U \mapsto U'$ is an isomorphism from the ordered set Ouv(E) onto the ordered set Ouv(X).

b) Let $\mathscr{O} \in \mathscr{U}$ be an ordered set admitting finite Inf and arbitrary Sup, and hence defining a category cat($\mathscr{O}$) having finite products and arbitrary coproducts. We say that a family of morphisms $U_i \to U$ with common target U is covering if U is the Sup of the U_i. Suppose that coproducts in cat($\mathscr{O}$) are universal, i.e. that in $\mathscr{O}$ the arbitrary Sup commute with Inf. This defines a topology on cat($\mathscr{O}$), making it a site, and hence a topos cat$(\mathscr{O})^\sim$, also denoted simply by $\mathscr{O}^\sim$. Show that $\mathscr{O}$ is canonically isomorphic to Ouv($\mathscr{O}^\sim$).

Show that the category $\mathbf{Homtop}(E, \mathscr{O}^\sim)$ is equivalent to the category associated with the ordered set of morphisms of ordered sets $\mathscr{O} \to \mathrm{Ouv}(E)$ that commute with finite Inf and arbitrary Sup. (Use the criterion 4.9.4.) If $\mathscr{O}'$ is an ordered set satisfying the same conditions as $\mathscr{O}$, deduce that $\mathbf{Homtop}(\mathscr{O}^\sim, \mathscr{O}'^\sim)$ is equivalent to the category associated with the ordered set of all morphisms of ordered sets $\mathscr{O} \to \mathscr{O}'$ that commute with finite Inf and arbitrary Sup.

c) Show that the topos E is equivalent to a topos of the form $\mathscr{O}^\sim$, with $\mathscr{O}$ as in 419
b), if and only if the family of opens of E is generating in E. It is equivalent to some Top(X) with $X \in \mathscr{U}$ if and only if it satisfies the preceding condition and has enough points. Assuming the first condition, express the second in terms of the structure of the ordered set $\mathscr{O} = \mathrm{Ouv}(E)$, using the last assertion of b).

d) Define a canonical morphism of topoi

$$E \longrightarrow \mathrm{Ouv}(E)^\sim,$$

which is universal (up to equivalence) for morphisms from E to topoi generated by their opens (cf. c)).

e) Let X' be a *small* subset of $X = \mathrm{Point}(E)$, equipped with the topology induced by that of X. Define a canonical morphism of topoi

$$\mathrm{Ouv}(E)^\sim \longrightarrow \mathrm{Top}(X'),$$

which is an equivalence when the family X' of points of E is conservative. Then deduce a canonical morphism of topoi

$$E \longrightarrow \mathrm{Top}(X').$$

Give a universal characterization (up to equivalence) of this morphism of topoi with respect to morphisms from the topos E to topoi of the form Top(Y). (Note in this connection that, up to equivalence, Top(X') does not depend on the chosen small conservative family of fiber functors, or equivalently (4.2), that X'_{sob} is independent, up to homeomorphism, of the choice of such a family.)

420 f) Suppose that the category $\mathbf{Point}(E)$ is not equivalent to a small category (cf. 7.3), and that E admits a small conservative family of fiber functors. Show that, if such a family is taken sufficiently large, the subset X' of $\mathbf{Point}(E)$ that it defines is not a sober topological space for the induced topology.

g) For the relations with Exercise 7.6, cf. 8.

8. Localization. Opens of a topos

8.1. Let C be a category, and let $T(X)$ be a relation involving an object X of C. The relation $T(X)$ is said to be *stable under base change* (in X) if, for every morphism $X' \to X$ of C, the relation $T(X)$ implies $T(X')$. This amounts to saying that the full subcategory R of C formed by the objects X for which $T(X)$ holds is a *sieve* (I 4.1).

Remark 8.1.1. Suppose that C is a $\mathcal{U}$-category (I 1.1), so that C may be identified with a full subcategory of $\widehat{C}$ (I 1.4). It is sometimes convenient to extend the definition of the relation T on $\mathrm{ob}\, C$ to a relation $\widehat{T}$ on an object F of $\widehat{C}$, by defining $\widehat{T}(F)$ to mean that $T(X)$ holds for every arrow $X \to F$ in $\widehat{C}$ with $X \in \mathrm{ob}\, C$. The relation $\widehat{T}(F)$ is clearly still stable under base change in F. Again denoting by R the subobject of the terminal object e of $\widehat{C}$ defined by the sieve R of C (I 4.2.1), the relation $\widehat{T}(F)$ may also be expressed as $\mathrm{Hom}_{\widehat{C}}(F, R) \neq \varnothing$, i.e. it means that the unique morphism $F \to e$ factors through the subobject R of e.

8.2. Now suppose that C is a *site*. A relation $T(X)$ in an argument $X \in \mathrm{ob}\, C$ is
421 said to be *stable under descent* if, for every covering family $(X_i : X_i \to X)_{i \in I}$ of an object X of C, the relation "$T(X_i)$ for every $i \in I$" implies $T(X)$. We say that T is a *relation of local nature* if it is stable under base change (8.1) and stable under descent. When C is a $\mathcal{U}$-category and T is already assumed stable under base change, i.e. is defined by a sieve R of C, which we identify with a subpresheaf of the terminal presheaf on C, one sees immediately that T is of local nature if and only if R is a *sheaf*. Thus *relations of local nature in an argument $X \in \mathrm{ob}\, C$ correspond exactly to the subsheaves of the terminal sheaf of C, i.e. to the subobjects of the terminal object of the topos $C^\sim$.* They therefore depend essentially only on the topos $C^\sim$ defined by the site C (like all important notions associated with a site!). In terms of the subsheaf R of the terminal sheaf, the relation $T(X)$ is expressed as $\mathrm{Hom}(\epsilon(X), R) \neq \varnothing$. It extends canonically to the relation $T^\sim(F) : \mathrm{Hom}(F, R) \neq \varnothing$ on the topos $C^\sim$.

Remark 8.2.1. With the notation introduced in 8.1.1, for a presheaf F on C, since $\mathrm{Hom}(F, R) \simeq \mathrm{Hom}(\underline{a}(F), R)$, the relation $\widehat{T}(F)$ is equivalent to the relation $\widehat{T}(\underline{a}(F))$.

Definition 8.3. *An open of a $\mathcal{U}$-topos E is a subobject of the terminal object e_E of E; if X is an object of E, an open of X sometimes means an open of the induced topos $E_{/X}$, i.e. a subobject of X.*

An open of a $\mathcal{U}$-site C is an open of the associated $\mathcal{U}$-topos $C^\sim$.

Since the terminal objects of E are canonically isomorphic, the ambiguity introduced
422 in 8.3 by the choice of e_E is harmless. One may also, in a more intrinsic but practically less convenient way, define the opens of E as the full subcategories R of E for which the relation $F \in \mathrm{ob}\, R$, for an object F of E, is of local nature. Likewise, the opens of a $\mathcal{U}$-site C correspond bijectively to the full subcategories of C for which the relation $F \in \mathrm{ob}\, R$, for an object of C, is of local nature; this notion is therefore essentially independent of

the chosen universe $\mathscr{U}$. Finally, by what was said in 8.2, a relation of local nature on a topos (or on a site) is essentially the same thing as an open of that topos (or site).

8.4. Examples of opens.

8.4.1. Let X be a topological space, and interpret $\mathrm{Top}(X)$ (2.1) as the category of étale spaces over X. Since the monomorphisms in $\mathrm{Top}(X)$ are plainly the injective maps, it follows that the opens of the topos $\mathrm{Top}(X)$ identify with the opens of the space X. More generally, the opens of a topos $\mathrm{Top}(X, G)$ (2.4) identify with the opens of X which are *invariant* under the action of G.

8.4.2. The opens of a $\mathscr{U}$-topos E form a $\mathscr{U}$-small set (I 7.4), ordered by inclusion, admitting arbitrary suprema and finite infima. Its least element is the initial object (or "empty object") ϕ_E, and its greatest element is the terminal object e_E of E. These two elements are identical only if E is an "empty topos" (2.2).

8.4.3. Let G be a monoid object in E, giving a topos B_G (2.6), the category of objects of E on which G acts on the left. The terminal object e_G of B_G is the terminal object 423
e_E of E with the trivial action of G. It follows that the ordered set of opens of B_G is canonically isomorphic to that of the opens of E. In particular, if E is the punctual topos, so that G is an ordinary monoid, the set of opens of B_G consists of exactly two elements, namely ϕ_{B_G} and e_G.

8.4.4. If E is a topos of the form $\widehat{C}$, where C is a $\mathscr{U}$-category, then (8.1) the ordered set of opens of E is isomorphic to the ordered set of sieves of C.

8.4.5. Let X be a sober nondiscrete topological space. Show that $\mathrm{TOP}(X)$ (2.5) has opens which do not arise from opens of X, i.e. from opens of $\mathrm{Top}(X)$, by inverse image under the canonical morphism (4.10.1) $\mathrm{TOP}(X) \to \mathrm{Top}(X)$.

8.5. Examples of relations of local nature. For every object X of the topos E, denote by $F \to F_X$ the *localization functor* $j_X^* : E \to E_{/X}, Y \mapsto Y \times X$ (5.2).

8.5.1. Let $u : F \to G$ be a morphism of E. The following relation in the argument $X \in \mathrm{ob}\, E$:

$u_X : F_X \to G_X$ is a monomorphism (resp. an epimorphism, resp. an isomorphism) is a relation of local nature. In particular, if G is a group object of E, the relation "G_X is the trivial group over X" is of local nature; the corresponding open of E is called the *cosupport of the group* G.

8.5.2. Let $u, v : F \rightrightarrows G$ be two morphisms of E. The relation in the argument 424
$X \in \mathrm{ob}\, E$

$$u_X = v_X : F_X \longrightarrow G_X$$

is a relation of local nature. In particular, if G is a group object of E, and u is a *section* of G (4.3.6), the relation in X

$$\text{the section } u_X \text{ of the group } G_X \text{ of } E_{/X} \text{ is zero}$$

is of local nature; the corresponding open of E is called the *cosupport of the section* u *of the group* G.

Remark 8.5.3. When $E = \mathrm{Top}(X)$, the cosupport of a sheaf of groups G, resp. of a section of such a sheaf G, is simply the open of X complementary to the *support of* G, resp. to the *support of* u, in the classical terminology [TF].

8.5.4. Let $P(f)$ be a relation in the argument $f \in \mathrm{fl}\, C$, where C is a site. When fiber products are representable in C, the relation $P(f)$ is said to be *stable under base change (resp. stable under descent, resp. of local nature) on the target* (or *on the base*, or "*downstairs*") if, for every morphism $f : X \to Y$ of C, the following relation in the argument $Y' \in \mathrm{ob}\, C_{/Y}$:

"the morphism $f' : X' = X \times_Y Y' \to Y'$ obtained from f by base change along the structural morphism $Y' \to Y$ satisfies $P(f')$"

is stable under base change, resp. stable under descent, resp. is of local nature. When the topology of E is defined by means of a pretopology, the relation $P(f)$ is said to be *of local nature on the source* (or *"upstairs"*) if, for every morphism $f : X \to Y$ of C
425 and every family $(g_i : X_i \to X)_{i\in I}$ in $\mathrm{Cov}(X)$, the relation $P(f)$ is equivalent to the relation "$P(fg_i)$ for every $i \in I$." When $\mathrm{Cov}(X)$ consists of all covering families of C, this condition also means that the following relation in the object X' of the induced site $C_{/X}$

$$P(fg), \text{ where } g : X' \to X \text{ is the structural morphism}$$

is of local nature. Note that if the relation $P(f)$ is of local nature upstairs, it is a fortiori of local nature downstairs.

8.5.5. Take, for example, $C = (\mathrm{Sch})$, the category of schemes in $\mathscr{U}$, with the faithfully flat quasi-compact topology (SGA 3 IV 6.3) or a coarser topology. Each of the following properties of a morphism is then of local nature downstairs:

surjective, radicial, universally open, universally closed, proper, quasi-compact, quasi-compact and dominant, universal homeomorphism, separated, quasi-separated, locally of finite type, locally of finite presentation, of finite type, of finite presentation, an isomorphism, a monomorphism, an open immersion, a closed immersion, a quasi-compact immersion, affine, quasi-affine, finite, quasi-finite, integral, flat, faithfully flat, neat, smooth, étale

(EGA IV 2.6.4, 2.7.1 and EGA IV 17.7.1). On the other hand, endow (Sch) with the pretopology for which, for every scheme $X \in \mathscr{U}$, $\mathrm{Cov}(X)$ consists of the families of morphisms $(f_i : X_i \to X)_{i\in I}$ which are surjective and for which the f_i are flat and locally of finite presentation. Each of the following properties is then *of local nature upstairs*:

locally of finite type, locally of finite presentation, of finite type, flat, neat, smooth, étale.

(EGA IV 17.7.5, 17.7.7).

426 Exercise 8.6. Let $f : E \to F$ be a morphism of $\mathscr{U}$-topoi. Consider the following relation in the argument $X \in \mathrm{ob}\, F$: the induced morphism $E_{/f^*(X)} \to F_{/X}$ is an equivalence of topoi. Prove that this relation is of local nature.

Exercise 8.7. (*Partitions of a topos, sum of topoi*). Let E be a $\mathscr{U}$-topos. A family $(e_i)_{i\in I}$ of opens of E, i.e. of subobjects of the terminal object e of E, is called a *partition of e* (or of the topos E) if the canonical morphism $\coprod_{i\in I} e_i \to e$ is an isomorphism, i.e. (II 4.6.2) if e is the supremum of the e_i and $i \neq j$ implies $e_i \cap e_j = \varnothing_E$.

a) Show that a family $(e_i)_{i\in I}$ of objects of E is a partition of E if and only if the functor

$$E \longrightarrow \prod_i E_{/e_i} \tag{8.7.1}$$

defined by the localization functors $E \to E_{/e_i}$ is an equivalence of categories.

b) Conversely, let $(E_i)_{i\in I}$ be a family of $\mathscr{U}$-topoi, with $I \in \mathscr{U}$. Prove that $E = \Pi_{i\in I} E_i$ is a $\mathscr{U}$-topos (called the *sum topos* of the family $(E_i)_{i\in I}$). Define a partition $(e_i)_{i\in I}$ of E and equivalences of categories $E_{/e_i} \approx E_i$ in such a way that the projection functors $E \to E_i$ identify with the localization functors $E \to E_{/e_i}$.

c) Let E be the sum topos of the family $(E_i)_{i\in I}$. Show that, for every $i \in I$, the projection functor $E \to E_i$ is of the form u_i^*, where

$$u_i : E_i \longrightarrow E \tag{8.7.2}$$

is a morphism of topoi. Prove that, for every $\mathcal{U}$-topos F, the functor $v \mapsto (v \circ u_i)_{i\in I}$ 427

$$\mathbf{Homtop}(E, F) \longrightarrow \prod_i \mathbf{Homtop}(E_i, F) \tag{8.7.3}$$

is an equivalence of categories.

d) Let $(e_i)_{i\in I}$ be a partition of the topos E. For every morphism of topoi $g : F \to E$, the family $(f_i)_{i\in I}$, where $f_i = g^*(e_i)$, is a partition of F, and the induced morphisms $g_i : E_{/e_i} \to F_{/f_i}$ make it possible to reconstruct g (up to unique isomorphism), the functor g^* identifying with the Cartesian product of the functors g_i^* (taking into account the equivalences of the type in 8.7 1)). Deduce a complete description of the category $\mathbf{Homtop}(F, E)$ in terms of categories of the form $\mathbf{Homtop}(F', E_i)$, where F' is a topos of the form $F_{/f}$ (with f a direct summand of the terminal object of F) and $E_i = E_{/e_i}$.

e) In particular, if F is connected and nonempty (cf. 4.3.5), i.e. if, for every partition $(f_i)_{i\in I}$ of F, there exists a unique $i \in I$ such that $f_i \neq \phi_F$, prove that the canonical functor

$$\coprod_{i\in I} \mathbf{Homtop}(F, E_i) \longrightarrow \mathbf{Homtop}(F, E) \tag{8.7.4}$$

induced by the morphisms of topoi (8.7.2) is an equivalence of categories. More particularly, the family of morphisms of topoi (8.7.2) induces an equivalence of categories

$$\coprod_{i\in I} \mathbf{Point}(E_i) \xrightarrow{\approx} \mathbf{Point}(E).$$

f) A subset I of the set of opens of E is called a *reduced partition* of E if the identity family $(e_i)_{i\in I}$ indexed by I is a partition and the e_i are different from ϕ_E. Show that the refinement relation (I 4.3.2) between reduced partitions makes the set P of reduced partitions into a $\mathcal{U}$-small ordered set in which the supremum of any two elements exists. This gives a strict projective system $(I_p)_{p\in P}$, regarded as a pro-set and denoted by $\pi_\circ(E)$. It is called the *pro-$\pi_\circ$ of the topos E*. 428

g) Prove that $\pi_\circ(E)$ pro-represents the functor

$$T \longmapsto f_* f^*(T) = \Gamma(E, T_E) : (\mathcal{U}\text{-}\mathrm{Ens}) \longrightarrow (\mathcal{U}\text{-}\mathrm{Ens})$$

associated with the canonical morphism $f : E \to P$ from E to the punctual topos P (4.3). In particular, this functor is representable if and only if $\pi_\circ(E)$ is essentially constant, i.e. isomorphic (as a pro-set) to an "ordinary" set, again denoted $\pi_\circ(E)$. The pro-set $\pi_\circ(E)$ consists of one point if and only if E is "connected and nonempty." It is empty if and only if E is the "empty topos" (2.2).

h) Suppose that E is quasi-compact, i.e. every covering of its terminal object e admits a finite subcovering. Show that $\pi_\circ(E)$ is then a profinite set and hence may be identified (using the familiar equivalence between pro-objects of the category of finite sets and compact totally disconnected spaces) with a compact totally disconnected space, also denoted $\pi_\circ(E)$.

i) Let X be a topological space, and let $\pi_\circ(X)$ be the set of connected components of X, regarded as a constant pro-set. Define a canonical morphism of pro-sets 429

$$\pi_\circ(X) \longrightarrow \pi_\circ(\mathrm{Top}(X)), \tag{8.7.5}$$

or equivalently a canonical map

$$\pi_\circ(X) \longrightarrow \varprojlim \pi_\circ(\mathrm{Top}(X)). \tag{8.7.6}$$

Show that the first morphism is an isomorphism if and only if X is homeomorphic to a sum space of connected spaces (which is the case, in particular, when X is locally connected).

j) Suppose that X is a quasi-compact scheme. Prove that (8.7.6) is bijective.

k) Establish the "functorial character" in X of the morphisms (8.7.5) and (8.7.6), first making precise the meaning of this expression.

l) (Compare 7.6 i)). Show that the following two conditions on the $\mathscr{U}$-topos E are equivalent: (i) The canonical morphism from E to the punctual topos is "essential" (Exercise 7.6 a)), i.e. the functor $I \mapsto I_E$ from ($\mathscr{U}$-Ens) to E commutes with products indexed by sets in $\mathscr{U}$. (ii) For every object X of E, the induced topos $E_{/X}$ has a $\pi_\circ(E_{/X})$ which is an ordinary set, i.e. X is isomorphic to the sum of a family of connected objects of E. In this case E is called a *locally connected topos* (compare 2.7.5).

Exercise 8.8. (*Dominant morphisms of topoi*).

a) Let $f : E' \to E$ be a morphism of topoi. Show that the following conditions are equivalent:
   (i) $f_*(\phi_{E'}) = \phi_E$ (where ϕ_E and $\phi_{E'}$ denote the initial objects of E and E', respectively).
430 (ii) For every object X of E which is "nonempty," i.e. not isomorphic to ϕ_E, the object $f^*(X)$ is nonempty in E'.
   (ii′) As in (ii), with X a subobject of the terminal object e_E of E, i.e. with X an open of the topos E.
   The morphism of topoi f is then called *dominant*.

b) Let $f : X' \to X$ be a continuous map of topological spaces, giving a morphism of topoi $\mathrm{Top}(f) : \mathrm{Top}(X') \to \mathrm{Top}(X)$ (4.1). Show that $\mathrm{Top}(f)$ is dominant if and only if f is dominant, i.e. if $f(X')$ is dense in X.

c) Show that if $f : E' \to E$ is a "conservative" morphism of topoi (6.4.0), i.e. if f^* is faithful, then f is dominant. Show that the converse need not hold, even for a localization morphism $E_{/U} \to E$, where U is an open of the topos E. (Show that in this case f is conservative only if U is the terminal object of E (and hence only if f is an equivalence of topoi), and use the example in b).)

d) Let $f : X' \to X$ be an arrow in a topos E. We call f a *dominant morphism* in the topos E if the corresponding morphism of induced topoi $E_{/X'} \to E_{/X}$ (5.5.2) is dominant. Show that this property depends only on the image $f(X')$ of X' in X, as an element of the ordered set $\mathrm{ouv}(X)$ of subobjects of X. Show that the morphism $E_{/X'} \to E_{/X}$ is conservative if and only if f is covering.

9. Subtopoi and gluing of topoi

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9.1. Subtopoi and embeddings of topoi.

DEFINITION 9.1.1. a) *A morphism of topoi $f : F \to E$ is called an embedding if the functor $f_* : F \to E$ is fully faithful (i.e. if the adjunction morphism $f^* f_* \to \mathrm{id}_F$ is an isomorphism).*

b) *A subtopos of a topos E is a strictly full subcategory E' of E which is a topos and for which the inclusion functor $\alpha : E' \to E$ is of the form i_*, where $i : E' \to E$ is a morphism of topoi (i.e. (3.1.1) α admits a left adjoint i^* which is left exact). The morphism i (which is determined up to unique isomorphism) is then called the inclusion morphism of the subtopos E' in E.*

REMARK 9.1.2. a) It is clear that the inclusion morphism of a subtopos is an embedding, and that a subcategory E' of a topos E is a subtopos if and only if it is the essential image of a direct-image functor f_* associated with a morphism of topoi $g : F \to E$ which is an embedding.

b) It follows immediately from the definitions that a morphism of topoi $f : F \to E$ is an embedding if and only if it is isomorphic to a composite morphism

$$F \xrightarrow{g} E' \xrightarrow{i} E,$$

where g is an equivalence of topoi and i is the inclusion morphism of a subtopos E' of E. The latter is uniquely determined by the preceding conditions as the essential image of f_*, and the preceding factorization is likewise unique up to unique isomorphism. In practice, of course, one should use g to identify F with the subtopos E' of E (compare 3.4.1).

c) An equivalence of topoi (3.4) $u : E \xrightarrow{\approx} E_1$ defines in the evident way a bijection between the set of subtopoi of E and the set of subtopoi of E_1. Indeed, since u_* is an equivalence of categories, it establishes a bijection between the set of 432
strictly full subcategories of E and the set of strictly full subcategories of E_1 (to $E' \subset E$ corresponds the essential image E'_1 of $u_*|E'$), and one sees immediately that E' is a subtopos of E if and only if E'_1 is a subtopos of E_1. Thus, in studying subtopoi, one may reduce to the case where E is of the form $\widetilde{C}$, with C a small site. In this case, II 5.5 shows that a strictly full subcategory E' of E is a subtopos if and only if its inclusion functor $\alpha : E' \to E$ admits a left adjoint which is left exact (which already implies that E' is a topos), and that these subtopoi are in one-to-one correspondence with the topologies T' on C finer than the given topology T of C: to such a T' one associates the strictly full subcategory $(C, T')^\sim$ of $C^\sim$ formed by the sheaves for the topology T'. This also shows, for *every* $\mathcal{U}$-topos E, that a strictly full subcategory E' of E is a subtopos if and only if its inclusion functor $\alpha : E' \to E$ admits a left adjoint which is left exact; *the ordered set $\mathfrak{S}(E)$ of subtopoi of E is $\mathcal{U}$-small, and every subset of $\mathfrak{S}(E)$ admits a supremum and an infimum.*

d) If E' and E'' are two subtopoi of a topos E, their intersection is a subtopos of E. Indeed, since this intersection is strictly full, one reduces to the case where E is of the form $C^\sim$. With the notation of c), E' and E'' then correspond to two topologies T' and T'' on C finer than T, and it suffices to see that *$E' \cap E''$ is the category of sheaves for the topology T''' which is the supremum of T' and T''*. We indicate the proof of this fact (which should have appeared as a corollary after II 4.4 or II 5.5!). Let E''' be the category of sheaves for T''', plainly contained in E' and E'', and hence in $E' \cap E''$. For every full subcategory D of E, let $t(D)$ be the finest topology among those for which the objects of D are sheaves (II 2.2). The inclusions $E''' \subset E' \cap E'' \subset E', E''$ plainly imply $t(E'), t(E'') \leq$ 433

$t(E' \cap E'') \le t(E''')$. On the other hand, (II 4.4.4) $t(E') = T'$, $t(E'') = T''$, and $t(E''') = T'''$, so the preceding inequalities and the definition of T''' as the supremum imply $T''' = t(E' \cap E'')$, and hence $E' \cap E'' \subset E'''$, C.Q.F.D. More generally, this argument shows that if T' is the supremum of an arbitrary family of topologies (T_i), the category E' of sheaves for T' is the intersection of the categories E_i of sheaves for the T_i. In particular, one obtains:

Proposition 9.1.3. Let E be a topos. For every family of subtopoi $(E_i)_{i \in I}$ of E, the intersection $\bigcap_{i \in I} E_i$ is a subtopos of E.

Proposition 9.1.4. *Let $i : E' \to E$ be an embedding of topoi, and let F be a topos. The functor*

(9.1.4.1) $$f \longmapsto i \circ f \,:\, \mathbf{Homtop}(F, E') \longrightarrow \mathbf{Homtop}(F, E)$$

is fully faithful, and its essential image consists of the morphisms of topoi $g : F \to E$ for which the essential image of g_* is contained in that of i_*.

Indeed, consider the evident commutative diagram

$$\begin{tikzcd}
\mathbf{Homtop}(F, E') \arrow[r] \arrow[d, hook] & \mathbf{Homtop}(F, E) \arrow[d, hook] \\
\mathbf{Hom}(F, E') \arrow[r] & \mathbf{Hom}(F, E),
\end{tikzcd}$$

where the vertical arrows are the functors $h \mapsto h_*$, which are fully faithful by the definition of **Homtop** (3.2). The second horizontal arrow $G \mapsto i_* G$ is also fully faithful, since i_* is, and therefore so is the first horizontal arrow. It remains to prove the characterization of the essential image of (9.1.4.1). The necessity of the condition is clear from the formula $(if)_* = i_* f_*$. Conversely, suppose that the essential image of g_* is contained in that of i_*, i.e. that one may write $g_* = i_* f_*$, with $f_* : F \to E'$ a functor. We must show that g belongs to the essential image of (9.1.4.1). This amounts to saying that f_* admits a left
434 adjoint which is left exact. Now it is clear that f_* admits $g^* i_*$ as a left adjoint, and this functor is left exact, being the composite of the left exact functors g^* and i_*, C.Q.F.D.

For a converse to 9.1.4, cf. Exercise 9.1.6.

Taking F to be the punctual topos (2.2), one obtains in particular from 9.1.4:

Corollary 9.1.5. *If $f : E' \to E$ is an embedding morphism of topoi, the corresponding functor*

(9.1.5.1) $$\mathrm{Point}(f) : \mathbf{Point}(E') \longrightarrow \mathbf{Point}(E)$$

is fully faithful.

Exercise 9.1.6. (*Embeddings of topoi and 2-monomorphisms*).

Let $i : E' \to E$ be a morphism of topoi.

a) If F and G are two topoi *over* E', define (with the notation of 5.14 a)) a canonical functor

(9.1.6.1) $$f \longmapsto f * i \,:\, \mathbf{Homtop}_{E'}(F, G) \longrightarrow \mathbf{Homtop}_E(F, G).$$

b) Suppose that i, as a 1-arrow of the 2-category of $\mathscr{V}$-$\mathscr{U}$-topoi (3.3.2), is a 2-monomorphism, i.e. that for every $\mathscr{U}$-topos F which is an element of $\mathscr{V}$, the functor (9.1.4.1) is fully faithful. Show that, for every pair of $\mathscr{U}$-topoi F, G (not necessarily in $\mathscr{V}$), the functor (9.1.6.1) is fully faithful.

c) Show that i is an embedding if and only if i is a 2-monomorphism. (For sufficiency, apply b) to induced topoi $E'_{/X'}$ and $E'_{/Y'}$, and use 5.14 c)).

d) Let $g : F \to E$ be a morphism of topoi, and suppose that the 2-fiber product (5.11) $F' = F \times^2_E E'$ exists (a condition which is always satisfied, as P. DELIGNE has shown[12]). Let $j : F' \to F$ be the projection morphism. Prove that if i is an embedding, then so is j. (Use c), and note that the stability of the notion of a 2-monomorphism under 2-base change is formal.) When E' is a subtopos of E and $i : E' \to E$ is the canonical inclusion morphism, the essential image of F' under j_* (which is a subtopos of F independent of the indeterminacy in the 435
construction of a 2-fiber product) is called the *subtopos of F which is the inverse image under $g : F \to E$ of the subtopos E' of E.*

e) With the notation at the end of d), choose as 2-fiber product the subtopos of F which is the inverse image of the subtopos E' of E. Let X be an object of F. Show that, for X to belong to F', it is necessary that it satisfy the following condition: for every object U of F, writing $g_U : F_{/U} \to E$ for the morphism of topoi induced by g, one has $g_{U*}(X|U) \in \operatorname{Ob} E'$, where $X|U = (X \times U \xrightarrow{\mathrm{pr}_2} U)$ is the "restriction of X to U." (NB: Note that if $i_U : F_{/U} \to F$ is the canonical inclusion, then $g_U = g \circ i_U$, whence $g_{U*}(X|U) = g_*(i_{U*}(X|U)) = g_*(\mathbf{Hom}(U, X))$, where $\mathbf{Hom}(U, X)$ is the object defined in 10.1 below.)
Problem. Conversely, is the preceding condition on X sufficient for X to belong to F'? This amounts to asking whether the strictly full subcategory of F formed by these objects X is a *subtopos* of F.

f) Let X be an object of E, let $X' = i^*(X)$, and let $i_{/X} : E'_{/X'} \to E_{/X}$ be the morphism of topoi induced by i ((5.10.1)). Show that if i is an embedding, then so is $i_{/X}$. (Use d) and 5.11.) Conversely, if $(X_\alpha)_\alpha$ is a family of objects covering the terminal object, and if $i_{/X_\alpha}$ is an embedding for every α, then so is i.

EXERCISE 9.1.7. (*Subtopoi and multiplicative systems of arrows*).

a) Let $i : E' \to E$ be a morphism of topoi, let S be the set of arrows u of E such that $i^*(u)$ is an isomorphism, and consider the canonical functor induced by i^* (cf. [2, Chap. I] or the proof of VI 6.2):

$$\rho : E[S^{-1}] \longrightarrow E'. \tag{9.1.7.1}$$

Show that i is an embedding if and only if the preceding functor ρ is an equivalence of categories, and that in this case S admits both a calculus of left fractions and a calculus of right fractions. (Use [2, Chap. I, 1.3].) In this case, the essential image of i_* can be recovered from S as consisting of those $X \in \operatorname{Ob}(E)$ such that for every $u : Y \to Y'$ in S, the map $\operatorname{Hom}(u, X) : \operatorname{Hom}(Y', X) \to$ 436
$\operatorname{Hom}(Y, X)$ is bijective.

b) We henceforth assume that i is an embedding. Show that, for every topos F, the functor

$$f^* \longmapsto f^* i^* : \mathbf{Hom}(E', F) \longrightarrow \mathbf{Hom}(E, F)$$

induces an equivalence between the category of the f^* arising from morphisms of topoi $f : F \to E'$ (i.e. which are left exact and admit a right adjoint) and the category of functors $g^* : E \to F$ arising from morphisms of topoi $g : F \to E$ and such that g^* takes arrows of S to isomorphisms.

[12]N.D.E.: See also [16], Proposition 4.47.

c) Let S be a subset of $Fl\ E$, where E is a $\mathcal{U}$-topos. Show that S is associated with a subtopos E' of E by the procedure of a) if and only if S satisfies the following conditions:

ST 1) S contains the invertible arrows and is stable under composition and base change.

ST 2) If a composite vu belongs to S, then $u \in S$ if and only if $v \in S$.

ST 3) If $u : X \to Y$ is such that there exists a covering family $Y_i \to Y$, $i \in I$, for which $u_i : X_i = X \times_Y Y_i \to Y_i$ belongs to S for every $i \in I$, then $u \in S$.

Show that this gives a one-to-one correspondence between the set of subtopoi E' of E and the set of subsets S of $Fl\ E$ satisfying conditions ST 1), ST 2), and ST 3). (Given S, associate with it a $\mathcal{U}$-topology T', finer than the canonical topology T of E, whose covering families $(X_i \to X)_{i \in I}$ are those for which the inclusion $X' \hookrightarrow X$ of the image of the X_i is $\in S$. Show that the category of sheaves for T' is equivalent to a subtopos E' of E, and use II 4.4 to prove that it recovers S.)

d) Show that S is determined by the subset S_0 of S consisting of the *monomorphisms*. Show that the map $S \mapsto S_0$ establishes a bijection between the set of subsets S of $Fl\ E$ satisfying conditions ST 1) through ST 3) of c) and the
437 set of subsets of $Fl\ E$ consisting of monomorphisms and satisfying the same conditions ST 1) through ST 3).

e) Let $(E'_i)_{i \in I}$ be a family of subtopoi of E, and let $(S_i)_{i \in I}$ be the corresponding family of subsets of $Fl\ E$. Show that $\bigcap_{i \in I} S_i$ satisfies conditions ST 1) through ST 3) of c), and hence corresponds to a subtopos E' of E, and that the latter is the *subtopos generated by the* E'_i, i.e. the *supremum* (cf. 9.1.2 c)) of the subtopoi E'_i.

f) Let A be a subset of $Fl\ E$. Show that there is a smallest subset S of $Fl\ E$ among those containing A and satisfying conditions ST 1) through ST 3), and that S corresponds to the largest subtopos E' of E such that the corresponding multiplicative system of $Fl\ E$ contains S. Show that S is equal to the union of the subsets A_n ($n \geq 0$) of $Fl\ E$, constructed inductively as follows: $A_0 = A$, and A_{n+1} consists of the arrows $u : X \to Y$ which are of one of the following four types (i) through (iv):

(i) u is invertible, or is the composite of two arrows of A_n, or is obtained from an arrow of A_n by base change.

(ii) u is one of the factors of a composite arrow whose other factor and whose composite both belong to A_n.

(iii) u is the coproduct of a small family of arrows $\in A_n$.

(iv) There exists an epimorphism $Y' \to Y$ such that $u' = X' = X \times_Y Y' \to Y'$, obtained from u by base change, belongs to A_n.

g) With the notation of e), let A be the union of the S_i, let S be the subset of $Fl\ E$ obtained from A by the procedure of f), and let E' be the corresponding subtopos of E. Show that E' is the intersection (= infimum) of the subtopoi E'_i.

Exercise 9.1.7.2. (*Canonical factorization of a morphism of topoi*).

a) Let $f : F \longrightarrow E$ be a morphism of topoi. Show that, up to isomorphism, one can find a factorization of f as

$$F \xrightarrow{f'} E' \xrightarrow{i} E, \tag{9.1.7.3}$$

where f' is *conservative* (i.e. (6.4.0) f'^* is conservative, or equivalently faithful) and i is an embedding, and that this factorization is unique up to equivalence 438
(cf. 9.1.4). To do this, introduce the set $S_f \subset Fl\ E$ of arrows u of E such that $f^*(u)$ is an isomorphism, prove that S_f satisfies the conditions of 9.1.7 c), take for E' the corresponding subtopos of E, and define f'^* by (9.1.7.1), which corresponds to a morphism of topoi f' by 9.1.7 b).

b) The subtopos of E defined by the embedding i is called the subtopos of E which is the *image of the morphism of topoi* f. Show that it is the smallest of the subtopoi E_1 of E such that f factors, up to isomorphism, through E_1, i.e. (9.1.4) the smallest of the subtopoi of E containing the image of f_* (or equivalently the essential image of f_*). For f to be an embedding, it is necessary and sufficient that f induce an equivalence f' of F with its image; for f to be conservative, it is necessary and sufficient that its image be all of E.

c) Suppose that $f = \mathrm{Top}(f_0)$ is associated with a continuous map of sober topological spaces $f_0 : Y \to X$ (4.1). Consider the canonical factorization of f_0 as

$$Y \xrightarrow{f'_0} X' = f_0(Y) \xrightarrow{i_0} X,$$

where i_0 is the inclusion. Show that the canonical factorization of f then identifies with the corresponding factorization

$$\mathrm{Top}(Y) \xrightarrow{\mathrm{Top}(f'_0)} \mathrm{Top}(X') \hookrightarrow^{\mathrm{Top}(i_0)} \mathrm{Top}(X).$$

Let Z be the smallest sober subspace of X containing $X' = f_0(Y)$, i.e. the set of points x of X such that $\{\overline{x}\} \cap X'$ is dense in $\{\overline{x}\}$. (Note that $Z = X'$ if the points of X are closed, or if X' is a *constructible* subset of the locally Noetherian space X.) Then f is conservative, i.e. the subtopos of $E = \mathrm{Top}(X)$ which is the image of $F = \mathrm{Top}(Y)$ under $f = \mathrm{Top}(f_0)$ is equal to E itself, if and only if $Z = X$, i.e. if and only if $f_0(Y)$ is *very dense* (EGA IV 10.13) in X. This makes precise the extent to which the notion of a conservative morphism of topoi may legitimately be regarded as the natural generalization of the notion of a *surjective* continuous map of topological spaces.

d) Let 439

$$\begin{tikzcd} Y \arrow[d] & Y' \arrow[l] \arrow[d] \\ X & X' \arrow[l] \end{tikzcd}$$

be a Cartesian diagram of topological spaces. If Y and X' are discrete spaces, and hence Y' is discrete, then the corresponding diagram of topoi is a 2-fiber product diagram (5.11). (NB. For a counterexample when Y and X' are no longer assumed discrete, with Y and X' subspaces of the Hausdorff space X, cf. 9.1.10 c)). Deduce an example of a 2-fiber product diagram of topoi

$$\begin{tikzcd} F \arrow[d, "f"'] & F' \arrow[l] \arrow[d, "f'"] \\ E & E' \arrow[l] \end{tikzcd}$$

such that f is conservative but f' is not conservative, more precisely with $E' =$ the punctual topos and $F' =$ the empty topos (2.2). (Take Y such that the image of Y in X is very dense and $\neq X$, and take X' to consist of a point not belonging to the image.)

e) Let $(E_i)_{i\in I}$ be a family of subtopoi of a topos E. Prove that E is the supremum of the E_i if and only if the family of embedding morphisms $E_i \to E$ is conservative (6.4.0).

Exercise 9.1.8. (*Subtopoi and points of E. The case of topological spaces*).
Let E be a topos.

a) Let Z be a full subcategory of $\mathbf{Point}(E)$ (or, what amounts to the same thing, a subset of Ob $\mathbf{Point}(E)$). Show that the set $S(Z)$ of arrows of E which are taken to bijections by the fiber functors corresponding to the $z \in \mathrm{Ob}\, Z$ satisfies the conditions of 9.1.7 c), and hence defines a subtopos E_Z of E. Show that this subtopos has enough points, and that Z is equivalent to a full subcategory of $\mathbf{Point}(E_Z)$. Show that every subtopos E' of E which has enough points can be obtained by the preceding procedure. (Take for Z the essential image
440 of the functor (9.1.5.1).) Show that one obtains a bijection between the set of subtopoi E' of E having enough points and a subset of the set of strictly full subcategories Z of $\mathbf{Point}(E)$ which are stable under direct factors (I 10.6) and under small filtered projective limits (cf. 7.3). (Problem: Which subcategories of $\mathbf{Point}(E)$ arise in this way? This is essentially, in another form, the problem already raised in 6.5.3, in view of the dictionary in 9.1.2 c). Note that, besides the necessary conditions just indicated, there are also subtler conditions on the topological nature of $\mathrm{Point}(E)$, such as sobriety; cf. e)).

b) Let $(Z_i)_{i\in I}$ be a family of full subcategories of $\mathbf{Point}(E)$, and let Z be the full subcategory which is the union of the Z_i. Show that the corresponding subtopos E_Z defined in a) is the subtopos of E generated by the E_{Z_i}. (Use 9.1.7 e).) Conclude that the subtopos of E generated by a family of subtopoi having enough points also has enough points.

c) Let $f : E' \to E$ be an embedding of topoi. Show that the map $U \mapsto f^*(U)$

$$\mathrm{Ouv}(E) \longrightarrow \mathrm{Ouv}(E') \tag{9.1.8.1}$$

is surjective, and conclude that the map

$$\mathrm{Point}(E') \longrightarrow \mathrm{Point}(E) \tag{9.1.8.2}$$

induced by f on the isomorphism classes of points is a homeomorphism from $\mathrm{Point}(E')$ onto a subspace of $\mathrm{Point}(E)$, for the topologies defined in 7.8 a). (For injectivity, use 9.1.5.)

d) Let $f : X' \to X$ be a continuous map, giving a morphism of topoi (4.1)

$$\mathrm{Top}(f) : \mathrm{Top}(X') \longrightarrow \mathrm{Top}(X).$$

Prove that the latter is an embedding if and only if the topology of X' is the inverse image under f of that of X (i.e. , when X' is a Kolmogorov space, for example a sober space, if and only if f induces a homeomorphism of X' onto a
441 subspace of X). (For the necessity, use c), and for the sufficiency use 9.1.7.2 c)).

e) Let X be a sober topological space, and let $E = \mathrm{Top}(X)$. Using a) and d), establish an isomorphism of ordered sets $X' \mapsto E_{X'}$ between the set of sober subspaces X' of X (i.e. , when X is Hausdorff, the set of *all* subsets of X) and the set of subtopoi of E having enough points. Show that $E_{X'}$ is equivalent to $\mathrm{Top}(X')$. If $(X'_i)_{i\in I}$ is a family of sober subspaces of X, show that there is a smallest sober subspace X' of X containing the X'_i, equal to the union of the

X'_i if I is finite or X is Hausdorff, and that $E_{X'}$ is the subtopos of E generated by the $E_{X'_i}$. (Use b)).

f) Give an example of a topos E of the form $\mathrm{Top}(X)$ and a subtopos E' which does not have enough points, and which is even nonempty (2.2) but has no points. (Take E of the form $\widehat{U}$ and $E' = U^{\sim}$, where U is the ordered set defined in 7.4. Or better, take the example in 9.1.10 b), which shows that one may take for X any nonempty compact space without isolated points.)

Exercise 9.1.9. (*The case of topoi defined by an ordered set*).

a) We use the yoga of 7.8 b) and c), which gives a dictionary between, on the one hand, topoi generated by their opens and morphisms of such topoi, and, on the other hand, ordered sets admitting arbitrary Sup, finite infima, and in which the infimum of two elements distributes over arbitrary Sup; morphisms between such ordered sets are increasing maps commuting with finite infima and arbitrary Sup.
b) Let E be a topos and $f : E' \to E$ an embedding of topoi. Show that, for every generating family $(X_i)_{i\in I}$ in E, the family $(f^*(X_i))_{i\in I}$ is generating. Deduce that if E is generated by its opens, then the same is true of E'. (Use 9.1.8 c).)
c) Let $f : E' \to E$ be a morphism of topoi generated by their opens, associated with a morphism $g : \mathcal{O} \to \mathcal{O}'$ of the corresponding ordered sets. Show that f is an embedding if and only if g is surjective, *and* g is surjective on the graphs 442
of the order relations, i.e. the order on $\mathcal{O}'$ is the quotient of that on $\mathcal{O}$.
d) Let E be a topos of the form $\mathcal{O}^{\sim}$ as in a). Deduce from b) and c) a bijective correspondence between the set of subtopoi of E and the set of equivalence relations R on $\mathcal{O}$ having the following properties, analogous to conditions ST 1) to ST 3) of 9.1.7 c):
   - Q0 1) The relation R is compatible with finite infima; in other words, if U and U' are equivalent under R, then so are $U \cap V$ and $U' \cap V$ for every $V \in \mathcal{O}$.
   - Q0 2) The relation R is compatible with arbitrary Sup; in other words, if U_i is equivalent to U'_i for every $i \in I$, then $\mathrm{Sup}_i\, U_i$ is equivalent to $\mathrm{Sup}_i\, U'_i$.

   Show that if subtopoi $E^!_i$ of E correspond to the equivalence relations R_i, then the subtopos generated by the $E^!_i$ corresponds to the intersection of the relations R_i.

Exercise 9.1.10. (*Intersections of subspaces; new topoi without points; nonexistence of a complement to a subtopos*).

Let X be a topological space, $E = \mathrm{Top}(X) = \mathrm{Ouv}(X)^{\sim}$ (2.1).

a) Let R be the following relation on $\mathrm{Ouv}(X)$: $(U, U') \in R \iff U \cap U'$ is dense in both U and U'. Show that R is an equivalence relation satisfying conditions Q0 1) and Q0 2) of 9.1.9 d), and therefore defines a subtopos E' of E. Show that, for a point $x \in X$, the corresponding point belongs to the essential image of $\mathbf{Point}(E')$ if and only if x is *thick*, i.e. belongs to the closure of the interior of $\overline{x}$. Show that E' is the empty topos (2.2) if and only if X is empty.
b) If X is sober, show that the category $\mathbf{Point}(E')$ is equivalent to the category defined by the ordered subset of X (for the specialization relation) consisting of the thick points of X. Show that if the points of X are closed (resp. if X is Noetherian), then the thick points of X are the isolated points of X, i.e. those for which $\{x\}$ is open (resp. are the maximal points of X). Conclude that if X is a nonempty Hausdorff topological space without isolated points, then the 443

subtopos E' of $E = \mathrm{Top}(X)$ is a non-"empty" topos having no points.

c) Let X' be a subset of X, and let $E_{X'}$ be the subtopos of E associated with X' by the construction of 9.1.8 a). Show that if X' is dense, then $E_{X'}$ contains E'. Show that if X is sober and X' and X'' are two sober subsets of X, then their intersection $X' \cap X''$ is sober and $E_{X' \cap X''} \subset E_{X'} \cap E_{X''}$, and that the inclusion can be strict [12]. (Take a nonempty Hausdorff space X having two disjoint dense subsets X', X''.) Show that the preceding inclusion is an equality if and only if the subtopos $E_{X'} \cap E_{X''}$ has enough points. Show that this is the case if X' or X'' is a *locally closed* subset of X.

d) Let X' be a subset of X, and let X'' be the complement of X'. A subtopos F of E contains the $E_{\{x\}}$ for $x \in X''$ if and only if it contains $E_{X''}$ (cf. 9.1.8 b)). Conclude that if X'' is a union of locally closed subsets of X (for example, if the points of X'' are closed), then the set of subtopoi F of E whose intersection with $E_{X'}$ is the empty subtopos $\mathrm{Top}(\varnothing)$ has a greatest element (which then deserves the name of *weak complement of* $E_{X'}$ in E; cf. 9.1.13) *only if* $E_{X'} \cap E_{X''}$ is the empty subtopos (a condition that is not satisfied if $X \neq \varnothing$ and if X' and X'' are both dense in X, with X, X', and X'' sober; cf. c)). Show that when this condition fails, the infimum of two elements in the ordered set of subtopoi of E is not distributive with respect to arbitrary Sup. (The editor does not know whether it is distributive with respect to finite Sup; cf. , however, 9.1.11 e) and 9.1.12 d)).

Exercise 9.1.11. (*Subtopoi of locally Noetherian topoi. The case of Noetherian spaces*).

a) Let $f : E' \to E$ be an embedding of topoi. Show that if X is a pre-Noetherian object of E (i.e. (VI 1.30) every increasing sequence of subobjects of X is sta-
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tionary), then $X' = f^*(X)$ is a pre-Noetherian object of E'. (By introducing the induced topoi $E'_{/X'}$ and $E_{/X}$ and using 9.1.6 e), first reduce to the case where X is the final object e of E, so that X' is the final object e' of E'. Then observe that every increasing sequence of subobjects of e' defines, upon applying f_*, an increasing sequence of subobjects of $f_*(e') = e$.) Conclude that if E admits a generating family consisting of pre-Noetherian objects (resp. if E is a Noetherian topos, resp. locally Noetherian (VI 2.11)), then E' satisfies the same condition. (Use 9.1.9 b).)

b) Let E be a locally Noetherian topos (VI 2.11). Show that every subtopos of E is of the form E_Z (9.1.8 a)), for a suitable full subcategory Z of $\mathbf{Point}(E)$. (Use a) and Deligne's theorem that every locally Noetherian topos has enough points.) Conclude that if E is of the form $\mathrm{Top}(X)$, where X is a sober locally Noetherian topological space, then $Z \mapsto E_Z$ is an isomorphism from the ordered set of sober subspaces of X to the ordered set of subtopoi of E. (Use 9.1.8 e).) The intersection of sober subspaces Z_i (is necessarily sober and) defines the intersection of the subtopoi E_{Z_i} (in contrast with what can occur in the general case (9.1.10 c)).

c) Let X be a sober locally Noetherian space, let X' be a subset of X, and let X'' be its complement. Show that the following conditions are equivalent:

(i) X' is constructible.

(ii) X' and X'' are sober.

(iii) X' is sober, and the subtopos $E_{X'} \approxeq \mathrm{Top}(X')$ of $E = \mathrm{Top}(X)$ admits a "complementary" subtopos (cf. 9.1.13).

[12] See, however, 9.1.11 b) below for the case where X is locally Noetherian.

(iv) The intersection of the subtopoi $E_{X'} \approx \mathrm{Top}(X')$ and $E_{X''} \approx \mathrm{Top}(X'')$ of $E = \mathrm{Top}(X)$ is the "empty" subtopos of E.

Show that if E' and E'' are two subtopoi of E, they are weak complements if and only if they are complements (9.1.13).

d) Let X be a sober Noetherian space. Show that the following conditions are equivalent: 445

(i) Every sober subspace X' of X is constructible, i.e. , according to c), its complement is sober.

(i bis) For every maximal point x of X, $X - \{x\}$ is sober, i.e. $\{x\}$ is constructible.

(ii) X is finite.

(iii) Every subtopos of $\mathrm{Top}(X)$ admits a complementary subtopos (cf. 9.1.13).

(iv) In the ordered set of subtopoi of $\mathrm{Top}(X)$, the infimum of two elements is distributive with respect to arbitrary Sup.

(Use the argument of 9.1.10 d), together with the intersection formula proved in b).)

e) Let X be a locally Noetherian space. Show that in the ordered set of subtopoi of $\mathrm{Top}(X)$, the infimum of two elements is distributive with respect to *finite* Sup. (Reduce to the case where X is sober, and use b).)

f) For every sober subspace Z of the sober locally Noetherian space X, let T_Z be the topology on the category $\mathrm{Ouv}(X)$ of open subsets of X for which a family of inclusions $U_i \to U$ is covering if and only if $U \cap Z = \bigcup_i U_i \cap Z$. Show that $Z \mapsto T_Z$ establishes a bijection (reversing inclusion relations) between the set of sober subspaces Z of X and the set of topologies on $\mathrm{Ouv}(X)$ finer than the canonical topology T_X. (Use b) and 9.1.2 c).)

g) Let X be a sober topological space. A subset X' of X is sober if and only if its complement is a union of locally closed subsets; if X is locally Noetherian, this also means that X' is "proconstructible" (EGA IV 1.9.4). Show that there is a topology $T_{\mathrm{cons}'}$ on X such that the closed subsets of $X_{\mathrm{cons}'} = (X, T_{\mathrm{cons}'})$ are the sober subsets of X. Assume henceforth that X is locally Noetherian, so that $X_{\mathrm{cons}'}$ is none other than the space X_{cons} of EGA IV 9.1.13. Deduce from b) and c) that the ordered set Σ of subtopoi of $\mathrm{Top}(X)$ is isomorphic to the ordered set of closed subsets of X_{cons}, and that, under this correspondence, the subtopoi admitting a complement correspond to the subsets that are both open and closed in X_{cons}. Show that X_{cons} is locally compact, and that it is compact if and only if X is Noetherian, i.e. the topos $\mathrm{Top}(X)$ is Noetherian. 446

h) Let E be a locally Noetherian topos. Suppose that it has the property considered in 9.1.14 b) (a condition that may perhaps always hold...). Show that the essential images of the canonical functors $\mathbf{Point}(E') \to \mathbf{Point}(E)$, where E' ranges over the set Σ of subtopoi of E, define a collection of subsets of the set $X = \mathrm{Point}(E)$ of isomorphism classes of points of E which is the collection of closed subsets for a *topology* T_{cons} *on* X, and that in this way one obtains an isomorphism of ordered sets between the set Σ of subtopoi of E and the set of closed subsets of X_{cons}. Conclude that in Σ, the Sup of two elements is distributive with respect to *arbitrary* infima. Show that X_{cons}, which is not necessarily $\mathscr{U}$-small, has an associated sober space (4.2.1) which is $\mathscr{U}$-small.

EXERCISE 9.1.12. (*Finite topoi*). A topos E is said to be *finite* if it is equivalent to a topos of the form $\widehat{C}$, with C a finite category.

a) (Dictionary). Let $\mathscr{V}$ be a universe such that $\mathscr{U} \in \mathscr{V}$. Let (Karfiness) be the 2-category defined as the full 2-subcategory of the category $(\mathrm{Cat})_{\mathscr{V}}$ consisting of categories that are elements of $\mathscr{V}$, are *Karoubian and equivalent to a finite category*, and let (Topfin) be the 2-category defined as the full 2-subcategory of ($\mathscr{V}$-$\mathscr{U}$-Top) (3.3.1) consisting of the *finite* $\mathscr{U}$-topoi that are $\in \mathscr{V}$. Show that there are mutually quasi-inverse 2-equivalences:

$$C \longmapsto \widehat{C} : (\mathrm{Karfiness}) \longrightarrow (\mathrm{Topfin}) \tag{9.1.12.1}$$

$$E \longmapsto \mathbf{Point}(E) : (\mathrm{Topfin}) \longrightarrow (\mathrm{Karfiness}). \tag{9.1.12.2.}$$

(Use 7.6 h), e).)

447 b) Show that a finite topos is Noetherian VI 2.11.

c) Show that every subtopos of a finite topos E is a finite topos. More precisely, if E is equivalent to $\widehat{C}$, where C is a finite Karoubian category (or, more generally, is equivalent to a finite category), show that one obtains an order isomorphism between the set of subtopoi E' of E and the set of strictly full subcategories C' of C that are Karoubian (or, equivalently, stable under taking direct factors in C), by associating with every C' the essential image of $f_* : \widehat{C}' \to \widehat{C}$, where $f : C' \to C$ is the inclusion functor. (Use 5.6 to verify that f_* is fully faithful, and b) and 9.1.11 b) for the fact that every subtopos E' of E is obtained in the indicated way.)

d) Let E be a finite topos. Construct on the finite set $X = \mathrm{Point}(E)$ of isomorphism classes of points of E a topology making it a sober topological space and such that the ordered set of subtopoi of E is canonically isomorphic to the set of closed subsets of X. (Take the "constructible topology" $\mathscr{C}_{\mathrm{cons}}$ on X, defined by the order relation $(x \le y) \Leftrightarrow (x$ is a direct factor of $y)$, for which the closure of a point x consists of the y such that $y \le x$.) In particular, the set Σ of subtopoi of E is finite, and the binary infimum operation $\wedge$ is distributive with respect to arbitrary Sup.

e) Let C be a category equivalent to a finite category. Show that, for every full subcategory C' of C, there exists a topology $T_{C'}$ on C for which a family $(X_i \to X)_{i \in I}$ is covering if and only if, for every $Y \in \mathrm{Ob}\, C'$, the family of maps $\mathrm{Hom}(Y, X_i) \to \mathrm{Hom}(Y, X)$ $(i \in I)$ is surjective. Show that the category of sheaves on $(C, T_{C'})$ is equivalent to $\widehat{C}'$, and that $C' \mapsto T_{C'}$ is a bijection (reversing the order relations) between the set of strictly full subcategories C' of C that are Karoubian (or, equivalently, stable under taking direct factors in C), and the set of topologies (II 1.1) on the category C. (Use *c*) and 9.1.2 c).)
448 In particular, if C is a site whose underlying category is equivalent to a finite category, then the topos $C^{\sim}$ is a finite topos.

f) Let $f : C' \to C$ be a functor of $\mathscr{U}$-categories, with C equivalent to a finite category. Show that the morphism of topoi $\widehat{f} : \widehat{C}' \to \widehat{C}$ (4.6) is conservative if and only if every object of C is isomorphic to a direct factor of an object in the image of f. (Reduce to the case where f is an inclusion $f : C' \hookrightarrow C$, with C' as in c).)

g) Let C be the category having two objects e (the final object) and a, and, besides the identity arrows, three arrows $f : a \to e$, $g : e \to a$, $p = gf$ (thus $p^2 = p$). Show that $E = \widehat{C}$ has, besides the subtopoi E and $\mathrm{Top}(\varnothing)$, exactly one subtopos E', namely $E' = \widehat{\{e\}}$; consequently, for every subtopos E'' of E with $E'' \neq \mathrm{Top}(\varnothing)$, one has $E' \cap E'' \neq \mathrm{Top}(\varnothing)$.

Exercise 9.1.13. (*Complementary subtopoi of a topos*).

Let E be a topos and let E' and E'' be two subtopoi. We say that E' and E'' are *complementary* if $E' \cap E'' = \mathrm{Top}(\varnothing)$ and $E' \vee E'' = E$ (where $\vee$ denotes the Sup in the ordered set of subtopoi of E).

Suppose that in the set Σ of subtopoi of E, the infimum of two elements is distributive over finite suprema.

a) If E' and E'' are complementary, then E'' is the largest among the subtopoi E''_1 such that $E' \cap E''_1 = \mathrm{Top}(\varnothing)$ (i.e. E'' is a weak complement of E', in the terminology of 9.1.10 d)).

b) Show that the following conditions on E are equivalent:

(i) If E' and E'' are two subtopoi of E such that E'' is a weak complement of E', then E'' is a complement of E'.

(ii) For every subtopos E' of E such that $E' \neq E$, there exists a subtopos E'' of E such that $E'' \neq \mathrm{Top}(\varnothing)$ and $E' \cap E'' = \mathrm{Top}(\varnothing)$.

(i bis) If E' and E'' are two subtopoi of E such that E'' is maximal among the subtopoi E''_1 satisfying $E' \cap E''_1 = \mathrm{Top}(\varnothing)$, then E'' is a complement of E'.

Show that even if E is a finite topos (9.1.12), these conditions need not hold 449
(9.1.12 g)). They do hold, however, if E is of the form $\mathrm{Top}(X)$, with X a locally Noetherian space. (Use 9.1.11 b)).

c) A subtopos E' of E is called *complemented* if it admits a complement (compare 9.1.12 c)). Prove that the set of complemented subtopoi is stable under finite infima and suprema and under complementation, and that complementation transforms infima into suprema and suprema into infima.

d) State the duals of observations a), b), and c) (which were in fact trivial general statements about an abstract ordered set Σ; note that the hypothesis made on Σ is itself self-dual).

9.1.14. *Open questions*[13]. Let E be a topos.

a) In the set of subtopoi of E, is the infimum of two elements distributive over finite suprema?

b) Let E' and E'' be two subtopoi of E such that $E = E' \vee E''$. Is every point of E then isomorphic to the image of a point of E' or a point of E''?

Note that if the answer to b) is affirmative for E and all its subtopoi, then the answer to a) is affirmative at least for subtopoi having enough points (and hence for all subtopoi if E is locally Noetherian (9.1.11 b)). The answer to a) (and a fortiori to b)) is nevertheless not known for all Noetherian topoi. The answers to a) and b) are affirmative, however, if E is of the form $\mathrm{Top}(X)$, with X a locally Noetherian space (9.1.11), or if E is a finite topos (9.1.12).

Note that a) can be reformulated as follows (applied to all subtopoi F of E):

a′) If E' and E'' are two subtopoi of a topos F such that the canonical morphism (cf. 8.7 b))

$$E' \amalg E'' \longrightarrow F \tag{9.1.14.1}$$

is conservative (which also means that $F = E' \vee E''$ (9.1.7.2 e)), then the same is true after 2-base change by $F_1 \hookrightarrow F$, where F_1 is a subtopos of F. 450

There is an analogous reformulation b′) of b), taking a base change $F_1 \to F$ by a *punctual* topos F_1 (2.2). Thus affirmative answers to a) and b) would follow

[13]N.D.E.: Positive answers to questions a) and d) were given in [17].

from an affirmative answer to the following question (where F is any subtopos of E):

c) If E' and E'' are two subtopoi of the topos F whose supremum is F, i.e. such that the morphism of topoi (9.1.14.1) is conservative, is this morphism even *universally conservative*, i.e. does it remain conservative after every 2-base change $F_1 \to F$?

This statement makes sense by a result of Deligne (see [16], Proposition 4.47) asserting the existence of 2-fiber products of topoi. See, however, the example in 9.1.7.2 d).

On the other hand, 9.1.11 h) shows that an affirmative answer to b) would, in the locally Noetherian case, provide an affirmative answer to the following question:

d) In the ordered set Σ of subtopoi of E, is the infimum of two elements distributive over *arbitrary* (not necessarily finite) suprema?

This also means that Σ may be interpreted as the ordered set of "closed subtopoi" of a suitable topos $E_{\mathrm{cons}'}$ (namely $\Sigma^{0\sim}$, where the opposite ordered set Σ^0 of Σ, regarded as a category, is endowed with its canonical topology, cf. 7.8 b)), which is generated by the subobjects of the terminal object and is therefore very close to an ordinary topological space. It would remain to study this topos $E_{\mathrm{cons}'}$, taking inspiration from the examples 9.1.11 g) and 9.1.12 d), and in particular to determine whether it is Noetherian when E is. This amounts to the following question for subtopoi F of E, which makes sense independently of d):

e) Let F be a Noetherian topos, and let $(E_i)_{i\in I}$ be a family of subtopoi of F whose intersection is $\mathrm{Top}(\varnothing)$. Is there a finite subfamily with the same property?

451 In a rather different direction, recall also the question raised in 9.1.6 e), which deserves clarification.

9.2. The case of open subtopoi.

9.2.1. Let U be an open of a topos E, i.e. a subobject of the terminal object e of E (8.3). Consider the localization morphism (5.2)

$$j : E_{/U} \longrightarrow E. \tag{9.2.2}$$

Since $U \to e$ is a monomorphism, the functor $j_!$ (which is interpreted as the forgetful functor) is fully faithful, and hence so is its biadjoint j_* (I 5.7). In other words, *the localization morphism j in (9.2.2) associated with an open U of the topos E is an embedding of topoi* (9.1.1 a)).

An embedding of topoi $F \to E$ is called an *open embedding* if it is isomorphic to the embedding of topoi defined by an open U of E. A subtopos E' (9.1.1 b)) of E is called an *open subtopos* if it is defined by an open of E, i.e. if the canonical inclusion morphism $E' \to E$ is an open embedding. Note that the open U of E associated with an open embedding $j : E' \to E$ is uniquely determined as the subobject $j_!(e')$ of e, where e' denotes the terminal object of E'. Thus one obtains a one-to-one correspondence between opens U of E and open subtopoi of E.

Remark 9.2.3. In accordance with the preceding definitions, the open subtopos of E associated with the open U of E is the strictly full subcategory of E which is the essential image of the direct-image functor

$$j_* : E_{/U} \longrightarrow E.$$

One should not confuse this subcategory of E with the strictly full subcategory which is the essential image of $j_!$, formed by the objects X of E for which the structural morphism $X \to e$ factors through U. These two subcategories plainly determine one another and are canonically equivalent. This is why some authors may have been tempted to call (and indeed have called) the strictly full subcategory which is the essential image of $j_!$ the "open subtopos defined by U," since its description in terms of U is simpler than that of the essential image of j_*. This terminology causes no difficulty so long as one does 452
not intend to study kinds of subtopoi other than open subtopoi. Since that is not our situation, we shall not follow the authors just mentioned.

For the reader's convenience, we summarize the most important special properties of open embeddings:

Proposition 9.2.4. *Let*

$$j : \mathscr{U} \longrightarrow E$$

be an open embedding of topoi (9.2.1). Then the functor $j_!$ left adjoint to j^* exists, so that there is a sequence of three adjoint functors

$$j_!, \quad j^*, \quad j_* :$$

$$\begin{tikzcd} \mathscr{U} \arrow[r, shift left=2, "j_!"] \arrow[r, shift right=2, "j_*"'] & E \arrow[l, "j^*"'] \end{tikzcd}$$

Moreover:

a) *The functors $j_!$ and j_* are fully faithful.*
b) *The functor $j_!$ commutes with fiber products, with products indexed by small sets $I \neq \phi$, and with projective limits indexed by small cofiltered categories (I 2.7).*
c) *For every object X of E, the adjunction morphism*

$$j_! j^*(X) \longrightarrow X$$

is a monomorphism.

Proof. We may suppose that j is the localization morphism defined by an open U of E. Then a) is recalled for reference, b) follows from the interpretation of $j_!$ as a forgetful functor and from the fact that $U \to e$ is a monomorphism. Finally, c) is immediate on observing that the morphism in question is obtained from the inclusion $U \to e$ by base change along $X \to e$, since a monomorphism is carried to a monomorphism by base change.

Remark 9.2.4.1. There exist embeddings of topoi $j : E' \to E$ for which $j_!$ exists but does not commute with products of two objects or with cofiltered projective limits (Exercise 9.5.9 c)).

9.2.5. Now let 453

$$g : E' \longrightarrow E$$

be an arbitrary morphism of topoi, and let $U' = g^*(U)$ be the inverse image of the open U of E. We saw in 5.11 that the diagram of topoi

$$\begin{tikzcd} E'/_{U'} \arrow[r, "j'"] \arrow[d, "g_U"'] & E' \arrow[d, "g"] \\ E/_U \arrow[r, "j"] & E \end{tikzcd}$$

is 2-Cartesian. It follows in particular that the inverse image under g of an open subtopos of E (cf. 9.1.6 d)) is an open subtopos of E', or equivalently that the notion of an open embedding of topoi (9.2.1) is stable under 2-base changes in the 2-category of $\mathscr{U}$-topoi (which are elements of a given universe $\mathscr{V}$).

Also note that the morphism of topoi $g : E' \to E$ factors, up to isomorphism, through $j : E_{/U} \to E$ if and only if $j' : E'_{/U'} \to E'$ is an equivalence of categories, which also means that the inclusion $U' \hookrightarrow e'$ (where e' is the terminal object of E') is an isomorphism. This makes 9.1.4 more precise by giving a simple characterization of the essential image of the functor considered in *loc. cit.*

9.2.6. Apply 9.2.5 when E' is of the form $E_{/V}$, where V is a second open of E, and g is the localization morphism. Then $U' = U \cap V$, and the 2-Cartesian product of $E_{/U}$ and $E_{/V}$ over E identifies with $E_{/U\cap V}$. It follows that *a finite intersection of open subtopoi of E is an open subtopos of E*; more precisely, the map (9.2.6.1)

$$U \longmapsto \text{the subtopos of } E \text{ defined by } U$$

commutes with finite intersections.

Exercise 9.2.7. Prove that the map (9.2.6.1) also commutes with *arbitrary* Sup. (Use 9.1.7.2 e)).

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9.3. Construction of the closed subtopos complementary to an open subtopos.

9.3.1. Let T be a topological space and U an open subset of T, so that $\mathrm{Top}(U)$ identifies with an open subtopos of $\mathrm{Top}(\mathsf{T})$ (notation of 2.1). Let Y be the closed topological subspace of T complementary to U. Up to equivalence, $\mathrm{Top}(Y)$ may then be regarded as a subtopos of $\mathrm{Top}(\mathsf{T})$, namely the subtopos formed by the objects F of $\mathrm{Top}(\mathsf{T})$ whose restriction to U is the terminal object of U. This description of a subcategory of $E = \mathrm{Top}(\mathsf{T})$ makes sense whenever one has a topos and an open U of it, and we shall see that it always gives a subtopos of E. Moreover, in the particular case first considered, it is immediate (with the terminology introduced in Exercise 9.1.13) that $\mathrm{Top}(Y)$ and $\mathrm{Top}(U)$ are complementary subtopoi (use 9.2.5 and 9.1.7.2 e)). The same will hold in the general case, and we shall see that this property uniquely characterizes the subtopos in question. It therefore deserves in every respect the name of the closed subtopos complementary to the given open U, or to the open subtopos defined by U. The details of the construction of this topos $\mathscr{F}$ and of the inverse-image functor $i^* : E \to \mathscr{F}$ will be given in the present section, while Section 9.4 develops its first properties.

9.3.2. Thus let E be a topos and U an open of E, as in 9.2.1, and consider the localization morphism

$$j : \mathscr{U} = E_{/U} \longrightarrow E, \tag{9.3.2.1}$$

which is an open embedding.

For every object X of E, put

$$X_{\complement U} = U \coprod_{U \times X} X, \tag{9.3.2.2}$$

where the amalgamation is formed with respect to the canonical projections

$$\mathrm{pr}_1 : U \times X \longrightarrow U, \quad \mathrm{pr}_2 : U \times X \longrightarrow X.$$

Thus there is a *cocartesian* diagram (I 10) in E, depending functorially on X: 455

$$\begin{tikzcd} U \times X \arrow[r, "\mathrm{pr}_2"] \arrow[d, "\mathrm{pr}_1"'] & X \arrow[d] \\ U \arrow[r] & X_{\complement U} \end{tikzcd} \tag{9.3.2.3}$$

Note that in the example considered in 9.3.1, $X_{\complement U}$ is canonically isomorphic to $i_* i^*(X)$ (where $i : Y \to T$ is the inclusion), and $X \to X_{\complement U}$ identifies with the adjunction morphism.

Proposition 9.3.3. *With the preceding notation, the following statements hold:*

a) *For every object X of E, the following conditions are equivalent:*
   (i) *There exists an object Y of E and an isomorphism $X \simeq Y_{\complement U}$.*
   (ii) *The canonical morphism $X \xrightarrow{m} X_{\complement U}$ is an isomorphism.*
   (iii) *The canonical morphism $\mathrm{pr}_2 : X \times U \to U$ is an isomorphism (i.e. $j^*(X)$ is a terminal object of $E_{/U}$).*

b) *For every morphism*

$$m : X \longrightarrow X'$$

in E, the following conditions are equivalent:
   (i′) *The diagram*

$$\begin{tikzcd} X \arrow[r, "m"] \arrow[d] & X' \arrow[d] \\ X_{\complement U} \arrow[r] & X'_{\complement U} \end{tikzcd}$$

   is Cartesian.
   (ii′) *The morphism $m \times \mathrm{id}_U : X \times U \to X' \times U$ is an isomorphism, i.e. $j^*(m)$ is an isomorphism.*

c) *The functor*

$$X \longrightarrow X_{\complement U}$$

from E to E is left exact, carries epimorphic families to epimorphic families, and commutes with filtered inductive limits and amalgamated sums. 456

Proof. First prove the proposition when E is of the form $\widehat{C}$, where C is a small category. By I.3.1 this reduces to the case where E is the "punctual" topos (Ens), where the proposition is evident. One then passes to the general case by the standard procedures of II 4, using the "associated sheaf" functor.

Proposition 9.3.4. *With the notation of 9.3.2, the strictly full subcategory $\mathscr{F}$ of E formed by the objects X of E such that $j^*(X)$ is a terminal object of $\mathscr{U}$* (cf. 9.3.3 a)) *is a subtopos of E: i.e. (9.1.1) it is a topos, and the inclusion functor $i_* : \mathscr{F} \to E$ is the direct-image functor of a morphism of topoi $i : \mathscr{F} \to E$, whose inverse-image functor is $X \mapsto i^*X = X_{\complement U}$. The adjunction morphism $X \to i_*i^*X$ is the canonical morphism $X \to X_{\complement U}$ of (9.3.2.3).*

Proof: For every object X of E, denote by $u(X)$ the canonical morphism $X \to X_{\complement U}$. The morphism $u(X)$ is functorial in X, and for every object X, $u(X_{\complement U})$ is an isomorphism. It follows formally and immediately from these two properties that the functor i^* is left adjoint to the functor i_* and that the adjunction morphism is $u(X)$. Since the functor $X \to X_{\complement U}$ is left exact, the functor i^* is left exact. It then follows from II 5.5 that $\mathscr{F}$ is a topos, and from Definition 3.1 that the pair (i^*, i_*) is a morphism of topoi.

9.3.5. The subtopos $\mathscr{F}$ of E described in 9.3.4 is called the *closed subtopos of E complementary to the open U of E*, or, equivalently, to the open subtopos of E corresponding to U. In accordance with 9.1.1 b), the morphism $i : \mathscr{F} \to E$ constructed in 9.3.4 is called the *inclusion morphism*. A full subcategory $\mathscr{F}$ of E is called a *closed subtopos* of E if there exists an open U of E for which $\mathscr{F}$ is the closed subtopos complementary to U. Note that this U is uniquely determined as $i_*(\phi_{\mathscr{F}})$, where $\phi_{\mathscr{F}}$ is the initial object of $\mathscr{F}$, equal to $i^*(\phi_E)$; it is also called *the open of E complementary to the closed subtopos $\mathscr{F}$*,
457 and the subtopos of E defined by U is also called the *open subtopos of E complementary to $\mathscr{F}$*. Finally, an embedding of topoi (9.1.1 a)) $i : F \to E$ is called a *closed embedding* if the corresponding subtopos of E is closed.

If G is a group object on E and $s \in \mathrm{Hom}(e_E, G)$, the *support* of G (resp. of s) is the closed subtopos of E complementary to the cosupport open of G (resp. of s) (8.5.1, 8.5.2).

9.3.6. With the notation of 9.3.2 and 9.3.4, one therefore has two morphisms of topoi

(9.3.6.1)
$$\mathscr{U} \xrightarrow{j} E \xleftarrow{i} \mathscr{F},$$

defining five functors which form two sequences of adjoint functors, $(j_!, j^*, j_*)$ and (i^*, i_*):

(9.3.6.2)
$$\begin{tikzcd} \mathscr{U} \arrow[r, shift left=2, "j_!"] \arrow[r, shift right=2, "j_*"'] & E \arrow[l, "j^*" description] \arrow[r, shift left, "i^*"] & \mathscr{F}. \arrow[l, shift left, "i_*"] \end{tikzcd}$$

The functor

(9.3.6.3)
$$\rho = i^* j_* : \mathscr{U} \longrightarrow \mathscr{F}$$

is called the *gluing functor* relative to the open U of E. More generally, if $\mathscr{U}$ and $\mathscr{F}$ are arbitrary topoi, a functor $\rho : \mathscr{U} \to \mathscr{F}$ is called a *gluing functor* if it is left exact and accessible (I.9.2), or equivalently (I 8) if it admits a pro-adjoint. The reasons for this terminology will appear in 9.4.1 d) and 9.5.4 below.

9.4. First properties of the closed subtopos $\mathscr{F}$ and of $i : \mathscr{F} \to E$.

Proposition 9.4.1. *The notation is that of 9.3.2 and 9.3.4.*

a) *The functor $i_* : \mathscr{F} \to E$ carries epimorphic families to epimorphic families. It commutes with the formation of amalgamated sums and filtered inductive limits.*
b) *For every object X of E such that the projection morphism $U \times X \to U$ is an epimorphism, the adjunction morphism $X \to i_*i^*X$ is an epimorphism.*

458 c) *The pair of functors* (i^*, j^*) *from* E *to* $\mathscr{F}$ *and* $\mathscr{U}$*, respectively, is conservative* (I 6.1)*, i.e. a morphism* m *of* E *is an isomorphism if and only if* $i^*(m)$ *and* $j^*(m)$ *are isomorphisms (equivalently, the pair* (i^*, j^*) *is faithful (6.4.0)).*
d) *The gluing functor* $\rho = i^* j_* : \mathscr{U} \to \mathscr{F}$ *is left exact and accessible* (I 9.2)*, or equivalently* (I 8) *it admits a pro-adjoint* $\sigma : \mathrm{Pro}\,\mathscr{F} \to \mathrm{Pro}\,\mathscr{U}$*.*

Proof: a) follows from 9.3.3 c). For b), it suffices to use the definition (9.3.2.2) of $X_{\complement U}$, taking into account that $i_* i^* X = X_{\complement U}$ (9.3.4). Assertion c) follows from 9.3.3 b). The functors i^* and j_* are left exact, and hence so is ρ. The functor j_* is a direct-image functor for a morphism of topoi and is therefore accessible (I 9.5); note that a topos is an accessible category (I 9.11.3). Since the functor i^* commutes with inductive limits (it is an inverse-image functor), the functor ρ, being a composite of two accessible functors, is accessible.

Proposition 9.4.2. *Let* E' *be a topos. Then the functor* $h \mapsto i \circ h$*,*

$$\mathrm{Homtop}(E', \mathscr{F}) \longrightarrow \mathrm{Homtop}(E', E),$$

is fully faithful, and its essential image consists of the morphisms of topoi $g : E' \to E$ *such that* $g^*(U)$ *is an initial object* $\phi_{E'}$ *of* E'*.*

By 9.1.4, the functor in question is fully faithful, and g belongs to its essential image if and only if, for every object X' of E', the object $g_*(X')$ of E belongs to $\mathscr{F}$, i.e. if

(*) $$j^*(g_*(X')) \simeq e_{\mathscr{U}} \text{ for every } X' \in \mathrm{Ob}\, E'.$$

Put $U' = g^*(U)$ and consider the diagram of topoi

$$\begin{tikzcd} E'_{/U'} \arrow[r, "j'"] \arrow[d, "g_U"'] & E' \arrow[d, "g"] \\ E_{/U} \arrow[r, "j"] & E \end{tikzcd},$$

There is plainly a canonical isomorphism 459

$$j^*(g_*(X')) \simeq g_{U*}(j'^*(X')).$$

The functor j'^* is also plainly essentially surjective, so condition (*) is equivalent to

(**) $$g_{U*}(Y') \simeq e_{\mathscr{U}} \text{ for every } Y' \in \mathrm{Ob}\, E'_{/U'}.$$

Using the adjunction between g_{U*} and g_U^*, one sees immediately that this means that $g_U^*(Y)$ is an initial object of $E'_{/U'}$ for every object Y of $E_{/U}$; or, using the fact that the initial object of $E'_{/U'}$ is strict (II 4.5), that it suffices for this to hold for $Y = e_{\mathscr{U}}$, the terminal object of $\mathscr{U}$. But this also means that the terminal object U' of $E'_{/U'}$ is initial, which is manifestly equivalent to U' being an initial object $\phi_{E'}$ of E', C.Q.F.D.

Corollary 9.4.3. *Let* $g : E' \to E$ *be a morphism of topoi, let* $U' = g^*(U)$*, let* $\mathscr{U}' = E'_{/U'}$*, and let* $\mathscr{F}'$ *be the closed subtopos of* E' *complementary to the open* U'*; finally, let* $i' : \mathscr{F}' \to E'$ *be the inclusion morphism. Then the morphism of topoi* $g \circ i' : \mathscr{F}' \to E$ *factors, up to isomorphism, through a morphism of topoi* $g_{\mathscr{F}} : \mathscr{F}' \to \mathscr{F}$*, and the* *corresponding diagram of topoi*

(9.4.3.1)
$$\begin{tikzcd} \mathscr{F}' \arrow[r, "i'"] \arrow[d, "g_{\mathscr{F}}"'] & E' \arrow[d, "g"] \\ \mathscr{F} \arrow[r, "i"] & E \end{tikzcd}$$

is 2*-Cartesian* (cf. 5.11).

This follows formally from 9.4.2 and the definitions.

With the terminology introduced in 9.1.6 d), one may therefore say in particular that the inverse image under a morphism of topoi $g : E' \to E$ of a *closed* subtopos $\mathscr{F}$ of E is a *closed* subtopos $\mathscr{F}'$ of E' (namely the closed subtopos complementary to the open $U' = g^*(U)$, where U is the open of E complementary to $\mathscr{F}$).

Corollary 9.4.4. *Let E be a topos, $\mathscr{U}$ an open subtopos of E, and $\mathscr{F}$ the complementary closed subtopos.*

460 a) *One has*

$$\mathscr{U} \cap \mathscr{F} \approxeq \mathrm{Top}(\varnothing) \quad , \quad \mathscr{U} \vee \mathscr{F} = E,$$

where $\vee$ *denotes the* Sup *in the set of all subtopoi of E* (9.1.2 c)). *(In the terminology introduced in 9.1.13, $\mathscr{U}$ and $\mathscr{F}$ are complementary subtopoi of E.)*

b) *$\mathscr{U}$ (resp. $\mathscr{F}$) is the largest among the subtopoi E' of E such that $E' \cap \mathscr{F} \approxeq$* Top($\varnothing$) *(resp. $\mathscr{U} \cap E' \approxeq$* Top($\varnothing$)*).*

Proof.

a) The first relation amounts to saying that if E' is a topos and $g : E' \to E$ is a morphism of topoi factoring, up to isomorphism, through both $\mathscr{U}$ and $\mathscr{F}$, then $E' \approxeq \mathrm{Top}(\varnothing)$. By 9.2.5 and 9.4.2, the hypothesis means that $g^*(U)$ is both an initial and a terminal object of E', which proves the assertion. For the second relation, note that by 9.1.7.2 e) it is equivalent to 9.4.1 c).

b) Suppose that $E' \cap \mathscr{F} \approxeq \mathrm{Top}(\varnothing)$. By 9.4.3, $E' \cap \mathscr{F}$ is equivalent to the subtopos of E' complementary to $g^*(U)$, where $g : E' \to E$ is the inclusion morphism. Thus the condition means that $U' = g^*(U)$ is a terminal object of E', i.e. (9.2.5) that $E' \subset \mathscr{U}$. Suppose that $\mathscr{U} \cap E' \approxeq \mathrm{Top}(\varnothing)$. By 5.11, $\mathscr{U} \cap E'$ is equivalent to $E'_{/U'}$, so the condition means that U' is an initial object of E', i.e. (9.4.2) that $E' \subset \mathscr{F}$.

Corollary 9.4.5. *Let $(\mathscr{F}_i)_{i \in I}$ be a family of closed subtopoi of the topos E, and for every $i \in I$ let U_i be the open of E complementary to $\mathscr{F}_i$. Then the intersection of the $\mathscr{F}_i$ as subtopoi of E (9.1.3) is a closed subtopos whose complementary open is $U = \mathrm{Sup}_i\, U_i$.*

This follows formally from the universal property of the intersection as a 2-product and from 9.4.2, taking into account that for a morphism of topoi $g : E' \to E$, $g^*(U) = \mathrm{Sup}_i\, g^*(U_i)$ is an initial object if and only if all the $g^*(U_i)$ are initial.

Similarly one proves:

Corollary 9.4.6. *With the notation of 9.4.5, suppose that I is finite. Then the supre-*
461 *mum of the subtopoi $\mathscr{F}_i$ of E* (9.1.2 c)) *is a closed subtopos of E, whose complementary open is $\bigcap_i U_i$.*

Exercise 9.4.7. (*Gluing of subtopoi*). Let E be a topos.

a) Let E' be a subtopos of E, let $(S_i)_{i\in I}$ be a family of objects of E covering the terminal object, and, for every $i \in I$, let S'_i be the inverse image of S_i in E', and let $E'_i \approx E'_{/S'_i}$ be the subtopos of $E_i = E_{/S_i}$ which is the inverse image, under the localization morphism $E_i \to E$, of the subtopos E' of E (9.1.6 d) and 5.11). Show that, for an object X of E, X belongs to E' if and only if, for every $i \in I$, the inverse image $X|S_i$ of X in E_i belongs to E'_i.

b) Suppose that E is of the form $C^\sim$, where C is a $\mathscr{U}$-site. Show that the presheaf on C

$$S \mapsto \text{set of subtopoi of the topos } (C_{/S})^\sim \approx E_{/\varepsilon(S)}$$

is a sheaf. For arbitrary E, deduce that there exists an object of E which represents the functor

$$S \mapsto \text{set of subtopoi of the topos } E_{/S};$$

c) With the notation of a), show that the subtopos E' of E is an open subtopos (resp. a closed subtopos) if and only if, for every $i \in I$, the same is true of the induced subtopos E'_i of E_i.

Exercise 9.4.8. (*Interior, exterior, closure, and boundary of a subtopos.*) Let E be a topos, E' a subtopos, and $i : E' \to E$ the inclusion morphism.

a) Show that $i_*(\phi_{E'})$ is the largest open U of E such that $E'|\mathscr{U} \approx \mathrm{Top}(\varnothing)$, where $\mathscr{U}$ is the open subtopos of E defined by U. This U, or $\mathscr{U}$, is called the *exterior* of the subtopos E' of E, and the closed subtopos of E complementary to U is called the *closure* of E'.

b) Show that there exists a largest open V of E such that the corresponding open subtopos $\mathscr{V}$ of E is contained in E'. (Use 9.2.7.) This V, or $\mathscr{V}$, is called the *interior* of the subtopos E' of E. The opens U and V of E are disjoint ($U \cap V = \phi_E$); the closed subtopos of E complementary to $\mathrm{Sup}(U, V) = \mathscr{U} \amalg \mathscr{V}$ is called 462 the *boundary* of the subtopos E' of E.

c) The subtopos E' is an open subtopos (resp. a closed subtopos) of E if and only if it is equal to its interior (resp. if it is equal to its closure). The boundary of E' is empty if and only if E' is both an open and a closed subtopos of E, or equivalently, if it is the open subtopos of E defined by an open V of E which is a *direct summand* of the terminal object e, i.e. such that there exists a subobject U of e with $e \simeq U \amalg V$. The exterior of E' is "empty" i.e. the closure is equal to E, if and only if $i : E' \to E$ is dominant (8.8).

d) Let S be an object of E, let $F = E_{/S}$ be the induced topos, and let $F' \simeq F_{/S'}$ be the subtopos of F induced by the subtopos E' of E. Show that the exterior (resp. the closure, resp. the interior, resp. the boundary) of F' is induced by the exterior (resp. ...) of E'.

Exercise 9.4.9. (*Locally closed subtopoi of a topos.*) A subtopos F of a topos E is called a *locally closed subtopos* of E if it is the intersection of an open subtopos and a closed subtopos.

a) Prove that the intersection of a finite family of locally closed subtopoi of E is locally closed. If $g : E' \to E$ is a morphism of topoi, prove that the inverse image (9.1.6 d)) of a locally closed subtopos of E is a locally closed subtopos of E'.

b) Let X be a topological space, and let $E = \mathrm{Top}(X)$. For every subset X' of X, consider the subtopos $T(X')$ of E which is the essential image of $\mathrm{Top}(X')$ in

Top(X). Show that if X' is a locally closed subset, $T(X')$ is a locally closed subtopos of E, the converse being true if, in addition, X and X' are assumed sober. Prove that $X' \mapsto T(X')$ defines an order isomorphism between the set of locally closed subsets X' of X and the set of locally closed subtopoi of E.

463 c) Let F be a subtopos of E. Show that every open subtopos of F is induced by an open subtopos of E. (Use 9.1.8 c).) Show that F is a locally closed subtopos of E if and only if it is an open subtopos of its closure (9.4.8 a)). Deduce, using 9.4.8 d), that the property for F of being a locally closed subtopos of E is a local property on E.

d) Let F be a locally closed subtopos of E. Show that, among all the ways of writing F as the intersection of an open subtopos $\mathscr{U}$ and a closed subtopos $\mathscr{F}$ of E, there is one for which $\mathscr{U}$ is the largest possible, and $\mathscr{F}$ the smallest possible. (Take $\mathscr{F}$ = the closure of F, and $\mathscr{U}$ = the largest open subtopos inducing on $\mathscr{F}$ the subtopos F.) Show that the formation of $\mathscr{U}$ and $\mathscr{F}$ is compatible with localization on E.

Exercise 9.4.10. Develop the notions of a *constructible, locally constructible, quasi-constructible, and locally quasi-constructible subtopos*, following the model of the familiar notions for topological spaces (EGA 0_{III} 9.1, EGA IV 10.1). For a topos of the form Top(X), one recovers a bijection between the set of constructible subsets (resp. ...) of X, and the set of constructible subtopoi (resp. ...) of Top(X).

9.5. The gluing theorem.

9.5.1. Let $f : A \to B$ be a functor between two categories. Denote by (B, A, f) the following category: the objects of (B, A, f) are the triples

$$(X, Y, u)$$

where X is an object of B, Y an object of A, and u a morphism from X to $f(Y)$; the morphisms between two objects (X, Y, u) and (X', Y', u') are the pairs (m, m'), where m is a morphism from X to X' and m' a morphism from Y to Y', such that the following diagram is commutative:

$$\begin{tikzcd} X \arrow[r, "u"] \arrow[d, "m"'] & f(Y) \arrow[d, "f(m')"] \\ X' \arrow[r, "u'"] & f(Y') \end{tikzcd}$$

464 the composition of morphisms is defined in the evident way.

9.5.2. Note that the category (B, A, f) depends functorially on the system of data A, B, f, in a sense whose precise formulation we leave to the reader. In particular, if f' is a second functor from A to B, then every morphism of functors

$$f \xrightarrow{u} f'$$

defines a functor $(B, A, f) \to (B, A, f')$, which is an isomorphism if $u : f \to f'$ is one.

9.5.3. Return to the situation of 9.3, and consider the category $(\mathscr{F}, \mathscr{U}, \rho)$ defined in 9.5.1. To each object X of E there corresponds an object $(i^*X, j^*X, h(X) : i^*X \to \rho j^*X)$ of $(\mathscr{F}, \mathscr{U}, \rho)$, where $h(X)$ is obtained by applying the functor i^* to the adjunction morphism $X \to j_*j^*(X)$ (one has $\rho = i^*j_*$ 9.3.6.3). Thus this defines a functor

$$\Phi : E \longrightarrow (\mathscr{F}, \mathscr{U}, \rho). \tag{9.5.3.1}$$

THEOREM 9.5.4. a) *Let E be a topos, let $\mathscr{U}$ be an open subtopos of E, and let $\mathscr{F}$ be the complementary closed subtopos. The functor*

$$\rho = i^* j_* : \mathscr{U} \longrightarrow \mathscr{F}$$

is left exact and accessible, and the functor Φ (9.5.3.1) is an equivalence of categories.

b) *Conversely, let $\mathscr{U}$ and $\mathscr{F}$ be two topoi, and let $\rho : \mathscr{U} \to \mathscr{F}$ be a gluing functor (9.3.6) (i.e. accessible (I 9.2) and left exact). The category $E = (\mathscr{F}, \mathscr{U}, \rho)$ is a topos. Let $\phi_{\mathscr{F}}$ be an initial object of $\mathscr{F}$, let $e_{\mathscr{U}}$ be a terminal object of $\mathscr{U}$, and let $U = (\phi_{\mathscr{F}}, e_{\mathscr{U}}, \phi_{\mathscr{F}} \to \rho(e_{\mathscr{U}})) \in \operatorname{Ob} E$. The functor $X \mapsto (\phi_{\mathscr{F}}, X, \phi_{\mathscr{F}} \to \rho(X))$ induces an equivalence of $\mathscr{U}$ with $E_{/U}$, hence with the open subtopos $\mathscr{U}'$ of E defined by U. The functor $Y \mapsto (Y, e, Y \to \rho(e))$ is a closed embedding (9.3.5), i.e. it induces an equivalence of $\mathscr{F}$ with a closed subtopos $\mathscr{F}'$ of E. The subtopoi $\mathscr{U}'$ and $\mathscr{F}'$ are complementary. Let $\rho' : \mathscr{U}' \to \mathscr{F}'$ be the gluing functor. The* 465
functors ρ and ρ' are compatible with the equivalences described above.

DEFINITION 9.5.5. *Let $\mathscr{U}$ and $\mathscr{F}$ be two topoi, and let $\rho : \mathscr{U} \to \mathscr{F}$ be a gluing functor (9.3.6). The topos $(\mathscr{F}, \mathscr{U}, \rho)$ is called the topos constructed from $\mathscr{U}$ and $\mathscr{F}$ by gluing by means of the functor ρ.*

9.5.6. *Proof of* 9.5.4 a) [13] The first assertion merely recalls 9.4.1 d). The pair of functors (i^*, j^*) is conservative, or equivalently, faithful (9.4.1 c)). A fortiori, the functor Φ is faithful. Let us show that it is fully faithful. It follows from 9.3.3 b) that, for every object X of E, the following diagram is cartesian:

(∗)
$$\begin{tikzcd} X \arrow[r] \arrow[d] & j_* j^* X \arrow[d] \\ i_* i^* X \arrow[r] & i_* i^* j_* j^* X; \end{tikzcd}$$

Let X and Y be two objects of E, and suppose that (m, m'):

$$\begin{tikzcd} i^* X \arrow[r] \arrow[d, "m"'] & i^* j_* j^* X \arrow[d, "i^* j_* m'"'] \\ i^* Y \arrow[r] & i^* j_* j^* Y; \end{tikzcd}$$

is a morphism from $\Phi(X)$ to $\Phi(Y)$. Using the functoriality of the fiber product and the diagrams (∗), one obtains a morphism $w : X \to Y$ which makes the following diagrams commutative:

$$\begin{tikzcd} X \arrow[r, "w"] \arrow[d] & Y \arrow[d] \\ j_* j^* X \arrow[r, "j_* m'"] & j_* j^* Y, \end{tikzcd} \qquad \begin{tikzcd} X \arrow[r, "w"] \arrow[d] & Y \arrow[d] \\ i_* i^* X \arrow[r, "i_* m"] & i_* i^* Y \end{tikzcd}$$

By applying the functors j^* and i^* respectively, one obtains $j^* w = m'$ and $i^* w = m$. Let 466

[13] For a gluing statement more general than 9.5.5, cf. Exercise[14] [N.D.E.: This exercise does not seem to exist.].

us now show that the functor Φ is essentially surjective. Let $(Y, X, u : Y \to i^* j_*(X))$ be an object of $(\mathscr{F}, \mathscr{U}, i^* j_*)$. From it one obtains a diagram:

$$\begin{tikzcd} & j_* X \arrow[d] \\ i_* Y \arrow[r, "i_* u"] & i_* i^* j_* X \end{tikzcd}$$

Hence, on taking the fiber product, one obtains a cartesian diagram:

$$(**) \qquad \begin{tikzcd} W \arrow[r] \arrow[d] & j_* X \arrow[d] \\ i_* Y \arrow[r, "i_* u"] & i_* i^* j_* X \end{tikzcd}$$

We leave it to the reader to show that $\Phi(W)$ is isomorphic to $(Y, X, u : Y \to i^* j_*(X))$ and that the diagram $(**)$ is isomorphic to the diagram $(*)$ associated with W.

9.5.7. *Proof* of 9.5.4 b). We shall confine ourselves to showing that $(\mathscr{F}, \mathscr{U}, \rho)$ is a topos. The proof of the other assertions is left to the reader. We verify properties a), b), c), d) of 1.1.2.

a) *Finite projective limits are representable*: This is clear because the categories $\mathscr{U}$ and $\mathscr{F}$ are topoi and ρ commutes with finite projective limits.

b) *Direct sums are representable, disjoint, and universal*: Let $(Y_i \xrightarrow{u_i} \rho(X_i))$, $i \in I$, be a family of objects of $(\mathscr{F}, \mathscr{U}, \rho)$. By the definition of inductive limits, there is a canonical morphism

$$\coprod_{i \in I} \rho(X_i) \longrightarrow \rho(\coprod_{i \in I} X_i),$$

467 and hence, by composing with $\coprod u_i$, a morphism

$$n : \coprod_{i \in I} Y_i \longrightarrow \rho(\coprod_{i \in I} X_i).$$

One verifies immediately that this last object is the direct sum of the given family and that this direct sum is disjoint and universal.

c) *Equivalence relations are effective and universal*:

Let

$$\begin{tikzcd} R_1 \arrow[r, shift left] \arrow[r, shift right] \arrow[d] & Y \arrow[d, "w"] \\ \rho(R_2) \arrow[r, shift left, "f(u)"] \arrow[r, shift right, "f(v)"'] & \rho(X) \end{tikzcd}$$

be an equivalence relation on the object $Y \xrightarrow{w} \rho(X)$. Then $R_1 \rightrightarrows Y$ is an equivalence relation on Y in $\mathscr{F}$, and $R_2 \overset{u}{\underset{v}{\rightrightarrows}} X$ is an equivalence relation on X in $\mathscr{U}$. The quotients Y/R_1, $\rho(X)/\rho(R_2)$, and X/R_2 are representable in $\mathscr{F}$ and $\mathscr{U}$ because these categories are topoi. Moreover, by the definition of inductive limits, the canonical morphism $\rho(X) \to \rho(X/R_2)$ factors through $\rho(X)/\rho(R_2)$. By functoriality of inductive limits, one obtains a canonical morphism $Y/R_1 \xrightarrow{t}$

$\rho(X/R_2)$. One verifies that this last object is the quotient of $Y \xrightarrow{w} \rho(X)$ by the equivalence relation in question, that this quotient is effective (I 10), and that all these properties are preserved under base change.

d) $(\mathscr{F}, \mathscr{U}, \rho)$ *admits a small generating family.* This property follows from I 9.25 (taking into account that ρ is accessible), by taking $B = \Delta_1 = (0 \xrightarrow{f} 1)$, $E_0 = \mathscr{F}$, $E_1 = \mathscr{U}$, and $f^* = \rho : E_1 \to E_0$, whence $\mathbf{Hom}_B(B, E) = (\mathscr{F}, \mathscr{U}, \rho)$.

One may also avoid using I 9.25 (whose proof as given was rather arduous!) and use the hypothesis on ρ in the form that ρ admits a pro-adjoint

$$\sigma : \mathrm{Pro}(\mathscr{F}) \longrightarrow \mathrm{Pro}(\mathscr{U}),$$

(this interpretation (I 8) being based on I 9, whose proof is distinctly more comprehensible). It suffices to note that if C' (resp. C'') is a generating subcategory of $\mathscr{U}$ (resp. $\mathscr{F}$), and if, for every $X'' \in \operatorname{Ob} C''$, one represents $\sigma(X'')$ as a projective system $(\sigma(X'')_i)_{i \in I(X'')}$, where $I(X'')$ is a small filtered ordered set, then the family of objects of E which are either of the form $(\phi_{\mathscr{F}}, X', \phi_{\mathscr{F}} \to \rho(X'))$ with $X' \in \operatorname{Ob} C'$, or of the form $(X'', \sigma(X'')_i, u_{X''} : X'' \to \rho(\sigma(X'')_i))$, where $X'' \in \operatorname{Ob} C''$, $i \in I(X'')$, and $u_{X''}$ corresponds under adjunction to the canonical morphism $\sigma(X'') \to \sigma(X'')_i$, is a generating family (plainly a small one) when C' and C'' are chosen small. The verification is indeed immediate and is left to the reader. 468

. Use the notation of 9.5.7 b), and let us explain how, up to canonical isomorphisms of functors, the system of functors (9.3.6.2) can be made explicit when $E = (\mathscr{F}, \mathscr{U}, \rho)$. The details verifying the assertions below (which are essentially mechanical applications of 9.5.4) are left to the reader.

. The functor

$$i^* : (\mathscr{F}, \mathscr{U}, \rho) \longrightarrow \mathscr{F}$$

is given by

$$i^*(X, Y, u : X \longrightarrow \rho(Y)) = X.$$

. The functor

$$i_* : \mathscr{F} \longrightarrow (\mathscr{F}, \mathscr{U}, \rho)$$

is given by $i_*(X) = (X, e, X \to \rho(e))$, where e is the terminal object of $\mathscr{U}$.

. The adjunction morphism

$$\mathrm{id} \longrightarrow i_* i^*$$

is isomorphic to the functorial morphism

$$\begin{array}{ccc} X & \xrightarrow{u} & \rho(Y) \\ {\scriptstyle id}\downarrow & & \downarrow \\ X & \longrightarrow & \rho(e) \end{array}$$

. The functor 469

$$j^* : (\mathscr{F}, \mathscr{U}, \rho) \longrightarrow \mathscr{U}$$

is given by $\quad j^*(X, Y, u : X \longrightarrow \rho(Y)) = Y.$

. The functor

$$j_! : \mathscr{U} \longrightarrow (\mathscr{F}, \mathscr{U}, \rho)$$

is given by $\quad j_!(Y) = (\phi_{\mathscr{F}}, Y, \phi_{\mathscr{F}} \longrightarrow \rho(Y))$
(where $\phi_{\mathscr{F}}$ is the initial object of $\mathscr{F}$).

. The functor

$$j_* : \mathscr{U} \longrightarrow (\mathscr{F}, \mathscr{U}, \rho)$$

is given by $\qquad j_*(Y) = (\rho(Y), Y, \mathrm{id} : \rho(Y) \longrightarrow \rho(Y)).$

. The adjunction morphism

$$j_! j^* \longrightarrow \mathrm{id}$$

is isomorphic to the functorial morphism

$$\begin{tikzcd} \phi_{\mathscr{F}} \arrow[r] \arrow[d] & \rho(Y) \arrow[d, "\rho(\mathrm{id})"] \\ X \arrow[r, "u"] & \rho(Y) \end{tikzcd} .$$

. The adjunction morphism

$$\mathrm{id} \longrightarrow j_* j^*$$

is isomorphic to the functorial morphism

$$\begin{tikzcd} X \arrow[r, "u"] \arrow[d, "u"'] & \rho(Y) \arrow[d, "\rho(\mathrm{id})"] \\ \rho(Y) \arrow[r, "\mathrm{id}"] & \rho(Y) \end{tikzcd}$$

EXERCISE 9.5.9. (*Exactness properties of a glued topos.*) Let $\mathscr{U}$, $\mathscr{F}$ be two topoi, let $\rho : \mathscr{U} \to \mathscr{F}$ be a gluing functor, let $E = (\mathscr{F}, \mathscr{U}, \rho)$ be the glued topos, and let $j : \mathscr{U} \to E$ and $i : \mathscr{F} \to E$ be the canonical morphisms of topoi, so that $\rho = i^* j_*$.

a) Prove that i^* commutes with (small) products (i.e. (1.8 or I 8) the functor i^* admits a left adjoint $i_!$) if and only if the same holds for ρ, i.e. (*loc. cit.*) if and only if ρ admits a left adjoint $\sigma : \mathscr{F} \to \mathscr{U}$ (or equivalently if and only if the
470 pro-adjoint of ρ, restricted to $\mathscr{F}$ (9.6.3), factors, up to isomorphism, through $\mathscr{U}$): one then has $\sigma = j^* i_!$. Thus the gluing functors ρ having this property correspond, up to isomorphism, to the functors $\sigma : \mathscr{F} \to \mathscr{U}$ which have a right adjoint, i.e. (1.6) which commute with $\mathscr{U}$-small inductive limits, or equivalently which are continuous. When $\mathscr{F}$ is equivalent to a topos $C^\sim$, where C is a $\mathscr{U}$-site, these functors therefore correspond, up to isomorphism, to continuous functors $C \to \mathscr{U}$ (III 1.2 and III 1.7).

b) Suppose that the functor $i_!$ is defined. Show that $i_!$ commutes with a specified type of small projective limits if and only if $\sigma : \mathscr{F} \to \mathscr{U}$ commutes with them. (For necessity, use the fact that j^* commutes with small projective limits; for sufficiency, use the equivalent description of E in terms of σ as consisting of the triples (Y, X, u) with $Y \in \mathrm{Ob}\,\mathscr{F}$, $X \in \mathrm{Ob}\,\mathscr{U}$, $u : \sigma(Y) \to X$.)

c) Deduce from a) and b) an example of a closed embedding of topoi $i : \mathscr{F} \to E$ such that $i_!$ exists, but commutes neither with fiber products nor with filtered projective limits. (It suffices to find a continuous functor between two topoi which has neither of these two exactness properties.)

EXERCISE 9.5.10. (*Gluing of topoi of the form* $\widehat{C}$.) Let C be a small category, and let $E = \widehat{C}$, which is a $\mathscr{U}$-topos.

a) Recall that the opens U of E correspond to *sieves* C' of C (I 4.1, 8.4.4), with $E_{/U}$ then being equivalent to $\widehat{C'}$, and the functor $j^* : E \to E_{/U}$ identifying with the restriction functor 5.4, i.e. the inclusion morphism $j : E_{/U} \to E$ being $\widehat{f} : \widehat{C'} \to \widehat{C}$ (4.6.1), where $f : C' \to C$ is the inclusion. Let C'' be the full subcategory of C complementary to C' (i.e. $\operatorname{Ob} C''$ is the complement of $\operatorname{Ob} C'$ in $\operatorname{Ob} C$), and let $g : C'' \to C$ be the inclusion. Then the closed subtopos $\mathscr{F}$ of $E = \widehat{C}$, complementary to the open subtopos defined by C', is canonically equivalent to $\widehat{C''}$, with the inclusion morphism $i : \mathscr{F} \to E$ identifying with $\hat{g} : \widehat{C''} \to \widehat{C}$.

b) Consider the functor 471

(9.5.10.1) $$h : C'^0 \times C'' \longrightarrow (\mathrm{Ens}), \quad h(X', X'') = \operatorname{Hom}_C(X', X''),$$

and note that C, together with its full subcategory C', can be reconstructed up to isomorphism from the categories C', C'' and the bifunctor h (the latter may be chosen arbitrarily). (Compare 9.7.1 below.) The datum of h is equivalent to that of a functor $C'' \to \widehat{C'} = \mathbf{Hom}(C'^0, (\mathrm{Ens}))$, or equivalently (III 1.2 and III 1.7) to that of a functor

(9.5.10.2) $$\sigma : \widehat{C}'' \longrightarrow \widehat{C}'$$

which is continuous, i.e. (1.6) commutes with small inductive limits, or equivalently, admits a right adjoint

(9.5.10.3) $$\rho : \widehat{C}' \longrightarrow \widehat{C''}.$$

Show that ρ is precisely the gluing functor associated with the open subtopos $\widehat{C}'$ of $E = \widehat{C}$.

c) Deduce from a) and b) that if one is given a gluing functor (9.5.10.3) between two presheaf categories, and hence a glued topos E, the following conditions are equivalent:
   (i) E is equivalent to a topos of the form $\widehat{C}$.
   (ii) The functor ρ commutes with small products (or equivalently, it admits a left adjoint (9.5.10.2)).
   (iii) The inclusion morphism $i : \widehat{C''} \to E$ is "essential," i.e. i^* commutes with products, or equivalently i^* admits a left adjoint $i_!$.

   When this is so, describe explicitly a category C such that $E \approx \widehat{C}$ using the bifunctor (9.5.10.1) defined by the restriction of (9.5.10.2) to C''.

d) Show that a topos obtained by gluing two punctual topoi need not be equivalent to a topos of the form $\widehat{C}$.

Exercise 9.5.11. (Morphism from a glued topos to a topos.)

a) Let $\rho : \mathscr{U} \to \mathscr{F}$ be a functor between two categories, let $E = (\mathscr{F}, \mathscr{U}, \rho)$ be the "glued" category of 9.5.1, and let F be an arbitrary category. Consider the functor $f \mapsto \rho \circ f$: 472

$$\overline{\rho} : \overline{\mathscr{U}} = \mathbf{Hom}(F, \mathscr{U}) \longrightarrow \overline{\mathscr{F}} = \mathbf{Hom}(F, \mathscr{F}),$$

and define an equivalence of categories

$$\mathbf{Hom}(F, E) \xrightarrow{\approx} (\overline{\mathscr{U}}, \overline{\mathscr{F}}, \overline{\rho}).$$

b) Let $f : F \to E$ be a functor, and consider its composites with the canonical functors $j^* : E \to \mathscr{U}$ and $i^* : E \to \mathscr{F}$. Show that f commutes with a specified type of inductive limits (assumed to be representable in $\mathscr{U}$ and $\mathscr{F}$, and hence in E) if and only if $j^* f$ and $i^* f$ commute with them. Show that if ρ commutes with a specified type of projective limits (which is then representable in E), then f commutes with them if and only if the same holds for $j^* f$ and $i^* f$.

c) Now suppose that $\mathscr{U}$ and $\mathscr{F}$ are topoi, and that ρ is a gluing functor. Let F be a topos and $f : F \to E$ a functor. Conclude from b) that f is an inverse-image functor associated with a morphism of topoi $E \to F$ if and only if the same holds for the functors $j^*f : F \to \mathscr{U}$ and $i^*f : F \to \mathscr{F}$. Conclude from this and from a) that there is an equivalence between the category $\mathbf{Homtop}(E, F)$ and the category of triples (f', f'', u), where f' (resp. f'') is an object of $\mathbf{Homtop}(\mathscr{U}, F)$ (resp. $\mathbf{Homtop}(\mathscr{F}, F)$), and where $u : f''^* \to \rho \circ f'^*$ is a homomorphism of functors $F \to \mathscr{F}$: if f is an object of $\mathbf{Homtop}(E, F)$ and (f', f'', u) is the corresponding triple, then $f' = f \circ j$, $f'' = f \circ i$.

d) Specify in what sense c) may be said to provide a characterization, up to equivalence (in a 2-category of topoi), of the topos E in terms of the triple $(\mathscr{U}, \mathscr{F}, \rho)$.

Exercise 9.5.12. (*Gluing of two topological spaces.*)

a) Let X be a topological space, let $\rho : \mathrm{Top}(X) \to (\mathrm{Ens}) = \mathrm{Top}(pt)$ be a gluing functor, i.e. (I 8) a pro-representable functor, let $A \in \mathrm{Ob}\,\mathrm{Pro}(\mathrm{Top}(X))$ be the object which pro-represents it, and let E be the topos obtained from ρ by gluing. Show that E is equivalent to a topos of the form $\mathrm{Top}(Y)$ if and only if A is
473 isomorphic to an object of $\mathrm{Pro}(\mathrm{Ouv}(X))$. Conclude that if X is nonempty, one can find ρ such that E is not equivalent to a topos of the form $\mathrm{Top}(Y)$.

b) More generally, consider a gluing functor $\rho : \mathrm{Top}(X) \to \mathrm{Top}(Y)$, where X, Y are two topological spaces. Deduce from it a map

$$\phi : X \longrightarrow \mathrm{Ob}\,\mathrm{Pro}(\mathrm{Top}(Y))$$

(the composite of $X \to \mathrm{Ob}\,\mathbf{Point}(\mathrm{Top}(X)) \to \mathrm{Ob}\,\mathrm{Pro}(\mathrm{Top}(X))$ (6.8.5) and the pro-adjoint σ of ρ). (N.B. for $x \in X$, $\phi(x)$ identifies with the *fiber at x of the gluing sheaf* (9.6.4) *defined by ρ*.) Show that if E is equivalent to a topos of the form $\mathrm{Top}(Z)$, then ϕ takes its values in the essential image of $\mathrm{Pro}(\mathrm{Ouv}(Y))$.

Problem. Is the converse true?

9.6. Gluing sheaf. We saw in 9.5.4 that the datum of a topos E equipped with an open U amounts essentially to that of two topoi $\mathscr{U}$ and $\mathscr{F}$ and a gluing functor

$$\rho : \mathscr{U} \longrightarrow \mathscr{F} \tag{9.6.1}$$

i.e. a functor admitting a pro-adjoint

$$\sigma : \mathrm{Pro}(\mathscr{F}) \longrightarrow \mathrm{Pro}(\mathscr{U}). \tag{9.6.2}$$

By the sorites on pro-adjoint functors (I 8), the functor ρ is determined up to unique isomorphism (and consequently the glued topos $E = (\mathscr{F}, \mathscr{U}, \rho)$ is determined up to equivalence of topoi) once the functor σ is known, or, equivalently, once the functor induced by σ is known:

$$\sigma_\circ : \mathscr{F} \longrightarrow \mathrm{Pro}(\mathscr{U}). \tag{9.6.3}$$

The functor σ is determined, up to unique isomorphism, by the property of commuting with small filtered projective limits and of extending the functor $\sigma_\circ$. On the other hand, for a given functor (9.6.3) to be isomorphic to a pro-adjoint, it is plainly necessary and sufficient that, for every object X of $\mathscr{U}$, the contravariant functor

$$Y \longmapsto \mathrm{Hom}_{\mathrm{Pro}(\mathscr{U})}(\sigma_\circ(Y), X)$$

474 on $\mathscr{F}$ be representable, or equivalently be a sheaf on $\mathscr{F}$ for the canonical topology (1.2

iii)). It follows at once that this also means that the opposite functor

$$\sigma_\circ^\circ : \mathscr{F}^\circ \longrightarrow \mathrm{Pro}(\mathscr{U})^\circ \subset \check{\mathscr{U}} = \mathbf{Hom}(\mathscr{U}, (\mathrm{Ens})) \tag{9.6.4}$$

is a *sheaf* on $\mathscr{F}$ with values in $\mathrm{Pro}(\mathscr{U})^\circ$ (II 6.1). Thus, *the datum of a gluing functor* (9.6.1) *is also equivalent* (up to isomorphism) *to that of a sheaf on* $\mathscr{F}$ *with values in* $\mathrm{Pro}(\mathscr{U})^\circ$. The sheaf thus associated with a gluing functor (or with a situation $(E, U, \mathscr{F})$ as in (9.3)) is called the *gluing sheaf* associated with the gluing functor ρ under consideration (resp. with the open U of the topos E).

Exercise 9.6.5. Refine 9.5.4 into a statement of 2-equivalence of 2-categories between the 2-category consisting of pairs (E, U), where E is a topos and U an open of the latter, and the 2-category consisting of triples $(\mathscr{U}, \mathscr{F}, \rho)$, where $\mathscr{U}$ and $\mathscr{F}$ are topoi and $\rho : \mathscr{U} \to \mathscr{F}$ is a gluing functor (the explicit description of the 2-categories in question being left to the reader). Give a variant of this statement involving the 2-category of triples $(\mathscr{U}, \mathscr{F}, \phi)$, where ϕ is now a sheaf on $\mathscr{F}$ with values in $\mathrm{Pro}(\mathscr{U})^\circ$.

9.7. Points of a glued topos.

9.7.1. Let C be a category, and let

$$u : D' \longrightarrow C \quad , \quad v : D'' \longrightarrow C$$

be two fully faithful functors such that the essential images C' and C'' of u and v satisfy

$$\mathrm{Ob}\, C' \cap \mathrm{Ob}\, C'' = \phi \quad , \quad \mathrm{Ob}\, C' \cup \mathrm{Ob}\, C'' = \mathrm{Ob}\, C.$$

Suppose, in addition, that

$$X' \in \mathrm{Ob}\, C' \quad , \quad X'' \in \mathrm{Ob}\, C'' \Rightarrow \mathrm{Hom}(X'', X') = \phi,$$

or, equivalently, that

$$\mathrm{Hom}(v(A''), u(A')) = \phi \text{ for } A' \in \mathrm{Ob}\, D', A'' \in \mathrm{Ob}\, D''.$$

It is then immediate that the category C can be reconstructed, up to canonical isomorphism, from the categories C' and C'' and the functor 475

$$(X', X'') \longmapsto \mathrm{Hom}(X', X'') : C'^\circ \times C'' \longrightarrow (\mathrm{Ens});$$

similarly, C can be reconstructed up to equivalence (this equivalence being defined up to unique isomorphism) from D', D'', and the functor

$$(A', A'') \longmapsto \mathrm{Hom}(u(A'), v(A'')) : D'^\circ \times D'' \longrightarrow (\mathrm{Ens}).$$

This observation applies in particular to the situation described in the following proposition, and shows that the category $\mathbf{Point}(E)$ is determined, up to equivalence, by the categories $\mathbf{Point}(\mathscr{U})$ and $\mathbf{Point}(\mathscr{F})$ and by a certain functor

$$\mathbf{Point}(\mathscr{U})^\circ \times \mathbf{Point}(\mathscr{F}) \longrightarrow (\mathrm{Ens})$$

derived from the gluing functor $\rho : \mathscr{U} \to \mathscr{F}$.

Proposition 9.7.2. *Let E be a topos, let U be an open of E, let $\mathscr{U} = E_{/U}$, and let $\mathscr{F}$ be the closed subtopos of E complementary to U. Consider the functors $p \mapsto p \circ j$ and $q \mapsto q \circ i$ induced by the inclusion morphisms $j : \mathscr{U} \to E$ and $i : \mathscr{F} \to E$:*

$$u : \mathbf{Point}(\mathscr{U}) \longrightarrow \mathbf{Point}(E), \quad v : \mathbf{Point}(\mathscr{F}) \longrightarrow \mathbf{Point}(E). \tag{9.7.2.1}$$

Then the following assertions hold:

a) *The preceding functors u and v are fully faithful, and every point of E belongs to the essential image of exactly one of the functors u and v.*

b) *Let p and q be two points of E such that p (resp. q) belongs to the essential image of u (resp. of v). Then*

$$\mathrm{Hom}(q,p)=\phi. \tag{9.7.2.2}$$

c) *Let p (resp. q) be a point of $\mathscr{U}$ (resp. of $\mathscr{F}$). There is then an isomorphism, functorial in (q,p):*

$$\mathrm{Hom}(u(p),v(q))\simeq \mathrm{Hom}(\mathrm{Pro}(\rho)(p),q)\simeq \mathrm{Hom}(p,\sigma(q)), \tag{9.7.2.3}$$

where

$$\sigma : \mathrm{Pro}(\mathscr{F})\longrightarrow \mathrm{Pro}(\mathscr{U})$$

is the functor (9.6.2)*, pro-adjoint to the gluing functor $\rho : \mathscr{U}\to\mathscr{F}$, and where, in writing*

$$\rho : \mathscr{U}\longrightarrow\mathscr{F},$$

476 *we have identified (up to equivalence) the categories* $\mathbf{Point}(\mathscr{U})$ *and* $\mathbf{Point}(\mathscr{F})$ *with full subcategories of* $\mathrm{Pro}(\mathscr{U})$ *and* $\mathrm{Pro}(\mathscr{F})$*, respectively (6.8.5).*

Proof

a) The full faithfulness of u and v is known (9.1.4). The assertion concerning the essential images of u and v follows from the criteria 9.2.5 and 9.2.4 for a point $P\to E$ to factor, up to isomorphism, through $\mathscr{U}$ resp. $\mathscr{F}$, taking into account the fact that the punctual topos P has only two opens, namely the initial object and the terminal object of P.

b) In view of a), the assertion means that if the point $p : P\to E$ factors through $\mathscr{U}$, then the same holds for every point $p' : P\to E$ such that $\mathrm{Hom}(p',p)\neq\phi$. Now, if $\alpha : p'\to p$, it induces $p^*(e_{\mathscr{U}})\to p'^*(e_{\mathscr{U}})$, and then $p^*(e_{\mathscr{U}})=e_p$ implies $p'^*(e_{\mathscr{U}})=e_p$, C.Q.F.D.

c) By the lemma 9.7.2.4 below, applied to the inclusions $j : \mathscr{U}\to E$ and $i : \mathscr{F}\to E$, there is a diagram of functors commuting up to canonical isomorphisms:

$$\begin{tikzcd}
\mathrm{Point}(\mathscr{U}) \arrow[r, "u"] \arrow[d, hook] & \mathrm{Point}(E) \arrow[d, hook] & \mathrm{Point}(\mathscr{G}) \arrow[l, "v"'] \arrow[d, hook] \\
\mathrm{Pro}(\mathscr{U}) \arrow[r, "\mathrm{Pro}(j_*)"] & \mathrm{Pro}(E) & \mathrm{Pro}(\mathscr{G}), \arrow[l, "\mathrm{Pro}(i_*)"']
\end{tikzcd}$$

where the vertical arrows denote the canonical fully faithful functors (6.8.5). Moreover, since i_* admits a left adjoint i^*, $\mathrm{Pro}(i_*)$ admits the left adjoint $\mathrm{Pro}(i^*)$ I 8.11.5, and one finally obtains the functorial isomorphisms

$$\begin{aligned}
\mathrm{Hom}(u(p),v(q)) \simeq{} & \mathrm{Hom}_{\mathrm{Pro}(E)}(\mathrm{Pro}(j_*)(p),\mathrm{Pro}(i_*)(q))\\
& \mathrm{Hom}_{\mathrm{Pro}(\mathscr{F})}(\mathrm{Pro}(i^*)(\mathrm{Pro}(j_*)(p)),q)\\
& \mathrm{Hom}_{\mathrm{Pro}(\mathscr{F})}(\mathrm{Pro}(i^*j_*)(p),q),
\end{aligned}$$

which is precisely the first formula in (9.7.2.3), the second then being immediate from the definition of σ as a pro-adjoint of ρ. This proves 9.7.2, subject to the following lemma.

Lemma 9.7.2.4. *Let $f : F\to E$ be a morphism of topoi. Then the diagram of functors*

(9.7.2.4.1)
$$\begin{tikzcd}
\mathrm{Point}(F) \arrow[r, "p\mapsto f\circ p"] \arrow[d] & \mathrm{Point}(E) \arrow[d] \\
\mathrm{Pro}(F) \arrow[r, "\mathrm{Pro}(f_*)"] & \mathrm{Pro}(E)
\end{tikzcd}$$

commutes up to canonical isomorphism (the vertical arrows denoting the fully faithful functors 6.8.5). 477

The verification is immediate from the definitions and is left to the reader.

COROLLARY 9.7.3. *Let E be a topos having enough points. Then every closed subtopos $\mathscr{F}$ of E also has enough points. More precisely, if $(p_i)_{i\in I}$ is a conservative family of points of E, then the family of points of $\mathscr{F}$ defined by the subfamily $(p_i)_{i\in J}$ consisting of those p_i which come from points of $\mathscr{F}$ is conservative. (Compare 6.7.3, 6.7.4.)*

Indeed, let $f : X \to Y$ be an arrow of $\mathscr{F}$ which is carried to an isomorphism by the fiber functors associated with the points under consideration of $\mathscr{F}$. Since $i^*i_* \simeq \mathrm{id}_{\mathscr{F}}$, it follows that $i_*(f) : i_*(X) \to i_*(Y)$ is carried to an isomorphism by the fiber functors associated with the p_i for $i \in J$; the same is true for the p_i with $i \in I - J$, i.e. those coming from points of U, since j^*i_* is the constant functor with value the terminal object. Therefore $i_*(f)$ is an isomorphism, and hence so is $f \simeq i^*i_*(f)$, C.Q.F.D.

REMARK 9.7.4. If E is a topos having enough points, it is clear that an open U of E is determined once the full subcategory of $\mathbf{Point}(E)$ which is the essential image of $\mathbf{Point}(E_{/U})$ is known; in fact, if C is a full subcategory of $\mathbf{Point}(E)$ defining a conservative family of points of E, it even suffices to know the subcategory $C_{/U}$ consisting of those elements of C which belong to the essential image of $\mathbf{Point}(U)$. It follows from this and from 9.7.2 a) that a closed subtopos $\mathscr{F}$ of E is determined once the essential image of $\mathbf{Point}(\mathscr{F})$ in $\mathbf{Point}(E)$ is known, or even merely its intersection with C.

EXERCISE 9.7.5. Let E be a topos. For every subtopos F of E, let $P(F)$ be the subset of Point(E) formed by the isomorphism classes of points of E which factor through F. Thus $P(F)$ is homeomorphic to Point(F) (9.1.8 c)). Show that if F is an open (resp. closed, resp. locally closed (9.4.9)) subtopos of E, then $P(F)$ is an open (resp. closed, resp. locally 478
closed) subset of Point(E). Show that when E has enough points, the same is true of every locally closed subtopos of E, and the map $F \mapsto P(F)$ induces a bijection between the set of open (resp. closed, resp. locally closed) subtopoi of E and the set of open (resp. closed, resp. locally closed) subsets of Point(E). Extend the preceding considerations to the case where Point(E) is replaced by any subspace X of Point(E) corresponding to a conservative family of points of E.

9.8. Complements concerning certain topoi associated with topological spaces. In this section we give a few complements in the form of exercises. The reader is advised to look through them in order to become accustomed to the viewpoint of topoi in various classical situations.

EXERCISE 9.8.1. (*Descent of sheaves on a topological space.*)
Let $f : X \to Y$ be a continuous map of topological spaces in $\mathscr{U}$, giving a functor

$$f^* : \mathrm{Top}(Y) \longrightarrow \mathrm{Top}(X).$$

a) The following conditions are equivalent:
  (i) f^* is faithful.
  (ii) f^* is conservative.
  (iii) $f(X)$ is a very dense subset (EGA IV 10.1.3) of Y.

  It suffices for f or f_{sob} to be surjective. Give an example where f^* is faithful and $f = f_{\mathrm{sob}}$ is not surjective (take X discrete, f injective, $f(X)$ the set of closed points of Y, and Y a nondiscrete Noetherian topological space). Give an example where f_{sob} is surjective but f is not, and another where f is surjective but f_{sob} is not.

b) Suppose that $f(X)$ is very dense in Y and that its topology is the quotient topology induced from X. Prove that the arrow f of (Esp) is a descent morphism for the fibered category of sheaves of sets on varying spaces, i.e. that the functor
479 f^* induces a *fully faithful* functor from $\mathrm{Top}(Y)$ to the category of objects of $\mathrm{Top}(X)$ equipped with descent data relative to $f : X \to Y$. Reduce to the case where f is surjective and follow the argument of VIII 9.1 given in the setting of étale topologies of schemes. Give a converse.

c) Suppose that $f(X)$ is very dense in Y, that its topology is the quotient topology induced from X, and that the fibers of f are connected. Prove that the functor f^* is fully faithful. (Follow the argument of SGA 1 IX 3.4.) Give a converse, at least if the points of Y are closed.

d) Suppose that Y is the union of open subsets Y_i over which X admits sections. Prove that f is then an *effective descent morphism* for the fibered category of sheaves of sets on varying topological spaces, i.e. that the functor considered in c) is in fact an equivalence of categories.

Exercise 9.8.2. (*Quotient topoi of topological spaces. Topological étendues.*)

a) Let

$$X_2 \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow}\; X_1 \;\underset{p_2}{\overset{p_1}{\rightrightarrows}}\; X_0 \tag{9.8.5.1}$$

be a semi-simplicial object truncated in degree 2 in the category (Esp) of topological spaces $\in \mathscr{U}$; the inverse-image functors then give a diagram of categories of sheaves

$$\mathrm{Top}(X_0) \;\underset{p_2^*}{\overset{p_1^*}{\rightrightarrows}}\; \mathrm{Top}(X_1) \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow}\; \mathrm{Top}(X_2).$$

(More precisely, one has a category cofibered over the category of standard simplices Δ_n, $0 \le n \le 2$.) Show that the category $\varprojlim$ of this diagram (i.e. the category of sheaves of sets on X_0 *equipped* with a descent datum relative to the diagram 9.8.2.1, i.e. with an isomorphism $p_1^*(F) \simeq p_2^*(F)$ satisfying...), is a $\mathscr{U}$-topos T. (For a more general statement on the stability of topoi under $\varprojlim$ operations, cf. VI 8.) Define a morphism of topoi $q : \mathrm{Top}(X_0) \to T$, and prove
480 that for every $\mathscr{U}$-topos E, $\mathbf{Homtop}(T, E)$ is equivalent, via q^*, to the category $\varprojlim$ of the following diagram of categories, deduced from (9.8.2.1):

$$\mathbf{Homtop}(\mathrm{Top}(X_0), E) \rightrightarrows \mathbf{Homtop}(\mathrm{Top}(X_1), E) \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow}\; \mathbf{Homtop}(\mathrm{Top}(X_2), E).$$

b) In particular, let R be an equivalence relation in an object X of (Esp), i.e. (I 10.9) a subobject R of $X \times X$ (N.B. the inclusion $R \hookrightarrow X \times X$ is continuous, but the topology of R is not necessarily induced by that of $X \times X$), such that, for every object Y of (Esp), $\mathrm{Hom}(Y, R)$ is the graph of an equivalence relation in $\mathrm{Hom}(Y, X)$. This gives a diagram of the form (9.8.2.1), with $X_0 = X$, $X_1 = R$, $X_2 = (R, p_2) \times_X R, p_1$, where p_1, p_2 are the two projections from R to X, and hence a topos, which we denote by $\mathrm{Top}(X)/R$, or even $\mathrm{Top}(X/R)$, or indeed X/R, by abuse of notation, and which plays the role of a quotient of X by R in the sense made precise in a). Suppose that the topology of R is induced by that of $X \times X$ and that, denoting by X/R the ordinary quotient topological space, X

locally admits sections over X/R. Prove that T is then equivalent to $\mathrm{Top}(X/R)$. (Use 9.8.1 d).)

c) Let $H \to X$ be an injective morphism of topological groups, i.e. a monomorphism of group objects in (Esp), and let R be the equivalence relation that it defines in X, i.e. $R = H \times X$, with $p_1 = \mathrm{pr}_1$ and p_2 defined by the action of H on X by left translations. Show that the topology of R is induced by that of $X \times X$ if and only if the topology of H is induced by that of X. Give examples in which this condition is not satisfied and in which the ordinary quotient topology on X/H is the indiscrete topology, with a) $H = \mathbb{Z} \times \mathbb{Z}$, $X = \mathbb{R}$, and b) $H = \mathbb{R}$, $X = \mathbb{T} \times \mathbb{T}$, where $\mathbb{T} = \mathbb{R}/\mathbb{Z}$ ("geodesics on the torus"). Prove that the two topoi so obtained are equivalent and are not equivalent to $\mathrm{Top}(X/H)$ (which is a final topos (2.2)), nor to any topos of the form $\mathrm{Top}(Y)$.

d) Let G be a topological group acting on a topological space X, whence, in the familiar way, a semi-simplicial object

$$\ldots G \times G \times X \;\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow}\; G \times X \;\substack{\longrightarrow\\ \longrightarrow}\; X.$$

Show that the topos T deduced from it by a) identifies with the category of spaces X' over X having G as a group of operators, such that $X' \to X$ is an étale map (compatible with the action of G). In particular, when G is discrete, one recovers the topos $\mathrm{Top}(X, G)$ of 2.3. 481

The preceding considerations justify the notation $\mathrm{Top}(X)/G$, or indeed $\mathrm{Top}(X/G)$ or even simply X/G, for the preceding topos T. One should bear in mind, however, that when G is a discrete group (to fix ideas) acting *properly* on X, so that the ordinary quotient space X/G has reasonably good properties, the natural morphism of topoi $T \to \mathrm{Top}(X/G)$ deduced from the universal characterization a) of T is an equivalence of topoi only if G acts freely, i.e. without fixed points, i.e. when $G \times X \to X \times X$ is a monomorphism. Thus, in the case of a "pre-equivalence relation" (or "groupoid" in the sense of (SGA 3 V 1)) which is not an equivalence relation, passage to the quotient in the sense of topoi (or "passage to the fine quotient") does not in general correspond (via the correspondence $Y \mapsto \mathrm{Top}(Y)$) to the usual topological quotient.

e) Let T be a $\mathcal{U}$-topos. Show that the following conditions are equivalent:
   (i) There exists a family $(S_i)_{i \in I}$ of objects of T covering the final object of T, such that for every $i \in I$ the induced topos $T_{/S_i}$ is equivalent to a topos of the form $\mathrm{Top}(X)$, with $X \in$ (Esp).
   (i′) There exists an $S \in \mathrm{Ob}\,T$ covering the final object such that the induced topos $T_{/S}$ is equivalent to a topos of the form $\mathrm{Top}(X)$.
   (ii) There exist a topological space X and an *étale* pre-equivalence relation R in X (in the sense of the category (Esp)) (i.e. one for which $p_1 : R \to X$ is an étale map), such that T is equivalent to $\mathrm{Top}(X/R)$, with the notation of b).

The topos T is then said to be *locally a topological space*, or alternatively a *topological étendue* (or simply an *étendue*, when no confusion is likely). 482

f) Show that the topos T constructed in a) has enough points and, more precisely, that it admits a conservative family of points parametrized by the (small) set X_0. In particular, every étendue has a small conservative family of points, and hence has enough points. When an étendue T is represented in the form $\mathrm{Top}(X/R)$ as in e) ii), prove that every point of T is isomorphic to the image of a point of

Top(X), and is therefore defined by an ordinary point of X_{sob} (or of X itself, when X is sober).

g) Show that the topos Top(X/G) of 2.3 is an étendue. (Use its description in d) or 7.1.10 c).) Show that, for every $x \in X$, the monoid of endomorphisms of the point of $\text{Top}(X, G) = \text{Top}(X)/G$ defined by x is canonically isomorphic to the stabilizer group G_x of x. In particular, the points of an étendue may have nontrivial automorphism groups. Determine **Point**(T) for the étendue T defined by an étale pre-equivalence relation in a space X, and show that every endomorphism of a point of T is an automorphism.

h) Let T be a topos. Show that the following conditions are equivalent:
   i) For every point p of T, every endomorphism of p (resp. every automorphism of p) is the identity.
   ii) For every topos S having enough points and every morphism of topoi $f : S \to T$, every endomorphism of f (resp. every automorphism of f) is the identity.

   Likewise, show that the following conditions are equivalent:
   i′) For two points p, q of T, there exists at most one morphism from p to q.
483 ii′) For two morphisms $f, g : S \rightrightarrows T$ from a topos S having enough points to T, there exists at most one morphism from f to g.

   When these latter conditions hold, the topos T is said to be *pointwise rigid*, or simply *rigid*. Show that if X is a topological space equipped with an étale pre-equivalence relation R, the quotient étendue $T = \text{Top}(X)/R$ is rigid if and only if R is an equivalence relation.

i) Let (X, G) be a topological space with a discrete group of operators, let H be a normal subgroup of G such that the induced action of H on X is proper and free, and let $X' = X/H$, $G' = G/H$, so that G' acts on X' in the evident way. Prove that the morphism of topoi

$$\text{Top}(X, G) \longrightarrow \text{Top}(X', G')$$

   induced by the canonical morphism $(X, G) \to (X', G')$ of spaces with operators (4.1.2) is an equivalence of topoi.

   Show that the set of open subtopoi of Top(X/G) identifies with the set of open subsets of X stable under the action of G, or equivalently with the set of open subsets of the quotient topological space X/G. Deduce that Top(X/G) is connected if and only if every subset of X which is both open and closed and stable under the action of G is either empty or equal to X. When the connected components of X are open, Top(X/G) is connected if and only if G acts transitively on the set $\pi_0(X)$ of connected components of X; more generally, $\pi_0(\text{Top}(X/G))$ (8.7 g)) is an essentially constant pro-set isomorphic to the quotient set $\pi_0(X)/G$.

   Show that when X is locally connected and locally simply connected, and $T = \text{Top}(X, G)$ is connected, i.e. G is transitive on $\pi_0(X)$, then among all the ways of representing T by means of a space with operators (X', G') (cf. above for some such representations), there is one, unique up to nonunique isomorphism, for which X' is connected and simply connected. Show that choosing such an (X', G') amounts to choosing a "universal covering" of the final object of T, and that G' is isomorphic to the group $\pi_1(T)$ relative to this universal covering (cf. 2.7.5).

484 (Hint: show that giving an equivalence of T with a Top(X, G) amounts to

giving a G_T-torsor P in T, and that, if (X, G) and P correspond, there is a canonical equivalence $\mathrm{Top}(X) \simeq T/P$.)

Deduce from this a trivial proof of the last assertion of c). Characterize the topoi T equivalent to a $\mathrm{Top}(X, G)$, where G is a discrete group acting on a locally connected and locally simply connected topological space while transitively permuting its connected components, as the connected, locally connected, and locally simply connected étendues whose universal covering P of the final object e_T of T is such that the induced topos $T_{/P}$ is equivalent to a topos of the form $\mathrm{Top}(X)$ (or, as we shall simply say by abuse of language, $T_{/P}$ "is" a topological space).

Give an example of an étendue, locally isomorphic to $\mathbb{R}$, which is connected and simply connected but is not a topological space. (Take the universal covering of the line $\mathbb{R}$ with doubled origin.)

j) A ringed topos $(T, \mathscr{O}_T)$ (cf. § 11) is called a *differentiable étendue* (resp. a *real-analytic étendue*, resp. a *complex-analytic étendue*) if there exist objects S_i of T covering the final object such that the induced ringed topos $(T_{/S_i}, \mathscr{O}_T/S_i)$ is equivalent to the ringed topos defined by a differentiable manifold with its sheaf of real C^∞ functions (resp. to the ringed topos defined by a real-analytic space, resp. to the ringed topos defined by a complex-analytic space). Give a constructive description of these ringed topoi of the type in e) ii) (taking for X, respectively, a differentiable manifold, a complex-analytic space, or a real-analytic space). Give examples of such ringed topoi which are not equivalent to ringed topoi associated with topological spaces, by returning to the examples considered in c).

Exercise 9.8.3. (*Topoi associated with local equivalence relations and with foliations*)[14]. 485

Let X be a topological space.

a) For every open subset U of X, let $\mathrm{Quot}(U)$ be the set of equivalence relations on U, and, for an open subset $V \subset U$, consider the natural map $\mathrm{Quot}(U) \to \mathrm{Quot}(V)$. This gives a presheaf Quot_X on X. Let Q_X, or simply Q, be the associated sheaf, and let r be a section of Q (a "*local equivalence relation on* X"). From the order relation on the set $\mathrm{Quot}(U)$ of equivalence relations on an open subset U of X, deduce on the presheaf Quot, and on the associated sheaf Q, an order structure, i.e. a structure of a presheaf resp. a sheaf with values in the category of ordered sets.

b) For every equivalence relation R on X, consider the section $\mathrm{loc}(R)$ of Q that it defines. Let $E(r)$ be the set of equivalence relations on X such that $\mathrm{loc}(R)$ is less fine than r, and let $\mathrm{glob}(r)$ be the least upper bound of the equivalence relations in $E(r)$ (its graph is the intersection of the graphs of the $R \in E(r)$). Let $(U_x)_{x \in X}$ be a family of open neighborhoods of the points $x \in X$, and, for every $x \in X$, let R_x be an equivalence relation on U_x whose germ at x is r_x. For every family $\mathscr{V}$ of open subsets V_x ($x \in X, x \in V_x \subset U_x$), let $R_{\mathscr{V}}$ be the equivalence relation on X generated by the family of relations $R_x | V_x$. Show that if $\mathscr{V}' \leq \mathscr{V}$ (in the obvious sense), then $R_{\mathscr{V}'} \geq R_{\mathscr{V}}$, and that $\mathrm{glob}(r)$ is the least upper bound of the increasing filtered family of equivalence relations $R_{\mathscr{V}}$. Give an example in

[14] This exercise is given subject to every reservation, since it was written hastily and insufficiently checked. Mr. N. Saavedra checked parts a) through e). The reader is requested to communicate any observations to us.

which glob(r) does not belong to $E(r)$, i.e. in which there is an $a \in X$ such that,
486 for every open neighborhood $W \subset U_a$ of a, there exist a $\mathscr{V} = (V_x)$ and two points y, z of W which are equivalent under R_a but not under $R_{\mathscr{V}}$. Show that, for every $R \in \mathrm{Quot}(X)$, one has $\mathrm{glob}\,\mathrm{loc}(R) \geq R$.

c) The local equivalence relation r is said to be *precoherent (resp. coherent)* if $\mathrm{glob}(r) \in E(r)$, i.e. if $\mathrm{loc}(\mathrm{glob}(r)) \leq r$ (resp. if, for every open subset U of X, the restriction of r to U is precoherent). It is said to be globally coherent if it is coherent and if, in addition, $r = \mathrm{loc}\,\mathrm{glob}(r)$. An equivalence relation R on X is said to be *locally coherent* if $r = \mathrm{loc}(R)$ is coherent, and *coherent* if, in addition, $R = \mathrm{glob}(r)$, i.e. if $R = \mathrm{glob}(\mathrm{loc}(R))$. Show that globally coherent local equivalence relations r on X correspond bijectively to coherent equivalence relations R on X, by $r \mapsto \mathrm{glob}(r)$ and $R \mapsto \mathrm{loc}(R)$.

d) Show that, in order for r to be coherent, it is enough that, for every $a \in X$ and every open neighborhood $W' \subset U_a$ of a, there exist an open neighborhood $W \subset W'$ of a such that every equivalence class of $R_a|W$ is contained in a connected component of an equivalence class of $R_a|W'$ (a fortiori, it is enough that there exist a fundamental system of open neighborhoods $W_a \subset U_a$ of a such that the fibers of the induced equivalence relation $R_a|W_a$ are connected). (Hint: reduce to proving precoherence in the case where r is defined by an $R \in \mathrm{Quot}(X)$, and observe in that case that the fibers of the equivalence relations $R_{\mathscr{V}}$ are relatively open—and hence also relatively closed—subsets of the fibers of R.) Show that, in order for a global equivalence relation R to be coherent, it is enough that it be locally coherent and that its fibers be connected; prove that this condition is necessary if the fibers are closed. Give an example of an equivalence relation with connected and locally connected fibers which is not locally coherent (?).

487 e) Let $f : X' \to X$ be a continuous map. Define a natural homomorphism of presheaves $f_*(\mathrm{Quot}_{X'}) \leftarrow \mathrm{Quot}_X$, inducing a homomorphism of sheaves $f_*(Q_{X'}) \leftarrow Q_X$, and hence a homomorphism $f^*(Q_X) \to Q_{X'}$. If r is a local equivalence relation on X, then f is said to be a "*fiber map*" for the local equivalence relation r on X if the inverse image of r is the coarse local equivalence relation on X'. For every topological space X', let $\mathrm{Homfib}_r(X', X)$ be the set of continuous maps from X' to X which are fiber maps relative to r. Show that, as X' varies, this defines a contravariant functor in X'.

f) Suppose that r is coherent. Show that the preceding functor is representable by a topological space X^r over X. Show that the universal fiber map $\Phi : X^r \to X$ is bijective, and that every $x \in X^r$ has an open neighborhood U such that the map $U \to X$ induced by Φ is a homeomorphism of U onto its image. Show that the connected components of X^r are open and correspond under the bijection Φ to the fibers of the equivalence relation $R = \mathrm{glob}(r)$. Give an example in which the restriction of Φ to the connected components of X^r does not induce homeomorphisms from those spaces onto their images in X.

g) The local equivalence relation r is said to be *open* if it is a section of the subsheaf Qouv_X of Q arising from the subpresheaf $\mathrm{Quotouv}_X$ of Quot_X whose value on every open subset U of X is the set of *open* equivalence relations on U. The local equivalence relation r is said to be *strictly open* if it is coherent and open or, equivalently, if it is coherent and if, for every open subset U of X, $\mathrm{glob}(r|U)$ is an open equivalence relation on U. An equivalence relation R on X is said to be *strictly open* if it is coherent and if $\mathrm{loc}(R)$ is a strictly open local equivalence relation. Prove that, for the open relation r to be strictly open, it is enough that it satisfy the condition considered in d); if r is locally defined by an R with closed fibers, then that condition is also necessary. **488**

h) Let F be a sheaf of sets on X. For every open subset U of X, let Quot(U, F) be the set of pairs consisting of an equivalence relation R_U on U and an equivalence relation $R_{F|U}$ on $F|U$ (F being regarded as an étale space over X), such that the structure map $p : F|U \to U$ is compatible with the equivalence relations R_U and $R_{F|U}$, and such that the corresponding diagram of topological spaces

$$\begin{tikzcd} F|U \arrow[r] \arrow[d, "p"'] & (F|U)/R_{F|U} \arrow[d, "q"] \\ U \arrow[r] & U/R_U \end{tikzcd}$$

is Cartesian, with q an étale map (so that $F|U$ is identified with the inverse image of the sheaf $(F|U)/R_{F|U}$ on U/R_U). Show that the Quot(U, F), for varying U, define a presheaf on X, whose associated sheaf will be denoted by $Q(F/X)$. Define a homomorphism of sheaves $Q(F/X) \to Q_X$. For a given section r of Q_X, an *r-structure* on the sheaf F is a section of $Q(F/X)$ lying above the given section r of Q_X. Define the category Top(X/r) of r-sheaves on X, and define a conservative and faithful functor, "forgetting the r-structure", Top$(X/r) \to$ Top(X). Prove that finite projective limits and finite inductive limits are representable in Top(X/r), and that the preceding functor commutes with those limits. Deduce that finite sums in Top(X/r) are disjoint and universal and that equivalence relations are universally effective. Prove that Top(X/r) has a small generating family. Give an example, with r coherent, in which Top(X/r) does not have infinite coproducts (?), and hence is not a topos.

i) Let $f : X \to Y$ be a continuous map compatible with $R = \mathrm{glob}(r)$. Define a functor

$$f_r^* : \mathrm{Top}(Y) \longrightarrow \mathrm{Top}(X/r)$$

whose composite with the inclusion functor Top$(X/r) \to$ Top(X) is f^*. Show **489**
that, if r is globally coherent and strictly open (i.e. if R is strictly open and $r =$ loc(R)), and if f is the canonical map $X \to Y = X/R$, then f_r^* is an equivalence of categories. Consequently, Top(X/r) is a $\mathscr{U}$-topos, and Top$(X/r) \to$ Top(X) is the inverse-image functor of a morphism of topoi.

j) Deduce from i) that, if r is strictly open, then Top(X/r) is a topos, and that the inclusion functor Top$(X/r) \to$ Top(X) is the inverse-image functor associated with a morphism of topoi

$$p : \mathrm{Top}(X) \longrightarrow \mathrm{Top}(X/r).$$

k) Define a functorial construction of the topos Top(X/r) and of the preceding morphism of topoi p with respect to the pair (X, r) consisting of a topological space X equipped with a strictly open local equivalence relation. Show that, if $X' \to X$ is an étale map and r' is the inverse image of r on X', then the induced morphism Top$(X'/r') \to$ Top(X/r) is equivalent to a localization morphism Top$(X/r)_{/E} \to$ Top(X/r), where E is an object of Top(X/r) (determined up to a unique isomorphism). The object E covers the final object of Top(X/r) if and only if the saturation under $R = \mathrm{glob}(r)$ of the image of X' in X is all of X.

l) Deduce from k) that $\mathrm{Top}(X/r)$ is a *rigid étendue* (9.8.2 h)). Show that, if X is sober, the set of isomorphism classes of points of $\mathrm{Top}(X/r)$, endowed with its canonical topology (7.8), is homeomorphic to the quotient topological space $X/\operatorname{glob}(r)$, and that, for any two points of $\mathrm{Top}(X/r)$ arising from points x and x' of X, there is at most one morphism from one to the other. Such a morphism exists if and only if x' and x are equivalent modulo $\operatorname{glob}(r)$ to points x_1' and x_1 such that x_1 is a generalization of x_1'.

490 m) Study the canonical morphism of topoi $\mathrm{Top}(X) \to \mathrm{Top}(X/r)$, observing that, for every object U of $\mathrm{Top}(X/r)$ such that the topos induced on U "is" an ordinary topological space, the induced morphism $\mathrm{Top}(X)_{/p^*(U)} \to \mathrm{Top}(X/r)_{/U}$ "is" a continuous map of ordinary topological spaces $X_U \to U$. Show that the fibers of the map $X_U \to U$ are homeomorphic to connected components of the space X^r of f).

n) Let X be a differentiable manifold equipped with a foliation, i.e. with a subsheaf F, locally a direct factor of the tangent sheaf of X, which is stable under brackets. Define on X a strictly open local equivalence relation associated with the foliation, and define on the quotient topos $T = \mathrm{Top}(X/r)$ a ring which makes it a *differentiable étendue* (9.8.2 j)). Define an equivalence between the category of locally free modules on the differentiable étendue T (also called differentiable vector bundles on T) and the category of locally free modules on X equipped with a connection relative to the given subbundle F of the tangent bundle. Give variants in the real-analytic and complex-analytic cases.

10. Sheaves of morphisms

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Proposition 10.1. *Let E be a topos and let X and Y be two objects of E; the functor $Z \mapsto \mathrm{Hom}_E(Z \times X, Y) \simeq \mathrm{Hom}_{E_{/Z}}(X_Z, Y_Z)$ is representable.*

Indeed, inductive limits are universal in E (II 4.3). Hence the functor $Z \mapsto Z \times X$ commutes with inductive limits. Consequently, the functor $Z \mapsto \mathrm{Hom}_E(Z \times X, Y)$ carries inductive limits in the argument Z to projective limits. It is therefore representable (1.2.1).

10.2. The object representing the functor $Z \mapsto \mathrm{Hom}_E(Z \times X, Y)$ is denoted by $\mathbf{Hom}_E(X, Y)$ (or more simply by $\mathbf{Hom}(X, Y)$) and is called the *sheaf of morphisms from X to Y*. It is a bifunctor in X and Y. We thus have a trifunctorial isomorphism

$$\mathrm{Hom}_E(Z, \mathbf{Hom}_E(X, Y)) \simeq \mathrm{Hom}_E(Z \times X, Y). \tag{10.2.1}$$

It then follows from formula (10.2.1) that the bifunctor $(X, Y) \mapsto \mathbf{Hom}(X, Y)$ carries inductive limits in the argument X (resp. projective limits in the argument Y) to projective limits in E.

Proposition 10.3. *Let $v : E \to E'$ be a morphism of topoi. For a variable object X of E and a variable object Y of E', there is a bifunctorial isomorphism*

$$v_* \mathbf{Hom}_E(v^* Y, X) \simeq \mathbf{Hom}_{E'}(Y, v_* X). \tag{10.3.1}$$

Proof. For every object Z of E', we have a sequence of isomorphisms, functorial in all the arguments:

$$\begin{array}{lll}
\mathrm{Hom}_{E'}(Z, v_* \mathbf{Hom}_E(v^*Y, X)) & \simeq \mathrm{Hom}_E(v^*Z, \mathbf{Hom}_E(v^*Y, X)) & \text{(adjunction)} \\
\mathrm{Hom}_E(v^*Z, \mathbf{Hom}_E(v^*Y, X)) & \simeq \mathrm{Hom}_E(v^*Z \times v^*Y, X) & (10.2.1) \\
\mathrm{Hom}_E(v^*Z \times v^*Y, X) & \simeq \mathrm{Hom}_E(v^*(Z \times Y), X) & (v^* \text{ is left exact}) \\
\mathrm{Hom}_E(v^*(Z \times Y), X) & \simeq \mathrm{Hom}_{E'}(Z \times Y, v_*X) & \text{(adjunction)} \\
\mathrm{Hom}_{E'}(Z \times Y, v_*X) & \simeq \mathrm{Hom}_{E'}(Z, \mathbf{Hom}_{E'}(Y, v_*X)) & (10.2.1)
\end{array}$$

The two sides of (10.3.1) therefore represent isomorphic functors, C.Q.F.D.

Corollary 10.4. *Let C be a $\mathscr{U}$-site, X a $\mathscr{U}$-presheaf on C, Y a $\mathscr{U}$-sheaf on C, and $\mathscr{V}$ a universe containing $\mathscr{U}$ such that C is $\mathscr{V}$-small. Let $C^{\wedge}_{\mathscr{V}}$ be the topos of $\mathscr{V}$-presheaves on C, and let $C^{\sim}_{\mathscr{U}}$ be the topos of $\mathscr{U}$-sheaves on C. The presheaf $\mathbf{Hom}_{C^{\wedge}_{\mathscr{V}}}(X, Y)$ is a $\mathscr{U}$-sheaf. Let $X \to \underline{a}X$ be the canonical morphism from X to its associated sheaf. The corresponding morphism $\mathbf{Hom}_{C^{\wedge}_{\mathscr{V}}}(\underline{a}X, Y) \to \mathbf{Hom}_{C^{\wedge}_{\mathscr{V}}}(X, Y)$ is an isomorphism. There is a canonical isomorphism $\mathbf{Hom}_{C^{\wedge}_{\mathscr{V}}}(\underline{a}X, Y) \simeq \mathbf{Hom}_{C^{\sim}_{\mathscr{U}}}(\underline{a}X, Y)$.* 492

10.4.1. Suppose first that C is a small site and that $\mathscr{V} = \mathscr{U}$. We then have a morphism of topoi $C^{\sim}_{\mathscr{U}} \to C^{\wedge}_{\mathscr{U}}$ (4.8), from which (10.3.1) gives the assertions in this case. To pass from this to the general case, note first that the "associated sheaf" functor is independent of the universe (II 3.6), and that the topos of $\mathscr{V}$-sheaves on C is equivalent to the topos of $\mathscr{V}$-sheaves on $C^{\sim}_{\mathscr{U}}$ (III 4 and 1). It therefore suffices to prove the following lemma:

Lemma 10.4.2. *Let E be a $\mathscr{U}$-topos, let $\mathscr{V}$ be a universe containing $\mathscr{U}$, let $E^{\sim}_{\mathscr{V}}$ be the topos of $\mathscr{V}$-sheaves on E, and let X and Y be two objects of E. There exists a canonical isomorphism $\mathbf{Hom}_E(X, Y) \simeq \mathbf{Hom}_{E^{\sim}_{\mathscr{V}}}(X, Y)$.*

10.4.3. The functor $\epsilon_E : E \to E_{\widetilde{\mathscr{V}}}$ is fully faithful and left exact. Consequently, for every object Z of E, there is an isomorphism

$$\mathrm{Hom}_{E_{\widetilde{\mathscr{V}}}}(\epsilon_E Z, \epsilon_E \mathbf{Hom}_E(X, Y)) \simeq \mathrm{Hom}_{E_{\widetilde{\mathscr{V}}}}(\epsilon_E Z \times \epsilon_E X, \epsilon_E Y).$$

But every object of $E_{\widetilde{\mathscr{V}}}$ is an inductive limit of objects coming from E; the assertion therefore follows from (10.2.1).

10.5. Let $v : E \to E'$ be a morphism of topoi. We shall define four bifunctorial morphisms:

$$\begin{array}{llll}
\Phi_v : v^* \mathbf{Hom}(X, Y) \longrightarrow \mathbf{Hom}(v^*X, v^*Y) & , & X \text{ and } Y \text{ objects of } E', \\
\Psi_v : v_* \mathbf{Hom}(X, Y) \longrightarrow \mathbf{Hom}(v_*X, v_*Y) & , & X \text{ and } Y \text{ objects of } E, \\
\Xi_v : \mathbf{Hom}(v_!X, Y) \longrightarrow v_* \mathbf{Hom}(X, v^*Y) & , & X \in \mathrm{ob}\, E, Y \in \mathrm{ob}\, E', \\
\Lambda_v : v_!(X \times v^*Y) \longrightarrow v_!(X) \times Y & & X \in \mathrm{ob}\, E, Y \in \mathrm{ob}\, E',
\end{array}$$

where, in order to define the morphisms Ξ_v and Λ_v, we suppose that the inverse-image functor v^* of v admits a left adjoint $v_!$. 493

10.5.1. *Definition of Ψ_v.* Let X be an object of E. There is an adjunction morphism $v^*v_*X \to X$, and hence, for every object Z of E', a bifunctorial morphism $v^*(Z \times v_*X) \to (v^*Z) \times X$. Thus, for every object Y of E, there is a trifunctorial morphism

$$\mathrm{Hom}_E((v^*Z) \times X, Y) \longrightarrow \mathrm{Hom}_E(v^*(Z \times v_*X), Y).$$

By adjunction this gives a trifunctorial morphism

$$\mathrm{Hom}_{E'}(Z, v_* \mathbf{Hom}(X, Y)) \longrightarrow \mathrm{Hom}_{E'}(Z, \mathbf{Hom}(v_*X, v_*Y));$$

and hence the morphism Ψ_v.

10.5.2. *Definition of Λ_v.* For every object X of E, there is an adjunction morphism $X \to v^*v_!X$, and hence, for every object Y of E', a bifunctorial morphism $X \times v^*Y \to v^*v_!(X) \times v^*Y \simeq v^*(v_!(X) \times Y)$; by adjunction this gives the morphism Λ_v.

10.5.3. *Definition of Ξ_v.* Let X be an object of E, and let Z and Y be two objects of E'. The morphism $\Lambda_v : v_!(X \times v^*Z) \to v_!(X) \times Z$ gives a trifunctorial morphism $\mathrm{Hom}_{E'}(v_!(X) \times Z, Y) \to \mathrm{Hom}_{E'}(v_!(X \times v^*Z), Y)$; by adjunction this gives a trifunctorial morphism

$$\mathrm{Hom}_{E'}(Z, \mathbf{Hom}(v_!X, Y)) \longrightarrow \mathrm{Hom}_{E'}(Z, v_* \mathbf{Hom}(X, v^*Y));$$

and hence the morphism Ξ_v.

10.5.4. *Definition of Φ_v.* The trifunctorial morphism of (10.5.3)

$$\mathrm{Hom}_{E'}(v_!(Z) \times X, Y) \longrightarrow \mathrm{Hom}_{E'}(v_!(Z \times v^*X), Y),$$

where Z is an object of E and X, Y are objects of E', gives, by adjunction, a trifunctorial morphism

$$\mathrm{Hom}_E(Z, v^* \mathbf{Hom}(X, Y)) \longrightarrow \mathrm{Hom}_E(Z, \mathbf{Hom}(v^*X, v^*Y));$$

and hence a definition of the morphism Φ_v. There is a second way to define Φ_v. Let X, Y, and Z be three objects of E'. The functor $v^* : E' \to E$ gives a trifunctorial morphism

$$\mathrm{Hom}_{E'}(Z \times X, Y) \longrightarrow \mathrm{Hom}_E(v^*(Z) \times v^*(X), v^*Y);$$

and hence, by adjunction, a trifunctorial morphism

$$\mathrm{Hom}_{E'}(Z, \mathbf{Hom}(X, Y)) \longrightarrow \mathrm{Hom}_{E'}(Z, v_* \mathbf{Hom}(v^*X, v^*Y)).$$

494 We therefore have a bifunctorial morphism

$$\mathbf{Hom}(X, Y) \longrightarrow v_* \mathbf{Hom}(v^*X, v^*Y);$$

and hence, by adjunction, a morphism which the reader may verify is indeed the morphism Φ_v defined above.

PROPOSITION 10.6. *Let $v : E \to E'$ be a morphism of topoi.*

1) *The morphism Ψ_v (10.5.1) is an isomorphism for every object Y of E if and only if the adjunction morphism $v^*v_*X \to X$ is an isomorphism. In particular, Ψ_v is an isomorphism for every pair of objects (X, Y) if and only if the functor $v_* : E \to E'$ is fully faithful, i.e. if v is an embedding of topoi.*
2) *The following conditions are equivalent:*
   (i) *For every pair (X, Y) of objects of E', the morphism Φ_v (10.5.4) is an isomorphism.*
   (ii) *The functor v^* admits a left adjoint $v_!$, and for every object X of E and every object Y of E', the morphism Ξ_v (10.5.3) is an isomorphism.*
   (iii) *The functor v^* admits a left adjoint $v_!$, and for every object X of E and every object Y of E', the morphism Λ_v (10.5.2) is an isomorphism.*

10.6.1. The first assertion follows immediately from (10.5.1). It also follows immediately from (10.5.2), (10.5.3), and (10.5.4) that (iii) ⇔ (ii) ⇔ ((i) and the functor v^* admits a left adjoint $v_!$). It remains to show that (i) implies that v^* admits a left adjoint, i.e. (1.8) that (i) implies that v^* commutes with small products. Let e' be the final object of E' and let I be a small set. Since v^* is left exact, $v^*(e')$ is a final object of E, and since v^* commutes with inductive limits, $v^*(\amalg_I e') \simeq \amalg_I v^*(e')$. For every object Y of E', we have $\mathbf{Hom}(\amalg_I e', Y) \simeq \Pi_I \mathbf{Hom}(e', Y) \simeq \Pi_I Y$ and $\mathbf{Hom}(\amalg_I v^*(e'), v^*Y) \simeq \Pi_I \mathbf{Hom}(v^*(e'), v^*Y) \simeq \Pi_I v^*Y$. The functorial morphism Φ_v therefore induces an iso- 495
morphism $v^* \Pi_I Y \simeq \Pi_I v^* Y$, which one checks is the canonical isomorphism. C.Q.F.D.

Corollary 10.7. *Let E be a topos, let Z be an object of E, let $j_Z : E_{/Z} \to E$ be the localization morphism (5.2), and let X, Y be two objects of E.*

1) *There is a bifunctorial isomorphism*

$$j_Z^* \mathbf{Hom}(X, Y) \simeq \mathbf{Hom}(j_Z^* X, j_Z^* Y).$$

2) *Let X' be an object of $E_{/Z}$. There is a bifunctorial isomorphism*

$$\mathbf{Hom}_E(j_{Z!}X', Y) \simeq j_{Z*}\mathbf{Hom}_{E/Z}(X', j_Z^* Y).$$

3) *Let Y' be an object of E/Z. When Z is open in E, there is a bifunctorial isomorphism*

$$j_{Z*}\mathbf{Hom}(X', Y') \simeq \mathbf{Hom}(j_{Z*}X', j_{Z*}Y').$$

Assertions 1) and 2) are proved by noting that the morphism Λ_{j_Z} is an isomorphism. For assertion 3), it suffices to note that j_{Z*} is fully faithful, since $j_{Z!}$ is fully faithful (I 5.7. a)).

Corollary 10.8. *Let E be a topos, let X, Y, and Z be three objects of E, and let $j_Z : E_{/Z} \to E$ be the localization morphism.*

1) *There is a canonical isomorphism*

$$j_{Z*} j_Z^* Y \simeq \mathbf{Hom}(Z, Y).$$

2) *There is a canonical isomorphism*

$$\mathrm{Hom}_E(Z, \mathbf{Hom}(X, Y)) \simeq \mathrm{Hom}_{E_{/Z}}(j_Z^* X, j_Z^* Y).$$

Let e_Z be the final object of $E_{/Z}$ (i.e. the object $\mathrm{id}_Z : Z \to Z$). It follows from (10.2.1) that there is a canonical isomorphism

$$\mathbf{Hom}_{E/Z}(e_Z, j_Z^* Y) \simeq j_Z^* Y;$$

hence, by (10.7 2), an isomorphism

$$j_{Z*} j_Z^* Y \simeq \mathbf{Hom}(j_{Z!} e_Z, Y).$$

Since $j_{Z!} e_Z = Z$, this proves 1). Let us prove 2). From (10.7) we obtain a canonical isomorphism 496

$$\mathrm{Hom}_{E_{/Z}}(e_Z, j_Z^* \mathbf{Hom}(X, Y)) \simeq \mathrm{Hom}_{E_{/Z}}(j_Z^* X, j_Z^* Y);$$

hence, by adjunction on the first member,

$$\mathrm{Hom}_E(Z, \mathbf{Hom}(X, Y)) \simeq \mathrm{Hom}_{E_{/Z}}(j_Z^* X, j_Z^* Y).$$

11. Ringed topoi, localization in ringed topoi

11.1.1. Let $\mathscr{U}$ be a universe. A *$\mathscr{U}$-ringed topos* is a pair (E, A) where E is a $\mathscr{U}$-topos and A is an object endowed with a ring structure. A $\mathscr{U}$-ringed site is a pair (C, A) where C is a $\mathscr{U}$-site and A is a $\mathscr{U}$-sheaf of rings on C. We omit mention of the universe when the context leaves no room for confusion. To a ringed site (C, A) is associated the ringed topos $(C^{\sim}, A)$. To a ringed topos (E, A) is associated the ringed site consisting of the site E and the sheaf of rings represented by A.

11.1.2. Let (E, A) be a ringed topos. We denote by ${}_A E$ (resp. E_A) the category of sheaves of unital left (resp. right) A-modules (we will also write A-Modules). The category ${}_A E$ (resp. E_A) is an abelian category (II 6.7). Let M and N be two sheaves of left (resp. right) A-modules. The commutative group of A-Module morphisms from M to N is denoted by $\operatorname{Hom}_A(M, N)$.

11.1.3. Let A, B, and C be three rings in a topos E, let M be a sheaf of $A-B$-bimodules, and let N be a sheaf of $A-C$-bimodules. Let e be a final object of E, and put $\operatorname{Hom}(e, B) = \Gamma(B)$ and $\operatorname{Hom}(e, C) = \Gamma(C)$. The $A-B$-bimodule structure on M gives a ring homomorphism from $\Gamma(B)^{\circ}$ to the ring of endomorphisms of the A-module M; similarly, the $A-C$-bimodule structure on N gives a ring homomorphism from $\Gamma(C)^{\circ}$ to the ring of endomorphisms of the A-module N. By functoriality, we deduce a $\Gamma(B)-\Gamma(C)$-bimodule structure on the commutative group $\operatorname{Hom}_A(M, N)$. In particular, when A is a sheaf of commutative rings, the group $\operatorname{Hom}_A(M, N)$ is canonically endowed with a $\Gamma(A)$-module structure.

497 11.1.4. Let E be a topos, let A and B be two rings in E, let $u : A \to B$ be a morphism of sheaves of rings, and let M be a B-module (a left module, to fix ideas). The sheaf M may be regarded as a sheaf of A-modules by means of u: for every object X of E, $M(X)$ is endowed with the $A(X)$-module structure obtained from its $B(X)$-module structure and the homomorphism $u(X) : A(X) \to B(X)$ by restriction of scalars. We thus obtain a functor "*restriction of scalars along u*"

$$\operatorname{Res}(u) : {}_B E \longrightarrow {}_A E.$$

11.1.5. In particular, when $A = \mathbf{Z}$ (the constant sheaf $\underline{\mathbf{Z}}$, that is, the sheaf associated to the constant presheaf $\mathbf{Z}$) and when $u : \underline{\mathbf{Z}} \to B$ is the unique canonical morphism, we obtain a functor from ${}_B F$ to the category ${}_{\underline{\mathbf{Z}}} E$, which is none other than the category of abelian sheaves, denoted by E_{Ab}. This functor is called the *underlying abelian-sheaf functor.*

Proposition 11.1.6. *The restriction-of-scalars functor along u commutes with inductive and projective limits. It is conservative.*

The proposition is true for the restriction-of-scalars functor for ordinary modules, that is, when E is the punctual topos. It is therefore true when E is the topos of presheaves on a small site. It then follows from the determination of inductive and projective limits by means of the associated-sheaf functor (II 6.4) that the proposition is true in general.

11.2.1. Let (E, A) be a ringed topos, let X be an object of E, and let $j_X : E_{/X} \to E$ be the localization morphism 5.2. By III 1.7 or 3.1.2, the sheaf $j_X^* A$ is canonically endowed with a ring structure. The sheaf of rings $j_X^* A$ is most often denoted by $A|X$, or, by abuse, simply by A. Unless otherwise stated, the topos $E_{/X}$ will be ringed by $A|X$. The sheaf $j_{X*} j_X^* A$ is canonically endowed with a ring structure. The adjunction morphism $A \to j_{X*} j_X^* A$ is a morphism of sheaves of rings.

498 11.2.2. Let , in addition, M be an A-module (a left module, to fix ideas). The sheaf $j_X^* M$ is endowed with an $A|X$-module structure (III 1.7; or 3.1.2); hence there is a functor

$$j_X^* : {}_A E \longrightarrow {}_{A|X} E_{/X}$$

called the restriction functor to $E_{/X}$, or also the restriction functor to X. The functor j_X^* commutes with inductive and projective limits (*loc. cit.*). In particular, it is exact. Now let N be an $A|X$-Module. The sheaf $j_{X*}N$ is a sheaf of $j_{X*}A|X$-Modules (III 1.7 or 3.1.2); hence, by restriction of scalars along the adjunction morphism $A \to j_{X*}(A|X)$ (11.1.4), it is an A-Module, again denoted by $j_{X*}N$. We have therefore defined a functor

$$j_{X*} : {}_{A|X} E_{/X} \longrightarrow {}_A E$$

which commutes with projective limits (11.1.6 and III 1.7). For every A-Module M, the adjunction morphism $M \to j_{X*}j_X^* M$ is a morphism of A-Modules. For every $A|X$-Module N and every A-Module M, the adjunction morphism $M \to j_{X*}j_X^* M$ defines a morphism, bifunctorial in M and N,

$$\mathrm{Hom}_{A|X}(j_X^* M, N) \longrightarrow \mathrm{Hom}_A(M, j_{X*}N). \tag{11.2.2.1}$$

The latter morphism is an isomorphism; that is, the functors j_X^* and j_{X*} for A-Modules are adjoint (III 1.7).

11.2.3. The functors j_X^* and j_{X*} for Modules commute with the “underlying set” functor (III 1.7 4)). They therefore commute with the restriction-of-scalars functors and, in particular, with the underlying abelian-sheaf functor.

Proposition 11.3.1. *Let (E, A) be a ringed topos and let X be an object of E. The functor*

$$j_X^* : {}_A E \longrightarrow {}_{A|X} E_{/X}$$

admits a left adjoint, denoted by $j_{X!} : {}_{A|X} E_{/X} \to {}_A E$ and called extension by zero. The extension-by-zero functor is exact and faithful, and commutes with inductive limits. The functors $j_{X!}$ for Modules commute with restriction-of-scalars functors and, in particular, 499 *they commute with the underlying abelian-sheaf functor.*

The existence of the functor $j_{X!}$ follows from III 1.7. Since $j_{X!}$ is a left adjoint, it commutes with inductive limits (I 2). To prove the other assertions, first suppose that E is the topos of presheaves of sets on a small category C containing the object X. We then know that $E_{/X}$ is equivalent to the category of presheaves on $C_{/X}$ and that, under this equivalence, the functor $j_X^* : E \to E_{/X}$ is simply composition with the forgetful functor $C_{/X} \to C$ (I 5.11). It then follows immediately from the explicit construction of $j_{X!}$ (I 5.1) that, for every $A|X$-Module N and every object Y of C, one has

$$j_{X!}N(Y) = \bigoplus_{u \in \mathrm{Hom}_C(Y,X)} N(u);$$

hence $j_{X!}$ is exact, and extension by zero commutes with restriction-of-scalars functors in this case. In the general case, we may suppose that E is the topos of sheaves on a small site C and that X comes from an object of C (1.2.1). The topos $E_{/X}$ is then equivalent to the topos of sheaves on $C_{/X}$ (endowed with the induced topology) (III 5.4), and the topos morphism $j_X : E_{/X} \to E$ comes from the forgetful functor $C_{/X} \to C$, which is continuous and cocontinuous (III 5.2). It then follows from III 1.7 that extension by zero for sheaves is obtained by composing extension by zero for presheaves with the associated-sheaf functor; hence the assertion of exactness and the compatibility with restrictions of scalars.

To prove faithfulness, it is equivalent to show that, for every $A|X$-Module N, the adjunction morphism

$$\mathrm{ad}_N : N \longrightarrow j_X^* j_{X!} N$$

is a monomorphism. Now, denoting by $j_{\widehat{X}!}$ the extension-by-zero functor for presheaves and by $\underline{a}$ the "associated sheaf" functor, we have a commutative diagram

$$\begin{tikzcd} N \arrow[r, "\mathrm{ad}_N"] \arrow[dr, "\mathrm{ad}_{\widehat{N}}"'] & j_X^* \underline{a} j_{\widehat{X}!} N \\ & \underline{a} j_X^* j_{\widehat{X}!} N, \arrow[u] \end{tikzcd}$$

500 where $\mathrm{ad}_{\widehat{N}}$ is the adjunction morphism for presheaves. The morphism $\mathrm{ad}_{\widehat{N}}$ is a monomorphism by what precedes, and therefore $\underline{a}(\mathrm{ad}_{\widehat{N}})$ is a monomorphism. Moreover, since the forgetful functor $C_{/X} \to C$ is continuous and cocontinuous, the canonical morphism $\underline{a} j_X^* \to j_X^* \underline{a}$ is an isomorphism (III 2.3). Consequently, ad_N is a monomorphism.

Remark 11.3.2. Let N be an $A|X$-Module. The sheaf of sets underlying $j_{X!}N$ is not, in general, isomorphic to the sheaf obtained by extending the sheaf of sets underlying N by the empty set (III 5.3 and 5.2). One must therefore take care not to confuse the *extension-by-zero* functor denoted by $j_{X!} : {}_{A|X}E_{/X} \to {}_A E$ in 11.3.1 with the extension-by-the-empty-set functor, also denoted by $j_{X!} : E_{/X} \to E$, in III 5.3 and IV 5.2. In most cases encountered in practice, the abuse of notation just noted causes no confusion. When confusion is nevertheless possible, we will use the notations $j_{X!}^{\mathrm{ab}}$ and $j_{X!}^{\mathrm{ens}}$ for the extension-by-zero and extension-by-the-empty-set functors, respectively.

Proposition 11.3.3. *Let (E, A) be a ringed topos and let X be an object of E. The A-Module $j_{X!}(A|X)$, most often denoted by A_X or $A_{X,E}$, is the free A-Module generated by X (II 6.5); that is, for every A-Module M, there is a canonical isomorphism, functorial in M:*

$$\mathrm{Hom}_E(X, M) \xleftarrow{\sim} \mathrm{Hom}_A(A_X, M).$$

Let $(X_i)_{i\in I}$ be a topologically generating family of E (II 3.0.1). The family $(A_{X_i})_{i\in I}$ is a generating family of the category of A-Modules.

Let e_x be the final object of the topos $E_{/X}$ ($e_X = X \xrightarrow{\mathrm{id}} X$). There is an isomorphism

$$\mathrm{Hom}_{A|X}(A|X, j_X^* M) \xrightarrow{\sim} \mathrm{Hom}_{E_{/X}}(e_X, j_X^* M);$$

501 deduced from the unit section $e_X \to A/X$; hence, by adjunction, there is an isomorphism

$$\mathrm{Hom}_A(j_{X!}^{\mathrm{ab}}(A|X), M) \xrightarrow{\sim} \mathrm{Hom}_E(j_{X!}^{\mathrm{ens}}(e_X), M).$$

Since $j_{X!}^{\mathrm{ens}}(e_X) = X$, we obtain the asserted isomorphism. The last assertion follows from II 6.6.

Remark 11.3.4. Let A and B be two sheaves of rings on a topos E, let X be an object of E, and let N be an $A|X$-$B|X$-bimodule. The abelian sheaf obtained by extending N by zero is canonically endowed with an A-B-bimodule structure, as follows immediately from its explicit description (11.3.1). In particular, the free A-module so obtained, A_X, is an A-bimodule.

Exercise 11.3.5. Let E be a topos, let X be an object of E, let $x : P \to E$ be a point of E (6.1), and let $(x_i)_{i\in I}$ be the family of points of $E_{/X}$ above x (in bijective correspondence with the fiber X_x (6.7.2)). Show that, for every abelian sheaf M on $E_{/X}$, the fiber $(j_{X!}M)_x$ is canonically isomorphic to $\oplus_i M_{x_i}$.

12. Operations on modules

Proposition 12.1. *Let (E, A) be a ringed topos, and let M and N be two left A-modules (resp. right A-modules). The functor on E which assigns to every object X of E the commutative group $\mathrm{Hom}_A(M, \mathbf{Hom}_E(X, N))$ is representable by an abelian sheaf denoted by $\mathbf{Hom}_A(M, N)$ (or sometimes $\mathbf{Hom}(M, N)$ when no confusion can result). For every object X of E, there is a canonical isomorphism*

$$\mathbf{Hom}_A(M, N)(X) \simeq \mathrm{Hom}_{A|X}(j_X^* M, j_X^* N). \tag{12.1.1}$$

There is a canonical isomorphism $\mathbf{Hom}_E(X, N) = j_{X*}j_X^* N$ (10.8); consequently, $\mathbf{Hom}_E(X, N)$ is endowed, functorially in X, with a structure of left A-module (resp. right A-module). The functor $X \longmapsto \mathbf{Hom}_E(X, N)$ transforms inductive limits in the argument X into projective limits of A-modules (10.2 and 11.2.3); consequently, the functor

$$X \longmapsto \mathrm{Hom}_Z(M, \mathbf{Hom}_E(X, N))$$

transforms inductive limits in the argument X into projective limits of commutative 502
groups, and hence into projective limits of the underlying sets. It is therefore representable (1.4 and 1.2), and the object representing it is endowed with the structure of an abelian sheaf. By definition, for every object X of E, there is a canonical isomorphism

$$\mathbf{Hom}_A(M, N)(X) = \mathrm{Hom}(X, \mathbf{Hom}_A(M, N)) \simeq \mathrm{Hom}_A(M, \mathbf{Hom}(X, N)) \tag{12.1.2}$$

and the isomorphism (12.1.1) follows from (12.1.2), from the isomorphism $\mathbf{Hom}(X, N) \approx j_{X*}j_X^* N$ (10.8), and from the adjunction formulas of 10.

12.2. The abelian sheaf $\mathbf{Hom}_A(M, N)$ is called the *sheaf of A-module homomorphisms from M to N*. It is a bifunctor in M and N. It follows from its definition and from (10.2) that it transforms projective limits in the argument N (resp. inductive limits in M) into projective limits of abelian sheaves. In particular, it is left exact in both arguments.

Proposition 12.3. *Let (E, A) be a ringed topos, let M and N be two right A-modules (resp. left A-modules), and let X be an object of E:*

a) *There are canonical isomorphisms:*

$$\mathbf{Hom}_A(M, N)(X) \approx \mathrm{Hom}_A(M, j_{X*}j_X^* N) \simeq \mathrm{Hom}_{A|X}(j_X^* M, j_X^* N) \simeq \mathrm{Hom}_A(j_{X!}j_X^* M, N).$$

b) *There is a canonical isomorphism:*

$$\Phi_X : j_X^* \mathbf{Hom}_A(M, N) \simeq \mathbf{Hom}_{A|X}(j_X^* M, j_X^* N).$$

Let, in addition, P be a right $A|X$-module (resp. a left $A|X$-module):

c) *There is a canonical isomorphism:*

$$j_{X*} \mathbf{Hom}_{A|X}(j_X^* M, P) \simeq \mathbf{Hom}_A(M, j_{X*}P).$$

d) *There is a canonical isomorphism:*

$$\Xi_X : \mathbf{Hom}_A(j_{X!}P, N) \simeq j_{X*} \mathbf{Hom}_{A|X}(P, j_X^* N).$$

The isomorphisms in a) follow from (12.1.2), from the isomorphism $\mathbf{Hom}_E(X,N) \simeq j_{X*}j_X^* N$ (10.8), and from the adjunction formulas of 10.

Proof of b). For every object Y of $E_{/X}$, there is the following sequence of isomorphisms:

$\mathrm{Hom}(Y, j_X^* \mathbf{Hom}_A(M,N)) \simeq \mathrm{Hom}(j_{X!}Y, \mathbf{Hom}_A(M,N))$ (adjunction for sheaves of sets)

$$\begin{aligned}
&\mathrm{Hom}(j_{X!}Y, \mathbf{Hom}_A(M,N)) \simeq \mathrm{Hom}_A(M, \mathbf{Hom}_E(j_{X!}Y,N)) && (12.2.1)\\
&\mathrm{Hom}_A(M, \mathbf{Hom}_E(j_{X!}Y,N)) \simeq \mathrm{Hom}_A(M, j_{X*}\mathbf{Hom}_{E_{/X}}(Y, j_X^*N)) && (10.7)\\
&\mathrm{Hom}_A(M, j_{X*}\mathbf{Hom}_{E_{/X}}(Y, j_X^*N)) \simeq \mathrm{Hom}_{A|X}(j_X^*M, \mathbf{Hom}_E(Y, j_X^*N)) && (11.2.2.1)\\
&\mathrm{Hom}_{A|X}(j_X^*M, \mathbf{Hom}_E(Y, j_X^*N)) \simeq \mathrm{Hom}(Y, \mathbf{Hom}_{A|X}(j_X^*M, j_X^*N)) && (12.2.1)
\end{aligned}$$

503 Proof of c). For every object Y of E, there is the following sequence of isomorphisms:

$$\begin{aligned}
&\mathrm{Hom}(Y, j_{X*}\mathbf{Hom}_{A|X}(j_X^*M,P)) \simeq \mathrm{Hom}(j_X^*Y, \mathbf{Hom}_{A|X}(j_X^*M,P)) && \text{(adjunction)}\\
&\mathrm{Hom}(j_X^*Y, \mathbf{Hom}_{A|X}(j_X^*M,P)) \simeq \mathrm{Hom}_{A/X}(j_X^*M, \mathbf{Hom}_{E_{/X}}(j_X^*Y,P)) && (12.2.1)\\
&\mathrm{Hom}_{A|X}(j_X^*M, \mathbf{Hom}_{E_{/X}}(j_X^*Y,P)) \simeq \mathrm{Hom}_A(M, j_{X*}\mathbf{Hom}_{E_{/X}}(j_X^*Y,P)) && (11.2.2.1)\\
&\mathrm{Hom}_A(M, j_{X*}\mathbf{Hom}_{E_{/X}}(j_X^*Y,P)) \simeq \mathrm{Hom}_A(M, \mathbf{Hom}_E(Y, j_{X*}P)) && (10.3)\\
&\mathrm{Hom}_A(M, \mathbf{Hom}_E(Y, j_{X*}P)) \simeq \mathrm{Hom}(Y, \mathbf{Hom}_A(M, j_{X*}P)) && (12.2.1)
\end{aligned}$$

Proof of d). For every object Y of E, there is the following sequence of isomorphisms:

$$\begin{aligned}
&\mathrm{Hom}(Y, \mathbf{Hom}_A(j_{X!}P,N)) \simeq \mathrm{Hom}_A(j_{X!}P, \mathbf{Hom}_E(Y,N)) && (12.2.1)\\
&\mathrm{Hom}_A(j_{X!}P, \mathbf{Hom}_E(Y,N)) \simeq \mathrm{Hom}_{A|X}(P, j_X^*\mathbf{Hom}_E(Y,N)) && (11.3.1)\\
&\mathrm{Hom}_{A/X}(P, j_X^*\mathbf{Hom}_E(Y,N)) \simeq \mathrm{Hom}_{A|X}(P, \mathbf{Hom}_{E_{/X}}(j_X^*Y, j_X^*N)) && (10.7)\\
&\mathrm{Hom}_{A/X}(P, \mathbf{Hom}_{E_{/X}}(j_X^*Y, j_X^*N)) \simeq \mathrm{Hom}(j_X^*Y, \mathbf{Hom}_{A/X}(P, j_X^*N)) && (12.2.1)\\
&\mathrm{Hom}(j_X^*Y, \mathbf{Hom}_{A|X}(P, j_X^*N)) \simeq \mathrm{Hom}(Y, j_{X*}\mathbf{Hom}_{A|X}(P, j_X^*N)) && \text{(adjunction)}
\end{aligned}$$

Corollary 12.4. *Let C be a $\mathscr{U}$-site, let A' be a $\mathscr{U}$-presheaf of rings on C, let M be a $\mathscr{U}$-presheaf of A'-modules (left modules, to fix ideas), and let N be a $\mathscr{U}$-sheaf of A'-modules (left modules). Denote by A the sheaf associated with A', so that the sheaves N and $\underline{a}M$ (the sheaf associated with M) are A-modules. The presheaf $X \longmapsto \mathrm{Hom}_{A'|X}(j_X^*M, j_X^*N)$ is a $\mathscr{U}$-abelian sheaf canonically isomorphic to $\mathbf{Hom}_A(\underline{a}M,N)$.*

Indeed, it follows from III 5.5 and III 2.3 that the localization functor to $C_{/X}$ commutes with the associated-sheaf functors (for C and $C_{/X}$). Consequently, there are isomorphisms, functorial in the variable object X of C:

$$\begin{aligned}
\mathrm{Hom}_{A'|X}(j_X^*M, j_X^*N) &\simeq \mathrm{Hom}_{A|X}(\underline{a}j_X^*M, j_X^*N)\\
&\simeq \mathrm{Hom}_{A|X}(j_X^*\underline{a}M, j_X^*N) \simeq \mathbf{Hom}_A(\underline{a}M,N)(X).
\end{aligned}$$

504 Corollary 12.5. *Let E be a topos, let A, B, and C be three sheaves of rings on E, let M be an A-B-bimodule, and let N be an A-C-bimodule. The abelian sheaf $\mathbf{Hom}_A(M,N)$ is canonically endowed with a B-C-bimodule structure. In particular, when A is a sheaf of commutative rings and M and N are A-modules, $\mathbf{Hom}_A(M,N)$ is canonically endowed with an A-module structure. The canonical isomorphisms b), c), and d) of 12.3 are isomorphisms of bimodules.*

We shall confine ourselves to giving indications. For every object X of E, one has $\mathbf{Hom}_A(M,N)(X) \simeq \mathrm{Hom}_{A|X}(j_X^*M, j_X^*N)$ (12.3). Consequently, the commutative group

$\mathbf{Hom}_A(M, N)(X)$ is canonically endowed with a $B(X)$-$C(X)$-bimodule structure (11.1.3), which is readily checked to be functorial in X. The other assertions are left to the reader.

Corollary 12.6. *Let (E, A) be a ringed topos, let X be an object of E, and let N be an A-module (a left module, to fix ideas). There are canonical isomorphisms of A-modules:*

$$\mathbf{Hom}_E(X, N) \simeq j_{X*} j_X^* N \simeq \mathbf{Hom}_A(A_X, N);$$

the A-module structure on $\mathbf{Hom}_A(A_X, N)$ comes from the bimodule structure on A_X (11.3.4).

The first isomorphism follows from 10.7, and the second from the isomorphism of A-modules $N \simeq \mathbf{Hom}_A(A, N)$ and from 12.3.

Proposition 12.7. *Let (E, A) be a ringed topos, let M be a right A-module, and let N be a left A-module. The functor which assigns to every abelian sheaf P the commutative group $\mathrm{Hom}_A(M, \mathbf{Hom}_{\underline{Z}}(N, P))$ is representable by an abelian sheaf denoted by $M \otimes_A N$ and called the tensor product over A of M and N.*

There is a canonical injection of $\mathrm{Hom}_A(M, \mathbf{Hom}_{\underline{Z}}(N, P))$ into the set $\mathrm{Hom}(M, \mathbf{Hom}(N, P)) \simeq \mathrm{Hom}(M \times N, P)$; returning to the definitions, one sees immediately that a morphism f of sheaves of sets from $M \times N$ to P comes from an element of $\mathrm{Hom}_A(M, \mathbf{Hom}_{\underline{Z}}(N, P))$ if and only if, for every object X of E, the map $f(X) : M(X) \times N(X) \to P(X)$ is an $A(X)$-bilinear map from $M(X) \times N(X)$ to $P(X)$. We call morphisms of sheaves of 505
sets from $M \times N$ to P having this property A-bilinear morphisms. Thus, the problem is to represent the functor of A-bilinear morphisms from $M \times N$ to P. To do this, one proceeds as in the ordinary case, i.e. the case where E is the punctual topos. Consider the eight morphisms:

$$\begin{array}{lll} \text{(i)}: & M \times M \times N \longrightarrow M \times N & 1 \le i \le 3 \\ \text{(i)}: & M \times N \times N \longrightarrow M \times N & 4 \le i \le 6 \\ \text{(i)}: & M \times A \times N \longrightarrow M \times N & 7 \le i \le 8 \end{array}$$

defined by the formulas:

$$\begin{array}{ll} (1) & (m_1, m_2, n) \longmapsto (m_1 + m_2, n) \\ (2) & (m_1, m_2, n) \longmapsto (m_1, n) \\ (3) & (m_1, m_2, n) \longmapsto (m_2, n) \\ (4) & (m, n_1, n_2) \longmapsto (m, n_1 + n_2) \\ (5) & (m, n_1, n_2) \longmapsto (m, n_1) \\ (6) & (m, n_1, n_2) \longmapsto (m, n_2) \\ (7) & (m, a, n) \longmapsto (ma, n) \\ (8) & (m, a, n) \longmapsto (m, an); \end{array}$$

here formula (i) describes the map (i)(X) for the variable objects X of E. Let L denote the "free generated abelian sheaf" functor. It is clear that the largest quotient (in the sense of abelian sheaves) of $L(M \times N)$ which equalizes $L(1)$ with $L(2) + L(3)$, $L(4)$ with $L(5) + L(6)$, and $L(7)$ with $L(8)$ represents the functor of A-bilinear morphisms from $M \times N$ to P.

12.8. One observes that the functor $P \longmapsto \mathrm{Hom}_A(N, \mathbf{Hom}_{\underline{Z}}(M, P))$ is also canonically isomorphic to the functor of A-bilinear maps from $M \times N$ to P. Consequently, the tensor product $M \otimes_A N$ also represents the functor $\mathrm{Hom}_A(N, \mathbf{Hom}_{\underline{Z}}(M, P))$. Thus there are isomorphisms, functorial in all the arguments:

$$\mathrm{Hom}_Z(M \otimes_A N, P) \simeq \mathrm{Hom}_A(M, \mathbf{Hom}_{\underline{Z}}(N, P)), \tag{12.8.1}$$

$$\mathrm{Hom}_Z(M \otimes_A N, P) \simeq \mathrm{Hom}_A(N, \mathbf{Hom}_{\underline{Z}}(M, P)). \tag{12.8.2}$$

506 **12.9.** It follows from (12.8), or alternatively from (12.8.1) and (12.2), that the functor $\otimes_A$ commutes with inductive limits in both arguments.

PROPOSITION 12.10. *Let C be a $\mathcal{U}$-site, let A' be a presheaf of rings on C, let M' (resp. N') be a right (resp. left) A'-module, and let A, M, and N be the sheaves associated with A', M', and N' respectively. The sheaf associated with the presheaf $X \longmapsto M'(X) \otimes_{A'(X)} N'(X)$ ($X \in \operatorname{ob} C$) is canonically isomorphic to the sheaf $M \otimes_A N$.*

The presheaf $X \longmapsto M'(X) \otimes_{A'(X)} N'(X)$ can be constructed from the presheaves M', N', and A' by the operations specified in the proof of 12.7. Since the "associated sheaf" functor commutes with finite projective and inductive limits and with the "free generated abelian object" functors, formation of the tensor product commutes with the "associated sheaf" functor.

PROPOSITION 12.11. *Let (E, A) be a ringed topos, let M be a right A-module, let N be a left A-module, and let X be an object of E.*

a) *There is a canonical isomorphism*

$$j_X^*(M \otimes_A N) \simeq (j_X^* M) \otimes_{A|X} (j_X^* N).$$

Let, in addition, P be a right $A|X$-module and Q a left $A|X$-module.

b) *There are canonical isomorphisms (projection formulas):*

$$j_{X!}(P \otimes_{A|X} j_X^* N) \simeq (j_{X!} P) \otimes_A N$$
$$j_{X!}(j_X^* M \otimes_{A|X} Q) \simeq M \otimes_A (j_{X!} Q).$$

Proof of a). For every abelian sheaf R on $E_{/X}$, there is the following sequence of isomorphisms, functorial in R:

$$\operatorname{Hom}_{\underline{Z}}(j_X^*(M \otimes_A N), R) \simeq \operatorname{Hom}_{\underline{Z}}(M \otimes_A N, j_{X*} R) \qquad (11.2.2.1)$$
$$\operatorname{Hom}_{\underline{Z}}(M \otimes_A N, j_{X*} R) \simeq \operatorname{Hom}_A(M, \mathbf{Hom}_Z(N, j_{X*} R)) \qquad (12.8.1)$$
$$\operatorname{Hom}_A(M, \mathbf{Hom}_Z(N, j_{X*} R)) \simeq \operatorname{Hom}_A(M, j_{X*} \mathbf{Hom}_{\underline{Z}}(j_X^* N, R)) \qquad (12.3)$$
$$\operatorname{Hom}_A(M, j_{X*} \mathbf{Hom}_{\underline{Z}}(j_X^* N, R)) \simeq \operatorname{Hom}_{A|X}(j_X^* M, \mathbf{Hom}_{\underline{Z}}(j_X^* N, R)) \qquad (11.2.2.1)$$
$$\operatorname{Hom}_{A/X}(j_X^* M, \mathbf{Hom}_{\underline{Z}}(j_X^* N, R)) \simeq \operatorname{Hom}_{\underline{Z}}(j_X^* M \otimes_A j_X^* N, R) \qquad (12.8.1)$$

507 Let us exhibit the first isomorphism in b). For every abelian sheaf R, there is a sequence of isomorphisms, functorial in R:

$$\operatorname{Hom}_Z(j_{X!}(P \otimes_{A|X} j_X^* N), R) \simeq \operatorname{Hom}_Z(P \otimes_{A|X} j_X^* N, j_X^* R) \qquad (11.3.1)$$
$$\operatorname{Hom}_Z(P \otimes_{A|X} j_X^* N, j_X^* R) \simeq \operatorname{Hom}_{A/X}(P, \mathbf{Hom}_Z(j_X^* N, j_X^* R)) \qquad (12.8.1)$$
$$\operatorname{Hom}_{A|X}(P, \mathbf{Hom}_Z(j_X^* N, j_X^* R)) \simeq \operatorname{Hom}_{A|X}(P, j_X^* \mathbf{Hom}_Z(N, R)) \qquad (12.3)$$
$$\operatorname{Hom}_{A|X}(P, j_X^* \mathbf{Hom}_Z(N, R)) \simeq \operatorname{Hom}_A(j_{X!} P, \mathbf{Hom}_Z(N, R)) \qquad (11.3.1)$$
$$\operatorname{Hom}_A(j_{X!} P, \mathbf{Hom}_Z(N, R)) \simeq \operatorname{Hom}_Z(j_{X!} P \otimes_A N, R) \qquad (12.8.1)$$

The second isomorphism is obtained similarly using (12.8.2).

COROLLARY 12.12. *Let E be a topos, let A, B, and C be three sheaves of rings on E, let M be a B-A-bimodule, and let N be an A-C-bimodule. The abelian sheaf $M \otimes_A N$ is canonically endowed with a B-C-bimodule structure. In particular, when A is a sheaf of commutative rings and M and N are A-modules, $M \otimes_A N$ is canonically endowed with an A-module structure. The isomorphisms of 12.11 are isomorphisms of bimodules.*

For every object X of E, j_X^*M is a right $A|X$-module on which the ring $B(X)$ acts on the left, and likewise j_X^*N is a left A/X-module on which $C(X)$ acts on the right. By functoriality, the abelian sheaf $j_X^*M \otimes_{A/X} j_X^*N \simeq j_X^*(M \otimes_A N)$ is endowed with the structure of a $B(X)$-$C(X)$-"object", and this structure varies functorially in X. Consequently, $M \otimes_A N$ is endowed with a B-C-bimodule structure. The $B(X)$-$C(X)$-object structure on $j_X^*(M \otimes_A N)$ is reflected in the evident way on the functor of "A-bilinear morphisms". One then observes that, when A is commutative and M and N are A-modules, the right and left A-module structures so obtained are "equal"; consequently, $M \otimes_A N$ is an A-module in this case. The last assertion is left to the reader.

Corollary 12.13. *Let (E, A) be a ringed topos, let M be a left A-module (resp. a right A-module), and let X be an object of E. With the notation A_X of 11.3.3, there is a canonical isomorphism* 508

$$A_X \otimes_A M \simeq j_{X!}j_X^*M \quad (\textit{resp. } M \otimes_A A_X \simeq j_{X!}j_X^*M).$$

This follows from the projection formulas (12.11).

Proposition 12.14. *Let E be a topos, let A and B be two sheaves of rings on E, let M be a right A-module, let N be an A-B-bimodule, and let P be a right B-module. There is a canonical isomorphism*

$$\mathrm{Hom}_B(M \otimes_A N, P) \simeq \mathrm{Hom}_A(M, \mathbf{Hom}_B(N, P)).$$

From (12.8.1) one obtains a canonical A-bilinear morphism $\mathbf{Hom}_Z(N, P) \times N \to P$; hence, by restricting it to $\mathbf{Hom}_B(N, P) \times N$ (which is a subsheaf), one obtains an A-bilinear morphism $\mathbf{Hom}_B(N, P) \times N \to P$, which is immediately seen to be B-linear in the second factor. Thus there is a canonical morphism of B-modules $\mathbf{Hom}_B(N, P) \otimes_A N \to P$; hence a canonical map, functorial in M:

$$\mathrm{Hom}_A(M, \mathbf{Hom}_B(N, P)) \longrightarrow \mathrm{Hom}_B(M \otimes_A N, P). \tag{12.14.1}$$

Let us show that this morphism of functors is an isomorphism. Since both sides transform inductive limits in M into projective limits, it is enough to show that (12.14.1) is an isomorphism when M ranges over a generating family of the category E_A. It is therefore enough to show that (12.14.1) is an isomorphism when $M = A_X$, where X is an object of E. The verification is then immediate.

13. Morphisms of ringed topoi

Definition 13.1. *Let (E, A) and (E', A') be two ringed topoi. A morphism of ringed topoi $u : (E, A) \to (E', A')$ is a pair (m, θ), where $m : E \to E'$ is a morphism of topoi (3.1) and $\theta : m^*A' \to A$ is a morphism of Rings.*

13.1.1. Since the functor $m^* : E' \to E$ is left adjoint to the functor $m_* : E \to E'$, giving a morphism of ringed topoi $(m, \theta) : (E, A) \to (E', A')$ amounts to giving a morphism of topoi $m : E \to E'$ and a morphism of sheaves of rings $\Theta : A' \to m_*A$.

13.2. To a morphism of ringed topoi $u = (m, \theta) : (E, A) \to (E', A')$, we associate two notable functors between the categories of Modules: 509

13.2.1. *The direct-image functor for Modules:* Let M be a left (resp. right) A-Module. The object m_*M is canonically endowed with an m_*A-Module structure; restriction of scalars along the canonical morphism $\Theta : A' \to m_*A$ therefore gives an A'-Module, denoted by $u_*(M)$ and called the *direct image of M under the morphism u.*

13.2.2. *The inverse-image functor for Modules:* Let N be a left (resp. right) A'-Module. The object m^*N of E is canonically endowed with the structure of a left (resp. right) m^*A'-Module. The left A-Module $A \otimes_{m^*A'} m^*N$ (resp. the right A-Module $m^*N \otimes_{m^*A'} A$), where A is endowed with the m^*A'-Module structure defined by $\theta : m^*A' \to A$, is denoted by u^*N and is called the *inverse image of the Module N under the morphism of ringed topoi u.*

13.2.3. Note that for every A-Module M, the sheaf of sets and the abelian sheaf underlying u_*M is the sheaf m_*M. Thus the notation u_* is most often also used for the functor m_*, the direct image for sheaves of sets. On the other hand, for an A'-Module N, the sheaf of sets or abelian sheaf underlying u^*N is in general neither equal nor isomorphic to m^*N (except when θ is an isomorphism). One must therefore distinguish between the inverse image for Modules and the inverse image for sheaves of sets or abelian sheaves. The notation $u^{-1} : E' \to E$ is most often used for the functor m^*, the inverse image for sheaves of sets. The functor u^{-1} is called the *"set-theoretic" inverse-image functor* under the morphism of ringed topoi u, as opposed to the inverse image "in the sense of Modules" u^*.

Definition 13.3. *Let (C, A) and (C', A') be two $\mathscr{U}$-ringed sites. A morphism of ringed sites $u : (C, A) \to (C', A')$ is a pair (m, θ), where m is a morphism from the site C to the*
510 *site C' (4.9) and $\theta : m^*A' \to A$ is a morphism of Rings.*

13.3.1. A morphism of ringed topoi is a morphism of $\mathscr{U}$-ringed sites. A morphism of ringed sites gives rise to a morphism between the corresponding ringed topoi (11.1.1 and 4.9.1).

Proposition 13.4. *Let $u : (E, A) \to (E', A')$ be a morphism of ringed topoi.*

a) *For a variable left (resp. right) A-Module M and a variable left (resp. right) A'-Module N, there are canonical bifunctorial isomorphisms (called adjunction isomorphisms):*

$$\mathrm{Hom}_A(u^*N, M) \simeq \mathrm{Hom}_{A'}(N, u_*M), \tag{13.4.1}$$

$$u_* \mathbf{Hom}_A(u^*N, M) \simeq \mathbf{Hom}_{A'}(N, u_*M). \tag{13.4.2}$$

b) *For a variable object X of E', there is a canonical isomorphism, functorial in X:*

$$u^*A'_X \xrightarrow{\sim} A_{u^{-1}(X)}, \tag{13.4.3}$$

where A'_X and $A_{u^{-1}(X)}$ denote the generated free Modules (11.3.3).

c) *When A' is commutative and the canonical morphism $u^{-1}A' \to A$ is central (resp. when the canonical morphism $u^{-1}A' \to A$ is an isomorphism), there is, for a variable right A'-Module M and a variable left A'-Module N, a canonical bifunctorial isomorphism:*

$$u^*M \otimes_A u^*N \simeq u^*(M \otimes_{A'} N) \tag{13.4.4}$$

$$u^*M \otimes_A u^*N \simeq u^{-1}(M \otimes_{A'} N). \qquad \text{(resp. (13.4.5))}$$

First, there is a canonical isomorphism $\mathrm{Hom}_A(u^*N, M) \simeq \mathrm{Hom}_{u^{-1}A'}(u^{-1}N, M)$ (12.14), and then an isomorphism $\mathrm{Hom}_{u^{-1}A'}(u^{-1}N, M) \simeq \mathrm{Hom}_{A'}(N, u_*M)$ (III 1.7); this gives (13.4.1). Let us exhibit the isomorphism (13.4.2). For every object X of E', there is the following sequence of functorial isomorphisms:

$$
\begin{array}{lll}
\mathrm{Hom}_{E'}(X, u_* \mathbf{Hom}_A(u^*N, M)) \simeq & \mathrm{Hom}(u^{-1}X, \mathbf{Hom}_A(u^*N, M)) & \text{(adjunction)},\\
\mathrm{Hom}_A(u^*N, \mathbf{Hom}_E(u^{-1}X, M)) \simeq & \quad '' \quad '' \quad '' \quad '' & (12.1),\\
\mathrm{Hom}_A(u^*N, \mathbf{Hom}_E(u^{-1}X, M)) \simeq & \mathrm{Hom}_{A'}(N, u_* \mathbf{Hom}_E(u^{-1}X, M)) & (13.4.1),\\
\mathrm{Hom}_{A'}(N, \mathbf{Hom}_{E'}(X, u_*M)) \simeq & \quad '' \quad '' \quad '' \quad '' & (10.3.1),\\
\quad '' \quad '' \quad '' \quad '' \simeq & \mathrm{Hom}_{E'}(X, \mathbf{Hom}_{A'}(N, u_*M)) & (12.1).
\end{array}
$$

Formula (13.4.5) then follows from formula (13.4.2) by adjunction (the definition of the tensor product (12.7)). Formula (13.4.4) follows from (13.4.5), using the commutativity of the tensor product when the base ring is commutative (12.8). Finally, to prove (13.4.3), consider the sequence of isomorphisms 511

$$
\begin{array}{lll}
\mathrm{Hom}_A(u^*A'_X, M) \simeq & \mathrm{Hom}_{A'}(A'_X, u_*M) & (13.4.1),\\
\mathrm{Hom}_{E'}(X, u_*M) \simeq & \quad '' \quad '' \quad '' & (11.3.3),\\
\quad '' \quad '' \quad '' \simeq & \mathrm{Hom}_E(u^{-1}X, M) & \text{(adjunction)},\\
\mathrm{Hom}_A(A_{u^{-1}X}, M) \simeq & \quad '' \quad '' \quad '' & (11.3.3).
\end{array}
$$

Corollary 13.5. *Let (E, A) be a ringed topos, let $x : P \to E$ be a point of E, and let $F \mapsto F_x$ be the associated fiber functor (6.1). The fiber functor at x carries the tensor product of A-Modules to the (ordinary) tensor product of A_x-modules.*

The tensor product in P is the ordinary tensor product of modules (P, the punctual topos, is the category of sets). The assertion therefore follows from (13.4.4).

Corollary 13.6. *Let $u : (E, A) \to (E', A')$ be a morphism of ringed topoi. The direct-image functor u_* for right or left Modules commutes with projective limits and, in particular, is left exact. The inverse-image functor u^* for right or left Modules commutes with inductive limits and, in particular, is right exact.*

This follows from formula (13.4.1) (I 2.11).

13.7. Note that the inverse-image functor u^* for Modules is not, in general, exact, whereas the set-theoretic inverse-image functor u^{-1} is exact. The exactness of u^* can nevertheless be asserted when the canonical morphism $u^{-1}A' \to A$ is flat (on the right or on the left) (V 1.8). This is so, in particular, when the canonical morphism $u^{-1}A' \to A$ 512
is an isomorphism.

13.8. Let C be a $\mathscr{U}$-site and let A' be a presheaf of rings on C; denote its associated sheaf by A. The category of presheaves of A'-modules which are sheaves is equivalent to the category of A-Modules. The notation $\mathrm{Hom}_{A'}(M, N)$ is sometimes used for the group $\mathrm{Hom}_A(M, N)$ (11.1.2). Likewise, and by abuse, the notations $\mathbf{Hom}_{A'}(M, N)$ and $M \otimes_{A'} N$ are used for the sheaves $\mathbf{Hom}_A(M, N)$ and $M \otimes_A N$. When $A' = k_E$ is the constant presheaf defined by an ordinary ring k, we also write $\mathbf{Hom}_k$, $\otimes_k$ instead of $\mathbf{Hom}_{k_E}$, $\otimes_{k_E}$.

Exercise 13.9. *Locally ringed topoi* (cf. [9] for further information in the direction considered below).

Let $(E, \mathscr{O}_E)$ be a commutatively ringed topos. For $X \in \mathrm{ob}\, E$ and $f \in \mathscr{O}_E(X)$, let X_f be the largest subobject of X on which f is invertible.

a) Show that, for $X' \in \mathrm{ob}\, E_{/X}$, $f_{X'}$ is invertible if and only if the structural morphism $X' \to X$ factors through X_f, and that, for $f, g \in \mathscr{O}_E(X)$, one has

$$X_{fg} = X_f \cap X_g.$$

b) Show that the following conditions (i) through (iii) are equivalent:

(i) For $X \in \mathrm{ob}\, E$ and $f, g \in \mathscr{O}_E(X)$, one has

$$X_{f+g} \subset \mathrm{Sup}(X_f, X_g).$$

(ii) For $X \in \mathrm{ob}\, E$ and $f, g \in \mathscr{O}_E(X)$ such that $f + g$ is invertible, one has $X = \mathrm{Sup}(X_f, X_g)$.

(iii) For $X \in \mathrm{ob}\, E$ and $f \in \mathscr{O}_E(X)$, one has

$$X = \mathrm{Sup}(X_f, X_{1-f}).$$

513 Show that these conditions imply the following condition (iv), and are equivalent to it if E has enough points:

(iv) For every point p of E, the fiber ring $\mathscr{O}_{E,p}$ is a *local ring*.

When the equivalent conditions (i) through (iii) hold, $(E, \mathscr{O}_E)$ is said to be a *locally ringed topos*; likewise, a ringed site is said to be a *locally ringed site* if the corresponding ringed topos is a locally ringed topos.

c) Let $(E, \mathscr{O}_E)$ and $(E', \mathscr{O}_{E'})$ be two locally ringed topoi, and let f be a morphism of ringed topoi from the first to the second. Show that the following condition (i) implies condition (ii), and is equivalent to it if E has enough points:

(i) For every $X' \in \mathrm{ob}\, E'$ and $s' \in \mathscr{O}_{E'}(X')$, setting $X = f^{-1}(X')$ and $s = f^*(s')$, one has

$$X_s = f^{-1}(X'_{s'}).$$

(ii) For every point p of E, setting $p' = f(p)$, the natural homomorphism on the fibers

$$\mathscr{O}_{E',p'} \longrightarrow \mathscr{O}_{E,p}$$

is a *local* homomorphism of local rings.

When condition (i) holds, f is said to be a *morphism of locally ringed topoi*, or also an *admissible morphism of locally ringed topoi* if there is a risk of confusion. We denote by **Homtoplocan**(E, E') the full subcategory of the category **Homtopan**(E, E') of all morphisms of ringed topoi from E to E' defined by the admissible morphisms from E to E'.

d) Suppose that $(E, \mathscr{O}_E)$ is the ringed topos associated with a ringed topological space $(X, \mathscr{O}_X)$ (2.1). Show that $(E, \mathscr{O}_E)$ is locally ringed if and only if $(X, \mathscr{O}_X)$ is locally ringed, i.e. if and only if, for every $x \in X$, the fiber $\mathscr{O}_{X,x}$ is a local ring. (Note that in criterion (iv) of b), it suffices to take p in a *conservative* family of
514 points of E.) Let $f : (X, \mathscr{O}_X) \to (X', \mathscr{O}_{X'})$ be a morphism of ringed spaces, where $\mathscr{O}_X$ and $\mathscr{O}_{X'}$ are sheaves of local rings. Show that the corresponding morphism of ringed topoi $\mathrm{Top}(X, \mathscr{O}_X) \to \mathrm{Top}(X', \mathscr{O}_{X'})$ is admissible if and only if the same is true of the morphism f, i.e. if and only if, for every $x \in X$,

$$\mathscr{O}_{X',f(x)} \longrightarrow \mathscr{O}_{X,x}$$

is a *local* homomorphism of local rings.

e) Let $(E, \mathscr{O}_E)$ and $(E', \mathscr{O}_{E'})$ be two locally ringed topoi such that E has enough points and $(E', \mathscr{O}_{E'})$ is equivalent to the ringed topos defined by a *scheme* $(X', \mathscr{O}_{X'})$;

prove that the category **Homtoplocan**(E, E') is equivalent to a *discrete* category, i.e. that it is a rigid groupoid (every morphism is an isomorphism, and the automorphism groups of the objects are the trivial groups).
(Hint: reduce to the case where $(E, \mathscr{O}_E)$ is the punctual topos ringed by a field.) Recall, on the other hand, that **Homtop**(E, E') is not in general equivalent to a discrete category, even if E and E' are topoi defined by schemes (and even if E is the punctual topos), cf. 4.2.3.

f) Let E be a locally ringed topos, and let P be the ringed topos defined by the ringed space $\operatorname{Spec} k$, where k is a field. Show that the admissible morphisms of locally ringed topoi $P \to E$ correspond to pairs (p, u) consisting of a point p of E and an injection of $k(p)$ into k, where $k(p)$ is the residue field of the local ring $\mathscr{O}_{E,p}$. Generalize this to a statement exhibiting the admissible morphisms from a punctual locally ringed topos to E (generalizing EGA I 2.4.4).

A *geometric point of a locally ringed topos* E is any admissible morphism into E from the locally ringed topos defined by a ringed space of the form $\operatorname{Spec}(k)$, where k is an *algebraically closed* field. Define the *category of geometric points of* E (with the understanding that the corresponding fields k belong to $\mathscr{U}$), denoted by **Ptgéom**(E), and a canonical functor **Ptgéom**$(E) \to$ **Point**(E). Show that this functor is faithful if and only if E has no point, i.e. **Point**(E) is the 515
empty category.

g) Let E be a $\mathscr{U}$-topos and let $\mathscr{V}$ be a universe such that $\mathscr{U} \in \mathscr{V}$. Define a 2-category $(\mathscr{V}\text{-}\mathscr{U}\text{-}\mathrm{Top})_{/E}$ in terms of the object E of the 2-category $(\mathscr{V}\text{-}\mathscr{U}\text{-}\mathrm{Top})$ (3.3.1), following 5.14 a). When E is ringed by a Ring A, define similarly a 2-category $(\mathscr{V}\text{-}\mathscr{U}\text{-}\mathrm{Topan})_{/E}$, and when E is locally ringed, use the notion of admissible morphism (cf. c)) to define a 2-category $(\mathscr{V}\text{-}\mathscr{U}\text{-}\mathrm{Topanloc})_{/E}$ having **Ptgéom**(E) (cf. f)) as a full subcategory.

14. Modules on a topos defined by gluing

14.1. Let (E, A) be a ringed topos, let U be an open subtopos of E, let F be the complementary closed subtopos, and let $j : U \to E$ and $i : F \to E$ be the canonical morphisms (9.3). The topos U is ringed by j^*A (11.2.1), and the morphism of topoi i, completed in the evident way, becomes a morphism of ringed topoi. In this section, the topos F will be ringed by the sheaf i^*A, denoted by $A|F$, and the morphism j, completed in the evident way, is then a morphism of ringed topoi.

14.2. We therefore have two morphisms of ringed topoi:

$$j : (U, A|U) \longrightarrow (E, A) \text{ and } i : (F, A|F) \longrightarrow (E, A);$$

whence five functors between the corresponding categories of modules (left modules, to fix ideas):

(14.2.1)
$$\begin{tikzcd} ({}_{A|U}U) \arrow[r, shift left=2, "j_!"] \arrow[r, shift right=2, "j_*"'] & ({}_AE) \arrow[l, "j^*"'] \arrow[r, shift left, "i^*"] & ({}_{A|F}F) \arrow[l, shift left, "i_*"] \end{tikzcd} \quad .$$

Each functor in diagram (14.2.1) is left adjoint to the functor lying below it.

14.3. We know that the functor $X \mapsto (j^*X, i^*X; i^*X \to i^*j_*j^*X)$ is an equiv- 516
alence from E to the topos (U, F, i^*j_*) (9.5.4). This equivalence induces an equivalence between the corresponding categories of modules, and hence the functor $P \mapsto (j^*P, i^*P; i^*P \to i^*j_*j^*P)$ is an equivalence, denoted by Φ, from the category ${}_AE$ to

the category $({}_{A|U}U, {}_{A|F}F, i^*j_*)$. The functors in (14.2.1), composed with Φ or Φ^{-1}, are then the functors:

$$\begin{aligned}
\Phi \circ j_* &: M \mapsto (M, i^*j_*M; i^*j_*M \xrightarrow{\mathrm{id}} i^*j_*M),\\
j^* \circ \Phi^{-1} &: (M, N; N \to i^*j_*M) \mapsto M,\\
\Phi \circ j_! &: M \mapsto (M, 0; 0 \to i^*j_*M),\\
\Phi \circ i_* &: N \mapsto (0, N; N \to 0),\\
i^* \circ \Phi^{-1} &: (M, N; N \to i^*j_*M) \mapsto N.
\end{aligned}$$

The reader may, as an exercise, make the various adjunction morphisms explicit.

14.4. Denote by

$$i^! : {}_AE \longrightarrow {}_{A|F}F \tag{14.4.1}$$

the functor defined by the formula

$$i^! \circ \Phi^{-1}(M, N; N \xrightarrow{u} i^*j_*M) = \mathrm{Ker}(u). \tag{14.4.2}$$

Proposition 14.5. *The functor $i^!$ is right adjoint to the functor i_*. The adjunction morphism $i_*i^! \to \mathrm{id}$ is a monomorphism. The functors $i^!$ (for varying rings) commute with restriction-of-scalars functors.*

It is clear that every morphism

$$(0, N; 0) \longrightarrow (M, N; N \xrightarrow{u} i^*j_*M)$$

factors uniquely through $(0, \ker(u); 0)$, which proves the adjunction property. The other assertions are trivial.

517 Proposition 14.6. *For every object P of ${}_AE$, there are exact sequences, functorial in P:*

$$0 \longrightarrow j_!j^*P \longrightarrow P \longrightarrow i_*i^*P \longrightarrow 0; \tag{14.6.1}$$

$$0 \longrightarrow i_*i^!P \longrightarrow P \longrightarrow j_*j^*P, \tag{14.6.2}$$

where the nontrivial arrows are the adjunction arrows.

First note that a sequence

$$(M', N'; u') \longrightarrow (M, N; u) \longrightarrow (M'', N''; u'')$$

in $({}_{A/U}U, {}_{A/F}F; i^*j_*)$ is exact if and only if the corresponding sequences

$$\begin{gathered}
M' \longrightarrow M \longrightarrow M''\\
N' \longrightarrow N \longrightarrow N''
\end{gathered}$$

are exact.

The sequences (14.6.1) and (14.6.2) are transformed by the equivalence Φ into sequences of the type in (14.3):

$$\begin{gathered}
0 \longrightarrow (M, 0; 0) \longrightarrow (M, N; u) \longrightarrow (0, N; 0) \longrightarrow 0\\
0 \longrightarrow (0, \ker(u); 0) \longrightarrow (M, N, u) \longrightarrow (M, i^*j_*M; \mathrm{id}).
\end{gathered}$$

The exactness of these sequences is immediate.

14.7. The exact sequence (14.6.2) gives a new interpretation of the functor $i_* i^!$. Indeed, let X be an object of E. Sequence (14.6.2) gives the exact sequence of abelian groups

$$0 \longrightarrow \mathrm{Hom}_E(X, i_* i^! P) \longrightarrow \mathrm{Hom}_E(X, P) \longrightarrow \mathrm{Hom}_E(X, j_* j^* P).$$

Also denote by U the open subobject of E which is the final object of the open subtopos U. It follows from the adjunction properties of the functors j_*, j^*, and $j_!^{\mathrm{ens}}$ that the group $\mathrm{Hom}_E(X, j_* j^* P)$ is canonically isomorphic to $\mathrm{Hom}_E(X \times U, P)$, and that the morphism $\mathrm{Hom}_E(X, P) \to \mathrm{Hom}_E(X, j_* j^* P)$ is precisely the morphism defined by the monomorphism $X \times U \to X$ (5). In other words (8.5.2):

Proposition 14.8. *The sections of $i_* i^! P$ on $E_{/X}$ are the sections of P on $E_{/X}$ whose* 518
support (9.3.5) *is contained in F. In other words, $i_* i^! P$ is the largest subsheaf of P whose support is contained in F.*

14.9. By abuse of language, one sometimes says that $i_* i^! P$ is the submodule of P defined by the sections of P supported in F. It follows from 8.5.3 that this terminology merely extends to general topoi terminology used for topoi of sheaves on topological spaces.

Proposition 14.10. 1) *The functor $P \mapsto i_* i^* P$ is isomorphic to the functor $P \mapsto i_* i^* A \otimes_A P$.*

2) *The functor $P \mapsto i_* i^! P$ is isomorphic to the functor $P \mapsto \mathbf{Hom}_A(i_* i^* A, P)$.*

The exact sequence (14.6.1), in the particular case where P is the A-module A, reads

$$0 \longrightarrow A_U \longrightarrow A \longrightarrow i_* i^* A \longrightarrow 0. \tag{14.10.1}$$

Here A_U is the free A-module generated by the open subobject U corresponding to the open subtopos U (9 and 11.3.3). For every A-module P, the A-modules $j_! j^* P$ and $j_* j^* P$ are canonically isomorphic respectively to $A_U \otimes_A P$ and $\mathbf{Hom}_A(A_U, P)$ (12.3 and 12.6). Moreover, under these isomorphisms, the canonical morphisms $j_! j^* P \to P$ and $P \to j_* j^* P$ arise from the monomorphism $A_U \to A$. The exact sequence 14.10.1 therefore gives two exact sequences (12.2 and 12.11):

$$j_! j^* P \longrightarrow P \longrightarrow i_* i^* A \otimes_A P \longrightarrow 0,$$
$$0 \longrightarrow \mathbf{Hom}_A(i_* i^* A, P) \longrightarrow P \longrightarrow j_* j^* P;$$

and the stated isomorphisms follow by comparison with (14.6.1) and (14.6.2).

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EXPOSÉ V

Cohomology in Topoi

J.-L. Verdier

Introduction

In this exposé we present the commutative and elementary cohomological invariants of topoi. In n° 1, we study flat modules and flat morphisms of ringed topoi. The proofs are given using the hypothesis, most often satisfied in practice, that the topoi have enough points (IV 6). These proofs are repeated in the general case in Appendix n° 8, where Deligne, by means of the technique of local inductive limits, also generalizes to topoi D. Lazard's theorem on the structure of flat modules. The theorems in this exposé are existence theorems for spectral sequences relating the various cohomological invariants (n° 3, 5, 6). It is known that, even for topological spaces, Čech cohomology does not in general coincide with sheaf cohomology [11]. In Appendix n° 7, we introduce a modified Čech computation which makes it possible to obtain the sheaf cohomology of an arbitrary topos by means of coverings. In that appendix we are led to use simplicial coverings (hypercoverings), whose homotopy invariants were studied in [1] (cf. also [17]). 2

The cohomological invariants introduced here are elementary in the sense that we do not use derived categories [12]. A reader familiar with this language will immediately translate the various statements of this exposé and will then be able to generalize them to complexes and hypercohomology. The language of derived categories is, moreover, used later in this seminar.

Here we restrict ourselves to commutative cohomology. For noncommutative H^1, which is used in this seminar, and for noncommutative H^2, we refer to [9]. The functors that we derive are additive; we say nothing about multiplicative structures (cf. [7]).

0. Generalities on abelian categories

In this section we recall several lemmas, most of which can be found in [11]. 3

Proposition 0.1. *Let $\mathfrak{A}$ be an abelian category having a generator. The following conditions are equivalent:*

i) *The category $\mathfrak{A}$ satisfies axiom $AB\,5$): Small direct sums are representable, and if (X_i), $i \in I$, is a small increasing filtered family of subobjects of an object X of $\mathfrak{A}$ and Y is a subobject of X, then*

$$(\sup_i X_i) \cap Y = \sup_i (X_i \cap Y).$$

ii) *Small pseudo-filtered inductive limits (I.2.7) are representable and commute with finite projective limits.*

Moreover, if the preceding conditions hold, small filtered inductive limits are universal (I 2.6).

Proof. It is clear that (ii) $\Rightarrow$ (i). To show that (i) $\Rightarrow$ (ii), and to prove the additional assertion, it suffices to use the fact that $\mathfrak{A}$ is a full subcategory of a category of modules $\mathscr{M}$ over a suitable ring, such that the inclusion functor $u : \mathfrak{A} \to \mathscr{M}$ admits an exact left adjoint v [5]. The verification is then immediate.

0.1.1. It is known [11] that an abelian category $\mathfrak{A}$ having a generator and satisfying axiom AB 5) has enough injectives, that is, every object embeds in an injective object. Moreover, by the result already cited [5], small products are representable in $\mathfrak{A}$ (axiom AB 3) *).

Proposition 0.2. *Let $\mathfrak{A}$ and $\mathfrak{B}$ be two abelian categories and let* $\mathfrak{A} \underset{u}{\overset{v}{\rightleftarrows}} \mathfrak{B}$ *be two adjoint functors (u is left adjoint to v). Consider the following two properties:*

i) *The functor u is exact.*

4 ii) *The functor v sends injective objects of $\mathfrak{B}$ to injective objects of $\mathfrak{A}$.*

1) *The implication* (i) $\Rightarrow$ (ii) *always holds.*
If, in addition, every nonzero object of $\mathfrak{B}$ is the source of a nonzero morphism into an injective object, then (ii) $\Leftrightarrow$ (i).

2) *Suppose that:*
   a) *the category $\mathfrak{B}$ has enough injectives;*
   b) *one of the two equivalent conditions* (i) *and* (ii) *above holds;*
   c) *the functor u is faithful.*

Then the category $\mathfrak{A}$ has enough injectives.

Proof. The proof is left to the reader.

Remark 0.2.1. The category of commutative groups has enough injectives. Applying Proposition 0.2, we deduce that every category of unital modules over a ring with identity has enough injectives. Applying the result of [5] (used in the proof of 0.1) together with 0.2, we then deduce that every abelian category having a generator and exact small filtered inductive limits has enough injectives; this gives a new proof of that fact.

Proposition 0.3. *Let $\mathfrak{A}$, $\mathfrak{B}$, and $\mathfrak{C}$ be three abelian categories, and let $u : \mathfrak{A} \to \mathfrak{B}$ and $v : \mathfrak{B} \to \mathfrak{C}$ be two left exact additive functors. Suppose that $\mathfrak{A}$ and $\mathfrak{B}$ have enough injective objects. The following two properties are equivalent:*

i) *There exists a spectral functor whose $E_2^{p,q}$ term is*
$$R^p v R^q u$$
and which converges to $R^{p+q} vu$ (suitably filtered).

ii) *The functor u sends injective objects of $\mathfrak{A}$ to objects that are acyclic for the functor v.*

Proof. (i) $\Rightarrow$ (ii) is immediate, since it suffices to apply the spectral functor to an
5 injective object. The implication (ii) $\Rightarrow$ (i) is proved in [11].

Proposition 0.4. *Let $\mathfrak{A}$ and $\mathfrak{B}$ be two abelian categories and let $u : \mathfrak{A} \to \mathfrak{B}$ be a left exact additive functor. Let M be a subset of the set of objects of $\mathfrak{A}$ having the following properties:*

1) *Every object of $\mathfrak{A}$ embeds in an element of M.*
2) *If $X \oplus Y$ belongs to M, then the object X belongs to M.*
3) *If*
$$0 \longrightarrow X' \longrightarrow X \longrightarrow X'' \longrightarrow 0$$

is an exact sequence and X' and X belong to M, then X'' belongs to M, and the sequence

$$0 \longrightarrow u(X') \longrightarrow u(X) \longrightarrow u(X'') \longrightarrow 0$$

is exact. Zero objects belong to M.

Then every injective belongs to M, and the objects of M are acyclic for u, that is, for every $q \neq 0$ and every object X of M, one has $R^q u(X) = 0$. (In particular, resolutions by objects of M can be used to compute the derived functors of u.)

For the proof, see [11] 3.3.1.

Proposition 0.5. *Let $\mathscr{U} \subset \mathscr{V}$ be two universes, let $\mathfrak{A}$ (resp. $\mathfrak{B}$) be a $\mathscr{U}$-category (resp. $\mathscr{V}$-category) which is abelian, satisfies axiom AB 5) relative to $\mathscr{U}$ (resp. to $\mathscr{V}$), and has a $\mathscr{U}$-small (resp. $\mathscr{V}$-small) generating family. Let $\epsilon : \mathfrak{A} \to \mathfrak{B}$ be an exact fully faithful functor. The following conditions are equivalent:*

1) *There exists a generating family $(X_i)_{i\in I}$ of $\mathfrak{A}$ such that the family $(\epsilon(X_i))_{i\in I}$ is generating in $\mathfrak{B}$.*

1′) *Every object of $\mathfrak{B}$ is isomorphic to a quotient of an object of the form $\bigoplus_{\alpha\in A}\epsilon(Y_\alpha)$, where $A \in \mathscr{V}$.*

Under these conditions, the following property holds:

2) *ϵ sends $\mathscr{U}$-small products to products (and thus commutes with $\mathscr{U}$-small projective limits).*

Moreover, under the equivalent conditions 1) or 1′), the following conditions are equivalent : 6

a) *For every object Y of $\mathfrak{A}$, every subobject of $\epsilon(Y)$ is isomorphic to the image under ϵ of a subobject of Y.*

a′) *There exists a generating family $(X_i)_{i\in I}$ of $\mathfrak{A}$ such that the family $(\epsilon(X_i))_{i\in I}$ is generating in $\mathfrak{B}$ and such that, for every $i \in I$, every subobject of $\epsilon(X_i)$ is isomorphic to the image under ϵ of a subobject of X_i.*

b) *Every object of $\mathfrak{B}$ is isomorphic to a subobject of an object of the form $\prod_{\alpha\in A}\epsilon(Y_\alpha)$, where $A \in \mathscr{V}$.*

c) *ϵ commutes with $\mathscr{U}$-small direct sums (and thus commutes with $\mathscr{U}$-small inductive limits).*

Moreover, under conditions 1) and a′), one has:

d) *ϵ sends injective objects to injective objects.*

Remark 0.5.1. When in 0.5 one has $\mathscr{U} = \mathscr{V}$, conditions 1) and a′) imply that ϵ is an equivalence of categories (since b), 2), and a) then hold).

0.5.2. We will confine ourselves to giving indications for the proof. The implications 1) $\Leftrightarrow$ 1′) and 1) $\Rightarrow$ 2) are left to the reader. It is clear that a) $\Rightarrow$ a′). We show that a′) implies d). Let M be injective in $\mathfrak{A}$. For every $i \in I$ and every subobject $Y \hookrightarrow X_i$ of X_i, the homomorphism $\mathrm{Hom}(X_i, M) \to \mathrm{Hom}(Y, M)$ is surjective. Consequently, by a′) and the full faithfulness of ϵ, for every subobject $U \hookrightarrow \epsilon(X_i)$, the homomorphism $\mathrm{Hom}(\epsilon(X_i), \epsilon(M)) \to \mathrm{Hom}(U, \epsilon(M))$ is surjective. Since the $\epsilon(X_i)$ form a generating family of $\mathfrak{B}$, $\epsilon(M)$ is injective [11].

We show that a′) $\Rightarrow$ b). By enlarging the family of the X_i if necessary, we may suppose that, for every $i \in I$, every quotient of X_i is isomorphic to an X_j for a suitable j. For every $i \in I$, let $X_i \hookrightarrow M_i$ then be a monomorphism into an injective object. The family $(\epsilon(X_i))_{i\in I}$ is stable under quotients, and for $i \in I$, the morphism $\epsilon(X_i) \hookrightarrow \epsilon(M_i)$

is, by d), a monomorphism into an injective object. It is then immediately verified that, since the family $\epsilon(X_i)$ is generating, the family $(\epsilon(M_i))_{i\in I}$ is cogenerating, whence b).

We show that b) ⇒ c). Let $(Z_\alpha)_{\alpha\in A}$ be a $\mathscr{U}$-small family of objects of $\mathfrak{A}$, and let us show that the canonical morphism

$$\bigoplus_{\alpha\in A}\epsilon(Z_\alpha) \longrightarrow \epsilon(\bigoplus_{\alpha\in A} Z_\alpha)$$

7 is an isomorphism. Since the family of objects $\epsilon(Y)$ is cogenerating, it suffices to show that, for every object Y of $\mathfrak{A}$, the homomorphism

$$\mathrm{Hom}(\epsilon(\bigoplus_a Z_a), \epsilon(Y)) \longrightarrow \mathrm{Hom}(\bigoplus_a \epsilon(Z_a), \epsilon(Y))$$

is an isomorphism, which follows from the full faithfulness of ϵ.

It remains to show that c) ⇒ a). Let Z be a subobject of $\epsilon(Y)$. By 1′), there exists a $\mathscr{U}$-small family Y_α of objects of $\mathfrak{A}$ and an epimorphism from $\bigoplus_\alpha \epsilon(Y_\alpha)$ onto Z. Since ϵ commutes with $\mathscr{U}$-small direct sums, there is therefore an epimorphism $\epsilon(Y') \to Z$. Denote by $u : \epsilon(Y') \to Z \to \epsilon(Y)$ the composite morphism. We have $Z = \mathrm{Im}(u)$. Since ϵ is fully faithful, we have $u = \epsilon(v)$, and consequently $Z = \mathrm{Im}(\epsilon(v)) = \epsilon(\mathrm{Im}(v))$.

Exercise 0.5.2. Let $\mathrm{Sex}_{\mathscr{V}}(\mathfrak{A})$ be the category of contravariant functors from $\mathfrak{A}$ to the category of $\mathscr{V}$-commutative groups which commute with $\mathscr{U}$-small inductive limits. Show that, under conditions 1) and a′), the canonical functor from $\mathfrak{B}$ to $\mathrm{Sex}_{\mathscr{V}}(\mathfrak{A})$ is an equivalence.

1. Flat modules

Definition 1.1. *Let* (E, A) *be a ringed topos* (IV 11.1.1). *A right (resp. left) A-module M is said to be flat if the functor $M\otimes_A\cdot$ (resp. $\cdot\otimes_A M$) from the category of left (resp. right) A-modules to the category of abelian sheaves of E is exact.*

Proposition 1.2. *Let M be a B-A-bimodule.*

1) *The following properties are equivalent:*
   i) *The module M is flat as a right A-module.*
   ii) *For every injective B-module I, the left A-module* $\mathbf{Hom}_B(M, I)$ *is injective.*
2) *A module M which is a pseudo-filtered inductive limit* (I 2.7.1) *of flat modules is flat.*
3) *Finally, if $M_\bullet = \cdots \to M_{i+1} \to M_i \to \cdots$ is an acyclic complex of flat modules ($M_i = 0$ for $i < i_0$), then, for every module F, the complex*

$$M_\bullet \otimes_A F = \cdots \longrightarrow M_{i+1}\otimes_A F \longrightarrow M_i \otimes_A F \longrightarrow \cdots$$

8 *is acyclic.*

Proof. By (IV 12.12), there is an adjunction isomorphism:

$$\mathrm{Hom}_B(M\otimes_A \cdot, \cdot) \xrightarrow{\sim} \mathrm{Hom}_A(\cdot, \mathbf{Hom}_B(M, \cdot)).$$

It is then enough to apply 0.2 to obtain the equivalence (i) ⇔ (ii). This adjunction isomorphism also shows that the tensor product commutes with inductive limits. The fact that pseudo-filtered inductive limits are exact (0.1) implies the second assertion. To show that the complex $M_\bullet\otimes_A F$ is acyclic, it is enough to show that, for every injective abelian sheaf I, the complex $\mathrm{Hom}^\bullet_{\mathbf{Z}}(M_\bullet \otimes_A F, I)$ is acyclic. By the adjunction formulas, this complex is isomorphic to $\mathrm{Hom}^\bullet_A(F, \mathbf{Hom}_{\mathbf{Z}}(M_\bullet, I))$. But, by the equivalence (i) $\Leftrightarrow$ (ii), the complex of sheaves

$$\mathbf{Hom}_{\mathbf{Z}}(M_\bullet, I)$$

is an acyclic complex whose objects are injective, whence the conclusion.

Proposition 1.3.1. *Let (E, A) be a ringed topos, let H be an object of E, and let M be a flat $A_{/H}$-module. Then $j_{H!}M$ is a flat A-module. In particular, A_H is flat on both the right and the left.*

Suppose, to fix ideas, that M is a right $A_{/H}$-module. For every left A-module P, there is a canonical isomorphism (IV 12)

$$j_{H!}M \otimes_A P \simeq j_{H!}(M \otimes_{A_{/H}} j_H^* P).$$

The functors $j_{H!}$ and j_H^* are exact (IV 11.3.1 and 11.12.2), and, by hypothesis, the functor $M \otimes_{A_{/H}} -$ is exact. Consequently, the functor $P \longmapsto j_{H!}M \otimes_A P$ is exact, and $j_{H!}M$ is flat.

Proposition 1.3.2. *(Projection formula for closed immersions):*
Let (E, A) be a ringed topos, and let $i : F \to E$ be a closed subtopos of E. Set $i^*A = A_{/F}$. For every right $A_{/F}$-module M and every left A-module P, there is a functorial 9
isomorphism

$$i_*(M \otimes_{A_{/F}} i^*P) \simeq (i_*M) \otimes_A P.$$

Let U be the open complement of F, and let $j : U \to E$ be the canonical immersion morphism. One has $j^*(i_*M \otimes_A P) \simeq 0 \otimes_{A_{/U}} j^*(P)$ (IV 12); consequently, $i_*M \otimes_A P$ has its support in F. Hence the adjunction morphism $i_*M \otimes_A P \to i_*i^*(i_*M \otimes_A P)$ is an isomorphism (IV 14). One has $i^*(i_*M \otimes_A P) \simeq i^*i_*M \otimes_{A_{/F}} i^*P$ (IV 12), and, since $i : F \to E$ is a closed immersion, $i^*i_*M \simeq M$ (IV 14); this gives the asserted isomorphism.

Corollary 1.3.3. *For every flat $A_{/F}$-module M, i_*M is flat.*

It follows from 1.3.2 and from (IV 14) that the functor $P \longmapsto (i_*M) \otimes_A P$ is exact.

1.4. Let (E, A) be a ringed topos, let $x : P \to E$ be a point of E (IV 6.1), and let $\mathrm{Vois}(x)$ be the category of neighborhoods of x (IV 6.8). To every object V of $\mathrm{Vois}(x)$ there corresponds an object of E, also denoted by V, and a point $x_V : P \to E/V$ of E/V. Moreover, to every morphism $u : V \to W$ in $\mathrm{Vois}(x)$ there corresponds an essentially commutative diagram of morphisms of topoi (IV 6.7)

(1.4.1)
$$\begin{tikzcd} P \arrow[r, "x_V"] \arrow[dr, "x_W"'] & E/V \arrow[d, "j_u"] \\ & E/W \end{tikzcd}.$$

Let N be an A_x-module (resp. a set). From the diagram (1.4.1), one obtains a diagram of A_x-modules (resp. sets):

$$\begin{tikzcd} x_W^* x_{W*} N \arrow[r, "\mathrm{ad}_W"] \arrow[d, "\phi(u)"'] & N \\ x_V^* x_{V*} N \arrow[ur, "\mathrm{ad}_V"'] & \end{tikzcd} \tag{1.4.2}$$

10 where ad_W and ad_V are the adjunction morphisms. Moreover, one immediately checks that $\phi(uv) = \phi(v)\phi(u)$. Thus there is a functor from the filtered category $\mathrm{Vois}(x)^\circ$ to the category of A_x-modules (resp. sets), and a homomorphism

$$\Lambda(N) : \varinjlim_{V \in \mathrm{Vois}(x)^\circ} x_V^* x_{V*} N \longrightarrow N. \tag{1.4.3}$$

Proposition 1.5. *For every A_x-module (resp. set) N, $\Lambda(N)$ is an isomorphism.*

This proposition is a special case of a proposition due to Deligne (8.2.6). Let us give a direct proof. By passing to the underlying sets, it is enough to prove the proposition when N is a set (I 2.8). The cofiltered category $\mathrm{Fl}(\mathrm{Vois}(x))$ of arrows of $\mathrm{Vois}(x)$ is fibered over the category $\mathrm{Vois}(x)$ by the functor $p : \mathrm{Fl}(\mathrm{Vois}(x)) \to \mathrm{Vois}(x)$ which assigns to an arrow its target. Let D be a category and $F : \mathrm{Fl}(\mathrm{Vois}(x)) \to D$ a pro-object (I 8.10). For every object V of $\mathrm{Vois}(x)$, denote by F_V the pro-object obtained by restricting the functor F to the fiber category $\mathrm{Vois}(x)/V$. To every morphism $m : U \to V$ of $\mathrm{Vois}(x)$, the base-change functor by m associates a morphism from the pro-object F_U to the pro-object F_V, and the canonical morphisms of pro-objects $F \to F_V$ determine a morphism of pro-objects

$$F \longrightarrow \varprojlim_{V \in \mathrm{Vois}(x)^\circ} F_V \tag{1.5.1}$$

which is immediately checked to be an isomorphism. Apply this observation to the pro-object of pointed sets

$$(U, V, m) \longmapsto x_V^*(m) = F(U, V, m). \tag{1.5.2}$$

One obtains an isomorphism of pro-objects

$$F \xrightarrow{\sim} \varprojlim_{\mathrm{Vois}(x)} \varprojlim_{\mathrm{Vois}(x)/V} x_V^*(m); \tag{1.5.3}$$

hence, for every set N, a bijection

$$\varinjlim_{\mathrm{Fl}(\mathrm{Vois}(x))^\circ} \mathrm{Hom}(x_V^*(m), N) \simeq \varinjlim_{\mathrm{Vois}(x)^\circ} \varinjlim_{(\mathrm{Vois}(x)/V)^\circ} \mathrm{Hom}(x_V^*(m), N). \tag{1.5.4}$$

11 But, for fixed V, $\varinjlim_{(\mathrm{Vois}(x)/V)^\circ} \mathrm{Hom}(x_V^*(m), N)$ is identified with $x_V^* x_{V*} N$. Indeed, by IV 6.8.1 one has $x_V^* x_{V*} N \simeq \varinjlim_{\mathrm{Vois}(x_V)^\circ} \mathrm{Hom}_{E/V}(W, x_{V*} N)$, and hence, by adjunction, $x_V^* x_{V*} N \simeq \varinjlim_{\mathrm{Vois}(x_V)^\circ} \mathrm{Hom}(x_V^*(W), N)$; moreover, the category $\mathrm{Vois}(x_V)$ is equivalent to the category $\mathrm{Vois}(x)/V$ (IV 6.7.2). Finally, the canonical transition maps $\varinjlim_{(\mathrm{Vois}(x)/U)^\circ} \mathrm{Hom}(x_U^*(m), N) \to$

$\varinjlim_{(\mathrm{Vois}(x)/V)^\circ} \mathrm{Hom}(x_V^*(n), N)$ in (1.5.4) are identified with the canonical maps $x_U^* x_{U*} N \to x_V^* x_{V*} N$ of (1.4.3), as the reader may verify. Thus there is a bijection

$$\varinjlim_{\mathrm{Fl}(\mathrm{Vois}(x))^\circ} \mathrm{Hom}(x_V^*(m), N) \simeq \varinjlim_{\mathrm{Vois}(x)^\circ} x_V^* x_{V*} N \tag{1.5.5}$$

and the map $\Lambda(N) : \varinjlim_{\mathrm{Vois}(x)^\circ} x_V^* x_{V*} N \to N$ of (1.4.3), composed with the bijection (1.5.5), comes from the maps $\mathrm{Hom}(x_V^*(m), N) \to N$ which assign to a map $r : x_V^*(m) \to N$ the image under r of the distinguished point of $x_V^*(m)$. To prove the proposition, it is then enough to observe that every object (U, V, m) of $\mathrm{Fl}(\mathrm{Vois}(x))$ is bounded below by (U, U, id_U), and that $x_U^*(\mathrm{id}_U)$ consists of a single element or, in other words, that the canonical morphism from the constant pro-object consisting of a single element to F is an isomorphism of pro-objects.

Proposition 1.6. *Let (E, A) be a ringed topos, and let M be an A-module.*

1) *If M is flat, then, for every point $x : P \to E$ of E, the A_x-module M_x is flat.*
2) *Let $(x_i)_{i\in I}$ be a conservative family of points of E such that, for every $i \in I$, M_{x_i} is a flat A_{x_i}-module.*

Then M is flat.

Lemma 1.6.1. *Let M be a flat module and let V be an object of E. The A/V-module $j_V^* M$ is flat.*

We must show that the functor $P \longmapsto P \otimes_{A/V} j_V^* M$ is exact. Since the extension-by-zero functor $j_{V!}$ is exact and faithful (IV 11.3.1), it is enough to show that the functor $P \longmapsto j_{V!}(P \otimes_{A/V} j_V^* M)$ is exact. There is a functorial isomorphism $j_{V!}(P \otimes_{A/V} j_V^* M) \simeq$ 12
$j_{V!}(P) \otimes_A M$ (IV 12.11); consequently, the functor $P \longmapsto j_{V!}(P) \otimes_A M$ is exact (IV 11.3.1).

1.6.2. Let us prove 1). To fix ideas, suppose that M is a flat left A-Module, and let $x : P \to E$ be a point of E. We shall show that the functor $N \mapsto N \otimes_{A_x} M_x$ from the category of right A_x-modules to the category of commutative groups is exact. For this it suffices to show that this functor is left exact. With the notation of 1.4, let V be a neighborhood of x, and let $j_V : E/V \to E$ be the localization morphism. For every A_x-module N, we have $x_V^* x_{V*} N \otimes_{A_x} M_x \simeq x_V^* x_{V*} N \otimes_{A_x} x_V^* j_V^* M \simeq x_V^*(x_{V*} N \otimes_{A/V} j_V^* M)$ (IV 12.11). Consequently, the functor $N \mapsto x_V^* x_{V*} N \otimes_{A_x} M_x$ is left exact. It follows that the functor $N \mapsto N \otimes_{A_x} M_x$, being a filtered inductive limit of left exact functors (1.5), is left exact.

Let us prove 2). Let $0 \to P' \to P \to P'' \to 0$ be an exact sequence of A-Modules, and let us show that the sequence $0 \to P' \otimes_A M \to P \otimes_A M \to P'' \otimes_A M \to 0$ is exact. It suffices to show that, for every $i \in I$, the sequence obtained by passing to the fibers at x_i is exact (IV 6). But the sequence of fibers at x_i is the sequence (IV 13.5)

$$0 \longrightarrow P'_{x_i} \otimes_{A_{x_i}} M_{x_i} \longrightarrow P_{x_i} \otimes_{A_{x_i}} M_{x_i} \longrightarrow P''_{x_i} \otimes_{A_{x_i}} M_{x_i} \longrightarrow 0,$$

and since M_{x_i} is flat, this sequence is exact.

Proposition 1.7. *Let (E, A) be a ringed topos, let $u : E' \to E$ be a morphism of topoi, and put $A' = u^* A$. Let M be a right (resp. left) A'-Module. The following conditions are equivalent:*

i) *The functor $N \mapsto M \otimes_{A'} u^* N$ (resp. $N \mapsto u^* N \otimes_{A'} M$) from the category of left (resp. right) A-Modules to the category of abelian sheaves on E' is exact.*
ii) *M is a flat A'-Module.*

It is clear that ii) $\Rightarrow$ i).

Let us prove that i) $\Rightarrow$ ii). We give the proof only in the case where E' possesses a conservative family of points $(x_i)_{i\in I}$. The general case is treated in (8.2.7). In this
13 particular case, which covers most applications, we reduce to the case where E' is the punctual topos and hence where the morphism u, which we then denote by x, is a point of E (1.6 and IV 11.3.1). We restrict ourselves to the case where M is a right A'-Module. The case where M is a left A'-Module reduces to this one by passing to opposite rings. Let V be a neighborhood of x, and let V_x be its fiber at x. With the notation of 1.4, there is an essentially commutative diagram of morphisms of topoi:

$$\begin{tikzcd} P \arrow[r, "j"] \arrow[dr, "x_V"'] & P/V_x \arrow[r, "j_{V_x}"] \arrow[d, "x/V"] & P \arrow[d, "x"] \\ & E/V \arrow[r] & E \end{tikzcd}$$

where x/V is the morphism deduced from x by localization over V (IV 5.10), and j is the morphism deduced from the marked point of V_x. Let us show that the functor $N' \mapsto M \otimes_{A'} x_V^* N'$ is exact. We have $x_V^* = j^*(x/V)^*$ and $\mathrm{id} = j^* j_{V_x}^*$; consequently, there is a canonical isomorphism

$$M \otimes_{A'} x_V^* N' \simeq j^*\big(j_{V_x}^* M \otimes_{A'/V_x} (x/V)^* N'\big).$$

Since j^* is exact and the functor $j_{V_x!}$ is exact and faithful (IV 11.3.1), it suffices to show that the functor

$$N' \longmapsto j_{V_x!}\Big(j_{V_x}^* M \otimes_{A'/V_x} (x/V)^* N'\Big)$$

is exact; consequently (IV 12.11), it suffices to show that the functor $N' \mapsto M \otimes_{A'} x^* j_{V!} N'$ is exact. But it follows from the definition of the inverse image of an induced topos (IV 5.10.2) that $j_{V_x!}(x/V)^* N'$ is equal to $x^* j_{V!} N'$, and since the functor $j_{V!}$ is exact (IV 11.3.1), the functor $N' \mapsto M \otimes_{A'} j_{V_x!}(x/V)^* N'$ is exact by hypothesis.

For every A'-module Q, there is a functorial isomorphism (1.5):

$$Q \simeq \varinjlim_{\mathrm{Vois}(x)^\circ} x_V^* x_{V*} Q.$$

By what precedes, the functor $Q \mapsto M \otimes_{A'} Q$ is a filtered inductive limit of left exact functors. It is therefore left exact, and hence M is flat.

Corollary 1.7.1. *Let $u : (E', A') \to (E, A)$ be a morphism of ringed topoi, and let M be a flat A-Module. Then $u^* M$ is a flat A'-Module.*

This follows from the criterion given in 1.7 and formula (IV 13.4.5).

14 Corollary 1.7.2. *Let $u : (E', A') \to (E, A)$ be a morphism of ringed topoi. The following conditions are equivalent:*

i) *The right (resp. left) $u^{-1}(A)$-Module A' is flat.*
ii) *The functor u^* from the category of left (resp. right) A-Modules to the category of left (resp. right) A'-Modules is exact.*

The equivalence follows from 1.7 and the definition of u^* (IV 13.2.1).

Definition 1.8. *A morphism of ringed topoi having the equivalent properties of 1.7.2 is called a morphism of ringed topoi which is left (resp. right) flat.*

Proposition 1.9. *Let $\mathscr{U} \subset \mathscr{V}$ be two universes, let C be a $\mathscr{U}$-site, let $C^\sim_{\mathscr{U}}$ and $C^\sim_{\mathscr{V}}$ be the categories of $\mathscr{U}$- and $\mathscr{V}$-sheaves of sets, respectively, let $\epsilon : C^\sim_{\mathscr{U}} \to C^\sim_{\mathscr{V}}$ be the canonical inclusion functor, and let A be a $\mathscr{U}$-sheaf of rings on C.*

1) *The functor ϵ commutes with $\mathscr{U}$-small inductive and projective limits of Modules. The functor ϵ is conservative and fully faithful on the categories of Modules.*
2) *For every object H of $C^\sim_{\mathscr{U}}$, there is a canonical isomorphism $\epsilon(A_H) \simeq \epsilon(A)_{\epsilon(H)}$.*
3) *Every injective left or right A-Module is carried by ϵ to an injective $\epsilon(A)$-Module.*
4) *Every flat right or left A-Module is carried by ϵ to a flat $\epsilon(A)$-Module.*
5) *The functor ϵ commutes with the formation of tensor products and sheaves of homomorphisms.*

The formation of the associated sheaf is independent of the universe (II 3.6). It then follows from the construction of inductive and projective limits in categories of sheaves (II 4.1 and 6.4) that the functor ϵ commutes with $\mathscr{U}$-small projective limits of sets or Modules, which gives 1). Let us prove 2). By taking a $\mathscr{U}$-small generating subcategory of $C^\sim_{\mathscr{U}}$, we may reduce to the case where C is $\mathscr{U}$-small (III 4.1). It then follows from 15
(II 6.5) that the free A-Module generated by H is obtained by forming the presheaf of free A-modules generated by H, a construction that commutes with enlargement of universes, and then forming the associated sheaf, an operation which likewise commutes with enlargement of universes (II 3.6). To prove 3), it suffices, by 0.5, to show that every $\mathscr{V}$-subsheaf of a $\mathscr{U}$-sheaf is a $\mathscr{U}$-sheaf, which is clear.

Let us prove 4). Let M be a flat A-Module. It suffices to show that, for every injective $\epsilon(A)$-Module J, the abelian sheaf $\mathbf{Hom}_{\epsilon(A)}(\epsilon(M), J)$ is injective (1.2). By 0.5, J is a subobject, hence a direct factor, of an object of the form $\prod_\alpha \epsilon(I_\alpha)$, where the I_α are injective. We may therefore suppose that $J = \epsilon(I)$, where I is an injective object. If the homomorphism

$$\epsilon(\mathbf{Hom}_A(M, I)) \longrightarrow \mathbf{Hom}_{\epsilon(A)}(\epsilon(M), \epsilon(I))$$

is an isomorphism, it follows from 3) that $\mathbf{Hom}_{\epsilon(A)}(\epsilon(M), \epsilon(I))$ is injective. It therefore remains to prove 5). The fact that formation of the tensor product commutes with the functor ϵ follows from IV 12.6 and the fact that formation of the associated sheaf commutes with enlargement of the universe (II 3.6). For every object X of $C^\sim_{\mathscr{U}}$ and every pair of A-Modules M, N of $C^\sim_{\mathscr{U}}$, we therefore have

$$\mathrm{Hom}_{C^\sim_{\mathscr{V}}}(\epsilon(X), \epsilon(\mathbf{Hom}_A(M, N))) \simeq \mathrm{Hom}_{C^\sim_{\mathscr{U}}}(X, \mathbf{Hom}_A(M, N)) \simeq \mathrm{Hom}_A(\mathbf{Z}_X \otimes_{\mathbf{Z}} M, N)$$

by the full faithfulness of ϵ, and then

$$\mathrm{Hom}_A(\mathbf{Z}_X \otimes_{\mathbf{Z}} M, N) \simeq \mathrm{Hom}_{\epsilon(A)}(\epsilon(\mathbf{Z}_X \otimes_{\mathbf{Z}} M), \epsilon(N)) \simeq \mathrm{Hom}_{\epsilon(A)}(\epsilon(\mathbf{Z}_X) \otimes_{\mathbf{Z}} \epsilon(M), \epsilon(N))$$

by the full faithfulness of ϵ and what precedes. Using 2) and IV 12.14, we finally obtain an isomorphism

$$\mathrm{Hom}_{C^\sim_{\mathscr{V}}}(\epsilon(X), \epsilon(\mathbf{Hom}_A(M, N))) \simeq \mathrm{Hom}_{C^\sim_{\mathscr{V}}}(\epsilon(X), \mathbf{Hom}_{\epsilon(A)}(\epsilon(M), \epsilon(N)));$$

hence the asserted isomorphism.

1.10. Let E be a topos and let $\mathscr{U} = (U_i \to X)_{i \in I}$ be a small family of morphisms with the same target. For every finite set $\Delta_n = \{0, \dots, n\}$, put

$$S_n(\mathscr{U}) = \coprod_{f : \Delta_n \to I} U_{f(0)} \times_X \cdots \times_X U_{f(n)},$$

the coproduct being taken over the set of maps from Δ_n to I. Let m, n be two integers, and let $g : \Delta_m \to \Delta_n$ be a map. We define a morphism 16

$$s(g) : S_n(\mathcal{U}) \longrightarrow S_m(\mathcal{U}) \tag{1.10.1}$$

as follows: for every $f : \Delta_n \to I$, the restriction of $s(g)$ to $U_{f(0)} \times_X \cdots \times_X U_{f(n)}$ is the composite of the canonical inclusion

$$U_{f(g(0))} \times_X \cdots \times_X U_{f(g(m))} \hookrightarrow S_m(\mathcal{U}),$$

and the unique morphism

$$s_f(g) : U_{f(0)} \times_X \cdots \times_X U_{f(n)} \longrightarrow U_{f(g(0))} \times_X \cdots \times_X U_{f(g(m))}$$

such that, for every $i \in \Delta_m$,

$$\mathrm{pr}_{f(g(i))}\, s_f(g) = \mathrm{pr}_{f(g(i))}$$

(pr_a denotes the a-th projection). We thus obtain a contravariant functor $\Delta_n \mapsto S_n$ from the category of finite sets to E; in other words, a semi-simplicial complex $S_\bullet(\mathcal{U})$ of objects of E. Note that this complex is canonically augmented toward X. Every functor from E to a category C carries $S_\bullet(\mathcal{U})$ to a simplicial object of C. In particular, if A is a Ring of E, the "free A-Module generated by" functor carries $S_\bullet(\mathcal{U})$ to a simplicial complex of A-biModules, augmented toward A_X and denoted by $A_\bullet(\mathcal{U})$. We have

$$A_n(\mathcal{U}) = \bigoplus_{f : \Delta_n \to I} A_{U_{f(0)} \times_X \cdots \times_X U_{f(n)}}. \tag{1.10.2}$$

This complex will often be written

$$\cdots \;\; \overset{s_0}{\underset{s_2}{\overset{\longrightarrow}{\underset{\longrightarrow}{\xrightarrow{s_1}}}}} \;\; \bigoplus_{i,j} A_{U_i \times_X U_j} \;\; \overset{s_0}{\underset{s_1}{\rightrightarrows}} \;\; \bigoplus_i A_{U_i} \longrightarrow A_X \tag{1.10.3}$$

where the arrows in (1.10.3) (except the last, which is the augmentation) represent the face operators of the simplicial complex $A_\bullet(\mathcal{U})$, that is, the morphisms corresponding to the increasing injective maps from Δ_n into Δ_{n+1} (s_i omits the integer i). To the complex $A_\bullet(\mathcal{U})$ we associate a differential complex augmented toward A_X:

$$\cdots \xrightarrow{d} \bigoplus_{i,j} A_{U_i \times_X U_j} \xrightarrow{d} \bigoplus_i A_{U_i} \longrightarrow A_X, \tag{1.10.4}$$

17 by putting

$$d = \sum (-1)^i s_i. \tag{1.10.5}$$

Proposition 1.11. *When the family $\mathcal{U} = (U_i \to X)_{i \in I}$ is epimorphic, the differential complex (1.10.4) is acyclic and gives a resolution of A_X.*

Denote by $\mathbf{Z}$ the constant sheaf with value $\mathbf{Z}$. By the definition of the "free A-Module generated by" functor, for every object H of E we have

$$A_H \simeq \mathbf{Z}_H \otimes_{\mathbf{Z}} A,$$

hence

$$A_\bullet(\mathcal{U}) \simeq \mathbf{Z}_\bullet(\mathcal{U}) \otimes_{\mathbf{Z}} A.$$

Since the components of $\mathbf{Z}_\bullet(\mathcal{U})$ are flat $\mathbf{Z}$-Modules (1.3.1), it suffices to prove the proposition when $A = \mathbf{Z}$.

Suppose first that E is the topos of sets. The augmented complex $S_\bullet(\mathscr{U})$ is then a direct sum of augmented complexes of the form

$$\cdots \overrightarrow{\overrightarrow{\overrightarrow{\rightarrow}}} S \times S \times S \overrightarrow{\overrightarrow{\rightarrow}} S \times S \rightrightarrows S \longrightarrow \{e\} \qquad (\{e\} \text{ a one-element set}).$$

Each of these augmented complexes is homotopically trivial. Thus $S_\bullet(\mathscr{U})$ is a homotopically trivial augmented complex, and its homology is therefore trivial, proving the proposition in this case.

Now let $p : \mathrm{Ens} \to E$ be a point of E. Since formation of the complex $\mathbf{Z}_\bullet(\mathscr{U})$ commutes with the inverse-image functors of morphisms of topoi, $p^*(\mathbf{Z}_\bullet(\mathscr{U})) \simeq \mathbf{Z}_\bullet(p^*(\mathscr{U}))$ is a resolution of $\mathbf{Z}_{p^*X} \simeq p^*(\mathbf{Z}_X)$; this proves the proposition when E has enough fiber functors (IV 4.6). This holds, in particular, when E is a topos of presheaves $C^\wedge$, since for every object X of C, $\Gamma(X, -)$ is a fiber functor. In the general case, E is equivalent to a topos of sheaves on a small site C (IV 1), and the epimorphic family $\mathscr{U} = (U_i \to X)_{i\in I}$ is the image, under the "associated sheaf" functor, of an epimorphic family $\mathscr{U}' = (U_i' \to X')_{i\in I}$. Consequently, $\mathbf{Z}_\bullet(\mathscr{U}) \simeq \underline{a}\mathbf{Z}_\bullet(\mathscr{U}')$ is a resolution of 18
$\underline{a}\mathbf{Z}_{X'} \simeq \mathbf{Z}_X$.

2. Čech cohomology. Cohomological notation

2.1. General notation.

2.1.1. Let (E, A) be a ringed topos, and let M, N be two A-Modules (left modules, to fix ideas). We denote by $\mathrm{Ext}^q_A(E; M, N)$ the value at N of the q-th right derived functor of the functor $\mathrm{Hom}_A(M, \cdot)$ [11]. The functors $\mathrm{Ext}^q_A(E; M, N)$, $q \in \mathbf{N}$, form a δ-functor in the variable N. They also form a contravariant δ-functor in the variable M. By definition, $\mathrm{Ext}^0_A(E; M, N) = \mathrm{Hom}_A(M, N)$.

2.1.2. Let X be an object of E. When $M = A_X$, the free A-Module generated by X (IV 12), we put $\mathrm{Ext}^q_A(E; A_X, N) = H^q(X, N)$. Notice that the ring A no longer appears in this new notation. This cannot cause confusion, since we will show (3.5) that the formation of $H^q(X, \cdot)$ commutes with restriction of scalars. The functor $H^q(X, \cdot)$ is the q-th right derived functor of $\mathrm{Hom}_A(A_X, \cdot) = \mathrm{Hom}_E(X, \cdot)$, also denoted by $\Gamma(X, \cdot)$. When $M = A$, we put $\mathrm{Ext}^q_A(E; A, N) = H^q(E, N)$.

2.2. Localization. Let X be an object of E, and let $j : E_{/X} \to E$ be the localization morphism (IV 8). The functor j^*_X for Modules is exact (4.11) and admits an exact left adjoint $j_{X!}$ (4.11). Consequently, it sends injective Modules to injective Modules. Thus, for a variable A-Module N of E and a variable $A|X$-Module M of $E_{/X}$, there are functorial isomorphisms

$$\mathrm{Ext}^q_{A|X}(E_{/X}; M, j^*_X N) \simeq \mathrm{Ext}^q_A(E; j_{X!}M, N). \tag{2.2.1}$$

In particular, there are canonical isomorphisms

$$H^q(E_{/X}, j^*_X N) \simeq H^q(X, N).$$

For every object X of E and every pair M, N of A-Modules, we put

$$\mathrm{Ext}^q_A(X; M, N) \simeq \mathrm{Ext}^q_{A|X}(E_{/X}; M|X, N|X). \tag{2.2.2}$$

By what precedes, the functors $\mathrm{Ext}^q_A(X; M, \cdot)$ are the derived functors of the functors 19
$N \mapsto \mathrm{Hom}_{A|X}(M|X, N|X)$. The functors $(M, N) \mapsto \mathrm{Ext}^q_A(X; M, N)$, $q > 0$, form a δ-functor with respect to each variable.

2.3. The case of presheaf topoi. Cohomology of a covering.

2.3.1. Let C be a small category endowed with a presheaf of rings A, let $C^\wedge$ be the topos of presheaves on C, and let X be a representable object of $C^\wedge$. The functor which associates to an A-Module M the group $\Gamma(X, M) = M(X)$ is exact (I 3). Consequently, $H^q(X, M) = 0$ for every $q > 0$ and every A-Module M, or, equivalently, A_X is a projective A-Module.

2.3.2. Let S be a presheaf on C. For every A-Module M, there is a canonical isomorphism (I 2):

$$\Gamma(S, M) \simeq \varprojlim_{C/S} M|S$$

Moreover, for every injective A-Module M, the $A|S$-Module $M|S$ is injective (2.2). Consequently, the groups $H^q(S, M)$ are the values at $M|S$ of the right derived functors of $\varprojlim_{C/S}$. Denoting these derived functors by $\varprojlim^q_{C/S}$, there are canonical isomorphisms

$$H^q(S, M) \simeq \varprojlim_{C/S}{}^q M|S.$$

In particular, there are canonical isomorphisms

$$H^q(C^\wedge, M) \simeq \varprojlim_{C}{}^q M.$$

2.3.3. Let X be an object of C, and let $\mathscr{U} = (U_i \to X)$, $i \in I$, be a family of morphisms of C such that, for every $i \in I$, $U_i \to X$ is pullbackable (I 10). Denote by $A_\bullet$ the simplicial complex studied in 1.10:

$$A_\bullet = \cdots \mathrel{\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow}} \coprod_{(i,j)\in I\times I} A_{U_i \times_X U_j} \rightrightarrows \coprod_{i\in I} A_{U_i}.$$

For every A-Module M, put $C^\bullet(\mathscr{U}, M) = \mathrm{Hom}_A(A_\bullet, M)$:

$$C^\bullet(\mathscr{U}, M) : \prod_{i\in I} M(U_i) \rightrightarrows \prod_{(i,j)\in I\times I} M(U_i \times_X U_j) \mathrel{\substack{\longrightarrow\\ \longrightarrow\\ \longrightarrow}} \cdots. \tag{2.3.3.1}$$

Put $H^q(\mathscr{U}, M) = H^q(C^\bullet(\mathscr{U}, M))$.

PROPOSITION 2.3.4.

1) *With the notation of 2.3.3, let $R \hookrightarrow X$ be the sieve generated by the family $(U_i \to X)$, $i \in I$. There is a canonical isomorphism*

$$H^q(\mathscr{U}, M) \simeq H^q(R, M).$$

2) *The functors $H^q(\mathscr{U}, \cdot)$ commute with restrictions of scalars.*

Since R is a subobject of X in $C^\wedge$, the fiber products $U_{i_1} \times_R U_{i_2} \times_R \cdots \times_R U_{i_n}$ and $U_{i_1} \times_X U_{i_2} \times_X \cdots \times_X U_{i_n}$ are canonically isomorphic. It follows from 1.11 that the complex $A_{\mathscr{U}\bullet}$ is a resolution of A_R, and from 2.3.1 that the components of $A_{\mathscr{U}\bullet}$ are projective Modules. The cohomology groups of $C^\bullet(\mathscr{U}, M)$ are therefore canonically isomorphic to $\mathrm{Ext}^q_A(C^\wedge; A_R, M)$, whence the isomorphism. Assertion 2) follows immediately from the description of $C^\bullet(\mathscr{U}, M)$ (2.3.3.1).

COROLLARY 2.3.5. *Let $\mathscr{U} = (U_i \to X, i \in I)$ and $\mathscr{U}' = (U'_j \to X, j \in J)$ be two families of morphisms with target X; let $\phi = (\phi : I \to J, m_i : U_i \to U'_{\phi(i)})$ and $\phi' = (\phi' : I \to J, m'_i : U_i \to U'_{\phi(i)})$ be two morphisms (over X) from $\mathscr{U}$ to $\mathscr{U}'$. The morphisms ϕ and ϕ' induce the morphisms ϕ^q and $\phi'^q : H^q(\mathscr{U}, M) \to H^q(\mathscr{U}', M)$. The* *morphisms ϕ^q and ϕ'^q are equal. In particular, if the families $\mathcal{U}$ and $\mathcal{U}'$ are equivalent (i.e. if there is a morphism from $\mathcal{U}$ to $\mathcal{U}'$ and a morphism from $\mathcal{U}'$ to $\mathcal{U}$), the A-Modules $H^q(\mathcal{U}, M)$ and $H^q(\mathcal{U}', M)$ are canonically isomorphic.*

Proof. Indeed, let R and R' be the sieves of X generated by the families $\mathcal{U}$ and $\mathcal{U}'$. The morphisms ϕ and ϕ' define the same morphism from R to R', and therefore induce two homotopic morphisms between the projective resolutions $A_{\mathcal{U}\bullet}$ and $A_{\mathcal{U}'\bullet}$.

Exercise 2.3.6. (Standard resolution)

a) Let C be a small category. For every integer $n > 0$, denote by $\mathrm{Fl}^n(C)$ the set of sequences of morphisms of C : $(u_1, \dots, u_n)$ such that, for every i, $0 < i < n$, the target of u_i equals the source of u_{i+1}, so that the morphisms u_i and u_{i+1} are composable. Define a semisimplicial set 21

$$ES(C) = \mathrm{ob}\ C \ \leftleftarrows\ \mathrm{Fl}^1(C)\ \overset{\longleftarrow}{\leftleftarrows}\ \mathrm{Fl}^2(C)\ \overset{\leftleftarrows}{\leftleftarrows}\ \mathrm{Fl}^3(C) \quad \cdots,$$

whose face operators $s_i : \mathrm{Fl}^n(C) \to \mathrm{Fl}^{n-1}(C)$ are as follows:

$$\begin{aligned} s_0(u_1, \dots, u_n) &= (u_2, \dots, u_n), \\ s_i(u_1, \dots, u_n) &= (u_1, \dots, u_{i+1}u_i, \dots, u_n), \quad 0 < i < n, \\ s_n(u_1, \dots, u_n) &= (u_1, \dots, u_{n-1}). \end{aligned}$$

b) Show that when C has an initial object or a final object, the complex $ES(C)$ is homotopically trivial.

c) For every object X of C, denote by ${}_{X\backslash}C$ the category of arrows with source X. Define a semisimplicial presheaf $ES(C)$ whose value at every object X of C is $ES({}_{X\backslash}C)$. Denote by $\mathbf{Z}_{ES(C)}$ the free abelian semisimplicial presheaf generated by $ES(C)$. It has a canonical augmentation $\mathbf{Z}_{ES(C)} \to \mathbf{Z}$ to the constant presheaf with value $\mathbf{Z}$. Show that, upon passing to the associated differential complex, the complex $\mathbf{Z}_{ES(C)}$ is a resolution of $\mathbf{Z}$ and that its components are projective abelian presheaves.

d) For every abelian presheaf M, put $ST^\bullet(M) = \mathrm{Hom}(\mathbf{Z}_{ES(C)}, M)$. Describe the components of $ST^\bullet(M)$ explicitly. Show that, for every integer $q \geq 0$, $H^q(ST^\bullet(M)) = H^q(C^\wedge, M)$.

e) Show that, for every sheaf S on C, the functors $H^q(S, \cdot)$ commute with restrictions of scalars.

f) Observe that when C is a group, $\mathbf{Z}_{ES(C)}$ is the standard resolution of the trivial module $\mathbf{Z}$, [3].

g) Show that, for every abelian presheaf M on C, $\mathbf{Z}_{ES(C)} \otimes M$ is a (left) resolution of M. Define the functor $\varinjlim_C$ on presheaves. Show that it is right exact. Denote its left derived functors by $H_q(C, M)$. Show that the components of the complex $\mathbf{Z}_{ES(C)} \otimes M$ are acyclic for $\varinjlim_C$. Deduce, denoting by $ST_\bullet(M)$ the complex $\varinjlim_C(\mathbf{Z}_{ES(C)} \otimes M)$, isomorphisms $H_q(ST_\bullet(M)) \simeq H_q(C, M)$.

h) Let $u : C^\wedge \to \mathrm{Ens}$ be the unique morphism from the topos $C^\wedge$ to the punctual topos. Denote by $u_! : C^\wedge_{\mathbf{Z}} \to \mathrm{Ab}$ the left adjoint of the functor $u^* : \mathrm{Ab} \to C^\wedge_{\mathbf{Z}}$. Show that for every abelian presheaf M, $u_! M = \varinjlim_C M$. 22

i) Henceforth denote by $ST^\bullet(C, M)$ and $ST_\bullet(C, M)$ the complexes denoted above by $ST^\bullet(M)$ and $ST_\bullet(M)$. A presheaf M on C is called locally constant if it sends all morphisms of C to isomorphisms. Associate to every locally constant presheaf M a locally constant presheaf $\check{M}$ on the category C° (the category opposite to C) such that, for every morphism u of C, $\check{M}(u) = M(u)^{-1}$. Find a canonical isomorphism between the complexes $ST^{\bullet}(C, M)$ and $ST^{\bullet}(C^{\circ}, \check{M})$ (resp. $ST_{\bullet}(C, M)$ and $ST_{\bullet}(C^{\circ}, \check{M})$).

j) Let C be a small category having an initial object (resp. a small filtered category). Show that, for every constant presheaf M, $H^q(C^{\wedge}, M) = 0$ for $q > 0$ (resp. $H_q(C, M) = 0$ for $q > 0$).

2.4. The case of small sites. Čech cohomology.

2.4.1. Let (C, A) be a ringed $\mathscr{U}$-site, let $C^{\sim}$ be the topos of sheaves on C, and let $\epsilon : C \to C^{\sim}$ be the canonical functor which associates to an object of C the sheaf associated to the presheaf represented by that object. By abuse of notation, for every object X of C and every sheaf of A-modules M, we put $H^q(X, M) = H^q(\epsilon(X), M)$ (cf. 2.3.1). Recall that when the topology of C is coarser than the canonical topology, as is always the case in practice, the functor ϵ is fully faithful and allows us to identify C with its image under ϵ.

2.4.2. Denote by $\mathscr{H}^0 : C^{\sim}_A \to C^{\wedge}_A$ the inclusion functor from sheaves of A-modules to the category of presheaves of A-modules. Thus, for every sheaf of A-modules M, one has by definition

$$\mathscr{H}^0(M)(X) = H^0(X, M) = M(X), \tag{2.4.2.1}$$

for every object X of C. The functor $\mathscr{H}^0$ is left exact. Its right derived functors are denoted by $\mathscr{H}^q$. Since, for every object X of C, the "sections over X" functor is exact in the category of presheaves, one has

$$\mathscr{H}^q(M)(X) = H^q(X, M), \tag{2.4.2.2}$$

23 for every object X of C and every sheaf of A-modules M, so that the presheaf $\mathscr{H}^q(M)$ is simply the presheaf $X \mapsto H^q(X, M)$.

2.4.3. Suppose that (C, A) is a ringed *small site*, so that $C^{\wedge}$ is a *topos* to which the results of 2.3 may be applied. Let X be an object of C, and let $R \hookrightarrow X$ be a covering sieve. For every *presheaf* of A-modules G, the groups $H^q(R, G)$ (which are thus computed in the topos $C^{\wedge}$) are called the *Čech cohomology groups of the presheaf G relative to the covering sieve R*. When $R \hookrightarrow X$ is the sieve generated by a covering family $\mathscr{U} = (U_i \to X)_{i \in I}$ of pullbackable morphisms, these groups can be computed by means of the complex $C^{\bullet}(\mathscr{U}, G)$ (2.3.3), called the *Čech complex of G relative to the covering family $\mathscr{U}$* (or relative to the covering $\mathscr{U}$). The groups $H^q(\mathscr{U}, G) = H^q(R, G)$ (2.3.4) are then called the *Čech cohomology groups of G relative to the covering family $\mathscr{U}$*.

2.4.4. Let M be a sheaf of A-modules on C. The groups $H^q(\mathscr{U}, \mathscr{H}^0(M))$ are most often denoted, by abuse, by $H^q(\mathscr{U}, M)$ and called the Čech cohomology groups of the sheaf M relative to the covering family $\mathscr{U}$.

2.4.5. Denote by $\check{\mathscr{H}}^0 : C^{\wedge}_A \to C^{\wedge}_A$ the natural extension to presheaves of A-modules of the functor L described in II. Thus, by definition, for a presheaf G and an object X of C,

$$\check{\mathscr{H}}^0(G)(X) = \varinjlim_{R \hookrightarrow X} G(R), \tag{2.4.5.1}$$

where the inductive limit is taken over the covering sieves of X. It follows from (2.4.5.1) that the functor $\check{\mathscr{H}}^0$ is left exact. The right derived functors of $\check{\mathscr{H}}^0$ are denoted by $\check{\mathscr{H}}^q$. Since the "sections over X" and "filtered inductive limit" functors are exact, it follows from (2.4.5.1) that

$$\check{\mathscr{H}}^q(G)(X) = \varinjlim_{R \hookrightarrow X} H^q(R, G), \tag{2.4.5.2}$$

where the limit is taken over the covering sieves of X.

The presheaves $\check{\mathscr{H}}^q$ are called the *Čech cohomology presheaves*. For every object X 24
of C, put

$$\check{H}^q(X, G) = \check{\mathscr{H}}^q(G)(X). \tag{2.4.5.3}$$

The groups $\check{H}^q(X, G)$ are called the *Čech cohomology groups*. When the topology of C is defined by a pretopology, as is most often the case in practice, it follows from 2.3.4 that

$$\check{H}^q(X, G) \simeq \varinjlim_{\mathscr{U}} H^q(\mathscr{U}, G), \tag{2.4.5.4}$$

where the inductive limit is taken over the pullbackable covering families preordered by the natural order relation on their corresponding sieves (2.3.5).

2.4.6. Let M be a sheaf of A-modules. By abuse of notation, put

$$\check{H}^q(X, M) = \check{H}^q(X, \mathscr{H}^0(M)), \qquad \check{\mathscr{H}}^q(M) = \check{\mathscr{H}}^q(\mathscr{H}^0(M)), \tag{2.4.6.1}$$

and the groups $\check{H}^q(X, M)$ are called the Čech cohomology groups of the sheaf M. Note that, although the functors $\check{H}^q$ are derived functors on the category of presheaves, they do not in general form a δ-functor on the category of sheaves.

2.5. Change of universe. Čech cohomology for $\mathscr{U}$-sites.

2.5.1. Let (C, A) be a ringed $\mathscr{U}$-site, and let $\mathscr{V}$ be a universe containing $\mathscr{U}$. The site (C, A) is then a $\mathscr{V}$-site, and there are a $\mathscr{U}$-topos $C^{\sim}_{\mathscr{U}}$, a $\mathscr{V}$-topos $C^{\sim}_{\mathscr{V}}$, and a canonical inclusion functor $\epsilon : C^{\sim}_{\mathscr{U}} \hookrightarrow C^{\sim}_{\mathscr{V}}$. The functor ϵ is exact and fully faithful on the categories of Modules, and sends injective Modules to injective Modules (1.9). Thus, for every pair of $\mathscr{U}$-sheaves of A-modules, there are canonical isomorphisms

$$\operatorname{Ext}^q_A(C^{\sim}_{\mathscr{U}}; M, N) \simeq \operatorname{Ext}^q_{\epsilon A}(C^{\sim}_{\mathscr{V}}; \epsilon M, \epsilon N), \quad q \geq 0. \tag{2.5.1.1}$$

In particular, for every $\mathscr{U}$-sheaf of sets R on C, there are canonical isomorphisms (2.1.1)

$$H^q(R, M) \simeq H^q(\epsilon R, \epsilon M), \quad q \geq 0, \tag{2.5.1.2}$$

and, even more particularly, for every object X of C, there are canonical isomorphisms 25
(2.4.1)

$$H^q(X, M) \simeq H^q(X, \epsilon M). \tag{2.5.1.3}$$

Loosely speaking, we may therefore say that sheaf cohomology does not depend on the choice of universes, and, when required for a proof or a construction, one may always enlarge the universe in order to compute the cohomology of a sheaf.

2.5.2. Let (C, A) be a ringed $\mathscr{U}$-site, and let $\mathscr{V}$ be a universe containing $\mathscr{U}$. Denote by $C^{\sim}_A$ the category of $\mathscr{U}$-sheaves of modules, and let $\hat{\epsilon} : C^{\wedge}_{A,\mathscr{U}} \to C^{\wedge}_{A,\mathscr{V}}$ be the inclusion functor from $\mathscr{U}$-presheaves to $\mathscr{V}$-presheaves of A-modules. The functor $\hat{\epsilon}$ is exact, and therefore the derived functors of the functor $\hat{\epsilon}\mathscr{H}^0 : C^{\sim}_A \to C^{\wedge}_{A,\mathscr{V}}($ $\quad$ $)$ are the functors $\hat{\epsilon}\mathscr{H}^q$, $q \geq 0$. By abuse of notation, we will again denote by $\mathscr{H}^q : C^{\sim}_A \to C^{\wedge}_{A,\mathscr{V}}$, $q \geq 0$, the functors $\hat{\epsilon}\mathscr{H}^q$. When C is $\mathscr{V}$-small, this enlargement of the universe has the following advantage: the category $C^{\wedge}_{\mathscr{U}}$ is not in general a $\mathscr{U}$-topos, and $\mathscr{U}$-presheaves of A-modules are not necessarily submodules of injective $\mathscr{U}$-presheaves, whereas the category of $\mathscr{V}$-presheaves is a topos (a $\mathscr{V}$-topos), and consequently every $\mathscr{V}$-presheaf of

A-modules is a subobject of an injective $\mathscr{V}$-presheaf (0.1.1). Thus, for every $\mathscr{V}$-presheaf of sets R (and in particular when R is a $\mathscr{U}$-presheaf) and every $\mathscr{U}$-sheaf of A-modules M, the groups $H^q(R, \mathscr{H}^q(M))$ are defined by 2.1.1, and it follows from (2.5.1.2) that these groups do not depend on the universe $\mathscr{V}$ under consideration. Likewise, for every pair of integers ≥ 0, denoted by p and q, the presheaves $\check{\mathscr{H}}^p(\mathscr{H}^q(M))$ are defined by 2.4.5, and it follows from (2.5.1.2) and (2.4.5.4) that these presheaves do not depend on the universe $\mathscr{V}$ used to define them.

3. The Cartan–Leray spectral sequence associated with a covering

26 Proposition 3.1. *Let (C, A) be a ringed $\mathscr{U}$-site, and let $\mathscr{V}$ be a universe containing $\mathscr{U}$. The functor $\mathscr{H}^0 : C_A^{\sim} \to C_{A,\mathscr{V}}^{\wedge}$ (2.5.2) sends injective A-modules to injective presheaves. For every integer $q > 0$ and every A-module M, the sheaf associated with the presheaf $\mathscr{H}^q(M)$ is zero.*

Denote by $\underline{a}_{\mathscr{V}}$ the "associated sheaf" functor for $\mathscr{V}$-presheaves, and by $\epsilon : C_A^{\sim} \hookrightarrow C_{A,\mathscr{V}}^{\sim}$ the inclusion functor from $\mathscr{U}$-sheaves to $\mathscr{V}$-sheaves. One has $\underline{a}_{\mathscr{V}}\mathscr{H}^0 = \epsilon$ (II 3.6), and, since the functors $\underline{a}_{\mathscr{V}}$ and ϵ are exact, one has $\underline{a}_{\mathscr{V}}\mathscr{H}^q = 0$ for every integer $q > 0$. For every $\mathscr{U}$-sheaf M and every $\mathscr{V}$-presheaf N, there is a functorial isomorphism $\mathrm{Hom}_{C_{A,\mathscr{V}}^{\wedge}}(N, \mathscr{H}^0(M)) \simeq \mathrm{Hom}_{C_{A,\mathscr{V}}^{\sim}}(\underline{a}_{\mathscr{V}}N, \epsilon M)$. When M is injective, ϵM is injective (1.9), and, since the functor $\underline{a}_{\mathscr{V}}$ is exact, the functor $N \longmapsto \mathrm{Hom}_{C_{A,\mathscr{V}}^{\wedge}}(N, \mathscr{H}^0(M))$ is exact. Consequently, $\mathscr{H}^0(M)$ is injective.

Theorem 3.2. *Let (C, A) be a ringed $\mathscr{U}$-site, let R be a $\mathscr{U}$-presheaf of sets on C, and let M be a sheaf of A-modules. There is a spectral sequence, functorial in R and M:*

$$H^{p+q}(\underline{a}R, M) \Longleftarrow E_2^{p,q} = H^p(R, \mathscr{H}^q(M)). \tag{3.2.1}$$

(When C is not $\mathscr{U}$-small, the term $H^p(R, \mathscr{H}^q(M))$ must be interpreted as the cohomology of the presheaf $\mathscr{H}^q(M)$ in the topos $C_{\mathscr{V}}^{\wedge}$, where $\mathscr{V}$ is a universe containing $\mathscr{U}$ such that C is $\mathscr{V}$-small (2.5.2).)

By the definition of the "associated sheaf" functor (cf. II), there is an isomorphism of functors $H^0(\underline{a}R, M) \simeq H^0(R, \mathscr{H}^0(M))$. The functor $\mathscr{H}^0$ sends injective objects to injective objects. The spectral sequence of composite functors (0.3) is the desired spectral sequence.

Corollary 3.3. *Let X be an object of C, and let $\mathscr{U} = (U_i \to X)$, $i \in I$, be a covering family such that, for every $i \in I$, the morphism $U_i \to X$ is pullbackable. Then there is a spectral sequence (called the Cartan–Leray spectral sequence):*

$$H^{p+q}(X, M) \Longleftarrow E_2^{p,q} = H^p(\mathscr{U}, \mathscr{H}^q(M)). \tag{3.3.1}$$

27 Let $R \hookrightarrow X$ be the sieve generated by $\mathscr{U}$. Since this sieve is covering, the sheaf associated with R is the sheaf associated with X (II 5.2). Taking 2.4.1 into account, one has $H^{p+q}(\underline{a}R, M) = H^{p+q}(X, M)$. The corollary then follows from 2.3.4.

Corollary 3.4. *There is a spectral sequence, functorial in the sheaf M and in the object X of C,*

$$H^{p+q}(X, M) \Longleftarrow E_2^{p,q} = \check{H}^p(X, \mathscr{H}^q(M)). \tag{3.4.1}$$

As X varies in $\mathbf{C}$, this spectral sequence gives the spectral sequence of presheaves

$$\mathscr{H}^{p+q}(M) \Longleftarrow E_2^{p,q} = \check{\mathscr{H}}^p(\mathscr{H}^q(M)). \tag{3.4.2}$$

These spectral sequences give functorial morphisms (edge morphisms):

$$\Phi^q(M) : \check{\mathscr{H}}^q(M) \longrightarrow \mathscr{H}^q(M), \tag{3.4.3}$$

$$\Phi^q_X(M) : \check{H}^q(X, M) \longrightarrow H^q(X, M). \tag{3.4.4}$$

The morphisms Φ^q and Φ^q_X are isomorphisms when q is 0 or 1. The morphisms Φ^2 and Φ^2_X are monomorphisms. More generally, when the presheaves $\mathscr{H}^i(M)$ vanish for $0 < i < n$, the morphisms $\Phi^q(M)$ and $\Phi^q_X(M)$ are isomorphisms for $0 \leq q \leq n$ and monomorphisms for $q = n + 1$.

The first spectral sequence is obtained by taking the inductive limit in the spectral sequence (3.2.1) over the covering sieves $R \hookrightarrow X$ of X (2.4.5.2). The second spectral sequence follows immediately from it (2.4.5.3). The sheaves associated with the presheaves $\mathscr{H}^q(M)$ are zero when $q > 0$ (3.1). Thus $\check{\mathscr{H}}^0\check{\mathscr{H}}^0\check{\mathscr{H}}^q(M) = 0$ for every $q > 0$ (II.3.4). Consequently, $\check{\mathscr{H}}^0\check{\mathscr{H}}^q(M) = 0$ for $q > 0$ (II 3.2). The assertions concerning the morphisms Φ^q and Φ^q_X follow immediately.

Corollary 3.5. *Let (E, A) be a ringed topos, and let M be an A-module (a left module, to fix ideas). Denote by $\mathscr{M}$ the underlying abelian sheaf of M. The functor $M \longmapsto \mathscr{M}$ is exact; consequently, for every object X of E, the canonical isomorphism*

$$H^0(X, M) \xrightarrow{\sim} H^0(X, \mathscr{M})$$

extends to morphisms 28

$$H^q(X, M) \longrightarrow H^q(X, \mathscr{M}) \qquad q \geq 0.$$

These morphisms are isomorphisms.

For every object Y of E, one has (2.4.5.4):

$$H^p(Y, M) = \varinjlim_{\mathscr{U}} H^p(\mathscr{U}, M) \quad , \quad H^p(Y, \mathscr{M}) = \varinjlim_{\mathscr{U}} H^p(\mathscr{U}, \mathscr{M}),$$

where the inductive limit is taken over the epimorphic families $\mathscr{U} = (Y_i \to Y)$, $i \in I$.

Consequently, the canonical homomorphism $\check{H}^p(Y, M) \to \check{H}^p(Y, \mathscr{M})$ is an isomorphism (2.3.4). Hence, by (2.4.5.3), the canonical homomorphism $\check{\mathscr{H}}^p(M) \to \check{\mathscr{H}}^p(\mathscr{M})$ is an isomorphism. If M is an injective A-module, then $\check{\mathscr{H}}^p(\mathscr{M}) = 0$ for $p > 0$; hence $\mathscr{H}^1(\mathscr{M}) = 0$ and, by induction on p, $\mathscr{H}^p(\mathscr{M}) = 0$ for $p > 0$ (3.4). Thus $H^p(X, \mathscr{M}) = 0$ for $p > 0$ (2.4.2.2) and 2.5. Consequently, the functor $M \longmapsto \mathscr{M}$ sends injective objects to objects acyclic for the functor $H^0(X, \cdot)$, which gives the asserted isomorphism.

Exercise 3.6. Let G be a topological group and let B_G be its classifying topos (IV 2.5). Denote by E_G the object of B_G consisting of the underlying topological space of G, endowed with the action of G by left translations. The canonical morphism from E_G to the final object e_G of B_G is an epimorphism; hence there is a covering $\mathscr{U} = (E_G \to e_G)$ and, for every abelian sheaf F of B_G, a spectral sequence

$$E_2^{p,q} = H^p(\mathscr{U}, \mathscr{H}^q(F)) \Longrightarrow H^{p+q}(B_G, F), \tag{3.6.1}$$

which we propose to study.

1) Show that, for every integer n, the topos

$$B_{G/E_G \times E_G \times \cdots \times E_G} \quad (n \quad \text{factors})$$

is canonically equivalent to the topos (IV 2.5)

$$\mathrm{TOP}(G \times G \times G \times \cdots \times G) \quad (n-1 \quad \text{factors}).$$

29 For every integer n, denote by F_n the sheaf on the topological space $G \times \cdots \times G$ (n factors) induced by F.

2) Deduce that the term $E_2^{p,q}$ of (3.6.1) is canonically isomorphic to the p-th cohomology group of a complex of the form

$$H^q(e, F_0) \longrightarrow H^q(G, F_1) \longrightarrow H^q(G \times G, F_2) \longrightarrow \cdots,$$

which is to be made explicit (e denotes the one-point topological space).

3) Deduce that the cohomology of the classifying topos B_G with values in locally constant sheaves is isomorphic to the corresponding singular cohomology of a classifying space, provided that, on the spaces of the form $G \times G \times \cdots \times G$, cohomology with coefficients in locally constant sheaves coincides with the corresponding singular cohomology (as is the case, for example, when G is locally contractible). (For classifying spaces, one may consult [15].)

4. Acyclic sheaves

Definition 4.1. Let (E, A) be a ringed topos, let F be an A-Module, and let S be a topologically generating family of E. The sheaf F is said to be S-acyclic if, for every object X of S and every integer $q > 0$, one has $H^q(X, F) = 0$. When S is equal to ob E, the S-acyclic sheaves are called *flasque sheaves.*

4.2. Let (C, A) be a $\mathscr{U}$-ringed site, and let F be a sheaf of A-modules on C. The sheaf F is said to be C-*acyclic* if it is S-acyclic, where S is the family of sheaves associated with the objects of C.

Proposition 4.3. *Let (C, A) be a $\mathscr{U}$-ringed site, and let F be a sheaf of A-modules. Denote by $\mathscr{H}^0 : C_A^{\sim} \to C_A^{\wedge}$ the inclusion functor from the A-Modules into the category of presheaves of A-modules (here these are $\mathscr{V}$-presheaves, where $\mathscr{V}$ is a universe such that C is $\mathscr{V}$-small). The following conditions are equivalent:*

30 i) *F is C-acyclic.*

ii) *For every object X of C and every covering sieve $R \hookrightarrow X$, one has*

$$H^q(R, \mathscr{H}^0(F)) = 0 \quad \textit{for every } q > 0.$$

iii) *For every object X of C, one has*

$$\check{H}^q(X, F) = 0 \quad \textit{for every } q > 0.$$

i ⇒ ii) : Since F is C-acyclic, the presheaves $\mathscr{H}^q(F)$ vanish for $q > 0$. The spectral sequence (3.2.1) then gives an isomorphism $H^q(R, \mathscr{H}^0(F)) \simeq H^q(X, F)$ for every q. Consequently, $H^q(R, \mathscr{H}^0(F)) = 0$ for $q > 0$.

ii ⇒ iii) : This is clear upon passing to the inductive limit.

iii ⇒ i) : By induction on q, one proves that the presheaves $\mathscr{H}^q(F)$ vanish for $q > 0$. For this, use (3.4.2).

4.4. It follows from criterion 4.3 ii) and 3.5 that the property of being S-acyclic depends only on the underlying abelian sheaf. In particular, a sheaf of A-modules is flasque if and only if its underlying abelian sheaf is flasque.

Corollary 4.5. *Let (E, A) be a ringed topos, and let F be an A-Module. The following properties are equivalent:*

i) *F is flasque;*

ii) *For every epimorphic family $\mathcal{X} = (X_i \to X)_{i \in I}$,*

$$H^q(\mathcal{X}, F) = 0 \quad \textit{for every } q > 0.$$

In 4.2, take C to be the topos E itself. The corollary then follows from the equivalence i) ⇔ ii) of 4.3.

4.6. Injective sheaves are flasque. Flasque sheaves are S-acyclic for every topologically generating family S. Flasque sheaves are not necessarily injective (take for E the topos of sets). S-acyclic sheaves are not necessarily flasque (Exercise 4.1.3).

Proposition 4.7. *Let (E, A) be a ringed topos, let F be a flasque A-Module, and let X be an object of E. For every subobject Y of X, the canonical homomorphism $H^0(X, F) \to H^0(Y, F)$ is surjective.* 31

Suppose that Y is a subobject of X such that the morphism $H^0(X, F) \to H^0(Y, F)$ is not surjective. Let Z be the object obtained by gluing two copies X_1 and X_2 of X along Y. The object Z is covered by the two subobjects X_1 and X_2, and one has $X_1 \times_Z X_2 = Y$. Denote the resulting covering by $\mathcal{X} = (X_1, X_2)$. One sees immediately that $H^1(\mathcal{X}, F) \neq 0$, a contradiction.

4.8. For general topoi, the property described in 4.7 does not characterize flasque sheaves (Exercise 4.15). It does, however, characterize them for topoi generated by their open subobjects, and in particular for topoi associated with topological spaces (Exercise 4.16). In the case of topological spaces, the terminology "flasque" adopted here agrees with that of [7] (Exercise 4.16).

Proposition 4.9. *Let $u : (E, A) \to (E', A')$ be a morphism of ringed topoi.*

1) *The functor u_* carries flasque A-Modules to flasque A'-Modules.*
2) *Let S and S' be topologically generating families of E and E', respectively, such that $u^*(S') \subset S$. The functor u_* carries S-acyclic A-Modules to S'-acyclic A'-Modules.*
3) *When u is a flat morphism (1.8), the functor u_* carries injective A-Modules to injective A'-Modules.*

Let X be an object of E', let $\mathcal{X} = (X_i \to X)_{i \in I}$ be an epimorphic family, let F be a flasque A-Module, and let $C^\bullet(\mathcal{X}, u_* F)$ be the Čech complex of the covering $\mathcal{X}$. Using the adjunction between u_* and u^*, and the fact that u^* commutes with fiber products, one obtains a canonical isomorphism

$$C^\bullet(\mathcal{X}, u_* F) \simeq C^\bullet(u^*(\mathcal{X}), F).$$

Moreover, u^* commutes with inductive limits, and hence $u^*(\mathcal{X}) = (u^* X_i \to u^* X)_{i \in I}$ is an epimorphic family. Since F is flasque, $H^q(u^*(\mathcal{X}), F) = 0$ for $q > 0$, and consequently

$$H^q(\mathcal{X}, u_* F) = H^q(C^\bullet(\mathcal{X}, u_* F)) = H^q(C^\bullet(u^*(\mathcal{X}), F)) = H^q(u^*(\mathcal{X}), F) = 0$$

for $q > 0$. Thus $u_* F$ is flasque.

Let us prove 2). When S' is stable under fiber products, an argument analogous to the preceding one proves 2). In the general case, we use the spectral sequence of
32 the morphism u (5.3.2). Assertion 2) will not be used before (5.3.2). Let F be an S-acyclic sheaf. The sheaves $R^q u_* F$ are the sheaves associated with the presheaves $X \mapsto H^q(u^* X, F)$. Since S' is a topologically generating family and F is S-acyclic, one has $R^q u_* F = 0$ for every $q > 0$. The spectral sequence (5.3.2) then gives, for every object X of E', an isomorphism $H^q(X, u_* F) \simeq H^q(u^* X, F)$; consequently, for every X in S', $H^q(X, u_* F) = 0$.

Let us prove 3). When u is flat, the inverse-image functor u^* for Modules is exact (1.8). Therefore its right adjoint u_* carries injective objects to injective objects (0.2).

Proposition 4.10. *Let F be an A-Module of the ringed topos (E, A), and let G be an injective A-Module.*

1) *The functor $F \mapsto \mathbf{Hom}_A(F, G)$ is exact.*
2) *The abelian sheaf $\mathbf{Hom}_A(F, G)$ is flasque.*

Proof. Let us prove the assertions in turn.

1) Let

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$

be an exact sequence. We must show that the sequence

$$0 \longrightarrow \mathbf{Hom}_A(F'', G) \longrightarrow \mathbf{Hom}_A(F, G) \longrightarrow \mathbf{Hom}_A(F', G) \longrightarrow 0$$

is exact; for this it suffices to show that, for every object H of E, the sequence

$$0 \longrightarrow \mathrm{Hom}_E(H, \mathbf{Hom}_A(F'', G)) \longrightarrow \mathrm{Hom}_E(H, \mathbf{Hom}_A(F, G)) \\ \longrightarrow \mathrm{Hom}_E(H, \mathbf{Hom}_A(F', G)) \longrightarrow 0$$

is exact. By (IV 6.12), the latter sequence is isomorphic to

$$0 \longrightarrow \mathrm{Hom}_A(A_H \otimes_A F'', G) \longrightarrow \mathrm{Hom}_A(A_H \otimes_A F, G) \longrightarrow \mathrm{Hom}_A(A_H \otimes_A F', G) \longrightarrow 0,$$

and the A-Module A_H is flat (1.3.1).

2) Let us show that $\mathbf{Hom}_A(F, G)$ is flasque. Let

$$\mathscr{X} = (X_i \longrightarrow X)_{i \in I}$$

be an epimorphic family of E, and let

$$C_\bullet(\mathscr{X}) : \cdots \quad \substack{\longrightarrow\\\longrightarrow\\\longrightarrow} \quad \coprod_{i,j} \mathbf{Z}_{X_i \times_X X_j} \quad \substack{\longrightarrow\\\longrightarrow} \quad \coprod_i \mathbf{Z}_{X_i}$$

33 be the complex defined in 1.5. This complex is a flat resolution of the object $\mathbf{Z}_X$, which is itself flat (1.4 and 1.3).

We then compute the groups:

$$H^q(\mathscr{X}, \mathbf{Hom}_A(F, G)) \xrightarrow{\sim} H^q\big(\mathrm{Hom}^\bullet_{\mathbf{Z}}(C_\bullet(\mathscr{X}), \mathbf{Hom}_A(F, G))\big) \xrightarrow{\sim} H^q\big(\mathrm{Hom}_A(C_\bullet(\mathscr{X}) \otimes_{\mathbf{Z}} F, G)\big).$$

The complex

$$C_\bullet(\mathscr{X}) \otimes_{\mathbf{Z}} F$$

is acyclic in degrees different from 0, because $C_\bullet(\mathscr{X})$ is a flat resolution of a flat module. Consequently, $\mathrm{Hom}_A(C_\bullet(\mathscr{X}) \otimes_{\mathbf{Z}} F, G)$ is acyclic in degrees different from 0.

Proposition 4.11. *Let (E, A) be a ringed topos, and let F be a flasque (resp. injective) sheaf of A-modules.*

1) *For every object X of E, the $A|X$-Module $j_X^* F$ is flasque (resp. injective).*
2) *For every closed subtopos Z of E, the sheaf of sections of F with support in Z* (IV 14) *is flasque (resp. injective).*

Assertion 1) follows from 4.5 when F is flasque. The functor j_X^* admits an exact left adjoint $j_{X!}$ (IV 11). Consequently, when F is injective, $j_X^* F$ is injective (0.2). Let us prove 2). Denote the inclusion morphism by $i : Z \to E$. The sheaf of sections of F with support in Z is the sheaf $i_* i^! F$ (IV 14). The functor $i_* i^!$ is right adjoint to the exact functor $i_* i^*$ (IV 14). It therefore carries injective sheaves to injective sheaves. Let U be the open complement of Z, let $j : U \to E$ be the inclusion morphism, and let F be a flasque sheaf. There is an exact sequence (IV 14):

$$0 \longrightarrow i_* i^! F \longrightarrow F \longrightarrow j_* j^* F. \tag{4.11.1}$$

For every object X of E, one has $j_* j^* F(X) = F(X \times U)$, and the morphism $F(X) \to j_* j^* F(X)$ induced by the last arrow of (4.11.1) comes from the canonical injection $X \times U \hookrightarrow X$. Since F is flasque, this morphism is surjective (4.7), and hence the last arrow of (4.11.1) is an epimorphism of presheaves. For every object X of E, the exact cohomology sequence deduced from (4.11.1) gives $H^q(X, i_* i^! F) = 0$ for $q > 0$, and hence $i_* i^! F$ is flasque.

4.12. The property of a sheaf being flasque (resp. injective) localizes (4.11.1). It 34
is not, in general, a property of local nature (Exercise 4.15). It is, however, a property of local nature for topoi generated by their open subobjects, and in particular for topoi associated with topological spaces (Exercise 4.16).

Exercise 4.13. Let X be a locally compact space, and let F be a c-soft sheaf [7]. Using the local nature of softness (*loc. cit.*), show that the restriction of F to every paracompact open subset of X is a soft sheaf. Show that the paracompact open subsets form a basis for the topology of X. Deduce that, if S denotes the family of paracompact open subsets of X, then F is S-acyclic. Show that the sheaf of continuous functions on X is S-acyclic but is not flasque when X is not discrete.

Problem 4.14. Study totally acyclic topoi, i.e. topoi such that $H^q(X, F) = 0$ for every $q > 0$, every object X, and every abelian sheaf F.

Exercise 4.15. Let G be a discrete group, and let B_G be its classifying topos (IV 2.4). Show that for every abelian sheaf F and every monomorphism $X \hookrightarrow Y$, the homomorphism $F(Y) \to F(X)$ is surjective. Let $E(G)$ be the group G regarded as a homogeneous space under itself. It is an object of B_G. The topos $B_{G/E(G)}$ is equivalent to the punctual topos (IV 8). The morphism $E(G) \to e$ (e the final object of B_G) is an epimorphism. For every abelian sheaf F on B_G, the sheaf $F|E(G)$ is flasque. Deduce that the property of being flasque or injective is not of local nature.

Exercise 4.16. A topos E is said to be generated by its open subobjects if the open subobjects of E (i.e. the subobjects of the final object e of E) form a generating family (I 7). Such a topos has the following property:

(P) Every epimorphic family $X_i \to X$, $i \in I$, admits a refinement by an epimorphic family $U_j \to X$, $j \in J$, in which the $U_j \to X$ are monomorphisms.

a) Are there topoi which have property (P) but are not generated by their open subobjects? Topoi associated with topological spaces are generated by their open subobjects. If E is generated by its open subobjects (resp. has (P)), then for every object X of E, $E_{/X}$ is generated by its open subobjects (resp. has (P)). 35

Every subtopos of a topos generated by its open subobjects is generated by its open subobjects. Property (P) is not a property of local nature.

b) Let E be a topos, and let $\mathrm{Ouv}(E)$ be the category of open subobjects of E, endowed with the induced topology. The inclusion functor $\mathrm{Ouv}(E) \to E$ is a morphism of sites, and hence gives a morphism of topoi $\Pi : E \to \mathrm{Ouv}(E)^{\sim}$. The functor Π^* is fully faithful. With respect to the 2-category of topoi generated by their open subobjects, the morphism Π has a universal property which the reader is invited to make explicit.

c) Let E be a topos generated by its open subobjects, and let $\mathrm{Point}(E)$ be the set of points of E up to isomorphism. Show that $\mathrm{Point}(E)$ is small. Put a topology on $\mathrm{Point}(E)$ and define a morphism $\mathrm{Top}(\mathrm{Point}(E)) \to \mathrm{Ouv}(E)^{\sim}$ making $\mathrm{Top}(\mathrm{Point}(E))$ a subtopos of $\mathrm{Ouv}(E)^{\sim}$. For every topological space X, show that every morphism from $\mathrm{Top}(X)$ to E gives a morphism from $\mathrm{Top}(X)$ to $\mathrm{Top}(\mathrm{Point}(E))$. Show that a topos generated by its open subobjects is equivalent to a topos $\mathrm{Top}(X)$ (X a topological space) if and only if it has enough points. Show that there exist topoi generated by their open subobjects which are not equivalent to any topos $\mathrm{Top}(X)$, where X is a topological space.

d) Let E be a topos having property (P). An abelian sheaf F on E is flasque if and only if, for every monomorphism $X \hookrightarrow Y$, the homomorphism $F(Y) \to F(X)$ is surjective.

e) Let E be a topos having property (P). Show that the property of a sheaf F being flasque is a property of local nature.

f) Let E be a topos generated by its open subobjects, and let e be the final object of E. An abelian sheaf F is flasque if and only if, for every open subobject U of e, the canonical morphism $F(e) \to F(U)$ is surjective.

36 Exercise 4.17. (Flasque sheaves and change of universe)

Let C be a $\mathscr{U}$-site (for example, a $\mathscr{U}$-topos), and let $\mathscr{V}$ be a universe containing $\mathscr{U}$. Denote the corresponding categories of sheaves by $C^{\sim}_{\mathscr{U}}$ and $C^{\sim}_{\mathscr{V}}$, and denote the canonical inclusion by $\epsilon : C^{\sim}_{\mathscr{U}} \to C^{\sim}_{\mathscr{V}}$. Let F be a flasque abelian $\mathscr{U}$-sheaf on C. We shall show that ϵF is flasque. First note that, for every object X of $C^{\sim}_{\mathscr{U}}$, one has $H^q(\epsilon X, \epsilon F) = 0$ for $q > 0$. Every object Y of $C^{\sim}_{\mathscr{V}}$ admits an epimorphic family $\epsilon X_i \to Y$, $i \in I$, in which the $\epsilon X_i \to Y$ are monomorphisms. It follows that

$$H^q(Y, \epsilon F) = displaystyle \varprojlim^{q}_{\epsilon X \hookrightarrow Y} F(X).$$

To show that these $\varprojlim^q$ vanish for $q \neq 0$, one may reduce to the topos $\mathrm{Ouv}(C_{\mathscr{V}/Y})^{\sim}$ and use the fact that, for this topos, a sheaf is flasque if it is so locally.

5. The $R^q u_*$ and the Cartan–Leray spectral sequence for a morphism of topoi

5.0. Let $u : (E, A) \to (E', A')$ be a morphism of ringed topoi, and let $u_* : E \to E'$ be the direct-image functor. The notation u_* will also denote the extension of the direct-image functor to modules. The functor u_* on modules (left modules, to fix ideas) is left exact. Its right derived functors are denoted by $R^q u_*$, $q \geq 0$.

Proposition 5.1. 1) *For every A-module M, the sheaf $R^q u_* M$ is the sheaf associated with the presheaf $X' \mapsto H^q(u^* X', M)$ ($X' \in \mathrm{ob}\, E'$).*

2) *Formation of the functors $R^q u_*$ commutes with restriction of scalars.*

3) *Formation of the functors $R^q u_*$ commutes with localization. More precisely, for every object X' of E', if $u_{/X'} : E_{/u^*X'} \to E'_{/X'}$ denotes the morphism deduced from u by localization* (IV 8)*, then, for every A-module M, there is a canonical isomorphism*

$$R^q(u_{/X'})_*(M|u^*X') \simeq R^q u_*(M)|X' \qquad q \geq 0. \tag{5.1.1}$$

Let $u_*^\wedge : E^\wedge \to E'^\wedge$ denote the direct-image functor for $\mathscr{U}$-presheaves ($u_*^\wedge M = M \circ u^*$). Since u^* and u_* are adjoint, there is an isomorphism

$$u_* \simeq \underline{a} u_*^{\wedge 0},$$

where $\underline{a}$ is the associated-sheaf functor for E'. Since the functors $\underline{a}$ and $u_*^\wedge$ are exact, 37
we have

$$R^q u_* \simeq \underline{a} u_*^\wedge \mathscr{H}^q,$$

which is another way of stating 1). Assertion 2) then follows from 1) and 3.5. By definition of the morphism $u_{/X'}$, there is a canonical isomorphism (5.1.1) for $q = 0$. The general case follows by noting that the localization functors (IV 8) are exact and take injective objects to injective objects (4.11).

Proposition 5.2. *Let $u : E \to E'$ be a morphism of topoi and let S' be a generating family of E'. The sheaves M which are acyclic for the functors $H^0(u^*X', \cdot)$, $X' \in S'$, are acyclic for the functor u_*. In particular, flabby sheaves are acyclic for u_*.*

This follows from 5.1, 1).

Proposition 5.3. *Let $u : E \to E'$ be a morphism of topoi and let M be an abelian group of E. There is a spectral sequence*

$$E_2^{p,q} = H^p(E', R^q u_* M) \Longrightarrow H^{p+q}(E, M). \tag{5.3.1}$$

More generally, for every object X' of E', there is a spectral sequence

$$E_2^{p,q} = H^p(X', R^q u_* M) \Rightarrow H^{p+q}(u^*X', M). \tag{5.3.2}$$

By definition of the direct- and inverse-image functors, there is an isomorphism $H^0(X', u_*M) \simeq H^0(u^*X', M)$. The functor u_* takes injective objects to flabby sheaves (4.6 and 4.9). The asserted spectral sequences are therefore spectral sequences of composite functors (0.3).

Proposition 5.4. *Let $u : E \to E'$ and $v : E' \to E''$ be two morphisms of topoi. There is a spectral sequence*

$$R^p v_* R^q u_* \Longrightarrow R^{p+q}(v \circ u)_*.$$

We have $v_* u_* \simeq (vu)_*$, and the functor u_* takes injective objects to flabby sheaves 38
(4.9), hence to objects acyclic for v_* (5.2). Thus there is a spectral sequence of composite functors (0.3).

6. Local Ext and cohomology with supports

6.0. Let (E, A) be a ringed topos, and let M be an A-module, a left module to fix ideas. The functor $N \longmapsto \mathbf{Hom}_A(M, N)$, from the category of left A-modules to the category of abelian groups, is left exact (IV 12). Its right derived functors are denoted by

$$\mathbf{Ext}^q_A(M, N). \tag{6.0.1}$$

In particular, one has

$$\mathbf{Ext}^0_A(M, N) = \mathbf{Hom}_A(M, N). \tag{6.0.2}$$

By definition, there are canonical isomorphisms (2.1 and IV 12)

$$H^0(E, \mathbf{Ext}^0_A(M, N)) = \mathrm{Ext}^0_A(E; M, N) = \mathrm{Hom}_A(M, N). \tag{6.0.3}$$

More generally, for every object X of E, one has (2.2)

$$H^0(X, \mathbf{Ext}^0_A(M, N)) = \mathrm{Ext}^0_A(X; M, N) = \mathrm{Hom}_{A|X}(M|X, N|X). \tag{6.0.4}$$

Proposition 6.1. 1) *Formation of the functors* $\mathbf{Ext}^q_A$ *commutes with localization. More precisely, for every object* X *of* E*, there are functorial isomorphisms*

$$\mathbf{Ext}^q_A(M, N)|X \simeq \mathbf{Ext}^q_{A|X}(M|X, N|X). \tag{6.1.1}$$

2) *The sheaf* $\mathbf{Ext}^q_A(M, N)$ *is isomorphic to the sheaf associated with the presheaf* $X \longmapsto \mathrm{Ext}^q_A(X; M, N)$.

3) *There is a spectral sequence*

$$\mathrm{Ext}^{p+q}_A(E; M, N) \Longleftarrow E_2^{p,q} = H^p(E, \mathbf{Ext}^q_A(M, N)). \tag{6.1.2}$$

39 *More generally, for every object* X *of* E*, there is a spectral sequence functorial in* X *and in the arguments* M *and* N*:*

$$\mathrm{Ext}^{p+q}_A(X; M, N) \Longleftarrow E_2^{p,q} = H^p(X, \mathbf{Ext}^q_A(M, N)). \tag{6.1.3}$$

By definition, (6.1.1) is an isomorphism when $q = 0$ (IV 12). The general case follows, taking into account the fact that the localization functors are exact and send injective modules to injective modules (2.2). The functor $N \longmapsto \mathbf{Hom}_A(M, N) = \mathbf{Ext}^0_A(M, N)$ sends injective modules to flasque sheaves, and hence to sheaves acyclic for $H^0(X, \cdot)$ (4.10). The spectral sequences (6.1.2) and (6.1.3) are the spectral sequences of composite functors, in view of (6.0.3) and (6.0.4). As X varies in E, the spectral sequence (6.1.3) gives a spectral sequence of presheaves and hence, upon passing to the associated sheaves, a spectral sequence of sheaves. Since the sheaves associated with the presheaves $X \longmapsto H^p(X, \cdot)$ are zero when $p \neq 0$ (3.1), this spectral sequence degenerates and gives the isomorphism asserted in 2).

Proposition 6.2. *The functors* $(M, N) \longmapsto \mathbf{Ext}^q_A(M, N)$*,* $q \geq 0$*, form a* δ*-functor in the variable* M *and in the variable* N*. Explicitly, for every exact sequence* $0 \to N' \to N \to N'' \to 0$ *(resp.* $0 \to M' \to M \to M'' \to 0$*), there are long exact sequences*

(6.2.1)

$$\cdots \longrightarrow \mathbf{Ext}^q_A(M, N') \longrightarrow \mathbf{Ext}^q_A(M, N) \longrightarrow \mathbf{Ext}^q_A(M, N'') \xrightarrow{\delta} \mathbf{Ext}^{q+1}_A(M, N') \longrightarrow \cdots$$

(resp.
(6.2.2)
$$\cdots \longrightarrow \mathbf{Ext}^q_A(M'', N) \longrightarrow \mathbf{Ext}^q_A(M, N) \longrightarrow \mathbf{Ext}^q_A(M', N) \xrightarrow{\delta} \mathbf{Ext}^{q+1}_A(M'', N) \longrightarrow \cdots).$$

It follows from the general properties of the functors Ext^q that, for every object X of E, the functors $(M, N) \longmapsto \mathrm{Ext}^q_A(X; M, N)$, $q \geq 0$, form a δ-functor in each argument. The assertion follows by allowing X to vary and taking the associated sheaf (6.1).

6.3. Let (E, A) be a ringed topos, let M be an A-module, let Z be a closed subtopos of E (IV 9), and let U be its open complement. Denote by $H^0_Z(E, M)$ the group of sections of M whose support is contained in Z (IV 14), and by $\mathscr{H}^0_Z(M)$ the subsheaf of M defined by the sections of M "with supports in Z" (IV 14). The functors $H^0_Z(E, \cdot)$ and $\mathscr{H}^0_Z(\cdot)$ are left exact (IV 14). Their derived functors are denoted by $H^q_Z(E, \cdot)$ and $\mathscr{H}^q_Z(\cdot)$ respectively, and are called the *cohomology groups* (resp. *sheaves) of M with supports in Z*[14]. **40**

There are canonical isomorphisms (IV 14)

$$H^0_Z(E, M) \simeq \mathrm{Hom}_A(A_Z, M) \simeq \mathrm{Ext}^0_A(E; A_Z, M), \tag{6.3.1}$$

$$\mathscr{H}^0_Z(M) \simeq \mathbf{Hom}_A(A_Z, M) \simeq \mathbf{Ext}^0_A(A_Z, M), \tag{6.3.2}$$

and hence, for every $q \geq 0$, there are isomorphisms

$$H^q_Z(E, M) \simeq \mathrm{Ext}^q_A(E; A_Z, M), \tag{6.3.3}$$

$$\mathscr{H}^q_Z(M) \simeq \mathbf{Ext}^q_A(A_Z, M). \tag{6.3.4}$$

Note that, since A_Z is a bimodule, the sheaves $\mathbf{Ext}^q_A(A_Z, M) \simeq \mathscr{H}^q_Z(M)$ are canonically endowed with A-module structures.

For every object X of E, denote by $Z_{/X}$ the closed subtopos of $E_{/X}$ obtained from Z by localization (it is the complement of $U \times X$). By definition, there are canonical isomorphisms

$$H^0(X, \mathscr{H}^0_Z(M)) \simeq H^0_{Z_{/X}}(E_{/X}, M|X). \tag{6.3.5}$$

Set

$$H^q_Z(X, M) \simeq H^q_{Z_{/X}}(E_{/X}, M|X). \tag{6.3.6}$$

Taking (6.3.2) into account, there are canonical isomorphisms

$$H^q_Z(X, M) \simeq \mathrm{Ext}^q_A(X; A_Z, M). \tag{6.3.7}$$

Proposition 6.4. 1) *Formation of the functors $\mathscr{H}^q_Z$ commutes with localization. More precisely, for every object X of E and every A-module M, there are canonical isomorphisms* **41**

$$\mathscr{H}^q_Z(M)|X \simeq \mathscr{H}^q_{Z_{/X}}(M|X). \tag{6.4.1}$$

2) *The sheaf $\mathscr{H}^q_Z(M)$ is the sheaf associated with the presheaf $X \longmapsto H^q_Z(X, M)$.*

[14]Compare with SGA 2 I for the case of ordinary topological spaces, as well as with the exposé of Hartshorne cited on p. 80 below.

3) *There is a spectral sequence*

$$H_Z^{p+q}(E, M) \Longleftarrow H^p(E, \mathscr{H}_Z^q(M)). \tag{6.4.2}$$

More generally, for every object X of E, there is a spectral sequence

$$H_Z^{p+q}(X, M) \Longleftarrow H^p(X, \mathscr{H}_Z^q(M)). \tag{6.4.3}$$

This is simply Proposition 6.1 translated by means of the dictionary in 6.3.

Proposition 6.5. *With the notation of 6.3, denote by $j : U \to E$ the canonical morphism. For every A-module M, there is an exact sequence of sheaves*

$$0 \longrightarrow \mathscr{H}_Z^0(M) \longrightarrow M \longrightarrow j_*(M|U) \longrightarrow \mathscr{H}_Z^1(M) \longrightarrow 0, \tag{6.5.1}$$

and, for $q \geq 2$, there are isomorphisms

$$\mathscr{H}_Z^q(M) \simeq \mathbf{Ext}_A^{q-1}(A_U, M) \simeq R^{q-1}j_*(M|U). \tag{6.5.2}$$

There is, in addition, a long exact sequence

$$\cdots \longrightarrow H_Z^q(E, M) \longrightarrow H^q(E, M) \longrightarrow H^q(U, M) \longrightarrow H_Z^{q+1}(E, M) \longrightarrow \cdots \tag{6.5.3}$$ [14]

and, more generally, for every object X of E, there is a long exact sequence

(6.5.4)

$$\cdots \longrightarrow H_Z^q(X, M) \longrightarrow H^q(X, M) \longrightarrow H^q(X \times U, M) \longrightarrow H_Z^{q+1}(X, M) \longrightarrow \cdots.$$

By the definition of A_Z, there is an exact sequence (IV 14)

$$0 \longrightarrow A_U \longrightarrow A \longrightarrow A_Z \longrightarrow 0. \tag{6.5.5}$$

42 The functors $\mathbf{Ext}_A^q(A, \cdot)$ vanish for $q > 0$. One has $\mathbf{Hom}_A(A_U, M) \simeq j_*(M|U)$ (IV 14), and hence (2.2) $\mathbf{Ext}_A^q(A_U, M) \simeq R^q j_*(M|U)$. Finally, $\mathbf{Ext}_A^q(A_Z, M) \simeq \mathscr{H}_Z^q(M)$ (6.3.4). The long exact sequence (6.2.2) gives, in this case, (6.5.1) and (6.5.2). The exact sequences (6.5.3) and (6.5.4) follow from the dictionary in 6.3 and from the long exact sequence of the δ-functor $\mathrm{Ext}_A^q(X; \cdot, M)$, $q \geq 0$, associated with (6.5.5).

Proposition 6.6. 1) *Flasque sheaves are acyclic for the functors $\mathscr{H}_Z^0$ and $H_Z^0(X, \cdot)$.*

2) *For every $q \geq 0$, the functors $\mathscr{H}_Z^q$ and $H_Z^q(X, \cdot)$ commute with restriction of scalars.*

Let M be a flasque sheaf. The exact sequence (6.5.4) gives the equalities $H_Z^q(X, M) = 0$ for every X and every $q \geq 2$, together with an exact sequence

$$0 \longrightarrow H_Z^0(X, M) \longrightarrow H^0(X, M) \longrightarrow H^0(X \times U, M) \longrightarrow H_Z^1(X, M) \longrightarrow 0.$$

But, since the sheaf M is flasque, the morphism $H^0(X, M) \to H^0(X \times U, M)$ is surjective (4). Consequently, $H_Z^1(X, M) = 0$, and M is acyclic for $H_Z^0(X, \cdot)$. Passing to the associated sheaves, one deduces that M is acyclic for the functor $\mathscr{H}_Z^0$ (6.4). It is clear that the functors $\mathscr{H}_Z^0$ and $H_Z^0(X, \cdot)$ commute with restriction of scalars; since flasque sheaves are acyclic for both functors, assertion 2) follows.

[14] This exact sequence makes precise the role of the global cohomological invariants $H_Z^q(E, M)$ as "cohomology groups of E *modulo* the open subtopos U, with coefficients in M."

PROPOSITION 6.7. *Let (E, A) be a ringed topos, let Z be a closed subtopos of E, let U be its open complement, and let M and N be two A-modules. There are isomorphisms, functorial in M and N and compatible with changes of closed subtopos:*

$$\mathscr{H}^0_Z(\mathbf{Hom}_A(M,N)) \simeq \mathbf{Hom}_A(M, \mathscr{H}^0_Z(N)) \simeq \mathbf{Hom}_A(M \otimes_A A_Z, N).$$

This follows from (IV 14), taking (6.3.2) into account.

6.8. Set

$$\mathbf{Hom}_{A,Z}(M,N) = \mathscr{H}^0_Z(\mathbf{Hom}_A(M,N)). \tag{6.8.1}$$

The functor $N \longmapsto \mathbf{Hom}_{A,Z}(M,N)$ is left exact. Its derived functors are denoted by $\mathbf{Ext}^q_{A,Z}(M,N)$: these are the **Ext** *sheaves with supports in* Z. Thus

$$\mathbf{Ext}^0_{A,Z}(M,N) = \mathbf{Hom}_{A,Z}(M,N). \tag{6.8.2}$$

Set $M_Z = M \otimes_A A_Z$. It follows from IV 14 that M_Z is the direct image on E of the 43
inverse image of M on Z. Thus, taking 6.7 into account, there are isomorphisms

$$\mathbf{Ext}^q_{A,Z}(M,N) \simeq \mathbf{Ext}^q_A(M_Z,N). \tag{6.8.3}$$

We now pass to the global invariants. Set

$$\mathrm{Hom}_{A,Z}(M,N) = \mathrm{Ext}^0_{A,Z}(E;M,N) = H^0_Z(E, \mathbf{Hom}_A(M,N)). \tag{6.8.4}$$

The group $\mathrm{Hom}_{A,Z}(M,N)$ is the subgroup of the group of morphisms from M to N whose support is in Z (IV 14), i.e. those which vanish on U. More generally, for every object X of E, set

$$\mathrm{Ext}^0_{A,Z}(X;M,N) = H^0_Z(X, \mathbf{Hom}_A(M,N)). \tag{6.8.5}$$

The functors $N \longmapsto \mathrm{Ext}^0_{A,Z}(X;M,N)$ are left exact. Their derived functors are denoted by $\mathrm{Ext}^q_{A,Z}(X;M,N)$. These are the Ext *groups with supports in* Z. Definitions (6.3.6), (6.8.1), and (6.8.5), together with the isomorphisms of 6.7, give isomorphisms

$$\mathrm{Ext}^0_{A,Z}(X;M,N) \simeq \begin{cases} H^0(X, \mathbf{Hom}_{A,Z}(M,N)), \\ \mathrm{Ext}^0_A(X;M,\mathscr{H}^0_Z(N)), \\ \mathrm{Ext}^0_A(X;M_Z,N). \end{cases} \tag{6.8.6}$$

The last isomorphism in (6.8.6) gives isomorphisms

$$\mathrm{Ext}^q_{A,Z}(X;M,N) \simeq \mathrm{Ext}^q_A(X;M_Z,N). \tag{6.8.7}$$

PROPOSITION 6.9. 1) *There are two spectral sequences, functorial in M and N and compatible with changes of closed subtopos:*

$$\mathbf{Ext}^{p+q}_{A,Z}(M,N) \Longleftarrow \begin{cases} E_2^{p,q} = \mathscr{H}^p_Z(\mathbf{Ext}^q_A(M,N)), \\ {}'E_2^{p,q} = \mathbf{Ext}^p_A(M, \mathscr{H}^q_Z(N)). \end{cases} \tag{6.9.1}$$

2) *There are three spectral sequences, functorial in X, M, and N and compatible* 44
with changes of closed subtopos:

$$\mathrm{Ext}^{p+q}_{A,Z}(X;M,N) \Longleftarrow \begin{cases} E_2^{p,q} = H^p_Z(X, \mathbf{Ext}^q_A(M,N)), \\ {}'E_2^{p,q} = H^p(X, \mathbf{Ext}^q_{A,Z}(M,N)), \\ {}''E_2^{p,q} = \mathrm{Ext}^p_A(X;M,\mathscr{H}^q_Z(N)). \end{cases} \tag{6.9.2}$$

3) *The sheaves* $\mathbf{Ext}^q_{A,Z}(M,N)$ *are canonically isomorphic to the sheaves associated with the presheaves* $X \longmapsto \mathrm{Ext}^q_{A,Z}(X;M,N)$.

When N is injective, the sheaf $\mathbf{Hom}_A(M,N)$ is flasque (4.10), and hence acyclic for $\mathscr{H}^0_Z$ (6.6). The first spectral sequence in (6.9.1) is a spectral sequence of composite functors (6.8.1). Likewise, when N is injective, $\mathscr{H}^0_Z(N)$ is injective (4.11), and the second spectral sequence in (6.9.1) is a spectral sequence of composite functors deduced from 6.7. The spectral sequences in (6.9.2) are spectral sequences of composite functors deduced from definition (6.8.5) and the first two isomorphisms in (6.8.6). Finally, by allowing X to vary in the second spectral sequence in (6.9.2) and taking the associated sheaves, one obtains a spectral sequence of sheaves which degenerates by 3.1 and gives the isomorphisms in 3).

Proposition 6.10. *The functors* $(M,N) \longmapsto \mathbf{Ext}^q_{A,Z}(M,N)$, $q \geq 0$, *and* $(M,N) \longmapsto \mathrm{Ext}^q_{A,Z}(X;M,N)$, $q \geq 0$, *are δ-functors in each of the variables M and N. Denoting by M_U the sheaf $M \otimes_A A_U$ (cf. IV 11), there is a long exact sequence*

(6.10.1)

$$\dots \xrightarrow{\delta} \mathbf{Ext}^q_{A,Z}(M,N) \longrightarrow \mathbf{Ext}^q_A(M,N) \longrightarrow \mathbf{Ext}^q_A(M_U,N) \xrightarrow{\delta} \mathbf{Ext}^{q+1}_{A,Z}(M,N) \longrightarrow \dots$$

and an analogous exact sequence for the Ext *groups.*

The functors $\mathbf{Ext}^q_A(\cdot,\cdot)$ and $\mathrm{Ext}^q_A(X;\cdot,\cdot)$ form δ-functors with respect to each variable (6.2), and the functor $M \longmapsto M_Z$ is exact because A_Z is flat (1.3.3). The first assertion therefore follows from the isomorphisms (6.8.3) and (6.8.7). The exact sequence (6.10.1), and the analogous exact sequence for the Ext groups, is the long exact sequence of the δ-functor

$$\mathbf{Ext}^q_A(\cdot,N),\ q \geq 0 (\textit{resp.}\ \ \mathrm{Ext}^q_A(X;\cdot,N),\ q \geq 0)$$

associated with the exact sequence $0 \to M_U \to M \to M_Z \to 0$, taking (6.8.3) and (6.8.7) into account.

6.11. Let us indicate briefly how these results may be extended to the case of families of supports. Let E be a topos. For every object X of E, denote by $\mathrm{Fer}(X)$ the set of closed subtopoi of the topos $E_{/X}$. By passage to complements, this set is in bijective correspondence with the set of open subtopoi of $E_{/X}$, i.e. with the set of subobjects of X (IV 8). The presheaf $\mathrm{Fer} : X \longmapsto \mathrm{Fer}(X)$ is in fact a $\mathscr{U}$-sheaf, as is immediately verified, and is therefore representable. In other words, there is an object of E, again denoted by Fer, and a closed subtopos Z_{Fer} of $E_{/\mathrm{Fer}}$ such that, for every X, every element of $\mathrm{Fer}(X)$ is obtained from Z_{Fer} by base change along a morphism $u_Z : X \to \mathrm{Fer}$ uniquely determined by Z.

Definition 6.12. *A family of supports of E is a subset Φ of the set of closed subtopoi of E having the following properties:*

(S1) *The union of a finite family of elements of Φ belongs to Φ.*
(S2) *Every closed subtopos of E contained in an element of Φ is an element of Φ.*

6.12.1. Let E be a topos, let Φ be a family of supports of E, and let X be an object of E. Denote by $\Phi(X)$ the smallest family of supports of $E_{/X}$ containing the closed subtopoi of $E_{/X}$ obtained from closed subtopoi in Φ by the base change $X \to e$ (e being the final object of E). The functor $X \longmapsto \Phi(X)$ is a subpresheaf of the sheaf Fer. It is therefore separated.

6.12.2. A family Φ of supports of E is said to be *of local character* if it has the following property:
(LC) For every epimorphic family $(X_i \to e)$, $i \in I$ (where e is the final object of E), the sequence of sets

$$\Phi \longrightarrow \prod_i \Phi(X_i) \rightrightarrows \prod_{i,j} \Phi(X_i \times X_j)$$

is exact.

Let $a\Phi$ be the sheaf associated with the presheaf $X \longmapsto \Phi(X)$ (6.12.1). Condition (LC) is equivalent to the condition that the canonical morphism $\Phi \to a\Phi(e)$ be a bijection (cf. the construction of the associated sheaf in II in the case of a separated presheaf). To verify (LC), one may therefore restrict to a final family of epimorphic families $(X_i \to e)$, $i \in I$. 46

Example 6.12.3. 1) Let Z be a closed subtopos of E. The set of closed subtopoi of E contained in Z is a family of supports of E. It is of local character.
2) Let T be a topological space and let Φ be a paracompactifying family of supports of T [7]. In general, the family Φ is not of local character.
3) Let T be a topological space and let p be an integer. The family Φ_p of closed subsets of T of Krull codimension $\geq p$ is a family of supports of T. It is of local character.
4) Let E be a topos with the following property: every epimorphic family $(X_i \to e)$, $i \in I$, is dominated by a finite epimorphic family. Then every family of supports of E is of local character. Indeed, using example 1), it is enough to show that a filtered inductive limit Φ_λ of families of local character is a family of local character. This follows immediately by taking the inductive limit in the sequence of sets

$$\Phi_\lambda \longrightarrow \prod_i \Phi_\lambda(X_i) \rightrightarrows \prod_{i,j} \Phi_\lambda(X_i \times X_j),$$

where $(X_i \to e)$, $i \in I$, is a finite epimorphic family. Indeed, by 6.12.2 and the hypothesis on E, it is enough to consider finite epimorphic families in verifying condition (LC); and, since filtered inductive limits commute with finite projective limits (I 2), exactness of these sequences of sets is preserved upon passage to the inductive limit.
5) Let E be a coherent topos (VI 2.3). Then, by the preceding example, every family of supports of E is of local character.

6.13. Let (E, A) be a ringed topos, let Φ be a family of supports of E, let N be an A-module (a left module, to fix ideas), and let X be an object of E. Set

$$\mathscr{H}^0_\Phi(N) = \varinjlim_{Z \in \Phi} \mathscr{H}^0_Z(N), \tag{6.13.1}$$

$$H^0_\Phi(E, N) = \varinjlim_{Z \in \Phi} H^0_Z(E, N). \tag{6.13.2}$$

Let M be a left A-module, and set 47

$$\mathbf{Ext}^0_{A,\Phi}(M, N) = \mathbf{Hom}_{A,\Phi}(M, N) = \varinjlim_{Z \in \Phi} \mathbf{Ext}^0_{A,Z}(M, N), \tag{6.13.3}$$

$$\mathrm{Ext}^0_{A,\Phi}(X; M, N) = \mathrm{Hom}_{A|X,\Phi(X)}(M|X, N|X) = \varinjlim_{Z \in \Phi} \mathrm{Ext}^0_{A,Z}(X; M, N). \tag{6.13.4}$$

There are canonical isomorphisms

$$\mathbf{Ext}^0_{A,\Phi}(M,N) \simeq \mathscr{H}^0_{\Phi}(\mathbf{Ext}^0_A(M,N)), \tag{6.13.5}$$

$$\mathrm{Ext}^0_{A,\Phi}(X;M,N) \simeq H^0_{\Phi}(X,\mathbf{Ext}^0_A(M,N)). \tag{6.13.6}$$

When Φ is the family of closed subtopoi contained in a fixed closed subtopos Z, there are isomorphisms

$$\begin{cases} \mathscr{H}^0_{\Phi} \simeq \mathscr{H}^0_Z, \\ H^0_{\Phi} \simeq H^0_Z, \\ \mathbf{Ext}^0_{A,\Phi}(\cdot,\cdot) \simeq \mathbf{Ext}^0_{A,Z}(\cdot,\cdot), \\ \mathrm{Ext}^0_{A,\Phi}(X;\cdot,\cdot) \simeq \mathrm{Ext}^0_{A,Z}(X;\cdot,\cdot). \end{cases} \tag{6.13.7}$$

The inductive limits defining the preceding functors are filtered. Consequently, these functors are left exact. Their right derived functors are denoted by

$$\mathscr{H}^q_{\Phi},\ H^q_{\Phi},\ \mathbf{Ext}^q_{A,\Phi}(\cdot,\cdot),\ \mathrm{Ext}^q_{A,\Phi}(X;\cdot,\cdot). \tag{6.13.8}$$

They too are computed by inductive limits of the functors $\mathscr{H}^q_Z$, H^q_Z, etc. as Z ranges through Φ.

From the isomorphisms (6.13.5) and (6.13.6), one obtains two spectral sequences by taking inductive limits in the first spectral sequence in (6.9.1) and the first spectral sequence in (6.9.2):

$$\mathbf{Ext}^{p+q}_{A,\Phi}(M,N) \Leftarrow E_2^{p,q} = \mathscr{H}^p_{\Phi}(\mathbf{Ext}^q_A(M,N)), \tag{6.13.9}$$

$$\mathrm{Ext}^{p+q}_{A,\Phi}(X;M,N) \Leftarrow E_2^{p,q} = H^p_{\Phi}(X,\mathbf{Ext}^q_A(M,N)). \tag{6.13.10}$$

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6.14. With the notation of 6.13, and without any hypothesis on Φ, there is a functorial morphism

$$\theta : H^0_{\Phi}(N) \longrightarrow H^0(E,\mathscr{H}^0_{\Phi}(N)). \tag{6.14.1}$$

This morphism is always injective, but is not in general an isomorphism. However, if Φ is *of local character*, the morphism θ is an isomorphism, as is immediately verified. Likewise, if Φ is *of local character*, there is an isomorphism

$$\theta' : \mathrm{Ext}^0_{A,\Phi}(E;M,N) \simeq H^0(E,\mathbf{Ext}^0_{A,\Phi}(M,N)). \tag{6.14.2}$$

PROPOSITION 6.15. *Let (E,A) be a ringed topos such that E is coherent* (VI), *let Φ be a family of supports of E, and let M and N be two A-modules. There are two spectral sequences:*

$$H^{p+q}_{\Phi}(E,N) \Leftarrow E_2^{p,q} = H^p(E,\mathscr{H}^q_{\Phi}(N)), \tag{6.15.1}$$

$$\mathrm{Ext}^{p+q}_{A,\Phi}(E;M,N) \Leftarrow E_2^{p,q} = H^p(E,\mathbf{Ext}^q_{A,\Phi}(M,N)). \tag{6.15.2}$$

These spectral sequences are deduced from the spectral sequence (6.4.2) and the second spectral sequence in (6.9.2) by taking the inductive limit over the closed subtopoi Z in Φ, taking into account that the cohomology of a coherent topos commutes with inductive limits of sheaves (IV 5).

6.16. We mention, without proof, that the second spectral sequence in (6.9.1) and the third spectral sequence in (6.9.2) (with X equal to the final object of E) can be extended to the case of families of supports, provided that the topos E is coherent and the module M is perfect [13].

Finally, one can also generalize the notion of families of supports by introducing presheaves of families of supports, sheaves of families of supports, and the corresponding cohomology groups and sheaves[14]. 49

7. Appendix: Čech cohomology

Developing an idea due to P. Cartier, we show in this section how, in an arbitrary topos, one can compute the cohomology of a sheaf by means of coverings. For the classical theory of paracompact spaces, we refer to [7]; for another method which applies to certain topoi, and in particular to the étale topos, see [14].

7.1. Skeleton and coskeleton.

7.1.0. Let $\mathbf{\Delta}$ be the category of standard simplices (the objects of $\mathbf{\Delta}$ are the ordered sets $\Delta_n = [0, \dots, n]$, and the morphisms are the increasing maps). Let E be a $\mathcal{U}$-topos, and let $\mathbf{\Delta}(E)$ be the category of presheaves on $\mathbf{\Delta}$ with values in E; in other words, the category of *semi-simplicial objects* of E. Denote by $\mathbf{\Delta}[n]$ the full subcategory of $\mathbf{\Delta}$ defined by the objects Δ_p, $p \leq n$, by $i_n : \mathbf{\Delta}[n] \to \mathbf{\Delta}$ the inclusion functor, and by $\mathbf{\Delta}E[n]$ the category of presheaves on $\mathbf{\Delta}[n]$ with values in E; in other words, the category of *semi-simplicial objects of E truncated at order n*. By I 5.1, there is a sequence of three adjoint functors (I 5.3)

$$\begin{tikzcd} \mathbf{\Delta}E[n] \arrow[r, shift left=3ex, "i_{n!}"] \arrow[r, shift right=3ex, "i_{n*}"'] & \mathbf{\Delta}E \arrow[l, "i_n^*"'] \end{tikzcd}$$

where i_n^* is the restriction functor to the category $\mathbf{\Delta}[n]$, and i_{n*} and $i_{n!}$ are its right and left adjoints, respectively. Note that $\mathbf{\Delta}E$ and $\mathbf{\Delta}E[n]$ are $\mathcal{U}$-topoi (IV 1.2), and that (i_n^*, i_{n*}) is a morphism of topoi (IV 3.1) which is an embedding of $\mathbf{\Delta}E[n]$ into $\mathbf{\Delta}E$ (I 5.6 and IV 9.1.1).

Definition 7.1.1. *The functor $i_{n!}i_n^*$ (resp. $i_{n*}i_n^*$) from $\mathbf{\Delta}_E$ to $\mathbf{\Delta}_E$ is denoted by* sk_n 50
(resp. cosk_n*) and is called the* skeleton functor of order n *(resp. the* coskeleton functor of order n*).*

The skeleton and coskeleton functors are used constantly in the theory of semi-simplicial sets [3].

Proposition 7.1.2. 1) *The adjunction morphism* $\mathrm{sk}_n \to \mathrm{id}$ *is a monomorphism. The canonical morphisms* $\mathrm{sk}_n\,\mathrm{sk}_m \to \mathrm{sk}_n$ *(resp.* $\mathrm{cosk}_n \to \mathrm{cosk}_n\,\mathrm{cosk}_m$*) are isomorphisms when $n \leq m$ (resp. $n \geq m$).*

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[14] cf. [12], Chap. IV, § 1 (Lecture Notes 20, Springer), for the development of these notions in the form of a fugue with variations.

2) *Let $u : E \to E'$ be a morphism of topoi. The functor u^*, extended to semi-simplicial and truncated semi-simplicial objects, commutes with the functors $i_{n!}$, i_n^*, sk_n, and cosk_n.*

The first assertion follows immediately from the definitions. To prove 2), it suffices to observe that, by I 5.1, the functors $i_{n!}, \dots, \mathrm{cosk}_n$ are computed by means of inductive limits and finite projective limits.

7.2. **An acyclicity lemma.**

7.2.0. Let M be an abelian group of E. The *homology* of a semi-simplicial object K of E with coefficients in M is defined in the usual way. First consider $A(K)$, the free abelian semi-simplicial group generated by K, and then form the tensor product $M \otimes_{\mathbf{Z}} A(K)$; this gives an abelian semi-simplicial group of E. One then considers the associated complex of abelian groups (formed by taking the alternating sum of the boundary operators) and takes its homology. The homology objects are denoted by $H_i(K, M)$. They are abelian groups of E. When M is the group $\mathbb{Z}_E$ (the free abelian group generated by the final object of E), we write simply $H_i(K)$. The groups $H_i(K)$ are called the *homology groups* of K. A semi-simplicial object K is said to be *acyclic* if, for every abelian group M of E, the groups $H_i(K, M)$ vanish for $i > 0$.

51 Formation of homology commutes with inverse images under morphisms of topoi . The purpose of this section is to prove the following lemma:

Lemma 7.2.1. *Let E be a topos, let $K_\bullet$ and $L_\bullet$ be two semi-simplicial objects of E, let $v_\bullet : K_\bullet \to L_\bullet$ be a morphism of semi-simplicial objects, and let n be an integer ≥ 0. Suppose that v has the following properties:*

1) *For every integer $p < n$ ($p \geq 0$), the morphism $v_p : K_p \to L_p$ is an isomorphism.*
2) *The morphism $K_n \xrightarrow{v_n} L_n$ is an epimorphism.*
3) *The canonical morphisms $K_\bullet \to \mathrm{cosk}_n(K_\bullet)$ and $L_\bullet \to \mathrm{cosk}_n(L_\bullet)$ are isomorphisms.*

Then, for every integer p, the morphism v_p is an epimorphism, and the morphism $v_\bullet$ induces an isomorphism on homology objects.

Remark. The proof in fact shows that, after adopting a suitable definition of homotopy in topoi, the morphism $v_\bullet$ is a homotopy equivalence.

Proof. We may suppose that E is the topos of sheaves on a small site C (IV 1), and hence that E is a subtopos of a topos E' having enough fiber functors (for example, the topos $C^\wedge$). Denote by $a^* : E' \to E$ the inverse-image functor of the embedding morphism $E \hookrightarrow E'$. Let

$$v_\bullet[n] : K_\bullet[n] \longrightarrow L_\bullet[n]$$

be the morphism of truncated semi-simplicial objects obtained by truncating $v_\bullet$ at order n. Denote by $L_\bullet[n]'$ the image (in E') of $K_\bullet[n]$ under $v_\bullet[n]$, and by $v_\bullet[n]' : K_\bullet[n] \to L_\bullet[n]'$ the morphism induced by $v_\bullet[n]$. Put $L! = i_{n*}L_\bullet[n]'$, $K! = i_{n*}K_\bullet[n]$, and $v! = i_{n*}v_\bullet[n]'$ (here i_{n*} is relative to E'). By 1), the morphism $v! : K! \to L!$ has properties 1), 2), and 3) of the lemma. Moreover, by 2), 3), and 7.1.2, the morphism $a^*v!$ is none other than $v_\bullet$. It therefore suffices to prove the lemma for $v!$, and hence we may suppose that E has enough fiber functors. Using these fiber functors, we may then reduce to the case where E is the category of sets.

52 First observe that the hypotheses of the lemma on the morphism $v_\bullet$ are stable under every base change $M_\bullet \to L_\bullet$ for which $M_\bullet$ is a coskeleton of order n. Consequently, the

fiber $P_\bullet$ of $v_\bullet$ at any base point of $L_\bullet$ is of the form $i_{n*}P_\bullet[n]$, where $P_\bullet[n]$ is a nonempty complex truncated at order n of the form

$$P_n \longrightarrow e \longrightarrow e \longrightarrow e \longrightarrow \cdots \longrightarrow e. \tag{7.2.2}$$

For every integer i, denote by $\delta\Delta_i$ the semi-simplicial set forming the boundary of the simplex Δ_i. Every morphism from $\delta\Delta_i$ to $P_\bullet$ extends to a morphism from Δ_i to $P_\bullet$. Indeed, this property is immediate for $i \geq n$, because $P_\bullet$ is a coskeleton of order n, and it is clear for $i < n$ from the description (7.2.2). Since $P_\bullet$ is a Kan complex, it is homotopically trivial [6]. Note that the morphism $v_\bullet$ is a fibration in the sense of Kan [6]. Since the fibers of this fibration are homotopically trivial, $v_\bullet$ induces an isomorphism on the homotopy groups of $K_\bullet$ and $L_\bullet$ at all base points of $K_\bullet$. By the Hurewicz theorem, this implies that $v_\bullet$ induces an isomorphism on homology [6]. This proves the second assertion of the lemma. To prove the first, observe that every morphism from a complex truncated at order n to $L_\bullet[n]$ lifts to a morphism to $K_\bullet[n]$. Consequently, by 3), every morphism from a complex to $L_\bullet$ lifts to a morphism to $K_\bullet$. Thus $v_\bullet$ is surjective.

7.3. **Hypercoverings.**

7.3.0. Let C be a site belonging to the universe $\mathcal{U}$, in which fiber products and products of two objects are representable. Denote by $C^\wedge$ the topos of $\mathcal{U}$-presheaves on C, and by $C^\sim$ the topos of $\mathcal{U}$-sheaves on C. An object K of $C^\wedge$ is called *semi-representable* if it is isomorphic to a coproduct of representable presheaves.

Let $SR(C)$ be the full subcategory of $C^\wedge$ defined by the semi-representable objects. In the category $SR(C)$, projective limits indexed by finite nonempty categories are representable, and the inclusion functor $SR(C) \hookrightarrow C^\wedge$ commutes with these finite projective limits. By a remark already made, it follows that if $K_\bullet$ is a semi-simplicial object truncated in degree n in $C^\wedge$, all of whose objects are semi-representable, then the ex- 53
tension $i_n^+(K_\bullet)$ has the same property. Thus, if $K_\bullet$ is a semi-simplicial object of $C^\wedge$ all of whose objects are semi-representable, then, for every n, the object $\operatorname{cosk}_n(K_\bullet)$ has the same property.

7.3.1. *Definitions and notation.*

. Let $p > 0$ be an integer. A semi-simplicial object $K_\bullet$ of $C^\wedge$ is called a *hypercovering of type p* of C, or, when no confusion can result, a hypercovering of type p, if it has the following properties:

- HR1) For every integer $n \geq 0$, K_n is semi-representable.
- HR2) The canonical morphism $K_\bullet \to \operatorname{cosk}_p(K_\bullet)$ is an isomorphism.
- HR3) For every integer $n \geq 0$, the canonical morphism of presheaves $K_{n+1} \to (\operatorname{cosk}_n(K_\bullet))_{n+1}$ is a covering morphism of presheaves (II 6.2). The canonical morphism of presheaves $K_0 \to e$, where e is the final object of $C^\wedge$, is a covering morphism of presheaves.

. A hypercovering of type p is a hypercovering of type q for every $q \geq p$. A semi-simplicial object $K_\bullet$ will be called a *hypercovering* (of C) if it has properties HR 1) and HR 3).

. Denote by HR_p (resp. HR), and call the category of hypercoverings of type p (resp. the category of hypercoverings), the following category:

- a) The objects of HR_p (resp. HR) are the hypercoverings of type p (resp. the hypercoverings).
- b) Let $K_\bullet$ and $L_\bullet$ be two objects of HR_p (resp. HR). A morphism of HR_p (resp. HR) with source $K_\bullet$ and target $L_\bullet$ is a morphism $v_\bullet : K_\bullet \to L_\bullet$ of semi-simplicial objects of $C^\wedge$.

. Let C be a site in which fiber products are representable, and let X be an object of C. A hypercovering of X (resp. a hypercovering of type p of X) means a hypercovering of the site C/X (resp. a hypercovering of type p of C/X). The category of hypercoverings
54 of X (resp. of hypercoverings of type p of X) is defined similarly . There is a sequence of inclusion functors

$$\cdots HR_p \hookrightarrow HR_{p+1} \hookrightarrow \cdots \hookrightarrow HR.$$

These functors are fully faithful. We denote by HR_∞ the inductive limit of the categories HR_p.

. For every integer $n \geq 0$, denote by Δ_n^c the standard semi-simplicial object of dimension n with values in the category of sets, i.e. the functor on $\mathbf{\Delta}$ represented by the ordered set $[0, n]$. Also denote by $\underline{\Delta}_n^c$ the presheaf on C, with values in the category of semi-simplicial sets, that is constant with value Δ_n. Thus the presheaf $\underline{\Delta}_n^c$ is a presheaf of semi-simplicial sets. The two canonical injections from Δ_0 into Δ_1 define two morphisms of semi-simplicial presheaves:

$$\underline{\Delta}_0^c \underset{e_1}{\overset{e_0}{\rightrightarrows}} \underline{\Delta}_1^c,$$

and consequently define, for every semi-simplicial presheaf $K_\bullet$, two canonical injections:

$$K_\bullet \underset{e_1}{\overset{e_0}{\rightrightarrows}} K_\bullet \times \underline{\Delta}_1^c.$$

Two morphisms $K_\bullet \underset{u_1}{\overset{u_0}{\rightrightarrows}} L_\bullet$ of semi-simplicial presheaves are called *homotopic morphisms* if there exists a morphism

$$v : K_\bullet \times \Delta_1^c \longrightarrow L_\bullet$$

such that the diagrams

$$\begin{tikzcd} K_\bullet \arrow[rr, "e_i"] \arrow[dr, "u_i"'] & & K_\bullet \times \Delta_1^c \arrow[dl, "v"] \\ & L_\bullet & \end{tikzcd} \qquad \text{i = 0,1}$$

55 are commutative. The morphism v is called a *homotopy joining* u_0 *to* u_1. The relation "u_0 and u_1 are two homotopic morphisms from $K_\bullet$ to $L_\bullet$" is not, in general, an equivalence relation. Nevertheless, the equivalence relation generated by this relation is compatible with composition of morphisms. This therefore allows us to define the category of semi-simplicial presheaves up to homotopy by passing to the quotient by the equivalence relation generated by the homotopy relation.

. In this way we define the categories $\mathscr{HR}_p$, $\mathscr{HR}_\infty$, and $\mathscr{HR}$, the *categories of hypercoverings up to homotopy.*

Theorem 7.3.2. *Let C be a site satisfying the conditions of 7.3.0.*

1) *The category $\mathscr{HR}_p^0$ (resp. $\mathscr{HR}^0$) is filtered* (I 2.7).
2) *Let $K_\bullet$ be an object of HR_p (resp. of HR), let n be an integer such that $0 \leq n \leq p$ (resp. $0 \leq n$), and let*

$$u : X \longrightarrow K_n$$

be a covering morphism of presheaves. There exists an object $L_\bullet$ of HR_p (resp. of HR) and a morphism $f : L_\bullet \to K_\bullet$ such that the morphism

$$f_n : L_n \longrightarrow K_n$$

factors through u.

3) *The associated semi-simplicial sheaves* (II 3.5) *of hypercoverings are acyclic (7.2.0) in strictly positive degrees. The 0-th homology sheaf is isomorphic to the sheaf associated to the constant presheaf $\mathbb{Z}$.*

Proof. We prove 3). Let $K_\bullet^\sim$ be the semi-simplicial sheaf associated to $K_\bullet$. The "associated sheaf" functor commutes with the functor cosk_n (7.1.2. 2)). It follows that the semi-simplicial sheaf $K_\bullet^\sim$ has the following properties:

a) The canonical morphism $K_0 \to e$ is an epimorphism of sheaves.
b) For every integer $n \geq 0$, the canonical morphism $K_{n+1} \to (\operatorname{cosk}_n(K_\bullet))_{n+1}$ is an epimorphism of sheaves.

For $n \geq 0$, put $L_{\bullet n} = \operatorname{cosk}_n K_\bullet^\sim$. Then, for every n, there is a sequence of morphisms of semi-simplicial sheaves

$$K_\bullet^\sim \xrightarrow{u_n} L_{\bullet n} \xrightarrow{v_n} L_{\bullet n-1} \cdots \longrightarrow L_{\bullet 0} \xrightarrow{v_0} e_\bullet,$$

and , for every j, $0 \leq j \leq n$, v_j has the properties of 7.2.1. Moreover, the morphism u_n 56
induces an isomorphism on the components of degree $\leq n$. It then follows from 7.2.1, by induction on n, that $K_\bullet$ is acyclic.

To prove 1) and 2), we introduce some terminology.

Definition 7.3.3. *Let $f : X_\bullet \to Y_\bullet$ be a morphism of semi-simplicial presheaves. We say that f is* special of type p *(resp.* special*) if:*

1) *The objects $X_\bullet$ and $Y_\bullet$ are canonically isomorphic to their coskeleta of order p, and* $\operatorname{cosk}_p(f) = f$ *(resp. no conditions are imposed on f, $X_\bullet$, or $Y_\bullet$).*
2) *For every integer n such that $0 \leq n \leq p$ (resp. $0 \leq n$), the morphism Φ_{n+1} appearing in the diagram below is covering:*

$$\begin{tikzcd}
X_{n+1} \arrow[rr, "f_{n+1}"] \arrow[dr, "\Phi_{n+1}"] \arrow[dd] & & Y_{n+1} \arrow[dd] \\
& P_{n+1} \arrow[ur] \arrow[dl] & \\
(\operatorname{cosk}_n X_\bullet)_{n+1} \arrow[rr, "(\operatorname{cosk}_n f)_{n+1}"'] & & (\operatorname{cosk}_n Y_\bullet)_{n+1}.
\end{tikzcd}$$

(The vertical arrows are defined by the canonical morphisms $X_\bullet \to \operatorname{cosk}_n X_\bullet$ and $Y_\bullet \to \operatorname{cosk}_n Y_\bullet$. The object P_{n+1} is the fiber product, and Φ_{n+1} is the unique arrow making the diagram commutative.)

3) *The morphism $f_0 : X_0 \to Y_0$ is covering.*

A semi-simplicial presheaf $K_\bullet$ is called special of type p (resp. special) if the canonical morphism

$$K_\bullet \longrightarrow e_\bullet \quad (e_\bullet \quad \text{is the final semi-simplicial object})$$

is special of type p (resp. special), i.e. if $K_\bullet$ satisfies conditions HR 2) *and* HR 3) *(resp.* HR 3)*) of 7.3.1.1.*

57 Lemma 7.3.4. 1) *The composite of two special morphisms (resp. special morphisms of type p) is a special morphism (resp. a special morphism of type p).*

2) *Let $K_\bullet$ be a special semi-simplicial presheaf (resp. special of type p), let $X_\bullet \xrightarrow{f} Y_\bullet$ be a special morphism (resp. special of type p), and let $u : K_\bullet \to Y_\bullet$ be a morphism of complexes. Then the fiber product $P_\bullet = K_\bullet \times_{Y_\bullet} X_\bullet$ is special (resp. special of type p).*

The proof of this lemma is left to the reader.

7.3.5. *Proof of assertion* 2) *of* 7.3.2. We may first suppose that X is semi-representable. For every semi-simplicial object $L_\bullet$, denote by $\mathrm{Hom}_{(X)}(L_\bullet, K_\bullet)$ the set of morphisms of semi-simplicial objects equipped with a factorization $L_n \to X \xrightarrow{u} K_n$. The functor $\mathrm{Hom}_{(X)}(\cdot, K_\bullet)$ is representable. Indeed, this functor carries inductive limits to projective limits, and the category $\mathbf{\Delta}(C^\wedge)$ is a topos. Denote by $P_\bullet$ an object representing this functor. For an object Z of $C^\wedge$, likewise denote by $j_n^+(Z)$ the object representing the functor $L_\bullet \mapsto \mathrm{Hom}(L_n, Z)$ on $\mathbf{\Delta}C^\wedge$, whose construction by $\varprojlim$ is made explicit in III 1.1. Observe here that this construction involves only finite projective limits. It follows immediately that j_n^+ carries semi-representable presheaves to semi-simplicial presheaves having semi-representable components, and covering morphisms $Z' \to Z$ to special morphisms (7.3.3).

By the definition of $P_\bullet$, there is a cartesian square:

$$\begin{tikzcd} P_\bullet \arrow[r] \arrow[d] & j_n^+(X) \arrow[d, "j_n^+(u)"] \\ K_\bullet \arrow[r] & j_n^+(K_n) \end{tikzcd} .$$

By what has just been observed, the semi-simplicial objects $j_n^+(X)$ and $j_n^+(K_n)$ are semi-representable, and the morphism $j_n^+(u)$ is a special morphism of type n. The result now follows by applying 3.4 and the sorites 7.3.0.

58 . Let $M_\bullet$ and $N_\bullet$ be two semi-simplicial presheaves. The functor

$$L_\bullet \longmapsto \mathrm{Hom}(L_\bullet \times M_\bullet, N_\bullet)$$

is representable. The semi-simplicial presheaf representing it will be denoted by

$$\mathbf{Hom}_\bullet(M_\bullet, N_\bullet).$$

The degree-n component presheaf of this object will be denoted by $\mathbf{Hom}_n(M_\bullet, N_\bullet)$. There is a canonical isomorphism

$$\mathbf{Hom}_n(M_\bullet, N_\bullet) \xrightarrow{\sim} \mathbf{Hom}_0(M_\bullet \times \Delta_n^c, N_\bullet).$$

Lemma 7.3.6. *Let*

$$\begin{tikzcd} M_\bullet \arrow[r, hook] & N_\bullet \\ \mathrm{sk}_n \Delta_{n+1}^c \arrow[u] \arrow[r, hook, "u"] & \Delta_{n+1}^c \arrow[u] \end{tikzcd} \qquad (u \text{ the canonical injection})$$

be a cocartesian diagram of semi-simplicial presheaves.

For every hypercovering $L_\bullet$, the morphism

$$\mathbf{Hom}_0(N_\bullet, L_\bullet) \longrightarrow \mathbf{Hom}_0(M_\bullet, L_\bullet)$$

is covering.

Proof. There is a cartesian diagram:

$$\begin{tikzcd}
\mathbf{Hom}_0(N_\bullet, L_\bullet) \arrow[r] \arrow[d] & \mathbf{Hom}_0(M_\bullet, L_\bullet) \arrow[d] \\
\mathbf{Hom}_0(\Delta^c_{n+1}, L_\bullet) \arrow[r, "(*)"] & \mathbf{Hom}_0(\mathrm{sk}_n\, \Delta^c_{n+1}, L_\bullet).
\end{tikzcd}$$

It therefore suffices to show that the morphism $(*)$ is covering. But $(*)$ is isomorphic to the morphism

$$L_{n+1} \longrightarrow (\mathrm{cosk}_n\, L_\bullet)_{n+1},$$

which is covering by hypothesis. C.Q.F.D.

7.3.7. Proof of assertion 1) of 7.3.2. The product of two objects of HR_p (resp. of HR) is again an object of HR_p (resp. of HR) (7.3.4). To prove that the category $\mathscr{H}\mathscr{R}^0_p$ (resp. $\mathscr{H}\mathscr{R}^0$) is filtered, it suffices to show that, given two morphisms in HR_p (resp. HR)

$$\begin{tikzcd}
K_\bullet \arrow[r, shift left, "u_0"] \arrow[r, shift right, "u_1"'] & L_\bullet
\end{tikzcd}$$

there exists a morphism $v : M_\bullet \to K_\bullet$ in HR_p (resp. in HR) such that the morphisms

$$\begin{tikzcd}
M_\bullet \arrow[r, shift left, "u_0 v"] \arrow[r, shift right, "u_1 v"'] & L_\bullet
\end{tikzcd}$$

are homotopic, i.e. such that there exists $w : M_\bullet \times \Delta^c_1 \to L_\bullet$ making the diagrams 59

(x)
$$\begin{tikzcd}
M_\bullet \arrow[r, "e_i"] \arrow[d, "v"'] & M_\bullet \times \Delta^c_1 \arrow[d, "w"] \\
K_\bullet \arrow[r, "u_i"] & L_\bullet
\end{tikzcd} \qquad i = 0, 1.$$

commute. For every semi-simplicial object $M_\bullet$ of $C^\wedge$ (not necessarily a hypercovering), let $F(M_\bullet)$ be the set of pairs (v, w), where $v : M_\bullet \to K_\bullet$ and $w : M_\bullet \times \Delta^c_1 \to L_\bullet$, such that the diagrams (x) commute. The functor $M_\bullet \mapsto F(M_\bullet)$ is contravariant in $M_\bullet$, carries inductive limits to projective limits, and is consequently represented by a semi-simplicial object $F_\bullet$. The object $F_\bullet$ is clearly the upper-left vertex of a cartesian diagram

$$\begin{tikzcd}
F_\bullet \arrow[r] \arrow[d] & \mathbf{Hom}_\bullet(\Delta^c_1, L_\bullet) \arrow[d, "\pi"] \\
K_\bullet \arrow[r, "{u_0, u_1}"'] & L_\bullet \times L_\bullet,
\end{tikzcd}$$

where π is defined by the two inclusions of Δ^c_0 into Δ^c_1. To prove the assertion, it therefore suffices to show that $F_\bullet$ is an object of HR_p (resp. of HR). For this, by 7.3.4 2) and the sorites 7.3.0, it suffices to show that

60 a) $\mathbf{Hom}_\bullet(\varDelta^c_1, L_\bullet)$ is semi-representable,

b) the morphism π is special of type p (resp. special).

First, it follows immediately that $\mathbf{Hom}_\bullet(\varDelta^c_1, L_\bullet)$ is a semi-representable semi-simplicial object. Indeed, its components are computed by finite projective limits from the L_n, and are therefore semi-representable. Let us now verify b). First one shows that, when $L_\bullet$ is special of type p, the morphism

$$\mathbf{Hom}_\bullet(\varDelta^c_1, L_\bullet) \longrightarrow \mathrm{cosk}_p \mathbf{Hom}_\bullet(\varDelta^c_1, L_\bullet)$$

is an isomorphism; this verifies property 1) of 7.3.3. Next, the morphism

$$\pi_0 : \mathbf{Hom}_0(\varDelta^c_1, L_\bullet) \longrightarrow L_0 \times L_0$$

is isomorphic to the canonical morphism

$$L_1 \xrightarrow{(d_0, d_1)} L_0 \times L_0,$$

which is covering by hypothesis ($L_\bullet$ is a hypercovering).

It remains to verify property 2) of 7.3.3, i.e. to verify that, for every n, in the diagram

$$\begin{tikzcd}
\mathbf{Hom}_{n+1}(\varDelta^c_1, L_\bullet) \arrow[rr, "\pi_{n+1}"] \arrow[dd] \arrow[dr, "\Phi_{n+1}"'] & & L_{n+1} \times L_{n+1} \arrow[dd] \\
& P_{n+1} \arrow[ur] \arrow[dl] & \\
(\mathrm{cosk}_n \mathbf{Hom}_\bullet(\varDelta^c_1, L_\bullet))_{n+1} \arrow[rr, "(\mathrm{cosk}_n \pi)_{n+1}"] & & (\mathrm{cosk}_n L_\bullet)_{n+1} \times (\mathrm{cosk}_n L_\bullet)_{n+1}
\end{tikzcd}$$

(P_{n+1} is the fiber product), the morphism Φ_{n+1} is covering. A simple "adjoint-functor chase" shows that

$$P_{n+1} \xrightarrow{\sim} \mathbf{Hom}_0\big(\mathrm{sk}_{n+1}(\varDelta^c_{n+1} \times \varDelta^c_1), L_\bullet\big),$$
$$\mathbf{Hom}_{n+1}(\varDelta^c_1, L_\bullet) \xrightarrow{\sim} \mathbf{Hom}_0(\varDelta^c_{n+1} \times \varDelta^c_1, L_\bullet),$$

61 and that the morphism Φ_{n+1} is isomorphic to the morphism

$$\mathbf{Hom}_0(\varDelta^c_{n+1} \times \varDelta^c_1, L_\bullet) \longrightarrow \mathbf{Hom}_0\big(\mathrm{sk}_{n+1}(\varDelta^c_{n+1} \times \varDelta^c_1), L_\bullet\big)$$

arising from the injection

$$\mathrm{sk}_{n+1}(\varDelta^c_{n+1} \times \varDelta^c_1) \hookrightarrow \varDelta^c_{n+1} \times \varDelta^c_1.$$

There is a sequence of subobjects of $\varDelta^c_{n+1} \times \varDelta^c_1$:

$$\mathrm{sk}_{n+1}(\varDelta^c_{n+1} \times \varDelta^c_1) = M_0 \hookrightarrow M_1 \hookrightarrow \cdots \hookrightarrow M_k = \varDelta^c_{n+1} \times \varDelta^c_1,$$

such that, for every $0 \leq i < k$, the morphism

$$M_i \hookrightarrow M_{i+1}$$

fits into a cocartesian diagram:

$$\begin{tikzcd} M_i \arrow[r, hook] & M_{i+1} \\ \mathrm{sk}_{n+1}\,\underline{\Delta}^c_{n+2} \arrow[u] \arrow[r, hook] & \underline{\Delta}^c_{n+2} \arrow[u] \end{tikzcd} .$$

(The nondegenerate simplices of dimension $n+2$ in $\Delta_{n+1}\times\Delta_1$ are added one after another.) By 7.3.6, the morphism Φ_{n+1} is a composite of covering morphisms and is therefore itself covering. This completes the proof of 7.3.2.

7.4. The isomorphism theorem.

7.4.0. Let F be a presheaf of abelian groups on a site C satisfying condition 7.3.0. For every hypercovering $K_\bullet$, put

$$H^q(K_\bullet, F) = H^q(\mathrm{Hom}_{C^\wedge}(K_\bullet, F)).$$

The $H^q(K_\bullet, F)$ form a δ-functor on the category of abelian presheaves on C, but they are not, in general, the derived functors of the functor $H^0(K_\bullet, F)$.

Now put 62

(7.4.0.1) $$\overset{r}{\check{H}}{}^q\,(C, F) = \varinjlim_{HR_r} H^q(\cdot, F) \ (r \text{ may be infinite}),$$

and

(7.4.0.2) $$\check{\check{H}}{}^q\,(C, F) = \varinjlim_{HR} H^q(\cdot, F).$$

The $\overset{r}{\check{H}}{}^q\,(C, F)$ (resp. $\check{\check{H}}{}^q\,(C, F)$) again form a δ-functor, because the category $\mathscr{HR}_r^0$ (resp. $\mathscr{HR}^0$) is filtered (7.3.2). Since the categories $\mathscr{HR}_r$ are not necessarily $\mathscr{U}$-categories, these δ-functors take values in the category of abelian $\mathscr{V}$-groups, for a suitable universe $\mathscr{V}$.

Suppose now that the presheaf F is a sheaf. Since the semi-simplicial sheaf associated with a hypercovering is acyclic (7.3.2), there is a spectral sequence, functorial in F and in $K_\bullet$:

(7.4.0.3) $$H^p(K_\bullet, \mathscr{H}^q(F)) \Rightarrow H^{p+q}(C^\sim, F).$$

Passing to the limit gives the spectral sequences

(7.4.0.4) $$\begin{gathered} \overset{r}{\check{H}}{}^p\,(C, \mathscr{H}^q(F)) \Rightarrow H^{p+q}(C^\sim, F), \\ \check{\check{H}}{}^p\,(C, \mathscr{H}^q(F)) \Rightarrow H^{p+q}(C^\sim, F). \end{gathered}$$

Theorem 7.4.1. *Let C be a site satisfying condition 7.3.0, and let F be an abelian sheaf on C.*

1) *The spectral sequences (7.4.0.4) define isomorphisms*

$$\overset{r}{\check{H}}{}^q\,(C, F) \xrightarrow{\sim} H^q(C^\sim, F), \quad q \le r+1,$$

and a monomorphism

$$\overset{r}{\check{H}}{}^{r+2}\ (C,F) \longrightarrow H^{r+2}(C^{\sim},F).$$

63 2) *There is an isomorphism of δ-functors:*

$$\overset{\infty}{\check{H}}{}^{q}\ (C,F) \xrightarrow{\sim} \check{\check{H}}{}^{q}\ (C,F) \xrightarrow{\sim} H^{q}(C^{\sim},F) \quad (\textit{for every } q).$$

3) *Let G be a presheaf of abelian groups, and let F be its associated sheaf. There are functorial isomorphisms*

$$\overset{r}{\check{H}}{}^{q}\ (C,G) \xrightarrow{\sim} H^{q}(C^{\sim},F), \quad q \le r-1,$$

and a monomorphism

$$\overset{r}{\check{H}}{}^{r}\ (C,G) \longrightarrow H^{r}(C^{\sim},F).$$

When G is a separated presheaf, the latter morphism is an isomorphism, and there is a monomorphism

$$\overset{r}{\check{H}}{}^{r+1}\ (C,G) \longrightarrow H^{r+1}(C^{\sim},F).$$

4) *There are isomorphisms*

$$\overset{\infty}{\check{H}}{}^{q}\ (C,G) \xrightarrow{\sim} \check{\check{H}}{}^{q}\ (C,G) \xrightarrow{\sim} H^{q}(C^{\sim},F) \quad (\textit{for every } q).$$

Proof. It suffices to show that, if N is an abelian presheaf whose associated sheaf is zero, then $\overset{r}{\check{H}}{}^{q}\ (C,N) = 0$ for $q \le r$ (resp. $\check{\check{H}}{}^{q}\ (C,N) = 0$ for every q); this follows immediately by using 3.2.

Remark 7.4.2. 1) Note that, for sites satisfying condition 7.3.0, hypercoverings of type 0 are ordinary coverings, and the functors $\overset{r}{\check{H}}{}^{q}$ for $r = 0$ are none other than the functors $\check{H}^{q}$ introduced in (2.4.5.1).

2) Let C be a site in which finite projective limits are representable, and suppose that for every object X of C and every covering family $(X_i \to X)$, $i \in I$, there exists an $i_0 \in I$ such that $X_{i_0} \to X$ is covering. One can then show that hypercoverings of type p whose components are representable are cofinal in $\mathscr{H}\mathscr{R}_p$. Likewise, hypercoverings $K_\bullet$ for which there exists an integer p such that
64 $K_\bullet$ is of type p, and whose components are representable, are cofinal in $\mathscr{H}\mathscr{R}$. (This may be proved by following the argument of 3.5.) Thus, in the case under consideration, these hypercoverings with representable components suffice to compute the cohomology of the sheaves associated with presheaves.

8. Appendix. Local direct limits (by P. Deligne)

The editor recommends that the reader should, as a rule, avoid reading this appendix. It presents a technique which sometimes makes it possible to extend to topoi without enough points assertions which the existence of points makes trivial. This technique yields a sheaf-theoretic variant of D. Lazard's theorem asserting that flat modules over a ring are direct limits of finite free modules.

8.1. Locally filtering categories.

8.1.0. If $\mathscr{B}$ is a category and $p : \mathscr{A} \to \mathscr{B}$ is a category over $\mathscr{B}$, we shall use the following notation:

- For $U \in \operatorname{Ob}\mathscr{B}$, $\mathscr{A}_U$ is the fiber category $p^{-1}(U)$.
- For $f : U \to V$ in $\mathscr{B}$ and $\lambda, \mu \in \operatorname{Ob}\mathscr{A}$ such that $p(\lambda) = U$ and $p(\mu) = V$, put

$$\operatorname{Hom}_f(\lambda, \mu) = p^{-1}(f), \quad \text{where } p : \operatorname{Hom}(\lambda, \mu) \longrightarrow \operatorname{Hom}(U, V).$$

Definition 8.1.1. *Let $\mathscr{S}$ be a site. A* locally cofiltering category (or locally left filtering category) *over $\mathscr{S}$ is a category $p : \mathscr{L} \to \mathscr{S}$ over $\mathscr{S}$ such that:*

($\mathscr{L}0$) *For every $f : U \to V$ in $\mathscr{S}$ and every $\mu \in \operatorname{Ob}\mathscr{L}_V$, there exists $\lambda \in \operatorname{Ob}\mathscr{L}_U$ such that $\operatorname{Hom}_f(\lambda, \mu) \neq \varnothing$;*

($\mathscr{L}1$) *For every $U \in \operatorname{Ob}\mathscr{S}$ and every finite family (λ_i) of objects of $\mathscr{L}_U$, there exist a covering $f_j : U_j \to U$ of U and objects $\mu_j \in \operatorname{Ob}\mathscr{L}_{U_j}$ such that $\operatorname{Hom}_{f_j}(\mu_j, \lambda_i) \neq \varnothing$ for all i and j;*

($\mathscr{L}2$) *For every $f : U \to V$ in $\mathscr{S}$ and every pair of parallel arrows $(\phi_0, \phi_1) : \lambda \rightrightarrows \mu$ over f, there exist a covering $f_j : U_j \to U$ of U and arrows ψ_j with target λ such that $p(\psi_j) = f_j$ and $\phi_0\psi_j = \phi_1\psi_j$.*

. This definition simplifies when $\mathscr{L}$ is fibered over $\mathscr{S}^*$: the axiom

($\mathscr{L}_*0$) is then satisfied, and ($\mathscr{L}1$), ($\mathscr{L}2$) may be stated as follows:

($\mathscr{L}'1$) For every $U \in \operatorname{Ob}\mathscr{S}$ and every finite family λ_i of objects of $\mathscr{L}_U$, locally on U there exists a μ mapping to all the λ_i.

($\mathscr{L}'2$) For every $U \in \operatorname{Ob}\mathscr{S}$, every pair of parallel arrows $(\phi_0, \phi_1) : \lambda \rightrightarrows \mu$ in $\mathscr{L}_U$ can, locally on U, be equalized by an arrow with target λ.

8.1.2. Let $\mathscr{S}$ be a site, let $\mathscr{L}$ be a category over $\mathscr{S}$, and let e be a stack over $\mathscr{S}$. Denote by $\Gamma\mathsf{e}$ the category of global sections $\mathbf{Hom}^{\mathrm{cart}}_{\mathscr{S}}(\mathscr{S}, \mathsf{e})$ of e. If $\mathscr{S}$ has a final object S, then $\Gamma\mathsf{e}$ "is none other than" e_S.

Definition 8.1.2.1. *The category of* local inverse systems of objects of e indexed by $\mathscr{L}$ *is the category $\mathbf{Hom}_{\mathscr{S}}(\mathscr{L}, \mathsf{e})$.*

Denote by c ("associated constant local inverse system") the composite functor

$$\Gamma\mathsf{e} = \mathbf{Hom}^{\mathrm{cart}}_{\mathscr{S}}(\mathscr{S}, \mathsf{e}) \hookrightarrow \mathbf{Hom}_{\mathscr{S}}(\mathscr{S}, \mathsf{e}) \xrightarrow{u \mapsto u \circ p} \mathbf{Hom}_{\mathscr{S}}(\mathscr{L}, \mathsf{e}).$$

When the dependence on $\mathscr{L}$ must be made explicit, we shall write $c_{\mathscr{L}}$ instead.

Definition 8.1.3. *The local inverse-limit functor, denoted by $\varprojlim_{\lambda \in \mathscr{L}}$, is the partially defined functor right adjoint to the functor c.*

Thus the functor

$$\varprojlim : \mathbf{Hom}_{\mathscr{S}}(\mathscr{L}, \mathsf{e}) \longrightarrow \Gamma\mathsf{e}$$

satisfies

$$\operatorname{Hom}(X, \varprojlim_{\lambda \in \mathscr{L}} X_\lambda) \simeq \operatorname{Hom}(c(X), (X_\lambda)_{\lambda \in \mathscr{L}}).$$

Definition 8.1.4. *An $\mathcal{S}$-functor $F : \mathcal{L} \to \mathcal{M}$ between locally cofiltering categories over $\mathcal{S}$ is said to be cofinal if it satisfies the following two conditions:*

65 ($C\mathfrak{F}1$) *For every $U \in \operatorname{Ob}\mathcal{S}$ and $\mu \in \operatorname{Ob}\mathcal{M}_U$, there exist a covering $f_j : U_j \to U$ of U and objects $\lambda_j \in \operatorname{Ob}\mathcal{L}_{U_j}$ such that $\operatorname{Hom}_{f_j}(F(\lambda_j), \mu) \neq \varnothing$.*

($C\mathfrak{F}2$) *For every $f : U \to V$ in $\mathcal{S}$ and every pair of parallel arrows $(\phi_0, \phi_1) : F(\lambda) \rightrightarrows \mu$ over f, there exist a covering $f_j : U_j \to U$ of U and arrows ψ_j with target λ in $\mathcal{L}$, such that $p(\psi_j) = f_j$ and $\phi_0 F(\psi_j) = \phi_1 F(\psi_j)$.*

The composite of two cofinal functors is cofinal; equivalences of locally filtering categories are cofinal (the proof is left to the reader).

Proposition 8.1.5. *Let $F : \mathcal{L} \to \mathcal{M}$ be a cofinal functor between locally cofiltering categories over $\mathcal{S}$, let e be a stack over $\mathcal{S}$, and let $(X_\mu)_{\mu\in\mathcal{M}}$ be a local inverse system indexed by $\mathcal{M}$. Then the morphism, functorial in $X \in \Gamma$e,*

$$F^* : \operatorname{Hom}(c_{\mathcal{M}}(X), (X_\mu)_{\mu\in\mathcal{M}}) \longrightarrow \operatorname{Hom}(c_{\mathcal{L}}(X), (X_{F(\lambda)})_{\lambda\in\mathcal{L}})$$

is an isomorphism, so that

$$F^* : \varprojlim_{\mathcal{M}} X_\mu \longrightarrow \varprojlim_{\mathcal{L}} X_{F(\lambda)}$$

is an isomorphism, the two sides being simultaneously defined or undefined.

The proof uses the following lemma, which is left to the reader:

Lemma 8.1.6. *If $F : \mathcal{L} \to \mathcal{M}$ is cofinal, then for every $f : U \to V$ in $\mathcal{S}$ and arrows over f, $\phi_i : F(\lambda_i) \to \mu$ $(i = 1, 2)$, there exist a covering $f_j : U_j \to U$ of U, objects $\lambda'_j \in \operatorname{Ob}\mathcal{L}_{U_j}$, and commutative diagrams*

$$\begin{tikzcd}
 & F(\lambda_1) \arrow[dr, "\phi_1"] & \\
F(\lambda'_j) \arrow[ur, dashed] \arrow[dr, dashed] & & \mu \\
 & F(\lambda_2) \arrow[ur, "\phi_2"'] &
\end{tikzcd}$$

$$U_j \xrightarrow{f_j} U \longrightarrow V$$

whose projection in $\mathcal{S}$ is (f_j, f).

For every $U \in \operatorname{Ob}\mathcal{S}$ and $\mu \in \mathcal{M}_U$, condition ($C\mathfrak{F}1$) gives a covering $f_j : U_j \to U$ of U and arrows $\phi_j : F(\lambda_j) \to \mu$ over the f_j. If

$$\psi = (\psi_\lambda) \in \operatorname{Hom}(c_{\mathcal{L}}X, (X_{F(\lambda)})_{\lambda\in\mathcal{L}}) : \quad \psi_\lambda : X|p(\lambda) \longrightarrow X_{F(\lambda)}$$

is the image of

$$\phi = (\phi_\mu) \in \operatorname{Hom}(c_{\mathcal{M}}X, (X_\mu)_{\mu\in\mathcal{M}}) : \quad \phi_\mu : X|p(\mu) \longrightarrow X_\mu,$$

then the diagrams

$$\begin{tikzcd}
 & X_{F(\lambda_j)} \arrow[dl, "\psi_{\lambda_j}"'] \arrow[dr, "\phi_j"] & \\
X|U_j \arrow[rr, "\phi_\mu|U_j"'] & & X_\mu|U_j
\end{tikzcd}$$

commute. Thus the ϕ_μ are locally determined by ψ; since e is a stack, it follows that F^* is injective. To prove that F^* is surjective, start with ψ and construct a ϕ such that $\psi = F^*\phi$. The preceding construction gives the composite arrows

$$\phi_{\mu,j} = \phi_j \circ \psi_{\lambda_j} : X|U_j \longrightarrow X_\mu|U_j.$$

Let us verify that these arrows glue, so that there exists $\phi_\mu : X|U \to X_\mu$ such that $\phi_\mu|U_j = \phi_{\mu,j}$. Consider a commutative diagram in

$$\begin{tikzcd}
 & U_{ij} \arrow[dl, "g_i"'] \arrow[dr, "g_j"] & \\
U_i \arrow[dr, "f_i"'] & & U_j \arrow[dl, "f_j"] \\
 & U &
\end{tikzcd}$$

To verify that $\phi_{\mu,j}|U_{ij} = \phi_{\mu,i}|U_{ij}$, it is enough to do so locally on U_{ij}; by 8.1.6 and ($\mathscr{L}0$) 66
this reduces us to the case where there is a commutative diagram

$$\begin{tikzcd}
 & F(\lambda_{ij}) \arrow[dl, "F(a_{ij})"'] \arrow[dr, "F(a_{ji})"] & \\
F(\lambda_i) \arrow[dr, "\phi_i"'] & & F(\lambda_j) \arrow[dl, "\phi_j"] \\
 & \mu &
\end{tikzcd}$$

over the preceding one. The diagram

$$\begin{tikzcd}
 & X_{F(\lambda_{ij})} \arrow[dl] \arrow[dr] \arrow[dd, bend right] & \\
X_{F(\lambda_i)} \arrow[dr] & X|U_{ij} \arrow[l] \arrow[r] \arrow[u] & X_{F(\lambda_j)} \arrow[dl] \\
 & X_\mu &
\end{tikzcd}$$

is then commutative, and there exists a family (ϕ_μ) such that, for every arrow $\phi : F(\lambda) \to \mu$ over $f : V \to U$, the diagram

$$\begin{tikzcd}
 & X_{F(\lambda)} \arrow[dr, "\phi"] & \\
X|V \arrow[ur, "\psi_\lambda"] \arrow[rr, "\phi_\mu|V"'] & & X_\mu|V
\end{tikzcd}$$

commutes. If $\sigma : \mu \to \mu'$ is an arrow of $\mathscr{M}$, then $(\sigma|V)(\phi_\mu|V) = (\sigma|V)\phi \cdot \psi_\lambda = (\sigma\phi|V) \cdot \psi_\lambda = \phi_{\mu'}|V$; hence $\phi = (\phi_\mu)$ is a morphism of functors such that $\psi = F^*\phi$, as required.

67 If E is an ordered set, we shall also denote by E the category whose set of objects is E and in which $\mathrm{Hom}(i, j)$ is a singleton or empty according as $i \leq j$ or not.

PROPOSITION 8.1.7. *Let $\mathscr{L}$ be a locally cofiltering category over a site $\mathscr{S}$. There exist an ordered set E and a functor F from E to $\mathscr{L}$ such that:*

(i) *E, regarded as a category over $\mathscr{S}$ by means of $pF : E \to \mathscr{S}$, is locally cofiltering.*
(ii) *The functor F is cofinal.*

Let $\mathscr{L}'$ be the locally cofiltering category which is the product of $\mathscr{L}$ with the category defined by the ordered set $\mathbf{Z}$. The projection from $\mathscr{L}'$ to $\mathscr{L}$ is cofinal, so it suffices to prove 8.1.7 for $\mathscr{L}'$.

If X is a finite set of arrows in $\mathscr{L}'$, denote by X^0 the set of endpoints of the arrows in X. Let E be the set of nonempty finite subsets X of $\mathrm{Fl}(\mathscr{L}')$ for which there exists $\lambda_X \in \mathrm{Ob}\,\mathscr{L}'$ satisfying:

a) no arrow of X, except 1_{λ_X}, has target λ_X;
b) if $\mu \in X^0$, then $1_\mu \in X$ and $\mathrm{Hom}(\lambda_X, \mu) \cap X \neq \varnothing$;
c) every diagram of the form

$$\begin{tikzcd}
\lambda_X \arrow[r] \arrow[dr] & \mu \arrow[d] \\
 & \mu'
\end{tikzcd}$$

consisting of arrows of X is commutative.

Order E by the relation opposite to inclusion. If $X \in E$ and $\mu \in X^0$, there is a unique arrow from λ_X to μ which belongs to X. If $X, Y \in E$ and $X \supset Y$, let $\lambda_{X,Y}$ be the unique arrow in X from λ_X to λ_Y. The assignments $X \mapsto \lambda_X$ and $(X \supset Y) \mapsto \lambda_{X,Y}$ form a functor from E to $\mathscr{L}'$.

68 LEMMA 8.1.8. *The category E is locally cofiltering over $\mathscr{S}$, and the functor $\lambda : E \to \mathscr{L}'$ is cofinal.*

We must verify the following axioms successively:

(i) *axiom* ($\mathscr{L}0$) Let $f : V \to U$ and $X \in E_U$. There exist $\lambda \in \mathrm{Ob}\,\mathscr{L}'_V$ and $\phi \in \mathrm{Hom}_f(\lambda, \lambda_X)$. Moreover, λ may be chosen so that $\lambda \notin X^0$ (this is where $\mathscr{L}' = \mathscr{L} \times \mathbf{Z}$ is used).

Let $X' = X \cup \{1_\lambda\} \cup Y$, where Y is the set of composite arrows $\lambda \xrightarrow{\phi} \lambda_X \xrightarrow{\psi} \mu$ ($\psi \in X$). Then $X' \in E$, $\lambda_{X'} = \lambda$, and $\mathrm{Hom}(X', X) \neq \varnothing$.

(ii) *axiom* ($\mathscr{L}1$) Let (X_i) be a finite family of objects of E_U. There exist a covering $f_j : U_j \to U$, objects $\lambda_j \in \operatorname{Ob} \mathscr{L}'_{U_j}$, and arrows $\phi_{ji} \in \operatorname{Hom}_{f_j}(\lambda_j, \lambda_{X_i})$. If $x \in X_a^0 \cap X_b^0$, then for each j there exist a covering $g_{jk} : U_{jk} \to U_j$ and arrows $\psi_{jk} : \lambda_{jk} \to \lambda_j$ such that $p(\psi_{jk}) = g_{jk}$ and which equalize the pair of arrows $\lambda_j \rightrightarrows x$, one belonging to X_a and the other to X_b. We may arrange that the λ_{jk} belong to none of the X_i^0. Replace the $(f_j, \lambda_j, \phi_{ji})$ by the $(f_j g_{jk}, \lambda_{jk}, \phi_{ji}\psi_{jk})$, and repeat this construction for every x in an intersection of two of the X_i. We obtain a new system $(f_j, \lambda_j, \phi_{ji})$; put

$$X_j = \bigcup_i X_i \cup \{1_{\lambda_j}\} \cup Y_j,$$

where Y_j is the set of composite arrows

$$\lambda_j \xrightarrow{\phi_{ji}} \lambda_{X_i} \xrightarrow{\psi} x \quad (\psi \in X_i).$$

Then $X_j \in E_{U_j}$, and these X_j satisfy ($\mathscr{L}1$).

(iii) *axiom* ($\mathscr{L}2$) Trivial, since there are no nontrivial parallel arrows!

(iv) *axiom* ($C\mathfrak{F}1$) Trivial, since the functor λ is surjective.

(v) *axiom* ($C\mathfrak{F}2$) Let $f : U \to V$ and let $(\phi_0, \phi_1) : \lambda_X \rightrightarrows \lambda$ be a pair of parallel arrows over f. There exist a covering $f_j : U_j \to U$ of U and arrows $\psi_j : \lambda_j \to \lambda_X$ such that $p(\psi_j) = f_j$, $\phi_0\psi_j = \phi_1\psi_j$, and $\lambda_j \notin X^0$. Let $X_j = X \cup \{1_{\lambda_j}\} \cup Y_j$, where Y_j is the set of composite arrows

$$\lambda_j \xrightarrow{\psi_j} \lambda_X \xrightarrow{\psi} \mu \quad (\psi \in X).$$

Then $X_j \in E_{U_j}$, and $F(\lambda_{X_j,X}) = \psi_j$ equalizes (ϕ_0, ϕ_1). 69

. Suppose that $\mathscr{S}$ is (the site of nonempty open subsets of) the one-point topological space. A locally cofiltering category over $\mathscr{S}$ is then simply a left filtering category (i.e. one for which $\mathscr{S}^0$ is filtering in the sense of I 2.7), and cofinal functors correspond to cofinal functors between left filtering categories (I 8). Proposition 8.1.7 may then be restated as follows.

Proposition 8.1.9. *For every cofiltering category $\mathscr{L}$, there exist a cofiltering ordered set E and a cofinal functor from E to $\mathscr{L}$.*

In this special case, the proof of 8.1.7 reduces essentially to the proof of 8.1.9 given in I 8.

8.1.10. Let $q : \mathscr{T} \to \mathscr{S}$ be a morphism of sites, and let $p : \mathscr{L} \to \mathscr{S}$ be a locally cofiltered category over $\mathscr{S}$. Define the category $\mathscr{L}^*$ as follows:

(i) An object of $\mathscr{L}^*$ is a quadruple (V, U, ϕ, λ) such that $V \in \operatorname{Ob} \mathscr{T}$, $U \in \operatorname{Ob} \mathscr{S}$, $\phi \in \operatorname{Hom}(V, q^*U)$, and $\lambda \in \operatorname{Ob} \mathscr{L}_U$.

(ii) An arrow $f : (V, U, \phi, \lambda) \to (V', U', \phi', \lambda')$ is a triple (f_1, f_2, f_3), where $f_1 \in \operatorname{Hom}(V, V')$, $f_2 \in \operatorname{Hom}(U, U')$, and $f_3 \in \operatorname{Hom}(\lambda, \lambda')$, such that $q^*f_2 \circ \phi = \phi' \circ f_1$ and $p(f_3) = f_2$.

The functor $f \mapsto f_1$ makes $\mathscr{L}^*$ a category over $\mathscr{T}$.

Lemma 8.1.11. *The category $\mathscr{L}^*$ is locally cofiltered over $\mathscr{T}$.*

Axiom ($\mathscr{L}0$) is trivial. Let us verify ($\mathscr{L}1$).

Let $L_i = (V, U_i, \phi_i, \lambda_i)$ be a finite family. The ϕ_i define a morphism from V to the sheaf $q^* a \prod U_i = a \prod q^* U_i$. Consequently, there are commutative diagrams

$$\begin{tikzcd}
V_j \arrow[r, "f_j"] \arrow[d, "\psi_j"] & V \arrow[d, "\phi_i"] \\
q^*U_j \arrow[r, "q^*(p_{ij})"'] & q^*U_i
\end{tikzcd}$$

70 such that the f_j cover V.

Choose λ_{ji} such that $\mathrm{Hom}_{p_{ij}}(\lambda_{ji}, \lambda_i) \neq \varnothing$. By ($\mathscr{L}1$), after refining U_j (and V_j), we may take $\lambda_{ji} = \lambda_j$ independently of i, and the $L_j = (V_j, U_j, \psi_j, \lambda_j)$ satisfy ($\mathscr{L}1$).

Let us verify ($\mathscr{L}2$). Let $(f_1, f_2) : L^1 \rightrightarrows L^2$, with $L^i = (V^i, U^i, \phi^i, \lambda^i)$ and $f_i = (f, g_i, h_i)$. The ϕ^i define a morphism from V^1 to the sheaf $q^* a \operatorname{Ker}(U^1 \rightrightarrows U^2) = a \operatorname{Ker}(q^*U^1 \rightrightarrows q^*U^2)$, so there are commutative diagrams

$$\begin{tikzcd}
V_j \arrow[r, "f_j"] \arrow[d, dash, "\psi_j"] & V^1 \arrow[r] \arrow[d, dash, "\phi^1"] & V^2 \arrow[d, dash, "\phi^2"] \\
q^*U_j \arrow[r, "q^*p_j"'] & q^*U^1 \arrow[r, shift left, "q^*g_1"] \arrow[r, shift right, "q^*g_2"'] & q^*U^2
\end{tikzcd} \qquad (g_1 p_j = g_2 p_j),$$

such that the f_j cover V^1. Let $\chi_j : \lambda_j \to \lambda^1$ satisfy $p(\chi_j) = p_j$. Applying ($\mathscr{L}2$) to $(h_1\chi_j, h_2\chi_j)$, and refining U_j (and V_j) if necessary, we may arrange that $h_1\chi_j = h_2\chi_j$; the $L_j = (V_j, U_j, \psi_j, \lambda_j)$ then satisfy ($\mathscr{L}2$).

PROPOSITION 8.1.12. *Let $t^* : \mathcal{S}^* \to \mathcal{S}$ be a locally cofiltered category over $\mathcal{S}$, and endow $\mathcal{S}^*$ with the topology induced from that of $\mathcal{S}$ by t^*. Then t^* is an equivalence of sites $t : \mathcal{S} \to \mathcal{S}^*$.*

By construction, the functor $t^* : \mathscr{F} \mapsto \mathscr{F} \circ t^*$ carries sheaves on $\mathcal{S}$ to sheaves on $\mathcal{S}^*$. By ($\mathscr{L}0$) and ($\mathscr{L}1$) applied to the empty family, every "sufficiently small" object of $\mathcal{S}$ belongs to the image of t^* (that is, the image of t^* is a *covering sieve* in $\mathcal{S}$). Hence, by the "comparison lemma" (III 4), the inclusion into $\mathcal{S}$ of the full subcategory defined by $t^*(\mathrm{Ob}\,\mathcal{S}^*)$ is an equivalence of sites. We may therefore suppose that t^* is surjective on objects.

Let $\mathscr{F}$ be a sheaf on $\mathcal{S}^*$.

(i) Let $(f_j : U_j \to U)$ be a covering in $\mathcal{S}$, let $\lambda^1, \lambda^2 \in \mathrm{Ob}\,\mathcal{S}^*_U$, and let $f^i_j \in \mathrm{Hom}_{f_j}(\mu_j, \lambda^i)$ for $i = 1, 2$. Then $\mathscr{F}(\lambda^1)$ and $\mathscr{F}(\lambda^2)$ have the same image in $\prod\limits_j \mathscr{F}(\mu_j)$.

71 Indeed, let $x = (x_j)$ belong to the image of $\mathscr{F}(\lambda^1)$. For x to belong to the

image of $\mathscr{F}(\lambda^2)$, it suffices that, for every commutative diagram

$$
\begin{tikzcd}
\mu \arrow[r] \arrow[d] & \mu_j \arrow[d] \\
\mu_k \arrow[r] & \lambda^2,
\end{tikzcd}
$$

x_j and x_k have the same image in $\mathscr{F}(\mu)$. Since $\mathscr{F}$ is a sheaf, it suffices to verify this after replacing μ by the objects of one of its coverings. By ($\mathscr{L}2$), we may then suppose that the diagrams analogous to the preceding one, with λ^2 replaced by λ^1, also commute, in which case the assertion is clear.

(ii) Let $f : U \to V$ be an arrow of $\mathcal{S}$, let $\lambda \in \operatorname{Ob} \mathcal{S}^*_U$, and let $\mu \in \operatorname{Ob} \mathcal{S}^*_V$. There then exist a covering $f_j : U_j \to U$, an object $\mu' \in \operatorname{Ob} \mathcal{S}^*_U$, and arrows in the diagram

$$
\begin{tikzcd}
 & \lambda & & \\
\lambda_j \arrow[ur, "\phi_j"] \arrow[dr, "\psi_j"'] & & & \mu \\
 & \mu' \arrow[urr, "\psi"'] & &
\end{tikzcd}
$$

lying over

$$U_j \xrightarrow{f_j} U \xrightarrow{f} V.$$

By (i), there is a unique arrow $\overline{f}$ making the following diagram commute:

$$
\begin{tikzcd}
\mathscr{F}(\mu) \arrow[r, "\overline{f}"] \arrow[d, "\psi"'] & \mathscr{F}(\lambda) \arrow[d, "\phi_j"] \\
\mathscr{F}(\mu') \arrow[r, "\psi_j"'] & \mathscr{F}(\lambda_j).
\end{tikzcd}
$$

The arrow $\overline{f}$ does not change if (f_j) is replaced by a finer covering $f_k : U_k \to U_{j(k)}$, if λ_j is replaced by λ_k equipped with $\phi_k \in \operatorname{Hom}_{f_k}(\lambda_k, \lambda_{j(k)})$, and if ϕ_j (respectively ψ_j) is replaced by $\phi_{j(k)} \circ \phi_k$ (respectively $\psi_{j(k)} \circ \phi_k$). 72

The reader may deduce that $\overline{f}$ depends only on f.

(iii) Let us prove that $\overline{f}$ is functorial in f. Thus let

$$W \xrightarrow{h} V \xrightarrow{g} U$$

be arrows in $\mathcal{S}$, and let λ^0, μ^0, and ν^0 belong respectively to Ob $\mathcal{S}^*_U$, Ob $\mathcal{S}^*_V$, and Ob $\mathcal{S}^*_W$. There exists a commutative diagram

$$\begin{tikzcd}
\nu^0 & & & \mu^0 & & \lambda^0 \\
& \nu^1 \arrow[urr] & & & \mu^1 \arrow[ur] & \\
& & \nu^2 \arrow[urr] & & & \\
& \nu_{jk} \arrow[uuul, "e^0_{jk}"] \arrow[uu, "e^1_{jk}"] \arrow[ur, "e^2_{jk}"'] \arrow[rr] & & \mu_j \arrow[uuu, "f^0_j"] \arrow[uuur, "f^1_j"'] & &
\end{tikzcd}$$

such that

a) $t^*(\nu^i)$, $t^*(\mu^i)$, $t^*(e^i_{jk})$, and $t^*(f^i_j)$ do not depend on i;

b) for fixed i, the e^i_{jk} cover W, and the f^i_j cover V.

First construct μ^1, ν^1, and ν^2 by $(\mathcal{L}0)$, and the μ_j as in (ii). Let (W'_{jk}) be a covering of W mapping to the covering $V_j = t^*(\mu_j)$ as required by the diagram. After refining (W'_{jk}) and applying $(\mathcal{L}0)$ and $(\mathcal{L}1)$, one obtains a diagram of the required type, satisfying a) and b), but not necessarily commutative. Refining once more and applying $(\mathcal{L}2)$ makes it commutative.

It remains only to contemplate the following commutative diagram:

$$\begin{tikzcd}
\mathcal{F}(\lambda^0) \arrow[r, "\bar{g}"] & \mathcal{F}(\mu^0) \arrow[rr, "\bar{h}"] \arrow[dr] & & \mathcal{F}(\nu^0) \arrow[ddd] \\
& & \mathcal{F}(\nu^1) \arrow[ddr] & \\
\mathcal{F}(\mu^1) \arrow[r] & \prod \mathcal{F}(\mu_j) \arrow[drr] & & \\
\mathcal{F}(\nu^2) \arrow[rrr] & & & \prod \mathcal{F}(\nu_{jk})
\end{tikzcd}$$

73 (iv) It is clear that $\overline{f}$ is functorial in $\mathcal{F}$. Thus every sheaf $\mathcal{F}$ on $\mathcal{S}^*$ is the direct image under t of a uniquely determined presheaf $\mathcal{F}_0$ on $\mathcal{S}$, and every morphism of sheaves $f : \mathcal{F} \to \mathfrak{g}$ is the image of a unique morphism of presheaves $f_0 : \mathcal{F}_0 \to \mathfrak{g}_0$.

It already follows that t_* is fully faithful; it remains to prove that $\mathcal{F}_0$ is a sheaf. If $(f_j : U_j \to U)$ is a covering, then this covering is the image of a covering in $\mathcal{S}^*$, so $\mathcal{F}_0(U)$ injects into $\prod_j \mathcal{F}_0(U_j)$. If elements $x_j \in \mathcal{F}_0(U_j)$ glue, then a fortiori they glue in $\mathcal{S}^*$, and hence come from an element of $\mathcal{F}_0(U)$. This completes the proof of 8.1.12.

. Let $q : \mathcal{T} \to \mathcal{S}$ and $p : \mathcal{L} \to \mathcal{S}$ be as in (8.1.10), let $\mathcal{L}^*$ be the category defined there, and let e be a stack on $\mathcal{T}$. The stack $q_*\mathsf{e}$ on $\mathcal{S}$ is the stack defined by $(q_*\mathsf{e})_U = \mathsf{e}_{q^*U}$.

If $(X_\lambda)_{\lambda \in \mathcal{L}}$ is a local projective system indexed by $\mathcal{L}$ and taking values in $q_*\mathsf{e}$, it defines a local projective system $(X_L)_{L \in \mathcal{L}^*}$ taking values in e by the formula

$$X_L = \phi^* X_\lambda quad \text{for } L = (V, U, \phi, \lambda).$$

The reader may verify the following:

Proposition 8.1.13.1. *With the preceding notation, one has*

$$\varprojlim_{\lambda \in \mathscr{L}} X_\lambda = \varprojlim_{L \in \mathscr{L}^*} X_L,$$

the two sides being simultaneously defined or undefined.

Let $\mathscr{S}$ be a site, and let $p^* : \mathscr{S}^* \to \mathscr{S}$ be a locally filtered category over $\mathscr{S}$. If e is a stack on $\mathscr{S}$, then $p_* \mathsf{e}$ is a stack on $\mathscr{S}^*$ (8.1.13.0). Moreover, the identity functor $\mathscr{S}^* \to \mathscr{S}^*$ is a locally filtered category over $\mathscr{S}^*$, endowed with the induced topology, and every local inductive system (X_λ) indexed by $\mathscr{S}^*$ defines a local inductive system over $\mathscr{S}^*$ indexed by $\mathscr{S}^*$.

One verifies easily:

Proposition 8.1.14. *With the preceding notation, one has* 74

$$\varprojlim_{\lambda \in \mathscr{S}^*/\mathscr{S}} X_\lambda = \varprojlim_{\lambda \in \mathscr{S}^*/\mathscr{S}^*} X_\lambda,$$

the two sides being simultaneously defined or undefined.

8.2. Local inductive limits in categories of sheaves.

8.2.0. Let $p : \mathscr{T} \to \mathscr{S}$ be a morphism of sites. Let e be the following stack on $\mathscr{S}$:

(i) An object of e is a pair $(U, \mathscr{F})$ consisting of an object U of $\mathscr{S}$ and a sheaf $\mathscr{F}$ on $p^* U$.

(ii) An arrow $f : (U, \mathscr{F}) \to (V, \mathscr{G})$ is a pair consisting of an arrow $f_0 : U \to V$ and a morphism of sheaves on $p^* U$:

$$f_1 : p^*(f_0)^* \mathscr{G} \longrightarrow \mathscr{F}.$$

(iii) The functor from e to $\mathscr{S}$ is given by $f \longmapsto f_0$.

Care must be taken that e_U is the *opposite category* of the category of sheaves on $p^* U$.

Let $\mathscr{L}$ be a locally cofiltered category (on the left) over $\mathscr{S}$.

Definition 8.2.1. *With the preceding notation, the category of local inductive systems, indexed by $\mathscr{L}$, of sheaves on $\mathscr{T}$ is the category* $\mathrm{Hom}_{\mathscr{S}}(\mathscr{L}, \mathsf{e})^\circ$.

The "projective limit" functor of 8.1.3 is called here the *"local inductive limit" functor*. It is a functor from $\mathrm{Hom}_{\mathscr{S}}(\mathscr{L}, \mathsf{e})^\circ$ to $(\Gamma \mathsf{e})^\circ = (\mathscr{T})^\sim$.

Theorem 8.2.2. *The "local inductive limit" functor from the category of inductive systems of sheaves on $\mathscr{T}$, indexed by $\mathscr{L}$, to the category of sheaves on $\mathscr{T}$ is defined everywhere, commutes with finite projective limits, and commutes with arbitrary inductive limits.*

The propositions 8.1.13, 8.1.14, and 8.1.12 reduce the assertion to the case $\mathscr{S} = \mathscr{T} =$ 75
$\mathscr{L}$. A local inductive system then consists of a sheaf $\mathscr{F}_U$ on $\mathscr{S}/U$ for every $U \in \mathrm{Ob}\, \mathscr{S}$, together, for every arrow $f : V \to U$ in $\mathscr{S}$, with an arrow $f^* \mathscr{F}_U \to \mathscr{F}_V$, functorial in f.

For such inductive systems, the $\varinjlim$ are computed as follows:

Lemma 8.2.3. *Let $U \longmapsto \mathscr{F}_U$ be a section of the category of presheaves on the objects of $\mathscr{S}$. Then*

$$a(U \longmapsto \Gamma(U, \mathscr{F}_U)) \simeq \varinjlim_{U} a\mathscr{F}_U.$$

By definition, the right-hand side represents the functor which assigns to every sheaf $\mathscr{F}$ on $\mathscr{S}$ the set of coherent systems of arrows $\phi_U : \mathscr{F}_U \to \mathscr{F}|U$. In turn, an arrow ϕ_U is a coherent system of arrows ϕ_g, one for every $g : V \to U$, where $\phi_g : \mathscr{F}_U(V) \to \mathscr{F}(V)$. Apply 8.1.11 to the identity functor of $\mathscr{S}$, thereby obtaining $\mathscr{S}^*$ locally filtered over $\mathscr{S}$. We have seen that

$$\varinjlim_{g\in\mathscr{S}^*} \mathscr{F}_U(V_g)^{\sim} \simeq \varinjlim_{U} a\mathscr{F}_U,$$

where V_g is the source of g and $^{\sim}$ denotes the "generated constant sheaf" functor. The functor $U \longmapsto 1_U$ from $\mathscr{S}$ to $\mathscr{S}^*$ is cofinal, and hence, again by 8.1.5,

$$\varinjlim_{U} \mathscr{F}_U(U)^{\sim} \simeq \varinjlim_{U} a\mathscr{F}_U.$$

Returning to the definitions, one finally finds

$$\varinjlim_{U} \mathscr{F}_U(U)^{\sim} = a(U \longmapsto \Gamma(U, \mathscr{F}_U)).$$

This shows that, in the case to which we have reduced, the functor $\varinjlim$ is defined everywhere and is computed as the composite of the "sheaf generated by a presheaf" functor and the functor $(\mathscr{F}_U) \longmapsto (U \longmapsto \Gamma(U, \mathscr{F}_U))$. Both of these functors commute with finite projective limits, and the functor $\varinjlim$ also commutes with arbitrary inductive limits by its definition as an adjoint functor.

76 Lemma 8.2.3 provides a systematic method for proving properties of the local-inductive-limit functor from the analogous properties of the "sheaf associated with a presheaf" functor. One obtains in this way:

PROPOSITION 8.2.4. *Let $\mathscr{T}$, $\mathscr{S}$, and $\mathscr{L}$ be as above, let $\mathscr{A}_\lambda$ be a local inductive system of sheaves of rings on $\mathscr{T}$, and let $\mathscr{L}_\lambda$ (resp. $\mathscr{M}_\lambda$) be a local inductive system of sheaves of right (resp. left) modules over $\mathscr{A}_\lambda$, all three indexed by $\mathscr{L}$.*

Then

$$\varinjlim(\mathscr{L}_\lambda \otimes_{\mathscr{A}_\lambda} \mathscr{M}_\lambda) \simeq \varinjlim \mathscr{L}_\lambda \underset{\varinjlim \mathscr{A}_\lambda}{\otimes} \varinjlim \mathscr{M}_\lambda.$$

(Reduce to the case $\mathscr{T} = \mathscr{S} = \mathscr{L}$, and note that both sides coincide with the sheaf generated by the presheaf

$$\mathscr{L}_U(U) \underset{\mathscr{A}_U(U)}{\otimes} \mathscr{M}_U(U).$$

)

PROPOSITION 8.2.5. *Let $\mathscr{T}_1 \xrightarrow{q} \mathscr{T}_2 \to \mathscr{S}$ be two morphisms of sites, and let (F_λ) be a local inductive system of sheaves on $\mathscr{T}_2$ indexed by a locally cofiltered category $\mathscr{L}$ over $\mathscr{S}$. Then*

$$q^* \varinjlim_{\lambda\in\mathscr{L}} F_\lambda \simeq \varinjlim_{\lambda\in\mathscr{L}} q^* F_\lambda.$$

This follows immediately from the fact that q^* has a right adjoint q_*.

Let $\mathscr{S}$ and $\mathscr{T}$ be two sites in which finite $\varprojlim$ are representable, let $p : \mathscr{T} \to \mathscr{S}$ be a morphism of sites such that $p^* : \mathscr{S} \to \mathscr{T}$ commutes with finite $\varprojlim$, and let $\mathscr{L}$ be the locally cofiltered category over $\mathscr{T}$ whose objects are triples (V, U, ϕ), where $V \in \operatorname{Ob}\mathscr{T}$, $U \in \operatorname{Ob}\mathscr{S}$, and $\phi \in \operatorname{Hom}(V, p^*U)$. It is obtained by applying 8.1.11 to $p : \mathscr{T} \to \mathscr{S}$.

PROPOSITION 8.2.6. *For every sheaf $\mathscr{F}$ on $\mathscr{T}$, one has*

$$\mathscr{F} = \varinjlim_{\lambda\in\mathscr{L}} \phi^*\phi_*\mathscr{F},$$

where, by abuse of notation, ϕ denotes the morphism induced by p from $\mathscr{T}/V$ to $\mathscr{S}/U$.

Let ϕ' be the inverse-image functor in the sense of presheaves. From 8.2.3 one obtains 77

$$\varinjlim_{\lambda\in\mathscr{L}} \phi'^*\phi_*\mathscr{F} = \varinjlim_{\lambda\in\mathscr{L}} a\phi'^*\phi_*\mathscr{F} = \varinjlim_{\lambda\in\mathscr{L}} \mathscr{F}(V)^\sim = \varinjlim_{V} \mathscr{F}(V)^\sim = \mathscr{F}.$$

PROPOSITION 8.2.7. *Let $p : T \to S$ be a morphism of topoi, let $\mathscr{O}_S$ be a sheaf of rings on S, and let $\mathscr{O}_T$ be its inverse image on T. A right $\mathscr{O}_T$-module $\mathscr{M}$ is flat over $\mathscr{O}_T$ if and only if the functor*

$$\mathscr{N} \longmapsto \mathscr{M} \underset{\mathscr{O}_T}{\otimes} p^*\mathscr{N} \tag{8.2.7.1}$$

from the category of left $\mathscr{O}_S$-modules to the category of abelian sheaves on T is exact.

The “only if” assertion is trivial; suppose, therefore, that the functor (8.2.7.1) is exact, and let us prove that $\mathscr{M}$ is flat.

We may suppose that p is defined by a morphism of sites $p : \mathscr{T} \to \mathscr{S}$ satisfying the hypotheses made in 8.2.6. Let $V \in \operatorname{Ob}\mathscr{T}$, $U \in \operatorname{Ob}\mathscr{S}$, and $\phi : V \to p^*U$. We again denote by ϕ the induced morphism of sites $\phi : \mathscr{T}/V \to \mathscr{S}/U$.

LEMMA 8.2.8. *The functor $\mathscr{N} \longmapsto \mathscr{M}|V \otimes_{\mathscr{O}_V} \phi^*\mathscr{N}$, from the category of $\mathscr{O}_S|U$-modules on $\mathscr{S}/U$ to that of $\mathscr{O}_T|V$-modules on $\mathscr{T}/V$, is exact.*

We reduce to taking $V = p^*U$. Let j and j' be the “inclusion morphisms” $j : \mathscr{S}/U \to \mathscr{S}$ and $j' : \mathscr{T}/p^*U \to \mathscr{T}$. One has

$$j'_!(\mathscr{M}|V \underset{\mathscr{O}_V}{\otimes} \phi^*\mathscr{N}) = \mathscr{M} \underset{\mathscr{O}_T}{\otimes} p^*j_!\mathscr{N},$$

which gives 8.2.8, since $j_!$ and $j'_!$ are exact and faithful.

Let now Q be a left $\mathscr{O}_T$-module. By (8.2.6), one has

$$Q = \varinjlim_{\phi:V\to p^*U} \phi^*\phi_*(Q|V),$$

so that (8.2.4)

$$\mathscr{M} \underset{\mathscr{O}_T}{\otimes} Q = \varinjlim_{\phi:V\to p^*U} \mathscr{M}|V \underset{\mathscr{O}_V}{\otimes} \phi^*\phi_*(Q|V).$$

By 8.2.8, the functors $Q \longmapsto \mathscr{M}|V \otimes_{\mathscr{O}_V} \phi^*\phi_*(Q|V)$ are left exact; therefore $\mathscr{M} \otimes_{\mathscr{O}_T} Q$ is left exact in Q, and hence exact. 78

COROLLARY 8.2.9. *Let $f : (T, \mathscr{O}_T) \to (S, \mathscr{O}_S)$ be a morphism of ringed topoi. Then the inverse image by f of a sheaf of flat right modules is a sheaf of flat modules.*

Factoring f through $(T, f^*\mathscr{O}_S)$ reduces us to the case $\mathscr{O}_T = f^*\mathscr{O}_S$. Let $\mathscr{M}$ then be a sheaf of flat modules on S. By 8.2.7, to verify that $f^*\mathscr{M}$ is flat, it is enough to verify the exactness of the functor

$$\mathscr{N} \longmapsto f^*\mathscr{M} \underset{\mathscr{O}_T}{\otimes} f^*\mathscr{N} = f^*(\mathscr{M} \underset{\mathscr{O}_S}{\otimes} \mathscr{N}),$$

which proves the assertion, since f^* is exact.

Lemma 8.2.10. *Let $(S, \mathscr{O}_S)$ be a ringed topos, let $\mathscr{F}$ be a sheaf of left modules locally of finite presentation, let $\mathscr{G}$ be a sheaf of bimodules, and let $\mathscr{H}$ be a flat sheaf of left modules. Then the canonical arrow*

$$\mathbf{Hom}(\mathscr{F}, \mathscr{G}) \otimes \mathscr{H} \longrightarrow \mathbf{Hom}(\mathscr{F}, \mathscr{G} \otimes \mathscr{H})$$

is an isomorphism.

The question is local on S, which allows us to suppose that $\mathscr{F}$ has a finite presentation

$$\mathscr{F}_1 \longrightarrow \mathscr{F}_0 \longrightarrow \mathscr{F} \longrightarrow 0.$$

The first row of the following diagram is exact, since $\mathscr{H}$ is flat:

$$\begin{tikzcd}
0 \arrow[r] & \mathbf{Hom}(\mathscr{F}, \mathscr{G}) \otimes \mathscr{H} \arrow[r] \arrow[d] & \mathbf{Hom}(\mathscr{F}_0, \mathscr{G}) \otimes \mathscr{H} \arrow[r] \arrow[d, "s"] & \mathbf{Hom}(\mathscr{F}_1, \mathscr{G}) \otimes \mathscr{H} \arrow[d, "s"] \\
0 \arrow[r] & \mathbf{Hom}(\mathscr{F}, \mathscr{G} \otimes \mathscr{H}) \arrow[r] & \mathbf{Hom}(\mathscr{F}_0, \mathscr{G} \otimes \mathscr{H}) \arrow[r] & \mathbf{Hom}(\mathscr{F}_1, \mathscr{G} \otimes \mathscr{H})
\end{tikzcd}$$

which proves the assertion. Taking $\mathscr{G} = \mathscr{O}_S$, one deduces from 8.2.10:

79 Lemma 8.2.11. *Every morphism from a sheaf of modules locally of finite presentation to a flat sheaf of modules factors locally on S through a finite free sheaf $\mathscr{O}_S^n$ $(n \in \mathbf{N})$.*

Theorem 8.2.12. (D. Lazard). *A sheaf of modules $\mathscr{M}$ on a ringed site $(S, \mathscr{O}_S)$ is flat if and only if it is a local inductive limit of finite free modules.*

The "if" assertion follows from 8.2.2 and 8.2.4.

For every $U \in \operatorname{Ob} \mathscr{S}$, let $\mathscr{L}_0(U)$ (resp. $\mathscr{L}_1(U)$) be the category of finite free sheaves of modules (resp. sheaves of modules of finite presentation) on U, equipped with a linear map to $\mathscr{M}|U$. For every arrow $f : V \to U$ in $\mathscr{S}$, let f^* be the restriction functor from $\mathscr{L}_i(U)$ to $\mathscr{L}_i(V)$. These functors define a category $\mathscr{L}_i$, fibered over $\mathscr{S}$, whose fibers are the opposite categories of the categories $\mathscr{L}_i(U)$.

The reader will verify that the category $\mathscr{L}_1$ is locally filtered over $\mathscr{S}$ and that

$$\mathscr{M} = \varinjlim_{(M,f)\in\mathscr{L}_1} M.$$

If $\mathscr{M}$ is flat, the inclusion of $\mathscr{L}_0$ in $\mathscr{L}_1$ is fully faithful and, by 8.2.11, satisfies $(C\mathfrak{F}1)$ of 8.1.4. It then automatically satisfies $(C\mathfrak{F}2)$, and $\mathscr{L}_0$ is locally filtered. By 8.1.5, one has

$$\mathscr{M} = \varinjlim_{(M,f)\in\mathscr{L}_0} M,$$

which proves 8.2.12.

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EXPOSÉ Vbis

Cohomological Descent Techniques

B. Saint-Donat

Introduction

1. Let X be a topological space and let $\mathscr{U} = (U_i)_{i\in I}$ be a covering of X, assumed either open or closed and locally finite. If $\mathfrak{F}$ is an abelian sheaf on X, the Leray spectral sequence 84

$$\check{H}^p\,(\mathscr{U}, \mathscr{H}^q(\mathfrak{F})) \Rightarrow H^*(X, \mathfrak{F}) \tag{1.1}$$

defined by $\mathscr{U}$ [[5] II (5.2.4) and (5.4.1)] may be described as follows.

The covering $\mathscr{U}$ defines a "Čech" resolution $\mathscr{C}^*(\mathscr{U}, \mathfrak{F})$, functorial in $\mathfrak{F}$ (ibid. (5.2.1)). On the other hand, for every $\mathfrak{F}$ there is a "canonical flasque" resolution $C^*(\mathfrak{F})$, functorial in $\mathfrak{F}$ (ibid. (4.3)). With this notation, when $\mathscr{U}$ is open the spectral sequence (1.1) is obtained from the double complex

$$\Gamma(X, \mathscr{C}^*(\mathscr{U}, C^*(\mathfrak{F}))). \tag{1.2}$$

When $\mathscr{U}$ is closed and locally finite, one considers the double complex

$$\Gamma(X, C^*(\mathscr{C}^*(\mathscr{U}, \mathfrak{F}))). \tag{1.3}$$

2. Let us seek a unified description of these double complexes. Let X_0 be the topological disjoint union of the U_i, and for $n \geq 0$ let X_n be the $(n+1)$-fold fiber product of X_0 with itself over X:

$$X_n = \coprod_{i_0,\ldots,i_n\in I} U_{i_0} \cap \cdots \cap U_{i_n} = \coprod_{\sigma\in \mathrm{Hom}([n],I)} \bigcap_{i=0}^{n} U_{\sigma(i)}. \tag{2.1}$$

The X_n form a simplicial system of topological spaces, and if j_n denotes the projection from X_n to X, then

$$\mathscr{C}^n(\mathscr{U}, \mathfrak{F}) = j_{n*}j_n^*(\mathfrak{F}). \tag{2.2}$$

Note that formation of the canonical flasque resolutions commutes with restriction to an open subset and with direct image by a closed immersion. Consequently: 85

(a) if $\mathscr{U}$ is open,

$$\mathscr{C}^q(\mathscr{U}, C^p(\mathfrak{F})) = j_{q*}j_q^* C^p(\mathfrak{F}) = j_{q*}C^p(j_q^*\mathfrak{F}); \tag{2.3}$$

(b) if $\mathscr{U}$ is closed and locally finite,

$$C^p(\mathscr{C}^q(\mathscr{U}, \mathfrak{F})) = C^p(j_{q*}j_q^*\mathfrak{F}) = j_{q*}C^p(j_q^*\mathfrak{F}). \tag{2.4}$$

Thus, to obtain a unified description of (1.2) and (1.3), it is enough to take the canonical flasque resolution of $j_q^*(\mathfrak{F})$ on X_q for every q, and then apply the functor j_{q*} to this resolution.

3. The preceding description remains meaningful for any simplicial system of topological spaces over X:

$$\begin{aligned} \Delta^0 &\longrightarrow \mathrm{Top}/X \\ [n] &\longmapsto X_n. \end{aligned} \tag{3.1}$$

Such a system need not be of the form (2.1). The double complex, coaugmented by $\mathfrak{F}$,

$$\mathfrak{F} \longrightarrow (j_{q*}C^p(j_q^*(\mathfrak{F})))_{p,q} \tag{3.2}$$

will not, however, in general define a resolution of $\mathfrak{F}$.

This work is devoted to finding sufficient conditions for (3.2) to define a resolution of $\mathfrak{F}$. In that case, the spectral sequence (1.1) generalizes to a spectral sequence

$$\check{H}^p\ (H^q(X_p, j_p^*(\mathfrak{F}))) \Rightarrow H^{p+q}(X, \mathfrak{F}), \tag{3.3}$$

called the “descent spectral sequence”.

86 In the case of “constant coefficients”, analogous spectral sequences were obtained by Segal (cf. [8]), by other methods and for other “cohomology theories”, such as K-theory. Segal works in the category of CW-complexes: he uses a “geometric realization” functor which associates a new topological space with a semi-simplicial complex of topological spaces; this new space is to be compared with the topos associated with a simplicial topos [cf. (1.2.12)].

4. In Section 5 we illustrate the criteria obtained by using resolution of singularities to construct, for every analytic space X over $\mathbf{C}$, a simplicial system of nonsingular analytic spaces over X,

$$[n] \longmapsto X_n,$$

such that (3.2) defines a resolution of $\mathfrak{F}$. Taking the constant sheaf $\mathbf{C}$ for $\mathfrak{F}$ gives in particular a spectral sequence

$$H^q(X_p, \mathbf{C}) \Longrightarrow H^{p+q}(X, \mathbf{C}) \tag{4.1}$$

which expresses the complex cohomology of X in terms of the complex cohomology of nonsingular analytic spaces. Moreover, if X is projective, all the X_p may be assumed projective; this is the essential ingredient which yields a kind of “Hodge theory” for X (cf. [2]).

5. The preceding constructions extend verbatim when the sheaf $\mathfrak{F}$ is replaced by a bounded-below complex of sheaves. They lead to “localization” techniques in derived categories.

It is known that, for X_0 given by (2.1), the arrow

$$j_0^* : D^b(X) \longrightarrow D^b(X_0)$$

87 is not faithful in general. We shall show that data more precise than $j_0^*(K^\bullet)$ (for $K^\bullet \in D^+(X)$), involving the X_n, sometimes make it possible to reconstruct the complex $K^\bullet$.

The resulting statements will be used in the appendix to Exposé XVII to extend the definition of the functor $Rf_!$ (where f is a separated morphism of finite type between schemes) to the case in which f is not assumed compactifiable.

For this application it is not possible to consider only topological spaces, which are replaced here by the étale sites of schemes. Moreover, to carry out the proofs it will be necessary to consider both simplicial systems of spaces and multisimplicial systems. This explains, justifies, or excuses the degree of hypergenerality from which we shall begin.

1. Preliminaries

1.1. Notation.

1.1.1. Throughout what follows, $\mathscr{U}$ is a universe such that $\mathbf{Z} \in \mathscr{U}$: *all the topoi considered will be $\mathscr{U}$-topoi.*

Let T and T' be two topoi: a morphism $\varphi : T \to T'$ consists of a pair of functors $\varphi_* : T \to T'$ and $\varphi^* : T' \to T$, endowed with an adjunction

$$\mathrm{Hom}_{T'}(.,\varphi_*.) \xrightarrow{\sim} \mathrm{Hom}_T(\varphi^*.,.),$$

such that φ^* is left exact (i.e. preserves finite projective limits).

Let $(T, \mathscr{O}_T)$ and $(T', \mathscr{O}_{T'})$ be two ringed topoi: a morphism from $(T, \mathscr{O}_T)$ to $(T', \mathscr{O}_{T'})$ is a pair (φ, θ), where $\varphi : T \to T'$ is a morphism of topoi and $\theta : \mathscr{O}_{T'} \to \varphi_*(\mathscr{O}_T)$ is a morphism of rings.

1.1.2. We will make constant use of the language of fibered categories as presented in [SGA 1 VI]; the reader may also consult [4]. Let us merely fix some notation: if $E \to B$ is a fibered (resp. cofibered) functor, then, for a morphism $m : i \to j$ in B, we denote by $m^* : E_j \to E_i$ (resp. $m_* : E_i \to E_j$) the associated "inverse image" (resp. "direct image") functor; each of these functors is defined up to a unique functorial isomorphism. If $\varphi : E \to E'$ is a B-functor, then, for every object i of B, we denote by $\varphi_i : E_i \to E'_i$ the restriction of φ to E_i. 88

1.1.3. Finally, we denote by Δ the following category: the objects of Δ are the ordered sets $[n] = \{0, 1, \dots, n\}$, and the morphisms of Δ are all weakly increasing maps. Δ^+ (resp. Δ^-) denotes the category whose objects are those of Δ and whose arrows are the monomorphisms (resp. the epimorphisms) of Δ. We will freely use, as needed, the classical notation introduced in connection with Δ [cf. [3] II 2].

1.2. D-topoi.

DEFINITION 1.2.1. *Let D be a category, and let $E \to D$ be a fibered and cofibered functor. We say that E is bifibered in topoi over D, or that E is a D-topos, if the following conditions hold:*

(a) *For every object i of D, the fiber category E_i is a topos.*
(b) *For every morphism $m : i \to j$ in D, there exists a morphism of topoi $f : E_j \to E_i$ such that $f_* = m^*$ and $f^* = m_*$.*

REMARK. In view of (a), condition (b) can equivalently be expressed by saying that the functor m_* is left exact[15].

When $D = \Delta$ (resp. $\Delta \times \Delta$), we speak of a *simplicial topos* (resp. a *double simplicial topos*) to mean a Δ-topos (resp. a $\Delta \times \Delta$-topos).

[15]N.D.E.: Care must be taken here: in a fibered category, $m_\star$ is left adjoint to $m^\star$ and is therefore tautologically right exact; cf. [SGA 1 VI.10]. This explains the formulas $f_* = m^*$ and $f^* = m_*$.

In practice, we will encounter D-topoi by means of the following considerations.

89 Definition 1.2.2. *Let $\mathscr{E}$ be a fibered and cofibered category over D. We say that $\mathscr{E}$ is* bifibered in duals of topoi over D *if $\mathscr{E}^\circ$ is a D°-topos.*

Remark 1.2.3. Explicitly, $\mathscr{E}$ is bifibered in duals of topoi over D if and only if the following conditions hold:

(a) For every object i of D, the opposite category $\mathscr{E}_i^\circ$ of the fiber category $\mathscr{E}_i$ is a topos.
(b) For every morphism $m : i \to j$ in D, the functor m^* is left exact[16].

1.2.4. The reader will find examples of categories bifibered in duals of topoi in paragraph 4.

1.2.5. Let $\mathscr{E} \to B$ be a category bifibered in duals of topoi over B, and let $X : D^\circ \to B$ be a functor. We leave it to the reader to verify that $(D^\circ \times_B \mathscr{E})^\circ$ is a D-topos, which we denote by $\overline{X}$.

1.2.6. We will often say that a functor $X : D^\circ \to B$ is a *D-object of B*, and we denote by X_i the image under this functor of an object i of D; the D-objects of B form a category denoted by $D^\circ B$. If S is an object of B, a D-object of B/S will be called a *D-object of B augmented toward S*. The datum of a D-object of B augmented toward S is trivially equivalent to that of a functorial morphism $X \to C^D_S$, where X is a D-object of B and C^D_S is the constant D-object of B with value S.

When $D = \Delta$, we speak of a *simplicial object* (resp. an *augmented simplicial object*).

We will also use *double simplicial objects* (taking $D = \Delta \times \Delta$).

90 1.2.7. Now suppose that the category B has finite fiber products.

Let $f : R \to S$ be an arrow in B: the bifunctor

$$([n], X) \rightsquigarrow \mathrm{Hom}_{\mathrm{Ens}}([n], \mathrm{Hom}_S(X, R))$$

from $\Delta^\circ \times (B/S)^\circ$ to Ens defines a functor

$$[n] \rightsquigarrow X_{[n]} = X_n$$

by taking as X_n a representative of the functor

$$Z \rightsquigarrow \mathrm{Hom}_{\mathrm{Ens}}([n], \mathrm{Hom}_S(Z, R))$$

$(X_n \simeq \overbrace{R \times_S R \times \cdots \times_S R}^{n+1 \text{ times}})$.

We denote by $[R|_f S]$ or $[R|S]$ the semi-simplicial object augmented toward S thus constructed.

Finally, if X and X' are two semi-simplicial objects of B (or of B/S), and if

$$u : X \longrightarrow X'$$

is a functorial morphism, then, for technical reasons, we introduce the object

$$[X|_u X']$$

computed in the category $\Delta^\circ B$. This object can be interpreted as a double simplicial object[17] of B, which we then denote by $[[X|_u X']]$: indeed, there is a canonical isomorphism

[16]N.D.E.: Cf. the preceding note. In other words, the family of functors $((m_*)^\circ, (m^*)^\circ)$ defines a compatible system of morphisms of topoi $\mathscr{E}_i^\circ \to \mathscr{E}_j^\circ$.

[17]N.D.E.: Explicitly, with the usual ordering of the indices, one has

$$[[X|_u X']]_{(n,m)} = \overbrace{X_m \times_{X'_m} X_m \times \cdots \times_{X'_m} X_m}^{n+1 \text{ times}}$$

of categories

$$\Delta^\circ(\Delta^\circ B) \xrightarrow{\sim} (\Delta \times \Delta)^\circ B.$$

We now return to the general notion of a D-topos.

1.2.8. Let E be a D-topos: we denote by $\Gamma(E)$ the category $\mathrm{Hom}_D(D, E)$. Let $f : D' \to D$ be a functor: the category $D' \times_D E$ is a D'-topos and, by composition with f, we obtain a functor

$$f^* : \Gamma(E) \longrightarrow \Gamma(D' \times_D E).$$

In the case where D' is reduced to an object i of D (with the identity of i as its only morphism), and where f is the canonical inclusion denoted by e_i, we may take $D' \times_D E$ to be the fiber category E_i, and e_i^* is then identified with the "*evaluation at i*" functor. 91

Proposition 1.2.9. *If D' is a $\mathcal{U}$-small category, the functor f^* has a right adjoint and a left adjoint (denoted by f_* and $f_!$, respectively).*

This follows from a slight generalization of Kan's lemma [III 1.1], which we will use repeatedly.

Lemma 1.2.10. *Let I, J, and A be three categories over the same category B. Suppose that I is $\mathcal{U}$-small and that A is fibered and cofibered over B. Let a B-functor $f : I \to J$ be given, and denote by f^* the functor*

$$\mathrm{Hom}_B(J, A) \longrightarrow \mathrm{Hom}_B(I, A)$$

defined by composition with f. If, in every fiber of A over B, $\mathcal{U}$-inductive limits (resp. projective limits) exist, then f^* has a left adjoint (resp. a right adjoint).

Proof. We give only the proof of the existence of the left adjoint, the resp. part of the lemma following by duality. We use the following fact, whose verification is left to the reader.

Lemma 1.2.10.1. *Let A be a bifibered category over a category B, let b be an object of B, and let $(F_\lambda)_{\lambda\in\Lambda}$ be a functor from a category Λ to the fiber A_b. For every G in $A_{b'}$, and every $u : b' \to b$ (resp. $u : b \to b'$) in B, there is a bijection*

$$\mathrm{Hom}_u(G, \varprojlim F_\lambda) \xrightarrow{\sim} \varprojlim \mathrm{Hom}_u(G, F_\lambda)$$

(resp. $\mathrm{Hom}_u(\varinjlim F_\lambda, G) \xrightarrow{\sim} \varprojlim \mathrm{Hom}_u(F_\lambda, G)$*) whenever the first member is defined.*

With this understood , let $I \amalg_f J$ be the category over B defined by 92

$$\mathrm{ob}(I \amalg_f J) = \mathrm{ob}(I) \amalg \mathrm{ob}(J)$$

$$\mathrm{Hom}(x, y) = \begin{cases} \mathrm{Hom}_I(x, y) & \text{if } x, y \in \mathrm{ob}(I) \\ \mathrm{Hom}_J(x, y) & \text{if } x, y \in \mathrm{ob}(J) \\ \mathrm{Hom}_J(f(x), y) & \text{if } x \in \mathrm{ob}(I) \text{ and } y \in \mathrm{ob}(J) \\ \varnothing & \text{if } x \in \mathrm{ob}(J) \text{ and } y \in \mathrm{ob}(I). \end{cases}$$

The category $\mathrm{Hom}_B(I \amalg_f J, A)$ is equivalent to the category of triples consisting of a B-functor F from I to A, a B-functor G from J to A, and a morphism φ of B-functors from F to $G \circ f = f(G)$.

For every object j of J, denote by I/j the category of objects of I "placed by f over j": the objects of I/j are the pairs (i, α), where i is an object of I and $\alpha : i \to j$ is an arrow in $I \amalg_f J/j$; the morphisms of I/j are those of $I \amalg_f J/j$. If p_I and p_J are the projections

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with the functorialities deduced from those on the factors.

of I and J onto B, and if (i,α) is an object of I/j, denote by $\alpha_* : A_{p_I(i)} \to A_{p_J(j)}$ the functor $p_J(\alpha)_*$.

To give a B-functor from $I \amalg_f J$ to A is equivalently to give $F \in \mathrm{Hom}_B(I,A)$, $G \in \mathrm{Hom}_B(J,A)$, and a morphism, functorial in j,

$$\psi_j : \varinjlim_{(i,\alpha)\in I/j} \alpha_*(F(i)) \longrightarrow G(j).$$

(The functoriality in j of the left-hand member follows from 1.2.10.1.) The left adjoint of f^* is therefore given by the formula

$$f_!(F)(j) = \varinjlim_{(i,\alpha)\in I/j} \alpha_*(F(i)).$$

93 Corollary 1.2.11. *Let E be a D-topos and let i be an object of D; the functor e_i^* has a left adjoint (resp. a right adjoint), given by*

$$e_{i!}(a)(j) = \coprod_{\alpha\in\mathrm{Hom}(i,j)} \alpha_*(a) \left(\text{resp. } e_{i*}(a)(j) = \prod_{\alpha\in\mathrm{Hom}(j,i)} \alpha^*(a)\right)$$

where a is an object of E over i.

Proposition 1.2.12. *Let D be a $\mathscr{U}$-small category and let E be a D-topos. Then the category $\Gamma(E)$ is a $\mathscr{U}$-topos.*

One verifies “fiber by fiber,” using 1.2.10.1, that the category $\Gamma(E)$ has the following properties:

a) Finite projective limits are representable.
b) Coproducts indexed by an element of $\mathscr{U}$ are representable. They are disjoint and universal.
c) Equivalence relations are universally effective.

It remains to show that $\Gamma(E)$ has a system of generators indexed by an element of $\mathscr{U}$. But if, for every object i of D, $(G_{i\lambda})_{\lambda\in\Lambda_i}$ is a system of generators of E_i (where Λ_i is a $\mathscr{U}$-small set), then the family $(e_{i!}(G_{i\lambda}))_{i,\lambda}$ is a system of generators of $\Gamma(E)$.

1.2.13. We now introduce the notion of a morphism between D-topoi. First, let us specify that if F and F' are two categories over the same category B and $T : F \to F'$ is a B-functor, a *B-left adjoint* to T means a functor $S : F' \to F$ left adjoint to T such that the canonical morphisms $1 \to T\circ S$ and $S\circ T \to 1$ are B-morphisms of functors. Under these conditions, one verifies immediately that T is cartesian and S is cocartesian.

94 Definition 1.2.14. *Let E and E' be two D-topoi. A morphism from E to E' is a pair of D-functors $\Phi_* : E \to E'$ and $\Phi^* : E' \to E$, equipped with a D-adjunction*

$$\mathrm{Hom}(\Phi^*(-),-) \xrightarrow[\xi]{\sim} \mathrm{Hom}(-,\Phi_*(-)),$$

such that, for every object i of D, the pair (Φ_{*i},Φ_i^*), equipped with the adjunction induced by ξ, is a morphism of topoi from E_i to E_i'.

Proposition 1.2.15. *Let E and E' be two D-topoi, and let $(\Phi_*,\Phi^*) : E \to E'$ be a morphism. If D is a $\mathscr{U}$-small category, then the pair $(\Gamma(\Phi_*),\Gamma(\Phi^*)) : \Gamma(E) \to \Gamma(E')$ is a morphism of topoi.*

This follows immediately from the preceding definition.

The following lemma, whose proof is left to the reader, provides a way of constructing morphisms of D-topoi.

Lemma 1.2.16. *Let E and E' be two bifibered categories over the same category D, and let Φ be a* cartesian *D-functor from E to E' such that, for every object i of D, $\Phi_i : E_i \to E'_i$ has a left adjoint. Then the choice, for every i, of a left adjoint to Φ_i canonically determines a D-functor $\Psi : E' \to E$ which is D-left adjoint to Φ.*

Scholium 1.2.17. Under the conditions of (1.2.16), suppose that E and E' are two D-topoi and that, for every object i of D, every left adjoint to Φ_i is left exact. Then, if $\Psi : E' \to E$ is a D-left adjoint to Φ, the pair $(\Phi, \Psi) : E \to E'$ is a morphism of D-topoi.

Now let $\mathscr{E} \to B$ be a category bifibered in duals of topoi over B, let X and X' be two D-objects of B, and let $\alpha : X \to X'$ be a functorial morphism. A choice of normalized cleavages for $\mathscr{E}$ and $\mathscr{E}^\circ$ then canonically determines a morphism $(\alpha_*, \alpha^*) : \overline{X} \to \overline{X}'$ of D-topoi such that, for every object i of D, $(\alpha_{*i}, \alpha_i^*) : \overline{X}_i \to \overline{X}'_i$ is equal to $(\alpha^\circ_{i*}, \alpha_i^{*\circ})$, 95
where $\alpha_i : X_i \to X'_i$ is the arrow of B given by α. For two different choices of cleavages for $\mathscr{E}$ and $\mathscr{E}^\circ$, there is a unique D-isomorphism between the resulting morphisms. (For the verification of these facts, the reader may consult [4], (1.17).)

1.3. Ringed D-topoi.

Definition 1.3.1. *A ringed D-topos is a pair (E, A), where E is a D-topos and A is a ring of $\Gamma(E)$.*

One then verifies that, for every object i of D, A_i is a ring of the topos E_i, and that, for every morphism $m : i \to j$, the canonical arrow $A_i \to m^*(A_j)$ is a ring homomorphism.

Definition 1.3.2. *Let (E, A) and (E', A') be two ringed D-topoi. A morphism from (E, A) to (E', A') is a pair (Φ, θ), where $\Phi : E \to E'$ is a morphism of D-topoi and $\theta : A' \to \Gamma(\Phi_*)(A)$ is a ring homomorphism.*

Remark 1.3.3. A morphism $\Phi : (E, A) \to (E', A')$ of ringed D-topoi induces a morphism $(\Gamma(\Phi), \theta) : (\Gamma(E), A) \to (\Gamma(E'), A')$ of ringed $\mathscr{U}$-topoi when D is a $\mathscr{U}$-small category (cf. 1.2.15).

1.3.4. Let $\mathscr{E} \to B$ be a category bifibered in duals of topoi over B, and let $\mathscr{O}$ be a ring of $\Gamma(\mathscr{E}^\circ) = \mathrm{Hom}_{B^\circ}(B^\circ, \mathscr{E}^\circ)$. If $X : D^\circ \to B$ is a D-object of B, the D-topos $\overline{X}$ (cf. (1.2.5)) is naturally ringed by $(\mathscr{O} \,.\, X^\circ) : D \to \mathscr{E}$; the resulting ringed D-topos is denoted by $(\overline{X}, \mathscr{O})$. If $\alpha : X \to X'$ is a functorial morphism, the morphism $(\alpha_*, \alpha^*) : \overline{X} \to \overline{X}'$ (cf. 1.2.17) canonically induces a morphism $(\overline{X}, \mathscr{O}) \to (\overline{X}', \mathscr{O})$ of ringed D-topoi, again denoted by α.

1.3.5. A ringed D-topos (E, A) canonically defines a category $\mathrm{Mod}(E, A)$ bifibered in abelian categories over D, whose fiber at an object i of D is the category $\mathrm{Mod}(E_i, A_i)$ 96
of modules on the ringed topos (E_i, A_i). With this notation, the category of modules of $\Gamma(E)$ over A, denoted by $\mathrm{Mod}(\Gamma(E), A)$, identifies with the category $\mathrm{Hom}_D(D, \mathrm{Mod}(E, A))$.

1.3.6. Let $\varphi = (\Phi, \theta) : (E, A) \to (E', A')$ be a morphism of ringed D-topoi. It defines two functors $\varphi_* : \underline{\mathrm{Mod}}(E, A) \to \underline{\mathrm{Mod}}(E', A')$ and $\varphi^* : \underline{\mathrm{Mod}}(E', A') \to \mathrm{Mod}(E, A)$ between the corresponding categories of modules:

- Let M be an object of $\mathrm{Mod}(E, A)$ over an object i of D. Then $\Phi_*(M)$ is a module over $\Phi_*(A_i)$, and the morphism $\theta_i : A'_i \to \Phi_*(A_i)$ makes it an A'_i-module, denoted by $\varphi_*(M)$. The functor φ_* is called the *direct-image functor under the morphism φ*.

- Let M' be an object of $\mathrm{Mod}(E', A')$ over an object i of D. Then $\Phi^*(M')$ is a module over $\Phi^*(A'_i)$, and $\varphi^*(M') = \Phi^*(M') \otimes_{\Phi^*(A'_i)} A_i$ is canonically endowed with an A_i-module structure. By means of (1.2.16), this defines a functor φ^* left adjoint to φ_*, called the *inverse-image functor under the morphism φ*.

. We say that φ is *flat* if the functor $\Gamma(\varphi^*)$ is exact.

PROPOSITION 1.3.7. *Let $f : D' \to D$ be a functor, and let (E, A) be a ringed D-topos. Then the canonical functor*

$$f^* : \mathrm{Mod}(\Gamma(E), A) \longrightarrow \mathrm{Mod}(\Gamma(E \times_D D'), A \circ f)$$

has both a right and a left adjoint if D' is a $\mathscr{U}$-small category. In particular, it is exact.

This follows immediately from (1.2.10) and the identification $\mathrm{Hom}_D(D, \mathrm{Mod}(E, A)) \simeq \mathrm{Mod}(\Gamma(E), A)$.

97 In accordance with the general notation, we denote the left (resp. right) adjoint to f^* by $f_!$ (resp. f_*).

1.3.8. The considerations below provide a convenient computational method for derived functors of the form

$$R\Gamma(\varphi_*) : D^+(\Gamma(E), A) \longrightarrow D^+(\Gamma(E'), A'),$$

where $\varphi : (E, A) \to (E', A')$ is a morphism of ringed D-topoi (cf. (1.3.6)). [If $(S, \mathscr{O}_S)$ is a ringed topos, we denote by $D^+(S, \mathscr{O}_S)$ the bounded-below derived category[18] of the category of $\mathscr{O}_S$-modules on S.]

1.3.9. This formal calculation may moreover be applied in other contexts, such as “descent in ℓ-adic cohomology” (cf. SGA 5).

PROPOSITION 1.3.10. *Let D be a $\mathscr{U}$-small category. If (E, A) is a ringed D-topos, denote by $I_{(E,A)}$ the class of objects of* $\mathrm{Mod}(\Gamma(E), A)$ *isomorphic to an object of the form $\prod_{i \in \mathrm{ob}(D)} e_{i*}(Q_i)$, where, for every object i of D, Q_i is totally acyclic* (cf. V 4.1)[18]*; $I_{(E,A)}$ has the following properties:*

(i) *For every object F of $I_{(E,A)}$ and every object i of D, $e_i^*(F)$ is totally acyclic.*
(ii) *Every object of* $\mathrm{Mod}(\Gamma(E), A)$ *embeds into an object of $I_{(E,A)}$.*
(iii) *$I_{(E,A)}$ is stable under finite direct sums.*
(iv) *For every morphism $\varphi : (E, A) \to (E', A')$ of ringed D-topoi, the functor $\Gamma(\varphi_*)$ carries every acyclic complex in $C^+(\Gamma(E), A)$ formed of objects of $I_{(E,A)}$ to an acyclic complex formed of objects of $I_{(E',A')}$.*

Proof

(i) In view of the explicit expression for e_i^* (cf. (1.2.1.1)), it suffices to prove the following lemma:

98 LEMMA 1.3.10.1. *Let $(S, \mathscr{O}_S)$ be a ringed topos, and let $(F_t)_{t \in T}$ be a family of totally acyclic $\mathscr{O}_S$-modules indexed by a $\mathscr{U}$-small set T. Then $\prod_{t \in T} F_t$ is totally acyclic.*

[18] N.D.E.: If $f : (S, \mathscr{O}_S) \to (T, \mathscr{O}_T)$ is a flat morphism of ringed topoi, we simply write f^* (and not $L^+(f^*)$) for the derived functor

$$f^* : D^+(T, \mathscr{O}_T) \to D^+(S, \mathscr{O}_S),$$

as in the original version.

[18] “Flasque” in the terminology of V.

Let X be an object of S. There is a spectral sequence

$$H^p\left(X, \prod_{t\in T}^{(q)} F_t\right) Rightarrow \prod_{t\in T} H^{p+q}(X, F_t) \tag{1.3.10.2}$$

where $\prod_{t\in T}^{(q)}$ denotes the q-th derived functor of the "product indexed by T" functor. Since $\prod_{t\in T}^{(q)} F_t$ is the sheaf associated with the presheaf $U \mapsto \prod_{t\in T} H^q(U, F_t)$, (1.3.10.2) degenerates and gives isomorphisms

$$H^n\left(X, \prod_{t\in T} F_t\right) \xrightarrow{\sim} \prod_{t\in T} H^n(X, F_t) \tag{1.3.10.3}$$

and hence, finally, $H^n(X, \prod_{t\in T} F_t) = 0$ for every $n > 0$.

(ii) Let F be an object of $\mathrm{Mod}(\Gamma(E), A)$, and put $F_i = e_i^*(F)$. For every i, choose a monomorphism $F_i \to Q_i$, where Q_i is totally acyclic in $\mathrm{Mod}(E_i, A_i)$. Then $e_{i*}(F_i) \to e_{i*}(Q_i)$ is a monomorphism (because e_{i*} has a left adjoint), and the canonical arrow

$$F \longrightarrow \prod_{i\in \mathrm{Ob}(D)} e_{i*}(F_i) \longrightarrow \prod_{i\in \mathrm{Ob}(D)} e_{i*}(Q_i) \tag{1.3.10.4}$$

is again a monomorphism, as is immediately verified.

(iii) The proof is left to the reader.

(iv) The reader may also verify that $\Gamma(\varphi_*)$ carries an object of $I_{(E,A)}$ to an object of $I_{(E',A')}$. Once this point is established, in view of (i), it suffices to verify the following lemma:

Lemma 1.3.10.5. *If $0 \to F \to G \to H \to 0$ is a short exact sequence in* $\mathrm{Mod}(\Gamma(E), A)$, 99
with $F, G \in I_{(E,A)}$, then the sequence

$$0 \longrightarrow \Gamma(\varphi_*)(F) \longrightarrow \Gamma(\varphi_*)(G) \longrightarrow \Gamma(\varphi_*)(H) \longrightarrow 0$$

is exact in $\mathrm{Mod}(\Gamma(E'), A')$.

It suffices to observe that $e_i^*(H)$ is acyclic for every object i of D, and that $\Gamma(\varphi_*)$ is computed fiber by fiber. This completes the proof of (1.3.10).

Corollary 1.3.11. *Let $\varphi : (E, A) \to (E', A')$ be a morphism of ringed D-topoi. The functor $R\Gamma(\varphi_*)$ can be computed by means of resolutions formed of objects of $I_{(E,A)}$.*

Let $L \subset K^+(\Gamma(E), A)$ be the full subcategory of closed complexes of objects of $I_{(E,A)}$. By ((1.3.10), (ii)), L is a triangulated subcategory, and the theorem (5.1) of [[7], Chapter 1] applies to it.

Corollary 1.3.12. *Let $\varphi : (E, A) \to (E', A')$ be a morphism of ringed D-topoi. For every object i of D, denote by $\varphi_i : (E_i, A_i) \to (E_i', A_i')$ the induced morphism of ringed topoi over i. Then the diagram*

$$\begin{tikzcd}
D^+(\Gamma(E), A) \arrow[r, "e_i^*"] \arrow[d, "R\Gamma(\varphi_*)"'] & D^+(E_i, A_i) \arrow[d, "R\varphi_{i*}"] \\
D^+(\Gamma(E'), A') \arrow[r, "e_i^*"'] & D^+(E_i', A_i')
\end{tikzcd}$$

is essentially commutative.

There is a morphism

$$e_i^* \circ R\Gamma(\varphi_*) \longrightarrow R\varphi_{i*} \circ e_i^*$$

which is verified to be an isomorphism by (1.3.10).

100 Remark 1.3.13. Denote by $\mathscr{A}_{(E,A)}$ the class of objects F of $\mathrm{Mod}(\Gamma(E), A)$ such that $e_i^*(F)$ is totally acyclic for every object i of D. It follows from the proof of (1.3.10) that $R\Gamma(\varphi_*)$ can be computed by means of resolutions formed of objects of $\mathscr{A}_{(E,A)}$; this is what is done, in particular, in the introduction by using the "canonical flasque resolution"[18]. By [(1.3.10)(i)], one has $I_{(E,A)} \subset \mathscr{A}_{(E,A)}$, but we will use $I_{(E,A)}$ explicitly in Section 2.

2. The method of cohomological descent

In this section, D is a $\mathscr{U}$-small category.

2.1. Generalities. Notation.

2.1.1. Let $(S, \mathscr{O}_S)$ be a ringed topos. The category $S \times D$, equipped with the canonical projection $S \times D \to D$, is a D-topos; moreover, the constant section with value $\mathscr{O}_S$ defines a ringed D-topos $(S \times D, \mathscr{O}_S)$, called the *constant ringed D-topos with value* $(S, \mathscr{O}_S)$. With this notation, the category $\mathrm{Mod}(\Gamma(S \times D), \mathscr{O}_S)$ is identified with the category of covariant functors from D to $\mathrm{Mod}(S, \mathscr{O}_S)$.

Define an exact functor

$$\epsilon^* : \mathrm{Mod}(S, \mathscr{O}_S) \longrightarrow \mathrm{Mod}(\Gamma(S \times D), \mathscr{O}_S)$$

by associating with every module F on S the constant functor with value F. The functor ϵ^* has a right adjoint ϵ_*, which associates with every functor $H : D \to \mathrm{Mod}(\mathscr{O}_S)$ its projective limit; the adjunction morphism $F \to \epsilon_*\epsilon^*(F)$ is the morphism from F to the projective limit of the constant functor with value F. Thus ϵ^* is fully faithful if and only if D is connected.

101 Definition 2.1.2. *Let (E, A) be a ringed D-topos. An augmentation of (E, A) is a morphism (of ringed D-topoi) from (E, A) to a constant ringed D-topos. A ringed D-topos equipped with an augmentation will be called an augmented ringed D-topos.*

2.1.3. Let $\mathscr{E} \to B$ be a category bifibered in duals of topoi, and let $\mathscr{O}$ be a ring in $\Gamma(\mathscr{E}^\circ) = \mathrm{Hom}_{B^\circ}(B^\circ, \mathscr{E}^\circ)$. Let S be an object of B. A functor $D^\circ \to B/S$, that is, a functorial morphism $X \xrightarrow{\theta} C_S^D$, where X is a functor $D^\circ \to B$ (cf. (1.2.6)), induces an augmentation

$$\theta : (\overline{X}, \mathscr{O}) \longrightarrow (\mathscr{E}_S^\circ \times D, \mathscr{O}_S) \qquad \text{(cf. (1.3.4))}.$$

which we shall denote by the same letter as the functorial morphism from which it arises.

2.2. Cohomological descent.

[18] See Exposé XVII for the generalization of this notion.

2.2.1. Let $\theta : (E, A) \to (S \times D, \mathcal{O}_S)$ be an augmented ringed D-topos. With the notation of (1.3.6) and (2.1.1), set

$$\overline{\theta}^* = \Gamma(\theta^*) \circ \epsilon^* : \mathrm{Mod}(S, \mathcal{O}_S) \longrightarrow \mathrm{Mod}(\Gamma(E), A)$$

and $\overline{\theta}_* = \epsilon_* \circ \Gamma(\theta_*) : \mathrm{Mod}(\Gamma(E), A) \to \mathrm{Mod}(S, \mathcal{O}_S)$.

The image of $\overline{\theta}^*$ lies in the full subcategory of $\mathrm{Mod}(\Gamma(E), A)$ consisting of the cocartesian sections of $\Gamma(D, \mathrm{Mod}(E, A))$. We shall denote this category by $\Gamma^{\mathrm{cocart}}(D, \mathrm{Mod}(E, A))$ [19]

Definition 2.2.2. *The morphism θ is said to be an* effective descent augmentation *if*

$$\overline{\theta}^* = \Gamma(\theta^*) \circ \epsilon^* : \mathrm{Mod}(S, \mathcal{O}_S) \longrightarrow \Gamma^{\mathrm{cocart}}(D, \mathrm{Mod}(E, A))$$

is an equivalence of categories.

2.2.3. With the notation of (2.2.1), suppose that θ is flat, so that $\overline{\theta}^*$ is exact and passes trivially to the derived categories; we again denote by $\overline{\theta}^*$ the functor thereby obtained. With this notation: **102**

Lemma 2.2.3.1. *There are two functorial morphisms*

$$\alpha : \mathrm{id}_{D^+(S, \mathcal{O}_S)} \longrightarrow R\overline{\theta}_* \circ \overline{\theta}^* \quad \textit{and} \quad \beta : \overline{\theta}^* \circ R\overline{\theta}_* \longrightarrow \mathrm{id}_{D^+(\Gamma(E), A)}$$

which make these two functors adjoint.

(cf. [9] (3.3)).

Definition 2.2.4. *The morphism θ is said to be a* 1-cohomological-descent augmentation *if*

1°) *θ is flat;*

2°) *$\overline{\theta}^* : D^+(S, \mathcal{O}_S) \to D^+(\Gamma(E), A)$ is fully faithful.*

Remark 2.2.5. Condition 2°) of the preceding definition can also be expressed by saying that the morphism $\alpha : \mathrm{id}_{D^+(S, \mathcal{O}_S)} \to R\overline{\theta}_* \circ \overline{\theta}^*$ in (2.2.3.1) is an isomorphism. This is what one proves in practice and what is useful.

Definition 2.2.6. *The morphism θ is said to be a* 2-cohomological-descent augmentation *(or an* effective cohomological-descent augmentation*) if θ is both an effective descent augmentation and a 1-cohomological-descent augmentation.*

The preceding terminology is justified by the following result:

[19] N.D.E.: In other words, an augmented ringed D-topos is the datum of a compatible system of morphisms of S-topoi

$$(m^*, m_*) : E_j \longrightarrow E_i,$$

where $m \in \mathrm{Hom}_D(i, j)$. The category of cocartesian sections is the category of descent data; its objects are families $F_i \in \mathrm{Ob}(E_i)$ equipped with cocycles of isomorphisms

$$F_j \xrightarrow{\sim} m^* F_i.$$

The functor which associates with every object F on S the constant functor with value F is the inverse image of a morphism of topoi

$$\epsilon : \Gamma(S \times D) \longrightarrow S,$$

inducing the morphism of ringed topoi of 2.1.3. The pair $(\overline{\theta}_*, \overline{\theta}^*)$ is therefore simply induced by the composite of morphisms of topoi $\overline{\theta} = \epsilon \circ \Gamma(\theta) : \Gamma(E) \to S$ (cf. 1.2.15).

Proposition 2.2.7. *Let θ be an effective cohomological-descent augmentation. Then the essential image of $\overline{\theta}^*$ is the full subcategory of $D^+(\Gamma(E), A)$ consisting of the complexes $F^\bullet$ such that, for every i, $H^i(F^\bullet)$ is a cocartesian section of* $\mathrm{Mod}(E, A)$.

For brevity, we shall call a complex $F^\bullet$ satisfying the preceding conditions a *cohomological descent datum.*

Proof. It is clear that, if $K^\bullet \in D^+(S, \mathcal{O}_S)$, then $\overline{\theta}^*(K^\bullet)$ is a cohomological descent datum. It is therefore enough to show that, for every cohomological descent datum $F^\bullet$, the canonical morphism

$$\beta(F^\bullet) : \overline{\theta}^* \circ R\overline{\theta}_*(F^\bullet) \longrightarrow F^\bullet$$

103 is an isomorphism in $D^+(\Gamma(E), A)$.

a) *The case where $F^\bullet$ is bounded*: we argue by induction on the length ℓ of the interval of integers i for which $H^i(F^\bullet) \neq 0$:
- for $\ell \leq 1$, the assertion is true because θ is of effective descent.
- suppose $\ell > 1$, and let n be the greatest integer such that $H^n(F^\bullet) \neq 0$. There is a distinguished triangle[19]

$$\begin{tikzcd} & H^n(F^\bullet)[-n] \arrow[dl] & \\ F'^\bullet \arrow[rr] & & F^\bullet \arrow[ul] \end{tikzcd}$$

such that the induction hypothesis applies to $F'^\bullet$. Since $\beta(H^n(F^\bullet)[-n])$ and $\beta(F'^\bullet)$ are isomorphisms, the same is true of $\beta(F^\bullet)$.

b) *The general case*: denote by $\tau_{\leq n}(F^\bullet)$ the complex

$$\ldots \longrightarrow F^{n-2} \longrightarrow F^{n-1} \longrightarrow \operatorname{Ker} d^n \longrightarrow 0 \longrightarrow 0 \ldots,$$

(cf. [7] (7.1)), so that, for every n, there is a commutative diagram

$$\begin{tikzcd} \overline{\theta}^* \circ R\overline{\theta}_*(\tau_{\leq n}(F^\bullet)) \arrow[r, "\sim"] \arrow[d] & \tau_{\leq n}(F^\bullet) \arrow[d] \\ \overline{\theta}^* \circ R\overline{\theta}_*(F^\bullet) \arrow[r] & F^\bullet \end{tikzcd}$$

and it is enough to see that the morphism

$$R\overline{\theta}_*(\tau_{\leq n}(F^\bullet)) \longrightarrow R\overline{\theta}_*(F^\bullet)$$

104 induces an isomorphism on H^i for $i \leq n$. But there is a triangle

$$\begin{tikzcd} & F''^\bullet \arrow[dl] & \\ \tau_{\leq n}(F^\bullet) \arrow[rr] & & F^\bullet \arrow[ul] \end{tikzcd}$$

[19]For every complex $K^\bullet$, $K^\bullet[-n]$ denotes the complex $T^{-n}(K^\bullet)$, where T is the translation functor.

with $H^i(F''^{\cdot}) = 0$ for $i \leq n$. It follows that $H^i(R\overline{\theta}_*(F''^{\cdot})) = 0$ for $i \leq n$, and then that $H^i(R\overline{\theta}_*(\tau_{\leq n}(F^{\cdot}))) \to H^i(R\overline{\theta}_*(F^{\cdot}))$ is an isomorphism for $i \leq n$, which completes the proof.

C.Q.F.D.

We shall now present an explicit method for computing

$$R\overline{\theta}_* = R\epsilon_* \circ R\Gamma(\theta)_*,$$

which is very useful in proving certain criteria for cohomological descent and in exploiting that notion.

2.3. A method for computing $R\epsilon_*$. We shall begin with two sorites.

2.3.1. Let D be a $\mathcal{U}$-small category, and let Ab be the category of abelian groups belonging to $\mathcal{U}$. Denote by r_D the functor $D \to (\mathrm{Hom}(D, \mathrm{Ab}))^\circ$ defined by

$$r_D(i)(j) = \mathbf{Z}^{\mathrm{Hom}(i,j)}$$

for every pair (i, j) of objects of D. For every pair (i, X) consisting of an object of D and an object of $\mathrm{Hom}(D, \mathrm{Ab})$, one then constructs a canonical and functorial isomorphism

$$\mathrm{Hom}(r_D(i), X) \xrightarrow{\sim} X(i).$$

. Let A be an additive category (cf. Tôhoku), and let F be a functor $D \to A$. Define a functor $\overline{F} : (\mathrm{Hom}(D, \mathrm{Ab}))^\circ \to \mathrm{Hom}(A^\circ, \mathrm{Ens})$ by

$$\overline{F}(X)(a) = \mathrm{Hom}_{\mathrm{Hom}(D,\mathrm{Ab})}(X, \mathrm{Hom}_A(a, F(.))),$$

so that the following diagram, in which h_A denotes the canonical functor, is essentially 105
commutative:

$$\begin{tikzcd}
D \arrow[r, "r_D"] \arrow[d, "F"'] & (\mathrm{Hom}(D, \mathrm{Ab}))^\circ \arrow[d, "\overline{F}"] \\
A \arrow[r, "h_A"'] & \mathrm{Hom}(A^\circ, \mathrm{Ens}).
\end{tikzcd}$$

We leave it to the reader to verify that $\overline{F}$ sends finite direct sums in $\mathrm{Hom}(D, \mathrm{Ab})$ to products and that, if $\mathbf{Z}$ denotes the constant functor $D \to \mathrm{Ab}$ with value $\mathbf{Z}$, then $\overline{F}(\mathbf{Z}) = \varprojlim F$.

Finally, the correspondence $F \rightsquigarrow \overline{F}$ defines a covariant functor

$$\mathrm{Hom}(D, A) \longrightarrow \mathrm{Hom}((\mathrm{Hom}(D, \mathrm{Ab}))^\circ, \mathrm{Hom}(A^\circ, \mathrm{Ens})).$$

. Denote by $\mathrm{Add}(D)$ the full subcategory of $(\mathrm{Hom}(D, \mathrm{Ab}))^\circ$ defined by the objects of the form $r_D(i)$, where i is an object of D, and their finite direct sums. The category $\mathrm{Add}(D)$ is additive, and the functor $r_D : D \to \mathrm{Add}(D)$ has the following universal property:

. *For every additive category A, the functor $G \longmapsto G \circ r_D$ induces an equivalence from the full subcategory of* $\mathrm{Hom}(\mathrm{Add}(D), A)$ *consisting of the additive functors* onto *the category* $\mathrm{Hom}(D, A)$.

The category $\mathrm{Add}(D)$, characterized up to equivalence by (2.3.1.3), will be called the *additive category generated by D*.

2.3.2. Let A be an abelian category, and let $K^{\cdot\cdot}$ be a double complex, regarded as an object of $C^+(C^+(A))$, with the first index corresponding to the first symbol C^+. Denote 106
by $(K^{\cdot\cdot})_s$ the *associated simple complex* (cf. Tôhoku (2.4), whose notation we shall retain). This defines a functor

$$(\quad)_s : C^+(C^+(A)) \longrightarrow C^+(A) \tag{2.3.2.1}$$

and we leave it to the reader to verify the following lemma:

LEMMA 2.3.2.2. *The functor* $(\quad)_s$ *defines a triangulated functor*

$$K^+(C^+(A)) \longrightarrow K^+(A)$$

which preserves quasi-isomorphisms; it therefore defines a triangulated functor

$$D^+(C^+(A)) \longrightarrow D^+(A),$$

again denoted by $(\quad)_s$.

REMARK. The same result holds for the functor $(\quad)_s : C(C^b(A)) \to C(A)$, since the spectral sequence under consideration is biregular by (EGA III (11.3.3)).

2.3.3. Now let $(S, \mathscr{O}_S)$ be a ringed topos, let D be a $\mathscr{U}$-small category, and let $Z^\cdot \in C^+(\mathrm{Add}(D))$ be a cochain complex such that $Z^n = 0$ for $n < 0$.

Every object F of $\mathrm{Mod}(\Gamma(S \times D), \mathscr{O}_S)$ (which is identified with a functor $D \to \mathrm{Mod}(S, \mathscr{O}_S)$ by (2.1.1)) defines, by the universal property of $\mathrm{Add}(D)$ (2.3.1.3), an additive functor

$$\widetilde{F} : \mathrm{Add}(D) \longrightarrow \mathrm{Mod}(S, \mathscr{O}_S).$$

The image complex

$$\varepsilon_{Z^\cdot *}(F) := \widetilde{F}(Z^\cdot)$$

is an object of $C^+(S, \mathscr{O}_S)$ which varies functorially with F. We have thus defined an additive functor

$$\varepsilon_{Z^\cdot *} : \mathrm{Mod}(\Gamma(S \times D), \mathscr{O}_S) \longrightarrow C^+(S, \mathscr{O}_S),$$

which is visibly exact.

This defines a triangulated functor [20]:

$$K^+(\varepsilon_{Z^\cdot *}) : K^+(\Gamma(S \times D), \mathscr{O}_S) \longrightarrow K^+(C^+(S, \mathscr{O}_S)). \tag{2.3.3.1}$$

which preserves quasi-isomorphisms. Moreover, it follows from (2.3.2.2) that the composite functor

$$(\quad)_s \circ K^+(\varepsilon_{Z^\cdot *}) : K^+(\Gamma(S \times D), \mathscr{O}_S) \longrightarrow K^+(S, \mathscr{O}_S)$$

107 sends acyclic objects to acyclic objects. It therefore defines a triangulated functor

$$R\varepsilon_{Z^\cdot *} : D^+(\Gamma(S \times D), \mathscr{O}_S) \longrightarrow D^+(S, \mathscr{O}_S) \tag{2.3.3.2}$$

satisfying the relation [21] $Q \circ (\quad)_s \circ K^+(\varepsilon_{Z^\cdot *}) = R\varepsilon_{Z^\cdot *} \circ Q$ [21]

[20]N.D.E.: K^+ denotes the category of bounded-below complexes up to homotopy.

[21]N.D.E.: In other words, $\varepsilon_{Z^\cdot *}$ derives trivially as $F^\cdot \rightsquigarrow \varepsilon_{Z^\cdot *}(F^\cdot)$.

[21]For every abelian category A, the letter Q denotes the canonical functor $K^+(A) \to D^+(A)$.

2.3.4. Retain the preceding data and set $A = C^+(S, \mathcal{O}_S)$. First, $\widetilde{F}$ and $\overline{F}$ are compatible in the sense of commutativity of the Yoneda-type diagram

$$\begin{tikzcd}
 & A \arrow[r, "h"] & \mathrm{Hom}(A^\circ, \mathrm{Ens}) \\
D \arrow[ur, "F"] \arrow[r] \arrow[dr, "r_D"'] & \mathrm{Add}(D) \arrow[u, "\widetilde{F}"'] \arrow[d, "\widetilde{r_D}"'] & \\
 & \mathrm{Hom}(D, \mathrm{Ab}) \arrow[uur, "\overline{F}"'] &
\end{tikzcd}$$

where h and r_D are respectively the Yoneda-type arrows defined by

$$h(a) = (\alpha \longmapsto \mathrm{Hom}_A(\alpha, a)) \quad \text{and} \quad r_D(d) \overset{2.3.1}{=} (\delta \longmapsto \mathrm{Hom}_D(d, \delta)),$$

that is, $h \circ \widetilde{F} = \overline{F} \circ \widetilde{r_D}$. It follows that

$$h(\varepsilon_{Z^\bullet *}(F)) = \overline{F}(\widetilde{r_D}(Z^\bullet)).$$

By the definition of a projective limit, $h(\epsilon_* F) = \varprojlim F$, and, as already observed, one has tautologically $\overline{F}(\mathrm{Z}) = \varprojlim F$; hence

$$\overline{F}(\mathrm{Z}) = h(\epsilon_*(F)).$$

Regard $Z^\bullet$ as a chain complex in $\mathrm{Hom}(D, \mathrm{Ab})$, that is, as a complex of the abelian category $\mathrm{Hom}(D, \mathrm{Ab})^\circ$ by means of the functor $\tilde{r}_D$. We may therefore speak of exact sequences and resolutions. Explicitly, a sequence in $\mathrm{Add}(D)$

$$Z \longrightarrow Z' \longrightarrow Z''$$

is exact if, for every object j of D, the sequence

$$\mathrm{Hom}_{\mathrm{Add}(D)}(Z'', [j]) \longrightarrow \mathrm{Hom}_{\mathrm{Add}(D)}(Z', [j]) \longrightarrow \mathrm{Hom}_{\mathrm{Add}(D)}(Z, [j])$$

is exact in Ab. Let $t : Z^0 \to \mathrm{Z}$ be a morphism from Z^0 to the constant functor with value Z such that the composite morphism

$$Z^1 \longrightarrow Z^0 \xrightarrow{t} \mathrm{Z} \tag{2.3.4.1}$$

is zero.

By functoriality (cf. (2.3.1.1)), one deduces a morphism $\varepsilon_*(F) \to \varepsilon_{Z^\bullet *}(F)^0$ such that the composite morphism

$$\varepsilon_*(F) \longrightarrow \varepsilon_{Z^\bullet *}(F)^0 \longrightarrow \varepsilon_{Z^\bullet *}(F)^1 \tag{2.3.4.2}$$

is zero.

There is then a canonical morphism of triangulated functors

$$Q \circ K^+(\varepsilon_*) \xrightarrow{j} Q \circ (\quad)_s \circ K^+(\varepsilon_{Z^\bullet *}) = R\varepsilon_{Z^\bullet *} \circ Q \tag{2.3.4.3}$$

Proposition 2.3.5. *Retain the conditions and notation of (2.3.3) and (2.3.4); suppose, in addition, that the augmented complex*

$$\ldots \longrightarrow Z^n \longrightarrow Z^{n-1} \longrightarrow \ldots \longrightarrow Z^1 \longrightarrow Z^0 \xrightarrow{t} \mathrm{Z}$$

defines a resolution of Z.

Then, for every complex $F^{\cdot}$ consisting of objects of $I_{(S\times D,\mathcal{O}_S)}$ (cf. 1.3.10),

$$j(F^{\cdot}) : Q \circ K^{+}(\varepsilon_{*})(F^{\cdot}) \longrightarrow (R\varepsilon_{Z^{\cdot}*} \circ Q)(F^{\cdot})$$

108 *is an isomorphism.*

Let i be an object of D, and let Q_i be a module on $(S, \mathcal{O}_S)$. We show that $j(e_{i*}(Q_i))$ is an isomorphism. It is enough to see that the sequence

$$0 \longrightarrow Q_i \longrightarrow \varepsilon_{Z^{\cdot}*}(e_{i*}(Q_i))^0 \longrightarrow \varepsilon_{Z^{\cdot}*}(e_{i*}(Q_i))^1 \longrightarrow \dots \tag{2.3.5.1}$$

is exact.

For this, it is enough to show that, for every object X of $\mathrm{Mod}(S, \mathcal{O}_S)$, the sequence (2.3.5.2)

$$0 \longrightarrow \mathrm{Hom}(X, Q_i) \longrightarrow \mathrm{Hom}(X, \varepsilon_{Z^{\cdot}*}(e_{i*}(Q_i))^0) \longrightarrow \mathrm{Hom}(X, \varepsilon_{Z^{\cdot}*}(e_{i*}(Q_i))^1) \longrightarrow \dots$$

is exact. An immediate calculation shows that (2.3.5.2) reduces to

$$\begin{aligned} 0 \longrightarrow \mathrm{Hom}(\mathbf{Z}, \mathrm{Hom}(X, Q_i)) \longrightarrow \mathrm{Hom}(\mathrm{Hom}_{\mathrm{Add}(D)}(Z^0, [i]), \mathrm{Hom}(X, Q_i)) \longrightarrow \\ \mathrm{Hom}(\mathrm{Hom}_{\mathrm{Add}(D)}(Z^1, [i]), \mathrm{Hom}(X, Q_i)) \longrightarrow \dots . \end{aligned} \tag{2.3.5.3}$$

By hypothesis, the sequence

$$0 \leftarrow \mathbf{Z} \leftarrow \mathrm{Hom}_{\mathrm{Add}(D)}(Z^0, [i]) \leftarrow \mathrm{Hom}_{\mathrm{Add}(D)}(Z^1, [i]) \leftarrow \dots$$

is an exact sequence of free modules and is therefore null-homotopic. It follows that (2.3.5.3) is exact.

If, for every object i of D, Q_i is a totally acyclic module, then, by applying the functor $\prod_{i\in\mathrm{Ob}(D)}$ to the complexes (2.3.5.1), one sees that the resulting complex is still acyclic by (1.3.10.1). Thus $j(\prod_{i\in\mathrm{Ob}(D)} e_{i*}(Q_i))$ is an isomorphism.

We leave it to the reader to deduce that $j(F^{\cdot})$ is an isomorphism when $F^n \in I_{(S\times D,\mathcal{O}_S)}$ for every n (for example, by spectral sequences).

109 Corollary 2.3.6. *Under the conditions of (2.3.5), the morphism j identifies $R\varepsilon_{Z^{\cdot}*}$ with the derived functor of ε_{*}.*

The subcategory of $K^{+}(\Gamma(S\times D), \mathcal{O}_S)$ defined by the complexes consisting of objects of $I_{(S\times D,\mathcal{O}_S)}$ satisfies the conditions of theorem (5.1) of [[7], I] for the functor $K^{+}(\varepsilon_{*})$, by (1.3.10) and (2.3.5). The corollary follows immediately from this observation.

Corollary 2.3.7. *Under the conditions of (2.3.5), let (E, A) be a ringed D-topos and let $\theta : (E, A) \to (S \times D, \mathcal{O}_S)$ be an augmentation. There is a canonical isomorphism*

$$R\overline{\theta}_{*} \xrightarrow{\sim} R\varepsilon_{Z^{\cdot}*} \circ R\Gamma(\theta_{*}).$$

This follows from [[7], I, 5.4].

Remark 2.3.8. We knew a priori that $R\varepsilon_{*}$ exists and that there is an isomorphism $R\overline{\theta}_{*} \xrightarrow{\sim} R\varepsilon_{*} \circ R\Gamma(\theta_{*})$, since $\Gamma(\theta_{*})$ and ε_{*} are induced by morphisms of ringed topoi (cf. (1.3.6) and (2.1.1)). Nevertheless, the preceding calculation will prove very useful and applicable in other contexts, since in practice the conditions of (2.3.5) will always be satisfied.

We shall now give the fundamental examples to which (2.3.5) can be applied.

2.3.9. In $\mathrm{Add}(\Delta)$, put $Z^n = [n]$ and define $d^n : Z^n \to Z^{n+1}$ by the formula

$$d^n = \sum_{i=0}^{n+1} (-1)^i \partial^i. \tag{2.3.9.1}$$

Here $\partial^i : [n] \to [n+1]$ is the ith face [cf. [3] II 2]. For $t : Z^0 \to \mathbf{Z}$, take the evident natural augmentation[22].

The condition of 2.3.4 is trivially satisfied. The condition of 2.3.5 follows from [5] I (3.7.4).

2.3.10. In $\mathrm{Add}(\Delta \times \Delta)$, consider the simple complex associated with the double complex 110

$$\begin{tikzcd}
\big([n],[m+1]\big) \arrow[r, "s_1^{n,m+1}"] & \big([n+1],[m+1]\big) \\
\big([n],[m]\big) \arrow[u, "s_2^{n,m}"] \arrow[r, "s_1^{n,m}"] & \big([n+1],[m]\big) \arrow[u, "s_2^{n+1,m}"']
\end{tikzcd} \tag{2.3.10.1}$$

with

$$\begin{cases} S_1^{n,m} = \displaystyle\sum_{i=0}^{n+1} (-1)^i (\partial^i, \mathrm{id}_{[m]}), \\ S_2^{n,m} = \displaystyle\sum_{j=0}^{m+1} (-1)^j (\mathrm{id}_{[n]}, \partial^j). \end{cases} \tag{2.3.10.2}$$

The reader may verify (using 2.3.9) that the complex so defined, equipped with the evident natural augmentation, satisfies the conditions of 2.3.5.

2.3.11. The multisimplicial case is treated similarly (with $\underbrace{\Delta \times \Delta \times \cdots \times \Delta}_{p \text{ times}}$).

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2.4. Relative cohomological descent.

2.4.0. Let $\mathscr{E} \to \boldsymbol{B}$ be a category bifibered in duals of topoi, and let $\mathscr{O}$ be a ring of $\Gamma(\mathscr{E}^\circ)$.

These data canonically define a category fibered and cofibered in abelian categories over $\boldsymbol{B}^\circ$, denoted by $\mathrm{Mod}(\mathscr{E}^\circ, \mathscr{O})$ (cf. (1.3.3) and (1.3.4)). Throughout what follows, the "direct-image" and "inverse-image" functors will always be taken with respect to the fibered and cofibered functor $\mathrm{Mod}(\mathscr{E}^\circ, \mathscr{O}) \to \boldsymbol{B}^\circ$.

Definition 2.4.0.1. *A morphism $f : T \to S$ in $\boldsymbol{B}$ is said to be* flat *(relative to $(\mathscr{E}, \mathscr{O})$) if $f_* : \mathrm{Mod}(\mathscr{E}^\circ_S, \mathscr{O}_S) \to \mathrm{Mod}(\mathscr{E}^\circ_T, \mathscr{O}_T)$ is an exact functor*[23].

[22]N.D.E.: Explicitly, if F is a simplicial complex with differential d_F, then $\varepsilon_{Z^\bullet *}(F)$ is the simple complex associated with the double complex $F^j(i)$ whose horizontal differential is the simplicial differential d^i and whose vertical differential is d^j_F.

[23]N.D.E.: This notation may cause confusion: one should probably write $(f^\circ)_\star$, which formally coincides with $(f^\star)^\circ$, where $f^\star$ is the fibered functor associated with $\mathscr{E} \to \boldsymbol{B}$. For example, for the category of sheaves on topological spaces fibered over the category of topological spaces (cf. 4.1), one has $(f^\circ)_\star(S, F) = (T, f^{-1}F) = (f^\star)^\circ(S, F)$, where $f : T \to S$ is a continuous map of topological spaces, F is a sheaf on S, and f^{-1} is the usual inverse image. One then recognizes the usual flatness conditions.

2.4.1. Let G be a fibered and cofibered subcategory of $\mathrm{Mod}(\mathscr{E}^\circ, \mathscr{O})$ such that, for every object S of B, G_S is a *thick* subcategory (cf. Tôhoku (1.11)) of $(\mathrm{Mod}(\mathscr{E}^\circ, \mathscr{O}))_S$. The reader will find examples of such situations in Section 4.

If S is an object of B, denote by $D^+(S)$ the derived category of $\mathrm{Mod}(\mathscr{E}^\circ_S, \mathscr{O}_S)$. Following [[7] (I § 4)], introduce the category $D^+_{G_S}(S)$, which identifies with the full subcategory of $D^+(S)$ formed by the objects $X^\cdot$ such that $H^n(X^\cdot) \in G_S$ for every n.

We assume throughout that the following condition holds.

. For every morphism $h : T \to S$ in B, the functor $Rh^* : D^+(T) \to D^+(S)$ has the property that $Rh^*(F) \in D^+_{G_S}(S)$ whenever $F \in G_T$[24].

By [[7] (I. (7.3). (ii))], this condition implies that $Rh^*(D^+_{G_T}(T)) \subset D^+_{G_S}(S)$.

112 2.4.2. Let D be a $\mathscr{U}$-small category. *Throughout what follows, we suppose given a complex $Z^\cdot$ of* $\mathrm{Add}(D)$ *satisfying the conditions of* (2.3.5) (in applications, one will in fact have $D = \Delta$ or $D = \Delta \times \Delta$).

If $X : D^\circ \to B$ is a D-object of B, denote by $\mathrm{Mod}_G(\Gamma(\overline{X}), \mathscr{O})$ the full subcategory of $\mathrm{Mod}(\Gamma(\overline{X}), \mathscr{O})$ formed by the objects F such that $e_i^*(F) \in G_i$ $(= G_{X_i})$ for every object i of D. The category $\mathrm{Mod}_G(\Gamma(\overline{X}), \mathscr{O})$ is a thick subcategory of $\mathrm{Mod}(\Gamma(\overline{X}), \mathscr{O})$. Denote by $K^+_G(\Gamma(\overline{X}), \mathscr{O})$ (resp. $D^+_G(\Gamma(\overline{X}), \mathscr{O})$) the full subcategory of $K^+(\Gamma(\overline{X}), \mathscr{O})$ (resp. $D^+(\Gamma(\overline{X}), \mathscr{O})$) formed by the objects $X^\cdot$ such that $H^n(X^\cdot) \in \mathrm{Mod}_G(\Gamma(\overline{X}), \mathscr{O})$ for every n.

2.4.3. Let X and X' be two D-objects of B, and let $\alpha : X \to X'$ be a functorial morphism. Recall (cf. (1.3.4)) that the corresponding morphism of ringed D-topoi is again denoted by $\alpha : (\overline{X}, \mathscr{O}) \to (\overline{X}', \mathscr{O})$.

Definition 2.4.3.1. *The morphism $\alpha : X \to X'$ is said to be* flat *if the corresponding morphism of ringed D-topoi is flat in the sense of (1.3.6.1).*

We then have the evident lemma:

Lemma 2.4.3.2. *The morphism $\alpha : X \to X'$ is flat if and only if, for every object i of D, $\alpha_i : X_i \to X'_i$ is a flat morphism* (cf. (2.4.0.1)).

2.4.4. A morphism $\alpha : X \to X'$ defines a functor

$$K^+(\Gamma(\alpha_*)) : K^+(\Gamma(\overline{X}), \mathscr{O}) \longrightarrow K^+(\Gamma(\overline{X}'), \mathscr{O}). \tag{2.4.4.1}$$

Lemma 2.4.4.2. *The functor*

$$K^+(\Gamma(\alpha_*))|_{K^+_G(\Gamma(\overline{X}),\mathscr{O})} : K^+_G(\Gamma(\overline{X}), \mathscr{O}) \longrightarrow K^+(\Gamma(\overline{X}'), \mathscr{O})$$

has a right derived functor, denoted by $R_G\Gamma(\alpha_*)$, and the canonical morphism

$$R_G\Gamma(\alpha_*) \longrightarrow R\Gamma(\alpha_*)|_{D^+_G(\Gamma(\overline{X}),\mathscr{O})}$$

113 *is an isomorphism.*

Moreover,

$$R_G\Gamma(\alpha_*)(D^+_G(\Gamma(\overline{X}), \mathscr{O})) \subset D^+_G(\Gamma(\overline{X}'), \mathscr{O}).$$

[24] N.D.E.: As above, it would be more correct to write $R(h^\circ)^* = R(h_*)^\circ$ rather than Rh^*, which would make the notation more consistent with the usual geometric intuition. One could equally well have taken the fibered and cofibered functors with respect to $\mathrm{Mod}(\mathscr{E}, \mathscr{O}) \to B$ rather than the opposite categories. The fibers, however, would then have been only duals of topoi.

The first part follows from (1.3.10) and [[7], (I. (5.2))]. To verify the last assertion, by [[7], (I. (7.3). (ii))], it suffices to show that $R\Gamma(\alpha_*)(F) \in D^+_G(\Gamma(\overline{X}'), \mathcal{O})$ whenever $F \in \mathrm{Mod}_G(\Gamma(\overline{X}), \mathcal{O})$. This follows from the explicit calculation of $R\Gamma(\alpha_*)$ given by (1.3.1.1) and from hypothesis (2.4.1.1).

2.4.5. Let S be an object of B. The ringed D-topos $(\overline{C^D_S}, \mathcal{O})$ (cf. (1.2.6)) identifies with the constant ringed D-topos $(\mathscr{E}^\circ_S \times D, \mathcal{O}_S)$, and we have the following lemma, which follows from [[7] (I. (7.3). (ii))]:

Lemma 2.4.5.1. *The functor $R\varepsilon_{Z^\cdot *}|_{D^+_G(\Gamma(\overline{C^D_S}),\mathcal{O})}$, denoted by $R_G\varepsilon_{Z^\cdot *}$, takes values in $D^+_{G_S}(S)$ and identifies with a right derived functor of*

$$K^+(\varepsilon_*)|_{K^+_G(\Gamma(\overline{C^D_S}),\mathcal{O})} : K^+_G(\Gamma(\overline{C^D_S}), \mathcal{O}) \longrightarrow K^+(S).$$

Proposition 2.4.6. *Let $\theta : X \to C^D_S$ be an augmented D-object of B. The functor $K^+(\overline{\theta}_*)|_{K^+_G(\Gamma(\overline{X}),\mathcal{O})}$ has a right derived functor, denoted by $R_G\overline{\theta}_*$, taking values in $D^+_{G_S}(S)$. Moreover, there is a canonical isomorphism*

$$R_G\overline{\theta}_* \xrightarrow{\sim} R_G\varepsilon_{Z^\cdot *} \circ R_G\Gamma(\theta_*).$$

The proof, from what precedes, is left to the reader.

We are now in a position to give the definitions concerning cohomological descent.

2.4.7. Let $\theta : X \to C^D_S$ be an augmented D-object of B. Suppose that θ is flat, so that the restriction to $D^+_{G_S}(S)$ of the functor $\overline{\theta}^*$ takes values in $D^+_G(\Gamma(\overline{X}), \mathcal{O})$; denote this restriction by $\overline{\theta}^*_G$. With this notation: 114

Lemma 2.4.7.1. *There are two functorial morphisms $\alpha : \mathrm{id}_{D^+_{G_S}(S)} \to R_G\overline{\theta}_* \circ \overline{\theta}^*_G$ and $\beta : \overline{\theta}^*_G \circ R_G\overline{\theta}_* \to \mathrm{id}_{D^+_G(\Gamma(\overline{X}),\mathcal{O})}$ which make these two functors adjoint.*

This follows immediately from (2.2.3.1) and (2.4.6).

Definition 2.4.8. *The morphism θ is called an augmentation of cohomological 1-descent relative to G if*

1°) *θ is flat.*

2°) *The functor $\overline{\theta}^*_G$ is fully faithful.*

Remark 2.4.9. *Condition 2°) of the preceding definition may equivalently be expressed by saying that the morphism α in (2.4.7.1) is an isomorphism.*

Definition 2.4.10. *Let $\theta : X \to C^D_S$ be an augmented D-object of B. The morphism θ is called an* augmentation of effective descent relative to G *if the functor*

$$\Gamma(\theta^*) \circ \varepsilon^* : G_S \longrightarrow \Gamma^{\mathrm{Cocart}}(D, G)$$

is an equivalence of categories.

Definition 2.4.11. *The morphism θ is called an* augmentation of cohomological 2-descent *(or of* effective cohomological descent*)* relative to G *if θ is both an augmentation of cohomological 1-descent and an augmentation of effective descent relative to G.*

The proof of (2.2.7) then gives the following result:

Theorem 2.4.12. *Let $\theta : X \to C^D_S$ be an augmentation of cohomological 2-descent relative to G. Then the essential image of $\overline{\theta}^*_G$ is the full subcategory of $D^+_G(\Gamma(\overline{X}), \mathcal{O})$ formed by the complexes $F^\bullet$ such that, for every n, $H^n(F^\bullet)$ is a cocartesian section of G.* 115

The following proposition, whose verification is left to the reader, translates Definition (2.4.10) into the language of ([4], 6):

Proposition 2.4.13. *The morphism $\theta : X \to C^D_S$ is an augmentation of effective descent relative to G if and only if the functor $D^\circ \to B/S$ over B defined by θ induces an equivalence between the category of cartesian sections of G° over B/S and the category of cartesian sections of G° over D°.*

Corollary 2.4.14. *Suppose that finite fiber products are representable in B, and let $f : R \to S$ be a morphism. The augmented semi-simplicial object $[R|_f S]$* (cf. (1.2.7)) *is of effective descent relative to G if and only if f is an effective descent morphism for the fibered category G° over B.*

This follows from (2.4.13) and ([4], 9).

Definition 2.4.15. *Suppose that finite fiber products are representable in B, and let $f : R \to S$ be a morphism. We say that f is a morphism of cohomological i-descent relative to G ($i = 1, 2$) if the augmented semi-simplicial object $[R|_f S]$ is of cohomological i-descent relative to G.*

Definition 2.4.16. *Suppose that finite fiber products are representable in B, and let $\theta : X \to C^D_S$ be an augmentation. We say that θ is an* augmentation of universal cohomological i-descent ($i = 1, 2$) relative to G *if, for every morphism $T \to S$, the augmentation*
116 *tion $X_T \xrightarrow{\theta_T} C^D_T$ obtained by base change is an augmentation of cohomological i-descent relative to G.*

The notion of a *morphism of universal cohomological i-descent relative to G* follows immediately from the two preceding definitions.

We will often use the terminology "augmentation of G-cohomological i-descent" in place of "augmentation of cohomological i-descent relative to G".

2.5. The descent spectral sequence.

2.5.1. Let A be an abelian category. Denote by $F(A)$ the category whose objects are the objects of A equipped with a discrete and codiscrete filtration, and whose morphisms are the filtered morphisms of A: the category $F(A)$ is an additive category (but is not abelian).

The category $K(F(A))$ of filtered complexes of A, whose filtration is discrete and codiscrete degree by degree, is a triangulated category, and the canonical functors

$$\mathrm{Gr}_n : K(F(A)) \longrightarrow K(A) \tag{2.5.1.1}$$

are triangulated. The set Σ of morphisms f of $K(F(A))$ such that $\mathrm{Gr}_n(f)$ is a quasi-isomorphism for every n is therefore a saturated multiplicative system [cf. [9]. § 2. n° 1].

By inverting the arrows of this system, one obtains a new triangulated category, denoted by $DF(A)$, and the functors Gr_n extend to functors

$$\mathrm{Gr}_n : DF(A) \longrightarrow D(A). \tag{2.5.1.2}$$

In what follows, we will consider only complexes bounded below: this naturally gives rise to the notation $K^+(F(A))$ and $D^+F(A)$.

2.5.2. Let B be a thick subcategory of A, and denote by $K^+_B(F(A))$ the full subcategory of $K^+(F(A))$ whose objects are the complexes $X^\bullet$ such that $H^i(\mathrm{Gr}_j(X^\bullet)) \in B$ for every pair of integers (i,j) : $K^+_B(F(A))$ is a *localizing triangulated subcategory* of $K(F(A))$ [cf. [7]. (I. § 5)]. We denote by $D^+_B F(A)$ the full subcategory of $DF(A)$ whose objects are the complexes bounded below $X^\bullet$ such that $H^i(\mathrm{Gr}_j(X^\bullet)) \in B$ for every pair (i,j): by (*loc. cit.*), $D^+_B F(A)$ identifies with the category of fractions $K^+_B(F(A))_{\Sigma\cap K^+_B(F(A))}$. 117

The "forgetful functor for filtrations"

$$\iota : K^+(F(A)) \longrightarrow K^+(A)$$

is triangulated. Moreover, it follows from [[5]. I. (4.7)] that $\iota(f)$ is a quasi-isomorphism if $f \in \Sigma$ and that $\iota(K^+_B(F(A))) \subset K^+_B(A)$. The functor ι thus extends to a triangulated functor

$$\iota : D^+F(A) \longrightarrow D^+(A) \tag{2.5.2.1}$$

such that $\iota(D^+_B F(A)) \subset D^+_B(A)$.

Finally, note that there is a canonical *spectral functor* $D^+_B F(A) \to B$, converging to $H^n \circ \iota$, whose E_1 term is given by $H^{p+q} \circ \mathrm{Gr}_p$ [cf. [5] I. (4.2)].

We now return to the notation of (2.3), *assuming that* $D = \Delta$: $Z^\bullet$ will denote the complex of $\mathrm{Add}(\Delta)$ defined by (2.3.9.1).

Proposition 2.5.3. *Let $(S, \mathcal{O}_S)$ be a ringed topos. The functor*

$$R\varepsilon_{Z^\bullet *} : D^+(\Gamma(S\times D), \mathcal{O}_S) \longrightarrow D^+(S, \mathcal{O}_S)$$

admits a canonical factorization

$$D^+(\Gamma(S\times D), \mathcal{O}_S) \xrightarrow{F^{R\varepsilon_{Z^\bullet *}}} D^+F(S, \mathcal{O}_S) \xrightarrow{\iota} D^+(S, \mathcal{O}_S).$$

Moreover, for every integer i, the following diagram is commutative:

$$\begin{tikzcd}
D^+(\Gamma(S\times D), \mathcal{O}_S) \arrow[r, "Re_i^*"] \arrow[d, "F^{R\varepsilon_{Z^\bullet *}}"'] & D^+(S, \mathcal{O}_S) \\
D^+F(S, \mathcal{O}_S) \arrow[ur, "\mathrm{Gr}_i"'] &
\end{tikzcd}$$

.

Consider the functor $C^+(C^+(S, \mathcal{O}_S)) \xrightarrow{(\ \ )_s} C^+(S, \mathcal{O}_S)$: if $X^{\bullet\bullet}$ is an object of $C^+(C^+(S, \mathcal{O}_S))$, 118 we equip $(X^{\bullet\bullet})_s$ with its second canonical filtration [cf. [5]. I.4.8]; this gives a factorization

$$(\ \ )_s : C^+(C^+(S, \mathcal{O}_S)) \longrightarrow C^+(F(S, \mathcal{O}_S)) \xrightarrow{\iota} C^+(S, \mathcal{O}_S) \tag{2.5.3.1}$$

which passes to the categories K^+:

$$(\ \ )_s : K^+(C^+(S, \mathcal{O}_S)) \longrightarrow K^+(F(S, \mathcal{O}_S)) \xrightarrow{\iota} K^+(S, \mathcal{O}_S) \tag{2.5.3.2}$$

because a homotopy in $C^+(C^+(S, \mathcal{O}_S))$ induces a filtered homotopy.

We then have a diagram

(2.5.3.3)
$$\begin{tikzcd}
D^+(\Gamma(S\times\Delta),\mathcal{O}_S) \arrow[rr, "R\epsilon_{Z^\bullet *}"] \arrow[dr, dashed] & & D^+(S,\mathcal{O}_S) \\
& D^+F(S,\mathcal{O}_S) \arrow[ur, "\iota"'] & \\
& K^+(F(S,\mathcal{O}_S)) \arrow[u] \arrow[dr, "\iota"] & \\
K^+(\Gamma(S\times\Delta),\mathcal{O}_S) \arrow[uuu] \arrow[ur] \arrow[rr, "(\ )_s\circ K^+(\epsilon_{Z^\bullet *})"'] & & K^+(S,\mathcal{O}_S) \arrow[uuu]
\end{tikzcd}$$

and one verifies that there is a unique functor

$$_F R\epsilon_{Z^\bullet *} : D^+(\Gamma(S\times\Delta),\mathcal{O}_S) \longrightarrow D^+F(S,\mathcal{O}_S)$$

119 making (2.5.3.3) commutative.

The rest of the proposition is evident.

Proposition 2.5.4. *Let (E,A) be a semi-simplicial ringed topos and*

$$\theta : (E,A) \longrightarrow (S\times\Delta,\mathcal{O}_S)$$

an augmentation. For every integer i, denote by

$$\theta_i : (E_i,A_i) \longrightarrow (S,\mathcal{O}_S)$$

the morphism of ringed topoi induced by θ over i.

The functor $R\overline{\theta}_* : D^+(\Gamma(E),A) \to D^+(S,\mathcal{O}_S)$ admits a canonical factorization

$$D^+(\Gamma(E),A) \xrightarrow{_F R\overline{\theta}_*} D^+F(S,\mathcal{O}_S) \xrightarrow{\iota} D^+(S,\mathcal{O}_S)$$

such that, for every integer i, the canonical diagram

$$\begin{tikzcd}
D^+(\Gamma(E),A) \arrow[r, "Re_i^*"] \arrow[d, "_F R\overline{\theta}_*"'] & D^+(E_i,A_i) \arrow[d, "R\theta_{i*}"] \\
D^+F(S,\mathcal{O}_S) \arrow[r, "\mathrm{Gr}_i"'] & D^+(S,\mathcal{O}_S)
\end{tikzcd}$$

is essentially commutative.

This follows from what precedes and from (1.3.1.2).

Proposition 2.5.5. *Let (E,A) be a semi-simplicial ringed topos and let*

$$\theta : (E,A) \longrightarrow (S\times\Delta,\mathcal{O}_S)$$

be an augmentation of cohomological 1-descent. Let

$$H : (S,\mathcal{O}_S) \longrightarrow (R,\mathcal{O}_R)$$

be a morphism of ringed topoi. Then there is a spectral functor from $D^+(S,\mathcal{O}_S)$ to $\mathrm{Mod}(R,\mathcal{O}_R)$*:*

$$E_1^{pq} = R^q(H\circ\theta_p)_*\circ\theta_p^* \longrightarrow R^{p+q}H_*$$

120 *called the* descent spectral functor.

Let $H \circ \theta : (E, A) \to (R \times \Delta, \mathcal{O}_R)$ be the augmentation canonically deduced from θ by composition with H; we have

$$\overline{(H \circ \theta)}_* = H_* \circ \overline{\theta}_* \tag{2.5.5.1}$$

so that the diagram

$$\begin{tikzcd}
D^+(S, \mathcal{O}_S) \arrow[r, "\overline{\theta}^*"] \arrow[dr, "RH_*"'] & D^+(\Gamma(E), A) \arrow[d, "R\overline{(H\circ\theta)}_*"] \\
 & D^+(R, \mathcal{O}_R)
\end{tikzcd} \tag{2.5.5.2}$$

is essentially commutative, since

$$R\overline{(H \circ \theta)}_* \xrightarrow{\sim} RH_* \circ R\overline{\theta}_* \text{ and } \mathrm{id}_{D^+(S,\mathcal{O}_S)} \xrightarrow{\sim} R\overline{\theta}_* \circ \overline{\theta}^*.$$

By (2.5.4), for every i we have an essentially commutative diagram

$$\begin{tikzcd}
D^+(S, \mathcal{O}_S) \arrow[r, "\overline{\theta}^*"] \arrow[ddr, "RH_*"'] & D^+(\Gamma(E), A) \arrow[r, "Re_i^*"] \arrow[d, "{}_F R\overline{(H\circ\theta)}_*"'] & D^+(E_i, A_i) \arrow[d, "R(H\circ\theta_i)_*"] \\
 & D^+F(R, \mathcal{O}_R) \arrow[r, "\mathrm{Gr}_i"'] \arrow[d, "\iota"'] & D^+(R, \mathcal{O}_R) \\
 & D^+(R, \mathcal{O}_R) &
\end{tikzcd} \tag{2.5.5.3}$$

The asserted spectral functor is obtained by taking the canonical spectral functor on 121
$D^+F(R, \mathcal{O}_R)$ and composing it with ${}_F R\overline{(H \circ \theta)}_* \circ \overline{\theta}^*$.

We leave it to the reader to verify the following proposition:

Proposition 2.5.6. *With the preceding notation, the E_2^{pq} term of the preceding spectral functor is*

$$\check{H}^p \circ R^q\Gamma((H \circ \theta)_*) \circ \overline{\theta}^*$$

where $\check{H}^p : \mathrm{Mod}(\Gamma(R\times\Delta), \mathcal{O}_R) \to \mathrm{Mod}(R, \mathcal{O}_R)$ is the functor that assigns to every functor $\Delta \xrightarrow{F} \mathrm{Mod}(R, \mathcal{O}_R)$ the homology object in degree p of the associated complex (denoted by $\varepsilon_{Z^\bullet *}(F)^\bullet$ in (2.3.3)).

We now return to the terminology introduced in (2.4): by the sorites (2.5.2), all the preceding considerations carry over word for word by placing the letter G as a subscript wherever this makes sense. The details are left to the reader; in particular, one obtains:

Proposition 2.5.7. *Let $\theta : X \to C_S^D$ be an augmentation of cohomological 1-descent* 122
relative to G, and let $h : S \to R$ be a morphism of B; there is a spectral functor from $D^+_{G_S}(S)$ to G_R:

$$E_1^{pq} = R_G^q(h \circ \theta)_* \circ \theta^*_{p,G} \longrightarrow R_G^{p+q} h_*$$

with $E_2^{pq} = \check{H}^p \circ R^q_G \Gamma((h \circ \theta)_*) \circ \overline{\theta}^\star_G$.

3. Descent criteria

3.0. Notation. Throughout what follows, we retain the notation of (2.4). We shall always assume *that finite fiber products are representable in* $\boldsymbol{B}$.

We shall also assume that $\boldsymbol{B}$ satisfies the following condition:

3.0.0. *Coproducts exist in* $\boldsymbol{B}$*, are disjoint and universal, and are families of effective* $\mathscr{E}$*-descent and effective* $\boldsymbol{G}^\circ$*-descent* [cf. [4] (9.23), (9.25), and (9.27)].

3.0.1. We now recall some classical notation concerning simplicial objects of a category [cf. [3], Chap. II, and (V, Appendix)].

Let E be a category *having finite projective limits.* $\Delta^\circ E$ denotes the category of simplicial objects of E (cf. (1.2.6)).

For an integer n, denote by Δ_n (resp. ${\Delta_n}^+$, ${\Delta_n}^-$) the full subcategory of Δ (resp. Δ^+, Δ^-) consisting of the objects $[p]$ such that $p \leq n$.

The restriction functor

$$i_n^* : \Delta^\circ E \longrightarrow \underset{n}{\Delta^\circ} E \tag{3.0.1.1}$$

123 has a right adjoint i_{n*}, since finite projective limits exist in E (cf. (1.2.10)). The functor $i_{n*} \circ i_n^*$ is denoted by cosq_n and is called the *coskeleton functor of order* n; by abuse of notation, we shall also use cosq_n for the functor i_{n*}.

For every integer p, denote by $\Delta_{n[p]}$ (resp. ${\Delta_{n[p]}}^+$) the full subcategory of $\Delta_{[p]}$ (resp. $\Delta^+_{[p]}$), the category of objects of Δ (resp. Δ^+) over $[p]$, defined by the objects $[q]$ over $[p]$ such that $q \leq n$. We leave it to the reader to verify that

$$(\operatorname{cosq}_n(X))_p = \varprojlim_{\Delta_{n[p]}} X_q \xrightarrow{\sim} \varprojlim_{\Delta^+_{n[p]}} X_q. \tag{3.0.1.2}$$

3.0.2. Let X and X' be two objects of $\Delta^\circ E$, and let $f, g : X \rightrightarrows X'$ be two morphisms. A *homotopy from* f *to* g consists of a map $h_n : \operatorname{Hom}_\Delta([n],[1]) \to \operatorname{Hom}_E(X_n, X'_n)$ for every n, satisfying the following two conditions:

$$h_n(\partial^1) = f_n \quad \text{and} \quad h_n(\partial^\circ) = g_n \quad \text{for every } n. \tag{3.0.2.1}$$

. For every arrow $[n] \to [p]$ in Δ, the following diagram is commutative:

$$\begin{tikzcd}
\operatorname{Hom}_\Delta\big([p],[1]\big) \arrow[r] \arrow[dd] & \operatorname{Hom}_E(X_p, X'_p) \arrow[dr] & \\
 & & \operatorname{Hom}_E(X_p, X'_p) \\
\operatorname{Hom}_\Delta\big([n],[1]\big) \arrow[r] & \operatorname{Hom}_E(X_n, X'_n) \arrow[ur] &
\end{tikzcd}.$$

N.B.: The relation thus introduced on the set of morphisms from X to X' is not an equivalence relation. This inconvenience can be remedied by introducing the notion of *composite homotopy* [cf. [3], Chap. III], but we shall not use that notion.

124 Lemma 3.0.2.3. *Let* X *and* X' *be two objects of* $\Delta^\circ E$*, let* $f, g : X \rightrightarrows X'$ *be two morphisms, and let* $F : E \to C$ *be a functor, where* C *is an abelian category. A simplicial*

homotopy from f to g induces a homotopy between the morphisms of cochain complexes canonically associated with $F(f)$ and $F(g)$.

This reduces to the case where C is the category of modules over a ring, and one applies [[5], I, (3.7.1)].

Lemma 3.0.2.4. *Let n be an integer, and let X and X' be two objects of $\Delta^\circ E$. Let f and g be two morphisms from X to X' such that $f_p = g_p$ for $p < n$. Then there exists a simplicial homotopy from $\operatorname{cosq}_n(f)$ to $\operatorname{cosq}_n(g)$.*

For $p < n$, define

$$h_p : \operatorname{Hom}_\Delta([p],[1]) \longrightarrow \operatorname{Hom}(X_p, X'_p)$$

to be the constant map with value $f_p = g_p$. Next define $h_n : \operatorname{Hom}_\Delta([n],[1]) \to \operatorname{Hom}(X_n, X'_n)$ by sending every element of $\operatorname{Hom}_\Delta([n],[1])$, except ∂°, to f_n, and ∂° to g_n. Finally, observe that

$$\operatorname{Hom}_\Delta([k],[1]) = \varprojlim_{\substack{[q]\to[k] \\ q\leq n}} \operatorname{Hom}([q],[1])$$

for $k > n$, which makes it possible to define h_k canonically.

3.1. Comparison of two augmentations from the viewpoint of 1-cohomological descent. Let S be an object of B. Denote by $\operatorname{Hom}_{\text{plat}}(D^\circ, B/S)$ the full subcategory of $\operatorname{Hom}(D^\circ, B/S)$ whose objects are the flat augmentations [cf. (1.2.6) and (2.4.3.1)].

Proposition 3.1.1. *Let $X \xrightarrow{u} C^D_S$ and $X' \xrightarrow{u'} C^D_S$ be two objects of $\operatorname{Hom}_{\text{plat}}(D^\circ, B/S)$. Let $f : X \to X'$ be a morphism over S (i.e. a morphism in $\operatorname{Hom}(D^\circ, B/S)$). There is naturally associated with it a functorial morphism*

$$\eta^f_G : R_G\bar{u}'_* \circ \bar{u}'^*_G \longrightarrow R_G\bar{u}_* \circ \bar{u}^*_G$$

such that the following diagram is commutative:

$$\begin{tikzcd}
& R_G\bar{u}'_* \circ \bar{u}'^*_G \arrow[dd, "\eta^f_G"] \\
\mathrm{id}_{D^+_{G_S}(S)} \arrow[ur] \arrow[dr] & \\
& R_G\bar{u}_* \circ \bar{u}^*_G.
\end{tikzcd}$$

Moreover, if $D = \Delta$, then η^f_G depends only on the homotopy class (cf. (3.0.2)) *of f in* 125
$\operatorname{Hom}(\Delta^\circ, B/S)$.

It is clear that, for the construction of η^f_G, we may suppose that $G = \operatorname{Mod}(\mathscr{E}^\circ, \mathscr{O})$.

Let I be the category defined by the graph

$$0 \longrightarrow 1. \tag{3.1.1.1}$$

Denote by $r_0 : D \to I \times D$ (resp. r_1) the fully faithful functor defined by $r_0(i) = (0, i)$ (resp. $r_1(i) = (1, i)$).

By virtue of the canonical isomorphism

$$\mathrm{Hom}((I\times D)^\circ, B/S) \xrightarrow{\sim} \mathrm{Hom}(I^\circ, \mathrm{Hom}(D^\circ, B/S)) \tag{3.1.1.2}$$

the data of (3.1.1) define a flat augmentation [25]

$$X f X' \xrightarrow{ufu'} C_S^{I\times D}. \tag{3.1.1.3}$$

Let F be an object of $\mathrm{Mod}(\mathscr{E}_S^\circ, \mathscr{O}_S)$. The canonical morphism

$$\mathscr{E}^*_{I\times D}(F) \longrightarrow \Gamma((ufu')_*)\circ \Gamma((ufu')^*)(\mathscr{E}^*_{I\times D}(F)) \tag{3.1.1.4}$$

may be interpreted as a commutative triangle in $\mathrm{Mod}(\Gamma(\overline{C_S^D}), \mathscr{O})$:

$$\begin{tikzcd}
 & \Gamma(u'_*)\circ\Gamma(u'^*)(\mathscr{E}^*_D(F)) \arrow[dd] \\
\mathscr{E}^*_D(F) \arrow[ur] \arrow[dr] & \\
 & \Gamma(u_*)\circ\Gamma(u^*)(\mathscr{E}^*_D(F))
\end{tikzcd} \tag{3.1.1.5}$$

126 and hence gives a functorial morphism $\overline{u}'_*\circ\overline{u}'^* \xrightarrow{\alpha_f} \overline{u}_*\circ\overline{u}^*$ such that the diagram

$$\begin{tikzcd}
 & \overline{u}'_*\circ\overline{u}'^* \arrow[dd, "\alpha_f"] \\
\mathrm{id}_{\mathrm{Mod}(\mathscr{E}_S^\circ, \mathscr{O}_S)} \arrow[ur] \arrow[dr] & \\
 & \overline{u}_*\circ\overline{u}^*
\end{tikzcd} \tag{3.1.1.6}$$

is commutative.

[25] N.D.E.: Explicitly, one has $(XfX')_{(0,i)} = X_i$ and $(XfX')_{(1,j)} = X'_j$; to $\alpha \in \mathrm{Hom}_D(i,j) = \mathrm{Hom}((0,i),(0,j))$ is associated $\alpha\circ f_i : (XfX')_{(0,i)} \to (XfX')_{(1,j)}$, and the endomorphisms of (ε, i), $\varepsilon = 0, 1$, act in the evident way.

One consequently obtains a functorial morphism such that the diagram

(3.1.1.7)
$$\begin{tikzcd}
 & K^+(\overline{u}'_*)\circ K^+(\overline{u}'^*) \arrow[dd, "K^+(\alpha_f)"] \\
\mathrm{id}_{K^+(\mathscr{E}^\circ_S,\mathscr{O}_S)} \arrow[ur] \arrow[dr] & \\
 & K^+(\overline{u}_*)\circ K^+(\overline{u}^*)
\end{tikzcd}$$

is commutative.
(Observe that $K^+(\alpha_f)$ can also be obtained formally by adjunction, using the isomorphism $K^+(\Gamma(f^*))\circ K^+(\overline{u}'^*) \xrightarrow{\sim} K^+(\overline{u}^*)$.)

Let $F^\bullet$ be a complex in $K^+(S)$, and let $\xi : K^+(\overline{ufu'}^*)(F^\bullet) \to J^\bullet$ be a quasi-isomorphism such that, for every integer n, J^n is totally acyclic objectwise (cf. (1.3.13)). By (*loc. cit.*), the canonical arrows

$$K^+(\overline{u}'^*)(F^\bullet) \longrightarrow r_1^*(J^\bullet) \quad \text{and} \quad K^+(\overline{u}^*)(F^\bullet) \longrightarrow r_0^*(J^\bullet)$$

are quasi-isomorphisms (cf. (1.2.8) for the notation). One therefore obtains a morphism 127

$$R\overline{u}'_*\circ\overline{u}'^*(F^\bullet) \xrightarrow{\eta^f(F^\bullet)} R\overline{u}_*\circ\overline{u}^*(F^\bullet)$$

such that the following diagram is commutative in $D^+(S)$:

(3.1.1.8)
$$\begin{tikzcd}
 & Q\circ K^+(\overline{u}'_*)(r_0^*(J^\bullet)) \arrow[rr, "\sim"] \arrow[dd] & & R\overline{u}'_*\circ Q(r_0^*(J^\bullet)) \arrow[dd] \\
Q\circ K^+(\overline{u}'_*)\circ K^+(\overline{u}'^*)(F^\bullet) \arrow[ur] \arrow[rr] \arrow[dd, "Q(K^+(\alpha_f)(F^\bullet))"'] & & R\overline{u}'_*\circ\overline{u}'^*(F^\bullet) \arrow[ur, "\sim"] \arrow[dd, dashed, "\eta^f(F^\bullet)"] & \\
 & Q\circ K^+(\overline{u}_*)(r_1^*(J^\bullet)) \arrow[rr, "\sim"] & & R\overline{u}_*\circ Q(r_1^*(J^\bullet)) \\
Q\circ K^+(\overline{u}_*)\circ K^+(\overline{u}^*)(F^\bullet) \arrow[ur] \arrow[rr] & & R\overline{u}_*\circ\overline{u}^*(F^\bullet) \arrow[ur] &
\end{tikzcd}$$

It is clear that $\eta^f(F^\bullet)$ is independent of the chosen resolution, and one verifies that it is functorial in $F^\bullet$ as the latter varies in $D^+(S)$.

To complete the proof of (3.1.1), it is enough to prove the following lemma:

LEMMA 3.1.1.9. *Let $X \xrightarrow{u} C^\Delta_S$ and $X' \xrightarrow{u'} C^\Delta_S$ be two objects of* $\mathrm{Hom}_{\mathrm{plat}}(\Delta^\circ, B/S)$. *Let $f, g : X \to X'$ be two morphisms in* $\mathrm{Hom}(\Delta^\circ, B/S)$. *If there exists a simplicial homotopy from f to g, then $\eta^f = \eta^g$.* 128

Let $h_n : \mathrm{Hom}_\Delta([n],[1]) \to \mathrm{Hom}_{B/S}(X_n, X'_n)$ be a simplicial homotopy from f to g. Let $\overline{I \times D}$ be the category obtained from $I \times D$ by adjoining the arrows $(1,n) \to (0,n)$ in bijective correspondence with $h_n(\mathrm{Hom}([n],[1]))$, together with the relations imposed by the compatibility of the h_n as n varies. The data of the lemma define a flat augmentation $\overline{XfX'} \xrightarrow{t} C_S^{\overline{I\times D}}$. Let $F^\cdot$ be a complex in $C^+(S)$, and let $J^\cdot$ be a resolution of $C^+(t^*)(F^\cdot)$ such that, for every integer n, J^n is totally acyclic objectwise. There exists a simplicial homotopy between

$$C^+(\Gamma(u'_*))(r_0^*(J^\cdot)) \rightrightarrows C^+(\Gamma(u_*))(r_1^*(J^\cdot)),$$

and hence a homotopy after passage to the total complexes (cf. (3.0.2.3)).

Definition 3.1.2. *With the notation of (3.1.1), we say that f is an* equivalence for G-1-cohomological descent *if η_G^f is an isomorphism.*

3.1.3. In what follows, denote by $L_i : \Delta \to \Delta \times \Delta$ (resp. $C_i : \Delta \to \Delta \times \Delta$) the canonical functor defined by $[n] \longmapsto [n] \times [i]$ (resp. $[n] \longmapsto [i] \times [n]$).

Proposition 3.1.4. *Let $X \xrightarrow{u} C_S^{\Delta\times\Delta}$ and $X' \xrightarrow{u'} C_S^{\Delta\times\Delta}$ be two objects of* $\mathrm{Hom}_{\mathrm{plat}}((\Delta \times \Delta)^\circ, B/S)$. *Let $f : X \to X'$ be a morphism over S. Suppose that $f * \mathrm{id}_{L_i} : X \circ L_i \to X' \circ L_i$ (resp. $f * \mathrm{id}_{C_i} : X \circ C_i \to X' \circ C_i$) is an equivalence for G-1-cohomological descent for every integer i. Then f is an equivalence for G-1-cohomological descent.*

By the description of η^f (cf. the proof of (3.1.1)), it is enough to show that a certain morphism of triple complexes induces an isomorphism on the cohomology of the total complexes. A standard spectral-sequence argument then gives the conclusion.

129 Proposition 3.1.5. *Let X and X' be two objects of* $\mathrm{Hom}_{\mathrm{plat}}(\Delta^\circ, B/S)$. *Let $f : X \to X'$ be a functorial morphism such that $f_n : X_n \to X'_n$ is a morphism of G-1-cohomological descent. Then the canonical morphism*

$$[[X|_f X']] \longrightarrow [[X'|_{\mathrm{id}} X']]$$

(cf. (1.2.7)) *is an equivalence for G-1-cohomological descent.*

This follows from (3.1.4) and the following lemma, whose proof is left to the reader:

Lemma 3.1.5.1. *Let $f : R \to S$ be a flat morphism, and let $Y \xrightarrow{u} C_S^\Delta$ and $Y \xrightarrow{v} C_R^\Delta$ be two* flat *augmentations such that the following diagram is commutative in* $\mathrm{Hom}(\Delta^\circ, B)$:

$$\begin{tikzcd} Y \arrow[rr, "v"] \arrow[dr, "u"'] & & C_R^\Delta \arrow[dl, "f"] \\ & C_S^\Delta & \end{tikzcd}$$

If v is an augmentation of G-1-cohomological descent, then it is an equivalence for G-1-cohomological descent.

In the applications, we shall combine (3.1.4) and (3.1.5) with the following result:

Proposition 3.1.6. *Let $X \xrightarrow{u} C_S^{\Delta\times\Delta}$ be a flat augmentation. For u to be an augmentation of G-1-cohomological descent, it is enough that*

$$X \circ L_i \xrightarrow{u * \mathrm{id}_{L_i}} C_S^\Delta \quad (\textit{resp.}\ \ X \circ C_i \xrightarrow{u * \mathrm{id}_{C_i}} C_S^\Delta)$$

be so for every integer i.

This follows from the explicit construction of $R_G\overline{u}_*$ (cf. (2.3.10)).

Corollary 3.1.7. *Let $X \xrightarrow{u} C_S^{\Delta}$ be an augmentation. Then u is of G-1-cohomological descent if and only if $[[X|_{\mathrm{id}}X]] \to C_S^{\Delta\times\Delta}$ is so.*

3.2. Localization criteria. 130

Proposition 3.2.1. *Let $Y \xrightarrow{v} C_S^{\Delta}$ be an augmentation of G-cohomological 1-descent. For a flat augmentation $X \xrightarrow{u} C_S^{\Delta}$ to be of G-cohomological 1-descent, it suffices that it become so after all base changes $Y_n \to S$.*

By means of the base changes $Y_n \to S$, one constructs a double semi-simplicial object $X \times_S Y$ augmented toward S. By (3.1.4) and (3.1.5.1), the canonical morphism

$$X \times_S Y \longrightarrow [[X|_{\mathrm{id}}X]]$$

is an equivalence for G-cohomological 1-descent. By (3.1.7), it therefore suffices to show that the augmentation of $X \times_S Y$ toward S is of G-cohomological 1-descent. This follows from the fact that the canonical morphism

$$X \times_S Y \longrightarrow [[Y|_{\mathrm{id}}Y]]$$

is an equivalence for G-cohomological 1-descent; again one uses (3.1.4), (3.1.5.1), and (3.1.7).

3.2.2. Let

(3.2.2.1)
$$\begin{tikzcd} Y' \arrow[r, "g'"] \arrow[d, "f'"'] & Y \arrow[d, "f"] \\ X' \arrow[r, "g"'] & X \end{tikzcd}$$

be a commutative diagram in $\boldsymbol{B}$, and let F be an object of $\mathrm{Mod}(\mathscr{E}_Y^\circ, \mathscr{O}_Y)$. There is a unique morphism

$$\varphi(F) : g_*(f^*(F)) \longrightarrow f'^*(g'_*(F))$$

such that the following diagram (in $\mathrm{Mod}(\mathscr{E}^\circ, \mathscr{O})$) commutes[26]:

(3.2.2.2)
$$\begin{tikzcd} & f^*(F) \arrow[r, "f^\circ"] \arrow[dl, "g^\circ"'] & F \arrow[d, "g'^\circ"] \\ g_*(f^*(F)) \arrow[dr, "\mathrm{Id}"'] & & g'_*(F) \\ & f'^*(g'_*(F)) \arrow[ur, "f'^\circ"'] & \end{tikzcd}$$

[26]N.D.E.: As above, one must regard the commutative diagram as lying in $\boldsymbol{B}^\circ$ and use the fibered and cofibered structure of $\mathrm{Mod}(\mathscr{E}^\circ, \mathscr{O}) \to \boldsymbol{B}^\circ$, which explains the unusual appearance of the base-change morphism. Under the (adjunction) arrows in the following diagram we have written their images in $\boldsymbol{B}^\circ$

131 If all the arrows of (3.2.2.1) are now assumed flat, one defines in the same way a morphism, called the *base-change* morphism[27]

$$\xi_G : g_{G*} \circ R_G f^* \longrightarrow R_G f'^* \circ g'_{G*}. \tag{3.2.2.3}$$

Definition 3.2.2.4. *The diagram* (3.2.2.1) is said to satisfy the base-change theorem relative to G if f, g, f', and g' *are flat morphisms and* ξ_G *is an isomorphism.*

3.2.3. Let $h : S \to S'$ be a *flat* morphism in B, let $X \xrightarrow{u} C^\Delta_S$ and $X' \xrightarrow{u'} C^\Delta_{S'}$ be two *flat* augmentations, and let $f : X \to X'$ be a functorial morphism such that the diagram (in $\mathrm{Hom}(\Delta^0, B)$)

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "u"'] & X' \arrow[d, "u'"] \\ C^\Delta_S \arrow[r, "h_c"] & C^\Delta_{S'} \end{tikzcd} \tag{3.2.3.1}$$

commutes. (Here h_c denotes the constant functorial morphism defined by h.)

Let $K^\cdot$ be a complex in $D^+_G(S')$. The morphism

$$\eta^f_G(K^\cdot) : R\overline{u}'_* \circ \overline{u}'^*(K^\cdot) \longrightarrow R(\overline{h_c \circ u})_* \circ (\overline{h_c \circ u})^*(K^\cdot)$$

induces, in view of the identities

$$R(\overline{h_c \circ u})_* \simeq Rh^*_c \circ R\overline{u}_* \quad \text{and} \quad (\overline{h_c \circ u})^* \simeq \overline{u}^* \circ h_{c*},$$

a morphism $\mathrm{ch}^{f,h}_G(K^\cdot)$, functorial in $K^\cdot$, such that the following diagram commutes[28]:

$$\begin{tikzcd} h_{c*} \circ R\overline{u}'_* \circ \overline{u}'^*(K^\cdot) \arrow[rr, "\mathrm{ch}^{f,h}_G(K^\cdot)"] & & R\overline{u}_* \circ \overline{u}^* \circ h_{c*}(K^\cdot) \\ & h_{c*}(K^\cdot) \arrow[ul, "h_{c*}(\alpha(K^\cdot))"] \arrow[ur] & \end{tikzcd} \tag{3.2.3.2}$$

.

132 Lemma 3.2.3.3. *Suppose that, for every integer i, the diagram*

[27]N.D.E.: Here again, it would be more correct to write $f^\circ, g^\circ, f'^\circ, g'^\circ$ instead of f, g, f', g'. If one considers the nonopposite fibered and cofibered structure, ξ_G becomes

$$g^*_G \circ R_G f_* \longrightarrow R_G f'_* \circ g'^*_G.$$

[28]N.D.E.: For readability, we have suppressed the reference to the thick category G in the derived functors. The same unusual derivations h_{c*} reappear, owing to the choice of the cofibered structure.

$$\begin{tikzcd} X_i \arrow[r, "f_i"] \arrow[d, "u_i"'] & X_i' \arrow[d, "u_i'"] \\ S \arrow[r, "h"] & S' \end{tikzcd}$$

satisfies the base-change theorem relative to G. Then $\mathrm{ch}_G^{f,h}$ is an isomorphism.

The morphism $\mathrm{ch}_G^{f,h}(K^\bullet)$ may be computed as follows. Consider the augmentation

$$XfX' \xrightarrow{(h_c \circ u)fu'} C_S^{I\times\Delta}$$

(cf. (3.1.1.2)), and choose a resolution $J^\bullet$ of $K^+(\overline{(h_c \circ u)fu'}^*)(K^\bullet)$ such that, for every integer n, J^n is totally acyclic object by object.

One obtains a quasi-isomorphism

$$\theta : K^+(\Gamma(f^*))(r_0^* J^\bullet) \longrightarrow r_1^*(J^\bullet)$$

and a morphism

$$t : K^+(\Gamma(h_c^*)) \circ K^+(\Gamma(u_*'))(r_0^*(J^\bullet)) \longrightarrow K^+(\Gamma(u_*))(r_1^*(J^\bullet))$$

which has the factorization

$$K^+(\Gamma(h_c^*)) \circ K^+(\Gamma(u_*'))(r_0^*(J^\bullet)) \xrightarrow{\varphi} K^+(\Gamma(u_*)) \circ K^+(\Gamma(f^*))(r_0^*(J^\bullet)) \xrightarrow{K^+(\Gamma(u_*))(\theta)} \\ \longrightarrow K^+(\Gamma(u_*))(r_1^*(J^\bullet)),$$

and $\mathrm{ch}^{f,h}(K^\bullet)$ is obtained by taking the image of t under the functor $R\epsilon_{Z^\bullet*}$. 133

The preceding factorization shows that t identifies "object by object" with the base-change morphisms associated with the diagrams

$$\begin{tikzcd} X_i \arrow[r, "f_i"] \arrow[d, "u_i"'] & X_i' \arrow[d, "u_i'"] \\ S \arrow[r, "h"] & S' \end{tikzcd}$$

It follows, by spectral sequences, that $\mathrm{ch}_G^{f,h}(K^\bullet)$ is an isomorphism, completing the proof.

This leads to a *second localization criterion*:

Proposition 3.2.4. *Let $X \xrightarrow{u} C_S^\Delta$ be an object of* $\mathrm{Hom}_{\mathrm{plat}}(\Delta^\circ, B/S)$. *For u to be an augmentation of G-cohomological 1-descent, it suffices that it become so after "enough for G"*[28] *flat base changes $h : S' \to S$ for which the cartesian diagrams*

$$\begin{tikzcd} X_i' \arrow[r] \arrow[d] & X_i \arrow[d] \\ S' \arrow[r, "h"] & S \end{tikzcd}$$

[28]The expression "enough for G" means that there exists a family $(S_\alpha \xrightarrow{h_\alpha} S)_{\alpha\in A}$ of flat morphisms satisfying the preceding conditions and such that the family of functors $(h_{\alpha*} : G_S \to G_{S_\alpha})_{\alpha\in A}$ is *conservative.*

satisfy the base-change theorem relative to G.

134 Proposition (3.2.4) follows immediately from (3.2.3.3), in view of diagram (3.2.3.2).

3.3. Properties of morphisms of cohomological descent. In this section, we assume that the *class of flat morphisms in B is stable under base change.*

Proposition 3.3.1. *Let $i = 1, 2$.*

a) *Every flat morphism which has a section is a morphism of universal G-cohomological 2-descent.*
b) *Let $f : X \to S$ be a flat morphism, let $g : S' \to S$ be a morphism of G-cohomological i-descent, put $X' = X \times_S S'$, and let $f' = f_{(S')} : X' \to S'$. Then f is of universal G-cohomological i-descent if and only if f' is.*
c) *If the composite of two morphisms $f : X \to Y$ and $g : Y \to Z$ is of universal G-cohomological i-descent, then g is of universal G-cohomological i-descent.*
d) *The composite of two morphisms of universal G-cohomological i-descent is a morphism of universal G-cohomological i-descent.*
e) *If $f : X \to X'$ and $g : Y \to Y'$ are two S-morphisms of universal G-cohomological i-descent, then $f \times_S g$ is of universal G-cohomological i-descent.*
f) *Let $(u_\alpha : X_\alpha \to Y_\alpha)_{\alpha \in A}$ be a family of morphisms of G-cohomological i-descent. Then $\amalg u_\alpha : \amalg_{\alpha \in A} X_\alpha \to \amalg_{\alpha \in A} Y_\alpha$ is a morphism of G-cohomological i-descent.*

135 *Proof.* For effective descent relative to G, we refer to [4].

a) Using a section s of f, compare the semi-simplicial objects augmented toward S, $[X|_f S]$ and $[S|_{\mathrm{id}} S]$. By (3.0.2.4), one obtains an equivalence for G-cohomological 1-descent[29].
b) This follows from (3.2.1).
c) This follows from a) and b).
d) Let $f : X \to Y$ and $g : Y \to S$ be two morphisms of universal G-cohomological 1-descent. Put $h = g \circ f$ and consider the fiber product

$$\begin{tikzcd} Y \arrow[d, "g"'] & R \arrow[l, "h'"'] \arrow[d, "g'"] \\ S & X \arrow[l, "h"] \end{tikzcd}$$

The morphism g' has a section $s : X \to R$ such that $h' \circ s = f$. By c), h' is of universal G-cohomological 1-descent, and the same holds for h by b).
e) This follows formally from b) and d).
f) Let $(Z_\lambda)_{\lambda \in A}$ be a family of objects of B. It follows from hypotheses (3.0.0) that there is a canonical equivalence

$$\mathrm{Mod}\left(\mathscr{E}^\circ_{\amalg_\lambda Z_\lambda}, \mathscr{O}_{\amalg_\lambda Z_\lambda}\right) \xrightarrow{\sim} \prod_\lambda \mathrm{Mod}(\mathscr{E}^\circ_{Z_\lambda}, \mathscr{O}_{Z_\lambda})$$

which induces an equivalence

$$G_{\amalg_\lambda Z_\lambda} \xrightarrow{\sim} \prod_\lambda G_{Z_\lambda}.$$

[29] N.D.E.: For this, note that $[X|_f S]$ and $[S|_{\mathrm{id}} S]$ are simply the coskeleta of order 0 of f and id, respectively, so that s and f induce inverse isomorphisms in cohomology by 3.1.1.

Moreover, a family $(Q_\lambda)_\lambda \in \prod_\lambda \mathrm{Mod}(\mathscr{E}^\circ_{Z_\lambda}, \mathscr{O}_{Z_\lambda})$ corresponds to a totally acyclic object if and only if Q_λ is totally acyclic for every λ.

Now let $(u_\lambda : X_\lambda \to Y_\lambda)_{\lambda\in\Lambda}$ be a family of morphisms of G-cohomological 1-descent. Since direct sums in B are disjoint and universal, for every n there is an identification 136

$$\left[\coprod_\lambda X_\lambda |_{\coprod u_\lambda} \coprod Y_\lambda\right]_n \simeq \coprod_\lambda [X_\lambda|_{u_\lambda} Y_\lambda]_n$$

which is functorial in n. It then follows from the preceding remarks and the explicit calculation of $R_G\theta_*$ (cf. (2.3)) that $\coprod_\lambda u_\lambda$ is a morphism of G-cohomological 1-descent; the details are left to the reader. The reader may also verify that $\coprod_\lambda u_\lambda$ *remains of G-cohomological 1-descent after every base change if the same is true of every u_λ.*

3.3.2. Associate with each object X of B the set of families of morphisms $(X_\alpha \to X)_{\alpha\in A}$ such that $\coprod_\alpha X_\alpha \to X$ is a morphism of universal G-i-cohomological descent. By 3.3.1, this defines a pretopology on B.

The topology generated by this pretopology is called the *topology of G-i-cohomological descent.* It follows from 3.3.1 c) that the covering morphisms for this topology are precisely the morphisms of universal G-i-cohomological descent. Finally, note that the topology of G-2-cohomological descent is coarser than the topology of G°-descent [cf. [4] (6.23)].

Theorem 3.3.3. *Suppose that* all arrows of B are flat. *Let S be an object of B. Every hypercovering of S for the topology of G-1-cohomological descent whose objects are all representable* (cf. V, Appendix) *defines an augmentation of universal G-1-cohomological descent.*

The proof proceeds in two steps; we specify that all coskeleta are to be computed in the category B/S.

Lemma 3.3.3.1. *Let X be an object of* $\mathrm{Hom}(\Delta^\circ, B/S)$. *For X to be of 1-cohomological descent (resp. universal 1-cohomological descent), it is enough that* $\mathrm{cosq}_n(X)$ *have this property for all sufficiently large n.* 137

Indeed, let $\alpha_n : X \to \mathrm{cosq}_n(X)$ be the canonical morphism. One checks that if $K^\cdot$ is a complex of $D^+_{G_S}(S)$ such that $H^j(K^\cdot) = 0$ for $j < N$, then $H^i(\eta_G^{\alpha_n})$ is an isomorphism for $i < N + n$, whence the assertion.

Lemma 3.3.3.2. *Let $n \geq 0$, let X and X' be two simplicial objects of B/S, and let $f : X \to X'$ be a morphism. Suppose that:*

(i) $X \to \mathrm{cosq}_{n+1}(X)$ *is an isomorphism;*
(ii) $X' \to \mathrm{cosq}_{n+1}(X')$ *is an isomorphism;*
(iii) $f_i : X_i \to X'_i$ *is an isomorphism for $i \leq n$;*
(iv) f_{n+1} *is a morphism of universal G-1-cohomological descent.*

Then, if X is of universal G-1-cohomological descent, the same is true of X'.

Lemmas 3.3.3.1 and 3.3.3.2 clearly prove Theorem 3.3.3 by induction.

Lemma 3.3.3.3. *Under the hypotheses of 3.3.3.2, all the morphisms f_p are morphisms of universal G-1-cohomological descent.*

This is trivial for $p \leq n+1$. For $p > n+1$, X_p (resp. X'_p) may be written as $\varprojlim_{\Delta^+_{n+1}[p]} X_q$ (resp. $\varprojlim_{\Delta^+_{n+1}[p]} X'_q$).

We may suppose that $p > n+1$. Write

$$\operatorname{cosq}_{n+1}(X)_p = \varprojlim_{\substack{[q]\to[p]\\ q\leq n+1}} X_q$$

as the kernel of the pair of arrows

$$\varPi X = \prod_{\substack{[q]\to[p]\\ q\leq n+1}} X_q \rightrightarrows \prod_{\substack{[i]\overset{\alpha}{\hookrightarrow}[j]\\ \searrow\ \swarrow \\ [p] \\ j\leq n+1}} X_i.$$

The component α_X indexed by $\alpha \in \operatorname{Hom}_{[p]}([i],[j])$ of this pair consists, on the one hand, of the projection $\varPi X \to X_i$ indexed by $[i]\to[p]$, and, on the other hand, of the composite

$$\varPi X \longrightarrow X_j \longrightarrow X_i$$

of the projection indexed by $[j]\to[p]$ with $\alpha \in \operatorname{Hom}(X_j, X_i)$.

It follows that the diagram

$$\begin{tikzcd} \operatorname{cosq}_{n+1}(X)_p \arrow[r] \arrow[d] & \varPi X \arrow[d] \\ \operatorname{cosq}_{n+1}(X')_p \arrow[r] & \varPi X' \end{tikzcd}$$

is Cartesian. Indeed, by the preceding description, it is enough to prove that for every *increasing injection* $\alpha \in \operatorname{Hom}_{[p]}([i],[j])$, the diagram

$$\begin{tikzcd} \operatorname{Ker}(\alpha_X : \varPi X \rightrightarrows X_i) \arrow[r] \arrow[d] & \varPi X \arrow[d] \\ \operatorname{Ker}(\alpha_{X'} : \varPi X' \rightrightarrows X'_i) \arrow[r] & \varPi X' \end{tikzcd}$$

is Cartesian. If $i = n+1$, then $\alpha : [i] \hookrightarrow [j]$ is the identity of $[n+1]$, so the pair of arrows imposes no condition. The respective kernels are $\varPi X$ and $\varPi X'$, which settles this case. Otherwise $i < n+1$ and $X_i \to X'_i$ is an isomorphism, so the diagram is Cartesian in this case as well (testing an equality in X_i or X'_i amounts to the same thing)[30].

Using 3.3.1, the relevant morphisms are therefore morphisms of universal G-1-cohomological descent. Let $[X'/X]^p$ be the $(p+1)$-fold fiber product of X' over X. By 3.3.3.3, it is enough to verify 3.3.3.2 after a base change $[X'/X]^p \to X$. After such a base change, the hypotheses of 3.3.3.2 still hold and, moreover, f_{n+1} admits a section. Lemma 3.0.2.4 then completes the proof.

For effective descent, we have the following result.

Proposition 3.3.4. *Let $X : \Delta^\circ \to B/S$ be a functor. Suppose that $X_0 \to S$ is a morphism of universal G°-2-descent, as are the morphisms $X_{n+1} \to (\operatorname{cosq}_n(X))_{n+1}$ for $n = 0, 1$. Then the functor X is G°-2-faithful and remains so after every base change (in other words, X defines an augmentation of universal effective descent in the sense of 2.4.11).*

By [4] (7.12), it is enough to show that $i_2^*(X)$ is G°-2-faithful; for this we use the following lemmas.

138 Lemma 3.3.4.1. *Let X and X' be two functors $\Delta_2^\circ \to B/S$, and consider a commutative*

[30] N.D.E.: The proof of Lemma 3.3.3.3 has been slightly rewritten.

diagram

$$\begin{tikzcd}
X_2' \arrow[r, shift left=2] \arrow[r] \arrow[r, shift right=2] & X_1' \arrow[l, shift left=1] \arrow[l, shift right=1] \arrow[r, shift left=1] \arrow[r, shift right=1] \arrow[d, equal] & X_0' \arrow[l] \arrow[r] \arrow[d, equal] & S \arrow[d, equal] \\
X_2 \arrow[r, shift left=2] \arrow[r] \arrow[r, shift right=2] & X_1 \arrow[l, shift left=1] \arrow[l, shift right=1] \arrow[r, shift left=1] \arrow[r, shift right=1] & X_0 \arrow[l] \arrow[r] & S
\arrow[from=1-1, to=2-1, "k"']
\end{tikzcd}$$

such that k is a morphism of G°-0-descent. If X is G°-2-faithful, then so is X'.

This is evident.

Lemma 3.3.4.2. *Let X and X' be two functors $\Delta_2^\circ \to B/S$, and consider a commutative diagram*

$$\begin{tikzcd}
X_2' \arrow[r, shift left=3, "\partial^0"] \arrow[r, "\partial^1"] \arrow[r, shift right=3, "\partial^2"'] & X_1' \arrow[l, shift left=1.5] \arrow[l, shift right=1.5] \arrow[r, shift left=2, "\partial^0"] \arrow[r, shift right=2, "\partial^1"'] \arrow[d, "f_1"] & X_0' \arrow[l, "\sigma^0"'] \arrow[r] \arrow[d, equal] & S \arrow[d, equal] \\
X_2 \arrow[r, shift left=3] \arrow[r] \arrow[r, shift right=3] & X_1 \arrow[l, shift left=1.5] \arrow[l, shift right=1.5] \arrow[r, shift left=2] \arrow[r, shift right=2] & X_0 \arrow[l] \arrow[r] & S
\arrow[from=1-1, to=2-1, "f_2"']
\end{tikzcd}$$

such that f_1 is of G°-1-descent and f_2 is of G°-0-descent. Suppose in addition that $X_2' \xrightarrow{\sim} (\mathrm{cosq}_1(X'))_2$. If X is G°-2-faithful, then so is X'.

Let $X_1'' = X_1' \times_{X_1} X_1'$, and let $\partial_0 : X_1'' \to X_1'$ and $\partial_1 : X_1'' \to X_1'$ be the two projections.

The definition of a coskeleton gives an arrow $\varphi : X_1'' \to X_2'$ such that

$$\begin{cases} \partial_0 \circ \varphi = \partial_0, \\ \partial_1 \circ \varphi = \partial_1, \\ \partial_2 \circ \varphi = \sigma_0 \partial_1 \partial_0 = \sigma_0 \partial_1 \partial_1. \end{cases}$$

Now take a descent datum on X': it gives an object on X_0, two objects on X_1, and an isomorphism between them on X_1'. The arrow φ shows that the two inverse images of these isomorphisms on X_1'' are equal. Since f_1 is of 1-descent, we obtain an isomorphism between the two objects on X_1; this isomorphism is a descent datum (because $X_2' \to X_2$ is of 0-descent), and the result follows.

Corollary 3.3.5. *Suppose that* all arrows of B are flat. *Every hypercovering of S for the topology of G-2-cohomological descent whose objects are representable defines an augmentation of universal G-2-cohomological descent.*

4. Examples

4.1. Sheaves of abelian groups on topological spaces.

4.1.0. In this section, Top denotes the category of topological spaces [elements of the fixed universe $\mathscr{U}$]. We define a category $\mathscr{E}$ bifibered in duals of topoi over Top by taking as objects the pairs (X, F), where X is a topological space and F is a sheaf of sets on X, and as morphisms the pairs $(f, \varphi) : (X, F) \to (Y, H)$, where $f : X \to Y$ is a continuous map and $\varphi : H \to f_*(F)$ is a morphism of sheaves. We take for $\mathscr{O}$ the section of $\mathscr{E}^0$ over Top^0 that associates with each topological space the constant sheaf

$\mathbf{Z}_X$: we set $G = \mathrm{Mod}(\mathscr{E}^0, \mathscr{O})$, so that, for every topological space X, G_X identifies with the category of sheaves of abelian groups on X.

Recall that a morphism $f : X \to Y$ is *separated* if the diagonal in $X \times_Y X$ is closed. A *proper* morphism is a separated and universally closed morphism (note that this definition is more restrictive than Bourbaki's).

139 The proof of the "base-change theorem" below is inspired by ([5] II(4.11.1)).

Theorem 4.1.1. *Let $f : X \to Y$ be a proper morphism, let F be an abelian sheaf on X, and consider a Cartesian diagram*

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"] & X \arrow[d, "f"] \\ Y' \arrow[r, "g"] & Y. \end{tikzcd}$$

Then the canonical map (XII 4.2) *or* (3.2.2.3)

$$g^* R^i f_* F \xrightarrow{\sim} R^i f'_* g'^* F$$

is an isomorphism[31].

This theorem is equivalent to the following corollary (the corollary is obtained by taking $Y' = (\text{Point})$; the theorem is obtained by applying the corollary to f and f').

Corollary 4.1.2. *Let $f : X \to Y$ be a proper morphism, let F be an abelian sheaf on X, and let $y \in Y$. The canonical map*

$$(R^i f_* F)_y \longrightarrow H^i(f^{-1}(y), F)$$

is bijective.

By definition, as U ranges over the neighborhoods of y, we have

$$(R^i f_* F)_y = \varinjlim H^i(f^{-1}(U), F).$$

Since f is closed, the $f^{-1}(U)$ form a fundamental system of neighborhoods of $f^{-1}(y)$, and the result follows from the following lemma, (4.1.3).

140 Lemma 4.1.3. *Let X be a topological space, let $K \subset X$, and let F be an abelian sheaf on X. Suppose that K is compact and that any two distinct points of K have disjoint neighborhoods in X. Then, as U ranges over the neighborhoods of K, we have*

$$\varinjlim H^i(U, F) \xrightarrow{\sim} H^i(K, F). \tag{4.1.3.1}$$

We first treat the case $i = 0$. In this case, it is clear that (4.1.3.1) is injective. For surjectivity, we use the following result.

Lemma 4.1.4. *Under the hypotheses of* (4.1.3)*, if A and B are two closed subsets of K and W is a neighborhood of $A \cap B$ in X, then there are neighborhoods U and V of A and B in X such that $U \cap V \subset W$.*

[31] N.D.E.: Unlike above, these are the derived functors of the usual functors, and not those arising from the fibered-category structure $G^\circ \to \mathrm{Top}^\circ$.

We first treat the case where A consists of a single point a. The assertion is trivial if $a \in B$ (take $U = W$). If $a \notin B$, then for every $b \in B$ there are disjoint open neighborhoods U_b and V_b of a and b in X. We take for V a finite union of the V_b that covers B, and for U the intersection of the corresponding U_b.

In the general case, for every $a \in A$, there are open neighborhoods U_a and V_a of a and B in X such that $U_a \cap V_a \subset W$. We take for U a finite union of the U_a that covers A, and for V the corresponding intersection of the V_a.

Return to (4.1.3). If $s \in H^\circ(K, F)$, there is an open covering U_i of K in X and there are sections $s_i \in H^\circ(U_i, F)$ such that $s_i = s$ on $U_i \cap K$. We may suppose that there are only finitely many U_i. Let (K_i) be a closed covering of K, with $K_i \subset K \cap U_i$. Let W_y be the open subset of $U_i \cap U_j$ on which $s_i = s_j$; we have $K_i \cap K_j \subset W_{ij}$. Apply (4.1.4) to all pairs (K_i, K_j) and to the W_{ij}: we find open subsets V_{ij} with $K_i \subset V_{ij} \subset U_i$ and $V_{ij} \cap V_{ji} \subset W_{ij}$. Let $U_i' = \bigcap_j V_{ij}$: then $K_i \subset U_i' \subset U_i$, and $s_i = s_j$ on $U_i' \cap U_j'$. The s_i therefore glue on the neighborhood of K given by the union of the U_i', and (4.1.3.1) is surjective (hence bijective) for $i = 0$[32]. 141

Lemma 4.1.5. *If F is flabby, the sheaf $F|K$ on K is soft.*

Let $A \subset K$ be closed in K. Every section of F on A extends to a neighborhood, by what precedes. Since F is flabby, it extends to X, and hence in particular to K.

We now prove (4.1.3). Let F^* be a flabby resolution of F. We have

$$\begin{aligned}\varinjlim H^i(U, F) &\overset{33}{=} \varinjlim H^i(\Gamma(U, F^*)) = H^i \varinjlim \Gamma(U, F^*) \overset{34}{=} H^i\Gamma(K, F^*)\\ &\overset{(4.1.5)}{=} H^i(K, F).\end{aligned}$$

Corollary 4.1.6. *Let $f : X \to Y$ be a proper surjective morphism of topological spaces. Then f is of G-1-cohomological descent.*

This follows from (4.1.2) and (3.2.4).

Corollary 4.1.7. *Let $\mathscr{U} = (U_i)_{i\in I}$ be a locally finite closed covering of a topological space X. Then the canonical morphism $f : \coprod_{i\in I} U_i \to X$ is a morphism of G-1-cohomological descent.*

Indeed, f is proper and surjective. Applying (2.5.6), we recover the spectral sequence of [[5]. II (5.2.4)] (cf. the Introduction).

Proposition 4.1.8. *Let $f : X \to Y$ be a surjective local homeomorphism of topological spaces. Then f is a morphism of universal G-1-cohomological descent.*

By localization on Y (cf. (3.2.4)), we are reduced to the case where f has a section.

Corollary 4.1.9. *Let $\mathscr{U} = (U_i)_{i\in I}$ be an open covering of a topological space X. Then the canonical morphism $f : \coprod_{i\in I} U_i \to X$ is a morphism of universal G-1-cohomological descent.* 142

The descent spectral sequence then gives [[5] II (5.4.1)] (cf. the Introduction).

Remark 4.1.10. The proofs of (VIII 9) show that "1-cohomological descent" may be replaced by "2-cohomological descent" in all the preceding statements.

[32]N.D.E.: The two meanings of the symbol i should not be confused.

[33]N.D.E.: $H^i(\Gamma(U, F^*))$ denotes here the i-th cohomology group of the complex $\Gamma(U, F^*)$.

[34]N.D.E.: By (4.1.3), applied with $\Gamma = H^0$

4.2. Quasi-coherent modules on schemes.

4.2.0. Let $(\mathrm{Sch})_s$ be the category whose objects are the schemes that are elements of $\mathscr{U}$ and whose arrows are the quasi-compact and quasi-separated morphisms. We define a category $\mathscr{E}$ bifibered in duals of topoi over $(\mathrm{Sch})_s$ by taking as objects the pairs (X, F), where X is a scheme and F a sheaf on X (for the Zariski topology), and as morphisms the pairs $(f, \varphi) : (X, F) \to (Y, G)$, where $f : X \to Y$ is a continuous map and $\varphi : G \to f_*(F)$ is a morphism of sheaves. For $\mathscr{O}$ we take the section of $\mathscr{E}^0$ over $(\mathrm{Sch})_s^0$ that associates with every scheme its structure sheaf; for $\boldsymbol{G}$ we take the subcategory of $\mathrm{Mod}(\mathscr{E}^0, \mathscr{O})$ such that, for every scheme X, $\boldsymbol{G}_X$ is the category of quasi-coherent modules on X.

Proposition 4.2.1. *Every faithfully flat, quasi-compact, and quasi-separated morphism $f : X \to Y$ is a morphism of universal $\boldsymbol{G}$-2-cohomological descent.*

By [SGA I (VIII.5.2)], f is a morphism of effective $\boldsymbol{G}^0$-descent. By [EGA IV (1.7.21)], we may apply (3.2.4) to the Cartesian square

$$\begin{tikzcd} X \times_Y X \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ X \arrow[r, "f"'] & Y \end{tikzcd}$$

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4.3. Étale sheaves on schemes.

4.3.0. Let (Sch) be the category of all schemes belonging to $\mathscr{U}$. We thus define a category $\mathscr{E}$ bifibered in duals of topoi over (Sch) by taking as objects the pairs $(X, \mathscr{F})$, where X is a scheme and $\mathscr{F}$ an étale sheaf on X, and as morphisms the pairs $(f, \varphi) : (X, \mathscr{F}) \to (Y, \mathscr{G})$, where $f : X \to Y$ is a morphism of schemes and $\varphi : \mathscr{G} \to f_*(\mathscr{F})$ is a morphism of sheaves.

Let n be an integer ≥ 0. We denote by $\mathscr{O}_n$ the section of $\mathscr{E}^0$ over $(\mathrm{Sch})^0$ that associates with every scheme X the constant sheaf $(\mathbf{Z}/n\mathbf{Z})_X$. Put $\boldsymbol{F}_n = \mathrm{Mod}(\mathscr{E}^0, \mathscr{O}_n)$, so that, for every scheme X, $\boldsymbol{F}_{n_X}$ identifies with the category of sheaves of $\mathbf{Z}/n\mathbf{Z}$-modules on X.

4.3.1. Now let $(\mathrm{Sch})_s$ be the subcategory of (Sch) whose objects are the schemes that are elements of $\mathscr{U}$ and whose arrows are the quasi-compact and quasi-separated morphisms. Denote by $\mathscr{E}_s$ the category bifibered in duals of topoi over $(\mathrm{Sch})_s$ deduced from $\mathscr{E}$ by the base change $(\mathrm{Sch})_s \to (\mathrm{Sch})$; we again denote by $\mathscr{O}_0$ the restriction of $\mathscr{O}_0$ to $(\mathrm{Sch})_s$. Let $\boldsymbol{G} \subset \mathrm{Mod}(\mathscr{E}_s^0, \mathscr{O}_0)$ be the fibered and cofibered subcategory of $\mathrm{Mod}(\mathscr{E}_s^0, \mathscr{O}_0)$ such that, for every scheme X, $\boldsymbol{G}_X$ is the category of torsion abelian sheaves on X. The reader will note that condition (2.4.1.1) is satisfied.

Proposition 4.3.2. *Let $f : X \to Y$ be a* proper surjective *morphism. Then f is a morphism of universal $\boldsymbol{F}_n$-2-cohomological descent for $n \geq 1$. Considered as a morphism of $(\mathrm{Sch})_s$, f is a morphism of universal $\boldsymbol{G}$-2-cohomological descent.*

144 By (VIII 9.4), f is a morphism of effective $\boldsymbol{F}^0$-descent, and by (XII.(5.1)) we may

apply (3.2.4) to the Cartesian diagram

$$\begin{tikzcd} X \times_Y X \arrow[r] \arrow[d] & X \arrow[d, "f"] \\ X \arrow[r, "f"'] & Y. \end{tikzcd}$$

PROPOSITION 4.3.3. *Let $f : X \to Y$ be a* surjective *morphism of schemes. Suppose that one of the following hypotheses is satisfied:*

a) *f is integral.*
b) *Y is locally Noetherian, and f is universally open and locally of finite presentation.*
c) *f is flat and locally of finite presentation.*

Then f is a morphism of F_n-2-cohomological descent for every $n \geq 0$.

a) Repeat the proof of (4.3.2), replacing (XII.(5.1)) by ((VIII 5.6)).
b) Since the question is local on Y, we may suppose that Y is Noetherian. Apply (EGA IV (14.5.10)). By (3.3.1) and a), we reduce to the case in which every point y of Y has a neighborhood U such that $f^{-1}(U) \to U$ has a section, and the result then follows from (3.2.4).
c) We must show that f is a morphism of F-1-*cohomological descent* (we know from ((VIII 9.4)) that f is a morphism of effective F^0-descent).

By localization on Y (cf. (3.2.4)), we may suppose that Y is affine, with ring A. We may then write $\operatorname{Spec} A = \varprojlim_{i \in I} \operatorname{Spec} A_i$, where I is a filtered ordered set and A_i is a Noetherian ring for every i, with $X = \varprojlim X_i$, where, for every i, X_i is a scheme *flat* and *locally of finite presentation* over $\operatorname{Spec} A_i$ such that $f_i : X_i \to \operatorname{Spec} A_i$ is *surjective*; moreover, the diagrams

$$\begin{tikzcd} X \arrow[r] \arrow[d, "f"'] & X_i \arrow[d, "f_i"] \\ Y \arrow[r, "u_i"] & \operatorname{Spec} A_i \end{tikzcd}$$

are Cartesian. 145

Let $\mathfrak{F}$ be an abelian sheaf on Y. We have $\mathfrak{F} = \varinjlim_i u_i^* u_{i*}(\mathfrak{F})$, so that, by a standard passage to the limit, we reduce to the case in which $\mathfrak{F}$ comes from a sheaf on $\operatorname{Spec} A_i$ for some i, and b) gives the conclusion.

REMARK 4.3.4. If $f : X \to Y$ is integral, then $R^q f_* = 0$ for $q > 0$. It follows readily that it is unnecessary to restrict to complexes bounded below when performing descent by means of f.

COROLLARY 4.3.5. *Let $(X_\alpha \xrightarrow{u_\alpha} X)_{\alpha \in A}$ be a covering family for the étale topology. Then $\coprod_{\alpha \in A} X_\alpha \to X$ is a morphism of universal F_n-2-cohomological descent for every $n \geq 0$.*

Indeed, there exists a factorization

$$\begin{tikzcd} & \coprod_{a\in A} X_\alpha \arrow[dr] & \\ X \arrow[ur] \arrow[rr, "h"] & & X \end{tikzcd}$$

where h is a surjective étale morphism.

5. Applications

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5.1. Construction of simplicial objects.

5.1.0. Up to and including 5.1.3, H is a category in which nonempty finite inverse limits exist and in which finite coproducts exist and are disjoint and universal. If $X : \Delta^0 \to H$ is a simplicial object of H, we denote by $d_i^k : X_k \to X_{k-1}$ (resp. by $s_i^k : X_k \to X_{k+1}$), for $i \in [k]$ and $0 \le k$, the arrows corresponding to the face operators $\partial_k^i : [k-1] \to [k]$ (resp. to the degeneracy operators $\sigma_k^i : [k] \to [k+1]$) (cf. [3] II 2).

Definition 5.1.1. *A simplicial object $X : \Delta^0 \to H$ of H is σ-split if there exists a family of subobjects NX_j of X_j ($j \ge 0$) such that the morphisms*

$$h_i : \coprod_{\substack{\mathrm{Hom}_{\Delta^-}([i],[j]) \\ j\le i}} NX_j \longrightarrow X_i$$

are isomorphisms.

Likewise, a k-truncated simplicial object $X : \Delta_k^0 \to H$ is σ-split if there exist NX_j ($0 \le j \le k$) satisfying the preceding condition for $i \le k$.

The NX_j are uniquely determined by X. Indeed, if X is σ-split, the degeneracy operators $s_\ell^{k-1} : X_{k-1} \to X_k$ are isomorphisms from X_{k-1} onto direct summands of X_k, and

$$NX_k = \bigcap_\ell \quad (\text{complement of } s_\ell^{k-1}(X_{k-1})).$$

For $k = 0$, $NX_0 = X_0$.

5.1.2. For a σ-split $(n+1)$-truncated simplicial object X of H, we denote by $\alpha_n(X)$ the triple consisting of

a) the restriction $i_n^*(X)$ of X to Δ_n^0;

147 b) NX_{n+1};

c) the evident map from NX_{n+1} to $(\mathrm{cosq}_n(X))_{n+1}$.

This triple (Y, N, β) satisfies the following condition:

$(*)$ Y is a σ-split n-truncated simplicial object of H, and β is a map from N to $(\mathrm{cosq}\, Y)_{n+1}$.

Proposition 5.1.3. *Let (Y, N, β) be a triple satisfying $(*)$ above.*

(i) *Up to unique isomorphism, there exists a unique σ-split $X : \Delta_{n+1}^0 \to H$ such that $\alpha_n(X) \simeq (Y, N, \beta)$.*

(ii) *To give a morphism f from X to a truncated simplicial object $Z : \Delta_{n+1}^0 \to H$ is equivalent to giving*

a) *a morphism $f' : Y \to i_n^*(Z)$;*

b) *a morphism $f'' : N \to Z_{n+1}$ such that the diagram*

$$\begin{tikzcd} N \arrow[r] \arrow[d, "f''"] & (\operatorname{cosq}_n Y)_{n+1} \arrow[d, "f'"] \\ Z_{n+1} \arrow[r] & (\operatorname{cosq}_n Z)_{n+1} \end{tikzcd}$$

commutes.

Let us construct X.

Set $Y'_{n+1} = (\operatorname{cosq}_n Y)_{n+1}$ and denote by

$$d'^{n+1}_i : Y'_{n+1} \longrightarrow Y_n \quad (\text{resp. } s'^n_i : Y_n \longrightarrow Y'_{n+1})$$

the face operators (resp. the degeneracy operators) obtained by functoriality.

Set $Y_{n+1} = N \coprod (\coprod_{\operatorname{Hom}_{\Delta^-_{\ell \leq n}}([n+1],[\ell])} NY_\ell)$, and let $\alpha : Y_{n+1} \to Y'_{n+1}$ be the arrow whose components are β and the arrows $NY_\ell \to Y_\ell \to Y'_{n+1}$ for every epimorphism $[n+1] \to [\ell]$ with $\ell \leq n$.

Let $s^n_i : Y_n \simeq \coprod_{\operatorname{Hom}_{\Delta^-_{\ell' \leq n}}([n],[\ell'])} NY_{\ell'} \to Y_{n+1}$ be the arrow induced by σ^i_n, so that 148

(I) $\alpha \circ s^n_i = s'^n_i$.

Set

(II) $d^{n+1}_i = d'^{n+1}_i \circ \alpha$.

To show that this defines an object of $\operatorname{Hom}(\Delta^0_{n+1}, H)$, it is enough, by [[3]. II (2.4)], to verify the formulas

a) $d^n_i \cdot d^{n+1}_j = d^n_{j-1} \cdot d^{n+1}_i \qquad i < j;$

b) $s^n_i \cdot s^{n-1}_j = s^n_{j+1} \cdot s^{n-1}_i \qquad i \leq j;$

c) $d^{n+1}_i \cdot s^n_j = \begin{cases} s^{n-1}_{j-1} \cdot d^n_i & i < j, \\ \mathrm{id} & i = j \text{ or } i = j+1, \\ s^{n-1}_j \cdot d^n_{i-1} & i > j+1. \end{cases}$

Formulas a) and c) are evident by (I) and (II), since they are known to hold if d^{n+1}_i is replaced by d'^{n+1}_i and s^n_i by s'^n_i.

Formula b) holds by construction.

We leave it to the reader to verify that the object X thus obtained satisfies 5.1.3.

Note that the functor constructed in the proof of 5.1.3, from the category of triples satisfying (5.1.2) $(*)$ to the category of $(n+1)$-truncated simplicial objects, is faithful but not full.

Proposition 5.1.3 provides a very convenient method for constructing simplicial objects of H by induction. In the remainder of this section, we formalize this observation in the form that will be used in 5.2 and 5.3.

5.1.4. Let $E \xrightarrow{\pi} B$ be a functor such that, for every object b of B, the fiber category E_b is nonempty and satisfies the conditions of 5.1.0. 149

Let Q be a property of an object of E, stable under isomorphism. Suppose that the following conditions hold:

a) *For every object b of B, there exists an object x of E_b satisfying Q.*

b) *For every object b of B and every pair (x, y) of objects of E_b satisfying Q, there exists an object z of E_b satisfying Q and morphisms $z \to x$, $z \to y$ in E_b.*

c) *For every morphism $b' \xrightarrow{t} b$ of B and every object x of E_b satisfying Q, there exists an object x' of $E_{b'}$ satisfying Q and a morphism $x' \to x$ over t.*

Let, on the other hand, P be a property of an arrow of E over an identity, stable under isomorphism. Suppose that the following conditions hold:

d) *For every object b of B and every object x of E_b, there exists an arrow $y \to x$ in E_b satisfying P.*

e) *For every object b of B, every diagram in E_b of the form*

$$\begin{tikzcd}
u \arrow[rr, "f"] & & x \\
& z \arrow[ur] \arrow[dr] & \\
v \arrow[rr, "g"'] & & y
\end{tikzcd}$$

in which f and g satisfy P, there is a completion to a commutative diagram in E_b

$$\begin{tikzcd}
& u \arrow[rr, "f"] & & x \\
w \arrow[ur] \arrow[dr] \arrow[rr, "h"] & & z \arrow[ur] \arrow[dr] & \\
& v \arrow[rr, "g"'] & & y
\end{tikzcd}$$

150 *in which h satisfies P.*

f) *For every morphism $b' \xrightarrow{t} b$ of B and every morphism $x \xrightarrow{f} y$ of E_b satisfying P, there exists a morphism $x' \xrightarrow{f'} y'$ of $E_{b'}$ satisfying P and a commutative diagram*

$$\begin{tikzcd}
x' \arrow[r, "\alpha"] \arrow[d, "f'"'] & x \arrow[d, "f"] \\
y' \arrow[r, "\beta"'] & y
\end{tikzcd}$$

in which α and β are morphisms over t.

Definition 5.1.5. *With the notation of 5.1.4, we say that an object X of* $\mathrm{Hom}(\Delta^0, E)$ *satisfies condition* (APQ) *if the following conditions hold:*

(i) *The image of $\pi \circ X$ is the constant simplicial object defined by an object b of B.*
(ii) *As an object of* $\mathrm{Hom}(\Delta^0, E_b)$, *$X$ is σ-split.*
(iii) *X_0 satisfies condition Q.*
(iv) *For every integer n, the arrow $NX_{n+1} \to (\mathrm{cosq}_n X)_{n+1}$ satisfies condition P.*

5.1.6. Let $E_{(APQ)}$ denote the subcategory of $\mathrm{Hom}(\Delta^0, E)$ whose objects are the simplicial objects satisfying (APQ) and whose morphisms are the morphisms $X \xrightarrow{\alpha} Y$ of $\mathrm{Hom}(\Delta^0, E)$ such that $\pi(\alpha_n) : \pi(X_n) \to \pi(Y_n)$ is the same morphism of $\boldsymbol{B}$ for every n. There is then a canonical functor $\overline{\pi} : E_{(APQ)} \to \boldsymbol{B}$.

Proposition 5.1.7. *The functor $\overline{\pi}$ is surjective. Moreover, for every object b of $\boldsymbol{B}$ and every pair (X, Y) of objects of $E_{(APQ),b}$, there exists a diagram* 151

$$\begin{tikzcd} & Z \arrow[dl] \arrow[dr] & \\ X & & Y \end{tikzcd}$$

in $E_{(APQ),b}$.

The proof follows immediately from 5.1.3, by virtue of the hypotheses made in 5.1.4.

Proposition 5.1.8. *Let $\boldsymbol{C}$ be a category and F a functor $E_{(APQ)} \to \boldsymbol{C}$. We introduce the following conditions on F:*

(i) *For every object b of $\boldsymbol{B}$, the restriction of F to $E_{(APQ),b}$ factors through the* coarse *category*[34] *defined by the objects of $E_{(APQ),b}$.*
(ii) *For every morphism $t : b' \to b$ of $\boldsymbol{B}$ and every pair $(X' \to X, Y' \to Y)$ of morphisms over t, the following diagram commutes:*

$$\begin{tikzcd} F(X') \arrow[r] \arrow[d, "\alpha'"'] & F(X) \arrow[d, "\alpha"] \\ F(Y') \arrow[r] & F(Y) \end{tikzcd}$$

where α and α' are the arrows determined by condition (i).

Then the functor $G \mapsto G \circ \overline{\pi}$ induces an equivalence between the category $\mathrm{Hom}(\boldsymbol{B}, \boldsymbol{C})$ *and the full subcategory of* $\mathrm{Hom}(E_{(APQ)}, \boldsymbol{C})$ *defined by the functors satisfying conditions* (i) and (ii).

The details of the proof are left to the reader. We indicate only how to prove that the preceding functor is essentially surjective. 152

Let $F : E_{(APQ)} \to \boldsymbol{C}$ be a functor satisfying conditions (i) and (ii). For every object b of $\boldsymbol{B}$, choose an object X of $E_{(APQ),b}$ and set $\overline{F}(b) = F(X)$; one then verifies that $\overline{F}(b)$ varies functorially with b.

5.2. Singular cohomology of a scheme of finite type over a field of characteristic zero.

In this section, k will denote a field of characteristic zero.

[34]A category is called *coarse* if, for every pair (x, y) of objects, $\mathrm{Hom}(x, y)$ consists of exactly one element.

5.2.0. We shall denote by Sch_f/k (resp. Ann) the category of schemes *locally of finite type* over k (resp. complex analytic spaces). All simplicial objects of Sch_f/k (resp. Ann) will be regarded as simplicial objects of Top by means of the canonical functor $\mathrm{Sch}_f/k \to \mathrm{Top}$ (resp. $\mathrm{Ann} \to \mathrm{Top}$). We retain the data of (4.1.0) concerning cohomological descent.

Following the usual notation, we write $S \to S^{\mathrm{an}}$ for the canonical functor $\mathrm{Sch}_f/\mathbf{C} \to \mathrm{Ann}$ defined in G.A.G.A.

5.2.1. Let X be a simplicial object of Sch/k (resp. Ann): the usual compatibilities in the construction of the de Rham complex of a morphism of schemes (resp. of analytic spaces) make it possible to construct a complex $\Omega^{\mathrm{Zar}\cdot}_{X/k}$ (resp. $\Omega^{\cdot}_{X/\mathbf{C}}$) of $D^+(\Gamma(\overline{X}, \mathbf{Z}))$ such that, for every integer i, $Re_i^*(\Omega^{\mathrm{Zar}\cdot}_{X/k})$ (resp. $Re_i^*(\Omega^{\cdot}_{X/\mathbf{C}})$) is the de Rham complex $\Omega^{\mathrm{Zar}\cdot}_{X/k}$ (resp. $\Omega^{\cdot}_{X_i/\mathbf{C}}$).

Let $X \xrightarrow{u} C^{\Delta}_S$ be an augmented simplicial object of Sch_f/k (resp. Ann). We define $H^{\mathrm{Zar}\cdot}_{DR}(u)$ (resp. $H^{\cdot}_{DR}(u)$ and $H^{\cdot}(u, \mathbf{C})$) to be the hypercohomology groups of the complex $R\overline{u}_+(\Omega^{\mathrm{Zar}\cdot}_{X/k})$ (resp. $R\overline{u}_*(\Omega^{\cdot}_{X/\mathbf{C}})$ and $R\overline{u}(\mathbf{C})$).

153 5.2.2. If $X \xrightarrow{u} C^{\Delta}_S$ is a semisimplicial object of $\mathrm{Sch}_f/\mathbf{C}$, there are three canonical morphisms:

$$H^{\mathrm{Zar}\cdot}_{DR}(u) \xrightarrow{\alpha} H^{\cdot}_{DR}(u^{\mathrm{an}})$$

$$H^{\cdot}(u^{\mathrm{an}}, \mathbf{C}) \xrightarrow{\beta} H^{\cdot}_{DR}(u^{\mathrm{an}})$$

$$H^{\cdot}(S^{\mathrm{an}}, \mathbf{C}) \xrightarrow{\gamma} H^{\cdot}(u^{\mathrm{an}}, \mathbf{C}).$$

PROPOSITION 5.2.3. *With the notation of (5.2.2), suppose that X_n is smooth for every n: then α and β are isomorphisms. If, moreover, $X^{\mathrm{an}} \xrightarrow{u^{\mathrm{an}}} C^{\Delta}_{S^{\mathrm{an}}}$ is a hypercovering of S^{an} for the topology of universal 1-cohomological descent, then γ is an isomorphism.*

The assertion that α is an isomorphism follows from [[6], Th. 1′]. The Poincaré lemma shows that β is an isomorphism. The final part of the proposition follows from (3.3.3).

5.2.4. We shall now apply (5.1.4) in the case where $B = \mathrm{Sch}_f/k$ and E is the category over B such that, for every scheme S, the fiber category E_S is the category of schemes of Sch_f/k over S, the arrows over an arrow of B being the obvious ones. We shall say that a morphism $h : X \to Y$ over S satisfies property P if X is smooth over k and if h is a composite of finitely many morphisms satisfying one of the following conditions:

(i) it is the coproduct of a family of étale surjective morphisms;

(ii) it is the coproduct of a family of proper surjective morphisms.

We shall say that a scheme X of E_S satisfies property Q if the structural morphism $X \to S$ satisfies property P.

We leave it to the reader to verify, using resolution of singularities, that all the conditions of (5.1.4) are satisfied.

154 The groups $H^{\mathrm{Zar}}_{DR}(u)$ define a contravariant functor from the category $E_{(APQ)}$ to the category of graded abelian groups, denoted by $H^{\mathrm{Zar}\cdot}_{DR}$.

PROPOSITION 5.2.5. *The functor $H^{\mathrm{Zar}\cdot}_{DR}$ satisfies the conditions of (5.1.8).*

By the Lefschetz principle, we are reduced to the case $k = \mathbf{C}$. In that case, for every morphism $S' \xrightarrow{t} S$ and every pair $(u' \to u, v' \to v)$ of morphisms over t, there is a commutative diagram:

$$\begin{tikzcd}
H_{DR}^{\mathrm{Zar}\,\cdot}(u') \arrow[r] \arrow[d, "\gamma^{-1}.\beta^{-1}.\alpha"'] & H_{DR}^{\mathrm{Zar}\,\cdot}(u) \arrow[d, "\gamma^{-1}.\beta^{-1}.\alpha"] \\
H^{\cdot}(S'^{\,\mathrm{an}}, \mathbf{C}) \arrow[r, "H^{\cdot}(t^{\mathrm{an}}{,}\mathbf{C})"] & H^{\cdot}(S^{\mathrm{an}}, \mathbf{C}) \\
H_{DR}^{\mathrm{Zar}\,\cdot}(v') \arrow[r] \arrow[u, "\gamma^{-1}.\beta^{-1}.\alpha"] & H_{DR}^{\mathrm{Zar}\,\cdot}(v) \arrow[u, "\gamma^{-1}.\beta^{-1}.\alpha"']
\end{tikzcd}$$

which completes the proof.

DEFINITION 5.2.6. *The contravariant functor defined by (5.1.8) from $H_{DR}^{\mathrm{Zar}\,\cdot}$ will be denoted by $H^{\cdot}(*, k)$. For every scheme S locally of finite type over k, the graded abelian group $H^{\cdot}(S, k)$ will be called the* singular cohomology of S.

5.2.7. We conclude this section with an illustration of the general principles introduced in it. Recall first that if S is a complex analytic space, the *de Rham cohomology* of S, denoted by $\mathbf{H}^{\cdot}_{DR}(S)$, is by definition the hypercohomology of the de Rham complex $\Omega^{\cdot}_{S/\mathbf{C}}$.

PROPOSITION 5.2.8. (Blum–Herrera). *Let S be a scheme locally of finite type over* $\mathbf{C}$. 155
The canonical arrow

$$H^{\cdot}(S^{\mathrm{an}}, \mathbf{C}) \longrightarrow \mathbf{H}^{\cdot}_{DR}(S^{\mathrm{an}})$$

has a retraction (in the category of graded abelian groups) that is functorial in S.

By (5.1.8), it is enough to consider the morphism $H^{\cdot}(+, \mathbf{C}) \circ \overline{\pi} \to \mathbf{H}^{\cdot}_{DR} \circ \overline{\pi}$ obtained from the preceding one by composition with $\overline{\pi}$.

Using, for every augmentation $X \xrightarrow{u} C^{\Delta}_{S}$, the canonical morphism $\mathbf{L}\overline{u}^{+}(\Omega^{\cdot}_{S/C}) \to \Omega^{\cdot}_{X/C}$, one defines a functorial morphism $\mathbf{H}^{\cdot}_{DR} \circ \overline{\pi} \to \mathbf{H}^{\cdot}_{DR}$ so that the following diagram is commutative:

$$\begin{tikzcd}
H^{\cdot}_{DR} & H^{\cdot}_{DR} \circ \overline{\pi} \arrow[l] \\
H^{\cdot}(*, \mathbf{C})_{\circ}\overline{\pi} \arrow[u, "\beta\circ\gamma", "\wr"'] \arrow[ur]
\end{tikzcd}$$

which gives the desired retraction.

REMARK 5.2.9. Statement (5.2.8) remains valid for every analytic space once resolution of singularities is available in the analytic setting.

5.3. Mixed Hodge theories. We give here a technical result that plays a key role in [2].

k is a field of characteristic zero.

5.3.1. Let $\mathrm{Sch}_{\mathrm{fs}}/k$ be the category of separated schemes of finite type over k; denote by $E \xrightarrow{\pi} \mathrm{Sch}_{\mathrm{fs}}/k$ the category over $\mathrm{Sch}_{\mathrm{fs}}/k$ defined as follows:

156 If S is an object of $\mathrm{Sch}_{\mathrm{fs}}/k$, an object of E_S is a triple $(\overline{X}, X, i)$, where $\overline{X}$ is a reduced projective scheme over k, X is a proper scheme over S, and $i : X \to \overline{X}$ is an open immersion over k such that $i(X)$ is dense in $\overline{X}$.

If $t : S' \to S$ is a morphism of schemes, $(\overline{X}, X, i)$ is an object over S, and $(\overline{X}', X', i')$ is an object over S', a morphism from $(\overline{X}, X, i)$ to $(\overline{X}', X', i')$ over t is a pair $(\overline{h}, h)$, where $\overline{h} : \overline{X}' \to \overline{X}$ is a morphism over k and $h : X' \to X$ is a morphism over S, such that the following diagram is commutative.

$$\begin{tikzcd}
X' \arrow[r, "i'"] \arrow[d, "h"'] & \overline{X}' \arrow[d, "\overline{h}"] \\
X \arrow[r, "i"] & \overline{X}.
\end{tikzcd}$$

N.B.: the properness hypotheses imply that $\overline{h}^{-1}(i(X)) = i(X')$.

Lemma 5.3.2. *For every object S of $\mathrm{Sch}_{\mathrm{fs}}/k$, the category E_S is nonempty, has nonempty projective limits, and has finite coproducts that are disjoint and universal.*

The fact that the category E_S is nonempty follows from Chow's lemma.

We shall describe only the construction of the direct product of two objects $(\overline{X}, X, i)$ and $(\overline{X}', X', i')$ of E_S.

The canonical arrow $j : X \times_S X' \to \overline{X} \times_k \overline{X}'$ is an immersion. Let T be the reduced closed subscheme of $\overline{X} \times_k \overline{X}'$ whose underlying space is the closure of the image of j. Then $(T, X \times_S X'_{\mathrm{red}}, j_{\mathrm{red}})$ has the required universal property.

157 5.3.3. Let S be an object of $\mathrm{Sch}_{\mathrm{fs}}/k$. We shall say that an object $(\overline{X}, X, i)$ of E_S satisfies property Q if $\overline{X}$ is smooth over k and $i(X)$ is the complement of a normal crossings divisor. We shall say that a morphism $(\overline{h}, h) : (\overline{X}', X', i') \to (\overline{X}, X, i)$ satisfies property P if $\overline{h}$ is surjective and $(\overline{X}', X', i')$ satisfies property Q.

Proposition 5.3.4. *With the notation of (5.3.3), all the conditions of (5.1.4) are satisfied.*

Conditions a), b), d), and e) are readily verified using resolution of singularities. Moreover, c) follows from f), so it remains to verify f).

We must show that every commutative diagram

$$\begin{tikzcd}
 & Z \arrow[r, "i_Z"] \arrow[d, "h"'] & \overline{Z} \arrow[d, "\overline{h}"] \\
 & X \arrow[r, "i_X"] \arrow[d] & \overline{X} \arrow[d] \\
S' \arrow[r, "t"] & S \arrow[r] & k
\end{tikzcd}$$

where $(\overline{X}, X, i_X)$ and $(\overline{Z}, Z, i_Z)$ are objects of E_S and $\overline{h}$ is surjective can be completed to a commutative diagram

$$\begin{tikzcd}
Z' \arrow[rr] \arrow[dr, "i_{Z'}"] \arrow[dd, "h'"'] & & Z \arrow[dr, "i_Z"] \arrow[dd, "h"' near end] & \\
& \overline{Z}' \arrow[rr] \arrow[dd] & & \overline{Z} \arrow[dd, "\overline{h}"] \\
X' \arrow[rr, "h'" near end] \arrow[dr, "i_{X'}"] \arrow[dd] & & X \arrow[dr, "i_X"] \arrow[dd] & \\
& \overline{X}' \arrow[rr] \arrow[dd] & & \overline{X} \arrow[dd] \\
S' \arrow[rr, "t" near end] \arrow[dr] & & S \arrow[dr] & \\
& k \arrow[rr, equal] & & k
\end{tikzcd}$$

where $(\overline{X}', X', i_{X'})$ and $(\overline{Z}', Z', i_{Z'})$ are objects of $E_{S'}$ and $\overline{h'}$ is surjective. 158

Consider a diagram

$$\begin{tikzcd}
T \arrow[r, "\varphi"] \arrow[d, hook, "i_T"'] & S' \arrow[r, "t"] & S \arrow[d] \\
\overline{T} \arrow[rr, "\psi"'] & & k
\end{tikzcd}$$

where φ is proper and ψ is projective. By base change we obtain a diagram

$$\begin{tikzcd}
Z \times_S T_{\mathrm{red}} \arrow[rr] \arrow[dr, "p"'] \arrow[dd] & & Z \arrow[dr] \arrow[dd] & \\
& \overline{Z} \times_k \overline{T}_{\mathrm{red}} \arrow[rr] \arrow[dd] & & \overline{Z} \arrow[dd] \\
X \times_S T_{\mathrm{red}} \arrow[rr] \arrow[dr, "q"'] \arrow[dd] & & X \arrow[dr] \arrow[dd] & \\
& \overline{X} \times_k \overline{T}_{\mathrm{red}} \arrow[rr] \arrow[d] & & \overline{X} \arrow[dd] \\
T \arrow[r] \arrow[dr] & S' \arrow[r] \arrow[d] & S \arrow[dr] & \\
& \overline{T} \arrow[rr] & & k
\end{tikzcd}$$

where p and q are immersions. It then suffices to replace $\overline{Z} \times_k \overline{T}_{\mathrm{red}}$ and $\overline{X} \times_k \overline{T}_{\mathrm{red}}$ by the closed images of p and q to obtain the required diagram. This completes the proof of (5.3.4). 159

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EXPOSÉ VI

Finiteness Conditions. Fibered Topoi and Sites. Applications to Questions of Passage to the Limit.

A. Grothendieck and J.-L. Verdier

0. Introduction

In the present exposé, we study two closely related types of questions. **165**

First, we undertake a systematic study of finiteness conditions for topoi, successively examining the notions of *quasi-compactness*, *quasi-separatedness*, and *coherence* (= quasi-compactness + quasi-separatedness). These are inspired by the familiar analogous notions from scheme theory. We study them first for an object X of a topos and for an arrow $u : X \to Y$ of a topos (§ 1), then for the topos E itself (§ 2), and finally for a morphism of topoi $f : E \to E'$ (§ 3). Section 4 studies these notions in the special case of a topos obtained by gluing two topoi (IV 9). For the subsequent sections, the essential facts to retain from these somewhat technical developments are the following: a *coherent topos* is a topos equivalent to one of the form $C^{\sim}$, where C is a small site in which finite projective limits are representable and every covering family admits a finite covering subfamily; and a *coherent morphism* $f : E \to E'$ of coherent topoi is a morphism that can be described (up to equivalence) as the morphism associated with a morphism of sites $f : C \to C'$, where C and C' are as above and the corresponding functor $f^* : C' \to C$ is left exact.

It is no doubt useful to note that, with the finiteness notions used here, and especially that of a coherent topos, we resolutely depart (and for the first time) from the topological spaces familiar to analysts and "topologists"[34]. Thus the topos associated (IV 2.1) with a separated topological space X is coherent only if X is a finite set! In other words, the present exposé is directed entirely toward the types of topoi arising from algebraic geometry and algebra. In fact, all the topoi used thus far in these disciplines (except, of course, in algebraic geometry by transcendental methods over the field of complex numbers!) turn out to be locally coherent.

Second, we develop two *theorems on passage to the inductive limit* for the cohomology of topoi. One (5.2) tells us that the functors $H^i(E, -)$ on a coherent topos commute **166**
with inductive limits of abelian sheaves. The other (8.7) says essentially that if a topos X is represented as a filtered projective limit of coherent topoi X_i, with coherent transition morphisms $f_{ij} : X_j \to X_i$, then the cohomology of X (with coefficients in an arbitrary abelian sheaf) is computed as the inductive limit of the cohomologies of the X_i. To give this statement a precise meaning, the main point is to specify the notion of a "projective limit of topoi," which is done in § § 7, 8. This is a very useful geometric notion, undoubtedly also in contexts other than that of projective limits of schemes (which had served as the model for M.Artin in his initial definition of this notion). We show, for example, in (8.4) how the various notions of being "localized at a point" (for a Noetherian space,

[34] The quotation marks indicate that this is a "…" unfamiliar with the notion of a topos (cf. IV 0.4 for the meaning of the word "topology").

for a scheme with its usual topology, or even with its étale topology) can be unified by defining the localization as the projective limit of the "neighborhoods" of that point—in perfect agreement with the intuitive idea of localization at a point.

The two limit-passage theorems will be stated again explicitly in the next exposé in the special setting of étale cohomology of schemes (VII 3.3 and 5.7); in that case, they will moreover be used constantly throughout the remainder of this Seminar, and practically everywhere étale cohomology has been used up to the present. The reader willing to accept these two theorems in the special case in which they are stated in Exposé VII, and interested exclusively in applications of étale cohomology, may therefore omit the present exposé. It is true that the notion of a *constructible object* of a topos (i.e. one whose structural morphism $X \to e$ to the final object e is coherent) will likewise play an important role throughout the rest of the Seminar (whereas it plays only a very secondary role in the present exposé, if only because, for a coherent topos, it coincides with the notion of a coherent object, which takes center stage here). But this notion is developed independently in Exposé IX, in the special case of étale sheaves on schemes, without using the general study made in the present exposé; in that case it also exhibits very useful special phenomena that served as the basis for its treatment in IX (beginning
167 with Definition IX 2.3). Since the text of IX predates the present text of Exposé VI by five years, yet remains quite usable for the reader, we have confined ourselves at the end of Exposé IX to proving the equivalence between the notion of constructibility used in IX and the one introduced here.

1. Finiteness conditions for the objects and arrows of a topos

Definition 1.1. *Let C be a site. An object X of C is called quasi-compact if, for every covering family $(X_i \to X)_{i\in I}$, there is a finite subset J of the index set I such that the subfamily $(X_i \to X)_{i\in J}$ is still covering.*

Since a $\mathcal{U}$-topos E is always regarded as a site (IV 1.1.1), the preceding definition applies in particular to an object of E. This case is moreover the general one, by the following proposition.

Proposition 1.2. *Let C be a $\mathcal{U}$-site, let $\epsilon : C \to C^{\sim}$ be the canonical functor, and let X be an object of C. Then X is a quasi-compact object of the site C if and only if $\epsilon(X)$ is a quasi-compact object of the topos $C^{\sim}$.*

Suppose that X is quasi-compact; we prove that $\epsilon(X)$ is as well. Let $(F_i \to \epsilon(X))_{i\in I}$ be an epimorphic family. For every $i \in I$, choose an epimorphic family $\epsilon(X_{ij}) \to F_i$, $j \in J_i$. The composite $\epsilon(X_{ij}) \to \epsilon(X)$ identifies with an element α_{ij} of $\epsilon(X)(X_{ij})$. In general one cannot assert that it comes from a morphism $X_{ij} \to X$, but, by the construction of the associated sheaf $\epsilon(X)$, one knows that, for fixed i and j, a covering family $Y_k \to X_{ij}$ can be found such that the images of α_{ij} in the $\epsilon(X)(Y_k)$ come from elements of $X(Y_k)$, i.e. from morphisms $Y_k \to X$. Thus, after refining the epimorphic family $\epsilon(X_{ij}) \to F_i$, we may suppose that the composites $\epsilon(X_{ij}) \to \epsilon(X)$ come from
168 morphisms $X_{ij} \to X$. As i and j vary, the family of $\epsilon(X_{ij}) \to \epsilon(X)$ is epimorphic, so (II 4.4) the family of $X_{ij} \to X$ is covering. By the hypothesis on X, it admits a finite covering subfamily, and the corresponding finite family of morphisms $\epsilon(X_{ij}) \to \epsilon(X)$ is again epimorphic. Since these factor through finitely many of the morphisms $F_i \to \epsilon(X)$, that finite family is already epimorphic. This proves that $\epsilon(X)$ is quasi-compact.

Conversely, the fact that quasi-compactness of $\epsilon(X)$ implies quasi-compactness of X is trivial, by *loc. cit.*

The preceding result therefore shows that the notion of quasi-compactness on sites reduces to the corresponding notion in topoi.

Proposition 1.3. *Let C be a site and let $(X_i \to X)_{i\in I}$ be a finite covering family, with the X_i quasi-compact. Then X is quasi-compact.*

To see this, we may assume, by 1.2, that C is a topos. Let $(X'_j \to X)$ be a covering, i.e. epimorphic, family in C. Then, for every $i \in I$, the family obtained by the base change $X_i \to X$ is epimorphic and hence (since X_i is quasi-compact) admits a finite epimorphic subfamily, corresponding to a finite subset $J_i \subset J$ of the index set J. Set $J' = \bigcup_{i\in I} J_i$. Then J' is a finite subset of J because I is finite, and the family $(X'_j \to X)_{j\in J'}$ is already epimorphic, since it is so locally, C.Q.F.D.

Corollary 1.4. *Let $(X_i)_{i\in I}$ be a family of objects of a topos E, and let X be their sum. Then X is quasi-compact if and only if all the X_i are quasi-compact and all but finitely many of the X_i are isomorphic to the "empty object" $\varnothing_E$ of E.* 169

Sufficiency follows immediately from 1.3 (since the sum does not change when the empty summands are omitted). For necessity, note that a finite subfamily $(X_i)_{i\in J}$ must cover X, which implies that, for $i \in I - J$, one has $X_i = X_i \times_X \amalg_{j=J} X_j = \amalg_{j=J} X_i \times_X X_j = \varnothing$ (II 4.5.1). On the other hand, to show that every X_i is quasi-compact, note that $X \simeq X_i \amalg X'_i$, and that every covering family $Y_j \to X_i$ of X_i defines a covering family $Y_j \amalg X'_i \to X_j \amalg X'_i = X$ of X. The latter admits a finite covering subfamily, which plainly implies that the corresponding finite family of $Y_j \to X_i$ is also covering; thus X_i is quasi-compact.

Remark 1.5.1. The property for an object X of a topos E of being quasi-compact depends only on the induced topos $E_{/X}$, and means that the final object of the latter is quasi-compact. Likewise, an object X of a site C is quasi-compact if and only if the final object of the induced site $C_{/X}$(III 5.1) is quasi-compact.

1.5.2. Let X be an object of a topos E. Then X is quasi-compact if and only if every filtered increasing family $(X_i)_{i\in I}$ of subobjects of X that covers X contains some X_i equal to X. This condition therefore depends only on the ordered set of subobjects of 170
X (i.e. the open subsets of the induced topos $E_{/X}$).

1.5.3. Consider a generating family $(X_i)_{i\in I}$ of E. It follows from the definitions and from 1.3 that, for an object X of E to be quasi-compact, it is necessary that it admit a covering family consisting of finitely many of the X_i, i.e. that X be isomorphic to a quotient object of a finite sum of objects of E; this condition is sufficient if the X_i are quasi-compact. Note also that, under the hypothesis made on E, after replacing the generating family under consideration by a subfamily if necessary, we may clearly suppose that $I \in \mathscr{U}$. Using the fact that the set of quotients of an object of E is small (I 7.5), we then find that the full subcategory E_{qc} of E formed by the quasi-compact objects of E is equivalent to a category $\in \mathscr{U}$, i.e. the cardinal of the set of isomorphism classes of objects of E_{qc} is $\in \mathscr{U}$.

Example 1.6.1. Let X be a topological space. An open subset U of X is a quasi-compact object of the site $\mathrm{Ouv}(X)$ of open subsets of X if and only if U is a quasi-compact space for the topology induced by X. More generally, a sheaf F on X is a quasi-compact object of the topos $\mathrm{Top}(X)$ if and only if the étalé space X' over X associated with F is quasi-compact. (This follows from the preceding assertion, from 1.5, and from the fact that $\mathrm{Top}(X)_{/F}$ is equivalent to $\mathrm{Top}(X')$ (IV 5.7)).

1.6.2. Consider on the category (Sch) of schemes (elements of $\mathcal{U}$) one of the topologies T_i of [6] 6.3 (for example, the Zariski topology, the étale topology, the fppf topology, or the fpqc topology). A scheme is quasi-compact for this topology if and only if it is a quasi-compact scheme in the usual sense. (With the notation of *loc. cit.*, only the fact that every family $\in P'$ is finite is used.) 171

Definition 1.7. *Let E be a topos and let $f : X \to Y$ be an arrow of E. We say that f is a quasi-compact morphism if, for every arrow $Y' \to Y$ with Y' quasi-compact, the object $X' = X \times_Y Y'$ is also quasi-compact. We say that f is quasi-separated if the diagonal morphism $X \to X \times_Y X$ is quasi-compact. We say that f is coherent if f is quasi-compact and quasi-separated.*

1.7.1. Unwinding the definition of a quasi-separated morphism $f : X \to Y$, we see that it also means that, for every pair of arrows $g_1, g_2 : X' \rightrightarrows X$ "over Y," i.e. such that $fg_1 = fg_2$, quasi-compactness of X' implies quasi-compactness of $\mathrm{Ker}(g_1, g_2)$.

Proposition 1.8. *Let E be a topos.*

(i) *Every isomorphism in E is a coherent morphism, i.e. is quasi-compact and quasi-separated. The composite of two quasi-compact (resp. quasi-separated, resp. coherent) morphisms is quasi-compact (resp. quasi-separated, resp. coherent).*

172 (ii) *Let $f : X \to Y$ and $g : Y' \to Y$ be morphisms in E, and let $f' : X' = X \times_Y Y' \to Y'$ be deduced from f by the base change g. If f is quasi-compact (resp. quasi-separated, resp. coherent), then so is f'.*

(iii) *Let $f : X \to Y$ and $g : Y \to Z$ (the arrow $Y \to Z$) be morphisms in E. If gf is quasi-separated, then f is quasi-separated.*

(iv) *Let $f : X \to Y$ and $g : Y \to Z$ be morphisms in E, with g quasi-separated. If gf is quasi-compact (resp. coherent), then f is quasi-compact (resp. coherent).*

The proofs are well known (Cf. EGA IV 1 or EGA I, 2nd edition). Statements (i) and (ii) in the quasi-compact case follow trivially from the definitions. In the quasi-separated case, stability under base change in (ii) follows immediately from the definition and the preceding case, as does the fact that every isomorphism is quasi-separated. Stability under composition $X \to Y \to Z$ follows from the diagram

$$(*)\qquad \begin{tikzcd} X \arrow[d, "\Delta_{X/Y}"'] \arrow[dd, bend right=60, "\Delta_{X/Z}"'] & \\ X \times_Y X \arrow[r] \arrow[d, "u"] & Y \arrow[d, "\Delta_{Y/Z}"] \\ X \times_Z X \arrow[r] & Y \times_Z Y \end{tikzcd}$$

173 whose square is Cartesian. By hypothesis, $\Delta_{Y/Z}$ is quasi-compact, and hence so is u, which is obtained from it by base change; moreover, $\Delta_{X/Y}$ is quasi-compact, and therefore so is the composite $u\Delta_{X/Y} = \Delta_{X/Z}$. This proves (i) and (ii), the coherent case following by combining the two preceding cases. To prove the first case considered in (iv),

consider the following diagram with Cartesian squares:

$$(**)\qquad \begin{tikzcd}
 & X \arrow[r, "gf"] & Z \\
X \arrow[r, "\Gamma_f=(\mathrm{id}_X,f)"] \arrow[d] \arrow[rr, bend right, "f"'] & X \times_Z Y \arrow[u, "\mathrm{pr}_1"] \arrow[r, "\mathrm{pr}_2"] \arrow[d, "f\times_Z \mathrm{id}_Y"] & Y \arrow[u, "g"'] \\
Y \arrow[r, "\Delta_{Y/Z}"] & Y \times_Z Y &
\end{tikzcd}.$$

By hypothesis, g is quasi-separated, so $\Delta_{Y/Z}$ is quasi-compact and hence Γ_f is quasi-compact by base change. Likewise, by hypothesis gf is quasi-compact, so pr_2 is also quasi-compact by base change. Consequently, $f = \mathrm{pr}_2\,\Gamma_f$ is quasi-compact as a composite of two quasi-compact morphisms. The corresponding case of (iv) follows immediately from the preceding case and (iii), which remains to be proved. For this, return to diagram $(*)$. The hypothesis that gf is quasi-separated means that $\Delta_{X/Z} = u\Delta_{X/Y}$ is quasi-compact, while the conclusion that f is quasi-separated means that $\Delta_{X/Y}$ is quasi-compact. Equivalently, the conclusion that f is quasi-separated again amounts to the quasi-compactness of $\Delta_{X/Y}$. This follows from (iv) once we have proved that u is quasi-separated. But u is plainly a monomorphism, and indeed we have the following trivial 174
result:

Corollary 1.8.1. *Every monomorphism is quasi-separated.*

Indeed, if $u : X \to Y$ is a monomorphism, then $\Delta_{X/Y}$ is an isomorphism and is therefore quasi-compact, C.Q.F.D.

Corollary 1.8.2. *Let $f : X \to Y$ and $f' : X' \to Y'$ be two S-morphisms in E. If f and f' are quasi-compact (resp. quasi-separated, resp. coherent), then so is $f \times_S f' : X \times_S X' \to Y \times_S Y'$.*

This follows in the usual way by combining (i) and (ii).

Remark 1.9.1. *Let E be any category, and let Q be a subset of* $\mathrm{ob}\,E$ *stable under isomorphism (thus, giving Q amounts to giving a strictly full subcategory of E), playing the role of the "quasi-compact" objects. Suppose that fiber products are representable in E. One can then paraphrase Definition 1.7 to define the notions of Q-quasi-compact, Q-quasi-separated, and Q-coherent morphisms. Proposition 1.8 and Corollary 1.8.1 then hold, as does 1.8.2 when products of two objects are representable in E.*

1.9.2. Let S be an object of E, and let $f : X \to Y$ be a morphism of $E_{/S}$. It follows immediately from Remark 1.5 that f is quasi-compact (resp. quasi-separated, resp. coherent) if and only if the corresponding morphism in E (obtained from f by the 175
"forget S" functor) has the same property. In particular, if $f : X \to Y$ is a morphism in E, then, by taking $S = Y$ above, we see that f is quasi-compact (resp. quasi-separated, resp. coherent) if and only if the structural morphism of the object (X, f) of the induced topos $E_{/Y}$, with target the final object of that topos, is quasi-compact (resp. quasi-separated, resp. coherent). One also says that (by means of the structural morphism f) *X is quasi-compact* (resp. *quasi-separated,* resp. *coherent) over Y*, an expression that may equally well be interpreted as referring to the topos E or to the induced topos $E_{/Y}$ (in which Y is regarded as a final object).

1.9.3. An object X of a topos E is called *constructible* if it is coherent over the final object. Every morphism from one constructible object to another is coherent (1.8 (iv)). The full subcategory E_{cons} of E formed by the constructible objects of E is stable under finite inverse limits, as follows immediately from 1.8.2 and 1.8 (ii).

1.9.4. The notions of 1.7 are of real interest only in topoi E that admit a generating family formed by quasi-compact objects. In that case, these notions are “local on the base” (IV 8.5.4), as we now show.

176 PROPOSITION 1.10. *Suppose that the topos E admits a generating subcategory C formed by quasi-compact objects, and let $f : X \to Y$ be a morphism in E.*

(i) *The morphism f is quasi-compact if and only if, for every morphism $S \to Y$ with $S \in \mathrm{Ob}\, C$, the object $X \times_Y S$ is quasi-compact.*
(ii) *Let $Y_i \to Y$, $i \in I$, be a covering family. The morphism f is quasi-compact (*resp. *quasi-separated,* resp. *coherent) if and only if, for every $i \in I$, the morphism $f_i : X_i = X \times_Y Y_i \to Y_i$ has the same property.*

Let us prove (i). Necessity is immediate from the definition. For sufficiency, we must prove that, for every quasi-compact object Y' and every morphism $Y' \to Y$, the object $X' = X \times_Y Y'$ is quasi-compact. There is a covering family $S_i \to Y'$ with the S_i in $\mathrm{Ob}\, C$, and since Y' is quasi-compact, this covering family may be taken to be *finite*. By hypothesis the objects $X'_i = X \times_Y S_i \simeq X' \times_{Y'} S_i$ are quasi-compact, and since they cover X' (the S_i covering Y'), X' is quasi-compact by 1.3.

Let us prove (ii). It is enough to treat the quasi-compact case, since the quasi-separated case follows from Definition 1.7, and the coherent case likewise follows by combining the two preceding cases. The necessity has already been proved in 1.8 (ii). It remains to show that if the f_i are quasi-compact, then f is quasi-compact, i.e. that, for every quasi-compact Y' and every morphism $Y' \to Y$, the object $X' = X \times_Y Y'$ is
177 quasi-compact. Since $(Y_i \to Y)$ is a covering family, there exist objects S_α over Y that cover Y' and such that each composite $S_\alpha \to Y' \to Y$ factors through some Y_i. Since C is generating, the S_α may be taken in $\mathrm{Ob}\, C$, and since Y' is quasi-compact, the covering family may be taken finite. The $X' \times_{Y'} S_\alpha$ then form a finite covering family of X', and by 1.3 we are reduced to proving that each $X' \times_{Y'} S_\alpha = X \times_Y S_\alpha$ is quasi-compact. Choosing a factorization of $S_\alpha \to Y$ through some Y_i, the object in question is also isomorphic to $X_i \times_{Y_i} S_\alpha$, which is indeed quasi-compact since X_i is quasi-compact over Y_i and S_α is quasi-compact.

COROLLARY 1.11. *Let $g : Y \to Z$ be a morphism in the topos E, and let $(f_i : X_i \to Y)_{i \in I}$ be a covering family of morphisms with target Y. Then:*

(i) *Suppose that I is finite. If the $g f_i$ are quasi-compact, then so is g.*
(ii) *Suppose that the f_i are quasi-compact. If the $g f_i$ are quasi-separated, then so is g. If I is finite and the $g f_i$ are coherent, then g is coherent.*

Case (i) follows immediately from the definitions and 1.3, without any hypothesis on E. To prove (ii), it is enough, by (i), to treat the nonparenthetical case. Consider diagram 1.8 (∗) with X replaced by some X_i. Since the family of the f_i is covering, so
178 is the family of the $f_i \times_Z \mathrm{id}_Y$ obtained from it by base change. Hence, by 1.10 (ii), to prove that $\Delta_{Y/Z}$ is quasi-compact, it is enough to prove that the Γ_{f_i} are quasi-compact. Since $g f_i$ is quasi-separated by hypothesis, so is $\mathrm{pr}_2 : X_i \times_Z Y \to Y$, which is obtained from it by base change. On the other hand, $\mathrm{pr}_2\, \Gamma_{f_i} = f_i$ is quasi-compact by hypothesis. Thus Γ_{f_i} is quasi-compact by 1.8 (iv), applied to the composite of Γ_{f_i} and pr_2, C.Q.F.D.

Corollary 1.12. *Let $(X_i)_{i\in I}$ be a finite family of objects of E, let X be the coproduct of the X_i, let $f_i : X_i \to Y$ be morphisms in E, and let $f : X \to Y$ be the corresponding morphism. The morphism f is quasi-compact (resp. quasi-separated, resp. coherent) if and only if the f_i have the same property; in the quasi-separated case, the hypothesis that I be finite may be omitted.*

Sufficiency is a special case of 1.11, taking into account that the canonical morphisms $X_i \to X$ form a finite covering family and are quasi-compact morphisms, as follows immediately from the definition and 1.4. Necessity follows from transitivity, 1.8 (i), taking into account that the morphisms $X_i \to X$ are even coherent (for they are quasi-separated by 1.8.1).

Definition 1.13. *Let X be an object of a topos E. We say that X is a* quasi-separated object of the topos *E if, for every quasi-compact object S of E, every morphism from S to X is quasi-compact, i.e. (1.7) if, for any two quasi-compact objects S, T of E and morphisms $S \to X$, $T \to X$, the fiber product $S \times_X T$ is always quasi-compact. We say that X is a coherent object of the topos E if X is quasi-compact and quasi-separated.*

Proposition 1.14. *Let $f : X \to Y$ be a morphism in a topos E.* 179

(i) *If Y is quasi-compact, then f quasi-compact implies that X is quasi-compact. If Y is quasi-separated, then X quasi-compact implies that f is quasi-compact. If Y is coherent, f is quasi-compact if and only if X is quasi-compact.*
(ii) *If Y is quasi-compact* (resp. *quasi-separated,* resp. *coherent) and f is quasi-compact* (resp. *quasi-separated,* resp. *coherent), then X is quasi-compact* (resp. *quasi-separated,* resp. *coherent).*

The first two assertions of (i) are contained immediately in the definitions of a quasi-compact morphism (1.7) and, respectively, a quasi-separated object (1.13); the third follows by combining the first two. The first case of (ii) is merely a restatement, and the third again follows by combining the first two. It remains to treat the quasi-separated case. Let S be a quasi-compact object of E and $g : S \to X$ a morphism; we prove that g is quasi-compact. Since Y is quasi-separated, fg is quasi-compact, and since f is quasi-separated, it follows (1.8 (iii)) that g is quasi-compact, C.Q.F.D.

Corollary 1.15. *Every subobject of a quasi-separated object of E is quasi-separated. Let $(X_i)_{i\in I}$ be a family of objects of E, with $I \in U$; then $X = \amalg_i X_i$ is quasi-separated* (resp. *coherent) if and only if the X_i are.*

The first assertion follows from 1.14 (ii) and 1.8.1, but is also immediate from the 180
definition. For the second assertion in the nonparenthetical case, since the $X_i \to X$ are monomorphisms, it remains only to prove sufficiency, which follows easily from the definitions and 1.3; the parenthetical case follows from the case just treated and 1.4.

Corollary 1.16. *Under the conditions of 1.10 (ii), if the Y_i are coherent, then f is quasi-compact if and only if the $X_i = X \times_Y Y_i$ are quasi-compact objects of E.*

Indeed, by the criterion 1.10 (ii), one must express that the $f_i : X_i \to Y_i$ are quasi-compact; by 1.14 (i), this is equivalent to the X_i being quasi-compact.

Corollary 1.17. *Let E be a topos admitting a generating family formed by quasi-compact objects, and let $(g_i : Y_i \to Y)_{i\in I}$ be a covering (resp. finite covering) family in E, with the Y_i coherent. The object Y is quasi-separated (resp. coherent) if and only if, for every $i \in I$, g_i is quasi-compact, or equivalently if, for every pair of indices $i, j \in I$, $Y_i \times_Y Y_j$ is a quasi-compact object of E.*

The parenthetical case follows immediately from the nonparenthetical case and 1.3. Necessity of the first condition is immediate from the definition; let us prove sufficiency.
181 Let $f : X \to Y$ be a morphism, with X quasi-compact; we prove that f is quasi-compact. Since the g_i are quasi-compact, the $X_i = X \times_Y Y_i$ are quasi-compact, which implies by 1.16 that f is quasi-compact. Finally, expressing the quasi-compactness of the g_i by the same criterion 1.16 gives the second criterion of 1.17. In particular:

Corollary 1.17.1. *Let E be as in 1.17, let X be a coherent object of E, let $\mathscr{R} \rightrightarrows X$ be an equivalence relation in X, and let $Y = X/\mathscr{R}$. Then the following conditions are equivalent:*

(i) *Y is coherent;*
(ii) *$f : X \to Y$ is quasi-compact;*
(iii) *$\mathscr{R}$ is quasi-compact.*

Corollary 1.18. *Let E be a topos admitting a generating subcategory C formed by quasi-compact objects, and let Y be an object of E. The object Y is quasi-separated if and only if every morphism $X \to Y$ with $X \in \operatorname{ob} C$ is quasi-compact.*

Necessity is immediate from the definition, since the objects $X \in \operatorname{ob} C$ are quasi-compact by hypothesis. To prove sufficiency, let Z be a quasi-compact object of E; we prove that every morphism $Z \to Y$ is quasi-compact. By 1.10 (i), it is enough to verify that, for every $X \to Y$ with $X \in \operatorname{ob} C$, the object $Z \times_Y X$ is quasi-compact. This follows from the fact that the $X \to Y$ are quasi-compact by hypothesis and that Z is quasi-compact.

182 Corollary 1.19. *Let E be a topos admitting a generating subcategory formed by quasi-compact objects, and let $(g_i : Y_i \to Y)_{i\in I}$ be a covering family in E, with the Y_i quasi-separated and the g_i quasi-compact. Then Y is quasi-separated.*

Indeed, let X be a quasi-compact object of E; we show that every morphism $f : X \to Y$ is quasi-compact. By 1.10 (ii), it is enough to prove that the induced morphisms $f_i : X_i = X \times_Y Y_i \to Y_i$ are quasi-compact. Since g_i is quasi-compact, X_i is quasi-compact (because X is), and since Y_i is quasi-separated, it follows that f_i is quasi-compact.

Remark 1.20.1. The reader accustomed to the use of the terms "quasi-compact" and "quasi-separated" in algebraic geometry will expect certain other relations between the "absolute" notions (concerning objects of the topos E) and the "relative" notions (concerning arrows of E) considered here; for example, if $f : X \to Y$ is a morphism in E such that X is quasi-compact and quasi-separated, then f is quasi-separated. Such properties hold only under restrictive finiteness conditions on the topos E, stronger than those considered in 1.10; we shall study these conditions in the next section.

1.20.2. It follows again from 1.5 that whether an object X of the topos E is quasi-separated depends only on the induced topos $E_{/X}$, and means that the final object of that topos is quasi-separated, i.e. that in $E_{/X}$ the product of two quasi-compact objects is quasi-compact.

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1.21. Case of sites. Let C be a $\mathscr{U}$-site, E the topos of $\mathscr{U}$-sheaves on C, f an arrow of C, and X an object of C. We say that f is a *quasi-compact morphism* (resp. quasi-separated morphism) *of the site* C if $\epsilon(f)$ is a quasi-compact morphism (resp. quasi-separated morphism) of the topos E; likewise, we say that X is a *quasi-separated object of the site* C if $\epsilon(X)$ is a quasi-separated object of the topos E (compare 1.2). When C is a $\mathscr{U}$-topos endowed with its canonical topology, the canonical functor $\epsilon : C \to E$ is an

equivalence of categories, and consequently the preceding definition is compatible with the earlier definitions 1.7 and 1.13. Note also that the definition under consideration does not change when the universe $\mathscr{U}$ is replaced by another universe $\mathscr{V}$ such that C is also a $\mathscr{V}$-site, at least when C admits a topologically generating family consisting of quasi-compact objects. To see this, we are immediately reduced to the case $\mathscr{U} \subset \mathscr{V}$, so that E is a $\mathscr{V}$-site, and to proving that, for every arrow f of the $\mathscr{U}$-topos E, f is quasi-compact (resp. quasi-separated) if and only if it defines a quasi-compact arrow in the topos E' of $\mathscr{V}$-sheaves on E, and likewise that, for every object X of E, X is quasi-separated if and only if the corresponding object of E' is quasi-separated. We already know (without any hypothesis on E) the analogous statement for a quasi-compact object (1.2). The case "f quasi-compact" follows from this, by the criterion 1.10 (i), applied successively in E and in E' to the same generating family $(X_i)_{i\in I}$ of E consisting of **184**
quasi-compact objects. This also gives the case "f quasi-separated", which means that the corresponding diagonal morphism is quasi-compact. Finally, in the case "X quasi-separated", we are again reduced to the case "X quasi-compact" by the criterion 1.18.

1.22. ***Examples.*** Return to the example 1.6.2 of the category (Sch) with a topology T_i. This is a $\mathscr{V}$-site for a suitable $\mathscr{V}$, and it admits the subcategory of affine schemes as a topologically generating category consisting of quasi-compact objects. We leave it to the reader to verify that an arrow $f : X \to Y$ of this site is quasi-compact (resp. quasi-separated) if and only if it is a quasi-compact (resp. quasi-separated) morphism of schemes in the usual sense ([3] or [4]); likewise, an object of the site is quasi-compact (resp. quasi-separated) if and only if it is a quasi-compact (resp. quasi-separated) scheme in the usual sense of *loc. cit.*. It is in fact appropriate to say, as we do here, *coherent morphism of schemes, coherent scheme*, instead of "quasi-compact and quasi-separated morphism of schemes", "quasi-compact and quasi-separated scheme", as is also done in EGA I, 2nd edition.

1.22.1. For a given scheme $X \in \mathscr{U}$, one may also consider the topos $\mathrm{Top}(X)$ associated with the underlying topological space, or the topos $\mathrm{Top}(X_{\mathrm{et}})$ (resp. $\mathrm{Top}(X_{\mathrm{fppf}})$) associated with the site of étale X-schemes (resp. X-schemes locally of finite presenta- **185**
tion) that are elements of $\mathscr{U}$, endowed with the étale topology (resp. fppf topology). It is a $\mathscr{U}$-topos admitting a generating family consisting of coherent objects, corresponding to the affine objects of the site of definition.

With this understood, the final object of this topos is quasi-compact (resp. quasi-separated, resp. coherent) if and only if the scheme X is quasi-compact (resp. quasi-separated, resp. coherent). Likewise, a morphism of the site of étalé spaces over X (resp. of the site X_{et} of étale schemes over X, resp. of the site X_{fppf} of schemes locally of finite presentation over X) is quasi-compact (resp. quasi-separated, resp. coherent) if and only if it is a morphism of schemes that is, in the usual sense of *loc. cit.*, quasi-compact (resp. quasi-separated, resp. coherent).

THEOREM 1.23. *Let E be a $\mathscr{U}$-topos and X an object of E.*

(i) *For X to be quasi-compact, it is necessary and sufficient that, for every filtered inductive system $(Y_i)_{i\in I}$ of E, with $I \in \mathscr{U}$, the natural map*

$$\varinjlim \mathrm{Hom}(X, Y_i) \longrightarrow \mathrm{Hom}(X, \varinjlim Y_i) \tag{1.23.1}$$

be injective.

(ii) *Suppose that E admits a generating subcategory C consisting of quasi-compact objects. A necessary condition for X to be coherent is that, for every datum as in* (i), *the map* (1.23.1) *be bijective.*

186 (i) Necessity. Suppose that X is quasi-compact. We must prove that if $i \in I$ and $f_i, g_i : X \rightrightarrows Y_i$ are such that the composites f, g of f_i, g_i with $Y_i \to Y = \varinjlim Y_j$ are equal, then there exists $j \geq i$ such that the composites f_j, g_j with $Y_i \to Y_j$ are already equal. Since filtered inductive limits in E commute with finite projective limits, and in particular with kernels, we have

$$\mathrm{Ker}(f, g) = \varinjlim_{j \geq l} \mathrm{Ker}(f_j, g_j).$$

By hypothesis $\mathrm{Ker}(f, g) = X$, so X is the limit of the filtered increasing family of its subobjects $\mathrm{Ker}(f_j, g_j)$. Since X is quasi-compact, it is equal to one of these subobjects; hence there exists $j \geq i$ such that $f_j = g_j$.

Sufficiency. To prove that X is quasi-compact, it is equivalent (1.5.2) to prove that, for every filtered increasing family $(X_i)_{i \in I}$ of subobjects of X whose $\varinjlim$ is X, there exists $i \in I$ such that $X_i = X$. Since every monomorphism in E is effective (II 4.0), it follows that

$$X_i = \mathrm{Ker}(f_i, g_i : X \rightrightarrows X \amalg_{X_i} X).$$

Let $Y_i = X \amalg_{X_i} X$, and consider the inductive system formed by the Y_i. If Y is its inductive limit, then $Y \simeq X \amalg_{\varinjlim X_i} X = X \amalg_X X = X$. Consequently, for a fixed $i \in I$, f_i, g_i give the same element of $\mathrm{Hom}(X, Y)$, so by hypothesis there exists $j \geq i$ such that $f_j = g_j$, i.e. $X_j = X$, C.Q.F.D.

187 (ii) It remains to prove that, under the stated conditions, every morphism $X \to Y$ comes from a morphism $X \to Y_i$ for a suitable $i \in I$. Since C generates E and the Y_i cover Y, we may cover X by objects X_α of C in such a way that each composite $X_\alpha \to X \to Y$ factors through one of the Y_i. Since X is quasi-compact, we may suppose that the family of the X_α is finite, and hence that the morphisms $X_\alpha \to Y$ factor through a fixed Y_i as morphisms $g_{\alpha i} : X_\alpha \to Y_i$. Since X is quasi-separated, the $X_\alpha \times_X X_\beta$ are quasi-compact. Moreover, for every (α, β), the composites of pr_1 and pr_2 on $X_\alpha \times_X X_\beta$ with $X_\alpha, X_\beta \to Y_i$ are

$$\begin{tikzcd}
X_\alpha \times_X X_\beta \arrow[dd] \arrow[dr] & & & \\
& X_\alpha \arrow[rr] \arrow[dl] & & Y_i \arrow[dl] \\
X \arrow[rr] & & Y &
\end{tikzcd}$$

such that their composites with $Y_i \to Y$ are equal. By (i), there therefore exists $j \geq i$ such that the composites with $Y_i \to Y_j$ are already equal. Since the set of pairs (α, β) is finite, j may be chosen independently of the pair. Consequently, the morphisms $g_{\alpha j}$ arise by composition from a morphism $g_j : X \to Y_j$, whose composite with $Y_j \to Y$ is moreover the given morphism $X \to Y$, since this is so after composition with the covering maps $X_\alpha \to X$. This completes the proof.

Remarks 1.23.2. The hypothesis on E in (ii) may be replaced by an additional hypothesis *on* X, namely that X admits a fundamental system of covering families by quasi-compact objects.

188 Corollary 1.24. *Let E be a topos, and let C be a strictly full subcategory of E consist-*

ing of coherent objects. Using the fact that the $\mathcal{U}$-inductive limits in E are representable, one obtains (I 8.6.1) *a canonical functor*

$$\mathrm{Ind}(C) \longrightarrow E. \tag{1.24.1}$$

This functor is fully faithful. If E admits a generating subcategory consisting of coherent objects, and if C is stable under direct factors, then the following conditions are equivalent:

(i) *The functor* (1.24.1) *is essentially surjective, i.e. it is an equivalence of categories.*
(ii) *C is a generating subcategory of E, stable under finite inductive limits.*
(iii) *The category C is equal to the full subcategory E_{coh} of E consisting of all coherent objects of E, and the necessary condition* 1.23 (ii) *for an object of E to be coherent is also sufficient.*

Let E_{PF} be the full subcategory of E defined in I 8.7.6. By I 8.7.5 a), statement 1.23 (ii) is equivalent to the second inclusion in

$$C \subset E_{\mathrm{coh}} \subset E_{PF},$$

and the composite inclusion $C \supset E_{PF}$ means that (1.24.1) is fully faithful. Since , by hypothesis, the subcategory E_{coh} of E is generating, the same is a fortiori true of E_{PF}; moreover, finite projective limits are representable in E, so that we may apply I 8.7.7. Condition (iii) of 1.24 may also be written $C = E_{\mathrm{coh}} = E_{PF}$ and is equivalent to the condition $C = E_{PF}$, which is precisely condition (ii bis) of *loc. cit.* (taking into account that, by hypothesis, E_{coh} and hence E_{PF} is a generating subcategory of E). Likewise, condition (ii) of 1.24 is precisely condition (iii bis) of *loc. cit.*. Thus the equivalence of conditions (i), (ii bis), and (iii bis) in *loc. cit.* gives the equivalence of conditions (i), (ii), and (iii) of 1.24, C.Q.F.D. 189

The proof (or, more precisely, the invocation of I 8.7.7 (ii bis)) also gives part a) of the

Corollary 1.24.2. *Let E be a topos admitting a generating subfamily consisting of coherent objects, and let E_{PF} be the strictly full subcategory of objects X of E such that the covariant functor* $\mathrm{Hom}(X, -)$ *represented by X commutes with filtered inductive limits. Then:*

a) *The natural functor*

$$\mathrm{Ind}(E_{PF}) \longrightarrow E$$

is an equivalence of categories.

b) *For an object X of E to belong to E_{PF}, it is necessary and sufficient that it be isomorphic to a direct factor (= image of a projector (I, 10.6)) of the cokernel of a double arrow $X_1 \rightrightarrows X_0$, with X_1 and X_0 coherent.* 190

c) *The subcategory E_{PF} of E is stable under finite inductive limits, and is equivalent to a small category.*

It remains to prove b) and c). The first assertion of c) is immediate from I 2.8, and implies the "if" part of b) by 1.23 (ii). For the "only if" part, use 1.23 (i), which shows that X is quasi-compact, whence the existence of an epimorphism $X_0 \to X$, with X_0 coherent, in view of the fact that E_{coh} generates E and is stable under finite sums. Let $R = X_0 \times_X X_0$, so that X identifies with the cokernel of $R \rightrightarrows X_0$. By a), we have $R = \varinjlim R_i$, the filtered inductive limit of the coker $(R_i \rightrightarrows X_0) = X_i$. By the hypothesis $X \in \mathrm{Ob}\, E_{PF}$, for sufficiently large i the morphism $u_i : X_i' \to X$ admits a section $v : X \to X_i'$. Let $w = vu_i : X_i' \to X_i'$, so that $w^2 = w$ and $X \simeq \mathrm{Coker}(w, \mathrm{id}_{X_i'} : X_i' \rightrightarrows X_i')$. Since R_i is again quasi-compact, cover it by $X_1 \to R_i$, with X_i coherent, whence $X_i \simeq \mathrm{Coker}(X_1 \rightrightarrows X_0)$, which proves b).

Finally, since $E_{PF} \subset E_{\text{qu.cpct}}$, the last assertion of c) follows from 1.5.3.

COROLLARY 1.25. *Let E be a topos such that the full subcategory C of E consisting of coherent objects is generating. Consider the canonical functor*

$$\varphi : \mathrm{Ind}(c) \longrightarrow E.$$

191 *The following conditions are equivalent:*

(i) *The functor φ is essentially surjective, i.e. every object of E is a filtered inductive limit of coherent objects.*

(i bis) *The functor φ is an equivalence of categories.*

(ii) *Every finite inductive limit in E of coherent objects is a coherent object.*

(ii bis) *The cokernel in E of a double arrow of coherent objects is a coherent object.*

(iii) *The converse of* 1.23 (ii) *holds.*

This is a special case of 1.24, taking into account that C is stable under finite sums (1.4), which implies the equivalence of (ii) and (ii bis), and taking into account the

LEMMA 1.25.1. *Let E be a topos. Every direct factor X' of a coherent object X of E is coherent.*

Indeed, X' is quasi-compact as a quotient of X (1.3), and quasi-separated as a subobject of X (1.15).

COROLLARY 1.26. *Let E be a topos satisfying the equivalent conditions* 1.25. *Then every induced topos does likewise.*

This follows immediately from criterion (ii), taking into account 1.20.2 and the fact that the existence of a generating subcategory of E consisting of coherent objects plainly implies the same property for an induced topos.

192 REMARK 1.27. Among the important examples of topoi satisfying the conditions of 1.25, let us mention Noetherian topoi (2.14), and the Zariski or étale topos (VII 1) of a scheme X (cf. IX, note p.42). But even when the topos E is *coherent* (2.3) (i.e. the full subcategory C of coherent objects is generating and stable under finite projective limits), it is not always true that C is *perfect* (2.9.1), i.e. satisfies the equivalent conditions of 1.25; cf. 1.28 below.

EXERCISE 1.28. Let E be a topos and $p_1, p_2 : X_1 \rightrightarrows X_0$ a double arrow in E. Define an increasing sequence of subobjects R_n ($n \geq 0$) of $X_0 \times X_0$ by the condition $R_0 = \mathrm{Sup}(\mathrm{Im}(p_1, p_2), \mathrm{Im}(p_2, p_1))$, and $R_n = \mathrm{pr}_{13}(\mathrm{pr}_{12}^{-1}(R_{n-1}), \mathrm{pr}_{23}^{-1}(R_{n-1}))$ for every $n \geq 1$, where pr_{ij} ($1 \leq i < j \leq 3$) denote the evident projections from the triple product of X_0 to the double product.

a) Show that one thus obtains an increasing sequence of subobjects of $X_0 \times X_0$, that $R = \varinjlim R_n$ is an equivalence graph (I 10.4) in X_0, and that the canonical morphism

$$X = \mathrm{Coker}(p_1, p_2) \longrightarrow X_0/R$$

is an isomorphism.

b) Deduce that if X_0 is quasi-compact and $X = \mathrm{Coker}(p_1, p_2)$ is quasi-separated
193 (hence coherent), then the sequence (R_n) is stationary. Conversely, if this sequence is stationary, and if X_0 is assumed coherent and X_1 quasi-compact, then X is coherent. (For this last point, write $R_n \simeq (R_{n-1}, \mathrm{pr}_2) \times_{X_0} (R_{n-1}, \mathrm{pr}_1)$ and deduce that R_n is quasi-compact over X_0 for every n, then use 1.19.)

c) Construct an example of two maps of sets $p_1, p_2 : X_1 \rightrightarrows X_0$ such that the sequence (R_n) is not stationary.
d) Let C be a small category in which finite inductive limits and finite projective limits are representable, and in which every morphism factors as an effective epimorphism followed by a monomorphism. Repeat the construction of the R_n for a double arrow (p_1, p_2) of C. Endowing C with the canonical topology, show that the canonical functor $\epsilon : C \to C^\sim$ is exact, and consequently commutes with the formation of the R_n.
e) Take for C a small full subcategory of (Ens), stable up to isomorphism under finite projective limits and finite inductive limits, and containing the sets X_0, X_1 of c). Deduce that the topos $C^\sim$ (which is a *coherent* topos (2.3), i.e. is generated by the subcategory of its coherent objects, which is stable under finite projective limits) does not satisfy the equivalent conditions of 1.25.
f) Let k be a field, and let C be the fppf site of schemes locally of finite presentation over k (SGA 3 IV 6.3). Show that there exists a reduced affine scheme X over k admitting an automorphism f that is not of finite order (take, for example, the affine plane E^2 endowed with the automorphism $f(x) = x$, $f(y) = x+y$). Show 194
that if $p_1, p_2 : X_1 \rightrightarrows X_0$ is a double arrow in C such that the corresponding double arrow in (Ens)$p_1(\overline{k}) : p_2(\overline{k}) : X_1(\overline{k}) \rightrightarrows X_0(\overline{k})$ gives a nonstationary sequence (R_n), then the same is true of the double arrow of $C^\sim$ defined by p_1, p_2. Thus, if X_0 is of finite type over k, the cokernel in $C^\sim$ of (p_1, p_2) is not coherent. Deduce that the cokernel in $C^\sim$ of the pair (f, id_X) is not coherent, and hence that the topos $C^\sim$ does not satisfy the equivalent conditions of 1.25. Deduce more generally that if S is a nonempty scheme and C is the fppf site of schemes locally of finite presentation over S, then the topos $C^\sim$ does not satisfy the conditions of 1.25. (Care must be taken that, contrary to appearances, $C^\sim$ is therefore Noetherian (2.11) only if S is empty, as follows from the preceding discussion and 2.14.)

Definition 1.30. *Let E be a topos. An object X of E is called a* pre-Noetherian object[34] *of the topos E if it satisfies the following two equivalent conditions:*

(i) *Every subobject of X is quasi-compact.*
(ii) *Every increasing sequence of subobjects of X is stationary.*

1.30.1. The equivalence of conditions (i) and (ii) is clear. Note that these conditions again depend only on the induced topos $E_{/X}$, and indeed only on the ordered set of opens of $E_{/X}$, which is isomorphic to the ordered set of subobjects of X. They are stable under passage to a subobject of X.

1.31. One proves, as in 1.3 and 1.4, the following facts: if $(X_i \to X)$ is a finite 195
covering family, with the X_i pre-Noetherian, then X is pre-Noetherian; in particular, a quotient of a pre-Noetherian object of E is pre-Noetherian. If $(X_i)_{i\in I}$ is a family of objects of E, then their coproduct X is a pre-Noetherian object of E if and only if all but finitely many of the X_i are isomorphic to $\varnothing_E$, and all the X_i are pre-Noetherian.

1.32. Clearly, a pre-Noetherian object of a topos E is quasi-compact. When E admits a generating family formed by pre-Noetherian objects, the converse holds (as follows immediately from 1.31): the pre-Noetherian objects of E are then precisely its quasi-compact objects.

[34] The stronger notion of a *Noetherian* object is given in 2.11 below.

Suppose that E admits a generating family $(X_i)_{i \in I}$ formed by quasi-compact objects. Then the following three conditions are equivalent:

(i) The X_i are pre-Noetherian.
(ii) Every quasi-compact object of E is pre-Noetherian.
(iii) Every monomorphism in E is quasi-compact.

(We have just proved the equivalence of (i) and (ii); (ii) ⇒ (iii) is immediate from the definitions, and (iii) ⇒ (ii) follows from Definition 1.30 (i).)

196 Note that (iii) implies that *every morphism of E is quasi-separated*; hence (1.14 (ii)) every object of E lying over a quasi-separated object of E is itself quasi-separated.

Example 1.33. (*Classifying topos of a group*). Let G be a group $\in \mathscr{U}$, and let B_G be its classifying topos (IV 2.4), i.e. the category of sets $\in \mathscr{U}$ on which G acts on the left. The nonempty connected objects of B_G (IV 4.3.5) are the nonempty G-sets on which G acts transitively, and every object E of B_G is the coproduct of its connected components, which are the minimal nonempty subobjects of E (namely, the orbits of G in E). Consequently, E is a quasi-compact object of B_G if and only if the set $\pi_0(E)$ of orbits of G in E is finite, and in that case E is even a pre-Noetherian object of B_G. These objects (and even the single object G_s, which is nonempty and connected) form a generating subcategory of B_G. The final object e of B_G is quasi-separated, i.e. the product of two quasi-compact objects of B_G is quasi-compact, if and only if the product $G_s \times G_s$ is quasi-compact, which is plainly equivalent to G being finite. More generally, one concludes that an object E of B_G is quasi-separated if and only if the stabilizers of its points are *finite* subgroups of G, i.e. if it is isomorphic to a coproduct of homogeneous spaces G/H, with H *finite*. Indeed, by 1.15, one may assume in this criterion that E is
197 connected; but if $x \in E$ has stabilizer group G_x, we know (IV 5.8) that B_G/E is equivalent to the topos B_{G_x}, and hence its final object is quasi-separated if and only if G_x is finite. This criterion shows that every object of B_G lying over a quasi-separated object is quasi-separated; a fortiori, the product of two quasi-separated objects of B_G is quasi-separated. It also shows that the coherent objects of B_G are the objects isomorphic to finite coproducts of homogeneous spaces G/H, with H *finite*. Since G_s is coherent, the coherent objects of B_G already form a generating subcategory of B_G. Moreover, this subcategory is stable under fiber products (as follows from the fact that every object lying over a quasi-separated object is quasi-separated). (In the terminology of 2.11 below, B_G is a *locally Noetherian* topos, but it is quasi-separated only when G is finite, in which case B_G is even Noetherian.)

It is immediate that a morphism $f : E' \to E$ is quasi-compact if and only if the induced map $\pi_0(f) : \pi_0(E') \to \pi_0(E)$ on the sets of orbits is proper, i.e. has finite fibers. We recover the fact that a monomorphism is quasi-compact (1.32), and hence that every morphism is quasi-separated. Thus the coherent morphisms in B_G are simply the quasi-compact morphisms, which have just been characterized.

Let E be an object of the topos B_G. One verifies easily that E is a filtered inductive limit of coherent objects of B_G if and only if the stabilizers of the points of E are ind-
198 finite groups (i.e. filtered inductive limits of their finite subgroups). In particular, if G is not ind-finite (for example, if $G = \mathbf{Z}$), the final object of the topos B_G is not an inductive limit of coherent objects. More precisely, the topos B_G satisfies the equivalent conditions of 1.25 if and only if its final object is a filtered inductive limit of coherent objects, or equivalently if the group G is ind-finite.

2. Finiteness conditions for a topos

PROPOSITION 2.1. *Let E be a $\mathcal{U}$-topos. The following conditions are equivalent:*

(i) *There exists a full generating subcategory C of E, formed by quasi-compact objects and stable under fiber products.*

(ii) *There exists a $\mathcal{U}$-site C such that every object of C is quasi-compact, fiber products are representable in C, and E is equivalent to the category of sheaves $C^{\sim}$.*

Suppose that these conditions hold. Then in (i) (resp. (ii)) *the category C may even be chosen to be $\mathcal{U}$-small. Moreover, for every $X \in \operatorname{ob} C$, the object X* (resp. $\epsilon(X)$)*of the topos E is* coherent.

The implication (i) ⇒ (ii) follows from IV 1.2.1, and the fact that in (i) (resp. (ii)) one may take C to be $\mathcal{U}$-small follows immediately from the argument of IV 1.2.3, applied to a small full generating subcategory in the topological sense (II 3.0.1) of C. It remains to prove (ii) ⇒ (i) and the last assertion of 2.1; these follow immediately from the following result.

COROLLARY 2.1.1. *Let C be a $\mathcal{U}$-site in which fiber products are representable and every object is quasi-compact, let $E = C^{\sim}$, and let $\varepsilon : C \to E$ be the canonical functor. Then, for every $X \in \operatorname{ob} C$, $\varepsilon(X)$ is a coherent object of E. The full subcategory of E formed by the coherent objects of E lying over some $\varepsilon(X)$ is stable under fiber products (and is plainly generating in E).* 199

We already know that $\epsilon(X)$ is quasi-compact (1.2); let us prove that it is quasi-separated. Let F, G be two quasi-compact objects of E, and let $F \to \varepsilon(X)$, $G \to \epsilon(X)$ be two morphisms. We must prove that the fiber product $F \times_{\epsilon(X)} G$ is quasi-compact. Covering F and G by finitely many objects of the form $\epsilon(Y_i)$ and, respectively, $\epsilon(Z_j)$, and proceeding as in 1.2 to reduce to the case in which the composites $\epsilon(Y_i) \to \epsilon(X)$ and $\epsilon(Z_j) \to \epsilon(X)$ come from morphisms $Y_i \to X$ and, respectively, $Z_j \to X$, we are reduced by 1.3 to the case in which the morphisms $F \to \epsilon(X)$ and $G \to \epsilon(X)$ are obtained by applying ϵ to morphisms $Y \to X$ and $Z \to X$. But since fiber products are representable in C and ϵ commutes with them, we are reduced to proving that $\epsilon(Y \times_X Z)$ is quasi-compact, which was already noted above for every object of the form $\epsilon(U)$.

Now let $G \to F$, $H \to F$ be morphisms between coherent objects of E, and suppose that there exists a morphism $F \to \epsilon(X)$ for some $X \in \operatorname{ob} C$. We prove that $G \times_F H$ is coherent. Since G and H are quasi-compact and F is quasi-separated, it already follows from the definitions that the fiber product is quasi-compact. It remains to see that it is 200 quasi-separated, or, equivalently (1.15), that $G \times_{\epsilon(X)} H$ is quasi-separated. This follows from the more general fact:

COROLLARY 2.1.2. *Under the conditions of 2.1.1, for every quasi-separated object G of E, every morphism $G \to \epsilon(X)$ is quasi-separated. If $H \to \epsilon(X)$ is a second morphism, with H quasi-separated, then $G \times_{\epsilon(X)} H$ is quasi-separated.*

The first assertion implies the second: by base change, it implies that $G \times_{\epsilon(X)} H \to H$ is quasi-separated, and since H is quasi-separated, $G \times_{\epsilon(X)} H$ is likewise quasi-separated (1.14 (ii)). To prove that $G \to \epsilon(X)$ is quasi-separated, we shall use the following lemma.

LEMMA 2.1.3. *Let E be a topos and $f : X \to Y$ a morphism in E. Suppose that E admits a generating family of quasi-compact objects, and that there exists a covering family $(g_i : X_i \to X)_{i \in I}$ of X such that the X_i are quasi-compact and, for every pair of indices $i, j \in I$, $X_i \times_Y X_j$ is quasi-separated. Then, if X is a quasi-separated object of E, f is a quasi-separated morphism.*

Indeed, the family $(g_i \times_Y g_j : X_i \times_Y X_j \to X \times_Y X)_{(i,j)\in I\times I}$ is covering. Therefore, to see that the morphism $\Delta_{X/Y} : X \to X \times_Y X$ is quasi-compact, it is enough to see that,
201 for every $(i, j) \in I \times I$, the morphism obtained from it by base change along $g_i \times_Y g_j$ is quasi-compact (1.10 (ii)). The latter is precisely the canonical morphism $X_i \times_X X_j \to X_i \times_Y X_j$. By hypothesis X is quasi-separated and the X_i are quasi-compact, so $X_i \times_X X_j$ is quasi-compact; and since $X_i \times_Y X_j$ is quasi-separated by hypothesis, the morphism $X_i \times_X X_j \to X_i \times_Y X_j$ is indeed quasi-compact, C.Q.F.D.

We can now complete the proof of 2.1.2 by proving that every morphism $f : G \to \epsilon(X)$, with G quasi-separated, is quasi-separated. To do this, cover G by objects $\epsilon(X_i)$ such that the composite morphisms $\epsilon(X_i) \to \epsilon(X)$ come from morphisms $X_i \to X$. By 2.1.3, since the $\epsilon(X_i)$ are quasi-compact, it is enough to verify that the fiber products $\epsilon(X_i) \times_{\epsilon(X)} \epsilon(X_j) = \epsilon(X_i \times_X X_j)$ are quasi-separated. We have already seen that every object of E of the form $\epsilon(U)$ is coherent. This proves 2.1.2 and completes the proof of 2.1.

Proposition 2.2. *Let E be a topos containing a full generating subcategory C formed by coherent objects. The following conditions on E are equivalent:*

(i) *Every quasi-separated object of E is quasi-separated over the final object.*

(i bis) *For every morphism $f : X \to Y$ in E, if X is quasi-separated, then f is quasi-separated.*

(i ter) *Every object X of C is quasi-separated over the final object of E, i.e. the diagonal morphism $X \to X \times X$ is quasi-compact.*

202 (ii) *The product of two quasi-separated objects of E is quasi-separated.*

(ii bis) *The product in E of two objects of C is quasi-separated.*

These conditions imply that the full subcategory E_{coh} of E formed by the coherent objects of E is stable under fiber products (and, a fortiori, that E satisfies the equivalent conditions of 2.1).

Clearly (i bis) ⇒ (i), and the converse follows from 1.8 (iii). We have (i) ⇒ (ii) by an argument already used: if X and Y are quasi-separated, then by hypothesis (i), X is quasi-separated over e, so $X \times Y$ is quasi-separated over Y by base change, and hence $X \times Y$ is quasi-separated by 1.14 (ii). The same argument shows that (i ter) ⇒ (ii bis), while (i) ⇒ (i ter) and (ii) ⇒ (ii bis) are immediate. It therefore remains to prove (ii bis) ⇒ (i). If X is a quasi-separated object of E, cover X by objects X_i of C (which is possible since C is generating). Since the X_i are quasi-compact by the hypothesis on C, we may apply 2.1.3 to the morphism $X \to e$; thus, to prove that this morphism is quasi-separated, we are reduced precisely to verifying that the $X_i \times X_j$ are quasi-separated, which is hypothesis (ii bis).

Remark 2.2.1. Condition (i ter) is also equivalent to the following condition (i quater): for every double arrow $g_1, g_2 : Y \rightrightarrows X$ in C, its kernel in E is quasi-compact (or equivalently (1.15) coherent).

203 To see this, it is enough to apply criterion 1.10 (i) to the diagonal morphism $X \to X \times X$ considered in (i ter).

Definition 2.3. *A topos E satisfying the equivalent conditions of 2.2* (resp. 2.1) *is called an* algebraic topos (resp. *a* locally algebraic topos, *or* locally coherent). *An object X of a topos E is called an* algebraic object of the topos E *if the induced topos $E_{/X}$ is algebraic. A topos E is called* quasi-separated (resp. coherent) *if it is algebraic and its final object is quasi-separated (resp. coherent).*

2.3.1. It is in algebraic topoi that the finiteness notions developed in n° 1 have fully satisfactory properties, analogous to those of the notions bearing the same names in the category of schemes. All locally algebraic (= locally coherent) topoi encountered in practice are in fact algebraic, so the practical interest of this notion seems rather limited for the present. Nevertheless, one can construct locally algebraic topoi that are not algebraic (2.17 f).

2.4. Here are some general remarks concerning the notions introduced in 2.3, which will convince the reader (or so we hope) that the terminology introduced there is coherent.

2.4.1. Under the conditions of 2.1 (i) resp. (ii), for every $X \in C$, X (resp. $\epsilon(X)$) is an algebraic (coherent) object of E (2.1.2 and 1.19.2); in other words, the induced topos **204**
$E_{/X}$ (resp. $E_{\epsilon(X)} \approx (C_{/X})^{\sim}$) is algebraic, hence *coherent*. Equivalently, *when C admits a final object, E itself is algebraic, hence coherent.*

2.4.2. If E is an algebraic topos, then every induced topos $E_{/X}$ is also algebraic (2.2 (i bis)). For a topos E to be locally algebraic = locally coherent, it is necessary and sufficient that there exist a family $(X_i)_{i\in I}$ of objects of E covering the final object, such that the X_i are algebraic (resp. coherent), i.e. such that the induced topoi $E_{/X_i}$ are algebraic (resp. coherent). Sufficiency follows immediately from the definitions, and necessity from the observations in 2.4.1. Of course, an algebraic topos is locally algebraic, and if E is a locally algebraic topos, every induced topos $E_{/X}$ is locally algebraic.

2.4.3. If X is an algebraic object of a topos E, then every object X' of E lying over X is again algebraic (cf. 2.4.2). In particular, every object of an algebraic topos is algebraic, and conversely, of course.

2.4.4. The full subcategory of E consisting of the *quasi-separated* objects that are *algebraic* is stable under fiber products and subobjects (1.15), and hence also under kernels. If E is algebraic, this category (which is then simply the category of quasi-separated **205**
objects of E) is also stable under products of two objects. If E is quasi-separated (and only then), this category is stable under arbitrary finite projective limits, or equivalently contains the final object of E.

The full subcategory C of E consisting of the *coherent* objects that are *algebraic* is stable under fiber products. If E is locally algebraic, i.e. locally coherent, saying that E is algebraic amounts to saying that C is also stable under kernels of double arrows (2.2.1); and saying that E is quasi-separated amounts to saying that, in addition, C is stable under products of two objects (cf. 1.17). Finally, saying that E is coherent amounts to saying that C is stable under arbitrary finite projective limits, or equivalently that it contains the final object of E.

2.4.5. Let E be a topos. For E to be locally coherent (resp. algebraic, resp. quasi-separated, resp. coherent), it is necessary and sufficient that it admit a full generating subcategory C consisting of quasi-compact objects (which will automatically be coherent (2.1)) and *stable* under fiber products (resp. under fiber products and kernels, resp. under fiber products and products of two objects, resp. under arbitrary finite projective limits); or equivalently that E be equivalent to a topos $C^{\sim}$, where C is a $\mathscr{U}$-site every object of which is quasi-compact, and in which fiber products and products of two objects (resp. all finite projective limits) are representable. (For necessity, take for C the **206**
category of objects of E that are coherent and *algebraic*, and apply 2.4; for sufficiency, apply 2.2 (i ter).) In this statement, one may of course also suppose that C is $\mathscr{U}$-small.

2.4.6. Let E be a topos. If E is algebraic, an object X of E is quasi-separated (resp. coherent) if and only if the induced topos $E_{/X}$ is so. When E is no longer assumed algebraic, one can only say that $E_{/X}$ is quasi-separated (resp. coherent) if and only if X is quasi-separated (resp. coherent) and *algebraic*. This is of little consequence, since we will probably not have to use these notions outside the case in which E is algebraic.

2.4.7. For the notion of a coherent (resp. quasi-separated) object of a topos E to have satisfactory properties, one must restrict to objects that are also algebraic, a condition automatically satisfied if E is algebraic. Since we will probably never have to work with locally algebraic topoi E that are not algebraic, the question of revising the terminology introduced in 1.13, possibly reserving it for algebraic objects alone, therefore does not arise with any great urgency!

Moreover, we have refrained from defining the notion of a quasi-compact topos. In the spirit of the present paragraph, this name would naturally be given to *algebraic* topoi whose final object is quasi-compact.

207 We hesitated to introduce such a usage, since if X is a topological space, it would not be true that X is quasi-compact if and only if the associated topos $\mathrm{Top}(X)$ is so. In this connection, note that $\mathrm{Top}(X)$ is locally coherent if and only if X admits a basis of opens consisting of quasi-compact opens U such that, for $U_j, U_k \subset U_i$, $U_j \cap U_k$ is again quasi-compact (and then $\mathrm{Top}(X)$ is even algebraic). This condition is satisfied for the topological spaces underlying schemes, but rarely for spaces encountered by analysts, even when the spaces themselves are compact. For ease of reference, let us state explicitly:

Proposition 2.5. *Let E be an algebraic topos and $f : X \to Y$ a morphism in E. If Y is a quasi-separated (resp. coherent) object of E, then, for X to be a quasi-separated (resp. coherent) object of E, it is necessary and sufficient that the morphism f be quasi-separated (resp. coherent).*

Sufficiency was already seen in 1.14 (ii), and holds without any hypothesis on E. Necessity in the non-parenthetical case holds even without supposing that Y is quasi-separated, and is simply Definition 2.3, via 2.2 i bis). It remains to prove that if Y is coherent and X is coherent, then f is coherent. Since we already know that f is quasi-separated, it is enough to see that f is quasi-compact, which follows from the fact that X is quasi-compact and Y quasi-separated (1.13).

208 Corollary 2.6. *Let E be an algebraic topos, $f : X \to Y$ a morphism in E, and $(Y_i \to Y)_{i\in I}$ a covering family, with the Y_i quasi-separated (resp. coherent). For f to be quasi-separated (resp. quasi-compact, resp. coherent), it is necessary and sufficient that, for every $i \in I$, $X_i = X \times_Y Y_i$ be a quasi-separated (resp. quasi-compact, resp. coherent) object of E.*

It is enough to combine 1.10 (ii) and 2.5 in the "quasi-separated" or "coherent" case; the "quasi-compact" case was already seen in 1.16. Note that the necessity of the condition stated in 2.6 also holds without any hypothesis on the topos E.

As a special case of 2.6, one obtains:

Corollary 2.7. *Let E be a coherent topos and X an object of E. For X to be a quasi-compact (resp. quasi-separated, resp. coherent) object of E, it is necessary and sufficient that it be quasi-compact (resp. quasi-separated, resp. coherent) over the final object (1.9.2).*

In particular, X is constructible (1.9.2) if and only if it is coherent.

Corollary 2.8. *Let E be a locally coherent topos and $f : X \to Y$ a morphism in E. For f to be quasi-separated, it is necessary and sufficient that, for every quasi-separated* *object Y' of $E_{/Y}$, $f^*(Y') = X \times_Y Y'$ be a quasi-separated object of $E_{/X}$ (or of E, which amounts to the same thing (1.20.2)).*

The "only if" part holds without any condition on E, since if Y' is quasi-separated, then $f' : X' = X \times_Y Y' \to Y'$ is quasi-separated by base change, and hence X' is quasi-separated (1.14 (ii)). For the converse, note that there exists a covering family $Y_i \to Y$ of Y by quasi-separated algebraic objects. By 1.10 (ii), to verify that f is quasi-separated it suffices to verify that the $f_i : X_i = X \times_Y X_i \to Y_i$ are so. But the hypothesis on f^* implies that the X_i are quasi-separated, and therefore so are the f_i, since $E_{/Y_i}$ is an algebraic topos. **209**

Remark 2.8.1. Statements 2.5 and 2.6 remain valid if the hypothesis "E algebraic" is replaced by the more general hypothesis "Y is algebraic", as is seen immediately by applying the original statement to the induced topos $E_{/Y}$, which is algebraic. Likewise, in 2.8 it is enough to suppose that $E_{/Y}$ (rather than E) is locally coherent.

Proposition 2.9. *Let E be a topos and C a subcategory of E. The following conditions are equivalent:*

- (i) *E is locally coherent (resp. coherent), satisfies the equivalent conditions of 1.25, and C is the full subcategory E_{coh} of E consisting of the coherent objects.*
- (ii) *C is a strictly full generating subcategory of E, stable under finite inductive limits and fiber products (resp. and under finite projective limits), and consists of quasi-compact objects.*

The implication (i) ⇒ (ii) follows immediately from the definitions and from formulation 1.25 (ii) of the conditions under consideration in 1.25. Let us prove the converse. The fact that C is a full generating subcategory consisting of quasi-compact objects and stable under fiber products (resp. under finite projective limits) implies, by definition, that E is locally coherent (resp. coherent). The additional fact that C is strictly full and stable under finite inductive limits then implies that $C = E_{\mathrm{coh}}$ (by 1.24 (ii) ⇒ (iii)), and that condition 1.25 (ii) is satisfied, C.Q.F.D. **210**

Definition 2.9.1. *A $\mathscr{U}$-topos E is called perfect if it is coherent (2.3) and satisfies the equivalent conditions of 1.25, i.e. if the subcategory E_{coh} of E consisting of the coherent objects of E is stable under finite inductive limits. We say that E is locally perfect if there exists a family $(X_i)_{i\in I}$ of objects of E covering the final object such that, for every $i \in I$, the induced topos $E_{/X_i}$ is perfect.*

2.9.2. It follows immediately from this definition that a perfect (resp. locally perfect) topos is coherent (resp. locally coherent). As examples of perfect (resp. locally perfect) topoi, let us mention the Noetherian (resp. locally Noetherian) topoi introduced in 2.11 below (cf. 2.14), and the Zariski and étale topoi of a coherent scheme (resp. an arbitrary scheme) (IX, note page 42). As a counterexample, let us mention the fppf topos **211**
of a scheme S, which is locally coherent (and even coherent if S is a coherent scheme, i.e. quasi-compact and quasi-separated) (VII 5.6), but is not locally perfect if $S \neq \varnothing$ (1.28 f)).

In a locally perfect topos, the notion of constructible object is particularly stable (2.9.3 below), which is one reason why the notion seems useful. Another reason is that, like the notion of a coherent or locally coherent topos, the notion of a perfect or locally perfect topos is stable under formation of the closed subtopos complementary to an open (4.6), and behaves particularly simply under the operation of gluing topoi (4.11, 4.14).

Proposition 2.9.3. *Let E be a locally perfect topos (2.9.1). Then the full subcategory E_{cons} of E consisting of the constructible objects of E (1.9.3) is stable under finite inductive limits (and also, of course, under finite projective limits, by (1.9.3)).*

Since the property of an object of E being constructible is local on E (1.10 (ii)), we are reduced to the case in which E is a perfect topos. But then $E_{\mathrm{cons}} = E_{\mathrm{coh}}$ (2.7), and the conclusion follows from Definition 2.9.1.

212 Corollary 2.9.4. *Let E be a locally perfect topos. Then E is perfect if and only if E is coherent. For every object X of E, the induced topos $E_{/X}$ is locally perfect.*

Problem 2.9.5. It is conceivable that perfect topoi are sufficiently special to admit a *structure theory* that is equally explicit (following a model proposed by M.Artin at the time of the oral seminar). Let us call a topos *finite* if it is equivalent to a topos of the form $\widehat{C}$, where C is a finite category (cf. Exercise 3.11), and call a topos *profinite* if it is a filtered projective limit of finite topoi (in the sense of 6 below, which applies by 3.12 c)). Since a finite topos is Noetherian (2.17 g)), hence perfect, and a filtered projective limit of perfect topoi is perfect (6), it follows that a profinite topos is perfect. The question is whether, conversely, every perfect topos is profinite. This is equivalent to asking whether every finite subcategory of E_{coh} is contained in a full subcategory F_0 of E_{coh} that is equivalent to a category F_{coh}, where F is a finite topos (cf. 3.11). (If the answer were negative, one would need to find manageable additional intrinsic conditions on a perfect topos that ensure that it is profinite.) Finally, it would remain to study the structure of profinite topoi in terms of a suitable notion of a "Karoubian profinite category", inspired by IV
213 7.6 h). This study has not yet been carried out even in the special case in which E is the Zariski or étale topos of a good coherent scheme X, and should then give invariants finer than the compact space X_{cons} (EGA IV 1.) associated with X.

In the same vein, let us mention the following question: Let E be a perfect topos, E' the topos induced on an open of E, and E'' the corresponding closed subtopos (IV 9.9). If E' and E'' are finite topoi, is the same true of E?

Proposition 2.10. *Let E be a topos. The following conditions are equivalent:*

(i) *E admits a full generating subcategory, stable under fiber products, consisting of pre-Noetherian objects of E.*
(ii) *E admits a full generating subcategory consisting of quasi-separated pre-Noetherian objects (i.e. coherent pre-Noetherian objects).*
(iii) *E is locally coherent (2.3), and every quasi-compact object of E is pre-Noetherian.*
(iii bis) *E is locally coherent and admits a generating family consisting of pre-Noetherian objects of E.*

The equivalence of (iii) and (iii bis) follows from 1.32, and it is trivial that (i) ⇒ (iii bis) and (iii) ⇒ (i); hence (i) is equivalent to (iii) and (iii bis). Finally, (i) ⇒ (ii) follows from the
214 last assertion of 2.1, and it remains to prove (ii) ⇒ (i). This implication follows from the fact that the full subcategory of E formed by the quasi-separated Noetherian objects, if it is generating, is stable under fiber products: such a fiber product is quasi-compact by the definitions, hence Noetherian by the first assertion of 1.32; and it is quasi-separated by the last assertion of 1.32.

Definition 2.11. *A topos E is called locally Noetherian if it satisfies the equivalent conditions of 2.10, and Noetherian if, in addition, its final object is coherent (or, equivalently (1.32), pre-Noetherian and quasi-separated). An object X of a topos E is called a Noetherian object of E if the induced topos $E_{/X}$ is a Noetherian topos.*

2.12. If E is a locally Noetherian topos, then so is every induced topos $E_{/X}$. Conversely, if one can find a family (X_i) covering the final object such that the $E_{/X_i}$ are locally Noetherian, then so is E. If C is a full generating subcategory of E consisting of pre-Noetherian objects and stable under fiber products (2.11 (i)), then for every $X \in C$, $E_{/X}$ is a Noetherian topos, i.e. the objects of C are even *Noetherian* (for, if E is locally Noetherian and $X \in \operatorname{Ob} E$, then $E_{/X}$ is Noetherian if and only if X is coherent). It follows that a topos E is locally Noetherian if and only if the final object of E can be covered by Noetherian objects X_i.

2.13. In a locally Noetherian topos E, every monomorphism is quasi-compact and 215
every morphism is quasi-separated (1.32). A fortiori, E is an *algebraic* topos (2.3), and not merely locally coherent, i.e. locally algebraic. Thus a topos E is Noetherian if and only if it is locally Noetherian and *coherent* (2.3). In other words, if E is a locally Noetherian topos, then the Noetherian objects of E are precisely the *coherent* objects of E.

2.14. Let E be a locally Noetherian topos. If E is quasi-separated, i.e. if its final object is quasi-separated, then *every* object of E is quasi-separated (1.32); hence the pre-Noetherian objects of E are the same as the Noetherian (i.e. coherent) objects of E. Since a quotient object of a pre-Noetherian object is pre-Noetherian, it follows that E satisfies the equivalent conditions of 1.25 (because it satisfies condition 1.25 (ii bis)). In particular, if C is the full subcategory of E consisting of the Noetherian (coherent) objects of E, then the natural functor

$$\operatorname{Ind}(C) \longrightarrow E$$

is an equivalence of categories. We conclude that *a Noetherian topos is perfect* (2.9.1), and hence that a locally Noetherian topos is locally perfect.

If one assumes only that E is locally Noetherian, it remains true that every object of E is the inductive limit of its pre-Noetherian subobjects (for, in a topos admitting a generating subcategory consisting of quasi-compact objects, every object is the filtered 216
inductive limit of its quasi-compact subobjects). But since a pre-Noetherian object of E is no longer necessarily coherent, this statement then has only very limited usefulness, and we know from (1.33) that the conditions of 1.25 are no longer necessarily satisfied.

Example 2.15. (*Topological spaces.*) Let X be a topological space. Then the following conditions are equivalent: (i) the topological space X is Noetherian (i.e. every open subset of X is quasi-compact, i.e. every increasing sequence of open subsets of X is stationary); (ii) the topos $\operatorname{Top}(X)$ is Noetherian; (iii) the final object X of $\operatorname{Top}(X)$ is Noetherian; (iv) the final object X of $\operatorname{Top}(X)$ is pre-Noetherian.

Indeed, trivially (ii) $\Leftrightarrow$ (iii) $\Rightarrow$ (iv) $\Leftrightarrow$ (i). On the other hand, (i) implies that the generating subcategory Ouv(X) of $\operatorname{Top}(X)$, which is stable under finite projective limits, consists of pre-Noetherian objects, whence (ii).

Likewise, one sees that the topological space X is locally Noetherian (i.e. is a union of Noetherian open subsets) if and only if the topos $\operatorname{Top}(X)$ is locally Noetherian.

2.15.1. Thus our Definitions 1.30 and 2.11 agree with the accepted terminology for topological spaces. On the other hand, in the general case we have chosen to give the term "Noetherian object" a stronger meaning than the more naive notion of 1.30, so that 217
all the properties associated with this notion (rather than merely the grammatical structure of the definition) agree fully with the intuition attached to *Noetherian* topological spaces and schemes.

The reader will find further examples in the following exercises.

EXERCISE 2.16. (*Spaces with operators and classifying topoi.*)

Let X be a topological space on which a discrete group G acts, giving (IV 2.3) a topos $E = \mathrm{Top}(X, G)$. The objects X' of this topos are interpreted as étalé spaces over X with operator group G, and we note that the induced topos $E_{/X'}$ is canonically equivalent to $\mathrm{Top}(X', G)$.

a) The final object of the topos E is quasi-compact if and only if the quotient topological space X/G is quasi-compact (use IV 8.4.1).
b) Let P be the object of E defined by the trivial étalé space $X \times G$ with the diagonal action of G. Show that the induced topos $E_{/P}$ is equivalent to the topos $\mathrm{Top}(X)$ (compare IV 5.8.3). Conclude that E is locally coherent (resp. locally Noetherian) if and only if $\mathrm{Top}(X)$ is locally coherent (cf. 2.4.7) (resp. locally Noetherian (cf. 2.15)). Show that if E is locally coherent, i.e. locally algebraic, it is in fact
218 algebraic, using the generating family formed by the objects T_U, where U is a coherent open subset of X, $T_U = G \times U$ with the action $g \cdot (u, g') = (u, gg')$ of G, and the structural morphism $p_U : T_U \to X$ defined by $p_U(u, g) = g \cdot u$.
c) Suppose that E is algebraic, i.e. that $\mathrm{Top}(X)$ is algebraic (2.4.7). Show that the structural morphism $P \to e$ is quasi-separated. Show that E is quasi-separated if and only if $\mathrm{Top}(X)$ is quasi-separated (or, equivalently, the topological space X is quasi-separated, i.e. the intersection of two quasi-compact open subsets of X is a quasi-compact open subset), *and* G is finite or X is empty. (Compare 1.33.)
d) Show that the structural morphism $P \to e$ is quasi-compact if and only if G is finite. Show that E is coherent (resp. Noetherian) if and only if $\mathrm{Top}(X)$ is so, *and* G is finite or X is empty.
e) Let E be a topos and G a group of E, giving a classifying topos B_G (IV 2.4). Study the finiteness conditions in B_G, using what precedes as a guide. Ask the same question starting from a strict pro-group $\mathscr{G} = (G_i)_{i \in I}$ of E. In particular, show that if E is coherent (resp. Noetherian) and the G_i are coherent, then $B_{\mathscr{G}}$ is coherent (resp. Noetherian).

EXERCISE 2.17. (*Topoi of the form $\widehat{C}$.*) Let C be a category equivalent to a category $\in \mathscr{U}$, and let $E = \widehat{C}$ be the category of presheaves on C. By means of the canonical functor $\varepsilon : C \to \widehat{C}$, we identify C with a full subcategory of E.

219 (a) C is a generating subcategory of E consisting of quasi-compact objects. These objects are pre-Noetherian (cf. 1.32) if and only if, for every object X of C, every sieve on X is generated by a finite family of morphisms $X_i \to X$ of C. The final object of E is quasi-compact (resp. pre-Noetherian) if and only if there is a final subcategory of C whose set of objects is finite (resp. if every sieve on C is generated by a subcategory of C whose set of objects is finite).
(b) The topos E admits a generating subcategory consisting of coherent (or, equivalently, quasi-separated) objects if and only if the objects of C are coherent in E, or equivalently if, for two arrows $Y \xrightarrow{f} X$, $Z \xrightarrow{g} X$ in C, the object $Y \times_X Z$ of E is quasi-compact, i.e. if one can find a finite family of commutative squares

in C

$$\begin{tikzcd}
& T_i \arrow[dl] \arrow[dr] & \\
Y \arrow[dr, "f"'] & & Z \arrow[dl, "g"] \\
& X &
\end{tikzcd},$$

such that every other commutative square in C

$$\begin{tikzcd}
& T \arrow[dl] \arrow[dr] & \\
Y \arrow[dr, "f"'] & & Z \arrow[dl, "g"] \\
& X &
\end{tikzcd},$$

comes from a morphism $T \to T_i$. (Note that if $(F_i)_{i\in I}$ is a generating family of E, then every $X \in \operatorname{ob} C$ is isomorphic to a direct factor of one of the F_i.) 220

(c) The topos E is locally coherent, i.e. is generated by a family of coherent and *algebraic* objects, if and only if the objects X of C are coherent and algebraic objects of E, i.e. if condition b) and the following condition hold: for every pair of arrows $f, g : Y \rightrightarrows Z$ in C *over an object X of C*, the kernel of this pair of arrows in E is a quasi-compact object of E, i.e. there is a finite family of commutative diagrams in C

$$\begin{tikzcd}
T_i \arrow[r] & Y \arrow[r, shift left, "f{,}g"] \arrow[r, shift right] & Z
\end{tikzcd}$$

such that every other commutative diagram in C

$$\begin{tikzcd}
T \arrow[r] & Y \arrow[r, shift left, "f{,}g"] \arrow[r, shift right] & Z
\end{tikzcd}$$

comes from a morphism $T \to T_i$. The topos E is algebraic if and only if it satisfies condition b) and the preceding condition, but with an arbitrary pair of arrows (f, g) of C (not necessarily over an object X of C). The topos E is quasi-separated if and only if it satisfies the conditions of b), and if, in addition, for two objects $X, Y \in \operatorname{ob} C$, the product $X \times Y$ in E is quasi-compact. The topos E is coherent if and only if it satisfies the two preceding conditions and there is a final subcategory of C whose underlying set is finite.

(d) Give an example in which the objects of the generating subcategory C of E are pre-Noetherian, but E admits no generating family consisting of coherent objects (i.e. b) fails, or the objects of C are not all coherent). For this, take for C the category having as objects X, T_i ($i = 0, 1, \dots$), and e (the final object), the only morphisms between these objects, besides the structural morphisms to e and the identity morphisms, being morphisms $u_i : T_i \to X$ and $v_i : X \to T_i$ subject to the conditions $u_i v_i = \mathrm{id}_X$, and finally set $p_i = v_i u_i$ (which necessarily satisfies $p_i^2 = p_i$). One verifies that the only sieves on X or on any T_i are the two trivial sieves, and that e admits exactly one nontrivial sieve. Thus, by a), the 221 objects of C are pre-Noetherian objects of E. Nevertheless, the product $X \times X$ in E is not quasi-compact, since it does not satisfy the criterion of b).

(e) Give an example in which the generating subcategory C of E consists of coherent objects of E, but E is not locally coherent. For this, take for C the category whose set of objects consists of the distinct objects e (the final object), Y, Z, and T_i ($i = 0, 1, \dots$), with the only morphisms between these objects, besides the morphisms to the final object e and the identity morphisms, being morphisms $f, g : Y \rightrightarrows Y$ subject to the conditions $fu_i = gu_i$. One verifies that C satisfies condition b) (with a little patience; one finds that all fiber products of objects
222 of C are isomorphic to $\varnothing_E$ or belong to C, with the exception of $Z \times_e Z$ and $Y \times_e Z$, which are covered by two elements of C), but clearly $\operatorname{Ker}(f, g)$ does not satisfy condition c).

(f) Give an example in which C is stable under fiber products (a fortiori, E is a locally coherent, i.e. locally algebraic, topos), but E is not an algebraic topos. (Take the example given in d), and the full subcategory C' of E consisting of the objects Y, Z, T_i ($i > 0$), and $\varnothing_E$.)

(g) Suppose that the category C is finite. Prove that the topos E is Noetherian; that its Noetherian (i.e. quasi-compact) objects are the contravariant functors $F : C^\circ \to (\text{Ens})$ such that, for every object X of C, $F(X)$ is a *finite* set; and that the fiber functors on E take Noetherian objects to finite sets (cf. IV 7.6. h)).

(h) Suppose that every morphism $f : X \to X$ of C factors as a composite ip, where i is a monomorphism and p has a right inverse. Let X be an object of C, let $I(X)$ be the set of its subobjects in the sense of C, regarded as an ordered set, and let $S(I(X))$ be the set of subsets U of $I(X)$ such that, for two elements X', $X'' \in I(X)$, the conditions $X' \in U$ and $X'' \leq X'$ imply $X'' \in U$. Show that the ordered set of subobjects in E of X is isomorphic to the set $S(I(X))$ (ordered by inclusion). Conclude that if $I(X)$ is finite, then X is a pre-Noetherian object of E. In particular, if (in addition to the factorization condition above) C is
223 stable under fiber products (resp. under finite projective limits), and if, for every $X \in \operatorname{ob} C$, the set $I(X)$ of subobjects of X in C is finite, then the topos E is locally Noetherian (resp. Noetherian).

(i) Let C be the category of finite sets (or, alternatively, of nonempty finite sets) $\in \mathscr{U}$, so that $E = \widehat{C}$ is the category of augmented simplicial sets (resp. of simplicial sets simpliciter). Show that E is a Noetherian topos. (Use h).) Taking a nonessentially constant pro-object $(X_i)_{i \in I}$ of C, show that E admits fiber functors that do not take Noetherian objects to Noetherian objects (i.e. to finite sets).

(j) Consider the following diagram of implications among properties of a topos E:

$$
\begin{tikzcd}
\text{E is Noetherian} \arrow[r, Rightarrow] \arrow[d, Rightarrow] & \text{E is locally Noetherian} \arrow[d, Rightarrow] \arrow[ddr, Rightarrow] & \\
\text{E is coherent} \arrow[r, Rightarrow] & \text{E is algebraic} \arrow[d, Rightarrow] & \\
& \begin{array}{c}\text{E is loc. coherent, i.e.}\\ \text{E}_{\text{cohalg}} \text{ generates E}\end{array} \arrow[d, Rightarrow] & \text{E}_{\text{pre-Noeth}} \text{ generates E} \arrow[ddl, Rightarrow] \\
& \text{E}_{\text{coh}} \text{ generates E} \arrow[d, Rightarrow] & \\
& \text{E}_{\text{qucpct}} \text{ generates E} &
\end{tikzcd}
.
$$

Show that every implication in this diagram is strict, and that among the notions considered there are no implications other than the composites of the implications in the preceding diagram. Here E_{coh}, E_{qucpct}, $E_{\text{prénoeth}}$, E_{cohalg} denote 224
respectively the full subcategories of E consisting of the coherent, resp. quasi-compact, resp. pre-Noetherian, resp. coherent and algebraic objects of E. (One may restrict attention to topoi of the form $\widehat{C}$, using the results stated in d), e), and f).)

(k) Answer the following question (to which the author of these lines does not know the answer): can $E = \widehat{C}$ be locally Noetherian without the $\mathrm{Hom}(X, Y)$ (for X, $Y \in \mathrm{ob}\, C$) being finite?

Exercise 2.18. Let E be a topos.

a) Let X be a pre-Noetherian object of E. Show that X is isomorphic to a finite sum of connected objects (IV 4.3.5) of E. (Prove by contradiction that an increasing sequence of partitions of X (IV 8.7) is stationary.)
b) Let X be an object of E and let $(f_i : X_i \to X)_{i \in I}$ be a covering family of X. Show that if each X_i is isomorphic to a sum of connected objects of E, then so is X. (Reduce to the case in which the X_i are connected, then to the case in which they are subobjects of X, and consider on the index set I the equivalence relation R generated by the relation $X_i \cap X_j \neq \phi_E$. Show that if $J = I/R$, then X is the sum of the $X(i) = \operatorname{Sup}_{i \in I} X_i$.)
c) Conclude from b) that if $(X_i)_{i \in I}$ denotes a generating family of E, then E is locally connected (IV 8.7. ℓ)) if and only if every X_i ($i \in I$) is isomorphic to a 225
sum of connected objects of E.
d) Conclude from a) and c) that if E admits a generating subcategory consisting of pre-Noetherian objects, in particular if E is locally Noetherian, then E is locally connected, and a fortiori is isomorphic to the sum topos (IV 8.7 b)) of a family of connected topoi (IV 8.7 e)), i.e. topoi whose final object is connected. (In particular, one can associate with E, for every morphism $f : P \to E$, where P is a "nonempty connected and simply connected" topos (IV 2.7.5), a fundamental pro-group $\pi_1(E, f)$. Note that if E is locally Noetherian and "nonempty," such

an f can always be found, with P the punctual topos, by a result of DELIGNE ([9]).

3. Finiteness conditions for a morphism of topoi

DEFINITION 3.1. *Let $f : E' \to E$ be a morphism of topoi. We say that f is* quasi-compact (resp. quasi-separated) *if, for every quasi-compact (resp. quasi-separated) object X of E, $f^*(X)$ is quasi-compact (resp. quasi-separated). We say that f is coherent if f is quasi-compact and quasi-separated.*

3.1.1. Note that if f has any one of the preceding properties, then every morphism of topoi isomorphic to f has the same property. It is also immediate that the composite of two quasi-compact (resp. quasi-separated, resp. coherent) morphisms of topoi is again quasi-compact (resp. quasi-separated, resp. coherent).

226 PROPOSITION 3.2. *With the notation of 3.1, let $(X_i)_{i\in I}$ be a generating family of quasi-compact (resp. coherent) objects of E. In the parenthetical case, suppose that E' admits a generating family formed by quasi-compact objects. Then f is quasi-compact (resp. coherent) if and only if, for every $i \in I$, $f^*(X_i)$ is quasi-compact (resp. coherent).*

Necessity is immediate. For sufficiency in the first case, note that, for every quasi-compact object X of E, there is a finite epimorphic family of morphisms $X_i \to X$. Hence there is a finite epimorphic family $f^*(X_i) \to f^*(X)$, with the $f^*(X_i)$ quasi-compact by hypothesis, and therefore $f^*(X)$ is quasi-compact (1.3). In the second case, it remains to see that if X is a quasi-separated object of E, then $f^*(X)$ is quasi-separated. The hypothesis on X may be expressed (1.17) by the existence of an epimorphic family of morphisms $X_i \to X$ such that the $X_i \times_X X_j$ are quasi-compact. We then have a covering family $f^*(X_i) \to f^*(X)$ such that the fiber products $f^*(X_i) \times_{f^*(X)} f^*(X_j)$ are quasi-compact (since, by the first case, f^* carries quasi-compact objects to quasi-compact objects). As the $f^*(X_i)$ are coherent, the conclusion follows again from 1.17.

COROLLARY 3.3. *Let C and C' be two $\mathcal{U}$-sites in which fiber products are representable and every covering family admits a finite covering subfamily, and let $g : C \to C'$ be a morphism of sites* (IV 4.9.1). *Then the morphism of topoi $f : C^\sim \to C'^\sim$ defined by g is coherent.*

227 Indeed, $\epsilon_C(C)$ (resp. $\epsilon_{C'}(C')$) is a generating family of $C^\sim$ (resp. $C'^\sim$) satisfying the corresponding conditions of 3.2 (2.1.1).

PROPOSITION 3.4. *Let E be a locally coherent topos (2.3), and let $f : X \to Y$ be an arrow of E, whence a morphism of induced topoi* (IV (5.5.2)) *$E_{/X} \to E_{/Y}$. This morphism of topoi is quasi-compact (resp. quasi-separated, resp. coherent) if and only if f is a quasi-compact (resp. quasi-separated, resp. coherent) morphism.*

The quasi-compact case follows immediately from the definitions (without any condition on E); the quasi-separated case is precisely 2.8; and the coherent case follows by combining the two preceding cases.

PROPOSITION 3.5. *Let $f : E' \to E$ be a morphism of locally coherent topoi. Let $(X_i)_{i\in I}$ be a generating family in E formed by coherent algebraic objects. The following conditions are equivalent:*

(i) *For every quasi-compact arrow u of E, $f^*(u)$ is quasi-compact.*

(i bis) *For every arrow $u : X_i \to X_j$, $f^*(u)$ is quasi-compact.*

(ii) *For every coherent object Y' of E', every coherent algebraic object Y of E, and every arrow $v : Y' \to f^*(Y)$, the corresponding morphism of topoi*

$$f_v : E'_{/Y'} \longrightarrow E_{/Y}$$

obtained by composing the localization morphism $E'_{/Y'} \to E'_{/f^*(Y)}$ deduced from v (IV (5.5.2)) *with the morphism $E'_{/f^*(Y)} \to E_{/Y}$ induced by f* (IV (5.10.1)) *is coherent.* 228

(ii bis) *The same condition as in* (ii), *but restricted to a collection of data $v_\alpha : Y'_\alpha \to f^*(Y_\alpha)$ such that the Y'_α are algebraic and cover the final object of E'.*

(ii ter) *(When a generating family $(X'_{i'})_{i' \in I'}$ in E' formed by coherent objects is given, together with, for every $i' \in I'$, an $i \in I$ and an arrow $v_{i'} : X'_{i'} \to f^*(X_i)$.) For every $i' \in I'$, the morphism of topoi $E'_{/X'_{i'}} \to E_{/X_i}$ defined by $v_{i'}$ is coherent.*

Since every arrow $X_i \to X_j$ is quasi-compact, it is clear that (i) ⇒ (i bis). Conversely, suppose (i bis) holds and let us prove (i). Let $u : X \to Y$ be a quasi-compact arrow of E. In terms of an epimorphic family $X_i \to Y$, the hypothesis on u says that the $X \times_Y X_i$ are quasi-compact (1.16), i.e. that for every i there exists a finite epimorphic family of morphisms $X_j \to X \times_Y X_i$ (1.3). We then have the corresponding epimorphic family $f^*(X_i) \to f^*(Y)$, such that for every i there is a finite epimorphic family $f^*(X_j) \to f^*(X \times_Y X_i) \simeq f^*(X) \times_{f^*(Y)} f^*(X_i)$, whose composite with the projection pr_2 to $f^*(X_i)$ is a quasi-compact morphism $f^*(X_j) \to f^*(X_i)$ by hypothesis (i bis). It follows (1.11 (i)) that, for every i, $\mathrm{pr}_2 : f^*(X) \times_{f^*(Y)} f^*(X_i)$ is quasi-compact, and hence (1.10 (ii)) that $f^*(u) : f^*(X) \to f^*(Y)$ is quasi-compact.

It remains to prove (i) ⇒ (ii) and (ii bis) ⇒ (i); the implication (ii) ⇒ (ii bis) is immediate, and (ii ter) is a special case of (ii bis). The implication (i) ⇒ (ii) is immediate: indeed, an object X of $E_{/Y}$ is quasi-compact (resp. quasi-separated) if and only if its structural morphism $X \to Y$ (resp. its diagonal morphism $X \to X \times_Y X$) is quasi-compact (2.7), and the same criterion of quasi-compactness and quasi-separatedness applies to $f^*_Y(X')$. Finally, let us prove (ii bis) ⇒ (i). Let $u : X \to Y$ be a quasi-compact arrow in E; we must show that $f^*(u)$ is quasi-compact. Since the Y'_α cover the final object of E', it is enough to prove that the $f^*(u)|Y'_\alpha$ are quasi-compact (1.10 (ii)). These morphisms are precisely the $f^*_{v_\alpha}(u|Y_\alpha)$, so we are reduced to proving the following result. 229

Corollary 3.6. *Let $f : E' \to E$ be a coherent morphism of locally coherent topoi. Then, for every quasi-compact arrow u of E, $f^*(u)$ is quasi-compact.*

By the implication (i bis) ⇒ (i) of 3.5 already proved, it is enough to show that, if $u : X \to Y$ is an arrow of E with X, Y coherent, then $f^*(u)$ is a quasi-compact morphism. This follows immediately from the hypothesis on f, which implies that $f^*(X)$ and $f^*(Y)$ are coherent as well.

Definition 3.7. *Let $f : E' \to E$ be a morphism of locally coherent topoi. We say that f is* locally coherent *if it satisfies the equivalent conditions of 3.5.* 230

Observe that "locally" means locally on the source.

3.7.1. This notion again depends only on the isomorphism class of the morphism of topoi f, and it is plainly stable under composition of morphisms of topoi. Moreover, by 3.5, if f is coherent then it is locally coherent; the converse holds if E' and E are coherent (by 3.5 (ii bis)). Criterion 3.5 (ii bis) immediately gives the following criterion:

Corollary 3.7. *Let $f : E' \to E$ be a morphism of locally coherent topoi, let $(Y_i)_{i \in I}$ and $(Y'_i)_{i \in I}$ be two families of objects of E and E', and, for every $i \in I$, let $v_i : Y'_i \to$*

$f^*(Y_i)$ be a morphism, whence a morphism of topoi

$$f_{v_i} : E'_{/Y'_i} \longrightarrow E_{/Y_i}.$$

Suppose that the Y'_i cover the final object of E'. Then f is locally coherent if and only if, for every $i \in I$, f_{v_i} is locally coherent.

In particular, applying this to the identity morphism of an induced topos gives:

Corollary 3.8. *Let E be a locally coherent topos, and let $v : X \to Y$ be an arrow of E. Then the morphism of induced topoi $E_{/X} \to E_{/Y}$ defined by v is locally coherent.*

231 Remark 3.9. *It seems that all morphisms of algebraic topoi encountered in practice are locally coherent (which is why the term "locally coherent" is probably not destined to be used very often). Equivalently, the morphisms between coherent topoi encountered in practice are coherent. Nevertheless, one can construct noncoherent morphisms between coherent topoi and, more particularly, a noncoherent morphism from the punctual topos* (IV 2.2) *to a Noetherian topos E,* cf. 2.17 i).

Examples 3.10. (*Topoi associated with schemes.*) Return to the example 1.22 of the category (Sch) with the topologies T_i of SGA IV 6.3. For every scheme X, let X_{T_i} be the $\mathscr{V}$-topos defined by the induced site $(\mathrm{Sch})_{/X}$. A morphism of schemes $f : X \to Y$ then defines a morphism of topoi $f_{T_i} : X_{T_i} \to Y_{T_i}$, and it follows from 3.4 and 1.22 that this morphism of topoi is quasi-compact (resp. quasi-separated, resp. coherent) if and only if the morphism of schemes f is quasi-compact (resp. quasi-separated, resp. coherent) in the usual sense of EGA IV 1. In any event, f_{T_i} is locally coherent by 3.8.

One may also associate with a scheme X the $\mathscr{U}$-topoi $\mathrm{Top}(X)$ and $\mathrm{Top}(X_{\text{ét}})$ as in 1.22.1, and to every morphism of schemes $f : X \to Y$ there are then associated morphisms $\mathrm{Top}(f)$ and $\mathrm{Top}(f_{\text{ét}})$ of the corresponding topoi. Let $T(f)$ denote either of
232 these two morphisms of topoi. One verifies immediately by 3.2 that this morphism of topoi is always locally coherent, and that it is quasi-compact (resp. quasi-separated, resp. coherent) if and only if the morphism of schemes f is quasi-compact (resp. quasi-separated, resp. coherent) in the usual sense.

These observations show once again that the terminology introduced in the present section is compatible with the accepted terminology of scheme theory, and therefore deserves to be accepted by even the most recalcitrant reader.

Exercise 3.11. (*Coherent topoi and pretopoi.*)

a) A $\mathscr{U}$-pretopos (or simply a pretopos) is a category C satisfying the following conditions:
   1) Finite projective limits are representable in C.
   2) Finite coproducts are representable in C, and they are disjoint and universal.
   3) Equivalence relations in C are effective, and every epimorphism in C is universally effective.
   4) C is equivalent to a category $\in \mathscr{U}$.

   Show that if E is a coherent $\mathscr{U}$-topos, then the full subcategory E_{coh} of E formed by its coherent objects is a $\mathscr{U}$-pretopos, and that the inclusion functor $E_{\mathrm{coh}} \to E$
233 is left exact, commutes with finite coproducts, and commutes with passage to the quotient by an equivalence relation. (Use 1.5.3, 1.15, and 1.17.1.) Show that the topology induced by E on C is the "precanonical" topology, i.e. the topology whose covering families $X_i \to X$ are those that admit a *finite* subfamily

covering for the canonical topology of C. Consequently, E can be recovered up to equivalence from the pretopos $C = E_{\mathrm{coh}}$ as the topos $C^{\sim}$ (with C endowed with its precanonical topology).

b) Let C be a $\mathscr{U}$-pretopos. Endow C with the precanonical topology. Show that every morphism f of C factors as $f''f'$, with f' an epimorphism and f'' a monomorphism. Show that the topos $E = C^{\sim}$ is coherent and that the canonical functor $\epsilon : C \to E$ induces an equivalence from C to the subcategory E_{coh} of E. (First show that ε is a fully faithful left exact functor commuting with finite coproducts and with passage to the quotient by an equivalence relation, and then that a coherent subobject in E of an object of C lies in the essential image of C.) Consequently, the $\mathscr{U}$-pretopos C can be recovered up to equivalence of categories from its associated $\mathscr{U}$-topos $E = C^{\sim}$.

c) Let C and C' be two $\mathscr{U}$-pretopoi. Show that a functor $\varphi : C' \to C$ is a morphism of sites from C to C' (for the precanonical topologies) if and only if φ is left exact and commutes with finite coproducts and with passage to a quotient by an equivalence relation.

d) With the notation of c), suppose that C and C' are associated with two coherent 234
$\mathscr{U}$-topoi E, E' as in a). Show that the canonical functor (IV 4.9.3)

$$\mathbf{Morsite}(C, C') \longrightarrow \mathbf{Homtop}(E, E')$$

induces an equivalence from its source (described explicitly in c)) to the full subcategory of its target formed by the *coherent* morphisms of topoi $E \to E'$. A quasi-inverse is obtained by associating with every coherent morphism of topoi $f : E \to E'$ the functor $E'_{\mathrm{coh}} = C' \to E_{\mathrm{coh}} = C$ induced by f^*.

e) Let C be a $\mathscr{U}$-pretopos and $E = C^{\sim}$. Express directly in terms of exactness properties of C the equivalent conditions of 1.25 (consult 1.28 b). Show that an object X of C is Noetherian in E if and only if every increasing sequence of subobjects of X in C is stationary, and hence that E is Noetherian if and only if every object of C satisfies the preceding condition.

Exercise 3.12. A topos E is called a *finite* topos if there exists a finite category C such that E is equivalent to $\widehat{C}$.

a) The topos E is finite if and only if E has enough essential points (IV 7.6 b)) and the category $\mathbf{Pointess}(E)$ of essential points of E is equivalent to a finite category. (Use IV 7.6 d) and h).)

b) Suppose that E is finite. Prove that every point of E is essential. Prove that 235
the 2-functors $E \mapsto \mathbf{Point}(E)$ and $C \mapsto \widehat{C}$ establish 2-equivalences between the 2-category of finite topoi E and the 2-category of Karoubian categories (IV 7.5 a)) C that are equivalent to finite categories, and that every morphism of finite topoi is essential (IV 7.6 a)). (Use a) and IV 7.6 h).)

c) Suppose that $E = \widehat{C}$, with C equivalent to a finite category, and let F be a coherent topos. Recall (IV (4.6.3.1)) that morphisms of topoi $f : E \to F$ correspond to functors $u : C \to \mathbf{Point}(F)$. Show that f is coherent if and only if f carries every coherent point of E (3.1) to a coherent point of F, or equivalently if, for every $X \in \mathrm{ob}\, C$, $u(X)$ is a *coherent* point of F. (Use 3.2 and 2.17 g).) Conclude that every morphism $f : E \to F$ from a finite topos to a finite topos is coherent.

d) Let X be a sober topological space (IV 4.2.1). The topos $\mathrm{Top}(X)$ (IV 2.1) is finite if and only if X is a finite set. (Use IV 7.1.6.)

4. Finiteness conditions in a topos obtained by gluing

The present section will not be used later in the Seminar.

4.1. Let E be a topos and U an open of E (i.e. a subobject of the final object of E),
236 and consider the corresponding open and closed subtopoi of E (IV 9)

$$E' = E_{/U}, E'' = E_{/U}.$$

We denote by

$$j : E' \longrightarrow E, i : E'' \longrightarrow E$$

the canonical morphisms of topoi. Recall (3.4) that for j to be a quasi-compact (resp. coherent) morphism of topoi, it is necessary and sufficient that the inclusion

$$j : U \longrightarrow e$$

of U into the final object of E (an inclusion that we shall also denote by j) be a quasi-compact (resp. coherent) morphism in E; moreover, $j : E' \to E$ is always quasi-separated (because $j : U \to e$ is so (1.8.1)). We propose to give criteria for the topos E'' to be coherent and for the morphism of topoi $i : E'' \to E$ to be coherent. First note:

Proposition 4.2. *With the notation of 4.1, the inclusion morphism $i : E'' \to E$ is quasi-compact, i.e. for every quasi-compact object X of E, the object $i^*(X)$ of E'' is quasi-compact; moreover, if X is pre-Noetherian (1.30), then $i^*(X)$ is pre-Noetherian.*

This follows from Definition 1.1 and the

237 Lemma 4.2.1. *The map $Y \mapsto i^*(Y)$ induces an isomorphism of ordered sets between the set of subobjects of X containing the subobject $X_U = X \times U$ of X, and the set of subobjects of $i^*(X)$.*

Note that since i_* is conservative (because it is fully faithful) and left exact (in particular, it commutes with fiber products), it follows immediately that a morphism $u : Y \to Z$ in E'' is a monomorphism if and only if $i_*(u) : i_*(Y) \to i_*(Z)$ is one. Consequently, on identifying (by i_*) E'' with the full subcategory of E consisting of objects Z satisfying the equivalent conditions of IV 9.7 1), the subobjects in E'' of the object Z of E'' identify with the subobjects Y of Z in E that belong to E'', or, equivalently, that contain $Z_U \xrightarrow{\sim} U$. For Z of the form $i_*i^*(X) = X_{\complement U}$ given by the amalgamated sum IV (9.5.1),

$$i_*i^*(X) = X_{\complement U} = X \amalg_{X_U} U,$$

the datum of a subobject Y of the latter in E is equivalent to the datum of a pair consisting of a subobject Y_1 of X and a subobject Y_2 of U, subject to the condition that the inverse images of these subobjects in X_U coincide. The condition that Y contain U then means that $Y_2 = U$, and the remaining condition on Y_1 is that it contain X_U. This therefore establishes a bijection between the set of subobjects of $i^*(X)$ in E'' and the set of
238 subobjects of X that contain X_U. It is clear that this is an isomorphism of ordered sets and that its inverse map is indeed the one asserted in 4.2.1.

Corollary 4.3. a) *Let X be an object of E. For X to be quasi-compact, it is sufficient that $i^*(X)$ and $j^*(X)$ be so, and this condition is also necessary if $j : U \to e$ is quasi-compact.*

b) *Let Y be an object of E''. For Y to be quasi-compact, it is sufficient that $i_*(Y)$ be so, and this condition is also necessary if U is quasi-compact.*

<u>Proof</u>

a) Sufficiency follows from Definition 1.1 and the fact that the pair (i^*, j^*) is conservative (IV 9.11 3)). Necessity follows from the fact that i and j are quasi-compact (by 3.14 and the hypothesis that j is quasi-compact).
b) Since $Y \simeq i^* i_*(Y)$, sufficiency follows from 4.2. Necessity follows from the sufficiency in a), taking into account that $i^* i_*(Y) \simeq Y$ and $j^* i_*(Y) \simeq U$.

Corollary 4.4. *Let Y be an object of E''. If U is quasi-compact or if E admits a generating family consisting of quasi-compact objects, then for Y to be quasi-separated (resp. coherent), it is* sufficient *that $i_*(Y)$ be so. If U is quasi-separated (resp. coherent) and if $j : U \to e$ is a quasi-compact morphism, then for Y to be quasi-separated (resp. coherent), it is necessary that $i_*(Y)$ be so.*

The parenthetical case follows from the non-parenthetical case, taking 4.3 b) into 239
account. Suppose that $i_*(Y)$ is quasi-separated, and let us prove that Y is so under one of the two stated hypotheses. Thus we must prove that, for two quasi-compact objects Y' and Y'' over Y, the fiber product $Y' \times_Y Y''$ is quasi-compact. When U is assumed quasi-compact, note that it is enough to prove that $i_*(Y' \times_Y Y'')$ is quasi-compact (4.3 b)); this follows from the fact that $i_*(Y')$ and $i_*(Y'')$ are so (4.3 b)), using here the quasi-compactness of U, the fact that i_* commutes with fiber products, and the hypothesis that $i_*(Y)$ is quasi-separated. When E is assumed to admit a generating family consisting of quasi-compact objects, $i_*(Y')$ is a filtered inductive limit of its quasi-compact subobjects ${X'}_\alpha$; hence $Y' \simeq i^* i_*(Y')$ is a filtered inductive limit of the subobjects $i^*({X'}_\alpha)$, and since Y' is quasi-compact, it is equal to one of the $i^*({X'}_\alpha)$. Likewise, Y'' is of the form $i^*({X''}_\beta)$, where ${X''}_\beta$ is a quasi-compact subobject of $i_*(Y'')$. But then $Y' \times_Y Y'' = i^*(X'_\alpha \times_{i_* Y} X''_\beta)$, and since $X'_\alpha \times_{i_*(Y)} X''_\beta$ is quasi-compact by the hypothesis that $i_*(Y)$ is quasi-separated, it follows by 4.2 that $Y' \times_Y Y''$ is also quasi-compact.

Conversely, suppose that U is quasi-separated and $j : U \to e$ is quasi-compact, and assume that Y is quasi-separated. Let us show that $i_*(Y)$ is also quasi-separated, i.e. that for two quasi-compact objects X', X'' over $i_*(Y)$, the fiber product $X' \times_{i_*(Y)} X''$ is quasi-compact. By 4.3 a), it is enough to prove that its image under i^* is quasi-compact (which follows from 4.2 and the hypothesis that Y is quasi-separated) *and* that its image 240
under j^* is quasi-compact. The latter is $j^*(X') \times j^*(X'')$, and is indeed quasi-compact because $j^*(X')$ and $j^*(X'')$ are so (j being quasi-compact) and U is quasi-separated.

Corollary 4.5. *Suppose that E admits a generating subcategory consisting of quasi-compact* (resp. *pre-Noetherian) objects. Then the same is true of E''.*

This follows from 4.2 and the

Lemma 4.5.1. *If $(X_\alpha)_\alpha$ is a generating family in E, then $(i^*(X_\alpha))_\alpha$ is a generating family in E''.*

Indeed, this means that the family of functors

$$Y \longrightarrow \mathrm{Hom}(i^*(X_\alpha), Y)$$

on E'' is conservative. But $\mathrm{Hom}(i^*(X_\alpha), Y) \simeq \mathrm{Hom}(X_\alpha, i_*(Y))$, and it is enough to use the fact that the functor i_* is conservative.

Proposition 4.6. *The notation is that of 4.1.*

a) *If E admits a generating family consisting of coherent objects, then the same is true of the closed subtopos E'', and the inclusion morphism $i : E'' \to E$ is coherent.*

b) *Suppose that the morphism $j : U \to e$ in E is quasi-compact. If E is locally coherent* (resp. *coherent,* resp. *quasi-separated,* resp. *locally Noetherian,* resp. *Noetherian,* resp. *locally perfect,* resp. *perfect), then the same is true of E'', and the inclusion morphism $i : E'' \to E$ is coherent.* 241

a) The first assertion will follow from the second and 4.5.1. In any event, 4.5 implies that E'' admits a generating family consisting of quasi-compact objects, so 3.2 applies and shows that it is enough to verify that, for every coherent object X of E, the object $i^*(X)$ of E'' is coherent. Using the compatibility of the formation of the topos complementary to an open with localization (IV 9.13 b)), we reduce to the case in which X is the final object of E, and hence to proving that the final object of E'' is coherent. Now 4.4 applies, so it is enough to prove that $i_*(e_{E''}) = e_E$ is coherent, which is indeed the case.

b) Under each of the hypotheses made in b), E admits a generating family consisting of coherent objects, so it already follows from a) that $i : E'' \to E$ is coherent. Moreover, 4.5.1 then implies that E'' admits as a generating subcategory the full subcategory E''_{coh} consisting of its coherent objects. Suppose that E is coherent, and let us show that E'' is also coherent, i.e. (2.4.5) that the subcategory E''_{coh} of E'' is stable under finite projective limits, knowing that this is so for the subcategory E_{coh} of E. This follows immediately from the parenthetical criterion 4.4 (which applies because U is now coherent, e being
242 coherent and $j : U \to e$ being quasi-compact), taking into account that i_* is left exact. By localization, again using IV 9.13 b)), we conclude that if E is locally coherent, then the same is true of E''. If E is quasi-separated, then E'' is also quasi-separated: indeed, its final object is quasi-separated by 4.4, since that of E is, and it remains to prove that the product of two quasi-separated objects of E'' is quasi-separated (knowing that this is so in E); this again follows from the criterion 4.4 and the fact that i_* commutes with products (taking into account that the subobject U of e is quasi-separated, since e is, so that 4.4 applies).

Since a topos is locally Noetherian (resp. Noetherian) if and only if it is locally coherent (resp. coherent) and admits a generating family consisting of pre-Noetherian objects (2.10 (iii bis)), it follows from the preceding discussion and 4.5 that if E is locally Noetherian (resp. Noetherian), then the same is true of E''. Now suppose that E is perfect, and let us prove that E'' is perfect. Since we already know that it is coherent, it remains to verify that it satisfies criterion 1.25 (i), i.e. that every object Y of E'' is a filtered inductive limit of coherent objects, knowing that the analogous statement is true in E. Since $i_*(Y)$ is a filtered inductive limit of coherent objects X_α, $Y \simeq i^*i_*(Y)$ is a filtered inductive limit of the coherent objects $i^*(X_\alpha)$. By localization (IV 9.13 b)), we conclude that if E is locally perfect, then the same is true of E''. This completes the proof of 4.6.

243 Remark 4.6.1. In fact, in 4.6 b), the hypothesis that $j : U \to e$ is quasi-compact is unnecessary, except perhaps in the case "E quasi-separated". Indeed, by the preceding proof it is enough to see that if E is coherent, then E'' is coherent. Now U is the filtered inductive limit of its quasi-compact subobjects U_α, and in 7 we shall verify that E'' then identifies with the projective-limit topos (in the sense of 7) of the closed subtopoi E''_i complementary to the $E_{/U_i}$, which by 4.6 b) are coherent topoi with coherent transition morphisms. It then follows that E'' is a coherent topos (7).

Corollary 4.7. *Suppose that the subcategory E_{coh} of E consisting of coherent objects is generating, and let X be an object of E.*

a) *If X is quasi-separated, then $i^*(X)$ and $j^*(X)$ are quasi-separated; this condition is also sufficient if $j : U \to e$ is quasi-compact.*
b) *Suppose that $j : U \to e$ is quasi-compact. Then X is coherent if and only if $i^*(X)$ and $j^*(X)$ are coherent.*
c) *Suppose that E is coherent and $j : U \to e$ quasi-compact. Then E is perfect (2.9.1) if and only if E' and E'' are perfect.*

a) We noted in 4.1 that $j : E' \to E$ is quasi-separated, and we know from 4.6 a) that the same is true of i, which gives necessity. For sufficiency, let X' and X'' be quasi-compact objects over X. We must prove that $X' \times_X X''$ is quasi-compact, and for this it is enough to prove that its images under i^* and j^* 244 are so (4.3 a)). Since i and j are quasi-compact, by 4.2 and the hypothesis that $j : U \to e$ is quasi-compact, the conclusion follows from the fact that i^* and j^* commute with fiber products and from the hypothesis that $i^*(X)$ and $j^*(X)$ are quasi-separated.
b) This follows by combining a) with 4.3 a).
c) Necessity was already seen (4.6 b)). Sufficiency follows from the fact that, since E is coherent, E' and E'' are coherent (4.6 b)), so that for any of the topoi under consideration, being perfect means that the category of its coherent objects is stable under finite $\varinjlim$. We therefore conclude by b).

Exercise 4.8. a) Solve the following question (to which the writer confesses, to his embarrassment, that he does not know the answer): with the notation of 4.1, if E is quasi-separated (resp. algebraic), is the same true of E''?
b) Suppose that E is equivalent to a topos of the form $\widehat{C}$, where C is a category equivalent to a category $\in \mathscr{U}$. Prove directly, using 2.17 c) and IV 9.24, that if E is locally coherent (resp. coherent, resp. algebraic, resp. quasi-separated, resp. locally Noetherian, resp. Noetherian), then the same is true of E''.

4.9. Keep the notation of 4.1, and recall (IV 9.16) that the datum of a situation (E, U) consisting of a topos E and an open U of E is essentially equivalent to that of a triple (E', E'', f), where E' and E'' are topoi and 245

$$f : E' \longrightarrow E'' \tag{4.9.1}$$

is a “gluing functor”, i.e. a functor that is *left exact and accessible*; starting from (E, U) as in 4.1, the associated gluing functor is given by

$$f = i^* j_*. \tag{4.9.2}$$

We propose to express, in terms of the data E', E'', and f, the fact that E is a coherent (resp. perfect) topos and that U is a quasi-compact object, or equivalently, that E and E' are coherent. We already know that this implies that E'' is also coherent (4.6 b)), so the question becomes the following: given two *coherent* topoi E' and E'' and a gluing functor f (4.9.1), under what conditions on f is the glued topos E coherent (resp. perfect)? We also know (4.7 c)) that E is perfect if and only if E is coherent and E' and E'' are perfect; thus the parenthetical case of the problem reduces to the non-parenthetical case. Let us nevertheless note that the solution (4.10) of the latter is more elegant if E' is already assumed perfect.

4.9.3. First note that if E is coherent, then the full subcategory E_{coh} of E consisting of the objects $X = (X', X'', u : X'' \to f(X'))$ of E that are coherent is reconstructed as the subcategory E_0 of E consisting of the X for which $X' = j^*(X)$ and $X'' = i^*(X)$ 246 are coherent (4.7 b)). A necessary condition for E to be coherent is therefore that the preceding subcategory E_0 be generating. This condition is also sufficient, because by 4.3 a), E_0 consists of quasi-compact objects; moreover, it is clear that E_0 is stable under finite projective limits, and we conclude by 3.4.5.

Recall now (IV 9.18) that the datum of a gluing functor (4.9.1) is equivalent to that of a *sheaf* on E'' with values in $\mathrm{Pro}(E')^0$:

(4.9.4) $$G \in \mathbf{Faisc}(E'', \mathrm{Pro}(E')^0), G : E''^0 \longrightarrow \mathrm{Pro}(E'^0),$$

or equivalently to that of a functor $g = G^0$,

(4.9.5) $$g : E'' \longrightarrow \mathrm{Pro}(E')$$

that *commutes with inductive limits*. If C is a generating subcategory of E'', endowed with the induced topology, we know (II 6.10) that the datum of G is also equivalent (up to unique isomorphism) to that of a sheaf on C with values in $\mathrm{Pro}(E')^0$

(4.9.6) $$G_C \in \mathbf{Faisc}(C, \mathrm{Pro}(E')^0), G_C : C^\circ \longrightarrow \mathrm{Pro}(E')^0,$$

or, equivalently, to that of a functor $g_C = G_C^0$,

(4.9.7) $$g_C : C \longrightarrow \mathrm{Pro}(E'),$$

247 satisfying the familiar left-exactness conditions (II 6.2.1)). Of course, g_C is simply the *restriction of* g (4.9.5) *to* C. The case of greatest interest to us is that in which we take $C = E''_{\mathrm{coh}}$, giving the objects

(4.9.8) $$G_0 \in \mathbf{Faisc}(E''_{\mathrm{coh}}, \mathrm{Pro}(E')^0), G_0 : E''^0_{\mathrm{coh}} \longrightarrow \mathrm{Pro}(E')^0,$$

(4.9.9) $$g_0 = G_0^0 : E''_{\mathrm{coh}} \longrightarrow \mathrm{Pro}(E'),$$

each of which is again equivalent to the datum of f.

4.9.10. Suppose that the full generating subcategory C of E'' consists of quasi-compact objects and is stable in E'' under finite sums, passage to the quotient by equivalence relations, and fiber products; this is the case, for example, for $C = E''_{\mathrm{coh}}$ (1.15 and 1.17.1). It is then immediate, if P is a category in which finite projective limits are representable (for example $P = \mathrm{Pro}(E')^0$), that a functor $G_C : C^\circ \to P$ is a sheaf with values in P if and only if the functor $g_C = G_C^0 : C \to P^0$ commutes with finite sums and with passage to the quotient by an equivalence relation. This specifies, in particular, which objects are the sheaves (4.9.6) (and thus expresses the gluing functors $f : E' \to E''$).

With these reminders in place, we can give the solution to the problem posed in 4.9:

248 Proposition 4.10. *Let E' and E'' be two coherent topoi, and let $f : E' \to E''$ be a gluing functor* (IV 9.10), *i.e. a left exact and accessible functor. Let E be the topos deduced from it by gluing* (IV 9.16), *and denote by E'_{coh} (resp. E'_{PF}) the strictly full subcategory of E' consisting of the coherent objects (resp. the objects X such that the covariant functor* $\mathrm{Hom}(X, -)$ *represented by X commutes with small filtered inductive limits). Consider the following conditions:*

(i) *E is coherent.*

(ii) *For every object X' of E', there exists a family of morphisms*

$$v_\alpha : Y'_\alpha \longrightarrow X'$$

with target X' and with coherent objects of E' as sources, such that the family of the

$$f(v_\alpha) : f(Y'_\alpha) \longrightarrow f(X')$$

in E'' is covering.

(ii bis) *f commutes with (small) filtered inductive limits, and* (ii) *holds for every* $X' \in \mathrm{Ob}\, E'_{PF}$.

(ii ter) *The canonical functor* (IV 9.20.1)

$$\pi : \mathbf{Point}(E'') \longrightarrow \mathrm{Pro}(E') \tag{4.10.1}$$

defined by the gluing functor f factors (up to isomorphism) through $\mathrm{Pro}(E'_{\mathrm{coh}})$ *(via the fully faithful functor* $\mathrm{Pro}(E'_{\mathrm{coh}}) \to \mathrm{Pro}(E')$ *arising from the inclusion* $E'_{\mathrm{coh}} \to E'$*):*

$$\pi_0 : \mathbf{Point}(E'') \longrightarrow \mathrm{Pro}(E'_{\mathrm{coh}}). \tag{4.10.2}$$

(iii) *The functor f commutes with (small) filtered inductive limits.* 249

(iii bis) *The functor g_0 (4.9.9) factors (up to isomorphism) through a functor*

$$E''_{\mathrm{coh}} \longrightarrow \mathrm{Pro}(E'_{PF}), \tag{4.10.3}$$

via the fully faithful functor $\mathrm{Pro}(E'_{PF}) \to \mathrm{Pro}(E')$ *deduced from the inclusion* $E'_{PF} \to E'$.

(iii ter) *The canonical functor π (4.10.1) factors (up to isomorphism) as*

$$\pi_1 : \mathbf{Point}(E'') \longrightarrow \mathrm{Pro}(E'_{PF}). \tag{4.10.4}$$

(iv) *The functor g_0 (4.9.9) factors (up to isomorphism) through a functor*

$$E''_{\mathrm{coh}} \longrightarrow \mathrm{Pro}(E'_{\mathrm{coh}}). \tag{4.10.5}$$

We then have the following diagram of implications:

$$\text{(iv)} \Rightarrow \text{(i)} \Leftrightarrow \text{(ii)} \Leftrightarrow \text{(ii bis)} \Leftrightarrow \text{(ii ter)} \Rightarrow \text{(iii)} \Leftrightarrow \text{(iii bis)} \Leftrightarrow \text{(iii ter)}. \tag{4.10.6}$$

Let us immediately record the

Corollary 4.11. a) *Suppose that E' is perfect. Then all the conditions considered in 4.10 are equivalent; in particular, E is coherent if and only if f commutes with filtered inductive limits, or equivalently if and only if the functor g_0 (4.9.9) factors (up to isomorphism) through* $\mathrm{Pro}(E'_{\mathrm{coh}})$. 250

b) *For E to be perfect, it is necessary and sufficient that E' and E'' be perfect and that f (resp. g_0) satisfy the condition stated in* a).

Indeed, a) follows from the fact that E' perfect means $E'_{\mathrm{coh}} = E'_{PF}$, so that (iii bis) implies (iv). Assertion b) follows, as is clear from the preliminary remarks of 4.9.

4.12. Proof of 4.10.

a) Let us spell out the condition equivalent to (i) obtained in 4.9.3, namely that the full subcategory E_0 of E is generating, i.e. that for every object $X = (X', X'', u : X'' \to f(X'))$ of E, the family of all morphisms

$$v = (v', v'') : Y = (Y', Y'', v : Y'' \longrightarrow f(Y')) \longrightarrow X,$$

with $Y \in \mathrm{Ob}\, E_0$, i.e. with Y' and Y'' coherent, is epimorphic. Since the pair of functors (i^*, j^*) is conservative, this also means that the family of corresponding morphisms $Y' \to X'$ is epimorphic in E', and that the family of corresponding morphisms $Y'' \to X''$ is epimorphic in E''. This is clear for the first family, as is seen by taking $Y \in \mathrm{Ob}\, E_0$ *over U*, i.e. such that $Y'' = \varnothing_{E''}$: one obtains the family of all arrows in E' with target X' and coherent source, which is indeed epimorphic because E' is coherent and hence the subcategory E'_{coh} is generating. It therefore remains to express the condition that the family $Y'' \to X''$ be epimorphic for every given object X of E.

251 b) Now suppose that condition (ii) is satisfied for the object X' under consideration. It follows that one can find an epimorphic family of morphisms $Y''_\beta \to X''$ whose composites with $u : X'' \to f(X')$ each lift to a morphism $Y''_\beta \to f(Y'_\alpha)$ for a suitable $\alpha = \phi(\beta)$. Since E''_{coh} is a generating subcategory of E'', we may moreover suppose the Y''_β coherent. The commutative diagrams

$$\begin{tikzcd} Y''_\beta \arrow[r] \arrow[d] & f(Y'_{\varphi(\beta)}) \arrow[d] \\ X'' \arrow[r] & f(X') \end{tikzcd}$$

then provide the required family of morphisms in E, giving an epimorphic family $Y''_\beta \to X''$. Thus (ii) $\Rightarrow$ (i).

Conversely, suppose (i), and let us prove (ii). Apply the condition made explicit in a) to the object

$$X = j_*(X') = (X', f(X'), \mathrm{id} = f(X') \xrightarrow{\sim} f(X')).$$

Since the $Y'' \to X'' \simeq f(X')$ form an epimorphic family, a fortiori so does the family of $f(v') : f(Y') \to f(X')$. The conclusion follows, since the Y' are coherent by hypothesis. Thus

$$\text{(i)} \Leftrightarrow \text{(ii)}.$$

252 c) Note now that condition (iv) also means that for every object X'' of E'', the pro-representable functor

$$g(X'') : X' \longmapsto \mathrm{Hom}(X'', f(X'))$$

on E' is pro-represented by a pro-object whose components belong to E'_{coh}. Equivalently, in the category $E'_{/g(X'')}$ formed by pairs consisting of an object X' of E' and an element $u \in g(X'')(X') = \mathrm{Hom}(X'', f(X'))$, $u : X'' \to f(X')$, there is a cofinal set of objects (X'_α, u_α), with $u_\alpha : X'' \to f(X'_\alpha)$ and $X'_\alpha \in \mathrm{Ob}\, E'_{\mathrm{coh}}$. Since the category $E'_{/g(X'')}$ is filtered, the cofinality condition simply means that for every object (X', u), $u : X'' \to f(X')$, of $E'_{/g(X'')}$, there is a morphism from one of the (X'_α, u_α) to this object, i.e. a morphism $v'_\alpha : X'_\alpha \to X'$ making the triangle

(∗)
$$\begin{tikzcd} & f(X'_\alpha) \arrow[d, "f(v'_\alpha)"] \\ X'' \arrow[ur, dash, "u_\alpha"] \arrow[r] & f(X'). \end{tikzcd}$$

commutative. Thus condition (iv) is equivalent to the following:

(iv bis) For every arrow $u : X'' \to f(X')$, with $X' \in \mathrm{Ob}\, E'$ and $X'' \in \mathrm{Ob}\, E''_{\mathrm{coh}}$, there is an arrow $v' : Y' \to X'$ in E', with Y' coherent, such that u factors through $f(v') : f(Y') \to f(X')$.

It is clear that condition (iv bis) implies (ii), since for fixed X' the family of $u : X'' \to f(X')$ whose source is coherent is epimorphic. A fortiori, the same is therefore true of the family of all $f(v'_\alpha)$ in the corresponding diagrams $(*)$. Thus

$$\text{(iv)} \Rightarrow \text{(i)}.$$

d) Let us spell out condition (ii ter). Let p be a point of E''. By definition, $\pi(p)$ is the pro-object of E' which pro-represents the functor $h : X' \longmapsto f(X')_p$, or, equivalently, the pro-object defined by the canonical functor $E'_{/h} \to E'$, where $E'_{/h}$ denotes the category of pairs (X', u) with $X' \in \operatorname{Ob} E'$ and $u \in h(X')$, i.e. $u \in f(X')_p$. To say that this pro-object is isomorphic to a pro-object coming from $\operatorname{Pro}(E'_{\text{coh}})$ means that the subcategory of $E'_{/h}$ formed by the (X', u) with $X' \in \operatorname{Ob} E'_{\text{coh}}$ is cofinal. In other words, for every object (X', u), $u \in f(X')_p$, of $E'_{/h}$, there is an object (X'_0, u_0) mapping to it, with X'_0 coherent; that is, there is a morphism $X'_0 \to X'$ (X'_0 coherent) such that $u \in \operatorname{Im}(f(X'_0) \to f(X'))$. To say that this holds for every point p means that, for every object X' of E', the family of maps $f(X'_0) \to f(X')$, with X'_0 coherent, is surjective on every stalk. As we shall see in the appendix (10) that a coherent topos E'' has enough points, the resulting condition also says that, for every $X' \in \operatorname{Ob} E'$, the family $f(X'_0) \to f(X')$ is epimorphic. This is precisely condition (ii), and hence

$$\text{(ii)} \Leftrightarrow \text{(ii ter)}.$$

e) The condition that f commute with filtered inductive limits (condition (iii)) 253
is equivalent, since the family of fiber functors of E'' is conservative, to the condition that the composite functors $h : X' \longmapsto f(X')_p$ do so. Let us prove that this means that the pro-object $\pi(p)$ which pro-represents this functor lies in the essential image of $\operatorname{Pro}(E'_{PF})$. This will prove the implications

$$\text{(iii)} \Leftrightarrow \text{(iii ter)} \Leftarrow \text{(ii ter)} \Rightarrow \text{(iii bis)}.$$

We are therefore led to prove the following lemma.

Lemma 4.12.1. *Let E' be a topos in which the subcategory E'_{coh} is generating, and let $X' \in \operatorname{Ob}\operatorname{Pro}(E')$. Then X' belongs to the essential image of* $\operatorname{Pro}(E'_{PF})$ *if and only if the functor $h : E' \to$* (Ens) *which it pro-represents commutes with filtered inductive limits.*

Since a filtered inductive limit of functors commuting with filtered inductive limits has the same property, sufficiency follows from the definition of E'_{PF}. For necessity, we must prove that if the functor h commutes with filtered inductive limits, then in the filtered category $E'_{/h}$ of pairs (X', u) with $u \in h(X')$, the subcategory formed by the pairs (X', u) with $X' \in \operatorname{Ob} E'_{PF}$ is cofinal. That is, for every (X', u) in $E'_{/h}$, there is a morphism $X'_0 \to X'$, with $X'_0 \in \operatorname{Ob} E'_{PF}$, such that $u \in \operatorname{Im}(h(X'_0) \to h(X'))$. But X' is known to be a filtered inductive 254
limit of objects X'_i of E'_{PF} (1.24.2 a)), so $h(X')$ is the filtered inductive limit of the $h(X'_i)$, which proves the assertion.

f) It remains only to prove the implications

$$\text{(ii bis)} \Rightarrow \text{(ii)} \quad \text{and} \quad \text{(iii)} \Leftrightarrow \text{(iii bis)}.$$

The first implication follows trivially from the fact that every object X' of E' is a filtered inductive limit of objects X'_i of E'_{PF}. For the second, it is enough to note that the family of functors

$$Y'' \longmapsto \mathrm{Hom}(X'', Y''),$$

for $X'' \in \mathrm{Ob}\, E''_{\mathrm{coh}}$, is conservative and consists of functors commuting with filtered inductive limits (1.23 (ii)). Consequently, the functor $f : E' \to E''$ commutes with filtered inductive limits if and only if the composite functors $X' \longmapsto \mathrm{Hom}(X'', f(X')) = \mathrm{Hom}_{\mathrm{Pro}(E')}(g_0(X''), X')$ do so. By 4.12.1, this is equivalent to the fact that the $g_0(X'')$ lie in the essential image of $\mathrm{Pro}(E'_{PF})$.
C.Q.F.D.

Remark 4.13. It is useful to state in sheaf-theoretic language the meaning of condition 4.10 (iv). Consider the commutative diagram of functors

(4.13.1)
$$\begin{tikzcd} \mathrm{Pro}(E'_{\mathrm{coh}})^{\circ} \arrow[r, hook, "\alpha"] \arrow[d, "\beta_0"] & \mathrm{Pro}(E')^{\circ}_{\beta} \arrow[d, "\beta"] \\ \check{E}'_{\mathrm{coh}} & \check{E}' \arrow[l, "\gamma"'] \end{tikzcd} ,$$

255 where the superscript circles denote opposite categories, and the second horizontal arrow γ is restriction of functors, all the other arrows denoting the familiar fully faithful functors. Note that the inclusion arrow α *does not in general commute with projective limits*, even finite ones. Thus, in general, a sheaf F on a site C (such as E'') defines a presheaf $\alpha \circ F$ on the same site, with values in $\mathrm{Pro}(E'')^0$, which need not be a sheaf. Nevertheless, if a *presheaf* F on C with values in $\mathrm{Pro}(E'_{\mathrm{coh}})^0$ is such that the corresponding presheaf αF with values in $\mathrm{Pro}(E')^0$ is a sheaf, then F itself is a sheaf. Indeed, the sheaf condition is a left-exactness property (II 6.2.1)), and the functors β, β_0, and γ commute with small projective limits.

On the other hand, the characterization in 4.9.10 of sheaves on E''_{coh} with values in an arbitrary category P shows that, for every functor $\alpha : P \to Q$ such that $\alpha^0 :$
256 $P^0 \to Q^0$ commutes with *finite coproducts* and with *passage to quotients by equivalence relations*, the functor $g \mapsto \alpha \circ g$ from $\mathrm{Hom}(E''^0_{\mathrm{coh}}, P)$ to $\mathrm{Hom}(E''^0_{\mathrm{coh}}, Q)$ takes sheaves to sheaves. We also know that in $\mathrm{Pro}(E'_{\mathrm{coh}})$ and in $\mathrm{Pro}(E')$, finite coproducts and quotients by equivalence relations are representable, and that the inclusion $E'_{\mathrm{coh}} \to E'$ commutes with them (1.15 and 1.17.1). The same is therefore true for the categories $\mathrm{Pro}(E'_{\mathrm{coh}})$ and $\mathrm{Pro}(E')$ and for the inclusion between them (I 8.9.5 b) and 8.9.7), which is precisely α^0, where α is the functor occurring in (4.13.1). It follows that if

(4.13.2)
$$\varphi : E''^0_{\mathrm{coh}} \longrightarrow \mathrm{Pro}(E'_{\mathrm{coh}})^0$$

is a functor, then φ *is a sheaf on* E''_{coh} *with values in* $\mathrm{Pro}(E'_{\mathrm{coh}})^0$ *if and only if* $\alpha \circ \varphi$ *is a sheaf on* E''_{coh} *with values in* $\mathrm{Pro}(E')^0$. Thus the category of gluing functors $f : E' \to E''$ satisfying condition 4.10 (iv) is canonically equivalent to the category of sheaves φ on E''_{coh} with values in $\mathrm{Pro}(E'_{\mathrm{coh}})^0$. Taking 4.10 into account, we conclude that such a sheaf φ canonically defines a gluing functor (up to unique isomorphism), and that the topos E obtained from the latter is *coherent*. Moreover (4.11), *if* E' *is perfect*, one thereby obtains *all* coherent topoi that can be obtained from E' and E'' by gluing.

257 Note also that E'_{coh} is equivalent to a small category and is stable under finite projec-

tive limits. Consequently (I 8.10.14), by the fully faithful functor β_0 of (4.13.1), $\mathrm{Pro}(E'_{\mathrm{coh}})^0$ is equivalent to the full subcategory of $\widehat{E'_{\mathrm{coh}}} = \mathrm{Hom}(E'_{\mathrm{coh}}, (\mathrm{Ens}))$ formed by the left exact functors. Thus the data of a sheaf (4.13.2) are equivalent to those of a functor

$$F : E''^{0}_{\mathrm{coh}} \times E'_{\mathrm{coh}} \longrightarrow (\mathrm{Ens}) \tag{4.13.3}$$

which is a sheaf in the first argument and left exact in the second; or, equivalently, to those of a functor

$$f_0 : E'_{\mathrm{coh}} \longrightarrow E'' \tag{4.13.4}$$

which is left exact. If f is the gluing functor satisfying condition 4.10 (iv) associated with φ, one immediately verifies that f_0 is canonically isomorphic to the restriction of $f : E' \to E''$. Thus *under the conditions being considered, f_0 determines f* (up to canonical isomorphism). To summarize the content of 4.11, we finally state:

Corollary 4.14. *When E' is a perfect topos, the category of gluing functors $f : E' \to E''$ which give a coherent topos E by gluing is equivalent, under $f \mapsto f_0 = f|E'_{\mathrm{coh}}$, to the category of left exact functors $f_0 : E'_{\mathrm{coh}} \to E''$, or equivalently to the category of sheaves* 258
φ on the site E''_{coh} (or, what amounts to the same thing, on the topos E'') with values in the category $\mathrm{Pro}(E'_{\mathrm{coh}})^0$ (cf. 4.9.10).

The first assertion can also be deduced from 4.10 by observing that $E' \approxeq \mathrm{Ind}(E'_{\mathrm{coh}})$ (1.25). Indeed, restriction gives an equivalence between the category of functors $f : E' \to P$ commuting with filtered inductive limits (where P is a category in which small filtered inductive limits are representable) and the category of arbitrary functors $f : E'_{\mathrm{coh}} \to P$. It is immediate that, under these conditions, if finite projective limits are representable in P, then f is left exact if and only if f_0 is.

Remarks 4.15.
a) The proof given for 4.10 shows that conditions equivalent to (ii ter) resp. (iii ter) are obtained by replacing the category $\mathbf{Point}(E'')$ by a subcategory which defines a *conservative* family of fiber functors.
b) Taking for E'' the point topos (IV 2.2), one sees immediately that conditions (ii ter) and (iii ter) are equivalent *only if* $E'_{\mathrm{coh}} = E'_{PF}$, i.e. if E' is perfect. Thus the condition "E is coherent" is equivalent to the condition "f commutes with filtered inductive limits" only if E' is perfect. On the other hand, taking more generally for E'' a coherent topos of the form $\widehat{C}$, one sees that *in this case condition* (i) *is equivalent to* (iv) *for every E'*. In fact, in the general case the editor has been unable to construct an example in which E is coherent without condition (iv) being satisfied. This is shameful!

5. Commutation of the functors $H^i(X,-)$ with filtered inductive limits 259

Theorem 5.1. *Let E' be an algebraic topos (2.3), let E be a locally coherent topos (2.3), and let $f : E' \to E$ be a coherent morphism of topoi (3.1). Then, for every integer q, the functors $R^q f_*$* (V 5.0) *commute with filtered inductive limits of abelian sheaves.*

Corollary 5.2. *Let E' be a coherent topos (2.3). For every integer q, the functor $H^q(E',-)$ commutes with filtered inductive limits of abelian sheaves.*

Corollary 5.3. *Let E be a topos (resp. an algebraic topos) (2.3), and let X be an algebraic and coherent (resp. coherent) object of E (2.3). For every integer q, the functor $H^q(X,-)$ commutes with filtered inductive limits of abelian sheaves.*

Corollary 5.2 follows from 5.1 by taking for E the point topos and for $f : E' \to E$ the unique morphism (IV 2.2). Let us show that 5.2 implies 5.3. Let $j_X : E_{/X} \to E$ be the localization morphism. There is a canonical isomorphism, functorial in the abelian sheaf F (V 2.2.1), $H^q(X,F) \simeq H^q(E_{/X}, j_X^* F)$. The functor j_X^* commutes with inductive limits, and the topos $E_{/X}$ is coherent (2.4.6), whence 5.3. Let us show that 5.3 implies 5.1. Let E_{cohalg} be the full subcategory of E defined by the algebraic and coherent objects of E. It is a generating category (2.4.5). For every abelian sheaf F, $R^q f_* F$ is the sheaf associated with the presheaf $X \mapsto H^q(f^*X, F)$ ($X \in \operatorname{ob} C$) (V 5.1). Since f is coherent, f^*X is coherent for every object X of C (3.1). By 5.3, the presheaf $X \mapsto H^q(f^*X, F)$ commutes with filtered inductive limits in the argument F, and 5.1 follows, since the associated-sheaf functor commutes with inductive limits. (Only the following property of the morphism f is used: there is a generating family S of E such that, for every X of S, $f^*(X)$ is algebraic and coherent.) It therefore remains to prove 5.2. We already know that the functor $H^\circ(E', -)$ commutes with filtered inductive limits of abelian sheaves (1.2.3). By a classical result of [7], to prove 5.2 it is enough to show that a filtered inductive limit of sheaves acyclic for the functor $H^\circ(E', -)$ is acyclic for $H^\circ(E', -)$. This follows from the next lemma, applied with $C = E'_{\mathrm{coh}}$.

260 Lemma 5.4. *Let E be a locally coherent topos (2.3), and let C be a full generating subcategory of E, stable under fiber products, all of whose objects are quasi-compact (1.1). A filtered inductive limit of C-acyclic sheaves* (V 4.2) *is a C-acyclic sheaf.*

Let $(F_i)_{i\in I}$ be a filtered inductive system of C-acyclic sheaves, and denote its inductive limit by F. Every object Y of C is coherent (2.1), and consequently $F(Y) = \varinjlim_i F_i(Y)$ (1.2.3). To show that F is C-acyclic, it is enough to show that $\check{H}^q(Y, F) = 0$ for every $q > 0$ and every Y of C (V 4.3). Let X be an object of C, and let $\mathscr{X} = (X_\alpha \to X)_{\alpha\in A}$ be a finite covering family by objects of C. We have $H^q(\mathscr{X}, F) = H^q(C^\cdot(\mathscr{X}, F))$ (V 2.4.3). By the preceding and the fact that $C^\cdot(\mathscr{X}, F)$ involves only finite products of groups (V 2.3.3), we have $C^\cdot(\mathscr{X}, F) = \varinjlim_i C^\cdot(\mathscr{X}, F_i)$. Hence

$$H^q(\mathscr{X}, F) = H^q\big(\varinjlim_i C^\cdot(\mathscr{X}, F_i)\big) = \varinjlim_i H^q(C^\cdot(\mathscr{X}, F_i)) = 0$$

for $q > 0$ (V 4.3). Consequently $\check{H}^q(X, F) = \varinjlim_{\mathfrak{X}} H^q(\mathfrak{X}, F) = 0$.

Corollary 5.5. *Let E be a coherent topos (2.3), and let $(X_i)_{i\in I}$ be a decreasing filtered family of coherent subobjects of the final object. Denote by Φ the family of closed subtopoi of E* (IV 9) *contained in the closed complement of one of the X_i. Then Φ is a family of supports of E* (V 6.12)*, and for every integer q the functors $H^q_\Phi(E, -)$ and $\mathscr{H}^q_\Phi$* (V 6.13) *commute with filtered inductive limits.*

Let Z_i be the closed complement of X_i. For every abelian sheaf F, there is an exact sequence (V 6.5):

$$0 \longrightarrow H^0_{Z_i}(E,F) \longrightarrow H^0(E,F) \longrightarrow H^0(X_i,F) \longrightarrow H^1_{Z_i}(E,F) \longrightarrow \dots \quad .$$

The functors $H^q(E,F)$ and $H^q(X_i,F)$ commute with filtered inductive limits in F (5.2, 5.3). Consequently $H^q_{Z_i}(E,F)$ does so as well, whence the analogous property for $H^q_\Phi(E,F)$ after taking the inductive limit over the Z_i (V 6.13). The assertion concerning the functors $\mathscr{H}^q_\Phi$ follows by localization and passage to the associated sheaf.

261 Definition 5.6. Let E be a ringed topos with ring A, and let q be an integer. An

A-module G is said to be of finite q-presentation if there is a family X_i of objects of E covering the final object of E such that, for every i, there is an exact sequence of modules on $E_{/X_i}$:

$$L_q \longrightarrow L_{q-1} \longrightarrow \cdots L_1 \longrightarrow L_0 \longrightarrow G_{/X_i} \longrightarrow 0 \tag{5.5.1}$$

where, for every p, $L_p = A_{/X_i}^{n_p}$, with n_p an integer.

Proposition 5.7. *Let G be a sheaf of finite q-presentation. For every $i \leq q-1$, the functor $\mathbf{Ext}^i_A(G,-)$ commutes with filtered inductive limits of A-modules.*

The problem is local on E (V 6.1). Thus, after localizing to the $E_{/X_i}$, we may suppose that there is an exact sequence of the form (5.5.1). Put $L^{\cdot} = L_q \to \cdots \to L_1 \to L_0$. We have $\mathbf{Ext}^i_A(G,F) \simeq \mathscr{H}^i(\mathbf{Hom}_A(L^{\cdot},F))$, and the functor $F \mapsto \mathbf{Hom}_A(L^{\cdot},F)$ commutes with inductive limits, which proves the proposition.

Corollary 5.8. *Let (E,A) be a ringed topos, let G be a sheaf of A-modules of finite q-presentation, and let X be an algebraic and coherent object of E. The functors $F \mapsto \mathrm{Ext}^i_A(X;G,F)$, for $\frac{i(i-1)}{2} \leq q-1$, commute with filtered inductive limits in F.*

This follows from 5.7 and 5.3 by the spectral sequence of V 6.1.3.

5.9. Let (E,A) be a ringed topos, let Φ be a family of supports of E (V 6.12), let G be an A-module of finite r-presentation (5.6), and let X be an algebraic and coherent object of E (2.3). Taking the inductive limit over the closed subtopoi in Φ in the last spectral sequences of V 6.9.1 and V 6.9.2, respectively, gives two spectral sequences (V 6.13):

$$\begin{cases} {}'E_2^{pq} = \varinjlim_{Z\in\Phi} \mathrm{Ext}^p_A(X;G,H^q_Z F) \Rightarrow \mathrm{Ext}^{p+q}_{A,\Phi}(X;G,F), \\ {}''E_2^{pq} = \varinjlim_{Z\in\Phi} \mathbf{Ext}^p_A(G,H^q_Z F) \Rightarrow \mathbf{Ext}^{p+q}_{A,\Phi}(G,F). \end{cases} \tag{5.9.1}$$

It follows from 5.7 and 5.8 that there are canonical isomorphisms

$$\begin{cases} {}'E_2^{pq} \simeq \mathrm{Ext}^p_A(X;G,H^q_\Phi F) \quad , \quad \frac{p(p-1)}{2} \leq r-1, \\ {}''E_2^{pq} \simeq \mathbf{Ext}^p_A(G,H^q_\Phi F) \quad , \quad p \leq r-1. \end{cases} \tag{5.9.2}$$

Corollary 5.10. *Use the hypotheses and notation of 5.5. Let A be a ring of E, and let* 262 *G be an A-module of finite q-presentation. For every integer i such that $\dfrac{i(i-1)}{2} \leq q-1$, the functors $F \mapsto \mathrm{Ext}^i_{A,\Phi}(E;G,F)$ and $F \mapsto \mathbf{Ext}^i_{A,\Phi}(G,F)$ (V 6.13) commute with filtered inductive limits.*

This follows from 5.5 and 5.7 by the first spectral sequences of V 6.9.1 and V 6.9.2, respectively.

6. Inductive and projective limits of a fibered category

6.0. We assume that the reader is familiar with the theory of fibered categories [5]. The essential purpose of this section is to fix the terminology and notation concerning fibered categories. Unless otherwise stated, all categories considered in this section belong to a fixed universe $\mathscr{U}$.

Let $\mathscr{F}$, $\mathscr{G}$, and $\mathscr{E}$ be three categories, and let $\pi : \mathscr{F} \to \mathscr{E}$ and $\pi' : \mathscr{G} \to \mathscr{E}$ be two functors. We denote by $\mathbf{Hom}_{\mathscr{E}}(\mathscr{F}, \mathscr{G})$ the category whose objects are the functors $u : \mathscr{F} \to \mathscr{G}$ such that the diagram

$$\begin{tikzcd}
\mathscr{F} \arrow[rr, dash, "u"] \arrow[dr, "\pi"'] & & \mathscr{G} \arrow[dl, "\pi'"] \\
& \mathscr{E} &
\end{tikzcd}$$

commutes, and whose morphisms are the $\mathscr{E}$-morphisms of functors, i.e. the morphisms of functors carried by π' to identity morphisms.

Let ξ be an object of $\mathscr{E}$. The objects X of $\mathscr{F}$ such that $\pi(X) = \xi$ are called *the objects of $\mathscr{F}$ over ξ*. Let $f : \xi \to \eta$ be a morphism of $\mathscr{E}$. The morphisms m of $\mathscr{F}$ such that
263 $\pi(m) = f$ are called the morphisms of $\mathscr{F}$ *over f*. Let $f : \xi \to \eta$ be a morphism of $\mathscr{E}$, let X be an object of $\mathscr{F}$ over ξ, and let Y be an object of $\mathscr{F}$ over η; we denote by $\mathrm{Hom}_f(X, Y)$ the set of morphisms from X to Y over f.

Definition 6.1. 1) *A morphism $m : X \to Y$ of $\mathscr{F}$ is called* Cartesian *if, for every morphism $p : Z \to Y$ over $\pi(m)$, there exists a unique morphism $q : Z \to X$ over the identity of $\pi(X)$ such that $mq = p$.*

2) *The category $\mathscr{F}$ is called* prefibered *by π over $\mathscr{E}$ if, for every morphism $f : \xi \to \eta$ of $\mathscr{E}$, every object of $\mathscr{F}$ over η is the target of a Cartesian morphism over f.*

3) *The category $\mathscr{F}$ is called* fibered *by π over $\mathscr{E}$ if it is prefibered and the composite of any two composable Cartesian morphisms of $\mathscr{F}$ is Cartesian.*

6.1.1. Let ξ be an object of $\mathscr{E}$. The *fiber category* of $\mathscr{F}$ at ξ, denoted by $\mathscr{F}_\xi$, is the subcategory of $\mathscr{F}$ whose objects are the objects of $\mathscr{F}$ over ξ and whose morphisms are the morphisms of $\mathscr{F}$ over id_ξ. For two objects X and Y of $\mathscr{F}_\xi$, we denote by $\mathrm{Hom}_\xi(X, Y)$ the set of morphisms from X to Y in $\mathscr{F}_\xi$.

6.1.2. Let $f : \xi \to \eta$ be a morphism of $\mathscr{E}$. An object Y over η is the target of a Cartesian morphism over f if and only if the set-valued functor on the fiber $\mathscr{F}_\xi$

$$Z \mapsto \mathrm{Hom}_f(Z, Y) \quad Z \in \mathrm{ob}(\mathscr{F}_\xi),$$

is representable. When the category $\mathscr{F}$ is prefibered, the functor

$$Y \mapsto \mathrm{Hom}_f(., Y) \quad Y \in \mathrm{ob}(\mathscr{F}_\eta),$$

with values in the category of presheaves of sets on $\mathscr{F}_\xi$, in fact takes its values in the category of representable functors on $\mathscr{F}_\xi$, and therefore defines, up to unique isomorphism,
264 a functor $f^* : \mathscr{F}_\eta \to \mathscr{F}_\xi$, called the *inverse image functor for f*. Suppose that $\mathscr{F}$ is prefibered and choose, for every morphism f of $\mathscr{E}$, a base-change functor f^*. The category $\mathscr{F}$ is then fibered over $\mathscr{E}$ if and only if, for every pair f, g of composable morphisms of $\mathscr{E}$, the composite functor f^*g^* is an inverse image functor. Let

$$C_{f,g} : f^*g^* \longrightarrow (gf)^* \tag{6.1.2.1}$$

be the canonical isomorphism. The isomorphisms $C_{f,g}$ satisfy a cocycle condition arising from the associativity of composition of morphisms in $\mathscr{E}$:

$$C_{f,hg}(f^* \circ C_{g,h}) = C_{gh,f}(C_{f,g} \circ h^*) \tag{6.1.2.2}$$

6.1.3. Conversely, suppose that for every object ξ of $\mathscr{E}$ one is given a category $\mathscr{F}_\xi$, for every morphism $f : \xi \to \eta$ a functor $f^* : \mathscr{F}_\eta \to \mathscr{F}_\xi$, and for every pair (f, g) of morphisms of $\mathscr{E}$ an isomorphism of functors $C_{f,g} : f^*g^* \to (fg)^*$, such that the $C_{f,g}$ satisfy condition (6.1.2.2). One can then construct, in an essentially unique way, a category $\mathscr{F}$ fibered over $\mathscr{E}$ whose fibers "are" the categories $\mathscr{F}_\xi$ and whose base-change functors may be chosen to be "equal" to the functors given in advance [5].

6.1.4. Let $\mathscr{F}$ and $\mathscr{G}$ be two categories over $\mathscr{E}$. We denote by

$$\mathbf{Hom}_{\mathrm{cart}/\mathscr{E}}(\mathscr{F}, \mathscr{G})$$

the full subcategory of $\mathbf{Hom}_{\mathscr{E}}(\mathscr{F}, \mathscr{G})$ defined by the functors that carry Cartesian morphisms of $\mathscr{F}$ to Cartesian morphisms of $\mathscr{G}$. The objects of $\mathbf{Hom}_{\mathrm{cart}/\mathscr{E}}(\mathscr{F}, \mathscr{G})$ are called *Cartesian functors*.

Let C and C' be two categories, and let S be a set of morphisms of the category C. We denote by $\mathrm{Hom}_{S^{-1}}(C, C')$ the set of functors from C to C' that carry the morphisms of S to isomorphisms.

Let (Cat) denote the category whose objects are the categories belonging to the universe and whose morphisms are the functors between these categories (the category (Cat) does not belong to the universe).

Let $\mathscr{F}$ be a category over $\mathscr{E}$, and let S be the set of Cartesian morphisms of $\mathscr{F}$. The category $\mathscr{F}$ defines two functors on (Cat) with values in the category of sets belonging to the universe: **265**

$$C \mapsto \mathrm{Hom}_{S^{-1}}(\mathscr{F}, C) \quad C \in \mathrm{ob(Cat)}$$

$C \mapsto \mathrm{Hom}_{\mathrm{Cart}/\mathscr{E}}(\mathscr{F}, C \times \mathscr{E}) = \{$Cartesian functors from $\mathscr{F}$ to $C \times \mathscr{E}\}$
(the category $C \times \mathscr{E}$ is regarded as a category over $\mathscr{E}$ by the second projection functor).

Proposition 6.2. *The functors* (Cat) $\to$ (Ens) :

$$C \mapsto \mathrm{Hom}_{S^{-1}}(\mathscr{F}, C)$$
$$C \mapsto \mathrm{Hom}_{\mathrm{Cart}/\mathscr{E}}(\mathscr{F}, C \times \mathscr{E})$$

are canonically isomorphic. They are representable.

Proof. To prove the first assertion, it is enough to observe that the Cartesian morphisms of $C \times \mathscr{E}$ are the morphisms of the form $m \times f$, where m is an isomorphism of C. To prove the second assertion, it is enough to prove that the functor $\mathrm{Hom}_{S^{-1}}(\mathscr{F}, .)$ is representable. This follows from [1]. We indicate only the idea of the proof. One formally adjoins to the morphisms of $\mathscr{F}$ inverses of the morphisms of S. One considers the free category generated by the objects of $\mathscr{F}$, the morphisms of $\mathscr{F}$, and the formal inverses of the morphisms of S (the path category). One then takes the quotient by the relations arising from the relations among morphisms of $\mathscr{F}$ and relations of the form

$$s^{-1}s = \mathrm{id} \quad , \quad ss^{-1} = \mathrm{id} \quad s \in S.$$

The category thus obtained is denoted by $\mathscr{F}(S^{-1})$; equipped with the canonical functor $Q : \mathscr{F} \to \mathscr{F}(S^{-1})$, it represents the functor $\mathrm{Hom}_{S^{-1}}(\mathscr{F}, .)$.

Definition 6.3. *When $\mathscr{F}$ is fibered over $\mathscr{E}$, the functor* $\mathrm{Hom}_{S^{-1}}(\mathscr{F}, .)$ *is denoted by $\varinjlim_{\mathscr{E}^\circ} \mathscr{F}$ and is called the inductive-limit functor of $\mathscr{F}$ over $\mathscr{E}^\circ$. The category representing it is likewise denoted by $\varinjlim_{\mathscr{E}^\circ} \mathscr{F}$ and is called the inductive-limit category of $\mathscr{F}$ over $\mathscr{E}^\circ$.*

6.4.0. Suppose that $\mathscr{F}$ is fibered over $\mathscr{E}$. For every morphism f of $\mathscr{E}$, choose a base-change functor over f, and let

$$Q : \mathscr{F} \longrightarrow \varinjlim_{\mathscr{E}^\circ} \mathscr{F}$$

266 be the canonical functor. Composing the inclusion functors of the fibers of $\mathscr{F}$ into $\mathscr{F}$ with Q gives, for every object ξ of $\mathscr{E}$, a functor

$$u_\xi : \mathscr{F}_\xi \longrightarrow \varinjlim_{\mathscr{E}} \mathscr{F}$$

and, for every morphism $f : \xi \to \eta$ of $\mathscr{E}$, a diagram that commutes up to canonical isomorphism:

$$\begin{tikzcd}
\mathscr{F}_\eta \arrow[dd, "f^*"'] \arrow[dr, "u_\eta"] & \\
& \varinjlim_{\mathscr{E}^0} \mathscr{F} \\
\mathscr{F}_\xi \arrow[ur, "u_\xi"'] &
\end{tikzcd}$$

The category $\varinjlim_{\mathscr{E}^\circ} \mathscr{F}$ then appears as the inductive limit *in the sense of pseudofunctors* [5] of the pseudofunctor $\mathscr{E}^\circ \to \mathrm{Cat}$ that associates $\mathscr{F}_\xi$ with every $\xi \in \mathrm{ob}$, and the functor $f^* : \mathscr{F}_\eta \to \mathscr{F}_\xi$ with every $f : \xi \to \eta$. Note, however, that even when $\xi \mapsto \mathscr{F}_\xi$ is an actual functor, the category $\varinjlim_{\mathscr{E}^\circ} \mathscr{F}$ is not in general the inductive limit in the sense of I 2 of the functor $\xi \mapsto \mathscr{F}_\xi$ (cf. 6.8).

Proposition 6.4. *Let $\mathscr{F}$ be a category fibered over $\mathscr{E}$. Suppose that the category $\mathscr{E}$ has the following properties:*

L1) *Every diagram*

$$\begin{tikzcd}
\cdot \arrow[dr] & \\
& \cdot \\
\cdot \arrow[ur] &
\end{tikzcd}$$

can be embedded in a commutative diagram

$$\begin{tikzcd}
& \cdot \arrow[dr] & \\
\cdot \arrow[ur] \arrow[dr] & & \cdot \\
& \cdot \arrow[ur] &
\end{tikzcd}$$

267 L2) *For every pair of morphisms $u, v : \cdot \rightrightarrows \cdot$ such that there exists a morphism t satisfying $tu = tv$, there exists a morphism w satisfying $uw = vw$.*

(Note that if the category $\mathscr{E}^\circ$ is pseudofiltered (I 2.7)*, it has properties* L1) *and* L2)*.) The set S of Cartesian morphisms of $\mathscr{F}$ then has the following properties:*

Fr1) *The set S is stable under composition. Isomorphisms belong to S.*

Fr2) *Every diagram*

$$\begin{tikzcd} \cdot \arrow[dr] & \\ & \cdot \\ \cdot \arrow[ur, "s"] & \end{tikzcd},$$

where s belongs to S, can be completed to a commutative diagram

$$\begin{tikzcd} & \cdot \arrow[dr] & \\ \cdot \arrow[ur, "t"] \arrow[dr] & & \cdot \\ & \cdot \arrow[ur, "s"'] & \end{tikzcd}$$

where t belongs to S.

Fr3) *For every pair of morphisms $u, v : \cdot \rightrightarrows \cdot$ such that there exists a morphism $s \in S$ satisfying $su = sv$, there exists a morphism $t \in S$ such that $ut = vt$.*

Proof. Left to the reader as an exercise.

Proposition 6.5. *Let $\mathscr{F}$ be a category, and let S be a set of morphisms of $\mathscr{F}$ having* 268
properties Fr1), Fr2), and Fr3) (6.4). *For every object X of $\mathscr{F}$, let $S(X)$ denote the category of morphisms of S with target X. The category $S(X)$ is cofiltered* (I 2.7). *The category $\mathscr{F}(S^{-1})$ (which represents the functor* $\mathrm{Hom}_{S^{-1}}(\mathscr{F}, \cdot)$*) may then be described as follows:*

a) *The objects of $\mathscr{F}(S^{-1})$ are the objects of $\mathscr{F}$.*

b) *Let X and Y be two objects of $\mathscr{F}(S^{-1})$. We have*

$$\mathrm{Hom}_{\mathscr{F}(S^{-1})}(X, Y) = \varinjlim_{S(X)} \mathrm{Hom}_{\mathscr{F}}(., Y).$$

c) *Let $f : X \to Y$ and $g : Y \to Z$ be two morphisms of $\mathscr{F}(S^{-1})$. Let*

$$\begin{tikzcd} \cdot \arrow[d, "t"'] \arrow[dr, "f'"] & \\ X & Y \end{tikzcd} \qquad t \in S$$

(resp.

$$\begin{tikzcd} \cdot \arrow[d, "t'"'] \arrow[dr, "g'"] & \\ Y & Z \end{tikzcd} \qquad t' \in S)$$

be a diagram in $\mathscr{F}$ whose image is f (resp. g). Let

$$\begin{tikzcd} \cdot \arrow[d, "t''"'] \arrow[dr, "f''"] & \\ \cdot \arrow[dr, "f'"'] & \cdot \arrow[d, "t'"] \\ & Y \end{tikzcd} \qquad t'' \in S$$

be a commutative diagram whose existence is ensured by Fr2). The image in $\mathrm{Hom}_{\mathscr{F}(S^{-1})}(X, Z)$ *of the diagram*

$$\begin{tikzcd} \cdot \arrow[d, "tt''"'] \arrow[dr, "g'f''"] & \\ X & Z \end{tikzcd}$$

is the composite morphism gf.

269 *Let*

$$Q : \mathscr{F} \longrightarrow \mathscr{F}(S^{-1})$$

be the canonical functor.

d) *The functor* $Q^* : \mathscr{F}(S^{-1})\hat{} \to \mathscr{F}\hat{}$ *(*$F \mapsto F \circ Q$ I 5.0*) is fully faithful and injective on objects.*
e) *The functor* $Q_! : \mathscr{F}\hat{} \to \mathscr{F}(S^{-1})\hat{}$ *(left adjoint to the functor* Q^* (I 5.1)*) is left exact. (Consequently, the same is true of* Q *when finite projective limits are representable in* $\mathscr{F}$*.)*

Proof. For the proof, we refer to [1].

Exercise 6.6. Let $\mathscr{F}$ be a category, and let S be a set of morphisms of $\mathscr{F}$ having properties Fr1), Fr2), and Fr3). For every object X of $\mathscr{F}$, let $S(X)$ denote the category of morphisms with target X.

a) The category $S(X)$ is cofiltered (I 2).
b) For every object X of $\mathscr{F}$, let $J_S(X)$ denote the set of sieves of X that contain a sieve generated by a morphism of S. The $J_S(X)$ define a topology T on $\mathscr{F}$.
c) For the topology T, the morphisms of S are bicovering (II 5.2).
d) For every object X of $\mathscr{F}$, let $\mathscr{J}^2(X)$ denote the category of bicovering morphisms with target X. The category $S(X)$ is cofinal in $\mathscr{J}^2(X)$ (I 8).
e) Let G be a presheaf of sets on $\mathscr{F}$, and let a be the associated-sheaf functor for the topology T. For every object X of $\mathscr{F}$, there is a canonical isomorphism

$$aG(X) \xrightarrow{\sim} \varinjlim_{S(X)} G(\cdot).$$

A presheaf is a sheaf if and only if it carries every morphism of S to an isomorphism.
f) Choose the associated-sheaf functor so that its restriction to $\mathscr{F}$ is injective on objects. Let $\underrightarrow{\mathscr{F}}$ denote the full subcategory of the category of sheaves of sets
270 on $\mathscr{F}$ defined by the sheaves associated with representable presheaves. The category $\underrightarrow{\mathscr{F}}$ is isomorphic to the category $\mathscr{F}(S^{-1})$ described in 6.5. The objects of $\underrightarrow{\mathscr{F}}$ form a generating family of the topos $\mathscr{F}^\sim$ of sheaves of sets on $\mathscr{F}$. The topology on $\underrightarrow{\mathscr{F}}$ induced by the canonical topology of $\mathscr{F}^\sim$ is the coarse topology (II 1.1.4).
g) Let $Q : \mathscr{F} \to \underrightarrow{\mathscr{F}}$ be the canonical functor. The functor Q is a morphism from the site $\underrightarrow{\mathscr{F}}$ (with the coarse topology) to the site $\mathscr{F}$ (with the topology T of b)) (IV 5.9), and induces an isomorphism on the corresponding topoi.
h) Let $u : \mathscr{F} \to C$ be a functor. The functor u is continuous if and only if it carries the morphisms of S to isomorphisms (III 1.1).

i) Let $u : \mathscr{F} \to C$ be a functor carrying the morphisms of S to isomorphisms. The functor u factors uniquely as

$$\begin{tikzcd} \mathscr{F} \arrow[rr, "u"] \arrow[dr, "Q"'] & & C \\ & \underrightarrow{\mathscr{F}} \arrow[ur, "v"'] & \end{tikzcd}$$

(Use III 1.4.)

j) Let $\mathrm{Pro}_S \mathscr{F}$ be the full subcategory of $\mathrm{Pro}\,\mathscr{F}$ (I 8) defined by the pro-objects whose transition morphisms belong to S. Show that the functor Q factors as

$$\mathscr{F} \xhookrightarrow{\ i\ } \mathrm{Pro}_S \mathscr{F} \xrightarrow{\ j\ } \mathscr{F}_{S^{-1}}$$

where j is the restriction of $\mathrm{Pro}\,Q$ to $\mathrm{Pro}_S \mathscr{F}$. Show that the functor j admits a fully faithful left adjoint

$$A : \mathscr{F}_{S^{-1}} \longrightarrow \mathrm{Pro}_S \mathscr{F}.$$

Show that, for every object X of $\mathscr{F}$, $AQ(X)$ is the pro-object

$$Y \longrightarrow X \in S(X) \mapsto Y.$$

Proposition 6.7. *Let $\mathscr{E}$ be a cofiltered category (I 8.7) and $\mathscr{F} \to \mathscr{E}$ a fibered category* 271
(6.1.3).

1) *Let ξ be an object of $\mathscr{E}$, and let $\mathscr{F}/\xi$ be the category fibered over $\mathscr{E}/\xi$ obtained by the base change $\mathscr{E}/\xi \to \mathscr{E}$. The canonical functor $\varinjlim_{\mathscr{E}/\xi} \mathscr{F}/\xi \to \varinjlim_{\mathscr{E}} \mathscr{F}$ is an equivalence of categories.*
2) *More generally, let $\mathscr{E}' \to \mathscr{E}$ be a cofinal functor* (I 8)*. Let $\mathscr{F}' \to \mathscr{E}'$ be the fibered category deduced from $\mathscr{F} \to \mathscr{E}$ by the base change $\mathscr{E}' \hookrightarrow \mathscr{E}$. The canonical functor*

$$\varinjlim_{\mathscr{E}'} \mathscr{F}' \longrightarrow \varinjlim_{\mathscr{E}} \mathscr{F}$$

is an equivalence of categories.

Proof. Exercise.

Exercise 6.8. 1) Let $g : \mathscr{E}^\circ \to \mathrm{Cat}$ be a functor and $\mathscr{G} \to \mathscr{E}$ the corresponding fibered category [5]. Show that there exists a canonical functor $\varinjlim_{\mathscr{E}^\circ} \mathscr{G} \to \varinjlim_{\mathscr{E}^\circ} g$. Show that this functor is not necessarily an equivalence of categories. (One may take for g the constant functor whose value is the final object of Cat, and for $\mathscr{E}$ the category associated with a group.)
2) Suppose that the category $\mathscr{E}^\circ$ is pseudo-filtered. Show that the canonical functor $\varinjlim_{\mathscr{E}^\circ} \mathscr{G} \to \varinjlim_{\mathscr{E}^\circ} g$ is then an equivalence of categories.

Proposition 6.9. *Let $\mathscr{F} \to \mathscr{E}$ be a fibered category (6.1). The functor* (Cat) $\to$ (Ens)

$$C \mapsto \mathrm{Hom}_{\mathrm{Cart}/\mathscr{E}}(C \times \mathscr{E}, \mathscr{F})$$

is represented by the category $\mathbf{Hom}_{\mathrm{Cart}/\mathscr{E}}(\mathscr{E}, \mathscr{F})$.

Let C be a category, let $F : C \times \mathscr{E} \to \mathscr{F}$ be a Cartesian $\mathscr{E}$-functor (6.1.4), and let X be an object of C. The functor $\xi \mapsto F((X, \xi))$ is a Cartesian functor from $\mathscr{E}$ to $\mathscr{F}$, denoted $F'(X)$, that depends functorially on X. Hence there is a map $F \mapsto F'$ from

$\mathrm{Hom}_{\mathrm{Cart}/\mathscr{E}}(C\mathscr{E}, \mathscr{F})$ to $\mathrm{Hom}(C, \mathbf{Hom}_{\mathrm{Cart}/\mathscr{E}}(\mathscr{E}, \mathscr{F}))$, functorial in C, which is immediately verified to be a bijection.

272 **6.10.** The category $\mathbf{Hom}_{\mathrm{Cart}/}(\mathscr{E}, \mathscr{F})$ is called the category of *Cartesian sections* of the fibered category $\mathscr{F}$. It is also sometimes called the *projective-limit* category of $\mathscr{F}$ over $\mathscr{E}^\circ$. It is then denoted $\varprojlim_{\mathscr{E}^\circ} \mathscr{F}$ [2].

6.11. Choose a splitting of the fibered category $\mathscr{F} \to \mathscr{E}$, i.e. for every arrow $f : \xi \to \eta$ of $\mathscr{E}$, choose a base-change functor $f^* : \mathscr{F}_\eta \to \mathscr{F}_\xi$. For every object ξ of $\mathscr{E}$, denote by

$$v_\xi : \varprojlim_{\mathscr{E}^\circ} \mathscr{F} \longrightarrow \mathscr{F}_\xi$$

the evaluation functor at ξ. For every morphism $f : \xi \to \eta$, there is a diagram of functors commutative up to canonical isomorphism:

$$\begin{tikzcd} \varprojlim_{\mathscr{E}^\circ} \mathscr{F} \arrow[r, dash, "v_\xi"] \arrow[dr, "v_\eta"'] & \mathscr{F}_\xi \arrow[d, "f^*"] \\ & \mathscr{F}_\eta . \end{tikzcd}$$

The category $\varprojlim_{\mathscr{E}^\circ} \mathscr{F}$ thus appears as the projective limit, *in the sense of pseudofunctors*, of $\xi \mapsto \mathscr{F}_\xi$. Even when $\xi \mapsto \mathscr{F}_\xi$ is an actual functor (i.e. even when the chosen splitting is a cleavage [5]), the categories $\varprojlim_{\mathscr{E}^\circ} \mathscr{F}$ (6.10) and $\varprojlim_{\mathscr{E}^\circ} \mathscr{F}_\xi$ (I 2) are not equivalent in general.

7. Fibered topoi and sites

7.1. Fibered topoi.

Definition 7.1.1. *Let $\mathscr{U}$ be a universe. A $\mathscr{U}$-topos fibered over a category I is a category fibered over I (6.1)*

$$p : F \longrightarrow I$$

whose fibers are $\mathscr{U}$-topoi and whose inverse-image functors[34] *relative to the morphisms $f : i \to j$ of I are functors $f^* : F_j \to F_i$ that are the inverse-image functors of morphisms of topoi $f\cdot : F_i \to F_j$* (IV 3.1.2).

7.1.2. Equivalently, a $\mathscr{U}$-topos fibered over a category I is a fibered category $p :$
273 $F \to I$ whose fibers are $\mathscr{U}$-topoi and whose inverse-image functors are exact and commute with inductive limits (IV 1.6).

7.1.3. For every morphism $f : i \to j$ of I, choose an inverse-image functor $f^* : F_j \to F_i$. For every pair $i \xrightarrow{f} j \xrightarrow{g} k$ of composable morphisms of I, there is then a canonical isomorphism $c_{f,g} : f^* g^* \to (gf)^*$ (6.1.2.1), and the $c_{f,g}$ have the cocycle property (6.1.2.2). In addition, for every $f : i \to j$, choose a right adjoint $f_* : F_i \to F_j$ to the functor $f^* : F_j \to F_i$. The functor $f_* : F_i \to F_j$ is defined up to canonical isomorphism by the property of being right adjoint to f^*. The general results on adjoint functors then give canonical isomorphisms

$$c'_{g,f} : g_* f_* \longrightarrow (gf)_*. \tag{7.1.3.1}$$

[34]For the fibered structure (6.1.2).

These have a cocycle property analogous to property 6.2.2.2, which we leave to the reader to make explicit. Applying the results of 6.1.3, we then obtain a category fibered over I°:

$$p' : F' \longrightarrow I^{\circ}. \tag{7.1.3.2}$$

It is readily verified that the fibered category $p' : F' \to I^{\circ}$ is independent, up to unique I°-isomorphism, of the various choices used to construct it[34]. The fiber at every object i of I° of the fibered category $p' : F' \to I^{\circ}$ is canonically isomorphic to the $\mathcal{U}$-topos F_i and is identified with it. The inverse-image functors of the fibered category $p' : F' \to I^{\circ}$ are *direct-image* functors of morphisms of topoi.

7.1.4. Let $p : F \to I$ be a fibered $\mathcal{U}$-topos. For every $f : i \to j$, choose a morphism of topoi (IV 3.1.2) $f\cdot : F_i \to F_j$ such that the inverse-image functor f^* (under $f\cdot$) is the inverse-image functor for the fibered structure (6.1). We then obtain cocycles $c_{f,g}$ relating these morphisms of topoi (7.1.3.1) and (7.1.3.2). The collection of the morphisms $f.$, $f \in \mathrm{Fl}(I)$, and the cocycles $c_{f,g}$ is called a *bisplitting* of the fibered topos $F \to I$. Choosing a bisplitting makes it possible to associate with the fibered topos $F \to I$ a pseudofunctor $i \mapsto F_i$ from the category I to the 2-category of $\mathcal{U}$-topoi. 274
Conversely, one can associate with it in an essentially unique manner (cf. [5]) a fibered $\mathcal{U}$-topos $F \to I$ endowed with a bisplitting such that the corresponding pseudofunctor is canonically isomorphic to $i \mapsto F_i$. In practice, a fibered $\mathcal{U}$-topos $F \to I$ will most often be denoted by $(F_i)_{i \in I}$, with the implicit assumption that a bisplitting has been chosen. Note, however, that the constructions carried out on fibered topoi (such as the associated total topos (7.4) and the projective limit (8.1)) depend only on the fibered topoi and not on any bisplittings that may have been chosen.

Definition 7.1.5. *Let $p : F \to I$ and $q : G \to I$ be two $\mathcal{U}$-topoi fibered over I. A morphism m of fibered topoi from (F, p) to (G, q) consists of the datum of an I-functor $m^* : G \to F$ (6.1.4) and, for every object i of I, the datum of a right adjoint $m_{i*} : F_i \to G_i$ to the functor $m_i^* : G_i \to F_i$ obtained by restricting m^* to the fibers at i. These data must moreover be subject to the following condition: for every object i of I, the pair of adjoint functors (m_i^*, m_{i*}) is a morphism from the topos F_i to the topos G_i.*

7.1.6. Let $m : (F, p) \to (G, q)$ be a morphism of $\mathcal{U}$-topoi fibered over I, and let $p' : F' \to I^{\circ}$ and $q' : G' \to I^{\circ}$ be the fibered categories associated by the construction of 7.1.3 with the fibered topoi (F, p) and (G, q), respectively. Choose bisplittings of F and G (7.1.4), denoted in both cases by $(f : i \to j) \mapsto f. : (f^*, f_*)$, from F_i to F_j and from G_i to G_j, respectively. Since $f^* : G \to F$ is an I-functor, for every morphism $f : i \to j$ there is a morphism of functors (transition morphism)

$$b_f : f^* m_j^* \longrightarrow m_i^* f^*, \tag{7.1.6.1}$$

and for every pair $i \xrightarrow{f} j \xrightarrow{g} k$ of composable morphisms, we have

$$m_i^*(c_{f,g}) b_f^* f^*(b_g) = b_{gf} c_{f,g}. \tag{7.1.6.2}$$

Passing to adjoint functors gives morphisms

$$b'_f : f_* m_{i*} \longrightarrow m_{j*} f_* \tag{7.1.6.3}$$

that satisfy relations analogous to (7.1.6.2). Since the functors f_* are the inverse-image 275

[34]The writer apologizes for the nonintrinsic character of this construction.

functors of the fibered categories (F', p') and (G', q') (7.1.3), one can construct an I°-functor

$$m_* : F' \longrightarrow G', \tag{7.1.6.4}$$

whose restrictions to the fibers are the functors m_{i*} and whose transition isomorphisms are the b'_f. It is verified, with a little patience, that the functor m_* so constructed does not depend on the choice of the morphisms of topoi f.[34]

7.1.7. Denote by $\mathrm{Homtop}_I(F, G)$ the set of morphisms of fibered topoi between the fibered topos (F, p) and the fibered topos (G, q). The set $\mathrm{Homtop}_I(F, G)$ is the set of objects of a category denoted $\mathbf{Homtop}_I(F, G)$: if m and n are two morphisms of fibered topoi, a morphism from m to n is an I-morphism of functors from n^* to m^*. By adjunction, one also obtains an I°-morphism from the functor m_* to the functor n_* (7.1.6). Thus, in the end, there are two functors

$$\mathbf{Homtop}_I(F, G) \longrightarrow \mathbf{Hom}_I(G, F)^\circ \tag{7.1.7.1}$$

$$\mathbf{Homtop}_I(F, G) \longrightarrow \mathbf{Hom}_{I^\circ}(F', G') \tag{7.1.7.2}$$

that associate with every morphism $(m^*, (m_{i*})_{i \in \mathrm{Ob}\, I})$, on the one hand the functor $m^* \in \mathrm{Hom}_I(G, F)$ and, on the other hand, the functor $m_* \in \mathrm{Hom}_{I^\circ}(F', G')$ obtained by gluing the m_{i*} as in (7.1.6). The functor $m^* : G \to F$ (7.1.7.1) is called the *inverse-image functor* under the morphism of fibered topoi $m : F \to G$; the functor $m_* : F' \to G'$ (7.1.7.2) is called the *direct-image functor* under the morphism of fibered topoi $m : F \to G$. It follows immediately from the definitions that the functors $m \mapsto m^*$ and $m \mapsto m_*$ are *fully faithful*, and that their essential images are, on the one hand, the set of I-functors from G to F that, fiber by fiber, are exact and commute with inductive limits and, on the other hand, the set of I^0-functors from F' to G' that, fiber by fiber, are exact and commute with inductive limits and, on the other hand, the set of I^0-functors from F' to G' that, fiber by fiber, possess an exact left adjoint. This justifies the abuses of language in which a morphism of fibered topoi is defined by its direct image or its inverse image.
276 In fact, such a morphism of fibered topoi is then defined only up to unique isomorphism.

Denote by $\mathbf{Homtop}_{\mathrm{Cart}/I}(F, G)$ the full subcategory of $\mathbf{Homtop}_I(F, G)$ generated by the morphisms m such that m_* (or, equivalently, m^*) is a Cartesian functor (6.1.4). Such morphisms are called *Cartesian morphisms*. Thus there are fully faithful functors induced by (7.1.7.1) and (7.1.7.2):

$$\mathbf{Homtop}_{\mathrm{Cart}/I}(F, G) \longrightarrow \mathbf{Hom}_{\mathrm{cart}/I}(G, F)^\circ, \tag{7.1.7.3}$$

$$\mathbf{Homtop}_{\mathrm{Cart}/I}(F, G) \longrightarrow \mathbf{Hom}_{\mathrm{cart}/I^\circ}(F', G'). \tag{7.1.7.4}$$

7.1.8. Let $p : F \to I$, $q : G \to I$, and $r : H \to I$ be three $\mathscr{U}$-topoi fibered over I, and let $m : F \to G$ and $n : G \to H$ be two morphisms of fibered $\mathscr{U}$-topoi. As in IV 3.3, define the composite of the two morphisms m and n, denoted $nm : F \to H$, and if $\mathscr{V}$ is a universe such that $\mathscr{U} \in \mathscr{V}$, define a category $(\mathscr{V}\text{-}\mathscr{U}\text{-}\mathrm{Top}_{/I})$ whose objects are the $\mathscr{U}$-topoi fibered over I with fiber categories that are elements of $\mathscr{V}$, and whose morphisms are the morphisms of $\mathscr{U}$-topoi fibered over I. Moreover, as in IV 3.3.2, the composition map

$$\mathrm{Homtop}_{/I}(F, G) \times \mathrm{Homtop}_{/I}(G, F) \longrightarrow \mathrm{Homtop}_{/I}(F, H)$$

[34]See the note at the bottom of page (page 112).

is the map induced on objects by a "functor of composition of morphisms":

$$\mathbf{Homtop}_{/I}(F,G)\,\mathbf{Homtop}_{/I}(G,H) \longrightarrow \mathbf{Homtop}_{/I}(F,H)$$

and the considerations of IV 3.3 and IV 3.4 extend without change to the case of fibered U-topoi. The reader will moreover have noticed that when I is a one-point category (one object and one identity morphism), a $\mathcal{U}$-topos fibered over I is "nothing but" a $\mathcal{U}$-topos, and a morphism of fibered $\mathcal{U}$-topoi is "nothing but" a morphism of $\mathcal{U}$-topoi; the preceding constructions then reduce to those of IV 3.

7.1.9. Let $p : F \to I$ be a fibered $\mathcal{U}$-topos and $\phi : J \to I$ a functor. The fibered category $p_J : F_J \to J$ deduced from (F,p) by the base change $\phi : J \to I$ is a $\mathcal{U}$-topos fibered over J. Let $p' : F' \to I^\circ$ and $(p_J)' : (F_J)' \to J^\circ$ be the fibered categories deduced respectively from (F,p) and (F_J,p_J) by the construction of 7.1.3. It is immediately verified that the category $((F_J)',(p_J)')$ is canonically isomorphic to the category $(p')_J : (F')_J \to J^\circ$ deduced from $p' : F' \to I^\circ$ by the base change $\phi : J^\circ \to I^\circ$. These 277
two categories fibered over J° will henceforth be identified and denoted $p'_j : F'_j \to J^\circ$. The base-change operation is functorial, and even 2-functorial, in its argument: for every pair $p : F \to I$ and $G \to I$ of $\mathcal{U}$-topoi fibered over I, there are two commutative diagrams of categories and functors:

(7.1.9.1)
$$\begin{tikzcd}
\mathbf{Homtop}_I(F,G) \arrow[r] \arrow[d] & \mathbf{Hom}_I(G,F)^\circ \arrow[d] \\
\mathbf{Homtop}_J(F_J,G_J) \arrow[r] & \mathbf{Hom}_J(G_J,F_J)^\circ,
\end{tikzcd}$$

(7.1.9.2)
$$\begin{tikzcd}
\mathbf{Homtop}_I(F,G) \arrow[r] \arrow[d] & \mathbf{Hom}_{I^\circ}(F',G') \arrow[d] \\
\mathbf{Homtop}_J(F_J,G_J) \arrow[r] & \mathbf{Hom}_{J^\circ}(F'_{J^\circ},G_{J^\circ}),
\end{tikzcd}$$

where the vertical functors are base-change functors, and the horizontal functors in (7.1.9.1) are respectively the functors "morphism $\mapsto$ inverse image" (7.1.7.1), while those in (7.1.9.2) are the functors "morphism $\to$ direct image" (7.1.7.2).

7.2. Fibered sites.

7.2.1. A *fibered $\mathcal{U}$-site C* over a category I is a fibered category $p : C \to I$ whose fibers are endowed with topologies making them $\mathcal{U}$-sites (IV 3.0.2), such that for every morphism $f : i \to j$ of I, the inverse-image functor $f^* : C_j \to C_i$ is a morphism from the site C_i to the site C_j (IV 4.9.1).

7.2.2. Let $p : C \to I$ and $q : D \to I$ be two fibered $\mathcal{U}$-sites over I. A *morphism of fibered $\mathcal{U}$-sites from C to D* is an I-functor (6.0) $m : D \to C$ such that, for every object i of I, the functor $m_i : D_i \to C_i$ is a morphism from the site C_i to the site D_i. Denote by $\mathrm{Morsite}_I(C,D)$ the set of morphisms of fibered sites from C to D. A *continuous functor from the fibered site D to the fibered site C* is an I-functor from D to C that induces continuous functors on the fibers (III 1). Denote by $\mathrm{Cont}_I(D,C)$ the set of continuous functors from the fibered site D to the fibered site C. Denote 278
by $\mathbf{Morsite}_I(C,D)$ the full subcategory of $\mathbf{Hom}_I(D,C)^\circ$ defined by the set of objects $\mathrm{Morsite}_I(C,D) \subset \mathrm{Hom}_I(D,C)$. Likewise, denote by $\mathscr{C}ont_I(D,C)$ the full subcategory of $\mathbf{Hom}_I(D,C)$ defined by the subset of objects $\mathscr{C}ont_I(D,C) \subset \mathrm{Hom}_I(D,C)$. Thus we have two fully faithful functors, injective on objects,

(7.2.2.1)
$$\mathbf{Morsite}_I(C,D) \hookrightarrow \mathscr{C}ont_I(D,C)^\circ \hookrightarrow \mathbf{Hom}_I(D,C)^\circ.$$

As in 7.1.7, one defines *cartesian* morphisms from the fibered site C to the fibered site D. We have a diagram of full subcategories

$$\begin{tikzcd} \mathbf{Morsite}_{\mathrm{Cart}/I}(C,D) \arrow[r, hook] \arrow[d, hook] & \mathbf{Hom}_{\mathrm{Cart}/I}(D,C)^{\circ} \arrow[d, hook] \\ \mathbf{Morsite}_{I}(C,D) \arrow[r, hook] & \mathbf{Hom}_{I}(D,C)^{\circ}. \end{tikzcd} \tag{7.2.2.2}$$

7.2.3. Morphisms of fibered $\mathscr{U}$-sites can be composed. This makes it possible to define the category ($\mathscr{V}$-$\mathscr{U}$-Site/I), whose objects are the $\mathscr{U}$-sites whose fibers belong to a universe $\mathscr{V}$ (of which $\mathscr{U}$ is an element). The category $\mathscr{V}$-$\mathscr{U}$-site is moreover endowed with a 2-category structure; in particular, composition of morphisms of fibered $\mathscr{U}$-sites is functorial with respect to morphisms between morphisms. For every functor $\phi : J \to I$ and every fibered site $p : C \to I$, the category $p_J : C_J \to J$ deduced from (C, p) by the base change $\phi : J \to I$ is canonically endowed with a fibered $\mathscr{U}$-site structure over J. The base-change operation is 2-functorial.

7.2.4. Let $p : C \to I$ be a fibered category whose fibers are endowed with topologies making them $\mathscr{U}$-sites. Suppose that projective limits are representable in the fiber categories. For $p : C \to I$ to be a fibered $\mathscr{U}$-site, it is necessary that the inverse-image functors be left exact and take covering families to covering families; this condition is also sufficient when the topologies of the fibers are coarser than the canonical topology (IV 4.9.2). All the fibered sites used in this Seminar will be of the type described above. Likewise, let $p : C \to I$ and $q : D \to I$ be two fibered sites such that finite projective
279 limits are representable in the fiber categories. A cartesian functor $m : D \to C$ that induces left exact functors on the fiber categories and takes covering families of the fiber categories to covering families is a morphism of fibered sites from C to D; the converse holds if the fiber topologies of C are coarser than the canonical topology (IV 4.9.2). All the morphisms of fibered sites used in this Seminar will be of the type described above.

7.2.5. A fibered $\mathscr{U}$-topos $p : F \to I$ is a fibered $\mathscr{U}$-site when its fibers are endowed with the canonical topology, as we shall always do from now on. Let $m : F \to G$ be a morphism of fibered $\mathscr{U}$-topoi. The inverse-image functor $m^* : G \to F$ (7.1.7) is a morphism from the fibered $\mathscr{U}$-site F to the fibered $\mathscr{U}$-site G. Thus we have a functor $(\mathscr{V}\text{-}\mathscr{U}\text{-}\mathrm{Top}_{/I}) \to (\mathscr{V}\text{-}\mathscr{U}\text{-}\mathrm{Site}_{/I})$, which is not faithful in general. But this functor extends naturally to a 2-functor which, for two fibered $\mathscr{U}$-topoi F and G over I, induces a functor

$$\mathbf{Homtop}_I(F,G) \longrightarrow \mathbf{Morsite}_I(F,G) \tag{7.2.5.1}$$

that is an *equivalence of categories*, as follows immediately from the definitions. Moreover, the functor (7.2.5.1), composed with the canonical inclusion $\mathbf{Morsite}_I(F,G) \hookrightarrow \mathbf{Hom}_I(G,F)^{\circ}$, is precisely the functor (7.1.7.1) that associates its inverse image with a morphism of fibered topoi.

7.2.6. Let $p : C \to I$ be a fibered $\mathscr{U}$-site. For every morphism $f : i \to j$ of I, choose a base-change functor $f^* : C_j \to C_i$. Passing to the categories of $\mathscr{U}$-sheaves, the functor f^* extends to a functor $\mathrm{Top}(f)^* : C_j^{\sim} \to C_i^{\sim}$, where $\mathrm{Top}(f) : C_i^{\sim} \to C_j^{\sim}$ is a morphism of topoi (IV 4.9.1.1), making the following diagram commutative up to isomorphism:

(7.2.6.1)
$$\begin{tikzcd} C_j \arrow[r, "f^*"] \arrow[d, "\varepsilon_j"'] & C_i \arrow[d, "\varepsilon_i"] \\ C_j^{\sim} \arrow[r, "\mathrm{Top}(f)^*"] & C_i^{\sim} \end{tikzcd}$$

Let

(7.2.6.2)
$$c_{f,g} : f^*g^* \longrightarrow (gf)^*$$

be the canonical isomorphisms (6.1.2.1), and let 280

(7.2.6.3)
$$b_f : \mathrm{Top}(f)^*\epsilon_j \longrightarrow \epsilon_i f^*$$

be the isomorphisms that "make" the diagrams (7.2.6.1) commutative. Using the fact that the functors $\mathrm{Top}(f)^*$ commute with inductive limits, one verifies immediately that, for every pair of composable morphisms $i \xrightarrow{f} j \xrightarrow{g} k$, there is a unique isomorphism

(7.2.6.4)
$$\mathrm{Top}(c_{f,g}) : \mathrm{Top}(f)^*\,\mathrm{Top}(g)^* \longrightarrow \mathrm{Top}(gf)^*$$

such that

(7.2.6.5)
$$\epsilon_i(c_{f,g})b_f\,\mathrm{Top}(f)^*(b_g) = b_{gf}\,\mathrm{Top}(c_{f,g}).$$

Moreover, the isomorphisms (7.2.6.4) automatically satisfy the cocycle condition of 6.2.2.2. We can therefore construct a fibered category

(7.2.6.6)
$$p^{\sim} : C^{\sim/I} \longrightarrow I$$

whose fibers are canonically isomorphic to the $\mathscr{U}$-topoi $C_i^{\sim}$ (6.1.3), and a cartesian functor

(7.2.6.7)
$$\epsilon_{C/I} : C \longrightarrow C^{\sim/I}$$

that induces on the fibers the functors $\epsilon_i : C_i \to C_i^{\sim}$. It follows immediately from the definitions that $p^{\sim} : C^{\sim/I} \to I$ is a fibered $\mathscr{U}$-topos over I, that $\epsilon_{C/I}$ is a cartesian morphism of fibered $\mathscr{U}$-sites from $C^{\sim/I}$ to C, and that the fibered $\mathscr{U}$-topos $C^{\sim/I} \to I$ and the cartesian functor $\epsilon_{C/I}$ are independent, up to canonical isomorphism, of the various choices used to construct them (the choices of the base-change functors f^* and of the extensions $\mathrm{Top}(f)^*$). The fibered $\mathscr{U}$-topos $C^{\sim/I}$ is called the *associated fibered $\mathscr{U}$-topos* of the fibered $\mathscr{U}$-site.

. One verifies that formation of the fibered $\mathscr{U}$-topos associated with a fibered site "commutes" with base-category change operations.

Proposition 7.2.7. *Let E be a fibered $\mathscr{U}$-topos over I, and let C be a fibered $\mathscr{U}$-site over I. The functor $m \mapsto m^*\epsilon_{C/I}$, which associates with every morphism of fibered topoi $m : E \to C^{\sim/I}$ the composite with $\epsilon_{C/I}$ of the associated inverse-image functor $m^* : C^{\sim/I} \to E$, induces an equivalence of categories preserving cartesian objects* 281

$$\mathbf{Homtop}_I(E, C^{\sim/I}) \xrightarrow{\approx} \mathbf{Morsite}_I(E, C).$$

When finite projective limits are representable in the fiber categories of C, the corresponding fully faithful functor

$$\mathbf{Homtop}_I(E, C^{\sim/I}) \longrightarrow \mathbf{Hom}_I(C, E)^{\circ}$$

has as its essential image the set of I-functors $g : C \to E$ that are left exact fiber by fiber and take covering families of the fiber categories to epimorphic families of the corresponding fiber categories.

This proposition merely generalizes Proposition IV 4.9.4 to the fibered context, and the proof given in IV 4.9.4 generalizes immediately.

7.3. Example.

Example 7.3.1. (*The fibered topos of arrows of a topos*)
Let E be a $\mathcal{U}$-topos, let $\mathrm{Fl}(E)$ be the category of arrows of E, and let $p : \mathrm{Fl}(E) \to E$ be the functor that assigns to an arrow its target. The category $(\mathrm{Fl}(E), p)$ is a fibered $\mathcal{U}$-topos over E (7.1.1 and IV 5.1). The fiber over every object X of E is the topos $E_{/X}$. The inverse-image functor $f^* : E_{/Y} \to E_{/X}$ for a morphism $f : X \to Y$ is the fiber-product functor. Thus, with every functor $\phi : I \to E$, one can associate the fibered $\mathcal{U}$-topos over I

$$p_I : \mathrm{Fl}(E)_I \longrightarrow I, \tag{7.3.1.1}$$

obtained by the base change $\phi : I \to E$, whose fiber over every object i of I is $E_{/\phi(i)}$ (7.1.9). If $\mathcal{V}$ is a universe of which E is an element, then, with the notation of 7.1.8, we have a map

$$\mathrm{Hom}(I, E) \longrightarrow \mathrm{ob}(\mathcal{V}\text{-}\mathcal{U}\text{-}\mathrm{top}_{/I}), \tag{7.3.1.2}$$

which associates with every functor $\phi \in \mathrm{Hom}(I, E)$ the fibered $\mathcal{U}$-topos over I deduced
282 from $(\mathrm{Fl}(E), p)$ by the base change $\phi : I \to E$. Let ϕ_1 and ϕ_2 be two functors from I to E, and let $m : \phi_1 \to \phi_2$ be a morphism of functors. Denote by $p_1 : \mathrm{Fl}(E)_1 \to I$ and $p_2 : \mathrm{Fl}(E)_2 \to I$ the fibered $\mathcal{U}$-topoi over I deduced from $p : \mathrm{Fl}(E) \to E$ by the base changes $\phi_1 : I \to E$ and $\phi_2 : I \to E$, respectively. For every object i of I, choose a morphism of topoi (IV 5.5.2)

$$\mathrm{loc}(m(i)) : E_{/\phi_1(i)} \longrightarrow E_{/\phi_2(i)}. \tag{7.3.1.3}$$

One then verifies easily that there is a unique[34] cartesian morphism of fibered topoi over I,

$$\mathrm{loc}(m) : \mathrm{Fl}(E)_1 \longrightarrow \mathrm{Fl}(E)_2, \tag{7.3.1.4}$$

that induces the morphisms $\mathrm{loc}(m(i))$ on the fiber topoi. The morphism $\mathrm{loc}(m) : \mathrm{Fl}(E)_1 \to \mathrm{Fl}(E)_2$ is determined up to canonical isomorphism by the morphism of functors $m : \phi_1 \to \phi_2$. If $n : \phi_2 \to \phi_3$ is a morphism of functors, we have an isomorphism

$$c(m, n) : \mathrm{loc}(n)\,\mathrm{loc}(m) \simeq \mathrm{loc}(nm), \tag{7.3.1.5}$$

and the morphisms $c(m, n)$ satisfy a cocycle property analogous to (6.1.2.2).

Example 7.3.2. (The fibered site of arrows of a site).
Let C be a $\mathcal{U}$-site in which *fiber products are representable*, let $\mathrm{Fl}(C)$ be the category of arrows of C, and let $p : \mathrm{Fl}(C) \to C$ be the functor that assigns to an arrow its target. The fibered category $(\mathrm{Fl}(C), p)$ is a fibered $\mathcal{U}$-site over C when the fiber categories are endowed with the induced topologies (II 5.2). Let $\epsilon_C : C \to C^\sim$ be the canonical functor from C to the category of $\mathcal{U}$-sheaves on C. Denoting by $\mathrm{Fl}(\epsilon_C)$ the natural extension of

[34] Uniqueness comes from requiring, with the notation of (IV 5.5.3), the formula $(gf)_! = g_! f_!$.

the latter functor to the arrow categories, we have a commutative diagram of categories and functors:

(7.3.2.1)
$$\begin{tikzcd} \mathrm{Fl}(C) \arrow[r, "\mathrm{Fl}(\varepsilon_C)"] \arrow[d, "p"'] & \mathrm{Fl}(C^{\sim}) \arrow[d, "p"] \\ C \arrow[r, "\varepsilon_C"] & C^{\sim} \end{tikzcd}$$

and hence a cartesian functor between categories fibered over C: 283

(7.3.2.2)
$$\begin{tikzcd} \mathrm{Fl}(C) \arrow[rr] \arrow[dr, "p"'] & & \mathrm{Fl}(C^{\sim})_C \arrow[dl, "p_{/C}"] \\ & C & \end{tikzcd}$$

where $p_{/C} : \mathrm{Fl}(C^{\sim})_C \to C$ is deduced from $\mathrm{Fl}(C^{\sim}) \to C^{\sim}$ by the base change $\epsilon_C : C \to C^{\sim}$. The functor $\mathrm{Fl}(C) \to \mathrm{Fl}(C^{\sim})_C$ of (7.3.2.2) is a morphism from the fibered site $\mathrm{Fl}(C^{\sim})_C$ to the fibered site $\mathrm{Fl}(C)$ (7.2.2); hence, by 7.2.7, there is a morphism of fibered topoi over C,

(7.3.2.3)
$$\mathrm{can} : \mathrm{Fl}(C^{\sim})_C \longrightarrow \mathrm{Fl}(C)^{\sim/C},$$

where $\mathrm{Fl}(C)^{\sim/C}$ is the fibered topos over C associated with the fibered site $\mathrm{Fl}(C) \to C$ (7.2.6). It follows from the definition of the associated fibered topos (7.2.6) and from II 5.5 that the morphism can of (7.3.2.3) is a C-equivalence of fibered $\mathcal{U}$-topoi over C, i.e. there is a morphism $\Theta : \mathrm{Fl}(C)^{\sim/C} \to \mathrm{Fl}(C^{\sim})_C$ of fibered $\mathcal{U}$-topoi over C such that the composites $\Theta \circ \mathrm{can}$ and $\mathrm{can} \circ \Theta$ are C-isomorphic to the identity.

Let $\phi : I \to C$ be a functor. By base change, we deduce a fibered site over I,

(7.3.2.4)
$$p_I : \mathrm{Fl}(C)_I \longrightarrow I$$

and, since formation of the associated fibered topos commutes with base change (7.2.6), we have an I-equivalence of fibered $\mathcal{U}$-topoi

(7.3.2.5)
$$\mathrm{can}_I : \mathrm{Fl}(C^{\sim})_I \longrightarrow \mathrm{Fl}(C)_I^{\sim/I}$$

where $\mathrm{Fl}(C^{\sim})_I \to I$ is the fibered $\mathcal{U}$-topos obtained by the base change $\epsilon_C \circ \phi : I \to C^{\sim}$.

Finally, let $\phi_1 : I \to C$ and $\phi_2 : I \to C$ be two functors, and let $m : \phi_1 \to \phi_2$ be a morphism of functors. Denote by $\mathrm{Fl}(C)_1$ and $\mathrm{Fl}(C)_2$ the fibered sites over I deduced from $p : \mathrm{Fl}(C) \to C$ by the base changes ϕ_1 and ϕ_2, respectively. As in 7.3.1, one constructs a morphism of fibered sites over I,

(7.3.2.6)
$$\mathrm{loc}(m) : \mathrm{Fl}(C)_1 \longrightarrow \mathrm{Fl}(C)_2.$$

Denote by $\mathrm{Fl}(C^{\sim})_1 \to I$ and $\mathrm{Fl}(C^{\sim})_2 \to I$ the fibered $\mathcal{U}$-topoi over I obtained by the base changes $\epsilon_C \circ \phi_1$ and $\epsilon_C \circ \phi_2$, respectively. We have a diagram of morphisms over I that is commutative up to isomorphism: 284

(7.3.2.7)
$$\begin{tikzcd} \mathrm{Fl}(C^{\sim})_1 \arrow[r, "\mathrm{loc}(\varepsilon_C * m)"] \arrow[d] & \mathrm{Fl}(C^{\sim})_2 \arrow[d] \\ \mathrm{Fl}(C)_1 \arrow[r, "\mathrm{loc}(m)"] & \mathrm{Fl}(C)_2 \end{tikzcd}$$

where the vertical morphisms are deduced by base change from the morphisms of (7.3.2.2).

Exercise 7.3.3. (*Small and big fibered sites*).

This exercise continues Exercise IV 4.10.6, whose hypotheses and notation we use. Denote by $\mathbf{M}$ the full subcategory of $\mathrm{Fl}(S)$ defined by the morphisms of $\mathscr{M}$, and let $p : \mathbf{M} \to S$ be the functor that assigns to a morphism its target.

6°) Show that $(\mathbf{M}, p)$ is a fibered site over S and that, when fiber products are representable in S, the inclusion functor $\mathbf{M} \to \mathrm{Fl}(S)$ is a cartesian morphism from the fibered site $\mathrm{Fl}(S) \to S$ to the fibered site $\mathbf{M} \to S$.

7°) Let $\mathbf{M}^{\sim/S} \to S$ be the fibered topos associated with $\mathbf{M} \to S$. Show that, with the notation of 7.3.2, there is a canonical morphism $g : \mathrm{Fl}(S^{\sim})_{/S} \to \mathbf{M}^{\sim/S}$ of fibered topoi over S which induces on each fiber over $X \in \mathrm{ob}(S)$ the morphism $(\mathrm{Prol}_X, \mathrm{Res}_X)$ (IV 4.10.6 4°)). Show that, although for every $X \in \mathrm{ob}(S)$ the morphism $g_X : S^{\sim}_{/X} \to S(X)^{\sim}$ admits a section (*loc. cit.*), the morphism g does not in general admit a section.

7.4. The total topology of a fibered topos or site.

Definition 7.4.1. *Let $p : C \to I$ be a fibered $\mathscr{U}$-site and, for every $i \in \mathrm{ob}(I)$, denote by $\alpha_{i!} : C_i \to C$ the inclusion functor of the fiber category C_i into C. The* total topology *on C is the coarsest topology on C for which the functors $\alpha_{i!}$, $i \in \mathrm{ob}(I)$, are continuous* (III 3.6). *The category C equipped with the total topology is called the* total site.

285 Proposition 7.4.2. *Let $p : C \to I$ be a fibered $\mathscr{U}$-site, let T be the total topology on C, and, for every $i \in \mathrm{ob}(I)$, let T_i be the topology of the fiber C_i.*

1) *Let X be an object of C over an object i of I. A family $(X_\beta \to X)_{\beta \in B}$ is covering for T if and only if it is refined by a covering family for T_i, $(Y_\gamma \to X)_{\gamma \in \Gamma}$, in C_i.*
2) *The topology T induces the topology T_i on each category C_i.*
3) *For every $i \in \mathrm{ob}(I)$, the functor $\alpha_{i!} : C_i \to C$ is cocontinuous.*

. Property 3) follows from 1) and III 2.1. Let us prove 1). For every object X of C, let $J(X)$ be the set of sieves of C/X described in 1). One verifies that the $J(X)$ have properties T 1), T 2), and T 3) of II 1.1, and hence that they define a topology T' on C. The topology T' is the coarsest topology on C for which the covering families for the topologies T_i are covering. Thus T' is coarser than T (III 1.6). To prove that $T' = T$, it is therefore enough to show that, for every object i of I, the functor $\alpha_{i!} : C_i \to C$ is continuous when C is equipped with the topology T'; equivalently, it is enough to show that the topology induced by T' on the category C_i is T_i, which will prove property 2) at the same time. Let T'_i be the topology on C_i induced by T', and let $u : R \hookrightarrow X$ be a covering sieve for T'_i. By III 3.2, the morphism $\alpha_{i!}(u) : \alpha_{i!}(R) \to \alpha_{i!}(X)$ is bicovering and in particular covering. By the definition of T', this means that the sieve $R \hookrightarrow X$ is covering for T_i. Thus T'_i is coarser than T_i. To prove that T'_i is finer than T_i, by the definition of the induced topology (III 3.1), it is enough to show that the functor $\alpha_{i!} : C_i \to C$ is continuous when C_i and C are equipped with T_i and T', respectively. In summary, by III 1.2 and 1.5, to prove properties 1) and 2), it is enough to prove the following lemma:

Lemma 7.4.2.2. *Under the hypotheses and with the notation of 7.4.2 and 7.4.2.1, let X be an object of C over i, and let $R \xrightarrow{u} X$ be a morphism of $\widehat{C_i}$. The following properties are equivalent:*

286 i) *The morphism $R \to X$ is bicovering for T_i.*
ii) *The morphism $\alpha_{i!}(u) : \alpha_{i!}(R) \to \alpha_{i!}(X)$ is bicovering for T'.*

i) $\Rightarrow$ ii): It is clear that if $R \to X$ is bicovering for T_i, then the morphism $\alpha_{i!}(u)$ is covering for T'. It remains to show that it is bicovering (II 5.2). Let Y be an object of C over an object j of I, let $m, n : Y \rightrightarrows \alpha_{i!}(R)$ be two morphisms of $C^\wedge$ such that $\alpha_{i!}(u)m = \alpha_{i!}(u)n$, and set $f = p(\alpha_{i!}(u)m)$. Since the functor $\alpha_{i!}$ commutes with inductive limits (I 5.4.3), we have (I 3.4):

$$\mathrm{Hom}_{C^\wedge}(Y, \alpha_{i!}(R)) \simeq \varinjlim_{C_i/R} \mathrm{Hom}_C(Y, \alpha_{i!}(.)).$$

For every object Z of C_i, with the notation of 6.1.1, we have

$$\mathrm{Hom}_C(Y, Z) \simeq \coprod_{w \in \mathrm{Hom}(j,i)} \mathrm{Hom}_w(Y, Z),$$

and therefore

$$\varinjlim_{C_i/R} \mathrm{Hom}(Y, \alpha_{i!}(.)) \simeq \coprod_{w \in \mathrm{Hom}(j,i)} \varinjlim_{C_i/R} \mathrm{Hom}_w(Y, \alpha_{i!}(.)).$$

For every $w \in \mathrm{Hom}(j, i)$, however, there is an inverse image functor $w^* : C_i \to C_j$, and $\mathrm{Hom}_w(Y, \alpha_{i!}(.)) \simeq \mathrm{Hom}_{C_j}(Y, w^*(.))$. Thus, still denoting by $w^* : C_i^\wedge \to C_j^\wedge$ the extension of w^* to presheaves (which, according to I 5.4, should be denoted $(w^*)_!$), and observing that the functor $w^* : C_i^\wedge \to C_j^\wedge$ commutes with inductive limits (I 5.4.3), we obtain an isomorphism

$$\mathrm{Hom}_{C^\wedge}(Y, \alpha_{i!}(R)) = \coprod_{w \in \mathrm{Hom}(j,i)} \mathrm{Hom}_{C_j^\wedge}(Y, w^*(R)).$$

Under this isomorphism, the morphisms $m, n : Y \rightrightarrows \alpha_{i!}(R)$ correspond to two morphisms $m', n' : Y \rightrightarrows f^*R$ such that $f^*(u)m' = f^*(u)n'$. Since the inverse image functor is a morphism of topoi, it is in particular continuous, and consequently $f^*(u)$ is bicovering (III 3.2). There is therefore a covering family $(Y_\delta \xrightarrow{v_\delta} Y)_\delta$ in C_j such that $m'v_\delta = n'v_\delta$ for every δ. It follows that $mv_\delta = nv_\delta$ for every δ, and hence that $\alpha_{i!}(u)$ is bicovering for T'.

ii) ⇒ i): Suppose that $\alpha_{i!}(R) \xrightarrow{\alpha_{i!}(u)} \alpha_{i!}(X)$ is bicovering. The functor $\alpha_{i!} : C_i \to C$ 287
is cocontinuous (III 2.1). Hence $\alpha_i^*\alpha_{i!}(u)$ is bicovering (III 2.3.2 and II 5.3 ii)). The functor $\alpha_{i!} : C_i \to C$ has property (PPF) of I 5.14. We therefore have a Cartesian diagram in $C_i^\wedge$ (I 5.14 3)):

$$\begin{tikzcd}
R \arrow[r] \arrow[d, "u"'] & \alpha_i^*\alpha_{i!}(R) \arrow[d, "\alpha_i^*\alpha_{i!}(u)"] \\
X \arrow[r] & \alpha_i^*\alpha_{i!}(X).
\end{tikzcd}$$

It follows that u is bicovering (II 5.2).

REMARK 7.4.3. 1) Proposition 7.4.2 can be stated and proved in a slightly more general setting than that of fibered sites, but apparently without useful application: instead of a fibered site $p : C \to I$, one considers a fibered category $p : C \to I$ whose fibers are equipped with topologies and whose inverse image functors are continuous for these topologies (whereas in the case of fibered sites one requires in addition that these inverse image functors be morphisms of sites (IV 4.9)).

2) One can prove that the total topology of a fibered site is the finest topology making the functors $\alpha_{i!} : C_i \to C$ cocontinuous (III 2).

3) Let $p : C \to I$ be a fibered $\mathscr{U}$-site over a category equivalent to a $\mathscr{U}$-small category. Then the total site C is a $\mathscr{U}$-site (II 3.0.2). Indeed, if, for every object i of I, $(X_\beta)_{\beta \in B_i}$ is a small topologically generating family of C_i, then the family $(X_\beta)_{\beta \in \coprod_i B_i}$ is topologically generating for C and is $\mathscr{U}$-small. The topos of sheaves on the total site is called the *total topos* and is denoted by $\mathrm{Top}(C)$.
4) Let $p : C \to I$ be a fibered $\mathscr{U}$-site over a small category. Since the functor $\alpha_{i!} : C_i \to C$ is cocontinuous, it gives rise to a morphism of topoi $\alpha_i = (\alpha_i^*, \alpha^{i*})$ from $C_i^\sim$ to $\mathrm{Top}(C)$ (IV 4.7). Since, moreover, the functor $\alpha_{i!} : C_i \to C$ is continuous, the functor $\alpha_i^* : \mathrm{Top}(C) \to C_i^\sim$ admits a left adjoint, again denoted
288 by $\alpha_{i!} : C_i^\sim \to \mathrm{Top}(C)$, which extends the functor $\alpha_{i!} : C_i \to C$ (III 1.2 iv)).

Proposition 7.4.4. *Let $p : C \to I$ be a fibered $\mathscr{U}$-site over a small category I.*

1) *A presheaf F on C is a sheaf for the total topology if and only if, for every object i of I, the presheaf $F \circ \alpha_{i!} = F|C_i$ is a sheaf on C_i.*
2) *A functor m from C to a site D is a continuous functor from the total site C to the site D if and only if, for every object i of I, the functor $m|C_i = m \circ \alpha_{i!} : C_i \to D$ is continuous.*

If F is a sheaf for the total topology, the presheaf $F \circ \alpha_{i!}$ is a sheaf on C_i because $\alpha_{i!} : C_i \to C$ is continuous. Conversely, suppose that $F \circ \alpha_{i!}$ is a sheaf on C_i for every i. Let X be an object of C over an object i of I. For every covering sieve R of C_i/X, we have $F(X) \simeq F \circ \alpha_{i!}(R)$ and hence $F(X) \simeq F(\alpha_{i!}(R))$. Let $R' \hookrightarrow X$ be the sieve of C/X that is the image of the canonical morphism of presheaves $\alpha_{i!}(R) \to X$. The morphism $\alpha_{i!}(R) \to R'$ is an epimorphism of presheaves, and therefore $F(R') \to F(\alpha_{i!}(R))$ is injective. Since $F(R') \to F(\alpha_{i!}(R))$ is also surjective, it is bijective, and hence $F(X) \to F(R')$ is bijective. Since the covering sieves of X are described by (7.4.2.1), the presheaf F is a sheaf: returning to the construction of the associated sheaf (II 3), one sees that F is isomorphic to its associated sheaf (II 3.3). The second assertion follows immediately from the first and the definition of continuity (III 1.1).

7.4.5. Let $p : C \to I$ be a fibered $\mathscr{U}$-site over a category I equivalent to a small category, and let

$$f : i \longrightarrow j$$

be an arrow of the index category I. Again denoting by $f : C_i^\sim \to C_j^\sim$ the morphism of topoi determined by the fibered-site structure, we have (with the notation of 7.4.3.4) a diagram of morphisms of $\mathscr{U}$-topoi

(7.4.5.1)
$$\begin{tikzcd} C_i^\sim \arrow[r, "\alpha_i"] \arrow[d, "f"'] & \mathrm{Top}(C) \\ C_j^\sim \arrow[ur, "\alpha_j"'] & \end{tikzcd}$$

289 where $\mathrm{Top}(C)$ is the total topos (7.4.3.3). The diagram (7.4.5.1) *is not commutative in general.* We shall define a canonical morphism of morphisms of topoi:

(7.4.5.2)
$$\rho_f : \alpha_i \longrightarrow \alpha_j \circ f.$$

It is enough to define such a morphism at the level of inverse images, i.e. to define a morphism of functors

(7.4.5.3)
$$\beta_f^* : f^* \circ \alpha_j^* \longrightarrow \alpha_i^*,$$

or, equivalently, using the adjunction between f^* and f_*, it is enough to define a morphism of functors

$$\gamma_f : \alpha_j^* \longrightarrow f_* \alpha_i^*. \tag{7.4.5.4}$$

Let Y be an object of C_j and F a sheaf on the total site C. By definition, $\alpha_j^* F(Y) = F(\alpha_{j!}(Y))$ and $f_* \alpha_i^* F(Y) = F(\alpha_{i!} f^*(Y))$, where in the second member $f^* : C_j \to C_i$ denotes the inverse image functor for the fibered structure. By the definition of this inverse image functor, however, there is a canonical Cartesian morphism

$$m_Y : \alpha_{i!} f^* \longrightarrow \alpha_{j!}; \tag{7.4.5.5}$$

hence, on applying the functor F, a map functorial in Y:

$$\gamma_f(F)(Y) = F(m_Y) : \alpha_j^* F(Y) \longrightarrow f_* \alpha_i^* F(Y), \tag{7.4.5.6}$$

which defines a morphism of sheaves on C_j:

$$\gamma_f(F) : \alpha_j^* F \longrightarrow f_* \alpha_i^* F; \tag{7.4.5.7}$$

whence the morphism of functors γ_f of (7.4.5.4).

Now let $f : i \to j$ and $g : j \to k$ be two composable morphisms of I. There is a canonical isomorphism 290

$$c'_{g,f} : g_* f_* \longrightarrow (gf)_*$$

and one immediately verifies the compatibility formula

$$c'_{g,f} \circ g_*(\gamma_f) \circ \gamma_g = \gamma_{gf}. \tag{7.4.5.8}$$

7.4.6. Introduce the fibered $\mathscr{U}$-topos $p^{\sim/I} : C^{\sim/I} \to I$ associated with $p : C \to I$ (7.2.6) and the corresponding category fibered over I°, $(p^{\sim/I})' : (C^{\sim/I})' \to I^\circ$ (7.1.3.2). The preceding considerations associate with every object F of $\mathrm{Top}(C)$ a family $F_i = (\alpha_i^* F)_{i \in \mathrm{ob}(I)}$ of objects of the $C_i^\sim$, together with a family of morphisms

$$\gamma_f F : F_j \longrightarrow f_* F_i, \quad f \in F1(I), \quad f : i \longrightarrow j,$$

subject to the compatibility conditions of (7.4.5.8). Thus with F we have associated an I°-functor from the category I° to the category $(C^{\sim/I})'$, i.e. an object $\Theta_C(F)$ of the category $\mathbf{Hom}_{I^\circ}(I^\circ, (C^{\sim/I})')$. Since this construction is functorial in F, we obtain a functor

$$\Theta_C : \mathrm{Top}(C) \longrightarrow \mathbf{Hom}_{I^\circ}(I^\circ, (C^{\sim/I})'). \tag{7.4.6.1}$$

Proposition 7.4.7. *Let $C \to I$ be a fibered $\mathscr{U}$-site, with I equivalent to a small category. The functor Θ_C (7.4.6.1) is an equivalence of categories.*

We shall content ourselves with describing a quasi-inverse to Θ_C. Let $\sigma \in \mathrm{Hom}_{I^\circ}(I^\circ, (C^{\sim/I})')$, i.e. an I°-functor $i \mapsto \sigma(i)$ from I° to $(C^{\sim/I})'$. For every object X of C over $p(X) = i \in \mathrm{ob}(I)$, there is a sheaf $\sigma(p(X)) \in \mathrm{ob}\, C_i^\sim$, and hence a set $\sigma(p(X))(X)$. Let $m : X \to Y$ be a morphism of C over the morphism $f : i \to j$ of I. The morphism m factors uniquely as a morphism $m' : X \to f^* Y$ of C_i followed by the canonical Cartesian morphism $f^* Y \to Y$. Likewise, the morphism $\sigma(f)$ factors uniquely as a morphism $\gamma_f(\sigma) : \sigma(p(Y)) \to f_* \sigma(p(X))$ followed by the canonical I°-Cartesian morphism $f_* \sigma(p(X)) \to \sigma(p(X))$. We therefore obtain a map from $\sigma(p(Y))(Y)$ to $\sigma(p(X))(X)$, by composing the maps 291

$$\sigma(p(Y))(Y) \xrightarrow{\gamma_f(\sigma)(Y)} f_* \sigma(p(X))(Y) = \sigma(p(X))(f^* Y) \xrightarrow{\sigma(p(X))(m')} \sigma(p(X))(X).$$

Thus with every object X of C we have associated a set $\sigma(p(X))(X)$, and with every morphism $m : X \to Y$ a map $\sigma(p(Y))(Y) \to \sigma(p(X))(X)$. One verifies that this indeed defines a presheaf on C, and that this presheaf does not depend on the choice of inverse image functors used in its construction. It follows from 7.4.4 that the presheaf $X \mapsto \sigma(p(X))(X)$ is a sheaf on the total site C, and it is clear that this sheaf depends functorially on the object σ of $\mathbf{Hom}_{I^\circ}(I^\circ, (C^{\sim/I})')$. Returning to the definition of Θ_C, it is also clear that the functor from $\mathbf{Hom}_{I^\circ}(I^\circ, (C^{\sim/I})')$ to $\mathrm{Top}(C)$ just constructed is a quasi-inverse to Θ_C.

Remark 7.4.8. It follows in particular from 7.4.7 that the category $\mathbf{Hom}_{I^\circ}(I^\circ, (C^{\sim/I})')$ is a $\mathcal{U}$-topos. This latter fact can be seen directly. Using I 9.21.10, one sees immediately that conditions a), b), and c) of IV 1.1.2 are satisfied, and the existence of a small generating category follows from I 9.25.

7.4.9. Let $p : C \to I$ and $q : D \to I$ be two $\mathcal{U}$-sites fibered over a small category, and let m be a morphism from the fibered site C to the fibered site D (7.2.2). Denote by $m^{\sim/I} : C^{\sim/I} \to D^{\sim/I}$ the corresponding morphism between the associated fibered $\mathcal{U}$-topoi. The morphism of fibered topoi $m^{\sim/I}$ is defined up to canonical isomorphism (7.2.7). It has a corresponding direct-image functor $m_*^{\sim/I} : (C^{\sim/I})' \to (D^{\sim/I})'$ (7.1.7), which, on passing to sections over I°, gives a functor

(7.4.9.1) $$\mathrm{Hom}_{I^\circ}(I^\circ, m_*^{\sim/I}) : \mathbf{Hom}_{I^\circ}(I^\circ, (C^{\sim/I})') \longrightarrow \mathbf{Hom}_{I^\circ}(I^\circ, (D^{\sim/I})').$$

Moreover, the functor m is a continuous functor between the total sites (7.4.4), and hence, by composition, gives a functor $m_* : \mathrm{Top}(C) \to \mathrm{Top}(F)$.

In summary, there is a diagram of functors

(7.4.9.2) $$\begin{tikzcd} \mathrm{Top}(C) \arrow[r, "\theta_C"] \arrow[d, "m_*"'] & \mathbf{Hom}_I^\circ(I^\circ, (C^{\sim/I})') \arrow[d, "\mathrm{Hom}_{I^\circ}(I^\circ, m_*^{\sim/I})"'] \\ \mathrm{Top}(D) \arrow[r, "\theta_D"] & \mathbf{Hom}_I^\circ(I^\circ, (D^{\sim/I})') \end{tikzcd}$$

292 Proposition 7.4.10. *The diagram (7.4.9.2) is commutative up to isomorphism. When m is Cartesian, m is a morphism from the total site C to the total site D.*

We leave it to the reader to verify that diagram (7.4.9.2) is commutative up to isomorphism. To show that m is a morphism between the total sites, it is enough, in view of 7.4.7, to show that the functor left adjoint to $\mathrm{Hom}_{I^\circ}(I^\circ, m_*^{\sim/I})$ is left exact. We first describe the inverse-image functor of the morphism $m^{\sim/I}$ (7.2.7); for every object i of I, denote by $m_i^* : D_i^\sim \to C_i^\sim$ the functor induced by $(m^{\sim/i})^*$ on the fibers at i. For every morphism $f : i \to j$ of I, choose morphisms of topoi, denoted in both cases by $(f^*, f_*) : C_i^\sim \to C_j^\sim$ and $(f^*, f_*) : D_i^\sim \to D_j^\sim$. Since $(m^{\sim/I})^*$ is a Cartesian I°-functor, for every $f : i \to j$ there is a canonical isomorphism

(7.4.10.1) $$m_i^* f_* \longrightarrow f^* m_j^*,$$

and since the functors f^* and f_* are adjoint, there are canonical morphisms (adjunction morphisms)

$$\begin{aligned} \phi &: \mathrm{id} \longrightarrow f_* f^* \\ \psi &: f^* f_* \longrightarrow \mathrm{id}\,. \end{aligned} \tag{7.4.10.2}$$

Now consider the following sequence of functorial morphisms:

$$m_j^* f^* \xrightarrow{(1)} f_* f^* m_j^* f_* \xrightarrow{(2)} f_* m_i^* f^* f_* \xrightarrow{(3)} f_* m_i^*, \tag{7.4.10.3}$$

where morphism (1) is deduced from ϕ, morphism (2) is deduced from the isomorphism (7.4.10.1)), and morphism (3 is deduced from ψ. By composing the various morphisms in the sequence (7.4.10.3), we obtain a functorial morphism 293

$$\mathrm{can}_f : m_j^* f_* \longrightarrow f_* m_i^*. \tag{7.4.10.4}$$

Now let τ be an object of $\mathbf{Hom}_{I^\circ}(I^\circ (D^{\sim/I})')$. Thus, for every object i of I, there is an object $\tau(i)$ of $D_i^\sim$, and for every morphism $f : i \to j$, a morphism

$$\gamma_f(\tau) : \tau(j) \longrightarrow f_* \tau(i), \tag{7.4.10.5}$$

where the family of the $\gamma_f(\tau)$ has a compatibility property analogous to (7.4.5.8)). Put $\sigma(i = m_i^*(\tau(i))$ and denote by

$$\gamma_f(\sigma) : \sigma(j) \longrightarrow f_* \sigma(i) \tag{7.4.10.6}$$

the composite morphism $m_j^* \tau(j) \xrightarrow{m_j^* \gamma_f(\tau)} m_j^* f_* \tau(i) \xrightarrow{\mathrm{can}_f(\tau(i))} f_* m_i^* \tau(i)$. One verifies that the $\gamma_f(\sigma)$ have the compatibility property of (7.4.5.8), and hence that an object of $\mathbf{Hom}_{I^\circ}(I^\circ, (C^{\sim/I})')$ has thus been defined. An "adjoint functor chasing" then shows that the functor from $\mathbf{Hom}_{I^\circ}(I^\circ, (D^{\sim/I})')$ to $\mathbf{Hom}_{I^\circ}(I^\circ, (C^{\sim/I})')$ just constructed is left adjoint to the functor $\mathrm{Hom}_{I^\circ}(I^\circ, m_*^{\sim/I})$. It remains to show that this left adjoint commutes with finite projective limits, which follows immediately from the fact that projective limits in the categories of sections are computed fiber by fiber, and that all the functors used to construct this left adjoint commute with finite projective limits.

Corollary 7.4.11. *Let $p : C \to I$ be a $\mathcal{U}$-site fibered over a small category, let $p^{\sim/I} : C^{\sim/I} \to I$ be the associated fibered $\mathcal{U}$-topos (7.2.6), and let $\epsilon_I : C \to C^{\sim/I}$ be the canonical morphism from the fibered $\mathcal{U}$-site $C^{\sim/I}$ to the fibered $\mathcal{U}$-site C. The morphism ϵ_I induces an equivalence $\mathrm{Top}(\epsilon_I) : \mathrm{Top}(C) \to \mathrm{Top}(C^{\sim/I})$ between the corresponding total topoi.*

This follows from the commutativity of diagram (7.4.9.2) and from the fact that the morphism ϵ_I gives an equivalence between the associated fibered $\mathcal{U}$-topoi.

Remark 7.4.11.1. It follows from 7.4.11 that in questions concerning the total topos one may replace fibered sites by the corresponding fibered topoi.

Lemma 7.4.12. *Let $p : C \to I$ be a site fibered over a small category I, and let i be a final object of I* (I 10). *The functor $\alpha_{i!} : C_i^\sim \to \mathrm{Top}(C)$ (7.4.3.4) is exact and fully faithful. The pairs of functors $\beta_i = (\alpha_{i!}, \alpha_i^*) : \mathrm{Top}(C) \to C_i^\sim$ and $\alpha_i = (\alpha_i^*, \alpha_{i*}) : C_i^\sim \to \mathrm{Top}(C)$ are morphisms of topoi. The composite morphism $\beta_i \alpha_i : C_i^\sim \to C_i^\sim$ is isomorphic to the identity morphism.* 294

For every object j of I, denote by $f_j : j \to i$ the unique morphism from j to i. Using the equivalence $\Theta_C : \mathrm{Top}(C) \xrightarrow{\sim} \mathbf{Hom}_{I^\circ}(I^\circ, (C^{\sim/I})')$ (7.4.7), one sees that for every object X of C_i, $\alpha_{i!(X)}$ is the section $j \to f_i^*(X)$, and that for every section $j \mapsto$

$\sigma(j) \in \mathrm{Hom}_{I^\circ}(I^\circ, (C^{\sim/I})')$, $\alpha_i^*(j \mapsto \sigma(j))$ is the object $\sigma(i)$. Hence $\alpha_i^*\alpha_{i!}$ is isomorphic to the identity. Therefore $\alpha_{i!}$ is fully faithful. Since the functors f_j^* are left exact, the functor $\alpha_{i!}$ is left exact. Thus β_i is a morphism of topoi, and since $\alpha_i^*\alpha_{i!}$ is isomorphic to the identity, the morphism $\beta_i\alpha_i$ is isomorphic to the identity morphism.

Theorem 7.4.13.1. *Let $p : C \to I$ and $q : D \to I$ be two $\mathcal{U}$-sites fibered over a small category I, and let $m : D \to C$ be an I-functor. For m to be a morphism from the fibered site C to the fibered site D (7.2.2), it is sufficient that m be a morphism from the total site C to the total site D.*

. We must show that if an I-functor $m : D \to C$ is a morphism from the total site C to the total site D, then m is a morphism from the fibered site C to the fibered site D, i.e. for every object i of I, the functor $m_i : D_i \to C_i$ induced by m on the fibers at i is a morphism from the site C_i to the site D_i. The proof of this last assertion occupies paragraphs 7.4.13.3 through 7.4.13.7.

. Let us show that, for every object i of I, the functor $m_i : D_i \to C_i$ is continuous. Indeed, there is a commutative diagram:

$$\begin{tikzcd} D_i \arrow[r, "m_i"] \arrow[d, "\alpha_{i!}"'] & C_i \arrow[d, "\alpha_{i!}"] \\ D \arrow[r, "m"] & C \end{tikzcd}$$

295 and it is enough to show that, for every covering sieve $R \hookrightarrow X$ of C_i/X, the morphism $m_{i!}(R) \to m_i(X)$ is bicovering (III 1.2 and 1.5). But by the commutativity of the diagram on page (page 133) and the fact that the functors $\alpha_{i!}$ and m are continuous, the morphism $\alpha_{i!}m_{i!}(R) \to \alpha_{i!}m_i(X)$ is bicovering. The assertion therefore follows from 7.4.2.2.

. *Reduction to the case in which C and D are fibered $\mathcal{U}$-topoi.* Since the functors m_i are continuous, they admit natural extensions to the categories of sheaves, which we denote by $m_i^* : D_i^\sim \to C_i^\sim$ (III 1.2 iv)). Let $D^{\sim/I}$ and $C^{\sim/I}$ be the fibered $\mathcal{U}$-topoi associated with the fibered sites D and C, respectively, and for every morphism f of I, denote by f^* the inverse-image functors for the four fibered categories C, D, $C^{\sim/I}$, and $D^{\sim/I}$. Since the functor m is an I-functor, for every $f : i \to j$ there is a canonical morphism

$$f^*mj \leftarrow m_i f^*.$$

Since the functors f^*, m_i, and m_j are continuous, by passing to the categories of sheaves we deduce canonical isomorphisms

$$f^*m_j^* \longleftarrow m_i^* f^*.$$

These isomorphisms have compatibility properties that make it possible to construct a Cartesian functor $m^* : D^{\sim/I} \to C^{\sim/I}$ inducing the functors m_i^* on the fiber topoi. Moreover, the diagram

$$\begin{tikzcd} D \arrow[r, "m"] \arrow[d, "\varepsilon_I"'] & C \arrow[d, "\varepsilon_I"] \\ D^{\sim/I} \arrow[r, "m^*"] & C^{\sim/I} \end{tikzcd}$$

is commutative up to isomorphism. Since the functors ϵ_I induce equivalences on the total topoi (7.4.11), and since m is a morphism between the total sites, the functor m^* :

$D^{\sim/I} \to C^{\sim/I}$ is a morphism between the total sites. We are therefore reduced to proving the assertion of 7.4.13.2 when D and C are fibered $\mathscr{U}$-topoi, which we assume henceforth.

. *For every object i of I, the functor $m_i : D_i \to C_i$ takes the final object of D_i to the final* 296
object of C_i. Denote by $m : D^\sim \to C^\sim$ the natural extension of m to the total topoi (III 1.2). Using the equivalences $\Theta_D : D^\sim \to \mathbf{Hom}_{I^\circ}(I^\circ, D')$ (7.4.7), one immediately verifies that m associates with every section $\sigma \in \mathrm{Hom}_{I^\circ}(I^\circ, D')$ the section $(i \mapsto m_i\sigma(i)) \in \mathrm{Hom}_{I^\circ}(I^\circ, C')$. Since m is a morphism of sites, the functor m is left exact and in particular takes the final object of $\mathbf{Hom}_{I^\circ}(I^\circ, D)$ to the final object of $\mathbf{Hom}_{I^\circ}(I^\circ, C')$. For every object i of I, denote by e_i a final object of D_i. A final object of $\mathbf{Hom}_{I^\circ}(I^\circ, D')$ is the section $i \mapsto e_i$. Thus the section $i \mapsto m_i(e_i)$ is a final object of $\mathbf{Hom}_{I^\circ}(I^\circ, C')$, and consequently, for every object i of I, $m_i(e_i)$ is a final object of C_i.

. *Reduction to the case in which i is a final object of I.* Let i be an object of I. The functor $\alpha_{i!} : D_i \to D$ factors as

$$D_i \xrightarrow{\overline{\alpha}_{i!}} D_{/e_i} \xrightarrow{\mathrm{loc}(e_i)} D,$$

where $\overline{\alpha}_i$ is the evident functor and $\mathrm{loc}(e_i)$ the forgetful functor. Denote by $m/i : D_{/e_i} \to D_{m(e_i)}$ the functor that associates with every object $u : X \to e_i$ of $D_{/e_i}$ the object $m(u) : m(X) \to m(e_i)$ of $C_{/m(e_i)}$. There is a commutative diagram

$$\begin{tikzcd}
D_i \arrow[r, "m_i"] \arrow[d, "\overline{\alpha}_{i!}"'] & C_i \arrow[d, "\overline{\alpha}_{i!}"] \\
D_{/e_i} \arrow[r, "m/i"] & C_{/m(e_i)}
\end{tikzcd}$$

Since e_i and $m(e_i)$ are final objects of D_i and C_i, respectively (7.4.13.5), the categories $D_{/e_i}$ and $C_{/m(e_i)}$ are $\mathscr{U}$-topoi fibered over $I_{/i}$, and the functor $m_{/i}$ is an $I_{/i}$-functor. Since the property of a functor being a morphism of sites is local (IV 4.9), the functor $m_{/i}$ is a morphism from the total site $C_{/m(e_i)}$ to the total site $D_{/(e_i)}$. We are therefore reduced to proving that m_i is a morphism of topoi when i is a final object of I, which we assume
henceforth. 297

. *End of the proof.* Denote by $\alpha_{i!} : D_i \to D^\sim$, $m^\sim : D^\sim \to C^\sim$, and $\alpha_{i!} : C_i \to C^\sim$ the natural extensions to the categories of sheaves of the functors $\alpha_{i!}$ and m. There is a diagram commutative up to isomorphism:

$$\begin{tikzcd}
D_i \arrow[r, "m_i"] \arrow[d, "\alpha_{i!}"'] & C_i \arrow[d, "\alpha_{i!}"] \\
D^\sim \arrow[r, "m^\sim"] & C.
\end{tikzcd}$$

Since i is a final object, the functor $\alpha_i^*\alpha_{i!} : C_i \to C_i$ is isomorphic to the identity (7.4.12), and consequently m_i is isomorphic to $\alpha_i^* m^\sim \alpha_{i!}$. Moreover, the functors α_i^*, $m^\sim$, and α_i^* are left exact (7.4.12). Consequently, m_i is left exact and is therefore a morphism from the site C_i to the site D_i.

Exercise 7.4.14. Let $X = (X_i)_{i \in I}$ be a topos fibered over I and let T be a topos. Show that the categories $\mathbf{Homtop}(X_i, T)$ are the fibers of a fibered category denoted

$\mathbf{Homtop}_I(X, T \times I)$. Show that the category of sections over I of $\mathbf{Homtop}_I(X, T \times I)$ is canonically equivalent to $\mathbf{Homtop}_I(\mathrm{Top}(X), T)$, where $\mathrm{Top}(X)$ is the total topos of X.

Exercise 7.4.15. Let $X = (X_i)_{i \in I}$ be a topos fibered over I. Define a Cartesian morphism m from X to the constant fibered topos with fiber (Ens). Show that the total topos of the constant fibered topos with fiber (Ens) is canonically equivalent to $I^{\hat{}}$. Deduce a morphism of topoi

$$\mathrm{Top}(m) : \mathrm{Top}(X) \longrightarrow I^{\hat{}}.$$

Let $F = (F_i)_{i \in I}$ (where $F_i = F|X_i$) be an object of $\mathrm{Top}(X)$ (7.4.7). Show that

$$\mathrm{Top}(m)(F) = (i \mapsto \Gamma(X_i, F_i)).$$

Deduce from 7.4.12 that if F is injective abelian, then F_i is injective abelian for every $i \in \mathrm{ob}\, I$. Deduce that

$$R^q \,\mathrm{Top}(m)(F) = (i \mapsto H^q(X_i, F_i)).$$

298 Show that the Cartan–Leray spectral sequence of the morphism m (V 5) is

$$H^{p+q}(\mathrm{Top}(X), F) \Leftarrow \varprojlim_I{}^{(p)} H^q(X_i, F|X_i).$$

8. Projective limits of fibered topoi

8.1. Generalities.

Definition 8.1.1. *Let $\mathcal{U}$ be a universe and let $p : F \to I$ be a fibered $\mathcal{U}$-topos. A pair (C, m) consisting of a $\mathcal{U}$-topos C and a cartesian morphism $m : C \times I \to F$ of fibered $\mathcal{U}$-topoi over I is called a projective limit of the fibered topos F if, for every $\mathcal{U}$-topos D, the functor*

$$\mathbf{Homtop}(D, C) \longrightarrow \mathbf{Homtop}_{\mathrm{cart}/I}(D \times I, F) \tag{8.1.1.1}$$

obtained by composing the base-change functor

$$\mathbf{Homtop}(D, C) \longrightarrow \mathbf{Homtop}_I(D \times I, C \times I)$$

with the functor of composition with m (7.1.8) is an equivalence of categories. (cf. 8.1.3.3 for a more “geometric” formulation).

8.1.2. We use the notation of 8.1.1. Let D be a $\mathcal{U}$-topos and let $m_1 : D \times I \to F$ be a morphism of fibered $\mathcal{U}$-topoi. Translating Definition 8.1.1, one immediately sees that there is a morphism of topoi $n : D \to C$ and an isomorphism of morphisms of fibered topoi over I, $\gamma : m_1 \xrightarrow{\sim} m \circ (n \times \mathrm{id}_I)$, making the diagram commutative:

$$\begin{tikzcd} D \times I \arrow[r, "m_1"] \arrow[d, "n \times \mathrm{id}_I"'] \arrow[dr, phantom, "\gamma"] & F \\ C \times I \arrow[ur, "m"'] & \end{tikzcd} \tag{8.1.2.1}$$

Moreover, if (n', γ') is another pair with the property above, there is a *unique* isomorphism $\eta : n \xrightarrow{\sim} n'$ such that

$$\gamma \circ m(\eta \times \mathrm{id}) = \gamma'. \tag{8.1.2.2}$$

It follows that if (D, m_1) is a projective limit of F, then n is an equivalence of $\mathscr{U}$-topoi (IV 3.4), and the additional datum of the isomorphism γ making diagram (8.1.2.1) commutative determines the pair (n, γ) up to canonical isomorphism. Existence of the projective limit will be proved in 8.2.3 when I is cofiltered. **299**

8.1.3. We denote by

$$\underleftarrow{\operatorname{limtop}}_{I} F \text{ or } \underleftarrow{\operatorname{limtop}}_{I} F_i \text{ or also } \underleftarrow{F}$$

a projective limit of the fibered topos F, by $\mu : (\underleftarrow{\operatorname{limtop}}_{I} F) \times I \to F$ the canonical morphism, and, for every object i of I, by

$$\mu_i : \underleftarrow{F} \simeq \underleftarrow{F} \times i \longrightarrow F_i$$

the morphism of topoi induced by μ on the fiber over i.

Choose a bisplitting of F (7.1.4), i.e. for every $f : i \to j$ a morphism of topoi $f\cdot = (f^*, f_*) : F_i \to F_j$ such that f^* is an inverse-image functor for the fibered structure. Then, for every pair $i \xrightarrow{f} j \xrightarrow{g} k$ of composable morphisms of I, we have an isomorphism of morphisms of topoi (7.1.3)

$$c_{f,g} : g.f. \simeq (gf). \quad .$$

Since μ is a cartesian morphism of fibered topoi, for every morphism $f : i \to j$ of I we have an isomorphism of morphisms of topoi (7.1.6)

$$b_f : f.\mu_i \xrightarrow{\sim} \mu_j, \tag{8.1.3.1}$$

and the family of the b_f satisfies relations analogous to (7.1.6.2).

. Let D be a $\mathscr{U}$-topos, let $m : D \times I \to F$ be a morphism of fibered topoi, let $m_i : D = D \times \{i\} \to F_i$ be the morphisms induced by m on the fibers, and, for every $f : i \to j$, let

$$b'_f : f.m_i \xrightarrow{\sim} m_j,$$

be the transition isomorphism. It follows from 8.1.2 that there is a morphism of topoi $n : D \to \underleftarrow{F}$ and a family of isomorphisms **300**

$$\gamma_i : m_i \xrightarrow{\sim} \mu_i n \quad i \in \operatorname{ob} I,$$

such that, for every $f : i \to j$,

$$b_f \circ f.(\gamma_i) = \gamma_j \circ b'_f.$$

Moreover, the morphism n and the family of the γ_i, $i \in \operatorname{ob}(I)$, are determined up to unique isomorphism (8.1.2).

Conversely, suppose we are given a topos $\underleftarrow{F}$, morphisms $\mu_i : \underleftarrow{F} \to F_i$, and isomorphisms b_f as in (8.1.3.1.), satisfying the usual compatibility conditions, and suppose all these data have the universal property described above. Then there is a unique morphism $\mu : \underleftarrow{F} \times I \to F$ of fibered topoi giving rise to the μ_i and the b_f, and $(\underleftarrow{F}, \mu)$ is a projective limit of the fibered topos F.

. Let $\mathscr{V}$ be a universe such that $\mathscr{U} \in \mathscr{V}$. The preceding considerations therefore allow projective limits of fibered topoi to be interpreted as "projective limits," in the sense of 2-categories, of the pseudofunctors with values in the 2-category of fibered $\mathscr{U}$-topoi that are elements of $\mathscr{V}$, deduced from the fibered structure by choosing the morphisms $f_\cdot$. The reader should take care, however, not to confuse this notion of "projective limit" in the sense of 2-categories (with commutation up to isomorphism) with the strict (or naive) notion of projective limit (strict commutation of diagrams). Even when the strict notion makes sense (when the pseudofunctor is a genuine functor), generalized "projective limits" and strict projective limits do not coincide in general.

8.1.4. Let $p : F \to I$ and $q : G \to I$ be two fibered $\mathscr{U}$-topoi, and let $m : F \to G$ be a morphism of fibered topoi. Suppose that the fibered topoi F and G have projective limits (8.1.1). It follows immediately from Definition 8.1.1 that there is a morphism of topoi

(8.1.4.1) $$\underleftarrow{m} : \underleftarrow{F} \longrightarrow \underleftarrow{G},$$

and an isomorphism of morphisms of fibered topoi $\gamma : m \circ \mu \xrightarrow{\sim} \mu \circ \underleftarrow{m} \times \mathrm{id}_I$, making the
301 diagram commutative:

(8.1.4.2)
$$\begin{tikzcd} \underleftarrow{F} \times I \arrow[r, "\mu"] \arrow[d, "\underleftarrow{m}\times\mathrm{id}_I"'] \arrow[dr, phantom, "\swarrow \gamma"] & F \arrow[d, "m"] \\ \underleftarrow{G} \times I \arrow[r, "\mu"] & G \end{tikzcd} .$$

Moreover, if $(\underleftarrow{m}', \gamma')$ is another pair with the property above, there is a *unique* isomorphism $\eta : \underleftarrow{m} \xrightarrow{\sim} \underleftarrow{m}'$ of morphisms of topoi such that

$$\gamma \circ \mu(\eta \times \mathrm{id}) = \gamma'.$$

8.1.5. Let $p : F \to I$ be a fibered $\mathscr{U}$-topos that is an element of $\mathscr{V}$, let $\phi : I' \to I$ be a functor that is an element of $\mathscr{V}$, and let $p_\phi : F_\phi \to I'$ be the fibered $\mathscr{U}$-topos deduced from (F, p) by the base change ϕ (7.1.9). Let $(\underleftarrow{F}, \mu)$ be a projective limit of F. By base change, we obtain a morphism of fibered topoi over I':

$$\mu' : \underleftarrow{F} \times I' \longrightarrow F_\phi.$$

Let $(\underleftarrow{F_\phi}, \mu_\phi)$ be a projective limit of F_ϕ. By the universal property of the projective limit (8.1.2), we then have a morphism of topoi

$$m_\phi : \underleftarrow{F} \longrightarrow \underleftarrow{F_\phi},$$

and an isomorphism of morphisms of fibered topoi over I':

$$\gamma : \mu' \xrightarrow{\sim} \mu_\phi \circ (m_\phi \times \mathrm{id}_{I'}).$$

The pair (m_ϕ, γ) is determined up to unique isomorphism.

8.1.6. Let $p : F \to I$ be a fibered $\mathscr{U}$-site over a small category I, let $F^{\sim/I} \to I$ be the associated fibered $\mathscr{U}$-topos (7.2.6), and let D be a $\mathscr{U}$-topos. First, we have a functor

(8.1.6.1) $$\mathbf{Homtop}_{\mathrm{cart}/I}(D \times I, F^{\sim/I}) \longrightarrow \mathbf{Morsite}_{\mathrm{cart}/I}(D \times I, F),$$

which associates with every m the composite of m^* with $\epsilon_{F/I}$ (7.2.7). Moreover, by the definition of the category $\mathbf{Morsite}_{\mathrm{cart}/I}(D \times I, F)$ (7.2.2.2), we have a functor

(8.1.6.2) $$\mathbf{Morsite}_{\mathrm{cart}/I}(D \times I, F) \longrightarrow \mathbf{Hom}_{\mathrm{cart}/I}(F, D \times I)^\circ.$$

Finally, by the definition of the inductive limit (6.3), we have a functor 302

(8.1.6.3) $$\mathbf{Hom}_{\mathrm{cart}/I}(F, D \times I)^\circ \longrightarrow \mathbf{Hom}(\varinjlim_{I^\circ} F, D)^\circ.$$

Denote by

(8.1.6.4) $$\Theta : \mathbf{Homtop}_{\mathrm{cart}/I}(D \times I, F^{\sim/I}) \longrightarrow \mathrm{Hom}(\varinjlim_{I^\circ} F, D)^\circ$$

the composite of these last three functors, and by

(8.1.6.5) $$\Pi : F \longrightarrow \varinjlim_{I^\circ} F,$$

the canonical functor.

Proposition 8.1.7. *The functor Θ is fully faithful. Its essential image is the set of functors $m : \varinjlim_{I^\circ} F \to D$ such that the functor*

$$(m \circ \Pi, p) : F \longrightarrow D \times I$$

is a morphism from the total site $D \times I$ to the total site F.

It follows from 7.2.7 that the functor (8.1.6.1) is an equivalence of categories, from 7.2.2 that the functor (8.1.6.2) is fully faithful, and from 6.3 that the functor (8.1.6.3) is an equivalence of categories. Consequently, Θ is fully faithful. A functor $m : \varinjlim_{I^\circ} F \to D$ is isomorphic to the image, under $\mathbf{Hom}_{\mathrm{cart}/I}(F, D \times I)^\circ \to \mathbf{Hom}(\varinjlim_{I^\circ} F, D)^\circ$, of the functor $(m \circ \Pi, p)$, and such a functor belongs to the essential image of (8.1.6.2) if and only if it is a morphism from the total site $D \times I$ to the total site F (7.4.13.1); hence the proposition.

8.2. Construction of the projective limit when the index category is cofiltered.

Lemma 8.2.1. *Let I and I' be two cofiltered categories* (I 8.1)*, and let $\phi : I' \to I$ be a functor such that $\phi^\circ : I'^\circ \to I^\circ$ is cofinal* (I 8.1)*. Let, moreover, $p : F \to I$ be a fibered $\mathcal{U}$-topos, let D be a $\mathcal{U}$-topos, let $m : D \times I \to F$ be a Cartesian morphism of fibered topoi, and let $m' : D \times I' \to F'$ be the morphism of fibered topoi deduced from m by the base change ϕ (7.1.9). The pair (D, m) is a projective limit of F if and only if (D, m') is a projective limit of F'.*

The proof is merely a rather lengthy routine verification. It is left to the patient 303
reader, who may use the viewpoint of projective limits of pseudofunctors (8.1.3.3).

Lemma 8.2.2. *Let $p : F \to I$ be a fibered $\mathcal{U}$-site over a small cofiltered category, let $\underline{F}$ be the inductive limit of the fibered category F (6.3.9), and let $\pi : F \to \underline{F}$ be the canonical functor. Equip $\underline{F}$ with the coarsest topology making π continuous* (III 1)*. Let D be a $\mathcal{U}$-site and $n : \underline{F} \to D$ a functor. The following properties are equivalent:*

i) *n is a morphism of sites $D \to \underline{F}$.*
ii) *$n \circ \pi$ is a morphism of sites (where F is equipped with the total topology).*
iii) *$(n \circ \pi, p) : F \to D \times I$ is a morphism from the total site $D \times I$ to the total site F.*
iv) *For every object i of I, the composite functor $F_i \xrightarrow{\alpha_{i!}} F \xrightarrow{n \circ \pi} D$ is a morphism of sites.*

The equivalence of i) and ii) follows from 6.6, and that of iii) and iv) follows from 7.4.13. Let us prove the equivalence of iii) and ii). For this it is enough to show that the first projection functor $\mathrm{pr}_1 : D \times I \to D$ is a morphism of sites. Denote by $D^\sim$ the topos of sheaves on D. It follows from 7.4.7 that $(D \times I)^{\sim}$ is equivalent to the topos $\mathbf{Hom}_{I^{\circ}}(I^{\circ}, D^{\sim} \times I^{\circ}) = \mathbf{Hom}(I^{\circ}, D^{\sim})$. Moreover, one sees immediately that the natural extension of $\mathrm{pr}_1 : D \times I \to D$ to sheaves (III 1.2) is the functor from $\mathbf{Hom}(I^{\circ}, D^{\sim})$ to $D^{\sim}$ that associates with the functor $i \mapsto \sigma(i)$ the object $\varinjlim_{I^{\circ}} \sigma(i)$. Since I° is filtered, inductive limits over I° are exact (II 4), and hence pr_1 is a morphism of sites (IV 4.9). It remains to prove the equivalence of ii) and iv). First treat the case in which D is a $\mathscr{U}$-topos and F a fibered $\mathscr{U}$-topos over I.

Let i be an object of I. Arguing as in 7.4.13.6, one sees that the functor $\alpha_{i!} : F_i \to F$ factors as a morphism $\overline{\alpha}_{i!} : F_i \to F/\alpha_{i!}(e_i)$ followed by the localization functor $F/\alpha_{i!}(e_i) \to F$, where e_i is a final object of F_i. It is therefore enough to show that the composite functor $F/\alpha_{i!}(e_i) \xrightarrow{n\circ\pi} D$ is a morphism of sites. But there is a diagram commutative up to isomorphism:

$$\begin{tikzcd}
F/\alpha_{i!}(e_i) \arrow[r, "n\circ\pi/\alpha_{i!}(e_i)"] \arrow[d] & D/n\circ\pi\alpha_{i!}(e_i) \arrow[d] \\
F \arrow[r, "n\circ\pi"] & D
\end{tikzcd}$$

304 Since the property of being a morphism of sites is local (IV 5.10), it is enough to show that $n \circ \pi \circ \alpha_{i!}(e_i)$ is a final object of D. For every morphism $f : j \to k$ of I, there exists a unique morphism $e_f : \alpha_{j!}(e_j) \to \alpha_{k!}(e_k)$ over f, and this morphism is Cartesian. Thus $j \mapsto \alpha_{j!}(e_j)$ is a Cartesian section of F over I. Consequently, the canonical morphism $n \circ \pi \circ \alpha_{i!}(e_i) \to \varinjlim_I n \circ \pi \circ \alpha_{j!}(e_j)$ is an isomorphism, because π carries Cartesian morphisms to isomorphisms. Moreover, $\varinjlim_I \alpha_{j!}(e_j)$, where the inductive limit is taken in $F^{\wedge}$, is a final object of $F^{\wedge}$. Since $n \circ \pi$ is a morphism of sites, it carries the final object of $F^{\sim}$ to a final object of D. Hence $\varinjlim_I n \circ \pi \circ \alpha_{j!}(e_j)$ is a final object of D, completing the proof in this case.

In the general case, the functor $(n\pi, p) : F \to D \times I$ is Cartesian and continuous (7.4.4). Moreover, $n \circ \pi$ is the composite functor

$$F \xrightarrow{(n\pi,p)} D \times I \xrightarrow{\mathrm{pr}_1} D.$$

Passing to the associated fibered $\mathscr{U}$-topoi gives a diagram commutative up to isomorphism:

$$\begin{tikzcd}
F \arrow[r, "(n\pi,p)"] \arrow[d, "\varepsilon_{F/I}"'] & D \times I \arrow[r, "\mathrm{pr}_1"] \arrow[d, "\varepsilon_D \times \mathrm{id}"] & D \arrow[d, "\varepsilon_D"] \\
F^{\sim/I} \arrow[r, "(n\pi,p)^{\sim/I}"] & D^{\sim} \times I \arrow[r, "\mathrm{pr}_1"] & D^{\sim}
\end{tikzcd}.$$

Since $(n\pi, p)^{\sim/I}$ is Cartesian, it is of the form $(n'\pi', p')$, where $p' : F^{\sim/I} \to I$ and $\pi' : F^{\sim/I} \to \varinjlim_{I^{\circ}} F_i^{\sim}$ are the canonical functors. Thus $\mathrm{pr}_1 \circ (n\pi, p)^{\sim/I} = n'$. Since the functors $\epsilon_{F/I}$ and ϵ_D induce equivalences on the topoi of sheaves (7.4.11), the functors n and n' induce isomorphic functors after passage to associated sheaves (III 1.2), and hence
305 n' is a morphism of sites. By what precedes, the functor $(n\pi, p)^{\sim/I}$ induces morphisms of topoi on the fibers. It is therefore a morphism from the total site $F^{\sim/I}$ to the total site $D^{\sim} \times I$ (7.4.13). Since the functors $\epsilon_{F/I}$ and $\epsilon_D \times \mathrm{id}$ induce equivalences on the total

topoi (7.4.11), the functor $(n\pi, p)$ is a morphism of fibered sites (7.4.13), which proves the assertion.

THEOREM 8.2.3. *Let $p : F \to I$ be a fibered $\mathcal{U}$-site over an essentially small cofiltered category* (I 8.1).

1) *Let $\underrightarrow{F}$ denote the site whose underlying category is the inductive limit of the fibered category $F \to I$* (6.3) *and whose topology is the coarsest topology making the canonical functor $\pi : F \to \underrightarrow{F}$ continuous. Then $\underrightarrow{F}$ is a $\mathcal{U}$-site and*
$$(\pi, p) : F \longrightarrow \underrightarrow{F} \times I$$
is a morphism from the fibered site $\underrightarrow{F} \times I$ to the fibered site F.
2) *Let $\mu : \underrightarrow{F}^{\sim} \times I \to F^{\sim/I}$ be the morphism of fibered $\mathcal{U}$-topoi deduced from (π, p) by passing to the associated fibered $\mathcal{U}$-topoi. The pair $(\underrightarrow{F}^{\sim}, \mu)$ is a projective limit of $F^{\sim/I}$.*

REMARK 8.2.4. 1) In view of the notation introduced in 8.1.3, one may set
$$\underset{\overleftarrow{I}}{\operatorname{limtop}}\, F^{\sim} = \underrightarrow{F}^{\sim}.$$

2) It follows from 7.4.4 that the topology on $\underrightarrow{F}$ is also the coarsest topology making the family of functors $\pi_j : F_j \to F \xrightarrow{\pi} \underrightarrow{F}$, $j \in \operatorname{ob} I$, continuous.

DEFINITION 8.2.5. *The site $\underrightarrow{F}$ introduced in 8.2.3 is called the projective-limit site of the fibered site F.*

As an exercise, the reader may, as in 8.2.1, define the notion of a projective limit of a fibered site and verify that the site $\underrightarrow{F}$ is indeed a projective limit of the site F. The second assertion of the theorem may therefore be paraphrased as follows: the topos of sheaves on the projective-limit site is equivalent to the projective limit of the associated fibered topos.

8.2.6. *Proof of the theorem: Reduction to the case of a small index category.* We shall **306**
give only indications. Let I' be a full subcategory of I such that I'° is cofinal in I°. Denote by $F' \to I'$ the fibered category deduced from F by the base change $I' \to I$, and by $\pi' : F' \to \underrightarrow{F}'$ the canonical functor. One first observes that the categories $\underrightarrow{F}$ and $\underrightarrow{F}'$ are equivalent and that the topology on $\underrightarrow{F}'$ deduced from the topology of $\underrightarrow{F}$ by this equivalence is the coarsest topology making the functor $\pi' : F' \to \underrightarrow{F}'$ continuous. Assertion 1) of the theorem follows when it has been proved in the case where I is small. Assertion 2) in the general case then follows from assertion 2) in the case where I is small by Lemma 8.2.1.

8.2.7. *End of the proof of the theorem.* The first assertion follows from Lemma 8.2.2, applied with $D = \underrightarrow{F}$ and $n = \mathrm{id}$. Let D be a $\mathcal{U}$-topos. There is a diagram commutative up to isomorphism:

$$\begin{tikzcd}
\mathbf{Homtop}(D,(\underrightarrow{F})^{\sim}) \arrow[r, "(1)"] \arrow[d, "(3)"'] & \mathbf{Morsite}(D,\underrightarrow{F}) \arrow[r, hook, "(2)"] \arrow[d, "(3)"'] & \mathbf{Hom}(\underrightarrow{F},D)^{\circ} \arrow[dd, "(5)"'] \\
\mathbf{Homtop}_{\mathrm{Cart}/I}(D\times I,(\underrightarrow{F})^{\sim}\times I) \arrow[r, "(1)"] \arrow[d, "(4)"'] & \mathbf{Morsite}_{\mathrm{Cart}/I}(D\times I,(\underrightarrow{F})\times I) \arrow[d] & \\
\mathbf{Homtop}_{\mathrm{Cart}/I}(D\times I,F^{\sim/I}) \arrow[r, "(1)"] & \mathbf{Morsite}_{\mathrm{Cart}/I}(D\times I,F) \arrow[r, hook, "(2)"] & \mathbf{Hom}_{\mathrm{Cart}/I}(F,D\times I)^{\circ},
\end{tikzcd}$$

where the functors of type (1) are equivalences (IV 4.9.4 and 7.2.7), the functors of type (2) are fully faithful (IV 4.9.1 and 7.2.2), the functors of type (3) are base-change functors, the functors of type (4) are composition functors (with μ, resp. (π, p)), and functor (5) is the equivalence deduced from the universal property of the inductive limit (6.3). To prove that the functor $\mathbf{Homtop}(D, (\underrightarrow{F})^\sim) \xrightarrow{(4)\circ(3)} \mathbf{Homtop}_{/I}(D \times I, F^{\sim/I})$ is an equivalence, it is enough to show that the functor

$$\mathbf{Morsite}(D, \underrightarrow{F}) \xrightarrow{(4)\circ(3)} \mathbf{Morsite}_{/I}(D \times I, F)$$

307 is an equivalence of categories. Equivalently, it is enough to show that the two fully faithful functors

$$\mathbf{Morsite}(D, \underrightarrow{F}) \xrightarrow{(5)\circ(2)} \mathbf{Homcart}_I(F, D \times I)^\circ$$

$$\mathbf{Morsite}_{/I}(D \times I, F) \xrightarrow{(2)} \mathbf{Homcart}_I(F, D \times I)^\circ$$

have the same essential images, which follows immediately from Lemma 8.2.2.

8.2.8. Let $p : F \to I$ be a fibered $\mathscr{U}$-topos over a category I, and let $(\underset{\overleftarrow{I}}{\mathrm{limtop}}\, F_i, \mu)$ be a projective limit of F. Thus there is a Cartesian morphism of fibered $\mathscr{U}$-topoi over I:

$$\mu : \underset{\overleftarrow{I}}{\mathrm{limtop}}\, F_i \times I \longrightarrow F,$$

whence, on passing to the categories fibered over I° by the direct image functors (7.1.3), a Cartesian functor

$$\mu_* : \underset{\overleftarrow{I}}{\mathrm{limtop}}\, F_i \times I^\circ \longrightarrow F'$$

which is the direct image functor for μ (7.1.7). By the definition of the projective limit of a fibered category (6.10), there is therefore a canonical functor

$$\Theta : \underset{\overleftarrow{I}}{\mathrm{limtop}}\, F_i \longrightarrow \underset{\overleftarrow{I, f_*}}{\lim}\, F_i = \mathbf{Hom}_{\mathrm{cart}/I^\circ}(I^\circ, F'). \tag{8.2.8.1}$$

Theorem 8.2.9. *Use the notation of 8.2.8, and suppose that I is an essentially small cofiltered category. Then the functor Θ is an equivalence of categories. Suppose that I is small. Set $\underset{\overleftarrow{I}}{\mathrm{limtop}}\, F_i = (\underrightarrow{F})^\sim$ (which is permitted by 8.2.3 and 8.2.4), and interpret the category $\mathbf{Hom}_{I^\circ}(I^\circ, F')$ as the category of sheaves on the total site F (7.4.7). Then the composite functor*

$$(\underrightarrow{F})^\sim = \underset{\overleftarrow{I}}{\mathrm{limtop}}\, F_i \xrightarrow{\Theta} \mathbf{Hom}_{\mathrm{cart}/I^\circ}(I^\circ, F') \hookrightarrow \mathbf{Hom}_{I^\circ}(I^\circ, F') \approxeq F^\sim,$$

is precisely the direct image functor for the morphism of sites defined by the functor

$$\pi : F \longrightarrow \underrightarrow{F}.$$

308 8.2.10. Some comments before the proof of Theorem 8.2.9. Besides solving the problem of the existence of projective limits when the index category is cofiltered, a solution which is curiously nontrivial in this theory, Theorems 8.2.3 and 8.2.9 provide two further pieces of information essential for working with projective-limit topoi in practice. First, Theorem 8.2.3 interprets the projective-limit topos as the topos of sheaves on a site whose underlying category is the inductive-limit category of the categories F_i

under the system of inverse image functors. In applications, one will be able to interpret this category and its topology concretely. Second, Theorem 8.2.9 asserts that a sheaf on the projective limit is determined by the system of its direct images in the topoi F_i (which gives a Cartesian section over I° of the category $(F^{\sim/I})'$). The theorem further asserts conversely that, when one is given for every object i of I a sheaf X_i on F_i and, for every morphism $f : i \to j$ of I, an isomorphism $X_j \simeq f_* X_i$, these isomorphisms being subject to compatibility conditions, there exists essentially a unique sheaf on the projective limit whose direct images give the system of the X_i.

8.2.11. Assertion 2) of 8.2.3 and the first assertion of 8.2.9 may be summarized by the striking formula

$$\varprojlim_{I,f_*} F_i^{\sim} \approx \operatorname{limtop}_{\overleftarrow{I,f.}} F_i^{\sim} \approx (\varinjlim_{I^\circ,f_*} F_i)^{\sim}. \tag{8.2.11.1}$$

8.2.12. *Proof of Theorem* 8.2.9. To prove the first assertion, reduce to the case where the category I is small by using 8.2.1 and an analogous lemma on projective limits of fibered categories. We leave the details to the reader. Suppose henceforth that I is a small category. We know that the functor

$$\pi : F \longrightarrow \varinjlim_{I^\circ} F_i$$

is a morphism of sites (8.2.2). The direct image functor

$$\pi_* : (\varinjlim_{I^\circ} F_i)^{\sim} \longrightarrow F^{\sim}$$

that it defines on the topoi is then fully faithful, and its essential image is the set of sheaves on F that carry the Cartesian morphisms of F to isomorphisms (6.6). These sheaves correspond, under the equivalence $F^{\sim} \simeq \mathbf{Hom}_{I^\circ}(I^\circ, F')$, to the Cartesian sections of F' over I° (7.4.7). Moreover, for every $j \in \mathrm{ob}(I)$, denote by $\alpha_{j!} : F_j \to F$ the inclusion functor, which gives rise to the three functors $(\alpha_{j!}, \alpha_j^*, \alpha_{j*}) : F_j \rightleftarrows F$ (7.4.3). It follows from the definition of the equivalence $F^{\sim} \simeq \mathbf{Hom}_{I^\circ}(I^\circ, F')$ (7.4.7) that a sheaf X on $\varinjlim_{I^\circ} F_i$ is carried by the functor $(\varinjlim_{I^\circ} F_i)^{\sim} \xrightarrow{\pi_*} F^{\sim} \simeq \mathbf{Hom}_{I^\circ}(I^\circ, F')$ to the Cartesian section $j \mapsto \alpha_j^* \pi_*(X)$. But the description in 8.2.3 of the functor $\mu : (\varinjlim_{I^\circ} F_i)^{\sim} \times I \to F$ shows that, for every object j of I, the direct image functor $\mu_{j*} : (\varinjlim_{I^\circ} F_i)^{\sim} \to F_j$ for the morphism $\mu_j : (\varinjlim_{I^\circ} F_i)^{\sim} \to F_j$ obtained from μ by passing to the fiber at j is precisely the functor $\alpha_j^* \pi_*$. Consequently, the functor $(\varinjlim_{I^\circ} F_i)^{\sim} \xrightarrow{\pi_*} F^{\sim} \simeq \mathbf{Hom}_{I^\circ}(I^\circ, F')$ is precisely the functor $(\varinjlim_{I^\circ} F_i)^{\sim} \xrightarrow{\Theta} \mathbf{Hom}_{\mathrm{cart}/I^\circ}(I^\circ, F') \hookrightarrow \mathbf{Hom}_{I^\circ}(I^\circ, F')$, which proves the theorem. 309

8.3. Topology of the projective-limit site: the case of coherent topoi.

8.3.1. In this section, $p : F \to I$ is a fibered $\mathscr{U}$-site over an essentially small cofiltered category, whose inverse-image functors $f^* : F_i \to F_j$ commute with fiber products. In addition, for every object X of a fiber category F_i, we are given a set $\mathrm{Cov}(X)$ of covering families for the topology of F_i which have properties (PT0) and (PT1) of II 1.3 and generate the topology of F_i. Finally, suppose that for every $f : i \to j$, every $Y \in \mathrm{ob}\, F_j$, and every family $(Y_\alpha \to Y)_{\alpha \in A}$ in $\mathrm{Cov}(Y)$, the family $(f^*(Y_\alpha) \to f^*(Y))_{\alpha \in A}$ belongs to $\mathrm{Cov}(f^*(Y))$. These conditions on $\mathrm{Cov}(X)$ are satisfied, for example, if one takes $\mathrm{Cov}(X) = \{\text{all covering families of } X \text{ in } F_i\}$. In this section we shall study the 310 projective-limit site of F under suitable finiteness hypotheses (8.3.13).

8.3.2. Let $\pi : F \to \underrightarrow{F}$ be the canonical functor, and let Y be an object of $\underrightarrow{F}$. Denote by $\mathrm{Cov}(Y)$ the set of families $(Y_\alpha \xrightarrow{n_\alpha} Y)_{\alpha\in A}$ of the following form:

There exist $i \in I$, an object X of F_i, a family $(X_\alpha \xrightarrow{n'_\alpha} X)_{\alpha\in A} \in \mathrm{Cov}(X)$, an isomorphism $\phi : \pi(X) \simeq Y$, and a family of isomorphisms $\phi_\alpha : \pi(X_\alpha) \simeq Y_\alpha$, $\alpha \in A$, such that for every $\alpha \in A$ the diagram

$$\begin{tikzcd}
\pi(X_\alpha) \arrow[r, "\phi_\alpha", "\sim"'] \arrow[d, "\pi(n'_\alpha)"] & Y_\alpha \arrow[d, "n_\alpha"] \\
\pi(X) \arrow[r, "\phi", "\sim"'] & Y
\end{tikzcd}$$

is commutative.

Proposition 8.3.3. 1) *The families belonging to* $\mathrm{Cov}(Y)$*, for* $Y \in \mathrm{ob}\,\underrightarrow{F}$*, generate the topology of the projective-limit site* (8.2.5)*.*

2) *The family* $Y \mapsto \mathrm{Cov}(Y)$*,* $Y \in \mathrm{ob}\,\underrightarrow{F}$*, has properties* (PT0) *and* (PT1) *of* II 1.3.

We shall need the following lemma.

Lemma 8.3.4. *Let i be an object of I, and let $n : X' \to X$ be a base-changeable morphism of F_i such that $f^*(n)$ is base-changeable for every $f : j \to i$. Then $\pi(n)$ is base-changeable. The functors*

$$\pi_j : F_j \hookrightarrow F \xrightarrow{\pi} \underrightarrow{F}, \quad j \in \mathrm{ob}\, I,$$

commute with fiber products.

Recall that every object Y of $\underrightarrow{F}$ is equal to an object $\pi(Z)$, $Z \in \mathrm{ob}\, F$, and that every morphism $m : Y \to \pi(X)$ is of the form

$$\pi(Z) \xrightarrow{\pi(s)^{-1}} \pi(Z') \xrightarrow{\pi(m')} \pi(X),$$

where s is a cartesian morphism of F. Consequently, to show that $\pi(n)$ is base-changeable,

311 it is enough to show that the fiber product of every diagram

$$\begin{tikzcd}
 & \pi(X') \arrow[d, "\pi(n)"] \\
\pi(Z') \arrow[r, "\pi(m')"] & \pi(X)
\end{tikzcd}$$

is representable. The morphism $m' : Z' \to X$ factors as $Z' \xrightarrow{m''} X'' \xrightarrow{s'} X$, where $p(m'')$ is the identity of $p(Z') = j$ and s' is cartesian over $p(s') = f$. Put $X''' = f^*(X')$

and $n' = f^*(n)$. We then have a commutative diagram

$$\begin{tikzcd}
X''' \arrow[r, "s''"] \arrow[d, "n'"] & X' \arrow[d, "n"] \\
X'' \arrow[r, "s'"] & X,
\end{tikzcd}$$

and hence a commutative diagram in $\underrightarrow{F}$:

$$\begin{tikzcd}
& \pi(X''') \arrow[r, "\sim"] \arrow[d, "\pi(n')"'] & \pi(X') \arrow[d, "\pi(n)"] \\
\pi(Z') \arrow[r, dash, "\pi(m'')"] & \pi(X'') \arrow[r, "\sim"] & \pi(X)
\end{tikzcd}$$

where n' is a base-changeable morphism of F_j. Thus, to show that $\pi(n)$ is base-changeable, it is enough to show that the product $\pi(Z')\times_{\pi(X'')}\pi(X''')$ is representable, and therefore it is enough to show that $\pi_j : F_j \hookrightarrow F \to \underrightarrow{F}$ commutes with fiber products.

Let

$$\begin{tikzcd}
Y' \arrow[r] \arrow[d] & X' \arrow[d] \\
Y \arrow[r] & X
\end{tikzcd}$$

be a cartesian diagram in F_j, and let W be an object of F over $k \in \operatorname{ob} I$. We have (6.5):

$$\mathbf{Hom}(\pi(W), \pi(Y')) \simeq \varinjlim_{\substack{l \xrightarrow{f} j \\ l \xrightarrow{g} k}} \mathbf{Hom}_{F_\ell}(g^*(W), f^*(Y'))$$

Since $f^* : F_j \to F_\ell$ commutes with fiber products, we have 312

$$\operatorname{Hom}(g^*(W), f^*(Y')) \simeq \operatorname{Hom}(g^*(W), f^*(Y)) \times_{\operatorname{Hom}(g^*(W), f^*(X))} \operatorname{Hom}(g^*(W), f^*(X')).$$

Using the commutation of filtered limits with fiber products (I 2) and formula (6.5), we obtain an isomorphism

$$\operatorname{Hom}(\pi(W), \pi(Y')) \simeq \operatorname{Hom}(\pi(W), \pi(Y)) \times_{\operatorname{Hom}(\pi(W), \pi(X))} \operatorname{Hom}(\pi(W), \pi(X')),$$

which shows that the diagram

$$\begin{tikzcd}
\pi(Y') \arrow[r] \arrow[d] & \pi(X') \arrow[d] \\
\pi(Y) \arrow[r] & \pi(X)
\end{tikzcd}$$

is cartesian in $\underrightarrow{F}$.

8.3.5. *Proof of* 8.3.3. We first prove the second assertion. It follows from 8.3.4 that the families in $\mathrm{Cov}(Y)$, $Y \in \underrightarrow{F}$, consist of base-changeable morphisms, whence property (PT0). To show that these families have property (PT1) (stability under base change), a series of reductions like those in the proof of 8.3.3 reduces us to proving the following assertion:

For every $i \in \mathrm{ob}\, I$, every $Y \in \mathrm{ob}\, F_i$, every $(Y_\alpha \to Y)_{\alpha \in A} \in \mathrm{Cov}(Y)$, and every morphism $u : Z \to Y$ of F_i, the family

$$(\pi(Z) \times_{\pi(Y)} \pi(Y_\alpha) \longrightarrow \pi(Z))_{\alpha \in A}$$

belongs to $\mathrm{Cov}(\pi(Z))$.

This assertion follows from the fact that the $\pi_i : F_i \hookrightarrow F \xrightarrow{\pi} \underrightarrow{F}$ commute with fiber products (8.3.4) and that the families $X \mapsto \mathrm{Cov}(X)$ of F_i are stable under base change. Let us prove the first assertion. It is clear that the families belonging to the $\mathrm{Cov}(Y)$, $Y \in \underrightarrow{F}$, are covering for the coarsest topology T making the functor π continuous (III 1). Thus the topology T' generated by these families is coarser than T. To show that it
313 is finer than T (and hence equal to T), it is enough to show that every sheaf M for T' is such that $M \circ \pi$ is a sheaf on F, or equivalently (8.2.4) that $M \circ \pi_i$ is a sheaf on F_i for every $i \in \mathrm{ob}\, I$. Since the functors $\pi_i : F_i \to \underrightarrow{F}$ commute with fiber products (8.3.4), M is a sheaf for T' if and only if (II 2.3 and I 2.12), for every $i \in \mathrm{ob}\, I$, every $Y \in \mathrm{ob}\, F_i$, and every $(Y_\alpha \to Y)_{\alpha \in A} \in \mathrm{Cov}(Y)$, the sequence of sets

$$M(\pi_i(Y)) \longrightarrow \prod_{\alpha \in A} M(\pi_i(Y_\alpha)) \rightrightarrows \prod_{(\alpha,\beta) \in A \times A} M\big(\pi_i(Y_\alpha \times_Y Y_\beta)\big)$$

is exact, i.e. (*loc. cit.*) if and only if $M \circ \pi_i$ is a sheaf on F_i for every $i \in \mathrm{ob}\, I$.

Proposition 8.3.6. *We use the notation and hypotheses of 8.3.1 and 8.3.2. If, for every $i \in \mathrm{ob}\, I$, $X \mapsto \mathrm{Cov}(X)$ is a pretopology on F_i (II 1.3), and if for every $X \in \mathrm{ob}\, F_i$ the covering families in $\mathrm{Cov}(X)$ are finite, then for every $Y \in \underrightarrow{F}$ the covering families in $\mathrm{Cov}(Y)$ are finite and $Y \mapsto \mathrm{Cov}(Y)$ is a pretopology on $\underrightarrow{F}$.*

8.3.7. The only thing to prove is that the families $Y \mapsto \mathrm{Cov}(Y)$ have property (PT2) of (II 1.3) (stability under composition). Let $i \in \mathrm{ob}\, I$, let Y be an object of F_i, and let $(Y_\alpha \to Y)_{\alpha \in A} \in \mathrm{Cov}(Y)$. In addition, let i_α, $\alpha \in A$, be a family of objects of I; for every α, let Z_α be an object of F_i; and let $(Z_{\alpha\beta} \xrightarrow{n_{\alpha\beta}} Z_\alpha)_{\alpha\beta \in B_\alpha}$ be a family in $\mathrm{Cov}(Z_\alpha)$. Finally, for every $\alpha \in A$, let an isomorphism $\phi : \pi(Z_\alpha) \xrightarrow{\sim} \pi(Y_\alpha)$ be given. Returning to the definition of the families in $\mathrm{Cov}(X)$ for $X \in \mathrm{ob}\, \underrightarrow{F}$ (8.3.2), we see that we must prove that the family

$$(\pi(Z_{\alpha\beta}) \xrightarrow{\pi(n_\alpha)\circ\phi_\alpha\circ\pi(n_{\alpha\beta})} \pi(Y))_{\alpha\beta \in \coprod_A B_\alpha}$$

belongs to $\mathrm{Cov}(\pi(Y))$. We shall do this after five reductions.

8.3.8. *Reduction to the case in which, for every $\alpha \in A$, we have $\phi_\alpha = \pi(m_\alpha)$, $m_\alpha : Z_\alpha \to Y_\alpha$. For every $\alpha \in A$, we have $\phi_\alpha = \pi(m_\alpha)\pi(s_\alpha)^{-1}$ (6.5), where s_α is a Cartesian* *morphism over $f_\alpha : i'_\alpha \to i_\alpha$. Thus there are commutative diagrams*

$$\begin{tikzcd}
Z_{\alpha\beta} \arrow[d, "n_{\alpha\beta}"'] & f^*(Z_{\alpha\beta}) \arrow[l, "s_{\alpha\beta}"'] \arrow[d, "f^*_\alpha(n_{\alpha\beta})"] \\
Z_\alpha & f^*_\alpha(Z_\alpha), \arrow[l, "s_\alpha"']
\end{tikzcd} \qquad \alpha\beta \in \coprod_A B_\alpha,$$

which give commutative diagrams 314

$$\begin{tikzcd}
\pi(Z_{\alpha\beta}) \arrow[r, "\sim"] \arrow[d, "\pi(n_{\alpha\beta})"] & \pi(f^*_\alpha(Z_{\alpha\beta})) \arrow[d, "\pi(f^*_\alpha(m_{\alpha\beta}))"] & \\
\pi(Z_\alpha) \arrow[r, "\pi(S_\alpha)^{-1}", "\sim"'] & \pi(f^*_\alpha(Z_\alpha)) \arrow[r, "\pi(m_\alpha)"] & \pi(Y_\alpha),
\end{tikzcd} \quad \alpha\beta \in \coprod_A B_\alpha.$$

Since the families $(f^*_\alpha(Z_{\alpha\beta}) \xrightarrow{f^*(n_\alpha)} f^*(Z_\alpha))$ belong to $\mathrm{Cov}(f^*_\alpha(Z_\alpha))$, we may reduce to the case in which $\phi_\alpha = \pi(m_\alpha)$ for every α.

8.3.9. *Reduction to the case in which, for every $\alpha \in A$, $i_\alpha = j$ and $p(m_\alpha) = k : j \to i$.* Put $g_\alpha = p(m_\alpha)$. Since I is cofiltered and A is finite, there exist an object j of I and morphisms $h_\alpha : j \to i_\alpha$ such that, for every pair (α, α'), we have $g_\alpha h_\alpha = g_{\alpha'} h_{\alpha'} = k$. Using the inverse-image functors h^*_α as in 8.3.8, we reduce to the stated case.

8.3.10. *Reduction to the case in which $i = j$* and *$k : j \to i$ is the identity.* For every α, there is a commutative diagram

$$\begin{tikzcd}
Z_\alpha \arrow[r, "m'_\alpha"] & k^*(Y_\alpha) \arrow[r, "t_\alpha"] \arrow[d, "k^*(n_\alpha)"'] & Y_\alpha \arrow[d, "n_\alpha"] \\
& k^*(Y) \arrow[r, "t"] & Y,
\end{tikzcd}$$

where the morphisms t and t_α, $\alpha \in A$, are Cartesian, m'_α lies over the identity of j, and $t_\alpha \circ m'_\alpha = m_\alpha$. Replacing the family $(Y_\alpha \to Y)\alpha \in A$ by the family $(k^*(Y_\alpha) \to k^*(Y))_{\alpha\in A} \in$ 315
$\mathrm{Cov}(k^*(Y_\alpha))$, and the morphisms m_α by the morphisms m'_α, we reduce to the stated case.

LEMMA 8.3.11. *Let j be an object of I and let $m : X' \to X$ be a morphism of F_j such that $\pi(m)$ is an isomorphism. There exists a morphism $\ell : j' \to j$ such that $\ell^*(m)$ is an isomorphism.*

Let us show that there exist a morphism $f : i \to j$ and a morphism $q : f^*(X) \to f^*(X')$ such that $f^*(m)q = \mathrm{id}_{f^*(X)}$ and such that $\pi(q)$ is an isomorphism. Indeed, there exists an isomorphism $n : \pi(X) \to \pi(X')$ such that $\pi(m) \circ n = \mathrm{id}_{\pi(X)}$. There consequently exist (6.5) two morphisms $X \xleftarrow{s_1} Y_1 \xrightarrow{q_1} X'$, where s_1 is Cartesian, such that $n = \pi(q_1)\pi(s_1)^{-1}$. Thus $\pi(m) \circ \pi(q_1) = \pi(s_1)$. The morphisms mq_1 and s_1 from Y_1 to X have the same image in $\underline{F}$. Consequently (cf. the description of the morphisms in the inductive limit (6.5)), there exists a Cartesian morphism $s_2 : Y_2 \to Y_1$ such that $mq_1 s_2 = s_1 s_2$. Put $q_1 s_2 = q_2$ and denote by s_3 the Cartesian morphism $s_1 s_2$. Thus

$mq_2 = s_3$. The morphism $q_2 : Y_2 \to X'$ factors in an essentially unique manner as $Y_2 \xrightarrow{q} Y_3 \xrightarrow{s_4} X'$, where s_4 is Cartesian and q lies over the identity of $p(Y_2)$. Thus there is a commutative diagram

(8.3.11.1)
$$\begin{tikzcd} Y_3 \arrow[d, "s_4"'] & Y_2 \arrow[l, "q"'] \arrow[d, "s_3"] \\ X' \arrow[r, "m"] & X.. \end{tikzcd}$$

Put $p(s_4) = p(s_3) = f : i \to j$. We have $f^*(X') \simeq Y_3$, $f^*(X') \simeq Y_2$, and $f^*(m) : Y_2 \to Y_3$ makes the following diagram commutative:

$$\begin{tikzcd} Y_3 \arrow[r, "f^*(m)"] \arrow[d, "s_4"'] & Y_2 \arrow[d, "s_3"] \\ X' \arrow[r, "m"] & X. \end{tikzcd}$$

316 Thus $s_3 = s_3(f^*(m) \circ q)$; since s_3 is Cartesian and $f^*(m) \circ q$ lies over the identity of $p(Y_2)$, we have $f^*(m) \circ q = \mathrm{id}_{Y_2}$. Moreover, the commutativity of diagram (8.3.11.1) shows that $\pi(q)$ is an isomorphism.

Now apply the preceding result to the morphism $q : f^*(X) \to f^*(X')$: there exist a morphism $f' : j' \to i$ and a morphism $q' : f'^*(X') \to f'^*(X)$ such that $f'^*(q)q' = \mathrm{id}_{f'^*(f'^*(X'))}$. But on putting $\ell^* = ff'$, there is an isomorphism $f'^*f^* \simeq \ell^*$. Thus there is a morphism $f'^*(q) : \ell^*(X) \to \ell^*(X')$ and an isomorphism $f'^*f^* \simeq \ell^*$. Thus there is a morphism $f'^*(q) : \ell^*(X) \to \ell^*(X')$ and a morphism $q' : \ell^*(X') \to \ell^*(X)$ such that $\ell^*(m)f'^*(q) = \mathrm{id}_{\ell^*(X)}$ and $f'^*(q)q' = \mathrm{id}_{\ell^*(X')}$. Consequently $\ell^*(m)$ is an isomorphism.

8.3.12. *End of the proof.* Since I is cofiltered and A finite, it follows from 8.3.11 that there exists an inverse-image functor ℓ^* that takes all the morphisms m_α, $\alpha \in A$, to isomorphisms. We are thus reduced to the case in which the m_α are isomorphisms. But then the covering families $(Z_{\alpha\beta} \to Z)_{\alpha\beta \in B_\alpha}$ are deduced by the base change $m_\alpha : Z_\alpha \to Y_\alpha$ from covering families $(Y_{\alpha\beta} \to Y_\alpha)_{\alpha\beta \in B_\alpha} \in \mathrm{Cov}(Y_\alpha)$. We are therefore reduced to the case in which $Z_\alpha = Y_\alpha$ and $\Phi_\alpha = \mathrm{id}_{\pi(Y_\alpha)}$. But in this case the family $(Y_{\alpha\beta} \xrightarrow{n_\alpha \circ n_{\alpha\beta}} Y)_{\alpha\beta \in \coprod_A B_\alpha}$ belongs to $\mathrm{Cov}(Y)$ (property (PT2)). Hence its image under π belongs to $\mathrm{Cov}(\pi(Y))$ (8.3.2).

THEOREM 8.3.13. *Let $p : F \to I$ be a $\mathcal{U}$-topos fibered over an essentially small cofiltered category. If, for every object i of I, the fiber topos F_i is coherent (2.3), and if, for every morphism $f : i \to j$, the morphism of topoi $f\cdot = (f^*, f_*) : F_i \to F_j$ is coherent (3.1), then the topos $\varprojlim\mathrm{top}_I F_i$ is coherent and, for every object i of I, the canonical morphism of topoi*

$$\mu_i : \varprojlim_I \mathrm{top}\, F_i \longrightarrow F_i$$

is coherent.

Moreover, if F_{coh} denotes the full subcategory of F defined by the objects of F that are coherent in their fiber (1.13), then the category F_{coh} is fibered over I, and $\underrightarrow{F}_{\mathrm{coh}}$ is canonically equivalent to the category of coherent objects of $\underset{I}{\underleftarrow{\mathrm{limtop}}}\ F_i$.

For every $i \in \mathrm{ob}\, I$, denote by $(F_i)_{\mathrm{coh}}$ the $\mathscr{U}$-site (endowed with the induced topology) of coherent objects of F_i (1.13). Since F_i is coherent, F_i is the topos of sheaves on $(F_i)_{\mathrm{coh}}$. For every morphism $f : i \to j$, the morphisms $f\cdot = (f^*, f_*)$ are coherent, and consequently $f^*(F_j)_{\mathrm{coh}} \subset (F_i)_{\mathrm{coh}}$. Thus there is a fibered $\mathscr{U}$-site $p_{\mathrm{coh}} : F_{\mathrm{coh}} \to I$ whose fiber sites are the sites $(F_i)_{\mathrm{coh}}$ and whose associated fibered topos is equivalent to $p : F \to I$. Consequently (8.2.3), 317

(8.3.13.1)
$$\underset{I,f.}{\underleftarrow{\mathrm{limtop}}}\ F_i \approxeq (\underrightarrow{F}_{\mathrm{coh}})^\sim$$

Since the topoi F_i are coherent, finite products and fiber products are representable in $(F_i)_{\mathrm{coh}}$ (2.2). Moreover, the functors $f^* : (F_i)_{\mathrm{coh}} \to (F_j)_{\mathrm{coh}}$ are left exact. Finally, since the objects of $(F_i)_{\mathrm{coh}}$ are quasi-compact, the finite covering families form a pretopology. It then follows from (8.3.4) that finite products and fiber products are representable in $\varprojlim_{I^\circ,F}(F_i)_{\mathrm{coh}}$ (addendum to Lemma 8.3.4: show that the canonical functors $(F_i)_{\mathrm{coh}} \to \underrightarrow{F}_{\mathrm{coh}}$ take the final object to the final object), and from (8.3.6) that the objects of $\underrightarrow{F}_{\mathrm{coh}}$ are quasi-compact. Consequently (2.4.5), $\underset{I}{\underleftarrow{\mathrm{limtop}}}\ (F_i)$ is coherent. Finally, the canonical morphisms $\mu_i : \underset{I}{\underleftarrow{\mathrm{limtop}}}\ (F_i) \to F_i$ are deduced, by passage to the topoi of sheaves, from the morphisms of sites (8.2.3)

$$(F_i)_{\mathrm{coh}} \longrightarrow \underrightarrow{F}_{\mathrm{coh}}$$

and hence (3.3) the morphisms μ_i are coherent.

Let us prove the last assertion. We have (8.3.13.1)

$$\underset{\mathscr{I}}{\underleftarrow{\mathrm{limtop}}}\ (F_i) \simeq (\underrightarrow{F}_{\mathrm{coh}})^\sim.$$

Let X be a coherent object of $(F_{\mathrm{coh}})^\sim$. Since X is quasi-compact, there exists a finite covering family $(Y_\alpha \to X)$, where $Y_\alpha \in \mathrm{ob}\,\underrightarrow{F}_{\mathrm{coh}}$ (1.1). On taking an index $i \in \mathrm{ob}\, I$ sufficiently small, we may suppose that the Y_α are the images under $\mu_i : F_{i\mathrm{coh}} \to \underrightarrow{F}_{\mathrm{coh}}$ of a family Y'_α; hence, by taking the direct sum of the Y'_α (1.15), we see that there exist an index $i \in \mathrm{ob}\, I$, an object $Y' \in F_{i\mathrm{coh}}$, and a surjective morphism $\mu_i(Y') \to X$. The equivalence relation $R = \mu_i(Y') \times_X \mu_i(Y')$ is then a coherent object because $(\underrightarrow{F}_{\mathrm{coh}})^\sim$ is 318
a coherent topos (2.2). Let us first show that R is an object of $\underrightarrow{F}_{\mathrm{coh}}$. By definition, R is a subobject of $\mu_i(Y' \times Y')$, and since R is coherent, we may, after changing the index i, find an arrow $m : Z \to (Y' \times Y')$ of $F_{i\mathrm{coh}}$ such that $\mathrm{Im}(\mu_i(m)) = R$. It follows that, since F_i is coherent, $\mathrm{Im}(m)$ is coherent in F_i (1.17.1), and hence belongs to $F_{i\mathrm{coh}}$. There consequently exist an index i and a diagram $R \overset{p_1}{\underset{p_2}{\rightrightarrows}} Y'$ of $F_{i\mathrm{coh}}$ such that $\mu_i(R) \overset{\mu_i(p_1)}{\underset{\mu_i(p_2)}{\rightrightarrows}} \mu_i(Y')$ is an equivalence relation and such that $X = \mu_i(Y')/\mu_i(R)$. Put $X_i = \mathrm{coker}(p_1, p_2)$ and $R' = Y' \times_{X'_i} Y'$. There is a canonical morphism in F_i, $u : R \to R'$. Moreover, $\mu_i^*(u)$ is an isomorphism. Thus, after changing the index i, we may suppose that $R = R'$, i.e. that R is an equivalence relation. But then X_i belongs to $F_{i\mathrm{coh}}$ (1.17.1), and consequently $X = \mu_i(X_i)$ belongs to $\underrightarrow{F}_{\mathrm{coh}}$.

Corollary 8.3.14. *Let I be an essentially small cofiltered category, and let $p : F \to I$ and $q : G \to I$ be two $\mathcal{U}$-topoi fibered over I such that, for every object i of I, the topoi F_i and G_i are coherent, and such that, for every morphism $f : i \to j$, the morphisms $f. : F_i \to F_j$ and $G_i \to G_j$ are coherent. In addition, let $m : F \to G$ be a Cartesian morphism of fibered $\mathcal{U}$-topoi such that, for every $i \in \mathrm{ob}(I)$, $m_i : F_i \to G_i$ is coherent. Then the morphism $\underleftarrow{m}$ deduced from m by passage to the projective limit is coherent.*

For every object i of I, the morphism m_i induces a functor $m^*_{i,\mathrm{coh}} : G_{i,\mathrm{coh}} \to F_{i,\mathrm{coh}}$, whence a Cartesian functor $m^*_{\mathrm{coh}} = G_{\mathrm{coh}} \to F_{\mathrm{coh}}$ that is a morphism from the fibered site F_{coh} to the fibered site G_{coh} (7.4.13). The functor $\varinjlim_{I^\circ} m^*_{\mathrm{coh}} : \varinjlim_{I^\circ} G_{\mathrm{coh}} \to \varinjlim_{I^\circ} F_{\mathrm{coh}}$ is a morphism of sites that, on passage to the corresponding topoi, gives the morphism $\underleftarrow{m}$. The assertion then follows from 8.3.13 and 3.3.

Exercise 8.3.15. With the notation of 8.3.13, show that if the F_i, $i \in I$, are algebraic
319 (2.3) and if the morphisms $f. : F_i \to F_j$, $f \in \mathrm{Fl}(I)$, are coherent, then $\underset{\overleftarrow{I}}{\mathrm{limtop}}\, F_i$ is algebraic and $(\underset{\overleftarrow{I}}{\mathrm{limtop}}\, F_i)_{\mathrm{coh}} \xrightarrow{\sim} \underrightarrow{F}_{\mathrm{coh}}$.

8.4. Example: local topoi.

8.4.1. Let $X = (X_i)_{i \in I}$ be a topos fibered over a category I. One deduces from it a topos fibered over the category $\mathrm{Pro}(I)$ (I 8.10) by the formula

$$X_\alpha = \underset{\overleftarrow{I_\alpha}}{\mathrm{limtop}}\, X_{i\alpha} \tag{8.4.1.1}$$

for every pro-object $\alpha = (i_\alpha)$ of I (to see this, one may generalize Exercise I 8.2.8 to pseudofunctors).

8.4.2. Let E be a topos, and let $p : \mathrm{Fl}(E) \to E$ be the fibered topos considered in 7.3.1. Extending it as in 8.4.1, one obtains a fibered topos $\overline{\mathrm{Fl}(E)} \to \mathrm{Pro}(E)$. The category $\mathrm{Point}(E)$ maps by a fully faithful functor into $\mathrm{Pro}(E)$ (IV 6.8.5), whence, by base change (7.1.9), a fibered topos $\mathrm{Loc}(E) \to \mathrm{Point}(E)$. The fiber $\mathrm{Loc}_p(E)$ at a point p of E is called the *localization of E at the point p*. By the preceding, this topos depends covariantly on the point p. By definition,

$$\mathrm{Loc}_p(E) = \underset{\overleftarrow{X \in \mathrm{Vois}(p)}}{\mathrm{limtop}}\; E_{/X}, \tag{8.4.2.1}$$

where $\mathrm{Vois}(p)$ is the category of neighborhoods of p (IV 6.8).

8.4.3. Let $m : p' \to p$ be a morphism of $\mathrm{Point}(E)$ (IV 6). For every $X \in \mathrm{Vois}(p)$ one thereby obtains a point p'_X of $E_{/X}$ and hence, by the universal property of $\underleftarrow{\mathrm{limtop}}$ 8.1, a point $\theta_p(m)$ of $\mathrm{Loc}_p(E)$. This defines a functor

$$\theta_p : \mathrm{Point}(E)_{/p} \longrightarrow \mathrm{Point}(\mathrm{Loc}_p(E)), \tag{8.4.3.1}$$

which is immediately seen to be an equivalence of categories. Thus $\mathrm{Point}(\mathrm{Loc}_p(E))$ is canonically equivalent to the category of generalizations of p (IV 7.1.8). There is moreover a canonical equivalence of topoi

$$\mathrm{can}. : \mathrm{Loc}_{\theta_p(m)}(\mathrm{Loc}_p(E)) \longrightarrow \mathrm{Loc}_{p'}(E),$$

which fits into a diagram commutative up to isomorphism

$$\begin{tikzcd}
\mathrm{Loc}_{\theta_p(m)}(\mathrm{Loc}_p(E)) \arrow[rr, "\mathrm{can}\,\cdot"] \arrow[dr, "\phi"'] & & \mathrm{Loc}_{p'}(E) \arrow[dl, "m"] \\
& \mathrm{Loc}_p(E) &
\end{tikzcd}$$

where ϕ is the canonical morphism of (8.4.2.1). 320

8.4.4. Localized topoi seem to be of interest only for topoi arising from algebraic geometry, or at least for topoi having suitable finiteness properties. Thus, when E is the topos of sheaves on a separated topological space, one sees immediately (using, for example, 8.2.9) that the localized topoi are point topoi. Moreover, in this case the category $\mathrm{Point}(E)$ is discrete and the situation described in 8.4.2 is trivial. On the other hand, let $E = \mathrm{Top}(X)$ be the topos of sheaves for the Zariski topology on a scheme X, let x be a point of X, let O_x be the local ring of X at x, and let $Y = \mathrm{spec}(O_x)$. One finds that

$$\mathrm{Loc}_x(E) = \mathrm{Top}(Y).$$

For other examples arising from the étale topology, we refer to Exposé VIII of this seminar.

8.4.5. When E is locally coherent (2.3), the localized topoi $\mathrm{Loc}_p(E)$ are coherent (in (8.4.2.1) one may restrict to the $X \in \mathrm{Vois}(p)$ which are algebraic and coherent, and then apply 8.3.13). For every morphism of points $m : p' \to p$, the morphism of topoi $m\cdot : \mathrm{Loc}_{p'}(E) \to \mathrm{Loc}_p(E)$ is coherent (8.3.14).

8.4.6. A *local topos* is a topos X such that the functor $\Gamma(X, -) =$ "sections over X" (IV 4.3) is a fiber functor (IV 6). The point corresponding to this fiber functor is called the *center* of the local topos. When E is locally coherent, its localized topoi are local topoi (1.2.3, 8.5.2, and 8.5.7). The center of $\mathrm{Loc}_p(E)$ is canonically isomorphic to $\theta_p(\mathrm{id}_p)$ ((8.4.3.1)). The image in E of the center of $\mathrm{Loc}_p(E)$ is isomorphic to p. The localized topos $\mathrm{Loc}_p(E)$, equipped with the canonical morphism $\mathrm{Loc}_p(E) \to E$, is the solution of 321
the universal (2-universal!) problem of mapping local topoi into E in such a way that the center maps "to" p.

8.4.7. A number of questions concerning local topoi have not been addressed by the editors. Thus, if X is a local topos, the final object e of X has a maximal open subtopos $U \neq e$. The complement Y of U is a local topos having no nontrivial open subtopos. Is such a topos a point topos? Let U be a topos and let $X = (U, \Phi : U \to (\mathrm{Ens}))$ be a topos obtained by gluing. What conditions must the gluing functor Φ satisfy for X to be a local topos?

8.5. Structure of the sheaves of a filtered projective limit of topoi.

8.5.1. In this section, $F = (F_i)_{i \in I}$ is a $\mathscr{U}$-topos fibered over a small cofiltered category I. Choose a bicleavage (7.1.4) of F, i.e. for every $f \in \mathrm{Fl}(I) : i \to j$, choose morphisms of topoi $f\cdot : F_i \to F_j$ such that $f^* : F_j \to F_i$ is the inverse-image functor for the fibered structure. There are then canonical isomorphisms $c_{f,g}$ satisfying a cocycle condition (7.1.3). Denote by $\underleftarrow{F}$ the projective limit of the fibered topos F (8.2.3), and for every $i \in \mathrm{ob}\, I$, denote by

$$\mu_i : \underleftarrow{F} \longrightarrow F_i, \tag{8.5.1.1}$$

the canonical morphism (8.1.3). Denote by $\mathrm{Top}(F)$ the total topos of F (7.4.3.3). By 8.2.9, the canonical functor $F \to \underleftarrow{F}$ defines a morphism of topoi

$$Q : \underleftarrow{F} \longrightarrow \mathrm{Top}(F). \tag{8.5.1.2}$$

The topos $\underleftarrow{F}$ is identified with $\mathbf{Hom}_{\mathrm{Cart}/I^\circ}(I^\circ, F')$ (8.2.9), and the topos $\mathrm{Top}(F)$ is identified with $\mathbf{Hom}_{I^\circ}(I^\circ, F')$ (7.4.7). Under these identifications, Q is the embedding morphism of $\mathbf{Hom}_{\mathrm{cart}/I^\circ}(I^\circ, F')$ into $\mathbf{Hom}_{I^\circ}(I^\circ, F')$ (8.2.9), and for every object M of $\underleftarrow{F}$ we have

$$Q_*(M) = (i \longmapsto \mu_{i*}(M)). \tag{8.5.1.3}$$

By 7.4.3.4, for every $i \in \mathrm{ob}\, I$ there is a morphism of topoi

$$\alpha_i : F_i \longrightarrow \mathrm{Top}(F). \tag{8.5.1.4}$$

322 The diagram

$$\begin{tikzcd} \underleftarrow{F} \arrow[rr, "\mu_i"] \arrow[dr, "Q"'] & & F_i \arrow[dl, "\alpha_i"] \\ & \mathrm{Top}(F) & \end{tikzcd} \tag{8.5.1.5}$$

is not commutative in general (even up to isomorphism). But the definition of the morphisms involved gives a canonical isomorphism

$$\alpha_i^* Q_* \simeq \mu_{i*}. \tag{8.5.1.6}$$

When i is a *final object* of I, α_i is an embedding having a retraction $\beta_i : \mathrm{Top}(F) \to F_i$ (7.4.12), and the diagram

$$\begin{tikzcd} \underleftarrow{F} \arrow[rr, "\mu_i"] \arrow[dr, "Q"'] & & F_i \arrow[dl, "\beta_i"] \\ & \mathrm{Top}(F) & \end{tikzcd} , \tag{8.5.1.7}$$

is commutative up to canonical isomorphism.

Proposition 8.5.2. *Let $j \mapsto M_j$ be an object of* $\mathrm{Top}(F)$. *There is a functorial isomorphism*

$$Q^*(j \mapsto M_j) \simeq \varinjlim_{I^\circ} \mu_j^* M_j. \tag{8.5.2.1}$$

An object of $\mathrm{Top}(F)$ (7.1.3) consists of the data of an assignment $j \mapsto M_j$, $M_j \in \mathrm{ob}\, F_j$, and, for every morphism $f : i \to j$, a morphism

$$\beta_f : M_j \longrightarrow f_* M_i \tag{8.5.2.2}$$

or, equivalently by adjunction, a morphism

$$\beta'_f : f^* M_j \longrightarrow M_i, \tag{8.5.2.3}$$

where the morphisms β'_f are subject to the condition that, for every pair of composable morphisms $i \xrightarrow{f} j \xrightarrow{g} k$, the following diagram be commutative:

(8.5.2.4)
$$\begin{tikzcd} f^*g^*M_k \arrow[r, "f^*\beta'_g"] \arrow[d, no head, "c^*_{g,f}"'] & f^*M_j \arrow[d, "\beta'_f"] \\ (gf)^*M_k \arrow[r, "\beta'_{gf}"] & M_i \end{tikzcd}$$

Recall moreover from (8.1.3.1) that, for every morphism $f : i \to j$, there is an isomorphism $b_f : f\cdot\mu_i \xrightarrow{\sim} \mu_j$, and that for every pair of composable morphisms $i \xrightarrow{f} j \xrightarrow{g} k$ there is a commutative diagram 323

(8.5.2.5)
$$\begin{tikzcd} g\cdot f\cdot\mu_i \arrow[r, "c_{g,f}"] \arrow[d, "g\cdot(b_f)"] & (gf)\cdot\mu_i \arrow[d, "b_{gf}"] \\ g\cdot\mu_j \arrow[r, "b_g"] & \mu_k \end{tikzcd}$$

Let $f : i \to j$ be a morphism of I. Denote by

(8.5.2.6)
$$t_f : \mu_j^*(M_j) \longrightarrow \mu_i^*(M_i)$$

the composite $\mu_j^*(M_j) \xrightarrow{b_f^*} \mu_i^*(f^*(M_j)) \xrightarrow{\mu_i^*(\beta'_f)} \mu_i^*(M_i)$. We leave it to the reader to verify that the commutativity of diagrams (8.5.2.4) and (8.5.2.5) implies that for every pair of composable morphisms $i \to j \to k$, one has

(8.5.2.7)
$$t_f t_g = t_{gf}.$$

We have therefore defined a functor $I^\circ \to \underline{F}$, whose inductive limit $\varinjlim_{I^\circ} \mu_i^*(M_i)$ can be considered. Let N be an object of $\underline{F}$. We have

(8.5.2.8)
$$\mathrm{Hom}(\varinjlim_{I^\circ} \mu_i^*(M_i), N) \simeq \varprojlim_{I^\circ} \mathrm{Hom}(\mu_i^*(M_i), N),$$

and hence, by adjunction, an isomorphism

(8.5.2.9)
$$\mathrm{Hom}(\varinjlim_{I^\circ} \mu_i^*(M_i), N) \simeq \varprojlim_{I^\circ} \mathrm{Hom}_{F_i}(M_i, \mu_{i*}(N)).$$

We now interpret the right-hand side of (8.5.2.9). For every morphism $f : i \to j$, denote by 324

$$d_f : \mathrm{Hom}_{F_i}(M_i, \mu_{i*}(N)) \longrightarrow \mathrm{Hom}_{F_j}(M_j, \mu_{j*}(N))$$

the transition map of the projective system occurring in (8.5.2.9). It follows from the definition of the morphisms t_f ((8.5.2.6)) that d_f associates with a morphism

$$u_i : M_i \longrightarrow \mu_{i*}(N)$$

the morphism

$$d_f(u_i) : M_j \longrightarrow \mu_{j*}(N),$$

obtained by composing the morphisms

$$M_j \xrightarrow{\beta_f} f_*(M_i) \xrightarrow{f_*(u_i)} f_*\mu_{i*}(N) \xrightarrow{b_f} \mu_{j*}(N).$$

Thus an element of the right-hand side of (8.5.2.9) is a family of morphisms $u_i : M_i \to \mu_{i*}(N)$, $i \in \mathrm{ob}\, I$, such that for every $f : i \to j$ the diagram

$$\begin{tikzcd}
M_j \arrow[r] \arrow[d, "u_j"'] & f_*(M_i) \arrow[d, "f_*(u_i)"] \\
\mu_{j*}(N) \arrow[r, "\sim"] & f_*\mu_{i*}(N)
\end{tikzcd}$$

is commutative, that is, a morphism from the section $(i \mapsto M_i) \in \mathbf{Hom}_{I^\circ}(I^\circ, F')$ to the section $Q_*(N)$. We therefore have a functorial isomorphism

$$\mathrm{Hom}(\varinjlim_{I^\circ} \mu_i^*(M_i), N) \simeq \mathrm{Hom}((i \mapsto M_i), Q_*(N));$$

the proposition follows by adjunction.

Proposition 8.5.3. *If, for every $f : i \to j \in \mathrm{Fl}(I)$, the functors $f_* : F_i \to F_j$ commute with small filtered inductive limits, then for every section $(i \mapsto M_i) \in \mathrm{Hom}^\circ_I(I^\circ, F')$ we have*

$$Q_*Q^*(i \mapsto M_i) \simeq i \mapsto \varinjlim_{f : j \to i} f_*M_j. \tag{8.5.3.1}$$

325 We leave it to the reader to describe explicitly, in terms of the β_f (8.5.2.2) and the $c_{f,g}$ (8.5.1), the transition morphisms of the inductive systems $(f : j \to i) \mapsto f_*(M_j)$ and the transition morphisms of the section $i \mapsto \varinjlim_{f:j\to i} f_*M_i$. Since the functors f_* commute with filtered inductive limits, the section $i \mapsto \varinjlim_{f:j\to i} f_*M_j$ is Cartesian. Moreover, there is a natural morphism u from $(i \mapsto M_i)$ to the section $(i \mapsto \varinjlim_{f:j\to i} f_*M_j)$, and it is clear that every morphism from $(i \mapsto M_i)$ to a Cartesian section factors uniquely through u, giving the isomorphism (8.5.3.1).

Corollary 8.5.4. *Under the hypotheses of 8.5.3, the functor Q_* commutes with small filtered inductive limits.*

Let $\alpha \mapsto N^\alpha$ be a small filtered inductive system of $\underleftarrow{F}$. Put $M^\alpha = Q_*(N^\alpha)$, so that M^α is a section $i \mapsto M_i^\alpha$. We have $N^\alpha \simeq Q^*Q_*(N^\alpha)$, and since Q^* commutes with inductive limits, we have $\varinjlim_\alpha N^\alpha \simeq Q^*(\varinjlim_\alpha M^\alpha)$. Inductive limits in $\mathrm{Top}(F) = \mathrm{Hom}_{I^\circ}(I^\circ, F')$ are computed fiber by fiber. Thus $\varinjlim_\alpha M \simeq (i \mapsto \varinjlim_\alpha M_i^\alpha)$. By (8.5.3.1), we have

$$Q_*(\varinjlim_\alpha N^\alpha) \simeq Q_*Q^*(\varinjlim_\alpha M^\alpha) \simeq i \mapsto \varinjlim_{f:j\to i} f_*(\varinjlim_\alpha M_j^\alpha).$$

Since the functors f_* commute with filtered inductive limits, it follows that

$$Q_*(\varinjlim_\alpha N^\alpha) \simeq i \mapsto \varinjlim_{f:j\to i} \varinjlim_\alpha f_*(M_j^\alpha) \simeq i \mapsto \varinjlim_\alpha \varinjlim_{f:j\to i} f_*(M_j^\alpha) \simeq \varinjlim_\alpha Q_*(N^\alpha).$$

Corollary 8.5.5. *If the functors $f_* : F_i \to F_j$ commute with small filtered inductive limits, then for every object i of I and every $M_i \in \mathrm{ob}\, F_i$ we have*

$$\mu_{i*}\mu_i^*(M_i) \simeq \varinjlim_{f:j\to i} f_*f^*(M_i). \tag{8.5.5.1}$$

After making the base change $I/i \to I$, we may suppose that i is a final object of I (8.2.1).
326 The morphism $\mu_i : \underleftarrow{F} \to F_i$ is then the composite of the morphisms (8.2.9) and (7.4.12):

$$Q : \underleftarrow{F} \longrightarrow F^{\sim},$$
$$\beta_i : (\alpha_{i!}, \alpha_i^*) : F^{\sim} \longrightarrow F_i.$$

Thus $\mu_i^*(M_i) \simeq Q^*\beta_i^*(M_i)$, where $\beta_i^*(M_i)$ is the section $(f : i \to j) \mapsto f^*(M_i)$ whose transition morphisms are deduced from the $c_{f,g}$. The assertion then follows from 8.5.3.

Corollary 8.5.6. *Under the hypotheses of 8.5.5, the functors*

$$\mu_{i*} : \underleftarrow{F} \longrightarrow F_i$$

commute with filtered inductive limits.

The functor μ_{i*} is the composite of the functor Q_*, which commutes with filtered inductive limits, and the functor "restriction to the fiber at i", which commutes with inductive limits.

Corollary 8.5.7. *Under the hypotheses of 8.5.3, let i be an object of I and X an object of F_i such that the functor* $\mathrm{Hom}(X, -)$ *on F_i commutes with filtered inductive limits. Then the functor* $\mathrm{Hom}(\mu_i^*(X), -)$ *commutes with filtered inductive limits, and for every object M_i of F_i we have*

$$\mathrm{Hom}(\mu_i^*(X), \mu_i^*(M_i)) \simeq \varinjlim_{f : j \to i} \mathrm{Hom}(f^*(X), f^*(M_i)). \tag{8.5.7.1}$$

For every object $(j \mapsto M_j)$ of $\mathrm{Top}(F)$, we have

$$\mathrm{Hom}(\mu_i^*(X), Q^*(j \mapsto M_j)) \simeq \varinjlim_{f : j \to i} \mathrm{Hom}(f^*(X), M_j). \tag{8.5.7.2}$$

The functor $\mathrm{Hom}(\mu_i^*(X), -)$ is isomorphic to the functor $\mathrm{Hom}(X, \mu_{i*}(-))$, and the latter commutes with filtered inductive limits (8.5.4). By (8.5.5.1), we have

$$\mathrm{Hom}(\mu_i^*(X), \mu_i^*(M_i)) \simeq \mathrm{Hom}(X, \mu_{i*}, \mu_i^*(M_i)) \simeq \mathrm{Hom}(X, \varinjlim_{f : j \to i} f_*f^*(M_i)),$$

which gives formula (8.5.7.1)). Likewise, by ((8.5.3.1), we have

$$\mathrm{Hom}(\mu_i^*(X), Q^*(j \mapsto M_j)) \simeq \mathrm{Hom}(X, \mu_{i*}Q^*(j \mapsto M_j)) \simeq \mathrm{Hom}(X, \varinjlim_{f : j \to i} f_*(M_j)),$$

which gives formula (8.5.7.2). 327

Proposition 8.5.8. *Let $G \to I$ be a fibered $\mathscr{U}$-topos and $m : F \to G$ a Cartesian morphism of fibered topoi (7.1.15). For G we use the notation introduced in 8.1.1. Let $\underleftarrow{m}$: $\underleftarrow{F} \to \underleftarrow{G}$ be the morphism deduced from m by passage to the projective limit (8.1.4). The diagram of topoi and morphisms of topoi*

$$\begin{tikzcd} \underleftarrow{F} \arrow[r, "Q"] \arrow[d, "\underleftarrow{m}"] & \mathrm{Top}(F) \arrow[d, "m^{\sim}"] \\ \underleftarrow{G} \arrow[r, "Q"] & \mathrm{Top}(G) \end{tikzcd} \tag{8.5.8.1}$$

is commutative up to isomorphism. For every object N of $\underleftarrow{F}$, we have

$$\underleftarrow{m}_*(N) \simeq \varinjlim_{I^\circ} \mu_j^* m_{j*}\mu_{j*}(N). \tag{8.5.8.2}$$

The commutativity of diagram (8.5.8.1) follows immediately from the definitions (8.2.8 and 8.5.1). We have $\underleftarrow{m}_*(N) \simeq Q^*Q_*\underleftarrow{m}_*(N)$ and, by the commutativity of (8.5.8.1), $\underleftarrow{m}_*(N \simeq Q^* m_*^\sim Q_*(N)$. Consequently, there is an isomorphism $\underleftarrow{m}_*(N) \simeq Q^* m_*^\sim(j \mapsto \mu_{j*}(N))$. The functor $m_*^\sim$ is simply the functor $\mathrm{Hom}_{I^\circ}(I^\circ, m_*)$ (7.4.10), and consequently there is a canonical isomorphism $\underleftarrow{m}_*(N) \simeq Q^*(j \mapsto m_{j*}\mu_{j*}(N))$. Formula (8.5.8.2) then follows from 8.5.2.

Proposition 8.5.9. *Under the hypotheses and with the notation of 8.5.8, suppose in addition that, for every object i of I, the functor $m_{i*} : F_i \to G_i$ commutes with filtered inductive limits, and that for every morphism $f : i \to j$ of I, the functor $f_* : F_i \to F_j$ commutes with filtered inductive limits. Then the functor $\underleftarrow{m}_* : \underleftarrow{F} \to \underleftarrow{G}$ commutes with filtered inductive limits, and for every object $(i \mapsto M_i)$ of $\mathrm{Top}(F)$ there is a canonical isomorphism*

$$\underleftarrow{m}_* Q^*(i \mapsto M_i) \simeq \varinjlim_{I^\circ} \mu_j^* m_{j*}(M_j). \tag{8.5.9.1}$$

328 The first assertion follows from (8.5.8.2)) and (8.5.6. By (8.5.8.2)) and ((8.5.3.1), there is a functorial isomorphism

$$\underleftarrow{m}_* Q^*(i \mapsto M_i) \simeq \varinjlim_{i} \mu_i^* m_{i*}(\varinjlim_{f:j\to i} f_*(M_j)).$$

Using the commutation of the m_{i*} with filtered inductive limits and the isomorphisms $m_{i*}f_* \simeq f_* m_{j*}$ (7.1.6), we obtain

$$\underleftarrow{m}_* Q^*(i \mapsto M_i) \simeq \varinjlim_{i} \mu_i^* (\varinjlim_{f:j\to i} f_* m_{j*}(M_j)).$$

Since the functors μ_i^* commute with inductive limits, it follows that

$$\underleftarrow{m}_* Q^*(i \mapsto M_i) \simeq \varinjlim_{i} \varinjlim_{f:j\to i} \mu_i^* f_* m_{j*}(M_j).$$

This last object can be interpreted as an inductive limit over the category $\mathrm{Fl}(I)^\circ$, where $\mathrm{Fl}(I)$ is the category of morphisms of I. Let $\phi : I \to \mathrm{Fl}(I)$ be the functor that associates with every object i of I the identity morphism of I. The functor $\phi^\circ : I^\circ \to \mathrm{Fl}(I^\circ)$ is cofinal, and consequently (I 8.1) there is a canonical isomorphism

$$\underleftarrow{m}_* Q^*(i \mapsto M_i) \simeq \varinjlim_{I^\circ} \mu_j^* m_{j*}(M_j).$$

Corollary 8.5.10. *Under the conditions of 8.5.9, for every object i of I and every object M_i of F_i, there is a canonical isomorphism*

$$\underleftarrow{m}_* \mu_i^*(M_i) \simeq \varinjlim_{f:j\to i} \mu_j^* m_{j*} f^*(M_i). \tag{8.5.10.1}$$

The proof is analogous to that of Corollary 8.5.5.

REMARK 8.5.11. Recall that the direct-image functor under a coherent morphism between coherent topoi commutes with filtered inductive limits (5.1). Consequently, Propositions 8.5.3 through 8.5.7, and 8.5.9, 8.5.10, apply when the fibered topoi under consideration are coherent and the morphisms of fibered topoi under consideration are coherent.

8.6. Ringed fibered $\mathscr{U}$-topoi.

8.6.1. A fibered $\mathscr{U}$-topos is called *ringed* if it is equipped with a sheaf of rings on the total site (7.4.1). Let $p : F \to I$ be a fibered $\mathscr{U}$-topos. Choose a bisplitting (7.1.4). 329
It follows from 7.4.7 that giving a sheaf of rings on F is equivalent to giving, for every object i of I, a ring A_i of F_i, and, for every morphism $f : i \to j$, a morphism $\phi_f : A_j \to f_* A_i$, the family of the ϕ_f being subject to the compatibility conditions made explicit in *loc. cit.*. The morphisms of topoi $f_. : F_i \to F_j$ are therefore morphisms of topoi ringed respectively by A_i and A_j (IV 11), and the ringed structure on F, together with the choice of the morphisms f, gives a pseudofunctor from I to the 2-category of ringed topoi. Conversely, given such a pseudofunctor, one can reconstruct the ringed fibered $\mathscr{U}$-topos from which it arises.

8.6.2. As in 8.1, one can define the notion of the projective limit of a ringed $\mathscr{U}$-topos $(F_i, A_i, i \in I)$. We shall suppose that this immediate generalization has been made, and we shall freely apply to this situation the language introduced in 8.1. When it exists, this projective limit is a ringed $\mathscr{U}$-topos (H, B), and, for every object i of I, we have morphisms of ringed topoi $\mu_i : (H, B) \to (F_i, A_i)$. If the underlying (unringed) fibered topos F admits a projective limit $\varprojlim\text{top}_I F_i$, and if the inductive system of rings $i \mapsto \mu_i^*(A_i)$ admits an inductive limit $\varinjlim_{I^\circ} \mu_i^* A_i$ in the category of rings of $\varprojlim\text{top}_I F_i$ (as is always the case if I is small), then the ringed topos $(\varprojlim\text{top}_I F_i, \varinjlim_{I^\circ} \mu_i^* A_i)$ is evidently a projective limit of the ringed fibered topos under consideration.

8.6.3. The formulas of 8.5, established for sheaves of sets, are valid for abelian sheaves. They are also valid for sheaves of modules, provided the inverse-image functors occurring in them are interpreted as inverse-image functors in the sense of sheaves of modules (IV 1.1). We leave the verification to the reader.

8.6.4. Let $(p : F \to I, A)$ be a ringed fibered $\mathscr{U}$-topos over a cofiltered category. We call $(p : F \to I, A)$ a ringed fibered $\mathscr{U}$-topos flat on the right (resp. on the left) if, for every morphism $f : i \to j$, the morphism $f_. : (F_i, A_i) \to (F_j, A_j)$ is a morphism of ringed topoi flat on the right (resp. on the left) (V 1.8). Suppose that I is essentially small. Then the morphism of ringed topoi $Q : (\varprojlim\text{top}_I F_i, \varinjlim_{I^\circ} \mu_i^*(A_i)) \to (\mathrm{Hom}_{I^\circ}(I^\circ, F'), A)$ (8.5.1) is flat on the right (resp. on the left), as follows from the definition and from 330
8.5.2; this holds without any flatness hypothesis on the transition morphisms $f_.$. If, in addition, $(p : F \to I, A)$ is a ringed fibered $\mathscr{U}$-topos flat on the right (resp. on the left), the morphisms $\mu_i : (\varprojlim\text{top}_I F_i, \varinjlim_{I^\circ} \mu_i^* A_i) \to (F_i, A_i)$ are flat on the right (resp. on the left). This follows from the fact that the inverse image of a flat module is flat (V 1.7.1), and that a filtered inductive limit of flat modules is flat. The same arguments prove the following fact: if $m : (F_i, A_i, i \in I) \to (G_i, A_i, i \in I)$ is a cartesian morphism of ringed fibered $\mathscr{U}$-topoi over an essentially small cofiltered category, and if, for every object $i \in \mathrm{ob}(I)$, the morphism m_i is flat on the right (resp. on the left), then the morphism $\underline{m}$ deduced from m by passage to the projective limit is flat on the right (resp. on the left).

8.7. Cohomology of sheaves on a projective limit of topoi.

8.7.1. In this section, I denotes a small cofiltered category, $(p : F \to I, A)$ and $(q : G \to I, B)$ two ringed fibered $\mathcal{U}$-topoi, and $m : (p : F \to I, A) \to (q : G \to I, B)$ a morphism of ringed fibered $\mathcal{U}$-topoi. We suppose that, for every morphism $f : i \to j$ of I, the derived functors $R^n f_*$, $n \in \mathbf{N}$, of the functor $f_* : \mathrm{Mod}(F_i, A_i) \to \mathrm{Mod}(F_j, A_j)$ on modules commute with filtered inductive limits, and that, for every object $i \in I$, the derived functors $R^n m_{i*}$ of the functor $m_{i*} : (F_i, A_i) \to (G_i, B_i)$ on modules[34] commute with filtered inductive limits. Recall that these hypotheses hold when the categories F_i, G_i, $(i \in \mathrm{ob}(I))$, are *coherent* and when the morphisms $f : F_i \to F_j$, $(f : i \to j \in F1(I))$, and $m_i : F_i \to G_i$, $(i \in I)$, are *coherent* (5.2). We use the notation of 8.5.1. In addition, we denote by $\underline{A}$ (resp. $\underline{B}$) the sheaf $\varinjlim \mu_i^*(A_i)$ (resp. $\varinjlim \mu_i^*(B_i)$). Finally, for modules, we use the notation μ_i^{-1} for inverse image in the sense of abelian sheaves, reserving μ_i^* for inverse image in the sense of modules (IV 11).

331 Lemma 8.7.2. *Let $j \mapsto M_j$ be an injective A-Module of* $\mathrm{Top}(F)$*. Then $Q^*(j \mapsto M_j)$ is an A-Module acyclic for $\underleftarrow{m}_*$, and, for every $i \in \mathrm{ob}(I)$, M_i is flasque.*

Let $i \in \mathrm{ob}(I)$, and let e_i be a final object of F_i. The fibered $\mathcal{U}$-topos $F_{/e_i} \to I_{/i}$ is deduced from $F \to I$ by the base change $I_{/i} \to I$. Since the sheaf $j \mapsto M_j$ is injective, it is flasque (V 4.6). Its restriction to the localized topos $F_{/e_i}$ is flasque (V 4.12). The restriction functor from $F_{/e_i}$ to F_i is a direct image functor for a morphism of topoi (7.4.12). It therefore carries flasque Modules to flasque Modules (V 5.2). Thus M_i is flasque.

Let us prove the first assertion of the lemma. Set $N' = Q^*(j \mapsto M_j)$. By (8.5.3.1), for every $i \in \mathrm{ob}(I)$ we have

$$\mu_{i*}(N') = \mu_{i*}Q^*(j \mapsto M_j) \simeq \varinjlim_{f : j \to i} f_*(M_j).$$

Using the hypothesis of 8.7.1, we see that the sheaf N' has the following property:

(P) For every object $i \in \mathrm{ob}(I)$, $\mu_{i*}(N')$ is m_{i*}-acyclic, and, for every morphism $f : i \to j$ of I, $\mu_{i*}(N')$ is f_*-acyclic.

It is enough to show that every object N' having property (P) is $\underleftarrow{m}_*$-acyclic. Since injectives have property (P), it is enough to verify the following two properties (V 0.4): for every exact sequence $0 \to N' \to N \to N'' \to 0$, where N' and N have property (P), (a) N'' has property (P), and (b) $m_*(N) \to m_*(N'')$ is an epimorphism. Verification of (a) is immediate. Let us verify (b). Set $K = \mathrm{coker}(Q_*(N') \to Q_*(N))$, so that K is the section $(i \mapsto K_i = \mathrm{coker}(\mu_{i*}(N') \to \mu_{i*}(N)))$. For every $f : i \to j$, there is a commutative diagram

$$\begin{tikzcd}
0 \arrow[r] & f_*\mu_{i*}(N') \arrow[r] & f_*\mu_{i*}(N) \arrow[r] & f_*(K_i) \arrow[r] & 0 \\
0 \arrow[r] & \mu_{j*}(N') \arrow[r] \arrow[u] & \mu_{j*}(N) \arrow[r] \arrow[u] & K_j \arrow[r] \arrow[u] & 0.
\end{tikzcd}$$

332 Since $\mu_{i*}(N')$ is f_*-acyclic, the top row is exact. Since $i \mapsto \mu_{i*}(N')$ and $i \mapsto \mu_{i*}(N)$ are Cartesian sections, the first two vertical morphisms are isomorphisms. Consequently, the canonical morphism $K_j \to f_*(K_i)$ is an isomorphism, and $i \mapsto K_i$ is a Cartesian section. Thus K is of the form $Q_*(K')$, and the canonical morphisms $Q_*(N) \to K$ and

[34] To fix ideas, we shall take left modules.

$K \to Q_*(N'')$ are of the form $Q_*(u)$ and $Q_*(v)$, respectively, because Q_* is fully faithful. Since Q^* is exact and Q^*Q_* is isomorphic to the identity, v is an isomorphism, and hence $Q_*(v)$ is an isomorphism. Thus the sequence $0 \to Q_*(N') \to Q_*(N) \to Q_*(N'') \to 0$ is exact, and consequently, for every object i of I, the sequence $0 \to \mu_{i*}(N') \to \mu_{i*}(N) \to \mu_{i*}(N'') \to 0$ is exact. Since $\mu_{i*}(N')$ is m_{i*}-acyclic, the sequence $0 \to m_{i*}\mu_{i*}(N') \to m_{i*}\mu_{i*}(N) \to m_{i*}\mu_{i*}(N'') \to 0$ is exact. Moreover, there is an isomorphism $\underleftarrow{m}_* \simeq \varinjlim_{I^\circ} \mu_j^{-1} m_{j*}\mu_{j*}$ (8.5.8.2). The functor μ_j^{-1} is exact, and filtered inductive limits are exact. Therefore the sequence $0 \to \underleftarrow{m}_*(N') \to \underleftarrow{m}_*(N) \to \underleftarrow{m}_*(N'') \to 0$ is exact.

Theorem 8.7.3. *The notation and hypotheses are those of 8.7.1. For every integer $n \in \mathbb{N}$, the derived functor $R^n\underleftarrow{m}_*$ commutes with filtered inductive limits, and for every A-Module $j \mapsto M_j$ there is a functorial isomorphism*

(8.7.3.1) $$R^n\underleftarrow{m}_* Q^*(j \mapsto M_j) \simeq \varinjlim_{I^\circ} \mu_j^* R^n m_{j*}(M_j).$$

We have $Q^{-1}(A) = \underleftarrow{A}$, and hence inverse image for Modules is isomorphic to inverse image for abelian sheaves. It follows from 8.5.8.2 that, for every section $j \mapsto M_j$, there is a canonical isomorphism $\varinjlim_{I^\circ} \mu_j^{-1}(M_j) \simeq \varinjlim_{I^\circ} \mu_j^*(M_j)$. Moreover, under the hypotheses of 8.7.1, there is a canonical isomorphism (8.5.9.1) $\underleftarrow{m}_* Q^*(j \mapsto M_j) \simeq \varinjlim_{I^\circ} \mu_j^{-1} m_{j*}(M_j)$. Since, for every injective $j \mapsto M_j$, the M_j are flasque and $Q^*(j \mapsto M_j)$ is acyclic for $\underleftarrow{m}_*$ (8.7.2), there is an isomorphism $R^n\underleftarrow{m}_*(Q^*(j \mapsto M_j)) \simeq \varinjlim_{I^\circ} \mu_j^{-1} R^n m_{j*}(M_j)$, which gives formula (8.7.3.1) by what precedes. The first assertion follows immediately from formula (8.7.3.1).

Corollary 8.7.4. *For $n \in \mathbb{Z}$, there are functorial isomorphisms* 333

(8.7.4.1) $$R^n\underleftarrow{m}_*(N) \simeq \varinjlim_{I^\circ} \mu_j^* R^n m_{j*}\mu_{j*}(N).$$

Indeed, N is isomorphic to $Q^*(j \mapsto \mu_{j*}(N))$.

Corollary 8.7.5. *For every object i of I, every integer n, and every A_i-Module M_i, there is a functorial isomorphism*

(8.7.5.1) $$R^n\underleftarrow{m}_* \mu_i^*(M_i) \simeq \varinjlim_{f:j\to i} \mu_j^* R^n m_{j*} f^*(M_i).$$

After making the base change $I_{/i} \to I$, we may assume that i is a final object of I (8.2.1). The object $\mu_i^*(M_i)$ is then isomorphic to $Q^*((f : j \to i) \mapsto f^*(M_i))$ (cf. 8.5.5).

Corollary 8.7.6. *Let i be an object of I and n an integer. There is an isomorphism, functorial in the section $j \mapsto M_j$:*

(8.7.6.1) $$R^n\mu_{i*} Q^*(j \mapsto M_j) \simeq \varinjlim_{f:j\to i} R^n f_*(M_j).$$

There is an isomorphism functorial in the $\underleftarrow{A}$-Module N:

(8.7.6.2) $$R^n\mu_{i*}(N) \simeq \varinjlim_{f:j\to i} R^n f_*\mu_{j*}(N).$$

There is an isomorphism functorial in the A_i-Module M_i:

$$R^n\mu_{i*}\mu_i^*(M_i) \simeq \varinjlim_{f:j\to i} R^n f_* f^*(M_i). \tag{8.7.6.3}$$

The functors $R^n\mu_{i*}$ commute with filtered inductive limits.

Making the base change $I_{/i} \to I$ reduces us to the case where i is a final object of I. Formulas (8.7.6.1) and (8.7.6.2) are then special cases of formulas (8.7.3.1), (8.7.4.1), and (8.7.5.1), respectively, obtained by taking for G the constant fibered topos with fiber F_i and for m the morphism $(f : F_j \to F_i, f \in \mathrm{ob}(I_{/i}))$. The last assertion follows from (8.7.6.2), taking 8.5.4 into account.

334 Corollary 8.7.7. *Let i be an object of I, and let X be an object of F_i such that the functors $H^n(X, -)$, $n \in \mathrm{N}$, commute with filtered inductive limits. For every n, there is a canonical isomorphism, functorial in the A_i-Module M_i:*

$$H^n(\mu_i^*(X), \mu_i^*(M_i)) \simeq \varinjlim_{f:j\to i} H^n(f^*(X), f^*(M_i)) \tag{8.7.7.1}$$

and the functors $H^n(\mu_i^*(X), -)$ commute with filtered inductive limits. In particular, if the functors $H^n(F_i, -)$ commute with filtered inductive limits, then the functors $H^n(\underleftarrow{F}, -)$ commute with filtered inductive limits, and there are canonical isomorphisms, functorial in the A_i-Modules M_i:

$$H^n(\underleftarrow{F}, \mu_i^*(M_i)) \simeq \varinjlim_{f:j\to i} H^n(F_j, f^*(M_i)). \tag{8.7.7.2}$$

It follows from V 5.3 that there are two spectral sequences

$$'E_2^{p,q} = \varinjlim_{f:j\to i} H^p(X, R^q f_* f^*(M_i)) \Rightarrow \varinjlim_{f:j\to i} H^{p+q}(f^*(X), f^*(M_i))$$

$$''E_2^{p,q} = H^p(X, R^q\mu_{i*}\mu_i^*(M_i)) \Rightarrow H^{p+q}(\mu_i^*(X), \mu_i^*(M_i)),$$

and from (8.5.7.1) that there is a morphism between these two spectral sequences. Since the $H^{p+q}(X, -)$ commute with filtered inductive limits, it follows from (8.7.6.2) that this morphism of spectral sequences is an isomorphism at the $E_2^{p,q}$ level. It therefore induces an isomorphism on the abutments, giving (8.7.7.1). For every $\underleftarrow{A}$-Module N, there is a spectral sequence (V 5.3)

$$E_2^{p,q} = H^p(X, R^q\mu_{i*}(N)) \Rightarrow H^{p+q}(\mu_i^*(X), N).$$

The assertion that the functors $H^n(\mu_i^*(X), -)$ commute with filtered inductive limits then follows from the analogous properties of the functors $H^n(X, -)$ and $R^n\mu_{i*}$ (8.7.6).

Remark 8.7.8. When the topoi F_j, $j \in \mathrm{ob}(I)$, are coherent and the transition morphisms are coherent, the hypothesis made on X (resp. F_i) in 8.7.7 holds when X (resp. F_i) is coherent (5.2). Moreover, we know that in this case $\mu_i^*(X)$ (resp. $\underleftarrow{F}$) is coherent (8.3.13).

335 Corollary 8.7.9. *Let i be an object of I, let M_i be a left A_i-Module, and let L_i be a right A_i-Module admitting a resolution of the form*

$$P_{i,k} \longrightarrow P_{i,k-1} \longrightarrow \cdots \longrightarrow P_{i,0} \longrightarrow L_i \longrightarrow 0$$

where, for every integer k, $P_{i,k}$ is isomorphic to a finite direct sum of objects of the form $A_{i|X}$ (IV 11.3.3), *with X satisfying the hypotheses of 8.7.7. If, in addition to the hypotheses* *of 8.7.1, the fibered topos F is right flat (8.6), then for $n \leq k-1$ there are canonical isomorphisms*

$$\mathrm{Ext}^n_{\underline{A}}(\underline{F}, \mu_i^*(L_i), \mu_i^*(M_i)) \simeq \varinjlim_{f:j\to i} \mathrm{Ext}^n_{A_j}(F_j, f^*(L_i), f^*(M_i)). \tag{8.7.9.1}$$

Denote by $P_{i,.}$ the resolution of L_i. Since F is right flat, for every $f : j \to i$, $f^*(P_{i,.})$ is a resolution of $f^*(L_i)$, and $\mu_i^*(P_{i,.})$ is a resolution of $\mu_i^*(L_i)$ (8.6). These resolutions give two spectral sequences converging respectively to the two sides of (8.7.9.1), together with a morphism between the two spectral sequences. At the $E_1^{p,q}$ level this morphism is an isomorphism by (8.7.7.1). It therefore induces an isomorphism on the abutments.

9. Appendix. Criterion for the existence of points

by P. Deligne

Proposition 9.0. *Every locally coherent topos S has enough points.*

Since the question is local on S, we may suppose that the topos S is defined by a site $\mathscr{S}$ in which finite projective limits are representable and such that every covering $f_i : U_i \to U$ admits a finite subcovering. It is enough to prove that if $f : \mathscr{F} \to \mathscr{G}$ is not a monomorphism, then there is a point x of $\mathscr{S}$ such that f_x is not injective. By hypothesis, there are $U \in \mathrm{ob}\,\mathscr{S}$ and $\mathscr{s},\mathscr{s}'\in\mathscr{F}(U)$ such that $\mathscr{s}\neq\mathscr{s}'$ and $f(\mathscr{s})=f(\mathscr{s}')$. Replacing $\mathscr{S}$ by $\mathscr{S}/U$, we reduce to the following lemma.

Lemma 9.1. *If $\mathscr{s}$ and $\mathscr{s}'$ are two distinct global sections of a sheaf $\mathfrak{F}$ on a site $\mathscr{S}$ satisfying the preceding hypotheses, then there is a point x of $\mathscr{S}$ such that $\mathscr{s}_x\neq\mathscr{s}'_x$.* 336

Let $P = (U_i)_{i\in I}$ be a projective system in $\mathscr{S}$, indexed by a filtered ordered set I. For every sheaf $\mathscr{H}$ on $\mathscr{S}$, put $P(\mathscr{H}) = \varinjlim \mathscr{H}(U_i)$, and for every $V \in \mathrm{Ob}\,\mathscr{S}$, put $P(V) = \varinjlim \mathrm{Hom}(U_i, V)$. The functors $P(\mathscr{H})$ and $P(V)$ commute with fiber products. If the functor $P(V)$ takes coverings to surjective families of maps, it is a morphism of sites from the point topos to $\mathscr{S}$, and $P(\mathscr{H})$ is the corresponding fiber functor.

If $P = (U_i)_{i\in I}$ and $Q = (V_j)_{j\in J}$ are two projective systems in $\mathscr{S}$, we say that Q refines P if I is a subset of J with the induced order and P is the restriction of Q to I. There are then morphisms of functors from $P(\mathscr{H})$ to $Q(\mathscr{H})$ and from $P(V)$ to $Q(V)$.

For every P as above, denote by $\mathscr{s}_P$ and $\mathscr{s}'_P$ the images of $\mathscr{s}$ and $\mathscr{s}'$ in $P(\mathscr{F})$. Lemma 9.1 follows from the next lemma, in which P is taken to be the projective system indexed by $\{0\}$ and consisting of the final object of $\mathscr{S}$.

Lemma 9.2. *For every P as above satisfying $\mathscr{s}_P\neq\mathscr{s}'_P$, there is a refinement Q of P satisfying $\mathscr{s}_Q\neq\mathscr{s}'_Q$ and such that, for every covering $f_i : V_i \to V$ and every $f \in Q(V)$, f belongs to the image of one of the $Q(V_i)$.*

We first prove:

Lemma 9.3. *Let $P = (U_i)_{i\in I}$ satisfy the hypotheses of Lemma 9.2, let $f_i : V_i \to V$ be a finite covering in $\mathscr{S}$, and let $f \in P(V)$. There is a refinement Q of P satisfying $\mathscr{s}_Q\neq\mathscr{s}'_Q$ and such that the image of f in $Q(V)$ belongs to the image of one of the $Q(V_i)$.*

There are $i_\circ \in I$ and $f' \in \mathrm{Hom}(U_{i_\circ}, V)$ which define f. For $i \geq i_\circ$, put $U_{i,k} = U_i \times_V V_k$, this fiber product being defined by f'. Since (V_k) is a covering of V, $(U_{i,k})$ is a covering of U_i; the following arrow is injective (for $i > i_\circ$):

$$\mathscr{F}(U_i) \longrightarrow \prod_k \mathscr{F}(U_{i,k}).$$

337 Passing to the limit, and using the commutation of *finite* products with filtered inductive limits, we see that

$$P(\mathcal{F}) \longrightarrow \prod_k \varinjlim \mathcal{F}(U_{i,k})$$

is injective. Hence there is a k such that $\mathcal{S}_P$ and $\mathcal{S}'_P$ have distinct images in $\varinjlim_{i>i_\circ} \mathcal{F}(U_{i,k})$. Put $I_\circ = \{i \mid i \geq i_\circ \text{ and } i \in I\}$ and $J = I \coprod I_\circ$. Order J by declaring $j' \leq j''$ if the images of j' and j'' in I satisfy $j' \leq j''$ and it is not the case that $j' \in I_\circ$ and $j'' \in I$. The U_i and $U_{i,k}$, indexed by J, form a refinement Q of P such that $\mathcal{S}_Q \neq \mathcal{S}'_Q$ and the image of f in $Q(V)$ belongs to the image of $Q(V_k)$.

Lemma 9.4. *Let $P = (U_i)_{i\in I}$ satisfy the hypotheses of Lemma 9.2. There is a refinement Q of P such that $\mathcal{S}_Q \neq \mathcal{S}'_Q$ and such that, for every finite covering (V_k) of an object V of $\mathcal{S}$ and every $f \in P(V)$, the image of f in $Q(V)$ belongs to the image of one of the $Q(V_k)$.*

Let E be the set of triples consisting of an object V of $\mathcal{S}$, a finite covering (V_k) of V, and an element $f \in P(V)$. Let $\leq$ be a well-ordering of E, and let $\overline{E}$ be the set obtained from E by adjoining a greatest element, denoted by ∞. By transfinite induction on $e \in \overline{E}$, we shall define refinements Q_e of P satisfying $\mathcal{S}_{Q_e} \neq \mathcal{S}'_{Q_e}$ and such that

(i) if $e' \leq e$, then Q_e refines $Q_{e'}$;
(ii) if $e = (V, (V_k), f)$, then the image of f in $Q_e(V)$ belongs to the image of one of the $Q_e(V_k)$.

Suppose the $Q_{e'}$ already defined for $e' < e$. If e is the first element of E, put $Q'_e = P$. If e has a predecessor $e-1$, put $Q'_e = Q_{e-1}$. Otherwise, let Q'_e be the projective system with index set $\bigcup_{e'<e} I_{e'}$ which refines the $Q_{e'}$ for $e' < e$. In this case,

$$Q'_e(\mathcal{F}) = \varinjlim_{e'<e} Q_{e'}(\mathcal{F}),$$

338 so that, in every case, Q'_e refines the $Q_{e'}$ for $e' < e$ and satisfies $\mathcal{S}_{Q'_e} \neq \mathcal{S}'_{Q'_e}$.

The required projective system Q_e is obtained by applying Lemma 9.3 to Q'_e and $e = (V, (V_k), f)$ (resp. by taking $Q_e = Q'_e$ if $e = \infty$).

The projective system Q_∞ satisfies Lemma 9.4.

Lemma 9.4 makes it possible to define, inductively on n, a sequence $Q_n = (U_i)_{i\in I_n}$ of projective systems in $\mathcal{S}$ such that

(i) $Q_0 = P$;
(ii) Q_{n+1} refines Q_n;
(iii) $\mathcal{S}_{Q_n} \neq \mathcal{S}'_{Q_n}$;
(iv) for every covering $f_k : V_k \to V$ and every $f \in Q_n(V)$, the image of f in $Q_{n+1}(V)$ belongs to the image of one of the $Q_{n+1}(V_k)$.

The projective system Q, whose index set is the union of the I_n and which extends the various Q_n, then satisfies Lemma 9.2. The proof also shows:

Corollary 9.5. *Let $\mathcal{S}$ be a site in which fiber products are representable and such that every covering in $\mathcal{S}$ admits a finite subcovering. Put $c = \sup(\aleph_0, \operatorname{card}(Fl\, \mathcal{S}))$. Then there is a very dense set X of points of $\mathcal{S}$ such that*

(i) $\operatorname{card}(X) \leq 2^c$;
(ii) *if $x \in X$ and $U \in \operatorname{Ob} \mathcal{S}$, then* $\operatorname{card}(U_x) \leq c$.

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EXPOSÉ VII

The étale site and topos of a scheme

A. Grothendieck

In the present exposé and the next, we develop the most elementary properties concerning étale topology and cohomology. The developments of the present exposé concern certain properties that are, for the most part, valid for other, very different topologies, such as the "fppf topology". In the next exposé, we shall develop properties rather special to the étale topology, arising from the very particular nature of étale morphisms. 342

Here we partly follow three oral exposés by J.E. Roos (which he had not been able to write up), notably in the proof of VIII 6.3.

Unless expressly stated otherwise, it will be understood that the schemes considered in the present exposé and the following ones are elements of the fixed universe $\mathscr{U}$.

1. The étale topology

1.1. We denote by (Sch) the category of schemes (elements of the fixed universe $\mathscr{U}$). Recall that a morphism $f : X \to Y$ in (Sch) is called *étale* if it is locally of finite presentation and flat, and if for every $y \in Y$, the fiber X_y is discrete and its local rings are finite *separable* extensions of $k(y)$. Equivalently, f is locally of finite presentation and, for every affine Y-scheme Y' (in the absolute sense) and every subscheme Y'_0 defined by a nilpotent ideal, the map 343

$$\mathrm{Hom}_Y(Y', X) \longrightarrow \mathrm{Hom}_Y(Y'_0, X)$$

is *bijective*. For the most important properties of this notion, I refer to EGA IV § § 17 and 18, and, in the meantime, to SGA 1 I and IV.

1.2. The *étale topology* on (Sch) is the topology generated by the pretopology for which, for every $X \in$ (Sch), the set Cov X consists of the families $(X_i \xrightarrow{u_i} X)_{i\in I}$ (indexed by an $I \in U$) such that the u_i are étale and $X = \bigcup_i u_i(X_i)$ (in the set-theoretic sense).

It is convenient to associate with every X the subcategory $\mathrm{Et}_{/X}$ of $(\mathrm{Sch})_{/X}$ consisting of the arrows $X' \to X$ that are étale. We endow it with the "induced topology" (III) from the étale topology of (Sch) (also called the "étale topology on X"), and denote the resulting site by $X_{\text{ét}}$ (the "étale site" of X) [34]. Note that every morphism of $\mathrm{Et}_{/X}$ is étale (SGA 1 I 4.8), whence a family $(X'_i \xrightarrow{u_i} X')_{i\in I}$ in $X_{\text{ét}}$ is covering if and only if it is *surjective*, i.e. $\bigcup_i u_i(X'_i) = X'$. One sees immediately (cf. 3.1) that $X_{\text{ét}}$ is a $\mathscr{U}$-site, so the results of Exposés I through VI apply. The topos $\widetilde{X_{\text{ét}}}$ of $\mathscr{U}$-sheaves on $X_{\text{ét}}$ is the *étale topos* of X.

1.3. In what follows, unless expressly stated otherwise, all the topological notions 344

[34]Note, however, that it would be preferable to denote by $X_{\text{ét}}$ the topos $\widetilde{X_{\text{ét}}}$ defined by the étale site of X. For practical reasons, in this Seminar we shall retain the notation introduced here (which is that of the original seminar).

considered in $\mathrm{Et}_{/X}$ or (Sch) are understood with respect to the étale topology. Moreover, in language and notation we shall commonly write X in place of $X_{\text{ét}}$ or $Et_{/X}$. Thus a "sheaf on X" (understood: for the étale topology) means a sheaf on the site $X_{\text{ét}}$. We denote by $\widetilde{X_{\text{ét}}}$ the category of these sheaves, which is a $\mathscr{U}$-topos. If F is an abelian sheaf on X, we denote its cohomology groups by $H^i(X, F)$; these would be denoted $H^i(X_{\text{ét}}, F)$ in Exposé V.

1.4. Let $f : X \to Y$ be a morphism of schemes. The inverse-image functor

$$f^* : Y' \mapsto Y' \times_Y X$$

induces a functor

$$f^*_{\text{ét}} : \mathrm{Et}_{/Y} \longrightarrow \mathrm{Et}_{/X} \tag{$*$}$$

that commutes with finite $\varprojlim$ and takes covering families to covering families; it is therefore a morphism of sites

$$f^*_{\text{ét}} : X_{\text{ét}} \longrightarrow Y_{\text{ét}},$$

and thus induces a functor on the categories of sheaves:

$$f^{\text{ét}}_* : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{Y_{\text{ét}}}$$

given by

$$f^{\text{ét}}_*(F) = F \circ f^*_{\text{ét}}.$$

Moreover, $f^{\text{ét}}_*$ admits a left adjoint

$$f^*_{\text{ét}} : \widetilde{Y_{\text{ét}}} \longrightarrow \widetilde{X_{\text{ét}}}$$

345 extending the functor considered in $(*)$ and commuting with arbitrary $\varinjlim$ and finite $\varprojlim$, i.e. $f^*_{\text{ét}}$ defines a *morphism of topoi*

$$f_{\text{ét}} : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{Y_{\text{ét}}}.$$

Clearly, for a composite of two morphisms of schemes

$$X \xrightarrow{f} Y \xrightarrow{g} Z$$

there are canonical isomorphisms

$$\begin{cases} (gf)^{\text{ét}}_* \simeq g^{\text{ét}}_* f^{\text{ét}}_* & , \quad \text{whence} \\ (gf)^*_{\text{ét}} \simeq f^*_{\text{ét}} g^*_{\text{ét}} & , \quad \text{i.e. there is an isomorphism} \\ (gf)_{\text{ét}} \simeq g_{\text{ét}} f_{\text{ét}} & , \end{cases}$$

(so that one obtains a "pseudofunctor" (SGA 1 VI 8) from the category (Sch) to the category (Top) of topoi $\in \mathscr{U}'$, where $\mathscr{U}'$ is the smallest universe such that $\mathscr{U} \in \mathscr{U}'$; compare IV).

1.5. Notation. In what follows, we shall often write f^* and f_* in place of $f^*_{\text{ét}}$ and $f^{\text{ét}}_*$. If $f : X \to Y$ is a morphism of schemes, the functors $R^i f^{\text{ét}}_*$ will therefore simply be denoted $R^i f_*$. In this notation, recall the Leray spectral sequence for f and an abelian sheaf F on X (V 5):

$$H^*(X, F) \Longleftarrow E_2^{p,q} = H^p(Y, R^q f_*(F)).$$

Likewise, if there are two morphisms $f : X \to Y$ and $g : Y \to Z$, there is a Leray spectral sequence (the spectral sequence of composite functors)

$$R^*(gf)_*(F) \Longleftarrow E_2^{p,q} = R^p g_*(R^q f_*(F)).$$

1.6. When $f : X \to Y$ is itself an étale morphism, X may be regarded as an object of $Y_{\text{ét}}$, and there is a canonical *isomorphism of sites* 346

$$X_{\text{ét}} \xrightarrow{\sim} (Y_{\text{ét}})_{/X}.$$

Under this isomorphism, the functor $f^*_{\text{ét}} : Y' \mapsto Y' \times_Y X$ of $(*)$ identifies with the internal base-change functor in the category $\mathrm{Et}_{/Y}$. It follows that the functor

$$f^* : \widetilde{Y_{\text{ét}}} \longrightarrow \widetilde{X_{\text{ét}}}$$

is then isomorphic to the functor "*restriction* to Y"

$$f^*(F) \simeq F \circ i_{X/Y},$$

where F is a sheaf on $Y_{\text{ét}}$ and $i_{X/Y} = X_{\text{ét}} \to Y_{\text{ét}}$ is the evident functor (which regards an étale scheme over X, $u : X' \to X$, as an étale scheme over Y via $fu : X' \to Y$).

1.7. Questions of universes. Note that if $X \neq \varnothing$, then $X_{\text{ét}}$ *is not* an element of the chosen universe $\mathcal{U}$. Nevertheless, as we have noted, it is easy to see (3.1) that one can find a full subcategory C of $X_{\text{ét}}$, *an element of* $\mathcal{U}$, satisfying the conditions of the "comparison lemma" (III), so that the restriction functor induces an equivalence $\widetilde{C} \to \widetilde{X_{\text{ét}}}$. Thus the *étale topos* is equivalent to a topos defined in terms of a site $C \in \mathcal{U}$. In other words, $X_{\text{ét}}$ is a U-site, and hence by *loc. cit.* the results of Exposés I through VI apply to this site.

2. Examples of sheaves

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a) Let $F \in \mathrm{Ob}(\mathrm{Sch})_{/X}$ be a scheme over X, and for every X' over X put

$$F(X') = \mathrm{Hom}_X(X', F).$$

The functor $(\mathrm{Sch})^{\circ}_{/X} \to (\mathrm{Ens})$ thus defined is a sheaf for the étale topology (and even for the finer fpqc topology studied in SGA 3 IV). In other words, the étale topology on (Sch) is coarser than the canonical topology. This is, essentially, the content of SGA 1 VIII 5.1 (cf. also SGA 3 IV 6.3.1). A fortiori, the restriction of this sheaf to $X_{\text{ét}}$ is a sheaf. It will again be denoted by F when there is no risk of confusion[34]. Note also that the functor thus obtained

$$(*) \qquad (\mathrm{Sch})_{/X} \longrightarrow \widetilde{X_{\text{ét}}}$$

commutes with finite $\varprojlim$ (this is immediate). This implies, for example, that when F is a group scheme (resp. ...) over X, the sheaf it defines is a sheaf of groups (resp. ...). Note that the functor $\mathrm{Et}_{/X} \to \widetilde{X_{\text{ét}}}$ induced by (*) is simply the canonical functor, associating with every $X' \in \mathrm{Ob}\, \widetilde{X_{\text{ét}}}$ the functor on $X_{\text{ét}}$ that it represents. It is therefore an isomorphism of the category $\mathrm{Et}_{/X}$ onto a full subcategory of the étale topos $\widetilde{X_{\text{ét}}}$, by which we generally identify an $X \in \mathrm{Ob}\, X_{\text{ét}}$ with the corresponding sheaf, denoted $\widetilde{X}$ or simply X.

Clearly, $H^{\circ}(X, F) = \mathrm{Hom}_{(\mathrm{Sch})_{/X}}(X, F)$.

Let us also give an interpretation of $H^1(X, F)$ when F is a group prescheme over X (commutative, if desired, so that the definition of $H^1(X, F)$ falls within Exposé V; for the general case one may consult J. Giraud's thesis[34]). Familiar ar- 348

[34] Care must be taken, however, that if $X \neq \varnothing$, the functor φ that associates with $F \in \mathrm{Ob}(\mathrm{Sch})_{/X}$ the corresponding sheaf $\varphi(F)$ is *not* fully faithful, nor even faithful, and that F cannot be reconstructed (up to isomorphism) from $\varphi(F)$. For example, if $S = \mathrm{Spec}\, k$, with k an algebraically closed field, knowledge of $\varphi(F)$ is equivalent merely to knowledge of the underlying set of F (by VIII 2.4). Compare IV

guments, which I shall not repeat here, then show that $H^1(X, F)$ is canonically isomorphic to the group of classes (up to isomorphism) of *sheaves of sets on $X_{ét}$ that are principal homogeneous under F*[34]. When F is *affine* over X, SGA 1 VIII 2.1 implies (again by standard arguments, cf. J. GIRAUD's thesis) that $H^1(X, F)$ is also the group of classes of *schemes P* over X on which F acts on the right and which are "principal homogeneous bundles over X for the étale topology", i.e. locally trivial for the étale topology[34].

REMARKS 2.1. Care must be taken that if, for every scheme Z over X, the associated sheaf on $X_{\text{ét}}$ is denoted by $\varphi_X(Z)$, and if $f : Y \to X$ is a morphism of schemes, there is an evident homomorphism (functorial in Z)

$$(*) \qquad f^*(\varphi_X(Z)) \longrightarrow \varphi_Y(Z_Y), \quad Z_y = Z \times_X Y,$$

i.e. an evident homomorphism

$$\varphi_X(Z) \longrightarrow f_*(\varphi_Y(Z_Y)),$$

namely the homomorphism, functorial in $X' \in \operatorname{Ob} X_{\text{ét}}$,

$$\operatorname{Hom}_X(X', Z) \longrightarrow \operatorname{Hom}_Y(X'_Y, Z_Y).$$

But note that *in general* $(*)$ *is not an isomorphism, i.e. the formation of the étale sheaf associated with a relative scheme does not commute with inverse-image functors.* Nevertheless, $(*)$ is an isomorphism in the special case in which Z is étale over X. Likewise,
349 the functor

$$\varphi_X : \mathrm{Sch}_{/X} \longrightarrow \widetilde{X_{\text{ét}}}$$

is not faithful if $X \neq \varnothing$[34]; nevertheless, its restriction to $X_{\text{ét}}$, which is simply the canonical functor $X_{\text{ét}} \to \widetilde{X_{\text{ét}}}$, is *fully faithful*, because by a) the topology of $X_{\text{ét}}$ is coarser than its canonical topology.

b) Note that $(*)$ also commutes with direct sums indexed by the $I \in \mathscr{U}$, as is readily verified. In particular, if for every set I we denote by I_X the constant X-scheme defined by I (SGA 3 I 1.8), the direct sum of I copies of X, then the associated sheaf is simply the constant sheaf $I_{X_{\text{ét}}}$ (the direct sum of I copies of the final sheaf on $X_{\text{ét}}$). Since the functor $I \mapsto I_X$ also commutes with finite $\varprojlim$, we see that it takes a group to a group, a commutative group to a commutative group, etc. If G is an ordinary commutative group, we shall simply write $H^i(X, G)$ in place of $H^i(X, G_X)$. Suppose, for example, that G is a finite group. Then G_X is finite, hence affine over X, and thus, using the final remark of a), we obtain an interpretation of $H^1(X, G)$ as the set *of classes of principal Galois coverings with group G* (SGA 1 V 2.7). When X is connected and endowed with a geometric point a, the "fundamental group" $\pi_1(X, a)$ (SGA 1 V 7) gives the canonical isomorphism

$$(**) \qquad H^1(X, G) \simeq \operatorname{Hom}(\pi_1(X, a), G),$$

(where G has been assumed commutative).

One may say that in passing from Zariski cohomology to the étale topology, "we have done what was needed" to obtain the "right" H^1 (which occurs in
350 the second member of $(*)$) for a finite constant coefficient group G. *It is a*

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[34]or F-torsors in the terminology of *loc. cit.*.

[34]or representable *F-torsors* in the now accepted terminology

[34]Cf. the note at the bottom of page (page 6)

remarkable fact, to be proved in the remainder of this seminar, that this also suffices to obtain the "right" $H^i(X, G)$ for every torsion coefficient group (at least if G is prime to the residual characteristics of X).

c) Let F be an $\mathcal{O}_X$-module on X, for the Zariski topology. Then (with the notation of SGA 3 I 4.6) define a functor

$$W(F) : (\mathrm{Sch})^{\circ}_{/X} \longrightarrow (\mathrm{Ens})$$

by

$$W(F)(X') = \Gamma(X', F \otimes_{\mathcal{O}_X} \mathcal{O}_{X'}).$$

This functor is again a sheaf for the étale topology (and even for the fpqc topology), as follows again from SGA 1 VIII 1.6 and SGA 3 IV 6.3.1. A fortiori, the restriction of $W(F)$ to $X_{\text{ét}}$ is again a sheaf, which we also denote by $W(F)$. By definition, therefore, $H^{\circ}(X, W(F)) = \Gamma(F) = H^{\circ}(X, F)$. But there is a stronger result if F is quasi-coherent; cf. 4.3.

3. Generators of the étale topos. Cohomology of an $\varinjlim$ of sheaves

Proposition 3.1. *Let X be a scheme, and let C be a full subcategory of $X_{\text{ét}}$ such that, for every affine scheme X' étale over X, C contains an object isomorphic to X'. Then C is a "family of topological generators" of the site $X_{\text{ét}}$, and hence a family of generators of the topos $\widetilde{X_{\text{ét}}}$. The restriction functor $\widetilde{X_{\text{ét}}} \to \widetilde{C}$ is an equivalence of categories (where C is equipped with the topology induced by that of $X_{\text{ét}}$).*

Immediate from the "comparison lemma." 351

Corollary 3.2. *Suppose that X is quasi-separated. Call the full subcategory C of $X_{\text{ét}}$ formed by the schemes étale over X that are of finite presentation over X, equipped with the topology induced by $X_{\text{ét}}$, the restricted étale site of X. Then:*

(i) *C is stable under fiber products and is a site of finite type if X is quasi-compact.*
(ii) *The restriction functor $\widetilde{X_{\text{ét}}} \to \widetilde{C}$ is an equivalence of categories.*

Assertion (i) is immediate, and (ii) follows from 3.1, since C satisfies the condition of 3.1 by the quasi-separatedness of X. Indeed, this implies that, if X' is quasi-compact, for example affine, then it is quasi-compact *over* X (EGA IV 1.2.4, with $Z = \operatorname{Spec} \mathbf{Z}$), and hence is of finite presentation over X if X' is locally of finite type over X.

Proposition 3.3. *Suppose that X is quasi-compact and quasi-separated. Then the functors $H^q(X_{\text{ét}}, F)$ on the category of abelian sheaves on $X_{\text{ét}}$ commute with $\varinjlim$.*

Indeed, by 3.2 (ii), one may replace $X_{\text{ét}}$ by C. Since X is quasi-compact, $X \in \operatorname{Ob} C$, and the conclusion follows from 3.2 (i) and VI 6.1.2 (3).

4. Comparison with other topologies

4.0. First note that the examples of sheaves on $X_{\text{ét}}$ considered in n° 2 are in fact naturally *restrictions* of sheaves defined on $(\mathrm{Sch})_{/X}$ equipped with its étale topology (or 352
even with the fpqc topology). More generally, let

$$u^* : X_{\text{ét}} \longrightarrow (\mathrm{Sch})_{/X}$$

be the inclusion functor. It is continuous (III) and commutes with finite $\varprojlim$, and hence defines a *morphism of sites*

$$u : (\mathrm{Sch})_{/X} \longrightarrow X_{\text{ét}},$$

whence a functor

$$u_* : \widetilde{(\mathrm{Sch})_{/X}} \longrightarrow \widetilde{X}_{\text{ét}}$$

on the associated categories of sheaves:

$$u_*(F) = F \circ u,$$

and a left adjoint[34] to the latter:

$$u^* : \widetilde{X}_{\text{ét}} \longrightarrow \widetilde{\mathrm{Sch}_{/X}}.$$

With this understood:

Proposition 4.1. (i) *The functor u^* is fully faithful; hence, for every sheaf G on $X_{\text{ét}}$, the canonical homomorphism $G \to u_* u^*(G)$ is an isomorphism.*

(ii) *Let G (resp. F) be an abelian sheaf on $X_{\text{ét}}$ (resp. $(\mathrm{Sch})_{/X}$). Then the following canonical homomorphisms (defined, for example, as edge homomorphisms of the*
353 *corresponding Leray spectral sequences) are isomorphisms:*

$$H^*(X_{\text{ét}}, G) \xrightarrow{\sim} H^*((\mathrm{Sch})_{/X}, u^* G)$$

$$H^*(X_{\text{ét}}, F \circ u) \xrightarrow{\sim} H^*((\mathrm{Sch})_{/X}, F).$$

Indeed, the functor u is a comorphism, and one can apply (III).

Thus (i) shows that, for the study of the cohomology of sheaves, it is essentially equivalent to work with the "small" étale site $X_{\text{ét}}$ or with the "big" étale site $(\mathrm{Sch})_{/X}$.

4.2. One should also introduce on (Sch) (and hence on the $(\mathrm{Sch})_{/X}$) various topologies other than the étale topology, the most useful of which are defined in SGA 3 IV 6.3. The coarsest among them is the *Zariski topology* (Zar), defined by the pretopology whose covering families are the surjective families of open immersions; it is coarser than the étale topology. The finest of these topologies is the *"faithfully flat quasi-compact" topology*, abbreviated (fpqc), which is the coarsest topology for which the covering families in the Zariski sense, as well as faithfully flat quasi-compact morphisms, are covering; the fpqc topology is finer than the étale topology. As already observed, the various examples of sheaves on $(\mathrm{Sch})_{/X}$ considered in n° 2 are in fact already sheaves for the fpqc topology.

4.2.1. One must bear in mind, however, that for an abelian sheaf F on $(\mathrm{Sch})_{/X}$ for the fpqc topology (or for a topology such as fppf, considered in SGA 3), the cohomology groups of F for the fpqc topology (resp. ...) are not always isomorphic to the cohomology
354 groups for the étale topology. This is so even if X is the spectrum of a field k and F is representable by an algebraic group over k, even for H^1. In general, one can show that the étale topology gives the "correct" cohomology groups for coefficient groups that are *étale*, or more generally *smooth*, group schemes over X[34], but this is no longer true

[34] Strictly speaking, since $(\mathrm{Sch})_{/X}$ is a $\mathscr{U}$-category but *not* a $\mathscr{U}$-site, one cannot invoke III for the existence of u^*. Nevertheless, one constructs u^* easily by

$$u^*(F)(X') = (p_{X'})^*_{\text{ét}}(F)(X') \quad (X' \in \operatorname{Ob} \mathrm{Sch}_{/X}),$$

where $p_{X'}$ denotes the structural morphism $X' \to X$. Moreover, to give meaning to 4.1 (ii) and justify the indicated proof of 4.1, one must introduce a universe $\mathscr{V}$ such that $\mathscr{U} \in \mathscr{V}$, and regard the sheaves occurring in the formulas of 4.1 (i) as sheaves with values in $\mathscr{V}$-(Ens). Alternatively, if one wishes to work only with $\mathscr{U}$-sites, one may replace $(\mathrm{Sch})_{/X}$ by a full subcategory of $(\mathrm{Sch})_{/X}$ stable under finite $\varprojlim$, containing the C of 3.1, and $\mathscr{U}$-small.

[34] Cf. A. Grothendieck, Le groupe de Brauer III, Th. 11.7, in "Dix exposés sur la topologie des schémas," North-Holland Publishing Co.

for group schemes such as *radicial* groups over S, for which one must replace the étale topology, still too coarse, by the fpqc or fppf topology.

4.2.2. As an example of the relations between cohomologies for different topologies, consider the Zariski and étale topologies. Denote by X_{Zar} the site of Zariski open subsets of X, so that there is a canonical inclusion functor

$$u^* : X_{\text{Zar}} \longrightarrow X_{\text{ét}}$$

which defines a morphism of sites

$$f : X_{\text{ét}} \longrightarrow X_{\text{Zar}},$$

whence corresponding direct image functors

$$f_* : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{X_{\text{Zar}}} \quad (f_*(F) = F \circ u),$$

and inverse image functors

$$f^* : \widetilde{X_{\text{Zar}}} \longrightarrow \widetilde{X_{\text{ét}}},$$

adjoint to one another. Geometrically, the pair (f_*, f^*) should be interpreted as a morphism of topoi

$$f : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{X_{\text{Zar}}}.$$

One obtains a homomorphism of cohomological functors

$$H^*(X_{\text{Zar}}, f_*(F)) \longrightarrow H^*(X_{\text{ét}}, F)$$

and a Leray spectral sequence 355

$$H^*(X_{\text{ét}}, F) \Longleftarrow H^p(X_{\text{Zar}}, R^q f_*(F)),$$

where F is an abelian sheaf on $X_{\text{ét}}$. This spectral sequence summarizes the general relations between étale and Zariski cohomology. Of course, for coefficient sheaves F such as constant coefficient sheaves, this spectral sequence is in general far from trivial, i.e. in general $R^q f_*(F) \neq 0$ (notably, take the case where X is the spectrum of a field). Nevertheless:

Proposition 4.3. *Let F be a sheaf of varying modules on the scheme X (a sheaf for the Zariski topology), whence a sheaf $F_{\text{ét}}$ on $X_{\text{ét}}$ (cf. 2, c) and a homomorphism of cohomological functors*

$$H^*(X, F) \longrightarrow H^*(X_{\text{ét}}, F_{\text{ét}}).$$

When F is quasi-coherent, the preceding homomorphism is an isomorphism.

Indeed, the first member is precisely $H^*(X_{\text{Zar}}, f_*(F_{\text{ét}}))$, and with the preceding notation it is enough to prove that

$$R^q f_*(F_{\text{ét}}) = 0 \quad \text{for} \quad q > 0.$$

By the method for computing the $R^q f_*$ and the fact that the affine open subsets form a basis for the topology of X, this follows immediately from:

Corollary 4.4. *If X is affine, then*

$$H^q(X_{\text{ét}}, F_{\text{ét}}) = 0 \quad \textit{for} \quad q > 0.$$

To see this, let $C = X_{\text{ét}}$, and let C' be the full subcategory formed by the *affine* X'. Then, by 3.1,

$$H^q(C, F_{\text{ét}}) \simeq H^q(C', F|C').$$

It is therefore enough to prove that $F|C'$ is C'-acyclic (V 4.1), or equivalently that 356
it satisfies the condition of V 4.3, i.e. that, for every $X' \in \operatorname{Ob} C'$ and every covering

family $\mathscr{R} = (X'_i \to X')_i$ in C', one has $H^q(\mathscr{R}, F) = 0$ for $q > 0$. One may suppose the family $(X'_i)_i$ finite and then, after replacing the X'_i by their coproduct, that the covering family consists of a single covering morphism $X'_i \to X'$, i.e. an (étale and) surjective morphism. We are thus reduced to proving that, if $f : X \to Y$ is a surjective étale morphism of affine schemes and F a quasi-coherent module on Y, then

$$(*) \qquad H^q(X/Y, F) = 0 \quad \text{for} \quad q > 0.$$

This is proved in TDTE I, B, 1.1 (in FGA, cf. reference [4] of IX).

Remarks 4.5. In fact, by *loc. cit.*, for $(*)$ it is enough that $X \to Y$ be surjective and flat (rather than étale). The preceding proof then shows that one still has isomorphisms analogous to that of 4.3, with the étale site $X_{\text{ét}}$ replaced by $(\mathrm{Sch})_{/X}$ equipped with any of the topologies finer than the Zariski topology considered in SGA 3 IV 6.3, for example the fpqc topology.

5. Cohomology of a projective limit of schemes

5.1. Let I be an increasing filtered preordered set,

$$\mathfrak{X} = (X)_{i \in I}$$

a projective system of schemes, the transition morphisms $u_i : X_i \to X_j$ $(i \geq j)$ being
357 *affine*. Recall (EGA IV 8...) that under these conditions, the projective limit

$$X = \varprojlim X_i$$

exists in the category of schemes (and even in the category of locally ringed spaces) and may be constructed as follows: one chooses $i_0 \in I$, so that for $i \geq i_0$ one has an X_{i_0}-isomorphism (EGA II 1.2 and 1.3)

$$X_i = \operatorname{Spec} \mathscr{A}_i,$$

where $\mathscr{A}_i$ is a quasi-coherent Algebra on X_i, the $\mathscr{A}_i$ $(i \geq i_0)$ thus forming an inductive system of such Algebras. Setting

$$\mathscr{A} = \varinjlim_i \mathscr{A}_i,$$

one may take

$$X = \operatorname{Spec} \mathscr{A}.$$

Moreover, denoting by esp S the underlying topological space of a scheme S, one proves (EGA IV 8...) that the canonical map

$$\operatorname{esp} X \longrightarrow \varprojlim \operatorname{esp}(X_i)$$

is a *homeomorphism* and that the canonical homomorphism of sheaves of rings

$$\varinjlim_i \gamma_i^{-1}(\mathscr{O}_{X_i}) \longrightarrow \mathscr{O}_X$$

is bijective, where $\gamma_i : X \to X_i$ is the canonical continuous map.

5.2. Informally, the content of EGA IV 8, 9 may be summarized by saying that, *when X_{i_0} is quasi-compact and quasi-separated*, then "every datum of a schematic nature
358 on X, of finite presentation over X", is equivalent to a datum of the same nature on one of the X_i, "for sufficiently large i". Thus one proves (EGA IV 8 …):

(a) if $i_1 \in I$ and X'_{i_1}, X''_{i_1} are two schemes of finite presentation over X_{i_1}, then, setting for every $i \geq i_1$

$$X'_i = X'_{i_1} \times_{X_{i_1}} X_i, \quad X' = X'_{i_1} \times_{X_{i_1}} X,$$

and defining X''_i, X'' in the same way, the canonical map

$$\varinjlim_{i \geq i_1} \mathrm{Hom}_{X_i}(X'_i, X''_i) \longrightarrow \mathrm{Hom}_X(X', X'')$$

is bijective.

(b) For every scheme X' of finite presentation over X, there exist an index $i_1 \in I$, a scheme X'_{i_1} of finite presentation over X_{i_1}, and an X-isomorphism

$$X' \simeq X'_{i_1} \times_{X_{i_1}} X.$$

5.3. The preceding result, and analogous results contained in EGA IV 8, may be expressed in the language of the $\varinjlim$ of fibered categories introduced in VI. For this purpose, consider the category $\mathscr{F}$ of morphisms $f : T \to S$ of finite presentation of schemes, and the "target functor"

$$\pi : \mathscr{F} \longrightarrow (\mathrm{Sch}),$$

which is plainly a *fibered functor*, its fiber categories being, moreover, equivalent to $\mathscr{U}$-small categories. Consider the inverse image of the fibered category $\mathscr{F}/(\mathrm{Sch})$ by the functor

$$i \mapsto X_i : \mathrm{cat}(I) \longrightarrow (\mathrm{Sch})$$

(where cat(I) is the category associated with I, with $\mathrm{Hom}(i, j) \neq \varnothing$ if and only if $i \geq j$), 359
giving a fibered category

$$\pi_{\mathfrak{X}} : \mathscr{F}_{\mathfrak{X}} \longrightarrow \mathrm{cat}(I),$$

whose fiber at each $i \in I$ is canonically isomorphic to the full subcategory $\mathscr{F}_{X_i}$ of $(\mathrm{Sch})_{/X_i}$ consisting of schemes of finite presentation over X_i, the base-change morphism $\mathscr{F}_i \to \mathscr{F}_j$ relative to $j \geq i$ being none other than $X'_i \mapsto X'_j = X'_i \times_{X_i} X_j$. This being so, by associating to every X'_i of finite presentation over X_i its inverse image on X

$$\varphi(X'_i) = X'_i \times_{X_i} X,$$

one obtains a natural functor

$$\varphi : \mathscr{F}_{\mathfrak{X}} \longrightarrow \mathscr{F}_X,$$

which plainly takes every cartesian morphism of $\mathscr{F}_{\mathfrak{X}}$ to an isomorphism, and hence, by definition, factors uniquely through a canonical functor

$$(*) \qquad \psi : \varinjlim_{\mathscr{F}_X/\mathrm{Cat}\, I} \mathscr{F}_{\mathfrak{X}} \longrightarrow \mathscr{F}_X.$$

In view of the description of the first member, the result of EGA IV 8 recalled above may then be stated by saying that $(*)$ is an *equivalence of categories*.

5.4. Statement a) of 5.2. above may be refined in various ways, by introducing some class $(\mathscr{M})$ of morphisms of schemes, stable under base change, and stating that for a given X_{i_1}-morphism $u_{i_1} : X'_{i_1} \to X''_{i_1}$, the corresponding morphism $u' : X' \to X''$ 360
is $\mathscr{E}(\mathscr{M})$ if and only if there exists $i \geq i_1$ such that the X_i-morphism $u_i : X'_i \to X''_i$ is $\mathscr{E}(\mathscr{M})$. The statement obtained in this way is true, for example, when $(\mathscr{M})$ is one of the following classes of arrows: proper morphisms (respectively projective, resp. quasi-projective, resp. affine, resp. affine, resp. finite, resp. quasi-finite, resp. radicial, resp. surjective, resp. flat, resp. smooth, resp. étale, resp. unramified). The reader will find the corresponding statements in EGA IV, §8 (for the first nine), §11 (for the flat case), and §17 (for the last three cases).

5.5. Now replace $\mathscr{F} \to$ (Sch) by the fibered subcategory $\mathscr{G}$ consisting of *étale morphisms* of finite presentation, whose fiber category $\mathscr{G}_S$, for every scheme S, is none other than the category of schemes étale and of finite presentation over S, and similarly form the fibered category $\mathscr{G}_{\mathfrak{X}}$ over $\operatorname{cat}(I)$, whose fiber category at every $i \in I$ is the category $\mathscr{G}_{X_i}$. By 3.2 (ii), the étale topoi $\widetilde{X_{i_{\text{ét}}}}$, $\widetilde{X_{\text{ét}}}$ are also canonically equivalent to $\widetilde{\mathscr{G}_{X_i}}$, $\widetilde{\mathscr{G}_X}$, (where each $\mathscr{G}_S$ is endowed with the topology induced by $S_{\text{ét}}$). Moreover, by 3.2 (i), each $\mathscr{G}_{X_i}$ is a site of finite type, and is moreover equivalent to a $\mathscr{U}$-small site. In view of the topologies on the $\mathscr{G}_{X_i}$, $\mathscr{G}$ becomes a *fibered site* over $\operatorname{cat}(I)$. This being so, we may state the following

Lemma 5.6. *The canonical functor*

$$\varinjlim_{\mathscr{G}_{\mathfrak{X}}/\operatorname{cat}(I)} \mathscr{G}_{\mathfrak{X}} \longrightarrow \mathscr{G}_X$$

361 *defines an equivalence, respecting the topologies, between the* $\varinjlim$ *site of the restricted étale sites (3.2) of the* X_i *and the restricted étale site of* $X = \varprojlim_i X_i$.

The fact that one obtains an equivalence of categories is one of the statements recalled in 5.4 (the one in which $(\mathscr{M})$ is the class of étale morphisms in (Sch)). The assertion concerning the topologies is obtained in the same way by taking $(\mathscr{M})$ = the class of surjective morphisms in (Sch).

The preceding result allows us to apply the results of VI to the present situation. In particular, we obtain

Theorem 5.7. *Let $\mathfrak{X} = (X_i)_{i\in I}$ be a projective system of schemes, with I an increasing filtered preordered set. Suppose that the transition morphisms $u_{ij} : X_j \to X_i$ are affine (so that the scheme $X = \varprojlim X_i$ is defined (5.1)), and that the X_i are quasi-compact and quasi-separated. Consider the fibered site $\mathscr{G}$ over $\operatorname{cat}(I)$ described in 5.5, whose fiber $\mathscr{G}_i$ at $i \in I$ is canonically isomorphic to the "restricted étale site" of X_i (consisting of schemes étale and of finite presentation over X_i). Let F be an abelian sheaf on $\mathscr{G}$ (i.e. (3) a functor $\mathscr{G}^0 \to$ (Ab) whose restriction F_i to each $\mathscr{G}_i$ is a sheaf, i.e. a sheaf on X_i). Let $F_\infty = \varinjlim_i u_i^*(F_i)$ be the induced sheaf on X, where $u_i : X \to X_i$ is the canonical morphism. Then the canonical homomorphisms*

$$\varinjlim_i H^n(X_i, F_i) \longrightarrow H^n(X, F_\infty)$$

are isomorphisms.

362 In view of 5.6, this follows by combining VI 8.7.3 and VI 8.5.2. We shall use 5.7 chiefly in the following particular case, which is also made explicit in

Corollary 5.8. *With the preceding hypotheses and notation for $(X_i)_{i\in I}$, X, u_{ij}, u_i, let $i_0 \in I$, and let F_{i_0} be an abelian sheaf on X_{i_0}. For every $i \geq i_0$, let $F_i = {u_{i_0 i}}^*(F_{i_0})$, and let moreover $F_\infty = u_i^*(F_{i_0})$. Under these conditions, the canonical homomorphisms*

$$\varinjlim_i H^n(X_i, F_i) \longrightarrow H^n(X, F_\infty)$$

are isomorphisms.

Here is, however, a sometimes useful case that follows from 5.7 and not from 5.8:

Corollary 5.9. *With the hypotheses and notation of 5.7 for $(X_i)_{i\in I}$, X, u_{ij}, u_i, let G_{i_0} be a commutative group scheme locally of finite presentation over X_{i_0} (where $i_0 \in I$ is given). For every $i \geq i_0$, let $G_i = G_{i_0} \times_{X_{i_0}} X_i$, and similarly let $G_\infty = G_{i_0} \times_{X_{i_0}} X$. Then the canonical homomorphisms*

$$\varinjlim_i H^n(X_i, G_i) \longrightarrow H^n(X, G_\infty)$$

are isomorphisms.

Indeed, we know that G_{i_0} defines a sheaf on $(\mathrm{Sch})_{/X_{i_0}}$ (cf. 2 (a)), and, supposing that i_0 is an initial object of I (which we may do), hence obtaining a canonical functor $\mathscr{G} \to (\mathrm{Sch})_{/X_{i_0}}$, composition gives a contravariant functor F on $\mathscr{G}$, whose restriction to $\mathscr{G}_i$ is canonically isomorphic to the sheaf on X_i defined by G_i. In particular, F is a 363
sheaf on $\mathscr{G}$, and we may apply 5.7 to it. The hypothesis that G_{i_0} is locally of finite presentation over X_{i_0} serves to ensure that the sheaf F_∞ of 5.7 is isomorphic, by the natural homomorphism $F_\infty \to \mathrm{faisc}(G_\infty)$, to the sheaf $\mathrm{faisc}(G_\infty)$ defined by G_∞ (as follows immediately from EGA IV 8.8.2 (i), by considering the values of the sheaves in question on the objects of the restricted étale site of G_∞ and applying 5.6). This completes the proof of 5.9.

Corollary 5.10. *With the hypotheses and notation of 5.7 for $(X_i)_{i\in I}$, X, u_{ij}, u_i, let F be a sheaf of commutative groups on X; then the canonical homomorphisms*

$$\varinjlim_i H^n(X_i, u_{i*}(F)) \longrightarrow H^n(X, F)$$

are isomorphisms.

The statements 5.7 through 5.10 generalize to corresponding statements for the Ext^i of sheaves of Modules, by VI 8.7.9; we shall not repeat them in the present particular case. On the other hand, for later reference we shall make explicit the variants of 5.8 and 5.9 in terms of the functors $R^n f_*$.

Corollary 5.11. *Let I be an increasing filtered preordered set, let $(X_i)_{i\in I}$ and $(Y_i)_{i\in I}$ be two projective systems of schemes with affine transition morphisms, $X = \varprojlim_i X_i$, $Y = \varprojlim_i Y_i$, $(f_i)_{i\in I}$ a projective system of quasi-compact and quasi-separated morphisms $f_i :$
$X_i \to Y_i$, $i_0 \in I$, and F_{i_0} a sheaf on X_{i_0}. For every $i \geq i_0$, let F_i be the sheaf on X_i* 364
that is the inverse image of F_{i_0}. Similarly, let $f : X \to Y$ be deduced from (f_i) and let F be the sheaf on X that is the inverse image of F_{i_0}. Under these conditions, the canonical homomorphism

$$\varinjlim_i u_i^*(R^n f_{i*}(F_i)) \longrightarrow R^n f_*(F)$$

is an isomorphism (where $u_i : X \to X_i$ is the canonical morphism).

We may suppose that i_0 is an initial object of I, and, since the question is local on Y_{i_0}, that Y_{i_0} is affine, so that the Y_i and the X_i are quasi-compact and quasi-separated. We are then under the hypotheses for applying VI 8.7.3 (where one reduces to 5.8, in view of the computation of the $R^n f_*$ as sheaves associated with presheaves). In the same way one proves:

COROLLARY 5.12. *The same statement as 5.11, except that now F_{i_0} denotes a commutative group scheme locally of finite presentation over X_{i_0}, and F_i, F its inverse images (in the sense of base change for schemes).*

Note that 5.9 (resp. 5.12) is not a particular case of 5.8 (resp. 5.11), cf. 2 a) remark 2.1.

COROLLARY 5.13. *With the notation of 5.11, let F be a sheaf of abelian groups on X. Then the canonical homomorphisms*

$$\varinjlim_i v_i^*(R^n f_{i*}(u_{i*}(F))) \longrightarrow R^n f_*(F)$$

are isomorphisms (where $u_i : X \to X_i$ and $v_i : Y \to Y_i$ are the canonical homomorphisms).

The proof is essentially the same as before.

365 REMARKS 5.14. a) The preceding limit-passage results remain valid for H^0 resp. f_*, for sheaves *of sets* (instead of abelian sheaves), and for H^1 resp. $R^1 f_*$ for sheaves of groups (not necessarily commutative), where the noncommutative H^1 and $R^1 f_*$ are defined as usual in terms of torsors (or principal homogeneous bundles) (cf. Giraud's thesis [34]). This can be checked directly in the general context of VI. It is certainly possible (and doubtless useful) also to give a variant for the noncommutative H^2 of "links" studied by Giraud.

b) The limit-passage results for fibered sites developed in VI assumed only that the base category was a filtered category, without it being necessary to suppose that it was associated with a filtered preordered set. The case of an arbitrary filtered index category is also the natural framework for the statements developed in the present n°. If we have worked in an overly restrictive framework, it was in order to be able to give correct references to EGA IV, where it is unfortunately assumed, in questions of passage to the limit (from §8 onward), that the index category is defined in terms of a filtered preordered set. We shall nevertheless assume henceforth that all the results used from EGA IV are valid for arbitrary filtered index categories (the proofs given in *loc. cit.* being valid, essentially without change, in this more general case). Thus we shall also use, without further comment, the results of the present n° in the case where I is replaced by an arbitrary filtered index category.

[34] cited p.(page 7) footnote (∗)

Fiber functors, supports, cohomological study of finite morphisms

A. Grothendieck

1. Topological invariance of the étale topos

Theorem 1.1. *Let* $f : X' \to X$ *be an* integral surjective radicial *morphism. Then the base-change functor* 367

$$f^* : X_{\text{ét}} \longrightarrow X'_{\text{ét}}$$

is an equivalence of sites (i.e. an equivalence of categories that is continuous, as is every quasi-inverse functor). Consequently, the functors

$$f_* : \widetilde{X'_{\text{ét}}} \longrightarrow \widetilde{X_{\text{ét}}}$$
$$f^* : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{X'_{\text{ét}}}$$

are equivalences of categories, quasi-inverse to one another.

The first assertion is well known (SGA 1 IX 4.10 and 4.11) when f is of finite presentation or when f is flat; the general case reduces to the case in which f is of finite presentation. Indeed, it is already known that the functor f^* is fully faithful (SGA 1 IX 3.4). It is enough to prove that f^* is essentially surjective, i.e. that every Z' étale over 368
X' "comes from" a Z étale over X. Using the fact that f is a universal homeomorphism and the full faithfulness of f^*, we are reduced to the case where X, X', and Z' are affine, the spectra of rings A, A', and B'. Writing A' as the inductive limit of its finitely generated A-subalgebras A'_i, B' comes from an étale algebra B'_i over some A'_i (cf. EGA IV 8), which reduces us to the case in which A' is integral and of *finite type* over A, hence finite over A. We then have an A-isomorphism $A' \simeq A''/J$, where $A'' \simeq A[t_1, \dots t_n]$ and J is an ideal of A''. Writing J as the inductive limit of its finitely generated subideals J_i, and setting $A'_i = A''/J_i$, we again have $A' = \varinjlim_i A'_i$, so B' comes from an étale algebra B'_i over some A'_i (*loco citato*). On the other hand, since $\operatorname{Spec} A' \to \operatorname{Spec} A$ is surjective, the same holds for the $\operatorname{Spec} A'_i \to \operatorname{Spec} A$; moreover, one easily verifies that since $\operatorname{Spec} A' \to \operatorname{Spec} A$ is radicial, the same holds for $\operatorname{Spec} A'_i \to \operatorname{Spec} A$ for sufficiently large i (apply EGA IV 1.9.9 to Spec A[34]). This reduces us to the case in which A' is one of the A'_i, hence is of finite presentation over A.

This proves the first assertion of 1.1. The assertion that f_* and f^* are mutually quasi-inverse equivalences follows at once.

Corollary 1.2. *Let F be an abelian sheaf on X, and let F' be its inverse image on X'. Then the canonical homomorphisms*

$$(*) \qquad H^i(X, F) \longrightarrow H^i(X', F')$$

[34] Let $S_i \subset S = \operatorname{Spec} A$ be the set of $s \in S$ such that the fiber of $X_i = \operatorname{Spec} A'_i$ at s is of separable rank 1. Then the S_i are *constructible* subsets of S (EGA IV 9.7.9) forming an increasing family; their union is S by the hypothesis on $S' = \operatorname{Spec} A' \to S$ and the fact that the fibers of S'_i are *noetherian* schemes. Thus $S = S_i$ for sufficiently large i, by EGA IV 1.9.9, C.Q.F.D..

are isomorphisms. Likewise, if F' is an abelian sheaf on X' and F its direct image on X, the canonical homomorphisms $(*)$ *are isomorphisms.*

369 EXAMPLES 1.3. Here are some frequently used applications of 1.1:

a) X' is a subscheme of X having the same underlying set as X, i.e. defined by a nil-ideal I.
b) X is a scheme over a separably closed field k, k' is an algebraic closure of k, and $X' = X \otimes_k k'$.
c) Let X be a geometrically unibranch scheme (for example, an algebraic curve over a field k whose only singularities are "cuspidal" singularities, thus excluding ordinary double points in particular). Then, if X' is the normalization of X_{red}, by definition $X' \to X$ is radicial, so 1.1 applies.

REMARKS 1.4. A morphism f as in 1.1 is a *universal homeomorphism*, i.e. it is a homeomorphism and remains so after every base extension. Conversely, if f is a universal homeomorphism, one proves that f satisfies the hypotheses of 1.1 [34], which explains the title of this section.

Let us point out explicitly, in connection with Example 1.3 c), that if X is an algebraic curve (over an algebraically closed field k, to fix ideas) having at least one singular point that is not "unibranch" (for example, an ordinary double point), then (with X' denoting the normalization of X_{red}) the conclusion of 1.2 already fails for H^1 with constant coefficients (compare SGA 1 I 11 a) and SGA 1 IX 5).

2. Sheaves on the spectrum of a field

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PROPOSITION 2.1. *Let k be a field, $\bar{k}$ a separable closure of k, $\pi = \mathrm{Gal}(\bar{k}/k)$ its topological Galois group, $X = \operatorname{Spec} k$, and $\bar{X} = \operatorname{Spec} \bar{k}$ (on which π acts on the left). Consider the canonical functor*

$$i : X_{\text{ét}} \longrightarrow \widetilde{X_{\text{ét}}}$$

(defined by $i(X')(X'') = \mathrm{Hom}_X(X', X'')$), and the canonical functor

$$j : \widetilde{X_{\text{ét}}} \longrightarrow \mathscr{B}_\pi$$

from the category of sheaves on $X_{\text{ét}}$ to the category $\mathscr{B}_\pi$ (cf. IV 2.7) *of (discrete) sets on which π acts continuously on the left, defined by*

$$j(F) = \varinjlim_{\alpha} F(\operatorname{Spec}(k_\alpha)),$$

where k_α ranges over the finite subextensions of $\bar{k}$. Then i and j are equivalences of categories. (Consequently, the topos $\widetilde{X_{\text{ét}}}$ is equivalent (IV 3.4) *to the "classifying topos" $\mathscr{B}_\pi$.)*

The composite functor

$$ji : X_{\text{ét}} \longrightarrow \widetilde{X_{\text{ét}}} \longrightarrow (\pi\text{-}\mathrm{Ens})$$

is an equivalence of categories by the fundamental theorem of Galois theory (in the form of SGA I V). Since $\mathscr{B}_\pi$ is plainly a topos (II 4.14), the same is therefore true of $X_{\text{ét}}$. Moreover, a family $(X_i \to X)$ of morphisms in $X_{\text{ét}}$ is covering if and only if it is surjective, i.e. its image in $\mathscr{B}_\pi$ is surjective, or, equivalently, covering for the canonical topology of $\mathscr{B}_\pi$. This shows that the topology of $X_{\text{ét}}$ is indeed its canonical topology. Consequently i is an equivalence, and hence so is j.

371 Corollary 2.2. *The functor j of 2.1 induces an equivalence between the category of abelian sheaves on $X = \operatorname{Spec} k$ and the category of Galois π-modules.*

Indeed, the latter are precisely the "abelian groups" of the topos $\mathscr{B}_\pi$. The same argument, by "transport of structure," gives:

Corollary 2.3. *Let F be an abelian sheaf on $X = \operatorname{Spec} k$, and let $M = j(F)$ be the associated Galois π-module. Then there is a canonical isomorphism of ∂-functors in F:*

$$H^*(X, F) \simeq H^*(\pi, M),$$

(where the right-hand side denotes Galois cohomology, studied, for example, in Serre's course C.G.*).*

Corollary 2.4. *Suppose that k is separably closed. Then the functor*

$$\Gamma : \widetilde{X_{\text{ét}}} \longrightarrow (\text{Ens})$$

is an equivalence of categories. If F is an abelian sheaf on $X = \operatorname{Spec} k$, then $H^i(X, F) = 0$ for $i \neq 0$. (In other words, the topos $\widetilde{X_{\text{ét}}}$ is a "punctual topos" (IV 2.2)*.)*

Corollary 2.5. *Let k' be a separably closed extension of a separably closed field k, let $X = \operatorname{Spec} k$, $X' = \operatorname{Spec} k'$, and let $u : X' \to X$ be the canonical morphism. Then the functor*

$$u^* : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{X'_{\text{ét}}}$$

is an equivalence of categories.

3. Fiber functors relative to geometric points of a scheme

3.1. Let X be a scheme. We shall call a *geometric point* of X any X-scheme ξ that is the spectrum of a *separably closed* field Ω. The datum of such a point is therefore equivalent to that of an ordinary point $x \in X$ together with a *separably closed* extension Ω of the residue field $k(x)$ of x. The case most frequent for us will be that in which Ω is a *separable closure* of $k(x)$, which we shall then generally denote by $\overline{k(x)}$; the corresponding geometric point of X is then denoted by $\overline{x}$. For a given $x \in X$, the $\overline{k(x)}$ (resp. $\overline{x}$) are determined up to *nonunique* k-isomorphism (resp. X-isomorphism). 372

Remarks 3.2. In most geometric questions, it is more natural to restrict to the case in which Ω is *algebraically closed.* The different convention used here is specific to the study of the étale topology. It is the convention we adopted in the definition of the fundamental group in SGA 1 V 7. Note, however, that the developments in *loco citato* are equally valid with the convention adopted here (because the property of Ω used there is that every étale covering of the spectrum of Ω is trivial, i.e. precisely that Ω is separably closed).

Definition 3.3. *Let X be a scheme, let ξ be a geometric point of X, and let $u : \xi \to X$ be its structural morphism. The (geometric)* fiber functor *relative to ξ, denoted $F \mapsto F_\xi$, is the composite functor*

$$\widetilde{X_{\text{ét}}} \longrightarrow (\text{Ens})$$

of

$$\widetilde{X_{\text{ét}}} \xrightarrow{u^*} \widetilde{\xi_{\text{ét}}} \xrightarrow{\Gamma_\xi} (\text{Ens}).$$

Equivalently, by 2.4, one may say that a geometric point ξ of X defines a point of the topos $X^{\sim}_{\text{ét}}$ (IV 6.1), whose associated fiber functor (*loc. cit.*) is simply being taken.

[34]Cf. EGA IV 8.11.6 if f is of finite presentation, and EGA IV 18.12.11 in the general case.

373 **3.4.** Since the functor Γ_ξ (the global-sections functor on ξ) is an *equivalence* of categories (12.4), knowing the fiber functors $F \mapsto F_\xi$ is essentially equivalent to knowing the inverse-image functors u^*. Moreover, it follows at once from 2.5 that if ξ is a geometric point of X for which there exists an X-morphism $\xi' \to \xi$ (i.e. ξ' corresponds to a separably closed extension Ω' of Ω), then the canonical homomorphism $F_\xi \to F_{\xi'}$ is a *functorial isomorphism*

$$F_\xi \simeq F_{\xi'}.$$

This shows that, for the study of fiber functors, one may (after replacing Ω by the separable algebraic closure of k in Ω) restrict oneself to the case where $\omega = k(\xi)$ is a separable closure $\overline{k(x)}$ of $k(x)$, and that the fiber functors corresponding to geometric points of X lying over the same $x \in X$ are isomorphic to one another (noncanonically).

In accordance with the conventions of 3.1, we shall denote by $F_{\overline{x}}$ a fiber functor corresponding to the choice of a separable closure $\overline{k(x)}$ of $k(x)$.

Let us point out an evident transitivity property (which shows the technical usefulness of not restricting oneself exclusively to geometric points defined by separable closures of residue fields): Let $f = X' \to X$ be a morphism of schemes, and let $u' : \xi' \to X'$ be a geometric point of X. Then $u = fu' : \xi' \to X$ is a geometric point of X, which we shall denote by ξ; relative to these geometric points, there is an isomorphism, functorial in $F \in \mathrm{Ob}\, \widetilde{X}_{\text{ét}}$:

$$f^*(F)_{\xi'} \simeq F_\xi.$$

374 Indeed, this follows from the transitivity formula for inverse-image functors

$$(fu)^* \xrightarrow{\sim} u^* f^*.$$

Theorem 3.5. *Let X be a scheme.*

a) *For every geometric point ξ of X, the fiber functor $F \mapsto F_\xi$, $\widetilde{X}_{\text{ét}} \to (\mathrm{Ens})$ commutes with arbitrary $\varinjlim$ (indexed by categories $\in \mathcal{U}$) and with finite $\varprojlim$[34]. In particular, it takes sheaves of groups (resp. abelian sheaves, etc.) to groups (resp. abelian groups, etc.).*

b) *As x ranges over the points of X, the fiber functors $F_{\overline{x}}$ form a "conservative" family of functors, i.e. if $u : F \to G$ is a homomorphism of sheaves on X, then u is an isomorphism if and only if, for every $x \in X$, the corresponding homomorphism $u_{\overline{x}} : F_{\overline{x}} \to G_{\overline{x}}$ is an isomorphism*[34].

Taking into account the exactness properties in a), let us immediately note the following formal consequences of b):

Corollary 3.6. *Let $u : F \to G$ be a homomorphism of sheaves. Then u is a monomorphism (resp. an epimorphism) if and only if, for every $x \in X$, $u_{\overline{x}} : F_{\overline{x}} \to G_{\overline{x}}$ is injective (resp. surjective).*

Corollary 3.7. *Let $u, v : F \to G$ be two morphisms of sheaves on X. Then $u = v$ if and only if, for every $x \in X$, one has $u_{\overline{x}} = v_{\overline{x}} : F_{\overline{x}} \to G_{\overline{x}}$. In particular, if u, v are two*
375 *sections of F, one has $u = v$ if and only if $u_{\overline{x}} = v_{\overline{x}}$ in $F_{\overline{x}}$ for every $x \in X$.*

Corollary 3.8. *Let $F \to G \to H$ be a sequence of two homomorphisms of sheaves on X. Then the sequence is exact if and only if, for every $x \in X$, the corresponding sequence $F_{\overline{x}} \to G_{\overline{x}} \to H_{\overline{x}}$ is exact.*

[34] Thus it is indeed a "fiber functor" in the sense of Exposé IV.

[34] Thus there are "enough fiber functors" of the form $F \mapsto F_{\overline{x}}$, in the terminology of IV 6.5.

We leave to the reader the task of deducing the corollaries [34], which will moreover be proved in part in the proof that follows.

Assertion a) follows from the exactness properties already noted in (VII 1.4) for every inverse-image functor, such as $u^* : \widetilde{X_{\text{ét}}} \to \widetilde{\xi_{\text{ét}}}$, and from the fact that $\widetilde{\xi_{\text{ét}}} \to (\text{Ens})$ is an equivalence. To prove b), let us note the following formula, which gives a procedure for *computing* fiber functors:

Proposition 3.9[34]. *Let $\xi \xrightarrow{u} X$ be a geometric point of X, and let C_ξ be the category of "pointed étale X-schemes at ξ", i.e. of pairs (X', u'), where $X' \in \operatorname{Ob} X_{\text{ét}}$ is a scheme étale over X, and u' is an X-morphism $\xi \to X'$. Then the opposite category C_ξ^0 is filtered and, for a varying sheaf F (resp. presheaf P) on X, there is a canonical functorial isomorphism*

$$F_\xi \simeq \varinjlim_{C_\xi^0} F(X'), \text{(resp. } (aP)_\xi \simeq \varinjlim_{C_\xi^0} P(X')),$$

(where aP = the sheaf associated with P).

Indeed, let us first note that for every presheaf P on $\xi_{\text{ét}}$, if aP is the associated sheaf, the natural homomorphism

$$P(\xi) \longrightarrow aP(\xi)$$

is an isomorphism, as follows, for example, immediately from the explicit computation of aP as $L(LP)$ (II 3.19). Let $\varphi : \widetilde{X_{\text{ét}}} \to \widetilde{\xi_{\text{ét}}}$ be the "inverse-image" functor. Then 376
$u^* = \varphi^*$ is nothing other (III 2.3. (d)) than the composite a $\varphi^{\boldsymbol{\cdot}} i$, where $i : \widetilde{X_{\text{ét}}} \to \widehat{X_{\text{ét}}}$ is the inclusion functor, and $a : \widehat{\xi_{\text{ét}}} \to \widetilde{\xi_{\text{ét}}}$ is the "associated sheaf" functor. By the preceding remark we find

$$u^*(F)(\xi) = (\varphi^{\boldsymbol{\cdot}} i(F))(\xi),$$

so it is enough to apply the explicit expression for $\varphi^{\boldsymbol{\cdot}}$ given in the relevant proof. The corresponding case, where one starts with a presheaf P on X, is proved in the same way, by using the functorial isomorphism $\varphi^* a(P) \simeq a\varphi^*(P)$. The fact that the category C_ξ is filtered (which remains true for *every* X-scheme ξ, not necessarily reduced to a geometric point) follows immediately from the fact that finite projective limits exist in C_ξ, which follows from the fact that $X_{\text{ét}}$ is a subcategory of $(\text{Sch})/X$ stable under finite projective limits.

Now let $u : F \to G$ be a homomorphism of sheaves on X such that, for every $x \in X$, $u_{\overline{x}} : F_{\overline{x}} \to G_{\overline{x}}$ is a monomorphism; let us prove that u is a monomorphism. For this, we must prove that if $X' \in \operatorname{Ob} X_{\text{ét}}$, $\varphi, \psi \in F(X')$ are such that $u(\varphi) = u(\psi)$, then $\varphi = \psi$. Replacing X by X' and using the transitivity of fiber functors, we may suppose $X = X'$. For every $x \in X$, we then have $u(\varphi)_{\overline{x}} = u(\psi)_{\overline{x}}$, i.e. $u_{\overline{x}}(\varphi_{\overline{x}}) = u_{\overline{x}}(\psi_{\overline{x}})$, hence $\varphi_{\overline{x}} = \psi_{\overline{x}}$ since $u_{\overline{x}}$ is a monomorphism. Using 3.9, it follows that there exists an $X'_x \in \operatorname{Ob} X_{\text{ét}}$, whose image contains x, such that φ and ψ have the same inverse image on X'_x. Since, as x varies in X, the X'_x form a covering family of X, it follows that $\varphi = \psi$.

This proves that if the $u_{\overline{x}}$ are monomorphisms, then so is u. Suppose in addition that 377
the $u_{\overline{x}}$ are epimorphisms; let us prove that the same holds for u, and hence that u is an isomorphism. It remains to prove that for every $X' \in \operatorname{Ob} X_{\text{ét}}$ and every $\psi \in G(X')$, there exists $\varphi \in F(X')$ such that $u(X')(\varphi) = \psi$. We may again suppose $X' = X$, and using 3.9, we see that, for every $x \in X$, there exists $X'_x \in \operatorname{Ob} X_{\text{ét}}$ such that the image of X'_x in X contains x, and a $\varphi_x \in F(X'_x)$ whose image in $G(X'_x)$ is the inverse image of

[34]Cf. IV 6.5.2.

[34]Cf. IV 6.8.3

ψ. Using the fact that u is a monomorphism, we see that the φ_x agree on the $X'_x \times_X X'_x$ $(x, y \in X)$ and therefore come from a uniquely determined $\varphi \in F(X)$; one then has $u(\varphi) = \psi$, since the inverse images on the X'_x agree. C.Q.F.D..

Scholium 3.10. Theorem 3.5 implies that the collection of "fiber functors" for the étale topology enjoys the same essential properties as in the usual theory of sheaves on a topological space. We shall use 3.5 and its corollaries very frequently, and for this reason we shall generally dispense with referring to them explicitly.

3.11. For every $x \in X$ we shall also, by abuse of language, denote by x the $X-$ scheme $\operatorname{Spec} k(x)$, for the usual structural morphism

$$i_x : x = \operatorname{Spec} k(x) \longrightarrow X.$$

It gives rise to a canonical functor

$$i_x^* : \widetilde{X_{\text{ét}}} \longrightarrow \widetilde{x_{\text{ét}}}.$$

If F is a sheaf on X, we shall put $F_x = i_x^*(F)$; thus it is a sheaf on $x = \operatorname{Spec} k(x)$ (and
378 as such may also be identified with an étale scheme over x, by virtue of 2.1). It depends functorially on F, and the functor $i_x^* : F \mapsto F_x$ still commutes with arbitrary $\varinjlim$ and finite $\varprojlim$, by virtue of VII 1.4.

If $\overline{x} = \operatorname{Spec} \overline{k(x)}$, the fiber functor $F \mapsto F_{\overline{x}}$ is canonically isomorphic to the composite of the functor $F \mapsto F_x$ and the functor j of 2.1 (taking into account that the latter is isomorphic to the fiber functor relative to the geometric point $\operatorname{Spec} \overline{k}$ of $\operatorname{Spec} k$). It follows immediately from this and from 3.5 that the system of functors $(F \mapsto F_x)$ $(x \in X)$ is likewise conservative. Moreover, if $k(x)$ is separably closed, i.e. $x = \overline{x}$, the notations F_x and $F_{\overline{x}}$ are, strictly speaking, in conflict (the first denoting an object of $\widetilde{x_{\text{ét}}}$, the second a set), but by virtue of 2.1 this conflict is inessential.

3.12. The preceding remarks show that if $\overline{x} = \operatorname{Spec} k(x)$, then, for every sheaf F on X, the group $\pi_x = \operatorname{Gal}(\overline{k(x)}/k(x))$ acts naturally (on the left) on the fiber set $F_{\overline{x}}$, so that $F \mapsto F_{\overline{x}}$ may in fact be regarded as a functor

$$\widetilde{X_{\text{ét}}} \longrightarrow \mathscr{B}_{\pi_x}$$

whose knowledge is essentially equivalent (thanks to 2.1) to that of the functor $F \mapsto F_x$.

This shows, in an important special case, that contrary to what happens for sheaves on topological spaces, "fiber functors" generally admit nontrivial automorphisms. In general, we shall determine in 7.9 all the homomorphisms from one fiber functor to another.

379 Remark 3.13. Here is a sometimes useful generalization of 3.5 b). Consider a subset E of X such that, for every étale morphism $f : X' \to X$, $E' = f^{-1}(E)$ is "very dense" in X' (EGA IV 10.1.3), i.e. for every f as above, every open subset U of X' which contains E' is equal to X'. Then the fiber functors $F_{\overline{x}}$ on $\widetilde{X_{\text{ét}}}$, for $x \in E$, already form a "conservative family" (and consequently Corollaries 3.6, 3.7, and 3.8 remain valid when one restricts oneself to choosing $x \in E$ in them). This follows immediately by repeating the proof given for 3.5 b). The preceding result applies in particular in the following situations:

a) X is a Jacobson scheme (EGA IV 10.4.1.), for example locally of finite type over a field, or over $\operatorname{Spec}(\mathscr{Z})$, and E is the set of closed points of X.

b) X is locally of finite type over a scheme S, and E is the set of points of X which are closed in their fiber.

c) X is locally noetherian, and E is the set of $x \in X$ such that $\overline{x}$ is a finite set (i.e. a semilocal scheme of dimension ≤ 1) cf. EGA IV 10.5.3 and 10.5.5.

4. Strictly local rings and schemes

4.1. Recall that, following Azumaya, a local ring A is said to be *henselian* if it satisfies the following equivalent conditions:

(i) Every finite A-algebra B is a direct product $\prod_i B_i$ of local rings (the B_i are then necessarily isomorphic to the $B_{\mathfrak{m}_i}$, where $\mathfrak{m}_i$ runs through the maximal ideals 380
of B).

(ii) Every A-algebra B which is the localization at a prime ideal $\mathfrak{p}$ lying over the maximal ideal $\mathfrak{m}$ of A of an A-algebra C of finite type, and such that $B/\mathfrak{m}B$ is finite over $k = A/\mathfrak{m}$, is finite over A.

(iii) The same as (ii), but with C assumed étale over A at $\mathfrak{p}$ and $k \to B/\mathfrak{m}B$ an isomorphism.

(iv) "Hensel's lemma" in its classical form holds for A, that is, for every monic polynomial $F \in A[T]$, writing $\dot{F} \in k[T]$ for the corresponding reduced polynomial, and for every factorization

$$\dot{F} = \dot{F}_1 \dot{F}_2$$

of $\dot{F}$ as a product of two relatively prime monic polynomials, $\dot{F}_1$ and $\dot{F}_2$ lift (necessarily uniquely) to $F_1, F_2 \in A[T]$ such that $F = F_1 F_2$.

For the proof of these equivalences and the general properties of henselian rings, see EGA IV 18.5.11. Recall that if A is any local ring, one proves (Nagata) that there exists a local homomorphism

$$A \longrightarrow A^h$$

with A^h *henselian*, which is *universal* for local homomorphisms from A to local henselian rings. The ring A^h is called the henselization of A. The construction given in EGA IV 18.6 (different from Nagata's) consists in setting

$$A^h = \varinjlim_i A_i,$$

where A_i runs through the A-algebras essentially of finite type and essentially étale over A (that is, localizations of an étale A-algebra) such that $A \to A_i$ is a 381
local homomorphism and the residue extension $k(A_i)/k(A)$ is trivial. This construction shows in particular that A^h is flat over A, and that if A is Noetherian, then so is its henselization.

Definition 4.2. *A local ring A is said to be strictly local* if it satisfies one of the following equivalent conditions*:*

(i) *A is henselian (4.1) and its residue field is separably closed.*

(ii) *Every local homomorphism $A \to B$, where B is a localization of an étale algebra over A, is an isomorphism.*

A scheme is called a strictly local scheme *if it is isomorphic to the spectrum of a strictly local ring.*

The equivalence of (i) and (ii) follows at once from 4.1. Note that, in geometric form, 4.2 (ii) (respectively 4.1 (iii)) says that for every étale Y-scheme X and every point x of the fiber X_y (respectively every point of X_y *rational over* $k(y)$), where y is the closed point of Y, there exists a section of X over Y passing through x (and this section is necessarily unique).

Definition 4.3. *Let X be a scheme, let $\mathscr{O}_{X_{\text{ét}}}$ be the sheaf on $X_{\text{ét}}$ defined by*

$$\mathscr{O}_{X_{\text{ét}}}(X') = \Gamma(X', \mathscr{O}_{X'})$$

(compare VII 2. c)*), and let $u : \xi = \operatorname{Spec}(\Omega) \to X$ be a geometric point of X. The* strictly
382 local ring of X at ξ, *denoted $\mathscr{O}_{X,\xi}$ or simply $\mathscr{O}_\xi$, is the fiber of the sheaf $\mathscr{O}_{X_{\text{ét}}}$ at the point ξ. Its spectrum is called the* strict localization of X at ξ.

By 3.9, the fiber $\mathscr{O}_{X,\xi}$ has the description

$$(*) \qquad \mathscr{O}_{X,\xi} = \varinjlim \Gamma(X', \mathscr{O}_{X'}),$$

where X' runs through the opposite filtered category of étale X-schemes pointed by ξ, that is, equipped with an X-morphism $\xi \to X'$. By transitivity of inductive limits, one may replace $\Gamma(X', \mathscr{O}_{X'})$ in the right-hand side by $\mathscr{O}_{X',x'}$, where x' is the image of ξ in X', and one obtains

$$(**) \qquad \mathscr{O}_{X,\xi} = \varinjlim_i A_i$$

where A_i runs through the algebras essentially of finite type and étale over $A = \mathscr{O}_{X,x}$, equipped with a k-homomorphism $k(A_i) \to \Omega$. (Here x denotes the image of ξ in X.) The transition homomorphisms between the A_i are, of course, the local homomorphisms of A-algebras $A_i \to A_j$ that make the corresponding triangle commute:

$$\begin{tikzcd} k(A_i) \arrow[r] \arrow[dr] & k(A_j) \arrow[d] \\ & \Omega. \end{tikzcd}$$

This implies that if there exists an admissible homomorphism from A_i to A_j, it is unique. Thus the inductive limit $(**)$ is a limit over an upward-filtered ordered index set.

383 4.4. The description $(**)$ shows that the strictly local ring of X at ξ depends essentially only on the usual local ring $A = \mathscr{O}_{X,x}$ $(x = u(\xi))$ and on the extension Ω of $k = k(A)$. It is also called the *strict henselization* of A (relative to the given separably closed extension Ω of k), and is denoted A^{hs}. We refer to EGA IV 18.8 for the general properties of A^{hs}. We merely point out that the construction again shows that A^{hs} is flat over A, and that it is Noetherian if A is. Moreover, one proves easily (*loc. cit.*) that the local homomorphism

$$A \longrightarrow A^{hs}$$

together with the k-homomorphism

$$k(A^{hs}) \longrightarrow \Omega$$

solves the universal problem, relative to the data of A and the extension Ω of k, of finding local homomorphisms

$$A \longrightarrow A',$$

where A' is *strictly local*, equipped with a k-homomorphism

$$k(A') \longrightarrow \Omega.$$

In particular, A^{hs} *is indeed a strictly local ring* (which justifies the terminology of 4.3).

4.5. Keep the notation of 4.3, and choose an affine open neighborhood U of x. Note that in 3.9 one may plainly restrict to those X' which are affine over U. These then form a pseudo-filtered inverse system of affine schemes, so that we are in the general situation 384
of VII 5 (see VI 5.12 b). By the expression (∗) for $\mathscr{O}_{X,\xi}$, there is an X-isomorphism

$$(***) \qquad \varprojlim_{\substack{X' \text{ étale} \\ \text{affine over } U, \\ X'_\xi\text{-pointed}}} X' \simeq \operatorname{Spec} \mathscr{O}_{X,\xi}$$

which makes precise the intuitive geometric meaning of $\operatorname{Spec}(\mathscr{O}_{X,\xi})$ as the "*limit of the étale neighborhoods of* ξ".

Proposition 4.6. *Let X be a strictly local scheme, and let x be its closed point, which is therefore a geometric point of X. Then the functor $F \mapsto \Gamma(X, F) : \widetilde{X_{\text{ét}}} \to (\text{Ens})$ is canonically isomorphic to the fiber functor $F \mapsto F_x$, and hence commutes with arbitrary* $\varinjlim$[34].

Indeed, by Definition 4.2 (ii), the final object X of the category of étale X-schemes pointed by x considered in 3.9 dominates every other object; hence $\varinjlim F(X')$ is canonically isomorphic to $F(X)$, as required.

In particular, the functor induced by Γ on the category of abelian sheaves is exact, and therefore:

Corollary 4.7. *Under the hypotheses of 4.6, for every abelian sheaf F on X one has*

$$H^q(X, F) = 0 \quad \text{for} \quad q \neq 0.$$

Corollary 4.8. *Let X be a scheme, let ξ be a geometric point of X, let $\overline{X} = \operatorname{Spec} \mathscr{O}_{X,\xi}$ be the corresponding strict localization, let F be a sheaf on X, and let $\overline{F}$ be its inverse image on $\overline{X}$. Then there is a functorial isomorphism*

$$F_\xi \simeq \Gamma(\overline{X}, \overline{F}).$$

Indeed, ξ may also be regarded as a geometric point of $\overline{X}$, and it suffices to apply 385
4.6 and the transitivity of fibers (3.4).

5. Application to the computation of the fibers of the $R^q f_*$

5.1. Let

$$f : X \longrightarrow Y$$

be a morphism of schemes and let F be an abelian sheaf on X. We wish to determine the $R^q f_*(F)$. By 3.5, this amounts, up to very little, to knowing the geometric fibers $R^q f_*(F)_{\overline{y}}$ for $y \in Y$. Now (V 5.1) $R^q f_*(F)$ is the sheaf associated with the presheaf

$$\mathscr{H}^q : Y' \longmapsto H^q(X \times_Y Y', F)$$

on Y, and hence (3.9)

$$R^q f_*(F)_{\overline{y}} = \varinjlim_{Y'} \mathscr{H}^q(Y'),$$

where the inductive limit is taken over the opposite filtered category of the étale Y-schemes pointed by $\overline{y}$. After choosing an affine open neighborhood U of y, one may

[34] not necessarily filtered!

restrict the limit on the right to the cofinal category of those Y' which are affine and lie over U. Thus one obtains

$$R^q f_*(F)_{\overline{y}} \simeq \varinjlim H^q(X \times_Y Y', F), \tag{5.1.1}$$

where Y' varies in the preceding category.

Put

$$\overline{Y} = \operatorname{Spec}(\mathscr{O}_{Y,\overline{y}}) = \varprojlim Y'$$

(see 4.5), and

$$\overline{X} = X \times_Y \overline{Y}.$$

386 Observe that in the inverse system $X \times_Y Y'$ (deduced from that of the Y' by the base change $X \to Y$), the transition morphisms are affine. We are therefore in the general situation of VII 5.1, and the construction of $\varprojlim$ in *loc. cit.* immediately gives a canonical isomorphism

$$\overline{X} = \varprojlim X \times_Y Y'.$$

Let $\overline{F}$ denote the sheaf on $\overline{X}$ that is the inverse image of F. We then obtain a canonical homomorphism

$$\varinjlim_{Y'} H^q(X \times_Y Y', F) \longrightarrow H^q(\overline{X}, \overline{F}),$$

and hence, by comparison with (5.1.1), a canonical homomorphism

$$R^q f_*(F)_{\overline{y}} \longrightarrow H^q(\overline{X}, \overline{F}), \tag{5.1.2}$$

which is plainly functorial in F.

Suppose now that $f : X \to Y$ is quasi-compact and quasi-separated. Then the same is true of $X \times_Y Y' \to Y'$, and since Y' is affine, the schemes $X \times_Y Y'$ are quasi-compact and quasi-separated. Applying (5.1.1) and the limit-passage theorem VII 5.8, one obtains:

Theorem 5.2. *Let $f : X \to Y$ be a quasi-compact and quasi-separated morphism of schemes, let F be an abelian sheaf on X, let y be a point of Y, and let $\overline{y}$ be the geometric point over y associated with a separable closure $k(\overline{y})$ of $k(y)$. Let $\overline{Y} = \operatorname{Spec}(\mathscr{O}_{Y,\overline{y}})$ be the corresponding strict localization, let $\overline{X} = X \times_Y \overline{Y}$, and let $\overline{F}$ be the inverse image of F on $\overline{X}$. Then the canonical homomorphism (5.1.2) is an isomorphism:*

$$R^q f_*(F)_{\overline{y}} \simeq H^q(\overline{X}, \overline{F}).$$

387 This statement practically reduces the determination of the fibers of a sheaf $R^q f_*(F)$ to the determination of the cohomology of a scheme over a strictly local scheme, and it will be used constantly below. Technically, 5.2 explains the important role of henselian and strictly local rings in the study of étale cohomology.

Remark 5.3. The statement remains valid for $R^0 f_*(F) = f_*(F)$, for an arbitrary sheaf of sets. Moreover, 5.2 also remains valid for $R^1 f_*(F)$ when F is a sheaf of groups (not necessarily commutative), with the usual definition of $R^1 f_*(F)$ (see Giraud's thesis).

5.4. Suppose now that f is a *finite* morphism. Then $\overline{f} : \overline{X} \to \overline{Y}$ is finite, and since $\overline{Y}$ is strictly local, it follows that $\overline{X}$ is a finite disjoint union of strictly local schemes $\overline{X}_i$. Hence, using 4.7, one finds

$$H^q(\overline{X}, \overline{F}) = 0 \quad \text{for} \quad q \neq 0. \tag{5.4.1}$$

Furthermore, the components $\overline{X}_i$ of $\overline{X}$ correspond to the points $\overline{x}_i$ of $\overline{X}$ above the closed point $\overline{y}$ of $\overline{Y}$, that is, to the points of $\overline{X}_{\overline{y}} = X_y \otimes_{k(y)} k(\overline{y})$. These points may also be regarded as geometric points of X, and $\overline{X}_i$ is then precisely the strict localization of X at $\overline{x}_i$. Thus (4.8)

$$H^0(\overline{X}_i, \overline{F}) \simeq F_{\overline{x}_i},$$

whence

$$H^0(\overline{X}, \overline{F}) = \prod_i F_{\overline{x}_i}. \tag{5.4.2}$$

This is an isomorphism functorial in the sheaf of sets F (there is no need here to restrict to abelian sheaves). Taking 5.2 and 5.3 into account, formulas (5.4.1) and (5.4.2) give: 388

Proposition 5.5. *Let $f : X \to Y$ be a finite morphism of schemes, and let y be a point of Y. Then for every sheaf F on X, there is an isomorphism, functorial in F,*

$$f_*(F)_{\overline{y}} \simeq \prod_{\overline{x} \in X_y \otimes_{k(y)} k(\overline{y})} F_{\overline{x}}.$$

Consequently, the formation of $f_*(F)$ commutes with every base change $Y' \to Y$; and if F is an abelian sheaf, then

$$R^q f_*(F) = 0 \quad \text{if} \quad q > 0.$$

Note that the first formula of 5.5 is in fact independent of 5.2 and of the general limit-passage theorem VII 5.7; together with 3.5, it implies that $F \mapsto f_*(F)$ is an *exact functor* on the category of abelian sheaves, and hence again that $R^q f_*(F) = 0$ for $q \neq 0$. Here is a slight variant of 5.5:

Corollary 5.6. *Let $f : X \to Y$ be an integral morphism. Then for every abelian sheaf F on X one has*

$$R^q f_*(F) = 0 \quad \text{for} \quad q \neq 0.$$

Moreover, the functor f_* on sheaves of sets commutes with every base change $Y' \to Y$.

Indeed, by 5.2 it is enough to prove that when Y is strictly local, $H^q(X, F) = 0$ for every abelian sheaf F on X. We then have $Y = \operatorname{Spec} A$, $X = \operatorname{Spec} B$, where B is an integral A-algebra; writing $B = \varinjlim B_i$, where B_i runs through the finite-type subalgebras of B (which are in fact finite over A), one has $X = \varprojlim_i X_i$, where $X_i = \operatorname{Spec} B_i$. By VII 5.13, it is enough to prove that $R^q f_*(F_i) = 0$ for $q \neq 0$, where F_i is an abelian sheaf on X_i; this follows from 5.5. The last assertion of 5.6 is proved by the same reduction to 5.5. 389

Corollary 5.7. *Let $f : X \to Y$ be an immersion. Then for every sheaf F on X, the canonical morphism*

$$f^* f_*(F) \longrightarrow F$$

is an isomorphism.

Since f factors as the composite of an open immersion and a closed immersion, and 5.7 is trivial for an open immersion (IV no. 3), it remains to consider the case where f is a closed immersion. We must prove that for every $x \in X$, the corresponding homomorphism

$$(f^* f_*(F))_{\overline{x}} \longrightarrow F_{\overline{x}}$$

is bijective. By transitivity of fibers (3.4), the left-hand side is precisely the fiber $f_*(F)_{\overline{x}}$; it is therefore enough to verify that the canonical homomorphism

$$f_*(F)_{\overline{x}} \longrightarrow F_{\overline{x}}$$

is bijective, which follows at once from 5.5.

Remark 5.8. Using 5.5 and arguing again as in 5.6, one easily proves that if $f : X \to Y$ is an integral morphism and F is a sheaf of groups on X (not necessarily commutative), then $R^1 f_*(F)$ is the terminal sheaf on Y (compare Remark 5.3).

6. Supports

390 Let U be a Zariski open subset of the scheme X. Then $I \in \mathrm{Ob}\, X_{\text{ét}}$, and in fact U is a subobject of the final object X of $X_{\text{ét}}$, hence defines a subobject $\widetilde{U}$ of the final object of $\widetilde{X}_{\text{ét}}$, i.e. an "open subset" of the étale topos $X_{\text{ét}}$ of X (IV 8.3).

Proposition 6.1. *The preceding map $U \mapsto \widetilde{U}$ is an isomorphism from the ordered set of open subsets (in the Zariski sense) of X onto the set of open subsets of the étale topos $\widetilde{X}_{\text{ét}}$.*

Since $X_{\text{ét}} \to \widetilde{X}_{\text{ét}}$ is fully faithful, it follows immediately that the map $U \mapsto \widetilde{U}$ preserves the order structures, i.e. $(U \subset V) \Leftrightarrow (\widetilde{U} \subset \widetilde{V})$; in particular, the preceding map is injective. To prove that it is surjective, consider a subsheaf F of the final sheaf $\widetilde{X}$, and consider the objects of $X_{\text{ét}/F}$, i.e. the schemes X' étale over X such that $F(X') \neq \varnothing$. Since $X' \to X$ is étale, it is an open map (EGA IV 2.4.6); in particular, its image (in the set-theoretic sense) $\mathrm{Im}(X')$ is open. Let U be the open subset which is the union of the $\mathrm{Im}(X')$ ($X' \in \mathrm{Ob}\, X_{\text{ét}/F}$). Since the family of the $X' \to U$ ($X' \in \mathrm{Ob}\, X_{\text{ét}/F}$) is surjective, hence covering, we conclude that $U \in \mathrm{Ob}\, X_{\text{ét}/F}$, hence $X_{\text{ét}/F'} = X_{\text{ét}/U}$, hence $F = U$, C.Q.F.D..

In view of 6.1, we may therefore speak unambiguously of an "open subset" of X, without specifying whether we mean this notion in the usual Zariski sense or in the sense of the étale topology.

Corollary 6.2. *Let U be an open subset of X, and let $j : U \to X$ be the canonical immersion. Then the functor*

$$j^* : \widetilde{X}_{\text{ét}} \longrightarrow \widetilde{U}_{\text{ét}}$$

391 *induces an equivalence of categories*

$$\widetilde{X}_{\text{ét}}/\widetilde{U} \longrightarrow \widetilde{U}_{\text{ét}}.$$

Indeed, one checks immediately that for every site C in which finite $\varprojlim$ exist, and for every subobject U of the final object e of C, if one considers the functor $j : C \to C_{/U}$ defined by $j(S) = S \times U$, then j is a morphism of sites and $j^* : \widetilde{C} \to \widetilde{C/U}$ induces an equivalence $\widetilde{C}/\widetilde{U} \to \widetilde{C/U}$. It then suffices to combine this general fact with the fact that $U_{\text{ét}}$ is canonically isomorphic to $X_{\text{ét}/U}$.

In pictorial terms, one can express 6.2 by saying that the operations of "restriction to an open subset," in the usual sense for schemes on the one hand and in the sense of topoi on the other, are compatible. Here is an analogous compatibility for the operations of "restriction to a closed subset":

Theorem 6.3. *Let X be a scheme, let Y be a closed subscheme of X, let $U = X - Y$ be endowed with the induced structure, and let $i : Y \to X$ and $j : U \to X$ be the canonical immersions. Then the functor*

$$i_* : \widetilde{Y_{\text{ét}}} \longrightarrow \widetilde{X_{\text{ét}}}$$

is fully faithful, and if $F \in \operatorname{Ob} \widetilde{X_{\text{ét}}}$, then F is isomorphic to a sheaf of the form $i_*(G)$ sss *$j^*(F)$ is isomorphic to the final sheaf $\widetilde{U}$ on U.*

Proof. Since i_* and i^* are adjoint to one another, the fact that i_* is fully faithful is also equivalent to the fact that the functorial homomorphism

$$i^*(i_*(G)) \longrightarrow G$$

is an isomorphism, which is precisely 5.7. On the other hand, if $G \in \operatorname{Ob} \widetilde{Y_{\text{ét}}}$, then one checks trivially, using 6.2, that $j^*(i_*(G))$ is the final sheaf on U. Conversely, if $F \in \operatorname{Ob} \widetilde{X_{\text{ét}}}$ is such that $j^*(F)$ is the final sheaf, let us prove that F is isomorphic to a sheaf of the form $i_*(G)$, or, what amounts to the same thing, that the canonical homomorphism 392

$$G \longrightarrow i_* i^* G$$

is an isomorphism. It is enough once again to check that for every $x \in X$, the homomorphism induced on the geometric fibers at $\overline{x}$ is an isomorphism. When $x \in U$, this is precisely the hypothesis made on G. When $x \in Y$, by transitivity of fibers we are reduced to checking that the homomorphism on the fibers at $\overline{x}$ induced by

$$i^*(G) \longrightarrow i^*(i_* i^*(G))$$

is an isomorphism; but the preceding homomorphism is an isomorphism by 5.7, applied to $F = i^* G$ and to i. This completes the proof of 6.3.

Corollary 6.4. *The functor i_* induces an equivalence from the category of abelian sheaves on Y to the category of abelian sheaves on X whose restriction to $U = X - Y$ is zero.*

In accordance with common usage, we shall therefore often identify an abelian sheaf on Y with an abelian sheaf on X which is zero on U.

6.5. By virtue of 6.1, we thus know (if X is a scheme) how to interpret the "open subsets" of the topos $\mathscr{E} = \widetilde{X_{\text{ét}}}$ as open subsets U of X in the usual sense, and by virtue of 6.3, under this identification the "residual topos" $\mathscr{E}_{\complement} U$ of IV 3 is canonically equivalent to the topos $\widetilde{Y_{\text{ét}}}$, where $Y = X - U$ is endowed with any induced structure making Y a scheme. We may consequently apply to this situation the results of IV 3, in particular the description IV 3.3 of sheaves F on X in terms of triples (F', F'', u), where F' is a sheaf on Y, F'' is a sheaf on U, and u is a homomorphism from F' to $i^* j_*(F')$. In what follows we shall freely use the notation $i^!$, $j_!$ of IV.3, which denotes functors 393

$$j_! : \widetilde{(U_{\text{ét}})}_{\text{ab}} \longrightarrow \widetilde{(X_{\text{ét}})}_{\text{ab}},$$
$$i^! : \widetilde{(X_{\text{ét}})}_{\text{ab}} \longrightarrow \widetilde{(Y_{\text{ét}})}_{\text{ab}},$$

where the subscript "ab" denotes the category of abelian sheaves. These functors give rise to the two exact sequences IV 3.7.

6.6. In view of the preceding developments and of the general terminology introduced in IV 8.5, it is appropriate, for an abelian sheaf F on X, or a section φ of such a sheaf, to introduce the notion of the *support of* F, resp. of φ, as the closed subset (in the usual, i.e. Zariski, sense) complementary to the largest open subset on which the object in question vanishes.

In the present situation, one should also replace the general notation V 4.3 $H^q(\mathscr{E}_{\complement U} F)$, $\mathscr{H}^q_{\complement U}(F)$ by the notation $H^q_Y(X, F)$, $\mathscr{H}^q_Y(F)$, the first denoting an abelian group and the second an abelian sheaf on Y (or equivalently an abelian sheaf on X which is zero on U). Thus, one has $\mathscr{H}^q_Y = R^{q_y!}$, etc.

We refer to V 6 for the general properties of the preceding functors.

7. Specialization morphisms of fiber functors

394 **7.1.** We saw in N° 4 how one associates, with every geometric point ξ of the scheme X, a strictly local x-scheme

$$\overline{X}(\xi) = \operatorname{Spec} \mathscr{O}_{X,\xi},$$

which in fact depends only on the geometric point above the image x of ξ defined by the separable closure $\overline{k(x)}$ of $k(x)$ in $\Omega = k(\xi)$. In what follows we shall often restrict ourselves to geometric points $\xi = \operatorname{Spec} \Omega$ which are *algebraic separable* over X, i.e. such that Ω is a separable closure of $k(x)$, i.e. $\xi = \overline{x}$. Let us call an X-scheme Z a *strict localization of* X if it is X-isomorphic to a scheme of the form $\operatorname{Spec} \mathscr{O}_{X,\overline{x}}$; thus one sees that

$$\xi \mapsto \operatorname{Spec} \mathscr{O}_{X,\xi} = \overline{X}(\xi)$$

is an equivalence from the category of geometric points *algebraic separable* over X to the category of X-schemes which are strict localizations of X, when only isomorphisms are taken as morphisms in both categories.

The last assertion is no longer correct when all X-morphisms are taken as morphisms, since in the second category there may be X-morphisms which are not isomorphisms. We then make the following

Definition 7.2. *Let ξ, ξ' be two geometric points of the scheme X. A specialization arrow from ξ' to ξ is any X-morphism between the corresponding strict localizations*

$$\overline{X}(\xi') \longrightarrow \overline{X}(\xi).$$

395 *We say that ξ is a specialization of ξ', or that ξ' is a generization of ξ, if there exists a specialization arrow from ξ' to ξ.*

Note that specialization arrows compose in the evident way, so that, with specialization arrows as morphisms, *the geometric points of X form a category*, equivalent (as is the full subcategory formed by the geometric points algebraic and separable over X) to the full subcategory of $(\mathrm{Sch})_{/X}$ formed by the strict localizations of X.

Lemma 7.3. *Let X be a scheme, let Z be an X-scheme which is isomorphic to a pseudo-filtered projective limit of X-schemes X_i which are étale, with affine transition morphisms* (VII 5.1)*, let ξ' be a geometric point of X, and let $Z' = \overline{X}(\xi')$ be the corresponding strict localization. Then:*

a) *The restriction map*

$$\operatorname{Hom}_X(Z', Z) \longrightarrow \operatorname{Hom}_X(\xi', Z)$$

is bijective.

b) *In order that the two sides be nonempty, it is necessary and sufficient that the image x' of ξ' in X lie in the image of Z.*
c) *Let T be a Z-scheme. In order that T be a strict localization of Z, it is necessary and sufficient that it be a strict localization of X.*

Proof.

a) We are immediately reduced to the case where Z is one of the X_i, i.e. where Z is étale over X, and by the base change $Z' \to X$ we may suppose that $Z' \xrightarrow{\sim} X$, whence the conclusion follows from 4.2 (ii).
b) Note that if x' is the image of a point z of Z, then $k(z)$ is necessarily a separable algebraic extension of $k(x')$; hence there exists a $k(x')$-homomorphism from the latter into $k(\xi')$, whence the conclusion.
c) The proof is an easy exercise left to the reader. 396

From 7.2 and 7.3 we conclude:

Proposition 7.4. *Let ξ, ξ' be two geometric points of the scheme X. Then the restriction map defines a bijection from the set $\mathrm{Hom}_X(\overline{X}(\xi'), \overline{X}(\xi))$ of specialization arrows from ξ' to ξ to the set of X-morphisms from ξ' to $\overline{X}(\xi)$.*

Corollary 7.5. *In order that ξ be a specialization of ξ', it is necessary and sufficient that the same hold for the images x, x' of ξ, ξ' in X, i.e. that x belong to the closure of $\{x'\}$.*

Since $\operatorname{Spec} \mathscr{O}_{X,\xi} \to \operatorname{Spec} \mathscr{O}_{X,x}$ is faithfully flat (4.4), it is surjective, and it is therefore enough to apply the second assertion of 7.3.

Corollary 7.6. *For every scheme X, let $\mathrm{Pt}(X)$ be the category of geometric points over X (or equivalently of geometric points algebraic separable over X), the morphisms being specialization arrows. Let ξ be a geometric point of a scheme X, and let $\overline{X}(\xi)$ be the corresponding strict localization. Then there is an equivalence of categories*

$$\mathrm{Pt}(\overline{X}(\xi)) \xrightarrow{\sim} \mathrm{Pt}(X)_{/\xi},$$

obtained by associating with every geometric point ξ' of $\overline{X}(\xi)$ the corresponding geometric point over X, together with the specialization morphism to ξ deduced from the structural morphism $\xi' \to \overline{X}(\xi)$ by 7.4.

In other words, *the datum of a generization ξ' of the geometric point ξ is essentially equivalent to the datum of a geometric point of $\overline{X}(\xi)$*. Note moreover that, by 7.3 c), the corresponding homomorphism $\overline{X}(\xi') \to \overline{X}(\xi)$ makes $\overline{X}(\xi')$ the strict localization of $\overline{X}(\xi)$ relative to ξ'. 397

7.7. Now interpret, for every geometric point ξ of X and every sheaf F on X, the fiber F_ξ as $\Gamma(\overline{X}(\xi), \overline{F}(\xi))$, where $\overline{F}(\xi)$ is the inverse image of F on $\overline{X}(\xi)$ (4.8). Then one sees that every specialization arrow

$$u : \xi' \longrightarrow \xi$$

induces a homomorphism, functorial in F:

$$u^* : F_\xi \longrightarrow F_{\xi'},$$

called the *specialization homomorphism* associated with the specialization arrow u. It is evident from the transitivity of inverse images of sheaves that, for a composite specialization arrow, one has:

$$(wu)^* = u^* w^*.$$

7.8. If $\mathscr{E}$ is a topos, recall from Exposé IV that a "fiber functor," or (by abuse of language) a "point" of the topos $\mathscr{E}$, is any morphism from the "final topos" (Ens) (isomorphic to the category of sheaves on a space reduced to one point!) to $\mathscr{E}$, i.e. any functor

$$\varphi = \mathscr{E} \longrightarrow (\mathrm{Ens})$$

which commutes with arbitrary inductive limits, and with finite projective limits. The set of fiber functors of $\mathscr{E}$ should be regarded as the set of objects of a full subcategory of $\mathbf{Hom}(\mathscr{E}, (\mathrm{Ens}))$, called the *category of fiber functors* of the topos E, whose opposite is called the *category of points of* $\mathscr{E}$, and is denoted $\mathrm{Pt}(E)$, cf. (IV 6.1).

398 When E is of the form $\widetilde{X}_{\text{ét}}$, where X is a scheme, we have defined in 3.3 and 7.7 a functor

$$(*) \qquad \mathrm{Pt}(X) \longrightarrow \mathrm{Pt}(\widetilde{X}_{\text{ét}}),$$

where the first member is defined in 7.6. With this understood, we have the following

Theorem 7.9. *Let X be a scheme. The preceding functor $(*)$ is an equivalence from the category of geometric points over X (with specialization arrows as morphisms) to the category of points of the étale topos $\widetilde{X}_{\text{ét}}$ (the* opposite *of that of fiber functors on $\widetilde{X}_{\text{ét}}$).*

Since this theorem will not be used later in the seminar, we confine ourselves here to a sketch of the proof, in which we shall allow ourselves certain liberties with questions of universes.

a) *The functor under consideration is fully faithful.* For every $X \in \mathrm{Ob}\, X_{\text{ét}}$, let $\overset{\vee}{X} : \widetilde{X}_{\text{ét}} \to (\mathrm{Ens})$ be defined by

$$\overset{\vee}{X}(F) = \mathrm{Hom}(X, F) = F(X).$$

Note that the fiber functor

$$\mathscr{E}_\xi : F \longrightarrow F_\xi$$

can be written

$$\mathscr{E}_\xi \simeq \varinjlim_i X_i^\vee,$$

where the X_i are affine schemes étale over X, indexed by a certain filtered category (cf. 4.3 and 4.5). Hence, for every functor $\varphi : \widetilde{X}_{\text{ét}} \to (\mathrm{Ens})$, one has:

$$\mathrm{Hom}(\mathscr{E}_\xi, \varphi) \simeq \varprojlim_i \mathrm{Hom}(X_i^\vee, \varphi),$$

and on the other hand it is well known that there is a bifunctorial isomorphism

$$\mathrm{Hom}(X_i^\vee, \varphi) \simeq \varphi(X_i).$$

399 When φ is of the form $\mathscr{E}_\xi$, hence

$$\varphi \simeq \varinjlim_j X'^\vee_j,$$

there is therefore a natural bijection

$$(*)\qquad \operatorname{Hom}(\mathscr{E}_\xi, \mathscr{E}_{\xi'}) \simeq \varprojlim_i \varinjlim_j \operatorname{Hom}_X(X'_j, X_i).$$

On the other hand, one has (in the category of schemes)

$$\overline{X}(\xi) = \varprojlim_i X_i, \qquad \overline{X}(\xi') = \varprojlim_j X'_j,$$

whence

$$\operatorname{Hom}_X(\overline{X}(\xi'), \overline{X}(\xi)) \simeq \varprojlim_i \operatorname{Hom}_X(\overline{X}(\xi'), X_i),$$

and on the other hand, since X_i is locally of finite presentation over X, one has

$$\operatorname{Hom}_X(\overline{X}(\xi'), X_i) \simeq \varinjlim_j \operatorname{Hom}_X(X_j, X_i),$$

whence

$$(**)\qquad \operatorname{Hom}_X(\overline{X}(\xi'), \overline{X}(\xi)) \simeq \varprojlim_i \varinjlim_j \operatorname{Hom}_X(X'_j, X_i).$$

Comparison of $(*)$ and $(**)$ gives the desired conclusion (subject to checking compatibilities, which is left to the reader).

b) *The functor under consideration is essentially surjective.*

7.9.1. First note that if C is a site in which finite $\varprojlim$ are representable, and $\widetilde{C} = \mathscr{E}$ is the corresponding topos, then every fiber functor φ on $\mathscr{E}$ can be represented as a "filtered inductive limit" of functors of the form

$$\check{Y}(F) = F(Y) = \operatorname{Hom}(\widetilde{Y}, F), Y \in \operatorname{Ob} C,$$

(where $\widetilde{Y}$ is the sheaf associated with Y), as follows [34]. Consider the category $C_{/\varphi}$ of 400
pairs (Y, ξ), with $Y \in \operatorname{Ob} C$, $\xi \in \varphi(\widetilde{Y})$ (the morphisms being defined in the evident way), and note that there is an evident homomorphism

$$(*)\qquad \varinjlim_{(Y,\xi)\in C^0_{/\varphi}} \check{Y} \longrightarrow \varphi$$

by observing that

$$\operatorname{Hom}(\varinjlim_{C_\varphi} \check{Y},) \simeq \varprojlim_{C_{/\varphi}} \operatorname{Hom}(\check{Y}, \varphi) \simeq \varprojlim_{C_{/\varphi}} \varphi(\check{Y}).$$

There is a canonical element in $\varprojlim_{C_{/\varphi}} \varphi(\widetilde{Y})$, obtained by associating with every (Y, ξ) the element ξ of $\varphi(Y)$. Using the fact that finite $\varprojlim$ exist in C, and that the functor $Y \mapsto \varphi(\widetilde{Y})$ commutes with them, one sees that in $C_{/\varphi}$ finite $\varprojlim$ exist; a fortiori $C^0_{/\varphi}$ is filtered. Using the fact that every sheaf F is an inductive limit of sheaves of the form $\widetilde{Y}$, and using the fact that φ commutes with inductive limits, one easily concludes that the homomorphism of functors $(*)$ is bijective and thus gives the asserted representation.

[34] This is a special case of IV 6.8.3.

7.9.2. Note also that if $\mathscr{E}$ is a topos and Y is an object of $\mathscr{E}$, hence defines a topos $\mathscr{E}_{/Y}$, then the datum of a fiber functor φ for $\mathscr{E}_{/Y}$ is equivalent to the datum of a pair (φ, ξ), where φ is a fiber functor for $\mathscr{E}$, and $\xi \in (Y)$. With the fiber functor $\psi : \mathscr{E}_{/Y} \to (\mathrm{Ens})$ one associates the pair (φ, ξ), where φ is the composite $Z \mapsto \varphi(Z) = \psi(Z \times Y)$, and where $\xi \in \varphi(Y) = \psi(Y \times Y)$ is the image of the unique element of $\psi(Y)$ under the diagonal morphism of $Y \times Y$.

7.9.3. We now return to the case of the site $C = \widetilde{X}_{\text{ét}}$, and let $\varphi : \widetilde{X}_{\text{ét}} \to (\mathrm{Ens})$
401 be a fiber functor on the étale topos of X. Considering the restriction of φ to the full subcategory of $\widetilde{X}_{\text{ét}}$ formed by the open subsets of this topos, or, what amounts to the same thing (6.1), by the open subsets U of X, one obtains a map

$$\varphi_0 : \mathscr{U}(X) \longrightarrow \{0, 1\}$$

which commutes with arbitrary suprema and finite infima. (Note that for $u \in \mathscr{U}(X)$, $\varphi(U)$ is either empty or consists of a single point, and one takes $\varphi_0(U) = 0$ or 1 according as one is in the former or the latter case.)

One easily sees, using the fact that every irreducible closed subset of X has one and only one generic point [34], that φ_0 is defined by means of a unique $x \in X$, by the condition

$$\varphi_0(U) = 1 \quad \text{sss} \quad x \in U,$$

i.e.

$$\varphi(U) = \varnothing \Leftrightarrow x \in U.$$

Let $F \in \mathrm{Ob}\, \widetilde{X}_{\text{ét}}$. Then the image of F in the final sheaf X is an open subset; and since

$$\varphi(F) \longrightarrow \varphi(U)$$

is surjective, one sees that $\varphi(F) \neq \varnothing$ sss $\varphi(U) \neq \varnothing$, i.e. sss $x \in U$. Let $C_{/\varphi}$ be the category of pairs (X', ξ) where $X' \in \mathrm{Ob}\, X_{\text{ét}}$, and $\xi \in \varphi(X')$. It was pointed out in 7.9.1 that $C^0_{/\varphi}$ is filtered and connected. Moreover, by 7.9.2 and by the preceding discussion applied to X' instead of X, if $(X', \xi) \in \mathrm{Ob}\, X_{\text{ét}}$, there is associated with it a unique point $x' \in X'$, such that for an étale morphism $X'' \to X'$, ξ lies in the image of $\varphi(X'') \to \varphi(X')$ sss x' lies in the image of X''. It follows that the scheme $\varprojlim$ of the X' over the category C of the (X', ξ) exists and is a strict localization Z of X, thus corresponding
402 to a geometric point ξ and a fiber functor $\mathscr{E}_\xi$. For every sheaf F, one then has

$$\mathscr{E}_\xi(F) \simeq \varinjlim_{C_\varphi} (F)(X').$$

In view of 7.9.1, we conclude that φ is isomorphic to $\mathscr{E}_\xi$, which completes the proof of 7.9.

8. Two spectral sequences for integral morphisms

PROPOSITION 8.1. *Let $f : X \to Y$ be a surjective integral morphism, and let F be an abelian sheaf on Y. For every integer i, let $\mathscr{H}^i(F)$ be the presheaf on* $(\mathrm{Sch})_{/Y}$ *defined by*

$$\mathscr{H}^i(F)(Z) = H^i(Z, F_Z),$$

where F_Z is the inverse image of F on Z. Then there exists a spectral sequence (functorial in F)

$$H^*(X, F) \Longleftarrow E_2^{pq} = H^p(X/Y, \mathscr{H}^q(F))$$

("descent spectral sequence").

[34] i.e. X is *sober* in the terminology of IV 4.2.1. The assertion made is a special case of IV 4.2.3.

Of course, the symbol $H^p(X/Y, G)$, for a presheaf G on $(\mathrm{Sch})_{/Y}$, denotes the p-th relative Čech cohomology group, defined by means of the complex of the $C^n(X/Y, G) = G(X^{n+1})$, where X^m denotes the m-th power in $(\mathrm{Sch})_{/Y}$. To prove 8.1, let $p_n : X^{n+1} \to Y$ be the projection, and set

$$A^n = p_{n*}(\mathscr{Z}_{X^{n+1}}),$$

where $\mathscr{Z}_{X^{n+1}}$ denotes the constant sheaf $\mathscr{Z}$ on X^{n+1}. Then the A^n are the components of a simplicial abelian sheaf on Y, hence of a complex of abelian sheaves A^* on Y. Note that there is an evident homomorphism

$$(*) \qquad \mathscr{Z}_Y \longrightarrow A^0.$$

With this understood, we have the following 403

Lemma 8.2. *The complex (A^*), endowed with the homomorphism $(*)$, is a resolution of $\mathscr{Z}_Y$. More generally, for every abelian sheaf F on Y, $A^* \otimes_{\mathscr{Z}} F$ is a resolution of F.*

Proof: We may suppose that Y is affine, hence $Y = \operatorname{Spec} A$, $X = \operatorname{Spec} B$, where B is an integral A-algebra. We have $B = \varinjlim_i B_i$, where B_i ranges over the finitely generated, hence finite, subalgebras of B, whence $X = \varprojlim X_i$, where $X_i = \operatorname{Spec} B_i$. Using VII 5.11, one sees that the augmented complex $A^*(X/Y)$ is then the inductive limit of the augmented complexes $A^*(X_i/Y)$. This reduces us to the case where f is *finite*. It suffices to prove that for every geometric point $\overline{y}$ of Y, the complex $A^*_{\overline{y}}$ is a resolution of $\mathscr{Z}_{\overline{y}}$, and likewise after tensoring with an F. But if $X_{\overline{y}}$ is the fiber of X at $\overline{y}$, it follows from 5.5 that the complex $A^*_{\overline{y}}$ is none other than the analogous complex $A^*(X_{\overline{y}}/\overline{y})$. This reduces us to the case where Y is the spectrum of a separably closed field k. Using 1.3 a) and b), we may even suppose that k is algebraically closed and X is reduced, hence that X is of the form I_Y, where I is a finite set. But then the complex $A^* \otimes F$ is identified with the trivial cochain complex of the index set I with coefficients in $F_{\overline{y}}$, and is therefore indeed a resolution of $F_{\overline{y}}$.

Thus, as a consequence of 8.2, we obtain a spectral sequence

$$H^*(Y, F) \Longleftarrow E_2^{pq} = H^p(H^q(Y, A^* \otimes F)),$$

and it remains to make the initial term explicit. There is a canonical homomorphism 404

$$A^n \otimes F \longrightarrow p_{n_*} p_n^*(F),$$

and the latter is an isomorphism, as one sees once again by reduction to the case where f is finite and passage to fibers. Using 5.5, one concludes that

$$H^q(Y, A^n \otimes F) \simeq H^q(X^{n+1}, p_n^*(F)) \Longrightarrow \mathscr{H}^q(F)(X^{n+1}),$$

which indeed gives the initial term stated in 8.1.

Remark 8.3. a) When one is given a *locally finite* covering of Y by closed subsets Y_i, and one sets $X = \coprod_i Y_i$, the canonical morphism $f : X \to Y$ is finite, and the initial term of the spectral sequence 8.1 can be made explicit as the cohomology of the complex defined by the $H^q(Y_{i_0 \ldots i_p}, F|Y_{i_0 \ldots i_p})$, where $Y_{i_0 \ldots i_p} = Y_{i_0} \cap \cdots \cap Y_{i_p}$. This is therefore the analogue of the Leray spectral sequence for a locally finite closed covering of an ordinary topological space (TF Chapter II 5.2.4). The latter can moreover also be generalized to a spectral sequence relative to a "finite" morphism, i.e. one which is proper with finite fibers, by proceeding as in 8.1.

b) Note the analogy between the spectral sequence of 8.1 and the Leray spectral sequence of a covering (X_i) of Y (in this case by X_i étale over Y); the latter would formally be obtained by writing the spectral sequence 8.1 for $X = \coprod_i X_i$. It is in fact likely that these two spectral sequences admit a common generalization, valid whenever one has a family of morphisms $(X_i \to Y)$ which is a "universally effective descent family" for the fibered category of étale sheaves over a
405 variable base scheme (cf. n° 9 below). The analogous question also arises in ordinary topology, concerning a common generalization of the two kinds of Leray spectral sequences of a covering, assumed either open or locally finite closed [34].

Proposition 8.4. *Let Y be a scheme, let π be a profinite group, let $(\pi_i)_i$ be the projective system of finite discrete quotient groups of π, let $(X_i)_{i\in I}$ be a projective system of principal coverings of Y, with groups π_i, the transition homomorphisms $X_j \to X_i$ being compatible with the homomorphisms $\pi_j \to \pi_i$ on the groups of operators, and let $X = \varprojlim X_i$ (cf. VII 5.1), so that the group π acts on the Y-scheme X. Let F be an abelian sheaf on Y. Then there is a* "Hochschild–Serre" *spectral sequence (functorial in F)*

$$H^*(Y,F) \Longleftarrow E_2^{pq} = H^p(\pi, \varinjlim_i H^q(X_i, F_{X_i})),$$

where F_{X_i} is the inverse image of F on X_i, and $H^p(\pi,-)$ denotes Galois cohomology.

The term E_2^{pq} written here is also (by the definition of $H^p(\pi,-)$ and the transitivity of inductive limits) isomorphic to

$$E_2^{pq} = \varinjlim_i H^p(\pi_i, H^q(X_i, F_{X_i})),$$

which shows that it is enough to find an inductive system of spectral sequences (depending on the index i)

$$H^*(Y,F) \Longleftarrow {}^iE_2^{pq} = H^p(\pi_i, H^q(X_i, F_{X_i})).$$

This reduces us to defining the spectral sequence in the case where π is finite. Then
406 $f : X \to Y$ is a covering morphism in $Y_{\text{ét}}$, so one may write the Leray spectral sequence of this morphism. In view of the canonical isomorphisms

$$X^n \simeq X \overset{G}{\times} G^n,$$

a well-known calculation then shows that for every presheaf $\mathscr{H}$ on $X_{\text{ét}}$ which transforms sums into products, $C^*(X/Y, \mathscr{H})$ is none other than the complex of homogeneous cochains of G with coefficients in $\mathscr{H}(X)$, whence the stated form of the initial term.

A variant of this proof, avoiding any calculation, consists in regarding $E \mapsto X \times_\pi E$ as a morphism from the site of finite π-sets to the étale site of Y, and writing the Leray spectral sequence of this morphism. Finally, when π is finite, one may also regard the Hochschild–Serre spectral sequence as a special case of 8.1 relative to the morphism $f : X \to Y$.

Corollary 8.5. *Suppose that Y is quasi-compact and quasi-separated. Then the spectral sequence of 8.4 takes the form*

$$H^*(Y,F) \Longleftarrow E_2^{pq} = H^p(\pi, H^q(X, F_X)).$$

[34] Since these lines were written, P. Deligne has developed a general theory of "descent spectral sequences," cf. Exp. V bis.

Indeed, by VII 5.8, there are then canonical isomorphisms

$$\varinjlim_i H^q(X_i, F_{X_i}) \simeq H^q(X, F_X).$$

Corollary 8.6. *Let Y be a local henselian scheme with closed point y, let F be an abelian sheaf on Y, and let F_0 be the induced sheaf on $Y_0 = \mathrm{Spec}(k(y))$. Then the canonical homomorphisms*

$$H^n(Y, F) \longrightarrow H^n(Y_0, F_0)$$

are isomorphisms. 407

Indeed, let $\overline{y}$ be a geometric point over y, corresponding to a separable closure $k(y) = k(\overline{y})$ of $k(y)$; since Y is henselian, the strict localization X of Y at $\overline{y}$ is the projective limit of the connected étale Galois coverings pointed at $\overline{y}$, denoted X_i, so that the hypotheses of 8.5 apply. Since $H^q(X, F_X) = 0$ for $q > 0$ by 4.7, one obtains isomorphisms

$$H^n(Y, F) \xleftarrow{\sim} H^n(\pi, F(X)),$$

where π is the "Galois group" of X over Y, isomorphic to that of $\overline{k(y)}$ over $k = k(y)$. Similarly (or by 2.3) one finds

$$H^n(Y_0, F_0) \xleftarrow{\sim} H^n(\pi, F_0(X_0)),$$

where $X_0 = X \times_{Y_0} Y \simeq \mathrm{Spec}(k(\overline{y}))$. But the restriction homomorphism $F(X) \to F_0(X_0)$ is an isomorphism by 4.8, from which the conclusion 8.6 follows immediately.

9. Descent of étale sheaves

This section will not be used later in this Seminar, and may be omitted on a first reading.

Proposition 9.1. *Let $f : X' \to X$ be a surjective (resp. universally submersive* (SGA I IX 2.1)*) morphism of schemes. Then the functor $f^* : X_{\text{ét}} \to X'_{\text{ét}}$ is faithful and "conservative"* (cf. 3.6) *(resp. induces a* fully faithful *functor from the category of sheaves on $X_{\text{ét}}$ to the category of sheaves on $X'_{\text{ét}}$ endowed with descent data relative to $f : X' \to X$).* 408

The first assertion follows immediately from the transitivity of fiber functors (3.4) and from (3.7); the second (which can also be stated by saying that f is a *descent morphism* relative to the fibered category of étale sheaves on variable schemes) also means that for two sheaves F, G on Y, the natural diagram

$$(*) \qquad \mathrm{Hom}(F, G) \to \mathrm{Hom}(F', G') \rightrightarrows \mathrm{Hom}(F'', G'')$$

is exact, where F' and G' (resp. F'' and G'') are the inverse images of F and G on X' (resp. on $X'' = X' \times_X X'$). Taking the final sheaf for F, the assertion gives the following

Corollary 9.2. *Let $f : X' \to X$ be a universally submersive morphism. Then for every sheaf F on X, the diagram*

$$\Gamma(X, F) \to \Gamma(X', F') \rightrightarrows \Gamma(X'', F'')$$

is exact, where $X'' = X' \times_X X'$, and where F', F'' are the inverse images of F on X', X''.

We shall prove 9.1 using the following

Lemma 9.3. *Suppose that f is universally submersive. Let F be a sheaf on X, denote by*
$S(F)$ the set of subsheaves of F, and define $S(F')$, $S(F'')$ analogously. Then the diagram 409
of natural maps

$$S(F) \to S(F') \rightrightarrows S(F'')$$

is exact.

Proof. The fact that $S(F) \to S(F')$ is injective follows immediately from the fact that the functor f^* is conservative; for if F_i $(i = 1, 2)$ are two subsheaves of F such that $f^*(F_1) = f^*(F_2)$, then the inclusions $F_3 = F_1 \cap F_2 \to F_i$ $(i = 1, 2)$ are isomorphisms, since they become so after applying the functor f^*, hence $F_1 = F_2$. It remains to prove that if G' is a subsheaf of F' such that $\mathrm{pr}_1^*(G') = \mathrm{pr}_2^*(G')$, then G' is the inverse image of a subsheaf of F. Note that one can find an epimorphism $F_1 \to F$ in $X_{\text{ét}}$, where F_1 is *representable* by a scheme étale over X. Introduce $F_2 = F_1 \times_F F_1$, and likewise F_1', F_1'', F_2', F_2'', sheaves giving rise to a diagram of sets

$$\begin{tikzcd}
S(F) \arrow[r] \arrow[d] & S(F') \arrow[r] \arrow[d] & S(F'') \arrow[d] \\
S(F_1) \arrow[r] \arrow[d, shift left] \arrow[d, shift right] & S(F_1') \arrow[r, shift left] \arrow[r, shift right] \arrow[d, shift left] \arrow[d, shift right] & S(F_1'') \arrow[d, shift left] \arrow[d, shift right] \\
S(F_2) \arrow[r] & S(F_2') \arrow[r, shift left] \arrow[r, shift right] & S(F_2''),
\end{tikzcd}$$

in which the columns are exact, by the exactness properties in the category of sheaves on X, X', X'', respectively. A standard diagram chase then shows that, to prove that the first row is exact, it is enough to prove this for rows 2 and 3. For row 2, this follows
410 from the fact that F_1 is representable, that $X' \to X$ is universally submersive, and
from SGA 1 IX 2.3. On the other hand, F_2 is also representable, since it is a subsheaf of $F_1 \times_X F_1$, which is representable, and one applies 6.1. Thus row 3 is also exact, C.Q.F.D.

We now prove 9.1, i.e. that every morphism $u' : F' \to G'$ compatible with the descent data on F', G', the morphism u' comes from a morphism $u : F \to G$. Let $H = F \times G$, hence $H' = F' \times G'$, $H'' = F'' \times G''$; then the *graph* of u' is a subsheaf Γ' of H', whose two inverse images are equal to the graph of the same morphism $u'' : F'' \to G''$. Hence, by 9.3, Γ' comes from a subsheaf Γ of H. I claim that Γ is the graph of a morphism $u : F \to G$, i.e. that the morphism $p : \Gamma \to F$ induced by pr_1 is an isomorphism: indeed, it becomes an isomorphism after the base change $X' \to X$, and we apply the already proved part of 9.1. The morphism $u : F \to G$ then has the required property, C.Q.F.D.

Theorem 9.4. *Let $f : X' \to X$ be a surjective morphism of schemes. Suppose that one of the following hypotheses holds:*

a) *f is integral.*
b) *f is proper.*
c) *f is flat and locally of finite presentation* [34].
d) *X is discrete (e.g. the spectrum of a field).*

Then f is an effective descent morphism *for the fibered category $\mathscr{F}$ over* (Sch) *of étale sheaves on variable schemes, i.e. the functor f^* induces an* equivalence *between the cate-*
411 *gory of sheaves on X and the category of sheaves on X' endowed with descent data relative*
to $f : X' \to X$.

[34] It is likely that this second hypothesis is in fact superfluous.

9.4.1. Let F' be a sheaf on X' endowed with descent data relative to $f : X' \to X$. Define a functor

$$G : X^0_{\text{ét}} \longrightarrow (\text{Ens})$$

by the formula

$$G(Y) = \operatorname{Ker}(F'(Y') \rightrightarrows F''(Y'')).$$

One sees immediately that G is a *sheaf* for the étale topology. On the other hand, there is a canonical injective homomorphism $G \to f_*(F')$, hence a homomorphism $f^*(G) \to F'$, and the latter is evidently compatible with the descent data. It remains to determine whether this homomorphism is an *isomorphism.* Note that G can also be defined by the exactness of

$$G \to f_*(F') \rightrightarrows g_*(F''),$$

so for every base change $Y \xrightarrow{h} X$ *which commutes with the formation of* $f_*(F')$ and $g_*(F'')$, $h^*(G) = G_Y$ is identified with the sheaf "descended" from $F'_{Y'}$ by $Y' \to Y$, and of course the homomorphism $f_Y{}^*(G_Y) \to (F_Y)'$ is the one deduced from $f^*(G) \to F'$ by base change. Now *suppose that the base change* $f : Y = X' \to X$ *commutes with the formation of* $f_*(F')$ and $g_*(F'')$, and note that since $Y' = X' \times_X X'$ has a section over $Y' = X'$, hence $Y' \to Y$ is an *effective* descent morphism for every fibered category, in particular for $\mathscr{F}$, it follows that $f^*_Y(G_Y) \to (F_Y)'$ is an isomorphism. Thus $f^*(G) \to F'$ becomes an isomorphism after the base change $Y' = X' \times_X X' \to X'$. Since the latter has a section, $f^*(G) \to F'$ is an isomorphism, and hence the descent datum under 412
consideration on F' is effective.

9.4.2. This proves Theorem 9.4 in case a), by 5.6. Case b) also follows from the fact that if f is proper, then f_* commutes with *every* base change $Y' \to Y$ (which will be proved in a later exposé (XII 5.1 (i))).

9.4.3. In case c), the question being local on X, we may suppose that X is affine and that there exists an affine scheme X'_1, of finite presentation, quasi-finite and faithfully flat over X, together with an X-morphism $X'_1 \to X'$, which reduces us to the case where f is quasi-finite. After localizing further on X, this time in the sense of the étale topology, one sees that there will exist an open subset X'_1 of X' such that $X'_1 \to X$ is finite and surjective, which reduces us to the case where f is finite and surjective, already treated in a).

9.4.4. In case d), we may suppose (by localizing on X) that X consists of a single point, and even, in view of 1.1, that it is the spectrum of a field k. Moreover, f is universally open (EGA IV 2.4.9), so the hypotheses of 9.1 hold, and the argument of 9.4.1 applies again, making a base change with $Y = \operatorname{Spec} k$, where $k = k(x)$, for an $x \in X$. But then it is still true that f_* commutes with the base change $Y \to X$, as we shall see later (XVI 1.4 and 1.5).

This completes the proof of 9.4.

EXPOSÉ IX

Constructible sheaves; cohomology of an algebraic curve

M. Artin

0. Introduction

1

Note that, beginning with the present exposé, in contrast to the preceding exposés and to the classical cohomological theory of topological spaces, we are obliged, for the validity of the essential statements, to restrict ourselves to *torsion* sheaves (and often, more precisely, to torsion sheaves prime to the residue characteristics).

Let us point out that in fact there is no "reasonable" theory of cohomology with integral (or even real) coefficients for varieties over an algebraically closed field of characteristic $p \neq 0$, analogous to the classical theory for the case $k = \mathbf{C}$ (as Serre observed). More precisely, there is no contravariant functor H^1, defined, say, on the category of smooth projective schemes over an algebraically closed field K of characteristic $p > 0$, with values in the category of finite-dimensional vector spaces over $\mathbf{R}$ (or a subfield of $\mathbf{R}$), "commuting with products," i.e. such that $H^1(X \times Y) \xleftarrow{\sim} H^1(X) \times H^1(Y)$, and such that $\dim H^1(X) = 2$ if X is a connected curve of genus 1. Indeed, it would follow that if X is an abelian variety over k, then the map $u \mapsto u^1$ from $\operatorname{End}(X)$ to $\operatorname{End}(H^1(X))$ is *additive*, hence a *representation* of the opposite ring of $\operatorname{End}(X)$. But in characteristic p there exist elliptic curves X whose endomorphism ring is a maximal order A in a 2
definite quaternion algebra over $\mathbf{Z}$ ([3], p.198), and such an algebra plainly has no representation in a vector space of dimension 2 over $\mathbf{R}$, since $A \otimes_{\mathbf{Z}} \mathbf{R}$ is a division algebra (the quaternion algebra).

1. The formalism of torsion sheaves

Let T be a topos and F an abelian sheaf on T. One may multiply by n in F ($n \in \mathbf{Z}$ an integer), this multiplication being induced by the analogous operation in (Ab). Denote by ${}_nF$ the kernel of this multiplication; thus ${}_nF(X)$, for $X \in \operatorname{Ob} T$, is the group of sections of $F(X)$ whose order divides n.

We denote by $\mathbf{P}$ the set of all prime numbers.

Definition 1.1. *Let p be a set of prime numbers. We say that F is a p-torsion sheaf, or a p-sheaf, if the canonical morphism*

$$\varinjlim_n {}_nF \longrightarrow F \text{ is bijective,}$$

where n ranges over the set of integers such that $\operatorname{ass} n \subset p$*, that is, such that the prime numbers dividing n belong to p. If $p = \mathbf{P}$ is the set of all prime numbers, we simply say that F is a torsion sheaf.*

Proposition 1.2. (i) *F is a p-torsion sheaf if and only if F is the sheaf associated with a presheaf taking values in p-torsion abelian groups; one may take*

$$P = \frac{\lim(\text{presheaf})}{\text{ass}\, n \subset p}\, {}_nF.$$

3 (ii) *If F is a p-torsion sheaf and if $X \in \operatorname{Ob} T$ is a quasi-compact object*[34]*, then $F(X)$ is a p-torsion group.*

(iii) *If T is locally of finite type* (VI 1.1)*, then F is a p-torsion sheaf if and only if $F(X)$ is a p-torsion group for every quasi-compact $X \in \operatorname{Ob} T$. In this case, one has*

$$\varinjlim_{\text{ass}\, n\subset p} H^q(T/X; {}_nF) \xrightarrow{\sim} H^q(T/X; F)$$

for every quasi-compact X and every q.

(iv) *If $u : T \to T'$ is a morphism of topoi and if F' is a p-torsion sheaf on T', then the inverse image u^*F is a p-torsion sheaf on T.*

(v) *If $u : T \to T'$ is a morphism of topoi, with T and T' locally of finite type and u quasi-compact*, and if F is a p-torsion sheaf on T, then the $R^q u_* F'$ are p-torsion sheaves on T for every q.*

Proof:

(i) Clearly, if a sheaf F is p-torsion, then F is the sheaf associated with the presheaf P whose group of sections $P(X)$ is the subgroup of p-torsion elements of $F(X)$. Conversely, let $F = \underline{a}P$ (II) be the sheaf associated with P, where P takes values in the category of abelian p-torsion groups. From the exact sequence

$$0 \to {}_nP \to P \xrightarrow{n} P$$

one deduces an exact sequence

$$0 \to \underline{a}({}_nP) \to F \xrightarrow{n} F,$$

4 whence $\underline{a}({}_nP) = {}_nF$. Since the functor $\underline{a}$ commutes with inductive limits, it follows from

$$\frac{\lim(\text{presheaf})}{\text{ass}\, n \subset p}\, {}_nP \xrightarrow{\sim} P$$

that one also has

$$\varinjlim_{\text{ass}\, n\subset p} {}_nF \xrightarrow{\sim} F.$$

(ii) Put $P = \varinjlim(\text{presheaf})\, {}_nF$. Then one has $P \subset F$, so P is a separated presheaf, and consequently

$$\underline{a}P(X) = \varinjlim(\ker(\prod_i P(X_i) \rightrightarrows)),$$

where, as usual, the limit is taken over the covering families $\{X_i \to X\}$. But since X is quasi-compact, it is enough to take finite covering families. For these families, $\prod_i P(X_i)$ is p-torsion; therefore so is the ker. It follows that $\underline{a}P(X) = F(X)$ is p-torsion.

[34] An object X of a topos is called quasi-compact if every covering family $\{X_i \to X\}$ can be refined by a finite covering family. A morphism u of topoi is quasi-compact if the inverse image of every quasi-compact object is again quasi-compact.

(iii) To verify that $\varinjlim_n F \to F$ is bijective, it is enough to do so on the values at $X \in \operatorname{Ob} T$ for every X in a family of generators. We are therefore reduced to (ii) for the first assertion. The second assertion is a consequence of

(iv) A consequence of (i), since $u^*(\underline{a}P') = \underline{a}(u \cdot P)$

(v) An easy consequence of (iii).

We shall also need a nonabelian variant. But there are technical problems in the definition (I do not know whether they are serious), and we shall therefore content ourselves with giving a definition for the étale topology of a scheme.

Definition 1.3. *Let p be a set of prime numbers. A group G is called an* ind-p-group 5
if every finite subset of G generates a finite subgroup of order n, with $\operatorname{ass} n \subset p$. This is equivalent to saying that the finite subgroups G_α of order n_α, with $\operatorname{ass} n_\alpha \subset p$, form a filtered set, and that $G = \varinjlim G_\alpha$. If $p = \mathbf{P}$, one says ind-finite group *instead of ind-p-group.*

One immediately verifies the following

Proposition 1.4. (i) *A subgroup as well as a quotient of an ind p-group is again an ind p-group.*

(ii) *A filtered inductive limit of ind-p-groups is an ind-p-group.*

(iii) *A finite projective limit of ind-p-groups is an ind-p-group.*

Definition 1.5. *Let X be a scheme. A sheaf F of groups on X is called a* sheaf of ind p-groups *(resp. of* ind-finite groups*) if, for every étale $U \to X$, with U quasi-compact, $F(U)$ is an ind p-group (resp. an ind-finite group).*

Proposition 1.6. *Let F be a sheaf of groups on the scheme X.*

(i) *F is a sheaf of ind-p-groups if and only if, for every geometric point ξ of X, the fiber F_ξ is an ind-p-group.*

(ii) *If $f : X \to X'$ is a morphism and F' is a sheaf of ind-p-groups on X', then f^*F' is a sheaf of ind-p-groups.*

(iii) *If $f : X \to X'$ is a quasi-compact morphism and F is a sheaf of ind-p-groups on X, then f_*F is a sheaf of ind-p-groups.*

Proof. (i) Suppose that F is a sheaf of ind-p-groups, and let ξ be a geometric point of X. Then $F = \varinjlim F(X')$, where X' ranges over a pseudo-filtered system of schemes étale 6
over X (VIII 3.9). It is evident that the affine X' (hence quasi-compact) form a cofinal set, and consequently that F is an inductive limit of ind p-groups, and hence is an ind-p-group by 1.4 (ii). Conversely, suppose that for every ξ the fiber F_ξ is an ind-p-group, and let $U \to X$ be étale, with U quasi-compact. Let $S \subset F(U)$ be a finite subset. For every geometric point ξ of U, the image of S in F_ξ generates a finite group. Since a finite group is finitely presented, one easily concludes that there exists a U_ξ étale over U, pointed at ξ, such that S generates a finite group in $F(U_\xi)$ (cf. VIII 4). A finite number of such U_ξ form a covering of U, say $\{U_1, \dots, U_n\}$. Then the image of S in $\prod F(U_i)$ generates a finite subgroup, and since $F(U) \subset \prod F(U_i)$, the result follows.

Assertion (ii) is trivial from (i), and (iii) is immediate.

2. Constructible sheaves

2.0. Let T be a topos. Recall that with every element $S \in \operatorname{Ob}(\mathrm{Ens})$ one associates the sheaf S_T associated with the presheaf P such that $P(X) = S$ for every $X \in \operatorname{Ob}(T)$. The object of T representing S_T is

$$\coprod_{s \in S} e = \text{"} S \times e \text{"},$$

where e is the final object of T. The sheaf S_T is called the *constant sheaf* with value S (IV). In the evident way one defines the notion of a *constant sheaf of groups, constant sheaf of A-modules* (A a ring), and *constant abelian sheaf*, with value M. A morphism $f_T : S_T \to S'_T$ of constant sheaves is said to be constant if it comes from a morphism of sets $f : S \to S'$.

A sheaf F is said to be *locally constant* if there exists a covering $\{e_i \to e\}$ of the final
7 object such that F becomes constant on each e_i. One defines analogously the notion of a locally constant *sheaf of groups* or sheaf of *A-modules*, and of a *locally constant morphism* $f : F \to G$, when F and G are locally constant sheaves. Finally, a locally constant sheaf of groups or of A-modules is said to be *of finite type* (resp. *of finite presentation*) if its local values are groups or A-modules of finite type (resp. of finite presentation).

LEMMA 2.1. (i) *Let $f : F \to G$ be a morphism of locally constant sheaves of sets on T, and suppose that the local values of F are finite sets. Then f is locally constant.*

(ii) *Let $f : F \to G$ be a morphism of locally constant sheaves of A-modules, with F of finite type. Then f is locally constant, and the kernel and cokernel of f are locally constant.*

(iii) *Let $0 \to F' \to F \to F'' \to 0$ be an exact sequence of sheaves of A-modules (resp. of groups), with F' and F'' locally constant, and F'' of finite presentation. Then F is locally constant.*

Proof of (i) and (ii). We reduce without difficulty to the case in which F and G are constant. Consider (ii): let $F = M_T$, $G = N_T$ (M, N being A-modules), and let $S \subset M$ be a finite set of generators of M. Then f is determined by its restriction to S_T, and f will be locally constant if $f|S_T$ is. Thus we are reduced to (i) for the first assertion of (ii), and the second assertion is a trivial consequence of the first.

It therefore remains to prove (i). Suppose that $F = S_T$, $G = S'{}_T$ are constant, and also denote by f the morphism of objects $f : S \times e \to S' \times e$ (notation as above). Let $f_s : e \to S' \times e$ ($s' \in S$) be the components of f: since S is finite, it plainly suffices to treat the case $f = f_s$, that is, the case in which S is a one-element set. Let $X_{s'} \to e$ ($s' \in S'$) be the family of morphisms that make the diagrams

$$\begin{tikzcd} X_{s'} \arrow[r] \arrow[d] & e \arrow[d, "i_{s'}"] \\ e \arrow[r, "f"] & S' \times e \end{tikzcd}$$

8 cartesian, where $i_{s'} : e \to S' \times e$ is the "inclusion" into the s'-th component. The family $\{i_{s'}\}$ is trivially covering, and hence so is $\{X_{s'} \to e\}$. One checks immediately (since coproducts in a topos are disjoint) that f becomes constant on $X_{s'}$, and consequently f is locally constant.

Proof of (iii). Consider the case of sheaves of A-modules. We may suppose F' and F'' constant, with F'' having value M'', an A-module of finite presentation. If M'' is free, and hence has a finite basis $(e_i)_{1 \le i \le n}$, then the e_i define sections of F'' and hence lift locally to sections of F; this shows that the extension F of F'' by F' splits locally, which proves that F is locally isomorphic to $F' \times F''$, and is therefore locally constant. In the general case, choose a surjective homomorphism $\varphi : L'' \to M''$, with L'' free of finite type; the kernel R of φ is of finite type, and setting $L = L''_T \times_{M''_T} F = L''_T \times_{F''} F$, one sees on the one hand that L is an extension of L''_T by G, and is therefore locally constant by what precedes, and on the other hand that there is an exact sequence $0 \to R_T \to L \to F \to 0$, which, by (ii), implies that F is locally constant.

From now on we restrict ourselves to the étale topoi of schemes.

Lemma 2.2. *Let X be a scheme and let F be a locally constant sheaf on X. Then F is represented by an étale Y/X. If, moreover, the fibers of F are finite (resp. and nonempty), then Y is an étale covering (resp. and is surjective).*

Proof. Descent (SGA 1 IX 4.1 if F has finite fibers, SGA 3 X 5.4 in the general case).

Definition 2.3. *A sheaf of sets (resp. of groups, resp. of A-modules) is said to be constructible if, for every affine open $U \subset X$, there exists a decomposition of U as a union of finitely many constructible* (EGA 0_{III} 9.1.2) *reduced locally closed subschemes U_i such that the sheaf induced by F on each U_i is locally constant, with finite value (resp. finite, resp. of finite presentation[34]).* 9

2.3.1. The finiteness hypothesis on the local values amounts to saying that for every geometric point ξ, the fiber F_ξ is finite (resp. finite, resp. of finite presentation). It follows immediately from 2.1 (i) that a sheaf of groups F is constructible if and only if the underlying sheaf of sets of F is constructible.

Care must be taken that if F is an abelian sheaf, it is constructible as a sheaf of groups (or equivalently, as a sheaf of sets) if and only if it is constructible as a sheaf of $\mathbf{Z}$-modules, and if *in addition* its fibers are finite. Thus, $\mathbf{Z}_X$ is constructible as a $\mathbf{Z}$-module, but not as a sheaf of groups.

Proposition 2.4. (i) *Let X be quasi-compact and quasi-separated. Then a sheaf (resp. sheaf of groups, resp. sheaf of A-modules) is constructible if and only if there exists a decomposition of X as a union of constructible locally closed subsets such that F becomes locally constant and finite (resp. finite, resp. of finite presentation) on each X_i.*

(ii) *To verify that F is constructible, it is enough to verify that $F|U_i$ is constructible for the U_i of a given open covering of X.*

(iii) *Let $F : Y \to X$ be a morphism. Then f^*F is constructible if F is.*

(iv) *The set of points at which the fiber of a constructible sheaf is nonempty (resp. nonzero) is a locally constructible subset* (EGA 0_{III} 9.1.2). 10

(v) *Let X be locally noetherian. Then a sheaf of sets (resp. ...) F on X is constructible if and only if for every $x \in X$ there exists a nonempty open subset of the closure $\overline{x}$ of x such that F induces a locally constant sheaf with finite value (resp. ...) on U.*

Proof. Plainly, if $f : Y \to X$ is a morphism and if there exists a decomposition of X as a union of constructible locally closed subsets X_i such that F becomes locally constant on each X_i, the same is true of Y and f^*F (cf. EGA IV 1.8.2). This proves the implication $\Leftarrow$ of (i). Now suppose that X is quasi-compact and quasi-separated, and let $\{U_i, \dots, U_n\}$ be an affine open covering of X such that $F|U_i$ is constructible for every i. To find a decomposition of X as in (i), it is enough to do so for U_n and for $Y = X - U_n$ (with the induced reduced structure), since U_n and Y are constructible in X by virtue

[34] When A is not assumed noetherian, for the purposes of Homological Algebra it is preferable to require here, instead of finite presentation alone for the A-module M, that it have a left resolution by free A-modules of finite type. The notion introduced in (2.3) should then be called: F is 1-constructible (more generally, one would define n-constructibility in the evident way, for every integer $n \geq 0$). For the present Exposé we shall provisionally retain the terminology in the form (2.3), which is chiefly reasonable when A is noetherian, in which case it coincides with that just indicated. Compare SGA 6 I for finiteness notions in nonnoetherian cases.

of the hypothesis that "X is quasi-separated". Now a decomposition exists for U_n by the definition of constructibility, and Y is the union of $n-1$ affine opens $V_i = Y \cap U_i$ $(i = 1, \dots, n-1)$ with $F|V_i$ constructible, whence result (i) follows by induction on n. The same reasoning proves (ii); indeed, the hypothesis implies that $F|U$ is constructible for every "sufficiently small" affine open U, and an arbitrary affine open V can be covered by finitely many such opens. Assertion (ii) shows that the notion of constructibility is local on X, and (iii), (iv) follow immediately. The implication $\Rightarrow$ of (v) is immediate and does not depend on the noetherian hypothesis. If X is noetherian, one proves the implication $\Leftarrow$ by the usual "noetherian induction".

Proposition 2.5. *Let X be quasi-compact and quasi-separated, and let F be a constructible sheaf of groups or of A-modules on X. Then there exists a finite filtration of F whose successive quotients are of the form $i_!G$, where $i : U \to X$ is the inclusion of a constructible locally closed subset and where G is locally constant and constructible on U. If X is noetherian, there exists such a filtration with the U irreducible.*

11 Proof. By 2.4 (i), there exists a decomposition of X as a finite union of constructible locally closed subsets X_i such that each $F|X_i$ is locally constant and finite (resp. of finite presentation). Write $X_i = U_i \cap \complement V_i$, where U_i and V_i are constructible opens, and let N be the number of opens of X in the subtopology T generated by $\{U_i, V_i\}$. We argue by induction on N: if W is a minimal nonempty element of T, it is plain that $F|W$ is locally constant and of finite presentation. Let $i : W \to X$ be the inclusion morphism and let $Y = X - W$. We have the exact sequence

$$0 \longrightarrow i_! i^* F \longrightarrow F \longrightarrow F|Y \longrightarrow 0,$$

and are thereby reduced to the same assertion for Y and the sheaf $F|Y$, where the induction hypothesis applies.

Proposition 2.6. *Suppose that A is a noetherian ring.*

(i) *A finite projective (resp. inductive) limit of constructible sheaves of A-modules or sets is constructible. In particular, the kernel, cokernel, and image of a morphism of constructible sheaves of A-modules are constructible.*

(i bis) *A finite projective limit of constructible sheaves of sets (resp. of groups) is constructible, and if $f : F \to G$ is a morphism of constructible sheaves of groups, the kernel, image, and cokernel (if $\mathrm{Im}(f)$ is a normal subgroup) are constructible. A finite inductive limit of constructible sheaves of sets is constructible.*

(ii) *Let $0 \to M' \to M \to M'' \to 0$ be an exact sequence of sheaves of A-modules (resp. of groups), with M', M'' constructible. Then M is constructible.*

(iii) *Let $M_1 \to M_2 \to M_3 \to M_4 \to M_5$ be an exact sequence of sheaves of A-modules. Then if M_i is constructible for $i = 1, 2, 4, 5$, the same holds for $i = 3$.*

12 *Proof.* (i) We leave the assertions for sets and groups to the reader. For sheaves of A-modules, it is enough to prove that the kernel and cokernel of a morphism $f : F \to G$ of constructible sheaves of A-modules are constructible. The assertion is local on X, and we may therefore suppose that X is affine, hence quasi-compact and quasi-separated. By 2.4. (i), we then reduce to the case in which F and G are locally constant, and conclude by using 2.1 (ii) and the noetherian hypothesis on A. Assertion (ii) is proved analogously using 2.1 (iii). Assertion (iii) follows immediately from (i) and (ii).

Proposition 2.7. *Let X be a quasi-compact and quasi-separated scheme, and let F be a sheaf of sets (resp. of A-modules). In order that F be constructible, it is necessary and sufficient that it be isomorphic to the cokernel of a pair of morphisms $H \rightrightarrows G$, where H*

and G are sheaves of sets representable by schemes étale and of finite presentation over X (resp. that F be isomorphic to the cokernel of a homomorphism $\mathscr{A}_{V,X} \to \mathscr{A}_{U,X}$, where U and V are two schemes étale and of finite presentation over X).

Sufficiency follows easily from 2.6 (i bis).

Suppose that F is a constructible sheaf of sets. Using the existence of infinite coproducts in the étale site over X, one sees that one can find an epimorphism $G \to F$, with G representable by a scheme étale over X, which may moreover be supposed to be a coproduct of affine schemes G_i ($i \in I$), and hence separated over S. For every finite subset J of the index set I, let G_J be the coproduct of the G_i for $i \in J$, and let F_J be its image in F. Since G_J and F are constructible, so is F_J (2.6 (i bis)); hence the set X_J of $x \in X$ such that the fibers of F and F_J at a geometric point of X over x are equal is locally constructible (2.4 (iv)). Since the family of X_J is increasing and has union X, and since X is quasi-compact, one of the X_J equals X (EGA IV 1.9.9). This shows, after replacing G by G_J, that we may suppose G affine, and hence separated over X, and quasi-compact over X since X is quasi-separated (EGA IV 1.2.4). Let then $H = G \times_F G$. It is a subsheaf of $G \times_S G$, which is representable, and is therefore itself representable (VIII 6.1). Since G is moreover constructible (2.6 (i bis)), it follows again from the preceding argument that 13
it is quasi-compact, and therefore (being separated over X) of finite presentation over X. This proves the first assertion of 2.7, and the second is proved by the same method (NB it is not necessary that A be noetherian). Let us note in passing that the proof also proves the following corollary:

Corollary 2.7.1. *Let X be a scheme and let F be a scheme étale over X. In order that the corresponding étale sheaf on X be constructible, it is necessary and sufficient that F be of finite presentation over X.*

Proposition 2.7 will be useful here chiefly for deducing certain limit-passage properties for constructible sheaves:

Corollary 2.7.2. *Let X be a quasi-compact and quasi-separated scheme. Then every sheaf of sets (resp. of ind-p-groups, resp. of A-modules) on X is the inductive limit of a filtered inductive system of constructible sheaves of sets (resp. ...).*

If F is a sheaf of sets, retain the notation of the proof of 2.7, and let L be the set of pairs (J, H'), where J is a finite subset of I and H' is a quasi-compact open subset of $H_J = G_J \times_F G_J$. Order L in the evident way, and observe that, since F is the limit of the F_J, and each F_J is the limit of the $\operatorname{Coker}(H' \to G_J)$ for the quasi-compact opens H' of H_J, F is the inductive limit of the inductive system of the $F_\ell = \operatorname{Coker}(H' \to F_J)$ for $\ell = (H', J) \in L$, proving the case of sets. The case of sheaves of A-modules is treated in essentially the same way, beginning by representing F as the cokernel of a homomorphism $\mathscr{A}_{V,X} \to \mathscr{A}_{U,X}$, with U, V schemes étale and separated over X. The case of ind-$\wp$-groups requires a little more attention. Retaining the preceding notation, denote by $L(G_J)$ the free group sheaf generated by the sheaf of sets G_J, by N_J the kernel of the homomorphism $L(G_J) \to F$ deduced from $G_J \to F$, and by L the set of pairs (J, N'), where N' is a subsheaf of *sets* of N_J which is "quasi-compact", i.e. such that there exists an epimorphism $P \to N'$, with P representable by a quasi-compact X-scheme. Order L by declaring that (J, N') is smaller than (J_1, N_1') if $J \subset J_1$ and if the canonical homomorphism $L(G_J) \to L(G_{J_1})$ maps N_1 into N_1'. It is immediate that L is 14
filtered, and that one obtains an inductive system of sheaves of groups F_ℓ, indexed by L, by taking, for $\ell = (J, N')$, $F_\ell =$ the quotient sheaf of groups of $L(G_J)$ by the invariant subsheaf of groups generated by N'. It is moreover immediate that F is isomorphic to

the inductive limit of the F_ℓ, and it remains to prove that there exists a subset L' of L, cofinal in L, such that for $\ell \in L'$, F_ℓ is a constructible sheaf of $\wp$-groups. For this, it is enough to prove that for every J one can find a subsheaf of sets N' of N_J which is quasi-compact, contains a given quasi-compact subsheaf of sets N'_0, and is such that the corresponding F_ℓ is a constructible sheaf of $\wp$-groups. Note first that it is enough for this that the fibers of F_ℓ be finite $\wp$-groups: then F_ℓ will automatically be constructible, as the reader will readily verify. On the other hand, the reader may also verify that the set $X_{N'}$ of points x of X such that the fiber of F_ℓ at a geometric point over x is a finite $\wp$-group is locally constructible. Moreover, the subsets $X_{N'}$ of X are increasing functions of N', and their union is all of X: this last point follows from the fact that the inductive limit of the F_ℓ, for $\ell = (N', J)$ with J fixed, is the subsheaf of groups F_J of F generated by the subsheaf of sets F_J, which is quasi-compact, and hence $\overline{F_J}$ has fibers that are finite $\wp$-groups. On the other hand, in a free group on a finite number of generators, if it is expressed as a filtered union of its finite subsets, one of these generates the whole subgroup. It follows readily that the union of the

$X_{N'}$ is indeed X. Using (EGA IV 1.9.9) once again, one finds that one of the $X_{N'}$ is equal to X, which completes the proof of 2.7.2.

. Note that the somewhat laborious proof in the case of group schemes brings to light the lack of a convenient sorites for constructibility conditions on sheaves of groups with fibers not necessarily finite; we apologize to the reader, inviting them to fill this gap by their own means if necessary, and point out by way of consolation that in 2.9 (iii) we obtain a simpler proof when X is Noetherian.

15 Corollary 2.7.3. *Let X be a quasi-compact and quasi-separated scheme, and let F be a constructible sheaf of sets (resp. of finite ind-groups, resp. of A-modules). Then the functor* $\mathrm{Hom}(F, G)$, *where G is a variable sheaf of sets (resp. ...), commutes with inductive limits in G. When F is a sheaf of A-modules, with A Noetherian, the functors* $\mathrm{Ext}^i(X; F, G)$ *in the A-module G also commute with inductive limits.*

This follows from 2.7 and (VII 3.3) by the usual arguments, which are left to the reader. Similarly, the conjunction of 2.7 and the results of (VII 5) easily gives:

Corollary 2.7.4. *Let I be a filtered category, and let $i \mapsto X_i : I^0 \to$* (Sch) *be a functor which transforms arrows into affine morphisms and objects into quasi-compact and quasi-separated schemes. For every scheme Y, let $F_c(Y)$ be the category of constructible sheaves of sets (resp. of ind-$\wp$-groups, resp. of A-modules) on Y. Then there is an equivalence of categories*

$$F_c(X) \xleftarrow{\approx} \varinjlim_i F_c(X_i),$$

where $X = \varprojlim_i X_i$ (VII 5) *and where the second member denotes the* $\varinjlim$ *category of fibered categories, defined in In particular, every constructible sheaf of sets (resp. ...) on X is isomorphic to the inverse image of a sheaf of the same kind on one of the X_i.*

Proposition 2.8. *Let $f : X \to Y$ be a surjective morphism locally of finite presentation, and let F be a sheaf of sets (resp. ...) on Y. Then F is constructible if and only if $f^*(F)$ is constructible.*

We shall use the following

Lemma 2.8.1. *Let $f : X \to Y$ be a surjective morphism locally of finite presentation, with Y quasi-compact and quasi-separated. Then there exists a finite partition of Y into*

16 *subschemes Y_i of finite presentation over Y (in particular, each Y_i is constructible in X), and, for every i, finite surjective morphisms $Y_i'' \xrightarrow{g_i} Y_i' \xrightarrow{h_i} Y_i$, with h_i étale and g_i locally free (i.e. flat and of finite presentation, in addition to the condition that g_i be finite) and radicial, and finally a Y-morphism $Y_i'' \to X$* [34].

Using the fact that Y is quasi-compact and quasi-separated, we are easily reduced to the case where Y is quasi-affine, hence separated (by using a finite affine open covering $(U_i)_{1\leq i\leq n}$ of Y and considering the $Y_i = U_i - U_i \cap \bigcup_{j<i} U_j$), and then to the case where Y is affine (by the same argument). The scheme X is a union of affine open subsets X_i, and the image T_i of X_i in Y is constructible (EGA IV 1.8.4); the union of the T_i is X since f is surjective, whence it follows that X is the union of a finite number of the T_i (EGA IV 1.9.9), so that, replacing X by the sum scheme of the corresponding X_i, we may suppose that X is affine. The standard limit procedure (EGA IV 8 8.1.2 c)) then allows us to suppose in addition that Y is Noetherian. The usual Noetherian induction, together with the limit procedure (EGA IV 8.1.2 a)), reduces us to the case where Y is the spectrum of a field k. Since then $X \neq \varnothing$, there exists a closed point x in X, corresponding to a finite extension k' of k, itself a radicial extension of a separable extension k_s' of k. We then take $Y' = \mathrm{Spec}(k_s')$, $Y'' = \mathrm{Spec}(k')$, which completes the proof of 2.8.1.

Necessity in 2.8 being trivial (2.4 (iii)), let us prove sufficiency. By the preceding lemma we are reduced to the case where f is of the form gh, where g and h satisfy the conditions stated for g_i, h_i in the lemma. We may therefore suppose successively either that f is finite radicial, a trivial case by (VIII 1.1), or that f is finite étale. Note moreover that, using the hypothesis that $f^*(F)$ is constructible and the definition of constructibility, we could suppose (after replacing X by an X-prescheme $X' = \coprod_i X_i$ of finite presentation and with surjective structural morphism) that $f^*(F)$ is already locally constant. But when f is étale and surjective, this implies that F is locally constant. Since in addition the hypotheses made on the fibers of $f^*(F)$ remain valid for those of F, with f surjective, it indeed follows that F is constructible, which completes the proof of 2.8.

Proposition 2.9. *Let X be a Noetherian scheme, let A be a Noetherian ring, and let p be a set of prime numbers.* 17

(i) *The category of sheaves of sets (resp. of ind p-groups, resp. of A-modules) on X is locally Noetherian, that is, it has a set of generators formed of Noetherian objects.*
(ii) *A sheaf F of sets (resp. of ind p-groups, resp. of A-modules) on X is constructible if and only if it is Noetherian. If F is a sheaf of A-modules, then F is constructible if and only if it is a quotient of a finite sum of sheaves of the form $A_{U/X}$, where $U \to X$ is étale and of finite type (notation of IV.2.4: $A_{U/X} = A_U$).*
(iii) *The constructible subsheaves of a sheaf F of sets (resp. ...) form an inductive system, and F is the inductive limit of these subsheaves.*

Proof. We leave the assertion for sets to the reader. First consider the case of a sheaf of A-modules. For (i), one must find a set of generators whose elements are Noetherian objects. But the sheaves of the form $A_{U/X}$, with $U \to X$ étale and of finite type, form a set of generators which are easily seen to be constructible, and it therefore suffices to prove the following lemma:

Lemma 2.10. *Let X be Noetherian. Then every constructible A-module is Noetherian.*

Proof of the lemma.

[34] This assertion is also given in EGA IV 17.16.4.

By Noetherian induction, we suppose the lemma true for every subscheme X' of X distinct from X. Choose a nonempty open subset X_0 of X such that F induces a locally constant sheaf F_0 on X_0.

Now let $\{F_i\}$, $i \in \mathrm{N}$, be an increasing sequence of subsheaves of F_0. Since, for a geometric point P which is generic for an irreducible component of X_0, the fiber of F_0
18 at P is of finite type, the sequence of fibers $(F_i)_P$ is stationary, and we may suppose that it is constant. Now let Q be a geometric point of X_0 which is a *specialization* of P (VIII 7.2). Since F_0 is locally constant on X_0, one sees immediately that the specialization morphism $F_{0_Q} \to F_{0_P}$ (VIII.7) is bijective, hence $(F_i)_Q \to (F_i)_P$ is *injective* for every i.

Let V/X be étale and pointed at P, and let $s_1, \dots, s_n$ be sections $\in F_1(V)$ which generate $(F_1)_P$. Then the s_i generate $(F_i)_Q$ for every i and every geometric point Q which is a specialization of P above V. The sequence of sheaves F_i is thus constant in a neighborhood of the image of P, say in $X' \neq \varnothing$. Let $Y = X - X'$; it therefore suffices to prove that the sequence of the $F_i|Y$ is stationary, which follows from the Noetherian induction hypothesis.

The assertions (ii) for A-modules are now immediate: since the $A_{U/X}$ are Noetherian generators, one plainly has F Noetherian $\Leftrightarrow$ F is a quotient of a finite sum of sheaves $A_{U/X}$. But $A_{U/X}$ is constructible, and consequently, in view of 2.6, F Noetherian $\Rightarrow$ F constructible; the converse implication having already been proved (2.10), this establishes (ii). Finally, assertion (iii) for A-modules is a property of locally Noetherian categories (cf. [1], Ch. IV).

Now consider the case of ind-p-groups. First note that a constructible sheaf of groups F is certainly Noetherian. Indeed, since the fibers are finite, we may use the same argument as for A-modules. We shall now prove assertion (iii) directly, which will imply (i)—indeed, it will show that constructible sheaves form a set of generators. Moreover, any Noetherian sheaf will then necessarily be constructible, whence (ii).

To prove (iii), it plainly suffices to prove that a subsheaf (of groups) S of a sheaf F of ind p-groups generated by a finite number of sections $s_i \in F(U_i)$, with $U_i \to X$ étale
19 and of finite type, is constructible [34]. (Since a constructible sheaf is Noetherian, it is generated by a finite number of sections.) By Noetherian induction, we may suppose that this is true for every closed subscheme Y of X distinct from X. Now the notion of a subsheaf generated by sections commutes with the inverse image of sheaves; indeed, let $L_{U/X}$ be the sheaf which represents the functor $F \mapsto F(U)$ in the category of groups. Then if $f : Y \to X$, one has $f^*(L_{U/X}) = L_{L \times_X U/Y}$. Since S is the image of the sum of the $L_{U_i/X}$, and since f^* commutes with sums, it is clear that f^*S is the subsheaf of f^*F generated by the sections $f^*(S_i)$.

We may therefore suppose that S induces a constructible sheaf on every closed subset Y distinct from X, and it thus suffices (2.4 (i)) to prove that S induces a constructible sheaf on a suitable nonempty open subset of X. Replacing X by a sufficiently small open subset, we may suppose that every U_i is an étale covering of X. It suffices to prove that S is then locally constant, with finite fibers, on a suitable nonempty open subset. This assertion is local on X for the étale topology, so we may suppose that every U_i is completely split, say $U_i = \amalg_{n_i} X$ (the sum of n_i copies of X). Let s_{ij} ($j = 1, \dots, n_i$) be the sections of $F(X)$ such that s_i $(s_{il}, \dots, s_{in_i})$. Plainly, S is none other than the subsheaf of F generated by the $s_{ij} \in F(X)$. Since $F(X)$ is an ind p-group (1.5), the s_{ij}

[34] Recall the definition of this sheaf: let $L_{U/X}$ be the sheaf representing the functor $F \mapsto F(U)$ in the category of groups. The section $s_i \in F(U_i)$ induces a morphism $L_{U_i/X} \to F$, and S is the image of the sum morphism $\amalg_i L_{U_i/X} \to F$. It is the smallest subsheaf which "contains" the sections s_i.

generate a finite subgroup 1.3. Thus we may suppose that the set of the s_i is a finite subgroup of $F(X)$; but then S is identical to the sheaf *of sets* generated by the s_i, which is constructible by 2.6 (i) as a sheaf of sets, hence as a sheaf of groups, C.Q.F.D.

In the following two propositions, we give criteria for a sheaf to be locally constant or constructible, using specialization homomorphisms (VIII 7.7).

Proposition 2.11. *Let X be a scheme, and let F be a constructible sheaf of sets (resp. of ind-finite groups, resp. of A-modules); in the case of sets, suppose in addition that F has* 20
finite fibers. Let $x \in X$, and let x be a geometric point over x. In order that F be locally constant in a neighborhood of x, it is necessary and sufficient that, for every geometric point $\overline{x}'$ which is a generization of $\overline{x}$, and every specialization arrow $\overline{x'} \to \overline{x}$ (VIII 7.2)*, the specialization homomorphism* (VIII 7.7) *$F_{\overline{x}} \to F_{\overline{x'}}$ be an isomorphism.*

The case of a sheaf of groups being a special case of that of a sheaf of sets, we confine ourselves to the latter; the case of a sheaf of modules is treated in essentially the same way. Necessity of the condition being trivial, we need only establish sufficiency. Since the fiber $I = F_{\overline{x}}$ is finite, after replacing X by a suitable scheme X' étale over X, we may suppose that one can find a homomorphism $u : I_X \to F$ from the constant sheaf I_X to F, inducing an isomorphism on the fibers at $\overline{x}$. Using the constructibility of both sheaves and 2.6, one easily sees that the set Z of points z of X such that the homomorphism induced on the fibers at a geometric point $\overline{z}$ over z is bijective is a locally constructible subset of X. The hypothesis immediately implies that it contains the generizations of x, hence (EGA IV 1.10.1) it is a neighborhood of x, which proves that F is locally constant (indeed, even constant in this case) in a neighborhood of x, C.Q.F.D.

Definition 2.12. [34]*: A scheme X is said to be path-connected if for every pair P, Q of geometric points of X, there exist geometric points $P = P_0, \dots, P_n$; $Q_1, \dots, Q_n = Q$ and specializations* (VIII 7.2) *$P_i \to Q_i$ and $P_{i-1} \to Q_i$ $(i = 1, \dots, n)$. One says that X is locally path-connected (for the étale topology) if for every étale $U \to X$ there exists an étale covering $\{U_i \to U\}$ of U such that U_i is path-connected for every i.*

One checks immediately that a locally Noetherian prescheme is locally path-connected.

Proposition 2.13. (i) *Let X be locally path-connected and let F be a sheaf of* 21
sets (resp. of groups, resp. of A-modules) on X. Suppose that the fibers of F are finite (resp. finite, resp. of finite presentation). Then F is locally constant if and only if, for every specialization $P \to Q$ of geometric points, the specialization morphism (VIII 7.7) *$F_Q \to F_P$ is bijective.*

(ii) *Suppose that X is locally Noetherian and that the fibers of F are finite (resp. finite, resp. of finite presentation). Then, if for every specialization $P \to Q$ the specialization morphism $F_Q \to F_P$ is injective, F is constructible.*

(iii) *Suppose that X is locally Noetherian and that the fibers of the sheaf of sets F are finite. Let $c : X \to \mathrm{N}$ be the function which associates with $x \in X$ the number of elements of the fiber $F_{\overline{x}}$ of F at a geometric point $\overline{x}$ above x. Then F is constructible if and only if c is a constructible function, i.e. if and only if $c^{-1}(n)$ is constructible for every $n \in \mathrm{N}$.*

Proof.

(i) Let P be a geometric point of X. We shall find an étale neighborhood of P, that is, an étale $U \to X$ pointed at P, such that F becomes constant on U. Let V' be

[34] The terminology is due to S. Lubkin.

an étale neighborhood of P such that $F(V) \to F_P$ is surjective. Such a V' exists by the finiteness hypothesis. Let $U \to V'$ be an étale neighborhood of P in V' such that U is path-connected. Then $F(U) \to F_P$ is still surjective. Choose a subset $S \subset F(U)$ which maps bijectively onto F_P. If F is a sheaf of groups (resp. of A-modules), one easily sees that, after changing U if necessary, one may find such a set which is moreover a subgroup (resp. submodule) of $F(U)$. This gives a morphism $S_U \to F|U$, where S_U is the constant sheaf with value S. One has

$$(S_U)_P \xrightarrow{\sim} F_P.$$

22 Since every specialization morphism on the fibers of F (resp. S_U) is bijective, and since U is path-connected, it follows that $S_U \xrightarrow{\sim} F|U$, whence the result.

(ii) The assertion is local, and we may therefore suppose that X is Noetherian. Let P be a geometric point above a maximal point of X. There exists an irreducible étale neighborhood, hence a path-connected one, $U \to X$ of P such that $F(U)$ maps surjectively onto F_P. It then maps surjectively onto F_Q for every geometric point Q of V, or, what amounts to the same thing, for every geometric point of the image U of V in X, since such a point is a specialization of P and $F_Q \to F_P$ is injective. Thus every specialization morphism in U is bijective, hence $F|U$ is locally constant. By Noetherian induction we may suppose that $F|X - U$ is constructible, whence the result.

(iii) It plainly suffices to treat the case where the function c is constant. Let $V \to X$ be an étale neighborhood of a geometric point P above a maximal point of X, such that $F(V)$ maps surjectively onto F_P. Let S be a subset of $F(V)$ such that $S \xrightarrow{\sim} F_P$. Then, for the constant sheaf S_V and every specialization $P \to Q$ of geometric points, one has a commutative diagram

$$\begin{tikzcd}
(S_V)_P \arrow[r, "\sim"] & F_P \\
(S_V)_Q \arrow[u, "\wr"] \arrow[r] & F_Q.\arrow[u]
\end{tikzcd}$$

Since $c(F_Q) = c(F_P)$, the arrow $F_Q \to F_P$ is bijective, so F is locally constant in a neighborhood of P by (i). By Noetherian induction, we are done.

The following result (ii) will be of technical use later; it will allow us to reduce certain verifications to the case of a constant sheaf.

23 PROPOSITION 2.14. *Let X be a scheme.*

(i) *For every finite morphism of finite presentation $f : X' \to X$, and every constructible sheaf of sets (resp. of ind-finite groups, resp. of A-modules) on X', $f_*(F)$ is constructible.*

(ii) *Suppose that X is quasi-compact and quasi-separated, and let F be a constructible sheaf of sets (resp. ...) on X. Then one can find a finite family $(p_i : X'_i \to X_i)$ of finite morphisms, for every i a constant constructible sheaf of sets (resp. ...) C_i on X'_i, and a monomorphism*

$$F \hookrightarrow \prod_i p_{i*}(C_i).$$

First prove (i). By definition, we may suppose that X is affine, and then, using 2.7.4, the standard limit procedure reduces us to the case where X is Noetherian. A further passage to the limit, again using 2.7.4 and also 2.4. (v), reduces us to the case where X is the spectrum of a field. The conclusion then reduces to the corresponding assertion concerning the fiber of $f_*(F)$ at a geometric point over X, which follows immediately from (VIII 5.5, 5.8).

Let us prove (ii). By 2.4 (i), there exists a partition of X into subschemes Y_i, with inclusion morphisms $u_i : Y_i \to X$ of finite presentation, such that the restriction F_i of F to every Y_i is a locally constant sheaf. It is then immediate that the canonical homomorphisms $F \to u_{i*}(F_i)$ define a homomorphism $F \to \prod_i u_{i*}(F_i)$ which is a monomorphism.

In the preceding construction, one could even arrange that every F_i be not only locally constant, but also *isotrivial,* i.e. that there exist a surjective *finite* étale morphism $q_i : Y_i' \to Y_i$ such that $q_i^*(F_i)$ is a constant sheaf on Y_i', say $C_{iY_i'}$. This follows easily, for example, from the definition of 2.8.1 and from (VIII 1.1). On the other hand, it follows 24
from the "Main Theorem" in the form (EGA IV 8.12.6) that there exists a finite morphism $p_i : Z_i \to X$ and an open immersion $v_i : Y_i' \to Z_i$ of X-schemes.

Since F_i embeds into $q_{i*}(q_i^*(F_i)) = q_{i*}(C_{iY_i'})$, F embeds into the product of the $u_{i*}(q_{i*}(C_{iY_i'})) = p_{i*}(v_{i*}(C_{iY_i'}))$. If every Z_i is normal, then Lemma (2.14.1) below implies that $v_{i*}(C_{iY_i'}) \simeq C_{iZ_i}$, hence F embeds into the product of the $p_{i*}(C_{iZ_i})$, and we are done. In the general case, introduce the normalization $\overline{Z}_i$ of Z_i, the inverse image $\overline{Y}_i'$ of Y_i' in $\overline{Z}_i$, and the immersion $\overline{v}_i : \overline{Y}_i' \to \overline{Z}_i$. One sees immediately that $\overline{Y}_i'$ contains the maximal points of $\overline{Z}_i$, and is therefore dense in $\overline{Z}_i$. If $r_i : \overline{Z}_i \to Z_i$ and $s_i : \overline{Y}_i' \to Y_i'$ are the projections, since the latter is surjective, it follows that $C_{iY_i'}$ embeds into $s_{i*}(C_{i\overline{Y}_i'})$, hence $v_{i*}(C_{iY_i'})$ embeds into $v_{i*}(s_{i*}(C_{i\overline{Y}_i'})) = r_{i*}(\overline{v}_{i*}(C_{i\overline{Y}_i'}))$, which is isomorphic to $r_{i*}(C_{i\overline{Z}_i})$ by (2.14.1) below. Thus F embeds into the product of the $p_{i*}(r_{i*}(C_{\overline{Z}_i})) = t_{i*}(C_{i\overline{Z}_i})$, where $t_i = p_i r_i$ is the canonical projection $\overline{Z}_i \to X$. When this morphism is finite for every i, we are done. In the general case, one can only say that $\overline{Z}_i$ is *integral* over X, hence of the form $\mathrm{Spec}(\mathscr{A}_i)$, where $\mathscr{A}_i$ is a quasi-coherent sheaf of $\mathscr{O}_X$-algebras which is integral over $\mathscr{O}_X$. Using (EGA IV 1.7.7), one sees that $\mathscr{A}_i$ is the inductive limit of its subalgebras $\mathscr{A}_{i,j}$ finite over X, hence $\overline{Z}_i$ is the projective limit (VII 5) of the preschemes $Z_{ij} = \mathrm{Spec}(A_{ij})$, finite over X. Applying (VII 5.11), one finds that $t_{i*}(C_{i\overline{Z}_i})$ is the inductive limit of the $t_{ij*}(C_{iZ_{ij}})$, where $t_{ij} : Z_{ij} \to X$ is the canonical projection. By (2.7.3), the homomorphism $F \to t_{i*}(C_{i\overline{Z}_i})$ therefore factors through a homomorphism $F \to t_{ij*}(C_{iZ_{ij}})$ for a suitable $j = j(i)$. We now set $X_i' = Z_{i,j(i)}$, and obtain an embedding as stated in 2.14. It remains to prove the following

Lemma 2.14.1. *Let X be a geometrically unibranch scheme (for example, a normal scheme), and let $i : U \to X$ be a quasi-compact open immersion. Then the closure Z of U, endowed with the induced reduced structure, is also geometrically unibranch; more* 25
precisely, for every $z \in Z$, one has $\mathscr{O}_{Z,z} \xleftarrow{\sim} \mathscr{O}_{X_{\mathrm{red}},z}$. Suppose on the other hand that i is dominant, and let C be a set, with C_U the constant sheaf on U defined by C. Then the natural homomorphism $C_X \to i_*(C_U)$ is an isomorphism. The same conclusions hold if the open immersion i is replaced by an inclusion $i : x \to X$, where x is a maximal point of X.

Using (EGA IV 2.3.11), we are reduced for the first assertion to the case where X is local, with closed point z; but then $X_{\text{réd}}$ is integral and U is nonempty, hence $Z = X_{\text{réd}}$, and the assertion is evident (thus, instead of assuming X geometrically unibranch, it would have sufficed here to suppose that the local rings of $X_{\text{réd}}$ are integral domains). For the second assertion, looking fiber by fiber, we are reduced by (VIII 5.3) *loc. cit.* to the case where X is strictly local, and then to proving that $H^0(X, C_X) \to H^0(U, C_U)$ is bijective. Here again, X is irreducible, and U is a nonempty open subset of it and hence irreducible, and the two sides of the preceding map are identified with C, whence the desired conclusion. (Here the notion "geometrically unibranch" has been used in the form of (EGA IV 18.8.14), as meaning that the strict localizations of the rings of $X_{\text{réd}}$ are integral domains.) The case of the inclusion of a maximal point (stated here for ease of future reference) is treated in exactly the same way.

Remarks 2.14.2. When X is Noetherian, the proof given (or an immediate deduction) shows that in 2.14 (ii), the X_i' may be assumed integral; if in addition X is assumed universally Japanese, i.e. the rings of its affine open subsets are universally Japanese (EGA 0_{IV} 23.1.1, IV 7.7.2), the X_i' may moreover be assumed normal.

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3. Kummer and Artin–Schreier theories

3.0. We denote by $(\mathrm{G}_m)_X$ the sheaf represented by the multiplicative group Spec $\mathscr{O}_X[t, t^{-1}]$ over X. Thus, for $U \to X$ étale, the sections of $(\mathrm{G}_m)_X$ over U are the invertible elements $\Gamma(U, \mathscr{O}_U^*)$ of $\Gamma(U, \mathscr{O}_U)$. Let $n \in \mathbf{N}$. The kernel of the n-th power map on $(\mathrm{G}_m)_X$ is the "sheaf of n-th roots of unity", denoted $(\boldsymbol{\mu}_n)_X$. Thus there is an exact sequence

$$0 \longrightarrow (\boldsymbol{\mu}_n)_X \longrightarrow (\mathrm{G}_m)_X \xrightarrow{n} (\mathrm{G}_m)_X. \tag{3.1}$$

If n is invertible on X (that is, if n is prime to every residue characteristic of X), the n-th power map is surjective *as a morphism of sheaves*:

Kummer theory 3.2. *If $n \in \mathbf{N}$ is invertible on X, there is an exact sequence*

$$0 \longrightarrow (\boldsymbol{\mu}_n)_X \longrightarrow (\mathrm{G}_m)_X \xrightarrow{n} (\mathrm{G}_m)_X \longrightarrow 0.$$

Indeed, let $u \in (\mathrm{G}_m)_X(U)$, with $U \to X$ étale. We must find an étale covering $\{U_i \to U\}$ such that u becomes an n-th power on every U_i. Since n is invertible on U, the equation

$$T^n - u = 0$$

is "separable" over U, that is,

$$U' = \operatorname{Spec} \mathscr{O}_U[T]/(T^n - u)$$

is étale over U. Since $U' \to U$ is surjective, it is a covering, and u becomes an n-th power on U'.

27 We point out that when n is invertible on X, $(\boldsymbol{\mu}_n)_X$ is a locally constant sheaf, in fact locally isomorphic to the sheaf $(\mathbf{Z}/n\mathbf{Z})_X$, and is constant in important cases. Kummer theory therefore gives information about the cohomology of X with values in certain constant sheaves. By definition,

$$H^0(X, (\mathrm{G}_m)_X) = \Gamma(X, \mathscr{O}_X^*),$$

and in dimension one we have:

Theorem 3.3. *("Hilbert's Theorem 90") There is an isomorphism*

$$H^1(X, (\mathrm{G}_m)_X) \xrightarrow{\sim} \operatorname{Pic} X,$$

where Pic X *is the group of isomorphism classes of invertible sheaves on* X.

Proof. This says that the canonical morphism

$$H^1(X_{\text{zar}}, (\mathbf{G}_m)_X) \longrightarrow H^1(X_{\text{ét}}, (\mathbf{G}_m)_X)$$

is bijective. Equivalently,

$$R^1\epsilon_*(\mathbf{G}_m)_X = 0,$$

where ϵ_* is direct image for the evident morphism of sites $\epsilon : X_{\text{et}} \to X_{\text{zar}}$ (VIII 4). We are therefore reduced to the case where X is affine. Since H^1 may be computed by Čech cohomology (V 2.5), and since for quasi-compact X it is enough to use quasi-compact coverings, this follows immediately from descent theory for sheaves (see FGA 190 or SGA VIII 1).

3.3.1. Suppose that X is Noetherian and has no embedded component, and let x_j be the maximal points of X. Let R_j be the local Artinian ring of X at x_j, and let i_j : Spec $R_j \to X$ be the inclusion. There is a canonical injection 28

$$(\mathbf{G}_m)_X \longrightarrow \prod_j i_{j*}(\mathbf{G}_m)_{\operatorname{Spec} R_j}$$

of sheaves on $X_{\text{ét}}$ (respectively X_{zar}). Its cokernel D is called the *sheaf of Cartier divisors* on $X_{\text{ét}}$ (respectively X_{zar})[34].

Corollary 3.4. *Let X be Noetherian and without embedded components, let $\epsilon : X_{\text{ét}} \to X_{\text{zar}}$ be the canonical morphism of sites, and let D be the sheaf of Cartier divisors on $X_{\text{ét}}$. Then $\epsilon_* D$ is the sheaf D_{zar} of Cartier divisors on X_{zar}, and in particular*

$$H^0(X_{\text{ét}}, D) \simeq H^0(X_{\text{zar}}, D_{\text{zar}}).$$

Proof. Applying ϵ_* to the exact sequence

$$0 \longrightarrow (\mathbf{G}_m)_X \longrightarrow \prod_j i_{j*}(\mathbf{G}_m)_{\operatorname{Spec} R_j} \longrightarrow D \longrightarrow 0$$

gives an exact sequence, since $R^1\epsilon_*(\mathbf{G}_m)_X = 0$ (3.3); the first assertion follows.

When X has characteristic $p \neq 0$, the following result sometimes replaces 3.2 in the study of p-torsion coefficients:

Artin–Schreier theory 3.5. *If X has characteristic $p > 0$, there is an exact sequence*

$$0 \longrightarrow (\mathbf{Z}/p\mathbf{Z})_X \longrightarrow \mathcal{O}_X \xrightarrow{\mathfrak{p}} \mathcal{O}_X \longrightarrow 0,$$

where $\mathfrak{p}$ is the morphism of additive groups defined by $\mathfrak{p}(f) = f^p - f$.

The kernel of $\mathfrak{p}$ is the sheaf of sections of $\mathcal{O}_X$ which locally belong to the prime field, and is therefore a constant sheaf. The morphism $\mathfrak{p}$ is surjective for the étale topology 29 because the equation $T^p - T - a$ ($a \in \Gamma(U, \mathcal{O}_U)$) is always "separable" in characteristic p, that is, defines an étale covering of U. In applying this exact sequence, we shall of course use VII 4.2.

Proposition 3.6. (i) *Let X be a scheme and let $i : x \to X$ be the inclusion of a point. Then*

$$H^1(X, i_*(\mathbf{Z})_x) = 0,$$

where $(\mathbf{Z})_x$ is the "constant sheaf" of value $\mathbf{Z}$ (see 2.0).

[34] In EGA IV 21, the word "divisors" is used instead of "Cartier divisors".

(ii) *If X is irreducible and if, for every geometric point $\overline{x}$ of X, the strict localization $X_{\overline{x}}$ of X at $\overline{x}$ is irreducible (one should say that "X is geometrically unibranch", see* EGA *IV 18.8.14), then*

$$H^1(X, (\mathbf{Z})_X) = 0.$$

Proof.

(i) The Leray spectral sequence

$$E_2^{pq} = H^p(X, R^q i_*(\mathbf{Z})_x) \Longrightarrow H^{p+q}(x, (\mathbf{Z})_x)$$

gives an injection $H^1(X, i_*(\mathbf{Z})_x) \to H^1(x, (\mathbf{Z})_x)$. Since $x = \operatorname{Spec} k(x)$ is the spectrum of a field, it is well known that $H^1(x, (\mathbf{Z})_x) = 0$. (This follows from the fact that $H^q(x, F)$ is torsion for $q > 0$ in any case (see VII 2.2 and CG), and from the exact sequences $0 \to \mathbf{Z} \xrightarrow{n} \mathbf{Z} \to \mathbf{Z}/n \to 0$.) This proves (i).

(ii) Let $i : x \to X$ be the inclusion of the generic point of X. Under the hypotheses of (ii), the canonical morphism $(\mathbf{Z})_X \to i_*(\mathbf{Z})_x$ is immediately seen to be bijective (and the converse is also true); thus (ii) is reduced to (i).

Remarks 3.7. In 3.6 (i) and (ii), $\mathbf{Z}$ may be replaced by any torsion-free abelian group. On the other hand, (ii) becomes false if the hypothesis that X is geometrically unibranch is omitted, as one sees, for example, by taking X to be an algebraic curve over a field K having an ordinary double point (compute $H^1(X, \mathbf{Z})$ in this case!).

4. The case of an algebraic curve

4.0. Let X be a Noetherian scheme of dimension 1. Let $R(X)$ be the ring of rational functions on X, and let $i : \operatorname{Spec} R(X) \to X$ be the inclusion. The ring $R(X)$ is Artinian, the product of the local rings of X at its maximal points, and the results of VIII 2 may be applied.

Let F be an abelian sheaf on $\operatorname{Spec} R(X)$, and consider the sheaves $R^q i_* F$, $q > 0$, on X. It follows at once from VIII 5.2 that the fiber of $R^q i_* F$ at a geometric point lying over a maximal point of X vanishes. Since X has dimension 1, such a sheaf is of a very special kind:

Lemma 4.1. *Let X be Noetherian and let F be a sheaf on X. The following conditions are equivalent (a sheaf satisfying them will be called a "skyscraper sheaf"):*

(i) *The fiber of F at every geometric point $\overline{x}$ of X lying over a nonclosed point $x \in X$ vanishes.*

(ii) *Every section $s \in F(X')$, where $X' \to X$ is étale of finite type, vanishes away from finitely many closed points of X.*

(iii) *There is an isomorphism $F \simeq \bigoplus_x i_{x*} i_x^* F$, where x runs through the closed points of X and $i_x : x \to X$ is the inclusion.*

30 *Proof.* Suppose (i) holds and let $z \in F(X')$, where $X' \to X$ is étale of finite type. The support of z (VIII 6.6) is then a closed subset of X' consisting of *closed* points, and is therefore finite because X' is Noetherian. Thus (i) implies (ii). The implication (ii) to (i) follows from the computation of fibers (VIII 3.9), and (iii) implies (i), since for every family (G_x) of sheaves on the closed points x of X, the sheaf $F = \bigoplus i_{x*}(G_x)$ on X satisfies (i): fiber functors commute with direct sums, and the support of $i_{x*}(G_x)$ is plainly contained in $\{x\}$.

It remains to prove that (ii) implies (iii). Consider the canonical morphism

$$F \longrightarrow \prod_x i_{x*}i_x^*F,$$

where x runs through the closed points of X. By (ii), it factors through

$$F \longrightarrow \bigoplus_x i_{x*}i_x^*F,$$

which gives the desired morphism. We leave to the reader the verification that it is bijective.

4.1.1. Apply the lemma to the sheaf $R^q i_* F$ introduced above. One has

$$R^q i_* F \xrightarrow{\sim} \bigoplus_x i_{x*}i_x^*(R^q i_* F),$$

where x runs through the closed points of X. By VIII 5, the fiber of $R^q i_* F$ at a geometric point $\overline{x}$ of X is

$$(R^q i_* F)_{\overline{x}} \simeq H^q(\operatorname{Spec} R(X_{\overline{x}}), F),$$

where $R(X_{\overline{x}}) \simeq R(X) \otimes_{\mathcal{O}_X} \mathcal{O}_{X_{\overline{x}}}$, $X_{\overline{x}}$ is the strict localization of X at $\overline{x}$ (VIII 4), and F also denotes the sheaf induced on $\operatorname{Spec} R(X_{\overline{x}})$. Suppose now that the residue field of every closed point x of X is separably closed. Then (see also VI 1)

$$\begin{aligned} H^p(X, R^q i_* F) &\simeq H^p\left(X, \bigoplus_x i_{x*}i_x^*(R^q i_* F)\right) \\ &\simeq \bigoplus_x H^p\left(X, i_{x*}i_x^*(R^q i_* F)\right) \\ &\simeq \bigoplus_x H^p\left(x, i_x^*(R^q i_* F)\right) \\ &\simeq \begin{cases} \bigoplus_x H^q(\operatorname{Spec} R(X_x), F) & \text{if } p = 0, \\ 0 & \text{if } p > 0. \end{cases} \end{aligned}$$

Thus:

Corollary 4.2. *Let X be a Noetherian scheme of dimension 1 such that, for every closed point x of X, the residue field $k(x)$ is separably closed. Let F be an abelian sheaf on* $\operatorname{Spec} R(X)$*, where $R(X)$ is the ring of rational functions on X. Consider the Leray spectral sequence*

$$E_2^{pq} = H^p(X, R^q i_* F) \Longrightarrow H^{p+q}(\operatorname{Spec} R(X), F).$$

One has

$$\begin{aligned} E_2^{pq} &= 0 \quad \text{if } p > 0 \text{ and } q > 0, \\ E_2^{0q} &= \bigoplus_x H^q(\operatorname{Spec} R(X_x), F) \quad \text{if } q > 0, \end{aligned}$$

where x runs through the closed points of X, and X_x is the strict localization of X at x.

The spectral sequence therefore gives an exact sequence 31

$$
\begin{aligned}
0 \longrightarrow H^1(X, i_*F) \longrightarrow H^1(\operatorname{Spec} R(X), F) \longrightarrow \bigoplus_x H^1(\operatorname{Spec} R(X_x), F)& \\
\longrightarrow H^2(X, i_*F) \longrightarrow H^2(\operatorname{Spec} R(X), F) \longrightarrow \bigoplus_x H^2(\operatorname{Spec} R(X_x), F)& \\
\longrightarrow \cdots .&
\end{aligned}
$$

4.3. Suppose moreover that the ℓ-cohomological dimension of Spec $R(X)$, for a given prime number ℓ, is at most 1 (that is, $H^q(\operatorname{Spec} R(X), F) = 0$ for $q > 1$ and every ℓ-torsion sheaf F on Spec $R(X)$). Let $K^1, \dots, K^m$ be the residue fields of the Artinian ring $R(X)$, that is, the residue fields of the maximal points of X. By the dictionary of VIII 2, the ℓ-cohomological dimension of Spec $R(X)$ is

$$\operatorname{cd}_\ell(\operatorname{Spec} R(X)) = \sup_i\{\operatorname{cd}_\ell(G(\overline{K}^i/K^i))\},$$

where $\operatorname{cd}_\ell(G)$ denotes the ℓ-cohomological dimension of the profinite group G (see CG).

Every residue field K_x of $R(\overline{X}(x))$, for a strict localization $\overline{X}(x)$ of X, is plainly a separable (possibly infinite) extension of one of the K^i, and hence

$$\operatorname{cd}_\ell(G(\overline{K}_x/K_x)) \le \operatorname{cd}_\ell(G(\overline{K}^i/K^i))$$

by CG II 4.1, Proposition 10. Thus $\operatorname{cd}_\ell(\operatorname{Spec} R(\overline{X}(x))) \le 1$ as well.

Applying the exact sequence of 4.2 to an ℓ-torsion sheaf F, one obtains

$$
\text{(4.3.1)} \qquad
\begin{cases}
0 \to H^1(X, i_*F) \to H^1(\operatorname{Spec} R(X), F) \to \\
\qquad \bigoplus_x H^1(\operatorname{Spec} R(\overline{X}(x)), F) \longrightarrow H^2(X, i_*F) \longrightarrow 0, \\
H^q(X, i_*F) = 0 \quad \text{for } q > 2.
\end{cases}
$$

32 Note that the hypothesis $\operatorname{cd}_\ell(\operatorname{Spec} R(X)) \le 1$ holds if X is an algebraic curve over a separably closed field (Tsen's theorem [6, Theorem 4.3]). Analogous exact sequences in this case were studied by Ogg[2] and Šafarevič.

4.4. Suppose that ℓ is invertible on X and take $F = (\mathrm{G}_m)_{R(X)}$. Applying 3.3 and [CG, Chapter I, Proposition 12], still under the hypothesis $\operatorname{cd}_\ell(\operatorname{Spec} R(X)) \le 1$, one finds

$$
\text{(4.4.1)} \qquad
\begin{cases}
H^1(\operatorname{Spec} R(X), \mathrm{G}_m) = 0, \\
H^q(\operatorname{Spec} R(X), \mathrm{G}_m) \underset{\ell}{=} 0 \quad (q \ge 2),
\end{cases}
$$

where the ℓ under the equality sign means that the group on the left, which is torsion in any case, has no ℓ-torsion.

The same result holds with $R(X)$ replaced by $R(\overline{X}(x))$. Since the sheaf G_m on Spec $R(X)$ plainly induces the sheaf G_m on Spec $R(\overline{X}(x))$, 4.2 gives

$$
\text{(4.4.2)} \qquad
\begin{cases}
H^1(X, i_*(\mathrm{G}_m)_{R(X)}) = 0, \\
H^q(X, i_*(\mathrm{G}_m)_{R(X)}) \underset{\ell}{=} 0 \quad \text{if } q > 1.
\end{cases}
$$

For every sheaf F on X, 4.1 shows that the kernel P and cokernel Q of the canonical homomorphism $F \to i_*i^*F$ are skyscraper sheaves. Since the residue fields at the closed points of X are assumed separably closed, one therefore has $H^j(X, P) = H^j(X, Q) = 0$ for $j > 0$. The long exact cohomology sequence immediately implies that

$$H^j(X, F) \longrightarrow H^j(X, i_*i^*F)$$

is an isomorphism for $j \geq 2$. Taking (4.4.2) into account, one obtains 33

(4.5) $$H^j(X, \mathbf{G}_m) \underset{\ell}{=} 0 \quad \text{for } j \geq 2,$$

valid when X is a Noetherian scheme of dimension 1 satisfying $\mathrm{cd}_\ell(R(X)) \leq 1$ and whose residue fields at closed points are separably closed. Kummer theory (3.2) and 3.3 now give:

Theorem 4.6. *Let X be Noetherian of dimension 1, and let n be an integer invertible on X. Suppose that $\mathrm{cd}_\ell(R(X)) \leq 1$ for every prime number ℓ dividing n, and that the residue field $k(x)$ is separably closed for every closed point x of X. Then*

$$H^q(X, (\boldsymbol{\mu}_n)_X) = 0 \quad \textit{if } q > 2,$$

and the cohomology for $q \leq 2$ is given by the exact sequence

$$\begin{aligned} 0 \longrightarrow H^0(X, (\boldsymbol{\mu}_n)_X) \longrightarrow \Gamma(X, \mathcal{O}_X^*) \xrightarrow{n} \Gamma(X, \mathcal{O}_X^*) \longrightarrow \\ H^1(X, (\boldsymbol{\mu}_n)_X) \longrightarrow \operatorname{Pic} X \xrightarrow{n} \operatorname{Pic} X \longrightarrow H^2(X, (\boldsymbol{\mu}_n)_X) \longrightarrow 0. \end{aligned}$$

Corollary 4.7. *Let X be a complete connected curve over a separably closed field k whose characteristic is prime to n. Then*

$$\begin{aligned} &H^0(X, (\boldsymbol{\mu}_n)_X) \simeq \boldsymbol{\mu}_n(k), \\ &H^1(X, (\boldsymbol{\mu}_n)_X) \simeq {}_n(\operatorname{Pic} X) = \{\textit{elements of } \operatorname{Pic} X \textit{ whose order divides } n\}, \\ &H^2(X, (\boldsymbol{\mu}_n)_X) \simeq (\operatorname{Pic} X)/n \simeq (\mathbf{Z}/n\mathbf{Z})^c, \end{aligned}$$

where c is the number of irreducible components of X. 34

For the last assertion, recall that the Picard scheme [TDTE V] of such an $X/\operatorname{Spec} k$ has a decomposition

(4.8) $$0 \longrightarrow A \longrightarrow \operatorname{Pic}_{X/k} \longrightarrow \mathbf{Z}^c \longrightarrow 0$$

where A is a connected algebraic group over $\operatorname{Spec} k$. The "multiplication by n" map on such an A is étale, hence surjective on k-valued points. It follows that

$$_n(\operatorname{Pic} X) \simeq {}_n(A(k))$$

and

$$(\operatorname{Pic} X)/n \simeq (\mathbf{Z}/n\mathbf{Z})^c.$$

If X is smooth over $\operatorname{Spec} k$, so that its Jacobian A is an abelian variety, the group $_n(\operatorname{Pic} X)$ is isomorphic to $(\mathbf{Z}/n)^{2g}$, where g is the genus of X. Of course, $\boldsymbol{\mu}_n(k) \simeq \mathbf{Z}/n\mathbf{Z}$ as well. Thus the ranks of the cohomology groups as $\mathbf{Z}/n$-modules are 1, $2g$, and 1, as expected.

5. The trace method

5.1. Let X be a scheme, let $f : X' \to X$ be an étale covering, and let F be a sheaf on X. Then the adjunction formulas give a *restriction* morphism

$$F \xrightarrow{\text{res}} f_* f^* F.$$

Since f is finite, one has $H^q(X, f_* f^* F) \simeq H^q(X', f^* F)$, whence a morphism, also called restriction:

(5.1.1) $$H^q(X, F) \xrightarrow{\text{res}} H^q(X', f^* F),$$

which is moreover the evident morphism. But when f is étale, there is also another 35 morphism, called the *trace morphism*,

$$f_*f^*F \xrightarrow{\mathrm{tr}} F, \tag{5.1.2}$$

which induces morphisms on cohomology

$$H^q(X', f^*F) \xrightarrow{\mathrm{tr}} H^q(X, F). \tag{5.1.3}$$

The trace morphism is characterized by the following two properties:

(i) It is local for the étale topology.
(ii) If X' is the disjoint sum of d copies of X, so that $f_*f^*F \simeq F^d$, the trace is the summation map $F^d \to F$.

Uniqueness follows trivially from (i), (ii), since every étale covering is locally constant. Once uniqueness is known, existence follows by the usual descent argument.

From (ii) one deduces the following formula:

(5.1.4) $\mathrm{tr} \circ \mathrm{res} : F \longrightarrow F$ is multiplication by the local degree of f.

If the degree is a constant d, it follows that the morphism $\mathrm{tr} \circ \mathrm{res} : H^q(X, F) \to H^q(X, F)$ is multiplication by d.

Corollary 5.2. *Let $f : X' \to X$ be an étale covering of constant degree d. If F is an r-torsion sheaf on X with $(r, d) = 1$, then $H^q(X', f^*F) = 0$ implies that $H^q(X, F) = 0$.*

36 Indeed, multiplication by d induces an isomorphism of F, hence an isomorphism on cohomology. If the latter isomorphism factors through zero, then $H^q(X, F) = 0$.

5.2.1. Now let $i : U \to X$ be an immersion, let $f : U' \to U$ be an étale covering of degree d, and suppose that f is induced by a finite morphism $g : X' \to X$ by base change, i.e. that one has the cartesian diagram

$$\begin{tikzcd} U' \arrow[r, "i'"] \arrow[d, "f"'] & X' \arrow[d, "g"] \\ U \arrow[r, "i"] & X \end{tikzcd}$$

Let F be a sheaf on U. One has

$$\begin{tikzcd} F \arrow[r, "\mathrm{res}"] \arrow[rr, bend right, "d"'] & f_*f^*F \arrow[r, "\mathrm{tr}"] & F, \end{tikzcd}$$

hence, since $i_*f_* = g_*i'_*$,

$$\begin{tikzcd} i_*F \arrow[r, "\mathrm{res}"] \arrow[rr, bend right, "d"'] & g_*{i'}_*f^*F \arrow[r, "\mathrm{tr}"] & i_*F, \end{tikzcd} \tag{5.3}$$

and since $i_!f_* \simeq g_*i'_!$ (which follows, for example, from (VIII 5.5)),

$$\begin{tikzcd} i_!F \arrow[r, "\mathrm{res}"] \arrow[rr, bend right, "d"'] & g_*{i'}_!f^*F \arrow[r, "\mathrm{tr}"] & i_!F, \end{tikzcd} \tag{5.4}$$

whence

(5.3′)
$$\begin{tikzcd} H^q(X, i_*F) \arrow[r, "\mathrm{res}"] \arrow[rr, bend right, "d'"'] & H^q(X', i'_* f^* F) \arrow[r, "\mathrm{tr}"] & H^q(X, i_*F) \end{tikzcd}$$

and 37

(5.4′)
$$\begin{tikzcd} H^q(X, i_!F) \arrow[r, "\mathrm{res}"] \arrow[rr, bend right, "d"'] & H^q(X', i'_! f^* F) \arrow[r, "\mathrm{tr}"] & H^q(X, i_!F); \end{tikzcd}$$

From (5.4) one deduces the following corollaries:

Proposition 5.5. *Let X be a quasi-compact and quasi-separated scheme, let $\ell \in \mathbf{P}$, and let H be a half-exact functor with values in* (Ab), *defined on the category of ℓ-sheaves on X, and compatible with filtered inductive limits. The following assertions are equivalent:*

(i) $H = 0$.

(ii) *$H(f_* i_!(\mathbf{Z}/\ell)) = 0$ for every finite $f : Y \to X$, and every open subscheme $i : U \to Y$ of Y, such that U is of finite presentation over X. If X is noetherian, one may even restrict to integral Y.*

Proof. By 2.7.2 and 2.5 it is enough to verify $H(F) = 0$ for F of the form $F = i, G$, where $i : U \to X$ is the inclusion of a subscheme of X of finite presentation over X, and where G is a locally constant "isotrivial" ℓ-sheaf on U. Let $U'' \to U$ be a principal covering with group g, such that g becomes constant on U''. Let A be the ℓ-group of values of G on U''. The group g acts on A, and this action determines the structure of G. Let $\hbar$ be an ℓ-Sylow subgroup of g, and let $U' = U''/\hbar$ be the covering of U corresponding to $\hbar$. The degree d of U' over U is prime to ℓ. Choose an extension of $f : U' \to U$ to a finite morphism $g : X' \to X$, which always exists (EGA IV 8.12.6), and let $i' : U' \to X'$ be the inclusion. Applying (5.4) and the fact that d is prime to ℓ, we are reduced to proving that $H(g_* i'_! f^* G) = 0$.

Now it is well known that the only abelian ℓ-torsion group that is simple under 38
an action of an ℓ-group $\hbar$ is $\mathbf{Z}/\ell$ with trivial action, and it follows that there exists a filtration of A as an $\hbar$-module whose successive quotients are isomorphic to $\mathbf{Z}/\ell$ with trivial action. This filtration gives a corresponding filtration of f^*G, and therefore, in order to prove that $H(g_* i'_! f^* G) = 0$, it is enough to prove that $H(g_* i'_! \mathbf{Z}/r) = 0$, whence the result.

Proposition 5.6. *Let X be a noetherian scheme, let $\ell \in \mathbf{P}$, and let φ be a function with values in $\mathbf{N}$, defined on the set of integral schemes finite over X. Suppose that $\varphi(Y_1) \leq \varphi(Y_2)$ whenever there exists an X-morphism $Y_1 \to Y_2$, and that the inequality is strict if Y_1 is a closed subscheme of Y_2, distinct from Y_2. For every Y finite over X, put $(Y) = \mathrm{Sup}(Y_i)$, where the Y_i are the irreducible components of Y, endowed with the induced reduced structure. Then the following assertions are equivalent:*

(i) *For every Y finite over X and every ℓ-sheaf F on Y, one has $H^q(Y, F) = 0$ if $q > \varphi(\mathrm{Supp}\, F)$.*

(ii) *For every integral Y finite over X, one has $H^q(Y, \mathbf{Z}/\ell) = 0$ if $q > \varphi(Y)$.*

Proof. Suppose that (ii) is true. We may apply the preceding proposition to each H^q, taking into account that cohomology commutes with inductive limits and with direct images under finite morphisms; we are thus reduced, for the verification of (i), to the case where Y is integral and finite over X, and F is of the form $i_!(\mathrm{Z}/\ell)_U$, where $i : U \to X$ is a nonempty open subscheme.

Now there is an exact sequence

$$0 \longrightarrow i_!(\mathrm{Z}/\ell)_U \longrightarrow (\mathrm{Z}/\ell)_Y \longrightarrow (\mathrm{Z}/\ell)_Z \longrightarrow 0$$

where $Z = Y - U$, hence $Z \neq Y$. Arguing by induction on the value of φ, we may
39 suppose that $H^q(\mathrm{Z}, \mathrm{Z}/\ell) = 0$ for $q > \varphi(Y) - 1$, whence $H^q(Y, i_!(\mathrm{Z}/\ell)) = 0$ for $q > \varphi(Y)$ by the exact cohomology sequence.

Corollary 5.7. *Let X be an algebraic curve over a separably closed field k of characteristic different from ℓ, with $\ell \in \mathbf{P}$. Then the cohomology of X with coefficients in a sheaf F of ℓ-torsion vanishes in degrees > 2. If X is affine, the cohomology vanishes in degrees > 1.*

For integral Y finite over X, put $\varphi(Y) = 2 \dim Y$ (resp. $\varphi(Y) = \dim Y$ if X, hence Y, is affine). By 7.6 it is enough to prove that $H^q(Y, \mathrm{Z}/\ell) = 0$ if $q > \varphi(Y)$, which is trivial if $\dim Y = 0$. It therefore remains to prove that $H^q(Y, \mathrm{Z}/\ell) = 0$ if $q > 2$ (resp. if $q > 1$ in the affine case) for an integral curve Y. Since $(\mathrm{Z}/\ell)_Y \simeq (\boldsymbol{\mu}_\ell)_Y$, this follows from 4.6 for $q > 2$.

Suppose that X is affine. Since radicial extensions do not affect cohomology (VIII 1.1), we are reduced to the case where k is algebraically closed. Let $f : X' \to X$ be the normalization of X, which is a finite morphism and an isomorphism outside a finite set of points, say S. There is an exact sequence of sheaves on X

$$0 \longrightarrow (\mathrm{Z}/n)_X \longrightarrow f_*(\mathrm{Z}/n)_{X'} \longrightarrow \epsilon \longrightarrow 0,$$

where ϵ is concentrated on S, so that $H^q(X, \epsilon) = 0$ for $q > 0$. By (VIII 5.6), we are reduced to proving that

$$H^q(X, f_*(\mathrm{Z}/n)_{X'}) \simeq H^q(X', \mathrm{Z}/n) = 0$$

if $q > 1$, that is, to the case where X is normal. We may also plainly suppose that X is connected. Let $i : X \to \overline{X}$ be an open immersion with $\overline{X}$ complete and nonsingular. Since X is affine, at least one point is missing from $\overline{X}$, and it follows immediately that the morphism

$$A(k) \longrightarrow \operatorname{Pic} X$$

40 is surjective, where A is the Jacobian of $\overline{X}$ (cf. (4.8)). Since $A(k)$ is divisible by $\ell \neq \operatorname{car} k$, the same is true of $\operatorname{Pic} X$, whence $H^2(X, \boldsymbol{\mu}_\ell) = 0$ by 4.6, C.Q.F.D.

Remark 5.8. Let us record, for later reference, that the argument of 5.5 in fact establishes the following result: Let X be a quasi-compact and quasi-separated scheme, let ℓ be a prime number, and let $\boldsymbol{C}$ be a strictly full subcategory of the category of constructible abelian ℓ-torsion sheaves, stable under direct factors and extensions. Suppose that for every finite morphism $f : Y \to X$ and every open subscheme $i : U \to Y$ of Y that is of finite presentation over X, one has $f_*(i_!(\mathbf{Z}/\ell\mathbf{Z})_U) \in \boldsymbol{C}$. Then $\boldsymbol{C}$ contains all constructible ℓ-torsion sheaves.

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EXPOSÉ X

Cohomological dimension: first results

M. Artin

1. Introduction

43

In this Exposé, unless expressly stated otherwise, “sheaf” means “abelian sheaf”.

Let X be a scheme and let $\ell \in \mathbf{P}$. “Recall” that the *ℓ-cohomological dimension* of X is the largest integer $\mathrm{cd}_\ell X = n$ (or $= \infty$ if no such number exists) such that $H^n(X, F) \neq 0$ for at least one ℓ-torsion sheaf F on X. If $X = \operatorname{Spec} A$, we write $\mathrm{cd}_\ell A = \mathrm{cd}_\ell \operatorname{Spec} A$. Taking into account the dictionary (VIII 2), we thus recover the “classical” notion of [CG] when $A = k$ is a field. Since we plainly have the formula

$$\mathrm{cd}_\ell(\coprod_i X_i) = \sup_i \mathrm{cd}_\ell X_i$$

the classical theory may, in the evident way, be extended to an arbitrary artinian ring. This theory is set out in (CG) = “Galois cohomology” by Serre. We shall add a few refinements in n° 2. 44

Consider, for example, the case of an irreducible algebraic variety X, of dimension n, over a separably closed field k, and fix $\ell \neq \operatorname{car} k$. A theorem to be found in n° 4 gives the upper bound $2n$ for its cohomological dimension; this is the “true” dimension of X (suggested by the topological analogy when $k = \mathbf{C}$). This result is fairly elementary once the theory for fields is known, and in fact the upper bound $2n$ can be improved except when X is complete: there is another upper bound, $2n-1$, which holds whenever X is “missing at least one point”. The latter is the correct “stable upper bound”, in the sense that every X of dimension n contains an open subset of cohomological dimension $2n-1$. A third upper bound is obtained for an affine variety, where one obtains n as the cohomological dimension. This is a generalization of one of the Lefschetz theorems for hyperplane sections. These last two assertions are deeper and can only be proved later (XIV).

Note that cohomological dimension is not a local notion (in contrast to what happens for paracompact spaces). For example, a henselian ring with separably closed residue field has cohomological dimension zero, but if the closed point is removed, one may obtain an arbitrarily large number, namely $2n-1$ if $n = \dim A$ in the most important cases; this also agrees with the topological analogy supplied by the cohomology of $U - \{x\}$, where U is a small open neighborhood of a point x of a complex analytic space.

2. Auxiliary results over a field

45

We have the following

Theorem 2.1. *Let k'/k be an extension of finite type of a field k, and let $\ell \in \mathbf{P}$. Then*

$$\mathrm{cd}_\ell k' \leqq \mathrm{cd}_\ell k + \operatorname{deg.tr.}(k'/k),$$

If the inequality is strict and if $\ell \neq \operatorname{car} k$, then $\ell = 2$, k is orderable, and k' is not orderable (i.e. -1 is a sum of squares in k').

Recall that when $\ell = \operatorname{car} k$ one has $\operatorname{cd}_\ell k' \leqq 1$ (CG II 2.2), so the question of equality in 2.1 scarcely arises. Moreover, $\operatorname{cd}_2 k = \infty$ if k is orderable. Indeed, the algebraic closure $\overline{k}$ of k contains a subextension $\tilde{k}$ (a maximal ordered subfield of $\overline{k}$) with $[\overline{k} : \tilde{k}] = 2$. Plainly, $\operatorname{cd}_2 \tilde{k} = \infty$, and $\operatorname{cd}_2 \tilde{k} \leqq \operatorname{cd}_2 k$ by (CG I Prop. 14).

Proof. Almost all of this is well known. The inequality, as well as equality if $\operatorname{cd}_\ell k < \infty$, are results of Tate (CG II 4.1.2). It remains to prove that $\operatorname{cd}_\ell k' < \infty$ and $\operatorname{cd}_\ell k = \infty$ imply that k is orderable and $\ell = 2$, and it is enough to treat the following two cases:

a) $[k' : k] < \infty$,
b) $k' = k(x)$ with x transcendental over k.

In case a) we may suppose k'/k separable. Then the Galois group $H = G(\overline{k}/k')$ is an open subgroup of $G = G(\overline{k}/k)$, and by a result of Serre [5] one has $\operatorname{cd}_\ell H = \operatorname{cd}_\ell G$ unless G contains an element of order ℓ. Since an algebraically closed field $\overline{k}$ cannot have a subfield $\tilde{k}$ of finite index unless $\ell = 2$ and $\tilde{k}$ is orderable [7], the result follows.

46 Consider case b). Let R be the henselization of $\operatorname{Spec} k[x]$ at the point $x = 0$, which is a discrete valuation ring, and let K be the fraction field of R. Then $\operatorname{cd}_\ell k(x) < \infty$ implies $\operatorname{cd}_\ell K < \infty$. It is therefore enough to prove that $\operatorname{cd}_\ell K < \infty$ implies $\operatorname{cd}_\ell k < \infty$, which is a consequence of the following theorem:

Theorem 2.2. [34]. *Let R be a henselian discrete valuation ring, with fraction field K and residue field k. Let $\ell \in \mathbf{P}$. We have*

$$\operatorname{cd}_\ell K = \operatorname{cd}_\ell k + 1$$

in the following two situations:

(i) *$\ell \neq \operatorname{car} k$.*
(ii) *$\ell \neq \operatorname{car} K$, k perfect, and $\operatorname{cd}_\ell k < \infty$.*

Remark. Naturally, if $\ell = \operatorname{car} K$ one has $\operatorname{cd}_\ell K \leq 1$ in every case (CG II 2.2). It therefore remains to study the relation of $\operatorname{cd}_\ell K$ with, perhaps, the module of absolute differentials Ω^1_k (EGA IV 20.4.3), in the unequal characteristic case when $\operatorname{car} k = \ell$, k imperfect.

Proof of the theorem. As for Theorem (2.1), most of it is due to Tate. Case (ii) is treated explicitly in (CG II Prop. 12) if R is complete, and we reduce to that case by means of the following lemma:

Lemma 2.2.1. *Let R be a henselian discrete valuation ring with fraction field K, and let $\widehat{K}$ be the fraction field of the completion $\widehat{R}$ of R. Then*

$$G(\overline{\widehat{K}}/\widehat{K}) \simeq G(\overline{K}/K).$$

47 Proof. This means that the functor $L \mapsto \widehat{L} = L \otimes_K \widehat{K}$ from the category of étale K-algebras to the category of étale $\widehat{K}$-algebras is an equivalence. Now every étale algebra $\widehat{K}'$ over $\widehat{K}$ is induced by a suitable étale algebra K' over K. Indeed, if $\widehat{K}' = \widehat{K}[\alpha]$ where α satisfies the separable irreducible equation $f(T) = 0$ with coefficients in $\widehat{R}$, it is enough to take as K' the algebra $K[\alpha_0]$, where α_0 is a root of an equation $f_0(T) = 0$ with coefficients in R, with $f_0 \equiv f(\widehat{m}^N)$, N large and $\widehat{m}$ the maximal ideal of $\widehat{R}$. This is an

[34]This theorem was proved recently by *Ax* [1].

application of the usual "Newton method" (Bourbaki, Alg. Comm. III 4.5). It remains to prove that the functor is fully faithful (it is plainly faithful), and this amounts to proving that if K' is a *field*, then so is $\widehat{K}'$. Let R' be the normalization of R in K', which is finite over R since K'/K is separable. The ring R' is a discrete valuation ring, and the completion of R' is $\widehat{R}' = R' \otimes_R \widehat{R}$. Hence $\widehat{R}'$ is a discrete valuation ring, and since K' is its total ring of fractions, it is indeed a field.

Consider assertion (i) of 2.2. If $\mathrm{cd}_\ell\, k < \infty$, we are again reduced by 2.2.1 to (CG II Prop. 12) (the hypothesis in the statement of that proposition that k be perfect is used in the proof only in the case of residue characteristic ℓ). It therefore remains to prove that $\mathrm{cd}_\ell\, K < \infty$ implies $\mathrm{cd}_\ell\, k < \infty$.

Let $\overline{G} = G(\overline{k}/k) = G(\widetilde{K}/K)$, where $\widetilde{K}$ is the maximal unramified extension of K (i.e. the fraction field of the discrete valuation ring $\widetilde{R}$ with separably closed residue field, the strict henselization of R), let $G = G(\overline{K}/K)$, $H = G(\overline{K}/\widetilde{K})$, so that $\overline{G} = G/H$. Let $\overline{G}' \subset \overline{G}$ be an ℓ-Sylow subgroup, and let k'/k, K'/K be the extensions corresponding to $\overline{G}'$. We have $\mathrm{cd}_\ell\, k < \infty$ if and only if $\mathrm{cd}_\ell\, k' < \infty$ (CG I Prop. 14), so by replacing R with its normalization in K', we reduce to the case in which $\overline{G}$ is an ℓ-group. It is then enough to prove that if $\mathrm{cd}_\ell\, G < \infty$, then $H^N(\overline{G}, \mathbf{Z}/\ell) = 0$ for sufficiently large N (CG I Prop. 20).

We have $\mathrm{cd}_\ell\, H = 1$ by the part of i) already proved, applied to $\widetilde{R}$. Moreover, we 48
have

$$H^1(H, \boldsymbol{\mu}_\ell) = H^1(\operatorname{Spec} \widetilde{K}, \boldsymbol{\mu}_\ell) = \widetilde{K}^*/\widetilde{K}^{*\ell} = \mathbf{Z}/\ell. \tag{2.2.2}$$

Indeed, $H^1(\operatorname{Spec} \widetilde{K}, \boldsymbol{\mu}_\ell) = \widetilde{K}^*/\widetilde{K}^{*\ell}$ by Kummer theory IX 3.2. We have an exact sequence

$$0 \longrightarrow \widetilde{R}^* \longrightarrow \widetilde{K}^* \longrightarrow \mathbf{Z} \longrightarrow 0,$$

and $\widetilde{R}^*$ is divisible by ℓ since $\widetilde{R}$ is strictly local and ℓ is invertible in $\widetilde{R}$, whence (2.2.2).

Consider the Hochschild–Serre spectral sequence

$$E_2^{pq} = H^p(\overline{G}, H^q(H, \boldsymbol{\mu}_\ell)) \Longrightarrow H^{p+q}(G, \boldsymbol{\mu}_\ell).$$

We have $E_2^{pq} = 0$ if $q > 1$. If, moreover, $\mathrm{cd}_\ell\, G < \infty$, the "boundary" morphism

$$H^{n-1}(\overline{G}, H^1(H, \boldsymbol{\mu}_\ell)) \longrightarrow H^{n+1}(G, \boldsymbol{\mu}_\ell)$$

must be bijective for sufficiently large n, and it is necessary to prove that then

$$H^{n-1}(\overline{G}, H^1(H, \boldsymbol{\mu}_\ell)) = H^{n-1}(\overline{G}, \mathbf{Z}/\ell) = 0.$$

Set $R' = R[x']$ with $x'^{\ell} = x$, where x is the local parameter of R, and let K' be the fraction field of R', $G' = G(\overline{K}/K')$, etc. The residue field of R' is k, since we have supposed that $\overline{G}$ is an ℓ-group, and hence that the ℓ-th roots of unity are in K. We have a diagram

$$\begin{tikzcd}
H' \arrow[r] \arrow[d, hook] & G' \arrow[r] \arrow[d, hook] & \overline{G} \\
H \arrow[r] & G \arrow[r] & \overline{G},
\end{tikzcd}$$

whence a morphism of Hochschild–Serre spectral sequences, which gives a commuta- 49 tive diagram

$$\begin{tikzcd}
H^{n-1}(\overline{G}, H^1(H', \boldsymbol{\mu}_l)) \arrow[dr] & \\
& H^{n+1}(G, \boldsymbol{\mu}_l) \\
H^{n-1}(\overline{G}, H^1(H, \boldsymbol{\mu}_l)) \arrow[uu, "\varepsilon"] \arrow[ur] &
\end{tikzcd}.$$

Since $\operatorname{cd}_\ell G < \infty$ implies $\operatorname{cd}_\ell G' < \infty$, all the morphisms in this diagram are isomorphisms for large n. But the morphism

$$\mathbf{Z}/r = H^1(H, \boldsymbol{\mu}_\ell) \longrightarrow H^1(H', \boldsymbol{\mu}_\ell) = \mathbf{Z}/\ell$$

which induces the arrow ϵ is plainly zero, and hence $H^{n-1}(\overline{G}, \mathbf{Z}/\ell) = 0$, C.Q.F.D.

THEOREM 2.3. *Let A be a noetherian henselian ring of dimension 1, and let $\ell \in \mathbf{P}$ be invertible in A. Let k be the residue field of A and let $R(A)$ be the ring of rational functions on* Spec A. *We have*

$$\operatorname{cd}_\ell R(A) \leq \operatorname{cd}_\ell k + 1,$$

and equality holds if $\ell \neq 2$ or if k is not orderable.

Proof. Replacing A by A/P for a minimal prime ideal P of A and applying VIII 1, we see that we may suppose A integral. Let K be its fraction field, and let R be its normalization, which is local and henselian, and hence a discrete valuation ring (it is noetherian by Krull–Akizuki (Bourbaki, Alg. Comm. Chap. 7, § 2, prop. 5)), and let k' be the residue field of R. It is known that $[k' : k] < \infty$ (*loc. cit.*). Now K is the fraction field of R, and therefore

$$\operatorname{cd}_\ell K = \operatorname{cd}_\ell k' + 1 \leq \operatorname{cd}_\ell k + 1$$

50 by 2.2, with equality under the last hypothesis of 2.3, by 2.1.

EXAMPLE. Equality does not hold for $\ell = 2$, when A is the henselization at the origin of $\mathbf{R}[x, y]/(x^2 + y^2)$.

COROLLARY 2.4. *Let A be a noetherian local ring with residue field k, and let $\ell \in \mathbf{P}$ be invertible in A. Suppose that $\ell \neq 2$ or that no residue field of* Spec A *is orderable. Then*

$$\operatorname{cd}_\ell R(A) \geq \operatorname{cd}_\ell k + \dim A,$$

where $R(A)$ is the ring of rational functions on Spec A.

Proof. We may suppose A integral, with fraction field $R(A) = K$. Let $x \in \mathfrak{m}_A$. Then $\dim A/(x) = \dim A - 1$. By induction on $\dim A$, we have

$$\operatorname{cd}_\ell R(A/(x)) \geq \operatorname{cd}_\ell k + \dim A - 1.$$

Let k' be a residue field of $R(A/(x))$ such that $\operatorname{cd}_\ell k' = \operatorname{cd}_\ell R(A/(x))$ (cf. VIII 1), and let P be the prime ideal that is the kernel of $A \to k'$. The ring A_P has dimension 1. It is therefore enough to prove the corollary for the local ring A_P; that is, we are reduced to the case in which $\dim A = 1$. Let $\widetilde{A}$ be the henselization of A. Every residue field of $R(\widetilde{A})$ identifies with a separable extension of $R(\widetilde{A}) = K$, and therefore

$$\operatorname{cd}_\ell R(A) = \operatorname{cd}_\ell K \geq \operatorname{cd}_\ell R(\widetilde{A}).$$

We are thus reduced to the case in which A is henselian of dimension 1, which follows from 2.3.

Corollary 2.5. *Let X be a noetherian scheme and let $\ell \in \mathbf{P}$ be invertible on X. Then* 51

$$\mathrm{cd}_\ell\, R(X) \geq \dim X.$$

Indeed, let $x \in X$ be a point such that $\dim X = \dim \mathcal{O}_{X,x}$ (or such that $\dim \mathcal{O}_{X,x}$ is very large if $\dim X = \infty$). Let A be the strict henselization of $\mathcal{O}_{X,x}$. Then $\mathrm{cd}_\ell\, R(X) \geq \mathrm{cd}_\ell\, R(\mathcal{O}_{X,x}) \geq \mathrm{cd}_\ell\, R(A)$ by the usual argument. Moreover, $\dim A = \dim X$ (resp. is very large). If $\ell = 2$, and if ℓ is invertible on X, then the equation $T^2 + 1 = 0$ is separable over A, so -1 is a square in A because A is strictly local; hence the residue fields of A are not orderable. Thus the hypotheses of 2.4 are satisfied for A in every case, and $\mathrm{cd}_\ell\, R(A) \geq \dim A$, whence the result.

3. Field of fractions of a strictly local ring

3.0. In the following sections we shall have essential need of knowing $\mathrm{cd}_\ell\, R(A)$, where A is a strictly local ring, that is, a henselian ring with separably closed residue field. Unfortunately, $\mathrm{cd}_\ell\, R(A)$ is not known in general, for example if $A = \widehat{\mathbf{Z}}_p[[x]] (p \neq \ell)$.

It is nevertheless conjectured that if A is noetherian and ℓ is invertible in A, then one has

$$\mathrm{cd}_\ell\, R(A) = \dim A \quad (\ell \text{ invertible in } A), \tag{3.1}$$

at least in the most important cases, for example if A is an *excellent* ring (EGA IV 7.8.2).

Note that if $\dim A = 1$, (3.1) is true by 2.3. Moreover, the inequality $\geq$ holds in any case by 2.5. We shall prove later (XIX), using resolution of singularities [3], that (3.1) is also true if A is excellent of residual characteristic zero. Here we shall prove (3.1) only 52
in the easy case of 3.2 below.

Naturally one has $\mathrm{cd}_\ell\, R(A) \leq 1$ if A has characteristic ℓ, so this case is trivial. It remains, moreover, to analyze the case of unequal characteristics with residual characteristic ℓ, and one does not expect (3.1) to be true in this case. It will certainly be necessary to take into account the rank of the module Ω^1_k (k the residue field), i.e. the degree $[k : k^\ell]$.

Proposition 3.2. *Let X be a prescheme of finite type over the spectrum of a field k or over the spectrum of a discrete valuation ring R, and let $\ell \in \mathbf{P}$ be invertible in k (resp. R). Then 3.1 is true for every ring A that is a strict localization of X.*

Proof. Take X integral (one reduces to this case as usual), and let $n = \dim X$.

Suppose that X is defined over a *separably closed* field k. Then for every ring A that is a strict localization of X at a *closed* point, one has $\dim A = n$. Since every residue field K of $R(A)$ is identified with an algebraic extension of $R(X)$, and since

$$\deg.\mathrm{tr}.(R(X)/k) = \dim X = n,$$

one has $\mathrm{cd}_\ell\, K \leq n$ by 2.1, whence $\mathrm{cd}_\ell\, R(A) \leq n$. The opposite inequality follows from 2.5.

Now suppose that X is of finite type over $\operatorname{Spec} R$, where R is a discrete valuation ring, and let A be the strict localization of X at a point x. If x does not lie above the closed point of $\operatorname{Spec} R$, one reduces immediately to the case where X is of finite type over a field (by localizing R). For the same reason, one may also suppose that the morphism $X \to \operatorname{Spec} R$ is dominant.

Suppose for the moment, in addition, that R is a *strictly local* discrete valuation ring 53
and that the point x is *closed* in X. Then one has $\dim A = \dim X = n$ (EGA IV 5.6.5).

Since every residue field K of $R(A)$ is identified with a separable extension of $R(X)$, one has

$$\mathrm{cd}_\ell R(A) \leq \mathrm{cd}_\ell R(X) \leq \mathrm{cd}_\ell L + \deg.\mathrm{tr}.(R(X)/L) = 1 + (n-1) = n$$

by 2.1 and 2.2, where L is the field of fractions of R; the result follows in this case, taking 2.5 into account.

The general case is deduced from these special cases by means of the following lemma:

Lemma 3.3. (i) *Let X be of finite type over the spectrum of a field k, and let A be a strict localization ring of X at a point x. Then there exist a separably closed field k', a scheme X' of finite type over* $\operatorname{Spec} k'$, *and a point x' closed in X' such that A is isomorphic to a strict localization ring of X' at x'.*

(ii) *Let X be of finite type over* $\operatorname{Spec} R$, *where R is a discrete valuation ring, let x be a point of X above the closed point of* $\operatorname{Spec} R$, *and let A be a strict localization of X at x. There exist a strictly local discrete valuation ring R', a scheme X' of finite type over* $\operatorname{Spec} R'$, *and a closed point x' of X' above the closed point of* $\operatorname{Spec} R'$, *such that A is isomorphic to a strict localization of X' at x'.*

Proof.

(i) We may take X affine.

Let Y be the closure of x in X, with the induced reduced structure, and let $d = \dim Y$. By the normalization lemma ([2] Chap. V § 3 n° 1), there exists a morphism

$$X \longrightarrow \operatorname{Spec} k[y_1, \dots, y_d] = T$$

54 inducing a *finite* morphism $Y \to T$. Consequently, the strict localization ring of $\mathcal{O}_{X,x}$ is isomorphic to the strict localization of $\mathcal{O}_{X',x'}$, where $X' = X \times_T \operatorname{Spec} \overline{R(T)}$, and where x' is a closed point of X'.

(ii) This case is treated similarly. We may suppose that X is affine, with ring A. Let Y be the closure of x in X with the induced reduced structure; it is a closed subscheme of the closed fiber X_0 of X over $\operatorname{Spec} R$, and let $d = \dim Y$. Then, lifting the elements $y_1, \dots, y_d$ of $A \otimes_R k$ considered in (i) to elements of A, one obtains a morphism

$$X \longrightarrow \operatorname{Spec} R[y_1, \dots, y_d] = T$$

inducing a *finite* morphism $Y \to T_0$. Let R' be the strict localization of $R[y_1, \dots, y_d] = B$ at the prime ideal $\mathfrak{m}B$. Then R' is a strictly local discrete valuation ring, and one may take $X' = X \times_T \operatorname{Spec} R'$, and x' a closed point of X' above x.

4. Cohomological dimension: the case where ℓ is invertible in $\mathcal{O}_X$

Theorem 4.1. *Let X be a noetherian scheme, let $\ell \in \mathbf{P}$, and let φ be a function on X with values in* $\mathbf{N}$ *and satisfying the following conditions:*

(i) *For every $x \in X$, $\mathrm{cd}_\ell\, k(x) \leq \varphi(x)$.*

(ii) *Let $x \in X$ and let $y \in Y =$ the closure of x, with $y \neq x$. Let K be the ring of rational functions of a strict localization of Y above y. One has $\mathrm{cd}_\ell K < \varphi(x) - \varphi(y)$.*

Then for every ℓ-torsion sheaf F on X, one has

$$H^q(X, F) = 0$$

if

$$q > \varphi(F) \underset{\mathrm{def}}{=} \operatorname{Sup}\{\varphi(x) \mid x \in X, F_{\overline{x}} \neq 0\}.$$

Proof. The hypotheses will also be satisfied by the restriction of φ to any subprescheme of X, so by noetherian induction we may suppose the theorem true for every closed subprescheme of X distinct from X. Let F be given, and write $F = \varinjlim F_i$, where the F_i are the constructible subsheaves of F (IX 2.9 (iii)). 55

For every F_i, the set of $x \in X$ such that $F_x \neq 0$ is constructible in X, and we may apply the induction hypothesis to every F_i whose support is not dense. We thus see that the induction hypothesis in fact implies that the theorem is true for every F that vanishes at at least one of the maximal points of X. We shall therefore suppose that F does not satisfy this condition.

Now let $i : \operatorname{Spec} R(X) \to X$ be the inclusion, where $R(X)$ is the ring of rational functions of X, and consider the following exact sequence, where K and C are defined as kernel and cokernel, respectively:

$$0 \longrightarrow K \longrightarrow F \longrightarrow i_* i^* F \longrightarrow C \longrightarrow 0.$$

Here K and C vanish at every maximal point of X, so by (ii), $\varphi(K)$ (resp. $\varphi(C)$) is strictly less than $n = \varphi(F)$, whence $H^q(X, K) = H^q(X, C) = 0$ for $q \geq n$. Thus, , in order to prove $H^q(X, F) = 0$ for $q > n$, it is enough, by the cohomology exact sequences, to prove $H^q(X, i_* i^* F) = 0$; hence we may suppose that F is of the form $i_* G$, for a suitable ℓ-sheaf G on $\operatorname{Spec} R(X)$.

Consider the Leray spectral sequence

$$E_2^{p,q} = H^p(X, R^q i_* G) \Longrightarrow H^{p+q}(\operatorname{Spec} R(X), G).$$

Now $R(X)_{\text{réd}}$ is the product of the fields $k(x)$ for x maximal in X, so $\operatorname{cd}_\ell R(X) \leq n$ by (i); therefore the abutment of the spectral sequence vanishes in dimension strictly greater than n. We wish to prove that $E_2^{p0} = 0$ for $p > n$. Examining the spectral sequence, 56
one sees that it is enough to prove that $E_2^{pq} = 0$ if $q > 0$ and if $p > n - q - 1$. Thus, by induction, it is enough to prove that $\varphi(R^q i_* G) < n - q$ for $q > 0$. But by VIII 5.2, the fiber of $R^q i_* G$ at a geometric point $\overline{y}$ above a point $y \in X$ is precisely $H^q(\operatorname{Spec} K, G_K)$, where K is the ring of rational functions of the strict localization of X at $\overline{y}$, and G_K is the induced sheaf on $\operatorname{Spec} K$. By (ii), one has $\operatorname{cd}_\ell K < n - \varphi(y)$, whence $(R^q i_* G)_{\overline{y}} = 0$ if $q \geq n - \varphi(y)$, which gives the desired result.

Corollary 4.2. *Let X be a noetherian scheme, let $\ell \in \mathbf{P}$, let $n \in \mathbf{N}$, and suppose that the following conditions are satisfied:*

(i) *For every point y of X of codimension c, one has $\operatorname{cd}_\ell k(y) \leq n - 2c$.*
(ii) *For every ring A that is a strict localization of an irreducible closed subscheme Y of X, one has $\operatorname{cd}_\ell R(A) \leq \dim A$.*

Then

$$\operatorname{cd}_\ell X \leq n.$$

More precisely, if F is an ℓ-sheaf that vanishes at every point y of codimension $< s$, one has

$$H^q(X, F) = 0 \text{ if } q > n - 2s.$$

Indeed, one may take $\varphi(y) = n - 2 \operatorname{codim} y$ and apply 4.1, whose condition (i) (resp. (ii)) follows from the corresponding condition (i) (resp. (ii)) of (4.2), taking into account for (ii) the inequality $\dim \mathscr{O}_{Y,y} + \dim \mathscr{O}_{X,x} \leq \dim \mathscr{O}_{X,y}$.

Corollary 4.3. *Let X be of finite type and dimension n over a field k, and let $\ell \neq \operatorname{car} k$. Then $\operatorname{cd}_\ell X \leq 2n + \operatorname{cd}_\ell k$.*

One applies the preceding corollary, replacing n there by $2n + cd_\ell k$, and using 2.1 and 3.2 to verify conditions (i) and (ii) of 4.2, taking into account for (i) the inequality 57

$$\deg.\operatorname{tr}.K(y)/k + \dim \mathcal{O}_{X,y} \leq \dim X.$$

Corollary 4.4. *Let X be noetherian, and let $\ell \in \mathbf{P}$ be invertible on X. Suppose that condition* (ii) *of* 4.2 *is satisfied, and moreover that $\ell \neq 2$ or that no residue field of X is orderable. Then one has*

$$\operatorname{cd}_\ell X \leq \operatorname{cd}_\ell R(X) + \dim X \leq 2\operatorname{cd}_\ell R(X).$$

Put $n = \operatorname{cd}_\ell R(X) + \dim X$ in 4.2, and apply 2.4, which implies condition (i) of 4.2, since

$$\operatorname{cd}_\ell k(y) \leq \operatorname{cd}_\ell R(\mathcal{O}_{X,y}) - \dim \mathcal{O}_{X,y} \leq \operatorname{cd}_\ell R(X) - \dim \mathcal{O}_{X,y} \leq n - 2\dim \mathcal{O}_{X,y}.$$

The second inequality in 4.4 follows from 2.5.

Example 4.5. The inelegant hypothesis on residue fields made in 4.4 is necessary, as is seen by taking $X = \operatorname{Spec}\mathbf{R}[x,y,z]/(x^2+y^2+z^2)$, where all the other conditions are satisfied, with $\dim X = \operatorname{cd}_\ell R(X) = \ell = 2$, but the sheaf $\mathbf{Z}/2$ concentrated at the point $(0,0,0)$ has nonzero cohomology in every degree, so $\operatorname{cd}_\ell X = +\infty$.

4.6. One is also tempted to try to impose hypotheses in 4.2 only on the closed points of X, but this is not possible. For example, there exist discrete valuation rings with algebraically closed residue field and with function field of arbitrarily large cohomological dimension[34]. The spectrum of such a ring will likewise have very large cohomological dimension.

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5. Cohomological dimension: the case $\ell = p$

5.0. Using Artin-Schreier theory (IX 3.5), one obtains a better upper bound than 4.3 for the p-cohomological dimension of a scheme X of characteristic p. Let $\operatorname{cdqc} X$ be the largest number n such that $H^n(X,F) \neq 0$ for at least one *quasi-coherent* sheaf of modules F on X (in the Zariski sense). We have the following

Theorem 5.1. *Let X be a Noetherian scheme of characteristic $p > 0$. Then*

$$\operatorname{cd}_p X \leq \operatorname{cdqc} X + 1.$$

In particular, the p-cohomological dimension of a Noetherian affine scheme X of characteristic $p > 0$ is at most 1.

Proof. Apply IX 5.5:

Taking account of the fact that the $R^q f_* F$, $q > 0$, vanish for a finite morphism f and an abelian sheaf F (VIII 5.5) (resp. a quasi-coherent sheaf F), we are reduced to proving that $H^q(X, i_!(\mathbf{Z}/p)_U) = 0$ for $i : U \to X$ an open immersion and for $q > \operatorname{cdqc} X$.

Let Y be a closed subscheme of X with support $X - U$, and let J be the coherent sheaf of ideals defining Y. The morphism $p : f \mapsto f^p - f$ (IX 3.5) induces a morphism $J \to J$. Examining the exact sequence IX 3.5 and the exact sequence

$$0 \longrightarrow i_!(\mathbf{Z}/p)_U \longrightarrow (\mathbf{Z}/p)_X \longrightarrow (\mathbf{Z}/p)_Y \longrightarrow 0,$$

[34] To see this, it is enough to note that if X is a smooth algebraic scheme over a field k, admitting a point $x \in X(k)$, then one can find a “formal curve arc” passing through x whose “algebraic envelope” is X, and whose function field therefore contains that K of X and induces on it a discrete valuation whose residue field is k.

one sees that there is in fact an exact sequence

$$(*) \qquad 0 \longrightarrow i_!(\mathbf{Z}/p)_U \longrightarrow J \longrightarrow J \longrightarrow 0.$$

Since J is coherent, one has $H^q(X, J) = 0$ for $q > \operatorname{cdqc} X$, whence the result by the 59
long exact cohomology sequence.

For a scheme quasi-projective over a separably closed field k, this upper bound can in certain cases be reduced by 1. In fact, we have

Corollary 5.2. *Let X be a scheme of finite type over a field k of characteristic $p > 0$. One has*

$$\operatorname{cd}_p X \leq \dim X + 1,$$

and if k is separably closed and X is quasi-projective [34], *one has*

$$\operatorname{cd}_p X \leq \dim X.$$

Indeed, the first assertion is clear, since $\operatorname{cdqc} X \leq \dim X$ ([6] 4.15.2). Let us treat the second. For such an X, the cohomology groups $H^n(X, F)$ ($n = \dim X$), for a coherent sheaf F, are finite-dimensional vector spaces over k. Let $V = H^n(X, J)$, where J is the sheaf of ideals occurring in the exact sequence $(*)$. The morphism $\varphi : V \to V$ deduced from $(*)$ is evidently of the form $\varphi = \epsilon - \mathrm{id}$ where ϵ satisfies $\epsilon(rv) = r^p \epsilon(V)$ for $r \in k$ and $v \in V$. The Jacobian of the map Φ of the affine scheme deduced from Φ is therefore $-I$, where I is the identity matrix. Consequently Φ is étale, hence surjective because it is additive, and it follows that Φ is surjective since k is separably closed, whence the conclusion immediately.

6. Cohomological dimension for a prescheme of finite type over Spec $\mathbb{Z}$ 60

Proposition 6.1. *Let X be quasi-finite over* $\operatorname{Spec} \mathbf{Z}$. *Suppose that $\ell \neq 2$ or that every residue field of X of characteristic zero is totally imaginary. Then* $\operatorname{cd}_\ell X \leq 3$.

This is a consequence of Theorem 2.2 (ii), of Corollary 4.2, and of the fact that a number field (totally imaginary if $\ell = 2$) has ℓ-cohomological dimension ≤ 2 (CG II 4.4).

A consequence of class field theory is that equality holds if the morphism $X \to \operatorname{Spec} \mathbf{Z}$ is proper and surjective.

Unfortunately, the results of n° 4 do not apply as they stand if X is only of finite type over $\operatorname{Spec} \mathbf{Z}$, because hypothesis (ii) of (4.2) is not in general satisfied for local rings of *residual characteristic ℓ*. One must therefore use result (5.1) to obtain the correct upper bound; we shall now reap the benefit of Theorem (5.1).

Theorem 6.2. *Let X be of finite type over* $\operatorname{Spec} \mathbf{Z}$, *and let $\ell \in \mathbf{P}$. Suppose, if $\ell = 2$, that no residue field of X is orderable. Then*

$$\operatorname{cd}_\ell X \leq 2 \dim X + 1.$$

More precisely, let F be a sheaf of ℓ-torsion such that for every point x of X such that $F_{\overline{x}} \neq 0$, one has

$$\dim\{\overline{x}\} \leq \begin{cases} (N-1)/2 & \text{if} \quad \ell \neq \operatorname{car} k(x) \\ N-1 & \text{if} \quad \ell = \operatorname{car} k(x) \end{cases}$$

($\{\overline{x}\}$ is the closure of $\{x\}$); then

$$H^p(X, F) = 0 \text{ if } p > N.$$

Proof. By induction on X, we may suppose the theorem true for every closed subscheme distinct from X. If F is as in the statement and we write $F = \varinjlim F_i$, where the F_i are the constructible subsheaves of F (IX 2.9 (iii)), then every F_i also satisfies the same hypotheses. It therefore suffices to prove the theorem when F is constructible. 61

In this case, the set E of $x \in X$ such that $F_{\bar{x}} \neq 0$ is a constructible subset, and the induction hypothesis implies that the theorem is true for every F such that E is not dense. Moreover, the theorem is a consequence of 5.1 for a sheaf concentrated on the fiber of X over the point (ℓ) of $\operatorname{Spec}\mathbf{Z}$, and of 4.3 for a sheaf concentrated on the fiber over (p), where $p \neq \ell$, taking account of the fact that $\operatorname{cd}_\ell \mathbf{Z}/p\mathbf{Z} = 1$ (CG II 2.2).

Take X reduced, and let $X = X_0 \cup X_1$, where X_1 is the closed subset which is the union of the irreducible components of X of characteristic ℓ, and where X_0 is the union of the other irreducible components, X_0 and X_1 being endowed with their induced reduced structures. Let $j_\nu : X_\nu \to X$ be the inclusions, let $d_\nu = \dim X_\nu$, and let $N = \sup\{2d_0+1, d_1+1\}$. For every constructible F on X, there is an exact sequence

$$(*) \qquad 0 \longrightarrow F \longrightarrow j_{0*}j_0{}^*F \oplus j_{1*}j_1{}^*F \longrightarrow C \longrightarrow 0,$$

where C is concentrated on $X_0 \cap X_1$, which has characteristic ℓ and dimension $\leq d_1 - 1$. It follows that

$$H^q(X, C) = 0 \text{ if } q > N-1,$$

because by 5.1 one has $\operatorname{cd}_\ell(X_0\cap X_1) \leq \dim(X_0 \cap X_1)+1 \leq d_1 \leq N-1$. Examining the relative long exact cohomology sequence of $(*)$, we are reduced to the case $X = X_0$ or $X = X_1$, and the latter is a consequence of 5.1, as already noted.

62 Suppose therefore that every maximal point of X has characteristic different from ℓ, and let $i_\nu : x_\nu \to X$ be the inclusions of the maximal points. Examining the exact sequence

$$0 \longrightarrow K \longrightarrow F \longrightarrow \bigoplus_\nu i_{\nu*}i_\nu{}^*F \longrightarrow C \longrightarrow 0$$

we are reduced by induction to the case $F = i_{\nu*}i_\nu{}^*F$, that is, where F is of the form i_*G for a suitable sheaf G on $\operatorname{Spec} k(x)$, with x a point of X. We may therefore suppose that X is irreducible and reduced, with generic point x, and that the characteristic of $k(x)$ is zero by 4.3.

Consider the spectral sequence

$$(**) \qquad H^p(X, R^q i_* G) \Longrightarrow H^{p+q}(\operatorname{Spec} k(x), G).$$

Let $s = \dim X$. We must prove that

$$(+) \qquad E_2^{p0} = 0 \text{ for } p > 2s+1.$$

Now $k(x)[i]$ has transcendence degree $s-1$ over $\mathbf{Q}[i]$ which is a field of cohomological dimension 2. Since $k(x)$ is not orderable if $\ell = 2$, it follows from 2.1 that $\operatorname{cd}_\ell k(x) = s+1$. This implies that the target of the spectral sequence vanishes for $n > s+1$. It therefore suffices, for (+), to prove that in $(**)$ one has

$$(++) \qquad E_2^{pq} = 0 \text{ if } p > 2s-q \text{ and } q > 0.$$

Let y be a point of X, and let K be the ring of fractions of a strict localization at a geometric point $\tilde{y}$ of X above y. Then K is a direct product of fields, each of transcendence

[34] Using Chow's lemma, it is easy to replace the condition "X quasi-projective" by "X separated."

degree $s-1$ over a strict localization of $\mathbf{Z}$, and hence, by 2.1 and 2.2, one has $\operatorname{cd}_\ell K \leq s$. Moreover, one may apply 3.2 if the residual characteristic of y is not ℓ. Thus one has

$$('') \qquad (R^q i_* G)_{\bar{y}} = 0 \begin{cases} \text{if } q > \operatorname{codim} y, \text{ i.e. } \dim \overline{y} > s-q, \text{ and } \ell \neq \operatorname{car} k(y), \\ \text{or if } q > s \text{ in all cases.} \end{cases}$$

In particular, one has $E_2^{pq} = 0$ if $q > s$. To verify (++) in the case $0 < q \leq s$, apply the induction hypothesis to the sheaves $R^q i_*(G)$. 63

According to the statement and (″), one must set $N' = 2s - q$ and verify the two inequalities

$$s - q \leq (N' - 1)_{/2}$$
$$s - 1 = \dim X_\ell \leq N' - 1,$$

where X_ℓ is the fiber of X over the point (ℓ) of $\operatorname{Spec} \mathbf{Z}$. They do indeed hold if $0 < q \leq s$. C.Q.F.D.

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EXPOSÉ XI

Comparison with classical cohomology: the case of a smooth scheme

M. Artin

1. Introduction

64

In Section 4 we shall prove that étale cohomology with values in a locally constant torsion sheaf agrees with classical cohomology for a scheme X smooth over Spec $\mathbf{C}$. The proof, which is essentially the one given in [1], is elementary once one has the theorem of Grauert recalled below (4.3 (iii)), which says that every finite covering of $X(\mathbf{C})$ comes from an étale covering of X; this is why we give it here, although we shall prove a more complete result later, using Hironaka's resolution of singularities theorem [3][34]. Furthermore, the good neighborhoods constructed in Section 3 could be useful in other situations. 65

Note that étale cohomology and classical cohomology do not agree for sheaves that are not torsion. For example, one has $H^1(X_{\text{et}}, \mathbf{Z}) = 0$ for every normal scheme X (IX 3.6 (ii)), but classical cohomology does not necessarily vanish.

2. Existence of sufficiently general hyperplane sections

The following result is classical, but a reference is lacking (pending EGA V):

Theorem 2.1. *Let k be a field and let $V \subset \mathbf{P}^n_k$ be a projective algebraic scheme purely of dimension r. Let V' be the open subscheme of points of V smooth over* Spec k*, and let P be a rational point of $\mathbf{P}^n$. Then:*

(i) *A "sufficiently general" linear subspace L of a given dimension d in $\mathbf{P}^n$ meets V transversely at every point of $L \cap V'$.*

(ii) *Let $\{H_1, \dots, H_s\}$ $(s \leq r)$ be a system of hypersurfaces of degree $d \geq 2$ passing through P, and let $Y = H_1 \cap \cdots \cap H_s$. If $\{H_1, \dots, H_s\}$ is sufficiently general, then Y meets V transversely at every point of $Y \cap V'$.*

The phrase "for a sufficiently general linear subspace" means that there exists a dense open subset U of the Grassmannian parametrizing the linear subspaces of dimension d, such that the assertion under consideration about L is true whenever the parameter of L lies in U. Thus this does not imply that such an L (defined over k) exists if k is not infinite. Likewise, the hypersurfaces of degree d passing through P are parametrized by 66 a projective space $\mathbf{P}^N$, and the phrase "for $\{H_1, \dots, H_s\}$ sufficiently general" means that there exists a dense open subset U of $(\mathbf{P}^N)^s$ such that the assertion under consideration is true whenever the parameter of $(H_1, \dots, H_s)$ lies in U.

Proof of (i): By induction one is reduced immediately to the case where L is a hyperplane. Let (V'_ν) be a finite open covering of V'. If assertion (i) is true when the

[34] Note, moreover, that Hironaka's resolution of singularities also gives a very simple proof of Grauert's result, and in this capacity may in any case be regarded as underlying the comparison theorems. See in this connection the exposé by Mme M. Raynaud in SGA 1 XII (North Holland Pub. Cie).

symbol V' is replaced by V'_ν for every ν, then it is also true for V', because the intersection of finitely many dense open subsets is again open and dense. We may therefore replace the situation by an affine algebraic scheme, say $V \subset \mathbf{E}^n_k = \operatorname{Spec} k[x_1, \dots, x_n]$. The hyperplane L will be the zero set of a linear equation

$$\ell(x) = a_0 + \sum_{i=1}^{n} a_i x_i.$$

Recall that a smooth scheme is locally a complete intersection. We may therefore replace V' by an open subset of points of V such that there exist polynomials $f_1(x), \dots, f_{n-r}(x)$ vanishing on V and indices $1 \le j_\nu \le n$ ($\nu = 1, \dots, n-r$) such that the determinant

$$\left| \frac{\partial f_i}{\partial x_{j_\nu}} \right| \qquad (i, \nu = 1, \dots, n-r)$$

is invertible. Say $j_\nu = \nu$ ($\nu = 1, \dots, n-r$). We may then solve the equations

$$y_j = \sum_{i=1}^{n-r} c_i \frac{\partial f_i}{\partial x_j}$$

67 for c_i as rational functions of the variables x_i, y_j ($i = 1, \dots, n-r; j = 1, \dots, n$), and these functions $c_i(x, y)$ are defined at every point of V'.

Now to say that L meets V transversely at a point Q of V' is equivalent to saying that the value of the matrix

$$\begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & & \vdots \\ \frac{\partial f_{n-r}}{\partial x_1} & & \frac{\partial f_{n-r}}{\partial x_n} \\ a_1 & \cdots & a_n \end{bmatrix}$$

at Q has rank $n - r + 1$. Thus, for L to meet V transversely at every point of V', it is necessary and sufficient that the system of $r + 1$ equations

$$\begin{aligned} a_j &= \sum_{i=1}^{n-r} c_i(x, a) \frac{\partial f_i}{\partial x_j} \qquad (j = n-r+1, \dots, n) \\ a_0 &+ \sum_{i=1}^{n} a_i x_i = 0 \end{aligned}$$

have no solution on V'. Such a condition is plainly constructible in (a): Let $X = V' \times A$, where A is the affine space with coordinates $a_0, \dots, a_n$, and let $Y \subset X$ be the closed subset of solutions of these equations. The condition is that the fiber of Y/A be empty,
68 which is indeed a constructible condition (EGA IV 9.5.1). It is therefore enough to verify that the fiber above the generic point of A is empty, which follows immediately from the form of the equations. This completes the proof of (i). (ii): Consider the rational map $\mathbf{P}^n \to \mathbf{P}^N$ given by the linear system of hypersurfaces of degree d ($d \ge 2$) passing through P. It is well known (and easily verified) that this is a "blowing up" of P, and therefore gives a locally closed immersion

$$f : \mathbf{P}^n - \{P\} \longrightarrow \mathbf{P}^N.$$

Let $V'' = V' - (\{P\} \cap V')$, let $\overline{V}'' = F(V'')$, and let L_i be the hyperplane of $\mathbf{P}^N$ corresponding to the hypersurface H_i ($i = 1, \dots, s$). Then V'' is isomorphic to $\overline{V}''$, and by the definition of f, to say that Y meets V'' transversely is equivalent to saying that $L_1 \cap \cdots \cap L_s$ meets $\overline{V}''$ transversely. Putting $\overline{V}$ = the closure of $\overline{V}''$, we are reduced to assertion (i) for $\overline{V}$, which implies (ii) with V'' in place of V'. It remains to consider the point P, if it is a smooth point of V. But it is evident that if $\{H_1, \ldots, H_s\}$ is sufficiently general and P is smooth on V, then Y meets V transversely at P, whence the result.

3. Construction of good neighborhoods

Definition 3.1. *An elementary fibration is a morphism of schemes $f : X \to S$ which can be embedded in a commutative diagram*

$$\begin{tikzcd} X \arrow[r, "j"] \arrow[dr, "f"'] & \overline{X} \arrow[d, "\overline{f}"] & Y \arrow[l, "i"'] \arrow[dl, "g"] \\ & S & \end{tikzcd}$$

satisfying the following conditions: 69

(i) *j is an open immersion, dense in every fiber, and $X = \overline{X} - Y$.*
(ii) *$\overline{f}$ is smooth and projective, with geometrically irreducible fibers of dimension* 1.
(iii) *g is an étale covering, and every fiber of g is nonempty.*

One easily checks that such an embedding of X, if it exists, is unique up to canonical isomorphism.

Definition 3.2. *A good neighborhood relative to S is an S-scheme X for which there exist S-schemes*

$$X = X_n, \ldots, X_0 = S$$

and elementary fibrations $f_i : X_i \to X_{i-1}$, $i = 1, \ldots, n$.

Proposition 3.3. [34] *Let k be an algebraically closed field, let $X/\operatorname{Spec} k$ be a smooth scheme, and let $x \in X$ be a rational point. There exists an open subset of X containing x which is a good neighborhood (relative to* $\operatorname{Spec} k$*).*

Proof. Take X irreducible. By induction on $\dim X = n$, it suffices to find a neighborhood U of x in X and an elementary fibration $f : U \to V$, with V smooth and of dimension $n-1$. Indeed, there will exist a neighborhood V' of $v = f(x)$ which is a good neighborhood, and we may take $U' = U \cap V'$ as a good neighborhood of x.

We may suppose that $X \subset \mathbf{E}^r$ is affine. Let X_0 be the closure of X in $\mathbf{P}^r$. Let $\overline{X}$ be the normalization of X_0 and let $Y = \overline{X} - X$, endowed with the induced reduced structure. In addition, let $S \subset \overline{X}$ be the closed subset of singular points. One has $S \subset Y$ and 70

$$\begin{aligned} \dim \overline{X} &= \dim X = n, \\ \dim Y &= n - 1, \\ \dim S &\leq n - 2. \end{aligned}$$

Embed $\overline{X}$ in a projective space $\mathbf{P}^N$ by means of an invertible sheaf L of the form $M^{\otimes r}$, with M very ample and $r \geq 2$. There then exist hyperplanes $H_1, \ldots, H_{n-1}$ of $\mathbf{P}^N$,

[34] This result extends without difficulty to a semilocal assertion.

where H_i is the zero set of

$$\sum_{\nu=0}^{N} a_{i\nu} x_\nu = 0,$$

which contain x and such that the intersection $L = H_1 \cap \cdots \cap H_{n-1}$ has dimension $N-n+1$ and meets $\overline{X}$ and Y transversely (2.1). The intersection $\overline{X} \cap L$ is a smooth connected curve (this follows from Bertini's theorem), and $Y \cap L$ has dimension 0. Choose in addition another hyperplane

$$H_0 : \sum_{\nu=0}^{N} a_{0\nu} x_\nu = 0$$

such that H_0 meets $X \cap L$ transversely and $H_0 \cap Y \cap L = \varnothing$.

Consider the projection $\mathbf{P}^N \to \mathbf{P}^{n-1}$ obtained by "introducing the projective coordinates"

$$y_i = \sum_{\nu=0}^{N} a_{i_\nu} x_\nu.$$

It is a rational map defined outside the center of projection $C = H_0 \cap \cdots \cap H_{n-1}$. Let $\epsilon : P' \to \mathbf{P}^N$ be the blowup of C, so that there is a diagram

$$\begin{tikzcd}
\mathbf{P}^N \arrow[dr] & & P' \arrow[ll, "\epsilon"'] \arrow[dl, "\pi"] \\
& \mathbf{P}^{n-1} &
\end{tikzcd}$$

71 where π is a morphism. Let $\overline{X}' \subset P'$ be the "proper" inverse image of $\overline{X}$, that is, the closure of $\epsilon^{-1}(\overline{X} - (\overline{X} \cap C))$. Since, by hypothesis, C meets $\overline{X}$ transversely, the morphism $\overline{X}' \to \overline{X}$ is a blowup of the finite set $\overline{X} \cap C$. Let $X' = X - \overline{X} \cap C$, which is also identified with an open subscheme of $\overline{X}'$, and let $Y' = \overline{X}' - X'$ be the closed subscheme with the induced reduced structure. There is a diagram $\mathbf{D}$ of morphisms

$$\begin{tikzcd}
X' \arrow[r, "j"] \arrow[dr, "f'"'] & \overline{X}' \arrow[d, "\overline{f}'"] & Y' \arrow[l, "i"'] \arrow[dl, "g'"] \\
& \mathbf{P}^{n-1} &
\end{tikzcd},$$

and I claim that there exists a neighborhood V of $v = f'(x)$ such that the restriction of $\mathbf{D}$ to V satisfies conditions (i), (ii), (iii) of (3.1), hence that $f'_{|V}$ is an elementary fibration. This will complete the proof.

Condition (i) is trivial. For (ii), note that $\overline{X} \cap L$ is a smooth curve by hypothesis, and one sees immediately that there is a one-to-one morphism $\overline{f}'^{-1}(v) \to \overline{X} \cap L$ induced by ϵ. Hence $\left(\overline{f}'^{-1}(v)\right)_{\mathrm{red}} \to \overline{X} \cap L$ is bijective. To check that $\overline{f}'$ is smooth over a neighborhood of v, it suffices, by Hironaka's lemma (SGA 1 II 2.6 [34]), to check that it is

[34] and EGA IV 5.12.10.

smooth at the generic point of $\overline{f}'^{-1}(v)$. At this point, $\overline{X}'$ is isomorphic to $\overline{X}$, and the morphism is smooth because L meets $\overline{X}$ transversely.

It remains to prove that g' is étale in a neighborhood of v (since Y has dimension $n-1$, it is evident that every fiber of g' is nonempty). One has $Y' = \epsilon^{-1}(Y) \amalg D_1 \amalg \cdots \amalg D_r$, where $\overline{X} \cap C = \{P_1, \ldots, P_r\}$ and D_i is the blowup of P_i in $\overline{X}$. Every D_i is identified 72
with $\epsilon^{-1}(P_i)$, and maps isomorphically onto $\mathbf{P}^{n-1}$ is defined at every point of Y, and the induced morphism on Y is none other than $g'_{|\epsilon^{-1}(Y)}$. It is étale over v, because L meets Y transversely, hence g' is étale over v.

4. The comparison theorem

4.0. Let X be a scheme locally of finite type over Spec C, and consider the usual topology on the space $X(\mathrm{C})$ of rational points of X. We shall denote by X_{cl} the site of *local isomorphisms* $f : U \to X(\mathrm{C})$, that is, of morphisms f of topological spaces having the following property:

For every $x \in U$ there exists an open neighborhood U_x such that $f_{|U_x}$ is a homeomorphism of U_x onto an open neighborhood of $f(x)$.

A family $\{U_\nu \to U\}$ of morphisms of X_{cl} is said to be *covering* if U is the union of the images of the U_ν.

Since an open immersion is a local isomorphism, we have a morphism (inclusion) of sites (IV) [34]

$$\delta : X_{\mathrm{cl}} \longrightarrow X(\mathrm{C}) \qquad (*). \tag{4.1}$$

Now for every local isomorphism $U \to X(\mathrm{C})$ there exists, by definition, a "covering" of U by open subsets of $X(\mathrm{C})$, and one deduces from ( ) that the topoi associated with the two sites are equivalent (by δ_*). In particular, one may replace $X(\mathrm{C})$ by X_{cl} when computing ordinary cohomology.

Now let $f : X' \to X$ be an étale morphism of schemes. Then $f(\mathrm{C}) : X'(\mathrm{C}) \to X(\mathrm{C})$ is a local isomorphism. This follows immediately from the Jacobian criterion (SGA 1 II 73
4.10) and the implicit function theorem. The functor $X' \mapsto X'(\mathrm{C})$ therefore gives a morphism of sites

$$\epsilon : X_{\mathrm{cl}} \longrightarrow X_{\mathrm{et}}, \tag{4.2}$$

where X_{et} is the étale site.

Recall the following results[34]:

THEOREM 4.3. (i) *X is connected and nonempty if and only if $X(\mathrm{C})$ is connected and nonempty.*

(ii) *The functor ϵ is fully faithful.*

(iii) *("Grauert–Remmert theorem"). The functor ϵ induces an equivalence between the category of finite coverings of $X(\mathrm{C})$, endowed with its usual topology as a locally compact space, and the category of étale coverings of X.*

We shall accept (i) as known. It follows, for example, from (GAGA) [4]. Assertion (ii) follows easily from (i). Indeed, to verify the surjectivity assertion contained in (ii), let $\varphi : X'(\mathrm{C}) \to X''(\mathrm{C})$ be an $X(\mathrm{C})$-morphism. We may suppose that all the schemes are connected and affine, and hence separated. Then the graph Γ of φ is a connected

[34] More precisely, the inclusion of categories $\mathrm{Ouv}(X(\mathrm{C})) \to X_{\mathrm{cl}}$ defines a morphism of sites (4.1) in the opposite direction.

[34] Which appear in the Exposé cited in the footnote on p. 2.

component of $X' \times_X X''(\mathbf{C})$, whence $\Gamma = Y(\mathbf{C})$ for some connected component Y of $X' \times_X X''$, by (i). Then Y is the graph of a morphism $f : X' \to X''$ inducing φ, i.e. the projection $Y \to X'$ is an isomorphism: indeed, $Y(\mathbf{C}) \to X(\mathbf{C})$ is an isomorphism, from which it follows easily that the same is true of $Y \to X'$ (which is étale, radicial, and surjective!). For (iii), in view of (ii), everything amounts to proving that every finite
74 covering of $X(\mathbf{C})$ is of the form $X'(\mathbf{C})$, where $X' \to X$ is étale.

For this, descent (SGA 1 IX 4.7) immediately reduces us to the case in which X is connected and normal. The problem is local on X (for the Zariski topology), and we may therefore suppose X affine. Let $f : X \to \mathbf{E}^n$ be a finite morphism, and let $T' \to X(\mathbf{C})$ be a connected finite covering. One checks immediately that the composite morphism $\eta' : T' \to \mathbf{E}^n(\mathbf{C})$ is an "analytic covering" in the sense of ([2] § 2, Def. 3). By ([2] § 2, Satz 8), extends to an "analytic covering" $\eta : T \to \mathbf{P}^n(\mathbf{C})$, and it is enough to prove that η is induced by a suitable morphism of schemes $f : Y \to \mathbf{P}^n$, with Y normal. Now the fundamental theorem ([2] § 13 Satz 42) asserts that there is a canonical normal analytic structure on T such that η induces a morphism of analytic spaces. Since η is then a finite morphism, the direct image of the structure sheaf on T is a sheaf $\mathfrak{a}$ of coherent analytic algebras on $\mathbf{P}^n$. It follows from GAGA [4] that $\mathfrak{a}$ is the analytic sheaf associated with an algebraic sheaf A on $\mathbf{P}^n$, and we take $Y = \operatorname{Spec} A$.

We can now prove the following result:

Theorem 4.4. *Let X be a smooth scheme over* $\operatorname{Spec}\mathbf{C}$.

(i) *There is an equivalence between the category of constructible locally constant torsion sheaves on X_{et} and the category of locally constant torsion sheaves with finite fibers on X_{cl}, the equivalence being given by the quasi-inverse functors $_*$ and *.*

(ii) *Let F be a locally constant torsion sheaf with finite fibers on X_{cl}, and also denote by F the induced sheaf $\epsilon_* F$ on X_{et}. Then*

$$R^q\epsilon_* F = 0 \quad \text{if } q > 0.$$

75 (iii) *With the notation of* (i)*, the canonical morphism*

$$H^q(X_{\mathrm{et}}, F) \longrightarrow H^q(X_{\mathrm{cl}}, F)$$

is bijective for every q. In particular,

$$H^q(X_{\mathrm{et}}, \mathbf{Z}/n) \xrightarrow{\sim} H^q(X_{\mathrm{cl}}, \mathbf{Z}/n)$$

for every q.

Proof. For (i), note that the sheaves in question are those that are representable by étale coverings of X (resp. by finite coverings of $X(\mathbf{C})$). Since there is an equivalence between the categories of these coverings (4.3), assertion (i) follows immediately. The isomorphism of (iii) follows from (ii) and from the Leray spectral sequence V 5.2 for ϵ. It remains to prove (ii).

Now $R^q\epsilon_* F$ is the sheaf associated with the presheaf $\mathscr{R}^q F$,, where $(\mathscr{R}^q F)(X') = H^q(X'_{\mathrm{cl}}, F)$ for $X' \to X$ étale. To prove that $R^q\epsilon_* F = 0$ if $q > 0$, we must therefore prove the following :

Lemma 4.5. *Let $\xi \in H^q(X_{\mathrm{cl}}, F)$ and $x \in X_{\mathrm{cl}} = X(\mathbf{C})$. There exists an étale morphism $X' \to X$ whose image contains x and such that the image of ξ in $H^q(X'_{\mathrm{cl}}, F)$ is zero.*

Proof of the lemma. Induction on $n = \dim X$; the problem is local on X for the étale topology, and we may therefore suppose F constant, and by dévissage we may even suppose that $F = \mathbf{Z}/n$. Moreover, replacing X by a Zariski open neighborhood of x, we

may suppose (3.3) that X admits an elementary fibration (3.1) $f : X \to S$. Retain the notation of (3.1): by direct calculation, one sees that $R^q j_* F = 0$ if $q > 1$, that $R^1 j_{\mathrm{cl}*} F$ is 76
the extension by 0 of a constant sheaf on Y_{cl}, and finally that $j_{\mathrm{cl}*} F$ is the constant sheaf $\mathbf{Z}/n$ on $\overline{X}$. Moreover, one may compute $R^q \overline{f}_{\mathrm{cl}*}(j_{\mathrm{cl}*} F)$ fiber by fiber. From the Leray spectral sequence

$$E_2^{pq} = R^p \overline{f}_{\mathrm{cl}*}(R^q j_{\mathrm{cl}*} F) \Longrightarrow R^{p+q} f_{\mathrm{cl}*} F$$

one deduces that $R^{p+q} f_{\mathrm{cl}*} F$ may likewise be computed fiber by fiber, and hence that

$R^0 f_{\mathrm{cl}*} \mathbf{Z}/n = \mathbf{Z}/n$,

$R^1 f_{\mathrm{cl}*} \mathbf{Z}/n$ is a locally constant torsion sheaf on S_{cl},

$R^q f_{\mathrm{cl}*} \mathbf{Z}/n = 0$ if $q > 1$.

The Leray spectral sequence

$E_2^{pq} = H^p(S_{\mathrm{cl}}, R^q f_{\mathrm{cl}*} F) \Longrightarrow H^{p+q}(X_{\mathrm{cl}}, F)$

thus reduces to the exact sequence

$$(*) \qquad \cdots \longrightarrow H^q(S_{\mathrm{cl}}, f_{\mathrm{cl}*} F) \longrightarrow H^q(X_{\mathrm{cl}}, F) \longrightarrow H^{q-1}(S_{\mathrm{cl}}, R^1 f_{\mathrm{cl}*} F) \longrightarrow$$

Let s be the image of x in S. By the induction hypothesis and by (i), for every class $\eta \in H^q(S_{\mathrm{cl}}, G)$ (G locally constant torsion, with finite fibers), there exists an "étale neighborhood" $S' \to S$ of s such that the image of η in $H^q(S'_{\mathrm{cl}}, G)$ is zero. Now the functors $R^q f_{\mathrm{cl}*}$ plainly commute with étale base changes, and if $S' \to S$ is an étale neighborhood of s, then $X' = X \times_S S' \to X$ is an étale neighborhood of x. When $q \geq 1$, we therefore conclude by means of the exact sequence $(*)$, first taking $G = R^1 f_{\mathrm{cl}*} F$ and then $G = f_{\mathrm{cl}*} F$. When $q = 1$, note that $H^1(X_{\mathrm{cl}}, \mathbf{Z}/n\mathbf{Z})$ (resp. $H^1(X, \mathbf{Z}/n\mathbf{Z})$) classifies 77
the principal étale coverings of X_{cl} (resp. X), with group $\mathbf{Z}/n\mathbf{Z}$; hence, by 4.3 (iii), the map $H^1(X, \mathbf{Z}/n\mathbf{Z}) \to H^1(X_{\mathrm{cl}}, \mathbf{Z}/n\mathbf{Z})$ is bijective, from which the effacement result 4.5 for $q = 1$ follows at once.

Variant 4.6. One may also prove (4.5) in the following elegant way, due to Serre: one notes that a connected good neighborhood (3.2) X satisfies the following two conditions:

(i) $\pi_n(X(\mathbf{C})) = 0$ if $n > 1$.
(ii) $\pi_1(X(\mathbf{C}))$ is an iterated extension of free groups of finite type.

Indeed, if $X \to S$ is an elementary fibration where S is a good neighborhood, we may suppose (i) and (ii) true for S by induction. Now $X(\mathbf{C}) \to S(\mathbf{C})$ is a locally trivial fiber space, as is readily seen, with fibers isomorphic to $X_0(\mathbf{C})$, where X_0 is an arbitrary fiber of X/S. Now X_0 is a connected nonsingular *noncomplete* curve, and it is well known that this implies that $\pi_n(X_0(\mathbf{C})) = 0$ if $n > 1$, and that $\pi_1(X_0(\mathbf{C}))$ is a free group. Then (i) and (ii) follow from the homotopy exact sequence

$$\longrightarrow \pi_0(X_0(\mathbf{C})) \longrightarrow \pi_n(X(\mathbf{C})) \longrightarrow \pi_n(S(\mathbf{C})) \longrightarrow \ldots$$

Thus $X(\mathbf{C})$ is a $K(\pi, 1)$ space, and this implies that every cohomology class $\xi \in H^n(X(\mathbf{C}), F)$ becomes zero on the universal covering of $X(\mathbf{C})$. It is enough to prove that ξ already becomes zero on a finite covering X' of X, which amounts to proving that $\pi_1 = \pi_1(X(\mathbf{C}))$ is a *good group* (cf. CG 1 16), that is, that the morphism from π_1 to 78
its completion $\pi_1 = \varprojlim_N \pi_1/N$ (where N runs through the set of invariant subgroups of π_1 of finite index) induces a bijection

$$H^q(\hat{\pi}_1, M) \xrightarrow{\sim} H^q(\pi_1, M)$$

for every continuous finite π_1-module M (where the symbol on the left denotes the cohomology of $\widehat{\pi}_1$ as a profinite group). Now an iterated extension of free groups of finite type is a good group (cf. CG I p. 15-16, exc. 1,2), whence the result.

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EXPOSÉ XII

The base-change theorem for a proper morphism

M. Artin

1. Introduction

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In this exposé we prove the fundamental theorem which says, in a slightly more precise form, that for a *proper* morphism $f : X \to Y$ of schemes and a torsion abelian sheaf F on X, the fiber of $R^q f_* F$ at a geometric point ξ of Y is isomorphic to the cohomology $H^q(X_\xi, F|X_\xi)$ of the fiber X_ξ.

Recall that the analogous result for a proper morphism of paracompact spaces is well known (Godement II 4.11) and easy. It is considerably more delicate for schemes. For example, in contrast with the case of paracompact spaces, the hypothesis that F be torsion is essential.

The proof proceeds in several stages. By more or less formal dévissage arguments, one reduces to the case where Y is Noetherian, the relative dimension of f is at most 1, and the sheaf F is constant. To handle this special case, one uses the specialization theorem for the fundamental group (5.9), which is a nonabelian variant of the theorem stated above and can be proved directly in the case under consideration (XIII 2).

2. An example

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We show that the hypothesis that F be torsion is necessary. Let X be a nonsingular surface over an algebraically closed field k, let Y be a nonsingular curve over k, and let $f : X \to Y$ be a proper morphism whose generic fiber is smooth and whose geometric fiber at a geometric point $\overline{y}$ of Y is irreducible and has an ordinary double point. Let F be the constant sheaf $\mathbf{Z}_X$.

Recall the following fact. For any point z of a scheme Z, let $i : z \to Z$ be the inclusion. Then $H^1(Z, i_* \mathbf{Z}_z) = 0$ (IX 3.6 (i)). Since X is normal, hence unibranch, it follows that $\mathbf{Z}_X \simeq i_* \mathbf{Z}_x$, where $i : x \to X$ is the inclusion of the generic point. Thus $H^1(X, \mathbf{Z}_X) = 0$ and likewise $R^1 f_* \mathbf{Z}_X = 0$. But for the geometric fiber $X_\circ$ of X/Y having the ordinary double point, I claim that $H^1(X_\circ, \mathbf{Z}_{X_\circ}) \simeq \mathbf{Z}$[34]; this gives the counterexample, since $\mathbf{Z}_{X_\circ} \simeq \mathbf{Z}_X | X_\circ$.

Let $i_\circ : x_\circ \to X_\circ$ be the inclusion of the generic point of $X_\circ$, and let Q be the singular point of $X_\circ$. The curve $X_\circ$ has two "branches" at Q, and it follows that the fiber of $i_{\circ *}(\mathbf{Z}_{x_\circ})$ at Q is isomorphic to $\mathbf{Z} \times \mathbf{Z}$. Away from Q, the curve $X_\circ$ is normal, hence $i_{\circ *}(\mathbf{Z}_{x_\circ}) \simeq \mathbf{Z}_{X_\circ}$ on $X_\circ - Q$. Therefore there is an exact sequence

$$0 \longrightarrow \mathbf{Z}_{X_\circ} \longrightarrow i_{\circ *}(\mathbf{Z}_{x_\circ}) \longrightarrow (\mathbf{Z})_Q \longrightarrow 0,$$

where the last term is extension by zero of the sheaf $\mathbf{Z}$ at the point Q. All three terms have H^0 isomorphic to $\mathbf{Z}$, and $H^1(X_\circ, i_{\circ *}(\mathbf{Z}_{x_\circ})) = 0$. The long exact cohomology sequence therefore gives $H^1(X_\circ, \mathbf{Z}_{X_\circ}) = \mathbf{Z}$. 81

[34] Compare, from the point of view of Π_1, with SGA 3 X (6, Examples) and (1.6).

3. Review of nonabelian H^1

We briefly recall some results on nonabelian cohomology that will be needed below. We omit most proofs, which are well known in various special cases and may be found in Giraud's thesis [1].

Let X be a scheme and F a sheaf of groups on X. For every étale X'/X, one defines a pointed set $H^1(X', F)$, covariantly functorial in F and contravariantly functorial in X'. The functor H^1 commutes with finite products of sheaves. If $f : X \to Y$ is a morphism, one defines a sheaf of pointed sets (that is, with a distinguished section) $R^1 f_* F$. It is the sheaf associated with the presheaf of pointed sets $\mathscr{R}^1 F$ which assigns to every étale Y'/Y the set $H^1(X \times_Y Y', F)$.

There are the usual exact cohomology sequences; we record one of them:

Proposition 3.1. *Let $0 \to F \xrightarrow{u} G$ be an injection of sheaves of groups, and let $C = G/F$ be the sheaf of homogeneous spaces under G. There is an exact sequence of sheaves of pointed sets*

$$0 \longrightarrow f_*F \longrightarrow f_*G \longrightarrow f_*C \xrightarrow{\partial} R^1 f_*F \xrightarrow{u^1} R^1 f_*G.$$

This exact sequence is obtained, of course, by passing to the associated sheaves in the exact sequence of presheaves whose value on an étale Y'/Y is

$$0 \longrightarrow F(X') \longrightarrow G(X') \longrightarrow C(X') \xrightarrow{\partial} H^1(X', F) \longrightarrow H^1(X', G),$$

82 where $X' = X \times_Y Y'$. The equivalence relation induced by ∂ is the one given by the action of f_*G on f_*C. As usual, the equivalence relation induced by u^1 may be made explicit using the notion of "twisting" [1].

Proposition 3.2. *Let $X \xrightarrow{f} Y \xrightarrow{g} Z$ be morphisms of schemes. For every étale Z'/Z, putting $Y' = Y \times_Z Z'$ and $X' = X \times_Z Z'$, there is an exact sequence*

$$0 \longrightarrow H^1(Y', f_*F) \longrightarrow H^1(X', F) \longrightarrow H^0(Y', R^1 f_*F).$$

Passing to the associated sheaves gives an exact sequence of pointed sheaves on Z:

$$0 \longrightarrow (R^1 g_*) f_*F \longrightarrow R^1(gf)_*F \longrightarrow g_*(R^1 f_*)F.$$

Proposition 3.3. *The family of functors of the form $H^1(X', F)$, for X' étale over X, and of the form $R^1 f_*F$, for an arbitrary morphism $f : X \to Y$, is effaceable: there exists an injection $F \xrightarrow{u} G$ such that all the induced morphisms $H^1(X', F) \to H^1(X', G)$ and $R^1 f_*F \to R^1 f_*G$ are zero. More precisely, there exists such an injection for which every $H^1(X', G)$, with X' étale over X, and every pointed sheaf $R^1 f_*G$ vanish. If X is Noetherian and F is a constructible sheaf of ind-$\mathbf{L}$-groups, then such an injection exists with G also a sheaf of ind-$\mathbf{L}$-groups.*

Proof. The first assertion is presumably true in an arbitrary topos, but for the étale topology it is convenient to perform the effacement using the "Godement resolution"

$$F \longrightarrow G = \prod_{x \in X} i_{x*} i_x^* F. \tag{3.4}$$

83 Here x runs through the points of X and $i_x : \overline{x} \to X$ is a geometric point over x. For

X'/X étale, one has

$$G(X') = \prod_{x\in X}(i_x^* F)(X' \times_X \overline{x}) = \prod_{x\in X}\left(\prod_{\overline{x}'/\overline{x}} F_{\overline{x}'}\right),$$

where the $\overline{x}'$ are geometric points of X'. For each point of X' there is plainly at least one $\overline{x}'$ above it; hence $F(X') \subset G(X')$ (VIII 3), and therefore $F \subset G$. Moreover, every étale $\overline{u}/\overline{x}$ is a disjoint union of spectra of separably closed fields, so $H^1(\overline{u}, \overline{G}) = 0$ for every sheaf $\overline{G}$ on $\overline{x}$. It follows from 3.2 that $H^1(X', i_{x*}i_x^* F) = 0$ for every étale X'/X. Since H^1 commutes with products, $H^1(X', G) = 0$, and consequently $R^1 f_* G = 0$ for every $f : X \to Y$, by the description of $R^1 f_* G$ as the sheaf associated with the presheaf $Y' \mapsto H^1(X \times_Y Y', G)$.

Suppose in addition that F is constructible and X is Noetherian. It is immediate that there is a finite set of points $\{x_1, \dots, x_n\}$ of X such that

$$F \longrightarrow \prod_{1\le i\le n} i_{x_i *} i_{x_i}^* F$$

is already injective. The sheaf $i_{x_i *} i_{x_i}^* F$ is a sheaf of ind-finite groups (IX 1.5), and hence so is the finite product (IX 1.4 and IX 1.6 (iii)). Replacing G by this sheaf gives the last assertion.

3.5. As in the case of abelian cohomology (treated in V 2.4), H^1 may be computed by the Čech procedure [1].

4. The base-change morphism

Let 84

$$\begin{array}{ccc} X & \xleftarrow{g'} & X' \\ {\scriptstyle f}\downarrow & & \downarrow{\scriptstyle f'} \\ S & \xleftarrow{g} & S' \end{array}$$

be a Cartesian square of morphisms of schemes. From it one obtains a morphism of functors $\varphi = \varphi_g$

$$g^* f_* \xrightarrow{\varphi} f'_* g'^* \tag{4.1}$$

as follows. Giving such a morphism is equivalent, by the adjunction between g^* and g_*, to giving a morphism of functors

$$f_* \longrightarrow g_* f'_* g'^*. \tag{4.1 adj}$$

We take this to be the morphism

$$f_* \xrightarrow{(\mathrm{id}\to g'_* g'^*)} f_* g'_* g'^* = g_* f'_* g'^*,$$

using

$$g_* f'_* = (g f')_* = (f g')_* = f_* g'_*.$$

There is another candidate for (4.1). Indeed, giving such a morphism is also equivalent to giving

$$f'^* g^* f_* \longrightarrow g'^*,$$

and one may take

$$f'^*g^*f_* \simeq g'^*f^*f_* \xrightarrow{(f^*f_*\to\mathrm{id})} g'^*.$$

85 I do not know whether these two choices always coincide [34]; in any event, we shall use only the first. We call it the *base-change morphism.*

Restricting (4.1) to abelian sheaves (respectively, to sheaves of groups) gives a morphism $\varphi^q = \varphi^q_g$ for every $q \geq 0$ (respectively, for $q = 0, 1$):

$$g^*(R^qf_*) \xrightarrow{\varphi^q} (R^qf'_*)g'^*. \tag{4.2}$$

Indeed, this is equivalent to giving

$$(R^qf_*) \longrightarrow g_*(R^qf'_*)g'^*, \tag{4.2 adj}$$

and there are the following "evident" morphisms:

$$\begin{aligned}(R^qf_*) \xrightarrow{a} (R^qf_*)g'_*g'^* &\xrightarrow{b} R^q(fg')_*g'^*\\ = R^q(gf')_*g'^* &\xrightarrow{c} g_*(R^qf'_*)g'^*.\end{aligned}$$

Remark 4.3. Suppose that g (and hence g') is a closed immersion. Then the morphisms b and c above are isomorphisms (VIII 5.7). It follows that (4.2) is essentially just the morphism a above, obtained by applying R^qf_* to the canonical morphism $F \to g'_*g'^*F$.

Proposition 4.4. *(Composition of base-change morphisms).*

(i) *Let*

$$\begin{tikzcd} X & X' \arrow[l, "g'"'] \arrow[d, "f'"] & X'' \arrow[l, "h'"'] \arrow[d, "f''"] \\ S & S' \arrow[l, "g"] & S'' \arrow[l, "h"] \arrow[from=1-1, to=2-1, "f"'] \end{tikzcd}$$

86 *be a Cartesian diagram of morphisms of schemes. Then the following diagram is commutative:*

$$\begin{tikzcd} (gh)^*f_* \arrow[rr, "\varphi_{gh}"] \arrow[d, "\wr"] & & f''_*(g'h')^* \arrow[d, "\wr"] \\ h^*g^*f_* \arrow[r, "h^*\varphi_g"] & h^*f'_*g'^* \arrow[r, "\varphi_h g'^*"] & f''_*h'^*g'^*. \end{tikzcd}$$

The same is true of the diagram obtained by replacing f_*, f'_*, and f''_* by R^qf_*, $R^qf'_*$, and $R^qf''_*$, in the natural domains of definition of the functors in question (cf. n° 3).

[34] Mr. Deligne resolved this uncertainty in the affirmative, in a considerably more general context; cf. Exposé XVII.

(ii) *Let*

$$\begin{tikzcd} Y \arrow[d, "e"'] & Y' \arrow[l, "g''"'] \arrow[d, "e'"] \\ X \arrow[d, "f"'] & X' \arrow[l, "g'"'] \arrow[d, "f'"] \\ S & S' \arrow[l, "g"'] \end{tikzcd}$$

be a Cartesian diagram of morphisms of schemes. Then the following diagram is commutative:

$$\begin{tikzcd} g^*(fe)_* \arrow[rr, "\varphi_g^{fe}"] \arrow[d, equal] & & (f'e')_* g''^* \arrow[d, equal] \\ g^* f_* e_* \arrow[r, "\varphi_g^f e_*"] & f'_* g'^* e_* \arrow[r, "f'_* \varphi_{g'}^e"] & f'_* e'_* g''^*, \end{tikzcd}$$

where the three horizontal arrows arise respectively from the base-change homomorphisms for fe, f, and e.

In the abelian case, the base-change morphisms for fe are induced by a morphism of spectral sequences

$$\begin{tikzcd} E_2^{p,q} = g^*(R^p f_*)(R^q e_*)F \arrow[r, Rightarrow] \arrow[d, "\varphi"'] & g^* R^{p+q}(fe)_* F \arrow[d, "\varphi"'] \\ (R^p f'_*)(R^q e'_*)g''^* F \arrow[r, Rightarrow] & R^{p+q}(f'e')_* g''^* F, \end{tikzcd}$$

which, on the initial terms, is induced by the base-change morphisms for e and f. For a sheaf of groups, there is a morphism of exact sequences

$$\begin{tikzcd} 0 \arrow[r] & g^*(R^1 f_*)e_* F \arrow[r] \arrow[d, "\varphi"] & g^* R^1(fe)_* F \arrow[r] \arrow[d, "\varphi"] & g^* f_*(R^1 e_*)F \arrow[d, "\varphi"] \\ 0 \arrow[r] & (R^1 f'_*)e'_* g''^* F \arrow[r] & R^1(f'e')_* g''^* F \arrow[r] & f'_*(R^1 e'_*)g''^* F. \end{tikzcd}$$

We leave the proof to the reader.

5. Statement of the main theorem and some variants

The theorem is as follows:

THEOREM 5.1. *(Proper base-change theorem). Let*

$$\begin{tikzcd} X \arrow[d, "f"'] & X' \arrow[l, "g'"'] \arrow[d, "f'"] \\ S & S' \arrow[l, "g"'] \end{tikzcd}$$

be a Cartesian diagram of schemes, with f proper.

(i) *Let F be a sheaf of sets on X. Then the morphism (4.1)*
$$\varphi : g^* f_* F \longrightarrow f'_* g'^* F$$
is bijective.

87 (ii) *Let F be a sheaf of groups (resp. of ind-finite groups) on X. Then the morphism (4.2)*
$$\varphi^1 : g^*(R^1 f_*)F \longrightarrow (R^1 f'_*)g'^* F$$
is injective (resp. bijective).

(iii) *Let F be a torsion abelian sheaf on X. Then the morphism (4.2)*
$$\varphi^q : g^*(R^q f_*)F \longrightarrow (R^q f'_*)g'^* F$$
is bijective for every $q \geq 0$.

We first state some immediate consequences of the theorem. If $S' = \xi$ is a geometric point of S, so that a sheaf on ξ is determined by its global sections, then $H^0(\xi, R^q f'_* G) = H^q(X', G)$ for every sheaf G on X'. This gives the following corollary (which is, moreover, essentially equivalent to 5.1):

Corollary 5.2. *Let $f : X \longrightarrow S$ be proper, let $\xi \longrightarrow S$ be a geometric point, and let X_ξ be the fiber of f at ξ.*

(i) *If F is a sheaf of sets on X, the canonical base-change morphism*
$$(f_* F)_\xi \longrightarrow H^0(X_\xi, F|X_\xi)$$
is bijective.

(ii) *If F is a sheaf of groups (resp. of ind-finite groups), the base-change morphism*
$$(R^1 f_* F)_\xi \longrightarrow H^1(X_\xi, F|X_\xi)$$
is injective (resp. bijective).

88 (iii) *If F is a torsion abelian sheaf, then for every $q \geq 0$ the base-change morphism*
$$(R^q f_* F)_\xi \longrightarrow H^q(X_\xi, F|X_\xi)$$
is bijective.

Taking X 4.3 and X 5.2 into account, we obtain:

Corollary 5.3. *Under the hypotheses of X 5.2, let n be the dimension of the fiber X_ξ. Then, for every torsion abelian sheaf F on X, the fiber $(R^q f_* F)_\xi$ vanishes for $q > 2n$. If F is a sheaf of p-torsion groups, where p is the characteristic of ξ, then the fiber of $R^q f_* F$ at ξ vanishes for $q > n$.*

Recall that the *relative dimension of X/S is* $\leq n$ if every fiber has dimension $\leq n$. We therefore have:

Corollary 5.3 bis. *Let $f : X \longrightarrow S$ be proper, with X/S of relative dimension $\leq n$. Then $R^q f_* F = 0$ for $q > 2n$ and every torsion abelian sheaf F on X. If S has characteristic p and F is a sheaf of p-torsion abelian groups, then $R^q f_* F = 0$ for $q > n$.*

Applying 5.1 to a morphism between spectra of separably closed fields gives:

Corollary 5.4. *(Invariance of cohomology under extension of the base field in the proper case). Let $k \subset K$ be separably closed fields and let X be a proper scheme over* Spec k. *Put $X' = X_K$. Let F be a sheaf on X and let F' be its inverse-image sheaf on X'. Then:*

(i) $H^0(X, F) \xrightarrow{\sim} H^0(X', F')$.

89 (ii) *If F is a sheaf of groups (resp. of ind-finite groups), the morphism*

$$H^1(X,F) \longrightarrow H^1(X',F')$$

is injective (resp. bijective).

(iii) *If F is a torsion abelian sheaf, then*

$$H^q(X,F) \xrightarrow{\sim} H^q(X',F')$$

for every q.

A slightly different variant of 5.2 is the following:

Corollary 5.5. *Let S be the spectrum of a henselian ring, and let $S' = s_\circ$ be its closed point. Let $f : X \longrightarrow S$ be proper. Put $X_\circ = X'$ and $F_\circ = g'^* F$. Then:*

(i) $H^0(X,F) \xrightarrow{\sim} H^0(X_\circ, F_\circ)$
for every sheaf of sets F.

(ii) *The morphism $H^1(X,F) \longrightarrow H^1(X_\circ, F_\circ)$*
is injective (resp. bijective) for every sheaf of groups (resp. of ind-finite groups) F.

(iii) $H^q(X,F) \xrightarrow{\sim} H^q(X_\circ, F_\circ)$
for every torsion abelian sheaf F and every q.

Indeed, this statement is just a special case of 5.2 when S is strictly local. Suppose that 5.2 is known. Let $\overline{S}$ be the strict localization of S, let $\overline{s}$ be the closed point of $\overline{S}$, and let G be the Galois group of $\overline{S}/S$ (which is also the Galois group of $\overline{s}/s_\circ$). Then G acts on the fibers of the R^q, and, with the evident notation, one has

$$H^0(X,F) \simeq H^0(\overline{X},\overline{F})^G,$$
$$H^0(X_\circ,F_\circ) \simeq H^0(\overline{X}_\circ,\overline{F}_\circ)^G$$

for a sheaf of sets F. Assertion (i) follows, since $H^0(\overline{X},\overline{F}) \xrightarrow{\sim} H^0(\overline{X}_\circ,\overline{F}_\circ)$ and this 90
isomorphism plainly commutes with the action of G.

If F is a sheaf of groups, one readily defines a morphism of exact sequences of pointed sets

(5.5.1)
$$\begin{array}{ccccccc}
0 \longrightarrow & H^1(G, H^0(\overline{X},\overline{F})) & \longrightarrow & H^1(X,F) & \longrightarrow & H^1(\overline{X},\overline{F})^G \\
& \downarrow & & \downarrow & & \downarrow \\
0 \longrightarrow & H^1(G, H^0(\overline{X}_\circ,\overline{F}_\circ)) & \longrightarrow & H^1(X_\circ,F_\circ) & \longrightarrow & H^1(\overline{X}_\circ,\overline{F}_\circ)^G,
\end{array}$$

in which the left-hand vertical arrow is an isomorphism and the right-hand one is injective by 5.2. Let $\alpha, \beta \in H^1(X,F)$ be classes having the same image in $H^1(X_\circ,F_\circ)$, and let F^α be the sheaf obtained by twisting F by α [1]. Let β' be the class in $H^1(X,F^\alpha)$ corresponding to β. Since the class α' corresponding to α is neutral, the image of β' in $H^1(X_\circ,(F^\alpha)_\circ)$ is neutral.

Consider the diagram deduced from (5.5.1) by replacing F with F^α. It shows that the image of β' in $H^1(\overline{X},\overline{F}^\alpha)^G$ is neutral, and hence that β' comes from an element of $H^1(G,H^0(\overline{X},\overline{F}^\alpha))$; this element must itself be neutral. Thus β' is neutral, that is, $\beta' = \alpha'$, whence $\beta = \alpha$. This proves the injectivity of the morphism in (ii).

Now suppose that F is a sheaf of ind-finite groups, and let us prove that the arrow in (ii) is surjective. It is enough to show that the functor $F \mapsto H^1(X_\circ,F_\circ)$ is effaceable.

Indeed, let $0 \longrightarrow F \xrightarrow{u} G$ be an injection which effaces this functor, and put $C = G/F$. There is a morphism of exact sequences (4.4)

$$\begin{tikzcd} H^0(X,C) \arrow[r] \arrow[d] & H^1(X,F) \arrow[r] \arrow[d, "\epsilon"'] & H^1(X,G) \arrow[d] \\ H^0(X_\circ,C_\circ) \arrow[r] & H^1(X_\circ,F_\circ) \arrow[r, "u^1_\circ"] & H^1(X_\circ,G_\circ) \end{tikzcd}$$

in which $u^1_\circ$ is zero; this indeed implies the surjectivity of ϵ.

91 Let us prove that the functor $F \mapsto H^1(X_\circ, F_\circ)$ is effaceable for sheaves F of ind-finite groups. Let $F \xrightarrow{u} C$ be the "Godement resolution" (3.4). We have

$$H^1(G, H^0(\overline{X}_\circ, \overline{C}_\circ)) \simeq H^1(G, H^0(\overline{X}, \overline{C})) \subset H^1(X, C) = 0.$$

Moreover, there is a commutative square

$$\begin{tikzcd} H^1(\overline{X},\overline{F}) \arrow[r, "\sim"] \arrow[d] & H^1(\overline{X}_\circ,\overline{F}_\circ) \arrow[d] \\ 0 = H^1(\overline{X},\overline{C}) \arrow[r] & H^1(\overline{X}_\circ,\overline{C}_\circ) \end{tikzcd}$$

(indeed, $H^1(\overline{X},\overline{C}) = \varinjlim H^1(X',C) = 0$, where X'/X runs through the étale morphisms), so its right-hand vertical arrow is zero. We therefore have a morphism of exact sequences

$$\begin{tikzcd} 0 \arrow[r] & H^1(G, H^0(\overline{X}_\circ,\overline{F}_\circ)) \arrow[r] \arrow[d] & H^1(X_\circ,F_\circ) \arrow[r] \arrow[d, "\beta"] & H^1(\overline{X}_\circ,\overline{F}_\circ)^G \arrow[d, "\alpha^G"] \\ & 0 \arrow[r] & H^1(X_\circ,C_\circ) \arrow[r] & H^1(\overline{X}_\circ,\overline{C}_\circ)^G \end{tikzcd}$$

in which α^G is zero. It follows that β is also zero, and hence that the functor is effaceable.

Finally, let us prove assertion (iii) of 5.5. There is a morphism of spectral sequences

$$\begin{tikzcd} E_2^{p,q} = H^p(G, H^q(\overline{X},\overline{F})) \arrow[r, Rightarrow] \arrow[d] & H^n(X,F) \arrow[d, "\varphi"] \\ H^p(G, H^q(\overline{X}_\circ,\overline{F}_\circ)) \arrow[r, Rightarrow] & H^n(X_\circ,F_\circ). \end{tikzcd}$$

By 5.2, the morphisms $H^q(\overline{X},\overline{F}) \longrightarrow H^q(\overline{X}_\circ,\overline{F}_\circ)$ are isomorphisms. Consequently the morphism of $E_2^{p,q}$ terms is an isomorphism, and therefore so is the morphism of abutments. This proves the result, C.Q.F.D.

92 **5.6.** Let us examine 5.5 (i) and (ii) when F is constant. Let $F = T_X$ be the constant sheaf of sets on X with value T. Then $F(X)$ is simply the set of locally constant maps from X to T. If $X_\circ$ is a subscheme of X and T has at least two elements, it follows that $F(X) \longrightarrow F(X_\circ)$ is bijective if and only if the map

$$U \longmapsto U_\circ = U \cap X_\circ$$

from the set $\Theta(X)$ of subsets of X that are both open and closed to the analogous set $\Theta(X_\circ)$ is bijective. If the set $\Pi_\circ(X_\circ)$ of connected components of $X_\circ$ is finite, this is equivalent to saying that the map $\Pi_\circ(X_\circ) \longrightarrow \Pi_\circ(X)$ is bijective. Thus 5.5 (i) may also be restated as follows (bearing in mind that in this case $X_\circ$ is of finite type over a field, hence Noetherian, so that $\Pi_\circ(X_\circ)$ is finite):

Corollary 5.7. *Let S be the spectrum of a henselian ring, let $s_\circ$ be its closed point, let $f : X \longrightarrow S$ be a proper morphism, and let $X_\circ$ be the closed fiber of X/S. Then the map*

$$\Pi_\circ(X_\circ) \longrightarrow \Pi_\circ(X)$$

induced by $X_\circ \longrightarrow X$ is bijective.

(In particular, this implies that $\Pi_\circ(X)$ is finite.) We shall prove 5.7 directly in the Noetherian case:

Lemma 5.8. *Statement 5.7 is true if S is Noetherian* [34].

Proof. In the Noetherian case, $\Pi_\circ(X)$ is certainly finite. Replacing X by a connected component, we may suppose that $\Pi_\circ(X)$ has one element, that is, that X is connected and nonempty. We must show that $X_\circ$ is also connected and nonempty. The image of X in S is closed because f is proper, and is nonempty because X is nonempty; it therefore contains $s_\circ$, which shows that $X_\circ$ is nonempty. Let

$$\begin{tikzcd}
X \arrow[dd, "f"'] \arrow[dr, "f'"] & \\
& S' \arrow[dl, "\pi"] \\
S &
\end{tikzcd}$$

be the Stein factorization of f (EGA III 4.3.3), so that S'/S is finite and S' is connected. 93
Every fiber of f' is known to be connected and nonempty, and the underlying space of $X_\circ$ is the union of the underlying spaces of the fibers over the closed points of S'. It remains only to show that S' is the spectrum of a *local* ring. Since S is henselian, the ring A' of S' is a product of local rings (VII 4.1 (i)); because S' is connected, A' is local. This proves 5.8.

5.9.0. Suppose now that $F = G_X$ is a constant sheaf of groups with value a finite group G. Then $H^1(X, G_X)$ classifies the principal Galois coverings of X with group G (VII 2.1). Corollary 5.5 (ii) says that the morphism $X_\circ \longrightarrow X$ induces a bijection between the sets of isomorphism classes of these coverings. By the definition of the Π_1, it follows that 5.5 (ii) for constant sheaves of finite groups (in the presence of (i)) is equivalent to:

Theorem 5.9. *(Specialization theorem for the fundamental group). With the notation of 5.5, suppose that X is connected and nonempty. Then $X_\circ$ is connected and nonempty; and, if $\overline{x}_\circ$ is a geometric point of $X_\circ$, the morphism of fundamental groups*

$$\Pi_1(X_\circ, \overline{x}_\circ) \longrightarrow \Pi_1(X, \overline{x}_\circ)$$

is an isomorphism.

On the other hand, by Galois theory (SGA V), statement 5.9 (in the case where $X_\circ$ 94
is connected, as we may suppose by 5.7) is equivalent to:

[34] Cf. EGA IV 18.5.19.

Theorem 5.9 bis. *Let S be the spectrum of a henselian ring, let $s_\circ$ be the closed point of S, let $f : X \longrightarrow S$ be a proper morphism, and let $X_\circ$ be the closed fiber of X/S. For a scheme Z, denote by* $\mathrm{Et}(Z)$ *the category of étale coverings of Z. Then the restriction functor*

$$\mathrm{Et}(X) \longrightarrow \mathrm{Et}(X_\circ)$$

is an equivalence of categories.

Remark 5.10. Recall that 5.9 was already proved (SGA 1 X 2.1) when S is the spectrum of a complete Noetherian local ring. Unfortunately, that result does not seem to yield a proof of 5.9 in general [34]. The proof of 5.9 given in this exposé and the next is very different from the one given in SGA 1 X.

5.11. One can give another "noncommutative" variant of 5.5 (ii), using J. Giraud's theory of 2-cohomology (cf. J. Giraud's thesis, in preparation). Suppose that S is Noetherian. Let L be a *band* on X, and suppose that L is represented locally (for the étale topology) by a sheaf of ind-finite groups. Consider the map

$$\varphi^{(2)} : H^2(X, L) \longrightarrow H^2(X_\circ, L_\circ),$$

where $L_\circ$ is the restriction of L to $X_\circ$. Then

$$(\varphi^{(2)})^{-1}(H^2(X_\circ, L_\circ)') = H^2(X, L)', \tag{5.11.1}$$

95 where the "primes" denote the subsets formed by the neutral elements.

In view of 5.5 (iii), this statement is also equivalent to the following:

For every gerbe $\mathscr{G}$ on X, with band L as above, let $\mathscr{G}_\circ$ denote its restriction to $X_\circ$. Then the restriction functor on the categories of sections

$$\mathscr{G}(X) \longrightarrow \mathscr{G}_\circ(X_\circ)$$

is an equivalence of categories.

Indeed, the fact that this functor is fully faithful follows from 5.5 (i), and the fact that it is essentially surjective when $\mathscr{G}(X) \neq \varnothing$, that is, when the class $\xi \in H^2(X, L)$ of $\mathscr{G}$ is neutral, follows from 5.5 (ii). It remains to express the fact that $\mathscr{G}_\circ(X_\circ) \neq \varnothing$ implies $\mathscr{G}(X) \neq \varnothing$, and this is precisely (5.11.1).

We shall not give the proof of (5.11.1) here. We merely point out that, for every Noetherian scheme X and every closed subscheme of X, one can in fact show that the validity of statements 5.5 (i), (ii) implies that of (5.11.1). The proof is fairly elementary and uses the descent result VIII 9.4 a).

6. First reductions

We use the abbreviation (f, F, g) for the data of 5.1 in any one of cases (i), (ii), or (iii) of that theorem.

Lemma 6.1. *For 5.1 to hold for the data (f, F, g), where f and F are fixed and g is arbitrary, it is enough that the theorem hold for $(\overline{f}, \overline{F}, \overline{g})$ for every S-scheme $\overline{S}$ that is the*
96 *strict localization of a scheme S' locally of finite type over S at a point closed in its fiber. Here $\overline{f}$ and $\overline{F}$ are obtained from f and F by the base change $\overline{S} \to S$, and $\overline{g} : \overline{s} \to \overline{S}$ is the inclusion of the closed point of $\overline{S}$.*

[34] The situation has changed since these lines were written; cf. Th. 3.1 in M. Artin, Algebraic approximation of structures over complete local rings (to appear).

Proof. We may plainly assume that S and S' are affine, say the spectra of rings A and A'. Then A' is the filtered direct limit of its finite-type A-subalgebras A'_i. Using the theory of passage to the limit (VII 5.11 and 5.14), we are reduced to proving 5.1 for the triples (f, F, g_i), where $g_i : \mathrm{Spec}(A'_i) = S'_i \to S$; this reduces us to the case in which g is of finite type. To prove that the base-change homomorphisms are isomorphisms, it is enough to prove that they induce isomorphisms on the fibers of the sheaves in question, and it is enough to consider geometric fibers at points s' of S' that are closed in their fibers over S (VIII 3.13). Let $\overline{s}'$ be a geometric point of S' corresponding to a separable closure of $k(s')$. Since s' is algebraic over the corresponding point s of S, $k(\overline{s}')$ is also a separable closure of $k(s)$. Denote by $\overline{s}$ the prescheme $\overline{s}'$, regarded as lying over S, that is, as a geometric point of S, and consider the strict localizations $\overline{S}$ and $\overline{S}'$ of S and S' at $\overline{s}$ and $\overline{s}'$, respectively. We then have a commutative square

$$\begin{tikzcd}
\overline{s}' \arrow[r] \arrow[d] & \overline{s} \arrow[d] \\
\overline{S}' \arrow[r] & \overline{S},
\end{tikzcd}$$

in which the first horizontal arrow is an isomorphism. In view of VIII 5.2 and 5.3, the homomorphism induced on the fibers at $\overline{s}'$ by the base-change homomorphism associated with (f, F, g) identifies with the homomorphism

$$H^i(\overline{X}, \overline{F}) \longrightarrow H^i(\overline{X}', \overline{F}')$$

induced by $\overline{X}' \to \overline{X}$, where $\overline{X}$ and $\overline{X}'$ denote the schemes obtained from X by the base 97
changes $\overline{S} \to S$ and $\overline{S}' \to S$, respectively. The commutative square above plainly gives a commutative square

$$\begin{tikzcd}
H^i(\overline{X}_{\overline{s}}, \overline{F}) \arrow[r] & H^i(\overline{X}'_{\overline{s}'}, \overline{F}') \\
H^i(\overline{X}, \overline{F}) \arrow[r] \arrow[u] & H^i(\overline{X}', \overline{F}') \arrow[u]
\end{tikzcd}$$

in which the first horizontal arrow is again an isomorphism. Consequently the second horizontal arrow is an isomorphism as soon as the two vertical arrows are. This is precisely what the hypothesis of 6.1 guarantees, C.Q.F.D.

6.1.1. Thus, to prove 5.1 in a situation $(F, f, -)$, we are reduced to studying the special case 5.5, with S strictly local. This leads us to study in general a morphism $h : Y \to X$ of schemes and to seek general conditions under which the corresponding homomorphisms

$$H^i(X, F) \longrightarrow H^i(Y, h^*(F))$$

are bijective. This subsection gives some easy auxiliary results on this situation. For convenience in later references, we state them in greater generality and precision than is needed for the proof of 5.1.

Lemma 6.2. *Let C and C' be two abelian categories, let $T^\bullet = (T^i)$ and $T'^\bullet = (T'^i)$ be two cohomological functors from C to C', and let $\varphi = (\varphi^i)$ be a morphism of cohomological* 98
functors from $T^\bullet$ to $T'^\bullet$. Assume that $T^\bullet$ and $T'^\bullet$ vanish in degrees < 0, and that T^i is effaceable for $i > 0$. Let n be an integer. The following conditions are equivalent:

a) *φ^i is an isomorphism for $i \leq n$ and a monomorphism for $i = n + 1$.*
b) *(If $n \geq -1$.) The homomorphism φ^i is a monomorphism for $i = 0$ and an epimorphism for $i \leq n$.*
c) *(If $n \geq 0$.) The homomorphism φ° is an isomorphism, and the T'^i are effaceable for $0 < i \leq n$.*

If these conditions hold and $A \in \mathrm{Ob}(C)$, then $\varphi^{n+1}(A) : T^{n+1}(A) \to T'^{n+1}(A)$ is an epimorphism (hence an isomorphism) if and only if $T'^{n+1}(A)$ is effaceable, that is, if there exists a monomorphism $A \to B$ such that the corresponding homomorphism $T'^{n+1}(A) \to T'^{n+1}(B)$ is zero. If $C' = (\mathrm{Ab})$, the category of abelian groups, and $\xi \in T'^{n+1}(A)$, then ξ lies in the image of $T^{n+1}(A)$ if and only if ξ is effaceable, that is, if there exists a monomorphism $A \to B$ such that the image of ξ in $T'^{n+1}(B)$ is zero.

Proof. It is clear that (a) implies (b) and (c). Since the hypotheses on $T^\cdot$ imply that the T^i for $i > 0$ are the right satellites of T°, the implication (c) $\Rightarrow$ (a) follows from the axiomatic characterization of satellite functors. It also follows, by induction on n, from the last assertion of 6.2, proved by the familiar argument that we do not repeat here (compare the proof of 6.6 below). The implication (b) $\Rightarrow$ (a) is immediate if $n = -1$. If $n \geq 0$, we proceed by induction on n: the induction hypothesis allows us to suppose that
99 φ^i is an isomorphism for $i \leq n - 1$ and a monomorphism for $i = n$. The hypothesis then makes it an isomorphism also for $i = n$, and it remains to show, using the fact that φ^n is an isomorphism, that it is a monomorphism for $i = n + 1$. Let $A \to B$ be a monomorphism that effaces T^{n+1}, giving an exact sequence $0 \to A \to B \to C \to 0$ in C and a morphism of exact sequences in C':

$$\begin{tikzcd}
T^n(B) \arrow[r] \arrow[d] & T^n(C) \arrow[r] \arrow[d] & T^{n+1}(A) \arrow[r] \arrow[d] & T^{n+1}(B) \arrow[d] \\
T'^n(B) \arrow[r] & T'^n(C) \arrow[r] & T'^{n+1}(A) \arrow[r] & T'^{n+1}(B),
\end{tikzcd}$$

where the first two vertical arrows are isomorphisms by what we have just proved, and $T^{n+1}(A) \to T^{n+1}(B)$ is zero by construction. The five lemma then implies that $T^{n+1}(A) \to T'^{n+1}(A)$ is a monomorphism, C.Q.F.D.

Corollary 6.3. *Suppose that $T^\cdot$ and $T'^\cdot$ arise from cohomological functors (denoted by the same symbols) on the right derived category $D^+(C)$ [3], with values in C'; the T^i for $i > 0$ are not necessarily assumed effaceable. Suppose that, for $K^\cdot \in D^+(C)$, the condition $H^i(K^\cdot) = 0$ for $i < 0$ implies $T^i(K^\cdot) = T'^i(K^\cdot) = 0$ for $i < 0$. Let $K^\cdot \in \mathrm{Ob}\, D^+(C)$ be a bounded-below complex in C such that $H^i(K^\cdot) = 0$ for $i < 0$.*

(i) *If condition* (a) *of 6.2 is satisfied, then the homomorphism $\varphi^i(K^\cdot) = T^i(K^\cdot) \to T'^i(K^\cdot)$ is an isomorphism for $i \leq n$ and a monomorphism for $i = n + 1$.*
(ii) *Let, in addition, $L^\cdot \in \mathrm{Ob}\, D^+(C)$ satisfy $H^i(L^\cdot) = 0$ for $i < 0$, and let $u : L^\cdot \to K^\cdot$ be a morphism in $D^+(C)$ that induces an injective homomorphism $H^\circ(u) : H^\circ(L^\cdot) \to H^\circ(K^\cdot)$. Then the homomorphism $T^{n+1}(L^\cdot) \to T'^{n+1}(L^\cdot)$*
100 *is surjective if and only if the image of $T'^{n+1}(L^\cdot)$ in $T'^{n+1}(K^\cdot)$ is contained in that of $T^{n+1}(K^\cdot)$. If $C' = (\mathrm{Ab})$ and $\xi \in T'^{n+1}(L^\cdot)$, then ξ lies in the image of $T^{n+1}(L^\cdot)$ if and only if its image in $T'^{n+1}(K^\cdot)$ lies in that of $T^{n+1}(K^\cdot)$.*

Recall that the hypothesis that $T^\cdot$ and $T'^\cdot$ arise from cohomological functors on $D^+(C)$ may be spelled out as follows: they arise from cohomological functors on the abelian category of bounded-below complexes of C that commute with degree shifts, send morphisms of complexes homotopic to zero to zero morphisms in C', and send every morphism $u : K^{\bullet} \to L^{\bullet}$ of complexes that is a quasi-isomorphism (that is, induces isomorphisms $H^i(K^{\bullet}) \to H^i(L^{\bullet})$) to an isomorphism in C'. In practice, all the cohomological functors from C to C' that one encounters arise in this way.

Proof of 6.3. We first prove (i). The assertion is clear if $H^i(K^{\bullet}) = 0$ for $i \leq n+1$, since the degree properties imposed on $T^{\bullet}$ and $T'^{\bullet}$ then give $T^i(K^{\bullet}) = T'^i(K^{\bullet}) = 0$ for $i \leq n+1$. On the other hand, let $m \geq 0$ be an integer such that $H^i(K^{\bullet}) = 0$ for $i < m$; such an integer exists, and for example $m = 0$ will do. We proceed by descending induction on m, the assertion having been proved when $m = n+2$. Thus assume the assertion proved for $m' > m$, and prove it for m. By the hypothesis on the degrees of $K^{\bullet}$, the complex $K^{\bullet}$ is quasi-isomorphic to a complex concentrated in degrees $\geq m$, so we may suppose that it is itself concentrated in those degrees. There follows an exact sequence of complexes

$$0 \longrightarrow H^m[-m] \longrightarrow K^{\bullet} \longrightarrow K'^{\bullet} \longrightarrow 0,$$

where $H^m = H^m(K^{\bullet})$, the notation $[-m]$ denotes a shift of $-m$ in the degrees of a complex, and $K'^{\bullet}$ is defined as the cokernel. The cohomology objects of $K'^{\bullet}$ are therefore 101
those of $K^{\bullet}$ in degrees $\geq m+1$ and are zero in degrees $\leq m$. In particular, the induction hypothesis applies to $K'^{\bullet}$. The preceding exact sequence gives rise to a morphism of exact sequences

$$\begin{CD}
T^{i-1}(K'^{\bullet}) @>>> T^{i-m}(H^m) @>>> T^i(K^{\bullet}) @>>> T^i(K'^{\bullet}) @>>> T^{i+1-m}(H^m)\\
@VVV @VVV @VVV @VVV @VVV\\
T'^{i-1}(K'^{\bullet}) @>>> T'^{i-m}(H^m) @>>> T'^i(K^{\bullet}) @>>> T'^i(K'^{\bullet}) @>>> T'^{i+1-m}(H^m).
\end{CD}$$

Taking $i \leq n$ in this diagram and using the induction hypothesis and $m \geq 0$, we find that the vertical arrows surrounding the middle vertical arrow are isomorphisms, hence in particular epimorphisms, and that the rightmost vertical arrow is a monomorphism. The five lemma therefore shows that the middle vertical arrow is an epimorphism. Taking $i \leq n+1$, we similarly find that the vertical arrows surrounding the middle one are monomorphisms, while the leftmost vertical arrow is an isomorphism, hence in particular an epimorphism. The five lemma then shows that the middle vertical arrow is a monomorphism. This proves 6.3 (i).

To prove the second assertion, 6.3 (ii), we similarly use the exact sequence

$$0 \longrightarrow L^{\bullet} \longrightarrow K^{\bullet} \longrightarrow K'^{\bullet} \longrightarrow 0,$$

where $K'^{\bullet}$ is defined as the mapping cylinder of $L^{\bullet} \to K^{\bullet}$ and again has cohomology in nonnegative degrees. The usual diagram chase in

$$\begin{CD}
T^n(K^{\bullet}) @>>> T^n(K'^{\bullet}) @>>> T^{n+1}(L^{\bullet}) @>>> T^{n+1}(K^{\bullet}) @>>> T^{n+1}(K'^{\bullet})\\
@VVV @VVV @VVV @VVV @VVV\\
T'^n(K^{\bullet}) @>>> T'^n(K'^{\bullet}) @>>> T'^{n+1}(L^{\bullet}) @>>> T'^{n+1}(K^{\bullet}) @>>> T'^{n+1}(K'^{\bullet})
\end{CD}$$

gives the last assertion of 6.3, since, by the first part already proved, the first two vertical 102
arrows are isomorphisms and the last two are monomorphisms. This completes the proof of 6.3.

Lemma 6.4. *Theorem 5.1 holds if f is finite.*

Indeed, this follows at once from VIII 5.5 and 5.8.

PROPOSITION 6.5. *Let $h : Y \to X$ be a quasi-compact and quasi-separated morphism of schemes. In* (i) *below, I denotes a fixed set having at least two elements; in* (ii) *and* (iii), $\mathbf{L}$ *denotes a nonempty subset of the set* $\mathbf{P}$ *of prime numbers; and in* (iii), *n denotes an integer.*

(i) *The following conditions are equivalent:*

a) *For every sheaf of sets F on X, the canonical map*

$$H^{\circ}(X, F) \longrightarrow H^{\circ}(Y, h^*(F))$$

is injective (respectively, bijective).

b) *For every finite morphism $X' \to X$, let $h' : Y' \to X'$ be the morphism obtained from h by the base change $X' \to X$. The canonical map*

$$H^{\circ}(X', I_{X'}) \longrightarrow H^{\circ}(Y', I_{Y'})$$

induced by h' is injective (respectively, bijective).

b′) *With the notation of* (b), *the map $U \mapsto h'^{-1}(U)$ from the set of subsets of X' that are both open and closed to the set of subsets of Y' that are both open and closed is injective (respectively, bijective).*

103 b″) *(If X is locally Noetherian.) With the notation of* (b), *the induced map on sets of connected components*

$$\Pi_{\circ}(h') : \Pi_{\circ}(Y') \longrightarrow \Pi_{\circ}(X')$$

is surjective (respectively, bijective). Equivalently, if X' is nonempty (respectively, connected and nonempty), then the same is true of Y'.

Moreover, if these conditions hold, then for every sheaf of groups F on X, the functor $P \mapsto h^*(P)$ from the category of torsors (homogeneous principal bundles) under F to the category of torsors under $h^*(F)$ is faithful (respectively, fully faithful). In particular, in the latter case, the canonical map

$$H^1(X, F) \longrightarrow H^1(Y, h^*(F))$$

is injective. Finally, under the same hypotheses, the functor $X' \mapsto Y' = X' \times_X Y$ from the category of étale coverings of X to the category of étale coverings of Y is faithful (respectively, fully faithful).

(ii) *The following conditions are equivalent:*

a) *For every sheaf F of ind-$\mathbf{L}$-finite groups on X, the canonical map*

$$H^i(X, F) \longrightarrow H^i(Y, h^*(F))$$

is bijective for $i = 0, 1$.

b) *For every finite morphism $X' \to X$, let $h' : Y' \to X'$ be the morphism obtained from h by base change, and let G be any ordinary finite $\mathbf{L}$-group. The canonical map*

$$H^i(X', G_{X'}) \longrightarrow H^i(Y', G_{Y'})$$

is bijective for $i = 0, 1$.

104 b′) *(If $\mathbf{L} = \mathbf{P}$.) With the notation of* (b), *the inverse-image functor under h' induces an equivalence from the category of étale coverings of X' to the category of étale coverings of Y'.*

b″) *(If X is Noetherian and $\mathbf{L} = \mathbf{P}$.) With the notation of* (b), *if X' is nonempty then so is Y'. If y' is a geometric point of Y' and x' is its image in X', the canonical map*

$$\Pi_i(Y', y') \longrightarrow \Pi_i(X', x') \quad , \qquad \textit{for } i = 0, 1,$$

is bijective.

(iii) *The following conditions are equivalent:*

a) *For every sheaf F of $\mathbf{L}$-torsion groups on X, the homomorphism*

$$H^i(X, F) \longrightarrow H^i(Y, h^*(F))$$

is an isomorphism for $i \leq n$ and a monomorphism for $i = n + 1$.

a′) *(If $n \geq -1$.) For every F as above, the homomorphism in question is injective for $i = 0$ and surjective for $i \leq n$.*

b) *(If $n \geq -1$.) For every finite morphism $X' \to X$, let $h' : Y' \to X'$ be the morphism obtained from h by the base change $X' \to X$. Then for every $\ell \in \mathbf{L}$ and $\nu > 0$, the canonical homomorphism*

$$H^i(X', (\mathrm{Z}/\ell^\nu \mathrm{Z})_{X'}) \longrightarrow H^i(Y', (\mathrm{Z}/\ell^\nu \mathrm{Z})_{Y'})$$

is injective for $i = 0$ and surjective for $i \leq n$.

In addition, we have the following supplement to the preceding statement:

Corollary 6.6. (i) *Suppose that the non-parenthesized condition in 6.5 (i) (a) holds. Let F be a sheaf of sets on X, and let $\xi \in H^\circ(Y, h^*(F))$. Suppose that* 105
there exists a monomorphism of sheaves of sets $F \to G$ such that the image of ξ in $H^\circ(Y, h^*(G))$ lies in the image of $H^\circ(X, G)$. Then ξ lies in the image of $H^\circ(X, F)$.

(ii) *Suppose that the parenthesized condition in 6.5 (i) (a) holds. Let F be a sheaf of groups on X, and let $\xi \in H^1(Y, h^*(F))$. Suppose that there exists a monomorphism of sheaves of groups $F \to G$ such that the image of ξ in $H^1(Y, h^*(G))$ lies in the image of $H^1(X, G)$. Then ξ lies in the image of $H^1(X, F)$.*

(iii) *Suppose that condition 6.5 (iii) (a) holds for a fixed value of n. Let F be a sheaf of $\mathbf{L}$-torsion groups on X, and let $\xi \in H^{n+1}(Y, h^*(F))$. Suppose that there exists a monomorphism $F \to G$ of sheaves of $\mathbf{L}$-torsion groups such that the image of ξ in $H^{n+1}(Y, h^*(G))$ lies in the image of $H^{n+1}(X, G)$. Then ξ lies in the image of $H^{n+1}(X, F)$.*

Proof of 6.5 and 6.6. We first prove 6.6. The case of 6.6 (iii) is a special case of the last assertion of 6.3, applied when C is the category of abelian $\mathbf{L}$-torsion sheaves on X, C' is the category (Ab), and $T^i(F) = H^i(X, F)$, $T'^i(F) = H^i(Y, h^*(F))$ (note that $T'^{\cdot}$ is indeed a cohomological functor in F, since the functor h^* is exact). Of course, the use of complexes as in 6.3 is unnecessary here (it will be convenient later, in 6.8); we use only the last assertion of 6.3.

To prove (i), consider the amalgamated sum $H = G \amalg_F G$, the colimit in the category of sheaves of sets on X of the diagram

$$\begin{array}{ccc} & & G \\ & \nearrow & \\ F & & \\ & \searrow & \\ & & G \end{array}$$

The usual exactness properties of the topos of sheaves of sets show, since $F \to G$ is a 106
monomorphism, that there is an *exact* diagram of sheaves of sets on X

$$F \longrightarrow G \rightrightarrows H$$

(this may be expressed by saying that every monomorphism in a topos is effective, and may be proved by reducing, as usual, to the category of sets). Since the functors H^0 and

h^* are left exact, we obtain a morphism of exact diagrams of sets

$$\begin{tikzcd} H^0(X,F) \arrow[r] \arrow[d] & H^0(X,G) \arrow[r, shift left] \arrow[r, shift right] \arrow[d] & H^0(X,H) \arrow[d] \\ H^0(Y,h^*(F)) \arrow[r] & H^0(Y,h^*(G)) \arrow[r, shift left] \arrow[r, shift right] & H^0(Y,h^*(H)), \end{tikzcd}$$

in which the vertical arrows are injective. An immediate diagram chase then proves assertion 6.6 (i).

To prove 6.6 (ii), we first prove the second assertion of 6.5 (i), namely that, under condition a) of 6.5 (i), for every sheaf of groups F on X, the functor $P \mapsto h^*(P)$ from the category of torsors under F to the category of torsors under $h^*(F)$ is faithful (resp. fully faithful), and hence, in the respective case, induces an injection on H^1. For this, if P and P' are two torsors under F, denote by $\mathrm{Isom}_F(P, P')$ the sheaf of F-isomorphisms from P to P'. It is immediate that the formation of this sheaf commutes with every base extension, and in particular with base extension by h. (This assertion is valid for every morphism of sites.) Applying the hypothesis on the effect of h^* on H^0 to the sheaf $\mathrm{Isom}_F(P, P')$, we find that the map

$$\mathrm{Isom}_F(P, P') \longrightarrow \mathrm{Isom}_F(h^*(P), h^*(P'))$$

107 is injective (resp. bijective), which says that the functor $P \mapsto h^*(P)$ is faithful (resp. fully faithful). (Note that every morphism in the category of torsors under a sheaf of groups is an isomorphism.)

On the other hand, note that if F is a subsheaf of groups of a sheaf of groups G, then giving a torsor P under F amounts to giving a torsor Q under G (namely the one obtained from P by extending the structure group along $F \to G$), *together with* a section of Q/F (which is indeed interpreted as a reduction of the structure group from G to F). With the notation of 6.6 (ii), the image of ξ in $H^1(Y, h^*(G))$ comes from an element of $H^1(X, G)$; the latter is represented by a torsor Q under G, so that the image of ξ is the class of the torsor $h^*(Q)$ under $h^*(G)$. The fact that this class comes from $\xi \in H^1(Y, h^*(F))$ means explicitly that ξ is the class of the $h^*(F)$-torsor defined by $h^*(Q)$ and a suitable section of $h^*(Q)/h^*(F)$. Since the functor h^* is right exact, this last sheaf is none other than $h^*(Q/F)$; and since h^* induces a bijection on the H^0 of sheaves of sets, the section in question of $h^*(Q)/h^*(F)$ comes from a section of Q/F. This last section defines an F-torsor by reduction of the structure group, and the class of this torsor in $H^1(X, F)$ plainly has ξ as its image in $H^1(Y, h^*(F))$. This proves 6.6 (ii) and completes the proof of 6.6.

We now prove 6.5, beginning with 6.5 (iii). Clearly a) implies a$'$), and the converse implication is a special case of 6.2 (again applied when C is the category of $\mathbf{L}$-torsion sheaves on X and C' the category of abelian groups), provided that one proves that the functors $H^i(X, F)$, for $i > 0$, are effaceable in C. This presents no difficulty; alter-
108 natively, we can avoid proving that result by noting that F is the filtered colimit of its subsheaves F_i for which there exists an integer $n > 0$ annihilating F_i (IX 1.1). Then $h^*(F)$ is the colimit of the $h^*(F_i)$, and, in view of the commutation of cohomology with filtered colimits (VII 3.3), which follows from the quasi-compactness and quasi-separatedness hypotheses on X and Y (hypotheses not used for 6.6), we are reduced to proving condition a) for the F_i. Thus we are reduced to the case in which there exists an integer $m > 0$ such that $mF = 0$. We may plainly suppose that every prime divisor of m belongs to $\mathbf{L}$, and can then take for C the category of sheaves of $\mathbf{Z}/m\mathbf{Z}$-modules on X. In this

category it is known that the functors $H^i(X, -)$ are effaceable for $i \geq 1$. This proves the equivalence of conditions a) and a′) in 6.5 (iii).

On the other hand, apply a) to the direct image of a **L**-torsion sheaf F' on a scheme X' finite over X; this direct image is a **L**-torsion sheaf by IX 1.2 (v) and IX 1.6 (iii). Using 6.4 and VIII 5.5, we find that the homomorphism

$$H^i(X', F') \longrightarrow H^i(Y', h'^*(F'))$$

is an isomorphism for $i \leq n$ and a monomorphism for $i = n + 1$, whence, a fortiori, condition b). It remains to prove that this condition implies a′). For this, use the fact (IX 2.7.2) that every **L**-torsion sheaf F on X is the filtered colimit of constructible **L**-torsion sheaves. The preceding colimit argument then shows that, in a′), it suffices to consider constructible F. By IX 2.14, there then exist finitely many finite morphisms $p_i : X'_i \to X$, constant **L**-torsion sheaves $G_{X'_i}$ on the X'_i, and an injective homomorphism

$$F \longrightarrow \prod_i p_{i*}(G_{X'_i}).$$

Now if $n = -1$, that is, if the desired conclusion reduces to the assertion that **109**

$$H^0(X, F) \longrightarrow H^0(Y, h^*(F))$$

is injective, then in this question we may at once replace F by a sheaf G into which F embeds, and hence may replace F by the product of the $p_{i*}(G_{X'_i})$. In view of 6.4, we are thus reduced to proving the injectivity assertion after replacing $h : Y \to X$ by $h' : Y' \to X'$ and F by a constant sheaf on X' whose value is an ordinary finite **L**-torsion group G. But G is isomorphic to a direct sum of groups of the form $\mathbf{Z}/\ell^\nu\mathbf{Z}$, with $\ell \in \mathbf{L}$. We are therefore indeed reduced to verifying b).

When $n \geq 0$, we proceed similarly. By induction on n, we may suppose that the desired conclusion (and hence a)) has been proved for $n - 1$ in place of n. Applying 6.6 (iii), we see that we may again replace F by a sheaf G into which F embeds, and we again take $G = p_*(G_{X'})$. In view of 6.4 and the vanishing of $R^i p_*$ for $i \geq 1$ when p is finite (VIII 5.5), we are again reduced to proving surjectivity for H^n after replacing $h : Y \to X$ by $h' : Y' \to X'$ and F by a constant sheaf on X' whose value is an ordinary finite **L**-torsion group. Injectivity is already known by the induction hypothesis and the implication a′) ⇒ a); by devissage on G, we are thus again reduced to the case in which G is of the form $\mathbf{Z}/\ell^\nu\mathbf{Z}$, namely the case considered in condition b) of 6.5 (iii). This proves case (iii) of 6.5.

Cases (i) and (ii) are proved in exactly the same way as regards the equivalence of conditions a) and b) in 6.5 (i) resp. 6.5 (ii): one uses IX 2.7.2 and the commutation of **110** H^0 and H^1 with filtered colimits of sheaves (VII 5.14) to reduce to the case in which F is constructible, and then uses IX 2.14, 6.6 (i), (ii), and VIII 5.8 to reduce to the special case considered in b). Moreover, in (i), the equivalence of conditions b), b′), and b″) is trivial and is recorded only for reference, as is the fact that they imply the faithfulness (resp. full faithfulness) of the functor $X' \mapsto Y'$ from the category of etale coverings of X to the category of etale coverings of Y. The same is true of the equivalence of conditions b), b′), and b″) in 6.5 (ii), which were added for good measure. The proof of 6.5 is now complete.

Remark 6.7. [34]

[34] The editor recommends omitting these remarks, as well as 6.13; they were surreptitiously introduced (together with various other more or less felicitous changes to the original text) by an irreverent collaborator.

1) The arguments proving 6.5 and 6.6 are plainly very general and essentially trivial, and it would be useful to extract them as abstract lemmas of the same kind as 6.2 (with the sheaves $p_*(G_{X'})$ playing the role of cogenerators of the category C in which one works). We leave this pleasant exercise to the reader, merely pointing out that essentially the same statement could be given for functors such as $R^i f_*$ in place of the functors H^i. The same remarks apply to 6.8 and 6.11 below.
2) When X is noetherian, the arguments given show that in conditions b) and their variants in 6.5 (i), (ii), and (iii), one may restrict to *integral* X'; when X is universally Japanese, one may even take them to be integral and normal. This
111 is also possible without the Japanese hypothesis on X, provided that *integral* morphisms $p : X' \to X$ are used in place of finite morphisms. None of these variants will be used later in the seminar.
3) The proof of 6.6 does not use the quasi-compactness and quasi-separatedness hypothesis on X. The same is true of the fact that in (i), (ii), and (iii), condition a) implies the other conditions, and of the mutual equivalence of conditions b), b′), and b″). Note also that the form b) of the conditions in 6.5 (i) shows that, for $\mathbf{L} \neq \phi$, they are special cases (when X is quasi-compact and quasi-separated) of 6.5 (iii), for $n = -1$ resp. $n = 0$. This remains true, moreover (if one restricts to the conditions in form a)), without the stated restriction on X. Indeed, for every sheaf of sets F and every prime number ℓ, one can find two abelian ℓ-torsion sheaves G and H, and a homomorphism $u : G \to H$ of the underlying sheaves of sets, such that F is isomorphic to the inverse image of the zero section (one takes for G the free $\mathbf{Z}/\ell\mathbf{Z}$-module generated by F, and for H the constant sheaf $\mathbf{Z}/\ell\mathbf{Z}$ on X). The same argument shows that (without any quasi-compactness or quasi-separatedness restriction) 6.5 (ii) a) (where it even suffices to take $i = 0$) implies the respective condition 6.5 (i) a). Since the latter already implies the injectivity of $H^1(X, F) \to H^1(Y, h^*(F))$ for every sheaf of groups F on X, in the statement of 6.5 (ii) a) it therefore suffices, for $i = 1$, to require the *surjectivity* of $H^1(X, F) \to H^1(Y, h^*(F))$. Unfortunately, 6.5 (ii) cannot be regarded as a special case of 6.5 (iii), which often forces us to repeat in the noncommutative case an argument already carried out in essence in the commutative case. Note also that, apart from 6.10, the results that follow are stated so as to remain valid
112 without any quasi-compactness or quasi-separatedness hypothesis.

Proposition 6.8. *(Descent lemma). With the general notation of 6.5, suppose in addition that a morphism $p : \overline{X} \to X$ is given. Let $\overline{h} : \overline{Y} \to \overline{X}$ denote the morphism obtained from h by the base change p, and let $p_Y : \overline{Y} \to Y$ denote the canonical morphism. We suppose either that p is surjective, or that h is a closed immersion and $X = p(\overline{X}) \cup h(Y)$.*

(i) *Suppose that $\overline{h} : \overline{Y} \to \overline{X}$ satisfies the injectivity (resp. bijectivity) condition of 6.5* (i) a). *In the respective case, suppose in addition that the base-change morphism*

$$h^*(p_*(\overline{F})) \longrightarrow p_{Y*}(\overline{h}^*(\overline{F})) \tag{$*$}$$

is injective for every sheaf of sets $\overline{F}$ on $\overline{X}$. Then the morphism $h : Y \to X$ also satisfies the injectivity (resp. bijectivity) condition of 6.5 (i) a).

(ii) *Suppose that $\overline{h} : \overline{Y} \to \overline{X}$ satisfies condition 6.5* (ii) a), *and that for every sheaf of ind-$\mathbf{L}$-groups $\overline{F}$ on $\overline{X}$ the base-change morphism $(*)$ is bijective. Then $h : Y \to X$ also satisfies condition 6.5* (ii) a).

(iii) *Suppose that* $\overline{h} : \overline{Y} \to \overline{X}$ *satisfies condition* 6.5 (iii) a), *and that for every abelian* $\mathbf{L}$*-torsion sheaf* $\overline{F}$ *on* $\overline{X}$ *the base-change homomorphism*

$$h^*(R^i p_*(\overline{F})) \longrightarrow R^i p_{Y*}(\overline{h}^*(\overline{F}))$$

is bijective for $i \leq n-1$ *and injective for* $i = n$*, where* $n \geq 1$ *is a given integer. Then* $h : Y \to X$ *also satisfies condition* 6.5 (iii) a).

Proof of 6.8. In the case where p is not assumed surjective, but h is a closed immersion 113
and $X = p(\overline{X}) \cup h(Y)$, consider the disjoint-union scheme $\overline{X}'$ of $\overline{X}$ and Y, and the morphism $p' : \overline{X}' \to X$ induced by p and h. Then p' is surjective; moreover, in each of the three cases (i), (ii), and (iii), it is immediate that the hypotheses on the pair (h, p) remain satisfied for the pair (h, p'). We are therefore reduced to the case in which p is surjective, which we assume from now on. In this case, for every sheaf F on X, the canonical homomorphism

$$F \longrightarrow p_*(p^*(F))$$

is injective. In case (i), to verify the injectivity of $H^0(X, F) \to H^0(Y, h^*(F))$, we are immediately reduced to the case in which F is replaced by $p_*(\overline{F})$, where $\overline{F} = p^*(F)$. Consider the commutative diagram

$$\begin{tikzcd}
H^0(X, p_*(\overline{F})) \arrow[r] \arrow[dd, "\wr"] & H^0(Y, h^*(p_*(\overline{F}))) \arrow[d] \\
 & H^0(Y, p_{Y*}(\overline{h}^*(\overline{F}))) \arrow[d, "\wr"'] \\
H^0(\overline{X}, \overline{F}) \arrow[r] & H^0(\overline{Y}, \overline{h}^*(\overline{F})),
\end{tikzcd}$$

where the left vertical arrow and the second right vertical arrow are the canonical isomorphisms, and the first right vertical arrow is induced by the base-change homomorphism for h. By hypothesis the second horizontal arrow is injective, and it follows immediately that the first is as well; this proves the non-respective assertion of (i). For the respective assertion, suppose that the second horizontal arrow is surjective. Since the base-change homomorphism is a monomorphism by hypothesis, the first right vertical arrow is a monomorphism, and hence so is the composite of the two right vertical ar- 114
rows. It follows immediately that the first horizontal arrow is also surjective, completing the proof of (i).

To prove (ii), what precedes and 6.5 (i) reduce us to proving the surjectivity of $H^1(X, F) \to H^1(Y, h^*(F))$. Applying 6.6 (ii), we are further reduced to proving it for F of the form $p_*(\overline{F})$, where $\overline{F}$ is a sheaf of ind-$\mathbf{L}$-groups on $\overline{X}$ (IX 1.6 (ii)). Writing the exact sequences for H^1 (variants of 3.2) for the morphisms p and p_Y, we obtain a morphism of noncommutative exact sequences:

$$\begin{tikzcd}
H^1(X, p_*(\overline{F})) \arrow[r] \arrow[d] & H^1(\overline{X}, \overline{F}) \arrow[r] \arrow[d] & H^0(X, R^1 p_*(\overline{F})) \arrow[d] \\
H^1(Y, p_{Y*}(\overline{h}^*(\overline{F}))) \arrow[r] & H^1(\overline{Y}, \overline{h}^*(\overline{F})) \arrow[r] & H^0(Y, R^1 p_{Y*}(\overline{h}^*(\overline{F}))),
\end{tikzcd}$$

where in each row the first arrow is injective and identifies the first term with the inverse image, under the second arrow, of the distinguished point of the third term. By the hypothesis on $\overline{h}$, the middle vertical arrow is bijective. The two outer vertical arrows factor, respectively, through

$$H^1(Y, h^*(p_*(\overline{F}))) \quad \text{and} \quad H^0(Y, h^*(R^1 p_*(\overline{F}))),$$

by composing the homomorphisms $H^i(X, -) \to H^i(Y, h^*(-))$ (for $i = 1$ and $i = 0$, respectively) with the homomorphisms obtained by applying $H^i(Y, -)$ (for those same values of i) to the base-change homomorphisms

$$h^*(p_*(\overline{F})) \longrightarrow p_{Y*}(\overline{h}^*(\overline{F}))$$

and, respectively,

$$h^*(R^1 p_*(\overline{F})) \longrightarrow R^1 p_{Y*}(\overline{h}^*(\overline{F})).$$

Since the first of these arrows is an isomorphism by hypothesis, the first vertical arrow in the diagram above identifies with the homomorphism $H^1(X, -) \to H^1(Y, h^*(-))$ that we wish to study. To prove that it is bijective, it therefore remains to prove that the last vertical arrow in the diagram is injective. By what we have just said, it is the composite of
115 two maps, the first of which is a canonical homomorphism $H^0(X, -) \to H^0(Y, h^*(-))$, and hence is bijective, while the second is injective because it is obtained by applying the left exact functor $H^0(Y, -)$ to the monomorphism

$$h^*(R^1 p_*(\overline{F})) \longrightarrow R^1 p_{Y*}(\overline{h}^*(\overline{F})).$$

The latter fact follows readily from the bijectivity hypothesis for the base-change homomorphism in the preceding dimension 0, for example by using the second assertion of 6.5 (i) and the usual calculation of the fibers of the higher direct images (VIII 5.3). This proves 6.8 (ii).

It remains to prove 6.8 (iii). For this, rather than using the injection $F \to p_*(p^*(F))$ directly as above, it is more convenient to use the homomorphism

$$F \longrightarrow Rp_*(\overline{F})$$

in the right derived category $D^+(C)$, where C denotes the category of abelian sheaves on X. Recall that, by definition, $Rp_*(\overline{F})$ is the complex $p_*(C(\overline{F}))$, where $C(\overline{F})$ is an injective resolution of $\overline{F}$ in the category of abelian sheaves on $\overline{X}$. If $K^\cdot$ is a complex on X, we denote by $H^i(X, K^\cdot)$ the hypercohomology groups of X with coefficients in $K^\cdot$, and use analogous notation on Y. By 6.3, we are reduced to proving the surjectivity of

$$H^n(X, Rp_*(\overline{F})) \longrightarrow H^n(Y, h^*(Rp_*(\overline{F}))). \tag{1}$$

By hypothesis, the base-change homomorphism

$$h^*(Rp_*(\overline{F})) \longrightarrow Rp_{Y*}(\overline{h}^*(\overline{F})) \tag{2}$$

induces an isomorphism on the cohomology sheaves in degrees $\leq n-1$ and a monomor-
116 phism in degree n. Equivalently, the “mapping cylinder” $K^\cdot$ of the preceding homomorphism has nonzero cohomology sheaves only in degrees $\geq n$. Hence $H^i(Y, K^\cdot) = 0$ for $i < n$, and the cohomology exact sequence shows that the homomorphisms

$$H^i(Y, h^*(Rp_*(\overline{F}))) \longrightarrow H^i(Y, Rp_{Y*}(\overline{h}^*(\overline{F}))) \tag{3}$$

induced by the homomorphism above are isomorphisms for $i \leq n-1$ and a monomorphism for $i = n$. Consider the composite of (1) and (3):

$$H^n(X, Rp_*(\overline{F})) \longrightarrow H^n(Y, h^*(Rp_*(\overline{F}))) \longrightarrow H^n(Y, Rp_{Y*}(\overline{h}^*(\overline{F}))) \tag{4}$$

The two outer terms are respectively isomorphic, by definition, to $H^n(\overline{X}, \overline{F})$ and $H^n(\overline{Y}, \overline{h}^*(\overline{F}))$, and the composite in (4) is precisely the homomorphism induced by $\overline{h}$, which is bijective by the hypothesis on $\overline{h}$. Since the second arrow in (4) is injective by what we have just seen, the first is also bijective. This completes the proof of 6.8.

Moreover, the preceding proof immediately gives the following slightly more precise and more general result:

Corollary 6.9. *The notation is that of 6.8. Suppose in addition that h satisfies the unparenthesized condition in 6.5* (i) *(respectively, the parenthesized condition in 6.5* (i)*, respectively, the condition in 6.5* (iii)*, with $n \geq -1$), and that, for every sheaf of sets $\overline{F}$ on $\overline{X}$, the base-change homomorphism*

$$h^*(p_*(\overline{F})) \longrightarrow p_{Y*}(\overline{h}^*(\overline{F}))$$

is injective (respectively, bijective; respectively, that for every sheaf of $\mathbf{L}$*-torsion groups $\overline{F}$ on $\overline{X}$, the base-change homomorphism*

$$h^* R^i p_*(\overline{F}) \longrightarrow R^i p_{Y*}(\overline{h}^*(\overline{F}))$$

is bijective for $i \leq n$ and injective for $i = n+1$). Let F be a sheaf of sets (respectively, of ind-$\mathbf{L}$*-groups, respectively, a sheaf of* $\mathbf{L}$*-torsion groups) on X, and let $\xi \in H^{n+1}(Y, h^*(F))$ (where $n = -1$ in case* (i) *and $n = 0$ in case* (ii)*). Put $\overline{F} = p^*(F)$. Then ξ lies in the image* 117
of $H^{n+1}(X, F)$ if and only if its inverse image in

$$H^{n+1}(\overline{Y}, p_Y^*(h^*(F))) = H^{n+1}(\overline{Y}, \overline{h}^*(\overline{F}))$$

lies in the image of $H^{n+1}(\overline{X}, \overline{F})$.

We shall use this sharpened form of 6.8 to prove the following result.

Corollary 6.10. *The notation is that of 6.5. For simplicity, suppose that the unparenthesized condition in 6.5* (i) *holds.*

(i) *The parenthesized condition in 6.5* (i) *holds (respectively, condition 6.5* (ii) a) *holds, respectively, condition 6.5* (iii) a) *holds) if and only if the following condition is satisfied. For every sheaf F of sets (respectively, of ind-*$\mathbf{L}$*-groups, respectively, of abelian* $\mathbf{L}$*-torsion groups) on X, every $\xi \in H^\circ(Y, h^*(F))$ (respectively, every $\xi \in H^i(Y, h^*(F))$ with $i = 0, 1$, respectively, every $\xi \in H^i(Y, h^*(F))$ with $i \leq n$), and every nonempty closed subset X' of X, if Y' denotes its inverse image in Y, there exists a morphism $p : \overline{X} \to X$ satisfying the following conditions:*

1°) *The image $p(\overline{X})$ is contained in X' and contains a nonempty open subset of X'.*

2°) *For every sheaf of sets $\overline{F}$ on $\overline{X}$, the base-change homomorphism*

$$h^*(p_*(\overline{F})) \longrightarrow p_{Y*}(\overline{h}^*(\overline{F}))$$

is an isomorphism (respectively, for every sheaf of ind-$\mathbf{L}$*-groups $\overline{F}$ on $\overline{X}$, the base-change homomorphism*

$$h^*(R^i p_*(\overline{F})) \longrightarrow R^i p_{Y*}(\overline{h}^*(\overline{F}))$$

is bijective for $i = 0, 1$; respectively, for every sheaf of abelian $\mathbf{L}$*-torsion groups* $\overline{F}$ *on* $\overline{X}$*, the base-change homomorphism is bijective for* $i \leq n$*).*

3°) *The inverse image of* ξ *in*

$$H^i(\overline{Y}, p_Y^*(h^*(F))) = H^i(\overline{Y}, \overline{h}^*(p^*(F)))$$

lies in the image of $H^i(\overline{X}, p^*(F))$*.*

118 (ii) *Let* $n \geq -1$*, and suppose that, for every sheaf of sets* F *(respectively, every sheaf of ind-*$\mathbf{L}$*-groups, respectively, every sheaf of abelian* $\mathbf{L}$*-torsion groups) on* X*, the homomorphism*

$$H^i(X, F) \longrightarrow H^i(Y, h^*(F))$$

is bijective for $i \leq n$ *and injective for* $i = n + 1$ *(in the first case we take* $n = -1$*, and in the second case* $n = -1$ *or* 0*). Let* $\xi \in H^{n+1}(Y, h^*(F))$*. Then* ξ *lies in the image of* $H^{n+1}(X, F)$ *if and only if it satisfies the condition stated in* (i) *above, with* $i = n + 1$*.*

Proof of 6.10. Using 6.5 and an immediate induction on n, one sees that 6.10 (i) follows from 6.10 (ii), which we now prove.

Suppose that ξ does not lie in the image of $H^{n+1}(X, F)$. Let Φ be the set of closed subsets X' of X such that the image of ξ in $H^{n+1}(Y', h^*(F)|Y')$ does not lie in the image of $H^{n+1}(X', F|X')$. By hypothesis, $X \in \Phi$, so Φ is nonempty. Order Φ by the relation $\supset$, and let us show that Φ is inductive. It is enough to prove that, if (X'_λ) is a totally ordered family of elements of Φ and X' is their intersection, then $X' \in \Phi$. Give the X'_λ and X' their induced reduced structures. The morphisms $X'_\lambda \to X$ are affine (being closed immersions), and hence the inverse system of the X'_λ satisfies the hypotheses considered in VII 5. Moreover, one immediately sees that X' is the inverse limit of the X'_λ. Likewise, Y' is the inverse limit of the inverse system of the Y'_λ, which are affine over Y. By VII 5.8, therefore,

$$\varinjlim H^{n+1}(X'_\lambda, F|X'_\lambda) \xrightarrow{\sim} H^{n+1}(X', F|X') \quad , \quad \varinjlim H^{n+1}(Y'_\lambda, h^*(F)|Y'_\lambda) \xrightarrow{\sim} H^{n+1}(Y', h^*(F)|Y').$$

119 It follows at once that, if $(\xi|Y') \in \operatorname{Im} H^{n+1}(X', F|X')$, then there would be an index λ such that $(\xi|Y'_\lambda) \in \operatorname{Im} H^{n+1}(X'_\lambda, F|X'_\lambda)$, which is absurd. Thus $X' \in \Phi$, as asserted. Consequently Φ is inductive and therefore contains an inclusion-minimal element, say X'. Since $\varnothing \notin \Phi$, the set X' is nonempty. Apply the hypothesis of 6.10 to ξ and X'. This gives a morphism $p : \overline{X} \to X$ satisfying conditions 1°)– 3°) in 6.10. Observe that we may even suppose that p is *surjective onto* X'. Indeed, let U be a nonempty open subset of X' contained in $p(\overline{X})$, and let X'_1 be its complement in X'. By construction, $X'_1 \notin \Phi$. Put $\overline{X}_1 = \overline{X} \amalg X'_1$, and let $p_1 : \overline{X}_1 \to X$ be the morphism defined by p and the inclusion $X'_1 \hookrightarrow X$. Using the relation $X'_1 \notin \Phi$, one sees immediately that p_1 still satisfies the same hypotheses as p; moreover, it is surjective onto X'. We may therefore suppose that p is surjective onto X'. Now apply 6.9, replacing $h : Y \to X$ there by $h' : Y' \to X'$ (with X' endowed, say, with its induced reduced structure), and replacing ξ by $\xi|Y'$. It follows from 6.9 that $(\xi|Y') \in \operatorname{Im} H^{n+1}(X', F|X')$, contrary to $X' \in \Phi$. This completes the proof of 6.10.

We shall use 6.10 to reduce 5.5 to the case in which f is projective. In the non-Noetherian case, however, this requires a non-Noetherian variant of Chow's lemma, which will be given in the next section.

PROPOSITION 6.11. *(Transitivity lemma). The notation is that of 6.8, but no surjectivity hypothesis is imposed on p. Suppose that h satisfies condition* (i) a) *(respectively,* (ii) a)*, respectively,* (iii) a)*) of 6.5. In case* (i)*, suppose in addition that, for every sheaf of sets $\overline{F}$ on $\overline{X}$, the base-change homomorphism of 6.8 is a monomorphism (respectively, an isomorphism); in case* (ii)*, suppose that $\overline{X}$ is Noetherian and that, for every sheaf of ind-**L**-groups $\overline{F}$, the base-change homomorphism*

$$h^*(R^i p_*(\overline{F})) \longrightarrow R^i p_{Y*}(\overline{h}^*(\overline{F}))$$

is bijective for $i = 0, 1$; and finally, in case (iii)*, suppose that the base-change homomor-* 120
phism is bijective for $i \leq n$ and injective for $i = n+1$. Then $\overline{h} : \overline{Y} \to \overline{X}$ also satisfies condition (i) *(respectively,* (ii)*, respectively,* (iii)*) of 6.5.*

Proof of 6.11. In case (i), consider the commutative diagram in the proof of 6.8 (i). Here the hypotheses imply that the first horizontal arrow and the upper right vertical arrow are injective (respectively, bijective), and hence so is the second horizontal arrow, as required. In case (ii), let $\overline{F}$ be a sheaf of ind-**L**-groups on $\overline{X}$. We must show that every element of $H^1(\overline{Y}, \overline{h}^*(\overline{F}))$ comes from an element of $H^1(\overline{X}, \overline{F})$. By 6.6 (ii), applied to $\overline{h}$, it is enough to find a monomorphism $\overline{F} \to \overline{G}$ of sheaves of groups on $\overline{X}$ that *effaces* $H^1(\overline{Y}, \overline{h}^*(\overline{F}))$. Consider the second row of the diagram used in the proof of 6.8 (ii). It is enough, successively, to find a monomorphism $\overline{F} \to \overline{G}$, with $\overline{G}$ a sheaf of ind-**L**-finite groups, that effaces the last term $H^\circ(Y, R^1 p_{Y*}(\overline{h}^*(\overline{F})))$, and then a monomorphism of sheaves of groups $\overline{G} \to \overline{H}$ that effaces $H^1(Y, p_{Y*}(\overline{h}^*(\overline{G})))$. As observed in the proof of 6.8 (i), the two terms to be effaced are respectively isomorphic, by the outer vertical arrows of the diagram in question, to $H^\circ(X, R^1 p_*(\overline{F}))$ and $H^1(X, p_*(\overline{F}))$, in view of the hypotheses made here on the base-change homomorphisms for p and h. These isomorphisms are plainly functorial in $\overline{F}$, so it is enough to efface the latter two terms. By 3.3, one can efface $R^1 p_*(\overline{F})$ (and, a fortiori, $H^\circ(X, R^1 p_*(\overline{F}))$) by embedding $\overline{F}$ into an ind-
L-group **G**. It remains to efface $H^1(X, p_*(\overline{F}))$. This is possible because, by the diagram 121
in question, the latter term maps into $H^1(\overline{X}, \overline{F})$ by a monomorphism functorial in $\overline{F}$, while $H^1(\overline{X}, \overline{F})$ is again effaceable by 3.3.

It remains to treat case (iii), and here again we read the proof of 6.8 backwards[34]. We must prove that the homomorphism

$$H^i(\overline{X}, \overline{F}) \longrightarrow H^i(\overline{Y}, \overline{h}^*(\overline{F}))$$

is an isomorphism for $i \leq n$ and a monomorphism for $i = n+1$. These homomorphisms are induced by a homomorphism of complexes

$$\mathbf{R}\Gamma_{\overline{X}}(\overline{F}) \longrightarrow \mathbf{R}\Gamma_{\overline{Y}}(\overline{h}^*(\overline{F})),$$

obtained from an injective resolution $C(\overline{F})$ and a homomorphism from $\overline{h}^*(C(\overline{F}))$ (which is a resolution of $\overline{h}^*(\overline{F})$) to an injective resolution of $\overline{h}^*(\overline{F})$. The homomorphism above also identifies with the composite

$$\mathbf{R}\Gamma_X(Rp_*(\overline{F})) \longrightarrow \mathbf{R}\Gamma_Y(h^*(Rp_*(\overline{F}))) \longrightarrow \mathbf{R}\Gamma_Y(Rp_{Y*}(\overline{h}^*(\overline{F}))),$$

[34]Instead of 6.3, one could also use the Leray spectral sequence for p and p_Y.

where the first homomorphism is the homomorphism induced by h^* for $Rp_*(\overline{F})$, and the second is obtained by applying $\mathbf{R}\Gamma_Y$ to the base-change homomorphism

$$h^*(Rp_*(\overline{F})) \longrightarrow Rp_{Y*}(\overline{h}^*(\overline{F})).$$

By the hypothesis on h and 6.3 (i), the first homomorphism therefore induces isomorphisms on the cohomology groups of the complexes in question in the required range. The same is true of the second by our hypothesis on the base-change homomorphisms, using the argument given in the proof of 6.8 (iii). The same conclusion thus holds for their composite, completing the proof of 6.11.

122 Remark 6.12. The argument given for 6.11 in cases (i) and (iii) gives the following more precise result. In order to conclude, for a *given* sheaf $\overline{F}$ on $\overline{X}$, that

$$H^i(\overline{X}, \overline{F}) \longrightarrow H^i(\overline{Y}, \overline{h}^*(\overline{F}))$$

is bijective for $i \leq n$ and injective for $i = n + 1$, it is enough to assume that

1°) the base-change homomorphism

$$h^*(R^i p_*(\overline{F})) \longrightarrow R^i p_{Y*}(\overline{h}^*(\overline{F}))$$

is an isomorphism for $i \leq n$ and a monomorphism for $i = n + 1$, and

2°) for every pair of integers $i, j \geq 0$, the homomorphism

$$H^i(X, R^j p_*(\overline{F})) \longrightarrow H^i(Y, h^*(R^j p_*(\overline{F})))$$

induced by h is an isomorphism if $i+j \leq n$ and a monomorphism if $i+j = n+1$.

There is plausibly an analogous result in case (ii), but its statement would necessarily involve noncommutative 2-cohomology. To avoid having to invoke that theory, in 6.11 (ii) we used a different argument, based on the effaceability properties of noncommutative H^1; unlike the argument above, however, it yields no result for a fixed $\overline{F}$.

Remarks 6.13. The most important case in which the conditions of 6.5 (i), (ii), and (iii) hold is that considered in 5.5; it will be proved in the next exposé using the reductions made in the present one. Another interesting case (in which h is no longer a closed immersion as in 5.5) will be studied in XV. Let us also mention the closely related case in which X is the spectrum of a Noetherian ring A, separated and complete for the topology defined by an ideal J, and $Y = \operatorname{Spec}(A/J)$ is the closed subscheme of X defined by J.
123 One then verifies condition 6.5 (ii) directly, in the form b") (with $\mathbf{L} = \mathbf{P}$), by EGA IV 18.3.2. It is very plausible that the conditions of 6.5 (iii) also hold for every n; that is, for every torsion sheaf F on X, the homomorphisms

$$H^n(X, F) \longrightarrow H^n(Y, F|Y)$$

are bijective. One can formulate considerably more general conjectures, related to a generalization of the construction of henselization to nonlocal schemes. First, no example is known in which h is a closed immersion and the conditions of 6.5 (ii) and (iii) (with $\mathbf{L} = \mathbf{P}$) fail when the parenthesized condition in 6.5 (i) holds, that is, when (X, Y) is a *henselian pair* in the terminology of EGA IV 18.5.5. One may therefore conjecture that every such pair satisfies the conditions of 6.5 (ii) and (iii) (with $\mathbf{L} = \mathbf{P}$, for every n).

On the other hand, one may ask under what conditions the following property, for a pair (X, Y) consisting of a quasi-compact and quasi-separated scheme X and a closed subscheme Y, implies that the pair is henselian:

($*$) For every scheme X' étale over X, put $Y' = X' \times_X Y$. The canonical map

$$\Gamma(X'/X) \longrightarrow \Gamma(Y'/Y)$$

is bijective (cf. EGA IV 18.5.4 b)). This condition is plainly weaker *a priori* than the henselian condition 6.5 (i), and is in fact strictly weaker: it holds, for example, when X is a normal irreducible projective scheme of dimension ≥ 2 over a field k and Y is a hyperplane section of X (as follows readily from SGA 2 XII 2.1, taking $S = \operatorname{Spec}(k)$), whereas the pair (X, Y) is plainly not henselian. It is possible, however, that when X is affine, condition $(*)$ implies that (X, Y) is a henselian pair. If so, for every affine scheme X and every closed subset Y of X, a process of étale localization along Y, analogous to 124
the henselization of local rings, ought to associate to it a *henselian* pair (X', Y), where X' is "pro-étale" over X. Another hypothesis on (X, Y) (suggested by results of Hironaka and Rossi on the contractibility of certain subvarieties of algebraic varieties) under which the henselian property might perhaps be deduced from the weaker property $(*)$ is the following: the immersion $Y \to X$ is regular of codimension 1, and the conormal sheaf $\mathrm{N}_{Y/X} = \mathbf{J}/\mathbf{J}^2$ (where $\mathbf{J}$ is the ideal defining Y in X) is an *ample* invertible sheaf on Y. (In the counterexample considered above, by contrast, this conormal sheaf was anti-ample, that is, its inverse was ample.)

7. A variant of Chow's lemma [(*)][34].

We shall need the following variant of (EGA II 5.6.1), stripped of all Noetherian and irreducibility hypotheses.

Lemma 7.1. *Let S be a quasi-compact and quasi-separated scheme, and let $f : X \to S$ be a separated morphism of finite type, with X nonempty. There exists a commutative diagram*

$$\begin{tikzcd} X \arrow[dr, "f"'] & & \overline{X} \arrow[ll, "\pi"'] \arrow[dl, "\overline{f}"] \\ & S & \end{tikzcd}$$

with π projective and $\overline{f}$ quasi-projective, and a nonempty open subset U of X, such that 125
π induces an isomorphism from $\overline{U} = U \times_X \overline{X} = \pi^{-1}(U)$ onto U.

Proof. We first make the following observation. Let $i : X' \to X$ be a closed S-subscheme of X, and let $V \subset X'$ be a subscheme that is open in X and dense in X'. If the lemma holds for X', then it holds for X. Indeed, let

$$\begin{tikzcd} X' \arrow[dr, "f'"'] & & \overline{X}' \arrow[ll, "\pi'"'] \arrow[dl, "\overline{f}'"] \\ & S & \end{tikzcd}$$

be a diagram as in the statement, and let $U' \neq \varnothing$ be an open subset of X' such that π' induces an isomorphism $\overline{U}' = \pi'^{-1}(U') \to U'$. Since V is dense in X', we have $U' \cap V \neq \varnothing$, and we may therefore replace U' by this open subset. Then U' is open in X, and we set $\pi = i\pi'$, $\overline{f} = \overline{f}'$, and $U = U'$.

Now X admits a finite covering by nonempty affine open subsets U_k, $k = 1, \ldots, n$. The hypotheses on S imply that every $U_k \to S$ is quasi-affine, and hence, a fortiori,

[34] A reader interested in the statements of Section 5 only in the Noetherian case may omit the present section.

quasi-compact. We prove the lemma for every X by induction on n. If the intersection U of the U_k is dense in X, we may copy the proof of (EGA II 5.6, B, C, D), with the necessary changes in notation. If the intersection is nonempty, we reduce to the case in which it is dense by replacing X by the closure X' of U (as in the observation above), and U_k by $U'_k = U_k \cap X'$. Finally, suppose that U is empty, and choose r, $1 \leq r < n$, such that

$$V = \bigcap_{k=1}^{r} U_k \neq \varnothing \qquad \text{but} \qquad V \cap U_{r+1} = \varnothing;$$

this is plainly possible since $U_1 \neq \varnothing$. We may now replace X by the closure X' of V and each U_k by $U_k \cap X'$, discarding the empty members. Since $U_{r+1} \cap X' = \varnothing$ (it is open in X' and disjoint from the dense open V), the result follows from the induction hypothesis.

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8. Final reductions

LEMMA 8.1. *To prove any one of the assertions* 5.1 (i), (ii), (iii), *it is enough to do so under the assumptions that f is* projective *and S is Noetherian.*

Suppose first that 5.1 has been proved when f is projective; we show how the same assertion follows for arbitrary f. Using 6.1, we are reduced to the situation of 5.5, with S strictly local. We now apply 6.10, taking $Y = X_\circ$. Given (ξ, X') as in 6.10, we apply Chow's lemma 7.1 to find an S-morphism $p : \overline{X} \to X'$, with $\overline{X}$ projective over S, such that $p(\overline{X})$ contains a nonempty open subset of X' (condition 1° of 6.10). Since p is projective, the base-change homomorphisms for (p, h) are isomorphisms by hypothesis (condition 2° of 6.10). Finally, since $\overline{X}$ is projective over S, the inclusion $\overline{Y} = \overline{X}_\circ \to \overline{X}$ induces, by hypothesis, an isomorphism on cohomology groups

$$H^i(\overline{X}, \overline{F}) \longrightarrow H^i(\overline{X}_\circ, \overline{h}^*(\overline{F})),$$

which is condition 3° of 6.10. Thus all the conditions of 6.10 are satisfied, and the conclusion follows.

We next show that, in order to verify any one of the three assertions 5.1 for every situation (f, F, g) with f projective, we may further assume that S is Noetherian. By 6.1, we may suppose that S is strictly local and that $S' \to S$ is the inclusion of the closed point; we are thus reduced to verifying 5.5 with S strictly local. Moreover, since X embeds into the projective scheme $\mathbf{P}^r_S$, compatibility 4.4 (ii) allows us to replace X by $\mathbf{P}^r_S$ and F by its direct image on $\mathbf{P}^r_S$. We are therefore reduced to the case $X = \mathbf{P}^r_S$.

Write $S = \operatorname{Spec}(A)$. The ring A is the filtered direct limit of Noetherian Henselian local subrings A_i for which the inclusions $A_i \to A$ are local: one may take the strict henselizations, at the prime ideals induced by the maximal ideal of A, of the finitely generated $\mathbf{Z}$-subalgebras of A. Under these conditions, the residue field k of A is likewise the filtered direct limit of the residue fields k_i of the A_i (EGA IV 5.13.1). Set $S_i = \operatorname{Spec}(A_i)$ and $X_i = \mathbf{P}^r_{S_i}$. Suppose that a direct system (F_i) of sheaves of the type occurring in the assertion under consideration is given on the X_i, as in VII 5.7, and let F be the sheaf it defines on X. Let $F_{i\circ}$ (respectively, $F_\circ$) be the sheaf induced by F_i on the fiber $X_{i\circ}$ over the closed point s_i of S_i (respectively, by F on the fiber $X_\circ$ over the closed point s of S).

We then have a commutative diagram

$$\begin{tikzcd} \varinjlim\limits_{i} H^n(X_i, F_i) \arrow[r] \arrow[d] & H^n(X, F) \arrow[d] \\ \varinjlim\limits_{i} H^n(X_{i\circ}, F_{i\circ}) \arrow[r] & H^n(X_\circ, F_\circ), \end{tikzcd}$$

whose two horizontal arrows are isomorphisms by VII 5.7. Consequently, to prove that the second vertical arrow is an isomorphism, it is enough to prove this for the first one; hence it is enough to prove that the homomorphisms

$$H^n(X_i, F_i) \longrightarrow H^n(X_{i\circ}, F_{i\circ})$$

are isomorphisms. On the other hand, every sheaf F of the type under consideration on X is isomorphic to the limit of a direct system of the preceding kind; for example, one may take $F_i = u_{i*}(F)$, where $u_i : X \to X_i$ is the canonical morphism, by 5.2 and IX 1.6 (iii). This completes the proof of 8.1.

Corollary 8.2. *Theorem* 5.1 (i) *is true.*

Indeed, it is enough to combine 8.1, criterion 6.5 (i) b″), and 5.8.

Lemma 8.3. *To prove any one of the assertions* 5.1 (i), (ii), (iii)*, it is enough to do so* 127
when f is projective of relative dimension ≤ 1 and S is Noetherian.

Indeed, 8.1 already shows that we may restrict ourselves to the case in which f is projective and S is Noetherian, and an argument used above further allows us to suppose that $X = \mathbf{P}^r_S$. We proceed by induction on r, noting that the theorem holds by hypothesis for $r = 1$. We may therefore suppose that $r \geq 2$ and that the theorem has been proved for $\mathbf{P}^{r-1}_S \to S$ (over every base S). Embed $Z = \mathbf{P}^{r-2}_S$ in $X = \mathbf{P}^r_S$ in the usual way, and let $p : \overline{X} \to X$ be the blowup of X along Z. A direct calculation (cf. EGA V) gives a natural morphism $p_1 : \overline{X} \to X_1 = \mathbf{P}^1_S$ that makes $\overline{X}$ an X_1-scheme locally isomorphic to $\mathbf{P}^{r-1}_{X_1}$ (it is the projective bundle over X_1 associated with a suitable locally free sheaf of rank r on X_1). On the other hand, the fibers of p have dimension ≤ 1, as do those of $f_1 : X_1 \to S$. We may therefore apply the relative-dimension-one hypothesis to p and f_1, and the induction hypothesis to p_1, in the commutative diagram

$$\begin{tikzcd} & \overline{X} \arrow[dl, "p"'] \arrow[dr, "p_1"] & \\ X \arrow[dr, "f"'] & & X_1 \arrow[dl, "f_1"] \\ & S & \end{tikzcd}$$

Since p is surjective, 6.8 reduces us to proving 5.1 for the composite $\overline{f} = fp = f_1p_1$. But we may already apply 5.1 to both f_1 and p_1. Taking 6.11 into account, we conclude that 5.1 also holds for their composite f_1p_1. This completes the proof of 8.3.

Lemma 8.4. *To prove* 5.1 (ii)*, it is enough to prove the "specialization theorem for the* 128
fundamental group" 5.9 bis *in the case where f is projective of relative dimension ≤ 1 and S is Noetherian and strictly local.*

This follows at once from 8.3, 6.1, and 6.5 (ii) (criterion b)).

The specialization theorem in the case of 8.4 will be proved directly in the next exposé (XIII 2).

Lemma 8.5. *To prove* 5.1 (iii)*, it is enough to prove* 5.1 (ii) *and, in addition, the following proposition (which will be proved in the next exposé* (XIII 3)*, or in* (EGA IV 21.9.12)*):*

Proposition 8.6. *Let S be Henselian and Noetherian, let $f : X \to S$ be projective of relative dimension ≤ 1, and let $X_\circ$ be the closed fiber of X/S. For every scheme Z, let* $\operatorname{Pic} Z = H^1(Z, (G_m)_Z)$ *denote the Picard group of Z. The restriction homomorphism*

$$\operatorname{Pic} X \longrightarrow \operatorname{Pic} X_\circ$$

is surjective.

Proof of 8.5. By 6.5 (iii) and 8.2, it is enough to prove that

$$\varphi^q : H^q(X, \mathbf{Z}/\ell^\nu\mathbf{Z}) \longrightarrow H^q(X_\circ, \mathbf{Z}/\ell^\nu\mathbf{Z})$$

is surjective for every $q \geq 1$ and every prime ℓ. For $q = 1$, surjectivity is already known from 5.1 (ii). Moreover, since $X_\circ$ is a projective scheme of dimension ≤ 1 over a separably closed field k, we have

$$H^q(X_\circ, \mathbf{Z}/\ell^\nu\mathbf{Z}) = 0$$

for $q > 2$ when $\ell \neq \operatorname{char} k$ (X 4.3), and for $q > 1$ when $\ell = \operatorname{char} k$ (X 5.2). Thus surjectivity is automatic for these values of q.

129 It remains to consider $q = 2$ for a prime $\ell \neq \operatorname{char} k$. Put $n = \ell^\nu$. The sheaf $(\mu_n)_S$ of n-th roots of unity is locally isomorphic to $(\mathbf{Z}/n\mathbf{Z})_S$, and, since S is strictly local, there is a noncanonical isomorphism

$$(\mu_n)_S \simeq (\mathbf{Z}/n\mathbf{Z})_S,$$

and hence $(\mu_n)_X \simeq (\mathbf{Z}/n\mathbf{Z})_X$.

Kummer theory IX 3.2, applied to X and $X_\circ$, gives a morphism of exact sequences

$$\begin{tikzcd}
\operatorname{Pic} X \arrow[r] \arrow[d, "\epsilon"] & H^2(X, (\mu_n)_X) \arrow[r] \arrow[d, "\varphi^2"] & H^2(X, (G_m)_X) \arrow[d] \\
\operatorname{Pic} X_\circ \arrow[r] & H^2(X_\circ, (\mu_n)_{X_\circ}) \arrow[r] & H^2(X_\circ, (G_m)_{X_\circ}).
\end{tikzcd}$$

Now $H^2(X_\circ, (G_m)_{X_\circ}) = 0$ by IX 4.6. Consequently, the surjectivity of ϵ implies that of φ^2, which proves the lemma.

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EXPOSÉ XIII

Base-change theorem for a proper morphism: conclusion of the proof

M. Artin

1. The projective and flat case

Recall the statement of XII 5.9 bis in the case under consideration here: 132

Proposition 1.1. *Let S be the spectrum of a Noetherian henselian ring, and let $f : X \to S$ be a projective and flat morphism. Let $X_\circ$ be the closed fiber of X/S. The restriction functor*

$$\mathrm{Et}(X) \longrightarrow \mathrm{Et}(X_\circ)$$

is an equivalence of categories.

As we have already observed, it follows from XII 6.5 (i) that the functor is fully faithful. It remains to prove that every étale covering $Y_\circ$ of $X_\circ$ is induced by an étale covering Y of X.

Lemma 1.2. *Let S be locally Noetherian, let $f : X \to S$ be projective and flat, and let $Y \subset X$ be a subscheme. The subfunctor $\epsilon = \prod_{X/S} Y/X$ of the terminal functor, "which expresses the condition $Y = X$", that is, such that for $S' \to S$ one has*

$$\epsilon(S') = \begin{cases} \phi & \text{if} \quad Y' \neq X', \\ \{\phi\} & \text{if} \quad Y' = X', \end{cases}$$

(where a prime denotes the effect of the base change $S' \to S$), is representable by a subscheme Z of S. If Y is open (resp. closed), then so is Z. 133

Proof. Let Y be closed in an open subset U of X, and let C be the closed complement of U. Since f is proper, the image of C in S is a closed subset D, and it is clear that the condition $U = X$ is represented by the open subset $S - D$ of S. We may therefore suppose that $U = X$ already, that is, that Y is a closed subscheme. Consider the surjective morphism $\mathcal{O}_X \xrightarrow{\alpha} \mathcal{O}_Y$ of $\mathcal{O}_X$-modules; saying that $Y = X$ amounts to saying that there exists a morphism $\mathcal{O}_Y \xrightarrow{\beta} \mathcal{O}_X$ such that the composite $\beta\alpha$ is the identity, that is, that there exists a section of $\mathbf{Hom}(\mathcal{O}_Y, \mathcal{O}_X)$ lifting the identity section of $\mathbf{Hom}(\mathcal{O}_X, \mathcal{O}_X)$. Now it follows from EGA III 7.7.8 that the functor which assigns to S'/S the set of sections of $\mathbf{Hom}(\mathcal{O}_Y, \mathcal{O}_X)$ (resp. $\mathbf{Hom}(\mathcal{O}_X, \mathcal{O}_X)$) is represented by a vector bundle V_1 (resp. V_2) over S. Let $\varphi : V_1 \to V_2$ be the morphism $u \mapsto u\alpha$ induced by $\mathcal{O}_X \to \mathcal{O}_Y$, and let $s : S \to V_2$ be the section corresponding to the identity $\mathcal{O}_X \to \mathcal{O}_X$. The inverse image of the section s under φ is identified with a closed subscheme Z of S, and it is clear that Z represents the functor ϵ, which proves the lemma.

Lemma 1.3. *Let S be a locally Noetherian scheme, let $f : X \to S$ be a projective and flat morphism, and let E be a locally free sheaf on X. Let*

$$QE : (\mathrm{Sch})/S \longrightarrow (\mathrm{Ens})$$

be the functor which assigns to S'/S the set $QE(S')$ of quotient sheaves A' of $E' = E \otimes_{\mathcal{O}_S} \mathcal{O}_{S'}$, endowed with the structure of an étale $\mathcal{O}_{X'}$-algebra. This functor is representable by a scheme locally of finite type over S.

134 *Proof.* The functor $\mathrm{Quot}(E)$, with $\mathrm{Quot}(E)(S')$ equal to the set of quotients of E' that are flat over S', is representable by a scheme locally of finite type (TDTE IV, exp. 221), say by Z/S. Let M_u be the "universal"quotient of $E \otimes_{\mathcal{O}_S} \mathcal{O}_Z$ over X_Z. The subfunctor of $\mathrm{Quot}(E)$ consisting of the quotients which are flat *over* X' is represented by an open subset U of Z. Indeed, let M' be a quotient of E', given by an S-morphism $S' \to Z$. Since M' is flat over S', saying that it is flat over X' amounts to saying that, for every point $s' \in S'$, the module $M'_{s'}$ is flat over the fiber $X'_{s'}$ (EGA IV 5.9), which is equivalent to $M_u|_{X_z}$ being flat over the fiber X_z at the point z which is the image of s'. But the set U of $z \in Z$ such that $M_u|_{X_z}$ is flat over X_z, that is, such that M_u is flat over X_Z at every point of X_z, is open in Z by EGA IV 11.1.5, proving the assertion.

It now suffices to represent the relative functor $QE/U[f]$. We are thus reduced to the following lemma, after replacing S by U and X by $X \times_S U$:

Lemma 1.4. *Let S be a locally Noetherian scheme, let $f : X \to S$ be a projective and flat morphism, and let M be a locally free $\mathcal{O}_X$-module. The functor* Alget : (Sch)/S → (Ens)*, which assigns to S'/S the set* Alget(S') *of algebra structures on M' that are étale over $\mathcal{O}_{X'}$, is representable by a scheme of finite type over S.*

Proof. For brevity, we shall say "representable", understanding "by a relative scheme of *finite type*". All the vector bundles encountered below will be defined by *coherent* sheaves, and hence will be of finite type. A bilinear composition law on M' is given by a section of the locally free sheaf $\mathbf{Hom}(M' \otimes M', M')$. Since $\mathbf{Hom}(M' \otimes M', M')$ is
135 flat over S, it follows from EGA III 7.7.6 that the functor "bilinear composition law"is represented by a vector bundle Z over S. We must check that the additional conditions that the law be associative, commutative, and possess a unit section are represented by a subscheme of Z. By considering the relative functor, we may replace S by Z, and may therefore suppose that a composition law on M has been given and prove that the subfunctor of S expressing the preceding conditions is representable by a subscheme of S.

For associativity, the condition is that two sections of $\mathbf{Hom}(M \otimes M \otimes M, M)$ become equal after the base change $S' \to S$. Let Y be the fiber product of these sections over the vector bundle corresponding to $\mathbf{Hom}(M \otimes M \otimes M, M)$. The scheme Y is identified with a closed subscheme of X, and associativity after $S' \to S$ is expressed by the condition $Y' = X'$. It therefore defines a subfunctor of S represented by a closed subscheme of S, by 1.2. For commutativity, the condition is that two sections of $\mathbf{Hom}(M \otimes M, M)$ become equal, which again defines a subfunctor of S represented by a closed subscheme, by 1.2. Finally, suppose, as we now may, that the law is already associative and commutative, and consider the condition that a unit section exist; such a section will then be unique. The functor whose value on S'/S is the set of sections of M' is represented by a vector bundle, by EGA III 7.7.6, and it remains only to represent the subfunctor of this bundle corresponding to the *unit sections* (taking account of the uniqueness just recalled). Passing to the relative situation as usual, we may suppose that a section ϵ of M has already been given, and prove that the subfunctor of S expressing the condition
136 that it become a unit section is representable. But the condition that ϵ be a unit is that the composite of the evident morphisms $M' \to M' \otimes M' \to M'$ be the identity, that is, that two specified sections of $\mathrm{Hom}(M, M)$ become equal. We conclude once again by 1.2.

Finally, suppose that a commutative unital algebra structure has been given on the locally free module M. The locus where it is étale is open in X (SGA I 4.5), and is therefore represented over S by an (open) subscheme. This completes the proof of 1.4, and hence of 1.3.

1.5. Let Z/S now be the object representing the functor considered in 1.3, and consider the condition that Z be smooth over S at a point z. Translating SGA I III 3.1 into the language of the functor QE, one obtains the following condition:

Let B be a local Artinian ring, let $\mathscr{M} = \operatorname{rad} B$, let J be an ideal of B such that $J\mathscr{M} = 0$, put $S' = \operatorname{Spec} B$ and $S'' = \operatorname{Spec} B/J$, and let $S' \to S$ be a morphism. We use the evident notation for the effects of the base changes $S' \to S$ and $S'' \to S$.

(∗) For every quotient A'' of E'', endowed with the structure of an étale $\mathscr{O}_{X''}$-algebra and corresponding to an S-morphism $S'' \to Z$ whose image is z, there exists a quotient A' of E', endowed with the structure of an étale $\mathscr{O}_{X'}$-algebra, which induces it.

Now, since X' and X'' have the same underlying space, an algebra A' inducing A'' is already determined up to unique isomorphism (SGA I 8.3). We may therefore express condition (∗∗) as follows: 137

(∗∗) Let A' be an étale algebra over $\mathscr{O}_{X'}$, and let A'' be the algebra induced by A' on X''. Every surjective homomorphism

$$E'' \xrightarrow{\varphi''} A''$$

corresponding to a morphism $S'' \to Z$ whose image is z lifts to a homomorphism (necessarily surjective by Nakayama's lemma)

$$E' \xrightarrow{\varphi'} A'.$$

From the exact sequence

$$0 \longrightarrow J \longrightarrow B \longrightarrow B/J \longrightarrow 0$$

we obtain an exact sequence of sheaves on X':

$$0 \longrightarrow J \otimes \mathbf{Hom}(E', A') \longrightarrow \mathbf{Hom}(E', A') \longrightarrow \mathbf{Hom}(E'', A'') \longrightarrow 0.$$

The obstruction to lifting φ'' lies in $H^1(X', J \otimes \mathbf{Hom}(E', A'))$. Let $S_\circ = \operatorname{Spec} B/\mathscr{M}$. Since $J\mathscr{M} = 0$, one has

$$J \otimes \mathbf{Hom}(E', A') \simeq J \otimes \mathbf{Hom}(E_\circ, A_\circ),$$

and it is clear that

$$H^1(X', J \otimes \mathbf{Hom}(E_\circ, A_\circ)) \simeq J \otimes H^1(X_\circ, \mathbf{Hom}(E_\circ, A_\circ)).$$

Taking account of the fact that cohomology commutes with extension of the base fields, we obtain:

Corollary 1.6. *Let Z/S be the scheme representing the functor of 1.3, let $X_Z = X \times_S Z$, and let A_Z be the ("general") quotient étale algebra of E_Z. Then Z/S is smooth at every point $z_\circ$ such that*

$$H^1(X_\circ, \mathbf{Hom}(E_\circ, A_\circ)) = 0,$$

where the subscript zero denotes restriction to the fiber $X_\circ$ of X_Z/Z at the point $z_\circ$. 138

We can now complete the proof of 1.1. With the notation of the proposition, let $Y_\circ/X_\circ$ be an étale covering, given by a sheaf of algebras $A_\circ$ that is étale over $\mathscr{O}_{X_\circ}$. For n sufficiently large, $A_\circ(n)$ is generated by its sections (EGA III 2.2.1). We thus have a surjection

$$\mathscr{O}_{X_\circ}^N \longrightarrow A_\circ(n) \longrightarrow 0,$$

and hence a surjection

$$\mathscr{O}_{X_\circ}^N(-n) = E_\circ \longrightarrow A_\circ \longrightarrow 0.$$

Choose n sufficiently large that we also have

$$(+) \qquad H^1(X_\circ, \mathbf{Hom}(E_\circ, A_\circ)) = 0,$$

which is possible by EGA III 2.2.1, since

$$\mathbf{Hom}(E_\circ, A_\circ) = [\mathbf{Hom}(\mathscr{O}_{X_\circ}^N, A_\circ)](n) = [A_\circ(n)]^N.$$

Put $E = \mathscr{O}_X^N(-n)$, and let Z/S be the scheme representing the functor considered in 1.3. The quotient $A_\circ$ of $E_\circ$ corresponds to a point $z_\circ \in Z$ in the closed fiber of Z/S, rational over $k(s)$; by 1.5 and (+), Z/S is smooth at $z_\circ$. Since S is the spectrum of a *henselian* ring, there is a section of Z/S through $z_\circ$; this follows, for example, immediately from the original definition (SGA II 1.1) of smoothness. It follows that $A_\circ$ is induced by an algebra A étale over X (a quotient of E), which completes the proof of 1.1.

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2. The case of relative dimension at most one

The assertion is the following.

Proposition 2.1. *With the notation of* XII 5.9 bis, *the functor* $\mathrm{Et}(X) \to \mathrm{Et}(X_\circ)$ *is an equivalence of categories if S is Noetherian and strictly local and if f is projective of relative dimension* ≤ 1.

By XII 8.5, *this will complete the proof of* XII 5.1 (ii), *which generalizes both 1.1 and 2.1.*

Lemma 2.2. *Let S be the spectrum of a Noetherian local ring A, and let X/S be projective of relative dimension $\leq n$. There exist a projective and flat S-scheme Z/S of relative dimension $\leq n$ and a closed immersion $X \hookrightarrow Z$ over S.*

Proof. Let $X \to \mathbf{P}_S^N$ be a projective embedding over S, and let $J \subset A[x_\circ, \dots, x_N]$ be the homogeneous ideal defining X. Put $k = A/\mathfrak{m}_A$, $S_\circ = \operatorname{Spec} k$, and $X_\circ = X \otimes_S k$. Then $X_\circ$ is defined by the homogeneous ideal $J_\circ = \operatorname{Im}(J \to k[x_\circ, \dots, x_N])$. Since $\dim X_\circ \leq n$, it is well known that there exist elements $f_1^\circ, \dots, f_{N-n}^\circ$ of $J_\circ$ defining a subscheme $Z_\circ$ of $\mathbf{P}_k^N$ of dimension n. Let f_i $(1 \leq i \leq N-n)$ be elements of J lifting the f_i°, and let Z be the subscheme of $\mathbf{P}_S^N$ defined by $f_1, \dots, f_{N-n}$. By EGA IV 11.3, Z is flat over S; and by EGA IV 13.1.5 it has relative dimension $\dim Z_\circ = n$. This proves 2.2.

Proof of 2.1. Let X/S be projective of relative dimension ≤ 1, with S Noetherian and strictly local. We may assume that X (and hence $X_\circ$) is connected. Let $X \to Z$ be a closed immersion over S, where Z is projective, flat, and of relative dimension ≤ 1; see
140 2.2. Let $s_\circ$ be the closed point of S. We have a corresponding diagram of categories

$$\begin{tikzcd}
\mathrm{Et}(Z) \arrow[r] \arrow[d, "a"] & \mathrm{Et}(X) \arrow[d, "b"] \\
\mathrm{Et}(Z_\circ) \arrow[r, "c"] & \mathrm{Et}(X_\circ).
\end{tikzcd}$$

We wish to prove that b is an equivalence. By XII 5.8, b is fully faithful, so it remains to prove that it is essentially surjective. The functor a is an equivalence by 1.1; it therefore suffices to prove that c is essentially surjective. Dropping the subscript $_\circ$, we are reduced to the following proposition.

Proposition 2.3. *Let k be a separably closed field, let Z be a k-scheme locally of finite type and of dimension ≤ 1, and let $X \hookrightarrow Z$ be a connected closed subscheme. Then every étale covering of X is induced by an étale covering of Z.*

Proof. We may suppose that Z and X have dimension 1, the case in which X is zero-dimensional being immediate from the hypothesis on k. Then $Z = X \cup Y$, where Y is the closed subscheme obtained as the closure of $Z - X$. Put $V = X \cap Y$; this is a discrete, that is, locally Artinian, scheme. The morphism $X \amalg Y \to Z$ is finite and surjective, hence is a strict descent morphism for the category of étale coverings (SGA IX 4.7). Clearly

$$(*) \qquad (X \amalg Y) \times_Z (X \amalg Y) = X \amalg Y \amalg V^{(1)} \amalg V^{(2)} \quad (V^{(i)} = V),$$

where pr_i sends $V^{(i)}$ to the component X of $X \amalg Y$ and sends $V^{(j)}$ to the component Y when $j \neq i$.

Let X'/X be an étale covering of degree n, and let Y''/Y be a covering of degree n (for example, the trivial covering of degree n), so that $X' \amalg Y''$ is a covering of $X \amalg Y$ 141
of degree n. It is enough to find descent data for this covering relative to $X \amalg Y \to Z$. Inspection of $(*)$ shows that such descent data amount simply to an isomorphism

$$X' \times_X V = V' \xrightarrow{\sim} V'' = Y'' \times_Y V.$$

Since k is separably closed, so are the residue fields of the points of V; consequently every étale covering of V is completely split. Thus $V' \simeq V''$, which proves the result.

Remark 2.4. One may avoid the delicate result from SGA IX and descent theory by a direct argument showing that the category of étale coverings of Z is equivalent to the category of triples (X', Y', θ), where X' (respectively Y') is an étale covering of X (respectively Y), and θ is a V-isomorphism $X' \times_X V \simeq Y' \times_Y V$. This result, valid for any scheme Z that is the union of two closed subschemes X and Y, is proved directly from the analogous description of quasi-coherent modules on Z (when $X = V(\mathfrak{J})$, $Y = V(\mathfrak{K})$, and $\mathfrak{J} \cap \mathfrak{K} = 0$).

3. An auxiliary result on the Picard group (*)

Proposition 3.1. *Let X be a Noetherian scheme and let $X_\circ$ be a closed subscheme of X such that* (a) $\dim X_\circ \leq 1$, *and* (b) *for every nonempty connected closed subset Y of X, $Y_\circ = Y \cap X_\circ$ is nonempty and connected.*

(i) *Let $Z_\circ$ be the union of the sets $\overline{\{x\}} \cap X_\circ$, for $x \in \mathrm{Ass}\,\mathcal{O}_X$, that consist of a single point. For every Cartier divisor $D_\circ$ on $X_\circ$ such that $\mathrm{supp}(D_\circ)$ does not meet $Z_\circ$, there exists a Cartier divisor D on X inducing $D_\circ$ (and if $D_\circ \geq 0$, one may take $D \geq 0$).*

(ii) *Suppose there is an affine open subset $U_\circ$ of $X_\circ$ containing the finite set $\mathrm{Ass}\,\mathcal{O}_{X_\circ} \cup Z_\circ$ (this condition is automatically satisfied if $X_\circ$ admits an ample invertible sheaf). Then $\mathrm{Pic}(X) \to \mathrm{Pic}(X_\circ)$ is surjective.*

Proof.

[34] This also appears in EGA IV 21.9.11 and 21.9.12 in a slightly more general form; in particular, in 3.2 it is enough for f to be *separated* rather than proper.

(i) Since $\dim X_\circ \leq 1$, every Cartier divisor $D_\circ$ on $X_\circ$ plainly has discrete support, and is therefore the difference of two effective Cartier divisors whose supports are contained in that of $D_\circ$. We are thus reduced in (i) to the case where $D_\circ$ is effective. For every $x \in \operatorname{Supp} D_\circ$ it is then defined at x by a non-zero-divisor $f_x \in \operatorname{rad}(\mathcal{O}_{X_\circ,x})$. The element f_x is induced by a section g_x of $\mathcal{O}_X$ on an open neighborhood U_x of x. I claim that, after shrinking U_x, the section g_x is a non-zero-divisor in $\mathcal{O}_{U_x}$, that is, $V(g_x) \cap \operatorname{Ass} \mathcal{O}_{U_x} = \varnothing$.

Indeed, conditions (a) and (b) imply that, for every $z \in \operatorname{Ass} \mathcal{O}_X$, the set $\overline{\{z\}} \cap X_\circ$ is a connected closed subset of $X_\circ$ and hence is either a single closed point or a connected union of one-dimensional irreducible components of $X_\circ$. Since $x \notin Z_\circ$, in the first case $x \notin \overline{\{z\}}$. In the second case, one cannot have both $x \in \overline{\{z\}}$ and $z \in V(g_x)$, for then g_x, and therefore f_x, would vanish on an irreducible component of $\overline{\{z\}} \cap X_\circ$, which is also an irreducible component of $X_\circ$; this is incompatible with f_x being a non-zero-divisor in $\mathcal{O}_{X_\circ,x}$. After
142 shrinking U_x, we may therefore suppose that $V(g_x) \cap \operatorname{Ass} \mathcal{O}_{U_x} = \varnothing$, so g_x is a non-zero-divisor.

Let Y_x be the closure of $V(g_x)$ in X. The point x is isolated in $Y_x \cap X_\circ \cap U_x$, hence also in $Y_x \cap X_\circ$. By condition (b), Y_x has a connected component Y'_x such that $Y'_x \cap X_\circ = \{x\}$. Moreover, $Y'_x \subset U_x$: indeed, $T_x = Y'_x - (Y'_x \cap U_x)$ is closed in X and satisfies $T_x \cap X_\circ = \varnothing$, so condition (b) gives $T_x = \varnothing$. Let D'_x be the Cartier divisor on X induced by Y'_x on U_x and by zero on $X - Y'_x$. Then $D = \sum_x D'_x$ is an effective Cartier divisor on X inducing $D_\circ$ on $X_\circ$.

(ii) The conclusion follows from (i) and the fact that every invertible module $\mathbf{L}_\circ$ on $X_\circ$ is isomorphic to the module defined by a Cartier divisor $D_\circ$ whose support does not meet $Z_\circ$. This last assertion means that $\mathbf{L}_\circ$ admits a section on an open set $V_\circ$ containing $\operatorname{Ass} \mathcal{O}_{X_\circ} \cup Z_\circ$ which vanishes at no point of that set. It is enough to prove this after replacing $X_\circ$ by the affine open $U_\circ$; there it follows from the familiar fact that an invertible module on a semilocal scheme is isomorphic to the structure sheaf. Finally, if $X_\circ$ admits an ample invertible sheaf, every finite subset of $X_\circ$ is contained in an affine open subset. This proves (ii).

We have the following corollary, which generalizes XII 8.6 and completes the proof of XII 5.1 (iii).

Corollary 3.2. *Let S be the spectrum of a Noetherian Henselian ring, and let $f : X \to S$ be a proper morphism of relative dimension ≤ 1. For every closed subscheme $X'_\circ$*
143 *of X having the same underlying space as the closed fiber $X_\circ$ of X/S, the canonical map* $\operatorname{Pic}(X) \to \operatorname{Pic}(X'_\circ)$ *is surjective.*

Indeed, apply the proposition to $(X, X'_\circ)$. Condition (a) holds by hypothesis, and condition (b) follows from XII 5.7. It remains to verify the existence of $U_\circ$ required in (ii); this follows because $X'_\circ$ is necessarily projective, as is every proper curve over a field.

Remark. A reader who does not wish to assume or reconstruct the fact that proper curves over a field are projective may use the corollary only under the additional hypothesis that f is projective; this suffices for the application intended here. Conversely, admitting the projectivity of proper algebraic curves over a field, one concludes that under the hypotheses of 3.2, X is necessarily projective over S. Indeed, choose an ample $\xi_\circ \in \operatorname{Pic}(X_\circ)$, lift it to some $\xi \in \operatorname{Pic}(X)$, and apply EGA III 4.7.1.

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EXPOSÉ XIV

Finiteness theorem for a proper morphism; cohomological dimension of affine algebraic schemes

M. Artin

This exposé contains two important theorems whose proofs use the base-change theorem for a proper morphism. 145

1. Finiteness theorem for a proper morphism

Theorem 1.1. *Let $f : X \to S$ be a proper morphism of finite presentation. Let A be a left Noetherian ring which is a $\mathbf{Z}/n$-algebra for some suitable $n \in \mathbf{N}$. Let F be a* constructible *sheaf of sets (resp. groups, resp. A-modules) on X. Then the sheaf f_*F (resp. $R^1 f_*F$, resp. $R^q f_*F$ for every $q \geq 0$) is also constructible.*

In particular, we have:

Corollary 1.2. *Let X be a proper scheme over the spectrum of a separably closed field k, and let F be a constructible torsion abelian sheaf on X. Then the groups $H^q(X, F)$ are finite for every $q \geq 0$.*

Remark 1.3. The theorem is false for abelian sheaves if the condition that f be proper is omitted, as is seen by taking $F = \mathbf{Z}/p$, $p = \operatorname{car} k$, $X = \mathbf{E}^1_k$, the affine line, and f either the inclusion of X in $\mathbf{P}^1_k$ or the structural morphism $X \to \operatorname{Spec} k$. Nevertheless, it is conjectured[35] that it is true if F has order prime to the residue characteristics of S and 146
S is "excellent" (EGA IV 7.8). When resolution of singularities is available (for example, in characteristic zero), this can be proved for an excellent equicharacteristic scheme S (XIX 5.1).

Proof of 1.1. The assertion is local on S, so we may take S affine. Then S is a limit of spectra of rings of finite type over $\mathbf{Z}$, and the data of the theorem can be obtained, by base change $S \to S_0$, from a proper morphism $f_0 : X_0 \to S_0$, with S_0 of finite type over $\mathbf{Z}$, and a constructible sheaf F_0 on X_0; see EGA IV 8 and IX 2.7.4. Since the inverse image of a constructible sheaf is again constructible (IX 2.4 (iii)), application of XII 5.1 to the base-change morphism $S_0 \leftarrow S$ reduces us to proving the constructibility of $f_{0*}F_0$ (resp. $R^q f_{0*}F_0$), whence the following lemma.

Lemma 1.4. *It is enough to verify 1.1 when S is of finite type over* $\operatorname{Spec} \mathbf{Z}$.

[35]N.D.E.: Ofer Gabber presented the following generalization of 1.1 at the conferences held in honor of Luc Illusie and Pierre Deligne in 2005. Let $f : X \to Y$ be a morphism of finite type between quasi-excellent Noetherian schemes (EGA IV 7.8), and let F be a constructible sheaf of $\mathbf{Z}/n$-modules on $X_{\text{ét}}$, with n invertible on Y. Then the sheaves $R^i f_* F$ on $Y_{\text{ét}}$ are constructible and vanish for all but finitely many i. There are analogous assertions for constructible sheaves of sets resp. groups of order invertible on Y.
Before Gabber's result, Theorem 1.1 was known in the following case. Let S be a regular scheme of dimension ≤ 1, let $f : X \to Y$ be a morphism of finite type between S-schemes, and let F be a sheaf of left A-modules on $X_{\text{ét}}$. Then the sheaves $R^i f_* F$ are constructible (SGA 4 1/2, Finitude 1.1).

LEMMA 1.5. *The set-theoretic assertion of 1.1 is true: that is, f_*F is a constructible sheaf of sets if F is one.*

Proof. We may suppose that S is Noetherian (1.4). Let $F \to G = \prod_i \pi_{i*}(C_i)$ be an injection, with $\pi_i : X_i \to X$ finite and C_i constant and constructible on X_i (IX 2.14). We have $f_*F \subset f_*G$; hence, by IX 2.9 (ii), it is enough to treat the case $F = G$. Now $f_*G = \prod_i (f\pi_i)_* C_i$, and a finite product of constructible sheaves on S is constructible (IX 2.6 (ibis)). Replacing X by X_i, we are reduced to the case where $F = D_X$ is a constructible constant sheaf with value D. By XII 5.1 (i), the fiber $(f_*F)_{\bar{s}}$ is isomorphic
147 to $D^{C(s)}$, where $C(s) = \pi_0(X_{\bar{s}})$; it is therefore finite. By IX 2.13 (iii), it is enough to prove that the function "number of elements in the fiber of f_*F" is constructible on S. This function is card(D^C), where C is the function "number of connected components in the geometric fiber", and C is constructible (EGA IV 9.7.9), which proves the result.

LEMMA 1.6. *Let S be a Noetherian scheme, let $f : X \to S$ be proper, let F be a constructible sheaf of groups (resp. A-modules) on X, and let q be an integer equal to 0 or 1 (resp. an integer ≥ 0). Let S_i $(1 \leq i \leq n)$ be subschemes of S such that $S = \bigcup_i S_i$, and let F_i and $f_i : X_i \to S_i$ be the "restrictions" of the data to S_i. If $R^q f_{i*} F_i$ is constructible for every i, then $R^q f_* F$ is constructible as well.*

Proof. The restriction of $R^q f_* F$ to S_i is isomorphic to $R^q f_{i*} F_i$ by XII 5.1, and the lemma follows from IX 2.8.

We now proceed by arguments analogous to those of XII 8:

LEMMA 1.7. *Let S be Noetherian, and let*

$$\begin{tikzcd} X \arrow[dr, "f"'] & & \overline{X} \arrow[ll, "\pi"'] \arrow[dl, "\overline{f}"] \\ & S & \end{tikzcd}$$

be a commutative diagram of proper morphisms, and let $j : U \to X$ be an open subscheme such that the open subscheme $\overline{U} = U \times_X \overline{X}$ maps isomorphically onto U. Let $Y = X - U$ be the reduced closed subscheme, and let $f_0 : Y \to S$ and $i : Y \to X$ be the canonical morphisms. Suppose that Theorem 1.1 holds[36] for $\overline{f}$, f_0, and π. Then it also holds for f.

148 *Proof.* First consider a sheaf of A-modules F on X. We have the exact sequence

$$0 \longrightarrow j_! j^* F \longrightarrow F \longrightarrow i_* i^* F \longrightarrow 0.$$

In view of the corresponding cohomology exact sequence for the $R^q f_*$ and IX 2.6, it is enough to verify constructibility for the two outer terms. Notice that $i_* i^* F$ is constructible by IX 2.4 (iii) and IX 2.14 (i), and hence so is $j_! j^* F$ (IX 2.6). Since i is a closed immersion,

$$(R^q f_*) i_* (i^* F) \simeq (R^q f_{0*})(i^* F),$$

by VIII 5.6; constructibility in this case therefore follows from the hypothesis on f_0. Moreover, if $\overline{j} : \overline{U} \to \overline{X}$ denotes the inclusion morphism and $\overline{F} = \pi^* F$, then

$$(*) \qquad j_! j^* F \simeq \pi_* \overline{j}_! \overline{j}^* \overline{F},$$

[36]N.D.E.: The hypothesis concerning $R^q \pi_*$ occurs in the proof only when $q = 0$. In that case constructibility is already known from 1.5.

as is seen by first defining a morphism from left to right and then proving it to be invertible fiber by fiber using XII 5.1. At every geometric point $\overline{y}$ of Y, the restriction of $\overline{j}_!\overline{j}^*\overline{F}$ to the fiber $\overline{X}_{\overline{y}}$ of $\overline{X}$ over X is zero; hence, by XII 5.1 (iii), the fiber of $(R^q\pi_*)(\overline{j}_!\overline{j}^*\overline{F})$ at $\overline{y}$ is zero for every $q \geq 0$. On the other hand, π induces an isomorphism over $X - Y$, so

$$(R^q\pi_*)(\overline{j}_!\overline{j}^*\overline{F}) = 0$$

for $q > 0$. The Leray spectral sequence

$$E_2^{p,q} = (R^p f_*)(R^q\pi_*)(\overline{j}_!\overline{j}^*\overline{F}) \Longrightarrow R^n\overline{f}_*(\overline{j}_!\overline{j}^*\overline{F})$$

together with $(*)$ therefore gives isomorphisms

$$(R^q\overline{f}_*)(\overline{j}_!\overline{j}^*\overline{F}) \simeq (R^q f_*)(j_!j^*F)$$

for $q \geq 0$. Since $\overline{j}_!\overline{j}^*\overline{F}$ is constructible (by the same argument as for $j_!j^*F$), the left-hand side is constructible by the hypothesis on $\overline{f}$. Thus $(R^q f_*)(j_!j^*F)$ is constructible, and consequently so is $R^q f_*F$. 149

Now suppose that F is a constructible sheaf of groups on X. Replace $\pi : \overline{X} \to X$ by $\pi \amalg i : (\overline{X} \amalg Y) \to X$. It is then enough to prove that if $\pi : \overline{X} \to X$ is *surjective* and 1.1 holds for π and $\overline{f}$, then it also holds for f. First let $\overline{F}$ be a constructible sheaf of groups on $\overline{X}$. Then $\pi_*\overline{F}$ is constructible on X (1.5), and there is an injection

$$(R^1 f_*)\pi_*\overline{F} \hookrightarrow (R^1\overline{f}_*)\overline{F}.$$

The right-hand side is constructible by hypothesis, and hence so is the left-hand side (IX 2.9 (ii)). Since π is surjective, there is an injection $F \to \pi_*\pi^*F = G$. The sheaf $\overline{F} = \pi^*F$ is constructible, so G and $R^1 f_*G$ are constructible as well, by what we have just proved. We are thus reduced to the following lemma.

Lemma 1.8. *Let S be Noetherian, let $f : X \to S$ be a proper morphism, and let $F \overset{u}{\hookrightarrow} G$ be an injection of constructible sheaves of groups on X. If $R^1 f_*G$ is constructible, then $R^1 f_*F$ is constructible as well.*

Proof. Let $C = G/F$, which is a constructible sheaf of homogeneous spaces under G (apply IX 2.6). Consider the exact sequence

$$(*) \qquad f_*C \xrightarrow{\delta} R^1 f_*F \xrightarrow{u^1} R^1 f_*G.$$

To prove that $R^1 f_*F$ is constructible, it is enough, by Noetherian induction and 1.6, to prove that if $S \neq \varnothing$, there exists a nonempty open subset of S on which $R^1 f_*F$ is constructible. We may therefore suppose that $R^1 f_*G$ is locally constant. Since constructibility is local for the étale topology, we may further suppose that S is irreducible and that $R^1 f_*G$ is constant (and constructible), with value $E = \{e_1, \dots, e_n\}$. For $i = 1, \dots, n$, let D_i be the subsheaf of $R^1 f_*F$ which is the inverse image of the section e_i under u^1. Since "coproducts are universal" in a topos, we have $R^1 f_*F = \amalg_i D_i$; it is therefore enough to prove that each D_i is constructible over a suitable nonempty scheme S' étale over S. 150

Recall that $R^1 f_*F$ is the sheaf associated with the presheaf $\mathscr{R}^1F$, where $\mathscr{R}^1F(S') = H^1(X \times_S S', F)$. Consequently, we may suppose (after replacing S by a nonempty scheme S' étale over

S) that D_i is the "empty sheaf" (hence constructible), *or* that there exists an element $\alpha \in H^1(X, F)$ inducing a section $\overline{\alpha}$ of D_i. In the latter case, let

$$f_* C^\alpha \xrightarrow{\delta} R^1 f_* F^\alpha \longrightarrow R^1 f_* G^\alpha$$

be the exact sequence deduced from $(*)$ by twisting with α. We have

$$R^1 f_* F^\alpha \simeq R^1 f_* F \quad , \quad R^1 f_* G^\alpha \simeq R^1 f_* G,$$

and the section of $R^1 f_* F^\alpha$ corresponding to $\overline{\alpha}$ is the unit section. Thus the subsheaf D_i^α of $R^1 f_* F^\alpha$ corresponding to D_i is the inverse image of the unit section of $R^1 f_* G^\alpha$; that is, D_i^α is the image of $f_* C^\alpha$ under δ. Since $f_* C^\alpha$ is constructible (1.5), so is D_i^α, and hence so is D_i (apply IX 2.6). This proves the lemma.

Lemma 1.9. *Let S be noetherian, and let*

$$\begin{tikzcd} X \arrow[dr, "f"'] & & \overline{X} \arrow[ll, "\pi"'] \arrow[dl, "\overline{f}"] \\ & S & \end{tikzcd}$$

151 *be a commutative diagram of proper morphisms. If Theorem 1.1 holds for f and π, then it also holds for $\overline{f}$.*

Proof. In the case of a constructible sheaf $\overline{F}$ of A-modules, we use the Leray spectral sequence

$$E_2^{p,q} = (R^p f_*)(R^q \pi_*)\overline{F} \Longrightarrow (R^n \overline{f}_*)\overline{F}.$$

By hypothesis, $E_2^{p,q}$ is constructible for every p, q; hence the abutment is constructible as well, as follows from IX 2.6.

Let $\overline{F}$ be a constructible sheaf of groups. We have the exact sequence (XII 3.2)

$$(*) \qquad 0 \longrightarrow (R^1 f_*)\pi_* \overline{F} \longrightarrow (R^1 \overline{f}_*)\overline{F} \longrightarrow f_*(R^1 \pi_*)\overline{F},$$

whose outer terms are constructible by the hypothesis and 1.5. We proceed as in the proof of 1.8: by noetherian induction and 1.6, it suffices to assume $S \neq \varnothing$ and prove constructibility over a suitable nonempty open subset. We may thus assume that $f_*(R^1\pi_*)\overline{F}$ is locally constant and, after an étale localization, even constant and constructible (IX 2.4 (i)), with value $E = \{e_1, \dots, e_n\}$. Let D_i be the subsheaf of $(R^1 \overline{f}_*)\overline{F}$ that is the inverse image of e_i, so that $(R^1 \overline{f}_*)\overline{F} = \coprod_i D_i$. It is enough to show that each sheaf D_i induces a constructible sheaf over some suitable nonempty connected étale S'/S. If D_i is not the "empty sheaf", we reduce to the case in which there exists an element $\alpha \in H^1(\overline{X}, \overline{F})$ inducing a section $\overline{\alpha}$ of D_i. Let

$$0 \longrightarrow (R^1 f_*)\pi_* \overline{F}^\alpha \longrightarrow (R^1 \overline{f}_*)\overline{F}^\alpha \longrightarrow f_*(R^1 \pi_*)\overline{F}^\alpha$$

be the exact sequence deduced from $(*)$ by twisting $\overline{F}$ with α. The section of $(R^1 \overline{f}_*)\overline{F}^\alpha$ corresponding to $\overline{\alpha}$ is the unit section. Consequently the subsheaf D_i^α of $(R^1 \overline{f}_*)\overline{F}^\alpha$
152 corresponding to D_i is the inverse image of the unit section of $f_*(R^1\pi_*)\overline{F}^\alpha$, whence $D_i^\alpha \simeq (R^1 f_*)\pi_* \overline{F}^\alpha$, which is indeed a constructible sheaf, C.Q.F.D.

Lemma 1.10. *It suffices to verify 1.1 when S is of finite type over* $\operatorname{Spec} \mathbf{Z}$, *and f is projective and of relative dimension ≤ 1.*

Proof. Suppose first that f is projective, and argue by induction on the relative dimension of f, which may be assumed finite after replacing S by an affine open subset. If f has relative dimension $\leq n$, with $n \geq 2$, then locally on S one can find a finite morphism $\pi : X \to \mathbf{P}^n_S$. By 1.9 and IX 2.14 (i), we reduce to the case in which f is the structural morphism of $\mathbf{P}^n_S$. Consider the commutative diagram of morphisms

$$\begin{tikzcd} & \overline{X} \arrow[dl, "\pi"'] \arrow[dd, "\overline{f}"] \arrow[dr, "a"] & \\ \mathbf{P}^n_S \arrow[dr, "f"'] & & \mathbf{P}^1_S \arrow[dl, "b"] \\ & S & \end{tikzcd}$$

already considered in the proof of XII 8.3, where $\overline{X}$ is obtained from $\mathbf{P}^n_S$ by blowing up the closed subscheme $Y \simeq \mathbf{P}^{n-2}_S$. The relative dimension of b is ≤ 1, and that of a is $\leq n-1$; therefore the theorem holds for a and b by the induction hypothesis, and hence for $\overline{f}$ by 1.9. Moreover, the relative dimension of π is ≤ 1, while that of $f_0 : Y \to S$ is $\leq n-1$. It therefore follows from 1.7 that 1.1 holds for f, and thus for every projective morphism. Now let f be an arbitrary proper morphism, and argue by noetherian induction on X: we may assume the theorem holds for every proper closed subscheme of X and that $X \neq \emptyset$. Let

$$\begin{tikzcd} X \arrow[dr, "f"'] & & \overline{X} \arrow[ll, "\pi"'] \arrow[dl, "\overline{f}"] \\ & S & \end{tikzcd}$$

be a diagram of the type supplied by Chow's lemma XII 7.1. Applying 1.7 to this diagram 153
yields 1.1 for f, and proves the lemma.

1.11. We can now complete the proof of 1.1. If F is a constructible sheaf of groups, one can find an injection $F \hookrightarrow G = \prod_i \pi_{i,*}C_i$, where each π_i is finite and each C_i is constant and constructible (IX 2.14); by 1.8, we may therefore assume that F is constant. Likewise, if F is a constructible sheaf of A-modules, then by applying IX 2.14 one can find a resolution $F \to G^{\cdot}$, $G^{\cdot} = \{0 \to G^0 \to G^1 \to G^2 \to \dots\}$, of F such that every G^ν is of the form $G^\nu = \prod_i \pi_{i,*}C_i$. The spectral sequence of the resolution $G^{\cdot}$,

$$E_2^{p,q} = H^p((R^q f_*)G^{\cdot}) \Longrightarrow (R^n f_*)F,$$

then reduces us to the case in which F is constant and constructible, with value M.

Recall that A is a $\mathbf{Z}/n$-algebra. Let ℓ be a prime number dividing n. We have the exact sequence

$$0 \longrightarrow \ell M \longrightarrow M \longrightarrow M/\ell M \longrightarrow 0,$$

and it is again enough to prove 1.1 for ℓM and $M/\ell M$.

An easy induction then reduces us to the case in which A is a $\mathbf{Z}/\ell$-algebra, $\ell \in \mathbf{P}$, and M is a finite-type A-module. Then M is a vector space over $\mathbf{Z}/\ell$, and the canonical morphism

$$H^q(Y, \mathbf{Z}/\ell) \otimes_{\mathbf{Z}/\ell} M \longrightarrow H^q(Y, M)$$

is bijective for every quasi-compact and quasi-separated Y (indeed, this is trivial if M
154 has finite rank, and both sides commute with filtered direct limits (VI 5.1 and VII 3.3)). Hence

$$[R^q f_*(\mathbf{Z}/\ell)] \otimes_{\mathbf{Z}/\ell} M \xrightarrow{\sim} R^q f_*(M).$$

It plainly suffices, therefore, to show that $R^q f_*(\mathbf{Z}/\ell)$ is constructible as a $\mathbf{Z}/\ell$-module; in other words, we are reduced to the case $A = \mathbf{Z}/\ell$, which we assume from now on.

Observe that the case in which f has relative dimension ≤ 0 (that is, f is finite), with F arbitrary, now follows from VIII 5.6 and 1.5.

We proceed by noetherian induction on S. By 1.6, we may assume $S \neq \varnothing$ and may replace S by any nonempty open subset. Moreover, we may assume that S is integral, with generic point s. The fiber X_s is an algebraic scheme of dimension ≤ 1; consequently there is a purely inseparable extension K' of $k(s) = K$ such that the normalization Y of $(X_s \otimes_K K')_{\mathrm{red}}$ is smooth over $\operatorname{Spec} K'$ [36]. After replacing S by a nonempty open subset, we may assume that there exists a finite, radicial, surjective morphism $S' \to S$ such that S'_s is K-isomorphic to $\operatorname{Spec} K'$, a smooth projective morphism $\overline{X}' \to S'$, and a finite surjective S'-morphism $\pi : \overline{X}' \to X' = X \times_S S'$ inducing $Y \to X_s \otimes_K K'$ on the fibers over the generic point s of S (EGA IV 9.6.1). Now, by VIII 1.1, making a purely inseparable extension is harmless. We may therefore replace S by S' and X by X'; that is, we may assume that f fits into a diagram

$$\begin{tikzcd} X \arrow[dr, "f"'] & & \overline{X} \arrow[ll, "\pi"'] \arrow[dl, "\overline{f}"] \\ & S & \end{tikzcd},$$

155 in which π is finite and surjective and $\overline{f}$ is smooth. Moreover, by construction, we may assume that there is a closed subscheme Y of X whose generic fiber is finite and such that π induces an isomorphism over $X - Y$. Replacing S once more by a nonempty open subset, we reduce to the case in which $f_0 : Y \to S$ is also finite.

Since F is constant, $\pi^* F$ is constant as well, and $F \to \pi_* \pi^* F$ is injective because π is surjective. If F is a sheaf of groups, it suffices to prove that $(R^1 f_*)(\pi_* \pi^* F) \simeq (R^1 \overline{f}_*)(\pi^* F)$ is constructible (1.8). If F is a sheaf of $\mathbf{Z}/\ell$-modules, we have an exact sequence

$$0 \longrightarrow F \longrightarrow \pi_* \pi^* F \longrightarrow C \longrightarrow 0,$$

where C is supported on Y, because π induces an isomorphism over $X - Y$. It follows that $R^q f_* C = R^q f_{0,*} C$ is constructible (and vanishes if $q > 0$), so it is once again enough to show that $(R^q f_*)(\pi_* \pi^* F) = (R^q \overline{f}_*)(\pi^* F)$ is constructible. Replacing f by $\overline{f}$ and F by $\pi^* F$, we are reduced to the case in which F is constant and constructible and f is smooth.

Let us first treat a constant sheaf of groups, $F = G_X$. All the fibers of $(R^1 f_*) G_X$ are finite. Indeed, by applying XII 5.2 (ii), we reduce, in order to prove this, to the case in which S is the spectrum of a separably closed field and X is smooth of dimension ≤ 1 over S. The result is known in this case (SGA 1 X 2.6). It therefore remains only to prove that the specialization morphisms are injective (IX 2.13 (ii)). For this we reduce to the case in which S is strictly local; by the base-change theorem XII 5.1 (ii), we may

[36] Cf. EGA IV 17.15.14.

even replace S by the spectrum of a strictly local discrete valuation ring. Moreover, we may assume X connected. Since $H^1(X,G)$ classifies the principal Galois coverings of 156
X with group G, it is enough to check that if $X' \to X$ is a connected étale covering, then its geometric fiber is connected as well. This follows readily from the fact that f is smooth and proper with geometrically connected fibers.

We now treat the case $F = (\mathbf{Z}/\ell)_X$. For $q = 0$ and $q = 1$, the result is already known (1.5 and the preceding argument). By XII 5.3 bis, only the case $q = 2$ remains; furthermore, if S has characteristic ℓ, then $R^2 f_*(\mathbf{Z}/\ell) = 0$. Replacing S by a suitable nonempty open subset, we therefore reduce to the case in which ℓ is invertible on S. Then, by XII 5.2 (iii) and X 5.2, the fiber of $R^2 f_*(\mathbf{Z}/\ell)$ at a geometric point $\overline{s}$ above $s \in S$ is isomorphic to $(\mathbf{Z}/\ell)^{c(s)}$, where $c(s)$ is the number of one-dimensional irreducible components of the geometric fiber $X_{\overline{s}}$. The function c is constructible on S (EGA IV 9.7.9), and the result follows (IX 2.13 (iii)).

Remark 1.11. The proof of 1.1 becomes much simpler if one uses the formalism of cohomology with proper supports, which will be developed in XVII as a direct consequence of the base-change theorem for a proper morphism XII 5.1. Moreover, the statement of 1.1 generalizes to the case in which one is given an arbitrary sheaf $\mathscr{A}$ of torsion rings on S and a complex $K^\cdot$ of $f^*(\mathscr{A})$-modules on X satisfying a "constructibility" condition which, in the special case of 1.1, would simply say that the cohomology sheaves $\mathscr{H}^i(K^\cdot)$ of $K^\cdot$ are constructible A-modules and vanish for sufficiently large i.

2. A variant of dimension

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2.1. Let X be a scheme. We shall write

$$d(x) = \dim \overline{\{x\}} \tag{2.1.1}$$

for $x \in X$, where $\overline{\{x\}}$ is the closure of $\{x\}$. If F is an abelian sheaf on X, we put

$$d(F) = \sup\{d(x) \mid x \in X \quad \text{et} \quad F_{\overline{x}} \neq 0\}. \tag{2.1.2}$$

We shall need a corrected notion in the local case:

Definition 2.2. *Let Y be the spectrum of a universally catenary Noetherian local ring (EGA IV 5.6.2), and let $f : X \to Y$ be a morphism of finite type. Let $x \in X$ and $y = f(x)$. We put*

$$\delta(x) = \dim \overline{\{y\}} + \text{deg.tr. } k(x)/k(y).$$

If F is an abelian sheaf on X, we put

$$\delta(F) = \sup\{\delta(x) \mid x \in X \quad \textit{et} \quad F_{\overline{x}} \neq 0\}.$$

Proposition 2.3. (i) *Let $i : X' \to X$ be an immersion and let $x' \in X'$. Then $\delta(i(x')) = \delta(x')$.*

(ii) *Let X_0 be the closed fiber of X/Y. We have*

$$\delta(x) = \begin{cases} d(x) & \text{if } \overline{\{x\}} \cap X_0 \neq \varnothing \\ d(x)+1 & \text{if } \overline{\{x\}} \cap X_0 = \varnothing; \end{cases}$$

(iii) *$\delta(x) = d(x)$ if f is proper.*

Proof. Assertion (i) follows immediately from the definition. Since $\overline{\{x\}} \cap X_0$ is nonempty 158
when f is proper, (iii) follows from (ii). For (ii), we may replace X by $\overline{\{x\}}$ and Y by $\overline{\{y\}}$

(with their induced reduced structures), as is permitted by (i). Let $x' \in X$ and $y' = f(x')$. We have the formula (EGA IV 5.6.1.1)

$$\dim \mathcal{O}_{X,x'} = \dim \mathcal{O}_{Y,y'} + \text{deg.tr. } k(x)/k(y) - \text{deg.tr. } k(x')/k(y').$$

Suppose that $X_0 \neq \varnothing$. It is then clear that $\sup_{x'} \{\dim \mathcal{O}_{X,x'}\} = \dim X$ is attained by taking x' to be a closed point of X_0, whence

$$\dim X = \dim \mathcal{O}_{X,x'} = \dim Y + \text{deg.tr. } k(x)/k(y) = \delta(x).$$

If $X_0 = \varnothing$, the supremum is plainly attained by taking x' to be a closed point in its fiber $f^{-1}(y')$, with $\dim \overline{\{y'\}} = 1$. Such points exist because the $y' \in Y$ with $\dim \overline{\{y'\}} = 1$ form a very dense set (EGA IV 10.1.3, 10.3.1, 10.4.1, 10.5.9). Thus

$$\begin{aligned} \dim X = \dim \mathcal{O}_{X,x'} &= \dim \mathcal{O}_{Y,y'} + \text{deg.tr. } k(x)/k(y) \\ &= \dim Y - 1 + \text{deg.tr. } k(x)/k(y) = \delta(x) - 1, \end{aligned}$$

which proves the result.

Proposition 2.4. *Let $f : X \to Y$ be a morphism of schemes of finite type over the spectrum of a field k. Let $x \in X$, let $y = f(x)$, and let y_0 be a specialization of y. Let $f' : X' \to Y'$ be the strict localization of f at a geometric point $\overline{y}_0$ above y_0, and let $x' \in X'$ be a point above x. Then*

$$\delta(x') = d(x) - d(y_0).$$

159 *Proof.* We have

$$\begin{aligned} d(x) &= d(y) + \text{deg.tr. } k(x)/k(y) \\ \delta(x') &= d(y') + \text{deg.tr. } k(x')/k(y'), \end{aligned}$$

where $y' = f'(x')$. Moreover,

$$\text{deg.tr. } k(x)/k(y) = \text{deg.tr. } k(x')/k(y'),$$

because the extensions $k(x')/k(x)$ and $k(y')/k(y)$ are algebraic. It therefore remains to prove that

$$d(y') = d(y) - d(y_0).$$

We may assume that y is maximal in Y, and hence that y' is maximal in Y'. Then

$$\begin{aligned} d(y') &= \dim \mathcal{O}_{Y',\overline{y}_0} \\ &= \dim \mathcal{O}_{Y,y_0} \quad \text{(EGA IV 6.1.3)} \\ &= \dim_{y_0} Y - d(y_0) \quad \text{(EGA IV 5.2.3.1)} \\ &= d(y) - d(y_0), \end{aligned}$$

which proves the result.

3. Cohomological dimension of affine algebraic schemes

Theorem 3.1 (d). [37] *Let $f : X \to Y$ be an affine morphism of schemes of finite type over a field k. Let F be a torsion sheaf on X such that $d(F) \leq d$* (cf. 2.1). *Then $d(R^q f_* F) \leq d - q$.*

In particular, taking $Y = \text{Spec}(k)$, we obtain:

160 Corollary 3.2. *Let X be an affine scheme of finite type over a separably closed field k. Then*[38]

$$\operatorname{cd} X \leq \dim X.$$

Corollary 3.3. *Let $X \subset \mathbf{P}_k^N$ be a projective scheme of dimension n over a separably closed field k, and let $Y = H \cap X$ be a "hyperplane section" of X. Then, for every torsion sheaf F on X, the canonical homomorphism*

$$H_Y^q(X, F) \longrightarrow H^q(X, F)$$

is bijective for $q > n+1$ and surjective for $q = n+1$.

The corollary follows immediately from 3.2 and the exact sequence for a closed subset (V 6.5.4), since $X - Y = U$ is affine of dimension $\leq n$. Notice that 3.3 is a generalization of one of Lefschetz's theorems on hyperplane sections. Indeed, if X is smooth over k and F is locally constant of order prime to the characteristic, the cohomology with support $H_Y^q(X, F)$ can be computed explicitly using the relative cohomological purity theorem (XVI 3) and the spectral sequence (V 6.4.3), and one finds that

$$H_Y^q(X, \mathbf{Z}/n) \simeq H^{q-2}(Y, \mu_n^{-1}) \quad \text{for all } q;$$

the homomorphism $H^{q-2}(Y, \mu_n^{-1}) \to H^q(X, \mathbf{Z}/n)$ furnished by 3.3 is precisely the "Gysin homomorphism", which will be studied later [38].

Theorem 3.1 (d) is equivalent[39] to the following statement:

Corollary 3.4 (d). *Let Y' be a strictly local scheme that is a strict localization of an algebraic scheme over a field. Let $f' : X' \to Y'$ be an affine morphism of finite type, and* 161
let F' be a torsion sheaf on X' such that $\delta(F') \leq d$ (cf. (2.2)). *Then $H^q(X', F') = 0$ for $q > d$.*

Indeed, suppose that 3.4 (d) is true. Let $f : X \to Y$ be an affine morphism of algebraic schemes over k, and let F be a torsion sheaf on X such that $d(F) \leq d$. Let y be a point of Y, and let ξ be a geometric point over y. To prove 3.1 (d), it is enough to prove that the stalk $(R^q f_* F)_\xi$ vanishes when $d(y) > d - q$.

Let $f' : X' \to Y'$ be the strict localization of f at the point ξ; that is, let Y' be the strict localization of Y at ξ, and let $X' = X \times_Y Y'$.

Let F' be the sheaf induced by F on X', so that $(R^q f_* F)_\xi = H^q(X', F')$ (VIII 5.2). By (2.4), we have

$$\delta(F') \leq d(F) - d(y) \leq d - d(y),$$

and applying 3.4 (d), we obtain $H^q(X', F') = 0$ if $q > d - d(y)$, which proves the result.

Conversely, suppose that 3.1 (d) is true, and let F' and $f' : X' \to Y'$ be as in the statement of 3.4 (d). Since cohomology commutes with filtered inductive limits (VII 3.3) and F' is the inductive limit of its constructible subsheaves (IX 2.9), we may suppose that F' is constructible. Now Y' is the strict localization of an algebraic scheme Y at

[37] N.D.E.: Theorem 1.1 is now known as the *affine Lefschetz theorem.*

[38] N.D.E.: Consider the analogous assertion in complex analytic geometry: let X be an affine variety over $\mathbf{C}$ and let F be a constructible sheaf on X. Then $H^m(X^{\mathrm{an}}, F) = 0$ for every $m > \dim X$, where X^{an} is the analytic space associated with X. Although this assertion follows from 3.2 by the comparison theorem XVI 4.1, there are also two entirely analytic proofs, due to Pierre Deligne and Hélène Esnault; see Esnault, H., Variation on Artin's vanishing theorem, *Advances in Math.* **198** (2005), 435–438.

[38] Cf. SGA 2 XIV for a systematic study of Lefschetz-type theorems.

[39] N.D.E.: More precisely, the argument in the text gives the following result. Assertion 3.4 (d') for $d' \leq d$ implies 3.1 (d), and assertion 3.1 (d) implies 3.4 (d). This suffices for the conclusion below.

a geometric point ξ over a *closed* point y_0 (X 3.3). Write $Y' = \varprojlim_\alpha Y_\alpha$, where (Y_α) is a filtered inverse system of affine schemes étale over Y (VIII 4.5). By IX 2.7.4, F' is induced by a constructible torsion sheaf F_α on $X_\alpha = X \times_Y Y_\alpha$ for sufficiently large α. Using 2.4 and
162 EGA IV 9.5.5, we see immediately that we may further suppose that $d(F_\alpha) = \delta(F') \leq d$. Replacing Y by Y_α, we are reduced to the case in which F' is induced by a sheaf F on X, with $d(F) \leq d$. Then $R^q f_* F = 0$ for $q > d$ by 3.1 (d), and hence in particular

$$(R^q f_* F)_\xi = H^q(X', F') = 0 \quad \text{if} \quad q > d,$$

which proves 3.4 (d).

As a special case of 3.4 (d), we have:

Corollary 3.5. *Let Y' be a strictly local scheme of dimension $\leq d$, which is a strict localization of an algebraic scheme over a field k. Let $U \subset Y'$ be an affine open subset. Then* $\operatorname{cd} U \leq d$.

Remark 3.6. It seems very plausible[40] that 3.4, and hence 3.5, remains valid for *every* local Noetherian scheme Y' (not necessarily a strict localization of an algebraic scheme), at least when Y' is *excellent* (EGA IV 7.8). This will be proved in (XIX 6) under the further assumption that Y has characteristic zero.

4. Proof of Theorem 3.1

The proof proceeds by induction on d.

Lemma 4.1. *3.1 (0) and 3.1 (1) hold.*

Proof. This is trivial for $d = 0$, in which case supp F is finite over k, and we apply (VIII 5.5). Let F and $f : X \to Y$ be as in the statement of 3.1, and suppose that $d(F) \leq 1$. Replacing F by a constructible subsheaf (IX 2.9 and VII 3.3), we reduce to the case in
163 which F is constructible. The support of F is then contained in a closed subscheme of X of dimension ≤ 1 (IX 2.3), and we may replace X by this subscheme; that is, we may assume that $\dim X \leq 1$. By IX 5.7, the cohomological dimension of an affine algebraic scheme of dimension 1 over a separably closed field is ≤ 1. It is enough, moreover, to treat the case in which k is separably closed. Indeed, let $\overline{f} : \overline{X} \to \overline{Y}$ be the morphism obtained from f by the base change $\operatorname{Spec} \overline{k} \to \operatorname{Spec} k$, where $\overline{k}$ is the separable closure of k. For a geometric point ξ of Y, the strict localization $f' : X' \to Y'$ of f at ξ identifies with a strict localization of $\overline{f}$, and consequently the fiber $(R^q f_* F)_\xi$ identifies with a fiber of $R^q \overline{f}_* \overline{F}$. Since f is affine, it follows that $R^q f_* F = 0$ for $q > 1$. It is also clear that $d(f_* F) \leq 1$. For $q = 1$, note that the hypothesis $\dim X \leq 1$ implies that there is a finite closed subset Z of Y such that f is finite over $Y - Z$. It follows that $R^1 f_* F$ vanishes on $Y - Z$ (VIII 5.6), and hence $d(R^1 f_* F) \leq 1$. This proves the lemma.

Let $d \geq 2$, and now suppose that 3.1 (d') (or, equivalently, 3.4 (d')) holds for $d' < d$.

Lemma 4.2. *To prove 3.4 (d) in general, it is enough to treat the case in which $X' = \mathbf{E}^1_{Y'}$, is affine one-space over Y' and f is the structural morphism.*

[40]N.D.E.: Ofer Gabber has succeeded in proving the following analogue of 3.1: let $f : X \to Y$ be an affine morphism of finite type, with Y quasi-excellent (EGA IV 7.8) and endowed with a dimension function δ_Y. Let δ_X be the associated rectified dimension function on X. If n is invertible on Y, then for every sheaf F of $\mathbf{Z}/n$-modules on X one has $\delta_Y(R^q f_* F) \leq \delta_X(F) - q$ for every $q \geq 0$. Here, as above, for a sheaf G we define $\delta(G) = \sup\{\delta(x) \mid x \in X \quad \text{et} \quad G_{\overline{x}} \neq 0\}$.

Proof. We may plainly assume that $X' = \mathbf{E}^N_{Y'}$ for a suitable N, since X' can be embedded in some $\mathbf{E}^N_{Y'}$. Suppose that $N > 1$ and that the result is known, for every Y', for $\mathbf{E}^r_{Y'}$ with $r < N$. Let F' be a torsion sheaf on $\mathbf{E}^N_{Y'}$ with $d(F') \leq d$. Consider the diagram 164

$$\begin{tikzcd} \mathbf{E}^N_{Y'} \arrow[rr, "g'"] \arrow[dr, "f'"'] & & \mathbf{E}^{N-1}_{Y'} \arrow[dl, "h'"] \\ & Y' & \end{tikzcd},$$

where g' and h' are the canonical projections, so that $\mathbf{E}^N_{Y'}$ is isomorphic to affine one-space over $\mathbf{E}^{N-1}_{Y'}$. Since every strict localization of $\mathbf{E}^{N-1}_{Y'}$ is plainly also the strict localization of an algebraic scheme, we may apply the induction hypothesis to the fibers of $R^q g'_* F'$, and obtain

$$(*) \qquad \delta(R^q g'_* F') \leq d - q.$$

Moreover, 3.4 (d') holds for h' and $d' \leq d$, by the induction hypothesis on N.

Consider the Leray spectral sequence

$$E_2^{p,q} = H^p(\mathbf{E}^{N-1}_{Y'}, R^q g'_*(F')) \Longrightarrow H^n(\mathbf{E}^N_{Y'}, F').$$

We have $E_2^{p,q} = 0$ if $p > d - q$, as follows by applying $(*)$ and 3.4 (d') to the morphism h'. Consequently the abutment vanishes for $n > d$, proving the lemma.

Consider the following additional statement, which is a special case of 3.5:

Statement 4.3 (d). *Let X' be strictly local of dimension $\leq d$, and the strict localization of an algebraic scheme over a field. Let $f \in \Gamma(X', \mathscr{O}_{X'})$, and let $U \subset X'$ be the open subset $U = X' - V(f)$. Then* $\operatorname{cd} U \leq d$.

Lemma 4.4. 4.3 (d) *implies* 3.4 (d).

Proof. Consider the diagram 165

$$\begin{tikzcd} \mathbf{E}^1_{Y'} \arrow[rr, "i"] \arrow[dr, "f'"'] & & \mathbf{P}^1_{Y'} \arrow[dl, "g'"] \\ & Y' & \end{tikzcd}$$

where Y' is as in the statement of 3.4, and where $\mathbf{P}^1_{Y'}$ is the projective line obtained from $\mathbf{E}^1$ "by adjoining the section Y'_∞ at infinity." Let F' be a torsion sheaf on $\mathbf{E}^1$ with $d(\operatorname{supp} F') \leq d$. We have the spectral sequence

$$H^p(\mathbf{P}^1, R^q i_* F') \Longrightarrow H^n(\mathbf{E}^1, F').$$

Now the $R^q i_* F'$, for $q > 0$, are supported on Y'_∞, which is Y'-isomorphic to Y'. Since Y' is strictly local, it follows that $H^p(\mathbf{P}^1, R^q i_* F') = 0$ if $p, q > 0$. Moreover, by the *proper base-change theorem* (XII 5.1), the cohomology of a torsion sheaf on $\mathbf{P}^1$ can be computed on the closed fiber, which is a scheme of cohomological dimension ≤ 2 (X 4.3). Thus $H^p(\mathbf{P}^1, i_* F') = 0$ if $p > 2$. Since we wish to prove that $H^n(\mathbf{E}^1, F') = 0$ for $n > d$, and since $d \geq 2$, we are reduced to proving that $H^0(\mathbf{P}^1, R^q i_* F') = 0$ for $q > d$. This group, which is also isomorphic to $H^0(Y'_\infty, R^q i_* F')$, is, by (VIII 4.6), isomorphic to the fiber of $R^q i_* F'$ at the point Q, where Q is the point at infinity of $\mathbf{P}^1$ in the closed fiber.

It is enough (VII 3.3 and IX 2.9) to treat the case in which F' is constructible. Since $\delta(\operatorname{supp} F') \leq d$, there is then a closed subscheme X' of $\mathbf{P}^1$, with $\dim X' \leq d$, which contains the support of F'. Let $\widetilde{X}'$ be the localization of X' at the geometric point Q, put

$$\widetilde{U} = \widetilde{X}' - \widetilde{X}' \times_{\mathbf{P}^1} Y'_\infty,$$

166 and let $\widetilde{F}$ be the sheaf on $\widetilde{U}$ induced by F' via the morphism $\widetilde{U} \to \mathbf{P}^1$. The fiber of $R^q i_* F'$ at Q is then precisely $H^q(\widetilde{U}, \widetilde{F})$ (VIII 5.2), and we apply 4.3 (d) to the strictly local scheme $\widetilde{X}'$ and the open subset $\widetilde{U}$.

The following lemma, together with 4.4, will complete the proof.

Lemma 4.5. 3.4 (d') *for* $d' < d$ *implies* 4.3 (d') *for* $d' \leq d$.

Proof. Suppose that 3.4 (d') holds for $d' < d$. By induction, we may suppose that 4.3 (d') holds for $d' < d$. Let X', f, and U be as in the statement 4.3 (d). Then X' is the strict localization, at a *closed* geometric point, of an algebraic scheme X_0 over a field k. After replacing X_0 by a neighborhood of x_0, we may take X_0 to be affine and of dimension $\leq d$. Then X' is the inverse limit of affine schemes X_i which are étale over X_0 (VIII 4.5).

Let F be a torsion sheaf on U. We must prove that $H^q(U, F) = 0$ for $q > d$, and once again it is enough to treat the case in which F is constructible. The data f, F, and U then arise from analogous data on one of the X_i (IX 2.7.4). Thus there is a sheaf F_i on $U_i = X_i - V(f_i)$ which induces F on U. Since $U = \varprojlim U_i$, we have (VII 5.8)

$$H^q(U, F) = \varinjlim H^q(U_i, F_i).$$

It is therefore enough to prove the following fact: for every i, there is a commutative diagram

$$\begin{tikzcd} U \arrow[rr] \arrow[dr] & & U'' \arrow[dl] \\ & U_i & \end{tikzcd}$$

167 in which $U \to U_i$ is the given arrow and $\operatorname{cd}(U'') \leq d$. Indeed, this implies that every element of $H^q(U_i, F_i)$, for $q > d$, has zero image in $H^q(U, F)$.

Suppose that $i = 0$, and consider the morphism $g : X_0 \to \mathbf{E}^1_k = Y_0$ defined by $f_0 \in \Gamma(X_0, \mathcal{O}_{X_0})$. Let Y'' be the strict localization of $\mathbf{E}^1_k$ at a geometric point above the origin. Thus Y'' is the spectrum of a discrete valuation ring. Put $X'' = X_0 \times_{Y_0} Y''$.

The morphism $X' \to X_0$ factors through X'', since X' is strictly local. Moreover, the open subset $U'' = X'' \times_{X_0} U_0$ is precisely the generic fiber of X''/Y''. It is therefore an affine algebraic scheme over the generic point of Y'', and plainly $\dim U'' \leq d - 1$.

Let us verify that $\operatorname{cd} U'' \leq d$, which will complete the proof. Let K be the residue field of Y'' at its generic point. This field has cohomological dimension 1; that is, $\operatorname{cd}(G) = 1$, where G is the Galois group of the separable closure $\overline{K}$ of K (X 2.2). Put $\overline{U}'' = U'' \times_{\operatorname{Spec} K} (\operatorname{Spec} \overline{K})$. By the induction hypothesis on d, we have $\operatorname{cd} \overline{U}'' \leq d - 1$ (this is 3.2 for a scheme of dimension $< d$). It then follows from the Hochschild–Serre spectral sequence (VIII 8.4) that $\operatorname{cd} U'' \leq d$:

$$H^p(G, H^q(\overline{U}'', \overline{F})) \Longrightarrow H^n(U'', F),$$

which proves Lemma 4.5.

EXPOSÉ XV

Acyclic morphisms

M. Artin

Introduction 168

Let $g : X \to Y$ be a morphism of schemes. In the first section we study conditions under which g is acyclic, that is, such that for every torsion sheaf F on Y one has $H^q(Y, F) \simeq H^q(X, g^*F)$, and such that this remains true when Y is replaced by a Y' étale over Y. A natural necessary condition for acyclicity is that the geometric fibers be acyclic, that is, have trivial cohomology for constant torsion sheaves. We shall see that this condition, together with a local condition called local acyclicity of g (1.11), implies that g is acyclic.

The fundamental theorem is that a smooth morphism is locally acyclic for torsion sheaves of order prime to the residue characteristics 2.1. This will imply the base-change theorem for a smooth morphism XVI 1.1, which will be developed, together with its first consequences, in the next exposé.

The present exposé is independent of Exposés XI to XIV, and in particular of the "base-change theorem for a proper morphism." The latter will again enter in an essential way, but in conjunction with the fundamental theorem of the present exposé, beginning with the next exposé.

1. Generalities on globally and locally acyclic morphisms

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Proposition 1.1. *Let $g : X \to Y$ be a morphism of schemes. The following assertions* (i) *to* (iii) *are equivalent:*

- (i) *For every étale morphism $Y' \to Y$ (where Y' may be taken affine over an affine open subset of Y) and every sheaf of sets F on Y', the canonical morphism $F \to g'_* g'^* F$ is injective (resp. bijective).*
- (ii) *For every étale morphism $Y' \to Y$ (where Y' may be taken affine over an affine open subset of Y) and every sheaf of sets F on Y', the canonical morphism $H^0(Y', F) \to H^0(X', g'^* F)$ is injective (resp. bijective).*
- (iii) *For every locally quasi-finite morphism $Y' \to Y$ and every sheaf of sets F on Y', the canonical morphism $H^0(Y', F) \to H^0(X', g'^* F)$ is injective (resp. bijective).*
- (ii bis) *(If g is quasi-compact and quasi-separated.) As in* (ii)*, but with F a constant sheaf of the form $I_{Y'}$, where I is a prescribed set with* card $I \geq 2$.

Proof. The equivalence of (i) and (ii) follows immediately from the definitions, while (iii) $\Rightarrow$ (ii) $\Rightarrow$ (ii bis) are trivial. We have (ii bis) $\Rightarrow$ (ii), since we may suppose Y affine, so that X and Y are quasi-compact and quasi-separated, and may then apply XII 6.5 (i). For the implication (i) $\Rightarrow$ (iii), we first record the following lemma:

Lemma 1.2. *Let $f : Y' \to Y$ be a locally quasi-finite morphism. Then f is "locally* 170 *finite on Y' for the étale topology," that is, there are commutative diagrams*

$$(*)\qquad \begin{tikzcd} Y'_i \arrow[r, "a_i"] \arrow[d, "b_i"] & Y' \arrow[d, "f"] \\ Y_i \arrow[r, "c_i"] & Y \end{tikzcd}$$

in which the a_i and c_i are étale, the b_i are finite, and the a_i form a covering of Y' [40].

Indeed, let y' be a point of Y' and let $y = f(y')$. It is enough to find a diagram $(*)$ such that y' lies in the image of Y'_i. Let $\widetilde{Y}$ be the spectrum of the henselization of the ring $\mathscr{O}_{Y,y}$ VIII 4.1, and let Z be the (ordinary) localization of $\widetilde{Y}' = Y' \times_Y \widetilde{Y}$ at the unique point lying over y' and the closed point $\tilde{y}$ of $\widetilde{Y}$. By VIII 4.1 (ii), $Z \to \widetilde{Y}$ is finite, and Z is an induced open subscheme of $\widetilde{Y}'$. Since $\widetilde{Y}$ is a limit of schemes Y_i étale over Y, one sees immediately from EGA IV 9 that, for a suitable Y_i, there is an open subscheme Z_i of $Y' \times_Y Y_i$ such that $Z = Z_i \times_{Y_i} \widetilde{Y}$; in $(*)$ it is then enough to take $Y'_i = Z_i$. C.Q.F.D.

Now suppose that (i) of 1.1 holds. It is clear that assertion (iii) of 1.1 is equivalent to the assertion obtained from (i) by replacing the étale morphism $Y' \to Y$ by a locally quasi-finite morphism. In this form, the assertion (for given Y and Y') is local on Y and Y' for the étale topology; hence the lemma reduces us to the case of a *fi-*
171 *nite* morphism $f : Y' \to Y$. Let $f' : X' \to X$ be the morphism obtained from f by base change along g. Then the base-change morphism $g^* f_* F \to f'_* g'^* F$ is bijective VIII 5.6. By (i), we have $f_*(F) \to g_* g^* f_* F$. Consequently, the morphism $H^0(Y', F) \xrightarrow{\epsilon} H^0(Y', g'_* g'^* F)$ in (iii) is the composite $H^0(Y', F) = H^0(Y, f_* F) \xrightarrow{\epsilon'} H^0(Y, g_* g^* f_* F) = H^0(X, g^* f_* F) \simeq H^0(X, f'_* g'^* F) = H^0(X', g'^* F)$. Since ϵ' is injective (resp. bijective), the same is true of ϵ. This completes the proof of 1.1.

Definition 1.3. *A morphism $g : X \to Y$ having one of the equivalent properties* (i) *to* (iii) *(resp. the corresponding parenthesized properties) of 1.1 is called a* (-1)-acyclic morphism *(resp. a* 0-acyclic morphism*). The morphism is said to be* universally (-1)-*acyclic (resp. 0-acyclic) if, for every base-change morphism $Y' \to Y$, the morphism $g' = g \times_Y Y' : X \times_Y Y' \to Y'$ is (-1)-acyclic (resp. 0-acyclic). Finally, a scheme X is said to be (-1)-acyclic (resp. 0-acyclic) if it is nonempty (resp. connected and nonempty).*

Corollary 1.4. *Let $g : X \to Y$ be a 0-acyclic morphism. Then for every locally quasi-finite morphism $Y' \to Y$ and every sheaf of groups F on Y', the canonical morphism*

$$H^1(Y', F) \longrightarrow H^1(X', g'^* F)$$

is injective.

This follows immediately from XII 6.5 (i).

172 Corollary 1.5. *Let $g : X \to Y$ be a morphism. If g is surjective (i.e. its fibers are nonempty, i.e. its fibers are (-1)-acyclic), then g is (-1)-acyclic. The converse holds if g is quasi-compact and quasi-separated.*

The first assertion is clear in the form 1.1 (ii), by considering the fibers of the sheaves in question. The second follows from the definition in the form 1.1 (iii) and the limit

[40] Cf. EGA IV 18.12.1.

argument of VII 5, taking into account that, for every $y \in Y$, $y = \operatorname{Spec} k(y)$ is the inverse limit of affine schemes locally quasi-finite over Y, namely the affine neighborhoods of y in $\overline{\{y\}}$ (endowed with the induced reduced structure). – Note that 1.5 implies that, when g is quasi-compact and quasi-separated, the (-1)-acyclicity of g already implies that g is *universally* (-1)-acyclic.

Proposition 1.6. *Let $g : X \to Y$ be a morphism of schemes, let $\mathbf{L} \subset \mathbf{P}$ be a set of prime numbers, and let $n \in \mathbf{N}$ be a natural number (resp. let $n = 1$). The following assertions* (i) *to* (iii) *(resp. the corresponding parenthesized assertions) are equivalent. If, moreover, g is quasi-compact and quasi-separated, then* (i) *to* (iii) *(resp. the corresponding parenthesized assertions) are also equivalent to* (iv) *(resp. the corresponding parenthesized assertion).*

(i) *For every étale morphism $Y' \to Y$ and every abelian $\mathbf{L}$-torsion sheaf (resp. sheaf of* ind-$\mathbf{L}$-*groups)* IX 1.5 *F on Y', one has*

$$\begin{cases} F \xrightarrow{\sim} g'_* g'^* F, \\ (R^q g'_*) g'^* F = 0 \quad \text{if} \quad 1 \le q \le n \quad (\text{resp. if } q = 1). \end{cases}$$

(ii) *For every étale morphism $Y' \to Y$ and every abelian $\mathbf{L}$-torsion sheaf (resp. sheaf of ind-$\mathbf{L}$-groups) F on Y', the canonical morphism* 173

$$H^q(Y', F) \longrightarrow H^q(X', g'^* F)$$

is bijective if $q \le n$ and injective if $q = n + 1$ (resp. is bijective if $q \le 1$).

(iii) *For every locally quasi-finite morphism $Y' \to Y$ and every abelian $\mathbf{L}$-torsion sheaf (resp. sheaf of ind-$\mathbf{L}$-groups) F on Y', the morphism*

$$H^q(Y', F) \longrightarrow H^q(X', g'^* F)$$

is bijective if $q \le n$ and injective if $q = n + 1$ (resp. is bijective if $q \le 1$).

(iv) *The morphism g is surjective, and for every locally quasi-finite morphism $Y' \to Y$, every $\nu > 0$, and every $\ell \in \mathbf{L}$, the canonical morphism*

$$H^q(Y', \mathbf{Z}/\ell^\nu) \longrightarrow H^q(X', \mathbf{Z}/\ell^\nu)$$

is surjective *for $0 \le q \le n$ (resp. for every locally quasi-finite morphism $Y' \to Y$ and every ordinary finite $\mathbf{L}$-group G, the canonical morphism*

$$H^i(Y', G) \longrightarrow H^i(X', G)$$

is surjective *for $i = 0, 1$).*

Proof of 1.6. The implications (ii) ⇐ (iii) ⇒ (iv) are trivial. Now $(R^q g'_*) g'^* F$ is the sheaf associated with the presheaf $R^q(Y'') = H^q(X'', g''^* F)$ ($Y'' \to Y'$ étale). If (ii) holds, we have $H^q(Y'', F) \xrightarrow{\sim} H^q(X'', g''^* F)$, so the associated sheaf is zero, i.e. (ii) ⇒ (i). The implication (i) ⇒ (ii) follows from 1.1 (i) (which corresponds to the case $n = 0$), together with the Leray spectral sequence 174

$$H^p(Y', (R^q g'_*) g'^* F) \Longrightarrow H^{p+q}(X', g'^* F)$$

in the abelian case, and the exact sequence

$$0 \longrightarrow H^1(Y', g'_* g'^* F) \longrightarrow H^1(X', g'^* F) \longrightarrow H^0(Y', (R^1 g'_*) g'^* F)$$

in the case of sheaves of groups.

(i) ⇒ (iii): Since (i) and (ii) are equivalent, it is clear that (iii) is equivalent to the statement obtained from (i) by replacing the word "étale" by the words "locally quasi-finite." In this form, for given Y and Y', it is an assertion local on Y' and on Y for the étale topology, so 1.2 reduces us to the case in which $f : Y' \to Y$ is a *finite* morphism.

Let $f' : X' \to X$ be the morphism . By VIII 5.5 (resp. VIII 5.8) (applied three times), we have

$$f_*(R^q g'_*)g'^* F \simeq R^q(fg')_* g'^* F \simeq R^q(gf')_* g'^* F \simeq (R^q g_*) f'_* g'^* F \simeq (R^q g_*) g^*(f_* F)$$

and the last of these is zero, for the values of q in question, if (i) holds. Thus (i) ⇒ (iii).

175 It remains to prove (iv) ⇒ (iii), and this follows immediately from XII 6.5.

Definition 1.7. *Let $n \in \mathbf{N}$ and $\mathbf{L} \subset \mathbf{P}$. A morphism $g : X \to Y$ satisfying one of the equivalent conditions* (i) *to* (iii) *(resp. the corresponding parenthesized conditions) of 1.6 is called an n-*acyclic morphism *(resp. a* 1-aspherical morphism*) for* $\mathbf{L}$*. We say that g is* acyclic *for* $\mathbf{L}$ *if it is n-acyclic for* $\mathbf{L}$ *for every n. We say that g is* universally n-acyclic *(resp.* universally 1-aspherical*) for* $\mathbf{L}$ *if, for every $Y' \to Y$, the morphism $g' = g \times_Y Y'$ is n-acyclic (resp. 1-aspherical) for* $\mathbf{L}$*. Finally, a scheme X is said to be n-*acyclic *(resp.* 1-aspherical*) for* $\mathbf{L}$ *if it is 0-acyclic in the sense of 1.3 and if, for every $\ell \in \mathbf{L}$ and every $1 \le q \le n$, one has $H^q(X, \mathbf{Z}/\ell) = 0$ (resp. and if, for every ordinary finite* $\mathbf{L}$*-group G, one has $H^1(X, G) = 0$).*

Remarks 1.8.
a) If g is quasi-compact and quasi-separated, then, in order for g to be universally n-acyclic, it is enough that g' be n-acyclic whenever $Y' \to Y$ is locally of finite presentation. We do not need this fact and leave its proof, which is an exercise in passage to the limit, to the reader.
b) We do not know whether a morphism that is n-acyclic for $\mathbf{L}$ is necessarily universally n-acyclic for $\mathbf{L}$. The same question arises for the local variants below 1.11.
c) When $n = 0$, we recover the notion of 1.3. Of course, 1.6 and 1.7 could have been formulated so as to include the case $n = -1$. The following proposition is
176 also valid in that case.

Proposition 1.9.
(i) *Let $X \xrightarrow{g} Y \xrightarrow{h} Z$ be morphisms of schemes, let* $\mathbf{L} \subset \mathbf{P}$*, and let $n \in \mathbf{N}$. Suppose that g is 0-acyclic (resp. 1-aspherical for* $\mathbf{L}$*, resp. n-acyclic for* $\mathbf{L}$*). Put $f = hg$. Then, for every sheaf of sets (resp. of ind* $\mathbf{L}$*-groups, resp. abelian* $\mathbf{L}$*-torsion sheaf) F on Z, there is an isomorphism $f_* f^* F \xrightarrow{\sim} h_* h^* F$ (resp. isomorphisms $(R^p f_*) f^* F \xrightarrow{\sim} (R^p h_*) h^* F$ for $p \le 1$, resp. for $p \le n$). In particular, h is 0-acyclic (resp. 1-aspherical for* $\mathbf{L}$*, resp. n-acyclic for* $\mathbf{L}$*) if and only if f is.*
(ii) *Let $\{g_\alpha : X_\alpha \to Y_\alpha\}$ be an inverse system of morphisms of schemes such that the morphisms $X_\alpha \to X_\beta$ and $Y_\alpha \to Y_\beta$ are affine and the g_α are quasi-compact and quasi-separated. Let $g : X \to Y$ be the inverse limit of the g_α VII 5.1. Then, if all the g_α are 0-acyclic (resp. 1-aspherical for* $\mathbf{L}$*, resp. n-acyclic for* $\mathbf{L}$*), the same is true of g.*

Proof. For (i), first observe that

$$h_* h^* F \xrightarrow{\sim} h_*(g_* g^*) h^* F \simeq f_* f^* F.$$

The second assertion of (i) is an immediate consequence of the Leray spectral sequence

$$(R^p h_*)(R^q g_*) f^* F \Longrightarrow (R^{p+q} f_*) f^* F.$$

Finally, the nonabelian assertion follows from the exact sequence XII 3.2.

177 Assertion (ii) follows readily from the theory of passage to the limit VII 5 and VI.

Corollary 1.9.1. *Let $g : X \to Y$ be a quasi-compact and quasi-separated morphism, and let $n \in \mathbf{N}$, $\mathbf{L} \subset \mathbf{P}$. If g is n-acyclic for $\mathbf{L}$ (resp. 1-aspherical for $\mathbf{L}$), then, for every geometric point $\overline{y}$ of Y algebraic over a point y of Y, the geometric fiber $X_{\overline{y}}$ is n-acyclic for $\mathbf{L}$ (resp. 1-aspherical for $\mathbf{L}$).*

Indeed, $\overline{y}$ is the inverse limit of affine schemes Y_i that are locally quasi-finite over Y, and since the $X_i = X \times_Y Y_i \to Y_i$ are n-acyclic for $\mathbf{L}$ (resp. ...), the same holds for $X_{\overline{y}} \to \overline{y}$ by 1.9 (ii), which gives the asserted conclusion.

Proposition 1.10. *Let $g : X \to Y$ be a morphism of schemes, let $\mathbf{L} \subset \mathbf{P}$, and let $n \in \mathbf{N}$. Then conditions* (i) *through* (iv) *below are equivalent.*

(i) *Let $\overline{y}$ be a geometric point of Y and $\overline{x}$ a geometric point of X lying over $\overline{y}$. Let $\widetilde{X}$ and $\widetilde{Y}$ be the strict localizations of X and Y at $\overline{x}$ and $\overline{y}$, respectively, and let $\tilde{g} : \widetilde{X} \to \widetilde{Y}$ be the morphism induced by g. Then $\tilde{g}$ is 0-acyclic (resp. 1-aspherical for $\mathbf{L}$, resp. n-acyclic for $\mathbf{L}$).*

(ii) *For every diagram with Cartesian squares*

$$(*)\qquad \begin{tikzcd} X \arrow[d, "g"] & X' \arrow[l, "j"'] \arrow[d, "g'"] & X'' \arrow[l, "j'"'] \arrow[d, "g''"] \\ Y & Y' \arrow[l, "i"] & Y'' \arrow[l, "i'"] \end{tikzcd}$$

in which i is etale and i' is etale and of finite presentation, and for every sheaf of sets (resp. of ind-$\mathbf{L}$-groups, resp. an abelian $\mathbf{L}$-torsion sheaf) F on Y'', the base-change morphism XII 4.1, XII 4.2 178

$$g'^{*}i'_{*}F \longrightarrow j'_{*}g''^{*}F$$

is bijective (resp. the morphisms

$$g'^{*}(R^q i'_{*})F \longrightarrow (R^q j'_{*})g''^{*}F$$

are bijective for $q = 0, 1$, resp. are bijective for $0 \leq q \leq n$ and injective for $q = n+1$).

(iii) *The same statement as* (ii), *except that the morphism i' is assumed only quasi-finite and quasi-separated.*

(iv) *The same statement as* (ii), *but with i assumed only locally quasi-finite and, on the other hand, i' a quasi-compact open immersion.*

Proof. The implication (iii) $\Rightarrow$ (ii) is trivial. Let us verify that (i) $\Rightarrow$ (iii). It is enough to check the morphism in (iii) fiber by fiber. Let $\overline{y}$ be a geometric point of Y' and $\overline{x}$ a geometric point of X' lying over $\overline{y}$. Let

$$\begin{tikzcd} \widetilde{X}' \arrow[d, "\tilde{g}'"] & \widetilde{X}'' \arrow[l, "\tilde{j}'"'] \arrow[d, "\tilde{g}''"] \\ \widetilde{Y}' & \widetilde{Y}'' \arrow[l, "\tilde{i}'"] \end{tikzcd}$$

be the Cartesian diagram deduced from $(*)$, where $\widetilde{X}'$ and $\widetilde{Y}'$ are the strict localizations of X' and Y' at $\overline{x}$ and $\overline{y}$, respectively, and where $\widetilde{Y}'' = Y'' \times_{Y'} \widetilde{Y}'$, $\widetilde{X}'' = X'' \times_{X'} \widetilde{X}'$. By VIII 5.2 and VIII 5.3, we have 179

$$H^q(\widetilde{Y}'', \widetilde{F}) \simeq (R^q i'_* F)_{\overline{y}} \simeq (g'^*(R^q i'_*)F)_{\overline{x}},$$

where $\widetilde{F}$ is the inverse image of F on $\widetilde{Y}''$, and

$$H^q(\widetilde{X}'', \tilde{g}''^* \widetilde{F}) \simeq ((R^q j'_*) g''^* F)_{\overline{x}}.$$

Since $\tilde{i}'$ is quasi-finite, we may apply 1.1 (iii) (resp. 1.5 (iii)) to the morphism

$$H^q(\widetilde{Y}', \widetilde{F}) \longrightarrow H^q(\widetilde{X}', \tilde{g}'^* \widetilde{F})$$

for the values of q under consideration, which proves (i) $\Rightarrow$ (iii).

(ii) $\Rightarrow$ (i). Assume (ii) and let us verify the acyclicity criterion 1.6 (ii) for a morphism $\tilde{g} : \widetilde{X} \to \widetilde{Y}$. Let $\widetilde{Y}' \to \widetilde{Y}$ be etale, with $\widetilde{Y}'$ affine, and let $\widetilde{F}$ be a sheaf of sets (resp. an abelian $\mathbf{L}$-torsion sheaf, resp. a sheaf of ind-$\mathbf{L}$-groups) on $\widetilde{Y}'$. We must prove that

$$H^q(\widetilde{Y}', \widetilde{F}) \xrightarrow{\sim} H^q(\widetilde{X}', \tilde{g}'^* \widetilde{F}),$$

where $\tilde{g}' : \widetilde{X}' \to \widetilde{Y}'$ is the morphism obtained from $\tilde{g}$ by base change. Now write $\widetilde{Y} = \varprojlim Y_\alpha$, where the Y_α are affine and etale over Y (assuming Y affine, as we may). We can "descend" the etale morphism $\widetilde{Y}' \to \widetilde{Y}$ to one of the Y_α, say to $Y'_0 \to Y_0$, and we have $\widetilde{Y}' = \varprojlim Y'_\alpha = Y \times_{Y_0} Y'_0$. Every sheaf $\widetilde{F}$ on $\widetilde{Y}'$ is the direct limit of the sheaves $f'^*_\alpha f'_{\alpha *} \widetilde{F}$, where $f'_\alpha : \widetilde{Y}' \to Y'_\alpha$ is the canonical morphism. By IX 1.2 and IX 1.6, the sheaves $f'_{\alpha *} \widetilde{F}$ are sheaves of sets (resp. abelian $\mathbf{L}$-torsion sheaves, resp. sheaves of ind-$\mathbf{L}$-groups). In
180 view of VII 3.3, it is therefore enough to treat the case where $\widetilde{F}$ descends to one of the Y'_α, say to a sheaf F_0 on Y'_0. Applying (ii) to the diagram

$$\begin{tikzcd}
X & X_0 \arrow[l] & X'_0 \arrow[l, "j'_0"'] \\
Y \arrow[from=u, "g"'] & Y_0 \arrow[l] \arrow[from=u, "g_0"'] & Y'_0, \arrow[l, "i'_0"'] \arrow[from=u, "g'_0"']
\end{tikzcd}$$

we deduce that $g_0{}^*(R^q i'_{0*})F_0 \xrightarrow{\sim} (R^q j'_{0*}) g_0'^* F_0$ for the applicable values of q. The geometric points $\overline{y}$ and $\overline{x}$ lift canonically to geometric points of Y_0 and X_0 (which we denote by the same symbols), and by VIII 5.2 and VIII 5.3 we have

$$H^q(\widetilde{Y}', \widetilde{F}) = ((R^q i'_{0*})F_0)_{\overline{y}} = (g_0{}^*(R^q i'_{0*})F_0)_{\overline{x}}$$

and

$$H^q(\widetilde{X}', \tilde{g}'^* \widetilde{F}) = ((R^q j'_{0*}) g_0'^* F_0)_{\overline{x}}$$

for the values of q under consideration, which proves the result.

It remains to show that conditions (i) through (iii) are equivalent to (iv). Using form (i), we see that these conditions are stable under a locally quasi-finite base change $Y' \to Y$; in form (ii), this immediately shows that they imply (iv). Conversely, (iv) implies (iii), since, in order to verify (iii), it follows at once—using, for example, the Leray spectral sequence or the descent result XII 6.8—that we may assume Y' and Y'' affine. But then, by the "Main Theorem" EGA IV 8.12.8, $Y'' \to Y'$ factors as $Y'' \xrightarrow{i'_1} Y'_1 \xrightarrow{i'_2} Y'$, with i'_1 an
181 open immersion and i'_2 a *finite* morphism. Using VIII 5.5, VIII 5.8 for i'_2, we are reduced to verifying (ii) for i'_1, which follows from (iv). This completes the proof of 1.10.

Definition 1.11. *A morphism $g : X \to Y$ of preschemes satisfying any one of the equivalent conditions* (i) *through* (iv) *of* 1.10 *is called a* locally 0-acyclic *morphism (resp.* locally 1-aspherical *for* $\mathbf{L}$*, resp.* locally n-acyclic *for* $\mathbf{L}$*). In the obvious way (compare 1.7), one defines the notions of a* locally acyclic *morphism and of a* universally locally 0-acyclic *morphism (resp. ...) for* $\mathbf{L}$.

The statements (i) through (iii) of 1.10 can of course be varied. Thus, in (iii) and (iv), it is enough to restrict to a constant sheaf F of the form G_I, with G a finite set (resp. a finite $\mathbf{L}$-group, resp. G of the form $\mathbf{Z}/\ell^{\nu}\mathbf{Z}$, with $\ell \in \mathbf{L}$, $\nu \in \mathbf{N}$); this follows readily from VIII 5.2, VIII 5.3, and XII 6.5. We also record the following variant:

Corollary 1.12. *Suppose that $g : X \to Y$ is locally of finite type. To verify local* 0*-acyclicity (resp. ...), it is enough to verify* 1.10 (i) *in the case where the geometric point $\overline{x}$ is* closed *in the geometric fiber $X_{\overline{y}}$.*

Indeed, in the proof of (i) ⇒ (iii) of 1.10, we verified (iii) fiber by fiber, and it was enough to do so for the fibers at geometric points $\overline{x}$ that are closed in their geometric fiber, since g is locally of finite type VIII 3.13 b). **182**

Proposition 1.13. (i) *Let $X \xrightarrow{g} Y \xrightarrow{h} Z$ be morphisms of schemes, let $\mathbf{L} \subset \mathbf{P}$, and let $n \in \mathbf{N}$. Suppose that g is surjective and locally* 0*-acyclic (resp. locally* 1*-aspherical for* $\mathbf{L}$*, resp. locally n-acyclic for* $\mathbf{L}$*). Then h is locally* 0*-acyclic (resp. ...) if and only if $f = hg$ is.*

(ii) *Let $\{g_\alpha : X_\alpha \to Y_\alpha\}$ be an inverse system of morphisms of preschemes such that the transition morphisms $X_\alpha \to X_\beta$ and $Y_\alpha \to Y_\beta$ are affine. Let g be the inverse limit of the $\{g_\alpha\}$ VII 5.1. If every g_α is locally* 0*-acyclic (resp. ...), then so is g.*

This follows by a chain of reasoning analogous to 1.9.

Remarks 1.14. To simplify his task, the author omitted from 1.10 through 1.13 a treatment of the case $n = -1$, and leaves it to the reader to verify that the statements, definitions, and proofs apply to this case essentially without change. Thus, with the notation of 1.10, condition (i) says that the induced local morphisms $\widetilde{X} \to \widetilde{Y}$ are (-1)-acyclic, i.e., by 1.5, *surjective*, and conditions (ii) through (iv) say that the base-change morphism $g'^*(i'_*(F)) \to j'_*(g''^*(F))$ is a *monomorphism*. Under these conditions, we shall therefore say that g is *locally* (-1)*-acyclic*. This condition is thus stable under locally quasi-finite base change. As an example, a flat morphism is locally (-1)-acyclic, since for such a morphism the surjectivity of the maps $\widetilde{X} \to \widetilde{Y}$ is a well-known consequence of their flatness. We also note that, when f is locally of finite presentation, f is locally (-1)-acyclic if and only if it is *universally open*, at least when Y is locally Noetherian; this follows readily from EGA IV 14.4.9, 1.10.3. For such a morphism, local (-1)-acyclicity consequently implies *universal* local (-1)-acyclicity. (It is moreover very likely[40] that, in these last statements, the hypothesis that Y be locally Noetherian is unnecessary; this is at any rate so if the set of irreducible components of Y is locally finite EGA IV 14.4.10.) **183**

Theorem 1.15. *Let $g : X \to Y$ be a quasi-compact and quasi-separated morphism, let $\mathbf{L} \subset \mathbf{P}$ be a set of prime numbers, and let n be a natural number (resp. $n = 1$). Suppose that g is $(n-1)$-acyclic for* $\mathbf{L}$ *and locally $(n-1)$-acyclic for* $\mathbf{L}$ 1.3, 1.7, 1.11, 1.14*. Let F be a sheaf on Y, which, for $n \geq 1$, is assumed to have the structure of an abelian* $\mathbf{L}$*-torsion sheaf (resp. the structure of a sheaf of groups), and let ξ be an element of $H^n(X, g^*(F))$. In order*

[40] This was in fact verified by M. Raynaud. Cf. the new edition of EGA IV 14!

that ξ come from an element of $H^n(Y,F)$ (which is then uniquely determined, thanks to the $(n-1)$-acyclicity hypothesis on g), it is necessary and sufficient that, for every geometric point $\overline{y}$ of Y algebraic over a point y of Y, the image $\xi_{\overline{y}}$ of ξ in $H^n(X_{\overline{y}}, F|X_{\overline{y}})$ lie in the
184 *image of $H^n(\overline{y}, F_{\overline{y}})$ (which, for $n \geq 1$, means that $\xi_{\overline{y}} = 0$).*

We note at once that 1.15 shows that, under the preliminary conditions imposed on g, the morphism g is n-acyclic for $\mathbf{L}$ (resp. 1-aspherical for $\mathbf{L}$) if and only if, for every $\overline{y}$ as above, $X_{\overline{y}}$ is n-acyclic for $\mathbf{L}$ (resp. is 1-aspherical for $\mathbf{L}$): the "if" direction follows directly from 1.15 and the definitions, while the "only if" direction is in any case immediate from 1.9.1. With this said, an immediate induction on n gives the following result.

Corollary 1.16. *Let g be a quasi-compact and quasi-separated morphism that is locally $(n-1)$-acyclic for $\mathbf{L}$, where n is a given natural number (resp. where $n = 1$). Then g is n-acyclic for $\mathbf{L}$ (resp. is 1-aspherical for $\mathbf{L}$) if and only if, for every geometric point $\overline{y}$ of Y algebraic over a point y of Y, the fiber $X_{\overline{y}}$ is n-acyclic for $\mathbf{L}$ (resp. 1-aspherical for $\mathbf{L}$).*

Corollary 1.17. *Let $g : X \to Y$ be a morphism of preschemes, let $\mathbf{L} \subset \mathbf{P}$ be a set of prime numbers, and let n be a natural number (resp. $n = 1$). In order that g be locally n-acyclic for $\mathbf{L}$ (resp. locally 1-aspherical for $\mathbf{L}$), it is necessary and sufficient that, for every geometric point $\overline{x}$ of X, if $\tilde{g} : \widetilde{X} \to \widetilde{Y}$ is the morphism between the strict localizations of X and Y (at $\overline{x}$ and the corresponding geometric point $\overline{y}$ of Y) induced by g, every geometric fiber $X_{\overline{z}}$ of g, with respect to a geometric point $\overline{z}$ of Y algebraic over a point z of Y, is n-acyclic for $\mathbf{L}$ (resp. 1-aspherical for $\mathbf{L}$).*

185 Necessity follows from criterion 1.10 (i) and 1.9.1. For sufficiency, we proceed by induction on n, which allows us, when $n \geq 1$, to suppose that g is already locally $(n-1)$-acyclic for $\mathbf{L}$. The same then holds for the morphisms $\tilde{g}$, and 1.16 implies that $\tilde{g}$ is n-acyclic (resp. 1-aspherical for $\mathbf{L}$); by criterion 1.10 (i), this implies that g is locally n-acyclic (resp. locally 1-aspherical) for $\mathbf{L}$.

Corollary 1.18. *With the notation of 1.17, if g is locally of finite type, then in the stated criterion it is enough to consider geometric points $\overline{x}$ whose underlying point x is closed in its fiber.*

This follows from the proof just given and from 1.12.

Proof of 1.15[40]. We first show that we may suppose Y strictly local. In the abelian case, with $n \geq 1$, the Leray spectral sequence for g and $g^*(F)$, together with $R^i g_*(g^*(F)) = 0$ for $0 < i \leq n-1$ and $F \xrightarrow{\sim} g_*g^*(F)$, which express the $(n-1)$-acyclicity hypothesis on g, yields a canonical exact sequence

$$0 \longrightarrow H^n(Y,F) \longrightarrow H^n(X, g^*(F)) \longrightarrow H^0(Y, R^n g_*(g^*(F))),$$

which shows that it is enough to prove that the image of ξ in the third term is zero. This can be checked fiber by fiber, and in view of VIII 5.2 we are reduced to the case where Y is strictly local. The case $n = 0$, with F a sheaf of sets, is handled in the same way: the hypothesis on g implies that $F \to g_*g^*(F)$ is injective, and ξ is identified with a section of the second sheaf that must be shown to come from a section of the first. This too can
186 be checked fiber by fiber, and one applies VIII 5.3. Essentially the same argument works in the noncommutative case, with $n = 1$, using the exact sequence XII 3.2

$$H^1(Y,F) \longrightarrow H^1(X, g^*(F)) \longrightarrow H^0(Y, R^1 g_*(g^*(F))),$$

and using VIII 5.3.

[40]The reader interested only in the Noetherian case is invited to consult 1.18.

We therefore suppose that Y is strictly local, and hence that X is quasi-compact and quasi-separated. The morphism $g : X \to Y$ satisfies the conditions of XII 6.10, which reduces us to the case in which, for every closed subset Y_1 of Y distinct from Y, the restriction of ξ to $X_1 = X \times_Y Y_1$ lies in the image of $H^n(Y_1, F|Y_1) \to H^n(X_1, g^*(F)|X_1)$, and to checking in this case that one can find a finite morphism $f : Y' \to Y$ whose image in Y contains a nonempty open subset of Y, and such that the inverse image ξ' of ξ on $X' = X \times_Y Y'$ lies in the image of $H^n(Y', F') \to H^n(X', g'^*(F'))$. We note that, in view of VIII 5.5 and VIII 5.8, condition 2°) of *loc. cit.* is automatically satisfied for this morphism $f : Y' \to Y$.

To find such an f, we may suppose that Y is reduced, and we consider a maximal point y of Y. Let $\overline{y}$ be the geometric point corresponding to a separable closure of $k(y)$, so that $\overline{y}$ is the filtered inverse limit of open subsets U_i of finite integral Y-schemes Y_i, each of which has an image in Y containing a neighborhood of y. Using the hypothesis on ξ, we see that there is an index i such that the inverse image of ξ on $X \times_Y U_i$ is zero if $n \geq 1$, resp. lies in the image of $H^0(U_i, F_{U_i})$ if $n = 0$.

We take Y' to be Y_i, and note that we may suppose that $U' = U_i$ is the inverse 187
image of an affine open neighborhood U of y in Y. We then see that the inverse image of ξ on $X \times_Y Z'$ (where $Z' = Y' - U'$ is the inverse image in Y' of $Z = Y - U$) lies in the image of $H^n(Z', F_{Z'})$. After replacing Y by Y', we are therefore reduced to the following case: there is an affine open subset U of Y, with complement Z, such that the restrictions of ξ to X_U and X_Z lie respectively in the image of $H^n(U, F_U)$ and in the image of $H^n(Z, F_Z)$; moreover, if $n \geq 1$, these restrictions may be assumed to be *zero*.

When $n \geq 1$, we shall show that under these conditions $\xi = 0$. To simplify notation, we write Y' in place of X, denote by U' and Z' the inverse images of U and Z in Y', and denote the inverse image $g^*(F)$ by F'. Let $i : U \to Y$, $j : Z \to Y$, $i' : U' \to Y'$, and $j' : Z' \to Y'$ be the inclusions,

$$\begin{tikzcd} U' \arrow[r, "i'"] \arrow[d, "g_U"] & Y' = X \arrow[r, "j'"] \arrow[d, "g"] & Z' \arrow[d, "g_Z"] \\ U \arrow[r, "i"] & Y \arrow[r, "j"] & Z, \end{tikzcd}$$

and consider the exact sequence of sheaves

$$0 \longrightarrow F'_{U',Y'} \longrightarrow F' \longrightarrow F'_{Z',Y'} \longrightarrow 0.$$

Since the image of ξ in $H^n(Y', F'_{Z',Y'}) = H^n(Z', F'_{Z'})$ is zero, ξ comes from an element of $H^n(Y', F'_{U',Y'})$, and the restriction of the latter to U' is equal to that of ξ, and is therefore zero. Since $F'_{U',Y'}$ is isomorphic to the inverse image of $F_{U,Y}$, we are thus reduced to the case of a sheaf of the form $F_{U,Y}$; hence we may suppose that $F_Z = 0$ (which takes 188
account of the hypothesis $\xi|Z' = 0$). Note that if $F \to G$ is a monomorphism of abelian $\mathbf{L}$-torsion sheaves (resp. of sheaves of groups), it is enough to prove that the image η of ξ in $H^n(Y', G')$ lies in the image of $H^n(Y, G)$, i.e. is zero XII 6.6. This allows us to replace F by $i_*(F_U)$, since the hypothesis $F_Z = 0$ implies that the canonical homomorphism $F \to i_*(F_U)$ is injective. Now, since g is locally $(n-1)$-acyclic for $\mathbf{L}$, and a fortiori 0-acyclic for $\mathbf{L}$ (because $n \geq 1$), the inverse image of $i_*(F_U)$ on Y' is identified with $i'_*(F'_{U'})$. We are therefore reduced to showing that the image η of ξ in $H^n(Y', i'_*(F'_{U'}))$ is zero.

The restriction of η to U' is zero, so it is enough to check that the homomorphism

$$(*)\qquad H^n(Y', i'_*(F'_{U'})) \longrightarrow H^n(U', F'_{U'})$$

is a monomorphism. In the abelian case, we use the Leray spectral sequence for i' and $F'_{U'}$:

$$E_2^{p,q} = H^p(Y', R^q i'_*(F'_{U'})) \Longrightarrow H^*(U', F'_{U'}),$$

and the relations

$$(**)\qquad E_2^{p,q} = 0 \text{ for } 0 < p \leq n-1 \quad , \quad q \leq n-1.$$

To verify the latter, note that the local $(n-1)$-acyclicity hypothesis on g for $\mathbf{L}$ implies that there are isomorphisms

$$R^q i'_*(F'_{U'}) \simeq g^*(R^q i_*(F_U)) \text{ for } q \leq n-1,$$

189 and, in view of the fact that g is $(n-1)$-acyclic for $\mathbf{L}$, so that $H^p(Y, G) \xrightarrow{\sim} H^p(Y', G')$ for $p \leq n-1$ and every abelian $\mathbf{L}$-torsion sheaf G on Y, we obtain $(**)$, since $H^p(Y, G) = 0$ for $p > 0$. In the same way we obtain

$$(***)\qquad E_2^{0,q} \simeq H^0(Y, R^q i_*(F_U)) \text{ for } \quad q \leq n-1.$$

The Leray spectral sequence under consideration then gives rise to an exact sequence that forms the second row of the following commutative diagram:

$$\begin{tikzcd}
H^0(Y, R^{n-1} i_*(F_U)) \arrow[r] \arrow[d, "\wr"] & H^n(Y, i_*(F_U)) = 0 \arrow[d] & \\
H^0(Y', R^{n-1} i'_*(F'_{U'})) \arrow[r] & H^n(Y', i'_*(F'_{U'})) \arrow[r] & H^n(U', F'_{U'}),
\end{tikzcd}$$

where the first vertical arrow is bijective by $(***)$. The diagram shows that $(*)$ is injective, which completes the proof in this case. In the nonabelian case, $n = 1$, we use the same diagram with $n = 1$:

$$\begin{tikzcd}
H^0(Y, i_*(F_U)) \arrow[r] \arrow[d, "\wr"] & H^1(Y, i_*(F_U)) = 0 \arrow[d] & \\
H^0(Y', i'_*(F'_{U'})) \arrow[r] & H^1(Y', i'_*(F'_{U'})) \arrow[r] & H^1(U', F'_{U'}),
\end{tikzcd}$$

XII 3.2, where the first vertical arrow is again bijective by the hypothesis of local and global 0-acyclicity on g; once again we conclude that $(*)$ is injective in this case.

Finally, consider the case $n = 0$. By virtue of , specifying a section ξ of F' amounts to specifying a section $\xi_{U'}$ of $F'_{U'}$ and a section $\xi_{Z'}$ of $F'_{Z'}$ such that the image of $\xi_{Z'}$ under the natural homomorphism

$$F'_{Z'} \longrightarrow j'^*(i'_*(F'_{U'}))$$

190 is the section of the right-hand side induced by $\xi_{U'}$. By hypothesis, there are sections η_Z and η_U of F_Z and F_U, respectively, that induce $\xi_{Z'}$ and $\xi_{U'}$. It remains to see that they satisfy the analogous compatibility condition, which says that they are the restrictions of a single section η of F. We must verify the equality of two sections of $j^*(i_*(F_U))$. Since $Z' \to Z$ is surjective (g being surjective because it is (-1)-acyclic by hypothesis), it is enough to check that the two corresponding sections of

$$g_Z{}^*(j^*(i_*(F_U))) = j'^*(g^*(i_*(F_U)))$$

are equal. Through the base-change homomorphism

$$(*) \qquad g^*(i_*(F_U)) \longrightarrow i'_*(g_U{}^*(F_U)) = i'_*(F'_{U'}),$$

the sheaf under consideration maps to $j'^*(i'_*(F'_{U'}))$, and this latter homomorphism is injective because $(*)$ is injective, by the hypothesis that g is locally (-1)-acyclic. It therefore suffices to verify that the two sections of $j'^*(i'_*(F'_{U'}))$ that are the images of the preceding two sections are equal. But these are immediately seen to be precisely the sections considered above, which are equal by hypothesis. This completes the proof.

Remark 1.18. *There is a refinement of 1.15 that gives a much simpler proof of 1.15 when Y is assumed to be locally Noetherian. In addition to the preliminary hypotheses on g and F, suppose that there is a finite set of geometric points $f_i : \overline{y}_i \to Y$ of Y such that the morphism*

$$F \longrightarrow \prod_i f_{i*}(f_i{}^*(F))$$

is injective (a hypothesis automatically satisfied if Y is Noetherian and F is constructible IX 2.14). I claim that under these conditions criterion 1.15 remains valid when one restricts to the geometric points $\overline{y}_i$ alone. Indeed, by XII 6.6, we may suppose that F itself is of the form $f_*(M_{\overline{y}})$, where M is a set resp. a group resp. an abelian $\mathbf{L}$-torsion group, and $f : \overline{y} \to Y$ is a geometric point. Let $f' : X' = X_{\overline{y}} \to X$ be the canonical morphism. Assuming, for definiteness, that $n \geq 1$, the local hypothesis on g implies that 191

$$g^*(f_*(M_{\overline{y}})) \simeq f'_*(M_{X'}) \quad , \quad R^i f'_*(M_{X'}) = 0 \quad \text{for} \quad 1 \leq i \leq n-1,$$

the second relation following from the fact that its left-hand side is isomorphic to $g^*(R^i f_*(M_{\overline{y}}))$, which is zero since manifestly $R^i f_*(M_{\overline{y}}) = 0$ for $i \geq 1$. From these relations we conclude that

$$H^n(X, g^*(F)) \simeq H^n(X, f'_*(M_{X'})) \longrightarrow H^n(X', M_{X'})$$

is injective, using the Leray spectral sequence for f'. This proves that an element ξ of $H^n(X, g^*(F))$ whose induced element in $H^n(X_{\overline{y}}, M_{X_{\overline{y}}})$ is zero is itself zero. To deduce from this a simplified proof of 1.15 in the case where Y is locally Noetherian, note that we may restrict to the case where Y is Noetherian; an easy passage to the limit, using IX 2.7.2 and VII 5, then reduces us to the case where F is also constructible, to which the statement just proved applies.

2. Local acyclicity of a smooth morphism

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The theorem is the following:

Theorem 2.1. *Let $g : X \to Y$ be a smooth morphism, and let $\mathbf{L} \subset \mathbf{P}$ be the complement of the set of residual characteristics of X. Then g is locally acyclic and locally 1-aspherical for $\mathbf{L}$.*

Let us first record the corollary:

Corollary 2.2. *Let $g : \mathbf{E}^m_Y \to Y$ be the structural morphism of affine m-space. Then g is acyclic and 1-aspherical for the set $\mathbf{L}$ complementary to the set of residual characteristics of Y.*

Indeed, induction on m and 1.9 (i) reduce us to the case $m = 1$. By 2.1, g is locally n-acyclic and locally 1-aspherical for $\mathbf{L}$, so the *local* condition of 1.16 is satisfied. It therefore remains only to prove that the geometric fibers of $\mathbf{E}^1_Y$ are acyclic and 1-aspherical for $\mathbf{L}$; that is, that Spec $k[t]$ is acyclic and 1-aspherical for $\mathbf{L}$ when k is a separably closed field of characteristic $p \notin \mathbf{L}$. The acyclicity assertion follows from IX 4.6, and the 1-asphericity assertion is well known. More generally, let X' be a normal irreducible Galois covering of $X = \mathbf{P}^1_k$, unramified away from the point ∞ and tamely ramified at that point. We prove that the degree n of the covering is 1. The Hurwitz formula gives

$$2(g'-1) = 2n(g-1) + \deg \mathfrak{d}_{X'/X},$$

where g and g' are the genera of X and X', and $\mathfrak{d}_{X'/X}$ is the different divisor. We have $g = 0$, while tame ramification implies $\deg \mathfrak{d}_{X'/X} \leq n-1$; hence $2(g'-1) \leq -(n+1)$, so $n \leq 2(1-g') - 1$. Since $g' \geq 0$, it follows that $n \leq 1$, and therefore $n = 1$, C.Q.F.D.

Proof of 2.1. The assertion is local on X and Y for the étale topology, so we may assume SGA 1 II 1.1 that g is the structural morphism of affine m-space $\mathbf{E}^m_Y$ and that Y is affine. By the transitivity of local acyclicity, 1.13 (i), and the identity $\mathbf{E}^m_Y = \mathbf{E}^{m-1}_{\mathbf{E}^1_Y}$, we are immediately reduced to $m = 1$, that is, to $X = \operatorname{Spec} \mathscr{O}_Y[t]$.

Apply 1.18. It remains to prove that the geometric fibers of the morphism of strictly local schemes $\tilde{g} : \widetilde{X} \to \widetilde{Y}$ associated with a pair of geometric points $\overline{x}, \overline{y}$, where $\overline{x}$ is closed in the fiber $X_{\overline{y}}$, are acyclic for $\mathbf{L}$ and 1-aspherical for $\mathbf{L}$. We may plainly assume that $Y = \widetilde{Y}$. Let $Y' \to Y$ be a finite, local, surjective morphism, so that Y' is strictly local, and let $\overline{y}'$ be the point of Y' above $\overline{y}$ and $\overline{x}'$ the point of X' above both $\overline{x}$ and $\overline{y}'$. Since $X' \to X$ is finite, the strict localization of X' at $\overline{x}'$ is precisely $\widetilde{X}' = \widetilde{X} \times_Y Y'$. Consequently every geometric fiber of $\widetilde{X}'/Y'$ is obtained from a geometric fiber of $\widetilde{X}/Y$ by an algebraic (necessarily radicial) extension of the base field; moreover, at least one geometric fiber of $\widetilde{X}'/Y'$ lies above each geometric fiber of $\widetilde{X}/Y$. Thus, by VIII 1.1, we may replace Y by Y'. Choosing Y' suitably reduces us to the case $k(\overline{x}) = k(\overline{y})$. Then $\overline{x}$ is a rational point of the fiber $X_{\overline{y}} = \operatorname{Spec} k(\overline{y})[t]$, say the point $t = a^0$. Lift a^0 to a section a of $\mathscr{O}_{\widetilde{Y}}$ and make the "translation" $t_1 = t - a$; we are reduced to the case in which $\overline{x}$ is the point $\{t = 0\}$ of $\operatorname{Spec} k(\overline{y})[t]$. Put $A = \Gamma(\widetilde{Y}, \mathscr{O}_{\widetilde{Y}})$, so that A is a strictly local ring.

Notation 2.3. Let A be a Henselian local ring. We denote by $A\{t\}$ the Henselization of $A[t]$ at the prime ideal generated by $\operatorname{rad} A$ and t.

Corollary 2.4. *Let A be Henselian and let A' be a finite A-algebra. Then*

$$A'\{t\} \simeq A' \otimes_A A\{t\}.$$

This follows immediately from VIII 4.1. Note that if A is strictly local, then so is $A\{t\}$. The proof of the theorem is reduced to the following lemma:

Lemma 2.5. *Let A be a Henselian ring of residual characteristic p. Then every geometric fiber of* $\operatorname{Spec} A\{t\}$ *over* $\operatorname{Spec} A$ *is acyclic and* 1*-aspherical for* $\mathbf{L} = \mathbf{P} - \{p\}$.

Since $A\{t\}$ is the Henselization of $A[t]$, a fiber $\operatorname{Spec}(A\{t\} \otimes_A K)$, where K is a residue
193 field of A, is an inverse limit of affine schemes X_i of finite type over the "line" $\operatorname{Spec} K[t]$, that is, a limit of affine algebraic curves. Hence every geometric fiber of $\operatorname{Spec} A\{t\}$ is a limit of affine curves $(X_i)_{\overline{K}}$ over the separably closed field $\overline{K}$. It follows from IX 5.7 and VII 5.7 that the cohomological dimension of the geometric fibers is at most 1. It is therefore enough to verify that the geometric fibers are 1-*aspherical* for $\mathbf{L}$; that is, that such a geometric fiber $\overline{Z}$ is connected and nonempty and that $H^1(\overline{Z}, G) = 0$ for every finite $\mathbf{L}$-group G.

Write $A = \varinjlim B_\alpha$, where the B_α are rings of finite type over $\mathbf{Z}$. The morphism $B_\alpha \to A$ factors through the Henselization A_α of B_α at the prime ideal induced by $\operatorname{rad} A$, whence $A = \varinjlim A_\alpha$. One readily checks that $A\{t\} = \varinjlim A_\alpha\{t\}$ as well. Let $\overline{y}$ be a geometric point of Spec A, let $X_{\overline{y}}$ be the geometric fiber of S = Spec $A\{t\}$ over Spec A at $\overline{y}$, let $\overline{y}_\alpha$ be the point of Spec A_α induced by $\overline{y}$, and let $X_{\alpha,\overline{y}_\alpha}$ be the geometric fiber of Spec $A_\alpha\{t\}$ over Spec A_α at $\overline{y}_\alpha$. We have $X_{\overline{y}} = \varprojlim X_{\alpha,\overline{y}_\alpha}$. By 1.7 (ii), applied to the morphisms $X_{\alpha,\overline{y}_\alpha} \to \overline{y}_\alpha$, the proof of 2.5 is therefore reduced to the case $A = A_\alpha$, that is, to the case in which A is the Henselization of an algebra of finite type over $\mathbf{Z}$ at a given prime ideal. We assume this from now on. Recall that such an algebra is *excellent* EGA IV 7.8.3 (iii), 7.8.6 (i), and hence that A_α is excellent as well (as follows readily from the definition; cf. EGA IV 18.7.6).

Let us verify 0-acyclicity. (For a more general result, see the Appendix.) Since $A\{t\}/A$ has the section "$t = 0$", a geometric fiber $\overline{Z}$ is nonempty. To see that it is connected, 194
we reduce, by replacing A by a quotient 2.4, to the case in which A is integral and $\overline{Z}$ is the geometric fiber over the generic point of Spec A. It is enough to prove that for every finite separable extension K' of the fraction field K of A, the scheme $Z_{K'}$, where $Z = \mathrm{Spec}(A\{t\} \otimes_A K)$, is connected. Let A' be the normalization of A in K' (a finite A-algebra, EGA IV 7.8.2) and let Z' be the generic fiber of Spec $A'\{t\}$ over A'. By 2.4, $A'\{t\} \simeq A' \otimes_A A\{t\}$, and hence $Z_{K'} \simeq Z'$. Replace A by A'. It is then enough to prove that Z is integral, and for this it is enough to prove that $A\{t\}$ is integral. Since A is normal, $A[t]$ is normal, and so is $A\{t\}$ SGA 1 I 9.5 (i). This proves the assertion.

The proof of 2.5, and hence of 2.1, is thus reduced to the following lemma, which will be proved in the next section:

Lemma 2.6. *Let A be a Henselian excellent ring, and let $\overline{Z}$ be a geometric fiber of* Spec $A\{t\}/$Spec A. *Then every principal Galois covering $\overline{Z}'$ of $\overline{Z}$ with group G of order prime to the residual characteristic p of A is trivial.*

3. Proof of the main lemma

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Replacing A by a finite A-algebra A' (cf. 2.4), assertion 2.6 is reduced to the case where A is integral and $\overline{Z}$ is the geometric generic fiber of Spec $A\{t\}$ over Spec A. Let K be the fraction field of A, and let Z = Spec $A\{t\} \otimes_A K$. If $\overline{Z}'/\overline{Z}$ is a principal Galois covering, it can be descended to some $Z_{K'}$, where K' is a suitable finite separable extension of K. Replacing A (2.4) by its normalization in K', which is a finite A-algebra EGA IV 7.8.3 (vi), we are reduced to proving the lemma in the following form:

Lemma 3.1. *With the notation of 2.6, suppose that A is normal, and let Z'/Z be a principal étale Galois covering, with group G of order prime to the residual characteristic p of A. Then Z' is induced by a Galois extension K' of K; that is,* (SGA 1 IX 6.2)*, the covering $\overline{Z}'$ of $\overline{Z}$ deduced from Z' is trivial.*

Proof. Let B' be the normalization of $A\{t\}$ in the ring of rational functions of Z', and consider the height-1 prime ideals P of $A\{t\}$ at which the $A\{t\}$-algebra B' is ramified. Since Z'/Z is étale, P does not lie over the generic point of Spec A; that is, $\mathfrak{p} = P \cap A \neq 0$. Thus $A\{t\}/\mathfrak{p}A\{t\} = (A/\mathfrak{p})\{t\}$ is integral and has dimension at most dim A. Since $A\{t\}$ is excellent EGA IV 7.8.6 (i) and 18.7.6, hence catenary, we have dim Spec $A\{t\}/P$ = (dim $A + 1) - 1$ = dim $A \geq$ dim $A\{t\}/\mathfrak{p}A\{t\}$, whence $P = \mathfrak{p}A\{t\}$, since $P \supset \mathfrak{p}A\{t\}$. 196

Let $\mathfrak{p}_1, \ldots, \mathfrak{p}_m$ therefore be the height-1 prime ideals of A such that $B'/A\{t\}$ is ramified at the ideals $\mathfrak{p}_iA\{t\}$, and let s be the least common multiple of the orders of the inertia groups SGA IV 2 $G_i \subset G$ for B' at the prime ideals lying above the $\mathfrak{p}_iA\{t\}$. This is an integer dividing the order of G, and is therefore prime to p. Let $f \in A$ be an element that vanishes to order 1 at every $\mathfrak{p}_i$ (such an f exists, as follows, for example, from Bourbaki, Alg. Comm., Chap. II, § 3, cor. to prop. 17). Replacing A 2.4 by its normalization A_1 in the finite extension $K_1 = K[x]/(x^s - f)$ of K, and replacing B' by the normalization of $A_1 \otimes_A B'$, Abhyankar's lemma SGA 1 X 3.6 reduces us to the case where B' is ramified at no height-1 prime ideal of $A\{t\}$.

The algebra $B'/A\{t\}$ is then also étale over the fiber of Spec $A\{t\}$ at every codimension-1 point of Spec A. Indeed, let $\mathfrak{p}$ be a height-1 prime ideal of A. Since A is normal, $A_{\mathfrak{p}}$ is a discrete valuation ring, hence regular, and so $A\{t\} \otimes_A A_{\mathfrak{p}}$ is regular as well—indeed, it is a limit of schemes étale over $A_{\mathfrak{p}}[t]$. By hypothesis, the algebra $B'/A\{t\}$ is étale at every point of codimension ≤ 1 of Spec $A\{t\} \otimes_A A_{\mathfrak{p}}$, hence it is étale everywhere by the Zariski–Nagata purity theorem SGA 1 X 3.1; for a proof, see for example SGA 2, X 3.4.

Replacing A 2.4 by a suitable finite A-algebra, we are reduced to the case where, in addition, the algebra B'/tB' over $A\{t\}/tA\{t\} \simeq A$ is completely split over the generic point of Spec A. It therefore suffices to prove that, under these conditions, the algebra
197 B' over $A\{t\}$ is "trivial," and, since $A\{t\}$ is Henselian, it suffices to prove that B' is étale over $A\{t\}$. Indeed, the fact that B'/tB' splits over the generic point of Spec A implies that it splits over Spec(A) SGA 1 I 10.1; thus the residual extensions of B' over the maximal ideal of $A\{t\}$ are trivial, and we apply EGA IV 18.5.14. We are therefore reduced to the following lemma:

Lemma 3.2. *Let B be an excellent normal local ring, and let $0 \neq t \in \operatorname{rad} B$ be such that B/tB is normal. Let B' be a finite integral normal B-algebra containing B, such that:*

(i) *The algebra B'/tB' over B/tB is completely split at the maximal points of* Spec(B/tB).
(ii) *The algebra B' over B is étale at the codimension-*1 *points of* Spec(B/tB).

Then B' is étale over B.

Proof. By (ii), the assertion is trivial for $\dim B \leq 2$, that is, for $\dim B/tB \leq 1$. Suppose that B'/B is not étale; let x be a maximal point of the set of points of Spec B/tB over which B'/B is not étale, and let C be the local ring of Spec B at x. Then C is again an excellent normal ring (EGA IV 7.8.2), we have $0 \neq t \in \operatorname{rad} C$, and (i) and (ii) hold for the C-algebra $C' = C \otimes_B B'$. We may therefore suppose that $B = C$; that is, B' is étale over B above every point of Spec B/tB, except possibly at the origin. Since B/tB is normal
198 (in fact, unibranchness would suffice), condition (i) implies that the algebra B'/tB' is completely split away from the origin. Moreover, we may suppose that $\dim B \geq 3$. We are thus reduced to the following lemma:

Lemma 3.3. *Let B be an excellent normal local ring, let $0 \neq t \in \operatorname{rad} B$, and suppose that* $\dim B \geq 3$. *Let B' be a finite integral normal B-algebra containing B, such that the B/tB-algebra B'/tB' is completely split away from the origin. Then B' is étale over B.*

Proof. Let $X = \operatorname{Spec} B$, let $X_1 = X - \{x_0\}$, where x_0 is the closed point, and let $i : U \to X$ be the open immersion whose image consists of the points $x \in X$ at which the algebra B'/B is étale. Then $(X - U) \cap V(t) = \{x_0\}$ has dimension 0. Since $V(t)$ is defined by *one* equation, it follows that $\dim(X - U) \leq 1$ (EGA 0_{IV} 16.2.5); consequently (since X is catenary), $x \in X - U \Rightarrow \dim \mathcal{O}_{X,x} \geq \dim B - 1 \geq 2$; that is, $\operatorname{codim}(X - U, X) \geq 2$.

Let $X' = \operatorname{Spec}(B')$, and let U' be the inverse image of U in X'. Since X is universally catenary, the relation $\operatorname{codim}(X - U, X) \geq 2$ implies $\operatorname{codim}(X' - U', X') \geq 2$ (EGA IV 5.6.10). Since X' is normal, we therefore have

$$B' = \Gamma(X', \mathcal{O}_{X'}) \xrightarrow{\sim} \Gamma(U', \mathcal{O}_{U'}).$$

Let $\mathscr{B}' = \widetilde{B}'$ be the sheaf of $\mathscr{O}_X$-algebras that defines X' over X, and let $\mathscr{C}$ be its restriction to U. The preceding formula is then equivalent to

$$\mathscr{B}' \xrightarrow{\sim} i_*(\mathscr{C}),$$

and 3.3 is consequently reduced to the following statement (in which B' no longer 199
occurs): Let B, t be as in 3.3, let $X = \operatorname{Spec}(B)$, let $Y = \operatorname{Spec}(B/tB)$, let U be an open neighborhood of $Y_1 = Y - \{x_0\}$ in $X_1 = X - \{x_0\}$, and let U' be an étale covering of U whose restriction to Y_1 is completely split. If $\mathscr{C}$ denotes the sheaf of $\mathscr{O}_U$-algebras defining U', then (with $i : U \to X$ denoting the canonical immersion) $i_*(\mathscr{C})$ is a coherent *étale* algebra on X (or, equivalently, in view of the relation $\operatorname{prof} \mathscr{O}_{X,x} \geq 2$ for $x \in X - U$ noted above and EGA IV 5.10.5, U' extends to an *étale* covering X' of X).

Let $\widehat{X} = \operatorname{Spec} \widehat{B}$, where $\widehat{B}$ is the completion of B. Then $\widehat{B}$ is normal (EGA IV 7.8.3 (v)) and $\widehat{B}/B$ is faithfully flat. Let

$$\begin{tikzcd} \widehat{U} \arrow[r, "i"] \arrow[d, "f_1"] & \widehat{X} \arrow[d, "f"] \\ U \arrow[r, "i"] & X \end{tikzcd}$$

be the Cartesian diagram deduced from $\widehat{X} \to X$. Since f is faithfully flat, the functor i_* commutes with base change by f EGA IV 2.3.1, and we are reduced EGA IV 17.7.1 to proving that $\hat{i}_*(\widehat{\mathscr{C}})$ is an étale algebra on $\widehat{X}$. We may therefore replace B by $\widehat{B}$; that is, we may suppose that B is *complete*. In this case we shall prove that U' is a completely split covering of U (which will plainly imply that it extends to an étale covering of X, and will complete the proof). This assertion is now an immediate consequence of "local
Lefschetz theory" SGA 2 X 2.1 (i) and 2.5, as is seen at once, taking account once again 200
of the relation $\operatorname{prof} \mathscr{O}_{X,x} \geq 2$ for $\dim\{x\} = 1$.

Remark 3.4. One can prove 3.2 and 3.3 under more general conditions, by following essentially the same proof: instead of assuming that B is excellent and normal, it is enough to suppose that the completion $\widehat{B}$ is normal (a condition stable under localization, by the results of EGA IV 7), or merely that the completion of B for the (tB)-adic topology is normal and is a quotient of a regular ring.

4. Appendix: A criterion for local 0-acyclicity 201

The results of this section will not be used later in the seminar.

Theorem 4.1. *Let $g : X \to Y$ be a morphism of schemes. Assume that g is flat with separable fibers, and that X and Y are locally noetherian respectively, that g is locally of finite presentation. Then g is universally locally* 0*-acyclic.*

We may clearly assume that X and Y are affine; in the second case, the standard method of EGA IV 8, using EGA IV 9.7.7, then reduces us to the case in which Y, and hence X, is noetherian. We may therefore confine ourselves to proving the first assertion. Since the hypothesis is stable under base change of finite type, the passage to the limit in 1.13 (ii) reduces us to proving that g is locally 0-acyclic. For this, 1.17 reduces us to proving that, if g is moreover assumed to be strictly local, then the geometric fibers of g are connected. This is precisely EGA IV 18.9.8.

Corollary 4.2. *Let $g : X \to Y$ be a morphism of schemes, with Y discrete. Then g is universally locally* 0*-acyclic.*

By VIII 1.1, we may assume that Y is the spectrum of a perfect field k. We may also assume that X is affine, and hence is a projective limit of affine schemes of finite type over k; in view of 1.13 (ii), this reduces us to the case in which X is of finite type over k.
202 Finally, we may clearly assume that X is reduced, and it is then separable over k because k is perfect. The conclusion now follows from 4.1.

Corollary 4.3. *Assume that g is surjective and satisfies the hypotheses of 4.1 or 4.2; in the non-parenthetical case of 4.1, assume in addition that g is either quasi-compact and quasi-separated or universally open. Then g is a morphism of universal effective descent for the fibered category of étale sheaves on varying preschemes.*

In the cases under consideration, g is universally submersive, and it follows from VIII 9.1 that g is a morphism of universal descent for the fibered category in question. Thus, for the question of effectivity, we may confine ourselves to the case in which Y is affine. Moreover, in each of the cases under consideration, X can be covered by a *finite* number of affine open subsets X_i whose images cover Y. Replacing X by the disjoint union of the X_i, we may therefore assume that X is affine, and hence a fortiori quasi-compact and quasi-separated over Y. We may then conclude by the argument of VIII 9.4.1, using the fact XVI 1.1 that, for a quasi-compact and quasi-separated morphism g, the formation of $g_*(F)$ (where F is an étale sheaf on X) commutes with every base change $Y' \to Y$ that is locally 0-acyclic.

Remarks 4.4. a) This gives the proof, which had been left pending, of VIII 9.4 d), as a special case of 4.3. Another, easier proof, not using the rather deli-
203 cate result of EGA IV 18.9.8, is obtained by observing that, by VIII 9.1, we may confine ourselves to the case of the morphism $\mathrm{Spec}(K) \to \mathrm{Spec}(k)$ induced by a field extension K/k. But $\mathrm{Spec}(K)$ is universally locally 0-acyclic over $\mathrm{Spec}(k)$, since it is even universally acyclic for $\mathbf{L} = \mathbf{P} - \{p\}$, $p = \mathrm{car}\, k$, as we shall see in XVI 1.5.

b) It is not known whether every quasi-compact and quasi-separated, surjective, locally (-1)-acyclic (or universally locally (-1)-acyclic) morphism is a morphism of effective descent for the fibered category of étale sheaves on varying preschemes. This statement seems quite plausible and would strengthen 4.3 and VIII 9.4 c), d). Compare the effectivity statement in Murre, Sém. Bourbaki n° 293, p. 17.

c) It is plausible that, under the hypotheses of 4.2, g is even locally acyclic for L and locally 1-aspherical for L, where L is the set of prime numbers distinct from the residue characteristics of Y. For this question we may clearly assume that Y is the spectrum of an algebraically closed field k, and it can be shown (SGA 1964/65) that the answer is affirmative whenever resolution of singularities is available for schemes of finite type over k. Thus the answer is affirmative when Y has characteristic zero, as follows by using the results of Hironaka [40].

[40] Cf. SGA 1 XIII.

EXPOSÉ XVI

Smooth base change theorem and applications

M. Artin

1. The smooth base change theorem

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Theorem 1.1. *Let* $\mathbf{L} \subset \mathbf{P}$, *let* $n \in \mathbf{N}$, *and let*

$$(*)\qquad \begin{array}{ccc} X & \xleftarrow{g'} & X' \\ \downarrow{\scriptstyle f} & & \downarrow{\scriptstyle f'} \\ Y & \xleftarrow{g} & Y' \end{array}$$

be a cartesian diagram, with f *quasi-compact and quasi-separated*[41], *and* g *universally locally* 0*-acyclic (resp. universally locally* 1*-aspherical for* $\mathbf{L}$, *resp. universally locally* n*-acyclic for* $\mathbf{L}$*)* (XV 1.11). *Then, for every sheaf* F *of sets (resp. of* ind-$\mathbf{L}$*-groups, resp. an abelian sheaf of* $\mathbf{L}$*-torsion) on* X, *the base-change morphism* (XII 4.1.2)

$$\varphi^q : g^*(R^q f_*)F \longrightarrow (R^q f'_*)g'^* F$$

is bijective for $q = 0$ *(resp.* $q \leq 1$, *resp.* $q \leq n$*).*

In particular, applying XV 2.1, we obtain the following.

Corollary 1.2. *(Smooth base change theorem). Suppose that the morphism* g *in* $(*)$ *is smooth, that* f *is quasi-compact and quasi-separated, and that* $\mathbf{L} \subset \mathbf{P}$ *is the complement of the set of residue characteristics of* Y. *Then, for every sheaf* F *of sets (resp. of* ind-$\mathbf{L}$*-groups, resp. an abelian sheaf of* $\mathbf{L}$*-torsion) on* X, *the base- change morphism* φ^q *above is bijective* 208
for $q = 0$ *(resp. for* $q = 0, 1$, *resp. for every* q*).*

Proof of 1.1. We first treat the case in which $f : X \to Y$ is quasi-projective: this is simply a combination of the proper base change theorem XII 5.1 and Definition XV 1.9. Indeed, we have a commutative diagram

$$\begin{array}{ccc} X & \xrightarrow{i} & \overline{X} \\ & {\scriptstyle f}\searrow \quad \swarrow{\scriptstyle \overline{f}} & \\ & Y & \end{array}$$

where $\overline{f}$ is projective, hence proper, and i is an open immersion. We know from XII 5.1 that the base change theorem holds for $\overline{f}$, whatever the morphism $g : Y' \to Y$. Since $g \times_Y \overline{X} : Y' \times_Y \overline{X} \to \overline{X}$ is again universally 0-acyclic (resp. ...), it follows immediately from (XII 4.4 (ii)) that it suffices to prove the theorem for the morphism i; that is, we are reduced to the case in which f is an open immersion. The theorem then follows from Definition XV 1.10 (ii).

[41]N.D.E.: That is, coherent in the terminology of the source.

We now treat the general case. The assertion is local on Y, and we may therefore suppose that Y is affine.

Lemma 1.3. *We may moreover suppose that X is affine.*

Proof. Arguing as in XII 6.1, in order to prove 1.1 for a given f, we are reduced to proving
209 it in the situation deduced from the given situation by the base changes $\widetilde{Y} \to Y$, $\widetilde{Y}' \to Y'$ for a local morphism $\tilde{g} : \widetilde{Y}' \to \widetilde{Y}$ between strict localizations of Y' and Y induced by g, and to proving that the homomorphisms

$$(*) \qquad H^q(\widetilde{X}, F) \longrightarrow H^q(\widetilde{X}', F') \quad (\text{where } F' = g'^*(F))$$

are bijective for $q \leq n$, where $\widetilde{X} = X \times_Y \widetilde{Y}$, $\widetilde{X}' = X' \times_{Y'} \widetilde{Y}'$. We are thus reduced to the case in which Y, Y', and g are strictly local, and to proving in this case the bijectivity of the preceding maps, assuming that 1.1 has been proved for affine f. Let (X_i) be a finite covering of X by affine open subschemes X_i, and let Z be the disjoint union of the X_i, which is therefore affine and is equipped with a surjective morphism $h : Z \to X$, whence a morphism $h' : Z' \to X'$. Note that h is separated, and that it is affine when f is separated. Applying the descent lemma XII 6.8, we see that, in order to prove the bijectivity of the maps $(*)$ for $q \leq n$ and every F, it suffices to prove the bijectivity of the corresponding maps $H^q(Z, h^*(F)) \to H^q(Z', h'^*(F'))$, provided that the base-change homomorphisms for $Z \to X$ and the base change $X' \to X$ are isomorphisms in dimensions $\leq n$. If f is separated, then h is affine, and this is so by hypothesis; this proves 1.1 when f is assumed separated. In the general case, we may then apply the preceding result to h, which is always separated, and again conclude that 1.1 holds for f, proving 1.3.

Remark. One could also invoke the Leray spectral sequence for the open covering
210 (X_i) of X, and its noncommutative variants, but this method, which does not differ essentially from using XII 6.8, has the disadvantage of forcing us once again to distinguish the usual three cases, work that has already been done in *loc. cit.*

To complete the proof of the theorem in the general case, it suffices to reduce the proof to the case in which $f : X \to Y$ is affine and of finite type, hence quasi-projective. By 1.3, we may suppose that X is affine. Then let $X = \varprojlim X_\alpha$, where the morphisms $f_\alpha : X_\alpha \to Y$ are affine and of finite type. The usual limit argument (VII 5) completes the proof.

Complements 1.4. a) We leave it to the reader to check that the same proof still yields a conclusion when g is assumed universally locally (-1)-acyclic (XV 1.14): in this case, the base-change homomorphism

$$\phi^0 : g^*(f_*(F)) \longrightarrow f'_*(g'^*(F))$$

is *injective* for every sheaf of sets F on X. On the other hand, suppose again that $n \geq 0$. One may supplement 1.1 with the following injectivity statement, which is also contained in the preceding proof: if g is universally locally 0-acyclic, then for every sheaf of groups F on X, the base-change homomorphism ϕ^1 of 1.1 is a monomorphism; if g is universally locally n-acyclic for $\mathbf{L}$, then the homomorphism ϕ^{n+1} of 1.1 is a monomorphism for every abelian sheaf of $\mathbf{L}$-torsion F on X.

211 b) It is probably also possible to obtain an analogous injectivity conclusion from the hypothesis that g is locally 1-aspherical for $\mathbf{L}$, by introducing the nonabelian

2-cohomology invariants from Giraud's thesis; compare XII 5.11 [41]; the same question also arises for (1.6) and (2.3) below.

Corollary 1.5. *Let K/k be a field extension, and let* $\mathbf{L} = \mathbf{P} - \{p\}$*, where* $p = \operatorname{car} k$*. Then* $\operatorname{Spec}(K) \to \operatorname{Spec}(k)$ *is universally locally acyclic for* $\mathbf{L}$ *and universally locally* 1*-aspherical for* $\mathbf{L}$*; if k and K are separably closed, then the preceding morphism is also universally acyclic for* $\mathbf{L}$ *and universally* 1*-aspherical for* $\mathbf{L}$*.*

We immediately note the following consequence of (1.5).

Corollary 1.6. *Let K/k be an extension of separably closed fields, let X be a scheme, and let F be a sheaf of sets (resp. of ind-$\mathbf{L}$-groups, resp. of abelian groups of $\mathbf{L}$-torsion) on X, where* $\mathbf{L} = \mathbf{P} - \{p\}$, $p = \operatorname{car} k$*. Then the map*

$$H^q(X, F) \longrightarrow H^q(X_K, F_K)$$

is bijective for $q = 0$ (resp. for $q \leq 1$, resp. for every q).

Indeed, this follows at once from the fact that $\operatorname{Spec}(K) \to \operatorname{Spec}(k)$ is universally acyclic for $\mathbf{L}$ and universally 1-aspherical for $\mathbf{L}$, and from the definitions. Note that, by VIII 2.4, the preceding homomorphism on the H^q may also be interpreted as being identified, with the base-change homomorphism

$$g^*(R^q f_*(F)) \longrightarrow R^q f'_*(g'^*(F)),$$

where $f : X \to \operatorname{Spec}(k)$, $f' : X_K \to \operatorname{Spec}(K)$, $g : \operatorname{Spec}(K) \to \operatorname{Spec}(k)$, and $g' : X_K \to$ 212
X are the evident morphisms. Thus, when X is quasi-compact and quasi-separated, the bijectivity of these homomorphisms may also be regarded, by 1.1, as a consequence of the fact that $\operatorname{Spec}(K) \to \operatorname{Spec}(k)$ is universally *locally* acyclic for $\mathbf{L}$ and universally locally 1-aspherical for $\mathbf{L}$. This latter fact therefore already implies 1.6 for X quasi-compact and quasi-separated, which plainly suffices to imply that $\operatorname{Spec}(K) \to \operatorname{Spec}(k)$ is universally acyclic for $\mathbf{L}$ and universally 1-aspherical for $\mathbf{L}$ (XV 1.7, XV 1.6 (i)).

This shows that, to prove 1.5, it suffices to prove the first assertion of 1.5. After passing to the perfect closure of k, which is permissible by VIII 1.1, we may then suppose that k is *perfect*. Every extension K of k is the filtered direct limit of its finitely generated k-subalgebras A, and since k is perfect, after localizing such an A, we may suppose it smooth, so that K appears as a filtered direct limit of smooth subalgebras. Since the latter are universally locally acyclic for $\mathbf{L}$ and universally locally 1-aspherical for $\mathbf{L}$ by XV 2.1, the conclusion follows from XV 1.13 (ii), C.Q.F.D.

Remark 1.7. The invariance theorem 1.5 should be compared with XII 5.4, where it was not necessary to suppose that the torsion sheaf under consideration is prime to the residue characteristics, but where, on the other hand, X must be assumed *proper* over k. Already in the case of $H^1(X, \mathbf{Z}/p\mathbf{Z})$, where X is an affine algebraic curve (for 213
example, the affine line) over an algebraically closed field k of characteristic $p > 0$, the analogue of the preceding statements becomes false, because of the phenomena of "wild ramification[42]" at infinity, which imply that, in such a case, the classification of principal étale covers with group $\mathbf{Z}/p\mathbf{Z}$ is essentially "continuous."

[41] This is indeed what is established in J. Giraud's book.

[42] N.D.E.: That is, "wild."

2. Specialization theorem for cohomology groups

Theorem 2.1. *Let* $\mathbf{L} \subset \mathbf{P}$*, let* $n \in \mathbf{N}$*, and let* $f : X \to Y$ *be a proper morphism of finite presentation* [42] *that is locally* 0*-acyclic (resp. locally* 1*-aspherical for* $\mathbf{L}$*, resp. locally* n*-acyclic for* $\mathbf{L}$*). Let* F *be a sheaf of sets (resp. of* ind-$\mathbf{L}$*-groups, resp. an abelian sheaf of* $\mathbf{L}$*-torsion) that is constructible and locally constant on* X*. Then the* $R^q f_* F$ *are constructible and* locally constant *for* $q = 0$ *(resp. for* $q = 0, 1$*, resp. for* $q \leq n$*), and for every geometric point* $\overline{y}$ *of* Y *one has*

$$(R^q f_* F)_{\overline{y}} \simeq H^q(X_{\overline{y}}, F_{\overline{y}}),$$

for the same values of q*.*

Remark. Note that the last assertion is simply the proper base-change theorem XII 5.2.

214 In view of XV 2.1, we immediately obtain the following corollary:

Corollary 2.2. *(Specialization theorem for cohomology groups). Let* $f : X \to Y$ *be a proper smooth morphism, and let* $\mathbf{L} \subset \mathbf{P}$ *be the complement of the set of residual characteristics of* Y*. Let* F *be a sheaf of sets (resp. of ind-*$\mathbf{L}$*-groups, resp. an abelian sheaf of* $\mathbf{L}$*-torsion) that is constructible on* X*. Then the* $R^q f_*(F)$ *are constructible and* locally constant *for* $q = 0$ *(resp. for* $q = 0, 1$*, resp. for every* q*), and for every geometric point* $\overline{y}$ *of* Y *one has*

$$R^q f_*(F)_{\overline{y}} \simeq H^q(X_{\overline{y}}, F_{\overline{y}})$$

for the same values of q*.*

Remark. Taking $n = 1$ and translating into the language of fundamental groups, we recover (SGA 1 X 3.8), which compares the fundamental groups of the fibers of a proper smooth morphism.

Proof of 2.1. Since we already know that the $R^q f_*(F)$ are constructible XIV 1.1, it suffices, by IX 2.11, to prove that for every specialization morphism $\overline{y}_1 \to \overline{y}_0$ of geometric points of Y, the corresponding specialization morphism (VIII 7.7)

$$R^q f_*(F)_{\overline{y}_0} \longrightarrow R^q f_*(F)_{\overline{y}_1} \tag{$*$}$$

is bijective for the values of q under consideration. For this we have the following slightly
215 more general result (in which it is unnecessary to assume that f is of finite presentation or that F is constructible):

Corollary 2.3. *Let* $\mathbf{L} \subset \mathbf{P}$*, let* $n \in \mathbf{N}$*, let* $f : X \to Y$ *be a proper morphism that is locally* 0*-acyclic (resp. locally* 1*-aspherical for* $\mathbf{L}$*, resp. locally* n*-acyclic for* $\mathbf{L}$*), and let* F *be a sheaf of sets (resp. of ind-*$\mathbf{L}$*-groups, resp. an abelian sheaf of* $\mathbf{L}$*-torsion) on* X *that is locally constant. Then for every specialization morphism* $\overline{y}_1 \to \overline{y}_0$ *of geometric points of* Y*, the corresponding specialization homomorphism* $(*)$ *above is bijective for* $q = 0$ *(resp. for* $q = 0, 1$*, resp. for* $q \leq n$*). Moreover, when* $n = 0$*, i.e. when* f *is locally* 0*-acyclic, for every locally constant sheaf of groups* F *on* X*, the homomorphism* $R^1 f_*(F)_{\overline{y}_0} \to R^1 f_*(F)_{\overline{y}_1}$ *is injective; for arbitrary* n*, and for every locally constant abelian sheaf of* $\mathbf{L}$*-torsion* F *on* X*, the specialization homomorphism* $(*)$ *is injective for* $q = n + 1$*.*

There exists an integral strictly local scheme Y' and a morphism $g : Y' \to Y$ that maps the closed point y_0' to y_0 and the generic point y_1' to y_1: for example, it is enough

[42] For a less restrictive hypothesis than properness that yields the same conclusions, cf. SGA 5 II 3 and SGA 1 XIII.

first to take the strict localization of Y at $\overline{y}_0$, and then its integral closed subscheme defined by a point lying above y_1. Moreover, after replacing Y' by its normalization in an algebraic closure of its function field, we may assume that Y' is normal and that $k(y_1')$ is algebraically closed, so that y_0' and y_1' are geometric points of Y'. Using the base-change theorem (XII 5.1) for (f, F, g), the proof of 2.3 is reduced, after replacing $Y, \overline{y}_0$, 216
$\overline{y}_1$ by Y', y_0', y_1', to proving 2.3 when Y is normal and strictly local and when $\overline{y}_0$ and $\overline{y}_1$ are, respectively, its closed point y_0 and its generic point y_1. Moreover, using XII 4, we see that the specialization homomorphism $(*)$ is then identified with the homomorphism

$$H^q(X, F) \longrightarrow H^q(X_1, g_1^*(F)),$$

where $X_1 = X_{y_1}$ and where $g : y_1 \to Y$ and $g_1 : X_1 \to X$ are the canonical morphisms. Our assertion now follows from the following lemma (in which f is no longer assumed proper):

Lemma 2.4. *With the preliminary notation of 2.3, drop the properness assumption on f, but assume instead that Y is normal and integral, with generic point y_1 such that $k(y_1)$ is separably closed, and let $X_1 = X_{y_1}$ and $g_1 : X_1 \to X$ be the canonical morphism. Then the homomorphism*

$$(**) \qquad H^q(X, F) \longrightarrow H^q(X_1, g_1^*(F))$$

is bijective for $q = 0$ (resp. bijective for $q = 0, 1$, resp. bijective for $q \le n$ and injective for $q = n+1$). Moreover, when $n = 0$, i.e. when f is locally 0-acyclic, for every sheaf of groups F on X, the homomorphism $(**)$ is injective for $q = 1$.

An immediate argument (XV 1.6 (i) $\Rightarrow$ (ii)) shows that it is enough to prove the relation

$$F \xrightarrow{\sim} g_{1_*} g_1{}^*(F),$$

and, for $n \ge 1$, the additional relations 217

$$R^q g_{1_*} g_1{}^*(F) = 0 \quad \text{for} \quad 1 \le q \le n.$$

These relations are local on X for the etale topology, and F is locally constant for the etale topology, so we may assume that F is constant and hence of the form G_X, where G is a set (resp. an ind-L-group, resp. an abelian group of L-torsion). Note also that the assumption that Y is normal implies $g_*(G_{y_1}) \xleftarrow{\sim} G_Y$ (IX 2.14.1), and the assumption that $k(y_1)$ is separably closed implies that the $R^q g_*(G_{y_1})$ vanish for $q \ge 1$. On the other hand, the assumption that f is locally 0-acyclic (resp. ...) implies that the formation of the $R^q g_*(G_{y_1})$ commutes with the base change $f : X \to Y$ for $q \le n$. Here we use the fact that we may assume Y affine and that y_1 then appears as the inverse limit of its affine open neighborhoods, which are quasi-compact in Y, so that Definition XV 1.11 and the theory of passage to the limit VII 5 apply. We conclude that indeed

$$G_X \xrightarrow{\sim} g_{1_*}(G_{X_1}) \quad , \quad R^q g_{1_*}(G_{X_1}) = 0 \quad \text{for} \quad 1 \le q \le n,$$

which completes the proof of 2.4, and hence that of 2.1.

Remark. The preceding proof, via 2.4, which is simpler than our original proof, is due to M. Lubkin.

Corollary 2.5. *Under the hypotheses of 2.1, assume that Y is connected, and let $q \le n$. Then the groups $H^q(X_{\overline{y}}, F)$, for geometric points $\overline{y}$ of Y, are all isomorphic to one another.* 218

3. The relative cohomological purity theorem

Definition 3.1. *Let S be a scheme. A* smooth S-pair (Y, X) *is a closed S-immersion $i : Y \to X$ of smooth S-preschemes, that is, a commutative diagram of preschemes*

$$\begin{tikzcd} Y \arrow[rr, "i"] \arrow[dr, "h"'] & & X \arrow[dl, "f"] \\ & S & \end{tikzcd}$$

such that f and h are smooth and i is a closed immersion. We write $U = X - Y$, denote by $j : U \to X$ the open immersion, and by $g : U \to S$ the structural morphism. The codimension *of (Y, X) at a point $y \in Y$ is the codimension at y of the fiber Y_s in X_s, where $s = h(y)$.*

Thus the codimension is a locally constant function on Y. We shall express the fact that this function is in fact constant, of value c, by saying that "(Y, X) has codimension c."

Let (Y, X) be a smooth S-pair and let $y \in Y$. Recall (SGA II 4.10) that there exist integers m and n, a neighborhood X' of y in X, and an étale morphism

$$\phi : X' \longrightarrow \mathbf{E}^n_S = \operatorname{Spec} \mathscr{O}_S[t_1, \dots, t_n]$$

219 such that $Y' = Y \cap X'$ is the inverse image of the closed subscheme $\mathbf{E}^m_S$ defined by the equations

$$t_{m+1} = \cdots = t_n = 0.$$

This result can be expressed by saying that *locally for the étale topology, every smooth S-pair (Y, X) is isomorphic to the standard pair $(\mathbf{E}^m_S, \mathbf{E}^n_S)$ for suitable m and n,* where

$$\mathbf{E}^r_S = \operatorname{Spec} \mathscr{O}_S[t_1, \dots, t_r].$$

In the present section, we propose to calculate the local cohomology sheaves $\mathbf{H}^q_Y(X, F) = (R^q i^!)F$ (V 6 and VIII 6.6) for a smooth S-pair (Y, X), with coefficients in a locally constant abelian torsion sheaf F prime to the residue characteristics. We begin with some preliminary considerations:

Proposition 3.2. *Let*

$$\begin{tikzcd} U \arrow[rr, "j"] \arrow[dr, "g"'] & & X \arrow[dl, "f"] \\ & S & \end{tikzcd}$$

be a commutative diagram in which f is smooth and j is an open immersion such that the fiber U_s is dense in X_s for every $s \in S$. Let F be a sheaf of sets on S. Then the canonical morphism

$$f^* F \longrightarrow j_* g^* F = j_* j^* f^* F$$

is bijective.

220 *Proof.* We readily reduce to the case where U is quasi-compact in X. Let $\bar{s}$ be a geometric point of S, let $\bar{x}$ be a geometric point of X lying over $\bar{s}$, let $\widetilde{f} : \widetilde{X} \to \widetilde{S}$ be the morphism of the strict localizations of X and S at the corresponding geometric points, and let $\widetilde{U} = U \times_X \widetilde{X}$. With the evident notation, we have

$$(f^* F)_{\bar{x}} = H^0(\widetilde{X}, \widetilde{f}^* \widetilde{F}) \simeq H^0(\widetilde{S}, \widetilde{F}).$$

Consequently, it suffices to prove that $\tilde{g} : \widetilde{U} \to \widetilde{S}$ is 0-acyclic, which will imply

$$(j_* g^* F)_{\overline{x}} \simeq H^0(\widetilde{U}, \tilde{g}^* \widetilde{F}) \simeq H^0(\widetilde{S}, \widetilde{F}).$$

Since $\tilde{g}$ is a limit of smooth morphisms, it is locally 0-acyclic (XV 2.1, XV 1.11), and by XV 1.12 it suffices to prove that the geometric fibers of $\widetilde{U}/\widetilde{S}$ are 0-acyclic, that is, connected and nonempty. Let $\overline{s}'$ be a geometric point of $\widetilde{S}$. Then the fiber $\widetilde{X}_{\overline{s}'}$ is regular, connected, and nonempty, and the hypothesis implies that $\widetilde{U}_{\overline{s}'}$ is a dense open subset of $\widetilde{X}_{\overline{s}'}$, hence connected and nonempty, which proves the result.

Corollary 3.2.1. *Under the hypotheses of 3.2, the functor $X' \mapsto X'|U$ from the category of étale coverings of X to the category of étale coverings of U is fully faithful. In particular, for every finite group G, the restriction map*

$$H^1(X, G) \longrightarrow H^1(U, G)$$

is injective.

This follows from 3.2, by a standard argument which, for ease of reference, we shall 221
spell out in a

Lemma 3.2.2. *Let $f : X \to Y$ be a morphism of schemes, and let $f^* : \mathrm{Et}(Y) \to \mathrm{Et}(X)$ be the inverse-image functor $Y' \mapsto Y' \times_Y X$ from the category $\mathrm{Et}(Y)$ of étale coverings of Y to $\mathrm{Et}(X)$. The following conditions are equivalent:*

(i) *f^* is faithful (resp. fully faithful).*

(ii) *For every étale covering Y' of Y, denoting by $f' : X' \to Y'$ the morphism deduced from f by the base change $Y' \to Y$, the map $U \mapsto f'^{-1}(U)$ from the set of clopen subsets of Y' to the set of clopen subsets of X' is injective (resp. bijective).*

(ii bis) *With the notation of* (ii), *for every constant sheaf of sets C on Y', the canonical map*

$$H^0(Y', C) \longrightarrow H^0(X', f^*C)$$

is injective (resp. bijective).

(iii) *With the notation of* (ii), *the canonical map*

$$\Gamma(Y'/Y) \longrightarrow \Gamma(X'/X)$$

is injective (resp. bijective).

The proof is left to the reader (cf. SGA 2, IX 3.1 and 3.2).

Remarks 3.2.3. One can generalize Proposition 3.2 considerably, both for sheaves 222
of sets, by weakening the hypotheses on f, and for sheaves of groups, with conclusions concerning the $R^q g_*(f^* F)$ (cf. SGA 2 XIV 1 and SGA 1 XIII). The analogous remark applies to 3.4 through 3.6.

Let us also "recall" the Zariski–Nagata purity theorem in its relative form:

Theorem 3.3. *(Relative purity theorem). Let*

$$\begin{tikzcd} U \arrow[rr, "j"] \arrow[dr, "g"'] & & X \arrow[dl, "f"] \\ & S & \end{tikzcd}$$

be a commutative diagram in which f is smooth and j is an open immersion such that, for every $s \in S$, $X_s - U_s$ has codimension ≥ 2 everywhere in X_s. Then the functor $X' \mapsto X'|U$

from the category of étale coverings of X to the category of étale coverings of U is an equivalence of categories.

Proof. We saw in 3.2.1 that the functor is fully faithful. The fact that it is essentially surjective will follow from the classical theorem. Given an étale covering U' of U, we must show that it extends to an étale covering X' of X. There is an immediate and easy reduction first to the case where U is quasi-compact in X, and then to the case where S is affine and of finite type over $\operatorname{Spec}\mathbf{Z}$, X is connected, and f is of finite type.

223 We proceed by induction on $n = \dim S$, the assertion being a consequence of the classical theorem (SGA 2 X 3.4) when $n = 0$. Since uniqueness is available, we can apply the descent technique of SGA 1 IX 4.7 to the normalization $S \leftarrow \overline{S}$ and thereby reduce to the case where S (and hence X) is normal ($\overline{S} \to S$ being finite (EGA IV 7.8.3 (iii) (vi)). Moreover, after enlarging U, we may suppose that U is maximal among the open subsets over which U' can be extended, and we must then prove that $U = X$. Otherwise, let x be a maximal point of $X - U$, and put $s = f(x)$. It is enough to prove that U' can be extended over a neighborhood of x. We may suppose that $\dim \mathcal{O}_{S,s} = n \geq 1$ and that the theorem has already been proved in dimension $< n$. Put $B = \mathcal{O}_{X,x}$, and let B' be the normalization of B in the ring of rational functions on the covering U' of U; thus B' is a finite B-algebra (*loc. cit.*). It plainly suffices to prove that B' is an étale B-algebra.

Let $A = \mathcal{O}_{S,s}$, $0 \neq t \in \max A$, $A_0 = A/tA$, and $B_0 = B/tB$. By the induction hypothesis, the restriction of the covering to $U_0 = U \otimes_A A_0$ extends to an étale covering of $\operatorname{Spec} B_0$, which we denote by $\operatorname{Spec} B_0^*$. After replacing X by a suitable étale covering of a neighborhood of x (which is permissible), we may suppose that $\operatorname{Spec} B_0^*$ is completely split over $\operatorname{Spec} B_0$. We are thereby reduced to the case where the B_0-algebra B_0' is completely split away from the closed point of $\operatorname{Spec} B_0$. The hypotheses of XV 3.3 are then satisfied, which proves the result.

Corollary 3.4. *Under the same hypotheses as in 3.3, we have $(R^1 j_*)g^* F = 0$ for every sheaf F of ind-finite groups on S.*

224 We reduce as usual to the case where S is noetherian, f is of finite type, and F is constructible. Let $0 \to F \to G$ be an injection of sheaves of ind-finite groups on S, and put $C = G/F$. It follows from 3.2 that the functor $j_* g^*$ is exact, so that $j_* g^* G \to j_* g^* C$ is surjective, and hence (XII 3.1) that $(R^1 j_*)g^* F \to (R^1 j_*)g^* G$ is injective. We can therefore replace F by G, and thus reduce, by IX 2.14 and VIII 5.5, to the case where $F = G$ is constant, with value an ordinary finite group G. Let $\overline{x}$ be a geometric point of X, let $\widetilde{X}$ be the strict localization of X at $\overline{x}$, and let $\widetilde{U} = \widetilde{X} \times_X U$. It then follows from 3.3 that

$$(R^q j_* g^* F)_{\overline{x}} = H^q(\widetilde{U}, G_{\widetilde{U}}) \simeq H^q(\widetilde{X}, G_{\widetilde{X}}),$$

and the latter is zero, which proves the result.

Lemma 3.5. *(Relative Abhyankar's lemma). Let (Y, X) be a smooth S-pair of codimension 1, such that Y is defined in X by an equation $t = 0$. Let $U = X - Y$, and let V/U be a principal Galois covering of order n prime to the residue characteristics of Y. Put $X' = \operatorname{Spec} \mathcal{O}_X[z]/(z^n - t)$, $U' = U \times_X X'$, and $V' = V \times_X X'$. Then the covering V' extends uniquely to an étale covering of X'.*

Proof. We immediately reduce to the case where S, X, and Y are affine and of finite type over $\operatorname{Spec}\mathbf{Z}$. Uniqueness is a special case of 3.2.1. We shall deduce existence from XV 1.12: let $\overline{s}$ be a geometric point of S and let $\overline{x}$ be a geometric point of Y lying over $\overline{s}$. Let
225 $\widetilde{f} : \widetilde{X} \to \widetilde{S}$ be the morphism of the corresponding strict localizations induced by f, let

$\widetilde{U} = U \times_X \widetilde{X}$, etc...By descent, in view of uniqueness, it plainly suffices to prove that $\widetilde{V}'$ extends to an étale covering of $\widetilde{X}'$, that is (since $\widetilde{X}'$ admits no nontrivial covering), that $\widetilde{V}'$ is a trivial covering of $\widetilde{U}'$.

Let
$\overline{U} = \varprojlim_m \widetilde{U}_m = \operatorname{Spec} \mathcal{O}_{\widetilde{U}}[t^{1/m}]$, where m runs through the set of integers prime to the residue characteristic of $\widetilde{X}$. Then $\overline{U}/\widetilde{U}$ is an "infinite Galois covering" with abelian group $G = \varprojlim_m \mathbf{Z}/m\mathbf{Z}$, and one sees immediately that it is enough to prove that $\overline{U}$ is 1-aspherical for the set $\mathbf{L} \subset \mathbf{P}$ complementary to the residue characteristic of $\widetilde{X}$.

Consider the morphism $\overline{f} : \overline{U} \to \widetilde{S}$. It is a limit of smooth morphisms, and is therefore locally 1-aspherical for $\mathbf{L}$ (XV 2.1, XV 1.11 (ii)). By XV 1.12, it is enough to prove that the geometric fibres of $\overline{U}/\widetilde{S}$ are 1-aspherical for $\mathbf{L}$. Let $\overline{s}$ be a geometric point of $\widetilde{S}$. The geometric fibre $\overline{U}_{\overline{s}}$ is the limit of the fibres $(\widetilde{U}_m)_{\overline{s}}$. Consequently, it is enough to prove that every principal Galois covering of some $(\widetilde{U}_m)_{\overline{s}}$, of order prime to the residue characteristics, induces a trivial covering of $\overline{U}_{\overline{s}}$. Replacing $\widetilde{X}$ by $\widetilde{X}_m = \operatorname{Spec} \mathcal{O}_{\widetilde{X}}[t^{1/m}]$ (which is still smooth over $\widetilde{S}$), we reduce to the case $m = 1$, that is, $\overline{U}_m = \widetilde{U}$. Let V be such a covering of $\widetilde{U}_{\overline{s}}$. Now $\widetilde{X}_{\overline{s}} = Z$ is a regular strictly local scheme, and $W = \widetilde{U}_{\overline{s}}$ is the open complement of a regular subset of codimension 1. We may therefore apply the usual form of Abhyankar's lemma (SGA 1 X 3.6) to conclude that V induces a covering V_m of W_m which extends to an étale covering of Z_m, for a suitable m. But since Z, and 226
hence Z_m, is strictly local, this covering is trivial; therefore V_m is a trivial covering of W_m, C.Q.F.D.

Remark 3.5.1. When the order n of G is not assumed prime to the residue characteristics of X, one can still prove the following: let m be the largest integer prime to the residue characteristics of Y which divides the order of G, and suppose that for every point $s \in S$, the covering V_s of $U_s = X_s - Y_s$ has, at the maximal points of Y_s, inertia groups whose orders are prime to the residue characteristics of Y. Then the conclusion of 3.5 remains valid on taking $X' = \operatorname{Spec} \mathcal{O}_X[z]/(z^m - t)$. More generally, one may also take for m any common multiple of the orders of the inertia groups just considered, provided these orders are prime to the corresponding residue characteristics.

Corollary 3.6. *(Cohomological purity in dimension 1). Let (Y, X) be a smooth S-pair of codimension 1, and let F be a constructible locally constant sheaf of groups on X, whose orders are prime to the residue characteristics of X. Then, with the notation of 3.1, $(R^1 j_*)j^*F$ is locally isomorphic, as a sheaf of pointed sets, to $i_*(F_Y/\operatorname{int}(F_Y))$, where $F_Y = F|Y$ and where $F_Y/\operatorname{int}(F_Y)$ denotes the quotient of F_Y by its action on itself through inner automorphisms.*

Remark. The isomorphism of Corollary 3.6 is not canonical.

Proof. We immediately reduce to the case where X is noetherian and connected. 227
Since the assertion is local on X for the étale topology, we may suppose that F is a constant sheaf, with value an ordinary finite group G of order m prime to the residue characteristics of X, and moreover that $\boldsymbol{\mu}_{mX} \simeq (\mathbf{Z}/m\mathbf{Z})_X$. Let $U = X - Y$ and let $U' = \operatorname{Spec} \mathcal{O}_U[t^{1/m}]$, which is an "essential" principal Galois covering of U with group $\mathbf{Z}/m\mathbf{Z}$, since $\boldsymbol{\mu}_{mX} \simeq (\mathbf{Z}/m\mathbf{Z})_X$. Principal coverings of U with group G which are trivialized over U' are classified locally on Y by $H = \operatorname{Hom}(\mathbf{Z}/m\mathbf{Z}, G)$ modulo $\operatorname{int}(G)$, which is a pointed set isomorphic to the underlying set of $G/\operatorname{int}(G)$. This gives a morphism from

the constant sheaf H_Y on Y to $(R^1 j_*)G_U$, and it is enough to prove that this morphism is bijective, which follows immediately from 3.5.

Theorem 3.7. (Cohomological purity). *Let (Y, X) be a smooth S-pair of codimension $c > 0$. Let F be an abelian sheaf on X locally isomorphic (for the étale topology) to a sheaf of the form $f^*(G)$, where G is a torsion sheaf on S, prime to the residue characteristics (for example, F may be a locally constant sheaf of finite groups whose orders are prime to the residue characteristics of X). Then, with the notation of 3.1, one has*

$$\mathrm{H}^q_Y(X, F) = (R^q i^!)F = 0 \qquad \textit{if} \quad q \neq 2c,$$

$$\text{(i.e.)} \qquad \begin{cases} F \xrightarrow{\sim} j_* j^* F \quad \textit{and} \\ (R^q j_*) j^* F = 0 \qquad\qquad \textit{if} \quad q \neq 0, \quad 2c-1, \end{cases}$$

228 *and*

$$\mathrm{H}^{2c}_Y(X, F) = i^*[(R^{2c-1} j_*) j^* F]$$

is a sheaf locally isomorphic to $i^*(F)$.

Proof. The assertion is local on X for the étale topology; we may therefore [42] suppose that F is constant, and hence, if desired, that $F \simeq f^*(\mathrm{Z}/n\mathrm{Z})_S$ for the constant sheaf $(\mathrm{Z}/n\mathrm{Z})_S$ on S.

First treat the case $c = 1$. (The assertion concerning $(R^1 j_*) j^* F$, at least in the case where F is locally constant, would follow immediately from 3.6, but we shall derive it again.) Recall that (Y, X) is locally isomorphic to the pair $(\mathrm{E}^{n-1}_S, \mathrm{E}^n_S)$, and that, upon replacing S by E^{n-1}_S, we reduce to the case where $X = \mathrm{E}^1_S = \operatorname{Spec} \mathscr{O}_S[t]$ and Y is the section $t = 0$ of X/S. This situation is locally isomorphic to the inclusion of the section at infinity in the projective line $\mathbf{P}^1_S$. We may therefore take $X = \mathbf{P}^1_S$, take Y to be the section at infinity, and put $U = \mathrm{E}^1_S = X - Y$.

Consider the Leray spectral sequence

$$(R^p f_*)(R^q j_*)(\mathrm{Z}/n) \Longrightarrow (R^{p+q} g_*)(\mathrm{Z}/n).$$

Its abutment is zero for $p+q > 0$, by XV 2.2. Moreover, the $R^q j_*$ are supported on Y when $q > 0$, and Y maps isomorphically onto S under f; on the other hand, $j_*(F|U) \xleftarrow{\sim} F$ by 3.2. It follows that $(R^p f_*)(R^q j_*) = 0$ if $p, q > 0$, that $f_*(R^q j_*)$ determines the sheaf $(R^q j_*)$ when $q > 0$, and finally that $R^p f_* j_*(F|U) = R^p f_*(F)$. The spectral sequence
229 therefore reduces to isomorphisms

$$f_*(R^q j_*)(F|U) \xrightarrow{\sim} (R^{q+1} f_*)(F) \quad \text{if} \quad q > 0.$$

But $(R^q f_*)(F)$ can be computed fibre by fibre (XII 5.2); it is therefore zero unless $q = 0, 2$, and is a sheaf locally isomorphic to G if $q = 2$, as follows easily from XII 5.2 and IX 4.7. Thus $(R^q j_*)(F|U) = 0$ if $q \neq 1$, and it is a sheaf locally isomorphic to $i^*(F)$ if $q = 1$.

The case $c > 1$ is now handled easily by induction: let (Y, X) be a smooth S-pair of codimension $c > 1$. It is clear that, locally on X, one can find a subscheme Z of X of codimension 1 which contains Y and is smooth over S, so that there is a commutative

[42] At least if F were assumed locally constant. The reader will convince himself that the proof which follows also applies, essentially, to the general case.

diagram

$$\begin{tikzcd} Y \arrow[rr, "u"] \arrow[dr, "i"'] & & Z \arrow[dl, "v"] \\ & X & \end{tikzcd},$$

in which each pair is a smooth S-pair (one ought to call (Y, Z, X) a smooth S-triple), the codimension of (Z, X) is 1, and that of (Y, Z) is $c - 1$.

Now there is a spectral sequence of composite functors

$$(R^p u^!)(R^q v^!)F \Longrightarrow (R^{p+q} i^!)F.$$

By the case already proved, $(R^q v^!)F = 0$ if $q \neq 2$ and is locally isomorphic to $v^*(F)$ if $q = 2$. The induction hypothesis, applied to $(R^q v^!)F$, then shows that $(R^p u^!)(R^q v^!)F$ is zero if $(p, q) \neq (2c-2, 2)$ and is locally isomorphic to $u^* v^*(F)$ if $(p, q) = (2c-2, 2)$. The 230
result for $(R^{p+q} i^!)F$ follows immediately, C.Q.F.D.

Corollary 3.8. *Let (Y, X) be a smooth S-pair of codimension c, let n be a natural number prime to the residue characteristics of X, and put*

$$T_{Y/X} = \mathrm{H}^{2c}_Y(\mathbf{Z}/n\mathbf{Z}),$$

which, by (3.7), is a sheaf on Y locally isomorphic (for the étale topology) to the sheaf $(\mathbf{Z}/n\mathbf{Z})_Y$. Let F be an abelian sheaf on X, annihilated by n, satisfying the condition stated in 3.7. Then (for the record) $\mathrm{H}^i_Y(F) = 0$ for $i \neq 2c$, and moreover there is a canonical isomorphism

$$\mathrm{H}^{2c}_Y(F) \simeq i^*(F) \otimes T_{Y/X}.$$

Proof. One easily defines a canonical homomorphism

$$\varphi : i^*(F) \otimes T_{Y/X} \longrightarrow \mathrm{H}^{2c}_Y(F)$$

that is, a homomorphism

$$i^*(F) \longrightarrow \mathrm{Hom}(\mathrm{H}^{2c}_Y(\mathbf{Z}/n\mathbf{Z}), \mathrm{H}^{2c}_Y(F)),$$

using the canonical homomorphism $F \to \mathrm{Hom}(\mathbf{Z}/n\mathbf{Z}, F)$. It remains to prove that φ is an isomorphism, which is easily checked by following the proof given for 3.7.

Corollary 3.8.1. *Under the conditions of 3.8, the formation of the $\mathrm{H}^i_Y(F)$ commutes* 231
with every base change $S' \to S$.

This is clear.

Corollary 3.9. *(Absolute cohomological purity theorem). Let X be a regular scheme, and let Y be a regular closed subprescheme of codimension c at every point. Suppose moreover that X is locally of finite type over a perfect field k. Then, if n is prime to the residue characteristics of X, one has*

$$\mathrm{H}^i_Y(\mathbf{Z}/n\mathbf{Z}) = 0 \text{ if } i \neq 2c,$$

and $\mathrm{H}^{2c}_Y(\mathbf{Z}/n\mathbf{Z})$ is a sheaf on Y locally isomorphic (for the étale topology) to the constant sheaf $\mathbf{Z}/n\mathbf{Z}$.

This is a special case of 3.7, since the hypothesis implies that X and Y are smooth over k.

REMARKS 3.10. a) In 3.8 one can make the structure of the sheaf $T_{Y/X}$ explicit: one finds that it is canonically isomorphic to the sheaf $(\boldsymbol{\mu}_n)_Y^{\otimes -c}$, where $\boldsymbol{\mu}_n$ denotes the sheaf of nth roots of unity (locally isomorphic to the constant sheaf $\mathbf{Z}/n\mathbf{Z}$). The isomorphism in question is studied, under the name of the local *fundamental class* of Y in X, in the slightly more general setting of relative complete intersections, in Exposé XVIII, which is devoted to duality.

232 b) It is very plausible that the conclusions of 3.9 and 3.10 a) remain valid without assuming the existence of a base field k, at least when X is an "excellent" prescheme. This will in any case be established in Exposé XIX when X is of characteristic zero, using Hironaka's resolution of singularities. As we shall see in SGA 5, the "absolute purity theorem" (as well as resolution of singularities) is needed in particular to establish the "local duality formula", which itself is one of the principal ingredients of the LEFSCHETZ–VERDIER formula.

4. Comparison theorem for the cohomology of algebraic schemes over C.

Let X be an algebraic scheme over $\operatorname{Spec}\mathbf{C}$. We shall compare the étale and classical sites of X, and we retain the notation of XI 4. The comparison theorem, whose general form is XI 4.4, is as follows:

THEOREM 4.1. *Let $f : X \to S$ be a morphism of finite type between schemes locally of finite type over* $\operatorname{Spec}\mathbf{C}$*, so that there is a commutative diagram of morphisms of sites*

$$\begin{tikzcd} X_{\mathrm{et}} \arrow[d, "f_{\mathrm{et}}"'] & X_{\mathrm{cl}} \arrow[l, "\varepsilon"'] \arrow[d, "f_{\mathrm{cl}}"] \\ S_{\mathrm{et}} & S_{\mathrm{cl}} \arrow[l, "\varepsilon"'] \end{tikzcd}$$

233 *Let F be a sheaf of sets (resp. of ind-finite groups, resp. a torsion abelian sheaf), and suppose that one of the following cases holds:*

(i) *f is proper,*
(ii) *F is constructible.*

Then the base-change morphisms XII 4.2 [42]

$$\varphi : \epsilon^*(R^q f_{\mathrm{et}*})F \longrightarrow (R^q f_{\mathrm{cl}\,*})\epsilon^* F$$

are bijective for $q = 0$ (resp. for $q = 0, 1$, resp. for $q \geq 0$).

Observe that the Grauert–Remmert theorem (XI 4.3 (iii)), which we used in the proof of XI 4.4, is a special case of 4.1.

234 For the proof, we first treat the case in which f is proper. In this case, the result will follow from GAGA and the proper base-change theorem, by an elementary argument that could have appeared in Exposé XIV. To treat the general case, we shall also use resolution of singularities [1] and 3.7.

We may clearly assume that S, and hence also X, is of finite type over $\mathbf{C}$.

The case in which f is proper. We recall the following elementary facts:

(a) The topological space X_{cl} is locally compact.

[42] In fact, we use the base-change homomorphism under more general conditions than those considered in Exposé XII, where we restricted ourselves to a *Cartesian* square of morphisms of *preschemes*. The reader will readily verify that the definition extends verbatim to an *essentially commutative* square of morphisms of *sites*. See the following exposé for further developments on this point.

(b) If $f : X \to S$ is proper, then $f_{\mathrm{cl}} : X_{\mathrm{cl}} \to S_{\mathrm{cl}}$ is a *proper* map of topological spaces.

Indeed, (a) is trivial, since locally X_{cl} is a closed subset of some $\mathbf{C}^n$. For (b), Chow's lemma reduces us to the case in which f is projective, and this case is trivial.

Let F be a sheaf of sets (resp. ...) on X_{et}. For both morphisms f_{et} and f_{cl}, formation of the R^q commutes with formation of the fibers (XII 5.2 and [2], 4.11.1). It is therefore enough to verify the theorem fiber by fiber, at the closed points of S_{et}.

In the following lemma, we do not assume that g is proper:

Lemma 4.2. *Let* 235

$$\begin{tikzcd} X & & \overline{X} \arrow[ll, "h"'] \arrow[dl, "\overline{g}"] \\ & S \arrow[from=ul, "g"'] & \end{tikzcd}$$

be a commutative diagram with h proper, and let U be an open subset of X such that h induces an isomorphism from the open subset $\overline{U} = U \times_X \overline{X}$ of $\overline{X}$ onto U. Let $j : U \to X$ and $\overline{j} : \overline{U} \to \overline{X}$ be the inclusions, let F be a sheaf of sets (resp. ...) on U, and let $\overline{F}$ be the induced sheaf on $\overline{U}$. If Theorem 4.1 is true for $(\overline{g}, \overline{j}_!\overline{F})$, then it is also true for $(g, j_!F)$.

Proof. The theorem is true for $(h, \overline{j}_!\overline{F})$, because we are reduced to checking it fiber by fiber, and there are then two cases: either the restriction of the sheaf to the fiber is zero, or the morphism h is an isomorphism. In both cases the result is trivial. We find that

$$\epsilon^* j_! F \simeq \epsilon^* h_{\mathrm{et}\,*}(\overline{j}_!\overline{F}) \simeq h_{\mathrm{cl}\,*}\epsilon^*(\overline{j}_!\overline{F})$$

and

$$(R^q h_{\mathrm{et}\,*})\overline{j}_!\overline{F} = (R^q h_{\mathrm{cl}\,*})\epsilon^*(\overline{j}_!\overline{F}) = 0 \text{ if } q > 0$$

for the values of q under consideration, and hence that

$$(R^q g_{\mathrm{et}\,*}) j_! F \simeq (R^q \overline{g}_{\mathrm{et}\,*})\overline{j}_!\overline{F}$$

and

$$(R^q g_{\mathrm{cl}\,*}) j_! F \simeq (R^q \overline{g}_{\mathrm{cl}\,*})\overline{j}_!\overline{F},$$

which proves Lemma 4.2. 236

Lemma 4.3. *To verify 4.1 for all data (f, F) with f proper, it is enough to do so when, in addition, f has relative dimension ≤ 1 and $S = \operatorname{Spec} \mathbf{C}$ is a point.*

Proof. We proceed by induction on the relative dimension. Suppose the result known in relative dimension $< n$, where $n > 1$. As we have observed, we may check it fiber by fiber; that is, we may suppose that $S = \operatorname{Spec} \mathbf{C}$ is a point, and hence that X has dimension $\leq n$. Moreover, we may assume that X is reduced. Let φ be a rational function on X that is nonconstant on every irreducible component of X, let U be a dense open subset of X on which φ is defined, and let $\overline{X} \subset X \times \mathbf{P}^1$ be the closure of the graph of the morphism

$\varphi : U \to \mathbf{E}^1$. We have a commutative diagram

$$\begin{tikzcd}
& \overline{X} \arrow[dl, "h"'] \arrow[dr, "a"] \arrow[dd, "\overline{g}"] & \\
X \arrow[dr, "g"'] & & \mathbf{P}^1 \arrow[dl, "b"] \\
& Y = \operatorname{Spec} \mathbf{C} &
\end{tikzcd}$$

in which the relative dimensions of a and b are $< n$. By the induction hypothesis, the theorem is true for $(a, .)$ and $(b, .)$, and it follows that it is also true for $(\overline{g}, .)$. Indeed, this
237 is trivial for $q = 0$; if F is a torsion abelian sheaf on $\overline{X}$, it follows from the morphism of Leray spectral sequences

$$\begin{tikzcd}
\varepsilon^*(R^p b_{\mathrm{et}*})(R^q a_{\mathrm{et}*})F \arrow[r, Rightarrow] \arrow[d] & \varepsilon^*(R^{p+q}\overline{g}_{\mathrm{et}*})F \arrow[d] \\
(R^p b_{\mathrm{cl}*}(R^q a_{\mathrm{cl}*}))\epsilon^* F \arrow[r, Rightarrow] & (R^{p+q}\overline{g}_{\mathrm{cl}*})\epsilon^* F;
\end{tikzcd}$$

finally, if F is a sheaf of ind-finite groups on $\overline{X}$, note that since $\epsilon^*(R^1\overline{g}_{\mathrm{et}\,*})$ is an effaceable functor, it is enough to prove (in view of the result for $q = 0$) that $(R^1\overline{g}_{\mathrm{cl}\,*})\epsilon^*$ is also effaceable (cf. XII 8.2). This follows from the exact sequence

$$0 \longrightarrow (R^1 b_{\mathrm{cl}\,*})a_{\mathrm{cl}\,*}\epsilon^* F \longrightarrow (R^1\overline{g}_{\mathrm{cl}\,*})\epsilon^* F \longrightarrow b_{\mathrm{cl}\,*}(R^1 a_{\mathrm{cl}\,*})\epsilon^* F,$$

and from the fact that $(R^1 b_{\mathrm{cl}\,*})\epsilon^*$ and $(R^1 a_{\mathrm{cl}\,*})\epsilon^*$ are effaceable by the induction hypothesis.

Now let F be a sheaf of sets on X. We have $F \hookrightarrow h_{\mathrm{et}\,*}h_{\mathrm{et}}{}^* F = G$. The exact sequence

$$F \to G \rightrightarrows G \amalg_F G, \tag{4.3.1}$$

reduces us (for $q = 0$) to proving 4.1 for G, and hence for a sheaf of the form $h_{\mathrm{et}\,*}\overline{F}$. For such a sheaf, however, the result follows from the theorem for $(\overline{g}, .)$ and for $(h, .)$; and h has relative dimension $< n$, as is immediate.

Let F be a torsion abelian sheaf on X. We have the exact sequence

$$0 \longrightarrow j_! j^* F \longrightarrow F \longrightarrow i_* i^* F \longrightarrow 0. \tag{$*$}$$

238 Since U is dense in X, we have $\dim Y < n$, so 4.1 is true for $(g|Y, i^* F)$, and hence for $(g, i_* i^* F)$. It is therefore enough, by the five lemma, to verify 4.1 for $(g, j_! j^* F)$, and this follows from the theorem for $\overline{g}$ (4.2).

Let F be a sheaf of ind-finite groups on X. It is enough to prove that the functor $(R^1 g_{\mathrm{cl}\,*})\epsilon^*$ is effaceable for the sheaf F. This follows from $(*)$ and from effaceability for $j_! j^* F$ and for $i_* i^* F$, that is, from the theorem for $(g, j_! j^* F)$ (4.2) and for $(g, i_* i^* F)$ (the induction hypothesis), C.Q.F.D.

Lemma 4.4. *To prove (4.1) when f is proper, it is enough to prove the following assertion: let X be a complete regular curve over* Spec $\mathbf{C}$*, and let F be a constant constructible sheaf of sets (resp. ...) on X. Then the morphisms*

$$H^q(X_{\mathrm{et}}, F) \longrightarrow H^q(X_{\mathrm{cl}}, F)$$

are bijective for the values of q under consideration.

Proof. Since it is enough to verify the theorem fiber by fiber, 4.3 reduces us to the case in which X is proper and has dimension ≤ 1. The case of dimension 0 is trivial, and it follows that the theorem is true for a finite morphism. Since f is proper, the cohomology of X_{et} and of X_{cl} commutes with filtered direct limits (VII 3.3 and [2], 4.12.1), and we are therefore reduced to the case in which F is constructible (IX 2.9 (iii)).

We apply IX 2.14. For a sheaf of sets, the exact sequence (4.3.1) and IX 2.14 show that we may take $F = \pi_* C$, where $\pi : X' \to X$ is finite, X' is normal and hence regular, 239
and C is a constant sheaf on X'. For a torsion abelian sheaf, we take a resolution of F by finite products of sheaves of the form $\pi_* C$ and apply the spectral sequence of a resolution, again reducing to the case $F = \pi_* C$. For a sheaf of ind-finite groups, recall that it is enough to prove effaceability of $(R^1 g_{\text{cl}\,*})\epsilon^*$ for the sheaf F, hence for a sheaf G into which F can be embedded; thus we are again reduced to the case $F = \pi_* C$. It is now clear from (VIII 5.5 and VIII 5.8) that

we may now replace X by X', which proves the lemma.

To treat this last case, it is plainly enough (in view of the cohomological-dimension results (IX 5.7) and [2], 4.14.1) to prove that for a complete regular curve one has

($q = 0$): X is connected and nonempty if and only if X_{cl} is connected and nonempty.

($q = 1$): The base-change morphism induces an equivalence between the category of étale coverings of X and the category of finite analytic étale coverings of X_{cl}.

($q = 2$): There is an isomorphism $\epsilon : H^2(X_{\text{et}}, \mu_n) \xrightarrow{\sim} H^2(X_{\text{cl}}, \mu_n)$.

The first two assertions follow immediately from GAGA. For the last, note that $H^2(X_{\text{et}}, \mathbf{G}_m) = 0$ (IX 4.5). Likewise, if $\mathcal{O}^*$ denotes the sheaf of invertible holomorphic functions on X_{cl}, then $H^2(X_{\text{cl}}, \mathcal{O}^*) = 0$. This follows from the exact sequence

$$0 \longrightarrow \mathbf{Z} \longrightarrow \mathcal{O} \xrightarrow{\exp} \mathcal{O}^* \longrightarrow 0$$

and from the fact that $H^2(X_{\text{cl}}, \mathcal{O}) = H^2(X_{\text{et}}, \mathcal{O}_X) = 0$ by GAGA. Consequently the 240
Kummer exact sequence IX 3.2 (respectively, the exact sequence on X_{cl}

$$0 \to \mu_n \to \mathcal{O}^* \xrightarrow{n} \mathcal{O}^* \to 0)$$

gives

$$H^1(X_{\text{et}}, \mathbf{G}_m)/n \xrightarrow{\sim} H^2(X_{\text{et}}, \mu_n)$$

$$(\textit{resp.} \quad H^1(X_{\text{cl}}, \mathcal{O}^*)/n \xrightarrow{\sim} H^2(X_{\text{cl}}, \mu_n)).$$

By GAGA, the canonical morphism

$$H^1(X_{\text{et}}, \mathbf{G}_m) = \operatorname{Pic} X \longrightarrow \operatorname{Pic} X_{\text{cl}} = H^1(X_{\text{cl}}, \mathcal{O}^*)$$

is bijective. This proves the result, and hence proves 4.1 when f is proper.

The case where F is constructible. The proof proceeds by induction on $\dim X$. Suppose the result known in dimensions $< n$, and prove it when $\dim X \leq n$.

Lemma 4.5. *We may suppose that $f : X \to S$ is a dense open immersion and that F is constant.*

Proof. As in the proof of 1.3, we may suppose that X and S are affine, and hence that f is quasi-projective. We then apply the argument of 4.4 to reduce to $F = \pi_* C$, where $\pi : X' \to X$ is finite and C is constant. Replacing X by X' therefore reduces us to the 241 case where F is constant. We may replace X by X_{red}. Let

$$\begin{tikzcd} X \arrow[rr, "h"] \arrow[dr, "f"'] & & \overline{X} \arrow[dl, "\overline{f}"] \\ & S & \end{tikzcd}$$

be a commutative diagram in which $\overline{X}$ is reduced, $\overline{f}$ is projective, and h is a dense open immersion. It is enough to verify the theorem for the two morphisms h and $\overline{f}$ (compare the proof of 4.3); hence it is enough to verify it for h, since $\overline{f}$ is proper. We are thus reduced to a dense open immersion.

Lemma 4.6. *We may further suppose that S is regular.*

Proof. Let $S' \xrightarrow{h} S$ be a "resolution of the singularities of S", that is, a surjective, proper, birational morphism with S' regular, and let

$$\begin{tikzcd} X' \arrow[r, "f'"] \arrow[d, "h'"] & S' \arrow[d, "h"] \\ X \arrow[r, "f"] & S \end{tikzcd}$$

be the Cartesian diagram obtained from h.

Let F be a constant sheaf of sets on X. We have $F \hookrightarrow G = h'_* h'^* F$, and it follows that, for $q = 0$, it is enough to verify 4.1 for G (compare (4.3.1)), that is, for a sheaf of the
242 form $h'_* F'$, where F' is constant on X'. We have a commutative diagram (in which the symbol "ét" is suppressed)

$$\begin{tikzcd} \varepsilon^* f_*(h'_* F') = \varepsilon^* h_* f'_* F' \arrow[r, "a"] \arrow[d, "d"] & h_{\mathrm{cl}*} \varepsilon^* f'_* F' \arrow[d, "b"] \\ f_{\mathrm{cl}*} \varepsilon^*(h'_* F') \arrow[r, "c"] & f_{\mathrm{cl}*} h'_{\mathrm{cl}*} \varepsilon^* F' = h_{\mathrm{cl}*} f'_{\mathrm{cl}*} \varepsilon^* F', \end{tikzcd}$$

and a and c are bijective by the result for a proper morphism. To prove that d is bijective, it is enough to prove that b is, that is, that

$$\epsilon^* f'_* F' \xrightarrow{\sim} f'_{\mathrm{cl}*} \epsilon^* F'.$$

We are therefore reduced to $S = S'$, with S regular.
(Remark: we have used, without mentioning it explicitly, the finiteness theorem for a proper morphism XIV 1.1.)

Let F be a constant torsion abelian sheaf on X. It is enough to treat $F = (\mathrm{Z}/n)_X$. The morphism h' is an isomorphism above a dense open subset $j : U \to X$; let $j' : U' \to X'$ be the open subset $h'^{-1}(U) \xrightarrow{\sim} U$, and put $Y = X - U$. Since U is dense, $\dim Y < n$, so the theorem holds for $(f, (\mathrm{Z}/n)_Y)$. The exact sequence

$$0 \longrightarrow j_!(\mathrm{Z}/n)_U \longrightarrow (\mathrm{Z}/n)_X \longrightarrow (\mathrm{Z}/n)_Y \longrightarrow 0$$

and the five lemma reduce us to proving the theorem for $j_!(\mathbf{Z}/n)_U$ and f, and hence, by 4.2, to proving it for $(hf', j'_!(\mathbf{Z}/n)_{U'})$. We have an exact sequence

$$0 \longrightarrow j'_!(\mathbf{Z}/n)_{U'} \longrightarrow (\mathbf{Z}/n)_{X'} \longrightarrow (\mathbf{Z}/n)_{Y'} \longrightarrow 0,$$

and the theorem holds for $(hf', (\mathbf{Z}/n)_{Y'})$ by the induction hypothesis. It is therefore 243
enough to prove it for $(hf', (\mathbf{Z}/n)_{X'})$. But the proper case gives the theorem for $(h, \cdot)$, and the Leray spectral sequence for the pair of morphisms h, f' shows that it is enough to prove it for $(f', \mathbf{Z}/n)$. This proves the lemma in the abelian case.

Finally, suppose that F is a constant sheaf of finite groups. Recall that it is enough to prove that the functor $(R^1 f_{\mathrm{cl}\,*})\epsilon^*(\cdot)$ is effaceable for F, hence for a sheaf G into which F embeds, and therefore for $G = h'_* h'^* F$, that is, for a sheaf of the form $h'_* F'$, where F' is constant on X'. We have

$$\epsilon^*(h'_* F') \xrightarrow{\sim} h'_{\mathrm{cl}\,*}(\epsilon^* F')$$

and

$$0 \longrightarrow (R^1 f_{\mathrm{cl}\,*})h'_{\mathrm{cl}\,*}\epsilon^*(\cdot) \longrightarrow (R^1(f_{\mathrm{cl}}h'_{\mathrm{cl}})_*)\epsilon^*(\cdot) = (R^1(h_{\mathrm{cl}}f'_{\mathrm{cl}})_*)\epsilon^*(\cdot).$$

Thus it is enough to prove that the last functor is effaceable for a constant sheaf. But the exact sequence

$$0 \longrightarrow (R^1 h_{\mathrm{cl}\,*})f'_{\mathrm{cl}\,*}\epsilon^*(\cdot) \longrightarrow (R^1(h_{\mathrm{cl}}f'_{\mathrm{cl}})_*)\epsilon^*(\cdot) \longrightarrow h_{\mathrm{cl}\,*}(R^1 f'_{\mathrm{cl}\,*})\epsilon^*(\cdot)$$

shows that the first term is effaceable, for example by the Godement resolution (XII 3.3), because h is proper. It is therefore enough to prove that the right-hand term is effaced by the Godement resolution, hence that $(R^1 f'_{\mathrm{cl}\,*})\epsilon^*(\cdot)$ is so effaced, and thus that the theorem holds for (f', F'), where F' is constant. This proves the lemma.

End of the proof. We now suppose that F is a constant sheaf and that $f : X \to S$ is 244
a dense open immersion, with S regular. The assertion for sheaves of sets now follows immediately from 3.2 and the analogous trivial lemma for the classical topology. Indeed, these show that $\epsilon^* f_* F$ and $f_{\mathrm{cl}\,*}\epsilon^* F$ are constant with the same value and hence are isomorphic under φ.

For $q > 0$, we first reduce further to the case $X = S - Y$ with Y regular, that is, to the case in which (Y, S) is a smooth $\operatorname{Spec}\mathbf{C}$-pair. Indeed, write $X = S - C_1$, where C_1 is given its induced reduced structure, and define recursively

Y_ν = the dense open subset of regular points of C_ν,

$C_{\nu+1} = C_\nu - Y_\nu$, with its induced reduced structure.

Since Y_ν is dense in C_ν, we have $\dim C_{\nu+1} < \dim C_\nu$, and hence a sequence

$$X = X_1 \subset X_2 = S - C_2 \subset X_3 = S - C_3 \subset \cdots \subset X_r = S,$$

where the inclusions $i_\nu : X_\nu \to X_{\nu+1}$ are open immersions and

$$X_\nu = X_{\nu+1} - Y_\nu, \qquad Y_\nu \text{ closed in } X_{\nu+1} \text{ and regular.}$$

The direct image $i_{\nu *}F$ of a constant sheaf on X is constant (3.2), and the $R^q i_{\nu *}F$ are constructible and concentrated on the subvarieties of Y of dimension at most $n-1$ (3.4, 3.6, 3.7) for the values of q under consideration. The induction hypothesis and the Leray spectral sequence (respectively, the exact sequence XII 3.2) therefore reduce us to the 245
case of i_ν, which proves the asserted reduction.

For a smooth pair (Y, S), however, the values of $(R^q f_*)F$ have been computed explicitly for the étale topology (3.4, 3.6, 3.7); an analogous and easy computation, which we leave to the reader, gives the corresponding results for the classical topology. Direct comparison proves the theorem for f.

5. The finiteness theorem for algebraic schemes in characteristic zero

THEOREM 5.1. *Let k be a field of characteristic zero, and let $f : X \to S$ be a morphism of finite type between schemes locally of finite type over k. Let F be a constructible sheaf of sets (resp. of finite groups, resp. a torsion abelian sheaf) on X. Then the $R^q f_* F$ are constructible for $q = 0$ (resp. for $q = 0, 1$, resp. for every q).*

The hypothesis that k have characteristic zero is imposed because we shall use the resolution-of-singularities theorem [1]. The proof is also valid in characteristic $p > 0$ if resolution of singularities is assumed, *provided that the orders of F are prime to p.* Thus in nonzero characteristic one obtains a result for $\dim X \leq 2$ by applying Abhyankar's
246 results [3]. In addition to the ever-present proper base-change theorem, another ingredient of the proof is the "absolute cohomological purity theorem" 3.9.

Recall that the theorem has already been proved for a proper morphism and arbitrary S (XIV 1.1). Moreover, in Exposé XIX we shall obtain (again in characteristic zero) the finiteness theorem under the much weaker hypotheses that S be an *excellent* scheme and f be of finite type, by proving for such schemes the absolute-purity theorem in the form indicated in 3.10.

In nonzero characteristic and dimension at least 3, for the time being we have only the following easy consequence of XI 3.3:

THEOREM 5.2. *Let X be a smooth algebraic scheme over* $\operatorname{Spec} k$*, where k is a separably closed field, and let F be a locally constant sheaf of finite abelian groups (resp. finite groups), of orders prime to the characteristic of k. Then $H^q(X, F)$ is a finite group (resp. a finite pointed set) for every q (resp. for $q = 0, 1$).*

Proof of 5.2. We may suppose that k is algebraically closed (VIII 1.1). We argue by induction on $n = \dim X$. By XI 3.3, there is a hypercovering of X by open subsets that are "good neighborhoods". The spectral sequence of a hypercovering immediately reduces us to the case in which X is a good neighborhood and hence admits an elementary fibration (XI 3.1)

$$\begin{tikzcd} X \arrow[rr, "j"] \arrow[dr, "f"'] & & \overline{X} \arrow[dl, "\overline{f}"] \\ & S & \end{tikzcd} .$$

247 By 3.2, 3.6, and 3.7, the sheaf $j_* F$ is locally constant on $\overline{X}$, $(R^1 j_*)F$ is locally constant on $Y = \overline{X} - X$, and $(R^q j_*)F = 0$ for $q > 1$. It follows from 2.1 that $(R^p \overline{f}_*)(R^q j_*)F$ is a locally constant sheaf on S for the values of q under consideration. The result follows from the induction hypothesis and the Leray spectral sequence (resp. the exact sequence XII 3.2).

The proof of 5.1 is very close to the proof of 4.1 in the constructible case. We merely indicate the main steps. Argue by induction on $\dim X = n$. First reduce to X and S affine, and hence f quasi-projective, by applying the method used in the proof of 1.3. Next apply IX 2.14 and VIII 5.5, VIII 5.8 to reduce to the case in which F is constant.

There is a commutative diagram

$$\begin{tikzcd} X \arrow[rr, "h"] \arrow[dr, "f"'] & & \overline{X} \arrow[dl, "\overline{f}"] \\ & S & \end{tikzcd},$$

in which h is a dense open immersion and $\overline{f}$ is proper, so the theorem holds for $\overline{f}$ (XIV 1.1). It is therefore enough to prove the theorem for (h, F), that is, for a dense open immersion and a constant sheaf. Let $h : S' \to S$ be a resolution of the singularities of 248
S, and let

$$\begin{tikzcd} X \arrow[d, "f"] & X' \arrow[l, "h'"'] \arrow[d, "f'"] \\ S & S' \arrow[l, "h"'] \end{tikzcd}$$

be the Cartesian diagram obtained from h. We reduce to proving the theorem for f', and hence to the case in which $S = S'$ is regular. The inductive method used at the end of the proof of 4.1 reduces us to the case in which $Y = S - X$ is nonsingular, that is, (Y, S) is a smooth pair. The proof is completed by applying 3.2, 3.4, 3.6, and 3.7.

Remark 5.3. The proof of 5.1 just outlined is also valid when one is given a left Noetherian ring A annihilated by an integer $n > 0$ (which must be assumed prime to the characteristic when the latter is not assumed to be zero; compare the remarks following 5.1) and considers constructible sheaves of left A-modules.

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[2] Godement, P. Théorie des faisceaux, Paris, 1958. **249**

[3] Abhyankar, S. Local uniformization on algebraic surfaces over ground fields of characteristic $p \neq 0$, Ann. of Math., vol. 63 (1956), p. 491.

[4] J.-P. Serre, Géométrie algébrique et géométrie analytique, Annales Inst. Fourier 1956, pp. 1–42 (cited GAGA).

EXPOSÉ XVII

Cohomology with proper supports

P. Deligne

Introduction

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In this exposé we develop the formalism of cohomology with proper support. The questions of variance have been treated with considerable care, whereas they were only made plausible in the oral seminar. This explains the length of the exposé, whose Sections 1 to 4 are devoted to categories and general topology. The reader is very strongly advised to read only Sections 5 and 6, where the geometric substance of the exposé is concentrated (§ 5: construction and variance of cohomology with proper supports, base-change, finiteness, finite tor-dimension, and comparison theorems; § 6: theory of the trace morphism for a quasi-finite flat morphism).

In Sections 1 and 2, we "recall" some results on derived categories. The § 1 n° 1 is devoted to the sign formalism. In the § 3, we treat the problem of gluing two variance formalisms. In the § 4, we introduce flat resolutions and study the special properties of Godement's flasque resolutions.

In this exposé, the higher direct-image functors with proper supports $R^q f_!$, and the functor $Rf_!$ from which they arise, are defined only when f is a compactifiable morphism (3.2.1). In the appendix, written by B. Saint-Donat, it is shown how to extend the definition to separated morphisms of finite type with quasi-compact quasi-separated target.[43] 253

The § 5 n° 5 (cohomology of a symmetric product) will not be used again in this seminar. It will be used in SGA 5 to refine the rationality theorem for L-functions.[44]

The § 6 n° 3 (the theory of the trace for *continuous* coefficients) will be used in Exposé XVIII only in the relatively easy case of smooth groups; a reader interested in the Poincaré duality theorem (global duality), and willing to accept a transcendental argument, may even dispense entirely with reading this § 6 n° 3, as well as most of the § 1 of XVIII.

0. Terminological preliminaries

0.1. The sign = placed between two groups of symbols denoting objects of a category will sometimes mean (by abuse of notation) that these objects are canonically

[42] The present exposé and the following one, written in 1968 and 1969, reproduce and supplement the oral lectures of A. Grothendieck (spring 1964). The author of this written version, who did not attend the oral seminar, drew in part on notes by A. Grothendieck.

[43] N.D.E.: The Nagata compactification theorem ensures that every separated morphism of finite type with quasi-compact quasi-separated target is compactifiable. See B. Conrad, "Deligne's notes on Nagata compactifications", *J. Ramanujan Math. Soc.* **22** (2007), 205–257, cited as [C] below, 4.1, as well as "Erratum for 'Deligne's notes on Nagata compactifications'", *J. Ramanujan Math. Soc.* **24** (2009), 427–428.

[44] N.D.E.: See SGA $4\frac{1}{2}$, L-functions modulo ℓ^n and modulo p.

isomorphic. The category and the canonical isomorphism should in principle have been defined beforehand. In a diagram, the sign $=$ will then denote the isomorphism itself.

0.2.0. Let $f : S \to S'$ be a morphism of sites (IV 4.9.3). If U and U' are objects of S and S', an f-morphism from U to U' will by definition be a morphism from U to f^*U'. For the étale sites of schemes, this recovers the usual notion.

254 0.2.1. If $\mathscr{F}$ and $\mathscr{F}'$ are sheaves on S and S', an f-morphism from $\mathscr{F}$ to $\mathscr{F}'$ will mean, equivalently,

(i) a morphism from $f^*\mathscr{F}'$ to $\mathscr{F}$;
(ii) a morphism from $\mathscr{F}'$ to $f_*\mathscr{F}$;
(iii) a function which, to every f-morphism φ from an object U of S to an object U' of S', assigns a function from $\mathscr{F}'(U')$ to $\mathscr{F}(U)$, compatibly with the composition of φ with an arrow of S or of S' [44].

It follows from (i), (ii), and (iii) that sheaves of sets form a fibered and cofibered category over the category of sites.

Likewise for sheaves of left modules on ringed sites. Likewise for étale sheaves on schemes; this is not immediately a special case of the preceding statement, since sites in fact form a 2-category and “the étale site of X” is only a pseudofunctor in X (VII 1.4).

The preceding definition makes sheaves of sets (resp. ...) on varying sites into a fibered category over the category of sites, whose fiber categories are the *opposite categories* of the usual categories of sheaves of sets (resp. ...).

0.3. Let I be a finite set, let ϵ be a function on I with values in $\{+1, -1\}$, let $(\mathscr{A}_i)_{i\in I}$ be a family of additive categories graded by translation functors T_i ([11] I 1.1.0), and let F be a multiadditive multifunctor from the categories $\mathscr{A}_i$ to an additive category $\mathscr{A}$ graded by the translation functor T. We suppose that F is covariant (resp. contravariant) in the
255 variables i for which $\epsilon_i = 1$ (resp. $\epsilon_i = -1$). Modifying and supplementing ([11] II 2.1.1), we shall say that F is a *graded functor* if we are given a family $(\varphi_i)_{i\in I}$ of isomorphisms of functors between $T\circ F$ and $F\circ T_i^{\varepsilon_i}$, such that the following diagrams are anticommutative for $i \neq j$ in I:

$$\begin{tikzcd}
F\circ(T_i^{\varepsilon_i}, T_j^{\varepsilon_j}) \arrow[r, "\varphi_j * T_i^{\varepsilon_i}"] \arrow[d, "\varphi_i * T_j^{\varepsilon_j}"'] \arrow[dr, phantom, "-"] & T\circ F\circ T_i^{\varepsilon_i} \arrow[d, "T*\varphi_i"] \\
T\circ F\circ T_j^{\varepsilon_j} \arrow[r, "T*\varphi_j"] & T\circ T\circ F.
\end{tikzcd}$$

We leave it to the reader to define the composite of two graded functors and verify that it is again a graded functor.

0.4. Let $\mathscr{A}$ be an additive category and let I be a finite set. For $i \in I$, denote by 1_i the i^{th} basis vector of $\mathbf{Z}^I$. An *I-fold complex* (resp. a *naive I-fold complex*) of $\mathscr{A}$ consists of

(a) a family $(K^k)_{k\in\mathbf{Z}^I}$ of objects of $\mathscr{A}$;

[44] These terminologies are not compatible with identifying the objects of S with the associated sheaves.

(b) for every $i \in I$ and every $k \in \mathbf{Z}^I$, an arrow $d_i^k : K^k \to K^{k+1_i}$, these arrows satisfying $d_i^{k+1_i} d_i^k = 0$ and, for $i \neq j$, $d_i^{k+1_j} d_j^k + d_j^{k+1_i} d_i^k = 0$ (resp. $d_i^{k+1_j} d_j^k = d_j^{k+1_i} d_i^k$).

0.5. A *pro-object* of a category $\mathscr{C}$ is a functor from $\mathscr{C}$ to (Ens) which is a filtered direct limit (over a small filtered category) of representable functors (cf. I 8.10). Every pro-object is a direct limit, over a small filtered ordered set, of representable functors. If X_i is an inverse system of objects of $\mathscr{C}$, indexed by a small filtered category, we denote by "$\varprojlim$" X_i the pro-object $\varinjlim h^{X_i}$. 256

0.6. The reader may dualize 0.5 to the case of ind-objects (Cf. I 8.2).

0.7. In accordance with the new terminology, what was formerly called a prescheme is now called a *scheme*, and what was formerly called a scheme is now called a *separated scheme*.

0.8. The category of *ringed schemes* is the category whose objects are schemes whose étale site is endowed with a sheaf of rings, an arrow from $(S, \mathscr{A})$ to $(T, \mathscr{B})$ being a pair consisting of a morphism of schemes f from S to T and a homomorphism φ of sheaves of rings from $f^*\mathscr{B}$ to $\mathscr{A}$. The morphism of schemes f is said to be *induced by* (f, φ).

0.9. If X is a scheme over Y, we denote by $(X/Y)^n$ the n-fold fiber product of X over Y.

0.10. Let S be a scheme. The *big étale site* (resp. the *big fppf site*, resp. the *big fpqc site*) of S is the site Sch$/S$, endowed with the étale topology (resp. the fppf topology, resp. the fpqc topology) (SGA 3 IV 6.3). Note carefully that this is not a $\mathscr{U}$-site ($\mathscr{U}$ being the fixed universe, and Sch$/S$ consisting of the schemes $\in \mathscr{U}$).

The *small fpqc site* (resp. the *small fppf site*) of S is the site of schemes flat over S (resp. flat and of finite presentation), endowed with the fpqc topology (resp. the fppf topology). When it is necessary to avoid ambiguity, the *small étale site* of S will mean the site S_{et} (VII 1.2). 257

0.11. In many cases, the author has allowed himself to speak of commutative diagrams of morphisms of sites where one should have spoken of essentially commutative diagrams, i.e. diagrams commutative up to isomorphism (cf. IV 3.2.2). The reader may verify that the arguments we give also apply to the general situation.

0.12. The terminology "*coherent scheme*" for "quasi-compact quasi-separated scheme" has been surreptitiously introduced here and there by A. Grothendieck.

0.13. A torsion sheaf F on a scheme S will be said to be *prime to the residual characteristics of* S if it is the direct limit of its subsheaves annihilated by integers n invertible on S.

1. Derived categories

1.1. Exact functors (the sign rules).

1.1.1. For the fundamental theorems concerning derived categories, I refer to Verdier [11] and [12]. In this subsection, particularly careful attention has been paid to questions of signs.

Recall that if $f : X \to Y$ is a morphism of complexes (in an additive category, which
258 will henceforth be understood), its *cone* $C(f)$ is defined by

(1.1.1.1) $$C(f)^n = X^{n+1} \oplus Y^n \quad d^n = -d_X^{n+1} + f^{n+1} + d_Y^n.$$

When $X = 0$ (resp. $Y = 0$), we have $C(f) = Y$ (resp. $C(f) = X[1]$), so that the commutative diagram

$$\begin{tikzcd}
0 \arrow[r, equal] \arrow[d] & 0 \arrow[r] \arrow[d] & X \arrow[r, equal] \arrow[d, "f"] & X \arrow[d] \\
X \arrow[r, "f"] & Y \arrow[r, equal] & Y \arrow[r] & 0
\end{tikzcd}$$

defines a "triangle"

$$X \xrightarrow{f} Y \to C(f) \xrightarrow{j} X[1].$$

In the category of complexes up to homotopy, a triangle is called *distinguished* if it is isomorphic to a triangle of this type, and *antidistinguished* if it becomes distinguished when the signs of its arrows are changed. One checks that the triangle defined by an exact sequence of complexes which is split in each degree ([11] p. 10, cf. the proof of 1.1.5.4) is antidistinguished and that, up to isomorphism, every antidistinguished triangle is obtained in this way.

**1.1.2.* Sign calculations are most easily carried out using the following "formal rules". Write an element of $C(f)^n$ in the form $1 \otimes x + y$ ($x \in X^{n+1}, y \in Y^n$), where 1 is a "cell" of dimension 1, and set $d(1) \otimes x = f(x)$. Thus

$$d(1 \otimes x + y) = d1 \otimes x - 1 \otimes dx + dy = -1 \otimes dx + f(x) + dy._*$$

259 Definition 1.1.3. (i) *A graded functor (0.3) from a triangulated category $\mathscr{A}$ to a triangulated category $\mathscr{B}$ is said to be exact if it sends distinguished triangles to distinguished triangles.*

(ii) *A contravariant graded functor from a triangulated category $\mathscr{A}$ to a triangulated category $\mathscr{B}$ is said to be* exact *if, for every distinguished triangle (X, Y, Z, u, v, w) of $\mathscr{A}$, the triangle $(F(Z), F(Y), F(X), F(v), F(u), TF(w))$ is distinguished (in this definition, FX is identified with $TFTX$ by means of the grading of F).*

(iii) *Let $(\mathscr{A}_i)_{i\in I}$ be a finite family of triangulated categories, let ϵ be a function from I to $\{+1, -1\}$, and let $\mathscr{A}$ be a triangulated category. A graded functor F from the product of the $\mathscr{A}_i$ to $\mathscr{A}$, covariant in those i for which $\epsilon(i) = 1$ and contravariant in those for which $\epsilon(i) = -1$, is said to be* exact *if, for every $i \in I$, the functors from $\mathscr{A}_i$ to $\mathscr{A}$ obtained from F by fixing all variables except the i-th are exact functors.*

A composite of exact functors is again an exact functor.

1.1.4. If $(K^{n_1\cdots n_p}, d_1, \ldots, d_p)$ is a multicomplex (0.4), its associated simple complex is defined by

(1.1.4.0) $$K^n = \prod_{\sum n_i = n} K^{(n_i)} \quad , \quad d = \sum d_i.$$

When, exceptionally, it is defined using a direct sum rather than a product, this fact will be stated explicitly.

Let I be a finite set. Following Cartan–Eilenberg and J. P. Serre, we shall explain
260 how every naive I-fold complex K has canonically associated with it an I-fold complex

(0.4).

For every total order $<$ on I, let $K(<)$ be the following I-fold complex:

$$\begin{cases} K(<)^{\underline{k}} = K^{\underline{k}} \\ d(<)_i^{\underline{k}} = (-1)^{\sum_{j<i} k_j} d_i. \end{cases} \tag{1.1.4.1}$$

If $<_1$ and $<_2$ are two total orders on I, the *canonical isomorphism* τ *between* $K(<_1)$ *and* $K(<_2)$ is, by definition, the one given by

$$\tau^{\underline{k}} = (-1)^{\epsilon(<_1,<_2)}, \quad \text{where} \quad \epsilon(<_1,<_2) = \sum_{\substack{i<_1 j \\ j<_2 i}} k_i k_j. \tag{1.1.4.2}$$

1.1.4.2.1. One checks that these canonical isomorphisms form a transitive system of isomorphisms among the $K(<)$, as $<$ ranges over the total orders on I; these $K(<)$ therefore determine an I-fold complex, unique up to a unique isomorphism, which in this subsection will be denoted by $K(*)$ and is called the *I-fold complex associated with the naive I-fold complex K*. Note that $K(*)^{\underline{k}}$ is *not canonically isomorphic to $K^{\underline{k}}$: only the choice of a total order on I makes it possible to identify these two objects.*

If K is a naive I-fold complex and $i \in I$, let $K[1_i]$ denote the following naive complex, whose differentials are denoted by $d_j[1_i]$:

$$\begin{cases} K[1_i]^{\underline{k}} = K^{\underline{k}+1_i} \\ d_j[1_i]^{\underline{k}} = d_j^{\underline{k}+1_i} \quad \text{if } i \neq j \\ d_i[1_i]^{\underline{k}} = -d_i^{\underline{k}+1_i}. \end{cases} \tag{1.1.4.3}$$

If K is an I-fold complex, set 261

$$\begin{cases} K[1_i]^{\underline{k}} = K^{\underline{k}+1_i} \\ d_j[1_i]^{\underline{k}} = -d_j[1_i]^{\underline{k}+1_i}. \end{cases} \tag{1.1.4.4}$$

If K is a naive I-fold complex and $<$ is a total order on I, so that $K(*)$ is given by (1.1.4.1), define an isomorphism σ between $K[1_i](*)$ and $K(*)[1_i]$ by the formula

$$\sigma^{\underline{k}} = (-1)^{\sum_{j<i} k_j} \operatorname{id}_{K^{\underline{k}+1_i}}. \tag{1.1.4.5}$$

. One checks that this isomorphism is independent of the choice of the total order $<$; it therefore defines an *isomorphism, called the canonical isomorphism, between* $K[1_i](*)$ and $K(*)[1_i]$.

. For any $i, j \in I$, if K is an I-fold complex or a naive I-fold complex, we identify $K[1_i][1_j]$ and $K[1_j][1_i]$ in the evident way.

For $i \neq j$ in I, the following diagram of canonical isomorphisms of type 1.1.4.6 or 1.1.4.7 is then *anticommutative*:

$$\begin{tikzcd} K[1_i][1_j](*) \arrow[r, equal] \arrow[d, leftrightarrow] & K[1_j][1_i](*) \arrow[r, leftrightarrow] & K[1_j](*)[1_i] \arrow[d, leftrightarrow] \\ K[1_i](*)[1_j] \arrow[r, leftrightarrow] & K(*)[1_i][1_j] \arrow[r, equal] & K(*)[1_j][1_i]. \end{tikzcd} \tag{1.1.4.7.1}$$

1.1.5. Let $F : \mathscr{A} \to \mathscr{B}$ be a contravariant additive functor between additive categories. Its *extension to complexes* $F : C(\mathscr{A}) \to C(\mathscr{B})$ is defined as follows. If K is a complex of $\mathscr{A}$, set 262

(1.1.5.1)
$$\begin{cases} F(K)^k = F(K^{-k}) \\ d^k_{F(K)} = (-1)^{k+1} F(d^{-k-1}). \end{cases}$$

The *canonical isomorphism* σ *between* $F(K)[1]$ *and* $F(K[-1])$ is defined by

(1.1.5.2)
$$\sigma^k = (-1)^{k+1} \operatorname{id}_{F(K^{-k-1})} .$$

If $F : \mathscr{A} \to \mathscr{B}$ and $G : \mathscr{B} \to \mathscr{C}$ are two contravariant additive functors, define the canonical isomorphism ρ between $G{\circ}F(K)$ *and* $G(F(K))$ by

(1.1.5.3)
$$\rho^k = (-1)^k \operatorname{id}_{G\circ F(K^k)} .$$

Lemma 1.1.5.4. (i) *For every contravariant additive functor* $F : \mathscr{A} \to \mathscr{B}$*, formulas* (1.1.5.1) *and* (1.1.5.2) *define an exact contravariant functor from* $K(\mathscr{A})$ *to* $K(\mathscr{B})$.

(ii) *For every pair* (G, F) *of composable contravariant additive functors, formula* (1.1.5.3) *defines an isomorphism of graded functors between the extension of* GF *to complexes and the composite of the extensions of* G *and* F *to complexes.*

263 Proof. A morphism of degree i, $f : K \to L$, may be regarded as any one of the following:

(a) a morphism from K to $L[i]$;
(b) a morphism from $K[-i]$ to L;
(c) a family $f^n : K^n \to L^{n+i}$ which commutes or anticommutes with the differentials according to the parity of i.

If $f : K \to L[i]$ is a morphism of degree i, then $F(f) : F(L)[-i] \to F(K)$ is again a morphism of degree i; for $i = 1$, we have

(1.1.5.5)
$$F(f)^n = (-1)^{n+1} F(f^{-n-1}).$$

If $0 \to K \xrightarrow{u} L \xrightarrow{v} M \to 0$ is an exact sequence of complexes split degree by degree, the choice of a splitting allows the differential of L to be written in the form

$$d_L = d_K + d_M + f,$$

where f is a morphism of degree one from M to K. Likewise, the dual exact sequence

$$0 \longrightarrow F(M) \longrightarrow F(L) \longrightarrow F(K) \longrightarrow 0$$

defines a morphism of degree one f^* from $F(K)$ to $F(M)$. We have $f^* = F(f)$. It follows that the image under F of a triangle (K, L, M, u, v, f) defined by a short exact sequence of complexes is again a triangle of the same type and, by 1.1.1, this proves (i).

Under the hypotheses of (ii), for f of degree one, we have $G(F(f))^n = (-1)^{n+1}(-1)^{-n} G{\circ} F(f)^n = -G \circ F(f)^n$; in particular, the grading isomorphism $G(F(K[1])) \simeq G(F(K))[1]$ is the isomorphism whose components are -1.

264 The diagram

$$\begin{tikzcd} G \circ F(K)[1] \arrow[r, leftrightarrow, "\rho[1]"] \arrow[d, equal] & G(F(K))[1] \arrow[d, leftrightarrow] \\ G \circ F(K[1]) \arrow[r, leftrightarrow, "\rho"] & G(F(K[1])) \end{tikzcd}$$

is therefore commutative, which proves (ii).

1.1.6. Let $s = +$ or $s = -$; let I be a finite set, let $\epsilon : I \to \{+, -\}$ be a function, let $(\mathscr{A}_i)_{i\in I}$ and $\mathscr{A}$ be additive categories, and let F be a multiadditive multifunctor from the categories $\mathscr{A}_i$ to $\mathscr{A}$, covariant in the arguments indexed by those i for which $\epsilon(i) = +$ and contravariant in the others. We shall define the extension $K^s(F)$ (or simply F) of F to complexes as an exact multifunctor from the categories $K^{s\epsilon(i)}(\mathscr{A}_i)$ ($s\epsilon(i) = \pm$, according to the sign rule) to $K^s(\mathscr{A})$, [44] having the same variance as F. We shall then explain in what sense this construction is compatible with the composition of functors. Other conventions for *degrees* are possible and will be used; that issue is independent of the questions of *sign* considered here.

. If F is contravariant in one variable, $K^s(F)$ is defined by the formulas of 1.1.5.

. If F is covariant, the functor $K^s(F)$ is defined, by passage to the quotient, from the composite of the evident functor from the product of the categories $C^s(\mathscr{A}_i)$ to the category of naive I-fold complexes, the functor (1.1.4) from the category of naive I-fold complexes to that of I-fold complexes (1.1.4.2.1), and the associated simple complex 265
functor (1.1.4.0):

$$K^s(F) = C^s(F)(*)_{s.a}.$$

Let $K_j \in K^s(\mathscr{A}_j)$ be complexes, and let $i \in I$; set $K_j[1_i] = K_j$ for $j \neq i$, and $K_i[1_i] = K_i[1]$. We then have the evident isomorphisms

$$C^s(F)(K_j[1_i]) \simeq C^s(F)(K_j)[1_i],$$

and

$$C^s(F)(K_j)(*)[1_i]_{s.a} \simeq C^s(F)(K_j)(*)_{s.a}[1],$$

whence, via (1.1.4.5), a *canonical isomorphism* σ between $K^s(F)(K_j[1_i])$ and $K^s(F)(K_j)[1]$. It follows from 1.1.4.7 that these isomorphisms define a grading of $K^s(F)$.

. In the general case, F is the composite of the canonical contravariant functors $\mathscr{A}_i \to \mathscr{A}_i^\circ$ (for $\epsilon(i) = -$) with a covariant functor F^+; define the functor $K^s(F)$ by composition using 1.1.6.1 and 1.1.6.2.

Proposition 1.1.7. *Under the preceding hypotheses, the functor $K^s(F)$ is an exact multiadditive functor.*

It suffices to verify 1.1.7 under the hypotheses of 1.1.6.1 (where the proposition is 1.1.5.4 (i)) or under those of 1.1.6.2. To verify that the functors deduced from $K^s(F)$ by fixing all variables except the i-th are exact, one may describe $K^s(F)$ explicitly in terms of an order on I for which i is the least element; a distinguished triangle of the standard form (1.1.1.1) is then sent to a distinguished triangle of the same form. 266

1.1.8. Let I and J be two finite sets, let $\Psi : I \to J$ be a surjective map, and let $\epsilon_I : I \to \{+, -\}$ and $\epsilon_J : J \to \{+, -\}$ be functions, and let us be given multiadditive multifunctors, covariant in the arguments for which $\epsilon = +$ and contravariant in the others:

$$G_j : \prod_{\Psi(i)=j} \mathscr{A}_i \longrightarrow \mathscr{B}_j \qquad (j \in J)$$

$$F : \prod_{j\in J} \mathscr{B}_j \longrightarrow \mathscr{C}.$$

[44] The construction also provides an extension of F to a multifunctor from the categories $C^{s\epsilon(i)}(\mathscr{A}_i)$ to $C^s(\mathscr{A})$.

Choose total orders on I and J such that Ψ is an increasing map. We then define the *canonical isomorphism ρ between the extension to complexes of $F \circ (G_j)$ and the composite of the extensions to complexes of F and of the G_j* by the formula

$$\rho^{\underline{k}} = (-1)^{A(\underline{k})} \,:\, \text{automorphism of } F \circ (G_j)(K_i^{k_i \epsilon_I(i) \epsilon_J(\Psi(i))}). \tag{1.1.8.1}$$

where

$$A(\underline{k}) = \sum_{\epsilon_I(i)=\epsilon_J(\Psi(i))=-} k_i + \sum_{\epsilon_J(j)=-} \sum_{\substack{\Psi(a)=\Psi(b)=j \\ a<b}} k_a k_b.$$

This formula generalizes (1.1.5.3); one verifies that ρ does not depend on the orders chosen on I and J, and is an isomorphism of graded functors (cf. 1.1.5.4 (ii)).

These isomorphisms ρ satisfy the evident cocycle condition for a triple composite.

267 1.1.9. Let I be a finite set, let i and j be two distinct elements of I, let $I' = I \backslash \{i, j\}$, let $(\mathscr{A}_i)_{i \in I}$ and $\mathscr{A}$ be additive categories, and let $F : (\mathscr{A}_i)_{i \in I} \to \mathscr{A}$ and $G : (\mathscr{A}_i)_{i \in I'} \to \mathscr{A}$ be multiadditive multifunctors. Suppose that $\mathscr{A}_i = \mathscr{A}_j$ and that F is contravariant in the i-th variable and covariant in the j-th. For X, $Y \in \operatorname{Ob} \mathscr{A}_i$, denote by $F_{X,Y}$ the multifunctor from $(\mathscr{A}_i)_{i \in I'}$ to $\mathscr{A}$ obtained by fixing the i-th and j-th arguments:

$$F_{X,Y}(\dots) = F(\dots, X, \dots, Y, \dots).$$

A *contraction morphism* $c : F \to G$ consists of, for every $X \in \mathscr{A}_i$, a morphism of functors

$$c_X : F_{X,X} \longrightarrow G,$$

such that, for every arrow $f : X \to Y$ of $\mathscr{A}_i$, the diagram

$$\begin{tikzcd}
F_{Y,X} \arrow[r] \arrow[d] & F_{X,X} \arrow[d, "c_X"] \\
F_{Y,Y} \arrow[r, "c_Y"'] & G
\end{tikzcd}$$

is commutative.

The conventions of 1.1.6 are motivated by the

Lemma 1.1.9.1. *Let $c : F \to G$ be a contraction morphism, and let $(K_\ell)_{\ell \in I}$ be complexes in the $\mathscr{A}_\ell$, with $K_i = K_j$. Let $<$ be a total order on I for which i is the predecessor of j. With the notation of 1.1.6, for $k \in \mathbf{Z}$, let*

$$c^k : K^s(F)(K_\ell)^k \longrightarrow K^s(G)(K_\ell)^k$$

268 *be the morphism whose restriction to $F(\dots K_i^{k_i} K_j^{k_j} \dots)$ is 0 if $k_i \neq k_j$, and is $c_{K_i^{k_i}}$ if $k_i = k_j$. Then $(c^k)_{k \in \mathbf{Z}}$ is a morphism of complexes.*

We leave to the reader the verification of this lemma, as well as the fact that the *contraction morphism* (c^k) does not depend on the order chosen on I.

The standard example of a contraction morphism is the composition law

$$\operatorname{Hom}(Y, Z) \otimes \operatorname{Hom}(X, Y) \longrightarrow \operatorname{Hom}(X, Z).$$

Examples 1.1.10. (i) Applying the definitions of 1.1.6 to the *tensor-product* functor gives the usual notion of the tensor product of two complexes, with differential

$$d(x \otimes y) = dx \otimes y + (-1)^{\deg x} x \otimes dy.$$

(ii) Apply the definitions of 1.1.6 to the functor $\mathrm{Hom}(X, Y)$, *regarding* Y *as the first variable and* X *as the second variable*, a convention that we shall always follow. This convention identifies, for two complexes X, Y, the k-th component of the complex $\mathrm{Hom}(X, Y)$ defined in 1.1.6 with $\prod_n \mathrm{Hom}(X^n, Y^{n+k})$, and gives the usual differential: for $f \in \mathrm{Hom}(X^n, Y^{n+k})$, one has

$$df = d \circ f - (-1)^k f \circ d.$$

(iii) The preceding conventions have the virtue that the following systems of arrows are morphisms of complexes, without any signs having to be inserted (1.1.9): 269

$$\mathrm{Hom}^{\cdot}(L, M) \otimes \mathrm{Hom}^{\cdot}(K, L) \longrightarrow \mathrm{Hom}^{\cdot}(K, M) : f \otimes g \mapsto f \circ g$$
$$\mathrm{Hom}^{\cdot}(L, M) \otimes L \longrightarrow M : f \otimes x \mapsto f(x).$$

**1.1.10.1.* The convention (1.1.4.5)–(1.1.6.2) is justified as follows for $F = \otimes$: an element of $((K \otimes L)[1])^n$ is written in the form (*1.1.2) (recalling that $(K \otimes L)[1] = C(K \otimes L \to 0)$ (1.1.1.1)) $\sum 1 \otimes (x^p \otimes y^q)$, an element of $(K[1] \otimes L)^n$ in the form $\sum (1 \otimes x^p) \otimes y^q$, and an element of $(K \otimes L[1])^n$ in the form $\sum x^p \otimes (1 \otimes y^q)$; the sign appears when x^p and 1 are interchanged. For $F =$ dual, the convention (1.1.5.2) is justified as follows: an element of $K^*[1]$ is written $1 \otimes \omega$; its value on an element $1' \otimes x$ of $K[-1]$ ($1'$ a cell of dimension -1) is $< 1 \otimes \omega, 1' \otimes x >= (-1)^{\deg \omega} < 1, 1' >< \omega, x >= (-1)^{\deg \omega} < \omega, x >$; this is the convention adopted.$_*$

REMARK 1.1.11. “Recall” that a *preadditive category* is a category whose sets of arrows $\mathrm{Hom}(X, Y)$ are endowed with abelian-group structures, the composition of morphisms being biadditive. An *additive functor* F between preadditive categories is a functor satisfying the formula

$$F(f + g) = F(f) + F(g)$$

for $f, g \in \mathrm{Hom}(X, Y)$. The additive functors from a preadditive category $\mathscr{A}$ to another $\mathscr{B}$ form a category denoted by $\mathbf{Hom}_{\mathrm{add}}(\mathscr{A}, \mathscr{B})$.

Let (D) be the unique preadditive category whose set of objects is indexed by $\mathbf{Z}$, $(X^n)_{n \in \mathbf{Z}}$, and for which 270

$$(1.1.11.1) \qquad \begin{cases} \mathrm{Hom}(X^n, X^n) = \mathbf{Z}, \text{ generated by the identity} \\ \mathrm{Hom}(X^n, X^{n+1}) = \mathbf{Z}, \text{ generated by an arrow denoted } d_n \\ \mathrm{Hom}(X^i, X^j) = 0 \text{ if } j - i \neq 0 \text{ or } 1. \end{cases}$$

Let D be the complex of (D) defined by $D^n = X^n$ and $d^n = d_n$.

For every preadditive category $\mathscr{A}$, the functor $G \mapsto G(D)$ from $\mathbf{Hom}_{\mathrm{add}}((D), \mathscr{A})$ to $C(\mathscr{A})$ is an isomorphism.

The construction of 1.1.6 extends to the case of a multiadditive functor from preadditive categories $\mathscr{A}_i$ to an additive category $\mathscr{A}$.

Let

$$F^{\cdot} = (F^n, d^n)_{n \in \mathbf{Z}}$$

be a complex of multiadditive functors from additive categories $\mathscr{A}_i$ to an additive category $\mathscr{A}$. One may regard $F^{\cdot}$ as a functor from (D) to a category of multifunctors, or equivalently as a multiadditive functor

$$F_1 : \prod \mathscr{A}_i \times (D) \longrightarrow \mathscr{A}.$$

The *extension of* $F^{\cdot}$ *to complexes* is defined by the formula

$$(1.1.11.2) \qquad F^{\cdot}(K_1 \ldots K_n) = F_1(K_1 \ldots K_n, D).$$

1.1.12. Let $\mathscr{A}$ and $\mathscr{B}$ be two additive categories and let

$$F^{\bullet} = (F^n, d^n)_{n\in\mathbf{Z}}$$

271 be a complex of additive functors from $\mathscr{A}$ to $\mathscr{B}$. Suppose that the functors F^i admit right adjoints F_i; let ${}^t d^i : F_{i+1} \to F_i$ be the transpose of d^i, and let $F_{\bullet}$ be the complex of additive functors from $\mathscr{B}$ to $\mathscr{A}$ whose components are the F_i and whose differential operator is

$$(d_{\bullet})^i = (-1)^{i+1}\,{}^t d^{-i-1} : F_{-i} \longrightarrow F_{-i-1}.$$

Let ^{t} be the *contravariant* functor from (D) to (D) given by

$${}^t X^n = X^{-n} \quad , \quad {}^t d^n = d^{-n-1}.$$

The following diagram is commutative up to canonical isomorphism:

$$\begin{tikzcd}
\mathscr{A}\times(D)\times\mathscr{B} \arrow[r, "{(F^n(A),\mathrm{Id})}"] \arrow[d, "{(\mathrm{Id},{}^t,\mathrm{Id})}"] & \mathscr{B}\times\mathscr{B} \arrow[r, "\mathrm{Hom}_{\mathscr{B}}"] & (\mathrm{Ab}) \arrow[d, equal] \\
\mathscr{A}\times(D)\times\mathscr{B} \arrow[r, "{(\mathrm{Id},(F_{\bullet})^n(B))}"] & \mathscr{A}\times\mathscr{A} \arrow[r, "\mathrm{Hom}_{\mathscr{A}}"] & (\mathrm{Ab}).
\end{tikzcd} \tag{1.1.12.1}$$

This expresses the adjunction formula

$$\mathrm{Hom}(F^n(A), B) \xleftrightarrow{\sim} \mathrm{Hom}(A, F_n(B)).$$

The general conventions of 1.1.3 therefore give an isomorphism

$$\mathrm{Hom}^{\bullet}(F^{\bullet}(K), L) \xleftrightarrow{\sim} \mathrm{Hom}^{\bullet}(K, F_{\bullet}(L)).$$

Definition 1.1.13. *If K is a complex in an abelian category $\mathscr{A}$, the* right truncation (resp. left truncation) *of K in dimension n, denoted by* $\tau_{\leq n}(K)$ (resp. $\tau_{\geq n}(K)$), *means the following complexes:*

$$\begin{aligned}
\tau_{\leq n}(K) &: \dots K^{n-2} \longrightarrow K^{n-1} \longrightarrow \mathrm{Ker}(d^n) \longrightarrow 0 \\
\tau_{\geq n}(K) &: 0 \longrightarrow \mathrm{coker}(d^{n-1}) \longrightarrow K^{n+1} \longrightarrow K^{n+2} \dots .
\end{aligned}$$

272 Proposition 1.1.14. (i) *$H^i(\tau_{\leq n}(K)) = 0$ for $i > n$, and $H^i(\tau_{\geq n}(K)) = 0$ for $i < n$.*

(ii) *The canonical morphism $\tau_{\leq n}(K) \to K$ (resp. $K \to \tau_{\geq n}K$) induces isomorphisms on the H^i for $i \leq n$ (resp. $i \geq n$).*

The functor $\tau_{\leq n}$ (resp. $\tau_{\geq n}$) sends homotopic morphisms to homotopic morphisms. It therefore follows from 1.1.14 that the functor of 1.1.13 remains meaningful in the derived category.

1.1.15. If K is a double complex, with first (resp. second) differential d' (resp. d''), denote by $\tau''_{\leq n}(K)$ the double subcomplex of K such that

$$\tau''_{\leq n}(K)^{pq} = \begin{cases} K^{pq} & \text{if } q < n \\ \mathrm{Ker}(d'') & \text{if } q = n \\ 0 & \text{if } q > n \end{cases}$$

and define $\tau''_{\geq n}(K)$, $\tau'_{\leq n}(K)$, and $\tau'_{\geq n}(K)$ similarly.

1.1.16. We shall also have occasion to consider the "stupid truncations" of a complex K, defined by

$$\sigma_{\leq n}(K)^i = \begin{cases} K^i & \text{if } i \leq n \\ 0 & \text{if } i > n \end{cases} \qquad \sigma_{\geq n}(K)^i = \begin{cases} K^i & \text{if } i \geq n \\ 0 & \text{if } i < n. \end{cases}$$

1.2. Derived functors. Let $\mathscr{A}$ be a triangulated category and S a saturated multi- 273
plicative system ([11] I 2.1.2). If $A \in \operatorname{Ob}\mathscr{A}$, denote by A^+ and A^- the ind- and pro-objects ((0.5) and (0.6))

$$A^+ = \text{“}\varinjlim_{A \xrightarrow{s} A'}\text{”}\, A' \qquad A^- = \text{“}\varprojlim_{A' \xrightarrow{s} A}\text{”}\, A',$$

where s ranges over the filtered category of arrows in S with source or target A, respectively.

The functors $A \mapsto A^+$ and $A \mapsto A^-$ make the elements of S invertible and define fully faithful functors from $\mathscr{A}(S^{-1})$ to the categories of Ind- and pro-objects of $\mathscr{A}$, respectively.

Definition 1.2.1. *Let F be an exact covariant multifunctor from triangulated categories $(\mathscr{A}_i)_{0<i\leq n}$ to the triangulated category $\mathscr{A}$, and let S_i and S be saturated multiplicative systems in the categories $\mathscr{A}_i$ and $\mathscr{A}$.*

(i) *The right derived functor RF of F (relative to the S_i and S) is the functor from the categories $\mathscr{A}_i(S_i^{-1})$ to the category of ind-objects of $\mathscr{A}(S^{-1})$ that makes the following diagram commutative.*

$$\begin{tikzcd}
(\mathscr{A}_i) \arrow[r, "(A_i)\mapsto F(A_i^+)"] \arrow[d] & \operatorname{Ind}\mathscr{A} \arrow[d] \\
\mathscr{A}_i(S_i^{-1}) \arrow[r, "RF"] & \operatorname{Ind}\mathscr{A}(S^{-1}).
\end{tikzcd}$$

(ii) *We say that RF is defined at (A_i) if the ind-object $RF(A_i)$ of $\mathscr{A}(S^{-1})$ is "essen-* 274
tially constant," i.e. comes from an object of $\mathscr{A}(S^{-1})$; this object is called the value *of RF at (A_i).*

(iii) *We say that F is* right derivable *if RF is defined everywhere; in that case we shall again denote by RF the resulting functor from the $\mathscr{A}_i(S_i^{-1})$ to $\mathscr{A}(S^{-1})$.*

(iv) *A family (A_i) of objects of the $\mathscr{A}_i$ is said to be* adapted to *RF if the canonical morphism from $F(A_i)$ to $RF(A_i)$ is an isomorphism.*

We leave to the reader the task of defining by duality the left derived functor LF of F (with values in $\operatorname{Pro}\mathscr{A}(S^{-1})$) and of extending the preceding definitions to cases in which F is covariant in some arguments and contravariant in others.

When RF is defined everywhere, the definition given here agrees with Verdier's ([11] p. 39), as is readily verified.

Proposition 1.2.2. *Let (A_1', A_1, A_1'') be a distinguished triangle in $\mathscr{A}_1$, and let $A_i \in \operatorname{Ob}\mathscr{A}_i$ $(i \neq 1)$.*

(i) *If $(A_1', (A_i))$ and $(A_1'', (A_i))$ are adapted to RF, then the family $(A_1, (A_i))$ is also adapted to RF.*

(ii) *If RF is defined at $(A_1', (A_i))$ and at $(A_1'', (A_i))$, it is also defined at $(A_1, (A_i))$, and the triangle $(RF(A_1', (A_i)), RF(A_1, (A_i)), RF(A_1'', (A_i)))$ is distinguished in $\mathscr{A}(S^{-1})$.*

275 Let us prove (ii) $\Rightarrow$ (i). Since F is exact, the arrow

$$(F(A_1', (A_i)), F(A_1, (A_i)), F(A_1'', (A_i))) \to (RF(A_1', (A_i)), RF(A_1, (A_i)), RF(A_1'', (A_i)))$$

is a morphism of distinguished triangles. By hypothesis this arrow is an isomorphism at two vertices; it is therefore an isomorphism at the third, as desired.

Lemma 1.2.2.1. *If $\varDelta = (X, Y, Z)$ is a distinguished triangle in a triangulated category $\mathscr{A}$ and S is a saturated multiplicative system in $\mathscr{A}$, then the triangle $\varDelta^+ = (X^+, Y^+, Z^+)$ in* $\operatorname{Ind}\mathscr{A}$ *is a limit of distinguished triangles in $\mathscr{A}$.*

Let $\mathscr{L}$ denote the category of morphisms of distinguished triangles with source $\varDelta$ whose arrows belong to S.

(a) If $\varDelta = (X, Y, Z)$ and $\varDelta' = (X', Y', Z')$ are two distinguished triangles in $\mathscr{A}$, and if $f : \varDelta \to \varDelta'$ is an $\mathscr{A}(S^{-1})$-morphism of triangles, there exist in $\mathscr{A}$ morphisms of triangles $s : \varDelta' \to \varDelta''$ and $f_1 : \varDelta \to \varDelta''$ such that the components of s belong to S and the diagram

(1)
$$\begin{tikzcd} \varDelta \arrow[d, "f"'] \arrow[dr, "f_1"] & \\ \varDelta' \arrow[r, "s"] & \varDelta'' \end{tikzcd}$$

is commutative in $\mathscr{A}(S^{-1})$.

There is a commutative diagram

(2)
$$\begin{tikzcd} X \arrow[d, "f"'] \arrow[dr, "f_1"] & \\ X' \arrow[r, "s_X"] & X_0'' \end{tikzcd} \qquad (s_X \in S, f_1 \in \operatorname{Fl}\mathscr{A})$$

276 and a morphism $\varDelta' \xrightarrow{s} \varDelta_0''$ in S, having s_X as a component. Argue similarly at the other vertices: one obtains a diagram of type (1), in which, however, f_1 is not yet a morphism of triangles *in $\mathscr{A}$*, since the diagrams

(3)
$$\begin{tikzcd} Z \arrow[r, "f_1"] \arrow[d, "w"'] & Z_0'' \arrow[d, "w''"] \\ X \arrow[r, "f_1"] & X_0'' \end{tikzcd}$$

may, for example, fail to be commutative *in $\mathscr{A}$*, being commutative only in $\mathscr{A}(S^{-1})$. There exists $t_X : X_0'' \to X_1''$ in S such that $t_X w'' f_1 = t_X f_1 w$ and $t : \varDelta_0'' \to \varDelta_1''$ having t_X as a component. Replace $\varDelta_0''$ by $\varDelta_1''$, s by ts, and f_1 by tf_1; in the resulting diagram of type (1), (3) is now commutative; proceeding similarly at the other vertices, one obtains (1). We leave it to the reader to verify similarly:

(b) If $f, g : \varDelta \rightrightarrows \varDelta'$ are two morphisms of distinguished triangles in $\mathscr{A}$, equal in $\mathscr{A}(S^{-1})$, then there exists a morphism of distinguished triangles whose arrows are in S, $s : \varDelta' \to \varDelta''$, such that $sf = sg$.

(c) The category $\mathscr{L}$ is filtering on the right.

(d) The proof of the lemma is completed by noting that

$$\varDelta^+ = \text{“}\varinjlim_{s \in \mathscr{L}}\text{”}\,(\text{target of } s).$$

Let us prove 1.2.2 (ii). Let X' and X'' be objects of $\mathscr{A}(S^{-1})$ representing $RF(A'_1, (A_i))$ 277
and $RF(A''_1, (A_i))$, and complete the morphism of degree 1 from X'' to X' to a distinguished triangle (X', X, X''). Replacing the A_i, A'_1, A_1, and A''_1 by the target of an arrow in S_i or S_1 with source A_i, A'_1, A_1, or A''_1, we reduce to the case where there is a commutative diagram

$$\begin{tikzcd} F(A''_1, (A_i)) \arrow[d] & X'' \arrow[l, "a''"'] \arrow[d] \\ F(A'_1, (A_i)) & X', \arrow[l, "a'"'] \end{tikzcd}$$

where a'' and a' are sections of arrows from $F(A''_1, (A_i))$ to X'' and from $F(A'_1, (A_i))$ to X'. By axiom $TR\,3$ of [11], this diagram can be completed to a morphism of triangles

$$\begin{aligned}(X'XX'') &\longrightarrow (F(A'_1, (A_i)), F(A_1, (A_i)), F(A''_1, (A_i))) \\ &\longrightarrow (RF(A'_1, (A_i)), RF(A_1, (A_i)), RF(A''_1, (A_i))).\end{aligned}$$

The composite arrow is a morphism of triangles whose source is a distinguished triangle and whose target is a limit (in $\operatorname{Ind}\mathscr{A}(S^{-1})$) of distinguished triangles (1.2.2.1). For any Y in $\mathscr{A}(S^{-1})$, the following diagram is commutative and its rows are exact:

$$\begin{tikzcd} \mathrm{Hom}(Y, X''[-1]) \arrow[r] \arrow[d] & \mathrm{Hom}(Y, X') \arrow[r] \arrow[d] & \mathrm{Hom}(Y, X) \arrow[r] \arrow[d] & \mathrm{Hom}(Y, X'') \arrow[r] \arrow[d] & \mathrm{Hom}(Y, X'[1]) \arrow[d] \\ \mathrm{Hom}(Y, RF(A''_1, A_i)[-1]) \arrow[r] & \mathrm{Hom}(Y, RF(A'_1, A_i)) \arrow[r] & \mathrm{Hom}(Y, RF(A_1, A_i)) \arrow[r] & \mathrm{Hom}(Y, RF(A''_1, A_i)) \arrow[r] & \mathrm{Hom}(Y, RF(A'_1, A_i)[1]) \end{tikzcd}$$

By assumption, all arrows other than the middle arrow are isomorphisms. The 5- 278
lemma shows that the middle arrow is also an isomorphism, for every Y, and hence that the distinguished triangle (X', X, X'') is isomorphic to the triangle $(RF(A'_1, (A_i)), RF(A_1, (A_i)), RF(A''_1, (A_i)))$. This completes the proof.

If $\mathscr{A}$ and $\mathscr{B}$ are two abelian categories and F is an additive covariant functor from $\mathscr{A}$ to $\mathscr{B}$, F defines exact functors from $K(\mathscr{A})$ to $K(\mathscr{B})$, from $K^+(\mathscr{A})$ to $K^+(\mathscr{B})$, and from $K^-(\mathscr{A})$ to $K^-(\mathscr{B})$ (1.1.7).

Definition 1.2.3. (i) *The* right derived functors of F *are the functors from* $D^+(\mathscr{A})$ *to* Ind $D^+(\mathscr{B})$ *and from* $D(\mathscr{A})$ *to* Ind $D(\mathscr{B})$ *right derived from the extensions of* F *to complexes* $K^+(\mathscr{A}) \to K^+(\mathscr{B})$ *and* $K(\mathscr{A}) \to K(\mathscr{B})$.

(ii) *An object A of $\mathscr{A}$ is said to be* right acyclic *for F, or, by abuse of language,* acyclic for RF, *if the complex concentrated at A in degree* 0 *is adapted to* RF.

We shall be interested mainly in the case where RF is defined everywhere; it may then be regarded as a functor from $D^+(\mathscr{A})$ to $D^+(\mathscr{B})$, or from $D(\mathscr{A})$ to $D(\mathscr{B})$, as the case may be.

Proposition 1.2.4. *The following diagram is commutative.*

$$\begin{tikzcd} D^+(\mathscr{A}) \arrow[r] \arrow[d, "RF"] & D(\mathscr{A}) \arrow[d, "RF"] \\ \mathrm{Ind}\ D^+(\mathscr{B}) \arrow[r] & \mathrm{Ind}\ D(\mathscr{B}). \end{tikzcd}$$

If K is a complex that is zero in degrees $< n$ and $s : K \to L$ is a quasi-isomorphism, 279
the composite morphism $K \xrightarrow{s} L \to \sigma_{\geq n}(L)$ (1.1.6) is again a quasi-isomorphism. The quasi-isomorphisms from K to a complex bounded below therefore form a cofinal (full)

subcategory of the category of all quasi-isomorphisms with source K. The proposition follows formally.

In particular, if the functor RF is defined everywhere on $D(\mathscr{A})$, it is defined everywhere on $D^+(\mathscr{A})$, and the derived functor $RF : D^+(\mathscr{A}) \to D^+(\mathscr{B})$ is the restriction of the derived functor $RF : D(\mathscr{A}) \to D(\mathscr{B})$.

1.2.5. We leave it to the reader to:

(i) Generalize Definition 1.2.3 to the case of a complex of multifunctors, covariant in some variables and contravariant in others;
(ii) Dualize Definition 1.2.5 (i) to define the left derived functors;
(iii) Extend Proposition 1.2.4 to the case of a bounded complex of multifunctors;
(iv) Dualize Proposition 1.2.5 (iii) by considering left derived functors and the inclusion of $D^-(\mathscr{B})$ into $D(\mathscr{B})$.

Proposition 1.2.6. *Let $\mathscr{A}$ and $\mathscr{B}$ be two abelian categories, let $\mathscr{P}$ be a subset of* $\mathrm{Ob}(\mathscr{A})$, *and let F be an additive functor from $\mathscr{A}$ to $\mathscr{B}$. Assume that*

(a) *every object of $\mathscr{A}$ is a quotient of an element of $\mathscr{P}$;*
(b) *for every short exact sequence*

$$0 \longrightarrow P \longrightarrow Q \longrightarrow R \longrightarrow 0,$$

where Q and R belong to $\mathscr{P}$, P also belongs to $\mathscr{P}$, and the 0*-sequence* $0 \to FP \to$
280 $FQ \to FR \to 0$ *is exact. Then every object of $\mathscr{P}$ is acyclic for LF.*

If $A \in \mathrm{Ob}\,\mathscr{A}$, the left resolutions of A by objects of $\mathscr{P}$ form a cofinal system in the filtering category of left resolutions of A. It therefore suffices to prove that if a complex

$$\cdots \longrightarrow A_2 \longrightarrow A_1 \longrightarrow A_0 \longrightarrow A \longrightarrow 0$$

is acyclic and has its objects in $\mathscr{P}$, its image is acyclic. Splitting it into short exact sequences

$$(1.2.6.i) \qquad 0 \longrightarrow \mathrm{Im}(d_{i-1}) \longrightarrow A_i \longrightarrow \mathrm{Im}(d_i) \longrightarrow 0 \qquad (i \geq 0)$$

one sees by induction on i that $\mathrm{Im}(d_i) \in \mathbf{P}$ and that the image of (1.2.6.i) under F is a short exact sequence. The image of the complex is therefore acyclic.

Proposition 1.2.7. *Let F be an additive functor from an abelian category to an abelian category $\mathscr{B}$. If every object of $\mathscr{A}$ is a quotient (resp. subobject) of an object acyclic for LF (resp. RF) (1.2.3), the derived functor $LF : D^-(\mathscr{A}) \to D^-(\mathscr{B})$ (resp. $RF : D^+(\mathscr{A}) \to D^+(\mathscr{B})$) is defined everywhere, and every complex bounded above (resp. below) of objects acyclic for LF (resp. RF) is adapted.*

It suffices to consider the left derived functor LF. If K is a complex bounded above, the quasi-isomorphisms $s : L \to K$ with L bounded above and consisting of acyclic objects form a cofinal system in the category of quasi-isomorphisms with target K. It suffices to show that, when K consists of acyclic objects, $F(s) : F(L) \to F(K)$ is a
281 quasi-isomorphism; then, in the general case, one will have $F(L) = LF(K)$. Let M be the cone of s. The complex consists of acyclic objects and $H^*(M) = 0$; one must prove that $H^*(F(M)) = 0$.

Let i be an integer. The adapted complexes form a triangulated subcategory, so that every bounded complex of acyclic objects, in particular $\tau_{\geq i}(M)$, is adapted. Since $H^j(\tau_{\geq i}(M)) = 0$ for $j > i$, one has, for $j > i$, $H^j(F(M)) = H^j(F(\tau_{\geq i}(M))) = H^j LF(\tau_{\geq i}(M)) = 0$.

Proposition 1.2.8. *Let $F^{\bullet}$ be a complex bounded below of functors from $\mathscr{A}$ to $\mathscr{B}$. If for every $K \in K^{+}(\mathscr{A})$ there exists a quasi-isomorphism $L \xrightarrow{s} K$, where $L \in K^{+}(\mathscr{A})$ is adapted to all the RF^{i}, then the derived functor $RF^{\bullet} : D^{+}(\mathscr{A}) \to D^{+}(\mathscr{B})$ exists, and every complex* bounded below *that is adapted to all the RF^{i} is adapted to $RF^{\bullet}$.*

The proof is left to the reader, as is that of the dual statement, or of the variants obtained by taking F contravariant, or by taking a bounded complex of functors and working in $D(\mathscr{A})$ and $D(\mathscr{B})$.

Definition 1.2.9. *Let F be a functor from $D^{b}(\mathscr{A})$ to $D(\mathscr{B})$, and let d be an interval in $\mathbf{Z}$. We say that F has* cohomological amplitude $\subset d$ *if, for every $A \in \operatorname{Ob}\mathscr{A}$, $H^{i}F(A) = 0$ for $i \notin d$.*

Very often, d will be an interval of the form $[0, x]$ or $[-x, 0]$, and we then say that F has cohomological resp. homological dimension $\leq x$. We say that F has finite dimension if there exists a finite interval d in $\mathbf{Z}$ such that F has amplitude $\subset d$.

Proposition 1.2.10. *Assume that the hypotheses of 1.2.7 hold and that the functor $RF : D^{+}(\mathscr{A}) \to D^{+}(\mathscr{B})$ (resp. $LF : D^{-}(\mathscr{A}) \to D^{-}(\mathscr{B})$) has finite dimension. Then every complex of objects acyclic for RF (resp. LF) is adapted, and the functor $RF : D(\mathscr{A}) \to D(\mathscr{B})$ (resp. $LF : D(\mathscr{A}) \to D(\mathscr{B})$) is defined everywhere.* 282

For the proof, we refer to Verdier [**11**]. There is an analogous statement for bounded complexes of functors. The superscript $+$ (resp. $-$) may also be replaced by b.

2. Categories fibered in derived categories

2.1. Introduction. The editor insists that the reader refrain from reading this §. Here we give the formalism of the trivial duality theorem and resolve the perplexity raised by Artin in XII 4.

For the notions of a category over another category, a fibered category, a cofibered category, and a cleavage, we refer to SGA 1 VI. Among other things, that exposé establishes an equivalence between the notions of normalized split fibered categories over a category $\mathscr{C}$, and pseudofunctors from $\mathscr{C}^{\circ}$ to (Cat).

Definition 2.1.1. *If X is a ringed site, $D(X)$ denotes the derived category of the abelian category of sheaves of left modules on X.*

When confusion must be avoided, we instead write $D(X, \mathscr{A})$, where $\mathscr{A}$ denotes the sheaf of rings.

If $f : X \to Y$ is a morphism of ringed sites, f defines derived functors 283

(1) $Rf_{*} : D^{+}(X) \to D^{+}(Y)$ and
(2) $Lf^{*} : D^{-}(Y) \to D^{-}(X)$,

and, on passing to the opposite categories, the functors

(1°) $Rf_{*} : D^{+}(X)^{\circ} \to D^{+}(Y)^{\circ}$
(2°) $Lf^{*} : D^{-}(Y)^{\circ} \to D^{-}(X)^{\circ}$,

"functorial" in f. More precisely, (1°) and (2°) define two pseudofunctors from the category of sites to that of triangulated categories, one covariant and the other contravariant. One thereby obtains two categories over the category of sites, one cofibered, with fibers $D^{+}(X)^{\circ}$ and direct-image functors Rf_{*}, and the other fibered, with fibers $D^{-}(X)^{\circ}$ and inverse-image functors Lf^{*}. If the functor f^{*} has finite homological dimension (i.e. has finite "tor-dimension"), then in (2) the superscript may be replaced by a $+$. Thus one obtains two categories over the category $\mathcal{S}_{\mathrm{tdf}}$ of ringed sites and morphisms of ringed sites of finite tor-dimension; the fibers of these categories are the categories $D^+(X)^\circ$. The "trivial duality theorem" of Verdier ([11] p. 48) states that these two categories are canonically isomorphic; they are therefore fibered and cofibered; the "direct-image" functors are the functors Rf_* (sic), and the "inverse-image" functors are the functors Lf^* (sic).

2.1.2. The language of fibered categories readily resolves the perplexity raised by Artin in XII 4: it suffices to apply the following proposition to the category fibered and
284 cofibered over the category of sites whose fibers are the opposites of the categories of sheaves on the corresponding sites (cf. 0.2).

Let $\mathscr{E}$ be a category over $\mathscr{B}$, let $f : y \to x$ be an arrow of $\mathscr{B}$, and let $F \in \mathrm{Ob}(\mathscr{E}_x)$. Recall that a pair (G, φ) $(G \in \mathrm{Ob}\,\mathscr{E}_y, \varphi \in \mathrm{Hom}_f(G, F))$ is a *strict inverse image* of F under f if, for all $g : z \to y$ and $H \in \mathrm{Ob}\,\mathscr{E}_z$, one has

$$\mathrm{Hom}_g(H, G) \xrightarrow[\sim]{\varphi} \mathrm{Hom}_{fg}(H, F).$$

Similarly for direct images. If $\mathscr{E}$ is fibered or cofibered over $\mathscr{B}$, this notion reduces to that of inverse image.

Proposition 2.1.3. *Let ϵ be a category over $\mathscr{B}$, let $y \in \mathrm{Ob}\,\mathscr{B}$, let $F \in \mathrm{Ob}\,\mathscr{E}_y$, and consider a commutative diagram in $\mathscr{B}$:*

(2.1.3.1)
$$\begin{tikzcd} y' \arrow[r, "g'"] \arrow[d, "f'"'] & y \arrow[d, "f"] \\ x' \arrow[r] & x. \end{tikzcd}$$

Assume that the direct and inverse images f_*F, g^*f_*F, g'^*F, and $f'_*g'^*F$ exist in the strict sense. Then there exists a unique arrow $\varphi \in \mathrm{Fl}\,\mathscr{E}_{x'}$, called the base-change arrow, *making the diagram*

(2.1.3.2)
$$\begin{tikzcd} & g'^*F \arrow[r] \arrow[dl] & F \arrow[d] \\ f'_*g'^*F \arrow[dr, "\varphi"'] & & f_*F \\ & g^*f_*F \arrow[ur] & \end{tikzcd}$$
.

285 *commutative. If, in addition, the direct images $g'_*g'^*F$ and $g'_*f'_*g'^*F$ exist in the strict sense, then the arrow φ coincides with the one defined in* XII 4. *If $f'^*f'_*g'^*F$ and $f'^*f'_*F$ exist in the strict sense, it also coincides with the 2^{nd} arrow defined in* XII 4.

In particular, the two base-change arrows of XII 4 *coincide.*

By hypothesis, the arrows in diagram (2.1.3.2) induce bijections

$$\mathrm{Hom}_{x'}(f'_*g'^*F, g^*f_*F) = \mathrm{Hom}_{f'}(g'^*F, g^*f_*F) = \mathrm{Hom}_{gf'}(g'^*F, f_*F).$$

Since $fg' = gf'$, $\mathrm{Hom}_{fg'}(g'^*F, f_*F)$ is identical to $\mathrm{Hom}_{gf'}(g'^*F, f_*F)$, and the arrows of diagram (2.1.3.2) induce maps

$$\mathrm{Hom}_y(F, F) \longrightarrow \mathrm{Hom}_{fg'}(g'^*F, f_*F) = \mathrm{Hom}_{x'}(f'_*g'^*F, g^*f_*F),$$

and the image of 1_F is the unique arrow of $\mathscr{E}_{x'}$ making (2.1.3.2) commutative.

By definition, in XII 4, the base change arrow ψ was the one making the following diagram commutative

$$
\begin{tikzcd}
f'_*g'^*F \arrow[d, "\psi"] & g'^*F \arrow[l] \arrow[r, equal] \arrow[d, "u_4"] & g'^*F \arrow[r] \arrow[d, "u_3"] & F \arrow[r, equal] \arrow[d, "u_2"] & F \arrow[d, "u_1"] \\
g'_*f_*F & f'^*g^*f_*F \arrow[l] \arrow[r, equal] & g'f^*f_*F \arrow[r] & f^*f_*F \arrow[r] & f_*F.
\end{tikzcd}
\tag{2.1.3.3}
$$

The horizontal arrows and u_1 are the canonical arrows; u_2 is therefore the adjunction arrow, $u_3 = g'^*(u_2)$, u_4 is deduced from the isomorphism between f'^*g^* and g'^*f^*, and the arrow ψ is deduced from u_4 by adjunction.

The composite map with source g'^*F and target f_*F is the one that appears in (2.1.3.2), and the same therefore holds in the following commutative diagram, whose left-hand square is taken from (2.1.3.3) 286

$$
\begin{tikzcd}
g'^*F \arrow[r, "u_4"] \arrow[d] & f'^*g^*f_*F \arrow[r] \arrow[d] & f_*F \arrow[d, equal] \\
f'_*g'^*F \arrow[r, "\psi"] & g^*f_*F \arrow[r] & f_*F.
\end{tikzcd}
$$

This means that ψ satisfies the defining property of φ.

The definition of φ is self-dual, so there is no need to prove the second part of 2.1.3.

2.2. Categories fibered in triangulated categories.

Definition 2.2.1. *An "additive category over a category $\mathscr{B}$" is a category $\mathscr{A}$ over $\mathscr{B}$ whose sets of arrows* $\mathrm{Hom}_f(X,Y)$ *(for $f \in \mathrm{Fl}(\mathscr{B})$, with X (resp. Y) over the source (resp. the target) of f) have been endowed with abelian group structures, in such a way that:*

(i) *For any $x \xrightarrow{f} y \xrightarrow{g} z$ in $\mathscr{B}$ and X, Y, Z in $\mathscr{A}$ lying over x, y, and z, respectively, the composition law*

$$\mathrm{Hom}_g(Y,Z) \times \mathrm{Hom}_f(X,Y) \xrightarrow{\circ} \mathrm{Hom}_{gf}(X,Z)$$

is biadditive.

(ii) *The fiber categories are additive.*

We shall not need here the fact that these abelian group structures, when they exist, are uniquely determined by $\mathscr{A}$, $\mathscr{B}$, and ρ.[45] A $\mathscr{B}$-functor $F : \mathscr{A}_1 \to \mathscr{A}_2$ between additive categories over $\mathscr{B}$ is said to be *additive* if it induces homomorphisms

$$F : \mathrm{Hom}_{\mathscr{A}_1,f}(X,Y) \longrightarrow \mathrm{Hom}_{\mathscr{A}_2,f}(X,Y).$$

Definition 2.2.2. *An "abelian category over a category $\mathscr{B}$" is an additive category $\mathscr{A}$ over $\mathscr{B}$ whose fibers are abelian, and such that for every arrow $f : x \to y$ of $\mathscr{B}$ the bifunctor* 287

$$\mathrm{Hom}_f(X,Y) : \mathscr{A}_x^\circ \times \mathscr{A}_y \longrightarrow (\mathrm{Ab})$$

is left exact in X and Y.

We shall not need here the fact that if $\mathscr{A}$ is a category fibered and cofibered over $\mathscr{B}$ (SGA 1 VI 6), it is abelian over $\mathscr{B}$ as soon as its fibers are abelian categories.

[45] N.D.E.: here ρ denotes the structural functor $\mathscr{A} \to \mathscr{B}$.

Definition 2.2.3. *A “triangulated category over a category $\mathscr{B}$” is an additive category $\mathscr{A}$ over $\mathscr{B}$, endowed with a functor $T : \mathscr{A} \to \mathscr{A}$, and whose fibers $\mathscr{A}_x$, for $x \in \operatorname{Ob}\mathscr{B}$, are endowed with sets $\varDelta_x$ of triangles, in such a way that:*

(i) *T is an additive $\mathscr{B}$-automorphism of $\mathscr{A}$.*
(ii) *Each fiber category $\mathscr{A}_x$, endowed with the functor T_x induced by T and with the set $\varDelta_x$, is a triangulated category* ([11] I 1.1.1).
(iii) *If the triangles (X, Y, Z, u, v, w) and (X', Y', Z', u', v', w') belong to $\varDelta_x$ and $\varDelta_{x'}$, respectively, every diagram*

$$\begin{tikzcd}
& Z \arrow[ddl, "w"'] & & & Z' \arrow[ddl, "w'"'] & \\
& & Y \arrow[ul, "v"'] \arrow[rrr, "g"] & & & Y' \arrow[ul, "v'"'] \\
X \arrow[urr, "u"] \arrow[rrr, "f"'] & & & X' \arrow[urr, "u'"] & &
\end{tikzcd}$$

in which $u'f = gu$ can be completed to a morphism of triangles $(f, g, h) : (X, Y, Z) \to (X', Y', Z')$.

288 T will be called the translation functor; we shall often write $A[1]$ rather than $T(A)$, with $A[n]$ then denoting $T^n(A)$ ($n \in \mathbf{Z}$).

The triangles belonging to the sets $\varDelta_x$ will be called *distinguished* or *exact*.

Proposition 2.2.4. *Let $\mathscr{A}$ be a triangulated category over $\mathscr{B}$, let x be an object of $\mathscr{B}$, and let (X, Y, Z, u, v, w) be a distinguished triangle of $\mathscr{A}_x$. For every $f : y \to x$ and every object $A \in \operatorname{Ob}\mathscr{A}_y$, the following infinite sequence is exact:*

$$\begin{aligned}\cdots \longrightarrow \operatorname{Hom}_f(A, X) \longrightarrow \operatorname{Hom}_f(A, Y) \longrightarrow \operatorname{Hom}_f(A, Z)& \\ \longrightarrow \operatorname{Hom}_f(A, X[1]) \longrightarrow \operatorname{Hom}_f(A, Y[1]) \longrightarrow \dots&\end{aligned}$$

There is an analogous statement for $f : x \to y$ and $A \in \operatorname{Ob}\mathscr{A}_y$.

We leave it to the reader to prove this proposition by paraphrasing Verdier [11] p. 4. We also leave it to the reader to give a meaning to the following proposition and to verify it:

Proposition 2.2.5. *Let $\mathscr{A}$ be an additive (resp. abelian, resp. triangulated) category over $\mathscr{B}$. For every functor $\mathscr{B}' \to \mathscr{B}$ with target $\mathscr{B}$, the category $\mathscr{A}' = \mathscr{A} \times_{\mathscr{B}} \mathscr{B}'$ is additive (resp. abelian, resp. triangulated) over $\mathscr{B}$.*

Example 2.2.6. The category of sheaves of modules on varying ringed sites is abelian over the category of ringed sites.

Example 2.2.7. Let $\mathscr{A}$ be an additive category over $\mathscr{B}$. The category $\boldsymbol{C}(\mathscr{A}/\mathscr{B})$ of complexes of $\mathscr{A}$ over $\mathscr{B}$ has as objects the complexes of the fiber categories. An arrow
289 f from (X^n, d) to (Y^n, d) is a system of arrows $f^n : X^n \to Y^n$ such that $df^n = f^{n+1}d$. We set $p(f) = p(f^n)$ (this arrow does not depend on n, since the $p(d)$ are identities). $\boldsymbol{C}(\mathscr{A}/\mathscr{B})$ is an additive category over $\mathscr{B}$. A “translation” functor is defined by setting

$$(K[1])^n = K^{n+1} \quad , \quad (d[1])^n = -d^{n+1}.$$

Example 2.2.8. Let $\mathscr{A}$ be an additive category over $\mathscr{B}$. *The category $\boldsymbol{K}(\mathscr{A}/\mathscr{B})$ of complexes of $\mathscr{A}$ over $\mathscr{B}$ modulo homotopy* has the same objects as $\boldsymbol{C}(\mathscr{A}/\mathscr{B})$. It is obtained by declaring the arrows homotopic to zero to be zero in $C(\mathscr{A}/\mathscr{B})$; in other words, for $f : x \to y$ in $\mathscr{B}$, $X^\cdot \in \operatorname{Ob} C(\mathscr{A}/\mathscr{B})_x$ and $Y^\cdot \in \operatorname{Ob} C(\mathscr{A}/\mathscr{B})_y$, we set

$$\operatorname{Hom}_f(X^\cdot, Y^\cdot) = H^0 \operatorname{Hom}^\cdot_f(X^\cdot, Y^\cdot)$$

The translation functor descends to the quotient.

If $F : \mathscr{B}' \to \mathscr{B}$ is a functor with target $\mathscr{B}$, and if $\mathscr{A}' = \mathscr{A} \times_{\mathscr{B}} \mathscr{B}'$, we leave it to the reader to identify $C(\mathscr{A}'/\mathscr{B}')$ with $C(\mathscr{A}/\mathscr{B}) \times_{\mathscr{B}} \mathscr{B}'$ and $K(\mathscr{A}'/\mathscr{B}')$ with $K(\mathscr{A}/\mathscr{B}) \times_{\mathscr{B}} \mathscr{B}'$. In particular, the fibers of the category $K(\mathscr{A}/\mathscr{B})$ are the categories $K(\mathscr{A}_x)$ and, as such, are triangulated.

Proposition 2.2.9. (i) *With the additional structures defined in 2.2.8, the category $K(\mathscr{A}/\mathscr{B})$ is triangulated over $\mathscr{B}$.*

(ii) *The formation of the additive (resp. triangulated) category $C(\mathscr{A}/\mathscr{B})$ (resp. $K(\mathscr{A}/\mathscr{B})$) is compatible with every change of base category $\mathscr{B}' \to \mathscr{B}$.*

Conditions (i) and (ii) of Definition 2.2.3 are verified fiber by fiber. Condition (iii) is 290
verified by paraphrasing Verdier [11] p. 4.

Remark 2.2.10. The preceding discussion also applies to the categories $K^+(\mathscr{A}/\mathscr{B})$, $K^-(\mathscr{A}/\mathscr{B})$, and $K^b(\mathscr{A}/\mathscr{B})$ obtained by restricting respectively to complexes bounded below, bounded above, or bounded.

Remark 2.2.11. Taking $\mathscr{B}$ to be the final category, one recovers the usual categories of complexes.

2.2.12. Let F be a pseudofunctor from a category $\mathscr{B}^\circ$ to (Cat). It is known (SGA 1 VI) that F defines a category fibered over $\mathscr{B}$, whose fibers are the categories $F(x)$. For any $f : x \to y$ in $\mathscr{B}$ and $X \in \operatorname{Ob} F(x)$, $Y \in \operatorname{Ob} F(y)$, we have by definition

$$\operatorname{Hom}_f(X, Y) = \operatorname{Hom}_{F(x)}(X, F(f)(Y)). \tag{2.2.12.1}$$

It is known (ibid.) that the preceding construction defines an equivalence between the 2-category of pseudofunctors from $\mathscr{B}^\circ$ to (Cat) and the 2-category of categories fibered over $\mathscr{B}$.

Let us start with a pseudofunctor from $\mathscr{B}^\circ$ to the 2-category whose objects are additive (resp. abelian, resp. triangulated) categories and whose 1-arrows are additive (resp. left exact, resp. triangulated) functors. One immediately verifies that the category defined by (2.2.12.1) is additive (resp. abelian, resp. triangulated). Conversely:

Proposition 2.2.13. *The preceding construction establishes an equivalence between* 291

(a) *The 2-category of pseudofunctors from $\mathscr{B}^\circ$ to the 2-category whose objects are additive (resp. abelian, resp. triangulated) categories and whose 1-arrows are additive (resp. left exact, resp. triangulated) functors.*

(b) *The 2-category of additive (resp. abelian, resp. triangulated) categories over $\mathscr{B}$ that are fibered over $\mathscr{B}$.*

The theory developed in SGA 1 VI shows that it suffices to verify the following point:

. Let $\mathscr{A}$ be an additive (resp. abelian, resp. triangulated) category over $\mathscr{B}$, let $f : x \to y$ be an arrow of $\mathscr{B}$, and suppose that an inverse image functor $f^* : \mathscr{A}_y \to \mathscr{A}_x$ exists; then f^* is additive (resp. left exact, resp. triangulated).

The functor f^* gives rise to isomorphisms

$$\operatorname{Hom}_f(X, Y) \simeq \operatorname{Hom}_x(X, f^*(Y)).$$

In the triangulated case, it remains to prove that f^* is triangulated for the isomorphism $Tf^* = f^*T$ (where T is the translation functor) that makes the following diagrams commutative:

$$\begin{tikzcd}
\mathrm{Hom}_f(X,Y) \arrow[r, equal] \arrow[d, equal] & \mathrm{Hom}_x(X, f^*Y) \arrow[dr, equal] & \\
\mathrm{Hom}_f(TX,TY) \arrow[dr, equal] & & \mathrm{Hom}_x(TX, Tf^*Y) \arrow[dl, equal] \\
& \mathrm{Hom}_x(TX, f^*TY) &
\end{tikzcd} \tag{2.2.13.2}$$

292 *Additive case*: Consider a diagram

$$\begin{tikzcd}
f^*X \arrow[r, "p"] \arrow[d, shift right, "f^*(u)"'] \arrow[d, shift left, "f^*(v)"] & X \arrow[d, shift right, "u"'] \arrow[d, shift left, "v"] \\
f^*X' \arrow[r, "p'"] & X'.
\end{tikzcd} \tag{2.2.13.3}$$

We have $p'f^*(u+v) = (u+v)p = up+vp = p'f^*(u)+p'f^*(v) = p'(f^*(u)+f^*(v))$ and hence $f^*(u+v) = f^*(u) + f^*(v)$.

Abelian case: by definition (2.2), the functor $\mathrm{Hom}(X, f^*Y) = \mathrm{Hom}_f(X,Y)$ is left exact in Y, for every X, whence f^* is left exact.

Triangulated case: let (Y', Y, Y'') be a distinguished triangle in $\mathscr{A}_y$ and let (f^*Y', f^*Y, X'') be a distinguished triangle in $\mathscr{A}_x$. By 2.2.3, the canonical morphism between the bases of these triangles can be completed to a morphism of triangles, giving rise to the following diagram:

$$\begin{tikzcd}
& X'' \arrow[rrr, "s"] \arrow[ddl] & & & f^*Y'' \arrow[rrr] \arrow[ddl] & & & Y'' \arrow[ddl] & \\
& & f^*Y \arrow[ul] \arrow[rrr, equal] & & & f^*Y \arrow[ul] \arrow[rrr] & & & Y \arrow[ul] \\
f^*Y' \arrow[urr] \arrow[rrr, equal] & & & f^*Y' \arrow[urr] \arrow[rrr] & & & Y' \arrow[urr] & &
\end{tikzcd}.$$

Let us prove that s is an isomorphism, which will complete the proof. For every X in $\mathrm{Ob}\,\mathscr{A}_x$, we have the commutative diagram:

$$\begin{tikzcd}
\mathrm{Hom}_x(X, f^*Y') \arrow[r] \arrow[d, equal] & \mathrm{Hom}_x(X, f^*Y) \arrow[r] \arrow[d, equal] & \mathrm{Hom}_x(X, X'') \arrow[r] \arrow[d] & \mathrm{Hom}_x(X, f^*Y'[1]) \arrow[r] \arrow[d, equal] & \mathrm{Hom}_x(X, f^*Y[1]) \arrow[d, equal] \\
\mathrm{Hom}_x(X, f^*Y') \arrow[r] \arrow[d, equal] & \mathrm{Hom}_x(X, f^*Y) \arrow[r] \arrow[d, equal] & \mathrm{Hom}_x(X, f^*Y'') \arrow[r] \arrow[d, equal] & \mathrm{Hom}_x(X, f^*Y'[1]) \arrow[r] \arrow[d, equal] & \mathrm{Hom}_x(X, f^*Y[1]) \arrow[d, equal] \\
\mathrm{Hom}_f(X, Y) \arrow[r] & \mathrm{Hom}_f(X, Y) \arrow[r] & \mathrm{Hom}_f(X, Y'') \arrow[r] & \mathrm{Hom}_f(X, Y'[1]) \arrow[r] & \mathrm{Hom}_x(X, Y[1]).
\end{tikzcd}$$

293 The first and last rows are exact by 2.2.4. By the five lemma, the map induced by s from $\mathrm{Hom}(X, X'')$ to $\mathrm{Hom}(X, f^*Y'')$ is always bijective, and s is therefore an isomorphism.

There are, of course, dual results for cofibered categories.

2.3. Trivial duality formula. Let $\mathscr{A}$ be a category over a base category $\mathscr{B}$, and let S be a set of arrows of $\mathscr{A}$, each of which projects onto an identity of $\mathscr{B}$. The category $\mathscr{A}(S^{-1})$ obtained from $\mathscr{A}$ by the calculus of fractions is a category over $\mathscr{B}$ (by the universal property).

Let $\mathscr{B}' \to \mathscr{B}$ be a functor; denote by $\mathscr{A}'$ the fiber product $\mathscr{A} \times_{\mathscr{B}} \mathscr{B}'$ and by S' the inverse image of S in $\mathscr{A}'$. The composite functor $\mathscr{A}' \to \mathscr{A} \to \mathscr{A}(S^{-1})$ makes the arrows of S' invertible and hence factors through $\mathscr{A}'(S'^{-1})$. We thus obtain a canonical $\mathscr{B}'$-functor as follows:

$$\mathscr{A}'(S^{-1}) \longrightarrow \mathscr{B}' \times_{\mathscr{B}} \mathscr{A}(S'^{-1}). \tag{2.3.1}$$

Lemma 2.3.2. *If $\mathscr{B}'$ is a subcategory of $\mathscr{B}$, and if whenever a composite arrow $h = f \circ g$ of $\mathscr{B}$ belongs to $\mathscr{B}'$, both f and g belong to $\mathscr{B}'$, then the functor (2.3.1) is an isomorphism.*

Note that (2.3.1) always induces a bijection on objects, and that, in the present case, $\mathscr{B}' \times_{\mathscr{B}} \mathscr{A}(S^{-1})$ is a subcategory of $\mathscr{A}(S^{-1})$. 294

Recall the "explicit" construction of $\mathscr{A}(S^{-1})$ [2, Chap. I]. Given two objects X and Y of $\mathscr{A}$, one considers the "formal composites" $f_1 \dots f_n$, where each f_i is either an arrow of $\mathscr{A}$ or the inverse of an arrow of S. The set $\mathrm{Hom}_{\mathscr{A}(S^{-1})}(X, Y)$ is the quotient of the set of these "formal composites" by the equivalence relation generated by the relations

$$\begin{aligned} f_1 \dots uv \dots f_n &\equiv f_1 \dots (u \circ v) \dots f_n && \text{for } u, v \text{ in } \mathrm{Fl}(\mathscr{A}) \\ f_1 \dots ss^{-1} \dots f_n &\equiv f_1 \dots f_n && \text{for } s \in S \\ f_1 \dots s^{-1}s \dots f_n &\equiv f_1 \dots f_n && \text{for } s \in S. \end{aligned}$$

If these constructions are applied to $\mathscr{A}$ and to the subcategory $\mathscr{A}'$, one checks that $\mathscr{A}'(S^{-1})$ is identical to the inverse-image subcategory of $\mathscr{A}(S^{-1})$ of $\mathscr{B}'$, since if a "formal composite" $f_1 \dots f_n$ has as its composite in $\mathscr{B}$ an arrow of $\mathscr{B}'$, the image of each f_i will also be an arrow of $\mathscr{B}'$.

Suppose now that $\mathscr{A}$ is a triangulated category over $\mathscr{B}$, and that the sets $S_x = S \cap \mathrm{Fl}(\mathscr{A}_x)$, for $x \in \mathrm{Ob}(\mathscr{B})$, are saturated multiplicative systems ([11] Chap. I § 2 n° 1).

Definition 2.3.3. *Under the preceding hypotheses, the category $\mathscr{A}/\mathscr{B}$ is said to be almost derivable (resp. derivable, resp. coderivable) relative to S if, for every functor $F : \mathscr{B}' \to \mathscr{B}$, the functor (2.3.1) is an isomorphism (resp. and if $\mathscr{A}(S^{-1})$ is fibered, resp. cofibered over $\mathscr{B}$).*

We denote by I the category of the following diagram: 295

$$I : \underset{0}{\times} \xrightarrow{F} \underset{1}{\times}. \tag{2.3.4}$$

By 2.2.13, to give a triangulated category $\mathscr{A}$ over I, fibered (resp. cofibered) over I, is equivalently to give its fibers $\mathscr{A}_0$ and $\mathscr{A}_1$ and the triangulated inverse-image functor $F^* : \mathscr{A}_1 \to \mathscr{A}_0$ (resp. and the triangulated direct-image functor $F_* : \mathscr{A}_0 \to \mathscr{A}_1$). If $A_0 \in \mathrm{Ob}\,\mathscr{A}_0$ and $A_1 \in \mathrm{Ob}\,\mathscr{A}_1$, then by (2.2.12.1),

$$\mathrm{Hom}_F(A_0, A_1) = \mathrm{Hom}(A_0, F^* A_1)$$

(resp. $\mathrm{Hom}_F(A_0, A_1) = \mathrm{Hom}(F_* A_0, A_1)$).

Proposition 2.3.5. *Let $\mathscr{A}$ be a triangulated category over I and let S be as in (2.3.3).*

(i) *$\mathscr{A}$ is almost derivable relative to S, and $\mathscr{A}(S^{-1})$ is triangulated over I.*

(ii) *If $\mathscr{A}$ is fibered (resp. cofibered), the functor F^* (resp. F_*) is right derivable (resp. left derivable) relative to S_0 and S_1 (1.2.1) if and only if $\mathscr{A}$ is derivable (resp. coderivable) over I.*

By Lemma 2.3.2, the fiber categories of $\mathscr{A}(S^{-1})$ are $\mathscr{A}_0(S_0^{-1})$ and $\mathscr{A}_1(S_1^{-1})$, where $S_i = S \cap \mathrm{Fl}(\mathscr{A}_i)$ $(i = 0, 1)$. With the notation of § 1 n° 2, the functors $A_0 \to A_0^-$ and $A_1 \to A_1^+$ identify $\mathscr{A}_0(S_0^{-1})$ and $\mathscr{A}_1(S_1^{-1})$ with categories of pro- and ind-objects of $\mathscr{A}_0$ and $\mathscr{A}_1$. This makes it possible to define a category over I, with fibers $\mathscr{A}_0(S_0^{-1})$ and $\mathscr{A}_1(S_1^{-1})$, by setting, for $A_0 \in \mathscr{A}_0$ and $A_1 \in \mathscr{A}_1$,

$$\mathrm{Hom}_F(A_0, A_1) = \mathrm{Hom}_F(A_0^-, A_1^+) = \varinjlim_{A_0' \xrightarrow[s]{} A_0} \varinjlim_{A_1 \xrightarrow[t]{} A_1'} \mathrm{Hom}(A_0', A_1'),$$

296 where s and t range over the elements of S_0 and S_1 with target (respectively source) A_0 (respectively A_1). The category $\mathscr{A}$ maps to this category, and one checks immediately that it satisfies the universal property of $\mathscr{A}(S^{-1})$, over I and after every base change.

To verify (i), it suffices to prove that this category, namely $\mathscr{A}(S^{-1})$, is triangulated, and it suffices to verify condition 2.2.3 (iii). Let (X_0, Y_0, Z_0) and (X_1, Y_1, Z_1) therefore be distinguished triangles of $\mathscr{A}_0$ and $\mathscr{A}_1$, giving rise to the distinguished triangles (X_0^-, Y_0^-, Z_0^-) and (X_1^+, Y_1^+, Z_1^+) of $\mathscr{A}_0(S_0^{-1})$ and $\mathscr{A}_1(S_1^{-1})$.

(2.3.5.1)
$$\begin{tikzcd} X_0^- \arrow[r, "u"] \arrow[d] & X_1^+ \arrow[d] \\ Y_0^- \arrow[r, "v"] & Y_1^+ . \end{tikzcd}$$

By Lemma 1.2.2.1, the triangles (X_0^-, Y_0^-, Z_0^-) and (X_1^+, Y_1^+, Z_1^+) are limits of distinguished triangles; there therefore exist morphisms of distinguished triangles

(2.3.5.2)
$$\begin{gathered} (X_0^-, Y_0^-, Z_0^-) \longrightarrow (X_0', Y_0', Z_0') \\ (X_1', Y_1', Z_1') \longrightarrow (X_1^+, Y_1^+, Z_1^+), \end{gathered}$$

297 and a diagram

$$\begin{tikzcd} X_0' \arrow[r, "u'"] \arrow[d] & X_1' \arrow[d] \\ Y_0' \arrow[r, "v'"] & Y_1' , \end{tikzcd}$$

compatible with (2.3.5.1). Since $\mathscr{A}$ is triangulated over I, this diagram can be extended to a morphism of triangles

$$(X_0', Y_0', Z_0') \longrightarrow (X_1', Y_1', Z_1').$$

Composing the latter with the morphisms (2.3.5.2), we see that diagram (2.3.5.1) can be extended to a morphism of triangles, which proves 2.2.3 (iii).

Let us verify (ii), assuming for example that $\mathscr{A}$ is fibered over I. If $A_1 \in \mathrm{Ob}(\mathscr{A}_1(S_1^{-1}))$, an inverse image of A_1 in $\mathscr{A}(S^{-1})$ is an object B of $\mathscr{A}_0(S_0^{-1})$ such that, in $\mathscr{A}_0(S_0^{-1})$,

$$\begin{aligned} \mathrm{Hom}_{\mathscr{A}_0(S_0^{-1})}(A_0, B) &= \mathrm{Hom}_F(A_0^-, A_1^+) = \mathrm{Hom}(A_0^-, F^*(A_1^+)) \\ &= \mathrm{Hom}_{\mathscr{A}_0(S_0^{-1})}(A_0, F^*(A_1^+)). \end{aligned}$$

Such an object B exists if and only if the ind-object $F^*(A_1^+)$ of $\mathscr{A}_0(S_0^{-1})$ is essentially constant (i.e. representable), and B then represents this ind-object, which is precisely $RF(A_1)$ (1.2.1); this proves the assertion.

Corollary 2.3.6. *With the notation of 2.3.3, if $\mathscr{A}$ is almost derivable over $\mathscr{B}$, then $\mathscr{A}(S^{-1})$ is a triangulated category over $\mathscr{B}$.*

First, the fiber category $\mathscr{A}(S^{-1})_x$ can be identified with $\mathscr{A}_x(S_x^{-1})$: apply the hypothesis to the base change $\{x\} \to \mathscr{B}$. This category is triangulated, equipped with a set 298
Δ_x of triangles, which gives a meaning to the preceding statement. To verify condition 2.2.3 (iii), it suffices to do so after every base change $I \to \mathscr{B}$, and the assertion follows from 2.3.5 (i).

It follows immediately from the proof of 2.3.5 (ii) that:

Theorem 2.3.7. (trivial duality formula).

Let $\mathscr{A}$ be a triangulated category fibered and cofibered over I, and denote by F_* and F^* the direct- and inverse-image functors. Let $S = S_0 \cup S_1$ be a multiplicative system as in 2.3.3. Suppose that LF_* is defined at $A_0 \in \operatorname{Ob} \mathscr{A}(S_0^{-1})$ and that RF^* is defined at $A_1 \in \operatorname{Ob} \mathscr{A}_1(S_1^{-1})$. Then

$$\operatorname{Hom}_{\mathscr{A}_0(S_0^{-1})}(A_0, RF^*A_1) = \operatorname{Hom}_{\mathscr{A}(S^{-1})}(A_0, A_1) = \operatorname{Hom}_{\mathscr{A}_1(S_1^{-1})}(LF_*A_0, A_1).$$

The positions of the symbols R, L, *, and $_*$ in 2.3.7 are only apparently anomalous: in fact, in the category 0.2 of sheaves on variable sites, fibered and cofibered, the fiber categories are the opposites of the usual categories of sheaves.

2.4. Categories fibered in derived categories.

2.4.0. In this n°, $\mathscr{A}$ denotes a triangulated category cofibered over $\mathscr{B}$, and S a multiplicative system of morphisms of $\mathscr{A}$, all lying over an identity, such that $S_x = S \cap \operatorname{Fl}(\mathscr{A}_x)$ is a saturated multiplicative system of $\mathscr{A}_x$ [11, I 1.2.1] for every $x \in \operatorname{Ob}(\mathscr{B})$.

We seek a criterion for $\mathscr{A}$ to be coderivable relative to S (2.3.3); this implies that 299
$\mathscr{A}(S^{-1})$ is triangulated over $\mathscr{B}$ when it is equipped with the set of triangles isomorphic to the image of a distinguished triangle of $\mathscr{A}$ (2.3.5).

Proposition 2.4.1. *Let $\mathscr{A}$, $\mathscr{B}$, and S be as above. For $\mathscr{A}$ to be coderivable relative to S, it suffices that the following condition hold:*

(∗) *$\forall x \in \operatorname{Ob} \mathscr{B}$, $\forall A \in \operatorname{Ob}(\mathscr{A}_x)$, $\exists s : A' \to A$ in S, $\forall g, f : x \xrightarrow{f} y \xrightarrow{g} z$, f_*A' is left adapted to the functor g_*.*

Let $\mathscr{A}^1$ denote the full subcategory of $\mathscr{A}$ consisting of the objects all of whose direct images are adapted to all direct-image functors. Then 1.2.2 shows that $\mathscr{A}^1$ is a triangulated subcategory of $\mathscr{A}$, evidently cofibered. Set $S^1 = S \cap \operatorname{Fl}(\mathscr{A}^1)$.

Lemma 2.4.1.1. *Suppose that S is stable under direct images. Then $\mathscr{A}/\mathscr{B}$ is coderivable and $\mathscr{A}(S^{-1})$ can be computed by a calculus of right fractions.*

Let us verify condition c) of [2, I 2.2], possibly the least trivial of the conditions. We are given f and s ($s \in S$) and must find

(2.4.1.2)
$$\begin{tikzcd} A' \arrow[r, dashed] & B' \\ A \arrow[u, "s"] \arrow[r, "f"'] & B, \arrow[u, dashed, "t"'] \end{tikzcd}$$

a commutative diagram (2.4.1.2) with $t \in S$. Let F be the image of f in $\mathscr{B}$. By hypothesis, $F_*(s)$ still belongs to S, so by Verdier there is a commutative diagram 300

$$\begin{tikzcd}
F_*(A') \arrow[r] & B' \\
F_*(A) \arrow[u, "F_*(s)"] \arrow[r, "f"'] & B \arrow[u, "t"']
\end{tikzcd}$$

and hence also a diagram (2.4.1.2).

Since $\mathscr{A}(S^{-1})$ is computed by a calculus of right fractions, it is immediate that its formation commutes with base change $c : \mathscr{B}' \to \mathscr{B}$.

This lemma applies in particular to $\mathscr{A}^1$ and S^1; the following lemma therefore implies 2.4.1:

Lemma 2.4.1.3. *Over $\mathscr{B}$, and after every base change $c : \mathscr{B}' \to \mathscr{B}$, the canonical functor $\Phi : \mathscr{A}^1((S^1)^{-1}) \to \mathscr{A}(S^{-1})$ is an equivalence of categories.*

Let us construct a quasi-inverse to Φ. For each $A \in \operatorname{Ob}\mathscr{A}$, choose $s_A \in S$ with target A and source $A^1 \in \operatorname{Ob}\mathscr{A}^1$, with $s_A = \mathrm{Id}_A$ when $A \in \operatorname{Ob}\mathscr{A}^1$. Let f be an arrow of $\mathscr{A}$ with target A, source B, and image F in $\mathscr{B}$. Since $s_B \in S$ and $F_*A^1 \in \operatorname{Ob}\mathscr{A}^1$, one can find B', $s \in S^1$ and u making the following diagram commutative:

(2.4.1.4)
$$\begin{tikzcd}
A \arrow[r, "f"] & B & B^1 \arrow[l, "s_B"'] \\
A^1 \arrow[u, "s_A"'] \arrow[r, "a"] & F_*A^1 \arrow[u] & B'. \arrow[l, "s"'] \arrow[u, "u"']
\end{tikzcd}$$

301 The arrow $f^1 = us^{-1}a$ of $\mathscr{A}^1((S^1)^{-1})$ then depends only on f.

Let f and g be two composable arrows, with images F and G in $\mathscr{B}$. To prove that $(g \circ f)^1 = g^1 \circ f^1$, consider the following diagram, which contains diagrams (2.4.1.4) for f and g:

$$\begin{tikzcd}
 & A \arrow[rr, "f"] & & B \arrow[rrr, "g"] & & & C & & \\
A^1 \arrow[ur] \arrow[rr, "a"] & & F_*A^1 \arrow[ur] \arrow[rrr, "a'"] & & & (GF)_*A^1 \arrow[ur] & & & C^1 \arrow[ull] \\
 & & & B^1 \arrow[uu] \arrow[rrr, "b"] & & & G_*B^1 \arrow[uu] & & \\
 & & B' \arrow[uu, "s"] \arrow[ur, "u"] \arrow[rrr, "b'"] & & & G_*B' \arrow[uu, "G_*s"] \arrow[ur] & & C'' \arrow[ll, "r"] \arrow[r, "w"] & C'. \arrow[ull, "t"'] \arrow[uu, "v"']
\end{tikzcd}$$

Since $t \in S$, there exist C'', $r \in S^1$, and w making the diagram commutative. By definition,
$g^1 \circ f^1 = vt^{-1}bus^{-1}a$ and $(g \circ f)^1 = vw(G_*(s)r)^{-1}a'a$, and the conclusion follows from the commutativity of the diagram.

We have thus defined a functor from $\mathscr{A}$ to $\mathscr{A}^1((S^1)^{-1})$; by the universal property of $\mathscr{A}(S^{-1})$, it factors through $\Psi : \mathscr{A}(S^{-1}) \to \mathscr{A}^1((S^1)^{-1})$. The composite $\Phi\Psi$ is isomorphic to the identity because its composite with the functor $J : \mathscr{A} \to \mathscr{A}(S^{-1})$ is isomorphic to J. One checks similarly that $\Psi\Phi$ is isomorphic to the identity.

Corollary 2.4.2. *With the notation of 2.4.1, $\mathscr{A}$ is coderivable relative to S provided that, for every finite set B of arrows of $\mathscr{B}$, condition $(*)$ holds when only direct images under the arrows of B are considered.*

For every category $\mathscr{B}'$ with a finite set of arrows, Proposition 2.4.1 is applicable after every base change $\mathscr{B}' \to \mathscr{B}$. We shall take for $\mathscr{B}'$ the categories of the following diagrams: **302**

$$\underset{(0)}{*} \qquad \underset{(1)}{* \longrightarrow *} \qquad \underset{(2)}{* \longrightarrow * \longrightarrow *} \qquad \underset{(3)}{* \longrightarrow * \longrightarrow * \longrightarrow *}$$

Consider the categories $\mathscr{A}_x(S_x^{-1})$ $(x \in \operatorname{Ob}\mathscr{B})$. Given $f \in \operatorname{Fl}(\mathscr{B})$ and A_1, A_2 in these categories over the source and target of f, define $\operatorname{Hom}_f(A_1, A_2)$ to be the analogous set defined after the base change of type (1) determined by f. Composition of morphisms is defined using base changes of type (3). In this way one constructs a category $\mathscr{A}(S^{-1})^*$ over $\mathscr{B}$, whose formation is compatible with every base change; using base changes of types (0), (1), and (2), one checks that it solves the universal problem defining $\mathscr{A}(S^{-1})$.

The following result is dual to 2.4.2.

Proposition 2.4.3. *Let $\mathscr{A}$ be a triangulated category fibered over $\mathscr{B}$, and let S be as in 2.4.1. For $\mathscr{A}$ to be derivable relative to S, it suffices that, for every finite set B of arrows of $\mathscr{B}$,*

$(*')$ $\forall x \in \operatorname{Ob}(\mathscr{B})$, $\forall A \in \operatorname{Ob}(\mathscr{A}_x)$, $\exists s : A \to A'$ *in* S, $\forall g, f : Z \xrightarrow{g} Y \xrightarrow{f} X$ *in* B, f^*A' *is right adapted to the functor* g^*.

3. Gluing fibered or cofibered categories

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3.1. Introduction. Let $f : Y \to X$ be a separated morphism of finite type from a scheme Y to a scheme X, or a continuous map from a locally compact space to a locally compact space. One can then define a "direct image with proper supports" functor, denoted $f_!$, from the category of abelian sheaves on Y to the category of abelian sheaves on X (cf. 6.1 in the case of schemes, and Verdier [13] for separated locally compact spaces). When Y and X are schemes of finite type over $\mathbf{C}$, these two definitions are compatible in the evident sense. Unfortunately, for schemes, unlike for locally compact spaces, the derived functors of $f_!$ are pathological; one can nevertheless define a reasonable "direct image with proper supports" functor directly from $D^+(Y)$ to $D^+(X)$. To define it, factor f as an open immersion followed by a proper morphism: $f = \overline{f}j$. The desired functor $Rf_!$ is then defined as the composite of "extension by zero" $j_!$ and the functor $R\overline{f}_*$, derived from direct image under $\overline{f}$.

These "direct image with proper supports" functors $Rf_!$ are usable only after one verifies for them a "variance formalism" that includes, for a composite $f = gh$, a formula

$$Rf_! = Rg_! Rh_!$$

which is best expressed in terms of cofibered categories. Such a formalism is available separately for "extensions by zero" and for direct images under proper morphisms, and **304**
the problem is to glue these formalisms. To avoid later being bogged down by pointless computations, we propose in this § to list the elementary verifications required to carry out such a gluing. We do so in a slightly more abstract setting than is strictly necessary, in the hope that the result may be useful again.

The construction of $Rf_!$ sketched above requires that the morphism f be factorable as an open immersion followed by a proper morphism. Following HARTSHORNE [6, III 8 p. 189], in n° 2 we introduce the appropriate notion of a compactifiable morphism.

3.2. Compactifiable morphisms.

DEFINITION 3.2.1. *A morphism f from an S-scheme X to a quasi-compact, quasi-separated S-scheme Y is said to be* S-compactifiable *if there exists an S-scheme P proper over S and a factorization of f as a separated quasi-finite morphism j from X to $P \times_S Y$ followed by the projection from $P \times_S Y$ to Y*

$$\begin{tikzcd} & & P \times_S Y \arrow[dl] \arrow[dr] & \\ X \arrow[urr, "j"] \arrow[r, "f"] \arrow[drr] & Y \arrow[dr] & & P \arrow[dl] \\ & & S & \end{tikzcd}.$$

When $S = Y$, we shall simply speak of a compactifiable morphism.

305 PROPOSITION 3.2.2. (i) *An S-compactifiable morphism is separated and of finite type.*

(ii) *If Y is quasi-compact and quasi-separated, a separated quasi-finite morphism with target Y is S-compactifiable.*

(iii) *The composite of two S-compactifiable morphisms is again S-compactifiable.*

(iv) *Let there be a diagram of quasi-compact, quasi-separated S-schemes*

$$\begin{tikzcd} X \times_Y Y' \arrow[r] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r] & Y. \end{tikzcd}$$

If f is S-compactifiable, then f' is S-compactifiable.

(v) *Let g be an S-compactifiable morphism;*[46] *for a composite morphism $g \circ f$ to be S-compactifiable, it is necessary that f be so.*

Assertions (i), (ii), and (iv) are immediate. To prove (iii) and (v), consider a composite $f = gh$, with g S-compactifiable.

Suppose that gh is S-compactifiable, whence a diagram

$$\begin{tikzcd} & P \times_S Y \arrow[r, "v"] \arrow[d] & P \times_S Z \arrow[d] \\ X \arrow[ur, "u"] \arrow[r, "h"] & Y \arrow[r, "g"] & Z. \end{tikzcd}$$

The composite vu is quasi-finite; hence u is quasi-finite, and (v) is proved.

[46] N.D.E.: it suffices to assume that g is a quasi-compact, quasi-separated morphism of S-schemes.

If h is compactifiable, we have a diagram

$$\begin{tikzcd}
 & & & Q\times_S P\times_S Z \arrow[d] & \\
 & Q\times_S Y \arrow[urr] \arrow[d] \arrow[ddr] & & P\times_S Z \arrow[d] \arrow[ddr] & \\
X \arrow[ur] \arrow[r, "h"] \arrow[ddrr] & Y \arrow[urr] \arrow[rr, "g"] \arrow[ddr] & & Z \arrow[ddl] & \\
 & & Q \arrow[d] & & P \arrow[dll] \\
 & & S & &
\end{tikzcd},$$

and $f = fh$ factors through a separated quasi-finite morphism to $(Q \times_S P) \times_S Z$. **306**

Nagata asserts in [**9**] that every morphism of finite type between separated integral noetherian schemes is compactifiable.[47] The editor admits to not having understood the proof.

Now fix a scheme S and denote by (S) the subcategory of the category of S-schemes whose objects are the quasi-compact, quasi-separated S-schemes and whose arrows are the S-compactifiable S-morphisms.[48]

Proposition 3.2.3. *In the category* (S)*:*

(i) *The open immersions (resp. the proper morphisms) define a subcategory* (S, i) *(resp.* (S, p)*) of* (S) *having the same objects as* (S)*.*
(ii) *Fiber products exist in* (S, p) *and are fiber products in* (S)*.*
(iii) *Every morphism* f *is the composite* $f = pj$ *of a proper morphism* p *and an open immersion.*

By 3.2.2 (iii), (iv), and (v), fiber products exist in (S) and are fiber products in the **307**
category of all schemes. Since the fiber product $X = X_1 \times_Y X_2$ of two schemes proper over Y is again proper over Y, and a scheme is proper over X if and only if it is proper over X_1 and X_2, condition (ii) is satisfied.

If f is a morphism of (S), by hypothesis it admits, in (S), a factorization $f = pj'$, where p is proper and j' is separated and quasi-finite. By EGA IV 18.12.13, j' admits a factorization $j' = qj$, where q is finite and j is an open immersion; moreover, by 3.2.2 (ii), q and j belong to (S). We then have $f = (pq)j$, which proves (iii). Assertion (i) follows from 3.2.2 (iii).

3.2.4. In the rest of this §, suppose given a category (S), equipped with two subcategories (S, i) and (S, p), such that

(i) $\mathrm{Ob}(S) = \mathrm{Ob}(S, i) = \mathrm{Ob}(S, p)$;
(ii) fiber products exist in (S, p) and are fiber products in (S);
(iii) $\forall f \in \mathrm{Fl}(S)$, $\exists p \in \mathrm{Fl}(S, p)$, $\exists j \in \mathrm{Fl}(S, i)$, $f = pj$.

[47] N.D.E.: more generally, every separated morphism of finite type whose target is quasi-compact and quasi-separated is compactifiable. See la N.D.E. (43) page 639.

[48] N.D.E.: Thanks to the compactification theorem, one may also work in the subcategory of the category of (absolute) schemes whose objects are the quasi-compact, quasi-separated schemes and whose arrows are the separated morphisms of finite type (i.e. compactifiable morphisms). Note that the stability of compactifiable morphisms under composition is equivalent to the general form of the compactification theorem.

The arrows of (S, i) (resp. of (S, p)) will be called open immersions (resp. proper morphisms).

Definition 3.2.5. (i) *A* compactification *of a morphism f of (S) (resp. of a pair (f, g), resp. of a triple (f, g, h) of composable morphisms) is a commutative diagram (3.2.5.1) (resp. (3.2.5.2), resp. (3.2.5.3)) whose horizontal arrows are open immersions and whose oblique arrows are proper.*

308 (ii) A morphism of compactifications *is a morphism of diagrams (3.2.5.1) (resp. ...) that is the identity on the 1st vertical column and is proper at every vertex*

$$\begin{tikzcd} Y \arrow[r, hook, "i"] \arrow[d, "f"'] & \overline{Y} \arrow[dl, "\overline{f}"] \\ X & \end{tikzcd}$$

(3.2.5.1)

$$\begin{tikzcd} Z \arrow[r, hook, "j"] \arrow[d, "g"'] & \overline{Z} \arrow[r, hook, "j'"] \arrow[dl, "\overline{g}"] & \overline{\overline{Z}} \arrow[dl, "\overline{\overline{g}}"] \\ Y \arrow[r, hook, "i"] \arrow[d, "f"'] & \overline{Y} \arrow[dl, "\overline{f}"] & \\ X & & \end{tikzcd}$$

(3.2.5.2)

$$\begin{tikzcd} \cdot \arrow[r, hook, "k"'] \arrow[d, "h"'] & \cdot \arrow[r, hook, "k'"'] \arrow[dl, "\overline{h}"] & \cdot \arrow[r, hook, "k''"'] \arrow[dl, "\overline{\overline{h}}"] & \cdot \arrow[dl, "\overline{\overline{\overline{h}}}"] \\ \cdot \arrow[r, hook, "j"'] \arrow[d, "g"'] & \cdot \arrow[r, hook, "j'"'] \arrow[dl, "\overline{g}"] & \cdot \arrow[dl, "\overline{\overline{g}}"] & \\ \cdot \arrow[r, hook, "i"'] \arrow[d, "f"'] & \cdot \arrow[dl, "\overline{f}"] & & \\ \cdot & & & \end{tikzcd}$$

(3.2.5.3)

Proposition 3.2.6. (i) *Every morphism (resp. every pair (f, g), resp. every triple (f, g, h) of composable morphisms) admits a compactification.*

(ii) *The category of compactifications of f is left filtering.*[49]

(iii) *If diagrams $(3.2.5.2)_i$ $(i = 1, 2)$ are compactifications of (f, g), there exists a compactification $(3.2.5.2)_3$ of (f, g) and morphisms of compactifications from $(3.2.5.2)_3$ to the compactifications $(3.2.5.2)_i$.*

(i) The case of a morphism is precisely 3.2.4 (iii). Let us treat the case of a triple; with the notation of (3.2.5.3), it suffices to construct successively compactifications of f, g, and h, of $i\overline{g}$ and $j\overline{h}$, and finally of $j'\overline{\overline{h}}$.

309 (ii) Let $(3.2.5.1)_1$ and $(3.2.5.1)_2$ be two compactifications of f. The projection p from $\overline{Y}_1 \times_S \overline{Y}_2$ to X is proper, and Y maps to it by (i_1, i_2), which (3.2.4 (iii)) can be compactified as $(i_1, i_2) = qj$. The compactification $f = (p \circ q)j$ dominates $(3.2.5.1)_1$ and $(3.2.5.1)_2$.

If r and s are two morphisms from $(3.2.5.1)_1$ to $(3.2.5.1)_2$, the kernel K of the pair of arrows $\overline{Y}_1 \underset{s}{\overset{r}{\rightrightarrows}} \overline{Y}_2$ maps by a proper morphism to $\overline{Y}_1$, and i_1 factors through K: write $i_1 = ki$. If $i = pi_3$ is a compactification of i, the existence of the compactification $f = (\overline{f}_1 kp)i_3$ shows that axiom L_2 is satisfied.

(iii) Apply axiom L_1 for filtering categories to the compactifications of f and g deduced from the $(3.2.5.2)_i$ $(i = 1, 2)$. We thereby obtain $\overline{Y}_3$ and $\overline{Z}_3$ giving rise to

[49] N.D.E.: "left filtering" means "cofiltered" (I 2.7). For axioms L_1 and L_2 in the proof of 3.2.6, cf. V 8.1.1.1.

commutative diagrams (3.2.5.1)$_i$ ($i = 1, 2$)

$$\begin{tikzcd}
 & & \overline{Z}_i \arrow[r, hook] \arrow[d] & \overline{\overline{Z}}_i \arrow[d] \\
\overline{Z}_3 \arrow[urr] \arrow[d, "\overline{g}_3"] & & Y \arrow[r, hook] & \overline{Y}_i \\
Y \arrow[r, "i_3"'] \arrow[urr, equal] & \overline{Y}_3 \arrow[urr] & &
\end{tikzcd},$$

and it remains to find a compactification of $i_3\overline{g}_3$ by $\overline{\overline{Z}}_3$, such that $\overline{\overline{Z}}_3$ can map by a proper morphism to $\overline{\overline{Z}}_i$, so as to give rise to a commutative cube ($i = 1, 2$). If the target of a morphism between compactifications of $i_3\overline{g}_3$ has this property, its source has it as well, and by (ii), it will suffice to consider separately the cases $i = 1$ and $i = 2$. Let T be the fiber product of $\overline{Y}_3$ and $\overline{\overline{Z}}_i$ over $\overline{Y}_i$: then T occurs **310**
in a commutative cube analogous to the desired one, except that the arrow k from $\overline{Z}_3$ to T might not be an open immersion. Writing $k = pj$ (p proper, j an open immersion) and repla¸cing T by the source $\overline{\overline{Z}}_3$ of p, we obtain the desired cube.

Remark 3.2.7. *Proposition 3.2.3 would still have been valid if, in Definition 3.2.1 of compactifiable morphisms, we had omitted the requirement that Y be quasi-compact and quasi-separated and required instead that j be a quasi-compact open immersion and not merely a quasi-finite morphism.*

3.3. Gluing.

Recall that the hypotheses of 3.2.4 are satisfied.

3.3.1. Suppose given:

α) for each $X \in \mathrm{Ob}(S)$, a category $F(X)$,
β) an (S, i)-cofibered category F_i and an (S, p)- cofibered category F_p,
γ) for each $X \in \mathrm{Ob}(S)$, isomorphisms between $F(X)$, $(F_i)_X$, and $(F_p)_X$,
δ) a normalized cleavage of F_i and a normalized cleavage of F_p.

By (SGA 1 VI), it "amounts to the same thing" to give instead α) and:

(i) For every proper morphism $p : X \to Y$, a functor $p_* : F(X) \to F(Y)$,
(ii) for every composable pair (p, q) of proper morphisms, an isomorphism of func- **311**
tors $c_{p,q} : p_*q_* \leftrightarrow (pq)_*$,
(i') and (ii'), analogous data for (S, i),

these data satisfying the following conditions:

(a) if p is an identity, then p_*, $c_{p,q}$, and $c_{q,p}$ are identities,
(b) for every composable triple of proper morphisms, we have

$$c_{p,qr} \circ (p_* * c_{q,r}) = c_{pq,r} \circ (c_{p,q} * r_*),$$

(a') and (b'), analogous conditions for (S, p).

Suppose *given* in addition,

(iii) for every commutative diagram D

$$\begin{tikzcd} X \arrow[r, "j'"] \arrow[d, "p'"'] & Y \arrow[d, "p"] \\ Z \arrow[r, "j"'] & T \end{tikzcd} \qquad j, j' \in \mathrm{Fl}(S, i);\ p, p' \in \mathrm{Fl}(S, p),$$

an isomorphism of functors $d(D) : p_* j'_* \leftrightarrow j_* p'_*$.

Proposition 3.3.2. *If data* (i) (ii) (i′) (ii′) (iii) *satisfy conditions* (a) (b) (a′) (b′) *as well as* (c) *and* (c′) *stated below, then there exists one and "essentially" only one category F cofibered over S, equipped with isomorphisms $F \times_S (S, i) = F_i$ and $F \times_S (S, p) = F_p$, compatible with datum 3.3.1 γ) and such that the isomorphisms $d(D)$ of* (iii) *are the composite isomorphisms $j_* p'_* = (jp')_* = (pj')_* = p_* j'_*$.*

(c) for every commutative diagram of the following type

$$\begin{tikzcd} X_1 \arrow[r, hook, "i"] \arrow[d, "p_1"] & X_2 \arrow[r, hook, "i'"] \arrow[d, "p_2"] & X_3 \arrow[d, "p_3"] \\ Y_1 \arrow[r, hook, "j"] & Y_2 \arrow[r, hook, "j'"] & Y_3, \end{tikzcd}$$

312 whose horizontal arrows are open immersions and whose vertical arrows are proper, the following triangle of isomorphisms d from (iii) is commutative:

$$\begin{tikzcd} p_{3*} i'_* i_* \arrow[rr] \arrow[dr, no head] & & j'_* j_* p_{1*} \arrow[dl, no head] \\ & j'_* p_{2*} i_* & \end{tikzcd} .$$

(c') For every commutative diagram of the following type

$$\begin{tikzcd} X_1 \arrow[r, "p"] \arrow[d, "i_1"] & X_2 \arrow[r, "p'"] \arrow[d, "i_2"] & X_3 \arrow[d, "i_3"] \\ Y_1 \arrow[r, "q"] & Y_2 \arrow[r, "q'"] & Y_3, \end{tikzcd}$$

whose horizontal arrows are proper and whose vertical arrows are open immersions, the following triangle of isomorphisms d from (iii) is commutative:

$$\begin{tikzcd} i_{3*} p'_* p_* \arrow[rr] \arrow[dr, no head] & & q'_* q_* i_{1*} \arrow[dl, no head] \\ & q'_* i_{2*} p_* & \end{tikzcd} .$$

313 Definition of f_*. Let f be a morphism from Y to X. Each compactification of f, with diagram (3.2.5.1), defines a composite functor $\overline{f}_* i_*$ from $F(Y)$ to $F(X)$. A morphism of

compactifications defines a commutative diagram

(3.3.2.2)
$$\begin{tikzcd}
Y \arrow[r, hook, "i_1"] \arrow[d, equal] & \overline{Y}_1 \arrow[d, "p"] \\
Y \arrow[r, hook, "i_2"] \arrow[d] & \overline{Y}_2 \arrow[dl, "\overline{f}_2"] \\
X &
\end{tikzcd}.$$

Datum (iii) therefore provides us with an isomorphism

$$\overline{f}_{2*}i_{2*} = \overline{f}_{2*} \circ (i_{2*} \circ 1_{Y*}) = \overline{f}_{2*} \circ (p_* \circ i_{1*}) = \overline{f}_{1*} \circ i_{1*}.$$

Axiom (c$'$) also ensures that the isomorphism associated with a composite of morphisms of compactifications is the composite of the isomorphisms associated with each of them. Since the category of morphisms of compactifications is cofiltered (3.2.6 (ii)), we thereby obtain a transitive system of isomorphisms between the functors associated with the various compactifications of f. Choosing one of them, we define the functor f_*.

Transitivity homomorphisms. Every compactification of a pair (f, g) of composable morphisms (diagram (3.2.5.2)) defines a compactification of f, one of g, one of $f \circ g = (\overline{f}\overline{\overline{g}})(j'j)$, and an isomorphism $c_{f,g} : (fg)_* \xrightarrow{\sim} f_*g_*$, the composite of the isomorphisms

$$(fg)_* = (\overline{g}\,\overline{\overline{g}})_*(j'j)_* = \overline{f}_*(\overline{\overline{g}}_*j'_*)j_* = \overline{f}_*(i_*\overline{g}_*)j_* = (\overline{f}_*i_*)(\overline{g}_*j_*) = f_*g_*$$

(where the middle isomorphism comes from (iii)). 314

Let us prove that this isomorphism does not depend on the compactification chosen for (f, g). By 3.2.6 (iii), it will suffice to compare the isomorphisms $c_{f,g}$ obtained from two compactifications $(3.2.5.2)_i$ $(i = 1, 2)$ such that there exists a morphism of compactifications from the first to the second. The arrows in these morphisms will be denoted by p.

Consider the following diagram of isomorphisms of functors. For readability, the $*$ have been omitted.

$$\begin{tikzcd}
\overline{f}_2\overline{\overline{g}}_2j'_2j_2 \arrow[rr, dash] \arrow[dr, dash] \arrow[dd, dash] \arrow[ddr, phantom, "(3)"] & & \overline{f}_2\overline{\overline{g}}_2pj'_1j_1 \arrow[rr, dash] \arrow[dl, dash] \arrow[dr, dash] \arrow[ddd, phantom, "(4)"] & & \overline{f}_1\overline{\overline{g}}_1j'_1j_1 \arrow[dl, dash] \arrow[dd, dash] \arrow[ddl, phantom, "(5)"] \\
& \overline{f}_2\overline{\overline{g}}_2j'_2pj_1 \arrow[d, dash] \arrow[u, phantom, "(1)"] & & \overline{f}_2p\overline{\overline{g}}_1j'_1j_1 \arrow[d, dash] \arrow[u, phantom, "(2)"] & \\
\overline{f}_2i_2\overline{g}_2j_2 \arrow[r, dash] & \overline{f}_2i_2\overline{g}_2pj_1 \arrow[dr, dash] & & \overline{f}_2pi_1\overline{g}_1j_1 \arrow[r, dash] \arrow[dl, dash] & \overline{f}_1i_1\overline{g}_1j_1 \\
& & \overline{f}_2i_21_Y\overline{g}_1j_1 & &
\end{tikzcd}.$$

a) The composite isomorphism along the 1st row is the isomorphism 3.3.2.1 between two definitions of $(fg)_*$. 315

b) The composite isomorphism along the last row is the isomorphism 3.3.2.1 between two definitions of $f_* g_*$.
c) The two outer vertical isomorphisms are the isomorphisms $c_{f,g}$ between $(fg)_*$ and $f_* g_*$ deduced from one or the other compactification.

The problem is therefore to prove that the outer boundary of the diagram is commutative. A commutative diagram of functors remains commutative when each of them is composed with the same functor. Bearing this in mind, the commutativity of (1) follows from (c) applied to the diagram

$$\begin{tikzcd}
Z \arrow[r, hook, "j_1"] \arrow[d, equal] & \overline{Z}_1 \arrow[r, hook, "j'_1"] \arrow[d] & \overline{\overline{Z}}_1 \arrow[d] \\
Z \arrow[r, hook, "j_2"] & \overline{Z}_2 \arrow[r, hook, "j'_2"] & \overline{\overline{Z}}_2 .
\end{tikzcd}$$

The commutativity of (3) and (5) is trivial; that of (2), which essentially concerns proper morphisms, is also trivial. It remains to consider the hexagon (4).

Consider the following commutative cube

$$\begin{tikzcd}
& \overline{Z}_2 \arrow[rr, hook, "j'_2"] \arrow[dd, "\overline{g}_2"] & & \overline{\overline{Z}}_2 \arrow[dd, "\overline{\overline{g}}_2"] \\
\overline{Z}_1 \arrow[ur] \arrow[rr, hook, "j'_1"] \arrow[dr, dashed] \arrow[dd, "\overline{g}_1"'] & & \overline{\overline{Z}}_1 \arrow[ur] \arrow[dr, dashed] \arrow[dd, "\overline{\overline{g}}_1"] & \\
& Y \arrow[rr, hook, "i_2"] \arrow[dl, equal] & & \overline{Y}_2 \\
Y \arrow[rr, hook, "i_1"'] & & \overline{Y}_1 \arrow[ur] & .
\end{tikzcd}$$

316 There are six ways to follow the edges of this cube from $\overline{Z}_1$ to $\overline{Y}_2$, each "adjacent" to two others; there is an isomorphism between the composite functors associated with two adjacent paths (i.e. paths separated by a single face), and the commutativity of (4) will follow from the commutativity of the resulting hexagon.

The moral of this kind of diagram is that an edge represents a functor, a face an isomorphism between composite functors, and a volume a compatibility condition (between the faces forming its boundary). We decompose the cube into two triangular prisms (along the "plane" $\overline{Z}_1\overline{\overline{Z}}_1 Y\overline{Y}_2$) in order to reduce to the hypotheses (c′).

More precisely, we complete the hexagon to form the following diagram of isomorphisms (for readability, the $*$ have been omitted):

$$
\begin{tikzcd}
 & & \overline{\overline{g}}_2 j_2' p \arrow[dll, dash] \arrow[drr, dash] & & \\
i_2 \overline{g}_2 p \arrow[dr, dash] \arrow[ddd, dash] & & & & \overline{\overline{g}}_2 p j_1' \arrow[dl, dash] \arrow[ddd] \\
 & i_2(g_2 p) \arrow[rr, dash] \arrow[dl, dash] & & (\overline{\overline{g}}_2 p) j_1' \arrow[dr, dash] & \\
i_2 1_Y \overline{g}_1 \arrow[drr, dash] & & & & p \overline{\overline{g}}_1 j_1' \arrow[dll, dash] \\
 & & p i_1 \overline{g}_1 & &
\end{tikzcd}
$$

The triangles are trivially commutative, and the pentagons are commutative by (c′). 317

Cocycle condition. Let us verify that the isomorphisms $c_{f,g}$ constructed in 3.3.2.3 satisfy a cocycle condition for a composable triple (f, g, h) of morphisms. Let us introduce a compactification of (f, g, h), with diagram (3.2.5.3), and consider the following diagram of isomorphisms of functors:

$$
\begin{tikzcd}
\overline{f} i \overline{g} j \overline{h} k \arrow[r, dash, "c_{f,g} * h_*"] \arrow[d, dash, "f_* * c_{g,h}"'] & \overline{f} \overline{\overline{g}} j' j \overline{h} k \arrow[d, dash] \arrow[ddr, dash, "c_{fg,h}"] & \\
\overline{f} i \overline{g} \overline{\overline{h}} k' k \arrow[r, dash] \arrow[drr, dash, "c_{f,gh}"'] & \overline{f} \overline{\overline{g}} j' \overline{\overline{h}} k' k \arrow[dr, dash] & \\
 & & \overline{f} \overline{\overline{g}} \overline{\overline{\overline{h}}} k'' k' k.
\end{tikzcd}
$$

The commutativity of the square is trivial, and that of the triangles follows from (c) and (c′). Since the commutativity of the outer boundary is what had to be proved, this completes the proof of 3.3.2.

3.3.3. Suppose that over (S, i) and (S, p) there are given, respectively, fibered categories equipped with normalized cleavages F_i and F_p, and suppose that F_i and F_p have the same fiber $F(X)$ for every $X \in \mathrm{Ob}\, S$.

According to SGA 1 VI, it "amounts to the same thing" to give instead:

(i bis) for every proper morphism $p : X \to Y$, a functor $p^* : F(Y) \to F(X)$,

(ii bis) for every composable pair (p, q) of proper morphisms, an isomorphism of functors $c_{p,q} : q^* p^* \leftrightarrow (pq)^*$,

(i′ bis) and (ii′ bis), analogous data for (S, i), these data satisfying the conditions (a bis) (b bis) (a′ bis) (b′ bis) dual to 3.3.1 (a) (b) (a′) (b′). 318

Suppose *given* in addition,

(iii bis) for every commutative diagram D

$$\begin{tikzcd} X \arrow[r, "j'"] \arrow[d, "p'"'] & Y \arrow[d, "p"] \\ Z \arrow[r, "j"] & T \end{tikzcd} \qquad j, j' \in \mathrm{Fl}(S, i);\ p, p' \in \mathrm{Fl}(S, p)$$

an isomorphism of functors $d(D) : j'^* p^* \leftrightarrow p'^* j^*$.

We leave to the reader the formulation of the conditions (c bis) and (c′ bis) dual to 3.3.2 (c) and (c′).

Arguments parallel to the preceding ones then prove the following proposition.

Proposition 3.3.4. *If data* (i bis) (ii bis) (i′ bis) (ii′ bis) (iii bis) *satisfy the conditions* (a bis) (b bis) (a′ bis) (b′ bis) (c bis) (c′ bis)*, there exists one and essentially only one category F cofibered over S and equipped with isomorphisms $F \times_S (S, i) = F_i$, $F \times_S (S, p) = F_p$, these isomorphisms being compatible with the identifications $F_{iX} = F_{pX}$ ($X \in \mathrm{Ob}\, S$) and such that the isomorphisms $d(D)$ of* (iii bis) *become the composite isomorphisms $p'^* j^* = (jp')^* = (pj')^* = j'^* p^*$.*

It should be noted, however, that 3.3.4 is not dual to 3.3.2, since the hypotheses imposed on (S, i) and (S, p) are not self-dual.

4. Resolutions. Application to the base change morphism

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4.1. Flat resolutions. Recall that if $f : (X, \mathscr{A}) \to (Y, \mathscr{B})$ is a morphism of ringed sites (or ringed topoi), the inverse image under f of a sheaf of left $\mathscr{B}$-modules $\mathscr{M}$ is the sheaf of left $\mathscr{A}$-modules $\mathscr{A} \otimes_{f^*\mathscr{B}} f^*\mathscr{M}$.

Proposition 4.1.1. *Let $f : (X, \mathscr{A}) \to (Y, \mathscr{B})$ be a morphism of ringed sites, let K be a sheaf of right $\mathscr{A}$-modules, and let $0 \to L \to M \to N \to 0$ be an exact sequence of sheaves of left $\mathscr{B}$-modules. If M and N are flat, then*

(i) *L is flat,*
(ii) *the following sequence is exact:*

$$0 \longrightarrow K \otimes_{\mathscr{A}} f^*L \longrightarrow K \otimes_{\mathscr{A}} f^*M \longrightarrow K \otimes_{\mathscr{A}} f^*N \longrightarrow 0.$$

Factor f into the arrows $(X, \mathscr{A}) \xrightarrow{u} (X, f^{-1}\mathscr{B}) \xrightarrow{v} (Y, \mathscr{B})$. The functor v^* is exact and takes flat modules to flat modules (V 1), so it suffices to prove 4.1.1 for u. The isomorphism

$$K \otimes_{\mathscr{A}} (\mathscr{A} \otimes_{f^{-1}\mathscr{B}} P) = K \otimes_{f^{-1}\mathscr{B}} P \tag{4.1.1.1}$$

then reduces us to the case where f is the identity.

Take an exact sequence $0 \to R \to P \to K \to 0$ with P flat. The rows and columns of the following diagram are exact:

$$\begin{tikzcd}
 & & 0 \arrow[d] & & \\
 & R \otimes L \arrow[r] \arrow[d] & P \otimes L \arrow[r] \arrow[d] & K \otimes L \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & R \otimes M \arrow[r] \arrow[d] & P \otimes M \arrow[r] \arrow[d] & K \otimes M \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & R \otimes N \arrow[r] \arrow[d] & P \otimes N \arrow[r] \arrow[d] & K \otimes M \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 & .
\end{tikzcd}$$

Apply the snake lemma to the first two columns: we obtain 320

$$0 \xrightarrow{j} K \otimes L \to K \otimes M,$$

which proves (ii). Now drop the assumption that P is flat. In the preceding diagram the three columns and two rows are short exact sequences; the first row is therefore exact, i.e. L is flat.

4.1.2. Let $\mathscr{S}$ be a subcategory of the category of ringed sites.[50] The sheaves of modules on the ringed sites in $\mathscr{S}$ form a category $\mathscr{M}$, abelian over $\mathscr{S}$ (2.2.2), fibered and cofibered, whose fibers are the opposites of the usual categories of sheaves of modules on a site. We denote by $K(\mathscr{S})$ the category $K(\mathscr{M})$ (2.2.8) and by $K^+(\mathscr{S})$ (resp. $K^-(\mathscr{S})$) the full subcategory of $K(\mathscr{S})$ whose fiber categories are the opposites of the categories $K^+(S)$ (resp. $K^-(S)$) for $S \in \operatorname{Ob} \mathscr{S}$. The categories $K(\mathscr{S})$, $K^+(\mathscr{S})$, and $K^-(\mathscr{S})$ are triangulated over $\mathscr{S}$, fibered and cofibered.

Let $f : S \to S'$ be an arrow of $\mathscr{S}$. Every injective object is acyclic (1.2.3) for the functor Rf_*, and every flat sheaf of modules is acyclic for the functor Lf^* (4.1.1 and 1.2.6), whence the derived functors (1.2.7) 321

$$Rf_* : D^+(S) \longrightarrow D^+(S') \tag{4.1.2.1}$$

$$Lf^* : D^-(S') \longrightarrow D^-(S). \tag{4.1.2.2}$$

If Rf_* (resp. Lf^*) has finite dimension, then in (4.1.2.1) (resp. (4.1.2.2)) one may replace the exponent $+$ (resp. $-$) by b, by $-$ (resp. $+$), or omit it (1.2.10).[51]

We denote by $D(\mathscr{S})$ (resp. $D^+(\mathscr{S})$, resp. $D^-(\mathscr{S})$) the category obtained from $K(\mathscr{S})$ (resp. $K^+(\mathscr{S})$, resp. $K^-(\mathscr{S})$) by inverting the quasi-isomorphisms.

Scholium 4.1.3. *("Trivial duality theorem").*

(a) *Assume that all the functors Rf_* (resp. Lf^*) have finite dimension. Then:*

(i) *The formation of $D(\mathscr{S})$ commutes with every change of base category $\mathscr{S}' \to \mathscr{S}$. The category $D(\mathscr{S})$ is triangulated over $\mathscr{S}$ (2.2.3), and its fibers identify with the opposites of the categories $D(S)$ for $S \in \operatorname{Ob} \mathscr{S}$.*

[50] N.D.E.: the theory is valid more generally for categories $\mathscr{S}$ consisting of ringed sites.

[51] N.D.E.: using Spaltenstein resolutions, the exponents may be omitted unconditionally. In the commutative case, see M. Kashiwara and P. Schapira, *Categories and Sheaves*, Springer, 2010, cited as [KS] below, 18.6.

(ii) *The category $D(\mathcal{S})$ is cofibered (resp. fibered) over $\mathcal{S}$. Let $f : S \to S'$; if Lf^* (resp. Rf_*) has finite dimension, then every object of $D(S')$ (resp. $D(S)$) admits an inverse image (resp. direct image) under f in the sense of the category $D(\mathcal{S})$ over $\mathcal{S}$. The inverse image and direct image functors identify with Lf^* and Rf_*.*

322 (iii) *Every object of $D^-(S')$ (resp. $D^+(S)$) admits an inverse image (resp. direct image) in the sense of $D(\mathcal{S})$, and the inverse image (resp. direct image) functor identifies with Lf^* (resp. Rf_*).*[52]

(b) (i) *The formation of $D^-(\mathcal{S})$ (resp. $D^+(\mathcal{S})$) commutes with every change of base category $\mathcal{S}' \to \mathcal{S}$. The category $D^-(\mathcal{S})$ (resp. $D^+(\mathcal{S})$) is triangulated over $\mathcal{S}$, and its fibers identify with the opposites of the categories $D^-(S)$ (resp. $D^+(S)$).*

(ii) *The category $D^-(\mathcal{S})$ (resp. $D^+(\mathcal{S})$) is fibered (resp. cofibered) over $\mathcal{S}$; the "inverse image" (resp. "direct image") functors identify with the derived functors Lf^* (resp. Rf_*).*

(iii) *If $f : S \to S'$ is such that Rf_* (resp. Lf^*) has finite dimension, every object of $D^-(S)$ (resp. $D^+(S')$) admits a direct image in the sense of $D^-(\mathcal{S})$ (resp. an inverse image in the sense of $D^+(\mathcal{S})$), and the "direct image" (resp. "inverse image") functor identifies with Rf_* (resp. Lf^*).*

It suffices to verify that in each case the hypotheses of 2.4.1 or 2.4.3 are satisfied, the other assertions following, after a base change $\{S\} \to \mathcal{S}$, or $I \to \mathcal{S}$, from 2.3.5 and 1.2.10. In case (b), complexes of flat (resp. flasque) sheaves satisfy the hypotheses of 2.4.3 (resp. 2.4.1). In case (a), we use 2.4.1 (resp. 2.4.3) so that we need only consider those f for which the dimension of Rf_* (resp. Lf^*) is bounded by a fixed N. Let us prove that there are "enough" complexes K whose components have their direct (resp. inverse) images all acyclic for the direct image (respectively, inverse image) functors under con-
323 sideration. Let K be a complex and let K^* (resp. K_*) be a Cartan-Eilenberg resolution of K by complexes with flasque components (resp. flat components).[53] If $n > N$, we see that $f_*\tau''_{<n}(K^*) = \tau''_{<n}f_*(K^*)$ (resp. $f^*\tau''_{>-n}(K_*) = \tau''_{>-n}f^*(K_*)$), where τ'' denotes a truncation with respect to the second differential. For $n > 2N$, it follows that $\tau''_{<n}(K^*)$ (resp. $\tau''_{>-n}(K_*)$) satisfies the hypothesis of 2.4.1 (resp. 2.4.3), and every complex K is the source (resp. target) of a quasi-isomorphism whose target (resp. source) is such a complex.

4.1.4. Consider a commutative square of ringed sites (4.1.4.1) (cf. 0.11)

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ S' \arrow[r, "g"] & S. \end{tikzcd} \tag{4.1.4.1}$$

We may consider the subcategory of the category of sites consisting of this diagram;[54] we then use 2.1.3 to define a "base change morphism" from Lg^*Rf_* to $Rf'_*Lg'^*$ whenever 4.1.3 is applicable:

Proposition 4.1.5. *The base change morphism of 2.1.3, from Lg^*Rf_* to $Rf'_*Lg'^*$, is defined in each of the following cases:*

[52]N.D.E.: the finite-dimension hypotheses in (a) are superfluous. Cf. the preceding N.D.E.

[53]N.D.E.: it is required only that the augmentation $K^p \to (K^*)^{p\bullet}$ (resp. $(K_*)^{p\bullet} \to K^p$) be a flasque (resp. flat) resolution for every p.

[54]N.D.E.: one must instead consider the category of ringed sites defined by this diagram, which is not strictly speaking a subcategory of the category of ringed sites in general. Cf. la N.D.E. (50) page 673.

(i) *We work in the categories* D^+, *and* Lg^* *and* Lg'^* *have finite dimension.*
(ii) *We work in the categories* D^-, *and* Rf_* *and* Rf'_* *have finite dimension.*
(iii) *We work in the unbounded derived categories, and* Rf_*, Rf'_*, Lg^*, *and* Lg'^* *have finite dimension, as do* Lf^* *and* Lf'^* *(or: as do* Rg_* *and* Rg'_**).*[55]

It should be noted that the last hypothesis in (iii) looks rather like a practical joke. 324
The editor does not know how to dispense with it except when the sites under consideration have enough points (4.2.12). The functorial properties of this base change morphism follow at once from the description in 2.1.3.

4.1.6. Let $(S, \mathscr{A})$ be a ringed site, and consider the "tensor product" functor

$$\otimes : \mathrm{Mod}(S, \mathscr{A}^\circ) \times \mathrm{Mod}(S, \mathscr{A}) \longrightarrow \mathrm{Mod}(S, C)$$

where C is the center of $\mathscr{A}$.

The recipe of 1.1.6, in applying which the simple complex associated with a double complex is defined using a sum, extends this functor to a functor still denoted $\otimes$:

$$\otimes : K(S, \mathscr{A}^\circ) \times K(S, \mathscr{A}) \longrightarrow K(S, C).$$

Proposition 4.1.7. *With the preceding notation:*

(i) *The tensor-product functor is left derivable (1.2.1) to a "total tensor product" functor*

$$\overset{L}{\otimes} : D^-(S, \mathscr{A}^\circ) \times D^-(S, \mathscr{A}) \longrightarrow D^-(S, C);$$

(ii) *a pair* (K, L) *of bounded-above complexes is adapted to* $\overset{L}{\otimes}$ *whenever either* K *or* L *has flat components.*

The proof is standard.

Notation 4.1.8. *If* $K \in D^-(S, \mathscr{A}^\circ)$ *and* $L \in D^-(S, \mathscr{A})$, *denote by* $K \overset{L}{\otimes} L$ *the total* 325
tensor product of K *and* L. *The sheaf* $H_k(K \overset{L}{\otimes} L) = H^{-k}(K \overset{L}{\otimes} L)$ *is denoted by* $\mathrm{Tor}_k(K, L)$ *and is called the* k^{th} *local hypertor of* K *and* L.

Let $K \in D^-(S, \mathscr{A}^\circ)$ and $L \in D^-(S, \mathscr{A})$. If $H^i(K) = 0$ for $i > k$ and $H^i(L) = 0$ for $i > \ell$, one checks that $H^i(K \overset{L}{\otimes} L) = 0$ for $i > k + \ell$ and that $H^{k+\ell}(K \overset{L}{\otimes} L) = H^k(K) \otimes H^\ell(L)$. Fix K and regard $K \overset{L}{\otimes} L$ as a functor of L. If Definition 1.2.9 of cohomological amplitude as an interval in $\mathbf{Z}$ is applied, the upper endpoint of this interval is trivial in nature: it is the largest integer i such that $H^i(K) \neq 0$. The lower endpoint is more interesting:

Definition 4.1.9. *A complex* $K \in \mathrm{Ob}\, D^b(S, \mathscr{A}^\circ)$ *is said to have* tor-dimension $\leq n$ *if the following equivalent conditions hold:*

(i) *For every* $\mathscr{A}$*-module* L, $\mathrm{Tor}_k(K, L) = 0$ *for* $k > n$.
(ii) *For every complex of* $\mathscr{A}$*-modules* L *such that* $H^i(L) = 0$ *for* $i < \ell$, *one has* $H^i(K \overset{L}{\otimes} L) = 0$ *for* $i < \ell - n$.
(iii) *There is a quasi-isomorphism with target* K *and source a bounded complex* K' *whose components are flat and vanish for* $i < -n$.

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[55]N.D.E.: the finite-dimension hypotheses in (iii) are superfluous. Cf. la N.D.E. (51) page 673.

4.1.10. Here are some possible variants:

(i) If $K \in \mathrm{Ob}\, D^b(S, \mathscr{A}^\circ)$ has finite tor-dimension and $L \in \mathrm{Ob}\, D(S, \mathscr{A})$, then the functor $\overset{L}{\otimes}$ is defined (1.2.1) at (K, L).
(ii) If $K \in \mathrm{Ob}\, K^b(S, \mathscr{A}^\circ)$ has components of finite tor-dimension, the functor $L \mapsto K \otimes L$ ($L \in K(S, \mathscr{A})$) is left derivable.
(iii) If every module has finite tor-dimension,[56] the functor $\otimes$ derives to a functor

$$\overset{L}{\otimes} : D(S, \mathscr{A}^\circ) \times D(S, \mathscr{A}) \longrightarrow D(S, \mathscr{C}).$$

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4.2. Godement flasque resolutions.
4.2.1. Let S be a topos and X a set. To give a family of points of S indexed by X is equivalently to give a morphism $k : \mathrm{Top}(X) \to S$ from the topos defined by the discrete topological space X to S. For this family to be conservative (IV 6.4.1), it is necessary and sufficient that, for every sheaf F on S, the adjunction arrow from F to $k_* k^* F$ be a monomorphism (cf. IV 6.4.0).

Definition 4.2.2. *Let $k : \mathrm{Top}(X) \to S$ be a conservative family of points of a topos S. The* Godement resolution, *or* canonical flasque resolution *(relative to X), of an abelian sheaf F on S, denoted $\mathscr{C}^*_X(F)$ (or simply $\mathscr{C}^*(F)$), is the following right resolution of F:*

1) *$\mathscr{C}^0(F) = k_* k^* F$ and $\epsilon : F \to \mathscr{C}^0(F)$ is the adjunction arrow.*
2) *Set $d^{-1} = \epsilon$; recursively, for $n \geq 0$, define*
$\mathscr{C}^{n+1}(F) = \mathscr{C}^0(\mathrm{coker}\, d^{n-1})$, and define d^n as the composite arrow

$$d^n : \mathscr{C}^n(F) \to \mathrm{coker}(d^{n-1}) \to \mathscr{C}^0(\mathrm{coker}(d^{n-1})).$$

Proposition 4.2.3. *Under the hypotheses of 4.2.2,*

(i) *$\mathscr{C}^n(F)$ is a flasque sheaf V 4.1,*
(ii) *$\mathscr{C}^n(F)$ is an exact functor of F,*
(iii) *the stalk at a point $x \in X$ of the complex $\mathscr{C}^*(F)$ is a canonically split resolution of F_x.*

327 The sheaves $\mathscr{C}^n(F)$ are flasque because they are direct images of sheaves (automatically flasque) on the discrete topological space X.

The functors k^* and k_* are exact. Hence the functor $\mathscr{C}^0$ is exact. We prove by induction on $n \geq 0$ that the functor $\mathscr{Z}^n(F) = \mathrm{coker}(d^{n-1})$ is exact, and therefore so is $\mathscr{C}^{n+1} = \mathscr{C}^0 \mathscr{Z}^n$. Set $\mathscr{Z}^{-1}(F) = F$. An exact sequence of sheaves

$$0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0$$

[56] N.D.E.: this condition is superfluous. In the case where $\mathscr{A}$ is commutative, see [KS], 18.6, cited in la N.D.E. (51) page 673.

defines, for $n \geq 0$, a diagram

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & \mathscr{Z}^{n-1}(F') \arrow[r] \arrow[d] & \mathscr{Z}^{n-1}(F) \arrow[r] \arrow[d] & \mathscr{Z}^{n-1}(F'') \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathscr{C}^{n}(F') \arrow[r] \arrow[d] & \mathscr{C}^{n}(F) \arrow[r] \arrow[d] & \mathscr{C}^{n}(F'') \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \mathscr{Z}^{n}(F') \arrow[r] \arrow[d] & \mathscr{Z}^{n}(F) \arrow[r] \arrow[d] & \mathscr{Z}^{n}(F'') \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 & ,
\end{tikzcd}$$

whose columns are exact, as are, by the induction hypothesis, the first two rows. The 3rd row is therefore exact, proving (ii).

The short exact sequences

$$0 \longrightarrow \mathscr{Z}^{n+1}(F) \longrightarrow \mathscr{C}^{n}(F) \longrightarrow \mathscr{Z}^{n}(F) \longrightarrow 0$$

are special cases of the short exact sequence

$$0 \to F \xrightarrow{\epsilon} \mathscr{C}^0(F) \to \mathscr{Z}^0(F) \to 0 \tag{4.2.3.1}$$

(make the substitution $F \mapsto \mathscr{Z}^{n-1}(F)$). To prove (iii), it remains to show that the 328
inverse-image sequence of (4.2.3.1) under k is canonically split. Indeed, the composite arrow

$$k^*F \xrightarrow{k^*(\epsilon)} k^*\mathscr{C}^0F = k^*(k_*k^*)F = (k^*k_*)k^*F \longrightarrow k^*F$$

is the identity.

Remark 4.2.4. If $(x_i)_{i\in X}$ is a conservative family of points of a site, this family corresponds to a conservative family of points of the associated topos, which again makes it possible to define canonical flasque resolutions.

Remark 4.2.5. If S is a ringed topos, the functors $\mathscr{C}^n$ carry modules to modules.

Remark 4.2.6. Let $\mathscr{S}$ be a category consisting of sites. Suppose that, for each $S \in \operatorname{Ob}\mathscr{S}$, a set S^d and a conservative family $k_S : \operatorname{Top}(S^d) \to S$ of points of S indexed by S^d are given. Suppose that S^d is functorial in S and that the diagrams

$$\begin{tikzcd}
\operatorname{Top}(S^d) \arrow[r, "f^d"] \arrow[d, "k_S"] & \operatorname{Top}(T^d) \arrow[d, "k_T"] \\
S \arrow[r, "f"] & T
\end{tikzcd}$$

are commutative. The corresponding canonical flasque resolutions are then "functorial in S."

If $\mathscr{S}$ is the category of topological spaces, one may take S^d to be the underlying set. 329
If $\mathscr{S}$ is the category of the étale sites of schemes S locally of finite type over a scheme S_0, and if, for each $s \in S_0$, $\overline{k(s)}$ is an algebraic closure of $k(s)$, one may take for S^d the set of points of S with values in one of the $\overline{k(s)}$.

The condition "of finite type" may be replaced by a cardinality condition if the algebraic closures $\overline{k(s)}$ are replaced by sufficiently large algebraically closed extensions.

4.2.7. One advantage of canonical flasque resolutions is that they make it possible to construct complexes adapted simultaneously to functors of "tensor-product" type and to functors of "direct-image" type. They thereby give a perhaps more comprehensible construction of the base-change morphisms (4.1.5).

LEMMA 4.2.8. *Let $f : (S, \mathscr{A}) \to (S', \mathscr{B})$ be a morphism of ringed sites, let $K \in \operatorname{Ob} K(S, \mathscr{A}^\circ)$ and $L \in \operatorname{Ob} K(S', \mathscr{B})$ be complexes, and let $k : \operatorname{Top}(X) \to S$ be a conservative set of points of S. Suppose that one of the following conditions holds:*

(a) *The cohomologies of K and L are bounded above;*
(b) *$\mathscr{A}$ has finite tor-dimension over $f^{-1}\mathscr{B}$, and $K \in \operatorname{Ob} D^b(S, \mathscr{A}^\circ)$ has finite tor-dimension;*
(c) *$L \in \operatorname{Ob} D^b(S', \mathscr{B})$ has finite tor-dimension.*

330 *Then the functor $(K, L) \mapsto K \otimes f^*L$ is derivable on the left (1.2.1) at (K, L); denote its derived functor by $\overset{\mathbf{L}}{\otimes} \mathbf{L}f^*$.*

(i) *If x is a point of S, then $(K \overset{\mathbf{L}}{\otimes} \mathbf{L}f^* L)_x \simeq K_x \overset{\mathbf{L}}{\otimes}_{\mathscr{B}_{f(x)}} L_{f(x)}$.*

(ii) *The pair (K, L) is left adapted to $\overset{\mathbf{L}}{\otimes} f^*$ (1.2.1) if and only if, for every $x \in X$, the pair $(K_x, L_{f(x)})$ is left adapted to the functor $\otimes_{\mathscr{B}_{f(x)}}$.*

(iii) *Under hypotheses (a) or (b), the functor $K \overset{\mathbf{L}}{\otimes} f^* L$ in L has cohomological amplitude $\subset d$ (1.2.9) if and only if, for every $x \in X$, the complex K_x of $\mathscr{B}_{f(x)}$-modules has tor-amplitude $\subset d$.*[57]

This lemma, whose proof is standard, expresses the "pointwise nature" of the functor under consideration.

4.2.9. Let L be a complex. The *Godement resolution* (relative to X) of L is the simple complex (*defined using sums*) associated with the double complex $\mathscr{C}^*L$ (1.1.4). The *Godement resolution truncated at order n*, $\tau''_{\leq n}\mathscr{C}^*L$, is the simple complex associated with the double complex obtained from $\mathscr{C}^*L$ by truncation (1.1.15) in the direction of the Godement differential. By 4.2.3 (iii), L is, point by point on X, homotopy equivalent to the complexes $\tau''_{\leq n}\mathscr{C}^*L$ and $\mathscr{C}^*L$; these complexes are therefore "as good as L" with respect to functors of "tensor-product" type (4.2.8).

PROPOSITION 4.2.10. *Let $(S, \mathscr{A})$ be a ringed site, let X be a conservative set of points of S, and let $N, M \in \mathbf{N} \cup \{\infty\}$, with N or M finite. Let A be the subcategory of $K(S)$ consisting of complexes L such that:*

(a) *If $N = \infty$, then $H^i(L) = 0$ for i sufficiently large; if $M = \infty$, then $H^i(L) = 0$ for i sufficiently small.*

331 (b) *For every $x \in X$ and every $\mathscr{A}_x$-module Q of tor-dimension $\leq N$, the complex L_x is left adapted to the functor $Q \otimes_{\mathscr{A}_x} *$.*

(c) *For every morphism of topoi $f : S \to E$ of cohomological dimension $\leq M$, the complex L is right adapted to the functor f_*.*

Then the inclusion of A into $K(S)$ induces an equivalence between the category obtained from A by inverting quasi-isomorphisms and the category $D^*(S)$, where $* = +$ if $N < M = \infty$, $* = -$ if $M < N = \infty$, and $*$ is blank if $N, M < \infty$.

[57] N.D.E.: hypotheses (a), (b), and (c) are all superfluous. Cf. la N.D.E. (51) page 673.

Let K be a complex satisfying (a). If $N = \infty$, then the canonical arrow $a : K_1 = \tau_{\leq i} K \to K$ is a quasi-isomorphism for i sufficiently large. If $N < \infty$, set $K_1 = K$. Let K_1^* be a flat Cartan–Eilenberg resolution and set $K_2 = \tau''_{\geq -N} K_1$ (truncation in the direction of the new differential). The composite arrow from K_2 to K is a quasi-isomorphism, and K_2 satisfies (b).

Let K be a complex satisfying (a) and (b). If $M = \infty$, then $N < \infty$; for i sufficiently large, $b : K \to \tau_{\geq i} K_1 = K_3$ is a quasi-isomorphism, and K_3 still satisfies (a) and (b). If $M < \infty$, set $K = K_3$. Let $K_4 = \tau''_{\leq M} \mathscr{C}^* K_3$. Then, by 4.2.3 (iii), K_4 satisfies (a), (b), and (c).

The proposition therefore follows from two successive applications of [11, I 2.4.2].

Variants 4.2.11. If $N < \infty$ (resp. if $M < \infty$, resp. if $N, M < \infty$), one could equally have worked with bounded-below (resp. bounded-above, resp. bounded) complexes.

4.2.12. Here is how these constructions may be used to define the *base-change morphism* of 4.1.4 under hypotheses more general than those of *loc. cit.*, when the sites under consideration have enough points. 332

Let there be a commutative diagram (cf. (0.11)) of sites

(4.2.12.1)
$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"] & X \arrow[d, "f"] \\ S' \arrow[r, "g"] & S, \end{tikzcd}$$

let $\mathscr{A}$ be a sheaf of rings on S, $\mathscr{A}'$ a sheaf of rings on S', and L a complex of $(\mathscr{A}', g^*\mathscr{A})$-bimodules on S'. Suppose that:

(i) the sites X and X' admit conservative families of points, $k : Q \to X$ and $k' : Q' \to X'$, fitting into a commutative diagram

(4.2.12.2)
$$\begin{tikzcd} Q' \arrow[r, "g'_0"] \arrow[d, "k'"] & Q \arrow[d, "k"] \\ X' \arrow[r, "g'"] & X. \end{tikzcd}$$

(ii) One of the following hypotheses holds:
   (a) The functors Rf_* and Rf'_* have finite cohomological dimension, and L is bounded above. Set $s = -$.
   (b) L is bounded below, with components of uniformly bounded tor-dimension (as $g^*\mathscr{A}$-modules). Set $s = +$.
   (c) L is bounded, and the hypotheses of (a) and (b) hold. Set $s =$ blank.

We denote by $L \overset{\mathrm{L}}{\otimes} \mathrm{L}g^* : D^s(S, \mathscr{A}) \to D^s(S', \mathscr{A}')$ the left derived functor (1.2.1) of the functor $K \mapsto L \otimes_{g^*\mathscr{A}} g^*K$ from $K^s(S, \mathscr{A})$ to $K^s(S', \mathscr{A}')$. 333

We shall define a base-change morphism

$$\varphi : L \overset{\mathrm{L}}{\otimes} \mathrm{L}g^* Rf_* K \longrightarrow Rf'_*(f'^* L \overset{\mathrm{L}}{\otimes} \mathrm{L}g'^* K) \tag{4.2.12.3}$$

for K in $D^s(X, \mathscr{A})$.[58]

[58] N.D.E.: Adjunctions in unbounded derived categories make it possible to define such a morphism without hypotheses (i) and (ii). Cf. la N.D.E. (51) page 673.

By 4.2.10, it suffices to define it, functorially in K, for the complexes belonging to the category A of 4.2.10, with

$$N \geq \sup_i \text{ (tor-dimension of } L^i \text{ over } g^*)$$
$$M \geq \text{cohomological dimension of } Rf_*.$$

Such a complex is adapted both to the functor Rf_* and to the functor $f'^* L \overset{\mathbf{L}}{\otimes}_{\mathscr{A}'} \mathrm{L}g'^*$ (4.2.8). The canonical morphisms

$$\alpha : f_*K \longrightarrow Rf_*K$$
$$\beta : f'^* L \overset{\mathbf{L}}{\otimes} \mathrm{L}g'^* K \longrightarrow f'^* L \otimes g'^* K$$

are therefore quasi-isomorphisms. To compute the left-hand side I of (4.2.12.3), it therefore suffices to take a left resolution (i.e. a quasi-isomorphism)

$$\gamma : K_1 \longrightarrow f_*K,$$

with K_1 adapted to $L \overset{\mathbf{L}}{\otimes} \mathrm{L}g^*$; we have

$$I \simeq L \otimes g^* K_1.$$

334 Similarly, to compute the right-hand side II of (4.2.12.3), it suffices to take a right resolution

$$\delta : f'^* L \otimes g'^* K \longrightarrow K_2,$$

with K_2 adapted to Rf'_*; then

$$II \simeq f'_* K_2.$$

We then define the base-change morphism as the composite

$$I \simeq L \otimes g^* K_1 \xrightarrow{\gamma} L \otimes g^* f_* K \xrightarrow{c} f'_*(f'^* L \otimes g'^* K) \xrightarrow{\delta} f'_* K_2 \simeq II,$$

where c is the usual base-change morphism at the level of actual complexes.

This base-change morphism has the following characteristic property, which in principle makes it easy to verify its compatibility properties.

Proposition 4.2.13. *Under the preceding hypotheses, let $K \in \operatorname{Ob} C^s(X, \mathscr{A})$, $K_1 \in \operatorname{Ob} C^s(S, \mathscr{A})$ and $K_2 \in \operatorname{Ob} C^s(X', \mathscr{A})$. Moreover, let u be an f-morphism from K to K_1, i.e. $u \in \operatorname{Hom}(K_1, f_*K) \simeq \operatorname{Hom}(f^*K_1, K)$, and let v be a morphism from $f'^* L \otimes g'^* K$ to K_2. From u, v, and the usual base-change morphism, one obtains a morphism of complexes $c_{u,v} : L \otimes g^* K_1 \to f'_* K_2$. Let u', v', and c'_{uv} denote the composite morphisms*

$$u' : K_1 \xrightarrow{u} f_* K \to Rf_* K$$
$$v' : L \overset{\mathbf{L}}{\otimes} \mathrm{L}g'^* K \to L \otimes g'^* K \xrightarrow{v} K_2$$
$$c'_{u,v} : L \overset{\mathbf{L}}{\otimes} \mathrm{L}g'^* K_1 \to L \otimes g'^* K_1 \xrightarrow{c_{uv}} f'_* K_2 \to Rf'_* K_2.$$

335 *The following diagram is commutative:*

$$
\begin{tikzcd}
L \overset{\mathbf{L}}{\otimes} \mathbf{L}g^* Rf_*K \arrow[r, "\varphi"] & Rf'_*(f'^*L \overset{\mathbf{L}}{\otimes} \mathbf{L}g'^*K) \arrow[d, "Rf'_*(v')"] \\
L \overset{\mathbf{L}}{\otimes} \mathbf{L}g^* K_1 \arrow[u, "L\overset{\mathbf{L}}{\otimes}\mathbf{L}g^*(u')"'] \arrow[r, "c'_{uv}"] & Rf'_*K_2.
\end{tikzcd}
\tag{4.2.13.1}
$$

Suppose first that $K_1 = f_*K$ and that $K_2 = f'^*L \otimes g'^*K$, and consider the following commutative diagram of complexes:

$$
\begin{tikzcd}
K^a \arrow[r] \arrow[d] & K \arrow[d] \\
K^c \arrow[r] & K^b
\end{tikzcd}
$$

such that K^a is adapted to $f'^*L \overset{\mathbf{L}}{\otimes} \mathbf{L}g'^*$, K^b is adapted to Rf_*, and K^c is adapted to both functors. To construct this diagram, we first construct K^a, then apply a canonical flasque resolution (truncated if necessary) to K^a and K.

Apply the functor f_* (resp. $L \otimes g'^*$) to this diagram, and resolve the complexes in the diagram $(K_1, K_1^a, K_1^b, K_1^c)$ (resp. $(K_2, K_2^a, K_2^b, K_2^c)$) so as to obtain a commutative cube, in which $\approx$ denotes a quasi-isomorphism:

$$
\begin{tikzcd}
 & K'_1 \arrow[rr, "\approx"] \arrow[dd] & & K_1 \arrow[dd] \\
K_1^{a'} \arrow[ur] \arrow[rr, "\approx"] \arrow[dd] & & K_1^a \arrow[ur] \arrow[dd] & \\
 & K_1^{b'} \arrow[rr, "\approx"] & & K_1^b \\
K_1^{c'} \arrow[ur, "\approx"] \arrow[rr, "\approx"] & & K_1^c \arrow[ur, "\approx"'] &
\end{tikzcd}
$$

where K'_1, $K_1^{a'}$, $K_1^{b'}$, and $K_1^{c'}$ are adapted to the functor $L \overset{\mathbf{L}}{\otimes} \mathbf{L}g^*$ (resp. 336

$$
\begin{tikzcd}
 & K_2 \arrow[rr, "\approx"] \arrow[dd] & & K'_2 \arrow[dd] \\
K_2^{a} \arrow[ur] \arrow[rr, "\approx"] \arrow[dd, "\approx"] & & K_2^{a'} \arrow[ur] \arrow[dd, "\approx"] & \\
 & K_2^{b} \arrow[rr, "\approx"] & & K_2^{b'} \\
K_2^{c} \arrow[ur] \arrow[rr, "\approx"] & & K_2^{c'} \arrow[ur] &
\end{tikzcd}
,
$$

where K'_2, $K_2^{a'}$, $K_2^{b'}$, and $K_2^{c'}$ are adapted to the functor Rf'_*). Using the usual base-change morphisms, one obtains a commutative prism of complexes over S.

$$
\begin{tikzcd}[column sep=small]
& L\otimes g^*K_1' \arrow[rr, "4"'] \arrow[dd, "2"] & & L\otimes g^*K_1 \arrow[rr, "4"'] \arrow[dd] & & f'_*K_2 \arrow[rr, "4"'] \arrow[dd] & & f'_*K_2' \arrow[dd] \\
L\otimes g^*K_1^{a'} \arrow[ur] \arrow[rr] \arrow[dd] & & L\otimes g^*K_1^{a} \arrow[ur] \arrow[rr] \arrow[dd] & & f'_*K_2^{a} \arrow[ur] \arrow[rr] \arrow[dd] & & f'_*K_2^{a'} \arrow[ur, "3"'] \arrow[dd, "3^{-1}", "\wr"'] & \\
& L\otimes g^*K_1^{b'} \arrow[rr] & & L\otimes g^*K_2^{b} \arrow[rr] & & f'_*K_2^{b} \arrow[rr] & & f'_*K_2^{b'} \\
L\otimes g^*K_1^{c'} \arrow[ur, "2^{-1}", "\wr"'] \arrow[rr, "1"'] & & L\otimes g^*K_2^{c} \arrow[ur] \arrow[rr, "1"'] & & f'_*K_2^{c} \arrow[ur] \arrow[rr, "1"'] & & f'_*K_{2'}^{c'} \arrow[ur] &
\end{tikzcd}
$$

337 and one concludes by noting that the morphisms in diagram (4.2.13.1) are identified with the following composites:

for φ: the composite of the morphisms ①;

for $L \overset{\mathrm{L}}{\otimes} \mathrm{L}g^*(u')$: the composite of the morphisms ②, one of which is the inverse of a quasi-isomorphism;

for $Rf'_*(v')$: the composite of the morphisms ③, one of which is the inverse of a quasi-isomorphism;

for c'_{uv}: the composite of the morphisms ④.

In the general case, set $K_1' = f_*K$ and $K_2' = f'^*L \otimes g^*K$. We then have a diagram

$$
\begin{tikzcd}
& L \overset{\mathrm{L}}{\otimes} \mathrm{L}g^*Rf_*K \arrow[r, "\varphi"] & Rf'_*(f'^*L \overset{\mathrm{L}}{\otimes} \mathrm{L}g'^*K) \arrow[d] \arrow[dr] & \\
L \overset{\mathrm{L}}{\otimes} \mathrm{L}g^*K_1 \arrow[r] \arrow[ur] & L \overset{\mathrm{L}}{\otimes} \mathrm{L}g^*K_1' \arrow[r] \arrow[u] & Rf'_*K_2' \arrow[r] & Rf'_*K_2,
\end{tikzcd}
$$

whose triangles are commutative, as is the inner rectangle by the preceding argument. The boundary is therefore commutative, and this proves 4.2.13.

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4.3. The base-change theorem. We shall reformulate in terms of derived categories the base-change theorem for proper morphisms XII 5.1 (iii).

Theorem 4.3.1. *Let*

$$
\begin{tikzcd}
X' \arrow[r, "g'"] \arrow[d, "f'"] & X \arrow[d, "f"] \\
S' \arrow[r, "g"] & S
\end{tikzcd}
$$

be a Cartesian diagram of schemes, with f proper. Let $\mathscr{A}$ and $\mathscr{A}'$ be sheaves of torsion rings on S and S', and let L be a bounded-above complex of $(\mathscr{A}', g^*\mathscr{A})$-bimodules. Assume that the dimensions of the fibers of f are bounded. Hypothesis 4.2.12 (a) *is then satisfied, and the base-change morphism (4.2.12.3), between functors from $D^-(X, f^*\mathscr{A})$ to $D^-(S', \mathscr{A}')$,*

$$\varphi_K : L \overset{\mathrm{L}}{\otimes} \mathrm{L}g^*Rf_*K \longrightarrow Rf'_*(f'^*L \overset{\mathrm{L}}{\otimes} \mathrm{L}g'^*K),$$

is an isomorphism.

The functors under consideration are triangulated and "way-out" [6, I 7], so, to verify that φ_K is an isomorphism, it suffices to verify that the $\varphi_{\mathscr{H}^i(K)}$ are isomorphisms. We may therefore assume that K is a sheaf of torsion $f^*\mathscr{A}$-modules concentrated in degree 0.

To verify that φ_K is an isomorphism, it suffices to do so after applying the forgetful 339
functor from the category of $\mathscr{A}'$-modules to that of $\mathbf{Z}/n\mathbf{Z}$-modules, where $n\mathscr{A} = n\mathscr{A}' = 0$.

This reduces us to the case $\mathscr{A}' = \mathbf{Z}/n\mathbf{Z}$, with $n\mathscr{A} = 0$. We may then replace L by an isomorphic complex in the derived category, bounded above and with components flat over $g^*\mathscr{A}$. Since the functors under consideration are triangulated and "way- out" in L, it suffices to verify that φ_K is an isomorphism after replacing L by one of its components. We may therefore assume that L is a sheaf of flat $g^*\mathscr{A}$-modules placed in degree 0. We must prove that the morphisms

$$\varphi^q : L \otimes g^* R^q f_* K \longrightarrow R^q f'_*(f'^* L \otimes g^* K)$$

are isomorphisms. Let s be a geometric point of S', let X_s be the fiber of X' at s, and let K_s be the inverse image of K on X_s. By XII 5.1 (iii), the fiber at s of φ^q is the morphism

$$\varphi^q_s : L_s \otimes H^q(X_s, K) \longrightarrow H^q(X_s, L_s \otimes K).$$

By *D*. Lazard's theorem [7], the flat $\mathscr{A}_s$-module L_s is the filtered direct limit of free $\mathscr{A}_s$-modules $L_{s,i}$; the functor $H^q(X_s, *)$ commutes with filtered direct limits (VII 3.3). The morphism φ^q is therefore the direct limit of the morphisms

$$\varphi^q_i : L_{s,i} \otimes H^q(X_s, K) \longrightarrow H^q(X_s, L_{s,i} \otimes K).$$

The latter are plainly isomorphisms; hence so is φ^q_s, and φ^q is an isomorphism.

VARIANTS 4.3.2. (i) When L is a bounded complex whose components have finite tor-dimension over $g^*\mathscr{A}$, hypotheses (c) of 4.2.12 are satisfied, and one may work in the unbounded derived category.[59] 340

(ii) If L is bounded below, with components whose tor-dimension is uniformly bounded over $g^*\mathscr{A}$, hypotheses (b) of 4.2.12 are satisfied, even if $\mathscr{A}$ and $\mathscr{A}'$ are not torsion; the base-change morphism φ_K ($K \in D^+(X, f^*\mathscr{A})$) will be an isomorphism if the sheaves $\mathscr{H}^i(K)$ are torsion.

5. The direct-image functors with proper supports

5.1. The fundamental construction. We shall follow the program set out in 3.1 and define the functor $Rf_!$ for a compactifiable morphism f (3.2). In the appendix we shall show how to treat the case of an arbitrary separated morphism of finite type.[60]

5.1.1. Let X be a topos, U an open subtopos of X (IV 8.3), and j the inclusion morphism from U to X. In IV 5.2 we defined an "*extension by the empty set*" functor from the category of sheaves of sets on U to that of sheaves of sets on X. This functor $j_!$ is left adjoint to the restriction functor j^*.

We also denote by $j_!$ the functor from the category of sheaves of pointed sets on U (resp. sheaves of groups, resp. sheaves of $j^*\mathscr{A}$-modules, for $\mathscr{A}$ a sheaf of rings on X) to the analogous category on X, left adjoint to the restriction functor j^* (IV 11.3.1). 341
These three functors $j_!$ are compatible with the forgetful functors from the category of

[59]N.D.E.: One may work in the unbounded derived category without these hypotheses. Cf. la N.D.E. (51) page 673.

[60]N.D.E.: Cf. la N.D.E. (43) page 639.

modules to that of groups, and from the category of groups to that of pointed sets [60]. If $i : F \to X$ is the inclusion of the closed complement of U, then $j^* j_!$ is (canonically isomorphic to) the identity and $i^* j_!$ is the constant functor with value the initial object (or the final object, which here amounts to the same thing). This also characterizes $j_!$. It follows that the three functors $j_!$ under consideration are exact and embed into the corresponding functors j_*. They are called the *extension by zero* functors. The formula $(j_1 j_2)^* = j_2^* j_1^*$ transposes into a transitivity isomorphism

$$(j_1 j_2)_! = j_{1!} j_{2!}.$$

The functors $j_!$ are compatible with base change:

Lemma 5.1.2. *Let $f : X \to Y$ be a morphism of topoi, let V be an open subtopos of Y, and let U be its inverse image in X:*

$$\begin{tikzcd} U \arrow[r, "f'"] \arrow[d, "j'"] & V \arrow[d, "j"] \\ X \arrow[r, "f"] & Y \end{tikzcd}$$

There is then a unique base-change isomorphism $f^* j_! \xrightarrow{\sim} j'_! f'^*$ making the diagram

(5.1.2.1)
$$\begin{tikzcd} f^* j_! \arrow[r, "\sim"] \arrow[d, hook] & j'_! f'^* \arrow[d, hook] \\ f^* j_* \arrow[r, "\alpha"] & j'_* f'^*, \end{tikzcd}$$

commutative, where α is the base-change morphism.

342 Proof. The base-change morphism $\varphi : j^* f_* \to f'_* j'^*$ is trivially an isomorphism. It transposes into a base-change isomorphism $\varphi' : j'_! f'^* \xrightarrow{\sim} f^* j_!$. To verify that diagram (5.1.2.1) is commutative, it suffices to see this after restriction to U, where it is trivial.

5.1.3. Under the hypotheses of 5.1.1, with X ringed by a sheaf of rings $\mathscr{A}$, the functor $j_!$ on modules is exact, hence passes trivially to the derived categories and defines a functor, still denoted $j_!$, from $D(U, j^*\mathscr{A})$ to $D(X, \mathscr{A})$. We still have transitivity isomorphisms

(5.1.3.1)
$$(j_1 j_2)! = j_{1!} j_{2!}.$$

At the level of derived categories, we still have an adjunction formula

(5.1.3.2)
$$\mathrm{Hom}_{D(X)}(j_! K, L) \xrightarrow{\sim} \mathrm{Hom}_{D(U)}(K, j^* L).$$

5.1.4. Let $f : X \to Y$ be a proper morphism, let $\mathscr{A}$ be a sheaf of *torsion* rings on Y, and endow X with the inverse-image sheaf of rings of $\mathscr{A}$. Assume that the dimensions of the fibers of f are bounded above by an integer n (as is the case if Y is quasi-compact). We then know that, for every sheaf of $f^*\mathscr{A}$-modules F on X, the sheaves $R^q f_* F$ vanish for $q > 2n$ (XII 5.3 bis). The functor f_* therefore admits a derived functor (1.2.10)

$$Rf_* : D(X, f^*\mathscr{A}) \longrightarrow D(Y, \mathscr{A}).$$

343 If f is the composite gh of two proper morphisms, there is a transitivity formula $Rf_* = Rg_* Rh_*$, which, as usual, can be expressed in terms of cofibered categories (Scholium 4.1.3).

[60]But not to that of sets!

5.1.5. Consider a commutative diagram of schemes

(5.1.5.1)
$$\begin{tikzcd} U \arrow[r, hook, "j_1"'] \arrow[d, "g"] & X \arrow[d, "f"] \\ V \arrow[r, hook, "j_2"'] & Y, \end{tikzcd}$$

in which j_1 and j_2 are open immersions and f, g are proper morphisms whose fiber dimensions are bounded above by an integer n. Let $\mathscr{A}$ be a sheaf of torsion rings on Y, and equip U, V, and X with the appropriate inverse images of $\mathscr{A}$. By (5.1.3.2), if $K \in D(U)$, defining a morphism

(5.1.5.2)
$$d : j_{2!}Rg_*K \longrightarrow Rf_*j_{1!}K$$

amounts to defining a morphism

$$d' : Rg_*K \longrightarrow j_2^*Rf_*j_{1!}K.$$

The fiber product $X' = V \times_Y X$ fits into a commutative diagram

(5.1.5.3)
$$\begin{tikzcd} U \arrow[r, hook, "k"] \arrow[d, "g"] & X' \arrow[r, hook, "j_2'"'] \arrow[d, "f'"] & X \arrow[d, "f"] \\ V \arrow[r, equal] & V \arrow[r, hook, "j_2"'] & Y, \end{tikzcd}$$

and, trivially, we have $j_2^*Rf_*j_{1!}K \simeq Rf'_*(j_2'^*j_{1!}K) \simeq Rf'_*k_!K$. 344

The morphism k is a proper open immersion; hence $k_! = k_*$, giving an isomorphism $Rf'_*k_!K \simeq Rf'_*k_*K \simeq Rg_*K$, which provides the desired isomorphism d' and the morphism d. Moreover, the restriction of d to V is an isomorphism.

Lemma 5.1.6. *The morphism (5.1.5.2) is an isomorphism.*

It remains to verify that the restriction of d to $Y - V = F$ is an isomorphism. Let $X'' = X \times_Y F$ and apply the proper base-change theorem (4.3.1) to the Cartesian square

$$\begin{tikzcd} X'' \arrow[r, "i'"'] \arrow[d, "f''"] & X \arrow[d, "f"] \\ F \arrow[r, "i"'] & Y, \end{tikzcd}$$

with $\mathscr{A}' = i^*\mathscr{A}$ and $L = \mathscr{A}'[0]$. We obtain: $i^*Rf_*j_{1!}K \simeq Rf''_*i'^*j_{1!}K \simeq Rf''_*0 \simeq 0$, so that both sides of (5.1.5.2) are zero outside V and d is an isomorphism.

5.1.7. Let S be a scheme ringed by a sheaf of torsion rings $\mathscr{A}$. For every S-scheme $\varphi : X \to S$, set $\mathscr{A}_X = \varphi^*\mathscr{A}$. Work in the category (S) of quasi-compact, quasi-separated S-schemes and S-compactifiable morphisms (3.2). By 3.2.3, this category, the subcategory (S, i) of open immersions, and the subcategory (S, p) of proper morphisms satisfy the hypotheses of 3.2.4.

We propose to apply the theory of 3.3.2 to the following data (notation of 3.3.1) 345

α) $F(X) = D(X, \mathscr{A}_X)$ for $\varphi : X \to S$ in S:

- (i) For $p : X \to Y$ proper, take the functor p_* to be the functor Rp_*.
- (ii) Use the usual transitivity isomorphisms $R(pq)_* = Rp_*Rq_*$.
- (i$'$) For an open immersion $i : X \to Y$, take the functor i_* to be the functor $i_!$.
- (ii$'$) Use the transitivity isomorphisms (5.1.3.1).

(iii) For every commutative diagram (5.1.5.1), take the isomorphism (5.1.5.2) for d.

The conditions (a) (b) (c) (a′) (b′) (c′) of 3.3.2 are verified, proving part (a) of:

THEOREM 5.1.8. (a) *In the category* (S) *defined in 5.1.7, there exists one, and essentially only one, way to define, for every arrow* $f : X \to Y$*, a functor*

$$Rf_! : D(X, \mathscr{A}_X) \longrightarrow D(Y, \mathscr{A}_Y),$$

and, for every composite morphism $f = gh : X \xrightarrow{h} Y \xrightarrow{g} Z$*, transitivity isomorphisms* $c_{g,h}$ *between* $Rf_!$ *and* $Rg_!\, Rh_!$ *such that*

(i) *The* $c_{f,g}$ *satisfy the "cocycle condition"*

$$(c_{f,g} * Rh_!) \circ c_{fg,h} = (Rf_! * c_{g,h}) \circ c_{f,gh}.$$

(ii) *When restricted to proper morphisms (resp. to open immersions), one recovers the variance theory of 5.1.4 (resp. 5.1.3).*

346 (iii) *For every diagram (5.1.5.1), the isomorphism* d *of (5.1.5.2) is the composite isomorphism* $c_{f,j_1} \circ c^{-1}_{j_2,g}$.

(b) *The functors* $Rf_!$ *are "way out"* [6, I 7] *and triangulated.*

Every functor $Rf_!$ is the composite of a functor Rg_*, for g proper with quasi-compact target, hence with fibers of bounded dimension, and of a functor $j_!$ for j an open immersion. These functors are way out and triangulated, and hence so is $Rf_!$.

. The transitivity isomorphisms of 5.1.8 (a) are often used via the "spectral sequence of composite functors" deduced from them, valid for $K \in \operatorname{Ob} D(X, \mathscr{A}_X)$:

$$E_2^{pq} = R^p g_! R^q h_!(K) \Longrightarrow R^{p+q} f_!(K). \tag{5.1.8.2}$$

To obtain it, note that by 5.1.8 b) and VERDIER [12], for every $L \in \operatorname{Ob} D(Y, \mathscr{A}_Y)$, there is a spectral sequence

$$E_2^{pq} = R^p g_!(\mathscr{H}^q(L)) \Longrightarrow R^{p+q} g_!(L), \tag{5.1.8.3}$$

and (5.1.8.2) is the special case of (5.1.8.3) with $L = Rh_! K$.

DEFINITION 5.1.9. (i) *We call the "* total direct-image functors with proper supports*" the functors constructed in 5.1.8. We shall omit the adjective "total" when there is no risk of confusion.*

(ii) *The* q^{th} direct-image functor with proper support, *denoted* $R^q f_!$*, is the composite functor* $\mathscr{H}^q \circ Rf_!$.

(iii) *If* X *is a scheme over a separably closed field* k *that is embeddable in a proper*
347 *scheme over* k*, identify sheaves of modules on* $\operatorname{Spec}(k)$ *with ordinary modules, and denote the functor* $R^q f_!$ *by one of the notations* $H^q_c(X, \ )$, $H^q_!(X, \ )$*, or simply* $H^q_c(\ )$ *if there is no risk of confusion. It is then called the "*q^{th} hypercohomology group with proper supports,*" or the "*q^{th} cohomology group with proper supports*" if it is applied to a complex concentrated in degree* 0.

. The notation $Rf_!$ is abusive in that *the functor* $Rf_!$ *is not the right derived functor of the functor* $f_!$ that will be defined in 6.1.2. Some prefer the notation $R_! f$.

5.1.10. The ingredients in the proof of 5.1.8 are the proper base-change theorem and the identity between j_* and $j_!$ when j is a proper open immersion. This makes it possible to state various variants of 5.1.8; it is left to the reader to give them a precise meaning and to verify them.

Variant 5.1.11. *In the category* (S)*, there exists one, and essentially only one, way to define, for every arrow* $f : X \to Y$*, a functor* $f_!$ *from the category of sheaves of pointed sets on* X *to the category of sheaves of pointed sets on* Y*, and, for every composite morphism* $f = gh$*, transitivity isomorphisms satisfying conditions* (i) (ii) (iii) *analogous to those of 5.1.8.*

See 6.1.2 for a direct description of this functor.

Variant 5.1.12. *For every compactifiable morphism* $f : X \to Y$*, one can define* 348
a functor $R^1 f_!$ *from the category of sheaves of ind-finite groups on* X *to the category of sheaves of pointed sets on* Y*, together with, for every compactification* $f = \overline{f} j$ *of* f*, an isomorphism* $R^1 f_! \simeq R^1 \overline{f}_* \circ j_!$*.*

No transitivity formula is postulated here; the only problem is to show that the functor $R^1 f_!$ is independent of the chosen compactification of f. Let $p : (\overline{f}_1, i_1) \to (\overline{f}_2, i_2)$ be a morphism of compactifications, i.e. a diagram

(5.1.12.1)
$$\begin{tikzcd}
X \arrow[r, "i_1"] \arrow[dr, "i_2"] \arrow[dd, "f"'] & \overline{X}_1 \arrow[d, "p"] \\
 & \overline{X}_2 \arrow[dl, "\overline{f}_2"] \\
Y &
\end{tikzcd}.$$

where $\overline{f}_1 = \overline{f}_2 \circ p$, with p proper. If G is a sheaf of groups on X, there is an exact sequence of pointed sets (with α injective)

$$0 \to R^1\overline{f}_{2*}(p_* i_{1!}G) \xrightarrow{\alpha} R^1\overline{f}_{1*}(i_{1!}G) \to \overline{f}_{2*}R^1 p_*(i_{1!}G),$$

and a morphism from $i_{2!}G$ to $p_* i_{1!}G$, whence, by composition, a morphism β from $R^1\overline{f}_{2*}i_{2!}G$ to $R^1\overline{f}_{1*}i_{1!}G$. Reasoning as in 3.3.2.1, one sees that the problem is to prove that this morphism is always an isomorphism when G is ind-finite.

Applying the proper base-change theorem to p and the sheaf of sets $i_{1!}G$, one checks 349
that $p_* i_{1!}G = i_{2!}G$, yielding the exact sequence

$$0 \to R^1\overline{f}_{2*}(i_{2!}G) \xrightarrow{\beta} R^1\overline{f}_{1*}(i_{1!}G) \to \overline{f}_{2*}R^1 p_* i_{1!}G.$$

The morphism β is injective (since α is). Applying the proper base-change theorem to p and the sheaf of groups G, assumed ind-finite, one finds that $R^1 p_* i_{1!}G = 0$. Thus β is an epimorphism, which completes the proof of 5.1.12.

Definition 5.1.13. *For every site* X*, denote by* $D_{\mathrm{tors}}(X)$ *the full subcategory of the derived category of the category of abelian sheaves on* X *consisting of the complexes whose cohomology sheaves are torsion.*

Variant 5.1.14. *In the category* (S)*, there exists one, and essentially only one, way to define, for every arrow* $f : X \to Y$*, a triangulated and "way out" functor* [6, I 7]

$$Rf_! : D^+_{\mathrm{tors}}(X) \longrightarrow D^+_{\mathrm{tors}}(Y)$$

and, for every composite morphism $f = gh$, transitivity isomorphisms satisfying conditions (i) (ii) (iii) *analogous to those of 5.1.4.*

VARIANT 5.1.15. *Work in the category of S-schemes ringed by sheaves of torsion rings, the arrows being morphisms of ringed schemes inducing (0.8) an S-compactifiable morphism of schemes. For every arrow*

$$f : (X, \mathscr{A}) \longrightarrow (Y, \mathscr{B}),$$

350 *let $Rf_!$ be the composite of the forgetful functor from the category of $\mathscr{A}$-modules to that of $f^{-1}\mathscr{B}$-modules, followed by the usual $Rf_!$. These functors have the variance properties of the usual functors $Rf_!$.*

Note, however, that the notation $Rf_!$ is reasonable here only if $\mathscr{A}$ is finite over $f^{-1}\mathscr{B}$, as one sees by taking $X = Y = \mathrm{Spec}(k)$, with k algebraically closed.

5.1.16. Let $f : X \to S$ be a compactifiable morphism, let $j : U \to X$ be an open immersion into X, let $i : F \to X$ be the closed complement, let $\mathscr{A}_S$ be a sheaf of torsion rings on S, let $\mathscr{A}_X$ be its inverse image on X, and let $K \in \mathrm{Ob}\, D(X, \mathscr{A}_X)$.

The exact sequence of complexes

$$0 \longrightarrow j_! j^* K \longrightarrow K \longrightarrow i_! i^* K \longrightarrow 0$$

gives rise to a distinguished triangle in $D(X, \mathscr{A}_X)$ by VERDIER [11, II 1.1.5]. Its image under $Rf_!$ is a distinguished triangle (5.1.8 (b))

$$R(fj)_! j^* K \longrightarrow Rf_! K \longrightarrow R(fi)_! i^* K \longrightarrow R(fj)_! j^* K[1] \ldots \tag{5.1.16.1}$$

which defines a long exact cohomology sequence

$$\ldots \longrightarrow R^q(fj)_! j^* K \to R^q f_! K \to R^q(fi)_! i^* K \xrightarrow{j} R^{q+1}(fj)_! j^* K \to \ldots . \tag{5.1.16.2}$$

In the special case where S is the spectrum of a separably closed field and K is a sheaf $\mathscr{G}$ concentrated in degree 0, the sequence (5.1.16.2) becomes

$$\ldots H^q_c(U, \mathscr{G}) \to H^q_c(X, \mathscr{G}) \to H^q_c(F, \mathscr{G}) \xrightarrow{j} H^{q+1}_c(U, \mathscr{G}) \ldots \tag{5.1.16.3}$$

351 5.1.17. Consider a diagram of S-compactifiable schemes

$$\begin{tikzcd} X \arrow[rr, "u"] \arrow[dr, "f"'] & & Y \arrow[dl, "g"] \\ & S & \end{tikzcd},$$

with *u proper*. Equip X and Y with the inverse images of a sheaf of torsion rings $\mathscr{A}_S$ on S. For $K \in D(Y, \mathscr{A}_Y)$, the evident (adjunction) morphism $K \to Ru_* u^* K$ defines, upon applying the functor $Rg_!$, a morphism

$$u^* : Rg_! K \longrightarrow Rg_! Ru_* u^* K = Rg_! Ru_! u^* K \simeq Rf_! u^* K \tag{5.1.17.1}$$

(*contravariance of cohomology with proper supports with respect to proper morphisms*).

When S is the spectrum of an algebraically closed field and K is the sheaf $\mathscr{G}$ placed in degree 0, this morphism becomes

$$u^* : H^q_c(Y, \mathscr{G}) \longrightarrow H^q_c(X, \mathscr{G}) \tag{5.1.17.2}$$

5.2. The base-change theorem.

5.2.1. Let $f : X \to Y$ be a morphism of topoi, let V be an open of Y, let $U = f^{-1}(V)$ be its inverse image in X, let $\mathscr{A}$ be a sheaf of rings on Y, let $\mathscr{A}'$ be a sheaf of rings on X, and let L be a bounded-above complex of $(\mathscr{A}', f^*\mathscr{A})$-bimodules:

(5.2.1.1)
$$\begin{tikzcd} U \arrow[r, "f'"] \arrow[d, hook, "j'"] & V \arrow[d, hook, "j"] \\ X \arrow[r, "f"'] & Y. \end{tikzcd}$$

There is then a base-change isomorphism (5.1.2.1) between functors from $D(V, \mathscr{A})$ to $D(X, f^*\mathscr{A}) : f^* j_! \xrightarrow{\sim} j'_! f'^*$. Moreover, there is an evident isomorphism between functors from $D^-(U, f'^*\mathscr{A})$ to $D^-(X, \mathscr{A}') : L \overset{\mathbf{L}}{\otimes}_{f^*\mathscr{A}} j'_! K \simeq j'_!(j'^* L \overset{\mathbf{L}}{\otimes}_{f^*\mathscr{A}} K)$. To construct it, it suffices to note that if a complex K is adapted to the functor $j'^* L \overset{\mathbf{L}}{\otimes}_{f^*\mathscr{A}}$, then the complex $j'_! K$ is adapted to the functor $L \overset{\mathbf{L}}{\otimes}_{f^*\mathscr{A}}$, and that there is an isomorphism of the preceding type at the level of complexes. Composing these two isomorphisms gives a base-change isomorphism between functors from $D^-(V, \mathscr{A})$ to $D^-(X, \mathscr{A}')$: 352

(5.2.1.2)
$$L \overset{\mathbf{L}}{\otimes}_{f^*\mathscr{A}} f^* j_! K \xrightarrow{\sim} j'_!(j'^* L \overset{\mathbf{L}}{\otimes}_{f^*\mathscr{A}} f'^* K).$$

If L is bounded with components of finite tor-dimension over $f^*\mathscr{A}$,[61] this isomorphism is still defined for $K \in \operatorname{Ob} D(V, \mathscr{A})$.

5.2.2. Let $f : X \to S$ and $g : S' \to S$ be morphisms of schemes, and suppose that f is the composite of an open immersion j and a proper morphism p whose fibers have dimension bounded by an integer n (this is automatic when S is quasi-compact). These morphisms fit into a diagram with cartesian squares

$$\begin{tikzcd} X' \arrow[rrr, "g'"] \arrow[dr, hook, "j'"] \arrow[ddd, "f'"'] & & & X \arrow[dr, hook, "j"] \arrow[ddd, "f"] & \\ & \overline{X}' \arrow[rrr, "\overline{g}"] \arrow[ddl, "p'"] & & & \overline{X} \arrow[ddl, "p"] \\ & & & & \\ S' \arrow[rrr, "g"'] & & & S & \end{tikzcd}$$

Let $\mathscr{A}$ be a sheaf of torsion rings on S, $\mathscr{A}'$ a sheaf of torsion rings on S', and L a bounded-above complex of $(\mathscr{A}', g^*\mathscr{A})$-bimodules. We ring X and $\overline{X}$ (resp. X' and $\overline{X}'$) by the inverse images $\mathscr{A}_X$ and $\mathscr{A}_{\overline{X}}$ (resp. $\mathscr{A}'_{X'}$ and $\mathscr{A}'_{\overline{X}'}$) of $\mathscr{A}$ (resp. $\mathscr{A}'$) on X and $\overline{X}$ (resp. X' and $\overline{X}'$). We then have base-change isomorphisms (5.2.1.2) between the functors $p'^* L \overset{\mathbf{L}}{\otimes} \overline{g}^* j_! K$ and $j'_!(f'^* L \overset{\mathbf{L}}{\otimes} g'^* K)$ from $D^-(X, \mathscr{A}_X)$ to $D^-(\overline{X}', \mathscr{A}'_{\overline{X}'})$, and base-change isomorphisms (4.3.1) between the functors $L \overset{\mathbf{L}}{\otimes} g^* Rp_* K$ and $Rp'_*(p'^* L \overset{\mathbf{L}}{\otimes} \overline{g}^* K)$ 353

[61] N.D.E.: this condition is superfluous. Cf. la N.D.E. (51) page 673.

from $D^-(\overline{X}, \mathscr{A}_{\overline{X}})$ to $D^-(S', \mathscr{A}')$. By composition, one obtains a base-change isomorphism between functors from $D^-(X, \mathscr{A}_X)$ to $D^-(S', \mathscr{A}')$:

$$L \overset{\mathrm{L}}{\otimes} g^* Rp_* j_! K \xrightarrow{\sim} Rp'_* j'_! (f'^* L \overset{\mathrm{L}}{\otimes} g'^* K). \tag{5.2.2.1}$$

When L is bounded with components of finite tor-dimension, the preceding arguments and 4.3.2 (i) define the base-change isomorphism for $K \in \mathrm{Ob}\, D(X, \mathscr{A}_X)$.[62]

In order to refer to 5.1.8, suppose that S is quasi-compact and quasi-separated. The functors $Rp_* j_! = Rf_!$ and $Rp'_* j'_! = Rf'_!$ then depend only on f and f', giving a meaning to the following statement.

Lemma 5.2.3. *The base-change isomorphism (5.2.2.1) does not depend on the chosen compactification $f = pj$ of X.*

Let $q : \overline{X}_1 \to \overline{X}_2$ be a morphism of compactifications

$$\begin{tikzcd} X \arrow[rr, "j_1"] \arrow[drr, "j_2"] \arrow[dd, "f"'] & & \overline{X}_1 \arrow[d, "q"] \arrow[dll, "p_1"] \\ & & \overline{X}_2 \arrow[dll, "p_2"] \\ S & & \end{tikzcd} \tag{5.2.3.1}$$

354 The base-change isomorphism c_1 relative to $\overline{X}_1$ is obtained by "composing" those relative to j_1 and p_1, or, equivalently, by composing those relative to j_1, q, and p_2. The isomorphism c_2 relative to $\overline{X}_2$ is obtained by composing those relative to j_2 and p_2. To verify that $c_1 = c_2$, it suffices to verify that the isomorphism c'_1 relative to j_2 is the composite c'_2 of those relative to j_1 and q. Use a prime to denote the arrows obtained from the arrows (5.2.3.1) by the base change g; c'_1 and c'_2 are isomorphisms

$$c'_1, c'_2 : L \overset{\mathrm{L}}{\otimes} \overline{g}_2^* * j_{2!} K \xrightarrow{\sim} j'_{2!}(j'^*_2 * L \overset{\mathrm{L}}{\otimes} g'^* K).$$

By (5.1.3.2), to verify that $c'_1 = c'_2$ it suffices to verify it after restricting to $\overline{X}'_2$, which is immediate.

The base-change isomorphisms relative to two compactifications between which a morphism exists are therefore equal, and the conclusion follows from 3.2.6 (ii).

Lemma 5.2.4. *Let $f = uv : X \to Y \to S$ be a composite of S-compactifiable morphisms. The base-change isomorphisms (5.2.2.1) relative to u, v, and f give rise to a commutative diagram (notation as in (5.2.4.1) below):*

$$\begin{tikzcd} L \overset{\mathrm{L}}{\otimes} g^* R(uv)_! K \arrow[rr, "\sim"] \arrow[d, equal] & & R(u'v')_!((u'v')^* L \overset{\mathrm{L}}{\otimes} g''^* K) \arrow[d, equal] \\ L \overset{\mathrm{L}}{\otimes} g^* Ru_! Rv_! K \arrow[r, "\sim"] & Ru'_!(u'^* L \overset{\mathrm{L}}{\otimes} g'^* Rv_! K) \arrow[r, "\sim"] & Ru'_! Rv'_!(v'^* u'^* L \overset{\mathrm{L}}{\otimes} g''^* K). \end{tikzcd}$$

355 Choose a compactification of (u, v) (3.2.5):

[62]N.D.E.: by combining the preceding N.D.E. and la N.D.E. (58) page 679, this isomorphism can be defined without any condition on L or the hypothesis that the dimension of the fibers of p is bounded.

(5.2.4.1)
$$\begin{tikzcd}
X' \arrow[r, "g''"] \arrow[d, "v'"'] & X \arrow[r, hook, "j_1"] \arrow[d, "v"'] & \overline{X} \arrow[r, hook, "j_2"] \arrow[dl, "p_1"'] \arrow[d] & \overline{\overline{X}} \arrow[dl, "p_3"] \\
Y' \arrow[r, "g'"] \arrow[d, "u'"'] & Y \arrow[r, hook, "j_3"] \arrow[d, "u"'] & \overline{Y} \arrow[dl, "p_2"] & \\
S' \arrow[r, "g"] & S & &
\end{tikzcd}$$
.

Lemma 5.2.4 says that the base-change isomorphism c_1 relative to f is the composite c_2 of those relative to u and v. The isomorphism c_1 is obtained by composing the base-change isomorphisms relative to j_1, j_2, p_3, and p_2. For c_2, one composes those relative to j_1, p_1, j_3, and p_2. It therefore suffices to compare the isomorphisms c_1', the composite of those relative to p_3 and j_2, and c_2', the composite of those relative to j_3 and p_1.

Consider the cartesian diagram

$$\begin{tikzcd}
\overline{X}' \arrow[rr, "\overline{g}''"] \arrow[dr, hook, "j_2'"] \arrow[dd, "p_1'"'] & & \overline{X} \arrow[dr, hook, "j_2"] \arrow[dd, "p_1"] & \\
& \overline{\overline{X}}' \arrow[rr, "\overline{\overline{g}}"] \arrow[dd, "p_3'"] & & \overline{\overline{X}} \arrow[dd, "p_3"] \\
Y' \arrow[rr, "g'"] \arrow[dr, hook, "j_3'"] \arrow[ddr, "u'"'] & & Y \arrow[dr, hook, "j_3"] \arrow[ddr, "u"] & \\
& \overline{Y}' \arrow[rr, "\overline{g}'"] \arrow[d, "p_2'"] & & \overline{Y} \arrow[d, "p_2"] \\
& S' \arrow[rr, "g"] & & S
\end{tikzcd}$$
,

$L' = p_2'^* L$, $K' = j_{1!}K$ and 356

$$c_1', c_2' : L' \overset{\mathbf{L}}{\otimes} \overline{g}'^* R(j_3 p_1)_! K' \xrightarrow{\sim} R(j_3' p_1')_!((j_3' p_1')^* L' \otimes \overline{g}''^* K').$$

By (5.1.3.2), to verify that $c_1' = c_2'$ it suffices to verify that $j_3'^*(c_1') = j_3'^*(c_2')$, which is immediate.

Lemma 5.2.5. *Let there be morphisms of schemes ringed by sheaves of torsion rings*

$$(S'', \mathscr{A}'') \xrightarrow{g_2} (S', \mathscr{A}') \xrightarrow{g_1} (S, \mathscr{A}),$$

and let $f : X \to S$ be a compactifiable morphism, giving a diagram:

$$\begin{tikzcd}
X'' \arrow[r, "g_2'"] \arrow[d, "f''"] & X' \arrow[r, "g_1'"] \arrow[d, "f'"] & X \arrow[d, "f"] \\
S'' \arrow[r, "g_2"] & S' \arrow[r, "g_1"] & S.
\end{tikzcd}$$

Ring X, X', and X'' by $f^*\mathscr{A}$, $f'^*\mathscr{A}'$, and $f''^*\mathscr{A}''$, respectively. Then the diagram 357

$$\begin{tikzcd}
L(g_1g_2)^*Rf_! \arrow[rr, "\sim"] \arrow[d, equal] & & Rf''_!\,L(g'_1g'_2)^* \arrow[d, equal] \\
Lg_2^*Lg_1^*Rf_! \arrow[r, "\sim"] & Lg_2^*Rf'_!\,Lg_1'^* \arrow[r, "\sim"] & Rf''_!\,Lg_2'^*Lg_1'^*
\end{tikzcd}$$

is commutative.

This lemma follows immediately from the analogous results for the functors $j_!$ and Rp_* with p proper. We leave it to the reader to formulate the precise version with modules.

The preceding statements are summarized in the following theorem.

THEOREM 5.2.6. *The functor $Rf_!$ commutes with base change.*

VARIANTS 5.2.7. (i) The functors $f_!$ (for sheaves of pointed sets: 5.1.11) and the functors $R^1f_!$ (for sheaves of groups: 5.1.12) commute with base change.
(ii) The functors $Rf_!$ (5.1.14) commute with base change.
(iii) Under the hypotheses of (5.2.2), if L is bounded and has components of finite tor-dimension,[63] the functors $Rf_! : D(X) \to D(S)$ commute with base change.

358 The conjunction of the following two Propositions 5.2.8 and 5.2.9 is essentially equivalent to Theorem 5.2.6.

PROPOSITION 5.2.8. *Let $f : X \to S$ be a compactifiable morphism and $\mathscr{F}$ a torsion sheaf on X. For every geometric point s of S, let X_s be the geometric fiber of X at s and let $\mathscr{F}_s$ be the induced sheaf. Then canonically*

$$(R^qf_!(\mathscr{F}))_s = H^q_c(X_s, \mathscr{F}_s).$$

COROLLARY 5.2.8.1. *If the fibers of f have dimension $\leq d$, the functor $Rf_!$ has cohomological dimension $\leq 2d$ (1.2.9): for every torsion sheaf $\mathscr{F}$, one has $R^if_!\mathscr{F} = 0$ for $i > 2d$. The functor $R^{2d}f_!$ is therefore right exact on the category of torsion sheaves.*

By base change (5.2.8), we reduce to the case where S is the spectrum of an algebraically closed field. If $\overline{X}$ is then a proper scheme over S containing X as a dense Zariski open subscheme, with $j : X \to \overline{X}$, then $\dim(\overline{X}) = \dim(X) \leq d$. Hence, for $\mathscr{F}$ a torsion sheaf on X and $i > 2d$, by X 4.3, 5.2

$$H^i_c(X, \mathscr{F}) = H^i(\overline{X}, j_!\mathscr{F}) = 0.$$

PROPOSITION 5.2.9. *Let $f : X \to S$ be a compactifiable morphism, let $\mathscr{A}$ be a sheaf of torsion rings on S, let K be a bounded-above complex of left $f^*\mathscr{A}$-modules, and let L be a bounded-above complex of right $\mathscr{A}$-modules. Then canonically*

$$L \overset{\mathbf{L}}{\underset{\mathscr{A}}{\otimes}} Rf_!(K) = Rf_!(f^*L \overset{\mathbf{L}}{\underset{f^*\mathscr{A}}{\otimes}} K).$$

359 This proposition formally implies the following result:

THEOREM 5.2.10. *Let $f : X \to S$ be a compactifiable morphism, let $\mathscr{A}$ be a sheaf of torsion rings on S, let $K \in D^-(X, f^*\mathscr{A})$, and let $d \in \mathbf{Z}$. If K has finite tor-dimension $\leq d$ (4.1.9), then $Rf_!K$ has finite tor-dimension $\leq d$.*

---

[63] N.D.E.: as observed in 5.2.2, these hypotheses are superfluous.

It suffices to verify that, for every sheaf L of right $\mathscr{A}$-modules on S,

$$\mathscr{H}^{-i}(L \overset{\mathbf{L}}{\otimes} Rf_! K) = 0 \quad \text{for} \quad i > d.$$

By 5.2.9, we know that

$$\mathscr{H}^{-i}(L \overset{\mathbf{L}}{\otimes} Rf_! K) = R^{-i} f_!(f^* L \overset{\mathbf{L}}{\otimes} K),$$

and, by hypothesis,

$$\mathscr{H}^{-i}(f^* L \overset{\mathbf{L}}{\otimes} K) = 0 \quad \text{for} \quad i > d.$$

The conclusion follows by observing that if a complex M of sheaves of modules on X satisfies $\mathscr{H}^{-i}(M) = 0$ for $i > d$, then one still has $\mathscr{H}^{-i}(Rf_! M) = 0$ for $i > d$.

For constant noetherian sheaves of rings, one has the following more general result:

THEOREM 5.2.11. *Let $f : X \to S$ be a morphism of sites, let A be a ring, let C be its center, and let $d \in \mathbf{Z}$. Suppose that*

(i) *the functor $Rf_* : D^+(X, C) \to D^+(S, C)$ has finite cohomological dimension;*
(ii) *A is right noetherian;*
(iii) *S has enough points.*

For every complex $K \in \operatorname{Ob} D^-(X, A)$ of sheaves of left A-modules, if K has finite tor- 360
dimension $\leq d$, then the complex $Rf_* K \in \operatorname{Ob} D^-(S, A)$ has tor-dimension $\leq d$.

PROOF. By (i), the functor f_* is derivable to functors

$$Rf_* : D(X, A) \longrightarrow D(S, A)$$

and

$$Rf_* : D(X, C) \longrightarrow D(S, C).$$

By 4.2.8, it suffices to verify that, for every point s of S, the complex $(Rf_* K)_s$ of A-modules has tor-dimension $\leq d$; for this it suffices that, for every finitely generated right A-module L,

$$H^{-i}(L \overset{L}{\otimes} Rf_* K) = 0 \quad \text{for} \quad i > d.$$

Reasoning as in 5.2.10, we see that it suffices for the "base-change arrow"

$$L \overset{L}{\otimes} Rf_* K \longrightarrow Rf_*(L \overset{L}{\otimes} K) \tag{5.2.11.1}$$

to be an isomorphism.

Let L^* be a resolution of L by finitely generated free A-modules. Since the functors under consideration are way-out to the right, it suffices to prove that (5.2.11.1) is an isomorphism after replacing L by any component of L^*, which reduces us to the immediate case where L is finitely generated and free.

REMARKS 5.2.12. (i) The hypothesis (iii) of 5.2.11 is probably unnecessary. 361

(ii) If the noetherian hypothesis on A is omitted, 5.2.11 remains valid when the functors $R^i f_*$ commute with filtered direct limits.
(iii) The hypothesis that S is ringed by a *constant* sheaf of rings is essential. A counterexample is obtained by taking for S the spectrum of a strictly local discrete valuation ring, for f the inclusion of the generic point η, for the sheaf of rings $\mathscr{A}$ the sheaf equal to $\mathbf{Z}/n\mathbf{Z}$ at the generic point and to $\mathbf{Z}/n^2\mathbf{Z}$ at the closed point s ($n > 1$ prime to the residue characteristics), and for the sheaf the module $f^*\mathscr{A}$.

5.3. Comparison theorem and finiteness theorem.

5.3.1. To a scheme X of finite type over $\mathbf{C}$ are associated its étale site X_{et}, its transcendental site X_{cl}, and a morphism of sites (XI 4)

$$\epsilon : X_{\mathrm{cl}} \longrightarrow X_{\mathrm{et}}.$$

If $j : U \to X$ is an open immersion, the commutative diagram

$$\begin{tikzcd} U_{\mathrm{cl}} \arrow[r] \arrow[d] & U_{\mathrm{et}} \arrow[d] \\ X_{\mathrm{cl}} \arrow[r] & X_{\mathrm{et}} \end{tikzcd}$$

identifies $U_{\mathrm{cl}}^{\sim}$ with the open subtopos of $X_{\mathrm{cl}}^{\sim}$ that is the inverse image of the open subtopos $U_{\mathrm{et}}^{\sim}$ of $X_{\mathrm{et}}^{\sim}$.

If $p : Y \to X$ is a proper morphism, the continuous map $p^{\mathrm{an}} : Y^{\mathrm{an}} \to X^{\mathrm{an}}$ is proper and separated. To see this, Chow's lemma reduces us to the case where p is projective.

362 Let $f : X \to S$ be a compactifiable morphism of schemes of finite type over $\mathbf{C}$, and let $f = pj$ be a compactification of f.

(5.3.1.1)
$$\begin{tikzcd} X_{\mathrm{cl}} \arrow[rrr, "\epsilon"] \arrow[dr, "j_{\mathrm{cl}}"] \arrow[dd, "f_{\mathrm{cl}}"'] & & & X \arrow[dr, "j"] \arrow[dd, "f"] & \\ & \overline{X}_{\mathrm{cl}} \arrow[rrr, "\epsilon"] \arrow[dl, "p_{\mathrm{cl}}"] & & & \overline{X} \arrow[dl, "p"] \\ S_{\mathrm{cl}} \arrow[rrr, "\epsilon"] & & & S & \end{tikzcd}.$$

The functor $p_{\mathrm{cl}\,*} j_{\mathrm{cl}!}$ is precisely the direct-image functor with proper supports $f_{\mathrm{cl}!}$ (see Verdier [13]). If S_{cl}, and hence X_{cl}, is separated, the functor $j_{\mathrm{cl}!}$ carries soft sheaves on X_{cl} to soft sheaves on $\overline{X}_{\mathrm{cl}}$ [4, II 3.5.5]. Since the question is local on S, it in any case carries flasque sheaves on X_{cl} to sheaves on $\overline{X}_{\mathrm{cl}}$ that are acyclic for the functor $p_{\mathrm{cl}\,*}$. Hence (cf. Verdier [13])

$$Rf_{\mathrm{cl}!} = Rp_{\mathrm{cl}\,*} j_{\mathrm{cl}!}.$$

5.3.2. By 5.1.2, there is a base-change isomorphism between functors from $D(X)$ to $D(\overline{X}_{\mathrm{cl}})$:

$$\epsilon^* j_! \xrightarrow{\sim} j_{\mathrm{cl}!} \epsilon^*.$$

363 By 4.2.12 or 4.1.5, there is a base-change morphism between functors from $D^+(\overline{X})$ to $D^+(S_{\mathrm{cl}})$:

$$\epsilon^* Rp_* \longrightarrow Rp_{\mathrm{cl}\,*} \epsilon^*.$$

By composition one obtains a *base-change morphism between functors from $D^+_{\mathrm{tors}}(X)$ to $D^+_{\mathrm{tors}}(S_{\mathrm{cl}})$* (5.1.13):

(5.3.2.1)
$$\epsilon^* Rf_! \longrightarrow Rf_{\mathrm{cl}!} \epsilon^*.$$

As in 5.2.3, one checks that it does not depend on the chosen compactification of f; it satisfies a transitivity formula of the type in 5.2.4.

Theorem 5.3.3. (Comparison theorem). *The morphism (5.3.2.1) is an isomorphism.*

By construction, it suffices to treat the case where f is proper. The functors under consideration are triangulated and "way-out" [6, I 7]. It therefore suffices to treat the case where K is a torsion sheaf placed in degree 0, and the result follows from XVI 4.1 (i).

VARIANTS 5.3.4. (i) If $\mathscr{A}$ is a sheaf of torsion rings on S_{et}, one similarly obtains an isomorphism (5.3.2.1) between functors from $D(X, f^*\mathscr{A})$ to $D(S, \epsilon^*\mathscr{A})$.

(ii) In low degrees one obtains the usual variants for sheaves of pointed sets and sheaves of ind-finite groups.

COROLLARY 5.3.5. *Let X be a separated scheme of finite type over* $\mathbf{C}$ *which admits a* 364
compactification in a proper scheme over $\mathbf{C}$*, and let F be a torsion sheaf on X. Then*

$$H^q_!(X, F) \xrightarrow{\sim} H^q_c(X_{\text{cl}}, \epsilon^* F).$$

THEOREM 5.3.6. *Let $f : X \to S$ be a compactifiable morphism of finite presentation and let A be a left noetherian torsion ring. The functor $Rf_! : D(X, A) \to D(S, A)$ carries complexes with constructible cohomology to complexes with constructible cohomology.*

The assertion is local on S, which may be assumed affine. By passage to the limit, we reduce to the case where S is noetherian. Complexes with constructible cohomology form a triangulated subcategory of $D(S, A)$. Since the functor $Rf_!$ is "way-out," we reduce to verifying that if $\mathscr{F}$ is a constructible sheaf of A-modules, then the sheaves $R^q f_! \mathscr{F}$ are constructible. Write $f = pj$, with p proper and j an open immersion. Then $R^q f_! \mathscr{F} = R^q p_*(j_! \mathscr{F})$. The sheaf $j_! \mathscr{F}$ is constructible, and the conclusion follows from XIV 1.1.

5.3.7. In low degrees, there are analogous statements for direct images with proper supports of a constructible sheaf of ind-finite sets or groups.

COROLLARY 5.3.8. *Let X be a separated scheme of finite type over a separably closed field k, and let $\mathscr{F}$ be a constructible torsion sheaf on X. Suppose that X can be compactified in a proper scheme over k.*[64] *Then the groups $H^q_c(X, \mathscr{F})$ are finite.* 365

5.4. Künneth formula.

5.4.1. Let there be a commutative diagram (0.11) of sites

(5.4.1.1)
$$\begin{tikzcd}
 & Z' \arrow[dl, "p'"'] \arrow[dr, "q'"] \arrow[dd, "h"] & \\
X' \arrow[dd, "f"'] & & Y' \arrow[dd, "g"] \\
 & Z \arrow[dl, "p"'] \arrow[dr, "q"] & \\
X \arrow[dr] & & Y \arrow[dl] \\
 & S &
\end{tikzcd}.$$

Suppose S is ringed by a sheaf of rings $\mathscr{A}$ with center $\mathscr{C}$; we again denote by $\mathscr{A}$ and $\mathscr{C}$ the inverse images of these sheaves on the sites of (5.4.1.1).

If K is a sheaf of right $\mathscr{A}$-modules on X and L a sheaf of left $\mathscr{A}$-modules on Y, set

$$K \underset{\mathscr{A}}{\boxtimes} L = p^* K \underset{\mathscr{A}}{\otimes} q^* L$$

(external tensor product). Similarly for K (resp. L) on X' (resp. Y'). Now let K be a sheaf of right $\mathscr{A}$-modules on X' and L a sheaf of left $\mathscr{A}$-modules on Y'. The base-change arrows (2.1.3, XII 4.1) define

$$(5.4.1.2) \qquad \begin{cases} p^* f_* K \longrightarrow h_* p'^* K \\ q^* g_* L \longrightarrow h_* q'^* L, \end{cases}$$

[64] N.D.E.: this hypothesis is automatically satisfied. See la N.D.E. (43) page 639.

366 and hence, by taking their tensor product, an arrow

$$f_*K \boxtimes_{\mathscr{A}} g_*L \longrightarrow h_*p'^*K \otimes_{\mathscr{A}} h_*q'^*L \longrightarrow h_*(K \boxtimes_{\mathscr{A}} L). \tag{5.4.1.3}$$

Suppose that the functors

$$\begin{aligned} Rf_* &: D^+(X', \mathscr{A}^\circ) \longrightarrow D^+(X, \mathscr{A}^\circ) \\ Rg_* &: D^+(Y', \mathscr{A}) \longrightarrow D^+(Y, \mathscr{A}) \\ Rh_* &: D^+(Z', \mathscr{C}) \longrightarrow D^+(Z, \mathscr{C}) \end{aligned}$$

have finite cohomological dimension and therefore extend to the unbounded derived categories. For $K \in \operatorname{Ob} D^-(X', \mathscr{A}^\circ)$ and $L \in \operatorname{Ob} D^-(Y', \mathscr{A})$, one is then led to define a *Künneth arrow*

$$Rf_*K \overset{\mathrm{L}}{\boxtimes}_{\mathscr{A}} Rg_*L \longrightarrow Rh_*(K \overset{\mathrm{L}}{\boxtimes}_{\mathscr{A}} L). \tag{5.4.1.4}$$

We define it only when the sites under consideration have enough points, so that 4.2.10 and 4.2.8 may be used. By 4.2.10, it suffices to define it for a pair (K, L) left adapted to the functor $(K, L) \mapsto K \boxtimes_{\mathscr{A}} L$, where K and L are bounded above and right adapted to the functors f_* and g_*.

Let K_1 be a flat resolution of f_*K, let L_1 be a flat resolution of g_*L, and let $u : K \overset{\mathrm{L}}{\boxtimes}_{\mathscr{A}} L \to M$ be a resolution of $K \overset{\mathrm{L}}{\boxtimes}_{\mathscr{A}} L$ adapted to the functor Rh_*. We have morphisms

$$\begin{aligned} Rf_*K \overset{\mathrm{L}}{\boxtimes}_{\mathscr{A}} Rg_*L \simeq K_1 \boxtimes_{\mathscr{A}} L_1 \longrightarrow f_*K \boxtimes_{\mathscr{A}} g_*L \longrightarrow \\ \longrightarrow h_*(K \boxtimes_{\mathscr{A}} L) \longrightarrow h_*M \simeq Rh_*(K \overset{\mathrm{L}}{\boxtimes}_{\mathscr{A}} L) \end{aligned}$$

and define (5.4.1.4) to be their composite.

367 This arrow has the following characteristic property, whose proof is entirely analogous to that of 4.2.13:

. Let $K \in \operatorname{Ob} C^-(X', \mathscr{A}^\circ)$ and $L \in \operatorname{Ob} C^-(Y', \mathscr{A})$. Set $K_1 = f_*K$, $L_1 = g_*L$, and $M_2 = K \boxtimes_{\mathscr{A}} L$. The following diagram is then commutative:

$$\begin{tikzcd} Rf_*K \overset{\mathrm{L}}{\boxtimes}_{\mathscr{A}} Rg_*L \arrow[rrr] & & & Rh_*(K \overset{\mathrm{L}}{\boxtimes}_{\mathscr{A}} L) \arrow[d] \\ K_1 \overset{\mathrm{L}}{\boxtimes}_{\mathscr{A}} L_1 \arrow[u] \arrow[r] & K_1 \boxtimes_{\mathscr{A}} L_1 \arrow[r] & h_*M_2 \arrow[r] & Rh_*M_2. \end{tikzcd}$$

There is an analogous commutativity (cf. 4.2.13) starting instead from morphisms $K_1 \to f_*K$, $L_1 \to g_*L$, and $K \boxtimes_{\mathscr{A}} L \to M_2$.

The commutativity in 5.4.1.5 makes it, in principle, easy to verify the compatibility properties of the Künneth morphism.

5.4.2. Let (5.4.1.1) be a commutative diagram of schemes, and let $\mathscr{A}$ be a torsion sheaf on S. Suppose that f and g are compactifiable (3.2.1), that $Z = X \times_S Y$, and that $Z' = X' \times_S Y'$. Factor f and g as a proper morphism and an open immersion; this gives a diagram

$$\begin{tikzcd}
X' \arrow[d, "j_1"] \arrow[dd, bend right=60, "f"'] & Z' \arrow[l, "p'"'] \arrow[r, "q'"] \arrow[d, "j_3"] & {} \arrow[d, "j_2"] \arrow[dd, bend left=60, "g"] \\
\overline{X}' \arrow[d, "\overline{f}"] & \overline{Z}' \arrow[l, "\overline{p}"'] \arrow[r, "\overline{q}"] \arrow[d, "\overline{h}"] & \overline{Y}' \arrow[d, "\overline{g}"] \\
X & Z \arrow[l, "p"'] \arrow[r, "q"] & Y
\end{tikzcd}$$

in which $\overline{Z}' = \overline{X}' \times_S \overline{Y}'$. For complexes K and L on X' and Y', there is an isomorphism 368
of an immediate nature

$$j_{3!}(K \boxtimes_{\mathscr{A}} L) \xrightarrow{\sim} j_{1!}K \boxtimes_{\mathscr{A}} j_{2!}L. \tag{5.4.2.1}$$

By "composing" the inverse of this isomorphism with the Künneth arrow (5.4.1.4) relative to $(\overline{f}, \overline{g}, \overline{h}, j_{1!}K, j_{2!}L)$, one obtains, *for $K \in \operatorname{Ob} D^-(X', \mathscr{A})$ and $L \in \operatorname{Ob} D^-(Y', \mathscr{A})$, a Künneth arrow*

$$Rf_!K \overset{\mathbf{L}}{\boxtimes}_{\mathscr{A}} Rg_!L \longrightarrow Rh_!(K \overset{\mathbf{L}}{\boxtimes}_{\mathscr{A}} L), \tag{5.4.2.2}$$

which one checks is independent of the chosen compactifications.

THEOREM 5.4.3. *Under the hypotheses of 5.4.2, the Künneth arrow for cohomology with proper supports (5.4.2.2) is an isomorphism.*

PROOF. (a) By construction, it suffices to verify the result when f and g (and hence also h) are proper.

(b) Let there be a commutative diagram

$$\begin{tikzcd}
X_1 \arrow[r, "p_1"] \arrow[d] & S_1 \arrow[d] & Y_1 \arrow[l, "q_1"'] \arrow[d] \\
X \arrow[r, "p"] & S & Y \arrow[l, "q"'].
\end{tikzcd} \tag{5.4.3.1}$$

and let $Z_1 = X_1 \times_{S_1} Y_1$, $X_1' = X' \times_X X_1$, $Y_1' = Y' \times_Y Y_1$, and $Z_1' = X_1' \times_{S_1} Y_1' = Z' \times_{Z_1} Z$, so that there is a morphism of diagrams (5.4.1.1),

$$(S_1, X_1, Y_1, Z_1, X_1', Y_1', Z_1') \longrightarrow (S, X, Y, Z, X', Y', Z')$$

all of whose arrows will be denoted uniformly by C.

We shall admit that: 369

LEMMA 5.4.3.2. *The following diagram is commutative:*

$$\begin{tikzcd}
C^*(Rf_!K \overset{L}{\boxtimes}_{\mathscr{A}} Rg_!L) \arrow[r, equal] \arrow[d] & C^*Rf_!K \overset{L}{\boxtimes}_{\mathscr{A}} C^*Rg_!L \arrow[r, "\sim"] & Rf_{1!}C^*K \overset{L}{\boxtimes}_{\mathscr{A}} Rg_{1!}C^*L \arrow[d] \\
C^*(Rh_!(K \overset{L}{\boxtimes}_{\mathscr{A}} L)) \arrow[r, "\sim"] & Rh_{1!}(C^*(K \overset{L}{\boxtimes}_{\mathscr{A}} L)) \arrow[r, equal] & Rh_{1!}(C^*K \overset{L}{\boxtimes}_{\mathscr{A}} C^*L).
\end{tikzcd}$$

In this diagram, the horizontal arrows are base-change arrows and the vertical arrows are the Künneth arrows for the two diagrams (5.4.1.1) under consideration.

To prove 5.4.3, it suffices to prove that the Künneth arrow is an isomorphism at every geometric point of Z. By 5.4.3.2, it suffices to verify this after any base change (5.4.3.1) such that S_1 is the spectrum of an algebraically closed field and p_1, q_1 are isomorphisms.

After such a base change, diagram (5.4.1.1) reduces to a diagram

(5.4.3.3)
$$\begin{tikzcd} & X \times_S Y \arrow[dl, "p"'] \arrow[dr, "q"] \arrow[dd, "f\times g"] & \\ X \arrow[dr, "f"'] & & Y \arrow[dl, "g"] \\ & S & \end{tikzcd}$$

and the Künneth arrow reduces to an arrow (5.4.4.1)

(5.4.3.4)
$$k : Rf_!K \overset{\mathbf{L}}{\otimes}_{\mathscr{A}} Rg_!L \longrightarrow R(f\times g)_!(K \overset{\mathbf{L}}{\boxtimes}_{\mathscr{A}} L).$$

370 (c) Let (5.4.3.3) be a diagram in which f and g are proper. We shall admit that:

Lemma 5.4.3.5. *The following diagram is commutative:*

$$\begin{tikzcd} Rf_!K \underset{\mathscr{A}}{\overset{\mathbf{L}}{\otimes}} Rg_!L \arrow[r] \arrow[d] & R(f\times g)_!(K \underset{\mathscr{A}}{\overset{\mathbf{L}}{\otimes}} L) \arrow[d, equal] \\ Rf_!(K \overset{\mathbf{L}}{\otimes}_{\mathscr{A}} f^*Rg_!L) \arrow[r] & Rf_!Rp_!(p^*K \overset{\mathbf{L}}{\otimes}_{\mathscr{A}} q^*L). \end{tikzcd}$$

In this diagram, the horizontal arrows are the arrow of Künneth and the base-change arrow. The vertical arrows are a base-change arrow by f and a transitivity isomorphism. The theorem 5.4.3 follows from this lemma and 4.3.1.

The diligent reader may verify that this lemma remains valid if f and g are not necessarily proper.

5.4.4. Let S be a scheme, let $f_i : X_i' \to X_i$ $(i \in I)$ be a finite family of compactifiable morphisms of S-schemes, let X be the product over S of the X_i, let X' be the product over S of the X_i', and let $f : X' \to X$ be the product of the f_i. If S is ringed by $\mathscr{A}$ as in 5.4.1, and the X_i and X_i' are ringed by the inverse image of $\mathscr{A}$, there is more generally an isomorphism

(5.4.4.1)
$$\overset{\mathbf{L}}{\underset{i\in I}{\boxtimes}} Rf_{i!}(K_i) \xrightarrow{\sim} Rf_!(\overset{\mathbf{L}}{\underset{i\in I}{\boxtimes}} K_i),$$

for $K_i \in \operatorname{Ob} D^-(X_i', \mathscr{A})$. Neither the definition of this isomorphism nor that of the two sides of (5.4.4.1) *requires* the choice of a total order on I.

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5.5. Cohomology of symmetric powers. The main result of this n° is Theorem 5.5.21, which in particular gives the cohomology with proper supports of a symmetric power of a scheme X over an algebraically closed field k, for suitable coefficients (for example $\mathbf{Z}/n\mathbf{Z}$), in terms of the cohomology with proper supports of X. The analogue of this theorem in semisimplicial topology is well known, at least for cohomology with coefficients in a commutative ring A; it can be verified at the level of semisimplicial cochains. Indeed, if X_* is a semisimplicial set, there is a canonical continuous bijection from the symmetric $n-th$ power $(X_*)^n/\mathfrak{S}_n$ of X_* to the n^{th} symmetric power of the geometric realization of X_*, and this bijection induces an isomorphism on cohomology.

5.5.1. Recall some properties (due to D. Lazard) of flat modules, some properties (due to Roby) of the functors TS^n and T^n, and some properties (due to Dold and Puppe) of the derived functors of nonadditive functors.

. *Every flat module over a ring A is a filtered direct limit of finitely generated free modules* (Lazard [7, I 1.2]). This result implies the following statements:

. The functor $\varinjlim$ is an equivalence from the category of ind-objects (I 8.1) of the category of finitely generated free (resp. finitely generated projective) A-modules to the category of flat A-modules.

. If A and B are two rings, every functor $T : \mathrm{Mod}(A) \to \mathrm{Mod}(B)$ that carries finitely 372
generated free modules to flat modules and commutes with filtered direct limits carries flat modules to flat modules (Lazard [7, I 1.5]).

. Every exact sequence of A-modules, with L'' flat,

$$0 \longrightarrow L' \longrightarrow L \longrightarrow L'' \longrightarrow 0$$

is a filtered inductive limit of split exact sequences (Lazard [7, I 2.3]).

5.5.2. Let A be a commutative ring and M an A-module.

. For every integer n, denote by $TS^n(M)$ (or $TS^n_A(M)$) the module of symmetric tensors of degree n of M. Denoting by $\mathfrak{S}_n$ the symmetric group of degree n, by definition one has

$$TS^n(M) = (\bigotimes^n M)^{\mathfrak{S}_n}.$$

If B is an A-algebra, then $TS^n(B)$ is a subalgebra of $\bigotimes^n B$. If, moreover, B is commutative, then $\mathrm{Spec}(TS^n(B))$ is by definition the n^{th} symmetric power of $\mathrm{Spec}(B)$ over $\mathrm{Spec}(A)$:

$$\mathrm{Spec}(TS^n_A(B)) = \mathrm{Sym}^n_{\mathrm{Spec}(A)}(\mathrm{Spec}(B) \underset{\text{dfn}}{=} (\mathrm{Spec}(B)/\mathrm{Spec}(A))^n/\mathfrak{S}_n.$$

. Recall (Roby [10, I 2 p. 219]) that a *polynomial law* f from an A-module M to an A-module N is the datum, for every commutative A-algebra A', of a map $f_{A'} : M \otimes_A A' \to N \otimes_A A'$, these maps being such that, for every homomorphism of A-algebras $u : A' \to A''$, the diagram

$$\begin{tikzcd}
M \otimes_A A' \arrow[r, "f_{A'}"] \arrow[d, "u"] & N \otimes_A A' \arrow[d, "u"] \\
M \otimes_A A'' \arrow[r, "f_{A''}"] & N \otimes_A A''
\end{tikzcd}$$

is commutative. 373

A polynomial law f is said to be *homogeneous of degree n* (Roby [10, I 8 p. 226]) if, for every $a' \in A'$, one has identically

$$f_{A'}(a'x) = a'^n f_{A'}(x).$$

Such a polynomial law will also be called an n^{ic} *map* (quadratic, cubic, ...); their set will be denoted by

$$\mathrm{Hom}\, n^{\text{ique}}(M, N).$$

More generally, if $(M_i)_{i\in I}$ is a finite family of A-modules and $\underline{n} = (n_i)_{i\in I}$ a family of integers, a multihomogeneous polynomial law of multidegree $\underline{n}$ from the modules M_i to N is a polynomial law from $\oplus M_i$ to N satisfying identically, for $a_i \in A'$,

$$f_{A'}((a_i x_i)_{i\in I}) = \prod_{i\in I} a_i^{n_i} \cdot f_{A'}((x_i)_{i\in I}).$$

We shall denote by $\mathrm{Hom}\,\underline{n}^{\mathrm{ique}}((M_i), N)$ the set of these polynomial laws.

. For every module M, the covariant functor from A-modules to sets

$$N \longmapsto \mathrm{Hom}\, n^{\mathrm{ique}}(M, N)$$

374 is represented by an A-module $\Gamma^n(M)$ (or $\Gamma^n_A(M)$), equipped with a *universal* n^{ic} *map* $\gamma^n : M \to \Gamma^n(M)$ (Roby [10, IV 1 p. 265]). Given $(M_i)_{i\in I}$ and $\underline{n}$ as in 5.5.2.2, the covariant functor

$$N \longmapsto \mathrm{Hom}\,\underline{n}^{\mathrm{ique}}((M_i), N)$$

is represented by the module $\bigotimes_{i\in I} \Gamma^{n_i}(M_i)$, equipped with the *universal multihomogeneous polynomial law (of multidegree* $\underline{n}$*)* $(m_i) \mapsto \bigotimes_{i\in I} \gamma^{n_i}(m_i)$.

. The functors TS^n and Γ^n commute with filtered inductive limits (for the latter, Roby [10, IV 7 p. 277]). They take free modules to free modules, and hence flat modules to flat modules (5.5.1.2).

. The n^{ique} map from M to $TS^n(M)$ given by $m \mapsto \bigotimes^n m$ determines a map φ, called canonical, from $\Gamma^n(M)$ to $TS^n(M)$, satisfying

$$\varphi(\gamma^n(m)) = \bigotimes^n m.$$

This map is an isomorphism when M is finite free (Roby [10, IV 5 p. 272]), and hence also when M is flat (5.5.2.4 and 5.5.1.1).

Let M be an A-module and P an exact sequence

$$P : P_1 \xrightarrow{d} P_0 \xrightarrow{\epsilon} M \longrightarrow 0.$$

Let $P_1' = P_1 \times P_0$, and let j_0 and j_1 be the morphisms from P_1' to P_0 with coordinates (d, Id) and $(0, \mathrm{Id})$. The sequence

$$\Gamma^n(P_1') \underset{\Gamma^n(j_1)}{\overset{\Gamma^n(j_0)}{\rightrightarrows}} \Gamma^n(P_0) \longrightarrow \Gamma^n(M) \longrightarrow 0$$

375 is then exact (Roby [10, IV 10 p. 284]). *When P is a flat presentation of M*, this yields a description of the functor Γ^n in terms of the functor TS^n

(5.5.2.5.1) $$TS^n(P_1') \rightrightarrows TS^n(P_0) \to \Gamma^n(M) \to 0.$$

. If $M = M' \oplus M''$, the n^{ique} map

$$m \longmapsto (\gamma^{n'} m \otimes \gamma^{n''} m)_{n'+n''=n}$$

defines an isomorphism, called canonical (Roby [10, III 9 p. 262])

(5.5.2.6.1) $$\Gamma^n(M) \xrightarrow{\sim} \bigoplus_{n'+n''=n} \Gamma^{n'}(M) \otimes \Gamma^{n''}(M).$$

If Σ is an exact sequence

$$\Sigma : 0 \longrightarrow M' \longrightarrow M \longrightarrow M'' \longrightarrow 0,$$

the module $\Gamma^n(M)$ is equipped with a canonical (decreasing) filtration and morphisms

$$\Gamma^i(M'') \otimes \Gamma^{n-i}(M') \longrightarrow \mathrm{Gr}^i(\Gamma^n(M)). \tag{5.5.2.6.2}$$

These data are characterized by being functorial in Σ and satisfying the evident compatibility with (5.5.2.6.1) when Σ is split. The arrows (5.5.2.6.2) are isomorphisms when Σ is split, and hence also when M'' is flat (5.5.1.4).

. The functor Γ^n commutes with extension of scalars (Roby [10, III 8 p. 262]) (the analogous property for TS^n is false in general).

. If $u : M_1 \times M_2 \to N$ is a bilinear map, then $\gamma^n \circ u$ is a bihomogeneous polynomial law of bidegree (n, n) and therefore determines (5.5.2.3) $u^n : \Gamma^n(M_1) \otimes \Gamma^n(M_2) \to \Gamma^n(N)$ satisfying $u^n(\gamma^n x \otimes \gamma^n y) = \gamma^n u(x, y)$. In particular, if B is an A-algebra, multiplication 376
$B \times B \to B$ induces a product on $\Gamma^n(B)$. This product makes $\Gamma^n(B)$ an A-algebra, and the canonical map from $\Gamma^n(B)$ to $TS^n(B)$ is an algebra homomorphism.

5.5.3. Let $\mathscr{A}$ be an abelian category. We shall denote by $\mathrm{ssimpl}^+(\mathscr{A})$ the category of co-semisimplicial objects[65] in $\mathscr{A}$, by $C^{\geq 0}(\mathscr{A})$ the category of differential complexes K with $K^i = 0$ for $i < 0$, and by $K^{\geq 0}(\mathscr{A})$ the image of $C^{\geq 0}(\mathscr{A})$ in $K(\mathscr{A})$. If $K \in \mathrm{Ob}\,\mathrm{ssimpl}^+(\mathscr{A})$, the associated normalized complex $N(K) \in \mathrm{Ob}\,C^{\geq 0}(\mathscr{A})$ is defined by

$$\begin{cases} N(K)^n = \text{the intersection of the kernels of the degeneracy operators} \\ \qquad\qquad \text{from } K^n \text{ to } K^{n-1} \\ d = \sum(-1)^i \partial_i. \end{cases} \tag{5.5.3.1}$$

If K also denotes the complex with components K^n and differential $d = \sum(-1)^i\partial_i$, then $N(K)$ is a direct-summand subcomplex of K, and $K/N(K)$ is homotopic to zero.

By Dold-Puppe [1], the functor N is an equivalence of categories

$$N : \mathrm{ssimpl}^+(\mathscr{A}) \longrightarrow C^{\geq 0}(\mathscr{A}).$$

This equivalence and its inverse take homotopic morphisms to homotopic morphisms. Moreover, for every $K \in \mathrm{Ob}\,\mathrm{ssimpl}^+(\mathscr{A})$, K^n is a direct sum of copies of $N(K)^i$ $(i \leq n)$; if $\mathbf{P}$ is a property of objects of $\mathscr{A}$, stable under finite direct sums and direct summands, it follows that the K^i have property $\mathbf{P}$ for $i \leq n$ if and only if the $N(K)^i$ have this property for $i \leq n$.

The equivalence N is used by Dold and Puppe to extend every functor (not neces- 377
sarily additive) $T : \mathscr{A} \to \mathscr{B}$ from $\mathscr{A}$ to an arbitrary category $\mathscr{B}$ to a functor $C^{\geq 0}(T)$ (or simply T) from $C^{\geq 0}(\mathscr{A})$ to $C^{\geq 0}(\mathscr{B})$

$$C^{\geq 0}(T) = NTN^{-1}. \tag{5.5.3.2}$$

The functor $C^{\geq 0}(T)$ passes to the quotient to define

$$T \quad \text{or} \quad K^{\geq 0}(T) : K^{\geq 0}(\mathscr{A}) \longrightarrow K^{\geq 0}(\mathscr{B}). \tag{5.5.3.3}$$

. If T is *of degree* $\leq n$, i.e. if the $(n+1)^{\text{th}}$ "cross-effect" vanishes (Dold-Puppe [1, 4.18 p. 232]), then, for every $K \in \mathrm{Ob}\,C^{\geq 0}(\mathscr{A})$ such that $K^i = 0$ for $i > m$, one has $[T(K)]^i = 0$ for $i > nm$ (Dold-Puppe [1, 4.23 p. 233]).

. In particular, if T is a finite-degree functor from the category of A-modules to that of B-modules, which takes flat objects to flat objects, then the image of a bounded flat complex is a bounded flat complex.

Lemma 5.5.4. *Let $\mathscr{A}$ be an abelian category.*

(i) *The quasi-isomorphisms in $K^{\geq 0}(\mathscr{A})$ admit a calculus of left and right fractions.*

[65] N.D.E.: In modern terminology they are called *cosimplicial* objects.

(ii) *The evident functor is an equivalence between the category obtained from $K^{\geq 0}(\mathscr{A})$ by inverting the quasi-isomorphisms and the category $D^{\geq 0}(\mathscr{A})$.*

378 The functor $\tau_{\geq 0} : C(\mathscr{A}) \to C^{\geq 0}(\mathscr{A})$ (1.1.13) passes to the quotient to define a functor $\tau_{\geq 0} : K(\mathscr{A}) \to K^{\geq 0}(\mathscr{A})$, left adjoint to the inclusion of $K^{\geq 0}(\mathscr{A})$ in $K(\mathscr{A})$. The functor $\tau_{\geq 0}$ takes quasi-isomorphisms to quasi-isomorphisms, and the adjunction arrow $X \to \tau_{\geq 0}(X)$ is a quasi-isomorphism if and only if $H^i(X) = 0$ for $i < 0$. Lemma 5.5.4 therefore follows from Verdier [11] and parts (i) and (ii) of the following lemma, whose proof is left to the reader:

Lemma 5.5.4.1. *Let K be a category, K' a full subcategory, S a set of morphisms of K equal to the inverse image of the set of isomorphisms of $K(S^{-1})$ (cf. [2, Chap. 1]), and S' its trace in K'. Suppose that the inclusion i of K' in K has a left adjoint τ, and that $\tau(S) \subset S'$.*

(i) *The evident functor from $K'(S'^{-1})$ to $K(S^{-1})$ is fully faithful, and has τ as a left adjoint.*

(ii) *If $K(S^{-1})$ admits a calculus of left fractions (resp. right fractions), then $K'(S'^{-1})$ has the same property.*

If the inclusion i does not necessarily have a left adjoint, and if τ denotes the not everywhere defined functor left adjoint to i, one has:

(iii) *If $K(S^{-1})$ admits a calculus of left fractions (resp. right fractions) and if, for every $u \in S$ with target (resp. source) in K', $\tau(u)$ is defined and belongs to S', then $K'(S'^{-1})$ admits a calculus of left fractions (resp. right fractions), and the functor $K'(S'^{-1}) \to K(S^{-1})$ is fully faithful.*

379 5.5.5. Let $T : \mathscr{A} \to \mathscr{B}$ be a functor between abelian categories. As in 1.2.1, one defines the derived functors of T

$$LT : D^{\geq 0}(\mathscr{A}) \longrightarrow \mathrm{Pro}(D^{\geq 0}(\mathscr{B}))$$
$$RT : D^{\geq 0}(\mathscr{A}) \longrightarrow \mathrm{Ind}(D^{\geq 0}(\mathscr{B}))$$

by the formulas

$$LT(K) = \underset{s:K'\to K}{\text{“}\varprojlim\text{”}}\ T(K')$$
$$RT(K) = \underset{s:K\to K'}{\text{“}\varinjlim\text{”}}\ T(K'),$$

where s ranges over the filtered category of quasi-isomorphisms with target (resp. source) K.

We say that T is *left (resp. right) derivable at K if $LT(K)$ $(RT(K))$* is representable. We say that K is *left (resp. right) adapted* to T (or, by abuse of language, adapted to LT (resp. RT)) if the following canonical arrow is an isomorphism:

$$LT(K) \xrightarrow{\sim} TK \text{ (resp. } TK \xrightarrow{\sim} RT(K)).$$

Proposition 5.5.6. *Let F be a functor from the category of modules over a ring A to an abelian category $\mathscr{B}$. Suppose that filtered inductive limits in $\mathscr{B}$ are representable and exact, and that F commutes with filtered inductive limits. Then every bounded complex $M^\bullet \in \mathrm{Ob}\, K^{\geq 0}(A)$ of flat modules is adapted to LF.*

380 The proposition follows immediately from the definitions, the following lemma, which will be proved in 5.5.6.4, and the fact that the quasi-isomorphisms $u : N^\bullet \to M^\bullet$,

with $N^{\boldsymbol{\cdot}} \in C^{b,\geq 0}(A)$ having flat components, form a cofinal set in the category of quasi-isomorphisms with target $M^{\boldsymbol{\cdot}}$.

LEMMA 5.5.6.1. *Let $u : N^{\boldsymbol{\cdot}} \to M^{\boldsymbol{\cdot}}$ be a quasi-isomorphism between flat complexes that are objects of $C^{b,\geq 0}(A)$. Then $F(u)$ is a quasi-isomorphism.*

LEMMA 5.5.6.2. *Let $\mathscr{A}$ be an additive category and d an interval in $\mathbf{Z}$ that is bounded above. Denote by C^d (resp. $C^{d,b}$, resp. K^d, resp. $K^{d,b}$) the category of complexes that vanish in degrees $i \notin d$ (resp. and are bounded, resp. up to homotopy). There is an equivalence*

$$\ell : \mathrm{Ind}(C^{d,b}(\mathscr{A})) \xrightarrow{\sim} C^d(\mathrm{Ind}\,\mathscr{A});$$

the inverse of this equivalence induces a functor

$$\ell' : K^d(\mathrm{Ind}\,\mathscr{A}) \longrightarrow \mathrm{Ind}(K^{d,b}(\mathscr{A})).$$

A) Let us prove that ℓ is fully faithful. Thus, let K be a bounded complex of objects of $\mathscr{A}$ and $L = \text{“}\varinjlim\text{”}\, L_\alpha$ an ind-complex. One has

$$\begin{aligned}\mathrm{Hom}(K, L) &\overset{\mathrm{dfn}}{=} \varinjlim \mathrm{Hom}(K, L_\alpha) = \varinjlim Z^0(\mathrm{Hom}^{\boldsymbol{\cdot}}(K, L_\alpha)) \simeq Z^0 \varinjlim_{\alpha} \mathrm{Hom}^{\boldsymbol{\cdot}}(K, L_\alpha)\\ &\xrightarrow{\sim} Z^0\,\mathrm{Hom}^{\boldsymbol{\cdot}}(K, \ell L) = \mathrm{Hom}(K, \ell L),\end{aligned}$$

the penultimate isomorphism following from the fact that K is finite, so that $\varinjlim$ commutes with passage from the double complex to the total complex. This proves the assertion.

B) We leave it to the reader to verify (rather laboriously) that ℓ is essentially sur- 381
jective by constructing, from left to right and starting in an arbitrarily chosen degree, “enough” bounded complexes mapping to a bounded-above complex of ind-objects K; and likewise to develop the following point:

C) If $\ell f : \ell \varinjlim K_\alpha \to \ell \varinjlim L_\beta$ is null-homotopic, $f = dH + Hd$, then, for every α, H lifts to $H^i_\alpha : K^i_\alpha \to L^{i-1}_{\beta(\alpha)}$ which, for β sufficiently large, defines a null-homotopy of $f_\alpha : K_\alpha \to L_{\beta(\alpha)}$.

. Let A be a ring, d an interval in $\mathbf{Z}$ bounded above, $P(A)$ the category of finite projective A-modules, and $\mathrm{Pl}(A)$ the category of flat A-modules. Denote by $D^{\mathrm{Tor}\subset d}(A)$ the image of $C^d(\mathrm{Pl}(A))$ in $D(A)$ and by $D^{\mathrm{parf}\subset d}(A)$ the image of $C^{d,b}(P(A))$ in $D(A)$. By 5.5.1.2, one has $\mathrm{Pl}(A) \simeq \mathrm{Ind}\,P(A)$. On the other hand, $K^{d,b}(P(A)) \simeq D^{\mathrm{parf}\subset d}(A)$. It then follows from 5.5.6.2, and from the fact that H^i commutes with filtered inductive limits, that there exists a commutative diagram of functors

$$\begin{tikzcd}
C^d(\mathrm{Pl}(A)) \arrow[d] & \mathrm{Ind}\,C^{d,b}(P(A)) \arrow[l, "\varinjlim"', "\sim"] \arrow[d] \\
K^d(\mathrm{Pl}(A)) \arrow[r, "\ell'"] \arrow[d] & \mathrm{Ind}\,K^{d,b}(P(A)) \arrow[d, "\simeq"] \\
D^{\mathrm{Tor}\subset d}(A) \arrow[r, "\ell''"] & \mathrm{Ind}\,D^{\mathrm{parf}\subset d}(A).
\end{tikzcd}$$

. Let us prove 5.5.6.1. With the notation of 5.5.6.3, if d is an interval $[0, k]$, then (because F commutes with $\varinjlim$) there are commutative diagrams 382

$$\begin{tikzcd} K^d(\mathrm{Pl}(A)) \arrow[rr, "\ell'"] \arrow[d, "K^{\geq 0}(F)"'] & & \mathrm{Ind}\, K^d(P(A)) \arrow[d, "\mathrm{Ind}(K^{\geq 0}(F))"] \\ K^+(\mathscr{B}) \arrow[d, "H^i"'] & & \mathrm{Ind}\, K^+(\mathscr{B}) \arrow[d, "\mathrm{Ind}(H^i)"] \\ \mathscr{B} & & \mathrm{Ind}\, \mathscr{B}. \arrow[ll, "\varinjlim"'] \end{tikzcd}$$

If u is as in 5.5.6.1, then by 5.5.6.3, $\ell'(u)$ is an isomorphism, and $H^i K^{\geq 0}(F)(u)$ is therefore an isomorphism, as required.

5.5.7. Let X be a scheme over a scheme S, and let k be an integer. Suppose that the symmetric power $\mathrm{Sym}^k_S(X)$ is defined. A reader who wishes not to make this hypothesis may work in the category of algebraic spaces (Artin[66]). Ring S by a sheaf $\mathscr{A}$ of *commutative* rings. Beginning with 5.5.19, $\mathscr{A}$ will be assumed to be *torsion*. By abuse of notation, we shall *still denote by* $\mathscr{A}$ the inverse image of $\mathscr{A}$ on any S-scheme.

Let $\mathscr{F}$ be a sheaf of modules on X; the sheaf $\overset{k}{\boxtimes}\mathscr{F}$, the external tensor product of k copies of $\mathscr{F}$, is a sheaf on X^k/S, equivariant under the action of the symmetric group
383 $\mathfrak{S}_k$ on X^k/S. Passing to invariants, it defines a sheaf $TS^k_{\mathrm{ext}}(\mathscr{F})$ on $\mathrm{Sym}^k_S(X)$, called the *external symmetric tensor power of* $\mathscr{F}$.

To reason "point by point," it is useful to work with the points of $\mathrm{Sym}^k_S(X)$ described as follows.

. Let s be a geometric point of S. A point x of $\mathrm{Sym}^k_S(X)$ with values in $k(s)$ defines a geometric point of $\mathrm{Sym}^k_S(X)$. Let $(x_1 \dots x_k)$ denote a point of $(X/S)^k$ with values in $k(s)$ whose image is x, let $y_1 \dots y_\ell$ be the distinct elements of the sequence $x_1 \dots x_k$, and let n_i $(1 \leq i \leq \ell)$ be the multiplicity of y_i in this sequence ($\sum n_i = k$).

With the notation of 5.5.7.1, and writing $\mathfrak{S}$ for a symmetric group, one has

$$TS^k_{\mathrm{ext}}(F)_x \simeq (\bigotimes_1^\ell (\bigotimes^{n_i} F_{y_i}))^{\prod_1^\ell \mathfrak{S}_{n_i}}, \tag{5.5.7.2}$$

where the tensor products are taken over $\mathscr{A}_S$. In particular, *for F flat*, by 5.5.2.5 one has

$$TS^k_{\mathrm{ext}}(F)_x \xleftarrow{\sim} \bigotimes_1^\ell \Gamma^{n_i} F_{y_i}, \tag{5.5.7.3}$$

using induction on ℓ, the flatness of the $\Gamma^{n_i} F_{y_i}$ (5.5.2.4), and the following fact: if G (resp. H) is a finite group acting on an A-module M (resp. N), and if M and N^H are flat, then $M^G \otimes_A N^H \xrightarrow{\sim} (M \otimes_A N)^{G\times H}$.

5.5.8. Let F be a module on X, and let P be a flat presentation of F:

$$P : P_1 \longrightarrow P_0 \longrightarrow F \longrightarrow 0.$$

384 Define P_1', j_0, and j_1 as in 5.5.2.5, and define $\Gamma^k_{\mathrm{ext}}(F)$ (the *external Γ functor*) to be the

[66] N.D.E.: the original text cites an unspecified work here, probably one of the two works of Artin cited in the notes on pages 730 and 743.

cokernel of the pair of arrows $(TS^k_{\mathrm{ext}}(j_0), TS^k_{\mathrm{ext}}(j_1))$. With the notation of 5.5.7.1, one deduces from (5.5.7.3) and 5.5.2.5 isomorphisms

(5.5.8.1)
$$\Gamma^k_{\mathrm{ext}}(F)_x \simeq \bigotimes_1^{\ell} \Gamma^{n_i}(F_{y_i}).$$

This formula shows that $\Gamma^k_{\mathrm{ext}}(F)$, which a priori depends functorially on P, depends, up to a unique isomorphism, only on F.

5.5.9. There is a functorial morphism

$$\alpha : \Gamma^k_{\mathrm{ext}}(F) \longrightarrow TS^k_{\mathrm{ext}}(F).$$

The pair (Γ, α) is characterized by the following two properties:

(i) For F flat, α is an isomorphism.

(ii) With the notation of 5.5.2.5, for every presentation

$$P_1 \xrightarrow{d} P_0 \xrightarrow{\epsilon} F \longrightarrow 0$$

of F, the sequence

$$\Gamma^k_{\mathrm{ext}}(P_1') \rightrightarrows \Gamma^k_{\mathrm{ext}}(P_0) \to \Gamma^k_{\mathrm{ext}}(F) \to 0$$

is exact.

Moreover:

(iii) The functors TS^k and Γ^k commute with filtered direct limits (by (5.5.7.2), (5.5.8.1), 5.5.2.4).

(iv) The functor Γ^k commutes with every extension $\mathscr{A} \to \mathscr{A}'$ of the base ring.

(v) The functor Γ^k commutes with every base change $S' \to S$: for every cartesian 385
square

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ S' \arrow[r, "g"] & S, \end{tikzcd}$$

one has

$$\mathrm{Sym}^k_g(g')^* \Gamma^k_{\mathrm{ext}} \simeq \Gamma^k_{\mathrm{ext}} g'^*.$$

((5.5.8.1) and 5.5.2.7).

(vi) The functor Γ^k commutes with inverse images under every *radicial* morphism $i : X' \to X$ (for example a closed immersion): one has

$$\mathrm{Sym}^k_S(i)^* \Gamma^k_{\mathrm{ext}} \simeq \Gamma^k_{\mathrm{ext}} i^*$$

(5.5.8.1).

(vii) For an *immersion* $j : X' \to X$, one has

$$\mathrm{Sym}^k_S(j)_! \Gamma^k_{\mathrm{ext}} \simeq \Gamma^k_{\mathrm{ext}} j_!$$

(5.5.8.1).

5.5.10. Let k_1, k_2, and $k = k_1 + k_2$ be three integers ≥ 0. There is then a finite morphism ( ):[67]

$$\vee_{k_1 k_2} : \mathrm{Sym}_S^{k_1}(X) \times_S \mathrm{Sym}_S^{k_2}(X) \longrightarrow \mathrm{Sym}_S^{k}(X).$$

If F_i $(i = 1, 2)$ is a module on $\mathrm{Sym}_S^{k_i}(X)$, denote by $F_1 \vee F_2$ the sheaf on $\mathrm{Sym}_S^{k}(X)$

$$F_1 \vee F_2 = \vee_{k_1 k_2 *}(F_1 \boxtimes F_2).$$

386 5.5.11. Let there be an exact sequence of modules on X

$$\Sigma : 0 \longrightarrow F' \longrightarrow F \longrightarrow F'' \longrightarrow 0.$$

With the notation of 5.5.7.1, by (5.5.8.1) and 5.5.2.6 there is a filtration on $\Gamma^k_{\mathrm{ext}}(F)_x$, and there are morphisms

$$\sum_{\sum_1^\ell i_\alpha = i} \bigotimes_1^\ell \Gamma^{i_\alpha}(F'_{y_\alpha}) \otimes \bigotimes_1^\ell \Gamma^{n_i - i_\alpha}(F''_{y_\alpha}) \longrightarrow \mathrm{Gr}^i(\Gamma^k_{\mathrm{ext}}(F)_x).$$

The first member is precisely $(\Gamma^i_{\mathrm{ext}}(F') \vee \Gamma^{k-i}_{\mathrm{ext}}(F''))_x$. One readily checks that these data, constructed point by point, arise from a (unique) filtration with $k+1$ steps on $\Gamma^k_{\mathrm{ext}}(F)$ and from (unique) morphisms

$$\Gamma^i_{\mathrm{ext}}(F') \vee \Gamma^{k-i}_{\mathrm{ext}}(F'') \longrightarrow \mathrm{Gr}^i(\Gamma^k_{\mathrm{ext}}(F)). \tag{5.5.11.1}$$

If the exact sequence Σ is split, one has more precisely (verification point by point from (5.5.2.6.1)):

$$\Gamma^k_{\mathrm{ext}}(F' \oplus F'') \simeq \bigoplus_{k_1 + k_2 = k} \Gamma^{k_1}_{\mathrm{ext}}(F') \vee \Gamma^{k_2}_{\mathrm{ext}}(F''). \tag{5.5.11.2}$$

If the exact sequence Σ is split point by point, or if F'' is flat, the morphisms (5.5.11.1) are isomorphisms (cf. 5.5.1.4).

5.5.12. Let $u : X \to Y$ be a morphism of S-schemes. Suppose that the symmetric powers $\mathrm{Sym}_S^k(X)$ and $\mathrm{Sym}_S^k(Y)$ exist. Let F be a module on X, let G be a module on Y, and let $v \in \mathrm{Hom}(u^*G, F) \simeq \mathrm{Hom}(G, u_*F)$ be a u-homomorphism from F to G (0.2). If $\prod^n u$ is the n^{th} cartesian power of u, the homomorphism v induces a $\prod^n u$-
387 homomorphism $\overset{n}{\boxtimes} v$ from $\overset{n}{\boxtimes} F$ to $\overset{n}{\boxtimes} G$. Passing to invariants, the latter defines

$$TS^n(v) : TS^n_{\mathrm{ext}}(F) \xrightarrow[\mathrm{Sym}_S^n(u)]{} TS^n_{\mathrm{ext}}(G).$$

It follows from Definition 5.5.8 that there exists a unique $\mathrm{Sym}^n(u)$-homomorphism $\Gamma^n_{\mathrm{ext}}(v)$, functorial in (F, G, v),

$$\Gamma^n_{\mathrm{ext}}(v) : \Gamma^n_{\mathrm{ext}}(F) \xrightarrow[\mathrm{Sym}_S^n(u)]{} \Gamma^n_{\mathrm{ext}}(G) \tag{5.5.12.0}$$

[67] N.D.E.: I have not found references for the finiteness of $\vee_{k_1 k_2}$ in general.

making the following diagram commutative (cf. (0.2) for the unusual direction of the arrows):

$$\begin{tikzcd}
\Gamma^n_{\text{ext}}(F) \arrow[rr, "\Gamma^n_{\text{ext}}(v)", "\text{over } \mathrm{Sym}^n_S(u)"'] & & \Gamma^n_{\text{ext}}(G) \\
TS^n_{\text{ext}}(F) \arrow[u, "\alpha", "\text{over Id}"'] \arrow[rr, "TS^n_{\text{ext}}(v)", "\text{over } \mathrm{Sym}^n_S(u)"'] & & TS^n_{\text{ext}}(G). \arrow[u, "\alpha", "\text{over Id}"']
\end{tikzcd}$$

We shall use this arrow mainly as a "symmetric Künneth arrow": for $G = u_* F$ and v the canonical arrow, the arrow $\Gamma^n_{\text{ext}}(v)$ identifies with a homomorphism of modules

(5.5.12.1) $$k^n : \Gamma^n_{\text{ext}}(u_* F) \longrightarrow \mathrm{Sym}^n(u)_* \Gamma^n_{\text{ext}}(F).$$

5.5.13. Consider the condition:

. $K \in \mathrm{Ob}\, K^{b,\geq 0}(X, \mathscr{A})$ and, for every geometric point x of X, the complex K_x is homotopy equivalent to a complex of flat modules $L_x \in \mathrm{Ob}\, C^{b,\geq 0}(\mathscr{A}_x)$.

Lemma 5.5.13.2. *The subcategory $D^{b,\mathrm{tor}\leq 0}(X, \mathscr{A})$ of $D^b(X, \mathscr{A})$ consisting of complexes* 388
of tor-dimension ≤ 0 4.1.9 is equivalent to the category obtained from the category of complexes $K \in \mathrm{Ob}\, K^{b,\geq 0}(X, \mathscr{A})$ satisfying (5.5.13.1) by inverting quasi-isomorphisms. This construction may be performed indifferently by a calculus of left or right fractions.

Let P be the subcategory of $K^-(X, \mathscr{A})$ consisting of complexes that are pointwise homotopy equivalent to a complex of flat modules. It is known (Verdier [11, I 2.4.2]) that $D^-(X, \mathscr{A})$ is obtained from P by inverting quasi-isomorphisms, using a calculus of right or left fractions. Apply 5.5.4.1 to the inclusion of $K^{b,\geq 0}(X, \mathscr{A}) \cap P$ into P. Indeed, if K has tor-dimension ≤ 0 and is pointwise homotopy equivalent to complexes of flat modules L_x, then $\tau_{\geq 0}(K)$ is pointwise homotopy equivalent to the $\tau_{\geq 0}(L_x)$, and the latter are still complexes of flat modules, so $\tau_{\geq 0}(K)$ satisfies 5.5.13.1.

5.5.14. By 5.5.6.1, (5.5.7.2) or (5.5.8.1) and 5.5.2.4, every quasi-isomorphism $v : K' \to K$ between complexes satisfying (5.5.13.1) induces quasi-isomorphisms

$$TS^k(v) : TS^k_{\text{ext}}(K') \xrightarrow{\approx} TS^k_{\text{ext}}(K)$$
$$\Gamma^k(v) : \Gamma^k_{\text{ext}}(K') \xrightarrow{\approx} \Gamma^k_{\text{ext}}(K).$$

It follows that the functors Γ^k and TS^k are derivable (5.5.5) on the subcategory $D^{b,\mathrm{tor}\leq 0}(X, \mathscr{A})$ of $D^b(X, \mathscr{A})$, and that complexes satisfying 5.5.13.1 are left adapted to these functors. By 5.5.2.4, Γ^k and TS^k induce functors 389

(5.5.14.1) $$LTS^k : D^{b,\mathrm{tor}\leq 0}(X, \mathscr{A}) \longrightarrow D^{b,\mathrm{tor}\leq 0}(\mathrm{Sym}^k_S(X), \mathscr{A})$$

(5.5.14.2) $$L\Gamma^k : D^{b,\mathrm{tor}\leq 0}(X, \mathscr{A}) \longrightarrow D^{b,\mathrm{tor}\leq 0}(\mathrm{Sym}^k_S(X), \mathscr{A}).$$

Moreover, by 5.5.9 (i), the canonical morphism α (5.5.9) induces an isomorphism between the functors (5.5.14.1) and (5.5.14.2):

(5.5.14.3) $$\alpha : L\Gamma^k \xrightarrow{\sim} LTS^k.$$

. The isomorphisms of 5.5.9 (iv), (v), (vi), and (vii) pass to the derived category and define isomorphisms:

(iv′) For an $\mathscr{A}$-algebra $\mathscr{A}'$, one has

$$L\Gamma^k_{\mathrm{ext}_{\mathscr{A}}}(K) \overset{\mathbf{L}}{\otimes} \mathscr{A}' \simeq L\Gamma^k_{\mathrm{ext}_{\mathscr{A}'}}(K \overset{\mathbf{L}}{\underset{\mathscr{A}}{\otimes}} \mathscr{A}').$$

(v′) For every base change $g : S' \to S$, one has

$$\mathrm{Sym}^k_g(g')^* L\Gamma^k_{\mathrm{ext}} K \simeq L\Gamma^k_{\mathrm{ext}}(g'^* K).$$

(vi′) For every *radicial* morphism $i : X' \to X$, one has

$$\mathrm{Sym}^k_S(i)^* L\Gamma^k_{\mathrm{ext}} K \simeq L\Gamma^k_{\mathrm{ext}} i^* K.$$

(vii′) For an *immersion* $j : X' \to X$, one has

$$\mathrm{Sym}^k_S(j)_! L\Gamma^k_{\mathrm{ext}} \simeq L\Gamma^k_{\mathrm{ext}} j_!.$$

5.5.15. Let there be an exact sequence of flat complexes in $C^{\geq 0}(X, \mathscr{A})$,

$$0 \longrightarrow K' \longrightarrow K \longrightarrow K'' \longrightarrow 0,$$

and apply the functor Γ^k_{ext} (5.5.3). One finds (as in 5.5.11) that the semisimplicial module
390 $\Gamma^k_{\mathrm{ext}}(N^{-1}K)$ on $\mathrm{Sym}^k_S(X)$ has a filtration whose successive quotients are the semisimplicial modules $\Gamma^{k'}_{\mathrm{ext}}(N^{-1}K') \vee \Gamma^{k''}_{\mathrm{ext}}(N^{-1}K'')$ $(k' + k'' = k)$.

Passing to the derived category, one finds that $L\Gamma^k_{\mathrm{ext}}(K)$ is represented by a filtered complex whose successive quotients represent the objects $L\Gamma^i_{\mathrm{ext}}(K') \overset{\mathbf{L}}{\vee} L\Gamma^{k-i}_{\mathrm{ext}}(K'')$ of the derived category (apply 5.5.2.4 and the Eilenberg–Zilber–Cartier theorem (isomorphism in $\mathscr{K}$ between a bisemisimplicial object and its diagonal: Dold-Puppe [1, 2.9 p. 213]).

In the derived category, this construction can already be made from an exact sequence of complexes satisfying 5.5.13.1.

* In more intrinsic terms, let Δ be a "*true triangle*" of objects of $D^{b,\mathrm{tor}\leq 0}(X, \mathscr{A})$, i.e. an object of the "filtered derived category" with $\mathrm{Gr}^i(\Delta) = 0$ for $i \neq 0, 1$ and $\mathrm{Gr}^i(\Delta) \in \mathrm{Ob}\, D^{b,\mathrm{tor}\leq 0}(X, \mathscr{A})$ (Deligne, in the Appendix to Verdier [12], or 2.5.1, 2.5.2).

The preceding construction then gives an object $L\Gamma^k_{\mathrm{ext}}(\Delta)$ of the filtered derived category of $\mathrm{Sym}^k_S(X)$, isomorphisms

$$\mathrm{Gr}^i(L\Gamma^k_{\mathrm{ext}}(\Delta)) \simeq L\Gamma^i_{\mathrm{ext}}(\mathrm{Gr}^1(\Delta)) \overset{\mathbf{L}}{\vee} L\Gamma^{k-i}_{\mathrm{ext}}(\mathrm{Gr}^0(\Delta))$$

and, denoting by $|\quad|$ the underlying complex of a filtered complex, an isomorphism

$$|L\Gamma^k_{\mathrm{ext}}(\Delta)| \simeq L\Gamma^k_{\mathrm{ext}}(|\Delta|).{}_*$$

Lemma 5.5.16. *Let N be an integer. The category $D^{b,\mathrm{tor}\leq 0}(X, \mathscr{A})$ is obtained from the category of complexes $K \in \mathrm{Ob}\, K^{b,\geq 0}(X, \mathscr{A})$ satisfying conditions* (i) *and* (ii) *below by inverting quasi-isomorphisms:*

391 (i) *condition (5.5.13.1).*

(ii) *The complex K is adapted to every direct-image functor of cohomological dimension $\leq N$, relative to an isomorphism of topoi $f : \widetilde{X}_{\mathrm{et}} \to T$.*

The existence of a calculus of right fractions follows from 5.5.13 and from the fact that, by truncated canonical flasque resolutions (4.2.9), every complex K satisfying (i) maps to a complex satisfying (i) and (ii).

5.5.17. Let $u : X \to Y$ be a morphism of S-schemes. Suppose that $\mathrm{Sym}^n_S(X)$ and $\mathrm{Sym}^n_S(Y)$ exist, that there is an integer N such that the functor u_* has dimension $\leq N$ on the category of $\mathscr{A}$-modules, and that the functor Ru_* carries bounded complexes of tor-dimension ≤ 0 to complexes of tor-dimension ≤ 0. The latter two conditions hold automatically if u is proper, $\mathscr{A}$ is torsion, and Y is coherent (i.e. quasi-compact and quasi-separated) (5.2.8.1, 5.2.10).[68]

Let $K \in \mathrm{Ob}\, K^{b,\geq 0}(X, \mathscr{A})$ satisfy conditions (i) and (ii) of 5.5.16, with N as above. By 5.5.14, the arrow

$$\beta : L\Gamma^n_{\mathrm{ext}}(K) \xrightarrow{\sim} \Gamma^n_{\mathrm{ext}}(K)$$

is a quasi-isomorphism. Similarly, the arrow

$$\gamma : u_*K \xrightarrow{\sim} Ru_*K$$

is a quasi-isomorphism.

Define the *symmetric Künneth arrow* from $L\Gamma^n_{\mathrm{ext}}Ru_*(K)$ to $R\,\mathrm{Sym}^n(u)_*L\Gamma^n_{\mathrm{ext}}(K)$ to 392
be the composite

$$(5.5.17.1)\quad \begin{aligned} k^n : L\Gamma^n_{\mathrm{ext}}Ru_*K \underset{\gamma}{\xleftarrow{\sim}} L\Gamma^n_{\mathrm{ext}}u_*K \to \Gamma^n_{\mathrm{ext}}u_*K \xrightarrow[5.5.12.1]{} \mathrm{Sym}^n_S(u)_*\Gamma^n_{\mathrm{ext}}(K) \\ \longrightarrow R\,\mathrm{Sym}^n_S(u)_*\Gamma^n_{\mathrm{ext}}(K) \underset{\beta}{\xleftarrow{\sim}} R\,\mathrm{Sym}^n_S(u)_*L\Gamma^n_{\mathrm{ext}}(K). \end{aligned}$$

It follows from 5.5.16 that the morphisms (5.5.17.1) define a morphism between functors from $D^{b,\mathrm{tor}\leq 0}(X, \mathscr{A})$ to $D^+(\mathrm{Sym}^n_S(Y), \mathscr{A})$:

$$(5.5.17.2)\qquad k^n : L\Gamma^n_{\mathrm{ext}}Ru_* \longrightarrow R\,\mathrm{Sym}^n_S(u)_*L\Gamma^n_{\mathrm{ext}},$$

also called the *symmetric Künneth morphism.*

This morphism is characterized by Proposition 5.5.18 below.

. Let S, X, Y, and u be as above, let $K \in \mathrm{Ob}\, C^{b,\geq 0}(X, \mathscr{A})$ be a complex of tor-dimension ≤ 0, let $K_1 \in \mathrm{Ob}\, C^{b,\geq 0}(Y, \mathscr{A})$, and let $K_2 \in \mathrm{Ob}\, C^b(\mathrm{Sym}^n_S(X), \mathscr{A})$. Let v be a u-homomorphism (0.2) from K to K_1, and let t be a homomorphism (in the usual sense, as are all those that follow) from $\Gamma^n_{\mathrm{ext}}(K)$ to K_2. There are morphisms

a) $\Gamma^n_{\mathrm{ext}}(v) : \Gamma^n_{\mathrm{ext}}(K_1) \to \mathrm{Sym}^n_S(u)_*\Gamma^n_{\mathrm{ext}}(K)$ (5.5.12.0)

b) $\mathrm{Sym}^n_S(u)_*(t) : \mathrm{Sym}^n_S(u)_*\Gamma^n_{\mathrm{ext}}(K) \to \mathrm{Sym}^n_S(u)_*K_2$,

whose composite

$$\varphi' : \Gamma^n_{\mathrm{ext}}(K_1) \longrightarrow \mathrm{Sym}^n_S(u)_*K_2,$$

defines, by composition,

$$\varphi : L\Gamma^n_{\mathrm{ext}}(K_1) \longrightarrow R\,\mathrm{Sym}^n(u)_*K_2.$$

Moreover, v and t induce 393

$$\begin{cases} v' : K_1 \longrightarrow Ru_*K \\ t' : L\Gamma^n_{\mathrm{ext}}K \longrightarrow K_2. \end{cases}$$

Then:

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[68] N.D.E.: the coherence of Y may be replaced by the weaker condition that the dimensions of the fibers of u are bounded.

Proposition 5.5.18. *Under the hypotheses of 5.5.17.3, the following diagram is commutative in $D^+(\mathrm{Sym}^n_S(Y), \mathscr{A})$:*

$$\begin{tikzcd} L\Gamma^n_{\mathrm{ext}} Ru_* K \arrow[r, "k^n", "\text{(symmetric Künneth)}"'] & R\,\mathrm{Sym}^n_S(u)_* L\Gamma^n_{\mathrm{ext}} K \arrow[d, "R\,\mathrm{Sym}^n_S(u)_*(t')"'] \\ L\Gamma^n_{\mathrm{ext}} K_1 \arrow[u, "L\Gamma^n_{\mathrm{ext}}(v')"] \arrow[r, "\varphi"] & R\,\mathrm{Sym}^n_S(u)_* K_2. \end{tikzcd}$$

The proof is entirely similar to that of 5.4.1.5 and is left to the reader.

5.5.19. *Henceforth $\mathscr{A}$ is assumed to be torsion.* Let $u : X \to Y$ be a compactifiable morphism of S-schemes:

$$\begin{tikzcd} & \overline{X} \arrow[dr, "\overline{u}"] & \\ X \arrow[ur, hook, "j"] \arrow[rr, "u"] \arrow[dr] & & Y \arrow[dl] \\ & S & \end{tikzcd}$$

394 Suppose that $\mathrm{Sym}^n_S(X)$, $\mathrm{Sym}^n_S(\overline{X})$, and $\mathrm{Sym}^n_S(Y)$ exist. For every module F on X, one then has

$$\Gamma^n_{\mathrm{ext}}(j_! F) \simeq \mathrm{Sym}^n_S(j)_! \Gamma^n_{\mathrm{ext}}(F), \tag{5.5.19.1}$$

and this isomorphism passes immediately to the derived categories (5.5.14.4 (vii')). Define the *symmetric Künneth morphism* between functors from $D^{b,\mathrm{tor}\le 0}(X, \mathscr{A})$ to $D^{b,\mathrm{tor}\le 0}(\mathrm{Sym}^n_S(Y), \mathscr{A})$[69]

$$k^n_u \quad \text{or} \quad k^n : L\Gamma^n_{\mathrm{ext}} Ru_! \longrightarrow R\,\mathrm{Sym}^n_S(u)_! L\Gamma^n_{\mathrm{ext}} \tag{5.5.19.2}$$

to be the composite

$$\begin{aligned} L\Gamma^n_{\mathrm{ext}} Ru_! \overset{\mathrm{dfn}}{\simeq} L\Gamma^n_{\mathrm{ext}} R\overline{u}_* j_! &\xrightarrow{(5.5.17.2)} R\,\mathrm{Sym}^n_S(\overline{u})_* L\Gamma^n_{\mathrm{ext}} j_! \\ &\xrightarrow[(5.5.14.4\,(\mathrm{vii'}))]{\sim} R\,\mathrm{Sym}^n_S(\overline{u})_* \mathrm{Sym}^n_S(j)_! L\Gamma^n_{\mathrm{ext}} \\ &\to R\,\mathrm{Sym}^n_S(u)_! L\Gamma^n_{\mathrm{ext}}. \end{aligned}$$

5.5.20. We leave to the reader the following verifications, which are easy in principle using 5.5.18.

. (Intrinsic character). The morphism (5.5.19.2) is independent of the choice of the compactification $\overline{u}$ of u.

. (Transitivity). Let there be a diagram of S-schemes

$$X \xrightarrow{u} Y \xrightarrow{v} Z,$$

395 with u and v quasi-projective; suppose that $\mathrm{Sym}^n_S(Z)$ exists (and hence also $\mathrm{Sym}^n_S(Y)$

[69]N.D.E.: How can one show that $\mathrm{Sym}^n_S(u)$ is of finite type in general? This condition is necessary for $R\,\mathrm{Sym}^n_S(u)_!$ to be defined.

and $\operatorname{Sym}^n_S(X)$ (SGA 1 V 1.7, EGA II 4.5.4)). The following diagram of functors is then commutative:

$$\begin{tikzcd}
L\Gamma^n_{\mathrm{ext}} R(vu)_! \arrow[rr, "k^n_{vu}"] \arrow[d, equal] & & R\operatorname{Sym}^n_S(vu)_! L\Gamma^n_{\mathrm{ext}} \arrow[d, equal] \\
L\Gamma^n_{\mathrm{ext}} Rv_! Ru_! \arrow[r, "k^n_v"] & R\operatorname{Sym}^n_S(v)_! L\Gamma^n_{\mathrm{ext}} Ru_! \arrow[r, "k^n_u"] & R\operatorname{Sym}^n_S(v)_! R\operatorname{Sym}^n_S(u)_! L\Gamma^n_{\mathrm{ext}}.
\end{tikzcd}$$

. The symmetric Künneth morphism is compatible with changes of base or space of the types 5.5.9 (iv), (v), and (vi). More precisely, one has the following compatibilities with the morphisms 5.5.14.4 (iv′), (v′), and (vi′).

(iv″) For every $\mathscr{A}$-algebra $\mathscr{A}'$ and every quasi-projective S-morphism $u : X \to Y$, the diagram

$$\begin{tikzcd}
L\Gamma^n_{\mathrm{ext}} Ru_!(K) \overset{\mathrm{L}}{\otimes}_{\mathscr{A}} \mathscr{A}' \arrow[r, dash, "\sim"] \arrow[d, "k^n_u"] & L\Gamma^n_{\mathrm{ext}}(Ru_!(K) \overset{\mathrm{L}}{\otimes}_{\mathscr{A}} \mathscr{A}') \arrow[r, "\sim"] & L\Gamma^n_{\mathrm{ext}} Ru_!(K \overset{\mathrm{L}}{\otimes}_{\mathscr{A}} \mathscr{A}') \arrow[d, "k^n_u"] \\
R\operatorname{Sym}^n_S(u)_! L\Gamma^n_{\mathrm{ext}}(K) \overset{\mathrm{L}}{\otimes}_{\mathscr{A}} \mathscr{A}' \arrow[r, "\sim"] & R\operatorname{Sym}^n_S(u)_!(L\Gamma^n_{\mathrm{ext}}(K) \overset{\mathrm{L}}{\otimes}_{\mathscr{A}} \mathscr{A}') \arrow[r, dash, "\sim"] & R\operatorname{Sym}^n_S(u)_! L\Gamma^n_{\mathrm{ext}}(K \overset{\mathrm{L}}{\otimes}_{\mathscr{A}} \mathscr{A}')
\end{tikzcd}$$

is commutative.

(v″) For every base change $g : S' \to S$, giving rise to a diagram with cartesian squares 396

$$\begin{tikzcd}
X' \arrow[rr, "g'"] \arrow[dr, "u'"] \arrow[dd] & & X \arrow[dr, "u"] \arrow[dd] & \\
& Y' \arrow[rr, "g''"] \arrow[dl] & & Y \arrow[dl] \\
S' \arrow[rr, "g"'] & & S &
\end{tikzcd}$$

,

the diagram of functors, entirely analogous to (iv″), is commutative:

$$\begin{tikzcd}
\operatorname{Sym}^n_g(g'')^* L\Gamma^n_{\mathrm{ext}} Ru_! \arrow[r, dash, "\sim"] \arrow[d, "k^n_u"] & L\Gamma^n_{\mathrm{ext}} g''^* Ru_! \arrow[r, "\sim"] & L\Gamma^n_{\mathrm{ext}} Ru'_! g'^* \arrow[d, "k^n_{u'}"] \\
\operatorname{Sym}^n_g(g'')^* R\operatorname{Sym}^n_S(u)_! L\Gamma^n_{\mathrm{ext}} \arrow[r, "\sim"] & R\operatorname{Sym}^n_S(u')_! \operatorname{Sym}^n_g(g')^* L\Gamma^n_{\mathrm{ext}} \arrow[r, dash, "\sim"] & R\operatorname{Sym}^n_S(u')_! L\Gamma^n_{\mathrm{ext}} g'^*.
\end{tikzcd}$$

(vi″) For every *radicial* base change $g : Y' \to Y$, giving rise to a cartesian diagram of S-schemes

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "u'"] & X \arrow[d, "u"] \\ Y' \arrow[r, "g"] & Y, \end{tikzcd}$$

the following diagram of functors is commutative:

$$\begin{tikzcd} \mathrm{Sym}^n_S(g)^* L\Gamma^n_{\mathrm{ext}} Ru_! \arrow[r, dash, "\sim"] \arrow[d, "k^n_u"] & L\Gamma^n_{\mathrm{ext}} g^* Ru_! \arrow[r, "\sim"] & L\Gamma^n_{\mathrm{ext}} Ru'_! g'^* \arrow[d, "k^n_{u'}"] \\ \mathrm{Sym}^n_S(g)^* R\,\mathrm{Sym}^n_S(u)_! L\Gamma^n_{\mathrm{ext}} \arrow[r, "\sim"] & R\,\mathrm{Sym}^n_S(u')_! g'^* L\Gamma^n_{\mathrm{ext}} \arrow[r, dash, "\sim"] & R\,\mathrm{Sym}^n_S(u')_! L\Gamma^n_{\mathrm{ext}} g'^* \end{tikzcd}$$

397 Theorem 5.5.21. (Symmetric Künneth formula). *Let S be a coherent scheme (= quasi-compact and quasi-separated) ringed by a sheaf $\mathscr{A}$ of commutative torsion rings whose stalks are noetherian rings. Ring each S-scheme by the inverse image of $\mathscr{A}$. Let $u : X \to Y$ be a morphism of quasi-projective S-schemes. For every integer $n \geq 0$ and every $K \in$* Ob $D^{b,\mathrm{tor}\leq 0}(X, \mathscr{A})$*, the symmetric Künneth arrow (5.5.17.2)*

$$k^n : L\Gamma^n_{\mathrm{ext}} Ru_! K \longrightarrow R\,\mathrm{Sym}^n_S(u)_! L\Gamma^n_{\mathrm{ext}} K$$

is an isomorphism in $D^{b,\mathrm{tor}\leq 0}(\mathrm{Sym}^n_S(Y), \mathscr{A})$.

The proof of this theorem will occupy the rest of 5.5.

Remark 5.5.21.1. The quasi-projectivity hypothesis serves only to ensure that u admits a compactification $\overline{u}$

$$X \xrightarrow{j} \overline{X} \xrightarrow{\overline{u}} Y$$

such that the schemes $\mathrm{Sym}^n_S(X)$, $\mathrm{Sym}^n_S(\overline{X})$, and $\mathrm{Sym}^n_S(Y)$ exist for every n. The language of algebraic spaces should make it possible to replace this hypothesis by the sole hypothesis that u is compactifiable.

398 The first part of the proof consists in reducing 5.5.21, by dévissage, to an “irreducible” case:

Lemma 5.5.22. *Theorem 5.5.21 follows from the following special case:*

($*$) *For every algebraically closed field k, every smooth projective curve X over k, and every prime number ℓ, the symmetric Künneth arrows (obtained for $Y = S =$ Spec(k), $\mathscr{A} = \mathbf{Z}/\ell$, $K = \mathbf{Z}/\ell$; see the explanations below)*

$$L\Gamma^n(R\Gamma(X, \mathbf{Z}/\ell)) \longrightarrow R\Gamma(\mathrm{Sym}^n_k(X), \mathbf{Z}/\ell) \tag{5.5.22.1}$$

are isomorphisms ($n \geq 0$).

In (5.5.22.1), the symbol Γ has an unfortunate double use: in $R\Gamma$ it denotes the “global sections” functor, whereas in $L\Gamma^n$ it denotes the functor of 5.5.2.3. The isomorphism $\Gamma^n_{\mathrm{ext}}(\mathbf{Z}/\ell) \simeq \mathbf{Z}/\ell$ is used in this formula.

Let S, X, Y, u, and $\mathscr{A}$ be as in 5.5.21.

Lemma 5.5.22.2. *Let there be an exact sequence of complexes in $C^{b,\geq 0}(X, \mathscr{A})$, of tor-dimension ≤ 0:*

$$0 \longrightarrow K' \longrightarrow K \longrightarrow K'' \longrightarrow 0.$$

If all the Künneth arrows k^n ($n \geq 0$) relative to two of the complexes K', K, and K'' are isomorphisms, then the Künneth arrows relative to the third are also isomorphisms.

We reduce to the case where u is projective and the complexes K', K, and K'' 399
satisfy conditions (i) and (ii) of 5.5.16, for N at least twice the fiber dimension of u. The symmetric Künneth arrows are then given by (5.5.17.1).

We prove by induction on n that if, for $i \leq n$ and two of the complexes K', K, and K'', the arrows k^i are isomorphisms, then the arrows k^i ($i \leq n$) are isomorphisms for the third.

To see this, note that by 5.5.15, the morphism

$$\text{(a)} \qquad k^n : L\Gamma^n_{\text{ext}} Ru_* K \longrightarrow R\,\mathrm{Sym}^n(u)_* L\Gamma^n_{\text{ext}} K$$

can be represented by a morphism of filtered complexes

$$k'^n : K_0 \longrightarrow K_1$$

with finite filtration; on the successive quotients this morphism induces arrows isomorphic to the composite arrows

$$(\mathrm{b}_i) \qquad L\Gamma^i_{\text{ext}} Ru_* K' \overset{\mathrm{L}}{\vee} L\Gamma^{n-i}_{\text{ext}} Ru_* K'' \longrightarrow R\,\mathrm{Sym}^n(u)_*(L\Gamma^i_{\text{ext}} K' \overset{\mathrm{L}}{\vee} L\Gamma^{n-i}_{\text{ext}} K'')$$

as follows:

$$\begin{aligned}
L\Gamma^i_{\text{ext}} Ru_* K' \overset{\mathrm{L}}{\vee} L\Gamma^{n-i}_{\text{ext}} Ru_* K'' &\xrightarrow{k^i \overset{\mathrm{L}}{\vee} k^{n-i}} R\,\mathrm{Sym}^i(u)_* L\Gamma^i_{\text{ext}} K' \overset{\mathrm{L}}{\vee} R\,\mathrm{Sym}^{n-i}(u)_* L\Gamma^{n-i}_{\text{ext}}(K'') \\
&\underset{\text{defn}}{=} \vee_{i,n-i*}(R\,\mathrm{Sym}^i(u)_* L\Gamma^i_{\text{ext}} K' \overset{\mathrm{L}}{\boxtimes} R\,\mathrm{Sym}^{n-i}(u)_* L\Gamma^{n-i}_{\text{ext}}(K'')) \\
&\xrightarrow[\sim]{5.4.3} \vee_{i,n-i*}(R\,\mathrm{Sym}^i(u) \times R\,\mathrm{Sym}^{n-i}(u))_*(L\Gamma^i_{\text{ext}} K' \overset{\mathrm{L}}{\boxtimes} L\Gamma^{n-i}_{\text{ext}} K'') \\
= R\,\mathrm{Sym}^n(u)_* \vee_{i,n-i*} (L\Gamma^i_{\text{ext}} K' &\overset{\mathrm{L}}{\boxtimes} L\Gamma^{n-i}_{\text{ext}} K'') \overset{\text{dfn}}{=} R\,\mathrm{Sym}^n(u)_*(L\Gamma^i_{\text{ext}} K' \overset{\mathrm{L}}{\vee} L\Gamma^{n-i}_{\text{ext}} K'').
\end{aligned}$$

By the induction hypothesis (and the Künneth formula), these arrows are quasi- 400
isomorphisms for $i \neq 0, n$. Any two of the following assertions therefore imply the third: (a) is a quasi-isomorphism; (b_0) is a quasi-isomorphism; (b_n) is a quasi-isomorphism. These are precisely the three arrows in question, C.Q.F.D.

. Assume assertion (∗) of 5.5.22 and prove 5.5.21. Let S, X, Y, $\mathscr{A}$, K, and u therefore be as in 5.5.21. The problem is local on S, which may be assumed affine. Every complex K is a successive extension of complexes of the form $i_! i^* K$ for i the inclusion in X of a locally closed affine subset. By dévissage (5.5.22.2), we may therefore suppose that K has the form $i_! K_1$, for an immersion $i : X_1 \to X$ with affine source. Compatibility 5.5.20.2 permits us to replace X by X_1 and K by K_1, and hence to suppose that u is affine. Factor u as a composite $u = u_1 \dots u_k$ of morphisms with fibers of dimension ≤ 1. Compatibility 5.5.20.2 shows that it suffices to prove that the arrows k^n relative to the u_i and to any complex K are isomorphisms. We are thus reduced to the case where S is *affine and u has relative dimension* ≤ 1.

. For k^n to be an isomorphism, it suffices that its restrictions to the geometric fibers of $\mathrm{Sym}^n_S(Y)$ be isomorphisms; compatibility 5.5.20.3 (v″) therefore reduces us to the case where S is the *spectrum of an algebraically closed field k*.

. For every rational point y of $\mathrm{Sym}^n_k(Y)$, there is a finite reduced subscheme Y' of 401
Y such that y belongs to the image of $\mathrm{Sym}^n_k(Y')$. Compatibility 5.5.20.3 (vi″) therefore permits us to suppose that Y *is a disjoint union of finitely many copies of* $\mathrm{Spec}(k)$.

. If $Y = \amalg Y_i$, then every complex K on X is a sum of complexes supported on one of the schemes $X_i = u^{-1}(Y_i)$. By dévissage (5.5.22.2) and compatibility 5.5.20.3 (vi″), it suffices to prove 5.5.21 for each morphism $u_i : X_i \to Y_i$. By 5.5.22.5, this reduces us to the case where

$$Y = S = \operatorname{Spec}(k),\ k \text{ algebraically closed, } \dim X \leq 1.$$

The sheaf $\mathscr{A}$ now reduces to a noetherian ring A. The following proposition will allow us to reduce to the case where A is a field.

Proposition 5.5.23. *Let A be a (commutative) noetherian ring, let $K \in \operatorname{Ob} D^-(A)$ be a bounded-above complex of A-modules, and let n be an integer. The following conditions are equivalent:*

(i) *K has tor-dimension $\leq n$*

(ii) *For every $x \in \operatorname{Spec}(A)$ and every $i < -n$, $H^i(K \overset{\mathrm{L}}{\underset{A}{\otimes}} k(x)) = 0$.*

One has (i) ⇒ (ii) trivially. Suppose (ii). To prove (i), we prove by noetherian induction on the closed subset F of $\operatorname{Spec}(A)$ that if a module N satisfies $N_x = 0$ for $x \notin F$, then $H^i(K \overset{\mathrm{L}}{\otimes}_A N) = 0$ for $i < -n$.

402 Let I be the unique ideal defining F such that A/I is reduced, and let N_k be the submodule of N annihilated by I^k. One has $N = \varinjlim N_k$. Since the Tor functors commute with filtered direct limits, we reduce to the case where there exists k such that $N = N_k$, so that N is a successive extension of modules annihilated by I; the long exact cohomology sequence reduces us to the case where N is already an A/I-module.

Let η be a maximal point of F, and define A/I-modules M_1 and M_2 by the exact sequence

$$0 \longrightarrow M_1 \longrightarrow N \xrightarrow{i} N \otimes k(\eta) \longrightarrow M_2 \longrightarrow 0. \tag{5.5.23.1}$$

One has $(M_j)_\eta = 0$ $(j = 1, 2)$. If, for every closed subset $G \subset F$ with $\eta \notin G$, we denote by $M_{j,G}$ the submodule of M_j consisting of the sections of M_j supported in G, then

$$M_j = \varinjlim_{\eta \notin G} M_{j,G}.$$

By the induction hypothesis and passage to the limit, one therefore has

$$H^i(K \overset{\mathrm{L}}{\otimes}_A M_j) = 0 \quad \text{for} \quad i < -n.$$

The module $N \otimes k(\eta)$ is a vector space over $k(\eta)$, hence a direct sum of copies of $k(\eta)$. By hypothesis, one has

$$H^i(K \overset{\mathrm{L}}{\otimes} (N \otimes k(\eta))) = 0 \quad \text{for} \quad i < -n.$$

If I is the image of the arrow i in (5.5.23.1), there are exact sequences

$$H^{i-1}(K \overset{\mathrm{L}}{\otimes} M_2) \longrightarrow H^i(K \overset{\mathrm{L}}{\otimes} I) \longrightarrow H^i(K \overset{\mathrm{L}}{\otimes} (N \otimes k(\eta)))$$

$$H^i(K \overset{\mathrm{L}}{\otimes} M_1) \longrightarrow H^i(K \overset{\mathrm{L}}{\otimes} N) \longrightarrow H^i(K \overset{\mathrm{L}}{\otimes} I),$$

403 from which it follows that $H^i(K \overset{\mathrm{L}}{\otimes} N) = 0$ for $i < -n$, C.Q.F.D.

Corollary 5.5.24. *For a morphism $u : K \to L$ in $D^-(A)$ to be an isomorphism, it suffices that for every $x \in \mathrm{Spec}(A)$, the morphism*

$$u_x : K \overset{\mathbf{L}}{\otimes}_A k(x) \longrightarrow L \overset{\mathbf{L}}{\otimes}_A k(x)$$

be an isomorphism in $D^-(k(x))$.

Let M be the cone (mapping cylinder) of u. By hypothesis, one has $M \overset{\mathbf{L}}{\otimes}_A k(x) = 0$ for every $x \in \mathrm{Spec}(A)$. By 5.5.23, one therefore has $M = 0$, and u is an isomorphism.

. Let us continue the proof of 5.5.22, interrupted at 5.5.23. By the compatibility 5.5.20.3 (iv″) and 5.5.24, the morphism k^n associated with $K \in \mathrm{Ob}\, D^{b,\mathrm{tor}\geq 0}(X, A)$ is an isomorphism if and only if the morphisms k^n associated with $K \overset{\mathbf{L}}{\otimes}_A k(x) \in \mathrm{Ob}\, D^{b,\mathrm{tor}\geq 0}(X, k(x))$ are isomorphisms, for $x \in \mathrm{Spec}(A)$. This reduces us to the case where A *is a field of characteristic* $\ell > 0$.

. The exact sequence (where C denotes a cone)

$$0 \longrightarrow K[-1] \longrightarrow CK[-1] \longrightarrow K \longrightarrow 0$$

shows that the morphisms k^n are isomorphisms for K if and only if they are so for $K[-1]$ (5.5.22.2).

. The exact sequences 404

$$0 \longrightarrow \sigma_{\geq n}K \longrightarrow K \longrightarrow \sigma_{\leq n}K \longrightarrow 0$$

show that, for the Künneth morphisms to be isomorphisms when K is a flat complex, it suffices that they be so for the components of K, placed in degree 0 (5.5.22.2 and 5.5.25.2).

. Thus let K be a sheaf of A-vector spaces. This sheaf is a filtered direct limit of constructible sheaves (IX 2.7.2), automatically flat since A is a field, and, by passage to the limit, we reduce to the case where K is *constructible* and concentrated in degree zero.

. If a complex K is a direct summand of a complex L, the morphism k^n associated with K is a direct summand of the morphism k^n associated with L, hence is an isomorphism if the latter is one. The trace method (IX 5.8) therefore applies, and it suffices to treat the case where F is of the form $j_! A_{X'}$, for $j : X' \to X$ quasi-finite. The compatibility 5.5.20.2 allows us to replace X by X', whence the reduction

$$X \textit{ has dimension} \leq 1 \textit{ and } K = A_X.$$

. We may assume that X is reduced. If $j : U \to X$ is the smooth locus of X, with complement $i : F \to X$, the exact sequence

$$0 \longrightarrow j_! j^* A_X \longrightarrow A_X \longrightarrow i_! i^* A_X \longrightarrow 0$$

reduces us by dévissage to the cases where X *is either a smooth curve or a finite set.* Moreover, the formula

$$A_X = A \overset{\mathbf{L}}{\otimes}_{\mathbf{Z}/\ell} (\mathbf{Z}/\ell)_X$$

reduces us by 5.5.20.3 (iv″) to the case where $A_X = \mathbf{Z}/\ell$.

If X is a curve, we embed X in the corresponding smooth projective curve, and a further dévissage reduces us to one of the following two cases: 405

(i) X is a smooth projective curve, $Y = S = \mathrm{Spec}(k)$ with k algebraically closed, $\mathscr{A} = \mathbf{Z}/\ell$, $K = \mathbf{Z}/\ell$.

(ii) X is a finite disjoint union of copies of $\mathrm{Spec}(k)$, $Y = S = \mathrm{Spec}(k)$ (k algebraically closed), $\mathscr{A} = \mathbf{Z}/\ell$, $K = \mathbf{Z}/\ell$.

A final dévissage of $(\mathbf{Z}/\ell)_X$ into the sheaves $(\mathbf{Z}/\ell)_{X'}$, for X' a connected component of X, reduces us in (i) and (ii) to assuming that X is connected. Case (ii) is then trivial. This proves 5.5.22.

Lemma 5.5.26. *Theorem 5.5.21 follows from the following two special cases of assertion* $(*)$ *of 5.5.22:*

Case 1. *Assume that* $k = \mathbf{C}$.

Case 2. *Assume that* k *has characteristic* $p > 0$ *and that* $\ell = p$.

With the notation of 5.5.22, denote by p the characteristic of k. If $p = 0$, the Lefschetz principle (relying here on compatibility 5.5.20.3 (v″) and on XII 5.4) allows us to reduce to the case where $k = \mathbf{C}$.

It remains to treat the case where $p > 0$ and $\ell \neq p$. The curve X lifts to a curve X' over the ring $W(k)$ of Witt vectors over k (SGA 1 III 7.4). Put $S = \operatorname{Spec}(W(k))$, and consider the diagram

$$\begin{tikzcd}
X'_\eta \arrow[r] \arrow[d, "f_\eta"] & X' \arrow[d, "f'"] & X \arrow[l] \arrow[d, "f"] \\
\eta \arrow[r] & S & \operatorname{Spec}(k), \arrow[l]
\end{tikzcd}$$

406 and let f_n, f'_n, and $f_{\eta n}$ be the respective projections of $\operatorname{Sym}^n_k(X)$, $\operatorname{Sym}^n_S(X')$, and $\operatorname{Sym}^n_\eta(X'_\eta)$ onto $\operatorname{Spec}(k)$, S, and η. It is known that f'_n is a smooth projective morphism (projectivity is clear; for smoothness, the simplest approach is to reduce to the case where $X' = \mathbf{P}^1_S$, in which case $\operatorname{Sym}^n_S(X') \simeq \mathbf{P}^n$, as follows from the theorem on elementary symmetric functions). By the specialization theorem XVI 2.2, the sheaves $R^i f'_{n*}(\mathbf{Z}/\ell)$ are locally constant on S. The same holds for the sheaves $L^i\Gamma^n(Rf'_*\mathbf{Z}/\ell)$. To verify that the morphism of (5.5.22.1) associated with X over k is an isomorphism, it therefore suffices, by 5.5.20.3 (v″), to prove the analogous statement for the geometric generic fiber $X_{\overline{\eta}}$ of X over S. The field $k(\overline{\eta})$ has characteristic 0, and, in view of the first reduction, this proves 5.5.26.

To prove 5.5.21 in the two special cases of 5.5.26, we compare the symmetric Künneth morphism on the one hand with its transcendental analogue (5.5.27 and 5.5.28) and on the other hand with its coherent analogue (5.5.29 through 5.5.37).

5.5.27. If $p : X \to S$ is a continuous map of topological spaces, with S ringed by a sheaf of commutative rings $\mathscr{A}$, the constructions of 5.5.7 and 5.5.8 apply unchanged to define a functor Γ^n_{ext} from modules on X to modules on $\operatorname{Sym}^n_S(X) = X^n/\mathfrak{S}_n$. The formal arguments of 5.5.7 through 5.5.18 carry over. Likewise, if S, X, and Y are locally compact Hausdorff topological spaces

$$\begin{tikzcd}
X \arrow[rr, "u"] \arrow[dr] & & Y \arrow[dl] \\
& S &
\end{tikzcd} ,$$

407 and if the fibers of u have bounded cohomological dimension, the formalism of 5.5.19 and 5.5.20 for the symmetric Künneth morphism in cohomology with proper supports carries over.

If $p : X \to S$ is a morphism of schemes of finite type over $\mathbf{C}$, the functors Γ^n and $L\Gamma^n$ commute with the functors of “passage to the transcendental setting”. Finally, if

X, Y, S, $\mathscr{A}$, and K are as in 5.5.21, with S separated and of finite type over $\mathbf{C}$, and if we denote generically by ϵ the morphisms of sites of type $\epsilon : X_{\mathrm{cl}} \to X_{\mathrm{et}}$, the following diagram is commutative:

(5.5.27.1)
$$\begin{tikzcd} \epsilon^* L\Gamma^n_{\mathrm{ext}} Ru_! K \arrow[r, "k^n"] \arrow[d] & \epsilon^* R\,\mathrm{Sym}^n_S(u)_! L\Gamma^n_{\mathrm{ext}} K \arrow[d] \\ L\Gamma^n_{\mathrm{ext}} Ru^{\mathrm{an}}_! \epsilon^* K \arrow[r, "k^{n\,\mathrm{an}}"] & R\,\mathrm{Sym}^n_S(u^{\mathrm{an}})_! L\Gamma^n_{\mathrm{ext}} \epsilon^* K. \end{tikzcd}$$

5.5.28. By the compatibility (5.5.27.1) and an elementary case of the comparison theorem (XI 4.4 or XVI 4.1), to treat Case 1 of 5.5.26, it suffices to verify that if X is a smooth complex projective curve, then the topological analogue of (5.5.22.1) is an isomorphism.

The topological space X^{an} is triangulable (this fact is trivial here, since X is a branched covering of the Riemann sphere). The topological analogue of the dévissage lemma 5.5.22.2 then reduces us to proving

(∗) If D is a closed simplex (of dimension 0, 1, or 2), the topological symmetric Künneth morphism 408

(5.5.28.1)
$$k^n : L\Gamma^n R\Gamma(D, \mathbf{Z}/\ell) \longrightarrow R\Gamma(\mathrm{Sym}^n(D), \mathbf{Z}/\ell)$$

is an isomorphism.

One has

$$R\Gamma(D, \mathbf{Z}/\ell) = \mathbf{Z}/\ell \text{ placed in degree } 0,$$

and hence

$$L\Gamma^n R\Gamma(D, \mathbf{Z}/\ell) = \mathbf{Z}/\ell \text{ placed in degree } 0.$$

On the other hand, $\mathrm{Sym}^n(D)$ is contractible, and again one has

$$R\Gamma(\mathrm{Sym}^n(D), \mathbf{Z}/\ell) = \mathbf{Z}/\ell \text{ placed in degree } 0.$$

The H^i of the two sides of (5.5.28.1) therefore vanish in nonzero degrees, and the morphism induced on H^0 identifies with the identity of $\mathbf{Z}/\ell$.

5.5.29. For the application we have in mind, it is useful to regard a quasi-coherent sheaf on a scheme X as a quasi-coherent sheaf of $\mathscr{O}$-modules on the étale site of X.

Let $p : X \to S$ be a quasi-projective morphism of schemes (cf. 5.5.21.1 [69]). If $\mathscr{F}$ is a sheaf of $\mathscr{O}_X$-modules on X_{et}, then

a) We denote by $\bigotimes^n \mathscr{F}$ the sheaf of modules on $(X/S)^n_{\mathrm{et}}$ given by the tensor product of the inverse-image $\mathscr{O}_{(X/S)^n}$-modules of $\mathscr{F}$ on $(X/S)^n_{\mathrm{et}}$;

b) the sheaf $\bigotimes^n \mathscr{F}$ is a $\mathfrak{S}_n$-equivariant sheaf. Passing to invariants, it defines a sheaf $TS^n_{\mathrm{ext}}(\mathscr{F})$ on $\mathrm{Sym}^n_S(X)$: if σ is the projection of $(X/S)^n$ onto $\mathrm{Sym}^n_S(X)$, one has

(5.5.29.1)
$$TS^n_{\mathrm{ext}}(\mathscr{F}) = (\sigma_* \bigotimes^n \mathscr{F})^{\mathfrak{S}^n}.$$

Assume that p is flat; as in 5.5.8, one then defines the functor Γ^m_{ext} to be the 0th left 409

[69] Everything that follows, including 5.5.34, remains valid under the sole assumption that X and Y admit invertible sheaves relatively ample over S (without assuming that X and Y are of finite type over S).

derived functor of TS^n_{ext}. With the notation of 5.5.2.5 and 5.5.8, if

$$P_1 \longrightarrow P_0 \longrightarrow \mathscr{F} \longrightarrow 0$$

is a flat presentation of $\mathscr{F}$, then the sequence

$$TS^n_{\mathrm{ext}}(P_1') \underset{j_1}{\overset{j_0}{\rightrightarrows}} TS^n_{\mathrm{ext}}(P_0) \longrightarrow \Gamma^n_{\mathrm{ext}}(\mathscr{F})$$

is exact. If x is a geometric point of $\mathrm{Sym}^n_S(X)$ as in 5.5.7.1, one has (cf. (5.5.8.1))

(5.5.29.2) $$\Gamma^n_{\mathrm{ext}}(\mathscr{F})_x = \mathscr{O}_x \otimes_C \Big(\bigotimes_1^{\ell} \Gamma^{n_i}_{\mathscr{O}_S}(\mathscr{F}_{y_i})\Big),$$

where $$C = \bigotimes_1^{\ell} \mathrm{Sym}^{n_i}_{\mathscr{O}_S}(\mathscr{O}_{y_i}).$$

Indeed, it suffices to verify this formula in the trivial case where the fibers of $\mathscr{F}$ are free. Note that $\mathscr{O}_x$ is flat over C, being the strict henselization of C at the closed point that is the image of x.

5.5.30. Assume that X and S are affine, p is flat, and $\mathscr{F}$ is quasi-coherent. Let $S = \mathrm{Spec}(A)$, $X = \mathrm{Spec}(B)$, and let $F = H^0(X, \mathscr{F})$, so that $\mathscr{F}$ is the sheaf $F^\sim$ defined by the B-module F. It follows from (5.5.29.1) that

$$TS^n_{\mathrm{ext}}(\mathscr{F}) = TS^n_A(F)^\sim,$$

and, on deriving, one therefore has

(5.5.30.1) $$\Gamma^n_{\mathrm{ext}}(\mathscr{F}) = \Gamma^n_A(F)^\sim.$$

410 The functor Γ^n_{ext} thus appears as a functor Γ^n "relative to A", as (5.5.29.2) already expressed locally.

5.5.31. We denote by $D^{b,\mathrm{tor}_S\leq 0}(X, \mathscr{O}_X)$ the subcategory of $D^b(X, \mathscr{O}_X)$ formed by the complexes of $\mathscr{O}_X$-modules which, as complexes of $p^{-1}(\mathscr{O}_S)$-modules, have tor-dimension ≤ 0.

Consider the condition

. $K \in \mathrm{Ob}\, C^{b,\geq 0}(X, \mathscr{O}_X)$, and, at every geometric point x of X, K_x is homotopic to a complex of $\mathscr{O}_{X,x}$-modules, in nonnegative degrees, flat over $\mathscr{O}_{S,p(x)}$.

Since the morphism p is flat, so that there are "enough" $\mathscr{O}_X$-modules flat over S, one checks as in 5.5.13.2 that $D^{b,\mathrm{tor}_S\leq 0}(X, \mathscr{O}_X)$ is obtained from the category of complexes $K \in \mathrm{Ob}\, K^{b,\geq 0}(X, \mathscr{O}_X)$ satisfying (5.5.31.1) by inverting quasi-isomorphisms.

It follows from (5.5.29.2) and 5.5.6.1 that if $u : K \to L$ is a quasi-isomorphism between complexes satisfying (5.5.31.1), then $\Gamma^n_{\mathrm{ext}}(u)$ (in the sense of Dold-Puppe) is again a quasi-isomorphism. The functor Γ^n_{ext} can therefore be derived to give

(5.5.31.2) $$L\Gamma^n_{\mathrm{ext}} : D^{b,\mathrm{tor}_S\leq 0}(X, \mathscr{O}_X) \longrightarrow D^{b,\mathrm{tor}_S\leq 0}(\mathrm{Sym}^n_S(X), \mathscr{O}_{\mathrm{Sym}^n_S(X)})$$

5.5.32. We denote by $D^{b,\mathrm{tor}_S\leq 0}_{\mathrm{qcoh}}(X, \mathscr{O}_X)$ the subcategory of $D^{b,\mathrm{tor}_S\leq 0}(X, \mathscr{O}_X)$ formed by complexes with quasi-coherent cohomology.

411 Standard arguments make it possible, as in 5.5.17, to define a symmetric Künneth arrow of the following type:

Let $u : X \to Y$ be a morphism of quasi-projective S-schemes flat over S; for $K \in \mathrm{Ob}\, D^{b,\mathrm{tor}_S\leq 0}_{\mathrm{qcoh}}(X, \mathscr{O}_X)$, the symmetric Künneth arrow k^n in $D^{b,\mathrm{tor}_S\leq 0}_{\mathrm{qcoh}}(\mathrm{Sym}^n_S(Y), \mathscr{O})$ is an arrow

(5.5.32.1) $$k^n : L\Gamma^n_{\mathrm{ext}} Ru_* K \longrightarrow R\,\mathrm{Sym}^n_S\, u_* L\Gamma^n_{\mathrm{ext}} K.$$

We leave it to the reader to verify the formal properties of (5.5.32.1) that will be used, notably the following one:

Lemma 5.5.33. *If S is of characteristic $p > 0$, and if $u : X \to S$ is proper and flat, the diagram of symmetric Künneth arrows (5.5.17.2) and (5.5.32.1) is commutative ($Y = S$, $K = \mathbf{Z}/p$ or $\mathcal{O}_X$):*

$$\begin{tikzcd} L\Gamma^n_{\mathbf{Z}/p} Ru_*(\mathbf{Z}/p) \arrow[r, "k^n"] \arrow[d] & R\,\mathrm{Sym}^n_S(u)_*(\mathbf{Z}/p) \arrow[d] \\ L\Gamma^n_{\mathcal{O}} Ru_*(\mathcal{O}) \arrow[r, "k^n"] & R\,\mathrm{Sym}^n_S(u)_*\mathcal{O} \end{tikzcd} .$$

Proposition 5.5.34. *Under the hypotheses of 5.5.32, the symmetric Künneth arrow (5.5.32.1) is an isomorphism.*

a) The question is local on S, which may be assumed affine.
b) For every point x of $\mathrm{Sym}^n_S(Y)$, there exists an affine open Y_1 of Y such that x lies in $\mathrm{Sym}^n_S(Y_1)$. By localization, we may therefore assume that Y is affine.
c) Every complex K of the type under consideration can be represented by a com- 412
plex K_0 of quasi-coherent modules on S, zero in degrees < 0. If $\mathcal{U}$ is an affine covering of X, then K admits a finite resolution by sums of complexes of the form $j_* j^* K$, where j is the inclusion of an affine open (alternating Čech complex). Using the quasi-coherent analogue of the dévissage lemma 5.5.22.2, this reduces us to the case where there exist an affine open $j : X_1 \hookrightarrow X$ of X and $K_1 \in D^{b,\mathrm{tor}_S \le 0}_{\mathrm{qcoh}}(X_1, \mathcal{O})$ such that $K = Rj_* K_1$.
d) In view of the quasi-coherent analogue of transitivity 5.5.20.2, it suffices to prove 5.5.34 for uj and j. Reduction b), applied to j, finally reduces us to the case where X, Y, and S are affine. We may then represent K by a complex $K_2 \in C^{b,\ge 0}(X, \mathcal{O}_X)$ with S-flat quasi-coherent components (4.1.9). The functors Ru_* and $L\Gamma^n$ can then be computed without having to resolve K_2 further, and it remains to note that, by (5.5.30.1), for $\mathcal{F}$ quasi-coherent on X, we have

$$\Gamma^n_{\mathrm{ext}} u_* \mathcal{F} \xrightarrow{\sim} \mathrm{Sym}^n_S(u)_* \Gamma^n_{\mathrm{ext}} \mathcal{F},$$

the two sides being identified with

$$\Gamma^n_A(F)^\sim \quad \text{for} \quad \mathcal{F} = F^\sim,$$

C.Q.F.D.

5.5.35. Let us treat case 2 of 5.5.26. With the notation of 5.5.22, the following diagram is commutative (5.5.33):

(5.5.35.1) $$\begin{tikzcd} L\Gamma^n(R\Gamma(X, \mathbf{Z}/p)) \arrow[r, "k^n", "①"'] \arrow[d] & R\Gamma(\mathrm{Sym}^n_k(X), \mathbf{Z}/p) \arrow[d] \\ L\Gamma^n(R\Gamma(X, \mathcal{O})) \arrow[r, "k^n", "②"'] & R\Gamma(\mathrm{Sym}^n_k(X), \mathcal{O}), \end{tikzcd}$$

and by 5.5.34, arrow ② is an isomorphism. The Frobenius endomorphism $F : x \to x^p$ of 413

$\mathscr{O}$ acts p-linearly on $R\Gamma(X, \mathscr{O})$ and on $R\Gamma(\mathrm{Sym}^n_k(X), \mathscr{O})$. It therefore also acts p-linearly on $L\Gamma^n(R\Gamma(X, \mathscr{O}))$. One checks that k^n and F commute.

Artin–Schreier theory provides exact sequences

$$0 \longrightarrow H^i(X, \mathbf{Z}/p) \longrightarrow H^i(X, \mathscr{O}) \xrightarrow{F-I} H^i(X, \mathscr{O}) \longrightarrow 0,$$

and similarly for $\mathrm{Sym}^n_S(X)$. To deduce that ① is an isomorphism from the fact that ② is one, it remains to verify the following

LEMMA 5.5.36. *Let k be a perfect field of characteristic $p > 0$, let $K \in C^{\geq 0}(\mathbf{Z}/p)$ be a complex of finite-dimensional $\mathbf{Z}/p$-vector spaces, let $L \in C^{\geq 0}(k)$ be a complex of finite-dimensional k-vector spaces, let F be a p-linear endomorphism of L, and let $u : K \to L$ be a morphism. Suppose that $H^*(Fu) = H^*(u)$, and that the sequences*

$$0 \longrightarrow H^i(K) \longrightarrow H^i(L) \xrightarrow{F-I} H^i(L) \longrightarrow 0$$

414 *are exact. Then the sequences*

$$0 \longrightarrow H^i(\Gamma^n K) \longrightarrow H^i(\Gamma^n L) \xrightarrow{F-I} H^i(\Gamma^n L) \longrightarrow 0$$

are exact.

Since $H^*(Fu) = H^*(u)$ and $\mathbf{Z}/p$ is a field, there exists a homotopy

$$Fu - u = dH + Hd.$$

Since the field k is perfect, we can write

$$FH_0 - H_0 = H.$$

Replacing u by $u - (dH_0 + H_0 d)$, we may assume that $Fu = u$.

The functor S that assigns to every k-vector space V equipped with a p-linear endomorphism F the $\mathbf{Z}/p$-vector space V^S of those $v \in V$ such that $Fv = v$ is an exact functor. The hypothesis therefore means that morphism

$$K \longrightarrow L^S$$

is a quasi-isomorphism. It remains to prove that the functor S commutes with the functor Γ^n. This follows readily from the following well-known lemma, whose proof is left to the reader:

LEMMA 5.5.37. *A finite-dimensional vector space V over a perfect field k, equipped with a p-linear endomorphism F, admits a unique decomposition*

$$V = V_0 + V_n$$

with $V^S \otimes k \xrightarrow{\sim} V_0$, $F(V_n) \subset V_n$, and $F|V_n$ nilpotent.

This completes the proof of 5.5.21.

6. The functor $f_!$

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We propose to give a direct construction of the functor $f_!$ of 5.1.11. We will then use this construction to define the "trace morphism" (6.2.3) for quasi-finite flat morphisms of finite presentation.

6.1. New construction of the functor $f_!$. Let $f : X \to S$ be a morphism locally of finite type. A subset P of X is said to be *proper* over S if it is the image of a subscheme of X proper over S.

Proposition 6.1.1. *Let $f : X \to S$ be a morphism locally of finite type and quasi-separated.*

(i) *A subset of X proper over S is locally closed, and is closed if f is separated.*
(ii) *If f is separated, the union of two subsets of X proper over S is proper over S.*
(iii) *For every S-scheme U, let $\Phi_X(U)$ denote the set of subsets of $U \times_S X$ proper over U. The functor Φ_X is a sheaf for the* fpqc *topology. If f is separated, Φ_X is a sheaf of filtered ordered sets for the inclusion relation.*

Assertions (i) and (ii) are trivial, and the second assertion of (iii) follows from (ii). The **416**
functor $U \mapsto \mathscr{P}(U \times_S X)$ is a sheaf for every topology whose coverings are surjective. It follows from EGA IV 2.3.14 and EGA IV 2.7.1 (vii) that Φ_X is a subsheaf thereof.

If $\mathscr{F}$ is an étale sheaf of pointed sets on a scheme X, recall (IV 8.5.2) that the support of a section s of $\mathscr{F}$ is the complement of the largest open set on which $s = 0$, where 0 denotes the distinguished section.

Definition 6.1.2. *Let $f : X \to S$ be a separated morphism locally of finite type, and let $\mathscr{F}$ be a sheaf of pointed sets on X. We denote by $f_!\mathscr{F}$ the subsheaf of $f_*\mathscr{F}$ whose sections over U (étale over S) are the sections of $\mathscr{F}$ over $U \times_S X$ with support proper over U.*

It follows from 6.1.1 that $f_!\mathscr{F}$ is indeed a sheaf.

One readily verifies that:

Lemma 6.1.3. *If f is the composite gh of two separated morphisms locally of finite type, there exists a unique isomorphism of functors $f_! \simeq g_!h_!$ making the following diagram commutative*

$$\begin{tikzcd} f_! \arrow[r, dash, "\sim"] \arrow[d, hook] & g_!h_! \arrow[d, hook] \\ f_* \arrow[r, equal] & g_*h_*. \end{tikzcd}$$

If f is an open immersion (resp. a proper morphism), the functor $f_!$ is the extension- **417**
by-zero functor (resp. the functor f_*). By 6.1.3, definitions 6.1.2 and 5.1.11 therefore agree on their common domain of validity.

The properties of the functor $f_!$ verified in § 5 remain valid in the more general case 6.1.2.

Proposition 6.1.4. *Let $f : X \to S$ be a separated morphism locally of finite type.*

(i) *The functor $f_!$ is left exact (i.e. commutes with finite projective limits) and commutes with filtered inductive limits.*
(ii) *If f is quasi-finite, the functor $f_!$ is an exact functor from the category of abelian sheaves on X to that of abelian sheaves on S.*

(iii) *For every Cartesian diagram*

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"] & X \arrow[d, "f"] \\ S' \arrow[r, "g"] & S \end{tikzcd}$$

there exists a base-change isomorphism making the diagram

$$\begin{tikzcd} g^* f_! \arrow[r, "\sim"] \arrow[d, hook] & f'_! g'^* \arrow[d, hook] \\ g^* f_* \arrow[r] & f'_* g'^*, \end{tikzcd}$$

commute, in which the second horizontal arrow is the base-change arrow XII 4.1.

418 Proof. For every $U \in \mathrm{Ob}\, S_{\mathrm{et}}$, we have

$$f_! \mathscr{F}(U) = \varinjlim_{P \in \Phi(U)} \Gamma_P(\mathscr{F}),$$

and for U quasi-compact and quasi-separated, the functor $\mathscr{F} \mapsto f_!\mathscr{F}(U)$ commutes with finite projective limits and filtered inductive limits, as a filtered inductive limit (6.1.1 (iii)) of functors having these properties.

This proves (i). Assertion (ii) follows at once from (iii) (take the geometric points of S as base changes).

Assertion (iii) is local on S, which may be assumed affine. Let I be the filtered ordered set of open subsets of X of finite type over S. If $U \overset{j}{\hookrightarrow} X$ is an open subset of X, $(fj)_!(\mathscr{F}|U) = f_!(j_!(\mathscr{F}|U))$ maps to $f_!(\mathscr{F})$. Moreover

$$f_!(\mathscr{F}) = \varinjlim_{U \in I} (fj)_!(\mathscr{F}|U).$$

This allows us to assume that X is of finite type over S and to apply Chow's lemma (EGA II 2^{nd} ed.)[70] to find a projective and surjective morphism $p^0 : X^0 \to X$ such that fp^0 is compactifiable. Let $X^1 = X^0 \times_X X^0$ and let p^1 be the natural arrow from X^1 to X. The sequence

$$0 \to \mathscr{F} \to p^0_* p^{0*} \mathscr{F} \rightrightarrows p^1_* p^{1*} \mathscr{F}$$

is exact (VIII 9.2).

Taking into account the functoriality of the base-change arrow and (i), it suffices to verify (iii) for the sheaves $p^0_* p^{0^*} \mathscr{F}$ and $p^1_* p^{1^*} \mathscr{F}$. By the change of notation $(p^1 \mapsto p^0)$,
419 we are thus reduced to assuming that $\mathscr{F}$ is of the form $p^0_* \mathscr{G}$. Assertion (iii) for the pair $(f, p^0_* \mathscr{G})$ is equivalent to assertion (iii) for the pairs $(p^0_*, \mathscr{G})$ and $(fp^0_*, \mathscr{G})$. We conclude by 5.2.7 (i), which applies because p^0 and fp^0 are compactifiable.

Counterexample 6.1.6. The functor $Rf_!$ does not in general agree with the derived functor of the functor $f_!$. Indeed, if X is affine over an algebraically closed field k, we have

$$f_! \mathscr{F} = \bigoplus_{x \in X(k)} H^0_x(\mathscr{F}),$$

[70] N.D.E.: see [C, 2.6], cited in la N.D.E. (43) page 639.

and the i^{th} derived functor of $f_!$ therefore takes the value at $\mathcal{F}$

$$\bigoplus_{x\in X(k)} H^i_x(\mathcal{F}).$$

On the other hand, if X is smooth, irreducible, and of dimension 1,

$$H^2_c(X, \mu_{\ell^n}) \simeq \mathbf{Z}/\ell^n$$

for ℓ prime to the characteristic of k.

6.2. The trace morphism in relative dimension 0. The main result of this n° is Theorem 6.2.3. The reader is advised not to read the technical preliminaries 6.2.1 and 6.2.2. In the proof of 6.2.3, these allow one to consider only finite morphisms f, a reduction that, via EGA IV 18.12, is in any event fairly evident.

6.2.1. Let $f : X \to S$ be a *quasi-finite separated* morphism. For every $U \in$ **420**
$\operatorname{Ob} S^{\text{et}}$, let $\Psi_f(U)$ denote the category of pairs consisting of a finite pointed set I, with distinguished point denoted 0, and a decomposition of f^*U into open and closed subsets $f^*U = \coprod_{i\in I} V_i$, with V_i finite over U for $i \neq 0$. A morphism from $\varphi = (V_i)_{i\in I}$ to $\varphi' = (V'_i)_{i\in I'}$ is a map σ from I' to I such that, for $i \in I$, one has $V_i = \bigcup_{j\in\sigma^{-1}(i)} V'_j$. We say that φ' *refines* φ if there exists a morphism from φ to φ'.

Let $\mathcal{F}$ be a sheaf of abelian groups on S, let $U \in \operatorname{Ob} S_{\text{et}}$, let $\varphi = (V_i)_{i\in I} \in \operatorname{Ob} \Psi_f(U)$, let $I^* = I\setminus\{0\}$, and let f_i be the projection of V_i onto U. We have

$$f_!f^*\mathcal{F}|U = f_{0!}f_0^*(\mathcal{F}|U) \oplus \bigoplus_{i\neq 0} f_{i*}f_i^*(\mathcal{F}|U),$$

hence a composite arrow

$$\tau_\varphi : \mathcal{F}^{I^*}|U \longrightarrow \bigoplus_{i\neq 0} f_{i*}f_i^*(\mathcal{F}|U) \longrightarrow f_!f^*\mathcal{F}|U. \tag{6.2.1.1}$$

If $\sigma : I \to J$ is a morphism of pointed sets, we denote by ${}^t\sigma : \mathcal{F}^{J^*} \to \mathcal{F}^{I^*}$ the arrow with coordinates ${}^t\sigma^i{}_j = \mathrm{Id}$ if $\sigma(i) = j$, and ${}^t\sigma^i{}_j = 0$ if $\sigma(i) \neq j$. If $\sigma : I' \to I$ is a morphism from φ to φ', the diagram

(6.2.1.2)
$$\begin{tikzcd} \mathcal{F}^{I^*}|U \arrow[rr, "{}^t\sigma"] \arrow[dr, "\tau_\varphi"'] & & \mathcal{F}^{I'^*}|U \arrow[dl, "\tau_{\varphi'}"] \\ & f_!f^*\mathcal{T}|U & \end{tikzcd}$$

is commutative. Moreover, ${}^t(\sigma\rho) = {}^t\rho\,{}^t\sigma$. The $\Psi_f(U)$ form a fibered category Ψ_f over **421**
S_{et}; the preceding constructions commute with localization and therefore define an S_{et}-functor $\varphi = (V_i)_{i\in I} \in \Psi_f(U) \mapsto \mathcal{F}^{I^*}|U$ from Ψ to the S_{et}-category of sheaves on étale open subschemes of S, and a morphism of functors $\tau_\varphi : \mathcal{F}^{I^*}|U \to f_!f^*\mathcal{F}|U$.

Proposition 6.2.2. (i) *The category Ψ_f is locally filtering on the right* (V 8.1.1) *over S_{et}.*

(ii) *The arrows* (6.2.1.1) *define an isomorphism*

$$\tau : \varinjlim_{\varphi=(V_i)_{i\in I}\in\Psi_f(U)} \mathcal{F}^{I^*}|U \xrightarrow{\sim} f_!f^*\mathcal{F}|U.$$

If S is strictly local with closed point s, the S-scheme X admits a decomposition $\varphi \in \Psi_f(S)$ into open and closed parts $X = \coprod_{i \in I} X_i$ indexed by a pointed set I such that the fiber of X_0 at s is empty, the X_i $(i \neq 0)$ are finite over S, and the fiber of X_i $(i \neq 0)$ at s consists of a single point (EGA IV 18.12.1). Such a decomposition is a final object of $\Psi_f(S)$. Moreover, by 6.1.4 (iii) (or because this is evident), the fiber at s of the arrow τ_φ (6.2.1.1) is an isomorphism.

In the general case, let x be a geometric point of S and let Ψ_x be the fiber of Ψ_f at x,

$$\Psi_x = \varinjlim_{U(x)} \Psi_f(U),$$

where U ranges over the étale neighborhoods of x. Let S_x be the strict localization of
422 S at x, $X_x = X \times_S S_x$, and $f_x : X_x \to S_x$. The standard limit arguments show that the category Ψ_x is equivalent to the category $\Phi_{f_x}(S_x)$, hence is filtering on the right. It follows that Ψ_f is locally filtering on the right. Moreover, the fiber at x of τ identifies with the arrow

$$\tau_x : \varinjlim_{\Psi_x} \mathscr{F}_x^{I_x} \longrightarrow (f_! f^* \mathscr{F})_x$$

and, by the preceding discussion, this arrow is an isomorphism.

Theorem 6.2.3. *There is a unique way to define, for every morphism $f : X \to S$ that is separated, flat, of finite presentation, and quasi-finite, and every abelian sheaf $\mathscr{F}$ on S, a trace morphism*

$$\mathrm{Tr}_f : f_! f^* \mathscr{F} \longrightarrow \mathscr{F}$$

such that the following conditions hold:

(Var 1) Tr_f *is functorial in* $\mathscr{F}$.

(Var 2) Tr_f *is compatible with every base change, i.e. for every cartesian diagram*

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"] & X \arrow[d, "f"] \\ S' \arrow[r, "g"] & S \end{tikzcd}$$

the following diagram of functors is commutative:

$$\begin{tikzcd} f'_! f'^* g^* \arrow[r, equal] \arrow[d, "\mathrm{Tr}_{f'} * g^*"] & f'_! g'^* f^* \arrow[r, equal] & g^* f_! f^* \arrow[d, "g^* * \mathrm{Tr}_f"] \\ g^* \arrow[rr, equal] & & g^* \end{tikzcd}.$$

423 (Var 3) Tr_f *is compatible with composition, i.e. if f is the composite $f = gh$ of two morphisms that are separated, flat, of finite presentation, and quasi-finite, the following diagram is commutative:*

$$\begin{tikzcd} f_! f^* \arrow[r, equal] \arrow[ddr, "\mathrm{Tr}_f"'] & g_! h_! h^* g_* \arrow[d, "g_! * \mathrm{Tr}_h * g^*"] \\ & g_! g^* \arrow[d, "\mathrm{Tr}_g"] \\ & \mathrm{Id}. \end{tikzcd}$$

(Var 4) *(Normalization). If f is finite of constant rank n, the composite homomorphism*

$$\mathscr{F} \to f_* f^* \mathscr{F} = f_! f^* \mathscr{F} \xrightarrow{\mathrm{Tr}_f} \mathscr{F}$$

is multiplication by n.

First note that conditions (Var 2) through (Var 4) imply that

Lemma 6.2.3.1. *If $X = X' \amalg X''$, then, setting $f' = f|X'$ and $f'' = f|X''$, the morphism Tr_f:*

$$f_! f^* \mathscr{F} = f'_! f'^* \mathscr{F} \oplus f''_! f''^* \mathscr{F} \longrightarrow \mathscr{F}$$

is the sum of $\mathrm{Tr}_{f'}$ and $\mathrm{Tr}_{f''}$.

$$\begin{tikzcd}
X' \arrow[dr, hook, "j'"] & & \\
& X \arrow[r, "f"] & S \\
X'' \arrow[ur, hook, "j''"'] & &
\end{tikzcd}$$

.

If one applies (Var 2) to j', for base change by j', and (Var 4) to the identity map, one sees that 424

$$\mathrm{Tr}_{j'} : j'_! j'^* f^* \mathscr{F} \longrightarrow f^* \mathscr{F}$$

is the canonical map defined by the decomposition of $f^* \mathscr{F}$ as the direct sum

$$f^* \mathscr{F} = j'_! j'^* f^* \mathscr{F} \oplus j''_! j''^* f^* \mathscr{F}.$$

If one applies (Var 3) to $f' = f \circ j'$, one sees that the following diagram is commutative

$$\begin{tikzcd}
f'_! f'^* \mathscr{F} \arrow[rr] \arrow[dr, "\mathrm{Tr}_{f'}"'] & & f_! f^* \mathscr{F} \arrow[dl, "\mathrm{Tr}_f"] \\
& \mathscr{F} &
\end{tikzcd},$$

which, together with the analogous statement for f'', is precisely 6.2.3.1.

Let $f : X \to S$ be a separated, flat, quasi-finite morphism of finite presentation. For every $U \in \mathrm{Ob}\, S_{\mathrm{et}}$, let $\Psi'_f(U)$ be the full subcategory of $\Psi_f(U)$ (6.2.1) consisting of the decompositions $(V_i)_{i \in I}$ such that, for $i \neq 0$, V_i is locally free of constant rank over U. The categories $\Psi'_f(U)$ form a fibered subcategory of Ψ_f, cofinal in Ψ_f. 425

Let $\mathscr{F}$ and $\mathscr{G}$ be abelian sheaves on S. By 6.2.2 (ii), giving a morphism $g : f_! f^* \mathscr{F} \to \mathscr{G}$ is equivalent to giving, for each decomposition $\varphi = (V_i)_{i \in I} \in \Psi'_f(U)$, a morphism $g_\varphi : \mathscr{F}^{I^*}|U \to \mathscr{G}|U$, compatible with the morphisms ${}^t\sigma$ of (6.2.1.2). If f_i is the projection from V_i to U, g_φ is the arrow whose coordinate arrows are the composites:

$$g^i_\varphi : \mathscr{F}|U \to f_{i*} f_i^* \mathscr{F}|U \to f_! f^* \mathscr{F}|U \xrightarrow{g} \mathscr{G}|U.$$

It follows that there exists a unique arrow Tr_f, compatible with étale localization and satisfying (Var 4) and 6.2.3.1. It is clear that this arrow satisfies (Var 1) and (Var 2).

To verify (Var 3), by base change one reduces to the easy case in which S is the spectrum of an algebraically closed field.

6.2.4. A *weighting* of a separated quasi-finite morphism $f : X \to S$ is the datum, for each geometric point x of X, of an integer $n(x)$ depending only on the point of X underlying x, such that:

(∗) *For every $U \in \operatorname{Ob} S_{\mathrm{et}}$ and every open and closed subspace X_1 of $f^{-1}(U)$ that is finite over U, the function assigning to each geometrical point s of U the number $\sum_{x \in X_1(s)} n(x)$ is locally constant.*

If $f = gh$ is the composite of two weighted morphisms, we weight f by setting $n_f(x) = n_h(x) \cdot n_g(h(x))$.

426 If the morphism f' is obtained by a base change g from a weighted morphism f, we weight f' by setting $n_{f'}(x) = n_f(g'(x))$.

The preceding arguments in fact prove:

Proposition 6.2.5. *There is a unique way to define, for every weighted, separated, quasi-finite morphism $f : X \to S$ and every abelian sheaf $\mathscr{F}$ on S, a trace morphism $\operatorname{Tr}_f : f_! f^* \mathscr{F} \to \mathscr{F}$ that is functorial in $\mathscr{F}$, compatible with base change and with composition, and satisfies:*

(Var 4′) *If f is finite and there exists an integer n such that, for every geometric point s of S, one has $\sum_{x \in X(s)} n(x) = n$, then the composite arrow*

$$\mathscr{F} \longrightarrow f_* f^* \mathscr{F} = f_! f^* \mathscr{F} \xrightarrow{\operatorname{Tr}_f} \mathscr{F}$$

is multiplication by n.

Example 6.2.6. Let S be a reduced noetherian geometrically unibranch scheme (for example, a normal scheme), and let $f : X \to S$ be a separated quasi-finite morphism. For every geometric point x of X with image s in S, let $\mathscr{O}_x$ (resp. $\mathscr{O}_s$) be the strictly local ring of X at x (resp. of S at s). The scheme $\operatorname{Spec}(\mathscr{O}_s)$ is reduced and has a unique maximal point η. Moreover, $\mathscr{O}_x$ is finite over $\mathscr{O}_s$. Set $n(x) = \operatorname{rg}_\eta(\mathscr{O}_x)$. One verifies that $n(x)$ is a weighting of f, whence a trace morphism Tr_f.

427 6.2.7. Let $u : X \to Y$ be a separated, quasi-finite, flat morphism of finite presentation. Since the trace morphism (6.2.3) is functorial, it defines, for every complex of abelian sheaves K on Y, a morphism of complexes

$$\operatorname{Tr}_u : u_! u^* K \longrightarrow K. \tag{6.2.7.1}$$

We shall use the terms *trace morphism, Gysin morphism* and *morphism u_** interchangeably for any morphism obtained from (6.2.7.1) or 6.2.3 by applying a functor; for example, the following ones.

Consider a commutative diagram

$$\begin{tikzcd} X \arrow[rr, "u"] \arrow[dr, "f"'] & & Y \arrow[dl, "g"] \\ & S & \end{tikzcd}$$

with g separated and locally of finite type. For every abelian sheaf $\mathscr{F}$ on Y, the trace morphism defines a morphism

$$\operatorname{Tr}_u \quad \text{or} \quad u_* : f_! u^* \mathscr{F} \simeq g_! u_! u^* \mathscr{F} \xrightarrow{g_!(\operatorname{Tr}_u)} g_! \mathscr{F}. \tag{6.2.7.2}$$

If g is compactifiable, and if K is a complex bounded below with torsion cohomology, or a complex of $g^*\mathscr{A}$-modules for $\mathscr{A}$ a sheaf of torsion rings on S, the trace morphism (6.2.7.1) likewise induces

$$\text{(6.2.7.3)} \qquad \mathrm{Tr}_u \quad \text{or} \quad u_* : Rf_! u^* K \simeq Rg_! u_! u^* K \xrightarrow{Rg_!(\mathrm{Tr}_u)} Rg_! K,$$

$$\text{(6.2.7.4)} \qquad \mathrm{Tr}_u \quad \text{or} \quad u_* : R^q f_! u^* K \simeq R^q g_! u_! u^* K \xrightarrow{R^q g_!(\mathrm{Tr}_u)} R^q g_! K.$$

For S the spectrum of an algebraically closed field, (6.2.7.4) is a morphism 428

$$\text{(6.2.7.5)} \qquad u_* : \mathrm{H}^q_c(X, u^*K) \longrightarrow \mathrm{H}^q_c(Y, K).$$

Thus one sees that *cohomology with proper supports is covariant* with respect to quasi-finite morphisms.

6.2.8. Let $u : X \to Y$ be as in 6.2.7, let $\Delta_n = [0, n]$ be the standard set with $(n+1)$ elements, and let X_n be the $(n+1)^{\text{fold}}$ iterated fiber product of X over Y. The X_n form a simplicial scheme augmented to Y (cf. VBIS). We denote by $u_n : X_n \to Y$ the augmentation morphism. If $\mathscr{F}$ is an abelian sheaf on Y, the sheaves $\mathscr{F}_n = u_{n!} u_n^* \mathscr{F}$ form, by (6.2.7.2) (applied after the change of notation $S \mapsto Y$, $X \mapsto X_n$, $Y \mapsto Y$ or X_m), a *simplicial* sheaf augmented to $\mathscr{F}$. By definition, the differential complex of "*nondegenerate chains*" of this simplicial sheaf has as its n^{th} component the quotient of $\mathscr{F}_n$, namely the cokernel of the direct sum of the degeneracy operators with target $\mathscr{F}_n$ and of the morphisms $\sigma - \epsilon(\sigma)$ for σ a symmetry operator.

Proposition 6.2.9. *Under the preceding hypotheses, if u is separated, étale, of finite type, and surjective, then for every abelian sheaf $\mathscr{F}$ on Y, one has*

(i) *The sheaves $\mathscr{F}_n$ form a (left) resolution of $\mathscr{F}$.*
(ii) *The differential complex of nondegenerate chains of the simplicial sheaf $(\mathscr{F}_n)$ is a (left) resolution of $\mathscr{F}$. If all the geometric fibers of u have at most d points, this resolution has length $\leq d$.*

By base change, it suffices to verify this statement when Y is the spectrum of an algebraically closed field, in which case 6.2.8 reduces to computing the simplicial homology of a simplex. 429

Let $(U_i)_{i \in I}$ be a finite open covering of the scheme Y, let $\mathscr{F}$ be an abelian sheaf on Y, and, for each subset P of I, let j_P be the inclusion of $U_P = \cap_{i \in P} U_i$ into Y. Suppose that a total order on I has been chosen. If $X = \coprod_{i \in I} U_i$ and if u is the projection from X to Y, the resolution 6.2.9 (ii) of $\mathscr{F}$ is given by

$$\text{(6.2.9.1)} \qquad (\mathscr{F}_n)_{n\,\mathrm{deg}} = \bigoplus_{|P| = n+1} j_{P!} j_P^* \mathscr{F}.$$

This resolution still makes sense when I is infinite.

6.2.10. Under the hypotheses of 6.2.9, let $\mathscr{A}$ be a sheaf of torsion rings on Y and let $K \in \mathrm{Ob}^-(Y, \mathscr{A})$. We regard X as ringed by $u^*\mathscr{A}$. Since resolution 6.2.9 (i) (resp. 6.2.9 (ii)) is functorial in $\mathscr{F}$, it defines a resolution of K by a double complex K_1. This resolution is called the extraordinary *Čech resolution* (resp. the *extraordinary alternating* Čech resolution) of K.

Consider a commutative diagram

$$\begin{tikzcd} X \arrow[rr, "u"] \arrow[dr, "f"'] & & Y \arrow[dl, "g"] \\ & S & \end{tikzcd} ,$$

with g compactifiable. With the notation of 6.2.8, if f_n is the projection from X_n to S,
430 one has $Rf_{n!} = Rg_! u_{n!}$. The complex K_1, filtered by the semisimplicial degree, therefore defines a spectral sequence

$$E_1^{-p,q} = R^q f_{p!}(u_p^* K) \Longrightarrow R^{-p+q} g_! K. \tag{6.2.10.1}$$

The differentials d_1 arise from the simplicial arrows, so that

$$E_2^{-p,q} = \check{\mathscr{H}}^p(R^q f_{p!}(u_p^* K)) \Longrightarrow R^{-p+q} g_! K. \tag{6.2.10.2}$$

Under the hypotheses of (6.2.9.1), one similarly obtains a spectral sequence

$$E_1^{-p,q} = \bigoplus_{|P|=p+1} R^q f_{p!}(j_p^* K) \Longrightarrow R^{-p+q} g_! K. \tag{6.2.10.3}$$

The spectral sequences (6.2.10.1) and (6.2.10.3) are called the *upper localization spectral sequences.*

Proposition 6.2.11. *If f is a separated étale morphism of finite type, the functor $f_!$ is left adjoint to the functor f^*, and its adjunction arrow is the trace morphism.*

It is known that if X is an object of a site S, the functor f^* restricting abelian sheaves to X has a left adjoint $f_!$. If F is an abelian sheaf on X, $f_! F$ is the sheaf generated by the presheaf

$$V \longrightarrow \bigoplus_{\xi \in \mathrm{Hom}(V,X)} F(\xi).$$

In the étale case, provisionally denote this functor by $f_!^0$. Its adjunction property immediately implies that its formation commutes with every base change. The trace morphism
431 defines a morphism t from $f_! \mathscr{F}$ to $f_!^0 \mathscr{F}$, compatible with every base change. It suffices to see that t is an isomorphism fiber by fiber, i.e. when S is the spectrum of a separably closed field, where this is trivial.

. One should note that this interpretation of $f_!^0$ is specific to the case of the étale site and does not extend to the fpqf site of a prescheme, for example.

6.3. Variant of the trace for continuous coefficients. In this no. we prove the assertions of Grothendieck [5, no. 6], and apply these results to the definition of a "trace morphism" for suitable fppf sheaves.

6.3.1. Let A be a commutative ring. If L and M are two finitely generated projective A-modules, the map $u \mapsto \bigwedge^m u$ from $\mathrm{Hom}(L, M)$ to $\mathrm{Hom}(\bigwedge^m L, \bigwedge^m M)$ is compatible with every base change and is $m^{\text{homogeneous}}$ (5.5.2.2). By 5.5.2.3, it defines a morphism

$$\bigwedge^m : \Gamma^m \mathrm{Hom}(L, M) \longrightarrow \mathrm{Hom}(\bigwedge^m L, \bigwedge^m M) \tag{6.3.1.1}$$

such that, after every base change, for $u \in \mathrm{Hom}(L, M)$ one has $\bigwedge^m(\gamma^m(u)) = \bigwedge^m u$. Since $\mathrm{Hom}(L, M)$ is flat, this morphism is identified (5.5.2.5) with a morphism

$$\bigwedge^m : TS^m \,\mathrm{Hom}(L, M) \longrightarrow \mathrm{Hom}(\bigwedge^m L, \bigwedge^m M). \tag{6.3.1.2}$$

N.B. Such a homomorphism can also be defined whenever 2 is invertible in A: a symmetric element of $\bigotimes^m \mathrm{Hom}(L, M)$ defines a symmetric homomorphism from $\bigotimes^m L$ to $\bigotimes^m M$, and this descends to the quotient to define a homomorphism from $\bigwedge^m L$ to $\bigwedge^m M$.

Given three finitely generated projective A-modules, the diagram 432

$$\begin{tikzcd} \Gamma^m \,\mathrm{Hom}(L, M) \otimes \Gamma^m \,\mathrm{Hom}(M, N) \arrow[r] \arrow[d, "\bigwedge^m \otimes \bigwedge^m"] & \Gamma^m \,\mathrm{Hom}(L, N) \arrow[d, "\bigwedge^m"] \\ \mathrm{Hom}(\bigwedge^m L, \bigwedge^m M) \otimes \mathrm{Hom}(\bigwedge^m M, \bigwedge^m N) \arrow[r] & \mathrm{Hom}(\bigwedge^m L, \bigwedge^m N), \end{tikzcd} \tag{6.3.1.3}$$

in which the horizontal arrows are deduced from the composition maps (and, for the first, from (5.5.2.8)), is commutative. This expresses the formula $\bigwedge^m(vu) = \bigwedge^m v \circ \bigwedge^m u$.

If L is a free A-module of rank n, (6.3.1.2) defines a canonical morphism

$$\det : \Gamma^n \,\mathrm{End}_A(L) = TS^n \,\mathrm{End}_A(L) \longrightarrow A, \tag{6.3.1.4}$$

which is an algebra homomorphism by (6.3.1.3). By definition, one has

$$\det(u^{\otimes n}) = \det(u; L). \tag{6.3.1.5}$$

If, in addition, L is a module over an A-algebra B, (6.3.1.4) defines by composition an algebra homomorphism

$$\det_L : TS^n_A(B) \longrightarrow A. \tag{6.3.1.6}$$

Lemma 6.3.2. *Suppose that L occurs in an exact sequence*

$$E : 0 \longrightarrow L' \longrightarrow L \longrightarrow L'' \longrightarrow 0$$

with L' (resp. L'') locally free of rank n_1 (resp. n_2). Let $n = n_1 + n_2$, let α be the "associated graded" morphism from the group $\mathrm{End}(E)$ of endomorphisms of the extension E to 433 *$\mathrm{End}(L') \oplus \mathrm{End}(L'')$, and, for two flat A-modules X and Y, let β be the morphism from $TS^n(X \oplus Y)$ to $TS^{n_1}(X) \otimes TS^{n_2}(Y)$ deduced from the isomorphism (5.5.2.6)*

$$TS^n(X \oplus Y) \simeq \bigoplus_{n'+n''=n} TS^{n'}(X) \otimes TS^{n''}(Y).$$

The following diagram is commutative:

$$\begin{tikzcd} TS^n(\mathrm{End}(E)) \arrow[r, "TS^n(\alpha)"] \arrow[d] & TS^n(\mathrm{End}(L') \oplus \mathrm{Ens}(L'')) \arrow[r, "\beta"] & TS^{n_1}(\mathrm{End}(L')) \otimes TS^{n_2}(\mathrm{End}(L'')) \arrow[d, "\det \otimes \det"] \\ TS^n(\mathrm{End}(L)) \arrow[rr, "\det"] & & A. \end{tikzcd}$$

By the universal property 5.5.2 of TS^n, it suffices to check that the two images of $u^{\otimes n}$ coincide for $u \in \mathrm{End}(E)$. If $\alpha u = (u', u'')$, these images are respectively $\det(u)$ and $\det(u').\det(u'')$, which proves the assertion.

LEMMA 6.3.3. *Let A be a commutative ring, B a commutative A-algebra, C a not necessarily commutative B-algebra, E a B-module that is free of rank n over A, and F a C-module that is free of rank m over B. Suppose that C is flat over A. By the universal property 5.5.2.2 of Γ^{nm} and 5.5.2.3, there is a unique arrow $\sigma : TS^{nm}_A(C) \to TS^n_A TS^m_B(C)$ such that, after every ring extension $A \to A'$, one has $\sigma(c^{\otimes nm}) = (c^{\otimes m})^{\otimes n}$. The following diagram is then commutative:*

$$\begin{tikzcd}
TS^n_A(B) \arrow[d, "\det_E"] & & TS^n_A TS^m_B(C) \arrow[ll, "TS^n_A(\det_F)"'] \\
A & & TS^{nm}_A(C). \arrow[ll, "\det_{E\otimes_B F}"] \arrow[u, "\sigma"']
\end{tikzcd}$$

434 PROOF. By 5.5.2.2 and 5.5.2.3, it suffices to show that the two images of an element $u^{\otimes nm}$ of $TS^{nm}_A(C)$ coincide. These images are $\det_E(\det_F(u))$ and $\det_{E\otimes_B F}(u)$; they are expressed in terms of A, B, E, F, and $u \in \mathrm{End}_B(F)$, and, to check that they coincide, one may first replace B by its (commutative) image B_1 in $\mathrm{End}(E)$ and F by $F \otimes_B B_1$. The formula to be proved is then the block-determinant formula (BOURBAKI, *Alg.* Chap. 8 § 12, Lemma 2, p. 140).

When C is no longer necessarily flat over A, the same proof gives, more generally, a commutative diagram of algebras

(6.3.3.1)
$$\begin{tikzcd}
\Gamma^n_A(B) \arrow[d, "\det_E"] & & \Gamma^n_A \Gamma^n_B(C) \arrow[ll, "\Gamma^n_A(\det_F)"'] \\
A & & \Gamma^{nm}_A(C). \arrow[ll, "\det_{E\otimes_B F}"] \arrow[u, "\sigma"']
\end{tikzcd}$$

6.3.4. If B is a commutative A-algebra, $\mathrm{Spec}(TS^n(B))$ is precisely the n^{th} symmetric power of $\mathrm{Spec}(B)$ over $\mathrm{Spec}(A)$, i.e. the quotient of $(\mathrm{Spec}(B)/\mathrm{Spec}(A))^n$ by the symmetric group (SGA 1 V 1).

435 If X is a scheme separated over S, a quasi-coherent $\mathcal{O}_X$-module $\mathcal{F}$ on X will be called *finite locally free* (resp. of rank n) *over* S if its support is finite over S and its direct image is locally free of finite rank (resp. of rank n) over S. By the *support* of $\mathcal{F}$ we shall again mean the *subscheme* of X defined by the annihilator of $\mathcal{F}$ (which is a quasi-coherent ideal, since $\mathcal{F}$ is quasi-coherent of finite type).

If $\mathcal{F}$ is finite locally free over S, of rank n and with support E, then E is affine over S; by globalizing over S, $\mathcal{F}$ defines, by means of (6.3.1.6), a section $\sigma_E(\mathcal{F})$ of $\mathrm{Sym}^n_S(E)$, hence, by functoriality, a section $\sigma(\mathcal{F})$ of the algebraic space [70] $\mathrm{Sym}^n_S(X)$,

(6.3.4.1) $$\sigma(\mathcal{F}) \in \Gamma(\mathrm{Sym}^n_S(X)/S).$$

In particular, if $f : S' \to S$ is finite locally free of rank n, then, for $\mathcal{F} = \mathcal{O}_{S'}$, one obtains a canonical section

(6.3.4.2) $$t \in \Gamma(\mathrm{Sym}^n_S(S')/S).$$

[70] M. ARTIN, Algebraization of formal moduli I, in *Global Analysis: Papers in Honor of K. Kodaira*, Princeton University Press, 1969, 21–71. We shall use the preceding construction only when $\mathrm{Sym}^n_S(X)$ exists as a scheme, which is the case when X is quasi-projective over S.

PROPOSITION 6.3.5. *Let X be a scheme separated over S, and let $\mathscr{F}_1$ and $\mathscr{F}_2$ be two $\mathscr{O}_X$-modules finite locally free of ranks n_1 and n_2 over S. Put $n = n_1 + n_2$. Let d be the canonical arrow $d : \mathrm{Sym}^{n_1}_S(X) \times \mathrm{Sym}^{n_2}_S(X) \to \mathrm{Sym}^n_S(X)$. If $\mathscr{F}$ is an extension of $\mathscr{F}_1$ by $\mathscr{F}_2$, then*

$$\sigma(\mathscr{F}) = d(\sigma(\mathscr{F}_1), \sigma(\mathscr{F}_2)).$$

One may replace X by the support (finite over S) of $\mathscr{F}_1 \oplus \mathscr{F}_2$ and suppose that S is 436
affine. Translated into ring-theoretic terms, the proposition then follows from 6.3.2.

LEMMA 6.3.6. *If X is a scheme separated over S, $\mathscr{F}$ an $\mathscr{O}_X$-module finite locally free of rank n over S, and $\mathscr{L}$ an invertible sheaf on X, then*

$$\sigma(\mathscr{F}) = \sigma(\mathscr{F} \otimes \mathscr{L}).$$

Replace X by the support (6.3.4) of $\mathscr{F}$. Locally over S, $\mathscr{L}$ is then isomorphic to $\mathscr{O}_X$, and the assertion is trivial.

The reader may verify:

PROPOSITION 6.3.7. *For every diagram of S-schemes separated over S*

$$\begin{tikzcd} X \arrow[rr, "u"] \arrow[dr, "f"'] & & Y \arrow[dl, "g"] \\ & S & \end{tikzcd},$$

and every $\mathscr{O}_X$-module $\mathscr{F}$ finite locally free of rank n over S, the $\mathscr{O}_Y$-module $u_*\mathscr{F}$ is finite locally free of rank n over S, and

$$(\mathrm{Sym}^n_S u)(\sigma(\mathscr{F})) = \sigma(u_*\mathscr{F}).$$

. Finally, note that the construction of the morphisms σ (6.3.4.1) and t (6.3.4.2) is *compatible with every base change*, which makes sense by 5.5.2.7.

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Application 1. The case of smooth curves.

6.3.8. Let $f : X \to S$ be a smooth quasi-projective curve over S, i.e. a smooth quasi-projective morphism purely of relative dimension 1. The theory of Hilbert schemes GROTHENDIECK [5]) shows that the fpqc sheaf $\mathrm{Div}^n_{X/S}$ which assigns to each S-scheme S' the set of relative divisors on $X' = X \times_S S'$ that are *finite* over S and of degree n is representable.

For each relative divisor E on X, finite of degree n over S, there is a canonical element (6.3.4.2) $t(E) \in \mathrm{Sym}^n_S(E)$, with image $\sigma(E)$ in $\mathrm{Sym}^n_S(X)$. Since this construction is compatible with every base change, it defines

$$\sigma : \mathrm{Div}^n_{X/S} \longrightarrow \mathrm{Sym}^n_S(X) : E \longrightarrow \sigma(\mathscr{O}_E). \tag{6.3.8.1}$$

Each section s of f defines a relative divisor finite over S of degree 1, namely the divisor $s(S)$. The morphism

$$(X/S)^n \longrightarrow \mathrm{Div}^n_{X/S} : (s_i)_{0<i\leq n} \longmapsto \sum s_i(S)$$

is symmetric, and hence induces a morphism

$$\alpha : \mathrm{Sym}^n_S(X) \longrightarrow \mathrm{Div}^n_{X/S}. \tag{6.3.8.2}$$

PROPOSITION 6.3.9. *The morphisms σ and α of 6.3.8 are inverse isomorphisms.*

PROOF. a) $\sigma\alpha = \mathrm{Id}$. It suffices to prove that the composite arrow

$$(X/S)^n \xrightarrow{\pi} \mathrm{Sym}^n_S(X) \xrightarrow{\alpha} \mathrm{Div}^n_{X/S} \xrightarrow{\sigma} \mathrm{Sym}^n_S(X)$$

is π. For $n = 1$, the assertion is clear; the general case follows from the 438

LEMMA 6.3.9.1. *With the notation of 6.3.5 and (6.3.8.1), one has*

$$\sigma(E + F) = d(\sigma(E), \sigma(F)).$$

This lemma follows from 6.3.5, 6.3.6, and the exact sequence

$$0 \longrightarrow \mathcal{O}(-F) \otimes \mathcal{O}_E \longrightarrow \mathcal{O}_{E+F} \longrightarrow \mathcal{O}_F \longrightarrow 0.$$

b) It is known that $\mathrm{Div}^n_{X/S}$ is smooth over S (there is no obstruction to infinitesimally lifting divisors on a curve (cf. SGA 3 XI 1)). The morphism $\alpha\pi : (X/S)^n \to \mathrm{Div}^n_{X/S}$ is a quasi-finite surjective morphism between smooth S-schemes whose source has equidimensional fibers. It is therefore flat (EGA 0_{IV} 17.3.5), so $\alpha\pi$ is an epimorphism of fppf sheaves, and hence so is α. In view of a), this completes the proof of 6.3.9.

LEMMA 6.3.10. *Let k be a commutative ring, M a free k-module, and u an endomorphism of M. Denote by $N_{\det(u-T)=0}$ the norm from $k[T]/\det(u - T)$ to k. Then, for $P \in k[T]$, one has*

$$N_{\det(u-T)=0}(P(T)) = \det(P(u), M).$$

This amounts to checking an algebraic identity between the entries of the matrix of u and the coefficients of P. To prove 6.3.10, it therefore suffices (by familiar arguments) to treat the case in which k is algebraically closed and u is semisimple, with eigenvalues a_i.

439 One then has

$$\begin{aligned} &\det(u - T) = \Pi(a_i - T) \\ &N_{\det(u-T)=0}(P(T)) = \Pi P(a_i) \quad \text{and} \\ &\det(P(u), M) = \Pi P(a_i). \end{aligned}$$

. Let $f : X \to S$ be a quasi-projective morphism of finite presentation, flat with purely one-dimensional Cohen–Macaulay fibers; these conditions are thus satisfied if X is a smooth quasi-projective curve over S. Let $\mathcal{F}$ be an $\mathcal{O}_X$-module finite locally free *over* S. Since $\mathcal{F}$ vanishes on a fiberwise dense open subset of X, there is a canonical arrow $\psi : \mathcal{O}_X \to \det^*(\mathcal{F})$ (SGA 6 XI or MUMFORD [8, Chap. 5 § 3]). We denote by $\delta(\mathcal{F})$ the divisor defined by the annihilator ideal of $\mathrm{coker}(\psi)$ (ibid.); this divisor is finite over S.

PROPOSITION 6.3.11.1. *Under the hypotheses of 6.3.11.0 and with the notation of 6.3.8 and (6.3.4.1), if $\mathcal{F}$ has rank n, then $\delta(\mathcal{F})$ has degree n and*

$$\sigma(\mathcal{F}) = \sigma(\delta(\mathcal{F})).$$

PROOF. a) *The case where f is smooth.* By a standard reduction, we may assume that S is an Artinian local scheme with algebraically closed residue field. The additivity (6.3.5) of σ allows us to suppose that the support of $\mathcal{F}$ is concentrated at a rational point x of X. One checks at once that if T is a uniformizer at x, defining $p : X \to \mathbf{A}^1_S$, then $p_*(\mathcal{O}_{\delta(\mathcal{F})}) \simeq \mathcal{O}_{\delta(p_*\mathcal{F})}$ (because the constructions in question commute with passage to the completion). By this and 6.3.7, we reduce to the case $X = \mathbf{A}^1_S$.

440 If $X = \mathbf{A}^1_S$ and S is affine, $S = \operatorname{Spec}(k)$, then $\mathscr{F}$ is identified with a locally free k-module M equipped with an endomorphism u (the action of T). As a $k[T]$-module, M has the resolution

$$0 \to k[T] \otimes_k M \xrightarrow{\varphi} k[T] \otimes_k M \xrightarrow{\epsilon} M \to 0,$$

where $\epsilon(P \otimes m) = P(u)(m)$ and $\varphi(1 \otimes m) = 1 \otimes um - T \otimes m$. From the formula $\varphi = u - T$, it follows that the divisor $\delta(\mathscr{F})$ has equation $\det(u - T, M) = 0$.

To check 6.3.11.1, it suffices to show that, after every base change and for every section φ of $\mathscr{O}_X$, one has

$$\det_k(\varphi, M) = \det_k(\varphi, \mathscr{O}_{\delta(\mathscr{F})})(= N_{\delta(\mathscr{F})/S}(\varphi)).$$

For $\varphi = P(T)$, this is precisely the assertion of 6.3.10.

b) The general case. We first state more explicitly the assertion to be proved, in the following more concrete form:

Corollary 6.3.11.2. *Under the hypotheses of 6.3.11.0 on $f : X \to S$, let M be a finite locally free module on X, and let u be an endomorphism of M such that $\det u$ is a section of $\mathscr{O}_X$ that is regular on every fiber X_s (whence $\operatorname{Coker} u = M/u(M)$ and $\operatorname{Coker} \det u = \mathscr{O}_X / \det u$ are both flat and of finite presentation over S, with quasi-finite supports). Suppose that the common support[71] of these sheaves is finite over S, so that they define finite locally free sheaves over S. Then, for every section φ of $\mathscr{O}_X$, the determinants over $\mathscr{O}_S$ of φ acting on $\operatorname{Coker} u$ and on $\operatorname{Coker} \det u$ are equal:*

$$\det_S(\varphi, M/u(M)) = \det_S(\varphi, \mathscr{O}_X / \det u). \tag{6.3.11.3}$$

We note that this statement implies the equality of the ranks of the two locally free modules under consideration over S, by homogeneity (when φ is replaced by $T\varphi$, with T an "indeterminate" section of $\mathscr{O}_S$). Moreover, replacing φ by $1 + T\varphi$, where T is such an indeterminate (more precisely, one makes a base change $S' = S[T] \to S$), one deduces from (6.3.11.3) the apparently more general formula 441

$$\operatorname{Pol}_S(\varphi, M/uM)(T) = \operatorname{Pol}_S(\varphi, \mathscr{O}_X / \det u)(T) \tag{6.3.11.4}$$

for the characteristic polynomials.

To prove (6.3.11.3), we shall reduce to the already proved smooth case. By étale localization on S, we reduce to the case where S is strictly local with closed point s; then, by additivity, we reduce to the case where the support of $M/u(M)$ has a single point on X_s, say x. After making a flat extension of the base, we may assume the residue field extension $k(x)/k(s)$ is trivial; and after replacing X by a sufficiently small neighborhood of x, and adding to φ a function that induces the zero operation on $M/u(M)$ and on $\mathscr{O}_X / \det u$ (which does not change the formula to be proved), one readily sees that we may assume that φ induces a section of $\mathscr{O}_{X_s}$ regular at x, and hence (after shrinking X) defines a quasi-finite flat morphism

$$p : X \longrightarrow X' = S[T].$$

[71] N.D.E.: the support of M/uM in the sense of 6.3.4 is a subscheme of the support of $\mathscr{O}/\det u$ defined by a nilideal.

By étale localization on X and on X', we may therefore assume that we have a *finite* and *flat* morphism p as above, with X' smooth over S, and $\varphi = p^*(\varphi')$, where φ' is a section of $\mathscr{O}_{X'}$. Set

$$\begin{aligned} M' &= p_*(M) \quad , \quad u' = p_*(u), \\ P &= \operatorname{Coker} u \quad , \quad Q = \operatorname{Coker} \det u, \\ P' &= \operatorname{Coker} u' \quad , \quad Q' = \operatorname{Coker} \det u'. \end{aligned}$$

442 We know that

$$\det u' = N_{X/X'} \det u, \quad \text{hence} \quad Q' = \operatorname{Coker} N_{X/X'}(\det u).$$

We trivially have

$$\det_S(\varphi, P) = \det_S(\varphi', P'),$$

and by formula (6.3.11.3), proved in the smooth case,

$$\det_S(\varphi', P') = \det_S(\varphi', Q'),$$

so it remains to prove the formula

$$\det_S(\varphi', Q') = \det_S(\varphi, Q),$$

which is *equivalent* to formula (6.3.11.3) to be established. We thus see that the validity of the latter depends only on the section $\det u$ of $\mathscr{O}_X$, or equivalently that it is equivalent to the same formula in the case where M is replaced by $\mathscr{O}_X$, and φ by $\det u$. But in the case $M = \mathscr{O}_X$, formula (6.3.11.3) is tautological! This completes the proof.

443 Remark 6.3.11.5. In the case where S is Artinian, the equality over S of the ranks of $\operatorname{Coker} u$ and $\operatorname{Coker} \det u$ (notation of 6.3.11.2) is a special case of EGA IV 21.10.17.3, of which (6.3.11.3) may be regarded as a "relative" version.

Proposition 6.3.12. *Let X and Y be two quasi-projective curves of finite presentation, flat with CohenMacaulay fibers over S, and let $u : X \to Y$ be a quasi-finite flat morphism. The following diagram*

$$\begin{tikzcd} \mathrm{Div}^n_{X/S} \arrow[r, "u_*"] \arrow[d, "\sigma_X"'] & \mathrm{Div}^n_{Y/S} \arrow[d, "\sigma_Y"'] \\ \mathrm{Sym}^n_S(X) \arrow[r, "\mathrm{Sym}^n(u)"] & \mathrm{Sym}^n_S(Y) \end{tikzcd}$$

is then commutative [71].

Since $u_*(D) = \delta(u_*\mathscr{O}_D)$, 6.3.12 follows from 6.3.7 and 6.3.11.1.

Application 2. The trace morphism.

6.3.13. Let G be an abelian sheaf on the large fppf site of a scheme S (0.10). Suppose that G satisfies the following condition.

. *For every affine flat morphism of finite presentation $f : X \to Y$ of S-schemes, and for every integer $n \geq 1$, the morphism*

$$\mathrm{Hom}_S(\mathrm{Sym}^n_Y(X), G) \longrightarrow \mathrm{Hom}_S((X/Y)^n, G)$$

444 *is injective and has as its image the set of symmetric morphisms from $(X/Y)^n$ to G.*

This condition is stable under base change.

Let $f : T \to S$ be a finite locally free morphism of rank n. Let pr_i be the i^{th} projection from $(T/S)^n$ to T, and let t be the section (6.3.4.2) of $\mathrm{Sym}^n_S(T)$. If $u \in G(T)$, then $\sum \mathrm{pr}_i^*(u)$ is a symmetric element of $G((T/S)^n)$ and therefore defines

$$u_n \in G(\mathrm{Sym}^n_S(T)).$$

We set

$$\mathrm{Tr}_f(u) = t^*(u_n).$$

When f is finite locally free, we define $\mathrm{Tr}_f(u)$, locally on S, by the preceding formula. This construction is compatible with every base change and therefore defines a *trace morphism*

$$\mathrm{Tr}_f : f_* f^* G \longrightarrow G. \tag{6.3.13.2}$$

6.3.14. Condition (6.3.13.1) is satisfied in the following cases:

a) If G is representable, by the definition of symmetric powers.
b) For G the inverse image on S_{fppf} of a sheaf on the small étale site of S (0.10).
c) For G defined by a quasi-coherent $\mathcal{O}_S$-module G_0.

Assertion c) reduces to (5.5.2.7).

Proposition 6.3.15. *For G satisfying (6.3.13.1), we have:* 445

(i) *The morphism Tr_f is functorial in G, and hence additive.*
(ii) *Its formation is compatible with every base change $S' \to S$.*
(iii) *If $T = \coprod_i T_i$, so that $f_* G \simeq \sum f_{i*} G$, then $\mathrm{Tr}_f = \sum \mathrm{Tr}_{f_i}$.*
(iv) *If T has constant rank n over S, the composite morphism*

$$G \longrightarrow f_* f^* G \xrightarrow{\mathrm{Tr}_f} G$$

is multiplication by n.
(v) *The trace morphism is compatible with composition* (cf. 6.2.3 Var 3).

Assertions (i) and (ii) are clear, and (iii) is an easy consequence of 6.3.5. With the notation of 6.3.13, if $u \in G(S)$, then u_n is n times the pullback of u, so that $\mathrm{Tr}_f(f^* u) = t^*(u_n) = nu$.

To prove (v), we reduce, by étale localization on S and using (iii), to proving that for

$$f = gh : S'' \xrightarrow{h} S' \xrightarrow{g} S,$$

with g and h finite locally free of *constant ranks* n and m, the diagram

$$\begin{tikzcd} G(S'') \arrow[rr, "\mathrm{Tr}_h"] \arrow[dr, "\mathrm{Tr}_f"'] & & G(S') \arrow[dl, "\mathrm{Tr}_g"] \\ & G(S) & \end{tikzcd} \tag{6.3.15.1}$$

[71]The morphism u_* is defined because we restrict ourselves to divisors *finite* over S. We take, for example, $u_*(D) = \delta(u_* \mathcal{O}_D)$ as the definition, a formula that agrees with EGA IV 21.5.5 when u is finite.

is commutative. For $u \in G(S'')$, consider the diagram

$$\begin{tikzcd}
\mathrm{Sym}^n_S(S') \arrow[rr, "\mathrm{Sym}^n(t)"] \arrow[dr, "\mathrm{Tr}_h(u)_n"] & & \mathrm{Sym}^n_S\,\mathrm{Sym}^m_{S'}(S'') \arrow[dl, "(u_m)_n"'] \arrow[dd] \\
& G & \\
S \arrow[uu, "t"] \arrow[ur, shift left, "\mathrm{Tr}_g\,\mathrm{Tr}_h(u)"] \arrow[ur, shift right, "\mathrm{Tr}_{gh}(u)"'] \arrow[rr, "t"] & & \mathrm{Sym}^{nm}_S(S''). \arrow[ul, "u_{nm}"']
\end{tikzcd} \tag{6.3.15.2}$$

446 The right vertical arrow is defined because S'' is flat over S; every triangle is commutative, and (v) will therefore be a consequence of the commutativity of the outer boundary.

Translating into ring-theoretic terms, this follows from the special case $B = E$ and $C = F$ of 6.3.3.

6.3.16. Now let $f : T \to S$ be a quasi-finite flat morphism of finite presentation. If $\mathscr{F}$ is an fppf sheaf on T, we define its direct image with proper supports $f_!\mathscr{F}$ on S as the presheaf that assigns to S'/S the set of sections of $\mathscr{F}$ on T' whose support is proper over S'. This presheaf is a sheaf by 6.1.1. With the notation of 6.1.1, it may also be expressed as a local inductive limit (V 8.2):

$$f_!\mathscr{F}(U) = \varinjlim_{V \in \Phi(U)} f_*(\mathscr{F}|V),$$

447 a formula that accounts for the fairly evident fact that the properties of $f_!$ reduce to those of f'_* when f' is finite. We thus obtain:

Proposition 6.3.17. *There is a unique way to define, for every $f : T \to S$ separated quasi-finite flat morphism of finite presentation and every fppf sheaf G on S satisfying (6.3.13.1), a trace morphism*

$$\mathrm{Tr}_f : f_!f^*G \longrightarrow G$$

satisfying:

(i) *For finite f, one recovers the morphism (6.3.13.2).*
(ii) *Conditions* (Var 1) *through* (Var 4) *of 6.2.3 are satisfied, mutatis mutandis.*

The quasi-finite morphism f induces a commutative diagram (cf. (0.11)) of sites (in which α_* is the "restriction" functor)

$$\begin{tikzcd}
T_{\mathrm{fppf}} \arrow[r, "\alpha"] \arrow[d, "f"] & T_{\mathrm{et}} \arrow[d, "f_{\mathrm{et}}"] \\
S_{\mathrm{fppf}} \arrow[r, "\alpha"] & S_{\mathrm{et}}.
\end{tikzcd}$$

For G on T_{fppf}, we have

$$f_!\alpha_* \simeq \alpha_* f_!.$$

448 We also have a "base-change" morphism

$$c : f^*_{\mathrm{et}}\alpha_* \longrightarrow \alpha_* f^*.$$

We leave it to the reader to verify the following compatibility between the trace morphisms 6.2.3 and 6.3.17. For G on S_{fppf} satisfying (6.3.13.1), the following diagram is commutative:

(6.3.17.1)
$$\begin{tikzcd} f_{\mathrm{et}!}f_{\mathrm{et}}^{*}\alpha_{*}G \arrow[r, "f_!c"] \arrow[d, "\mathrm{Tr}_f"', "(6.2.3)"] & f_{\mathrm{et}!}\alpha_{*}f^{*}G \arrow[r, leftrightarrow, "\sim"] & \alpha_{*}f_{!}f^{*}G \arrow[d, "\alpha_{*}\mathrm{Tr}_f"', "(6.3.17)"] \\ \alpha_{*}G \arrow[rr, equal] & & \alpha_{*}G. \end{tikzcd}$$

In more concrete terms, this means that if G is an fppf sheaf on S whose restriction to the étale site defines an étale sheaf $\alpha_{*}G$, and if u is a properly supported section of the inverse image $f_{\mathrm{et}}^{*}(\alpha_{*}G)$, with image u_1 in $f^{*}G$, then the elements $\mathrm{Tr}_f(u)$ (in the sense of 6.2.3) and $\mathrm{Tr}_f(u_1)$ (in the sense of 6.3.17) agree in $G(S)$.

Example 6.3.18. Let $f : S' \to S$ be a finite locally free morphism of rank n and let $u \in \Gamma(S', \mathscr{O}_{S'}) = \mathrm{Hom}_S(S', \mathbf{E}^1_S)$.[72] The morphism u defines a morphism

$$\mathrm{Sym}(u) : \mathrm{Sym}^n_S(S') \longrightarrow \mathrm{Sym}^n_S(\mathbf{E}^1_S) = \mathrm{Spec}(S[\sigma_1 \dots \sigma_n],$$

where the σ_i are the elementary symmetric functions (cf. Bourbaki, *Alg.* 5, App. I no. 2[73]). We then have (by an easy calculation left to the reader):

(6.3.18.1)
$$[\mathrm{Sym}(u) \circ t]^{*}(\sigma_i) = \sigma_i(u),$$

where $t \in \Gamma\,(\mathrm{Sym}^n_S(S')/S)$ is the canonical element (6.3.4.2), and where $\sigma_i(u)$ denotes 449
the i^{th} coefficient of the characteristic polynomial of multiplication by u on $f_{*}\mathscr{O}_{S'}$. It follows that the "trace" maps

$$\begin{cases} \mathbf{G}_a(S') \longrightarrow \mathbf{G}_a(S) \\ \mathbf{G}_m(S') \longrightarrow \mathbf{G}_m(S) \end{cases}$$

are respectively the usual *trace* and *norm*; indeed, the morphism $\mathrm{Sym}^n_S(\mathbf{E}^1_S) \to \mathbf{E}^1_S$ induced by the additive law (resp. multiplicative law) on $\mathbf{E}^1_S$ is given by the function σ_1 (resp. σ_n) on $\mathrm{Sym}^n_S(\mathbf{E}^1_S) \simeq \mathrm{Spec}(S[\sigma_1, \dots, \sigma_n])$.

It then follows from (6.3.17.1) that, when n is invertible on S, the diagram of étale sheaves

(6.3.18.2)
$$\begin{tikzcd} 0 \arrow[r] & f_{*}\mu_n \arrow[r] \arrow[d, "\mathrm{Tr}_f"', "(6.2.3)"] & f_{*}\mathbf{G}_m \arrow[r] \arrow[d, "N_f"] & f_{*}\mathbf{G}_m \arrow[r] \arrow[d, "N_f"] & 0 \\ 0 \arrow[r] & \mu_n \arrow[r] & \mathbf{G}_m \arrow[r] & \mathbf{G}_m \arrow[r] & 0 \end{tikzcd}$$

is commutative.

Proposition 6.3.19. *Consider a commutative diagram*

$$\begin{tikzcd} X \arrow[rr, "u"] \arrow[dr, "p"'] & & Y \arrow[dl, "q"] \\ & S & \end{tikzcd}$$

in which X and Y are smooth quasi-projective curves over S and u is a quasi-finite mor- 450
phism.

[72] N.D.E.: Here $\mathbf{E}^1_S$ denotes the affine line over S.

[73] N.D.E.: reference to be checked. Probably IV § 6 no. 1 in the new version.

(i) *If D is a divisor on X finite over S (see note* p. 735*) and g is a section of G_Y on Y, then*

$$\mathrm{Tr}_{D/S}(u^* g) = \mathrm{Tr}_{(u_* D)/S}(g).$$

(ii) *If E is a divisor on Y finite over S, if g is a section of G_X on X, and if u is finite, then*

$$\mathrm{Tr}_{E/S}(\mathrm{Tr}_{X/Y}(g)) = \mathrm{Tr}_{(u^* E)/S}(g).$$

Proof. (i) Let g_n be the section of G on $\mathrm{Sym}^n_S(Y)$ obtained from g (6.3.13). By 6.3.12, we have

$$\mathrm{Tr}_{D/S}(u^* g) \overset{\text{dfn}}{=} \sigma(D)^*(u^* g)_n = \sigma(D)^*(\mathrm{Sym}^n_S(u)^* g_n) = (\mathrm{Sym}^n_S(u) \circ \sigma(D))^*(g_n)$$
$$= \sigma(u_* D)^*(g_n) = \mathrm{Tr}_{u_* D}(g_n).$$

(ii) Formula (ii) follows from 6.3.15 (ii) and (v):

$$\mathrm{Tr}_{E/S}(\mathrm{Tr}_{X/Y}(g)) = \mathrm{Tr}_{E/S}(\mathrm{Tr}_{u^* E/E}(g)) = \mathrm{Tr}_{u^* E/S}(g).$$

Application 3. Trace of a torsor.

6.3.20. Let G be a sheaf on the large fppf site of S that satisfies (6.3.13.1). Let $u : X \to Y$ be a finite locally free morphism of S-schemes, and let K be a G_X-torsor on X. Suppose that, locally for the fppf topology on Y, the torsor K on X is trivial. This is the case if K is locally trivial for the étale topology.

451 Proposition 6.3.21. *Under the hypotheses of 6.3.20, there exists, up to unique isomorphism, exactly one torsor $\mathrm{Tr}_u(K)$ under G_Y on Y, equipped with a morphism Tr_u or $\mathrm{Tr}^K_u : u_* K \to \mathrm{Tr}_u(K)$, which, for g a local section (in the fppf sense) of G on Y and k a local section of $u_* K$, satisfies*

$$\mathrm{Tr}^K_u(gk) = \mathrm{Tr}_u(g) \cdot \mathrm{Tr}^K_u(k).$$

The problem is local on Y; we may therefore assume that K is trivial: $K = G_X$. One solution to the problem is to take $K = G_Y$ and as trace morphism the morphism (6.3.13.2). One immediately verifies that it is the only one.

The trace functor 6.3.21 is defined in particular when G satisfies the following condition

For every S-scheme X, every torsor under G_X on X is locally trivial for the étale topology. This condition is satisfied when G is a smooth group, when G is the inverse image of a sheaf on the small étale site of S, or when G is defined by a quasi-coherent sheaf on S.

By construction one has

(6.3.21.2) $$\mathrm{Tr}_u(K) = u_*(K) \times_{u_* G_X} G_Y.$$

6.3.22. To define a trace functor under more general hypotheses, the key point is, for $f : X \to T$ a quasi-projective, flat morphism of finite presentation of S-schemes, to define a reasonable functor $K \mapsto K_n$ from the G_X-torsors on X to the G_Z-torsors on
452 $Z = \mathrm{Sym}^n_T(X)$. We first have the following lemma, where f_n denotes the projection from $\mathrm{Sym}^n_T(X)$ to T; it is proved in the same way as 6.3.21:

Lemma 6.3.22.1. *With the preceding notation, and if, locally on T, the torsor K is trivial, then, up to unique isomorphism, there exists a unique torsor K_n on $\mathrm{Sym}^n_T(X)$, equipped with a morphism $\alpha : f_* K \to f_{n*}(K_n)$, such that for g a local* (fppf) *section of $f_* G_X$ and k a local section of $f_* K$, one has $\alpha(gk) = g_n \cdot \alpha(k)$ (notation of 6.3.13 for g_n; the definition of g_n here uses the flatness hypothesis etc.).*

The general case will be handled by a somewhat contrived descent argument. First of all:

Proposition 6.3.23. *Consider a commutative diagram*

(6.3.23.1)
$$\begin{tikzcd} X \arrow[rr, "u"] \arrow[dr] & & Y \arrow[dl] \\ & T & \end{tikzcd}$$

of flat, finitely presented, quasi-projective T-schemes, with u étale and surjective. Then the semisimplicial system of schemes $\mathrm{Sym}^n_T((X/Y)^{k+1})$ is a hypercover for the étale topology of the scheme $\mathrm{Sym}^n_T(Y)$.

Proof. a) We shall use the following auxiliary result.

. *Let (X, Y, T) be as above, let $t \in T$, and let Z be a T-scheme that is the disjoint union of finitely many T-schemes $\mathrm{Spec}(k_i)$, where k_i is a finite extension of k_t,*[74] *and consider a commutative diagram of T-schemes* 453

(6.3.23.3)
$$\begin{tikzcd} & Z \arrow[dl, hook', "x"'] \arrow[dr, hook, "y"] & \\ X_t \arrow[rr, "u"] & & Y_t, \end{tikzcd}$$

such that x and y are closed immersions. Let $z \in \mathrm{Sym}^n_T(Z)$, and let $x(z)$ and $y(z)$ be its images in $\mathrm{Sym}^n_T(X)$ and $\mathrm{Sym}^n_T(Y)$. Then $\mathrm{Sym}^n_T(u)$ is étale at $x(z)$.

The formation of the scheme $\mathrm{Sym}^n_T(X)$ is compatible with every base change $T' \to T$ (5.5.2.7). By reduction to the noetherian case, it follows that $\mathrm{Sym}^n_T(X)$ is of finite presentation over T; moreover, it is flat over T (5.5.2.4). By the fiberwise flatness criterion, to prove (6.3.23.2) we reduce to the case where T is the spectrum of a field. The completion of $\mathrm{Sym}^n_T(X)$ at z then has as its affine algebra the inverse limit of the affine algebras localized at $x(z)$ of the schemes $\mathrm{Sym}^n_T(Z_k)$, where Z_k ranges over the infinitesimal neighborhoods of Z in X. By hypothesis, u induces isomorphisms between the infinitesimal neighborhoods of Z in X and those in Y. The completion of $\mathrm{Sym}^n_T(u)$ at $x(z)$ and $y(z)$ is therefore an isomorphism, and this proves (6.3.23.2).

It follows from 6.3.23.2 that the scheme given by the disjoint union, over $z \in \mathrm{Sym}^n_T(Z)$, of the henselizations at $y(z)$ of $\mathrm{Sym}^n_T(Y)$ depends only on Z and on the T-scheme X^h_Z that is the disjoint union of the henselizations of X at the various points of Z. We shall denote this scheme by $\mathrm{Sym}^{h\,n}_T(X^h_Z)$; it is functorial in X^h_Z.

b) By passing to the limit, to prove 6.3.23, it suffices to show that for $y \in \mathrm{Sym}^n_T(Y)$, closed in its fiber, the pullback to the henselization of $\mathrm{Sym}^n_T(Y)$ at that point of the simplicial scheme $\mathrm{Sym}^n_T((X/Y)^{k+1})$ is a hypercover for the étale topology. This problem is local on T for the étale topology; we thereby reduce to the case where there exists a diagram (6.3.23.3) with $y = x(z)$ (after killing the residue-field extensions by a separable extension of k_t). The semisimplicial system under consideration is then obtained from the system 454

$$\mathrm{Sym}^{h\,n}_T((X^h_{u^{-1}(Z)}/Y^h_Z)^{k+1}), \quad \text{augmented to} \quad \mathrm{Sym}^{h\,n}_T(Y^h_Z)$$

[74] N.D.E.: Here k_t denotes the residue field of T at t.

by base change, so it suffices to consider the latter. The semisimplicial system of $(X^h_{u^{-1}(Z)}/Y^h_Z)^{k+1}$ is semisimplicially homotopic to the constant system Y^h_Z, because $X^h_{u^{-1}(Z)}/Y^h_Z$ admits a section. The system $\mathrm{Sym}^{h\,n}_T((X^h_{u^{-1}(Z)}/Y^h_Z)^{h+1})$ is therefore likewise homotopic to $\mathrm{Sym}^{h\,n}_T(Y^h_Z)$, and in particular is a hypercover of it. This proves 6.3.23.

6.3.24. Let $f : Y \to T$ be a flat, quasi-projective morphism of finite presentation between S-schemes, and let K be a G_Y-torsor on Y. Consider the following problem:

. Define, for every morphism $u : X \to Y$, with X flat, quasi-projective, and of finite presentation over T, a torsor $(u^*K)_n$ on $\mathrm{Sym}^n_T(X)$, and for every $v : X' \to X$ an isomorphism $c_v : (v^*u^*K)_n \simeq \mathrm{Sym}^n_T(v)^*(u^K)_n$, such that

455 (i) the isomorphisms c_v are compatible with composition;
(ii) when u^*K is locally trivial on T, one recovers the torsor of 6.3.22.1 and its functoriality.

By 6.3.23, the existence and uniqueness of a solution to 6.3.24.0, for every torsor obtained from K by an inverse image of the type considered in 6.3.24.0, is a local question for the étale topology on Y. In particular, if G satisfies (6.3.21.1), the problem 6.3.24.0 has a unique solution.

Now suppose that G satisfies:

(6.3.24.1). *For every integer n and every diagram of affine S-schemes*

$$\begin{tikzcd} X \arrow[rr, "u"] \arrow[dr, "f"'] & & Y \arrow[dl, "g"] \\ & T & \end{tikzcd},$$

with f and g flat and of finite presentation and u finite, locally free, and surjective, the truncated semisimplicial scheme $(\mathrm{Sym}^n_T((X/Y)^{k+1}))_{0\le k\le 2}$, augmented to $\mathrm{Sym}^n_T(Y)$, is a descent diagram (Giraud [3]) *for the category of G-torsors.*

6.3.24.1 bis. It follows in the standard way from 6.3.23 that condition (6.3.24.0) implies the analogous condition in which one assumes only that f and g are flat, quasi-projective, and of finite presentation, and that u is flat and surjective. Indeed, locally for the étale topology, u admits finite locally free quasi-sections, and one concludes by 6.3.23 and Giraud [3], applied to a diagram forming a square of descent data.

456 Given a G_Y-torsor K on a T-scheme Y that is flat, quasi-projective, and of finite presentation, there exists by definition a faithfully flat morphism of finite presentation $u : X \to Y$ such that u^*K is trivial.[75] If u_i denotes the projection from $(X/Y)^{i+1}$ to Y ($i \ge 0$), the torsors $(u_i^*K)_n$ (6.3.22.1) on the schemes $\mathrm{Sym}^n_T((X/Y)^{i+1})$ form a descent datum ($i = 0, 1, 2$) on this semisimplicial scheme augmented to $\mathrm{Sym}^n_T(Y)$. Suppose that G satisfies the condition:

(6.3.24.2). *The descent data constructed above are all effective* (cf. 6.3.24.0).

Then we denote by K_n the torsor on $\mathrm{Sym}^n_T(Y)$ (determined up to unique isomorphism) defined by this descent datum.

6.3.25. Suppose that G satisfies (6.3.21.1) or conditions (6.3.24.0) and (456). We leave it to the reader to make sense of the following formalism and to verify it:

[75] N.D.E.: One may assume that T is affine. Then there exists u as stated, which may even be assumed affine so that the preceding paragraph can subsequently be applied.

. *Base change.* Let

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ T' \arrow[r, "g"'] & T. \end{tikzcd}$$

be a commutative square. Denote by g'_n the arrow from $\mathrm{Sym}^n_{T'}(X')$ to $\mathrm{Sym}^n_T(X)$. For X a torsor under G on X, one has

$$g'^*_n K_n \simeq (g'^* K)_n.$$

Now let $f : X \to T$ be a morphism, and let K be a G_X-torsor on X. 457
. *Change of group.* Let $r : G_1 \to G_2$ be a morphism between groups both satisfying (6.3.21.1) or (6.3.24.0), (456). One has

$$r(K_n) \simeq r(K)_n.$$

. *Additivity in K.* One has

$$(K' + K'')_n \simeq K'_n + K''_n.$$

. *Additivity in n.* Let n_1 and n_2 be two integers, $n = n_1 + n_2$, and let

$$\alpha : \mathrm{Sym}_T{}^{n_1}(X) \times_T \mathrm{Sym}_T{}^{n_2}(X) \longrightarrow \mathrm{Sym}^n_T(X)$$

be the canonical arrow; then

$$\alpha^* K_n \simeq \mathrm{pr}_1^* K_{n_1} + \mathrm{pr}_2^* K_{n_2}.$$

. *Multiplicativity in n.* Let n and m be two integers, and let α be the canonical arrow from $\mathrm{Sym}^n_T(\mathrm{Sym}^m_T(X))$ to $\mathrm{Sym}^{nm}_T(X)$ (cf. 6.3.3). One has

$$\alpha^*(K_{nm}) \simeq (K_m)_n.$$

6.3.26. Suppose that G satisfies (6.3.21.1) or (6.3.24.0), (456). If $f : X \to T$ is a finite locally free morphism of rank n between S-schemes, if t is the section (6.3.4.3) of $\mathrm{Sym}^n_T(X)$, and if K is a G_X-torsor on X, we define a G_T-torsor $\mathrm{Tr}_f(K)$ on T by the formula

$$\mathrm{Tr}_f(K) = t^*(K_n). \tag{6.3.26.1}$$

If f is locally free, we define $\mathrm{Tr}_f(K)$ locally on T by the same formula. This definition, via a canonical and functorial isomorphism, is equivalent to the definition 6.3.21 on their common domain of validity (the definition 6.3.21 providing a solution to the descent problem posed). It follows immediately from (6.3.25.1), (6.3.25.2), and (6.3.25.3) that the functor Tr_f is *compatible with base changes $T' \to T$, compatible with changes of group $r : G_1 \to G_2$, and compatible with addition of torsors.* 458

The commutativity of the outer boundary of diagram (6.3.15.2), together with (6.3.25.1) and (6.3.25.4), yields *compatibility with composition of morphisms*:

$$\mathrm{Tr}_{fg}(K) \simeq \mathrm{Tr}_f(\mathrm{Tr}_g(K)).$$

6.3.27. In the particular case of relative curves, one also has:

Construction 6.3.27.1. *If $f : X \to S$ is a smooth quasi-projective curve over S, D_1 and D_2 are two relative divisors finite over S, and K is a G-torsor on X, then, setting $\mathrm{Tr}_{D/S}(K) = \mathrm{Tr}_{D/S}(K|D)$, there is a canonical isomorphism*

$$\mathrm{Tr}_{D_1+D_2}(K) \simeq \mathrm{Tr}_{D_1}(K) + \mathrm{Tr}_{D_2}(K).$$

This isomorphism follows immediately from the isomorphism (6.3.25.4) and 6.3.9.1. Finally, the arguments of 6.3.19 also give:

Construction 6.3.27.2. *Consider a commutative diagram*

$$\begin{tikzcd} X \arrow[rr, "u"] \arrow[dr, "p"'] & & Y \arrow[dl, "q"] \\ & S & \end{tikzcd},$$

459 *in which X and Y are smooth quasi-projective curves over S and u is a quasi-finite morphism.*

(i) *If D is a divisor on X finite over S and K is a torsor under G on Y, one has* (cf. note p. 735)*:*
$$\mathrm{Tr}_{D/S}(u^*K) \simeq \mathrm{Tr}_{(u_*D)/S}(K).$$
(ii) *If E is a divisor on Y finite over S, K is a torsor under G on X, and u is finite, one has*
$$\mathrm{Tr}_{E/S}(\mathrm{Tr}_{X/Y}(K)) \simeq \mathrm{Tr}_{u^*E/S}(K).$$

Proposition 6.3.28. *Let A and B be two (commutative) R-algebras and let $u : A \to B$ be a homomorphism making B a faithful, finite-type, locally free A-module. Then the co-semisimplicial complex of algebras $(TS^n_R(\bigotimes_A^{k+1} B))_{k\geq 0}$ is, as a differential co-semisimplicial $TS^n_R(A)$-module, a split resolution of $TS^n_R(A)$.*

Proof. The complex of modules associated with the co-semisimplicial module $TS^n_R(\bigotimes_A^{k+1} B)$ depends only on the A-module B, equipped with a module homomorphism $u : A \to B$. As an A-module, B can be written $B = A \oplus L$, and this makes it possible to write down an explicit homotopy operator.

Recall that:

Proposition 6.3.29. *Let A be a ring, let B^* be a co-semisimplicial A-algebra, and let $c : A \to B^*$ be the structure morphism. For the inverse-image functor on modules (= extension of scalars) ϵ^* to be faithful (resp. fully faithful), it suffices that the sequence of A-modules*

$$0 \longrightarrow A \longrightarrow B^0$$

460 *(resp. $0 \to A \to B^0 \xrightarrow{j_0 - j_1} B^1$) be exact, and remain so after tensoring with any A-module.*

The proof is immediate and is left to the reader.
It follows in the standard way from (6.3.28) and (6.3.29) that

Corollary 6.3.30. *Condition (6.3.24.0) is satisfied by every affine group G.*

(Use the fact that a torsor under G is representable by a scheme affine over T.)

It should not be difficult to extend 6.3.30 to every group G representable by an algebraic space [75]. I conjecture that condition (456) is satisfied by every group representable by a flat algebraic space. We have the tongue-in-cheek statement:

PROPOSITION 6.3.31. *Let the following sequence be exact on S:*

$$0 \to G \xrightarrow{r} G' \xrightarrow{u} H \to 0.$$

Suppose that G' and H, and hence G, satisfy (6.3.13.1), that G' satisfies (6.3.21.1), and that G satisfies (6.3.24.0). Then G satisfies (456).

This applies, for example, to an affine group that is the kernel of an epimorphism of representable smooth groups.

PROOF OF 6.3.31. Let K be a G-torsor on X. By 6.3.23, the question is local on X 461
for the étale topology; it therefore suffices to treat the case where $r(K)$ is trivial. The G-torsors K such that $r(K)$ is trivial are identified with trivial G'-torsors K' equipped with an isomorphism $k : H \xrightarrow{\sim} u(K')$. Put $Z = \mathrm{Sym}^n_T(X)$. The morphism of trivial torsors k induces $k_n : H_Z \xrightarrow{\sim} u(K'_n)$, and the corresponding G-torsor on $\mathrm{Sym}^n_T(X)$ represents the descent datum under consideration.

7. Appendix

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7.0. Preliminaries. In this appendix we shall use the results of VBIS, whose notation we retain.

7.0.1. Let $\mathrm{Sch}_!$ be the category whose objects are quasi-compact, quasi-separated schemes and whose morphisms are separated morphisms of finite type. We denote by $\mathscr{E}$ the bifibred category over $\mathrm{Sch}_!$, with fibre categories opposite to topoi, defined by étale sheaves on schemes (Vbis 4.3.0). Fix once and for all an object R of $\mathrm{Sch}_!$, and denote by $\mathscr{E}_R$ the bifibred category over $\mathrm{Sch}_{!R}$ deduced from $\mathscr{E}$. Finally, let $\mathscr{A}$ be a ring in the topos $\underline{\Gamma}(\mathscr{E}^\circ_R)$ satisfying the following conditions:

(i) $\mathscr{A}$ is a *cocartesian* section of $\mathscr{E}^\circ_R$ over $\mathrm{Sch}^\circ_{!R}$.

(ii) For every scheme X in $\mathrm{Sch}_{!R}$, $\mathscr{A}_X$ is a *torsion* sheaf.

With this understood, we shall work with the category $\mathrm{Mod}(\underline{\Gamma}(\mathscr{E}^\circ_R), \mathscr{A})$ (1.3.5).

Our aim is to define, for every morphism $f : X \to Y$ in $\mathrm{Sch}_{!R}$, a functor $Rf_! : D(X, \mathscr{A}_X) \to D(Y, \mathscr{A}_Y)$, agreeing with the previously defined functor when f is compactifiable, and making the categories $D(X, \mathscr{A}_X)$, for varying X, into a cofibred category over $\mathrm{Sch}_{!R}$.[76]

7.0.2. Although all the results of this appendix are stated in the setting specified in 7.0.1, they could be transcribed, with obvious modifications, into the following setting: take as a section of $\underline{\Gamma}(\mathscr{E}^\circ_R)$ the constant section whose value is the constant sheaf $\mathbf{Z}$, and, for every scheme X, restrict to the derived category $D^*_{\mathrm{tors}}(X)$ defined by complexes of sheaves of abelian groups whose cohomology sheaves are torsion sheaves (4.3.1).

LEMMA 7.0.3. *Let there be given a commutative diagram of schemes* 463

[75] Algebraic Spaces by M. ARTIN. Yale University, Whittemore Lectures, May 1969.

[76] N.D.E.: Every morphism in $\mathrm{Sch}_!$ is compactifiable by Nagata's compactification theorem (see la N.D.E. (43) page 639). This appendix shows how to avoid using the compactification theorem in the definition of $Rf_!$ by relying on Chow's lemma (7.0.6).

(7.0.3.1)
$$\begin{tikzcd} X' \arrow[r, "j'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "j"'] & Y. \end{tikzcd}$$

where f and f' are proper and j and j' are immersions. Then, for every complex $K^{\cdot}$ in $D^{+}_{\mathrm{tors}}(X')$, the canonical arrow[77]

$$j_! \circ R^+ f'_*(K^{\cdot}) \longrightarrow R^+ f_* \circ j_!(K^{\cdot})$$

is an isomorphism.

By the "lemma on way-out functors" ([6] I 7.1), it suffices to show that, for every integer i and every torsion sheaf F, the arrow

(7.0.3.2)
$$j_! \circ R^i f'_*(F) \longrightarrow R^i f_* \circ j_!(F)$$

is an isomorphism.

By replacing j and j' with their closed images, we are reduced to the case where j and j' are open immersions with everywhere dense images. The properness of f and f' then implies that diagram (7.0.3.1) is cartesian. Under these conditions, the proper base-change theorem, in the form XII 5.2, shows that, for every geometric point ξ of Y, the arrow $(j_! \circ R^i f'_*(F))_\xi \to (R^i f_* \circ j_!(F))_\xi$ is an isomorphism.

Construction 7.0.4. Let D be a category, and let X and Y be two D-objects in $\mathrm{Sch}_{!R}$ such that, for every arrow $\alpha \to \beta$ in D, the morphisms $X_\beta \to X_\alpha$ and $Y_\beta \to Y_\alpha$ are proper. Let $j : X \to Y$ be a morphism such that, for every object α of D, $j_\alpha : X_\alpha \to Y_\alpha$ is an immersion. The isomorphism (7.0.3.2) for $i = 0$ satisfies the usual compatibility conditions and allows one to construct an *exact* functor:

$$j_! : \mathrm{Mod}(\underline{\Gamma}(\overline{X}), \mathscr{A}) \longrightarrow \mathrm{Mod}(\underline{\Gamma}(\overline{Y}), \mathscr{A})$$

464 (cf. 1.2.5 for the notation) which induces a triangulated functor, again denoted $j_!$:

$$j_! : D(\underline{\Gamma}(\overline{X}), \mathscr{A}) \longrightarrow D(\underline{\Gamma}(\overline{Y}, \mathscr{A}))$$

Then, for every object α of D, there is a commutative diagram, where $j_{\alpha!}$ denotes the usual "extension by 0" functor:

$$\begin{tikzcd} D(\underline{\Gamma}(\overline{X}), \mathscr{A}) \arrow[r, "j_!"] \arrow[d, "Re^*_\alpha"'] & D(\underline{\Gamma}(\overline{Y}), \mathscr{A}) \arrow[d, "Re^*_\alpha"] \\ D(X_\alpha, \mathscr{A}_{X_\alpha}) \arrow[r, "j_{\alpha!}"'] & D(Y_\alpha, \mathscr{A}_{Y_\alpha}). \end{tikzcd}$$

Construction 7.0.5. *Let there be given a commutative diagram in* $\mathrm{Sch}_{!R}$

(7.0.5.1)
$$\begin{tikzcd} X' \arrow[r, "i'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ Y' \arrow[r, "i"] & Y \end{tikzcd}$$

[77] N.D.E.: The definition of this arrow is similar to that of (5.1.5.2), up to an easy dévissage.

where i and i' are immersions, f is a proper morphism, and f' is a proper surjective *morphism. It defines a commutative diagram of simplicial objects in* $\mathrm{Sch}_{!R}$

$$\begin{tikzcd} {[X'|Y']} \arrow[r, "j"] \arrow[d, "u_{f'}"'] & {[X|Y]} \arrow[d, "u_f"] \\ C^{\Delta}_{Y'} \arrow[r, "i"'] & C^{\Delta}_{Y} \end{tikzcd} \tag{7.0.5.2}$$

where $j_n : [X'|Y']_n \to [X|Y]_n$ is an immersion for every n. With this notation, there is a canonical isomorphism

$$i_! \xrightarrow{\sim} R^+\overline{u}_{f*} \circ j_! \circ L^+\overline{u}^*_{f'}.$$

By replacing i and i' with their closed images, we may suppose that i and i' are open immersions with everywhere dense images. Diagram (7.0.5.1) is then cartesian and f is *surjective.* Moreover, for every integer n, j_n is an open immersion and the diagram 465

$$\begin{tikzcd} {[X'|Y']_n} \arrow[r, "j_n"] \arrow[d] & {[X|Y]_n} \arrow[d] \\ Y' \arrow[r, "i"'] & Y \end{tikzcd}$$

is cartesian.

Consequently, $L^+\overline{u}^*_f \circ i_!$ is isomorphic to $j_! \circ L^+\overline{u}^*_{f'}$. Using the fact that u_f is a cohomological-descent augmentation (4.3.2), we obtain a sequence of isomorphisms

$$i_! \xrightarrow{\sim} R^+\overline{u}_{f*} \circ L^+(\overline{u}^*_f) \circ i_! \xrightarrow{\sim} R^+\overline{u}_{f*} \circ j_! \circ L^+\overline{u}^*_{f'}$$

which gives the desired isomorphism.[78]

Finally, we shall use the following variant of Chow's lemma, whose proof will appear in EGA II, 2nd edition,[79] and which is an improvement of (XII 7.1):

LEMMA 7.0.6. *Let S be a quasi-compact and quasi-separated scheme and let $f : X \to S$ be a separated morphism of finite type. Then there exists a commutative diagram*

$$\begin{tikzcd} X' \arrow[r, "p"] \arrow[d, "f'"'] & X \arrow[dl, "f"] \\ S & \end{tikzcd}$$

satisfying the following conditions:

(i) *p is proper and* surjective*;*
(ii) *there is a dense open subset U of X such that p induces an isomorphism from $p^{-1}(U)$ onto U;*
(iii) *f' is quasi-projective.*

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7.1. Definition of $R^+f_!$.

[78]N.D.E.: More directly, without assuming f' surjective, there is still a sequence of canonical arrows

$$i_! \longrightarrow i_! R^+\overline{u}_{f'*} L^+\overline{u}^*_{f'} \xrightarrow{\sim} R^+\overline{u}_{f*} j_! L^+u^*_{f'},$$

of which the second is an isomorphism (7.0.3). When f' is surjective, the first is also an isomorphism by cohomological descent, and the composite agrees with the isomorphism constructed in the text.

[79]N.D.E.: See [C, 2.6], cited in la N.D.E. (43) page 639.

Definition 7.1.1. *Let $f : X \to S$ be a morphism of schemes. A* pseudo-compactification *of f consists of a commutative diagram*

(7.1.1.0)
$$\begin{tikzcd} X_1 \arrow[r, "j_1"] \arrow[d, "p_1"'] & T_1 \arrow[ddl, "f_1"] \\ X \arrow[d, "f"'] & \\ S & \end{tikzcd}$$

where p_1 is proper and surjective, j_1 is an immersion, and f_1 is proper.

7.1.2. Let $f : X \to S$ be a morphism in $\mathrm{Sch}_{!R}$. By Chow's lemma (7.0.6), f has a pseudo-compactification (p_1, j_1, f_1). Diagram (7.1.1.0) defines a morphism

$$[X_1|X] \xrightarrow{j_1} [T_1|S]$$

again denoted by j_1, such that j_{1n} is an immersion for every n. We then put $R^{+(1)}f_! = R^+\overline{u}_{f_1*} \circ j_{1!} \circ L^+\overline{u}^*_{p_1}$, thereby obtaining a *triangulated* functor from $D^+(X, \mathscr{A}_X)$ to $D^+(S, \mathscr{A}_S)$.

Construction 7.1.3. *For every pair $((p_1, j_1, f_1), (p_2, j_2, f_2))$ of pseudo-compactifications of f, there is an isomorphism $\alpha_{12} : R^{+(1)}f_! \to R^{+(2)}f_!$ such that, for every other pseudo-compactification (p_3, j_3, f_3), one has $\alpha_{23} \circ \alpha_{12} = \alpha_{13}$.*

a) Define a bisimplicial system $[[X_1, X_2|X]]$ by setting

$$[[X_1, X_2|X]]_{[n]\times[m]} = [X_1|X]_n \times [X_2|X]_n$$

Similarly, set

$$[[X_1|X]]_{[n]\times[m]} = [X_1|S]_n,$$

all the morphisms $[[X_1|X]]_{[n]\times[m]} \to [[X_1|X]]_{[n]\times[m']}$ induced by morphisms $[m'] \to [m]$ being equal to the identity. Define the bisimplicial systems $[[T_1, T_2|S]]$ and $[[T_1|S]]$ in the same way, so that there is a commutative diagram

$$\begin{tikzcd} & {[[T_1, T_2|S]]} \arrow[rr] \arrow[ddrr, "u_{f_1f_2}"] & & {[[T_1|S]]} \arrow[dd, "u_{f_1}"] \\ {[[X_1, X_2|X]]} \arrow[ur, "j_{12}"] \arrow[rr] \arrow[ddrr, "u_{p_1p_2}"'] & & {[[X_1|X]]} \arrow[ur, "j_1"] \arrow[dd, "u_{p_1}"] & \\ & & & C_S^{\Delta\times\Delta} \\ & & C_X^{\Delta\times\Delta} \arrow[ur] & \end{tikzcd}$$

467 where j_{12} is an immersion degree by degree. The computational procedure described in Vbis 2.3 makes it possible to construct a morphism

$$R^{+(1)}f_! \xrightarrow{\alpha_1} R^+\overline{u_{f_1f_2}}_* \circ j_{12!} \circ L^+\overline{u_{p_1p_2}}^* = R^{+(1,2)}f_!$$

which is obtained from a morphism of triple complexes of sheaves on S.

b) α_1 *is an isomorphism*:

To prove this, one may fix the first index n coming from $\big[[T_1, T_2]\big]_{[n]\times[m]}$ in the preceding morphism of triple complexes and consider the resulting morphism of double complexes. One checks that this morphism is precisely the isomorphism defined in Lemma 7.0.5 for the diagram

$$\begin{tikzcd}
{[X_1|X]_n \times X_2} \arrow[r] \arrow[d] & {[T_1|S]_n \times T_2} \arrow[d] \\
{[X_1|X]_n} \arrow[r] & {[T_1|S]_n}
\end{tikzcd}$$

c) In the same way, construct an isomorphism

$$R^{+(2)} f_! \xrightarrow{\alpha_2} R^+ \overline{u_{f_1 f_2}}_* \circ j_{12!} \circ L^+ \overline{u_{p_1 p_2}}^*$$

and set $\alpha_{12} = \alpha_2^{-1} \circ \alpha_1$.

To verify the relation $\alpha_{23} \circ \alpha_{12} = \alpha_{13}$, one uses a trisimplicial object. We leave it to the reader to check that the resulting diagram is commutative: 468

$$\begin{tikzcd}
R^{+(2,3)} f_! \arrow[drr] & & R^{+(2)} f_! \arrow[ll] \arrow[rr] & & R^{+(1,2)} f_! \arrow[dll] \\
& & R^{+(1,2,3)} f_! & & \\
R^{+(3)} f_! \arrow[uu] \arrow[drr] & & & & R^{+(1)} f_! \arrow[uu] \arrow[dll] \\
& & R^{+(1,3)} f_! \arrow[uu] & &
\end{tikzcd}$$

Construction 7.1.4. *Let $Z \xrightarrow{g} Y$ and $Y \xrightarrow{f} X$ be two morphisms in $\mathrm{Sch}_{!R}$. For every triple $((p_1, j_1, f_1), (p_2, j_2 g_2), (p_3, j_3, (fg)_3))$, where (p_1, j_1, f_1) is a pseudo-compactification of f, (p_2, j_2, g_2) a pseudo-compactification of g, and $(p_3, j_3, (fg)_3)$ a pseudo-compactification of fg, there is an isomorphism $R^{+(1)} f_! \circ R^{+(2)} g_! \xrightarrow{\sim} R^{+(3)} (fg)_!$ compatible with the previously defined isomorphisms α_{ij}.*

We proceed in several steps.

. Let (p_1, j_1, f_1) (resp. (p_2, j_2, f_2)) be a pseudo-compactification of $f : Y \to X$ (resp. of $g : Z \to Y$):

(7.1.4.1.1)
$$\begin{tikzcd}
& & Z \arrow[d, "g"] & Z_2 \arrow[l, "p_2"'] \arrow[r, hook, "j_2"] & T_2 \arrow[dll, "g_2"] \\
T_1 \arrow[drr, "f_1"'] & Y_1 \arrow[l, hook', "j_1"'] \arrow[r, "p_1"] & Y \arrow[d, "f"] & & \\
& & X & &
\end{tikzcd}$$

The morphism $Z' = Z \times_Y Y_1 \to Y_1$ obtained from g by base change is equipped with a pseudo-compactification:

(7.1.4.1.2)
$$\begin{tikzcd}
T'_2 = T_2 \times_Y Y_1 \arrow[drr, "g'_2"'] & & Z_2 \times_Y Y_1 = Z'_2 \arrow[ll, hook', "j'_2"'] \arrow[d, "p'_2"] \\
& & Z \times_Y Y_1 = Z' \arrow[d, "g'"] \\
& & Y_1
\end{tikzcd}$$

469 Now choose a pseudo-compactification of $j_1 \circ g'_2$[80]

(7.1.4.1.3)
$$\begin{tikzcd}
T_{12} \arrow[dd, "g''_2"'] & T''_2 \arrow[l, hook', "j'_1"'] \arrow[d, "p"] \\
& T_2 \times_Y Y_1 = T'_2 \arrow[d, "g'_2"] \\
T_1 & Y_1 \arrow[l, hook', "j_1"']
\end{tikzcd}$$

which defines a pseudo-compactification of the composite fg; denote the corresponding functor by $R^{+(12)}(fg)_!$.

. *The preceding data make it possible to define an isomorphism* $R^{+(12)}(fg)_! \xleftarrow{\sim} R^{+(1)}f_! \circ R^{+(2)}g_!$:

There is a commutative diagram, where $Z''_2 = T''_2 \times_{T'_2} Z'_2$ and all the arrows are naturally deduced from (7.1.4.1.1), (7.1.4.1.2), and (7.1.4.1.3).

(7.1.4.2.1)
$$\begin{tikzcd}
{[T_{12}|X]} \arrow[dr, "g''_2"'] & & {[T''_2|Y]} \arrow[ll, "j'_1"'] \arrow[dr, "g'_2 p"] & {[Z''_2|Z]} \arrow[l, "j''_2"'] \arrow[r, "r"] \arrow[d, "l"] & C^{\Delta}_Z \arrow[d, "g"] \\
& {[T_1|X]} \arrow[drrr, "u_{f_1}"] & & {[Y_1|Y]} \arrow[ll, "j_1"'] \arrow[r, "u_{p_1}"'] & C^{\Delta}_Y \arrow[d, "f"] \\
& & & & C^{\Delta}_X
\end{tikzcd}$$

470 We shall define a natural arrow:

$$\beta : R^{+(1)}f_! \circ R^{+(2)}g_! \longrightarrow R^{+(12)}(fg)_!$$

which we shall show to be an isomorphism.

For this we shall use two lemmas:

[80]N.D.E.: In the original text a compactification was chosen, which is possible by the compactification theorem. Since the purpose of the appendix is to avoid that theorem (cf. la N.D.E. (76) page 743), the editor has taken the liberty of replacing it with a pseudo-compactification and slightly modifying 7.1.4.2. Another solution would be to restrict to pseudo-compactifications (p_1, j_1, f_1) for which f_1 is *projective*.

LEMMA 7.1.4.2.2. *Let there be given a commutative diagram:*

$$\begin{tikzcd} X_1 \arrow[r, "j_1"] \arrow[d, "p_1"'] & T_1 \arrow[d, "h'"] \\ X \arrow[r, "j"'] \arrow[d, "f"'] & T \arrow[dl, "h"] \\ S & \end{tikzcd}$$

where h and h' are proper, p_1 is proper and surjective, and j and j_1 are immersions. Then the canonical arrow

$$R^+h_* \circ j_! \longrightarrow R^{+(1)}f_!$$

is an isomorphism.

Indeed, by (7.0.5), the arrow

$$j_! \longrightarrow R^+\overline{u}_{h'*} \circ j_{1!} \circ L^+\overline{u}^*_{p_1}$$

is an isomorphism. The result follows by applying R^+h_*.

LEMMA 7.1.4.2.3. *With the notation of diagram (7.1.4.2.1), the canonical arrows*

$$R^{+(1)}f_! \circ R^+\overline{u}_{p_1*} \longrightarrow R^+\overline{u}_{f_1*} \circ j_{1!}$$
$$R^{+(2)}g_! \circ R^+\overline{r}_* \longrightarrow R^+\overline{u}_{p_1*} \circ R^+(g_2'p)_* \circ j''_{2!}$$

are isomorphisms.

We prove only the first isomorphism, leaving it to the reader to deduce the second. 471
With the evident notation and the notation of 7.1.3 a), there is a commutative diagram

$$\begin{tikzcd} & {[Y_1|Y]} \arrow[rr, "j_1"] \arrow[dd, "\overline{u}_{p_1}"] & & {[T_1|X]} \arrow[dd, "\overline{u}_{f_1}"] \\ {[[Y_1,Y_1|Y]]} \arrow[ur, "k"] \arrow[rr, "i"] \arrow[dd, "q"'] & & {[[T_1,T_1|X]]} \arrow[ur] \arrow[dd] & \\ & Y \arrow[rr, "f"] & & X \\ {[Y_1|Y]} \arrow[ur, "\overline{u}_{p_1}"] \arrow[rr, "j_1"'] & & {[T_1|X]} \arrow[ur, "\overline{u}_{f_1}"'] & \end{tikzcd}$$

where q and k are the two projections from $[[Y_1, Y_1|Y]]$ to $[Y_1|Y]$. By cohomological descent, $R^+\overline{u}_{f_1*} \circ j_{1!}$ is identified with $R^+\overline{u}_{f_1*} \circ j_{1!} \circ R^+q_* L^+q^*$, which, by 7.0.3 applied to the front and top faces of the cube above, is identified with $R^+\overline{u}_{f_1*} \circ j_{1!} \circ R^+k_* \circ L^+q^*$. By the base-change theorem, this last functor is identified with $R^+\overline{u}_{f_1*} \circ j_{1!} \circ L^+\overline{u}_{p_1*} \circ R^+\overline{u}_{p_1*}$, which by definition is $R^{+(1)}f_! \circ R^+\overline{u}_{p_1*}$.[81]

[81] N.D.E.: the proof in the original text is slightly different and uses 7.1.4.2.2.

Lemma (7.1.4.2.3) having been proved, there is a sequence of isomorphisms

$$\begin{aligned}
R^{+(12)}(fg)_! &= R^{+}\overline{u}_{f_1*} \circ R^{+}g_2'' \circ j_{1!}' \circ j_{2!}'' \circ L^{+}\overline{r}^{*} \\
&\xleftarrow{\sim} R^{+}\overline{u}_{f_1*} \circ j_{1!} \circ R^{+}(g_2'p)_* \circ j_{2!}' \circ L^{+}\overline{r}^{*} \\
&\xleftarrow{\sim} R^{+(1)}f_! \circ R^{+}\overline{u}_{p_1*} \circ R^{+}(g_2'p)_* \circ j_{2!}' \circ L^{+}\overline{r}^{*} \\
&\xleftarrow{\sim} R^{+(1)}f_! \circ R^{+(2)}g_! \circ R^{+}\overline{r}_* \circ L^{+}\overline{r}^{*} \\
&\xleftarrow{\sim} R^{+(1)}f_! \circ R^{+(2)}g_!
\end{aligned}$$

472 whose composite gives β.

. If $(p_3, j_3, (fg)_3)$ is a pseudo-compactification of fg, an isomorphism

$$R^{+(1)}f_! \circ R^{+(2)}g_! \xrightarrow{\gamma_{123}} R^{+(3)}fg_!$$

is defined by composing $\alpha_{(12)3}$ with β.

. Now let $(p_{1'}, j_{1'}, f_{1'})$ and $(p_{2'}, j_{2'}, g_{2'})$ be two other pseudo-compactifications of f and g, together with a diagram (7.1.4.1.3); one checks that the following diagram is commutative:

$$\begin{tikzcd}
R^{+(12)}(fg)_1 \arrow[d, "\alpha_{(12)(1'2')}"'] & R^{+(1)}f_1 \circ R^{+(2)}g_! \arrow[l, "\beta"'] \arrow[d, "\alpha_{11'} * \alpha_{22'}"] \\
R^{+(1'2')}(fg)_! & R^{+(1')}f_! \circ R^{+(2')}g_! \arrow[l, "\beta'"]
\end{tikzcd}$$

by considering a diagram of the type (7.1.4.2.1) in which the simplicial objects are replaced by bisimplicial objects.

It follows that the following diagram is commutative:

$$\begin{tikzcd}
R^{+(3)}(fg)_! \arrow[d, "\alpha_{33'}"'] & R^{+(1)}f_! \circ R^{+(2)}g_! \arrow[l, "\gamma_{123}"'] \arrow[d, "\alpha_{11'} * \alpha_{22'}"] \\
R^{+(3')}(fg)_! & R^{+(1')}f_! \circ R^{+(2')}g_! \arrow[l, "\gamma_{1'2'3'}"']
\end{tikzcd}$$

for every pseudo-compactification $(p_{3'}, j_{3'}, (fg)_{3'})$ of fg, which completes the proof of 7.1.4.

7.1.5. By choosing a pseudo-compactification for every morphism f in $\mathrm{Sch}_{!R}$, in such a way that the trivial pseudo-compactification is associated with an identity morphism, one defines, for every morphism $f : X \to Y$, a triangulated functor

$$R^{+}f_! : D^{+}(X, \mathscr{A}_X) \longrightarrow D^{+}(Y, \mathscr{A}_Y)$$

473 and, for every pair (f, g) of morphisms, an isomorphism

$$R^{+}fg_! \xrightarrow[\sim]{c_{fg}} R^{+}f_! \circ R^{+}g_!$$

Moreover, if f is an identity, $R^{+}f_!$ is the identity functor and c_{fg} is the identity.

Proposition 7.1.6. *The correspondence $f \to R^{+}f_!$ defines a normalized pseudofunctor from $\mathrm{Sch}_{!R}$ to the 2-category of categories. In other words, the categories $D^{+}(X, \mathscr{A}_X)$, for X an object of $\mathrm{Sch}_{!R}$, can be organized into a cofibered category over $\mathrm{Sch}_{!R}$ by setting, for every morphism $f : X \to Y$ in $\mathrm{Sch}_{!R}$,*

$$\mathrm{Hom}_f(K, L) = \mathrm{Hom}(R^{+}f_!K, L)$$

with $K \in D^+(X, \mathscr{A}_X)$ and $L \in D^+(Y, \mathscr{A}_Y)$.

It remains to see that the identification $R^+fg_! \xrightarrow[\sim]{c_{fg}} R^+f_! \circ R^+g_!$ is compatible with triple composites. Let $h : T \to Z$, $g : Z \to Y$, and $f : Y \to X$ be three morphisms in $\mathrm{Sch}_{!R}$; by what precedes, to verify this compatibility it suffices to verify it after an arbitrary choice of pseudo-compactifications for h, g, and f.

There is a diagram:

(7.1.6.1)
$$\begin{tikzcd}
T_5 \arrow[dr] & T_4 \arrow[l, hook'] \arrow[dr] & T_3 \arrow[l, hook'] \arrow[dr] & T' \arrow[l, hook'] \arrow[r, "p_3"] \arrow[d] & T \arrow[d, "h"] \\
& T_3 \arrow[dr] & T_2 \arrow[l, hook'] \arrow[dr] & Z' \arrow[l, hook'] \arrow[r, "p_2"] \arrow[d] & Z \arrow[d, "g"] \\
& & T_1 \arrow[drr] & Y_1 \arrow[l, hook'] \arrow[r, "p_1"] & Y \arrow[d, "f"] \\
& & & & X
\end{tikzcd}$$

where all the oblique arrows are proper and p_1, p_2, and p_3 are proper and surjective.

A diagram of the type (7.1.4.4) constructed from (7.1.6.1) then gives the conclusion.

Proposition 7.1.7. *If $f : X \to S$ is a compactifiable morphism, the functor $Rf_!$* 474
coincides with the one defined in Exposé XVII.

Indeed, there is a commutative diagram:

$$\begin{tikzcd}
X_1 \arrow[r, "j_1"] \arrow[d, "p_1"'] & T_1 \arrow[d, "h'"] \\
X \arrow[r, "j"] \arrow[d, "f"'] & T \arrow[dl, "h"] \\
S &
\end{tikzcd}$$

to which one applies (7.1.4.2.2).

Definition 7.1.8. *If $f : X \to Y$ is a morphism in* $\mathrm{Sch}_!$, *the functor $R^+f_!$ just defined is called the* direct-image functor with proper supports *(in the sense of derived categories). The functor $H^q \circ R^+f_!$, denoted $R^qf_!$, is called the q^{th}* direct-image functor with proper supports.

With this notation, we leave it to the reader to verify the

Construction 7.1.9. *Let (π_1, j_1, f_1) be a pseudo-compactification of $f : X \to Y$, and denote by π_1^n the canonical morphism $[X_1|X]_n \to X$. Then, for every complex $K^\cdot$ in $D^+(X, \mathscr{A}_X)$, there are two biregular spectral sequences:*

(7.1.9.1) $$E_1^{pq} = R^qf_!(K^p) \Longrightarrow R^{p+q}f_!(K^\cdot)$$

(7.1.9.2) $$E_1^{pq} = R^q(f \circ \pi_1^p)_!(\pi_1^{p*}(K^\cdot)) \Longrightarrow R^{p+q}f_!(K^\cdot)$$

(Take, for example, the canonical flabby resolution of $j_{1!} \circ L^+\overline{u}_{p_1}^*(K^\cdot)$ (cf. XVII (4.2)), and then its direct image under u_{f_1}.)

475 **7.2. Some properties of $R^+ f_!$.**

PROPOSITION 7.2.1. *Let $f : X \to Y$ be a morphism in $\mathrm{Sch}_{!R}$ of relative dimension $\leq d$. Then the functor $R^+ f_!$ has cohomological dimension $\leq 2d$.*

It is enough to show that, for every sheaf F of $\mathscr{A}_X$-modules on X, one has $R^i f_!(F) = 0$ for $i > 2d$. (It then follows by a standard argument that, for every complex $K^\cdot$ in $D^+(X, \mathscr{A}_X)$ such that $H^i(K^\cdot) = 0$ for $i > k$, one has $R^i f_!(K^\cdot) = 0$ for $i > k + 2d$.)

Let $\mathscr{U} = (U_i)_{i \in I}$ be a finite covering of X by affine open subsets. One defines (6.2.9) a left resolution $K^\cdot_{\mathscr{U}} \to F$ of F by putting

$$K^n_{\mathscr{U}} = \bigoplus_{|P| = -n+1} j_{P!} j_P^*(F)$$

where, for every subset P of I, j_P is the inclusion of $\cup_{i \in P} U_i$ into X.

From (7.1.9.1) one obtains a spectral sequence

$$E_1^{pq} = \bigoplus_{\substack{|P| = -p+1 \\ P \subset I \text{ nonempty}}} R^p(f j_P)_! j_P^*(F) \Longrightarrow R^{p+q} f_!(F)$$

and the fact that $f j_P$ is compactifiable, a case covered by Exposé XVII, shows that $E_1^{pq} = 0$ for $p > 0$ and $q > 2d$; it follows that $R^n f_!(F) = 0$ for $n > 2d$, as required.

PROPOSITION 7.2.2. *Let $f : X \to Y$ be a morphism in $\mathrm{Sch}_{!R}$ and let $K^\cdot$ be an object of $D^b(X, \mathscr{A}_X)$ of finite tor-dimension. Then $R^+ f_!(K^\cdot)$ has finite tor-dimension.*

As above, let $\mathscr{U} = (U_i)_{i \in I}$ be a finite covering of X by affine open subsets. The functoriality of $K^n_{\mathscr{U}}$ makes it possible to construct a bounded left resolution of $K^\cdot$ by complexes of finite tor-dimension, since the latter property can be checked pointwise (4.1.9). We denote the double complex so obtained by $K^{\cdot\cdot}_{\mathscr{U}}$, the *second* index being the one corresponding to $K^\cdot$. With the conventions of (Vbis 2.3.2), for every integer n there is an exact sequence, where the operation $\sigma_{\leq n}$ is taken in $C^+(C^+(\mathrm{Mod}(X, \mathscr{A}_X))$:

$$0 \longrightarrow (-1)^{n+1} K^{n+1,\cdot}_{\mathscr{U}} \longrightarrow (\sigma_{\leq n+1}(K^{\cdot\cdot}_{\mathscr{U}}))_s \longrightarrow (\sigma_{\leq n}(K^{\cdot\cdot}_{\mathscr{U}}))_s \longrightarrow 0$$

476 hence a distinguished triangle in $D^+(X, \mathscr{A}_X)$:

$$\begin{tikzcd} & K^{n+1,\cdot}_{\mathscr{U}} \arrow[dr] & \\ (\sigma_{\leq n}(K^{\cdot\cdot}_{\mathscr{U}}))_s \arrow[ur] & & (\sigma_{\leq n+1}(K^{\cdot\cdot}_{\mathscr{U}}))_s \arrow[ll] \end{tikzcd}$$

Using the fact that $R^+ f_!(K^{n+1,\cdot}_{\mathscr{U}})$ has finite tor-dimension for every n (5.2.10), it follows that $R^+ f_!((\sigma_{\leq 0}(K^{\cdot\cdot}_{\mathscr{U}}))_s) = R^+ f_!(K^\cdot)$ has finite tor-dimension.

REMARK 7.2.3. The preceding argument does not show that $R^+ f_!(K^\cdot)$ has tor-dimension $\leq k$ whenever $K^\cdot$ has tor-dimension $\leq k$ (cf. 7.3.7).

PROPOSITION 7.2.4. *Let there be given a Cartesian diagram in $\mathrm{Sch}_{!R}$:*

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"'] & X \arrow[d, "f"] \\ S' \arrow[r, "g"'] & S \end{tikzcd}$$

Then, for every complex $K^{\cdot}$ in $D^+(X, \mathscr{A}_X)$, the canonical morphism

$$g^* \circ R^+ f_!(K^{\cdot}) \xrightarrow{\mathrm{ch}_{K^{\cdot}}} R^+ f'_! \circ g'^*(K^{\cdot})$$

is an isomorphism.

Let (π_1, j_1, f_1) be a pseudo-compactification of f; by base change it gives a pseudo-compactification (π'_1, j'_1, f'_1) of f' and a commutative diagram:

$$\begin{tikzcd}
 & T'_1 \arrow[rr] \arrow[dddl, "f'_1"] & & T_1 \arrow[dddl, "f_1"] \\
X'_1 \arrow[ur, "j'_1"] \arrow[rr] \arrow[d, "\pi'_1"'] & & X_1 \arrow[ur, "j_1"'] \arrow[d, "\pi_1"] & \\
X' \arrow[rr, "g'"'] \arrow[d, "f'"'] & & X \arrow[d, "f"'] & \\
S' \arrow[rr, "g"'] & & S &
\end{tikzcd}$$

which makes it possible to define the arrow ch_K. 477

By (7.1.9.2), there is then a morphism of biregular spectral sequences:

$$\begin{tikzcd}
g^* \circ R^q(f \circ \pi_1^p)_!(\pi_1^{p*}(K^{\cdot})) \arrow[r] \arrow[d, "u_1^{pq}"] & g^* \circ R^{p+q} f_!(K^{\cdot}) \arrow[d, "\mathrm{ch}_K"] \\
R^q(f' \circ \pi_1'^p)_!(\pi_1'^{p*}(g'^*(K^{\cdot}))) \arrow[r] & R^+ f'_! \circ g'^*(K^{\cdot})
\end{tikzcd}$$

and $\mathrm{ch}_{K^{\cdot}}$ is an isomorphism because the same is true of the u_1^{pq} by (5.2.6).

7.3. **Definition of $Rf_!$.** We now propose to extend the functor $R^+ f_!$ "in a reasonable way" to a functor $Rf_! : D(X, \mathscr{A}_X) \to D(Y, \mathscr{A}_Y)$, for every morphism f in $\mathrm{Sch}_{!R}$.

7.3.1. Let (p_1, j_1, f_1) be a pseudo-compactification of f, and let F be an $\mathscr{A}_X$-module. Taking the canonical flabby resolution of $j_{1!} \circ L\bar{u}^*_{p_1}(F)$ (cf. 4.2), then its direct image under f_1, and then the simple complex associated with the double complex so obtained gives a complex of $\mathscr{A}_Y$-modules:

$$0 \longrightarrow \mathscr{X}^0(F) \xrightarrow{S^1(F)} \mathscr{X}^1(F) \ldots \mathscr{X}^n(F) \xrightarrow{S^n(F)} \mathscr{X}^{n+1}(F) \ldots . \tag{7.3.1.1}$$

functorial in F and isomorphic in $D^+(Y, \mathscr{A}_Y)$ to $R^+ f_!(F)$. 478

By 7.2.1, $H^i(\mathscr{X}^{\cdot}(F)) = 0$ for $i > 2d$, where d is an integer such that the relative dimension of f is $\leq d$; hence the complex

$$0 \longrightarrow \mathscr{X}^0(F) \longrightarrow \mathscr{X}^1(F) \longrightarrow \ldots \longrightarrow \mathscr{X}^{2d-1}(F) \longrightarrow \operatorname{Ker} S^{2d}(F) \longrightarrow 0 \tag{7.3.1.2}$$

is canonically isomorphic in $D^+(Y, \mathscr{A}_Y)$ to the complex (7.3.1.1). Moreover:

LEMMA 7.3.2. *The functors $\mathscr{X}^0, \mathscr{X}^1, \ldots, \mathscr{X}^{2d-1}$, and* Ker S^{2d} *are exact in F.*

The exactness of the functors $\mathscr{X}^n$ follows from the exactness of the canonical flabby resolution and from the fact that the direct image under a morphism of a short exact sequence of flabby sheaves is again a short exact sequence.

To prove the exactness of $\operatorname{Ker} S^{2d}$, for a short exact sequence $0 \to F' \to F \to F'' \to 0$ of sheaves on X, consider the diagram

$$\begin{tikzcd}
\operatorname{Ker} S^{2d}(F') \arrow[r] \arrow[d] & \mathscr{X}^{2d+1}(F') \arrow[r] \arrow[d] & \mathscr{X}^{2d+2}(F') \dots \arrow[d] \\
\ker S^{2d}(F) \arrow[r] \arrow[d] & \mathscr{X}^{2d+1}(F) \arrow[r] \arrow[d] & \mathscr{X}^{2d+2}(F) \dots \arrow[d] \\
\ker S^{2d}(F'') \arrow[r] & \mathscr{X}^{2d+1}(F'') \arrow[r] & \mathscr{X}^{2d+2}(F'') \dots
\end{tikzcd}$$

and conclude by using the homology exact sequence associated with the exact sequence

$$0 \longrightarrow \sigma_{\geq 2d+1}(\mathscr{X}^{\cdot}(F')) \longrightarrow \sigma_{\geq 2d+1}(\mathscr{X}^{\cdot}(F)) \longrightarrow \sigma_{\geq 2d+1}(\mathscr{X}^{\cdot}(F'')) \longrightarrow 0.$$

We now have the following general lemma:

Lemma 7.3.3. *Let A and B be two abelian categories, and let* $\mathrm{Je}^{\cdot}$ *be an object of* $C^b(\mathscr{F}ex(A, B))$, *where* $\mathscr{F}ex(A, B)$ *denotes the category of exact functors from A to B. Associating with every complex $K^{\cdot}$ in $C(A)$ the simple complex associated with the double complex* $\mathrm{Je}^{\cdot}(K^{\cdot})$ *gives a triangulated functor $K(A) \to K(B)$ that preserves quasi-isomorphisms.*

479 The functor in question may be written as a composite

$$C(A) \longrightarrow C^b(C(B)) \xrightarrow{(\ \ )_s} C(B),$$

where $(\ \ )_s$ is the associated-simple-complex functor, and one uses (Vbis 2.3.2.2. Remark); the details are left to the reader.

Exercise 7.3.4. Show that, for every complex $K^{\cdot}$ in $D^+(X, \mathscr{A}_X)$, the canonical arrow

$$\tau_{\leq 2d}\mathscr{X}^{\cdot}(K^{\cdot}) \longrightarrow \mathscr{X}^{\cdot}(K^{\cdot})$$

induces a quasi-isomorphism on the associated simple complexes.

For this, observe first that the canonical arrow

$$\varinjlim_{k \geq 2d} (\tau_{\leq k}\mathscr{X}^{\cdot}(K^{\cdot}))_s \longrightarrow (\mathscr{X}^{\cdot}(K^{\cdot}))_s$$

is an isomorphism, and then that the cohomology sheaves of a complex commute with filtered direct limits.

7.3.5. Let $R^{(1)}f_! : D(X, \mathscr{A}_X) \to D(Y, \mathscr{A}_Y)$ be the functor defined by (7.3.1.2) using 7.3.3. In view of 7.3.4, it can be computed directly from (7.3.1.1), and one can check that it "does not depend" on the chosen pseudo-compactification. One would then have to verify statements 7.1.3, 7.1.4, and 7.1.6 after deleting the superscripts $+$, which the editor did not have the courage to do.

Definition 7.3.6. *The functor $Rf_! : D(X, \mathscr{A}_X) \to D(Y, \mathscr{A}_Y)$ constructed in 7.3.5 is again called the* direct-image functor with proper supports.

7.3.7. One can then show that the functor $Rf_!$ commutes with base change and that statements (5.2.9) and (5.2.10) remain valid for every morphism in $\mathrm{Sch}_{!R}$, which justifies the constructions just made.

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EXPOSÉ XVIII

The global duality formula

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0. Introduction

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0.1. The present exposé is devoted to Poincaré duality. This duality is constructed on the model of Verdier [1], who treats the case of locally compact Hausdorff topological spaces. We show a priori that the functor $Rf_!$ of Exposé XVII admits a right adjoint $Rf^!$ and establish many of its properties, before computing it more explicitly in the case of smooth morphisms.

The possibility of such a construction seems to rest on the following properties of the functor $Rf_!$:

a) For the existence of an adjoint $Rf^!$ and its formal properties:
   (I) the functor $Rf_!$ commutes with base change;
   (II) the functors $R^i f_!$ commute with filtered inductive limits; they vanish for sufficiently large i;
   (III) the functor $Rf_!$ can be computed "at the level of complexes", i.e. starting from a functor between categories of complexes.
b) For the computation of this adjoint when f is smooth:
   (IV) the computation of $Rf_!((\mathbf{Z}/n\mathbf{Z})_U)$ for "small" U.

0.2. In § 3 nº 1, which is independent of §§ 1, 2, we construct the functor $Rf^!$; constructions specific to the situation are used there, rather than only (I), (II), and (III). This makes this nº rather unpleasant; the author confesses that he did not always fully understand what was happening.

§ 2 is devoted to point (IV); it is purely formal from § 1, where the cohomology of curves is studied. 482

In § 1, we prove substantially more than is needed for § 2, also studying the case of continuous coefficients. This section was profoundly influenced by Serre [14]. The points useful in what follows are:

a) nº 1.1, devoted to the trace morphism in the case of curves, and generalized in 2.9;
b) the vanishing theorem 1.6.9. A reader willing to accept a transcendental argument may read nº 1.6, beginning with 1.6.6 (2nd proof), independently of nº 1.2 to 1.5[82]. The essential ingredient in 1.6.9 is either form of Poincaré duality for smooth curves over an algebraically closed field, together with the acyclicity lemma XV 2.6 (a preliminary to the smooth base-change theorem XVI 1.1). Its conclusion is generalized in 2.14.

Number 3.2 collects the consequences of § 2.

[82] N.D.E.: On a first reading, one cannot too strongly recommend following this route: after § 1.1, proceed directly to § 2 and § 3.

0.3. Verdier's method in Poincaré duality makes it possible to define the functor $Rf^!$ for compactifiable f. The price of this generality is that the compatibilities to be verified are unfamiliar and sometimes, even for the most trivial ones, such as 3.2.3, require outlandish proofs. The author admits that he has not proved as many of them as he should have.

An interested reader will have no difficulty in extending, using the appendix to Exposé XVII, the definition of $Rf^!$ to separated morphisms of finite type whose target is a coherent scheme (i.e. a quasi-compact quasi-separated scheme).

0.4. The main result 3.2.5 is stated in concrete terms that would make it possible to return formally to the viewpoint taken in the oral seminar, where the functor $Rf^!$ was
483 defined only for smoothable morphisms, by means of a factorization through a smooth morphism and an immersion. An outline of the proof from the oral seminar can be found in Verdier [16].

1. Cohomology of curves

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1.1. The trace morphism.

1.1.1. Let n be an integer ≥ 1 and let X be a scheme on which n is invertible. The sheaf $\boldsymbol{\mu}_n$ (IX 3.1) on X is then a sheaf of modules locally free of rank 1 over the constant sheaf of rings $\mathbf{Z}/n$. For $i \in \mathbf{Z}$, we denote by $\mathbf{Z}/n(i)$ its i^{th} tensor power:

$$\mathbf{Z}/n(i) = \mathbf{Hom}(\mathbf{Z}/n, \mathbf{G}_m)^{\otimes i}. \tag{1.1.1.1}$$

If $n = dn'$, exponentiation by d gives an isomorphism

$$\boldsymbol{\mu}_n \otimes_{\mathbf{Z}/n} \mathbf{Z}/n' \xrightarrow{\sim} \boldsymbol{\mu}_{n'},$$

and hence isomorphisms, called canonical:

$$\mathbf{Z}/n(i) \otimes_{\mathbf{Z}/n} \mathbf{Z}/n' \xrightarrow{\sim} \mathbf{Z}/n'(i). \tag{1.1.1.2}$$

If F is an abelian sheaf such that $nF = 0$, set

$$F(i) = F \otimes_{\mathbf{Z}/n} \mathbf{Z}/n(i). \tag{1.1.1.3}$$

If already $n'F = 0$, the isomorphism (1.1.1.2) allows us to identify $F \otimes_{\mathbf{Z}/n} \mathbf{Z}/n(i)$ with $F \otimes_{\mathbf{Z}/n'} \mathbf{Z}/n'(i)$, so that $F(i)$ depends only on F and not on the choice of n. More generally, if F is a torsion sheaf prime to the residual characteristics of X (XVII 0.13), and if ${}_mF$ is the kernel of multiplication by m on F, we define $F(i)$ to be the inductive limit, as m tends multiplicatively to infinity through the integers invertible on X:

$$F(i) = \varinjlim_m {}_mF(i). \tag{1.1.1.4}$$

The functor $F \mapsto F(i)$ is exact. If $f : Y \to X$ is a morphism of schemes (resp. a quasi-compact quasi-separated morphism of schemes), and if F is a torsion sheaf on
485 X (resp. on Y), prime to the residual characteristics of X, then $f^*(F(i)) \simeq (f^*F)(i)$ (resp. $(f_*F)(i) \simeq f_*(F(i))$). If $\mathscr{A}$ is a sheaf of rings on X, annihilated by an integer n invertible on X, this extends to an exact functor $K \mapsto K(i)$ from $D(X, \mathscr{A})$ to $D(X, \mathscr{A})$. This functor commutes with inverse-image functors and direct-image functors.

The functors $F \mapsto F(i)$ are sometimes called "*Tate twist*" functors.

Definition 1.1.2. *A* flat curve *over a scheme* S *is a flat, separated morphism* $f : X \to S$ *of finite presentation whose fibers are of pure dimension one.*

The expression "of pure dimension one" does not exclude the empty scheme. Contrary to EGA II 7.4.2, a curve over a field is allowed here to be empty, but is required to be separated.

When S is the spectrum of a field, we shall simply speak of a curve over S; a curve over a field is quasi-projective[83].

1.1.3. Let X be a curve over an algebraically closed field k of characteristic exponent p prime to an integer $n \geq 1$. First suppose that X is reduced, and let $\overline{X}$ be a complete curve over k in which X is a dense *open subscheme*.

The complement $Y = \overline{X} - X$ is of dimension 0. The exact cohomology sequence (XVII 5.1.16.3) therefore induces an isomorphism

$$H^2_c(X, \mathbf{Z}/n(1)) \xrightarrow{\sim} H^2(\overline{X}, \mathbf{Z}/n(1)). \tag{1.1.3.1}$$

We also have an isomorphism IX 4.7 (given by Kummer theory)

$$H^2(\overline{X}, \mathbf{Z}/n(1)) = \mathrm{Pic}(\overline{X})/n = (\mathbf{Z}/n)^c \tag{1.1.3.2}$$

where c denotes the set of irreducible components of X, or of $\overline{X}$, which amounts to the same thing.

If X is no longer necessarily reduced, and if c is the set of its irreducible components, then for $i \in c$ let n_i denote the multiplicity of X at the generic point η_i of the i^{th} irreducible component of X: 486

$$n_i = \lg(\mathcal{O}_{X,\eta_i}).$$

We know that the restriction map from $H^2_c(X, \mathbf{Z}/n(1))$ to $H^2_c(X_{\mathrm{red}}, \mathbf{Z}/n(1))$ is an isomorphism. Thus, via (1.1.3.1) and (1.1.3.2) applied to X_{red}, there is a canonical isomorphism between $H^2_c(X, \mathbf{Z}/n(1))$ and $(\mathbf{Z}/n)^c$. We denote by Tr_X, or simply by Tr, the composite map

$$\mathrm{Tr} : H^2_c(X, \mathbf{Z}/n(1)) \xrightarrow{\sim} (\mathbf{Z}/n)^c \xrightarrow{t} \mathbf{Z}/n, \tag{1.1.3.3}$$

where

$$t((a_i)_{i \in c}) = \sum n_i a_i.$$

If F is an abelian group annihilated by n, there is an isomorphism

$$H^2_c(X, F(1)) \overset{\sim}{\longleftrightarrow} F \otimes H^2_c(X, \mathbf{Z}/n(1))$$

and hence, via (1.1.3.3), a *trace morphism*

$$\mathrm{Tr} : H^2_c(X, F(1)) \longrightarrow F. \tag{1.1.3.4}$$

[83] N.D.E.: In EGA II 7.4.10, this statement is proved only under the additional assumption that X is normal, and it is announced (EGA II 7.4.12) that a proof in the general case will appear in EGA V, which has not been published. The reader may fill this gap as an exercise. In any case, this has no bearing on the construction of the trace morphism, since it can be carried out first locally (on quasi-projective open subschemes) and the trace morphisms can then be glued as in the proof of 1.1.6.

If $n = n'd$, the diagram

(1.1.3.5)
$$\begin{tikzcd}
0 \arrow[r] & \boldsymbol{\mu}_n \arrow[r] \arrow[d, "x^d"'] & \mathbf{G}_m \arrow[r, "x^n"] \arrow[d, "x^d"'] & \mathbf{G}_m \arrow[r] \arrow[d, equal] & 0 \\
0 \arrow[r] & \boldsymbol{\mu}_n \arrow[r] & \mathbf{G}_m \arrow[r, "x^{n'}"] & \mathbf{G}_m \arrow[r] & 0
\end{tikzcd}$$

is commutative, and therefore so is the diagram

(1.1.3.6)
$$\begin{tikzcd}
H^2_c(X, \mathbf{Z}/n(1)) \arrow[r, "\sim"] \arrow[d] & (\mathbf{Z}/n)^c \arrow[d] \\
H^2_c(X, \mathbf{Z}/n'(1)) \arrow[r, "\sim"] & (\mathbf{Z}/n')^c
\end{tikzcd}$$

487 by the definition in IX 4.7 of the isomorphism (1.1.3.2).

The map (1.1.3.4) thus depends only on F, and not on the choice of the integer n prime to p such that $nF = 0$. By passage to the limit, we define it for every torsion abelian group F prime to p.

The following lemma follows immediately from the definitions.

Lemma 1.1.4. *Under the preceding hypotheses, if, for $i \in c$, U_i is a nonempty open subscheme of X contained in its i^{th} irreducible component, the diagram*

$$\begin{tikzcd}
\bigoplus_{i\in c} H^2_c(U_i, F(1)) \arrow[rr, "\sim"] \arrow[dr, "\bigoplus_{i\in c} Tr_{U_i}"'] & & H^2_c(X, F(1)) \arrow[dl, "Tr_X"] \\
& F &
\end{tikzcd}$$

is commutative.

Lemma 1.1.5. *Let X and Y be two curves over an algebraically closed field k, let $u : X \to Y$ be a quasi-finite flat morphism, and let n be an integer prime to the characteristic exponent p of k.*

The following diagram, in which Tr_u is the map (XVII 6.2.3), is then commutative:

$$\begin{tikzcd}
H^2_c(X, \mathbf{Z}/n(1)) \arrow[rr, "Tr_u"] \arrow[dr, "Tr_X"'] & & H^2_c(Y, \mathbf{Z}/n(1)) \arrow[dl, "Tr_Y"] \\
& \mathbf{Z}/n &
\end{tikzcd} .$$

Lemma 1.1.4 allows us to reduce to the case where X and Y are irreducible, with X_{red} and Y_{red} smooth as well. If e (resp. f) is the multiplicity of X (resp. Y) at the generic point, and if $e = df$, the triangles marked $+$ in the following diagram are commutative:

$$\text{(1.1.5.1)}\qquad \begin{tikzcd}
H^2_c(X, \mathbf{Z}/n(1)) \arrow[rr, "\sim"] \arrow[dr, "\mathrm{Tr}_X"] \arrow[dd, "\mathrm{Tr}_u"'] & \arrow[d, phantom, "+" near start] & H^2_c(X_{\mathrm{red}}, \mathbf{Z}/n(1)) \arrow[dl, "e.\mathrm{Tr}_{X_{\mathrm{red}}}"'] \arrow[dd, "d.\mathrm{Tr}_{u_{\mathrm{red}}}"] \\
& \mathbf{Z}/n & \\
H^2_c(Y, \mathbf{Z}/n(1)) \arrow[rr, "\sim"] \arrow[ur, "\mathrm{Tr}_Y"] & \arrow[u, phantom, "+" near start] & H^2_c(Y_{\mathrm{red}}, \mathbf{Z}/n(1)), \arrow[ul, "f.\mathrm{Tr}_{Y_{\mathrm{red}}}"']
\end{tikzcd}$$

Its outer boundary is commutative, as follows from the commutativity of the diagram 488

$$\begin{tikzcd}
u_! \mathbf{Z}/n_X(1) \arrow[r, "\sim"] \arrow[d, "\mathrm{Tr}_u"] & u_{\mathrm{red}!} \mathbf{Z}/n_{X_{\mathrm{red}}}(1) \arrow[d, "d.\mathrm{Tr}_{u_{\mathrm{red}}}"] \\
\mathbf{Z}/n_Y(1) \arrow[r, "\sim"] & \mathbf{Z}/n_{Y_{\mathrm{red}}}(1).
\end{tikzcd}$$

It suffices to verify this last compatibility at the generic point of Y, which is trivial.

To establish the commutativity of the left-hand triangle in (1.1.5.1), it therefore remains only to establish that of the right-hand triangle. Hence we may now assume in addition that X and Y are reduced[84].

Let $\overline{X}$ and $\overline{Y}$ be the complete smooth curves containing X and Y as dense open subschemes (EGA II 7.4.11). The morphism u extends to a flat morphism $\overline{u} : \overline{X} \to \overline{Y}$ (EGA II 7.4.9), and by 1.1.4 it suffices to verify 1.1.5 for $\overline{u}$.

By (XVII 6.3.18.2), the diagram of sheaves on $\overline{Y}$ below, in which N denotes the norm,

$$\begin{tikzcd}
0 \arrow[r] & \overline{u}_* \boldsymbol{\mu}_n \arrow[r] \arrow[d, "\mathrm{Tr}_{\overline{u}}"] & \overline{u}_* \mathbf{G}_m \arrow[r, "x^n"] \arrow[d, "N"] & \overline{u}_* \mathbf{G}_m \arrow[r] \arrow[d, "N"] & 0 \\
0 \arrow[r] & \boldsymbol{\mu}_n \arrow[r] & \mathbf{G}_m \arrow[r, "x^n"] & \mathbf{G}_m \arrow[r] & 0,
\end{tikzcd}$$

is commutative, and hence so is the diagram

$$\begin{tikzcd}
\mathrm{Pic}(\overline{X})/n \arrow[r, "\sim"] \arrow[d, "N"] & H^2_c(\overline{X}, \mathbf{Z}/n(1)) \arrow[d, "\mathrm{Tr}_{\overline{u}}"] \\
\mathrm{Pic}(\overline{Y})/n \arrow[r, "\sim"] & H^2_c(\overline{Y}, \mathbf{Z}/n(1)),
\end{tikzcd}$$

and it remains to observe that if $\mathscr{L}$ is an invertible sheaf of degree 1 on $\overline{X}$, then its norm 489
again has degree 1 (this fact following from the compatibility EGA IV 21.10.17, taking into account the definition EGA IV 21.5.5 of the norm of a divisor).

[84] N.D.E.: This paragraph was added in this edition, and the next two were modified.

Proposition 1.1.6. *There is a unique way to define, for every compactifiable flat curve $f : X \to S$ and every torsion sheaf F on S prime to the residual characteristics of S, a trace morphism*

$$\mathrm{Tr}_f : R^2 f_!(f^* F(1)) \longrightarrow F,$$

functorial in F, whose formation is compatible with every base change (cf. XVII 6.2.3, (VAR 2))*, and which, when S is the spectrum of an algebraically closed field, coincides with the trace morphism (1.1.3.4).*

The morphism Tr_f is prescribed fiber by fiber; its uniqueness is therefore clear.

Let $p : \mathbf{P}^1_S \to S$ be the projective line over S, and let n be an integer invertible on S. We have $\mathbf{Pic}_S(\mathbf{P}^1_S) = \mathbf{Z}_S$, whence Kummer theory (IX 3.2) gives a morphism $\mathbf{Z}/n \to R^1 p_* \mathbf{Z}/n(1)$ which one checks fiber by fiber (XII 5.2) to be an isomorphism. Its inverse Tr_p induces the trace morphism (1.1.3.4) fiber by fiber.

Let $f : X \to S$ be a flat curve over S such that there exists a quasi-finite flat S-morphism u from X to $\mathbf{P}^1_S$. By 1.1.5, the composite morphism $\mathrm{Tr}_p \circ \mathrm{Tr}_u$ (XVII 6.2.3) has, at every geometric point of S, the trace morphism (1.1.3.4) as its fiber. In particular, it does not depend on the choice of u.

490 Suppose that X is a union of open subschemes U_i such that there exists a quasi-finite flat morphism from U_i to $\mathbf{P}^1_S$. If α_i (resp. α_{ij}) is the inclusion of U_i (resp. $U_{ij} = U_i \cap U_j$) in X, the sequence

$$\sum R^2(f\alpha_{ij})_! \mathbf{Z}/n(1) \rightrightarrows \sum R^2(f\alpha_i)_! \mathbf{Z}/n(1) \to R^2 f_! \mathbf{Z}/n(1) \to 0.$$

is exact, by (XVII 6.2.9) and the right exactness of $R^2 f_!$ (since $R^3 f_! = 0$). One checks fiber by fiber, using 1.1.5, that the sum of the trace morphisms $\sum_i R^2(f\alpha_i)_! \mathbf{Z}/n(1) \to \mathbf{Z}/n$ factors through a trace morphism Tr_f from $R^2 f_! \mathbf{Z}/n(1)$ which induces (1.1.3.4) fiber by fiber.

For a general X, let $j : X' \to X$ be the largest open subset of X such that $f|X'$ is a Cohen–Macaulay morphism (cf. EGA IV 12.1.1 (vi)). Every point x of X' has a neighborhood U_x in X' for which there exists a quasi-finite S-morphism u_x from U_x to $\mathbf{P}^1_S$. This morphism is automatically flat (EGA IV 11.3.10 and 15.4.2), and X' therefore satisfies the preceding hypotheses. Moreover, X' is fiberwise dense in X; the morphism from $R^2(fj_!)\mathbf{Z}/n(1)$ to $R^2 f_! \mathbf{Z}/n(1)$ is consequently an isomorphism, as can be checked fiberwise, and the composite arrow

$$\mathrm{Tr}_f : R^2 f_! \mathbf{Z}/n(1) \xleftarrow{\sim} R^2(fj)_! \mathbf{Z}/n(1) \xrightarrow{\mathrm{Tr}_{fj}} \mathbf{Z}/n$$

induces (1.1.3.4) on every fiber.

If F satisfies $nF = 0$, one has (XVII 5.2.6)

$$R^2 f_! \mathbf{Z}/n(1) \otimes F \xrightarrow{\sim} R^2 f_!(f^* F(1)) \qquad (1.1.6.1)$$

and Tr_f is defined to be the composite arrow

$$\mathrm{Tr}_f : R^2 f_!(f^* F(1)) \xleftarrow{\sim} R^2 f_! \mathbf{Z}/n(1) \otimes F \to F.$$

In the general case, Tr_f is defined as the filtered colimit of the trace morphisms relative to the sheaves ${}_nF$, for n invertible on S. These arrows induce (1.1.3.4) on every fiber and therefore commute with every base change, C.Q.F.D.

491 One checks fiberwise, using 1.1.4, 1.1.5, and (1.1.6.1), the following results:

LEMMA 1.1.7. *Let $f : Y \to S$ be a compactifiable flat curve over S, let $u : X \to Y$ be a quasi-finite flat morphism of finite presentation, and let F be a torsion sheaf on S, of torsion prime to the residue characteristics of S. The following diagram is commutative:*

$$\begin{tikzcd}
R^2(fu)_!(fu)^*F(1) \arrow[r, equal] \arrow[d, "\mathrm{Tr}_{fu}"] & R^2f_!(u_!u^*)f^*F(1) \arrow[r, "\mathrm{Tr}_u"] & R^2f_!f^*F(1) \arrow[d, "\mathrm{Tr}_f"] \\
F \arrow[rr, equal] & & F,
\end{tikzcd}$$

LEMMA 1.1.8. *Let $u : Y \to S$ be a quasi-finite flat morphism of finite presentation, let $f : X \to Y$ be an S-compactifiable flat curve (XVII 3.2.1), and let F be as in 1.1.7. The following diagram is commutative*[85]*:*

$$\begin{tikzcd}
R^2(uf)_!(uf)^*F(1) \arrow[r, equal] \arrow[d, "\mathrm{Tr}_{uf}"] & u_!R^2f_!f^*u^*F(1) \arrow[r, "\mathrm{Tr}_f"] & u_!u^*F \arrow[d, "\mathrm{Tr}_u"] \\
F \arrow[rr, equal] & & F
\end{tikzcd}$$

One similarly verifies from (1.1.3.2):

LEMMA 1.1.9. *If $f : X \to S$ is a compactifiable smooth curve and the geometric fibers of f are irreducible, then the trace morphism (1.1.6) is an isomorphism.*

1.2. 1-acyclicity of projective space. In this section, the fppf topology is in principle always used (XVII 0.10).

1.2.1. Let ϵ be a locally free module on a scheme S, everywhere of rank ≥ 2, let $p : \mathbf{P}(\epsilon) \to S$ be the corresponding projective bundle, and let G be an abelian sheaf on the big fppf site of S (XVII 0.10). Each pair (K_0, φ) consisting of a torsor K_0 under G and a homomorphism $\varphi : \mathbf{G}_{mS} \to G$ defines a p^*G-torsor $K(K_0, \varphi)$ on $\mathbf{P}(\epsilon)$, the sum of K_0 and the image under φ of the $\mathbf{G}_m$-torsor $\mathscr{O}(1)$[86]: 492

(1.2.1.1) $$K(K_0, \varphi) = K_0 + \varphi\mathscr{O}(1).$$

The pairs (K_0, φ) are made into a category by setting

$$\mathrm{Hom}((K_0, \varphi), (K_0', \varphi')) = \begin{cases} \varnothing & \text{if} \quad \varphi \neq \varphi' \\ \mathrm{Hom}(K_0, K_0') & \text{if} \quad \varphi = \varphi'. \end{cases}$$

The torsor $K(K_0, \varphi)$ then depends functorially on (K_0, φ). It is compatible with localization and therefore defines a functor from the stack over S of pairs (K_0, φ) to the stack over S of p^*G-torsors on $\mathbf{P}(\epsilon)$ (= the stack whose sections over U/S are the p^*G-torsors

[85] N.D.E.: This compatibility merits a few arguments. To verify it, one may first reduce by base change to the case where S is the spectrum of an algebraically closed field k. One may then suppose that Y is connected and nonempty, in which case $Y \to S$ has a (unique) section $S \to Y$ that is a closed immersion defined by a nilpotent ideal. The arguments in the proof of 1.1.6 allow us to suppose that $X = \mathbf{P}^1_Y$. It remains only to observe that the multiplicity of Y equals the multiplicity of X at its generic point. See also the proof of the commutativity of (2.9.3).

[86] N.D.E.: In this section, we implicitly identify line bundles and $\mathbf{G}_m$-torsors by the correspondence that associates with a line bundle the sheaf of its invertible sections, on which $\mathbf{G}_m$ acts by multiplication. A reader concerned with details may wish to know that, following Grothendieck's convention, the projective scheme $\mathbf{P}(\epsilon)$ is the bundle of hyperplanes in ϵ, and likewise the punctured vector bundle $\mathbf{V}(\epsilon)^* \to \mathbf{P}(\epsilon)$ is an open subset of $\mathbf{V}(\epsilon)$, which is the spectrum of the symmetric algebra of ϵ.

on $U \times_S \mathbf{P}(\epsilon)$). These stacks are stacks in groupoids (= every morphism over a U/S is an isomorphism).

THEOREM 1.2.2. *Under the hypotheses of 1.2.1, if G is a flat commutative group scheme of finite presentation over S, then the functor K of 1.2.1 is an equivalence.*

. This assertion is equivalent to the conjunction of

$$G \xrightarrow{\sim} p_* p^* G \quad \text{and}$$

$$\mathbf{Hom}(\mathrm{G}_m, G) \xrightarrow{\sim} R^1 p_* p^* G.$$

Indeed, a morphism of stacks in groupoids is an equivalence if and only if it induces an isomorphism on the sheaf generated by the presheaf of isomorphism classes of objects (the 2nd formula) and an isomorphism on the sheaf of automorphisms of any local object (the 1st formula).

We shall give two proofs (1.2.4 and 1.2.5) of full faithfulness[87].

493 LEMMA 1.2.3. *Let $f : X \to S$ be a proper flat morphism. Suppose that $f_* \mathscr{O}_X = \mathscr{O}_S$ universally and $R^1 f_* \mathscr{O}_X = 0$ universally. Every S-morphism from X to a finite-type S-group scheme G then factors through f and a section of G; more precisely, $G \xrightarrow{\sim} f_* f^* G$.*

It suffices to prove the 1st assertion.

By the rigidity lemma (MUMFORD [10, 6.1. p. 115]; note that MUMFORD does not use the Noetherian hypothesis imposed on S), it suffices to treat the case where S is the spectrum of an algebraically closed field. Suppose first that G is an abelian scheme, and let G^* be the dual abelian scheme. Since $G = G^{**}$, giving $\varphi : X \to G$ is equivalent to giving an invertible sheaf $\mathscr{L}$ on $X \times G^*$, trivialized along $X \times \{e\}$ and algebraically equivalent to zero on the fibers of the projection pr_1 from $X \times G^*$ to X. If $x : S \to X$ is a rational point of X, then $\mathscr{L} \sim \mathrm{pr}_2^*(x \times G^*)^* \mathscr{L} \otimes \mathscr{M}$, with $\mathscr{M}$ trivialized along $x \times G^*$ and $X \times \{e\}$. Giving $\mathscr{M}$ is equivalent to giving a morphism of schemes from G^* to $\mathbf{Pic}_{X/S}$ that takes e to e. Since $H^1(X, \mathscr{O}_X) = 0$, one has $\mathbf{Pic}^0_{X/S} = 0$[88], $\mathscr{M} \sim \mathscr{O}$, and $\mathscr{L} \sim \mathrm{pr}_2^* \mathscr{L}'$, with $\mathscr{L}'$ algebraically equivalent to zero on G^*, hence defining $g \in G(S)$ such that $\varphi = g \circ f$.

In the general case, G is an extension of an abelian scheme A by an affine group G_0. The preceding result reduces us to the case $A = 0$, and then

$$\mathrm{Hom}_S(\mathrm{Spec}(H^0(X, \mathscr{O}_X), G) \xrightarrow{\sim} \mathrm{Hom}_S(X, G).$$

LEMMA 1.2.4. *Under the hypotheses of 1.2.1, if G is a group scheme of finite presentation over S or is defined by a quasi-coherent sheaf on S, then the functor (1.2.1.1) is fully faithful.*

A. We prove that, if $K(K_0, \varphi)$ is isomorphic to $K(K_0', \varphi')$, then $\varphi = \varphi'$. For quasi-coherent G, one has $\varphi = \varphi' = 0$, and there is nothing to prove. For G of finite
494 presentation, by (SGA 3 IX 5.1), it suffices to prove the assertion when S is the spectrum of an algebraically closed field k. Let T be the largest subtorus of G, and put $H = G/T$. The cohomology exact sequence gives

$$0 \to H^0(\mathbf{P}(\epsilon), T) \to H^0(\mathbf{P}(\epsilon), G) \xrightarrow{\alpha} H^0(\mathbf{P}(\epsilon), H) \xrightarrow{\delta} H^1(\mathbf{P}(\epsilon), T) \xrightarrow{\beta} H^1(\mathbf{P}(\epsilon), G).$$

[87]N.D.E.: With regard to essential surjectivity, the reader should be warned that the proof will use the general proof technique explained in the introduction to the EGA .

[88]N.D.E.: This follows from the canonical isomorphism $\mathrm{Lie}(\mathbf{Pic}_{X/S}) = H^1(X, \mathscr{O}_X) = 0$, obtained by comparing both groups with the kernel of $\mathrm{Pic}(X \times_k \mathrm{Spec}(k[\epsilon])) \to \mathrm{Pic}(X)$ (where $S = \mathrm{Spec}(k)$). For further details on the properties of Picard schemes, see [6].

By 1.2.3, $H^0(\mathbf{P}(\epsilon), G) \sim G(k)$ and $H^0(\mathbf{P}(\epsilon), H) \sim H(k)$, so α is surjective and δ is zero. If $Y(T) = \mathrm{Hom}(\mathbf{G}_m, T)$, then $T \sim G_m \otimes Y(T)$ and $H^1(\mathbf{P}(\epsilon), T) \sim Y(T)$. Thus the injectivity of β expresses the injectivity of the canonical arrow defined by $\mathcal{O}(1)$, $\mathrm{Hom}(\mathbf{G}_m, G) \simeq \mathrm{Hom}(\mathbf{G}_m, T) \to H^1(\mathbf{P}(\epsilon), G)$, proving the assertion.

B. Proving that the functor (1.2.1.1) is fully faithful is local on S. We may therefore suppose that $\mathbf{P}(\epsilon)/S$ has a section s and choose an isomorphism $s^*\mathcal{O}(1) \sim \mathcal{O}$[89]. If $K(K_0, \varphi)$ is isomorphic to $K(K_0', \varphi')$, then $\varphi = \varphi'$ by A, and $K_0 \sim s^*K(K_0, \varphi) \sim s^*K(K_0', \varphi') \sim K_0'$. By the formula

$$\mathrm{Hom}(K(K_0, \varphi), K(K_0', \varphi')) \sim \mathrm{Hom}(K(K_0 - K_0', \varphi - \varphi'), G_{\mathbf{P}(\epsilon)}), \tag{1.2.4.1}$$

it remains only to show that

$$\mathrm{Hom}_{\mathrm{tors}\,.}(G, G) \xrightarrow{\sim} \mathrm{Hom}_{\mathrm{tors}\,.}(p^*G, p^*G).$$

This follows from 1.2.3.

1.2.5. Here is another, more Künneth-like, proof of full faithfulness. The question is local on S; by (1.2.4.1), it therefore suffices to check that

$$\mathrm{Hom}_{\mathbf{P}(\epsilon)}(G, K(G, \varphi)) \begin{cases} = \varnothing & \text{if} \quad \varphi \neq 0 \\ \xleftarrow{\sim} G(S) & \text{if} \quad \varphi = 0. \end{cases}$$

This is precisely what the following lemma says[90]:

Lemma 1.2.6. *Let ϵ be a locally free module everywhere of rank ≥ 2 on S, let G be a group scheme of finite presentation over S, let $\varphi : \mathbf{G}_m \to G$ be a group homomorphism, let $\mathbf{V}(\epsilon)$ be the vector bundle defined by ϵ, let $\mathbf{V}(\epsilon)^*$ be the corresponding punctured vector* 495 *bundle, and let $\Psi : \mathbf{V}(\epsilon)^* \to G$ be an S-morphism satisfying the following identity between S-morphisms from $\mathbf{G}_m \times \mathbf{V}(\epsilon)^*$ to G:*

$$\Psi(\lambda v) = \Psi(v)\varphi(\lambda).$$

Then $\varphi = e$, and Ψ factors through the projection p from $\mathbf{P}(\epsilon)$ to S.

It suffices to prove the second assertion. We reduce at once to the case where S is Noetherian and then, by (SGA 3 IX 5.1) and the rigidity lemma (Mumford [14, 6.1 p. 115]) applied to $\mathbf{P}(\epsilon)$, to the case where S is the spectrum of a field. Let n be an integer, let W_n and G_n be the n^{th} infinitesimal neighborhoods of the zero section in $\mathbf{V}(\epsilon)$ and G, let $\mathfrak{g}_n$ be the affine algebra of G_n, and let $P_n(\Psi) : \mathbf{V}(\epsilon)^* \to \mathbf{Hom}_S(W_n, G)$ be the principal part of n^{th} order of Ψ[91]. Every section of $\mathbf{Hom}_S(W_n, G)$ defines, by right

[89] N.D.E.: This additional necessary datum, which was missing from the original edition, has been added in this edition.

[90] N.D.E.: This translation merits some explanation. The line bundle $\mathcal{O}(1)$ on $\mathbf{P}(\epsilon)$ becomes canonically trivialized after pullback by the canonical projection $\mathbf{V}(\epsilon)^* \to \mathbf{P}(\epsilon)$. After pullback to $\mathbf{V}(\epsilon)^*$, the G-torsor $K(G, \varphi)$ is therefore canonically trivialized as well. *Via* this identification, an isomorphism $G \to K(G, \varphi)$ thus induces an automorphism of the trivial G-torsor over $\mathbf{V}(\epsilon)^*$, that is, an S-morphism $\mathbf{V}(\epsilon)^* \to G$. To descend from $\mathbf{P}(\epsilon)$, this morphism must satisfy a compatibility like the one stated, and the converse follows by faithfully flat descent. Strictly speaking, it seems to the reviewer that the correct condition should be $\Psi(\lambda v) = \varphi(\lambda)^{-1}\Psi(v)$, but this has no effect, since one can apply the lemma to Ψ^{-1} instead of Ψ. The condition $\Psi(\lambda v) = \varphi(\lambda)\Psi(v)$ in the original edition also gives a correct statement (since φ can be replaced by φ^{-1}), but it has been changed so that the proof need not be modified.

[91] N.D.E.: More concretely, $P_n(\Psi)$ is the morphism corresponding by adjunction to the morphism $\mathbf{V}(\epsilon)^* \times_S W_n \to G$ (which we also denote by P_n) obtained by composing Ψ with the morphism $\mathbf{V}(\epsilon)^* \times_S W_n \to \mathbf{V}(\epsilon)^*$ induced on these subschemes by addition on $\mathbf{V}(\epsilon)$. In terms of the functors of points of these

translation by the inverse of the image of the zero section, a morphism from W_n to G_n that takes 0 to e; this defines

$$\Psi_1 = P_n(\Psi).\Psi^{-1} \,:\, \mathbf{V}(\epsilon)^* \longrightarrow \mathbf{Hom}_S(W_n, G_n).$$

The hypothesis implies that Ψ_1 commutes with homotheties, which act on the right-hand side by transport of structure from their action on $W_n \subset \mathbf{V}(\epsilon)$[92]. The affine algebra of W_n is $\sum_{i=0}^n \mathrm{Sym}^i(\epsilon)$. By homogeneity, Ψ_1 is therefore defined by a section, on $\mathbf{P}(\epsilon)$, of the $\mathcal{O}_{\mathbf{P}(\epsilon)}$-module

$$\sum_{i=0}^{n} p^* \,\mathbf{Hom}(\mathfrak{g}_n, \mathrm{Sym}^i(\epsilon))(-i).$$ [93]

The components of index $i > 0$ of this module have no section other than the zero section, and it follows that

$$\Psi_1 = 0.$$

Since this holds for every n, it implies that Ψ factors through a section of G.

This second proof extends to prove the following analytic variant of 1.2.4. When G is affine, the proof of 1.2.2 can be given independently of this variant.

496 1.2.7. Let L be the fraction field of a complete discrete valuation ring $\mathcal{O}$, let U_m be the rigid analytic "generic fiber" group of the formal completion of $\mathbf{G}_m$, and let $\mathcal{O}(1)^\wedge$ be the U_m-torsor on $\mathbf{P}^r_L$ obtained by formally completing $\mathcal{O}(1)$. If G is a rigid analytic group over L, the functor $(K_0, \varphi) \mapsto p^*K_0 + \varphi\mathcal{O}(1)^\wedge$ (cf. 1.2.1) is a fully faithful functor from the category of pairs (K_0, φ) consisting of a G-torsor on $\mathrm{Spec}(L)$ and $\varphi \in \mathrm{Hom}(U_m, G)$ to the category of p^*G-torsors on $\mathbf{P}^r_L$[94].

Lemma 1.2.8. *Let k be a field of characteristic exponent p, and let $f : X \to \mathrm{Spec}(k)$ be a proper scheme over k satisfying $H^0(X, \mathcal{O}_X) = k$ and $\mathbf{Pic}^\tau_X = 0$[95]. For every finite-type commutative affine group scheme G over k containing no tori, one has $R^1 f_*(f^*G) = 0$.*

It suffices to show that after every base change $c : S \to \mathrm{Spec}(k)$, the functor f^* from the category of G-torsors on S to that of G-torsors on X_S is an equivalence. This problem is local on $\mathrm{Spec}(k)$, so that, in order to solve it, one may reduce to the case where X has a rational point, where G is a successive extension of the groups $\mathbf{G}_a$, α_p, $\boldsymbol{\mu}_p$, $\mathbf{Z}/p$, and $\mathbf{Z}/\ell_i$, with ℓ_i prime to p, and where $\mathbf{Z}/\ell_i \sim \boldsymbol{\mu}_{\ell i}$.

S-schemes, this can be interpreted more concretely by saying that, if v is a section of $\mathbf{V}(\epsilon)^*$ and t a section of W_n, then $P_n(\Psi)(v, t) = \Psi(v + t)$.

[92] N.D.E.: With the notation above, Ψ_1 can be characterized by the relation $\Psi_1(v, t).\Psi(v) = \Psi(v+t)$. It follows at once that, if λ is a section of $\mathbf{G}_m$, then $\Psi_1(\lambda v, \lambda t) = \Psi_1(v, t)$, which is the asserted commutation with homotheties provided that $\mathbf{G}_m$ acts correctly on $\mathbf{Hom}_S(W_n, G_n)$, namely, that a section λ of $\mathbf{G}_m$ acts by right composition with multiplication by the *inverse* of λ on W_n.

[93] N.D.E.: To verify this assertion in detail, one may begin by defining the pullback of this section to $\mathbf{V}(\epsilon)^*$; one then obtains tautologically a section $s = (s_0, \dots, s_n)$ of the trivial bundle obtained by tensoring the structure sheaf with the vector space $\sum_{i=0}^n \mathbf{Hom}(\mathfrak{g}_n, \mathrm{Sym}^i(\epsilon))$. It satisfies the relation $s_i(v) = \lambda^i s_i(\lambda v)$. This is indeed descent data giving the desired section on $\mathbf{P}(\epsilon)$. One can also argue that the sections s_i extend to all of $\mathbf{V}(\epsilon)$ (since the origin is a closed subset of codimension ≥ 2 in $\mathbf{V}(\epsilon)$). By the density of $\mathbf{G}_m$ in $\mathbf{A}^1$, the extended sections s_i satisfy the relation $s_i(v) = \lambda^i s_i(\lambda v)$ for $\lambda \in \mathbf{A}^1$. Taking $\lambda = 0$ gives $s_i = 0$ for $i > 0$.

[94] N.D.E.: The reader may find an introduction to the language of rigid analytic geometry in [13]. The nonexistence of nontrivial sections of the sheaves $\mathcal{O}(n)$ for $n < 0$ in this context follows, for example, from a rigid version of GAGA; see [9].

[95] N.D.E.: A definition of $\mathbf{Pic}^\tau$ in a more general setting will be found in 1.6.4.

We know that $\mathrm{Lie}(\mathbf{Pic}_X) = H^1(X, \mathcal{O}_X) = 0$. By 1.2.3 and the existence of a section, the functors f^* under consideration are fully faithful; it therefore suffices to prove that $R^1 f_*(f^*G) = 0$, or, by dévissage, that

$$R^1 f_* \mathbf{G}_a = R^1 f_* \alpha_p = R^1 f_* \boldsymbol{\mu}_p = R^1 f_* \mathbf{Z}/p = R^1 f_* \mathbf{Z}/\ell_i = 0.$$

For $\mathbf{G}_a$, this follows from $H^1(X, \mathcal{O}_X) = 0$. The case of α_p (resp. $\mathbf{Z}/p$) follows from it via 1.2.3, by the long exact cohomology sequence defined by the exact sequence

$$0 \longrightarrow \alpha_p \longrightarrow \mathbf{G}_a \xrightarrow{x^p} \mathbf{G}_a \longrightarrow 0$$

(resp. $$0 \longrightarrow \mathbf{Z}/p \longrightarrow \mathbf{G}_a \xrightarrow{x - x^p} \mathbf{G}_a \longrightarrow 0 \qquad (p \neq 1)).$$

The case of $\mathbf{Z}/\ell_i$ (resp. of $\boldsymbol{\mu}_p$) follows from the isomorphism $\mathbf{Z}/\ell_i \sim \boldsymbol{\mu}_{\ell_i}$, from 1.2.3, and 497
from the long exact sequence deduced from the Kummer exact sequence

$$0 \longrightarrow \boldsymbol{\mu}_n \longrightarrow \mathbf{G}_m \longrightarrow \mathbf{G}_m \longrightarrow 0.$$

LEMMA 1.2.9. *Let S be a local Artinian scheme, let $\overline{s}$ be a geometric point of S, let $f : X \to S$ be a proper flat morphism, and let G be a flat commutative group scheme of finite type over S. Assume that f is smooth or that G is affine. If $H^0(X_{\overline{s}}, \mathcal{O}) = k(\overline{s})$, if $\mathbf{Pic}^\tau_{X_{\overline{s}}} = 0$, and if $G_{\overline{s}}$ contains no subgroup $\mathbf{G}_m$, then the functor f^* from the category of G-torsors on S to that of f^*G-torsors on X is an equivalence.*

a. The problem in question is local on S when the finite flat morphisms $u : S' \to S$ are taken as localization morphisms. This makes it possible to assume that f admits a section x[96]. The functor f^* is then fully faithful, by 1.2.3 and the existence of x, and it remains to show that every torsor trivialized along x is trivial.
b. Suppose that S is the spectrum of a field. If G is affine, the assertion follows from 1.2.8. If X is smooth and G is an extension of an abelian variety A by an affine group G_0, denote by G^n the inverse image of ${}_nA$ in G. The group $H^1(X, A)$ is torsion (RAYNAUD [12, XIII 2.6]), and is therefore the union of the images of the $H^1(X, {}_nA)$. The group $H^1(X, G)$ is consequently the union of the images of the $H^1(X, G^n)$. Since the groups G^n are affine, the conclusion again follows from 1.2.8.
c. Let us prove the general case (assuming the existence of a section x) by induction on the length of S. Let $i : S_0 \hookrightarrow S$ be a closed subscheme defined by an ideal I of square zero and length 1.

By the induction hypothesis, the G-torsors P on X for which x^*P is trivial are deformations of the trivial G-torsor P_0 on $X \times S_0$. If G is smooth, and if L is the Lie algebra of G_s, these deformations are classified by

$$H^1(X_s, f^*(L \otimes I)) = 0,$$

which proves 1.2.9 in this case. 498

For G flat over S, this argument generalizes in terms of the *Lie complex* L of G_s. If $L \cap (G_s/k(s))$ denotes the relative cotangent complex of G_s, this complex $L \in D^b(k(s))$ is the dual of $Le^*(L \cap (G_s/k(s)))$. We have $H^i(L) = 0$ for $i \neq 0, 1$(ILLUSIE [7, VII 2.4.1]).

The deformations over S (resp. over X) of the trivial G-torsor over S_0 (resp. over X_0) are classified by $H^1(L \otimes I)$ (resp. by $H^1(X_s, f^*(L \otimes I))$ (ILLUSIE [7, VII 2.4.4 (ii)]).

[96]N.D.E.: The existence of a section after such a base change may be deduced from 2.3 and 2.4.

By hypothesis, we have $H^1(X_s, \mathscr{H}^0(f^*(L\otimes I))) = 0$ (because $H^1(X_s, \mathscr{O}) = 0$) and $H^1(L\otimes I) \sim H^0(X_s, \mathscr{H}^1(f^*L\otimes I)))$. We conclude that

$$H^1(L\otimes I) \xrightarrow{\sim} H^1(X_s, f^*(L\otimes I)),$$

which means that every deformation over X of the trivial G-torsor over X_0 is the inverse image of one and only one deformation over S of the trivial G-torsor over S_0. If this deformation is trivial along a section, it is therefore trivial, which proves 1.2.9.

1.2.10. Let us prove 1.2.2 when S is Artinian. Let T be the largest subtorus of G; since full faithfulness is already known (1.2.4), the problem is local on S for the étale topology: we may assume that T is diagonalizable, $T \sim \mathrm{G}_m^n$. By 1.2.3 and 1.2.9, the exact cohomology sequence gives

$$R^1 f_* T \xrightarrow{\sim} R^1 f_* G,$$

and the assertion follows, since it is trivial for $G = T$ and $\mathrm{Hom}(\mathrm{G}_m, T) \xrightarrow{\sim} \mathrm{Hom}(\mathrm{G}_m, G)$ (cf. 1.2.2.1).

1.2.11. Let us prove 1.2.2 when S is complete Noetherian local. Let s be a section of p[97].

If K is a G-torsor on $\mathrm{P}(\epsilon)$, by 1.2.10 there exists a unique pair $(\widehat{\varphi}, \widehat{\Psi})$ consisting of a formal

499 homomorphism $\widehat{\varphi} : \mathrm{G}_m \to G$ and a formal isomorphism $\widehat{\Psi} : \widehat{p^*s^*K} + \widehat{\varphi}\widehat{\mathscr{O}(1)} \xrightarrow{\sim} \widehat{K}$[98][99]. If G is affine, then $\widehat{\varphi}$ is algebraizable (SGA 3 IX 7.1). Let us temporarily admit that, as will be proved in (1.2.13), $\widehat{\varphi}$ is always algebraizable as $\varphi : \mathrm{G}_m \to G$.

If $K_0 = s^*K$, and if $K_1 = p^*K_0 + \varphi\mathscr{O}(1)$, we have a formal isomorphism $\widehat{\Psi} : \widehat{K_1} = \widehat{K(K_0, \varphi)} \xrightarrow{\sim} \widehat{K}$, i.e. a formal trivialization of $K - K_1$. By formal GAGA, this section is algebraic, which proves 1.2.11. More precisely, if G is quasi-affine, then $K - K_1$ is representable and one uses EGA III 5.1.4. In the general case, one proceeds in the same way with the algebraic space $K - K_1$.

1.2.12. *Algebraicity of $\widehat{\varphi}$: preliminary verification over a trait.* Suppose that S is a complete trait, $S = \mathrm{Spec}(V)$, and let L be the fraction field of V.

a) Over L, there exists a unique triple $(K_{0L}, \varphi_L, \Psi_L)$:

$$\Psi_L : K(K_{0L}, \varphi_L) \xrightarrow{\sim} K|\,\mathrm{Spec}(L).$$

b) By (1.2.10) (1.2.11), and with the notation of 1.2.7, we have a rigid analytic map $\widehat{\varphi_L} : U_m \to \widehat{G}$, a $\widehat{G}$-torsor $\widehat{K_0}$, and a rigid analytic isomorphism

$$\widehat{\Psi} : p^*\widehat{K_0} + \widehat{\varphi_L}\widehat{\mathscr{O}(1)} \xrightarrow{\sim} \widehat{K}.$$

By the uniqueness assertion 1.2.7, $\widehat{\varphi_L}$ is identified with the map deduced from φ_L. Thus the formal morphism $\widehat{\Psi}$ of 1.2.11 induces on $\boldsymbol{\mu}_n$ a map that coincides with the map deduced from the algebraic map $\varphi_L : \mathrm{G}_m \to G$.

[97] N.D.E.: As above, one must also choose an isomorphism $s^*\mathscr{O}(1) \sim \mathscr{O}$.

[98] N.D.E.: One must require this isomorphism to become the obvious isomorphism after applying s^*.

[99] N.D.E.: What is meant here is that, if A is the ring of S, I its maximal ideal, and $S_n = \mathrm{Spec}(A/I^n)$, a formal morphism consists of a compatible system of morphisms between objects over S_n for every n. The question of algebraicity is whether this compatible system is obtained by inverse image from similar data over S.

1.2.13. *Algebraicity of* $\widehat{\varphi}$. The formal map $\widehat{\varphi} : \mathbf{G}_m \to G$ of 1.2.11 induces, for ℓ invertible on S, maps

$$\varphi_\ell : T_\ell(\widehat{\mathbf{G}_m}) = T_\ell(\mathbf{G}_m) \longrightarrow T_\ell(G).$$

By (1.2.12) and (EGA II 7.1.9), for every point t of S, there exists an algebraic morphism $\varphi_t : \mathbf{G}_{m_t} \to G_t$ such that $T_\ell(\varphi_t) = T_\ell(\varphi)_t$. By (SGA 3 XV 4.1 bis), $\widehat{\varphi}$ is therefore algebraizable. 500

1.2.14. Let us prove 1.2.2. By passage to the limit, we reduce to the case where S is Noetherian, and then where S is Noetherian local with affine coordinate ring A. Let s be a section of p. Let K be a G-torsor on $\mathbf{P}(\epsilon)$. Over $S' = \operatorname{Spec}(\widehat{A})$, there exists one and only one homomorphism $\varphi' : \mathbf{G}_{mS'} \to G_{S'}$ such that the inverse image K' of K on $\mathbf{P}(\epsilon) \times_S S'$ is of the form $K(K'_0, \varphi')$. This homomorphism φ' descends to $\varphi : \mathbf{G}_m \to G$. To prove the essential surjectivity of the functor 1.2.1, we may replace K by $K - \varphi\mathscr{O}(1)$, i.e. assume that $\varphi = 0$. Over S', there then exists one and only one isomorphism $\Psi : p'^* s^* K' \xrightarrow{\sim} K'$, inducing the identity on the section s. This morphism descends to S (fpqc descent of morphisms of fppf torsors), and this proves 1.2.2.

Variants 1.2.15. *The conclusion of 1.2.2 still holds in the following cases.*

(i) *G is defined by a quasi-coherent module on S.*
(ii) *G is the inverse image of a torsion sheaf on the small étale site of S.*

In case (i), it is standard that the arrows (1.2.2.1) are isomorphisms, taking into account the fact that the fppf cohomology of G coincides with its cohomology for the Zariski topology.

In case (ii), the arrows (1.2.2.1) are isomorphisms by the base change theorem for a proper morphism and the simple connectedness of $\mathbf{P}^r_k$ for k algebraically closed.

1.3. Integration of torsors.

1.3.1. Let S be a scheme and G an abelian sheaf on the big fppf site of S, satisfying the following three conditions.

.

(i) *For every projective space $p : \mathbf{P}^r_S \to S$, with $r \geq 1$, one has, on the big* fppf *site of S* (XVII 0.10) 501

$$G \xrightarrow{\sim} p_* p^* G \quad \text{and}$$

$$\mathbf{Hom}(\mathbf{G}_m, G) \xrightarrow{\sim} R^1 p_* p^* G$$

(ii) *For every n and every smooth affine S-morphism $f : X \to T$, if K_1 and K_2 are two G-torsors on* $\operatorname{Sym}^n_T(X)$, *the set* $\operatorname{Hom}(K_1, K_2)$ *is identified with the set of symmetric morphisms from the inverse image of K_1 on $(X/T)^n$ to the inverse image of K_2 on $(X/T)^n$.*

This condition is equivalent to the conjunction of:

(ii′) $\operatorname{Hom}_S(\operatorname{Sym}^n_T(X), G) \xrightarrow{\sim} \operatorname{Hom}_S((X/T)^n, G)^{\sigma_n}$, and
(ii″) If the inverse image on $(X/T)^n$ of a torsor K on $\operatorname{Sym}^n_T(X)$ admits a symmetric section, then K is trivial.
(iii) *G satisfies either condition* (XVII 6.3.21.1), *or conditions* (6.3.24.0) *and* (456).

The conclusion of 1.2.2 therefore holds (cf. 1.2.2.1), as does the formula of XVII 6.3.26.

Condition (i) is discussed in 1.2.2 and 1.2.15. Condition (ii) holds whenever G is representable and of finite presentation, or is the inverse image of a sheaf on the small étale site of S, or is flat quasi-coherent. Condition 1.3.1.1 therefore holds in each of the following cases.

.

(a) G is smooth and of finite presentation (1.2.2 and XVII 6.3.21.1).
(b) G is the inverse image of a torsion sheaf on the small étale site of S (1.2.15 (ii) and XVII 6.3.21.1).
(c) G is affine, and the kernel of an epimorphism of smooth groups (1.2.2 and XVII 6.3.3.1).
(d) G is defined by a flat quasi-coherent sheaf on S (1.2.15 (i) and XVII 6.3.21.1).

502 1.3.2. Let $f : X \to S$ be a smooth projective curve over a scheme S, with nonempty geometrically connected fibers. Let $\mathscr{L}$ be an invertible sheaf on X that is "sufficiently" relatively ample in the following sense.

. For every geometric point s of S, if X_s has genus g, then

$$\deg_{X_s}(\mathscr{L}) > 2g-2,$$
$$\deg_{X_s}(\mathscr{L}) \geq 1 \quad \text{if} \quad g=0, \quad \text{et} \quad \deg_{X_s}(\mathscr{L}) \geq 2 \quad \text{if} \quad g=1$$

We then have $R^1 f_* \mathscr{L} = 0$, and $f_*\mathscr{L}$ is locally free, its formation commutes with every base change, and it has rank ≥ 2 everywhere. If a section s of $f_*\mathscr{L}$ vanishes at no point, the scheme D_s of zeros of s, regarded as a section of $\mathscr{L}$ on X, is a relative divisor (EGA IV 21.15.2), finite locally free over S since X/S has relative dimension 1. We have canonically

$$s : \mathscr{O}(D_s) \xrightarrow{\sim} \mathscr{L}.$$

The divisor D_s depends on s only up to multiplication by a section of $\mathscr{O}_S^*$. The preceding construction therefore gives us a relative divisor $Z_{\mathscr{L}}$ on the scheme obtained from X by the base change

$$p : \mathbf{P}(f_*(\mathscr{L})^\vee) \longrightarrow S.$$

Let K be a torsor under G_X, and let $K|Z_{\mathscr{L}}$ be its inverse image on $Z_{\mathscr{L}}$. The trace of this torsor (XVII 6.3.26), from $Z_{\mathscr{L}}$ to $\mathbf{P}(f_*(\mathscr{L})^\vee)$, is a torsor $K_{\mathscr{L}}$ on $\mathbf{P}(f_*(\mathscr{L})^\vee)$:

$$K_{\mathscr{L}} = \mathrm{Tr}_{Z_{\mathscr{L}}/\mathbf{P}}(K|Z_{\mathscr{L}}).$$

By 1.2.2, there exists a triple, unique up to unique isomorphism, consisting of a G-torsor $\langle \mathscr{L}, K]$ on S, a morphism $\varphi_{\mathscr{L},K} : \mathbf{G}_m \to G$, and an isomorphism

$$K_{\mathscr{L}} \simeq p^*\langle \mathscr{L}, K] + \varphi_{\mathscr{L},K}\mathscr{O}(1).$$

503 1.3.3. If the composition of torsors is written additively, we have canonically

$$\langle \mathscr{L}, K_1 + K_2] \simeq \langle \mathscr{L}, K_1] + \langle \mathscr{L}, K_2] \tag{1.3.3.1}$$
$$\varphi_{\mathscr{L},K_1+K_2} = \varphi_{\mathscr{L},K_1} + \varphi_{\mathscr{L},K_2}. \tag{1.3.3.2}$$

Let $\mathscr{L}_1$ and $\mathscr{L}_2$ be two invertible sheaves on X satisfying (1.3.2.1), and let $\mathscr{L} = \mathscr{L}_1 \otimes \mathscr{L}_2$. The tensor product of sections defines a morphism

$$\begin{tikzcd}
& \mathbf{P}(f_*(\mathscr{L}_1)^\vee)\times_S \mathbf{P}(f_*(\mathscr{L}_2)^\vee) \arrow[dl, "\mathrm{pr}_1"'] \arrow[dd, "p_{12}"] \arrow[dr, "\mathrm{pr}_2"] \arrow[rr, "\pi"] & & \mathbf{P}(f_*(\mathscr{L})^\vee) \arrow[dd, "p"] \\
\mathbf{P}(f_*(\mathscr{L}_1)^\vee) \arrow[dr, "p_1"'] & & \mathbf{P}(f_*(\mathscr{L}_2)^\vee) \arrow[dl, "p_2"] & \\
& S \arrow[rr, equal] & & S.
\end{tikzcd}$$

We have $\pi^* Z_{\mathscr{L}} \simeq Z_{\mathscr{L}_1} + Z_{\mathscr{L}_2}$, whence a isomorphism (XVII 6.3.27.1)

(1.3.3.3) $$\pi^* K_{\mathscr{L}} = \mathrm{pr}_1^* K_{\mathscr{L}_1} + \mathrm{pr}_2^* K_{\mathscr{L}_2}.$$

There is a canonical isomorphism

$$\pi^* \mathscr{O}(1) \simeq \mathrm{pr}_1^* \mathscr{O}(1) \otimes \mathrm{pr}_2^* \mathscr{O}(1),$$

so that (1.3.3.3) may be rewritten as

$$\begin{aligned}\pi^*(p^*\langle \mathscr{L}, K] + \varphi_{\mathscr{L},K}\mathscr{O}(1)) &= p_{12}^*\langle \mathscr{L}, K] + \mathrm{pr}_1^* \varphi_{\mathscr{L},K}\mathscr{O}(1) + \mathrm{pr}_2^* \varphi_{\mathscr{L},K}\mathscr{O}(1)\\ &= p_{12}^*\langle \mathscr{L}_1, K] + p_{12}^*\langle \mathscr{L}_2, K] + \mathrm{pr}_1^* \varphi_{\mathscr{L}_1,K}\mathscr{O}(1) + \mathrm{pr}_2^* \varphi_{\mathscr{L}_2,K}\mathscr{O}(1).\end{aligned}$$

By applying 1.2.2 twice, one obtains a canonical isomorphism

(1.3.3.4) $$\langle \mathscr{L}_1 \otimes \mathscr{L}_2, K] = \langle \mathscr{L}_1, K] + \langle \mathscr{L}_2, K]$$

and an identity 504

$$\varphi_{\mathscr{L}_1 \otimes \mathscr{L}_2, K} = \varphi_{\mathscr{L}_1,K} = \varphi_{\mathscr{L}_2,K}.$$

Thus one sees that $\varphi_{\mathscr{L},K}$ depends only on K, and one sets

(1.3.3.5) $$\deg(K) = \varphi_{\mathscr{L},K}.$$

Consequently,

(1.3.3.6) $$K_{\mathscr{L}} \simeq p^*\langle \mathscr{L}, K] + \deg(K)\mathscr{O}(1).$$

(1.3.3.7) $$\deg(K_1 + K_2) = \deg(K_1) + \deg(K_2).$$

Let $\mathscr{U}$ be an invertible sheaf that is locally a direct summand of $f_*\mathscr{L}$, defining a section u of $\mathbf{P}((f_*\mathscr{L})^\vee)$. Let D_u be the relative divisor defined by the local invertible sections of $\mathscr{U}$.

$$u^*\mathscr{O}(1) \simeq \mathscr{U}^\vee$$

It therefore follows from (1.3.3.6) that

(1.3.3.8) $$u^* K_{\mathscr{L}} = \mathrm{Tr}_{D_u/S}(K) \simeq \langle \mathscr{L}, K] - \deg(K)\mathscr{U}$$[100].

We leave it to the reader to verify the following compatibility:

[100] N.D.E.: The notation $\mathrm{Tr}_{D_u/S}(K)$ is introduced below in (1.3.5.0).

Proposition 1.3.4. *The following diagram of the isomorphisms (1.3.3.1) and (1.3.3.4) is commutative:*

$$\begin{tikzcd}
\langle \mathscr{L}_1 \otimes \mathscr{L}_2, K_1 + K_2] \arrow[r, phantom, "\simeq"] \arrow[dd, phantom, "\wr"] & \langle \mathscr{L}_1 \otimes \mathscr{L}_2, K_1] + \langle \mathscr{L}_1 \otimes \mathscr{L}_2, K_2] \arrow[d, phantom, "\wr"] \\
& \langle \mathscr{L}_1, K_1] + \langle \mathscr{L}_2, K_1] + \langle \mathscr{L}_1, K_2] + \langle \mathscr{L}_2, K_2] \arrow[d, equal] \\
\langle \mathscr{L}_1, K_1 + K_2] + \langle \mathscr{L}_2, K_1 + K_2] \arrow[r, phantom, "\simeq"] & \langle \mathscr{L}_1, K_1] + \langle \mathscr{L}_1, K_2] + \langle \mathscr{L}_2, K_1] + \langle \mathscr{L}_2, K_2].
\end{tikzcd}$$

The isomorphisms (1.3.3.1) and (1.3.3.4) are moreover compatible with the associativity and commutativity isomorphisms in a sense whose precise formulation we leave to the reader. Formula (1.3.3.4) then makes it possible to extend by "linearity" the definition
505 of $\langle \mathscr{L}, K]$ when $\mathscr{L}$ no longer necessarily satisfies condition (1.3.2.1).

1.3.5. For every effective relative divisor D on X, set

$$\mathrm{Tr}_{D/S}(K) = \mathrm{Tr}_{D/S}(K|D). \tag{1.3.5.0}$$

If $\mathscr{O}(D)$ satisfies (1.3.2.1), the section 1 of $\mathscr{O}(D)$ is a section of $f_*\mathscr{O}(D)$, and as such generates an invertible sheaf that is locally a direct summand of $f_*\mathscr{O}(D)$, of which it defines a trivialization. The isomorphism (1.3.3.8) therefore gives us an isomorphism

$$\langle \mathscr{O}(D), K] \simeq \mathrm{Tr}_{D/S}(K). \tag{1.3.5.1}$$

One immediately verifies that the diagrams

$$\begin{tikzcd}
\langle \mathscr{O}(D), K_1 + K_2] \arrow[r, "\sim"] \arrow[d, "\wr"'] & \mathrm{Tr}_{D/S}(K_1 + K_2) \arrow[d, "\wr"] \\
\langle \mathscr{O}(D), K_1] + \langle \mathscr{O}(D), K_2] \arrow[r, "\sim"] & \mathrm{Tr}_{D/S}(K_1) + \mathrm{Tr}_{D/S}(K_2)
\end{tikzcd} \tag{1.3.5.2}$$

and

$$\begin{tikzcd}
\langle \mathscr{O}(D_1 + D_2), K] \arrow[r, "\sim"] \arrow[d, "\wr"] & \mathrm{Tr}_{D_1+D_2/S}(K) \arrow[d, "\wr\ (XVII\ 6.3.27.1)"] \\
\langle \mathscr{O}(D_1), K] + \langle \mathscr{O}(D_2), K] \arrow[r, "\sim"] & \mathrm{Tr}_{D_1/S}(K) + \mathrm{Tr}_{D_2/S}(K)
\end{tikzcd} \tag{1.3.5.3}$$

are commutative. By linearity, this makes it possible to define the isomorphism (1.3.5.1) for every relative divisor.

1.3.6. Let D and E be two effective relative Cartier divisors, and let f be a rational function such that

$$\mathrm{div}\,(f) = D - E,$$

i.e. an isomorphism between $\mathscr{O}(D)$ and $\mathscr{O}(E)$. The isomorphism f between $\mathscr{O}(D)$ and $\mathscr{O}(E)$ defines, by (1.3.5.1), an isomorphism

$$\langle f, K] : \mathrm{Tr}_{D/S}(K) \xrightarrow{\sim} \mathrm{Tr}_{E/S}(K). \tag{1.3.6.1}$$

506 By (1.3.5.2) for the second formula, one has

$$\langle fg, K] = \langle f, K].\langle g, K]$$
$$\langle f, K_1 + K_2] = \langle f, K_1] + \langle f, K_2].$$

If λ is a section of $\mathscr{O}_S^*$, then, for every invertible sheaf $\mathscr{L}$ on X, multiplication by λ is an automorphism of $\mathscr{L}$. By transport of structure, this automorphism induces on the two sides of (1.3.3.8) automorphisms which correspond to one another under the isomorphism (1.3.3.8). It follows that

$$\langle \lambda, K] = \deg(K)(\lambda) \ ;$$

if $\deg(K) \neq 0$, the isomorphism $\langle f, K]$ between $\mathrm{Tr}_{D/S}(K)$ and $\mathrm{Tr}_{E/S}(K)$ thus depends on the choice of f.

Let $\lambda \in \Gamma(S, \mathscr{O}_S^*)$, $g \in \Gamma(S, G)$, let $\mathscr{L}$ be an invertible sheaf on X, and let K be a G-torsor on X. If $\langle \lambda, g]$ is the automorphism of $\langle \mathscr{L}, K]$ obtained by transport of structure from the automorphisms λ of $\mathscr{L}$ and g of K, then $\langle \lambda, g]$, identified with a section of G, is given by $\langle \lambda, g] = \langle \lambda, K] + \langle \mathscr{L}, g]$; since $\langle \mathscr{O}(D), g] = \mathrm{Tr}_{D/S}(g)$, the preceding formula gives

$$\langle \lambda, g] = \deg(K)(\lambda) + \deg(\mathscr{L}).g. \tag{1.3.6.2}$$

1.3.7. Let $f : X \to S$ be a smooth projective curve over S, and let G be a group over S satisfying 1.3.1.1. If $X \xrightarrow{f_0} T \xrightarrow{\pi} S$ is the Stein factorization of f, then π is an étale covering of S, and the sheaf $f_* G_m$ is the torus $\prod_{T/S} \mathrm{G}_m$. Locally on S (for the étale topology), X is the disjoint union of a family of curves $(X_i)_{i \in I}$ over S with nonempty geometrically connected fibers. If K is a torsor under G and $\mathscr{L}$ an invertible sheaf on X, one then sets

$$\langle \mathscr{L}, K] = \sum_i \langle \mathscr{L} | X_i, K | X_i]$$

and defines the degree of $K : f_* \mathrm{G}_m = \mathrm{G}_m^I \to G$ to have coordinates $\deg(K|X_i) : \mathrm{G}_m \to G$. The local constructions globalize and give functors and morphisms 507

$$\langle \mathscr{L}, K]_{X/S} = \mathrm{Tr}_{T/S}(\langle \mathscr{L}, K]_{X/T}) \tag{1.3.7.1}$$

$$\deg(K) : f_* \mathrm{G}_m = \prod_{T/S} \mathrm{G}_m \longrightarrow G : \deg_{X/S}(K) = \prod_{T/S} \deg_{X/T}(K). \tag{1.3.7.2}$$

For an automorphism λ of $\mathscr{L}$, defined by an element λ_0 of $\Gamma(T, \mathscr{O}_T^*)$, one has

$$\langle \lambda, K] = \deg(K)(\lambda). \tag{1.3.7.3}$$

The total degree of K is defined as the composite

$$\deg\mathrm{tot}(K) : \mathrm{G}_m \longrightarrow f_* \mathrm{G}_m \xrightarrow{\deg(K)} G.$$

The preceding results give:

Formulary 1.3.8. For every smooth projective curve $f : X \to S$, there is a functor $\langle \ , \ ]$, whose formation is compatible with every base change, which associates to an invertible sheaf $\mathscr{L}$ on X and a torsor K under $f^* G$ a torsor $\langle \mathscr{L}, K]$ under G on S. This functor is endowed with the following additional structures and satisfies the following conditions:

. $\langle \mathscr{L}, K]$ is biadditive in $\mathscr{L}$ and K; the biadditivity isomorphisms are compatible with the associativity and commutativity isomorphisms and with base changes, satisfy the compatibility (1.3.4), and are compatible with the isomorphisms (1.3.8.2) below (via XVII 6.3.25.3 and XVII 6.3.27.1).

. For every effective relative divisor E on X/S, there is a canonical isomorphism

$$\langle \mathscr{O}(E), K] \simeq \mathrm{Tr}_{E/S}(K|E).$$

. The degree (1.3.7.2) satisfies, for $\lambda \in f_*\mathrm{G}_m$ an automorphism of $\mathscr{L}$,

$$\langle \lambda, K] = \deg(K)(\lambda).$$

For $\lambda \in \mathrm{G}_m(S)$, the total degree satisfies

$$\langle \lambda, K] = \deg\mathrm{tot}(K)(\lambda).$$

508 Likewise, for $g \in G(S)$ one has $\langle \mathscr{L}, g] = \deg(\mathscr{L}).g$.

One deduces a canonical isomorphism:

$$\langle f^*\mathscr{L}, K] \simeq \deg\mathrm{tot}(K)(\mathscr{L}). \tag{1.3.8.4}$$

It follows immediately from the definitions that, for every homomorphism $\varphi : G \to G'$, there is a canonical isomorphism, compatible with the preceding data:

$$\langle \mathscr{L}, \varphi(K)] \simeq \varphi(\langle \mathscr{L}, K]). \tag{1.3.8.5}$$

. Let $u : X \to Y$ be a flat morphism between smooth projective curves over S. There are then isomorphisms compatible with biadditivity and base changes:

- For $\mathscr{L}$ an invertible sheaf on X and K a G_Y-torsor on Y, one has

$$\langle \mathscr{L}, u^*K] \xrightarrow{\sim} \langle N_{X/Y}(\mathscr{L}), K].$$

- For $\mathscr{L}$ an invertible sheaf on X and K a G_X-torsor on X, one has

$$\langle u^*\mathscr{L}, K] \xrightarrow{\sim} \langle \mathscr{L}, \mathrm{Tr}_{X/Y} K].$$

Moreover, these isomorphisms satisfy the evident compatibility for a composite uv of morphisms of curves; for $\mathscr{L} = \mathscr{O}(E)$, they identify with the isomorphisms (XVII 6.3.27.2)

$$\mathrm{Tr}_{E/S}(u^*K) \xrightarrow{\sim} \mathrm{Tr}_{u_*E/S}(K)$$

$$\mathrm{Tr}_{u^*E/S}(K) \xrightarrow{\sim} \mathrm{Tr}_{E/S}\mathrm{Tr}_{X/Y} K.$$

It remains to prove 1.3.8.6. One reduces to the case in which X and Y have nonempty geometrically connected fibers, and notes that the preceding explicit formulas give, for every section s of $\mathscr{L}$ that does not vanish as a section of the direct image of $\mathscr{L}$, an isomorphism Ψ_s of the desired kind, homogeneous in s, and therefore independent of s when $\mathscr{L}$ satisfies (1.3.2.1). The general case follows.

1.3.9. Under the hypotheses of 1.3.7, we saw in (1.3.8) that every G-torsor K on X
509 defines a functor $\langle *, K]$, denoted below by $F(*)$, endowed with the following additional data and satisfying the following conditions:

(i) F is a morphism of fppf stacks over S (XVII 0.10 and Giraud [4])
   - whose source is the stack whose T-objects, for an S-scheme T, are the invertible sheaves on X_T;
   - whose target is the stack of G-torsors.

(ii) The functor F is endowed with additivity isomorphisms $F(\mathscr{L}_1 \otimes \mathscr{L}_2) \xrightarrow{\sim} F(\mathscr{L}_1) + F(\mathscr{L}_2)$, which are functorial, compatible with the associativity and commutativity isomorphisms, and compatible with base changes.

Theorem 1.3.10. *Under the hypotheses of 1.3.7, the functor $\Phi : K \mapsto \langle *, K]$ is an equivalence between the category of G-torsors on X and the category of morphisms of stacks considered in (1.3.9).*

One reduces easily to the case in which X has nonempty connected fibers.

Let F satisfy 1.3.9 (i) and (ii). For every S-scheme T and every section t of X_T/T, denote also by t the relative divisor $t(T)$. Then $F(\mathscr{O}(t))$ is a G_T-torsor on T. We denote by $\Psi(F)$ the torsor thus obtained on X, for t the "universal point" of X with values in X, defined by the identity map from X to X. For every section t as above, one has

$$F(\mathscr{O}(t)) \simeq t^*\Psi(F). \tag{1.3.10.1}$$

The isomorphism 1.3.8.2 gives an isomorphism of functors $\Psi \circ \Phi \simeq \mathrm{Id}$. We construct an isomorphism $\Phi \circ \Psi \simeq \mathrm{Id}$.

For every section t of X_T over T, the isomorphism (1.3.10.1) is an isomorphism

$$F(\mathscr{O}(t)) \underset{\alpha}{\xrightarrow{\sim}} \langle \mathscr{O}(t), \Psi(F)].$$

By additivity, every family $t_1, \dots, t_n$ of sections of X_T over T defines an isomorphism

$$F(\mathscr{O}(\sum t_i)) \underset{\alpha}{\xrightarrow{\sim}} \langle \mathscr{O}(\sum t_i), \Psi(F)],$$

and by 1.3.9 (ii), this isomorphism is independent of the order of the t_i.

Take $T = (X/S)^n$ and for $t_1 \dots t_n$ the universal n^{tuple} of sections. If $T' = \mathrm{Sym}^n_S(X)$, there is then on $X_{T'}$ a divisor $\sum t_i$ whose inverse image is the divisor $\sum t_i$ on X_T. On T, α is a symmetric isomorphism 510

$$\alpha : F(\mathscr{O}(\sum t_i)) \xrightarrow{\sim} \langle \mathscr{O}(\sum t_i), \Psi(F)]$$

between torsors that are inverse images of torsors on T'. By hypothesis (1.3.1.1) (ii), this isomorphism comes from a unique isomorphism on T':

$$\alpha : F(\mathscr{O}(\sum t_i)) \xrightarrow{\sim} \langle \mathscr{O}(\sum t_i), \Psi(F)] \qquad (\text{on } T').$$

By XVII 6.3.9, one thus has, for every relative divisor D on X, an isomorphism

$$\alpha_D : F(\mathscr{O}(D)) \xrightarrow{\sim} \langle \mathscr{O}(D), \Psi(F)], \tag{1.3.10.2}$$

additive in D, whose formation is compatible with every base change.

Let $\mathscr{L}$ be a sufficiently ample invertible sheaf on X in the sense of 1.3.2.1. Every nowhere-zero section s of $f_*\mathscr{L}$ defines, with the notation of 1.3.2, an isomorphism

$$s : \mathscr{O}(D_s) \xrightarrow{\sim} \mathscr{L},$$

and hence, by (1.3.10.2), an isomorphism

$$\alpha_s : F(\mathscr{L}) \xrightarrow{\sim} \langle \mathscr{L}, \Psi(F)]. \tag{1.3.10.3}$$

Let us prove that this isomorphism is independent of s. Indeed:

(i) The construction of α_s commutes with every base change.

(ii) There exists a homomorphism $\varphi : \mathbf{G}_m \to G$ such that

$$\alpha_{\lambda s} = \varphi(\lambda).\alpha_s$$

(by the functoriality of both sides of (1.3.10.3)).

One necessarily has $\varphi = 0$, since otherwise the torsor on $\mathbf{P}((f_*\mathscr{L})^\vee)$ obtained by pulling back $F(\mathscr{L}) - \langle \mathscr{L}, \Psi(F)]$ would have degree $\neq 0$. The α_s therefore define a
511 morphism from $\mathbf{P}((f_*\mathscr{L})^\vee)$ to G, and this morphism is constant by hypothesis (1.3.1.1) (i). We may therefore set $\alpha_{\mathscr{L}} = \alpha_s$.

The isomorphisms $\alpha_{\mathscr{L}}$ thus form an isomorphism of functors between the restrictions of the functors F and $\langle x, \Psi(F)]$ to sufficiently ample sheaves. This isomorphism is compatible with the additivity isomorphisms, which allows it to be extended by linearity to all invertible sheaves and completes the construction of

$$\alpha : \Phi \circ \Psi \simeq \mathrm{Id}\,.$$

Example 1.3.11. $G = \mathbf{G}_m$. Under the hypotheses of 1.3.7, and in the special case where $G = \mathbf{G}_m$, we shall write $\langle\ ,\ \rangle$ rather than $\langle\ ,\ ]$; if $\mathscr{L}$ and $\mathscr{M}$ are two invertible sheaves on X, then $\langle \mathscr{L}, \mathscr{M}\rangle$ is an invertible sheaf on S. Moreover, there are isomorphisms (1.3.8.2)

$$\langle \mathscr{O}(D), \mathscr{M}\rangle \simeq N_{D/S}(\mathscr{M}).$$

If D and E are disjoint effective relative divisors, then $\mathscr{O}(E)|D$ is canonically isomorphic to $\mathscr{O}$, whence an isomorphism

$$\sigma_{D,E} : \langle \mathscr{O}(D), \mathscr{O}(E)\rangle \simeq N_{D/S}(\mathscr{O}(E)) \simeq N_{D/S}(\mathscr{O}) \simeq \mathscr{O}. \tag{1.3.11.1}$$

Proposition 1.3.12. *Let $(D_i)_{i=1,2}$ and E be effective relative divisors, with D_i disjoint from E $(i = 1, 2)$. Let f be a rational function satisfying* $\operatorname{div}(f) = D_1 - D_2$, *i.e. an isomorphism $f : \mathscr{O}(D_1) \xrightarrow{\sim} \mathscr{O}(D_2)$. The following diagrams of isomorphisms are commutative:*

(1.3.12.1)
$$\begin{tikzcd}
\langle \mathscr{O}(D_1), \mathscr{O}(E)\rangle \arrow[r, "\sigma_{D_1,E}"] \arrow[d, "{\langle f, \mathscr{O}(E)\rangle}"] & \mathscr{O} \arrow[d, "N_{E/S}(f)"] \\
\langle \mathscr{O}(D_2), \mathscr{O}(E)\rangle \arrow[r, "\sigma_{D_2,E}"] & \mathscr{O}
\end{tikzcd}$$

(1.3.12.2)
$$\begin{tikzcd}
\langle \mathscr{O}(E), \mathscr{O}(D_1)\rangle \arrow[r, "\sigma_{E,D_1}"] \arrow[d, "{\langle \mathscr{O}(E), f\rangle}"] & \mathscr{O} \arrow[d, "N_{E/S}(f)"] \\
\langle \mathscr{O}(E), \mathscr{O}(D_2)\rangle \arrow[r, "\sigma_{E,D_2}"] & \mathscr{O}
\end{tikzcd}$$

512 The formula (1.3.12.2) simply expresses the commutativity of the diagram

$$\begin{tikzcd}
N_{E/S}(\mathscr{O}(D_1)) \arrow[r, "\sim"] \arrow[d, "N_{E/S}(f)"] & \mathscr{O} \arrow[d, "N_{E/S}(f)"] \\
N_{E/S}(\mathscr{O}(D_2)) \arrow[r, "\sim"] & \mathscr{O}.
\end{tikzcd}$$

For every S-scheme T and every invertible sheaf $\mathscr{L}$ on X_T, set $F(\mathscr{L}) = N_{E_T/T}(\mathscr{L})$. The hypotheses of 1.3.10 are satisfied by this functor, so there exists a unique invertible sheaf $\mathscr{M}$ equipped with an isomorphism of functors compatible with biadditivity:

$$N_{E_T/T}(\mathscr{L}) \xrightarrow{\sim} \langle \mathscr{L}, \mathscr{M}\rangle.$$

Moreover, if $d : T \to X_T$ is a section of X over T, there is a canonical isomorphism

$$d^*\mathscr{M} \xrightarrow{\sim} N_{E_T/T}(\mathscr{O}(d)).$$

Taking for d the universal section of X, $\Delta : X \to X_X$, we obtain

$$\mathscr{M} \xrightarrow{\sim} N_{E_X/X}(\mathscr{O}(\Delta)).$$

Locally on S, E is a sum of sections $E = \sum t_i$ (XVII 6.3); for such an E, one has

$$N_{E_X/X}(\mathscr{O}(\Delta)) = \otimes t_i^*\mathscr{O}(\Delta) = \sum \mathscr{O}(t_i) = \mathscr{O}(E),$$

and this isomorphism, being independent of the order of the t_i, defines a canonical isomorphism[101]

(1.3.12.3) $$N_{E_X/X}(\mathscr{O}(\Delta)) \sim \mathscr{O}(E),$$

so that

(1.3.12.4) $$N_{E_T/T}(\mathscr{L}) \xrightarrow{\sim} \langle \mathscr{L}, \mathscr{O}(E)\rangle.$$

The isomorphism (1.3.12.3) is compatible with biadditivity and base changes and is characterized, in addition, by the fact that for every section d of X_T over T, the isomorphism 513

$$N_{E/T}(\mathscr{O}(d)) \xrightarrow{\sim} \langle \mathscr{O}(d), \mathscr{O}(E)\rangle \xrightarrow{\sim} d^*\mathscr{O}(E)$$

is the pullback of the universal isomorphism (1.3.12.3). In particular, for d disjoint from E, the diagram

$$\begin{tikzcd} N_{E/T}(\mathscr{O}(d)) \arrow[r, "\sim"] \arrow[d, equal] & \langle \mathscr{O}(d), \mathscr{O}(E)\rangle \arrow[r, "\sim"] & d^*\mathscr{O}(E) \arrow[d, equal] \\ \mathscr{O} \arrow[rr, equal] & & \mathscr{O} \end{tikzcd}$$

is commutative. By additivity[102], it follows that, for D disjoint from E, the diagram

(1.3.12.5) $$\begin{tikzcd} N_{E/T}(\mathscr{O}(D)) \arrow[r, "\sim"] \arrow[d, equal] & \langle \mathscr{O}(D), \mathscr{O}(E)\rangle \arrow[r, "\sim"] & N_{D/T}(\mathscr{O}(E)) \arrow[d, equal] \\ \mathscr{O} \arrow[rr, equal] & & \mathscr{O} \end{tikzcd}$$

is commutative. The arrow $\sigma_{E,D}$ is therefore the composite of the inverse of 1.3.12.4 and an evident arrow

$$\langle \mathscr{O}(D), \mathscr{O}(E)\rangle \xleftarrow{\sim} N_{E/S}(\mathscr{O}(D)) \longrightarrow \mathscr{O}$$

and (1.3.12.1) follows at once, since 1.3.12.4 is functorial.

We deduce from 1.3.12 (cf. Serre [14, Ch. III, prop. 7, p. 46])

[101]N.D.E.: The argument seems to assume that, locally, the t_i are well defined up to permutation, which is not true, as can be seen in the case where $S = \operatorname{Spec} k[\varepsilon]$ is the spectrum of the algebra of dual numbers, $X = \mathbf{P}^1_S$, and E is the closed subscheme of $\mathbf{A}^1_S = \operatorname{Spec} \mathscr{O}_S[T] \subset X$ defined by the equation $(T-\varepsilon)(T+\varepsilon) = T^2$. It seems instead necessary to apply the "cancelled descent" condition 1.3.1.1 (ii) to obtain the result in the situation of the universal divisor on X after base change by $\operatorname{Sym}^n_S X \to S$, before deducing it for the divisor E.

[102]N.D.E.: The same objection raised above applies here as well. One can argue in the same way by considering "the universal divisor disjoint from E," defined after base change by $\operatorname{Sym}^n_S(X - E) \to S$.

Corollary 1.3.13. *Let* $(D_i)_{i=1,2}$ *and* $(E_i)_{i=1,2}$ *be effective relative divisors, with* D_i *disjoint from* E_j, *let* f *be a rational function satisfying* $\operatorname{div}(f) = D_1 - D_2$, *and let* g *be a rational function satisfying* $\operatorname{div}(g) = E_1 - E_2$. *Then*

$$N_{E_1/S}(f).N_{E_2/S}(f)^{-1} = N_{D_1/S}(g).N_{D_2/S}(g)^{-1}$$

or, briefly,

$$f(\operatorname{div}(g)) = g(\operatorname{div}(f)). \tag{1.3.13.1}$$

This formula, in the form

$$N_{D_2/S}(g)N_{E_1/S}(f) = N_{E_2/S}(f)N_{D_1/S}(g),$$

expresses, via (1.3.12.1) and (1.3.12.2), the commutativity of the diagram

$$\begin{tikzcd}
\langle \mathcal{O}(D_1), \mathcal{O}(E_1)\rangle \arrow[r, "\langle \mathcal{O}(D_1), g\rangle"] \arrow[d, "\langle f, \mathcal{O}(E_1)\rangle"] & \langle \mathcal{O}(D_1), \mathcal{O}(E_2)\rangle \arrow[d, "\langle f, \mathcal{O}(E_2)\rangle"] \\
\langle \mathcal{O}(D_2), \mathcal{O}(E_1)\rangle \arrow[r, "\langle \mathcal{O}(D_2), g\rangle"] & \langle \mathcal{O}(D_2), \mathcal{O}(E_2)\rangle.
\end{tikzcd}$$

514 Corollary 1.3.14. *For* $G = \mathbf{G}_m$, *the degree (1.3.7) coincides with the usual degree of invertible sheaves.*

It suffices to compare formulas (1.3.7.3) and (1.3.12.1) for f the pullback of $f_0 \in \Gamma(S, \mathcal{O}_S^*)$.

1.3.15. For any sufficiently ample $\mathscr{L}$ and $\mathscr{M}$, there exist locally on S disjoint divisors D and E and isomorphisms

$$\alpha : \mathscr{L} \longrightarrow \mathcal{O}(D) \quad \text{et} \quad \beta : \mathscr{M} \longrightarrow \mathcal{O}(E).$$

Denote by τ the isomorphism making the diagram

$$\begin{tikzcd}
\langle \mathscr{L}, \mathscr{M}\rangle \arrow[r, "\langle \alpha, \beta\rangle"] \arrow[d, "\tau", "\wr"'] & \langle \mathcal{O}(D), \mathcal{O}(E)\rangle \arrow[r, "\sigma_{D,E}"] & \mathcal{O} \arrow[d, equal] \\
\langle \mathscr{M}, \mathscr{L}\rangle \arrow[r, "\langle \beta, \alpha\rangle"] & \langle \mathcal{O}(E), \mathcal{O}(D)\rangle \arrow[r, "\sigma_{E,D}"] & \mathcal{O}.
\end{tikzcd}$$

It follows from 1.3.12 that this isomorphism is independent of the particular choice of D, E, α, and β; it therefore globalizes to a canonical isomorphism

$$\tau : \langle \mathscr{L}, \mathscr{M}\rangle \xrightarrow{\sim} \langle \mathscr{M}, \mathscr{L}\rangle. \tag{1.3.15.1}$$

This isomorphism satisfies $\tau^2 = 1$; it is compatible with biadditivity, since (1.3.11.1) is, and extends by linearity to arbitrary invertible sheaves.

It follows from (1.3.12.5) that the morphism (1.3.12.4) identifies with the composite

$$N_{E_T/T}(\mathscr{L}) \xleftarrow{\sim} \langle \mathcal{O}(E), \mathscr{L}\rangle \xrightarrow{\sim\tau} \langle \mathscr{L}, \mathcal{O}(E)\rangle$$

(verify this for $\mathscr{L} = \mathcal{O}(D)$ with D disjoint from E).

515 Formulary 1.3.16. *For every smooth projective curve $f : X \to S$, there is a functor $\langle\ ,\ \rangle$, whose construction commutes with every base change, that associates with two invertible sheaves $\mathscr{L}$ and $\mathscr{M}$ on X an invertible sheaf $\langle \mathscr{L}, \mathscr{M}\rangle$ on S. This functor is endowed with the following additional structures and satisfies the following conditions:*

. With the notation of 1.3.8, for $G = \mathbf{G}_m$, one has

$$\langle \mathscr{L}, \mathscr{M}\rangle = \langle \mathscr{L}, \mathscr{M}],$$

in particular, $\langle \mathscr{L}, \mathscr{M}\rangle$ is biadditive in $\mathscr{L}$, $\mathscr{M}$ in the sense specified in 1.3.8.1, and this biadditivity is compatible with that of the right-hand side of a canonical isomorphism (functorial in $\mathscr{M}$)

$$\langle \mathscr{O}(E), \mathscr{M}\rangle \xrightarrow{\sim} N_{E/S}(\mathscr{M}).$$

. There is a symmetry isomorphism

$$\tau : \langle \mathscr{L}, \mathscr{M}\rangle \xrightarrow{\sim} \langle \mathscr{M}, \mathscr{L}\rangle,$$

functorial in $\mathscr{L}$ and $\mathscr{M}$, compatible with base changes and the biadditivity isomorphisms, and satisfying $\tau^2 = 1$. It makes the following diagram commutative, for disjoint divisors D and E:

$$\begin{tikzcd}
N_{D/S}(\mathscr{O}(E)) \arrow[d, "\wr"] & \langle \mathscr{O}(D), \mathscr{O}(E)\rangle \arrow[l] \arrow[r, "\tau"] & \langle \mathscr{O}(E), \mathscr{O}(D)\rangle \arrow[r] & N_{E/S}(\mathscr{O}(D)) \arrow[d, "\wr"] \\
\mathscr{O} \arrow[rrr, equal] & & & \mathscr{O}.
\end{tikzcd}$$

These formulas imply formulas (1.3.12.1), (1.3.12.2), (1.3.13.1), and

. For a, $b \in \Gamma(S, \mathscr{O}_S^*)$, denoting by $\langle a, b\rangle$ the automorphism of $\langle \mathscr{L}, \mathscr{M}\rangle$, identified with a section of $\mathbf{G}_m$, obtained by functoriality from the automorphisms a and b of $\mathscr{L}$ and $\mathscr{M}$, one has

$$\langle a, b\rangle = a^{\deg(\mathscr{M})} b^{\deg(\mathscr{L})}.$$

. If $\mathscr{L}_0$ (resp. $\mathscr{M}_0$) is an invertible module on S, there is a canonical isomorphism

$$\langle f^*\mathscr{L}_0, \mathscr{M}\rangle \xrightarrow{\sim} \mathscr{L}^{\otimes \deg(\mathscr{M})}$$

(resp. $$\langle \mathscr{L}, f^*\mathscr{M}_0\rangle \xrightarrow{\sim} \mathscr{M}^{\otimes \deg(\mathscr{L})}$$).

. If $u : X \to Y$ is a finite flat morphism of smooth projective curves over S, then, for 516
$\mathscr{L}$ an invertible sheaf on X and $\mathscr{M}$ an invertible sheaf on Y, the isomorphisms 1.3.8.6

$$\langle \mathscr{L}, u^*\mathscr{M}\rangle \sim \langle N_{X/Y}\mathscr{L}, \mathscr{M}\rangle$$
$$\langle u^*\mathscr{M}, \mathscr{L}\rangle \sim \langle \mathscr{M}, N_{X/Y}\mathscr{L}\rangle$$

are deduced from one another by the symmetry (1.3.16.2) (verify this for $\mathscr{L} = \mathscr{O}(E)$ and $\mathscr{M} = \mathscr{O}(F)$ with E and F disjoint).

. For an invertible sheaf $\mathscr{L}$ on X, the symmetry arrow

$$\tau : \langle \mathscr{L}, \mathscr{L}\rangle \longrightarrow \langle \mathscr{L}, \mathscr{L}\rangle$$

is multiplication by $(-1)^{\deg(\mathscr{L})}$.

1.3.17. It remains to prove 1.3.16.6. We shall give two proofs, the first using the theory of the determinant (SGA 6 XI)[103].

Before proceeding to each of the proofs, one may observe that if $\mathscr{L}_1$ and $\mathscr{L}_2$ are two invertible sheaves on X and $\mathscr{L} = \mathscr{L}_1 \otimes \mathscr{L}_2$, then the automorphism $\tau_{\mathscr{L}} : \langle \mathscr{L}, \mathscr{L} \rangle \longrightarrow \langle \mathscr{L}, \mathscr{L} \rangle$ (identified with an invertible function on S) is the product of the automorphisms $\tau_{\mathscr{L}_1}$ and $\tau_{\mathscr{L}_2}$. In the situation where S is the spectrum of a field (and X is geometrically connected), one may therefore suppose that $\mathscr{L} = \mathscr{O}(D)$ with D a nonzero effective divisor. Since $\tau^2 = \mathrm{Id}$, after replacing $\mathscr{L}$ by a sufficiently large odd power (or writing it as a quotient of two ample bundles), one may also make $\mathscr{L}$ as ample as desired, if necessary.

The arguments below will also require reducing to the case where S is the spectrum of a field. For this, it suffices to reduce to the case where S is reduced, and this can be done by showing that fppf-locally on S, the datum $(X, \mathscr{L})$ comes from a similar datum over a reduced base. Before making this reduction, one may reduce to the case where $X \to S$ is equipped with a section $s : S \to X$ and where the geometric fibers of $X \to S$ are connected and have some genus g. Using [10, § 5.2], one can obtain a smooth projective curve $X_0 \to S_0$ of genus g, with S_0 smooth of finite type over $\operatorname{Spec}(\mathbf{Z})$, such that *locally*, every relative curve as above is obtained from it by base change. Using the section $s_0 : S_0 \to X_0$, one can rigidify the Picard scheme $S_1 = \mathbf{Pic}_{X_0/S_0} \to S_0$ and thus obtain a "universal" line bundle $\mathscr{L}_1$ on $X_1 = X_0 \times_{S_0} S_1$. Locally, every relative curve as above equipped with a line bundle comes from $(X_1, \mathscr{L}_1)$ over S_1, which is indeed reduced[104].

1st proof: Let $u : T \to S$ be a finite locally free morphism. If $\mathscr{L}$ is an invertible sheaf on T, there is an isomorphism

$$N_{T/S}(\mathscr{L}) \simeq \det(u_* \mathscr{L}) \det(u_* \mathscr{O}_T)^{-1} \tag{1.3.17.1}$$

uniquely determined by the following three conditions:

a) its construction commutes with base changes;
b) for $\mathscr{L} = \mathscr{O}_T$, it is the identity morphism of $\mathscr{O}_S$;
c) it is functorial in $\mathscr{L}$.

Let D and E be two effective relative divisors on X, let i be the inclusion of D in X, and let u be the projection of D onto S. The short exact sequence

$$0 \longrightarrow \mathscr{O}(-E) \longrightarrow \mathscr{O} \longrightarrow \mathscr{O}_E \longrightarrow 0$$

allows the isomorphism

$$\langle \mathscr{O}(D), \mathscr{O}(-E) \rangle \simeq N_{D/S}(\mathscr{O}(-E)) \simeq \det(u_* i^* \mathscr{O}(-E)) \det(u_* i^* \mathscr{O})^{-1}$$

517 as an isomorphism

$$\langle \mathscr{O}(D), \mathscr{O}(-E) \rangle = \det Ru_*(Li^* \mathscr{O}_E)^{-1} \simeq \det Rf_*(\mathscr{O}_D \overset{\mathbf{L}}{\otimes} \mathscr{O}_E)^{-1}.$$

or

$$\langle \mathscr{O}(D), \mathscr{O}(E) \rangle \simeq \det Rf_*(\mathscr{O}_D \overset{\mathbf{L}}{\otimes} \mathscr{O}_E). \tag{1.3.17.2}$$

[103] N.D.E.: That exposé was not written, as explained in [3], which also contains a warning concerning the multiplicativity property of the determinant in distinguished triangles: it requires care, without which counterexamples arise. A reference on this subject is [8].

[104] N.D.E.: The preceding two paragraphs were added in this edition.

For disjoint D and E, one has $\mathscr{O}_D \overset{\mathbf{L}}{\otimes} \mathscr{O}_E = 0$, and the resulting isomorphism of $\langle \mathscr{O}(D), \mathscr{O}(E)\rangle$ with $\mathscr{O}_S$ is the one already considered. It follows that, for disjoint D and E, the symmetry τ and the symmetry isomorphism $\mathscr{O}_D \overset{\mathbf{L}}{\otimes} \mathscr{O}_E \xrightarrow{\sim} \mathscr{O}_E \overset{\mathbf{L}}{\otimes} \mathscr{O}_D$ are compatible via (1.3.17.2). By specialization[105], this remains true for arbitrary D and E.

When $D = E$ is defined by an ideal I, one has

$$
\begin{aligned}
&\mathscr{H}^0(\mathscr{O}_D \overset{\mathbf{L}}{\otimes} \mathscr{O}_D) = \mathscr{O}_D && \text{symmetry } = +\mathrm{Id}\\
&\mathscr{H}^{-1}(\mathscr{O}_D \overset{\mathbf{L}}{\otimes} \mathscr{O}_D) = I/I^2 && \text{symmetry } = -\mathrm{Id}\\
&\mathscr{H}^i(\mathscr{O}_D \overset{\mathbf{L}}{\otimes} \mathscr{O}_D) = 0 && \text{for } i \neq 0, -1.
\end{aligned}
$$

Consequently, the map

$$\tau : \langle \mathscr{O}(D), \mathscr{O}(D)\rangle \xrightarrow{\sim} \langle \mathscr{O}(D), \mathscr{O}(D)\rangle$$

is $(-1)^{\operatorname{rk} I/I^2} = (-1)^{\deg(\mathscr{O}(D))}$, which proves 1.3.16.6.

2^{nd} *proof*: By reduction to a "universal" case, we may suppose that S is reduced. This case reduces to that in which S is the spectrum of an algebraically closed field k. In this case, let X_1, X_2, Y_1, and Y_2 be four divisors on X, let f and g be two rational functions, and suppose that

$$
\begin{aligned}
&\operatorname{div}(f) = X_2 - X_1\\
&\operatorname{div}(g) = Y_2 - Y_1\\
&(X_1 \cup Y_2) \cap (X_2 \cup Y_1) = \varnothing.
\end{aligned}
$$

The maps f and g then define[106]

$$\langle f, g\rangle : \langle \mathscr{O}(X_2), \mathscr{O}(Y_2)\rangle \longrightarrow \langle \mathscr{O}(X_1), \mathscr{O}(Y_1)\rangle.$$

The two sides are canonically isomorphic to k (1.3.11.1), so that $\langle f, g\rangle$ is identified with 518
an element of k^*.

Lemma 1.3.18. *One has*

$$\langle f, g\rangle = \prod_{x \in X_1 \cup Y_2} (-1)^{v_x(f)v_x(g)} (f^{v_x(g)}/g^{v_x(f)})(x).$$

Let us show that (1.3.18) implies (1.3.16.6). We may suppose that $\mathscr{L} \sim \mathscr{O}(X) \sim \mathscr{O}(Y)$ with X and Y effective, and that $X \cap Y = \varnothing$[107]. Let f be a rational function such that

[105] N.D.E.: It seems that here one must first reduce to the case where S is the spectrum of a field, as explained above. It is then possible to make D and E occur in two families of divisors parametrized by an integral scheme in such a way that the divisors in the two families are disjoint over the generic point. This may be obtained, for example, by considering the universal situation over which the condition that D and E be disjoint holds on a dense open subset of $\operatorname{Sym}^n_S X \times \operatorname{Sym}^m_S X$, which is integral.

[106] N.D.E.: Compared with the original edition, numerous "signs" have been modified here and there through the end of this paragraph.

[107] N.D.E.: We saw above that we could suppose that $\mathscr{L}$ was very ample.

$\operatorname{div}(f) = Y - X$[108]. The commutative diagram

$$\begin{tikzcd}
\langle \mathscr{L}, \mathscr{L} \rangle \arrow[r, "\tau"] \arrow[d, equal] & \langle \mathscr{L}, \mathscr{L} \rangle \arrow[d, equal] \\
\langle \mathscr{O}(X), \mathscr{O}(Y) \rangle \arrow[r, "\langle f^{-1}, f \rangle"] & \langle \mathscr{O}(Y), \mathscr{O}(X) \rangle
\end{tikzcd}$$

shows that

$$\tau = \langle f^{-1}, f \rangle = \prod_{x \in Y} (-1)^{-v_x(f)^2} = (-1)^{\deg(\mathscr{L})}.$$

Let us prove 1.3.18. We reduce to supposing that X_i and Y_i are very ample. There are then factorizations $f = f_1 f_2$; $g = g_1 g_2$ with

$$\begin{aligned} \operatorname{div}(f_1) &= X' - X_1 & \qquad \operatorname{div}(g_1) &= Y' - Y_1 \\ \operatorname{div}(f_2) &= X_2 - X' & \qquad \operatorname{div}(g_2) &= Y_2 - Y', \end{aligned}$$

the divisors X', Y', and $X_1 \cup X_2 \cup Y_1 \cup Y_2$ being pairwise disjoint.

Denote by $\langle\ ,\ \rangle^*$ the right-hand side of (1.3.18), and let us prove that

$$\langle f, g \rangle^* = \langle f_1, g_1 \rangle^* \cdot \langle f_2, g_2 \rangle^*.$$

By Serre [14, Ch. III, prop. 6, p. 44], one has

$$\prod_x (-1)^{v_x(f_1) v_x(g_2)} (f_1^{v_x(g_2)} / g_2^{v_x(f_1)})(x) = 1. \tag{1.3.18.1}$$

It follows that

$$\begin{aligned} \langle f_1, g_1 \rangle^* \cdot \langle f_2, g_2 \rangle^* &= g_1(X_1) f_1(Y') g_2(X') f_2(Y_2) \cdot \\ &\quad \prod_{X_1 \cup X' \cup Y' \cup Y_2} (-1)^{v_x(f_1) v_x(g_2)} (f_1^{v_x(g_2)} \cdot g_2^{-v_x(f_1)})(x) \\ &= \prod_{x \in X'} g_2(x)^{-v_x(f_2)} \cdot g_2(x)^{-v_x(f_1)} \cdot \prod_{x \in Y'} f_1(x)^{v_x(g_1)} \cdot f_1(x)^{v_x(g_2)} \cdot \\ &\quad \prod_{X_1 \cup Y_2} (-1)^{v_x(f) v_x(g)} ((f_1 f_2)^{v_x(g_2)} \cdot (g_1 g_2)^{-v_x(f_1)} = \langle f_1 f_2, g_1 g_2 \rangle^*. \end{aligned}$$

519 One has, by functoriality of $\langle\ ,\ \rangle$, $\langle f, g \rangle = \langle f_1, g_1 \rangle \langle f_2, g_2 \rangle$, and it is readily checked that $\langle f_i, g_i \rangle = \langle f_i, g_i \rangle^*$[109]. The lemma follows.

1.4. Strictly commutative Picard categories. Some results from "general topology" will be needed to express the results of 1.3 by a "universal coefficient formula."

The results of this section are suggested in a letter from A. Grothendieck to J.L. Verdier (1966).

The notions of associativity and commutativity for bifunctors (see below) were introduced by Mac Lane.

[108] N.D.E.: Care should be taken not to confuse X with the ambient curve, which is also denoted by X.

[109] N.D.E.: This is the special case in which one assumes in addition that $X_1 \cap Y_2 = \varnothing$; it follows from 1.3.12.1 and 1.3.12.2.

1.4.1. Let C be a category, let $F : C \times C \to C$ be a functor, and let

$$\sigma : F(F(X, Y), Z) \xrightarrow{\sim} F(X, F(Y, Z))$$

be an isomorphism of trifunctors. The pair (F, σ) is said to be an *associative functor*, or σ is said to be an *associativity datum* on F, if the following condition holds:

(Ass) For every family $(X_i)_{i \in I}$ of objects of C, denoting by $e : I \to M(I)$ the canonical map from I into the free monoid (without identity) generated by I, there exist a map $\mathscr{F} : M(I) \to \operatorname{Ob} C$, isomorphisms $a_i : \mathscr{F}(e_i) \xrightarrow{\sim} X_i$, and isomorphisms $a_{g,h} : \mathscr{F}(gh) \xrightarrow{\sim} F(\mathscr{F}(g), \mathscr{F}(h))$ such that the diagrams

$$\begin{tikzcd}
\mathscr{F}(f(gh)) \arrow[r, "\sim", "a_{f,gh}"'] \arrow[d, equal] & F(\mathscr{F}(f), \mathscr{F}(gh)) \arrow[r, "\sim", "a_{g,h}"'] & F(\mathscr{F}(f), F(\mathscr{F}(g), \mathscr{F}(h))) \\
\mathscr{F}((fg)h) \arrow[r, "\sim", "a_{fg,h}"'] & F(\mathscr{F}(fg), \mathscr{F}(h)) \arrow[r, "\sim", "a_{f,g}"'] & F(F(\mathscr{F}(f), \mathscr{F}(g)), \mathscr{F}(h)) \arrow[u, "\wr"', "\sigma"]
\end{tikzcd} \tag{1.4.1.1}$$

are commutative.

We shall not need the fact that axiom (Ass) is equivalent to the "pentagon axiom," namely that the following diagram is commutative:

$$\begin{tikzcd}
 & F(F(X, Y), F(Z, T)) \arrow[dl] \arrow[dr] & \\
F(X, F(Y, F(Z, T))) \arrow[d, leftrightarrow] & & F(F(F(X, Y), Z), T) \arrow[d, leftrightarrow] \\
F(X, F(F(Y, Z), T)) \arrow[rr, leftrightarrow] & & F(F(X, F(Y, Z)), T).
\end{tikzcd}$$

Now suppose given functorial isomorphisms 520

$$\sigma_{X,Y,Z} : F(F(X, Y), Z) \xrightarrow{\sim} F(X, F(Y, Z))$$
$$\tau_{X,Y} : F(X, Y) \xrightarrow{\sim} F(Y, X).$$

We shall say that σ and τ make F an *associative* and *strictly commutative* functor if the following condition holds.

(Ass.s.Com) For every family $(X_i)_{i \in I}$ of objects of C, denoting by $e : I \to N(I)$ the canonical map from I into the free abelian monoid (without identity) generated by I, there exist a map $\mathscr{F} : N(I) \to \operatorname{Ob} C$, isomorphisms $a_i : \mathscr{F}(e_i) \xrightarrow{\sim} X_i$, and isomorphisms $a_{g,h} : \mathscr{F}(gh) \xrightarrow{\sim} F(\mathscr{F}(g), \mathscr{F}(h))$ such that diagram (1.4.1.1) is commutative, as is the diagram

$$\begin{tikzcd}
\mathscr{F}(gh) \arrow[r, "a_{g,h}"] \arrow[d, equal] & F(\mathscr{F}(g), \mathscr{F}(h)) \arrow[d, "\tau"] \\
\mathscr{F}(hg) \arrow[r, "a_{h,g}"] & F(\mathscr{F}(h), \mathscr{F}(g)).
\end{tikzcd} \tag{1.4.1.2}$$

We shall not need the fact that this axiom is equivalent to the conjunction of:

1) the pentagon axiom;
2) $\tau_{X,X} : X + X \to X + X$ is the identity;

3) $\tau_{Y,X} \circ \tau_{X,Y} : X + Y \to Y + X \to X + Y$ is the identity;
4) the hexagon axiom: the diagram

$$\begin{tikzcd}
& X+(Y+Z) \arrow[dl, dash, "\sigma"'] \arrow[dr, dash, "\tau"] & \\
(X+Y)+Z \arrow[d, dash, "\tau"] & & X+(Z+Y) \arrow[d, dash, "\sigma"] \\
Z+(X+Y) \arrow[dr, dash, "\sigma"'] & & (X+Z)+Y \arrow[dl, dash, "\tau"] \\
& (Z+X)+Y &
\end{tikzcd}$$

521 is commutative.

Note that condition 2) is not very often satisfied in practice (hence the terminology *strictly* commutative).

DEFINITION 1.4.2. *A* strictly commutative Picard category $\mathscr{P}$ *is a nonempty category all of whose morphisms are isomorphisms, equipped with a functor* $+ : \mathscr{P} \times \mathscr{P} \to \mathscr{P} : (X, Y) \mapsto X + Y$ *and functorial isomorphisms*

$$\sigma : (X+Y)+Z \xrightarrow{\sim} X+(Y+Z)$$
$$\tau : X+Y \longrightarrow Y+X$$

making $+$ *an associative and strictly commutative functor, and such that, moreover, for every* $X \in \operatorname{Ob}\mathscr{P}$*, the functor* $Y \mapsto X + Y$ *is an equivalence of categories.*

The reader will verify:

LEMMA 1.4.3. *If* $(X_i)_{i\in I}$ *is a family of objects of a strictly commutative Picard category* $\mathscr{P}$*, there exist a map* $\Sigma : \mathbf{Z}^{(I)} \to \operatorname{Ob}\mathscr{P}$*, isomorphisms* $a_i : \Sigma(e_i) \xrightarrow{\sim} X_i$*, and* $a_{\underline{n},\underline{m}} : \Sigma(\underline{n}+\underline{m}) \xrightarrow{\sim} \Sigma(\underline{n}) + \Sigma(\underline{m})$ *such that the diagrams of types (1.4.1.1) and (1.4.1.2) are commutative. The system* $(\Sigma, (a_i), (a_{\underline{n},\underline{m}}))$ *is unique up to unique isomorphism, and is functorial in* $(X_i)_{i\in I}$*.*

1.4.4. A strictly commutative Picard category admits, up to unique isomorphism, exactly one *neutral object*, which may here be defined as a pair (e, φ) consisting of an object e and an isomorphism $\varphi : e+e \xrightarrow{\sim} e$. If (e, φ) is a neutral object, there is a unique isomorphism of functors

$$\alpha_s : e + X \xrightarrow{\sim} X$$

making the diagram

$$\begin{tikzcd}
e+(e+X) \arrow[r, leftrightarrow] \arrow[d, "e+\alpha_s"] & (e+e)+X \arrow[d, "\varphi"] \\
e+X \arrow[r, equal] & e+X.
\end{tikzcd}$$

commutative.

522 One similarly defines $\alpha_d : X + e \xrightarrow{\sim} X$, and φ is a special case of both α_s and α_d; τ interchanges α_s and α_d. The group $\operatorname{Aut}(e)$ is abelian, and for every $X \in \operatorname{Ob}\mathscr{P}$, the functors $X+$ and $+X$ induce the same isomorphism between $\operatorname{Aut}(e)$ and $\operatorname{Aut}(X)$.

Definition 1.4.5. *A strictly commutative Picard stack $\mathscr{P}$ on a site $\mathscr{S}$ is a stack in groupoids $\mathscr{P}$ on $\mathscr{S}$ (Giraud [4]), equipped with a functor $+ : \mathscr{P} \times_{\mathscr{S}} \mathscr{P} \to \mathscr{P}$ and isomorphisms of functors*

$$\begin{aligned} \tau_{x,y} &: x + y \longrightarrow y + x \\ \sigma_{x,y,z} &: (x + y) + z \longrightarrow x + (y + z) \end{aligned}$$

which, for every $U \in \operatorname{Ob} \mathscr{S}$, make $\mathscr{P}(U)$ a strictly commutative Picard category.

In what follows, we shall simply speak of a *Picard stack*, without specifying "strictly commutative." By 1.4.4, every Picard stack $\mathscr{P}$ admits a global "neutral object" e, and $\mathbf{Aut}(e)$ is an abelian sheaf.

1.4.6. An *additive functor* $F : \mathscr{P}_1 \to \mathscr{P}_2$ between Picard stacks on $\mathscr{S}$ is an $\mathscr{S}$-functor (necessarily Cartesian) equipped with an isomorphism of functors

$$F(x + y) \xrightarrow{\sim} F(x) + F(y)$$

making the diagrams

$$\begin{tikzcd} F(x+y) \arrow[r] \arrow[d, "F(\tau)"] & F(x) + F(y) \arrow[d, "\tau"] \\ F(y+x) \arrow[r] & F(y) + F(x) \end{tikzcd}$$

and

$$\begin{tikzcd} F((x+y)+z) \arrow[r] \arrow[d, "F(\sigma)"] & F(x+y) + F(z) \arrow[r] & (F(x) + F(y)) + F(z) \arrow[d, "\sigma"] \\ F(x+(y+z)) \arrow[r] & F(x) + F(y+z) \arrow[r] & F(x) + (F(y) + F(z)). \end{tikzcd}$$

commutative.

An *morphism of additive functors* $u : F \to G$ is a morphism of $\mathscr{S}$-functors (automatically an isomorphism) making the diagram 523

$$\begin{tikzcd} F(x+y) \arrow[r, "u_{x+y}"] \arrow[d] & G(x+y) \arrow[d] \\ F(x) + F(y) \arrow[r, "u_x + u_y"] & G(x) + G(y). \end{tikzcd}$$

commutative.

1.4.7. If $\mathscr{P}_1$ and $\mathscr{P}_2$ are two Picard stacks on $\mathscr{S}$, the Picard stack $\operatorname{HOM}(\mathscr{P}_1, \mathscr{P}_2)$ is the following Picard stack:

a) if $U \in \operatorname{Ob} \mathscr{S}$, its objects over U are the additive functors from $\mathscr{P}_1|U$ to $\mathscr{P}_2|U$; its morphisms are morphisms of additive functors;

b) the sum of two additive functors F_1 and F_2 is defined by the formula $(F_1 + F_2)(x) = F_1(x) + F_2(x)$; the structural isomorphism is the one making the diagram

$$\begin{tikzcd}
(F_1+F_2)(x+y) \arrow[r, "\text{structural}"] \arrow[dd, equal] & (F_1+F_2)(x)+(F_1+F_2)(y) \arrow[dr, equal] & \\
& & F_1(x)+F_2(x)+F_1(y)+F_2(y) \arrow[dl, leftrightarrow, "\tau"] \\
F_1(x+y)+F_2(x+y) \arrow[r] & F_1(x)+F_1(y)+F_2(x)+F_2(y) &
\end{tikzcd}$$

commutative;

c) the associativity and commutativity isomorphisms are defined by means of the analogous isomorphisms in $\mathscr{P}_2$.

1.4.8. If $\mathscr{P}_1$, $\mathscr{P}_2$, and $\mathscr{P}$ are three Picard stacks, a biadditive functor from $\mathscr{P}_1 \times \mathscr{P}_2$ to $\mathscr{P}$ is an $\mathscr{S}$-functor F from $\mathscr{P}_1 \times_S \mathscr{P}_2$ to $\mathscr{P}$, equipped with isomorphisms of functors

$$F(x_1 + y_1, x_2) \xrightarrow{\sim} F(x_1, x_2) + F(y_1, x_2)$$

$$F(x_1, x_2 + y_2) \xrightarrow{\sim} F(x_1, x_2) + F(x_1, y_2),$$

such that

a) For fixed x_1 (resp. x_2), $F(x_1, *)$ (resp. $F(*, x_2)$) satisfies the compatibility conditions of 1.4.6.

524 b) For $U \in \operatorname{Ob} \mathscr{S}$, $x_1, y_1 \in \operatorname{Ob} \mathscr{P}_1(U)$ and $x_2, y_2 \in \operatorname{Ob} \mathscr{P}_2(U)$, the diagram

$$\begin{tikzcd}
F(x_1+y_1,x_2+y_2) \arrow[r] \arrow[dd] & F(x_1+y_1,x_2)+F(x_1+y_1,y_2) \arrow[dr] & \\
& & F(x_1,x_2)+F(y_1,x_2)+F(x_1,y_2)+F(y_1,y_2) \arrow[dl, leftrightarrow, "\tau"] \\
F(x_1,x_2+y_2)+F(y_1,x_2+y_2) \arrow[r] & F(x_1,x_2)+F(x_1,y_2)+f(y_1,x_2)+f(y_1,y_2) &
\end{tikzcd}$$

is commutative.

For example, the canonical functor

$$\operatorname{HOM}(\mathscr{P}_1, \mathscr{P}_2) \times \mathscr{P}_1 \longrightarrow \mathscr{P}_2 : (F, x) \longmapsto F(x)$$

is biadditive.

As in 1.4.7, one proves that the biadditive functors from $\mathscr{P}_1 \times \mathscr{P}_2$ to $\mathscr{P}$ form a Picard stack $\operatorname{HOM}(\mathscr{P}_1, \mathscr{P}_2; \mathscr{P})$.

We shall see in (1.4.20) that there exists a "2-universal" biadditive functor: $\mathscr{P}_1 \times \mathscr{P}_2 \to \mathscr{P}_1 \otimes \mathscr{P}_2$; more precisely, there exist a Picard stack $\mathscr{P}_1 \otimes \mathscr{P}_2$ and a biadditive functor $\otimes : \mathscr{P}_1 \times \mathscr{P}_2 \to \mathscr{P}_1 \otimes \mathscr{P}_2$ such that, for every Picard stack $\mathscr{P}$, the functor defined by $\otimes$:

$$\operatorname{HOM}(\mathscr{P}_1 \otimes \mathscr{P}_2, \mathscr{P}) \longrightarrow \operatorname{HOM}(\mathscr{P}_1, \mathscr{P}_2; \mathscr{P}) \tag{1.4.8.1}$$

is an equivalence. This stack is unique up to a unique equivalence (itself unique up to a unique isomorphism).

1.4.9. Let $u : \mathcal{S}_1 \to \mathcal{S}_2$ be a morphism of sites. The direct image $u_*\mathcal{P}$ of a Picard stack $\mathcal{P}$ on $\mathcal{S}_1$ is the Picard stack on $\mathcal{S}_2$ defined by

$$u_*\mathcal{P}(V) = \mathcal{P}(u^*V).$$

Its objects over $V \in \mathrm{Ob}\,\mathcal{S}_2$ are the objects of $\mathcal{P}$ over u^*V; its morphisms, its addition law, and its compatibility isomorphisms are those of $\mathcal{P}$. The direct image u_* is a 2-functor.

1.4.10. A *Picard prestack* $\mathcal{P}$ on a site $\mathcal{S}$ is a prestack $\mathcal{P}$ on $\mathcal{S}$ (Giraud [4]), equipped 525
with a functor $+ : \mathcal{P} \times_{\mathcal{S}} \mathcal{P} \to \mathcal{P}$ and associativity and commutativity isomorphisms σ and τ, such that, for every $U \in \mathrm{Ob}\,\mathcal{S}$, $\mathcal{P}(U)$ is a (strictly commutative) Picard category. If $\mathcal{P}_1$ and $\mathcal{P}_2$ are two Picard prestacks, one defines in the evident way (cf. 1.4.6, 1.4.7) the *additive functors* from $\mathcal{P}_1$ to $\mathcal{P}_2$ and the Picard prestack $\mathrm{HOM}(\mathcal{P}_1, \mathcal{P}_2)$ that they form. If $\mathcal{P}$ is a Picard prestack, and $j : \mathcal{P} \to a\mathcal{P}$ is the stack that it generates, there exists, up to an essentially unique equivalence, one and only one pair consisting of a Picard-stack structure on $a\mathcal{P}$ and an additive-functor structure on j; equipped with these data, the pair $(j, a\mathcal{P})$ is called *the Picard stack generated by* $\mathcal{P}$. For every Picard stack $\mathcal{P}_1$, one has an equivalence

$$\mathrm{HOM}(a\mathcal{P}, \mathcal{P}_1) \xrightarrow{\sim} \mathrm{HOM}(\mathcal{P}, \mathcal{P}_1). \tag{1.4.10.1}$$

1.4.11. We shall denote by $C^{[-1,0]}(\mathcal{S})$ the category of complexes of abelian sheaves K on $\mathcal{S}$ such that $K^i = 0$ for $i \notin [-1, 0]$. To every complex $K \in \mathrm{Ob}\,C^{[-1,0]}(\mathcal{S})$

$$K : d : K^{-1} \longrightarrow K^0$$

is associated the following Picard prestack $\mathrm{pch}(K)$:

(I) For $U \in \mathrm{Ob}\,\mathcal{S}$, one has $\mathrm{Ob}\,\mathrm{pch}(K)(U) = K^0(U)$.
(II) If $x, y \in K^0(U)$, an arrow from x to y is an element f of $K^{-1}(U)$ such that $df = y - x$.
(III) The composition law for arrows is the addition law in $K^{-1}(U)$.
(IV) The functor $+$ is given by the addition law in $K^0(U)$ and $K^{-1}(U)$.
(V) The associativity and commutativity isomorphisms are provided by the zero element of $K^{-1}(U)$.

We shall denote by $\mathrm{ch}(K)$ the Picard stack generated (1.4.10) by the Picard prestack $\mathrm{pch}(K)$. The sheaf $\mathcal{H}^0(K)$ is interpreted as the following associated sheaf

$$\mathcal{H}^0(K) \simeq a\,(U \longmapsto \text{group of isomorphism classes of objects of } \mathrm{ch}(K)(U)). \tag{1.4.11.1}$$

On the other hand, 526

$$\mathcal{H}^{-1}(K) \simeq \mathbf{Aut}(e) \tag{1.4.11.2}$$

1.4.12. Every morphism of complexes $f : K \to L$ induces an additive functor $\mathrm{pch}(f) : \mathrm{pch}(K) \to \mathrm{pch}(L)$, hence an additive functor

$$\mathrm{ch}(f) : \mathrm{ch}(K) \longrightarrow \mathrm{ch}(L).$$

It follows from (1.4.11.1) and (1.4.11.2) that $\mathrm{ch}(f)$ is an equivalence if and only if f is a quasi-isomorphism. From (1.4.10.1) one deduces that, for two morphisms $f, g : K \to L$, one has

$$\mathrm{Hom}(\mathrm{ch}(f), \mathrm{ch}(g)) \xrightarrow{\sim} \mathrm{Hom}(\mathrm{pch}(f), \mathrm{pch}(g)).$$

A morphism of *functors* $h : \mathrm{pch}(f) \to \mathrm{pch}(g)$ is a morphism of sheaves $h : K^0 \to L^{-1}$ such that

$$g(x) - f(x) = dh(x) \tag{1.4.12.1}$$

and such that, for every triple (x, y, u) with $y - x = du$, one has

$$h(y) + f(u) = g(u) + h(x) \tag{1.4.12.2}$$

That h is a morphism of additive functors means that

$$h(x + y) = h(x) + h(y). \tag{1.4.12.3}$$

Condition (1.4.12.3) makes it possible to rewrite (1.4.12.2) as

$$g(u) - f(u) = h(du); \tag{1.4.12.2$'$}$$

conditions (1.4.12.1) through (1.4.12.3) therefore mean that h is a homotopy from f to g, and

$$\mathrm{Hom}(\mathrm{ch}(f), \mathrm{ch}(g)) \simeq \{H : \text{ homotopy } K \longrightarrow L | g - f = dH + Hd\} \tag{1.4.12.4}$$

Lemma 1.4.13. (I) *For every Picard stack $\mathscr{P}$, there exists a complex K such that $\mathscr{P} = \mathrm{ch}(K)$.*

(II) *For every additive functor $F : \mathrm{ch}(K) \to \mathrm{ch}(L)$, there exist a quasi-isomorphism $k : K' \to K$ and a morphism $\ell : K' \to L$ such that $F \simeq \mathrm{ch}(\ell)\, \mathrm{ch}(k)^{-1}$.*

527 Let us prove (I). Let $(k_i, U_i)_{i \in I}$ be a family such that

(a) $k_i \in \mathrm{Ob}\, \mathscr{P}(U_i)$;

(b) every object of $\mathscr{P}$ is locally isomorphic to an inverse image of one of the k_i.

Set $K^0 = \bigoplus_{i \in I} \mathbf{Z}_{U_i}$. It follows from 1.4.3 that there exists an additive functor F from $\mathrm{ch}(0 \to K^0)$ to $\mathscr{P}$ taking the basis e_i of $\mathbf{Z}_{U_i}$ to k_i. Let K^{-1} be the sheaf of pairs consisting of a local section x of K^0 and an isomorphism $t : F(0) \xrightarrow{\sim} F(x)$. Addition on K^{-1} is defined by $(x_1, t_1) + (x_2, t_2) = (x_1 + x_2, t)$, where t makes the diagram

$$\begin{tikzcd}
F(0) + F(0) \arrow[r, "t_1+t_2"] \arrow[d, leftrightarrow] & F(x_1) + F(x_2) \arrow[d, leftrightarrow] \\
F(0+0) \arrow[r, "t"] & F(x_1 + x_2).
\end{tikzcd}$$

commutative. Define $d : K^{-1} \to K^0$ by $d(x, t) = x$; d is additive.

If $y - x = dt$, there exists a unique morphism $F(t)$ making the diagram

$$\begin{tikzcd}
F(x) \arrow[r, "F(t)"] \arrow[d, leftrightarrow] & F(y) \arrow[d, leftrightarrow] \\
F(0) + F(x) \arrow[r, "t+F(x)"] & F(y-x) + F(x)
\end{tikzcd}$$

commutative, and one verifies that this construction defines an equivalence

$$F : \mathrm{ch}(K) \longrightarrow \mathscr{P}.$$

Let us prove (II). Thus let $F : \mathrm{ch}(K) \to \mathrm{ch}(L)$ be an additive functor. There then exists a family $(k_i, \ell_i, \alpha_i, U_i)_{i\in I}$ such that

(c) $k_i \in K^0(U_i)$, $\ell_i \in L^0(U_i)$, and α_i is a morphism from $F(k_i)$ to ℓ_i;

(d) the homomorphism $k^0 : \bigoplus_{i\in I} \mathbf{Z}_{U_i} \to K^0$ with coordinates K_i is an epimorphism.

Set $K'^0 = \bigoplus_{i\in I} \mathbf{Z}_{U_i}$, and let $\ell^0 : K'^0 \to L^0$ be the homomorphism with coordinates ℓ_i. Since the functor F is additive, there exists a unique family of isomorphisms $\alpha_x : Fk^0(x) \xrightarrow{\sim} \ell^0(x)$ (x a local section of K'^0) such that

(e) for e_i the section 1 of $\mathbf{Z}_{U_i}$, one has $\alpha_{e_i} = \alpha_i$;

(f) The diagrams 528

$$\begin{tikzcd}
Fk^0(x+y) \arrow[r, "\alpha_{x+y}"] \arrow[d, leftrightarrow] & \ell^0(x+y) \arrow[d, equal] \\
Fk^0(x) + Fk^0(y) \arrow[r, "\alpha_x + \alpha_y"] & \ell^0(x) + \ell^0(y)
\end{tikzcd}$$

are commutative.

Let $K'^{-1} = K^{-1} \times_{K^0} K'^0$; by (d), the evident arrow is a quasi-isomorphism of complexes $k : K' \to K$. If $x, y \in K'^0(U)$, and if $t \in K'^{-1}(U)$ satisfies $y - x = dt$, then there exists a unique section $\ell^{-1}(t; x, y)$ of $L^{-1}(U)$ defining the morphism that makes the diagram

(g)
$$\begin{tikzcd}
F(k^0(x)) \arrow[r, "F(k^{-1}(t))"] \arrow[d, "\alpha_x"] & F(k^0(y)) \arrow[d, "\alpha_y"] \\
\ell^0(x) \arrow[r, "\ell^{-1}(t;x,y)"] & \ell^0(y).
\end{tikzcd}$$

commutative. The fact that F is a functor gives

(h) $\ell^{-1}(t; x, y) + \ell^{-1}(t'; y, z) = \ell^{-1}(t + t'; x, z)$.

The functoriality of the additivity isomorphism of F gives

(i) $\ell^{-1}(t_1 + t_2; x_1 + x_2, y_1 + y_2) = \ell^{-1}(t_1; x_1, y_1) + \ell^{-1}(t_2; x_2, y_2)$.

From (h), one obtains

$$\ell^{-1}(0; x, x) + \ell^{-1}(0; x, x) = \ell^{-1}(0, x, x) \qquad \text{i.e.}$$
$$\ell^{-1}(0; x, x) = 0.$$

It then follows from (i) that $\ell^{-1}(t; x, y)$ depends only on t:

$$\ell^{-1}(t; x, y) = \ell^{-1}(t)$$

and that $\ell^{-1}(t)$ is additive in t. One concludes that $\ell = (\ell^{-1}, \ell^0)$ is a morphism of complexes and, by (f) and (g), the arrows α form an isomorphism of additive functors $F \circ \mathrm{ch}(k) = \mathrm{ch}(\ell)$.

1.4.14. Let $\mathrm{Ch}^\flat(\mathcal{S})$ be the category whose objects are the small Picard stacks on $\mathcal{S}$, and whose arrows are the isomorphism classes of additive functors. The construction ch 529
defines a functor from $C^{[-1,0]}(\mathcal{S})$ (1.4.11) to $\mathrm{Ch}^\flat(\mathcal{S})$. Let $D^{[-1,0]}(\mathcal{S})$ be the subcategory of the derived category of $\mathcal{S}$ formed by the complexes K such that $H^i(K) = 0$ for $i \neq 0$ or -1. It follows from 1.4.12 and 1.4.13 that

Proposition 1.4.15. *The functor* ch *induces an equivalence of categories*

$$\mathrm{ch} : D^{[-1,0]}(\mathcal{S}) \xrightarrow{\sim} \mathrm{Ch}^\flat(\mathcal{S}).$$

We shall denote by $\flat$ the inverse equivalence of ch. For every Picard stack $\mathcal{P}$ on $\mathcal{S}$, $\mathcal{P}^\flat \in \mathrm{Ob}\, D^{[-1,0]}(\mathcal{S})$ determines $\mathcal{P}$ up to the isomorphism class of an equivalence.

Lemma 1.4.16. *Let $L \in \mathrm{Ob}\, C^{[-1,0]}(\mathcal{S})$ be such that L^{-1} is injective.*

(I) $\mathrm{pch}(L)$ *is already a stack.*

(II) *For $K \in \mathrm{Ob}\, C^{[-1,0]}(\mathcal{S})$, every additive functor $F : \mathrm{ch}(K) \to \mathrm{ch}(L)$ is isomorphic to a functor $\mathrm{ch}(f)$ for f a morphism from K to L.*

(I) For any $U \in \mathrm{Ob}\, \mathcal{S}$ and $x \in \mathrm{Ob}\, \mathrm{ch}(L)(U)$, the sheaf of pairs consisting of a local section ℓ of $L^0|U$ and an isomorphism between x and ℓ is a principal homogeneous space under L^{-1}; it therefore has a section, and $j : \mathrm{pch}(L) \to \mathrm{ch}(L)$ is surjective on the isomorphism classes of objects, hence is an equivalence.

(II) Let L' be a complex of injectives, equipped with a quasi-isomorphism $\varphi : L \underset{\approx}{\to} L'$, such that φ^i is an isomorphism for $i \leq -1$. By 1.4.13 and [17], there exists a diagram commutative up to homotopy

$$\begin{tikzcd} L \arrow[r, "\varphi", "\approx"'] & L' \\ K' \arrow[u, "f_0"] \arrow[r, "\Psi", "\approx"'] & K \arrow[u, "f_1"'] \end{tikzcd}$$

such that $F \simeq \mathrm{ch}(f_0)\,\mathrm{ch}(\Psi)^{-1}$. Moreover, $K = \tau_{\leq 0} K$ and $L = \tau_{\leq 0} L'$, so that f_1 factors through $f : K \to L$, and $F \simeq \mathrm{ch}(f)$.

The lemma refines 1.4.15 by the following consequence of 1.4.12.

530 Corollary 1.4.17. *The construction* ch *defines an equivalence of* 2*-categories between*

a) *the* 2*-category of Picard stacks on* $\mathcal{S}$.

b) *the* 2*-category whose objects and* 1*-arrows are the objects and arrows of the full subcategory of $C^{[-1,0]}(\mathcal{S})$ formed by the complexes L with L^{-1} injective, and whose* 2*-arrows are homotopies between arrows:* $\mathrm{Hom}(f,g) = \{H | g - f = \mathrm{dH} + \mathrm{Hd}\}$.

Construction 1.4.18. One has

$$\mathrm{HOM}(\mathcal{P}_1, \mathcal{P}_2)^\flat \simeq \tau_{\leq 0} R\,\mathbf{Hom}(\mathcal{P}_1^\flat, \mathcal{P}_2^\flat).$$

Let $K, L \in \mathrm{Ob}\, C^{[-1,0]}(S)$, with L^{-1} injective. By 1.4.16 (II) and 1.4.12, one has

$$\mathrm{ch}(\tau_{\leq 0}\, \mathbf{Hom}(K, L)) \xrightarrow{\sim} \mathrm{HOM}(\mathrm{ch}(K), \mathrm{ch}(L)) \tag{1.4.18.1}$$

and one deduces Construction 1.4.18 from this functorially.

Construction 1.4.19. For $f : \mathcal{S}_1 \to \mathcal{S}_2$ a morphism of sites, one has

$$(f_* \mathcal{P})^\flat \simeq \tau_{\leq 0}\, R f_*(\mathcal{P}^\flat).$$

Let $L \in \mathrm{Ob}\, C^{[-1,0]}$ with L^{-1} injective. By 1.4.16 (I), one has

$$\mathrm{ch}(f_* L) \xrightarrow{\sim} f_*\, \mathrm{ch}(L) \tag{1.4.19.1}$$

and from this one deduces a functorial construction 1.4.19.

We shall not need to use the following

Construction 1.4.20. The tensor product promised in 1.4.8 exists, and

$$(\mathscr{P}_1 \otimes \mathscr{P}_2)^\flat \simeq \tau_{\geq -1}\mathscr{P}_1^\flat \overset{\mathbf{L}}{\otimes} \mathscr{P}_2^\flat.$$

One checks, as in 1.4.12 and 1.4.13 (II), that for all K_1, K_2, and L in $C^{[-1,0]}(\mathscr{S})$, one has

(a) Every morphism $f \in \mathrm{Hom}(K_1 \otimes K_2, L) \simeq \mathrm{Hom}(\tau_{\geq -1}(K_1 \otimes K_2), L)$ defines a biadditive functor $\mathrm{ch}(f) \in \mathrm{Hom}(\mathrm{ch}(K_1), \mathrm{ch}(K_2); \mathrm{ch}(L))$; moreover, morphisms of functors identify with homotopies;

(b) For every biadditive functor $F : \mathrm{ch}(K_1), \mathrm{ch}(K_2) \to \mathrm{ch}(L)$, there exist quasi-isomorphisms $k_i : \mathrm{ch}(K_i') \to \mathrm{ch}(K_i)$ and $f : K_1' \otimes K_2' \to L$ such that $F \simeq \mathrm{ch}(f) \circ (\mathrm{ch}(k_1)^{-1}, \mathrm{ch}(k_2)^{-1})$.

If now K_1^0 or K_2^0 is flat, as is ${K'}_1^0$ or ${K'}_2^0$, then $k_1 \otimes k_2 : \tau_{\geq -1}{K'}_1 \otimes {K'}_2 \to$ 531
$\tau_{\geq -1}K_1 \otimes K_2$ is a quasi-isomorphism, and it follows that $\mathrm{ch}(\tau_{\geq -1}(K_1 \overset{\mathbf{L}}{\otimes} K_2))$ satisfies the universal property (1.4.8.1), whence the existence of $\otimes$ and of a functorial construction 1.4.20.

1.4.21. Let G be an abelian sheaf, and let $G[1]$ be the complex with value G placed in degree -1. The prestack $\mathrm{pch}(G[1])$ identifies with the prestack of trivial torsors under G, and hence $\mathrm{ch}(G[1])$ "is none other than" the stack of torsors under G, with its usual addition law.

1.4.22. Let $K \in C^{[-1,0]}(\mathscr{S})$, and let G be an abelian sheaf. The extensions E of K by G form a Picard stack, $\mathrm{EXT}(K, G)$, under Baer addition.

$$E : 0 \to G[0] \xrightarrow{\alpha} E \xrightarrow{\beta} K \to 0.$$

To every extension E one associates an additive functor from $\mathrm{ch}(K)$ to $\mathrm{ch}(G[1])$ which, when applied to a local section x of K^0, gives the torsor $\beta^{-1}(x)$. The reader will verify the following.

Proposition 1.4.23. *The construction outlined above is an equivalence of Picard stacks*

$$\mathrm{EXT}(K, G) \xrightarrow{\sim} \mathrm{HOM}(\mathrm{ch}(K), \mathrm{ch}(G[1])).$$

1.4.24. Let F be an object of the site $\mathscr{S}$. We shall also write F for the sheaf represented by F, and write $\mathbf{Z}^{(F)}$ for the abelian sheaf it generates. Let C be a Picard stack on $\mathscr{S}$. One checks easily (cf. 1.4.3) that giving any one of the following amounts to the same thing:

a) an additive functor $H : \mathrm{ch}(\mathbf{Z}^{(F)}) \to C$;
b) a morphism of *stacks*:
   (stack of local sections of F, with only identity morphisms) $\to C$;
c) an object of $C(F)$.

Suppose (in order to be able to apply Definition 1.4.9) that fiber products exist in $\mathscr{S}$, and let f be the canonical morphism of sites $f : \mathscr{S}/F \to \mathscr{S}$. The preceding construction gives an equivalence

$$\mathrm{HOM}(\mathrm{ch}(\mathbf{Z}^{(F)}), C) \xrightarrow{\sim} f_* f^* C. \tag{1.4.24.1}$$

The isomorphism obtained from (1.4.24.1) by the construction 532

$$\tau_{\leq 0} R\,\mathbf{Hom}(\mathbf{Z}^{(F)}, C^\flat) \xrightarrow{\sim} \tau_{\leq 0} Rf_* f^* C^\flat,$$

may also be deduced from the more general isomorphism

$$R\,\mathbf{Hom}(\mathbf{Z}^{(F)}, K) \xrightarrow{\sim} Rf_*f^*K \quad (K \in \operatorname{Ob} D^+(\mathcal{S})). \tag{1.4.24.2}$$

1.5. The universal coefficient formula.

1.5.1. Let $f : X \to S$ be a smooth projective curve over S. We denote by $\mathrm{PIC}(X/S)$ the Picard stack on the big fppf site of S (XVII 0.10), namely the direct image (1.4.9) of the stack of invertible sheaves on X. We have

$$\mathrm{PIC}(X/S) = f_*\,\mathrm{ch}\,\mathbf{G}_m[1] \tag{1.5.1.1}$$

$$\mathrm{PIC}(X/S)^\flat = \tau_{\le 0}Rf_*(\mathbf{G}_m[1]). \tag{1.5.1.2}$$

Each section t of X defines an invertible sheaf $\mathcal{O}(t)$ on X. With the notation of 1.4.24 (cf. 1.4.24 $a \leftrightarrow b$), this construction, being compatible with every base change, defines a morphism of Picard stacks on S

$$\mathrm{ch}\,\mathbf{Z}^{(X)} \longrightarrow \mathrm{PIC}(X/S). \tag{1.5.1.3}$$

If C is any Picard stack on S, the morphisms (1.5.1.3) and (1.4.24.1) define an additive functor

$$\mathrm{HOM}(\mathrm{PIC}(X/S), C) \longrightarrow \mathrm{HOM}(\mathrm{ch}(\mathbf{Z}^{(X)}), C) \xrightarrow{\sim} f_*f^*C. \tag{1.5.1.4}$$

Upon applying the construction $\flat$, the functor (1.5.1.3) defines a morphism[110]

$$\mathbf{Z}^{(X)} \longrightarrow \tau_{\le 0}Rf_*\mathbf{G}_m[1] \longrightarrow Rf_*\mathbf{G}_m[1], \tag{1.5.1.5}$$

and the analogue of the functor (1.5.1.4) is, by (1.4.24.2), a morphism

$$R\,\mathbf{Hom}(\tau_{\le 0}Rf_*(\mathbf{G}_m[1]), K) \longrightarrow Rf_*f^*K. \tag{1.5.1.6}$$

If C is the stack $\mathrm{ch}(G[1])$ of torsors under a group G satisfying (1.3.1.1), Theorem
533 1.3.10 asserts that the functor (1.5.1.4) is an equivalence, the inverse equivalence being the one that associates with a torsor K under G_X the additive functor $\langle *, K]$. The assertion that (1.5.1.4) is an equivalence amounts to saying that the morphism deduced from (1.5.1.4) or (1.5.1.6)

$$\tau_{\le 1}R\,\mathbf{Hom}(\tau_{\le 0}Rf_*(\mathbf{G}_m[1]), G) \longrightarrow \tau_{\le 1}Rf_*f^*G \tag{1.5.1.7}$$

is an isomorphism.

Theorem 1.5.2. (universal coefficient formula).
Let $f : X \to S$ be a smooth projective curve over S, let C be a Picard stack on S_{fppf}, and let $K = C^\flat$ (1.4.15). Suppose that K is locally isomorphic (in $D^+(S_{\mathrm{fppf}})$) to complexes of the form $G_{-1} \to G_0$, where the sheaves G_i satisfy (1.3.1.1). Then:

(I) *The functor (1.5.1.4)*

$$\mathrm{HOM}(\mathrm{PIC}(X/S), C) \longrightarrow f_*f^*C \tag{1.5.2.1}$$

is an equivalence of Picard stacks.

[110] N.D.E.: Here the shift functor [1] is to be understood to have higher precedence than the truncation $\tau_{\le 0}$.

(II) *The morphism deduced from (1.5.1.6) or (1.5.1.4)*

(1.5.2.2) $$\tau_{\leq 0} R\,\mathbf{Hom}(\tau_{\leq 0} Rf_*(\mathbf{G}_m[1]), K) \longrightarrow \tau_{\leq 0} Rf_* f^* K$$

is an isomorphism.

This theorem makes $\tau_{\leq 0} Rf_*(\mathbf{G}_m[1])$ play the role played by homology in the classical universal coefficient formula.

It is clear that, for every stack C, one has $(I) \leftrightarrow (II)$, and both (I) and (II) are local in nature on S. If C is of the form $\mathrm{ch}(G[1])$, then the theorem follows from 1.3.10, as noted above. It remains to show that, if K is of the form

$$K : d : G_{-1} \longrightarrow G_0,$$

and if (1.5.2.2) is an isomorphism for $G_{-1}[1]$ and $G_0[1]$, then (1.5.2.2) is an isomorphism for K. To verify this, it suffices to compare the cohomology long exact sequence

$$0 \longrightarrow R^{-1} f_* K \longrightarrow f_* G_{-1} \longrightarrow f_* G_0 \longrightarrow R^0 f_* K \longrightarrow R^1 f_* G_{-1} \longrightarrow R^1 f_* G_0$$

with the analogous long exact sequence for the left-hand side of (1.5.2.2), and to apply the five lemma. 534

1.5.3. Let u be a flat morphism of smooth projective curves over S

$$\begin{tikzcd} X \arrow[rr, "u"] \arrow[dr, "f_1"'] & & Y \arrow[dl, "f_2"] \\ & S & \end{tikzcd}.$$

The diagram of additive functors

(1.5.3.1) $$\begin{tikzcd} \mathbf{Z}^{(X)} \arrow[r] \arrow[d, "u"] & \mathrm{PIC}(X/S) \arrow[d, "N_{X/Y}"] \\ \mathbf{Z}^{(Y)} \arrow[r] & \mathrm{PIC}(Y/S) \end{tikzcd}$$

is then essentially commutative.

For every Picard stack C on S, the diagram

(1.5.3.2) $$\begin{tikzcd} \mathrm{HOM}(\mathrm{PIC}(Y/S), C) \arrow[r] \arrow[d, "\mathrm{HOM}(N_{X/Y}, C)"] & f_{2*} f_2^* C \arrow[d, "u^*"] \\ \mathrm{HOM}(\mathrm{PIC}(X/S), C) \arrow[r] & f_{1*} f_1^* C \end{tikzcd}$$

is therefore essentially commutative, and for $K \in \mathrm{Ob}\, D^+(S)$, the diagram

(1.5.3.3) $$\begin{tikzcd} R\,\mathbf{Hom}(\tau_{\leq 0} Rf_{2*}\mathbf{G}_m[1], K) \arrow[r] \arrow[d, "R\,\mathbf{Hom}(N_{X/Y}, K)"] & Rf_{2*} f_2^* K \arrow[d, "u^*"] \\ R\,\mathbf{Hom}(\tau_{\leq 0} Rf_{1*}\mathbf{G}_m[1], K) \arrow[r] & Rf_{1*} f_1^* K \end{tikzcd}$$

is commutative.

Now let G be an abelian sheaf on S satisfying (1.3.1.1). The trace XVII 6.3 gives us an additive functor

$$\mathrm{Tr}_u : f_{1*}f_1^* \operatorname{ch}(G[1]) \longrightarrow f_{2*}f_2^* \operatorname{ch}(G[1]) \tag{1.5.3.4}$$

535 and 1.3.8.6 gives us an isomorphism making the diagram essentially commutative

$$\begin{tikzcd} \mathrm{HOM}(\mathrm{PIC}(X/S), \operatorname{ch}(G[1])) \arrow[d, "\mathrm{HOM}(u^*, \operatorname{ch}(G[1]))"] & f_{1*}f_1^* \operatorname{ch}(G[1]) \arrow[l] \arrow[d, "\mathrm{Tr}_u"] \\ \mathrm{HOM}(\mathrm{PIC}(Y/S), \operatorname{ch}(G[1])) & f_{2*}f_2^* \operatorname{ch}(G[1]). \arrow[l] \end{tikzcd} \tag{1.5.3.5}$$

The diagram

$$\begin{tikzcd} \tau_{\leq 1} R\,\mathbf{Hom}(\tau_{\leq 0} Rf_{1*}\mathbf{G}_m[1], G) \arrow[d, "R\,\mathbf{Hom}(u^*, G)"'] & \tau_{\leq 1} Rf_{1*}f_1^* G \arrow[l, "\sim"'] \arrow[d, "Tr_u"] \\ \tau_{\leq 1} R\,\mathbf{Hom}(\tau_{\leq 0} Rf_{2*}\mathbf{G}_m[1], G) & \tau_{\leq 1} Rf_{2*}f_2^* G \arrow[l, "\sim"'] \end{tikzcd} \tag{1.5.3.6}$$

is therefore commutative.

1.5.4. With the notation of 1.5.2, let G be an abelian sheaf on S satisfying 1.3.1.1. Locally on S for the étale topology, one has (noncanonically)

$$\tau_{\leq 0} Rf_* \mathbf{G}_m[1] \simeq f_* \mathbf{G}_m[1] + R^1 f_* \mathbf{G}_m.$$

At the level of Picard stacks, one indeed defines a section $\operatorname{ch}(R^1 f_* \mathbf{G}_m) \to f_* \operatorname{ch}(\mathbf{G}_m[1])$[111] by associating with each isomorphism class of invertible sheaves the invertible sheaf belonging to that class, rigidified along suitable sections of X/S.

The isomorphism (1.5.2.2) therefore gives an isomorphism

$$\mathbf{Hom}(R^1 f_* \mathbf{G}_m, G) \xrightarrow{\sim} \mathbf{Hom}_S(X, G) \tag{1.5.4.1}$$

and an exact sequence

$$0 \longrightarrow \mathbf{Ext}^1(R^1 f_* \mathbf{G}_m, G) \longrightarrow R^1 f_* G \longrightarrow \mathbf{Hom}(f_* \mathbf{G}_m, G) \longrightarrow 0. \tag{1.5.4.2}$$

The morphisms in (1.5.4.1) and (1.5.4.2) are interpreted as follows:

536 a) The morphism (1.5.4.1) and the first morphism in (1.5.4.2) are defined by the canonical map from X to $\mathbf{Pic}_{X/S}$.
b) The second morphism in (1.5.4.2) is the degree (1.3.7.2).

Locally on S for the étale topology, one has (noncanonically) $\mathbf{Pic}_{X/S} \sim \mathbf{Pic}^0_{X/S} \times \mathbf{Z}^k$; hence

$$\mathbf{Ext}^1(R^1 f_* \mathbf{G}_m, G) \xrightarrow{\sim} \mathbf{Ext}^1(\mathbf{Pic}^0_{X/S}, G)$$

and (1.5.4.2) may also be written

$$0 \longrightarrow \mathbf{Ext}^1(\mathbf{Pic}^0_{X/S}, G) \longrightarrow R^1 f_* G \longrightarrow \mathbf{Hom}(f_* \mathbf{G}_m, G) \longrightarrow 0. \tag{1.5.4.3}$$

[111] N.D.E.: Recall that this stack identifies with $\mathrm{PIC}(X/S)$, cf. (1.5.1.1).

1.5.5. When, in (1.5.2.2), one takes $K = \mathbf{G}_m[1]$, Theorem 1.5.2 appears as a self-duality theorem for $\tau_{\le 0}(Rf_*\mathbf{G}_m[1])$. Any complex representing $\tau_{\le 0}Rf_*\mathbf{G}_m[1]$ admits a canonical three-step filtration, whose successive quotients are, up to unique quasi-isomorphism,

$$\mathbf{Pic}_{X/S}/\mathbf{Pic}^0_{X/S} \quad , \quad \mathbf{Pic}^0_{X/S} \quad , \quad (f_*\mathbf{G}_m)[1].$$

The preceding self-duality corresponds to a duality (with values in $\mathbf{G}_m$) between $\mathbf{Pic}_{X/S}/\mathbf{Pic}^0_{X/S}$ and $f_*\mathbf{G}_m$, and to a self-duality (with values in $\mathbf{G}_m[1]$) on $\mathbf{Pic}^0_{X/S}$ (self-duality of the Jacobian).

1.5.6. The situation is distinctly worse for smooth curves over S : $f : X \to S$ that are not assumed proper over S. The additive group seems to behave uncontrollably. The preceding arguments nevertheless extend to the case of smooth curves obtained from a proper curve over S by removing a part finite over S, when G is an étale torsion sheaf whose torsion is prime to the residue characteristics of S. One may hope that étale p-torsion sheaves also behave reasonably; see Serre [14, Ch. VI n° 11 p. 126].

The role of Theorem 1.2.2 will here be played by the acyclicity theorem XV 2.2, which 537
has the following as a corollary.

Lemma 1.5.7. *Let $p : X \to S$ be the projection of a standard affine space over S, and let G be an abelian torsion sheaf on S_{et}, of torsion prime to the residue characteristics of S. We also write G for the inverse image of G on the site S_{fppf}. Then the functor p^* is an equivalence of categories between the category of torsors under G on S_{et} (or on S_{fppf}, which amounts to the same thing, by Grothendieck* [5, Corollaire 11.9]*), and the category of torsors under G_X on X.*

1.5.8. Let $\overline{f} : \overline{X} \to S$ be a proper morphism of finite presentation whose fibers are purely one-dimensional, let Y be a closed subscheme of $\overline{X}$ finite over S, defined by an ideal $\mathfrak{m}$, and suppose that $X = \overline{X} - Y$ is a smooth curve over S. Let furthermore G be a torsion sheaf on S_{et}, of torsion prime to the residue characteristics of S. We shall also write G for the sheaf on the big fppf site of S (XVII 0.10) obtained as the inverse image of G.

Let $\mathbf{G}_m(\mathfrak{m})$ be the subsheaf of $\mathbf{G}_{m\overline{X}}$ consisting of sections equal to 1 on Y. A torsor under $\mathbf{G}_m(\mathfrak{m})$ identifies with an invertible sheaf on $\overline{X}$ trivialized on Y. Every relative divisor on X, finite over S, defines such an invertible sheaf. Locally on S for the étale topology, there exist such divisors giving rise to relatively ample invertible sheaves.

Lemma 1.5.9. *Under the hypotheses of 1.5.8, with S affine*[112]*, let $\mathscr{O}_{\overline{X}}(1)$ be a relatively ample invertible sheaf on $\overline{X}$, trivialized along Y, and let $\mathscr{M}$ be an invertible sheaf on $\overline{X}$, trivialized along Y. For every point $s \in S$, there exists an integer n_0 such that, for every integer $n \ge n_0$ and every section m_s of $\mathscr{M}(n) \otimes k(s)$ on the fiber $\overline{X}_s$ which is equal to 1 on Y_s, there exists a section m of $\mathscr{M}(n)$ which is equal to 1 on Y and lifts m_s.*

We reduce to the case where S is Noetherian (since a family of surjective morphisms remains a family of surjective morphisms after base change[113]). Let $\mathscr{N}$ be the coherent

[112]N.D.E.: It is in fact necessary to suppose that S is local with closed point s. If one wishes to retain the original hypothesis, then in the conclusion one should require that m be defined only over the inverse image of an open neighborhood of s in S, as occurs below in 1.5.10.1.

[113]N.D.E.: Eliminating this Noetherian hypothesis is somewhat more subtle than might appear at first sight, because the question is affine rather than linear. If $\overline{X}$, Y, and $\mathscr{M}$ are obtained by a base change $S \to S_0$ (which is assumed to be a local morphism between local schemes) from corresponding data $\overline{X}_0$, Y_0, and $\mathscr{M}_0$ over a Noetherian S_0 (cf. EGA IV 8), and if the desired assertion is known in this Noetherian

538 sheaf of local sections of $\mathscr{M}$ vanishing on $\overline{X}_s$ and Y. For each n, the local solutions to the problem form a principal homogeneous space under $\mathscr{N}(n)$, so that the obstruction to lifting m_s lies in $H^1(X, \mathscr{N}(n)) = H^0(S, R^1\overline{f}_*\mathscr{N}(n))$. But, for n sufficiently large, $R^1\overline{f}_*\mathscr{N}(n) = 0$ (EGA III 2.2.1).

1.5.10. Let $\mathscr{L}$ be an invertible sheaf on $\overline{X}$ trivialized along Y, and let $s \in S$. Consider the conditions

. Every section of $\mathscr{L} \otimes k(s)$ on X_s which is equal to 1 on Y_s lifts to a section of $\mathscr{L}$ on the inverse image U of a neighborhood of s, equal to 1 on $Y \times_S U$.

. There exist four sections m_i of $\mathscr{L} \otimes k(s)$ on X_s, equal to 1 on Y, and linearly independent at the generic point of every irreducible component of X_s.

Consider the following conditions, concerning respectively a section m of $\mathscr{L}$, a pair (m_1, m_2), and a triple (m_1, m_2, m_3) of sections of $\mathscr{L}$.

. m is equal to 1 on Y, and the subscheme $Z(m)$ of X defined by $m = 0$ is a relative divisor over S (which means that it is quasi-finite over S, and hence finite over S, since $X = \overline{X} - Y$ is smooth over S).

. Let λ be the canonical coordinate on the affine line $g : \mathrm{E}^1_s \to S$. The section $\lambda m_1 + (1-\lambda)m_2$ of $g'^*\mathscr{L}$ satisfies (1.5.10.3).

. Similarly, $\lambda m_1 + \mu m_2 + (1-\lambda-\mu)m_3$ satisfies (1.5.10.3).

We leave it to the reader to verify, by a general-position argument[114], the following

LEMMA 1.5.11. *Let $\mathscr{L}$ be an invertible sheaf on X, trivialized along Y, and let $\overline{s}$ be a geometric point of S with image s in S. If $\mathscr{L}$ satisfies (1.5.10.1) and (1.5.10.2) at s, then:*

(I) *If n sections m_i of $\mathscr{L}$ satisfy (1.5.10.3), there exist an étale neighborhood U of $\overline{s}$ and a section k of $\mathscr{L}|\overline{f}^{-1}(U)$ such that the pairs (m_i, k) satisfy (1.5.10.4).*

539 (II) *Let m_1, m_2, k_1, k_2 be four sections of $\mathscr{L}$ such that the pairs (m_i, k_j) satisfy (1.5.10.4). There then exist an étale neighborhood U of $\overline{s}$ and a section k of $\mathscr{L}|\overline{f}^{-1}(U)$ such that the triples (m_i, k_j, k) satisfy (1.5.10.5).*

When $k(s)$ is infinite, one could simply take U to be a Zariski neighborhood.

1.5.12. Now let $\mathscr{L}$ be an invertible sheaf on $\overline{X}$ rigidified along Y, and let K be a G-torsor on X, where G is as in 1.5.7. We propose to define a G-torsor $\langle \mathscr{L}, K]$ on S, generalizing the one studied in 1.3, and additive in $\mathscr{L}$.

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situation, then one can obtain the desired result over S at the cost of an unpleasant diagram chase, during which one may have to consider a variant of the desired assertion in which the sections are required to be equal to 0 instead of 1 on Y_s or Y.

[114]N.D.E.: It should be observed here that, in view of (1.5.10.1) and the fact that the second part of condition (1.5.10.3) can be checked in the local situation by considering only the fiber over s, the lemma now involves only sections of the restriction of $\mathscr{L}$ to the fiber X_s. For example, for (I), condition (1.5.10.2) provides an *affine* space V of dimension 3 of sections of $\mathscr{L} \otimes k(s)$ on X_s which are equal to 1 on Y_s, and no section belonging to this space V vanishes at any generic point of X_s. We seek to determine $k \in V$ such that, for every i, (m_i, k) satisfies (1.5.10.2), that is, such that the affine line joining the points m_i and k does not pass through the origin (in any of its incarnations associated with the different irreducible components of X_s); in other words, it suffices that k not belong to the line joining m_i and the origin. One thus obtains a conjunction of conditions satisfied on nonempty open subsets of V. To be completely rigorous and to account for the fact that the irreducible components need not be geometrically irreducible, when one must take a point of V in an extension (if $k(s)$ is finite), it is in fact necessary to introduce a finite Galois extension $s' \to s$ such that the irreducible components of $X_{s'}$ are geometrically irreducible. The dense open subset of $V_{s'}$ obtained by arguing on $X_{s'}$ is Galois-invariant and hence defines the desired dense open subset of V.

Let (m_1, m_2) be a pair of sections of $\mathscr{L}$ that satisfies (1.5.10.4). With the notation of 1.5.10.3 and 1.5.10.4, the trace, from $\lambda m_1 + (1-\lambda)m_2 = 0$ to $\mathbf{E}^1_S$, of the torsor g'^*K is a torsor $K_{1,2}$ on $\mathbf{E}^1_S$. Its pullback by the section $\lambda = 0$ (resp. $\lambda = 1$) is the torsor $K_2 = \mathrm{Tr}_{Z(m_2)/S}(K)$ (resp. $K_1 = \mathrm{Tr}_{Z(m_1)/S}(K)$). By 1.5.7, the torsor $K_{1,2}$ is canonically the pullback of a torsor under G on S; hence there is a canonical isomorphism between K_1 and K_2.

If m_1, m_2, and k are three sections of $\mathscr{L}$ such that the pairs (m_i, k) satisfy (1.5.10.4), composition again gives an isomorphism

$$\Psi_k : K_1 = \mathrm{Tr}_{Z(m_1)/S}(K) \xrightarrow{\sim} K_2 = \mathrm{Tr}_{Z(m_2)/S}(K). \tag{1.5.12.1}$$

If E is a relative divisor on X, finite over S, the sections $m_i' = m_i \otimes 1$ of $\mathscr{L} \otimes \mathscr{O}(E)$ still satisfy (1.5.10.3), and we have (for $i = 0, 1$)

$$K_i' = \mathrm{Tr}_{Z(m_i')/S}(K) \simeq K_i + \mathrm{Tr}_{E/S}(K) \tag{1.5.12.2}$$

and the isomorphism $\Psi_{k\otimes 1} : K_1' \to K_2'$ is deduced from Ψ_k:

$$\Psi_{k\otimes 1} = \Psi_k + \mathrm{id}_{\mathrm{Tr}_{E/S}}(K). \tag{1.5.12.3}$$

To verify that Ψ_k does not depend on k, it suffices to do so in an étale neighborhood of every point s of S, and, by 1.5.12.3 and 1.5.9, we reduce to the case in which condi- 540
tions (1.5.10.1) and (1.5.10.2) hold. With the notation of 1.5.11 (II), $\Psi_{\lambda k_i + (1-\lambda)k} - \Psi_{k_i}$ is a morphism from $\mathbf{E}^1_S$ to G, zero at $\lambda = 1$, and hence zero; thus $\Psi_{k_i} = \Psi_k$, and therefore $\Psi_{k_1} = \Psi_{k_2}$.

Let m_1 and m_2 be two sections of $\mathscr{L}$ satisfying 1.5.10.3, and let $K_i = \mathrm{Tr}_{Z(m_i)/S}(K)$. To define an isomorphism $\Psi_{m_2,m_1} : K_1 \to K_2$, it suffices to do so in an étale neighborhood of every point s of S, compatibly with base change. If E is a relative divisor on X, finite over S, such that $\mathscr{L} \otimes \mathscr{O}(E)$ satisfies (1.5.10.1) and (1.5.10.2) (see 1.5.9), and if the pairs $(m_i \otimes 1, k)$ satisfy (1.5.10.4), we set (cf. (1.5.12.3))

$$\Psi_{m_2,m_1} + \mathrm{Tr}_{E/S}(K) = \Psi_k.$$

This defines Ψ_{m_2,m_1}, and one verifies using (1.5.11. (I)) that $\Psi_{m_3,m_2}\Psi_{m_2,m_1} = \Psi_{m_3,m_1}$. These constructions are compatible with base changes. If there exists a section m of $\mathscr{L}$ satisfying (1.5.10.3), we set

$$\langle \mathscr{L}, K] = \mathrm{Tr}_{Z(m)/S}(K)$$

and, by what precedes, this torsor is independent, up to canonical isomorphism, of the choice of m.

In the general case, note that to define $\langle \mathscr{L}, K]$ it suffices to define it in an étale neighborhood of every point s of S, compatibly with base change. If, as above, $\mathscr{L} \otimes \mathscr{O}(E)$ satisfies (1.5.10.1) and (1.5.10.2), then $\mathscr{L} \otimes \mathscr{O}(E)$ has sections satisfying (1.5.10.3), and we set

$$\langle \mathscr{L}, K] = \langle \mathscr{L} \otimes \mathscr{O}(E), K] - \mathrm{Tr}_{E/S}(K).$$

1.5.13. Under the hypotheses of 1.5.8, if x and y are two sections of $\overline{f}_*\mathbf{G}_m(\mathfrak{m})$, then $\lambda x + (1-\lambda)y$ is again a section for λ in an open subset with nonempty fibers of the affine line over S.

This can be expressed by saying that

. "$\overline{f}_*\mathbf{G}_m(\mathfrak{m})$ is path-connected."

Every morphism from an open subset with nonempty fibers of the affine line to G is constant. There is therefore no nontrivial homomorphism from $\overline{f}_*\mathbf{G}_m(\mathfrak{m})$ to G, and the automorphisms of $\mathscr{L}$ act trivially on $\langle \mathscr{L}, K]$. The torsor $\langle \mathscr{L}, K]$ thus depends only on K and on the section of $R^1\overline{f}_*\mathbf{G}_m(\mathfrak{m})$ defined by $\mathscr{L}$. By globalization, this makes it possible to define a symbol $\langle \lambda, K]$ for λ a section of $R^1\overline{f}_*\mathbf{G}_m(\mathfrak{m})$. The torsor $\langle \lambda, K]$ is additive in λ, and therefore corresponds to an extension $e(K)$ of $R^1\overline{f}_*\mathbf{G}_m(\mathfrak{m})$ by G (see 1.4.23 or SGA 7 VII 1.1.6 and 1.2). 541

The formation of $e(K)$ is compatible with every base change and with addition of torsors, and hence gives an additive functor

$$e : \overline{f}_* \operatorname{ch}(G[1]) \longrightarrow \operatorname{EXT}(R^1\overline{f}_*\mathbf{G}_m(\mathfrak{m}), G).$$

Let j be the canonical map from X into $R^1\overline{f}_*\mathbf{G}_m(\mathfrak{m})$, $j : t \to$ (class of $\mathscr{O}(t)$). Pullback by j is an additive functor

$$j^* : \operatorname{EXT}(R^1\overline{f}_*\mathbf{G}_m(\mathfrak{m}), G) \longrightarrow \overline{f}_* \operatorname{ch}(G[1])$$

and trivially $j^*e \sim \mathrm{Id}$. One verifies as in 1.3.10 and 1.5.2 that, for the fppf topology,

Proposition 1.5.14. *Under the hypotheses of 1.5.8, the functors e and j of 1.5.13 are quasi-inverse equivalences. It follows from 1.4.19 and 1.4.23 that there is an isomorphism*

$$\operatorname{Ext}^1(R^1\overline{f}_*\mathbf{G}_m(\mathfrak{m}), G) \xrightarrow{\sim} H^1(X, G).$$

1.5.15. Let $f_i : X_i \to S$ ($i = 1, 2$) be two compactified curves as in 1.5.8.

Let $\overline{u} : \overline{X}_1 \to \overline{X}_2$ be a morphism such that Y_1 is a subscheme of the inverse image of Y_2 and which induces a finite flat morphism from X_1 to X_2. Then, for $\mathscr{L}$ an invertible sheaf on $\overline{X}_2$ rigidified along Y_2 and K a torsor on X_1, we have

$$\langle \mathscr{L}, \operatorname{tr}_{X_1/X_2}(K)] \xrightarrow{\sim} \langle \overline{u}^*\mathscr{L}, K],$$

542 and we deduce the commutativity of the diagram

$$\begin{tikzcd}
\operatorname{Ext}^1(R^1\overline{f}_{1*}\mathbf{G}_m(\mathfrak{m}_1), G) \arrow[r, "\sim"] \arrow[d, "\operatorname{Ext}^1(\overline{u}^*, \mathrm{Id})"] & H^1(X_1, G) \arrow[d, "\operatorname{Tr}_{X_1/X_2}"] \\
\operatorname{Ext}^1(R^1\overline{f}_{2*}\mathbf{G}_m(\mathfrak{m}_2), G) \arrow[r, "\sim"] & H^1(X_2, G).
\end{tikzcd} \tag{1.5.15.1}$$

If $\overline{u} : \overline{X}_1 \to \overline{X}_2$ is finite and flat, and satisfies $Y_1 = \overline{u}^{-1}Y_2$ (as schemes), then, for $\mathscr{L}$ an invertible sheaf on X_1 rigidified along Y_1 and K a torsor on X_2, we have

$$\langle \mathscr{L}, u^*K] \xrightarrow{\sim} \langle N_{\overline{X}_1/\overline{X}_2}\mathscr{L}, K]$$

and we deduce the commutativity of the diagram

$$\begin{tikzcd}
\operatorname{Ext}^1(R^1\overline{f}_{2*}\mathbf{G}_m(\mathfrak{m}_2), G) \arrow[r, "\sim"] \arrow[d, "\operatorname{Ext}^1(N_{\overline{X}_1/\overline{X}_2}, \mathrm{Id})"] & H^1(X_2, G) \arrow[d, "u^*"] \\
\operatorname{Ext}^1(R^1\overline{f}_{1*}\mathbf{G}_m(\mathfrak{m}_1), G) \arrow[r, "\sim"] & H^1(X_1, G).
\end{tikzcd} \tag{1.5.15.2}$$

One could of course state this result more precisely in terms of Picard stacks.

1.6. A vanishing theorem. The present section, from 1.6.6 (2^{nd} proof) to the end, can be read independently of sections 2 to 5.

LEMMA 1.6.1. *Let X be a scheme and Y a closed subscheme of X with complement U, defined by an ideal $\mathfrak{m}$.*

$$U \overset{j}{\hookrightarrow} X \overset{i}{\hookleftarrow} Y.$$

Let $\mathbf{G}_m(\mathfrak{m})$ be the fpqc *sheaf whose sections on an X-scheme X_1 are the sections of $\mathscr{O}^*_{X_1}$ equal to 1 on the inverse image of Y. For $n \geq 1$ invertible on X, the sequence*

$$0 \to j_!\boldsymbol{\mu}_n \to \mathbf{G}_m(\mathfrak{m}) \xrightarrow{n} \mathbf{G}_m(\mathfrak{m}) \to 0$$

is an exact sequence of sheaves on the big étale site (XVII 0.10) of X.

Let i be the morphism of big étale sites $i : Y \to X$ induced by i: 543

$$i^*(X_1/X) = X_1 \times_X Y.$$

The functor i_* is exact. In the diagram

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & j_!\boldsymbol{\mu}_n \arrow[r] \arrow[d] & \mathbf{G}_m(\mathfrak{m}) \arrow[r] \arrow[d] & \mathbf{G}_m(\mathfrak{m}) \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \boldsymbol{\mu}_n \arrow[r] \arrow[d] & \mathbf{G}_m \arrow[r] \arrow[d] & \mathbf{G}_m \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & i_*\boldsymbol{\mu}_n \arrow[r] \arrow[d] & i_*\mathbf{G}_m \arrow[r] \arrow[d] & i_*\mathbf{G}_m \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd} \tag{1.6.1.1}$$

the columns are exact, as are the last two rows (IX 3.2). The first row is therefore exact.

1.6.2. Let k be a field and

$$G_1' \longrightarrow G_2' \longrightarrow G \longrightarrow G_2'' \longrightarrow G_1''$$

an exact sequence of abelian sheaves on the big fppf site of $\operatorname{Spec}(k)$. Recall that if G_1' and G_2' are representable and of finite type, and G_2'' and G_1'' are representable and locally of finite type, then G is representable and locally of finite type[115].

[115]N.D.E.: In the case where $G_1' = G_2' = 0$, this is clear. In the case where $G_1'' = G_2'' = 0$, using SGA 3 V 10.1.1 (new edition), one can reduce to the case where $G_1' \to G_2'$ is a monomorphism, which is treated in (SGA 3 $\mathrm{VI_A}$ 3.2). We are thus reduced to the case where $G_1' = G_1'' = 0$. One can then regard the sheaf G as a torsor under G_2' over the scheme G_2''. If G_2' is assumed affine (the situation to which one is reduced, by (SGA 3 $\mathrm{VI_B}$ 11.17), if before making the reductions G_1' and G_2' were assumed affine), then the representability of G follows immediately from faithfully flat descent. This special case is sufficient for the applications made in this paragraph. If G_2' is quasi-projective, the representability of G can be obtained by using (SGA 1 VIII 7.7) and arguments similar to those of [11, Lemma 3.3.1]. In fact, every group scheme G of finite type over a field is quasi-projective: one may suppose that G is connected; this is then [2, Corollary 1.2] if G is smooth, and (M. Raynaud) the general case reduces to it because, by (SGA 3 $\mathrm{VII_A}$ 8.3), there exists a finite morphism from G to a smooth group scheme.

If P is a connected commutative algebraic group of finite type over k, and n is an integer invertible in k, the sequence

$$0 \to {}_nP \to P \xrightarrow{n} P \to 0 \tag{1.6.2.1}$$

is exact. If G is a finite étale commutative group over k, killed by n, the first group and the last arrow of the exact sequence

$$\mathrm{Hom}(P, G) \to \mathrm{Hom}({}_nP, G) \xrightarrow{j} \mathrm{Ext}^1(P, G) \xrightarrow{n} \mathrm{Ext}^1(P, G)$$

are zero, whence an isomorphism

$$j : \mathrm{Hom}({}_nP, G) \xrightarrow{\sim} \mathrm{Ext}^1(P, G). \tag{1.6.2.2}$$

544 1.6.3. Under the hypotheses of 1.6.1, suppose that X is proper over the spectrum of a field k

$$p : X \longrightarrow k.$$

The exact sequence

$$0 \longrightarrow \mathbf{G}_m(\mathfrak{m}) \longrightarrow \mathbf{G}_m \longrightarrow i_*\mathbf{G}_m \longrightarrow 0$$

defines an exact sequence of fppf sheaf cohomology on $\mathrm{Spec}(k)$, which is easily interpreted directly as

$$0 \to p_*\mathbf{G}_m(\mathfrak{m}) \to p_*\mathbf{G}_{m_X} \to (pi)_*\mathbf{G}_{m_Y} \xrightarrow{j} R^1p_*\mathbf{G}_m(\mathfrak{m}) \to \mathbf{Pic}_{X/k} \to \mathbf{Pic}_{Y/k}\,. \tag{1.6.3.1}$$

The sheaves $p_*\mathbf{G}_{m_X}$ and $(pi)_*\mathbf{G}_{m_Y}$ are representable by group schemes of finite type and affine. The sheaves $\mathbf{Pic}_{X/k}$ and $\mathbf{Pic}_{Y/k}$ are representable by group schemes locally of finite type. By 1.6.2, the sheaves $p_*\mathbf{G}_m(\mathfrak{m})$ and $R^1p_*\mathbf{G}_m(\mathfrak{m})$ are representable. By (1.5.13.1), the former is represented by a connected group scheme (of finite type); the latter is locally of finite type.

Definition 1.6.4. *Under the preceding hypotheses, we denote by* $\mathbf{Pic}_{\mathfrak{m},X/k}$ *the group scheme representing* $R^1p_*\mathbf{G}_m(\mathfrak{m})$. *We denote by* $\mathbf{Pic}^0_{\mathfrak{m},X/k}$ *its identity component and by* $\mathbf{Pic}^\tau_{\mathfrak{m},X/k}$ *the inverse image in* $\mathbf{Pic}_{\mathfrak{m},X/k}$ *of the torsion subgroup of* $\mathbf{Pic}_{\mathfrak{m},X/k}/\mathbf{Pic}^0_{\mathfrak{m},X/k}$.

The scheme $\mathbf{Pic}_{\mathfrak{m},X/k}$ represents the fppf sheaf generated by the presheaf which associates with every scheme S over k the set of isomorphism classes of invertible modules on $S \times_k X$, trivialized along $S \times_k Y$ (1.5.8).

The Kummer exact sequence 1.6.1 gives, for k algebraically closed and n invertible in k, an isomorphism

$$H^1_c(U, \boldsymbol{\mu}_n) \xrightarrow{\sim} {}_n\mathbf{Pic}^\tau_{\mathfrak{m},X/k}(k). \tag{1.6.4.1}$$

545 1.6.5. Let X be a smooth curve over an algebraically closed field k, let $\overline{X}$ be the smooth projective curve containing X as a dense open subscheme, let Y be a closed subscheme of $\overline{X}$ such that $X = \overline{X} - Y$, let $\mathfrak{m}$ be the ideal defining Y, let n be an integer invertible in k, and let G be a finite abelian group killed by n.

The group $\mathbf{Pic}_{\mathfrak{m},X/k}$ is noncanonically the product of the subgroup $\mathbf{Pic}^0_{\mathfrak{m},\overline{X}/k}$ and a free abelian group. By (1.6.4.1), (1.6.2.2), and 1.5.14, we therefore have a canonical composite isomorphism

$$\begin{aligned}\mathrm{Hom}(H^1_c(X,\boldsymbol{\mu}_n),G) &\xleftarrow{\sim} \mathrm{Hom}({}_n\,\mathbf{Pic}^{\tau}_{\mathfrak{m},\overline{X}/k}(k),G) = \mathrm{Hom}({}_n\,\mathbf{Pic}^{0}_{\mathfrak{m},\overline{X}/k},G)\\ &\xrightarrow{\sim} \mathrm{Ext}^1(\mathbf{Pic}^0_{\mathfrak{m},\overline{X}/k},G) \xleftarrow{\sim} \mathrm{Ext}^1(\mathbf{Pic}_{\mathfrak{m},\overline{X}/k},G) \xrightarrow{\sim} H^1(X,G)\end{aligned}$$

If $u : X \to Y$ is a finite morphism between smooth curves over k, we deduce from XVII 6.3. and (1.5.15.2) the commutativity of the diagram

(1.6.5.1)
$$\begin{tikzcd} \mathrm{Hom}(H^1_c(Y,\boldsymbol{\mu}_n),G) \arrow[r, "\sim"] \arrow[d, "\mathrm{Hom}(\mathrm{Tr}_u{,}G)"] & H^1(Y,G) \arrow[d, "u^*"] \\ \mathrm{Hom}(H^1_c(X,\boldsymbol{\mu}_n),G) \arrow[r, "\sim"] & H^1(X,G). \end{tikzcd}$$

Lemma 1.6.6. *Let X be a smooth curve over an algebraically closed field k, let n be an integer invertible in k, and let $u : X' \to X$ be a tame étale covering. The morphism*

$$\mathrm{Tr}_u : H^1_c(X',\mathbf{Z}/n) \longrightarrow H^1_c(X,\mathbf{Z}/n)$$

is then (noncanonically[116]) isomorphic to the transpose of the morphism

$$u^* : H^1(X,\mathbf{Z}/n) \longrightarrow H^1(X',\mathbf{Z}/n).$$

1^{st} proof. *Since $\boldsymbol{\mu}_n$ is isomorphic to $\mathbf{Z}/n$, 1.6.6 is the special case of (1.6.5.1) with $G = \mathbf{Z}/n$.*

2^{nd} proof. If k is the field $\mathbf{C}$ of complex numbers, then, by the comparison theorems, the morphisms Tr_u and u^* are identified with the analogous morphisms defined by the continuous map of topological spaces $u : X'(\mathbf{C}) \to X(\mathbf{C})$:

$$\begin{aligned}\mathrm{Tr}_u &: H^1_c(X'(\mathbf{C}),\mathbf{Z}/n) \longrightarrow H^1_c(X(\mathbf{C}),\mathbf{Z}/n) \quad \text{and}\\ u^* &: H^1(X(\mathbf{C}),\mathbf{Z}/n) \longrightarrow H^1(X'(\mathbf{C}),\mathbf{Z}/n).\end{aligned}$$

These two homomorphisms are transposes of one another under Poincaré duality. **546**

By the Lefschetz principle, the assertion 1.6.6 remains true for k of characteristic 0 (the cohomology groups under consideration are indeed invariant under extension of the algebraically closed base field).

If k has characteristic $p > 0$, let $W(k)$ be the ring of Witt vectors over k. Let $\overline{X}$ be the smooth projective completion of X. By (SGA 1 III 7.4), the curve $\overline{X}$ lifts to a smooth projective curve $\overline{X}_1$ over $W(k)$. Since $\overline{X}$ is smooth, every point $s_i \in S = \overline{X} - X$ lifts to a section s'_i of $\overline{X}$ over $W(k)$; set $S_1 = \bigcup_i s'_i(\mathrm{Spec}(W(k)))$ and $X_1 = \overline{X}_1 - S_1$.

Recall (SGA 1 XIII 2.4) that the covering X' of X lifts to an étale covering X'_1 of X_1, and that X'_1 is obtained from a smooth projective curve $\overline{X}'_1$ over $W(k)$ by removing the

[116]N.D.E.: The isomorphism is canonical if an isomorphism $\boldsymbol{\mu}_n \simeq \mathbf{Z}/n$ has been chosen.

union S_1' of finitely many disjoint sections.

$$\begin{tikzcd}
 & X_1 \arrow[dd, "f_1"] & & X \arrow[ll] \arrow[dd] \\
X_1' \arrow[ur, "u_1"] \arrow[dr, "f_1'"'] & & X' \arrow[ll] \arrow[ur, "u"'] \arrow[dr] & \\
 & \operatorname{Spec}(W(k)) & & \operatorname{Spec}(k). \arrow[ll]
\end{tikzcd}$$

The étale sheaves $R^1 f_{1*}(\mathrm{Z}/n)$, $R^1 f'_{1*}(\mathrm{Z}/n)$, $R^1 f_{1!}(\mathrm{Z}/n)$, and $R^1 f'_{1!}(\mathrm{Z}/n)$ are locally
547 constant and their formation is compatible with base change, as follows from the exact sequences relating the cohomology of $\overline{X}_1$, X_1, and S_1, or of $\overline{X}'_1$, X_1', and S_1'.

The special or generic geometric fibers of the morphisms

$$\mathrm{Tr}_{u_1} : R^1 f'_{1!}\mathrm{Z}/n \longrightarrow R^1 f_{1!}\mathrm{Z}/n$$

or

$$u_1^* : R^1 f_{1*}\mathrm{Z}/n \longrightarrow R^1 f'_{1*}\mathrm{Z}/n$$

are therefore isomorphic, and this reduces us to the case already treated in which k has characteristic zero.

Lemma 1.6.7. *Let X be a connected smooth curve over an algebraically closed field k of characteristic exponent p, and let n be an integer prime to p. There exists a principal covering* (= finite étale Galois (surjective)) *$u : X' \to X$, of degree dividing a power of n, such that the trace morphism*

$$\mathrm{Tr}_u : H^1_c(X', \mathrm{Z}/n) \longrightarrow H^1_c(X, \mathrm{Z}/n)$$

is zero.

Recall that the group $H^1(X, \mathrm{Z}/n)$ is finite (IX 4.6 or XVI 5.2). Let $u : X' \to X$ be the largest connected abelian principal covering of X whose Galois group is killed by n (its Galois group is denoted by $H_1(X, \mathrm{Z}/n)$). By construction, the homomorphism

$$u^* : H^1(X, \mathrm{Z}/n) \longrightarrow H^1(X', \mathrm{Z}/n)$$

is zero. By 1.6.6, the homomorphism

$$\mathrm{Tr}_u : H^1_c(X', \mathrm{Z}/n) \longrightarrow H^1_c(X, \mathrm{Z}/n)$$

is therefore zero.

Recall the following lemma (EGA IV 15.6.5):

Lemma 1.6.8. *Let $f : X \to S$ be a smooth morphism and let x be a section of f. Then there exists a Zariski open subset U of X, containing x, such that the geometric fibers of $f|U$ are connected.*

548 Fundamental Lemma 1.6.9. *Let $f : X \to S$ be a compactifiable smooth curve (1.1.2)* (XVII 3.2)*, let x be a geometric point of X, let s be its image in S, and let $n \geq 1$ be an integer*

invertible on S. There exist an étale neighborhood V of s in S and an étale neighborhood U of x in $f^{-1}(V) = V \times_S X$

(1.6.9.1)
$$\begin{tikzcd}
x \arrow[r] \arrow[d, dash] & U \arrow[r, "u"] \arrow[dr, "f'"'] & f^{-1}(V) \arrow[r] \arrow[d, "f_v"] & X \arrow[d, "f"] \\
s \arrow[rr] & & V \arrow[r] & S,
\end{tikzcd}$$

such that $R^0 f'_! \mathbf{Z}/n = 0$, the arrow XVII 6.2

$$\mathrm{Tr}_u : R^1 f'_! \mathbf{Z}/n \longrightarrow R^1 f_{v!} \mathbf{Z}/n$$

is zero, and the trace morphism

$$\mathrm{Tr}_{f'} : R^2 f'_! \mathbf{Z}/n(1) \longrightarrow \mathbf{Z}/n$$

is an isomorphism.

For $R^0 f'_! \mathbf{Z}/n = 0$, it suffices that f' be quasi-affine; for $\mathrm{Tr}_{f'}$ to be an isomorphism, it suffices that the geometric fibers of f' be connected (1.1.9); by 1.6.8, these properties of f' are easy to obtain locally on S for the étale topology. It remains to prove, for X with geometrically connected fibers, the existence of a diagram (1.6.9.1) for which the arrow Tr_u under consideration is zero: it will suffice to shrink U and V so that the other two required properties of f' hold.

The problem is local on S in a neighborhood of s; by XVII 5.2.6 (base change), we immediately reduce to the case where S is Noetherian. By passage to the limit, and using XVII 5.3.6 (constructibility), we reduce to the case where S is strictly local Noetherian, with closed point the image of s.

Let t be a geometric point of S. By 1.6.7, there exists a principal covering of degree prime to the characteristic exponent p of $k(s)$, $u : X_t^1 \to X_t$, such that the arrow

$$\mathrm{Tr}_u : H_c^1(X_t^1, \mathbf{Z}/n) \longrightarrow H_c^1(X_t, \mathbf{Z}/n)$$

is zero. By the acyclicity theorem XV 2.1 (or XV 2.6), there exists an étale neighborhood 549
$v : U \to X$ of x in X such that X_t^1 splits over U_t, so that $v_t : U_t \to X_t$ factors through X_t^1. We therefore still have

$$\mathrm{Tr}_{v_t} = 0 : H_c^1(U_t, \mathbf{Z}/n) \longrightarrow H_c^1(X_t, \mathbf{Z}/n).$$

For every geometric point t of S, there therefore exists an étale neighborhood of x in X

$$\begin{tikzcd}
U \arrow[r, "v"] \arrow[dr, "f'"'] & X \arrow[d, "f"] \\
& S
\end{tikzcd}$$

such that the fiber at t of

(1.6.9.2)
$$\mathrm{Tr}_v : R^1 f'_! \mathbf{Z}/n \longrightarrow R^1 f_! \mathbf{Z}/n$$

is zero (XVII 5.2.6).

As U varies, the images of the arrows (1.6.9.2) form a filtered decreasing system of subsheaves of the constructible sheaf $R^1 f_! \mathbf{Z}/n$, which becomes zero at any given geometric point of S. By Noetherian induction, one verifies that such a system is always stationary, with stationary value zero. Hence there exists U such that (1.6.9.2) is zero, and this completes the proof of 1.6.9.

2. The trace morphism

550 This § is devoted to the proof of Theorems 2.9 and 2.12. The proof of 2.9 uses only no. 1 of § 1.

Lemma 2.1. *Let $f : X \to S$ be a compactifiable morphism* (XVII 3.2.1) *of relative dimension $\leq d$, and let $j : U \hookrightarrow X$ be an open subscheme of finite type of X, whose complement has fiberwise dimension $< d$ (for example, U is fiberwise dense in X). If F is a torsion sheaf on X, the canonical arrow*

$$R^{2d}(fj)_! j^* F \xrightarrow{\sim} R^{2d} f_! F$$

is an isomorphism.

Let i be the inclusion of $Y = X - U$ into X. The exact sequence (XVII 5.1.16.2) gives us an exact sequence

$$R^{2d-1}(fi)_! i^* F \longrightarrow R^{2d}(fj)_! j^* F \longrightarrow R^{2d} f_! F \longrightarrow R^{2d}(fi)_! i^* F$$

whose outer terms are zero by (XVII 5.2.8.1).

Lemma 2.2. *Let $f : X \to S$ be a compactifiable morphism of relative dimension $\leq d$, and let $u_i : X_i \to X$ be a family of separated étale morphisms of finite type; let $u_{ij} : X_{ij} = X_i \times_X X_j \to X$. For every torsion sheaf F on X, the sequence*

$$\bigoplus_{i,j} R^{2d}(fu_{ij})_! u_{ij}^* F \rightrightarrows \bigoplus_i R^{2d}(fu_i)_! u_i^* F \to R^{2d} f_! F \to 0$$

of arrows (XVII 6.2.7.2) *is exact.*

This follows immediately from the right exactness of the functor $R^{2d} f_!$ and from XVII 6.2.9.

Lemma 2.3. *Let $f : X \to S$ be a flat morphism of finite presentation. The open subset of X on which f is Cohen–Macaulay* (EGA IV 6.8.1 and 12.2.1 (vii) *is relatively of finite presentation and fiberwise dense.*

551 The question is local on S, which we reduce to assuming Noetherian, whence the 1[st] assertion. The second is checked fiberwise, hence with S the spectrum of a field. The local rings of X at the maximal points of X are Artinian, hence Cohen–Macaulay, which proves the assertion since "Cohen–Macaulay" is an "open" property.

Lemma 2.4. *Let $f : X \to S$ be a flat morphism of finite presentation and Cohen–Macaulay, with S quasi-separated. Every point of X has a (Zariski) open neighborhood U such that there exists a quasi-finite flat S-morphism of finite presentation from U to the standard vector bundle $\mathbf{E}_S^d$.*

This is a special case of EGA IV 15.4.3.

Lemma 2.5. *Let $\underline{u} = (u_1, \dots, u_d)$ and $\underline{v} = (v_1, \dots, v_d)$ be two systems of parameters of a Cohen–Macaulay local ring A of dimension d and maximal ideal $\mathfrak{m}$. There exists a sequence $\underline{w}_0, \dots, w_n$ of systems of parameters of A such that $\underline{w}_0 = \underline{u}$, $\underline{w}_n = \underline{v}$, and each $\underline{w}_{i+1}$ is obtained from $\underline{w}_i$ by changing only one of the parameters.*

We argue by induction on d, the assertion being vacuous for $d < 2$; we therefore suppose $d \geq 2$. For an element $x \in \mathfrak{m}$ to be A/u_1-regular (resp. A/v_1-regular), it is necessary and sufficient that x belong to no member of $\mathrm{Ass}(A/u_1)$ (resp. $\mathrm{Ass}(A/v_1)$); since no finite union of prime ideals $\neq \mathfrak{m}$ is equal to $\mathfrak{m}$, there exists $x \in \mathfrak{m}$ that is both A/u_1- and A/v_1-regular, and hence there also exist systems of parameters $\underline{a} = (u_1, x, \dots)$ and $\underline{b} = (v_1, x, \dots)$. We conclude by applying the induction hypothesis to $\underline{u} - \{u_1\}$ and $\underline{a} - \{u_1\}$ in A/u_1, to $\underline{a} - \{x\}$ and $\underline{b} - \{x\}$ in A/x, and to $\underline{b} - \{v_1\}$ and $\underline{v} - \{v_1\}$ in A/v_1.

Lemma 2.6. *Let X be a Cohen–Macaulay scheme of finite type over an algebraically closed field k, and let u, $v : X \to \mathbf{E}_k^d$ be two quasi-finite flat morphisms from X to a standard vector bundle. Every closed point x of X has a (Zariski) open neighborhood U such that there exists a sequence of morphisms $w_0, \dots, w_n : U \to \mathbf{E}_k^d$ such that $w_0 = u|U$,* 552
$w_n = v|U$, and w_{i+1} differs from w_i in only one coordinate.

Suppose first that $u(x) = v(x) = 0$. The germs at x of quasi-finite flat morphisms from X to $\mathbf{A}_k^d$, sending x to 0, then identify with the systems of parameters of the local ring $\mathscr{O}_{X,x}$, and 2.6 follows from 2.5. In the general case, one first joins u to $u - u(x)$ (resp. v to $v - v(x)$) by the chain of morphisms

$$u - (u_1(x), \dots, u_i(x), 0, \dots, 0)(\text{resp. } v - (v_1(x), \dots, v_i(x), 0 \dots 0)).$$

2.7. Let there be a pair of composable S-compactifiable morphisms $X \xrightarrow{g} Y \xrightarrow{f} S$. Suppose that f has relative dimension $\leq d$ and g has relative dimension $\leq e$. For every torsion sheaf F on X, the composition spectral sequence

$$R^p f_! R^q g_! F \Longrightarrow R^{p+q}(fg)!F \tag{2.7.1}$$

then gives us (by the maximum-cycle argument) an isomorphism

$$R^{2d} f_! R^{2e} g_! F \simeq R^{2(d+e)}(fg)_! F. \tag{2.7.2}$$

2.8. Let

$$a^d : \mathbf{E}_S^d \longrightarrow S$$

be the standard vector bundle over a coherent (i.e. quasi-compact, quasi-separated) scheme S. If F is a torsion sheaf on S, prime to the residual characteristics of S, we define as follows, by induction on d, a trace morphism

$$\mathrm{Tr}_{a^d} : R^{2d} a^d_! F(d) \xrightarrow{\sim} F. \tag{2.8.1}$$

For $d = 0$, Tr_{a^0} is the identity. For $d = 1$, Tr_{a^1} is the isomorphism (1.1.6) (cf. 1.1.9). Finally, there is a canonical isomorphism

$$\mathbf{E}_S^{d+1} = \mathbf{E}_S^1 \times_S \mathbf{E}_S^d = \mathbf{E}^1_{\mathbf{E}_S^d}$$

and we define $\mathrm{Tr}_{a^{d+1}}$ as the composite of the isomorphisms $\mathrm{Tr}_{a^d_S}$ and $Ra^d_{S!}(\mathrm{Tr}_{a^1_{\mathbf{E}_S^d}})$ (cf. diagram 553
Var 3 of 2.9).

The formation of the source (and the target) of (2.8.1) is compatible with every base change. Every algebraic group over S acting on $\mathbf{E}_S^d$ therefore acts on these sheaves; a connected algebraic group necessarily acts trivially. The group of coordinate permutations, contained in the affine group, therefore acts trivially on the source (and the target) of (2.8.1).

2.8.2. The isomorphism (2.8.1) is invariant under permutation of the coordinates.

This point could also be verified by interpreting (2.8.1) in terms of cup products (XVII 5.4.2.2 and XVII 5.4.3.5).

THEOREM 2.9. *Consider triples (f, d, F) consisting of a compactifiable morphism (XVII 6.3) $f : X \to Y$, an integer d, and a torsion sheaf F on Y prime to the residual characteristics of Y, where the morphism f satisfies the condition $(*)_d$: there exists an open subset U of X such that the restriction of f to U is a flat morphism of finite presentation with fibers of dimension $\leq d$, and such that the fibers of $X - U$ have dimension $< d$.*

There is one and only one way to associate with every triple (f, d, F) as above a trace morphism

$$\mathrm{Tr}_f : R^{2d} f_! f^* F(d) \longrightarrow F$$

such that the following conditions are satisfied:

(Var 1) *(Functoriality). The trace morphism is functorial in F.*

(Var 2) *(Compatibility with base change). For every Cartesian square*

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"] & X \arrow[d, "f"] \\ Y' \arrow[r, "g"] & Y \end{tikzcd}$$

554 *with Y' quasi-compact and quasi-separated, the diagram*

$$\begin{tikzcd} R^{2d} f'_! f'^* g^* F(d) \arrow[r, equal] \arrow[d, "Tr_{f'}"] & R^{2d} f'_! g'^* f^* F(d) & g^* R^{2d} f_! f^* F(d) \arrow[l, "\sim"'] \arrow[d, "g^*(Tr_f)"] \\ g^* F \arrow[rr, equal] & & g^* F \end{tikzcd}$$

in which the horizontal isomorphism is the base-change arrow XVII 5.2.2.1, is commutative.

(Var 3) *(Compatibility with composition). For every pair of composable Z-compactifiable morphisms*

$$X \xrightarrow{g} Y \xrightarrow{f} Z,$$

integers d and e such that f satisfies $(*)_d$ and g satisfies $(*)_e$, and torsion sheaf F on Z, the composite morphism fg satisfies $(*)_{d+e}$ and the diagram

$$\begin{tikzcd} R^{2d} f_! R^{2e} g_! g^* f^* F(d)(e) \arrow[r, "R^{2d} f_!(\mathrm{Tr}_g)"] \arrow[d, leftrightarrow, "\wr"'] & R^{2d} f_! f^* F(d) \arrow[d, "\mathrm{Tr}_f"] \\ R^{2(d+e)} (fg)_! (fg)^* F(d+e) \arrow[r, "\mathrm{Tr}_{fg}"] & F \end{tikzcd}$$

in which the vertical isomorphism is the arrow (2.7.2), is commutative.

(Var 4) *(Normalizations).*

(I) *If f is finite locally free of rank n and $d = 0$, the composite morphism*

$$F \longrightarrow f_* f^* F = f_! f^* F \xrightarrow{\mathrm{tr}_f} F$$

is multiplication by n.

555 (II) *If f is the standard affine line $f : \mathbf{E}^1_Y \to Y$, and if $d = 1$, the trace morphism*

$$\mathrm{Tr}_f : R^2 f_! f^* F(1) \longrightarrow F$$

is the isomorphism (2.8.1).

a) For $d = 0$, condition $(*)_d$ says that f is quasi-finite and flat of finite presentation. By (XVII 6.2.3), for $d = 0$, the trace morphism must coincide with the morphism constructed in *loc. cit.*

When f is the projection of standard affine space $\mathbf{A}^d_S$ onto S, it follows from (Var 4) (II) and (Var 3) that Tr_f must be the isomorphism (2.8.1).

b) Let $f : X \to Y$ be a compactifiable morphism satisfying the condition $(*)_d$: there exists a quasi-finite flat Y-morphism of finite presentation u from X to the standard vector bundle $a^d : \mathbf{E}^d_Y \to Y$.

Let $t(f, u)$ be the "composite" morphism of the trace morphisms (2.8.1) and (SGA XVII 6.2.3) (cf. (Var 3))

$$t(f, u) = \mathrm{Tr}_{a^d} \circ R^{2d} a^d_! (\mathrm{Tr}_u).$$

This morphism $t(f, u)$ satisfies (Var 1) and (Var 2). By base change, to verify that $t(f, u)$ depends only on f and d, and not on u, it suffices to verify this when Y is the spectrum of an algebraically closed field k. By 2.2, the question is local on X, so by 2.6, it suffices to show that $t(f, u) = t(f, v)$ when u and v differ in only one coordinate, which, by 2.8.2, may be assumed to be the first. Setting $E = \mathbf{E}^{d-1}_Y$, we thus have a commutative diagram

$$\begin{tikzcd}
 & & \mathbf{E}^1_E \arrow[dr, "a"] & & \\
X \arrow[urr, "u"] \arrow[rrrr, "v"] \arrow[rrr, "x"'] \arrow[dr, "f"'] & & & & \mathbf{E}^1_E \arrow[dl, "a"'] \\
 & & & E \arrow[dll, "w"] & \\
 & Y & & &
\end{tikzcd}$$

By construction (2.8), 556

$$\mathrm{Tr}_{wa} = \mathrm{Tr}_w \circ R^{2(d-1)} w_! (\mathrm{Tr}_a) \,;$$

we deduce that $t(f, u)$ is the "composite" of Tr_w and $t(x, u)$:

$$\begin{aligned}
t(f, u) &= \mathrm{Tr}_{wa} \circ R^{2d} (wa)_! (\mathrm{Tr}_u) \\
&= \mathrm{Tr}_w \circ R^{2(d-1)} w_! (\mathrm{Tr}_a) \circ R^{2d} (wa)_! (\mathrm{Tr}_u) \\
&= \mathrm{Tr}_w \circ R^{2(d-1)} w_! (\mathrm{Tr}_a \circ R^2 a_! (\mathrm{Tr}_u)).
\end{aligned}$$

Moreover (1.1.7), $t(x, u)$ is precisely the morphism Tr_x of 1.1.6; hence $t(x, u) = t(x, v) = \mathrm{Tr}_x$ and $t(f, u) = t(f, v)$.

We denote by Tr_f the morphism $t(f, u)$ constructed above. By XVII 6.2.3, this trace morphism satisfies (Var 1), (Var 2), (Var 3) for $e = 0$, and (Var 4). Whenever it is defined, it necessarily coincides with the trace morphism sought, if the latter exists.

c) Let $f : X \to Y$ be a compactifiable Cohen–Macaulay morphism, flat of finite presentation and pure of relative dimension d. Let $(U_i)_{i\in I}$ be a cover of X by quasi-compact open subsets such that there exists a quasi-finite flat Y-morphism from U_i to E^d_Y (2.4). By 2.2 and b), there is then a unique morphism Tr_f making the following diagram commutative, where the notation of 2.2 is used:

$$\begin{tikzcd}
\bigoplus_{i,j} R^{2d}(fu_{ij})_!(fu_{ij})^*F(d) \arrow[r, shift left] \arrow[r, shift right] & \bigoplus_i R^{2d}(fu_i)_!(fu_i)^*F(d) \arrow[r] \arrow[dr, "\bigoplus_i \mathrm{Tr}_{fu_i}"'] & R^{2d}f_!f^*F(d) \arrow[d, "\mathrm{Tr}_f"] \\
 & & F
\end{tikzcd}$$

This trace morphism does not depend on the chosen open cover (compare the two with the sum cover); it satisfies the conditions (Var 1), (Var 2), (Var 3) for $e = 0$, and (Var 4), as follows immediately from b).

d) Let $f : X \to Y$ be a compactifiable morphism satisfying $(*)_d$, and let $j :$
557 $U \hookrightarrow X$ be the open subset promised by $(*)_d$. By 2.3, after replacing U by a smaller open subset, we may suppose that fj is Cohen–Macaulay and pure of dimension d.
By 2.1, the canonical morphism

$$R^{2d}(fj)_!(fj)^*F(d) \xrightarrow{\sim} R^{2d}f_!f^*F(d)$$

is an isomorphism, and we define Tr_f to be the composite

$$\mathrm{Tr}_f : R^{2d}f_!f^*F(d) \xleftarrow{\sim} R^{2d}(fj)_!(fj)^*F(d) \xrightarrow{\mathrm{Tr}_{fj}} F.$$

It is immediate, by (Var 3) for $e = 0$ established in c), that Tr_f does not depend on the choice of U. The trace morphism satisfies (Var 1), (Var 2), (Var 3) for $e = 0$, and (Var 4), and is the only one that can satisfy the imposed conditions. It remains to prove that it satisfies (Var 3), and for this purpose the preceding arguments reduce us to considering only pairs (f, d), (g, e) satisfying $(*)_d$ and $(*)_e$.

The reader who wishes to polish the arguments that follow will first verify the

Lemma 2.9.1. *Let (f, g, h) be three composable T-compactifiable morphisms, satisfying $(*)_k$, $(*)_\ell$, and $(*)_m$.*

$$X \xrightarrow{h} Y \xrightarrow{g} Z \xrightarrow{f} T.$$

If the pairs (f, g) and (g, h) satisfy (Var 3)*, then (f, gh) satisfies* (Var 3) *if and only if (fg, h) satisfies* (Var 3).

The condition in the conclusion also says that Tr_{fgh} is the "composite" of Tr_f, Tr_g, and Tr_h.

e) Let there be a commutative diagram:

$$\begin{tikzcd}
X \arrow[r, "v"] \arrow[dr, "g"'] & \mathbf{E}^e_Y \arrow[r, "u'"] \arrow[d, "c"] & \mathbf{E}^{d+e}_Z \arrow[d, "b"] \\
& Y \arrow[r, "u"] \arrow[dr, "f"'] & \mathbf{E}^d_Z \arrow[d, "a"] \\
& & Z
\end{tikzcd}$$

The trace morphism relative to fg is the "composite" of the trace morphisms relative to ab and $u'v$; by construction for ab, and by XVII 6.2.3 for $u'v$, this is also the "composite" of the trace morphisms relative to a, b, u', and v. On the other hand, the "composite" of the trace morphisms relative to f and g is the "composite" of those relative to a, u, c, and v. We must therefore verify that, for every Cartesian diagram 558

(2.9.2)
$$\begin{tikzcd}
\mathbf{E}^e_S \arrow[r, "u'"] \arrow[d, "b'"] & \mathbf{E}^e_T \arrow[d, "b"] \\
S \arrow[r, "u"] & T
\end{tikzcd}$$

with u quasi-finite and flat of finite presentation, the diagram of trace morphisms (2.8.1) and XVII 6.2.)

(2.9.3)
$$\begin{tikzcd}
R^{2e}b_!u'_!u'^*b^*F(e) \arrow[r, "R^{2e}b_!(\mathrm{Tr}_{u'})"] \arrow[d, equal] & R^{2e}b_!b^*F(e) \arrow[r, "\mathrm{Tr}_b"] & F \arrow[d, equal] \\
u_!R^{2e}b'_!b'^*u^*F(e) \arrow[r, "u_!(\mathrm{Tr}_{b'})"] & u_!u^*F \arrow[r, "\mathrm{Tr}_u"] & F
\end{tikzcd}$$

is commutative.

By base change, it suffices to verify this when T is the spectrum of an algebraically closed field k. If S is a sum of schemes S_i, we are reduced to verifying the commutativity of (2.9.3) only for the diagrams (2.9.2) with base $u_i : S_i \to T$. This makes it possible to suppose that S is the spectrum of a local Artinian k-algebra A, of finite degree $n = [A : k]$. If we then identify sheaves on T (resp. on $\mathbf{E}^e_T$) with sheaves on S (resp. on $\mathbf{E}^e_S$) by the functor u^* (resp. u'^*) (VIII 1.1), the morphism Tr_b identifies with the morphism $\mathrm{Tr}_{b'}$ (compatibility of (2.8.1) with base change), while Tr_u and $\mathrm{Tr}_{u'}$ identify with multiplication by n, whence the assertion[117]. This completes the proof of 2.9.

We have seen in the course of the proof that: 559

Proposition 2.10. *The trace morphisms* XVII 6.2.3, XVIII 1.1.6, *and* (2.8.1) *are special cases of the trace morphism* 2.9.

Remark 2.10.1. It can be shown without difficulty that the trace morphism of 2.9 is an isomorphism if and only if the geometric fibers of f have exactly one irreducible

[117]N.D.E.: It was also possible to reduce this compatibility to the case $e = 1$, which is obtained by combining 1.1.7 and 1.1.8. The arguments given here are moreover similar to those that were passed over in silence in 1.1.8.

component of dimension d, and its "multiplicity" is prime to n. We shall not give here the direct proof of this fact, which will follow immediately from the "global duality theorem" of § 3.

2.11. Let $f : X \to S$ and $g : Y \to S$ be two S-compactifiable morphisms satisfying $(*)_d$ and $(*)_e$, respectively.

$$\begin{tikzcd} & X \times_S Y \arrow[dl, "\mathrm{pr}_1"'] \arrow[dr, "\mathrm{pr}_2"] \arrow[dd, "f\times g"] & \\ X \arrow[dr, "f"'] & & Y \arrow[dl, "g"] \\ & S & \end{tikzcd}$$

The Künneth isomorphism (XVII 5.4.3)

$$Rf_!(\mathbf{Z}/n)\overset{\mathbf{L}}{\otimes}_{\mathbf{Z}/n}Rg_!(\mathbf{Z}/n) \xrightarrow{\sim} R(f\times g)_!(\mathbf{Z}/n)$$

induces an isomorphism

$$R^{2d}f_!\mathbf{Z}/n \otimes R^{2e}g_!\mathbf{Z}/n \xrightarrow{\sim} R^{2(d+e)}(f\times g)_!(\mathbf{Z}/n) \tag{2.11.1}$$

It is clear that $f\times g$ satisfies $(*)_{d+e}$; we have the compatibility:

Proposition 2.12. *The following diagram*

$$\begin{tikzcd} R^{2d}f_!\mathbf{Z}/n(d)\otimes R^{2e}g_!\mathbf{Z}/n(e) \arrow[r, "\sim"] \arrow[d, "\mathrm{Tr}_f\otimes\mathrm{Tr}_g"] & R^{2(d+e)}(f\times g)_!\mathbf{Z}/n(d+e) \arrow[d, "\mathrm{Tr}_{f\times g}"] \\ \mathbf{Z}/n \arrow[r, equal] & \mathbf{Z}/n, \end{tikzcd} \tag{2.12.1}$$

560 *in which the first horizontal arrow is a twisted form of (2.11.1), is commutative.*

To verify this, it suffices to use (Var 3) and the asymmetric interpretation of the Künneth arrow used in the proof of XVII 5.4.3 (XVII 5.4.3.5).

2.13. Let $f : X \to S$ be an S-compactifiable morphism satisfying $(*)_d$, let $\mathscr{A}$ be a sheaf of rings on S, and let n be an integer invertible on S such that $n\mathscr{A} = 0$.

For every $K \in D(S, \mathscr{A})$, we have (XVII 5.2.6)

$$K\overset{\mathbf{L}}{\otimes}_{\mathbf{Z}/n}Rf_!\mathbf{Z}/n(d) \xrightarrow{\sim} Rf_!(f^*K(d)). \tag{2.13.1}$$

The trace morphism

$$\mathrm{Tr}_f : R^{2d}f_!\mathbf{Z}/n(d) \longrightarrow \mathbf{Z}/n$$

may also be interpreted as a morphism

$$\mathrm{Tr}_f : Rf_!\mathbf{Z}/n(d)[2d] \longrightarrow \mathbf{Z}/n.$$

Tensoring this morphism with K defines, via (2.13.1), a trace morphism

$$\mathrm{Tr}_f : Rf_!(f^*K(d)[2d]) \longrightarrow K. \tag{2.13.2}$$

As we shall see in 3.2.5, the following theorem is essentially equivalent to the global duality theorem in étale cohomology.

Theorem 2.14. *Let $f : X \to S$ be a smooth compactifiable morphism purely of relative dimension d, and let n be an integer ≥ 1 invertible on S. For every geometric point x of X with image s in S, there exist an étale neighborhood V of s in S and an étale neighborhood U of x in $f^{-1}(V)$*

(2.14.1)
$$\begin{tikzcd} U \arrow[r, "j"] \arrow[dr, "f'_V"'] & f^{-1}(V) \arrow[r] \arrow[d, "f_V"] & X \arrow[d, "f"] \\ & V \arrow[r] & S \end{tikzcd}$$

such that the morphism XVII 6.2.7.4 561

$$R^i f'_{V!}\mathbf{Z}/n \longrightarrow R^i f_{V!}\mathbf{Z}/n$$

is zero for $i < 2d$, and the morphism

$$\mathrm{Tr}_{f'_V} : R^{2d} f'_{V!}\mathbf{Z}/n(d) \longrightarrow \mathbf{Z}/n$$

is an isomorphism.

We first prove the following.

Lemma 2.14.2. *Let $\mathscr{A}$ be an abelian category, let k be an integer ≥ 0, let $(K_i)_{0\leq i\leq 2k}$ be objects of $D^b(\mathscr{A})$ such that $H^p(K_i) = 0$ for $p \notin [0,k]$, let $f_i : K_i \to K_{i+i}$ $(0 \leq i \leq 2k-1)$ be morphisms, and let f be their composite. If $H^p(f_i) = 0$ for $p < k$, then there exists in $D^b(\mathscr{A})$ a morphism φ from the complex $H^k(K_0)[-k]$, consisting of $H^k(K_0)$ placed in degree k, to K_{2k}, making the diagram*

$$\begin{tikzcd} K_0 \arrow[r, "f"] \arrow[d] & K_{2k} \\ \sigma_{\geq k}(K_0) \arrow[r, equal] & H^k(K_0)[-k]. \arrow[u, "\varphi"'] \end{tikzcd}$$

commute.

The lemma is trivial for $k = 0$. We prove it by induction on k. Applying the induction hypothesis to the complexes $\sigma_{\geq 1}(K_i)[1]$ gives a morphism $\varphi' : H^k(K_0)[-k] \to \sigma_{\geq 1}K_{2k-2}$ making the following diagram commute, where $f' = f_{2k-3} \cdots f_0$:

$$\begin{tikzcd} K_0 \arrow[r, "f'"] \arrow[d] & \sigma_{\geq 1}K_{2k-2} \\ \sigma_{\geq k}(K_0) \arrow[r, equal] & H^k(K_0)[-k]. \arrow[u, "\varphi'"'] \end{tikzcd}$$

For every complex L, the distinguished triangle

$$\begin{tikzcd} & \sigma_{\geq 1}(L) \arrow[dl, "+1"'] & \\ \sigma_{\leq 0}(L) \arrow[rr] & & L \arrow[ul] \end{tikzcd}$$

gives, for every complex M, an exact sequence 562

$$\cdots \to \mathrm{Hom}(M, \sigma_{\leq 0}(L)) \to \mathrm{Hom}(M, L) \to \mathrm{Hom}(M, \sigma_{\geq 1}L) \xrightarrow{j} \mathrm{Ext}^1(M, \sigma_{\leq 0}L).$$

In particular, we have a diagram

(2.14.3)
$$\begin{tikzcd}[column sep=small]
\mathrm{Hom}(\sigma_{\geq k}K_0,\sigma_{\leq 0}K_{2k-2}) \arrow[r] \arrow[d] & \overset{\text{⑤}}{\mathrm{Hom}(\sigma_{\geq k}K_0,K_{2k-2})} \arrow[r] \arrow[d] & \overset{\text{③}}{\mathrm{Hom}(\sigma_{\geq k}K_0,\sigma_{\geq 1}K_{2k-2})} \arrow[r] \arrow[d] & \overset{\text{④}}{\mathrm{Ext}^1(\sigma_{\geq k}K_0,\sigma_{\leq 0}K_{2k-2})} \arrow[d] \\
\overset{\text{⑦}}{\mathrm{Hom}(K_0,\sigma_{\leq 0}K_{2k-2})} \arrow[r] & \overset{\text{①}}{\mathrm{Hom}(K_0,K_{2k-2})} \arrow[r] & \overset{\text{②⑥}}{\mathrm{Hom}(K_0,\sigma_{\geq 1}K_{2k-2})} \arrow[r] & \mathrm{Ext}^1(K_0,\sigma_{\leq 0}K_{2k-2})
\end{tikzcd}$$

which maps to the analogous diagrams for K_{2k-1} and K_{2k}. By hypothesis, the morphism f' at ① has an image at ② that lifts to ③. The obstruction ④ to the element ③ lifting to ⑤ has zero image in diagram (2.14.3) for K_{2k-1}. In this new diagram, ③ therefore lifts to ⑤, and the difference between ① and the image of ⑤ has zero image at ⑥, and hence is the image of an element ⑦. This element ⑦ has zero image in diagram 2.12.3 for K_{2k}; in this new diagram, f is therefore the image of ⑤, which solves the problem.

We prove that, for a given value of d, assertion 2.14 implies the following.

Corollary 2.14.4. *Under the hypotheses of 2.14, there exists a diagram (2.14.1) such that the morphism XVII 6.2.4, in $D^b(V, \mathbf{Z}/n)$, admits a factorization*

$$\begin{tikzcd}
Rf'_{V!}\mathbf{Z}/n(d) \arrow[rr] \arrow[dr, "t"'] & & Rf_{V!}\mathbf{Z}/n(d) \\
& \mathbf{Z}/n[-2d], \arrow[ur, "\varphi"'] &
\end{tikzcd}$$

where t is the arrow

$$t : Rf'_{V!}\mathbf{Z}/n(d) \to \sigma_{\geq 2d}Rf'_V\mathbf{Z}/n(d) = H^{2d}(Rf'_V\mathbf{Z}/n(d))[-2d] \xrightarrow{\mathrm{Tr}} \mathbf{Z}/n[-2d].$$

563 Set $f'_0 = f$ and construct, inductively, $4d$ diagrams (2.14.1)

$$\begin{tikzcd}
U'_{i+1} \arrow[r] \arrow[dr, "f'_{i+1}"'] & f_i'^{-1}(V'_{i+1}) \arrow[r] \arrow[d] & U'_i \arrow[d, "f'_i"] \\
& V'_{i+1} \arrow[r] & V'_i
\end{tikzcd}$$

satisfying 2.14. Set $V = V'_{4d}$ and $U_i = U'_i \times_{V'_i} V$. We obtain a diagram

$$\begin{tikzcd}
U_{4d} \arrow[r] \arrow[drrrr, "f_{4d}"] & \cdots \arrow[r] & U_1 \arrow[r] \arrow[drr, "f_1"] & & U_0 \arrow[r] \arrow[d, "f^0"] & X \arrow[d, "f"] \\
& & & & V \arrow[r, dash] & S,
\end{tikzcd}$$

with $U_0 = f^{-1}(V)$, such that the arrows

$$R^i f_{k+1!}\mathbf{Z}/n(d) \longrightarrow R^i f_{k!}\mathbf{Z}/n(d)$$

are zero (for $i < 2d$) and the arrow

$$\mathrm{Tr} : R^{2d} f_{4d!}\mathbf{Z}/n(d) \longrightarrow \mathbf{Z}/n$$

is an isomorphism. By 2.14.2, the corollary holds for $U = U_{4d}$.

We are now ready to prove 2.14 by induction on d. The case $d = 0$ is trivial, and the case $d = 1$ is precisely 1.6.9. We may therefore suppose that $d \geq 2$.

Since the problem is local on X and S, we may suppose that f admits a factorization $f = gh$, with g and h smooth and compactifiable, purely of relative dimensions d' and d'', and that $d', d'' < d = d' + d''$. Apply 2.14.4 to g and to the morphisms deduced from h by base change, so as to obtain a diagram

$$\begin{tikzcd}
U \arrow[r, "u"] \arrow[dr, "h'_0"] & f^{-1}W \arrow[r] \arrow[d, "h_0"] & X \arrow[d, "h"] \\
V \arrow[r, "v"] \arrow[dr, "g'_0"] & g^{-1}W \arrow[r] \arrow[d, "g_0"] & Y \arrow[d, "g"] \\
& W \arrow[r] & S,
\end{tikzcd}$$

such that the morphisms (XVII 6.2.7.3) admit factorizations 564

$$Rh'_{0!}\mathbf{Z}/n(d'') \xrightarrow{\mathrm{Tr}} \mathbf{Z}/n[-2d''] \xrightarrow{\varphi} Rh_{0!}\mathbf{Z}/n(d'')$$
$$Rg'_{0!}\mathbf{Z}/n(d') \xrightarrow{\mathrm{Tr}} \mathbf{Z}/n[-2d'] \xrightarrow{\Psi} Rg_{0!}\mathbf{Z}/n(d')$$

and such that $\mathrm{Tr}_{h'_0}$ and $\mathrm{Tr}_{g'_0}$ are isomorphisms.

Let h_1 and h'_1 be the morphisms deduced from h_0 and h'_0 by base change to V, and let $f' = g'_0 h'_1$, $f_W = g_0 h_0$.

The following diagram is commutative:

$$\begin{tikzcd}
Rf'\mathbf{Z}/n(d) \arrow[rr] \arrow[d, equal] & & Rf_{W!}\mathbf{Z}/n(d) \arrow[d, equal] \\
Rg'_{0!}Rh'_1\mathbf{Z}/n(d) \arrow[r] & Rg_{0!}Rh'_{0!}\mathbf{Z}/n(d) \arrow[r] & Rg_{0!}Rh_{0!}\mathbf{Z}/n(d),
\end{tikzcd}$$

as is the diagram

$$\begin{tikzcd}
Rg'_{0!}Rh'_{1!}\mathbf{Z}/n(d) \arrow[d, equal] \arrow[ddd, bend right] & & \\
Rg'_{0!}v^*Rh'_{0!}\mathbf{Z}/n(d) \arrow[r, "Rg'_{0!}v^*(t)(d')"'] \arrow[dd] & Rg'_{0!}\mathbf{Z}/n(d')[-2d''] \arrow[d, "t[-2d'']"] & \\
& \mathbf{Z}/n[-2d] \arrow[d, "\Psi[-2d'']"] & \\
Rg_{0!}Rh'_{0!}\mathbf{Z}/n(d) \arrow[r, "Rg_{0!}(t)(d')"'] \arrow[rr, bend right] & Rg_{0!}\mathbf{Z}/n(d')[-2d''] \arrow[r, "Rg_{0!}(\varphi)(d')"'] & Rg_{0!}Rh_{0!}\mathbf{Z}/n(d)
\end{tikzcd}$$

For the diagram 565

$$\begin{tikzcd} U_1 \arrow[r] \arrow[dr, "f'"'] & f^{-1}W \arrow[r] \arrow[d, "f_W"] & X \arrow[d, "f"] \\ & W \arrow[r, dash] & S, \end{tikzcd}$$

the canonical arrow from $Rf'_!\mathbf{Z}/n(d)$ to $Rf_W\mathbf{Z}/n(d)$ therefore factors through $\mathbf{Z}/n[-2d]$; hence the arrows from $R^i f'_!\mathbf{Z}/n(d)$ to $R^i f_W\mathbf{Z}/n(d)$ are zero for $i < 2d$. Finally, $\mathrm{Tr}_{f'}$ is an isomorphism, being the composite of the isomorphisms $\mathrm{Tr}_{g'_0}$ and $R^{2d'} g'_{0!}\,\mathrm{Tr}_{h'_1}$.

This completes the proof of 2.14.

3. The global duality theorem

566 **3.1. The functor $Rf^!$.** We shall need a variant of the canonical flasque resolutions.

Lemma 3.1.1. *Let X be a coherent scheme (i.e. quasi-compact and quasi-separated). The functor $\varinjlim$ induces an equivalence between the category of Ind-objects of the category of constructible sheaves of sets (resp. of groups, resp. of abelian groups) and the category of sheaves of sets (resp. of Ind-finite groups, resp. of torsion abelian groups).*

This follows from IX 2.7.2 and IX 2.7.3.

3.1.2. Let $\mathscr{F}$ be a torsion abelian sheaf on a scheme X, the filtered direct limit of constructible sheaves $(\mathscr{F}_i)_{i\in I}$, and let X_0 be a conservative set of points of X. The *modified canonical flasque resolution* of $\mathscr{F}$ is defined by the formula

$$C^*_\ell(\mathscr{F}) = \varinjlim_i C^*(\mathscr{F}_i),$$

where the $C^*(\mathscr{F}_i)$ are the canonical flasque resolutions of XVII 4.2.2. By 3.1.1, for X coherent, this definition is independent of the choice of the $\mathscr{F}_i$. It also commutes with localization; this fact allows us to define $C^*_\ell(\mathscr{F})$ by globalization when $\mathscr{F}$ is a torsion sheaf on an arbitrary scheme X, not necessarily coherent.

The complex $C^*_\ell(\mathscr{F})$ is a resolution of $\mathscr{F}$, functorial in $\mathscr{F}$, whose formation commutes with localization and filtered direct limits; the functors C^n_ℓ are exact.

It follows that, for a sheaf of rings $\mathscr{A}$ on X, the functors C^n_ℓ carry $\mathscr{A}$-Modules to $\mathscr{A}$-Modules; they likewise carry torsion sheaves to torsion sheaves. It is well known that

567 Lemma 3.1.3. *Let $\mathscr{A}$ be an abelian $\mathscr{U}$-category having a small generating set and such that direct limits indexed by a small filtered ordered set are representable in $\mathscr{A}$, and are exact. A functor F from $\mathscr{A}^\circ$ to* (Ens) *is representable if and only if it carries small direct limits (arbitrary ones) to inverse limits.*

Cf. for example SGA 4 I 8.12.5, where $\mathscr{A}$ is not assumed abelian, nor are filtered $\varinjlim$ assumed exact.

We shall apply this lemma to the category $\mathrm{Mod}(X, \mathscr{A})$ of sheaves of modules on a ringed site $(X, \mathscr{A})$. In this special case, if the functor F is represented by the sheaf $\mathscr{F}$, then

$$\mathscr{F}(U) = F(\mathscr{A}_U). \tag{3.1.3.1}$$

This description of $\mathscr{F}$ gives a direct verification of 3.1.3 in the case at hand.

Recall that if $f : X \to S$ is a compactifiable morphism (XVII 3.2.), then S is quasi-compact and quasi-separated (sic), and the dimensions of the fibers of f are bounded.

Theorem 3.1.4. *Let $f : X \to S$ be a compactifiable morphism and let $\mathscr{A}$ be a sheaf of torsion rings on S. Then the functor*

$$Rf_! : D(X, f^*\mathscr{A}) \longrightarrow D(S, \mathscr{A}) \tag{3.1.4.1}$$

admits a partial right adjoint

$$Rf^! : D^+(S, \mathscr{A}) \longrightarrow D^+(X, f^*\mathscr{A}) : \tag{3.1.4.2}$$

for $K \in \mathrm{Ob}\, D(X, f^*\mathscr{A})$ and $L \in \mathrm{Ob}\, D^+(S, \mathscr{A})$, there is a functorial isomorphism

$$\mathrm{Hom}(Rf_!K, L) \simeq \mathrm{Hom}(K, Rf^!L). \tag{3.1.4.3}$$

Endowed with the translation morphism making the diagram

$$\begin{tikzcd}
\mathrm{Hom}(Rf_!(K[1]), L) \arrow[r, leftrightarrow, "\sim"] \arrow[d, leftrightarrow, "\wr"'] & \mathrm{Hom}(K[1], Rf^!L) \arrow[r, dash, "\sim"] & \mathrm{Hom}(K, (Rf^!L)[-1]) \arrow[d, leftrightarrow, "\wr"] \\
\mathrm{Hom}((Rf_!K)[1], L) \arrow[r, dash, "\sim"] & \mathrm{Hom}(Rf_!K, L[-1] \arrow[r, leftrightarrow, "\sim", "\mathrm{adj.}"'] & \mathrm{Hom}(K, Rf^!(L[-1])),
\end{tikzcd} \tag{3.1.4.4}$$

commute, this functor $Rf^!$ is a triangulated functor. 568

N.B. The notation $Rf^!$ is abusive in that $Rf^!$ is not in general the derived functor of a functor $f^!$.

Proof Let d be an integer such that every fiber of f has dimension $< d$, and choose a compactification

$$\begin{tikzcd}
X \arrow[r, hook, "j"] \arrow[d, "f"'] & \overline{X} \arrow[dl, "\overline{f}"] \\
S. &
\end{tikzcd}$$

If K is a complex of $f^*\mathscr{A}$-Modules on X, then

$$Rf_!K \simeq R\overline{f}_*(j_!K).$$

For every sheaf F on X, the components F^i of the complex $\tau_{\leq 2d}C^*_\ell j_!F$ satisfy

$$R^k\overline{f}_*F^i = 0 \quad \text{for} \quad k > 0. \tag{3.1.4.5}$$

Indeed, for $i \neq 2d$, F^i is a direct limit of flasque sheaves, and for $i = 2d$ and $k > 0$, we have (XVII 5.2.8.1)

$$R^k\overline{f}_*F^{2d} = R^{k+2d}f_!F = 0.$$

The simple complex associated with the truncated modified canonical flasque resolution double complex of K is a resolution of K. By (3.1.4.5), we therefore have (notation of XVII 1.1.15)

$$R\overline{f}_*(j_!K) \xleftarrow{\sim} \overline{f}_*\tau''_{\leq 2d}C^*_\ell j_!K. \tag{3.1.4.6}$$

Denote by $f_!^{\boldsymbol{\cdot}}$ the functor from sheaves of modules on X to complexes of sheaves 569
on S defined by

$$f_!^{\boldsymbol{\cdot}}(F) = \overline{f}_*\tau_{\leq 2d}C^*_\ell j_!F. \tag{3.1.4.7}$$

LEMMA 3.1.4.8. (i) *The functors $f^i_!$ are exact and commute with filtered direct limits; we have $f^i_! = 0$ for $i \notin [0, 2d]$.*

(ii) *Since the functor $f^{\cdot}_!$ is bounded with exact components, its derived functor* (XVII 1.2.10) *is obtained trivially and is denoted $Rf^{\cdot}_!$, or simply $f^{\cdot}_! : D(X, f^*\mathscr{A}) \to D(S, \mathscr{A})$. The arrow (3.1.4.6) is an isomorphism of functors*

$$Rf^{\cdot}_! \xrightarrow{\sim} Rf_!.$$

The functors C^i_ℓ are exact and commute with filtered direct limits (3.1.2). They send every sheaf to an acyclic sheaf. The functors $\overline{f}_*$ and $j_!$ commute with filtered direct limits. It follows that

a) the functors $\overline{f}_* C^i_\ell j_!$, and hence also the functors $f^i_!$, commute with filtered direct limits;
b) for $i < 2d$, the functors $f^i_! = \overline{f}_* C^i_\ell j_!$ are exact. For $i = 2d$, one verifies that $f^i_!$ is exact by (3.1.4.5).

The remaining assertions of 3.1.4.8 are trivial.

REMARK 3.1.4.9. The definition of $f^{\cdot}_!$ is independent of the sheaf of rings $\mathscr{A}$, and still makes sense for any abelian sheaf. Assertion 3.1.4.8 (i) remains valid in the categories of torsion abelian sheaves on X and S.

By 3.1.4.8 and 3.1.3, the functor

$$f^i_! : \mathrm{Mod}(X, f^*\mathscr{A}) \longrightarrow \mathrm{Mod}(S, \mathscr{A})$$

has a right adjoint $f^!_i$; since $f^i_!$ is exact, $f^!_i$ sends injectives to injectives.

With the sign convention of (XVII 1.1.12), the functors $f^!_i$ form a complex of left exact functors, zero in cohomological degree $n \notin [-2d, 0]$:

$$f^!_{\cdot} : \mathrm{Mod}(S, \mathscr{A}) \longrightarrow C^b(X, f^*\mathscr{A}). \tag{3.1.4.10}$$

570 The functor $f^!_{\cdot}$ derives (XVII 1.2.10) to a triangulated functor

$$Rf^! : D^+(S, \mathscr{A}) \longrightarrow D^+(X, f^*\mathscr{A}), \tag{3.1.4.11}$$

right adjoint to the functor $Rf_!$ ($= Rf^{\cdot}_! = Lf^{\cdot}_!$) ("trivial duality formula"). More precisely, if $I \in \mathrm{Ob}\, C^+(S, \mathscr{A})$ is a complex of sheaves of injective $\mathscr{A}$-modules, the $\mathscr{A}$-modules $f^!_i I^n$ are injective, and there is an isomorphism of triple complexes (XVII 1.1.12)

$$\mathrm{Hom}^{\cdot}(f^{\cdot}_! K, I) \xrightarrow{\sim} \mathrm{Hom}^{\cdot}(K, f^!_{\cdot} I). \tag{3.1.4.12}$$

Passing to H^0 of the associated simple complex (computed using products) gives the adjunction statement. We leave it to the reader to verify the compatibility (3.1.4.4).

REMARK 3.1.5. Suppose that the functor $Rf^!$ has finite cohomological amplitude. By XVII 1.2.10, the functor $f^!_{\cdot}$ then admits a derived functor

$$Rf^! : D(S, \mathscr{A}) \longrightarrow D(X, \mathscr{A}). \tag{3.1.5.1}$$

This derived functor is again right adjoint to the functor $Rf_!$ (3.1.4.1). Indeed, we have

$$\mathrm{Hom}_{D(S)}(Rf_!K,L) \simeq \varinjlim_{L \xrightarrow{\sim} L'} \mathrm{Hom}_{K(S)}(f_!^{\cdot}K,L') \simeq \varinjlim_{\substack{K' \xrightarrow{\sim} K \\ L \xrightarrow{\sim} L'}} \mathrm{Hom}_{K(S)}(f_!^{\cdot}K',L')$$

$$\mathrm{Hom}_{D(X)}(K,Rf^!L) \simeq \varinjlim_{K' \xrightarrow{\sim} K} \mathrm{Hom}_{K(X)}(K',Rf^!L) \simeq \varinjlim_{\substack{K' \xrightarrow{\sim} K \\ L \xrightarrow{\sim} L'}} \mathrm{Hom}_{K(X)}(K',f_{\cdot}^!L')$$

and, by adjunction,

$$\mathrm{Hom}_{K(S)}(f_!^{\cdot}K',L') = H^0\,\mathrm{Hom}(f_!^{\cdot}K',L') \simeq H^0\,\mathrm{Hom}(K',f_{\cdot}^!L') = \mathrm{Hom}_{K(X)}(K',L').$$

Definition 3.1.6. *The triangulated functor*

$$Rf^! : D^+(S,\mathscr{A}) \longrightarrow D^+(X,f^*\mathscr{A})$$

(or, under the hypotheses of 3.1.5, 571

$$Rf^! : D(S,\mathscr{A}) \longrightarrow D(X,f^*\mathscr{A})),$$

right adjoint to the functor $Rf_!$ (3.1.4.1), is called the exceptional inverse-image functor.

Proposition 3.1.7. *Suppose that the compactifiable morphism $f : X \to S$ has relative dimension $\leq d$.*

(i) *If $L \in \mathrm{Ob}\,D^+(S,\mathscr{A})$ satisfies $H^i(L) = 0$ for $i \leq k$, then $H^iRf^!L = 0$ for $i \leq k-2d$.*
(ii) *For $L \in \mathrm{Ob}\,D^b(S,\mathscr{A})$, we have*

$$\dim\mathrm{inj}(Rf^!L) \leq \dim\mathrm{inj}(L).$$

The functor $Rf^!$ is the right derived functor of $f_{\cdot}^! : \mathrm{Mod}(S,\mathscr{A}) \to C^b\,\mathrm{Mod}(X,f^*\mathscr{A})$. Assertion (i) follows from the fact that $(f_{\cdot}^!)^i = 0$ for $i < -2d$, and assertion (ii) follows from the fact that $(f_{\cdot}^!)^i = 0$ for $i > 0$, and that the functors $f_i^!$ send injectives to injectives.

Proposition 3.1.8. *Let $f : X \to S$ be a quasi-finite compactifiable morphism.*

(i) *The functor $f_! : \mathrm{Mod}(X,f^*\mathscr{A}) \to \mathrm{Mod}(S,\mathscr{A})$ admits a right adjoint $f^! : \mathrm{Mod}(S,\mathscr{A}) \to \mathrm{Mod}(X,f^*\mathscr{A})$, and $Rf^!$ is the derived functor of $f^!$.*
(ii) *If f is a closed immersion, then $f^!$ is the functor of "sections with support in X."*
(iii) *If f is étale, then $f^!$ identifies with the functor f^*, with*

$$\mathrm{Tr}_f : f_!f^*F \longrightarrow F$$

as adjunction morphism.

Assertion (i) follows immediately from the special case $d = 0$ of the proof of 3.1.4: we have $f_!^{\cdot} = f_!$ and $f_{\cdot}^! = f^!$. Equivalently, the existence of an adjoint functor $f^!$ follows from 3.1.3 and from the fact that $f_!$ is exact and commutes with filtered direct limits; 572
the adjunction $(f_!, Rf^!)$ is a "trivial duality formula."

Assertion (ii) is clear, and (iii) is XVII 6.2.11.

3.1.9. Let $f : X \to S$ be a compactifiable morphism and $\mathscr{A}$ a sheaf of torsion rings on S, and equip X with the inverse-image sheaf of rings of $\mathscr{A}$. Let $K \in \operatorname{Ob} D^-(X, f^*\mathscr{A})$, $L \in \operatorname{Ob} D^+(X, f^*\mathscr{A})$, and represent L by a bounded-below complex of injective sheaves, so that

$$R\,\mathbf{Hom}(K, L) \simeq \mathbf{Hom}(K, L).$$

The complex $\mathbf{Hom}(K, L)$ has flasque components, and therefore

$$Rf_* R\,\mathbf{Hom}(K, L) \simeq f_* \mathbf{Hom}(K, L). \tag{3.1.9.1}$$

Choose a compactification of f and an integer d as in the proof of 3.1.4, giving a functor $f_!^{\boldsymbol{\cdot}}$ (3.1.4.7).

$$\begin{tikzcd}
X_V \arrow[rr, hook, "k_X"] \arrow[dr, hook, "j_V"'] \arrow[dd, "f_V"'] & & X \arrow[dr, hook, "j"] \arrow[dd, "f"] & \\
& \overline{X}_V \arrow[rr, "k_{\overline{X}}"] \arrow[dl, "\overline{f}_V"] & & \overline{X} \arrow[dl, "\overline{f}"] \\
V \arrow[rr, "k"] & & S. &
\end{tikzcd} \tag{3.1.9.2}$$

For every $V \in \operatorname{Ob} S_{\text{et}}$, there is an isomorphism[118]

$$k^* f_!^{\boldsymbol{\cdot}} \xrightarrow{\sim} f_{V!}^{\boldsymbol{\cdot}} k_X^*, \tag{3.1.9.3}$$

so that

$$\mathbf{Hom}^{\boldsymbol{\cdot}}(f_!^{\boldsymbol{\cdot}} K, f_!^{\boldsymbol{\cdot}} L)(V) \simeq \mathbf{Hom}^{\boldsymbol{\cdot}}(f_{V!}^{\boldsymbol{\cdot}}(k_X^* K), f_{V!}^{\boldsymbol{\cdot}}(k_X^* L)).$$

The functoriality of $f_{V!}^{\boldsymbol{\cdot}}$ gives us

$$\mathbf{Hom}^{\boldsymbol{\cdot}}(K, L)(X_V) = \operatorname{Hom}^{\boldsymbol{\cdot}}(k_X^* K, k_X^* L) \longrightarrow \operatorname{Hom}^{\boldsymbol{\cdot}}(f_{V!}^{\boldsymbol{\cdot}}(k_X^* K), f_{V!}^{\boldsymbol{\cdot}}(k_X^* L)),$$

hence an arrow, at the level of complexes,

$$f_* \mathbf{Hom}^{\boldsymbol{\cdot}}(K, L) \longrightarrow \mathbf{Hom}^{\boldsymbol{\cdot}}(f_!^{\boldsymbol{\cdot}} K, f_!^{\boldsymbol{\cdot}} L). \tag{3.1.9.4}$$

573 By (3.1.9.1), this arrow derives to

$$Rf_* R\,\mathbf{Hom}(K, L) \longrightarrow R\,\mathbf{Hom}(Rf_! K, Rf_! L). \tag{3.1.9.5}$$

Now let $K \in \operatorname{Ob} D^-(X, \mathscr{A})$ and $L \in \operatorname{Ob} D^+(S, \mathscr{A})$. The adjunction arrow from $Rf_! Rf^! L$ to L, together with (3.1.9.4), gives a composite arrow

$$Rf_* R\,\mathbf{Hom}(K, Rf^! L) \longrightarrow R\,\mathbf{Hom}(Rf_! K, Rf_! Rf^! L) \longrightarrow R\,\mathbf{Hom}(Rf_! K, L),$$

the local form on S of the adjunction isomorphism:

$$Rf_* R\,\mathbf{Hom}(K, Rf^! L) \longrightarrow R\,\mathbf{Hom}(Rf_! K, L). \tag{3.1.9.6}$$

Proposition 3.1.10. *The arrow 3.1.9.6 is an isomorphism.*

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[118] N.D.E.: This morphism is given by the construction of XVII 4.2.6, but there is no reason for it to be an isomorphism unless coherent choices of points have first been made on the topoi under consideration. For this to be the case, the square occurring in XVII 4.2.6 must in some sense be Cartesian. More precisely, one should fix here a family of points $(p)_{p \in P}$ of the topos S_{et}, and for every $V \in \operatorname{Ob} S_{\text{et}}$ one considers the modified canonical flasque resolution using the family of points associated by the correspondence of IV 6.7 with the pairs (p, ξ) where $p \in P$ and $\xi \in V_p$.

Let $V \in \operatorname{Ob} S_{\text{et}}$ and retain the notation of (3.1.9.2). The ("transitivity") isomorphism

$$k_! Rf_{V!} \simeq Rf_! k_{X!},$$

whose existence is the basis of the theory of the functor $Rf_!$, transposes to a *localization isomorphism* (3.1.8. (iii))

$$k_X^* Rf^! \simeq Rf_V^! k^*, \tag{3.1.10.1}$$

represented at the level of complexes by a morphism

$$k_X^* f^!_{\cdot} \longrightarrow f^!_{V\cdot} k^*.$$

To verify that (3.1.9.6) is an isomorphism, it suffices to verify that for every $V \in \operatorname{Ob} S_{\text{et}}$, (3.1.9.6) induces an isomorphism

$$\begin{aligned} \operatorname{Hom}(k_X^* K, k_X^* Rf^! L) &\simeq H^0(V, Rf_* R\,\mathbf{Hom}(K, Rf^! L)) \\ &\longrightarrow H^0(V, R\operatorname{Hom}(Rf_! K, L)) \simeq \mathbf{Hom}(k^* Rf_! K, k^* L) \end{aligned} \tag{3.1.10.2}$$

For $V = X$, this is precisely (3.1.4.3). The general case follows from the following compatibility, *which the editor has not verified*:

Lemma 3.1.10.3. *The diagram* 574

$$\begin{tikzcd} \operatorname{Hom}(k_X^* K, k_X^* Rf^! L) \arrow[r, "(3.1.10.2)"] \arrow[d, "\wr"', "(3.1.10.1)"] & \operatorname{Hom}(k^* Rf_! K, k^* L) \arrow[d, "\wr"', "\text{(localization)}"] \\ \operatorname{Hom}(k_X^*, K, Rf_V^! k_X^* L) & \operatorname{Hom}(Rf_! k^* K, k^* L) \arrow[l, "\text{(3.1.4.3 on V)}"', "\sim"] \end{tikzcd}$$

is commutative.

3.1.11. Consider a Cartesian square

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"] & X \arrow[d, "f"] \\ S' \arrow[r, "g"] & S, \end{tikzcd} \tag{3.1.11.1}$$

with f compactifiable, $\mathscr{A}$ a sheaf of torsion rings on S, $\mathscr{B}$ a sheaf of torsion rings on S', and M a bounded-above complex of $(f^*\mathscr{A}, \mathscr{B})$-bimodules. At the level of complexes, there is then a morphism, functorial in K ($K \in \operatorname{Ob} C(X, f^*\mathscr{A})$)

$$M \otimes_{\mathscr{A}} g^* f_!^{\cdot} K \longrightarrow f_!'^{\cdot}(f'^* M \otimes_{\mathscr{A}} g'^* K), \tag{3.1.11.2}$$

which transposes to a morphism, functorial in L ($L \in \operatorname{Ob} C(S', \mathscr{B})$)

$$g'_* \mathbf{Hom}_{\mathscr{B}}(f'^* M, f'^{!}_{\cdot} L) \longrightarrow f^!_{\cdot} g_* \mathbf{Hom}_{\mathscr{B}}(M, L). \tag{3.1.11.3}$$

Passing to derived functors gives a morphism, functorial in L, from $D^+ (S', \mathscr{B})$ to $D^+(X, f^*\mathscr{A})$:

$$Rg'_* R\,\mathbf{Hom}_{\mathscr{B}}(f'^* M, Rf'^! L) \longrightarrow Rf^! Rg_* R\operatorname{Hom}_{\mathscr{B}}(M, L), \tag{3.1.11.4}$$

which makes the following diagram commutative, where $K \in \mathrm{Ob}\, D^-(X, f^*\mathscr{A})$ and $L \in \mathrm{Ob}\, D^+(S', \mathscr{B})$:

$$\begin{tikzcd}
\mathrm{Hom}(K, Rg'_* R\,\mathbf{Hom}_{\mathscr{B}}(f'^*M, Rf'^!L)) \arrow[r, "①"] \arrow[d, leftrightarrow] & \mathrm{Hom}(K, Rf^! Rg_* R\,\mathbf{Hom}_{\mathscr{B}}(M,L)) \arrow[r, leftrightarrow, "\mathrm{adj}"'] & \mathrm{Hom}(Rf_! K, Rg_* R\,\mathbf{Hom}_{\mathscr{B}}(M,L)) \arrow[d, leftrightarrow] \\
\mathrm{Hom}(g'^*K, R\,\mathbf{Hom}_{\mathscr{B}}(f'^*M, Rf'^!L)) \arrow[d, leftrightarrow] & & \mathrm{Hom}(g^* Rf_! K, R\,\mathbf{Hom}_{\mathscr{B}}(M,L)) \arrow[d, leftrightarrow] \\
\mathrm{Hom}(f'^*M \otimes_{\mathscr{A}} g'^*K, Rf'^!L) \arrow[r, leftrightarrow, "\mathrm{adj}"'] & \mathrm{Hom}(Rf'^{\cdot}_!(f'^*M \otimes^L_{\mathscr{A}} g'^*K), L) \arrow[r, "②"] & \mathrm{Hom}_{\mathscr{B}}(M \otimes^{\mathbf{L}}_{\mathscr{A}} g^* Rf_! K, L).
\end{tikzcd}$$

575 In this diagram, ② (deduced from the arrow XVII 5.2.2.1) is an isomorphism (XVII 5.2.6). The arrow ① is therefore an isomorphism for every K, and this implies that (3.1.11.4) is an isomorphism.

The proper base change theorem thus transposes to the following

Proposition 3.1.12. *The arrow (3.1.11.4) is an isomorphism.*

This isomorphism is called the *induction isomorphism.*

Here are three special cases.

Corollary 3.1.12.1. *Let $f : X \to S$ be a compactifiable morphism, let $\mathscr{A}$ and $\mathscr{B}$ be two sheaves of torsion rings on S, and let $\varphi : \mathscr{A} \to \mathscr{B}$ be a homomorphism. Denote by ρ the restriction-of-scalars functor from $\mathscr{B}$ to $\mathscr{A}$. Then, for $K \in \mathrm{Ob}\, D^+(S, \mathscr{B})$, we have*

$$\rho Rf^! K \xrightarrow{\sim} Rf^!(\rho K).$$

It suffices in (3.1.11.3) to take $S = S'$, $X = X'$, $\mathscr{A} = \mathscr{A}$, $\mathscr{B} = \mathscr{B}$, $M = \mathscr{B}$.

Thus the sheaf of rings plays no essential role in the construction of $Rf^!$.

Corollary 3.1.12.2. *Let $f : X \to S$ be a compactifiable morphism, let $\mathscr{A}$ be a sheaf of torsion rings on S, and let $K \in \mathrm{Ob}\, D^-(S, \mathscr{A})$, $L \in \mathrm{Ob}\, D^+(S, \mathscr{A})$. We have the* induction formula

$$R\,\mathbf{Hom}(f^*K, Rf^!L) \xrightarrow{\sim} Rf^! R\,\mathbf{Hom}(K, L)$$
$$(\text{take } S = S', X = X').$$

Corollary 3.1.12.3. *For every square 3.1.11 and every $L \in \mathrm{Ob}\, D^+(S', \mathscr{A})$, we have*

$$Rg'_* Rf'^! L \xrightarrow{\sim} Rf^! Rg_* L$$
$$(\text{take } \mathscr{B} = f^*\mathscr{A} = M).$$

576 3.1.13. For a composite gh of S-compactifiable morphisms

$$X \xrightarrow{h} Y \xrightarrow{g} Z$$

the composition isomorphism

$$c_{g,h} : Rg_! Rh_! \xrightarrow{\sim} R(gh)_!$$

transposes to a composition morphism[119]

$$c^!_{g,h} : Rh^! Rg^! \xrightarrow{\sim} R(gh)^! \tag{3.1.13.1}$$

[119] N.D.E.: The transposed morphism actually goes in the other direction, but this causes no problem since it is an isomorphism.

satisfying the usual cocycle condition for a triple composite. In other words, the *categories $D^+(X, f^*\mathscr{A})$ form the fibers of a fibered and cofibered category over the category whose objects are the S-schemes $f : X \to S$*, and whose arrows are the S-compactifiable morphisms, *the direct-image* (resp. *inverse-image*) *functors being the functors $Rf_!$* (resp. $Rf^!$). For every commutative square of compactifiable morphisms,

$$\begin{tikzcd} X' \arrow[r, "g"'] \arrow[d, "f'"] & X \arrow[d, "f"] \\ S' \arrow[r, "g'"'] & S, \end{tikzcd}$$

we therefore have a cobase-change morphism (XVII 2.1.3)

$$ch^! : Rf'_! Rg^! \longrightarrow Rg^! Rf_!. \tag{3.1.13.2}$$

3.1.14. Consider a Cartesian square

$$\begin{tikzcd} X' \arrow[r, "g'"] \arrow[d, "f'"] & X \arrow[d, "f"] \\ S' \arrow[r, "g"] & S, \end{tikzcd}$$

with f compactifiable, and let $\mathscr{A}$ be a sheaf of torsion rings on S. Endow S', X, and X' with the inverse images of $\mathscr{A}$ as their sheaves of rings, again denoted by $\mathscr{A}$ by abuse of notation. The base-change isomorphism 577

$$g^* Rf_! \xrightarrow{\sim} Rf'_! g'^* \tag{3.1.14.1}$$

allows us to regard the categories $D(S, \mathscr{A})$, $D(X, \mathscr{A})$, $D(S', \mathscr{A})$, and $D(X', \mathscr{A})$ as the fibers of a cofibered category over the category defined by the commutative diagram

$$\begin{tikzcd} X \arrow[d] & X \arrow[l] \arrow[d] \\ X & X \arrow[l] \end{tikzcd}$$

the direct-image functors being the functors $Rf_!$, $Rf'_!$, g^*, and g'^* (sic). When we restrict to the categories D^+, we even obtain a fibered and cofibered category over this diagram, the direct-image functors being the functors $Rf^!$, $Rf'^!$, Rg_*, and Rg'_* (sic). We deduce from this

a) a "transitivity isomorphism" (sic) for the "inverse-image" functors

$$Rg_* Rf^! \xrightarrow{\sim} Rf^! Rg_*$$

already obtained in 3.1.12.3 (with the same definition by transposition of (3.1.14.1)).

b) a "base-change morphism" (XVII 2.1.3)

$$g'^* Rf^! \longrightarrow Rf'^! g^* \tag{3.1.14.2}$$

which, by *loc. cit.*, can be described in the following two ways:

a) by adjunction, giving a morphism (3.1.14.2) amounts to giving a morphism

$$Rf'_! g'^* Rf^! \longrightarrow g^*.$$

The base-change theorem provides such a morphism as the composite

$$Rf'_! g'^* Rf^! \xleftarrow{\sim} g^* Rf_! Rf^! \xrightarrow{g^* * \text{adj.}} g^*.$$

578 b) by adjunction, giving a morphism (3.1.14.2) amounts to giving a morphism

$$Rf^! \longrightarrow Rg'_* Rf'^! g^*.$$

The isomorphism (3.1.12.3) provides such a morphism as the composite

$$Rf^! \xrightarrow[Rf^! * \text{adj.}]{} Rf^! Rg_* g^* \xleftarrow{\sim} Rg'_* Rf'^! g^*.$$

From their interpretations in terms of a fibered and cofibered category, it is clear that *the morphisms* (3.1.12.3), (3.1.13.2), *and* (3.1.14.2) *each satisfy two compatibilities of the type* XII 4.4 *when* f *or* g *is a composite of two morphisms*[120].

The following results from general topology will be used in the following subsection.

3.1.15. Let $f : X \to S$ be a morphism of sites. Denote by $\Gamma(f)$ the following site:

a) The category $\Gamma(f)$ is the category of triples (U, V, φ) with

$$U \in \mathrm{Ob}(X), V \in \mathrm{Ob}\, S, \quad \varphi \in \mathrm{Hom}(U, f^* V).$$

b) A family of morphisms $(u_i, v_i) : (U_i, V_i, \varphi_i) \to (U, V, \varphi)$ is covering if, in X, the morphisms $u_i : U_i \to U$ cover U.

We then have:

. The functor from $\Gamma(f)$ to X given by

$$(U, V, \varphi) \longmapsto U$$

is an equivalence of sites $\gamma : X \to \Gamma(f)$.

. The functor f^*_Γ from S to $\Gamma(f)$

$$V \longmapsto (f^* V, V, \mathrm{Id}_{f^* V})$$

is a morphism of sites, denoted $f_\Gamma : \Gamma(f) \to S$, or simply f by abuse of notation. It satisfies $f_\Gamma \gamma \simeq f$. The functor f^*_Γ has a left adjoint $f_{\Gamma\cdot}$:

$$f_{\Gamma\cdot} : (U, V, \varphi) \longmapsto V.$$

579 The functor $f_{\Gamma\cdot}$ is therefore a comorphism of sites from $\Gamma(f)$ to S[121], and defines the same morphism of topoi as the morphism of sites f_Γ.

(3.1.15.3)
$$\begin{tikzcd}
X & & \Gamma(f) \arrow[ll, "\approx"'] \arrow[dl, shift left, "f_{\Gamma\cdot}"] \\
& S \arrow[ul, "f^*"] \arrow[ur, shift left, "f^*_\Gamma"] &
\end{tikzcd}$$
.

[120] N.D.E.: These constructions can also be interpreted in the context of exchange structures and cross functors, cf. [1].

[121] N.D.E.: Since this terminology is not familiar, it should be specified here that these are the notions appearing in III 2 and IV 4.7 (one should bear in mind that in the original edition, at the location of this second reference, misprints made it impossible to understand in precisely which direction the arrows went).

We deduce the following rule for computing the inverse image of a sheaf[122]. Let $\mathscr{F}$ be a sheaf on S, let $f^{\bullet}_{\Gamma}\mathscr{F}$ be the presheaf on $\Gamma(f)$ given by

$$f^{\bullet}_{\Gamma}\mathscr{F}(U,V,\varphi) = \mathscr{F}(V),$$

and let $af^{\bullet}_{\Gamma}\mathscr{F}$ be the associated sheaf. Then

$$f^{*}\mathscr{F}(U) \simeq af^{\bullet}_{\Gamma}\mathscr{F}(U,V,\varphi).$$

3.1.16. Let $f : X \to S$ be a morphism of sites, $\mathscr{A}$ a sheaf of rings on S, $\mathscr{B}$ a sheaf of rings on X, and d an integer. Let K be an object of the following type:

. K associates to every object U of X a complex $K(U)$ of $(\mathscr{A}, f_{|U*}(\mathscr{B}|U))$-modules on S, zero in degrees $> d$, and $K(U)$ is a *covariant* functor of U.

For every sheaf of $\mathscr{A}$-modules $\mathscr{F}$ on S, denote by $K^!\mathscr{F}$ the complex of sheaves on X generated by the complex of presheaves of $\mathscr{B}$-modules

$$U \longmapsto \mathrm{Hom}_{\mathscr{A}}(K(U),\mathscr{F}).$$

Denote by $RK^!$ the derived functor of $K^!$

$$RK^! : D^+(S,\mathscr{A}) \longrightarrow D^+(X,\mathscr{B}).$$

If $U : K_1 \to K_2$ is a morphism of objects (3.1.11.1), then U induces morphisms of functors **580**

$${}^t u : K^!_2 \longrightarrow K^!_1 \quad \text{et} \quad RK^!_2 \longrightarrow RK^!_1.$$

Under the preceding hypotheses, we have:

Proposition 3.1.17. (i) *For every point x of X and every $L \in \mathrm{Ob}\, D^+(S,\mathscr{A})$, denoting by $V(x)$ the category of neighborhoods of x, we have*

$$\mathscr{H}^n(RK^!L)_x \simeq \varinjlim_{U\in \mathrm{Ob}\, V(x)} \mathrm{Ext}^n(K(U),L). \tag{3.1.17.1}$$

(ii) *We therefore have a spectral sequence*

$$E_2^{pq} = \varinjlim_{U\in \mathrm{Ob}\, V(x)} \mathrm{Ext}^p(\mathscr{H}^{-q}(K(U)),L) \Longrightarrow \mathscr{H}^n(K^!L)_x. \tag{3.1.17.2}$$

(iii) *If X^0 is a conservative set of points of X, and $u : K_1 \to K_2$ is a morphism of objects (3.1.11.1), then, in order that*

$${}^t u : RK^!_2 \longrightarrow RK^!_1$$

be an isomorphism, it is necessary and sufficient that, for every n and every $x \in X^0$, the morphism of pro-objects of $\mathrm{Mod}(S,\mathscr{A})$

$$u_x : \text{“}\varprojlim_{U\in \mathrm{Ob}\, V(x)}\text{”}\ \mathscr{H}^n K_1(U) \longrightarrow \text{“}\varprojlim_{U\in \mathrm{Ob}\, V(x)}\text{”}\ \mathscr{H}^n K_2(U) \tag{3.1.17.3}$$

be an isomorphism.

To prove (i), take L to be a complex of injectives, so that (3.1.17.1) is immediate; it is clear that (i) $\Rightarrow$ (ii).

[122] N.D.E.: Here III 2.3 or IV 4.7.3 is being used.

The sufficiency of condition (3.1.17.3) in (iii) follows immediately from (3.1.17.2). We shall not need the fact that it is necessary. If ${}^t u$ is an isomorphism, then, by (3.1.16.1), for every injective sheaf I on $(S, \mathscr{A})$, we have

$$\operatorname{Hom}(\text{“}\varprojlim_{V(x)}\text{”}\,\mathscr{H}^n K_2(U), I) \xrightarrow{\sim} \operatorname{Hom}(\text{“}\varprojlim_{V(x)}\text{”}\,\mathscr{H}^n K_1(U), I),$$

and we conclude by noting that in every abelian category with enough injectives (here, $\operatorname{Mod}(S, \mathscr{A})$), the functors $\operatorname{Hom}(*, I)$, for I an injective $\mathscr{A}$-module, form a conservative family of functors on $\operatorname{Pro}(\mathscr{A})$.

581 Example 3.1.18. Let $f : X \to S$ be a compactifiable morphism of schemes, and let $\mathscr{A}$ be a sheaf of torsion rings on S. Endow X with $f^*\mathscr{A}$ as its sheaf of rings. Choose an integer d that bounds the dimension of the fibers of f, as well as a compactification of f, and set ((3.1.4.7))

$$K(U) = f_!^{\boldsymbol{\cdot}}((f^*\mathscr{A})_U). \tag{3.1.18.1}$$

By formula (3.1.3.1) for the adjoint of $f_!^{\boldsymbol{\cdot}}$, we have an isomorphism

$$f^!_{\boldsymbol{\cdot}} \simeq K^! : \operatorname{Mod}(S, \mathscr{A}) \longrightarrow C^b \operatorname{Mod}(X, f^*\mathscr{A}). \tag{3.1.18.2}$$

and hence

$$Rf^! = RK^! : D^+(S, \mathscr{A}) \longrightarrow D^+(X, f^*\mathscr{A}). \tag{3.1.18.3}$$

Example 3.1.19. Let $f : X \to S$, $\mathscr{A}$, $\mathscr{B}$, and d be as in 3.1.16. Let K be an object of the following type.

. For every $W = (U, V, \varphi) \in \operatorname{Ob} \Gamma(f)$ (3.1.15)[123], $K(W)$ is a complex of $(\mathscr{A}_V, \varphi_* B_{|U})$-modules on V, functorial in W: for every morphism $(u, v) : W_1 = (U_1, V_1, \varphi_1) \to W_2 = (U_2, V_2, \varphi_2)$ it is given by

$$\begin{aligned} K(u, v) &: K(W_1) \longrightarrow v^* K(W_2), \quad \text{i.e.} \\ K(u, v) &: v_! K(W_1) \longrightarrow K(W_2). \end{aligned}$$

With every object K of type (3.1.19.1) is associated an object K' of type (3.1.16.1) relative to $f_\Gamma : \Gamma(f) \to S$, given by

$$K'(U, V, \varphi) = j_! K(U, V, \varphi) \tag{3.1.19.2}$$

where j is the “inclusion morphism” of V into S. We set

$$K^! \underset{\text{defn}}{=} K'^! \quad \text{et} \quad RK^! \underset{\text{defn}}{=} RK'^!.$$

Example 3.1.20. With the notation of 3.1.19, take $\mathscr{B} = f^*\mathscr{A}$ and set

$$K(U, V, \varphi) = \mathscr{A}_V \quad (\text{in degree } 0) \tag{3.1.20.1}$$

582 By 3.1.15.4, we then have isomorphisms of functors

[123] N.D.E.: By definition, φ is an X-morphism $U \to V \times_S X$. To understand this example properly, one must adopt the equivalent viewpoint that interprets φ as an S-morphism $\varphi : U \to V$.

$$K^! \simeq f^* \tag{3.1.20.2}$$

$$RK^! \simeq f^*. \tag{3.1.20.3}$$

3.2. Poincaré duality.

3.2.1. Let f be a compactifiable flat morphism, purely of relative dimension d (or, more generally, satisfying condition $(*)_d$ of 2.9):

$$\begin{tikzcd} X \arrow[r, hook, "j"] \arrow[d, "f"'] & \overline{X} \arrow[dl, "\overline{f}"] \\ S & \end{tikzcd}.$$

Let n be an integer ≥ 1 invertible on S, and suppose that S is ringed by a sheaf of rings $\mathscr{A}$ such that $n\mathscr{A} = 0$. By 3.1.18, the functor $Rf^! : D^+(S, \mathscr{A}) \to D^+(X, \mathscr{A})$ can be computed by the procedure of 3.1.16, for K given by (3.1.18.1).

Let K' be the functor of type (3.1.19.1) that associates to $(U, V, \varphi) \in \operatorname{Ob} \Gamma(f)$ the complex of sheaves $\mathbb{Z}/n(-d)[-2d]$ on V, consisting of $\mathbb{Z}/n(-d)$ placed in degree $2d$, and let K'' be the corresponding object of type (3.1.16.1) under (3.1.19.2), relative to $f : \Gamma(f) \to S$.

We may regard K and K'' as objects of 3.1.16.1 relative to $\Gamma(f) \to S$. Moreover, for every $(U, V, \varphi) \in \operatorname{Ob} \Gamma(f)$, the trace morphism 2.9 defines a morphism of sheaves on V

$$\operatorname{Tr} : R^{2d}\varphi_!\mathbb{Z}/n \longrightarrow \mathbb{Z}/n(-d),$$

whence, denoting by j the morphism from V to S, a morphism of functors

$$\begin{aligned} K(U) = \overline{f}_*\tau_{\leq 2d}C^*_\ell\mathbb{Z}/n_U \longrightarrow \mathscr{H}^{2d}(K(U))[-2d] \simeq \\ R^{2d}f_!(\mathbb{Z}/n)_U \simeq j_!R^{2d}\varphi!\mathbb{Z}/n \xrightarrow{\operatorname{Tr}_\varphi} j_!\mathbb{Z}/N(-d) = K''(U, V, \varphi)). \end{aligned} \tag{3.2.1.1}$$

A twisted form of 3.1.20 shows that 583

$$RK''^! = Rf^*(d)[2d].$$

By 3.1.16, we therefore have a composite morphism

$$t_f : Rf^*(d)[2d] \simeq RK''^! \longrightarrow Rk^! \simeq Rf^!. \tag{3.2.1.2}$$

Remark 3.2.2. If the functor $Rf^!$ has finite cohomological dimension, then t_f is still defined as a morphism between functors from $D(S, \mathscr{A})$ to $D(X, f^*\mathscr{A})$ (definition in 3.1.5).

There is another method for constructing an arrow (3.2.1.2), by adjunction from (2.13.2)

$$\operatorname{Tr} : Rf_!(f^*L(d)[2d]) \longrightarrow L.$$

These two arrows coincide:

Lemma 3.2.3. *For* $L \in \operatorname{Ob} D^+(S, \mathscr{A})$ *(resp.* $D(S, \mathscr{A})$ if the functor $Rf^!$ has finite cohomological dimension*), the diagram*

$$\begin{tikzcd}
Rf_!(f^*L(d)[2d]) \arrow[r, "Rf_!(t_f)"] \arrow[d, "\mathrm{Tr}_f"] & Rf_!Rf^!L \arrow[d, "\text{adjunction}"] \\
L \arrow[r, equal] & L
\end{tikzcd}$$

is commutative.

We first give a variant of the definition of t_f, in terms of the functor $f^!_{\cdot}$ (3.1.4.10). Let $U \in \operatorname{Ob} X_{\mathrm{et}}$, $V \in \operatorname{Ob} S_{\mathrm{et}}$, and $\varphi \in \operatorname{Hom}_f(U, V)$. By definition,

$$f^!_{\cdot}(L)(U) = \operatorname{Hom}(f_!^{\cdot}\mathscr{A}_U, L)\ ;$$

the trace morphism defines a morphism

$$\mathrm{Tr}_f\ :\ f_!^{\cdot}\mathscr{A}_U \longrightarrow \mathscr{A}_V(-d)[-2d]$$

whose transpose is

$$t_\varphi\ :\ f^!_{\cdot}(L)(U) \longleftarrow L(d)[2d](V).$$

584 The morphisms t_φ define a morphism

$$t_f\ :\ f^*L(d)[2d] \longrightarrow f^!_{\cdot}L$$

and we leave it to the reader to verify that, in the derived category, it coincides with (3.2.1.2). This description shows that the following diagrams are commutative at the level of complexes, for L a complex of modules:

$$\begin{tikzcd}
\operatorname{Hom}^{\cdot}(\mathscr{A}_V, L) \arrow[r] \arrow[d, "\mathrm{Tr}_f"] & \operatorname{Hom}^{\cdot}(f^*\mathscr{A}_V(d)[2d], f^*L(d)[2d]) \arrow[d, "t_f"] \\
\operatorname{Hom}(f_!^{\cdot}f^*\mathscr{A}_V(d)[2d], L) \arrow[r, dash, "\sim"] & \operatorname{Hom}^{\cdot}(f^*\mathscr{A}_V(d)[2d], f^!_{\cdot}L).
\end{tikzcd}$$

By functoriality of the arrows in this diagram in $\mathscr{A}_V$, it follows that for every sheaf, or every complex of sheaves K, the following diagram is commutative:

$$\begin{tikzcd}
\operatorname{Hom}^{\cdot}(K, L) \arrow[r] \arrow[d, "\mathrm{Tr}_f"] & \operatorname{Hom}^{\cdot}(f^*K(d)[2d], f^*L(d)[2d]) \arrow[d] \\
\operatorname{Hom}^{\cdot}(f_!^{\cdot}f^*K(d)[2d], L) \arrow[r, leftrightarrow, "\sim"] & \operatorname{Hom}^{\cdot}(f^*K(d)[2d], f^!_{\cdot}L).
\end{tikzcd}$$

Taking $K = L$ and $\mathrm{Id}\ :\ L \to L$, we obtain 3.2.3 at the level of complexes.

I would be grateful to anyone who has understood this proof if they would explain it to me.

3.2.4. We deduce at once from this lemma and from the definition (3.1.13) of the composition isomorphisms by adjunction that if f is the composite of two flat, finitely presented, compactifiable morphisms that are purely of relative dimensions d_1 and d_2 : $f = gh$, and if $d = d_1 + d_2$, then the diagram

$$\begin{tikzcd} h^*(g^*L(d_2)[2d_2])(d_1)[2d_1]) \arrow[r, dash, "\sim"] \arrow[d, "t_h * t_g"] & (gh)^*L(d)[2d] \arrow[d, "t_{gh}"] \\ Rh^! \, Rg^! \, L \arrow[r, dash, "\sim"] & R(gh)^! L \end{tikzcd}$$

is commutative. 585

THEOREM 3.2.5 (Poincaré duality). *Let $f : X \to S$ be a compactifiable smooth morphism* (thus S is quasi-compact and quasi-separated (sic))*, let d be the locally constant function on X "relative dimension of f", let $n \geq 1$ be an integer invertible on S, and let $\mathscr{A}$ be a sheaf of rings on S satisfying $n\mathscr{A} = 0$. Then the functor*

$$Rf_! : D(X, f^*\mathscr{A}) \longrightarrow D(S, \mathscr{A})$$

admits as right adjoint the following functor from $D(S, \mathscr{A})$ to $D(X, f^*\mathscr{A})$:

$$K \longmapsto f^*K(d)[2d]$$

with the morphism (2.13.2)

$$\mathrm{Tr}_f : Rf_! f^*K(d)[2d] \longrightarrow K$$

as adjunction arrow:

$$\mathrm{Hom}(K, f^*L(d)[2d]) \xrightarrow{\sim} \mathrm{Hom}(Rf_!K, L).$$

Proof. We reduce to the case where d is constant. By the description 3.2.1 of the arrow t_f (3.2.1.2), Theorem 2.14, together with criterion 3.1.17 (iii), implies that the arrow t_f is an isomorphism (for $L \in D^+(S, \mathscr{A})$)

$$t_f : f^*L(d)[2d] \xrightarrow{\sim} Rf^!L.$$

In particular, the functor $Rf^!$ has finite cohomological dimension and is therefore defined on the unbounded derived category. The morphism t_f remains an isomorphism for $L \in D(S, \mathscr{A})$[124].

By definition, $Rf^!$ is the right adjoint of the functor $Rf_!$; by 3.2.3, the adjunction 586
arrow from $Rf_!Rf^!K$ to K is identified, via t_f, with Tr_f, and this completes the proof of 3.2.5.

Henceforth we identify, by means of t_f, the functors $f^*L(d)[2d]$ and $Rf^!L$.

[124]N.D.E.: More precisely, the fact that $Rf^!$ has finite cohomological dimension makes it possible to show that one can define the functor $Rf^!$ on the unbounded derived category in such a way that, for a complex $I^\bullet$ of injective sheaves, $Rf^!I^\bullet$ is the total complex associated with the double complex $f^!_\cdot I^\bullet$. Then, for every $n \in \mathbf{Z}$, the morphism induced on the n-th cohomology sheaf by t_f evaluated at $I^\bullet$ is identified with the same morphism for a stupid truncation $I^{\geq k}$ for k sufficiently small, which proves the assertion.

3.2.6. Let S be the spectrum of an algebraically closed field k, let X be a compactifiable scheme, smooth over k and purely of relative dimension d, let n be an integer invertible in k, and let F be a locally constant constructible sheaf of $\mathbf{Z}/n$-modules, i.e. a "coefficient system" killed by n. Denoting by $^\vee$ a $\mathbf{Z}/n$-dual, we have

$$R\,\mathbf{Hom}(F, \mathbf{Z}/n) = F^\vee \quad \text{(in degree 0)}.$$

This formula follows from the fact that $\mathbf{Z}/n$ is an injective $\mathbf{Z}/n$-module. Isomorphism 3.2.5.1, or rather the isomorphism deduced from it,

$$R\Gamma\ R\,\mathbf{Hom}(F, \mathbf{Z}/n(d)[2d]) \xrightarrow{\sim} R\,\mathrm{Hom}(R\Gamma_c(F), \mathbf{Z}/n),$$

is therefore written here as

$$R\Gamma(F^\vee(d)[2d]) \xrightarrow{\sim} R\,\mathrm{Hom}(R\Gamma_c(F), \mathbf{Z}/n). \tag{3.2.6.1}$$

Passing to cohomology groups, and using again the fact that $\mathbf{Z}/n$ is an injective $\mathbf{Z}/n$-module, this gives

$$H^{2d-i}(X, F^\vee(d)) \xrightarrow{\sim} H^i_c(X, F)^\vee, \tag{3.2.6.2}$$

which is the usual form of Poincaré duality.

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EXPOSÉ XIX

Cohomology of excellent preschemes of equal characteristic

M. Artin

In this exposé we develop the theory of étale cohomology for excellent preschemes **588**
(EGA IV 7.8.1) of equal characteristic, *assuming the theorem on resolution of singularities for such preschemes*, in the following form: **589**

Let $X = \operatorname{Spec} A$ be an excellent local scheme of equal characteristic, and let $U \subset X$ be a regular open subset. There exist a proper birational morphism $f : X' \to X$ and an open immersion $U \to X'$ over X such that X' is regular and $Y = X - U$ is everywhere of codimension 1, with normal crossings.

The proofs are therefore valid, "for the time being," only in characteristic zero [4]. One could also deduce from them results for preschemes of characteristic $p > 0$ in low dimensions [1].

Under the resolution hypothesis, we shall obtain more or less satisfactory results for preschemes of equal characteristic and for coefficients prime to the characteristic (cf. 3.2, 4.1, 5.1, 6.1). On the other hand, almost nothing is known about the cohomology of general preschemes of unequal characteristic, even in dimension 2, where resolution is available [2]. For example, the purity theorem 2.1 is not known for any complete ring of unequal characteristic of dimension > 1. This result, for the formal power series rings $k[[x_1, \dots, x_n]]$ (1.2), is one of the main tools in the theory. Let us also note that the resolution theorem is not used in the proof of 1.2. *This theorem is therefore proved in arbitrary characteristic.*

We shall often use without explicit mention the properties of excellent schemes de- **590**
veloped in EGA IV 7.8, such as stability under extensions of finite type (EGA IV 7.8.3(ii)) and under strict localizations. The latter fact follows from resolution and EGA IV 7.9.5.

1. Purity for the ring $k[[x_1, \dots, x_n]]$

Recall the following corollary of XVI 3.7:

Corollary 1.1. *Let A be a strictly local ring, and denote by $A\{x\}$ the strict localization of $A[x]$ at the prime ideal generated by* $\operatorname{rad} A$ *and x. Let $U = \operatorname{Spec} A\{x\}[1/x]$. Then, for n prime to the residual characteristic of A, we have*

$$H^q(U, \boldsymbol{\mu}_n) \simeq \begin{cases} \boldsymbol{\mu}_n(A) & \text{if} \quad q = 0 \\ \mathbf{Z}/n & \text{if} \quad q = 1 \\ 0 & \text{if} \quad q > 1. \end{cases}$$

Indeed, let $X = \operatorname{Spec} A[x]$, and let $Y = V(x)$ be the zero locus of x in X. Then (Y, X) is a smooth pair (XVI 3.1) over $\operatorname{Spec} A$. By (V 4, VIII 5), $H^q(U, \boldsymbol{\mu}_n)$ is identified with the fiber of the sheaf $\mathscr{H}^{q+1}_Y(X, \boldsymbol{\mu}_n)$ at a geometric point above the point $(\operatorname{rad} A, x)$ of X, and the corollary therefore follows immediately from (XVI 3.7). Note that $\boldsymbol{\mu}_n$ and $\mathbf{Z}/n$ are both cyclic groups of order n. We have used them here to give the canonical formulas **591**
(cf. 3.4).

THEOREM 1.2. *Let k be a separably closed field, and let $U = \operatorname{Spec} k[[x_1, \dots, x_r]]\,[1/x_r]$. Then, for n prime to the characteristic of k, we have*

$$H^q(U, \mu_n) = \begin{cases} \mu_n(A) & \text{if} \quad q = 0 \\ \mathbf{Z}/n & \text{if} \quad q = 1 \\ 0 & \text{if} \quad q > 1. \end{cases}$$

REMARK. The proof applies equally to the ring of convergent power series when $k = \mathbf{C}$. Moreover, one easily avoids the hypothesis that k be separably closed by stating the result with a little more care. But to establish the general purity theorem 3.2 (under the resolution-of-singularities hypothesis, of course), we need the result only when k is separably closed, and these other assertions are corollaries of that theorem 3.2.

In dimensions 0 and 1, the result can be proved for every regular strictly local ring:

LEMMA 1.3. *Let A be a regular strictly local ring and let $x \in \operatorname{rad} A$ be a local parameter. Let $U = \operatorname{Spec} A[1/x]$. Then $H^0(U, \mu_n) = \mu_n(A)$, and there are canonical isomorphisms*

$$\mathbf{Z}/n \simeq H^0(U, \mathbf{G}_m)/(H^0(U, \mathbf{G}_m))^n \simeq H^1(U, \mu_n),$$

592 *where the generator of $\mathbf{Z}/n$ is the "residue" of $x \in H^0(U, \mathbf{G}_m)$.*

Proof. For $q = 0$, the assertion is that U is connected and nonempty. This is true. For $q = 1$, apply the Kummer exact sequence (IX 3.7). We have $\operatorname{Pic} U = 0$ because A is regular and factorial, and U is an open subset of $X = \operatorname{Spec} A$. It follows that we have the exact sequence

$$H^0(U, \mathbf{G}_m) \xrightarrow{n} H^0(U, \mathbf{G}_m) \to H^1(U, \mu_n) \to 0,$$

and it remains to prove that $H^0(U, \mathbf{G}_m)/H^0(U, \mathbf{G}_m))^n$ is isomorphic to $\mathbf{Z}/n$ and generated by the residue of x. But there is an obvious exact sequence

$$\begin{tikzcd} 0 \arrow[r] & H^0(X, \mathbf{G}_m) \arrow[r] \arrow[d, phantom, "\wr"] & H^0(U, \mathbf{G}_m) \arrow[r] \arrow[d, phantom, "\wr"] & \mathbf{Z} \arrow[r] & 0 \\ & A^* & A[1/x]^* & & \end{tikzcd}$$

where the image of a section $a \in H^0(U, \mathbf{G}_m)$ in $\mathbf{Z}$ is the order of vanishing of a along $\{x = 0\}$. The group A^* is divisible by n, because A is strictly local and n is prime to the residual characteristic. Indeed, if $u \in H^0(X, \mathbf{G}_m)$, the equation $Y^n - u = 0$ defines a finite étale extension of A, which is therefore completely split. It follows that $H^1(U, \mu_n) \simeq \mathbf{Z}/n$, proving the lemma.

Proof of 1.2. It remains to treat the case $q > 1$. Recall the terminology of (V, Appendix). We have

$$H^q(U, \mu_n) = \varinjlim_{\mathbf{U}} H^q(\mathbf{U}, \mu_n)$$

593 where $\mathbf{U} = \left\{ U_0 \leftleftarrows U_1 \mathrel{\substack{\leftarrow\\[-1ex]\leftarrow\\[-1ex]\leftarrow}} \dots \right\}$ ranges over the category (filtered up to homotopy) of *hypercoverings* of U. Let $\mathbf{U}$ be such a hypercovering, and take the U_i to be separated and of finite type. Since we wish to prove that $H^q(U, \mu_n) = 0$, it suffices to find a hypercovering $\mathbf{V}$ and a morphism $\mathbf{V} \to \mathbf{U}$ such that the induced morphism

$$H^q(\mathbf{U}, \mu_n) \longrightarrow H^q(\mathbf{V}, \mu_n)$$

is the zero morphism.

Let $\{W_1, \dots, W_m\}$ be the set of connected components of all the U_i for $i \leq q+1$, and let B_j be the normalization of $A = k[[x_1, \dots, x_r]]$ in the field $R(W_j)$ of rational functions on W_j. Then, since $W_j \to U$ is étale, the scheme W_j is an open subset of Spec B_j, say the open complement of Spec C_j, where C_j is the appropriate reduced quotient ring of B_j.

Choose in addition an element b_j of B_j such that b_j generates the separable extension $R(W_j)$ of $k((x_1, \dots, x_r))$. Let $f_j(Y) \in A[Y]$ be the monic irreducible polynomial of b_j over A, so that B_j is isomorphic to the normalization of the ring $A[Y]/(f_j)$. Finally, let $d_j \in A$ be the discriminant of $f_j(Y)$, which is a nonzero element of A.

Now we may replace the x_i for $i < r$ by other elements of A, the only condition being that $\{x_1, \dots, x_r\}$ form a system of parameters of A.

It is easy to see that, after changing these elements if necessary, we may suppose 594
that the following hypotheses are satisfied:

1.4.

(i) For every j such that d_j is not a unit, $x_1, \dots, x_{r-1}$ and d_j generate an ideal primary for rad A.

(ii) For every j, the images of $x_1, \dots, x_{r-1}$ in C_j generate an ideal primary for rad C_j.

We then have the following lemma:

Lemma 1.5. *Let $A^0 = k[[x_1, \dots, x_{r-1}]]\ \{x_r\}$ be the henselization of $k[[x_1, \dots, x_{r-1}]]$ $[x_r]$ at the point $(x_1, \dots, x_r)$, and let $X^0 =$ Spec A^0 and $U^0 =$ Spec A^0 $[1/x_r]$. Under the conditions of 1.4 above, there exists a hypercovering $\mathbf{U}^0$ of U^0 (of order $q+1$) and an isomorphism $\mathbf{U} \times_{X^0} X \simeq \mathbf{U}$.*

Assume the lemma. The ring A^0 is excellent (EGA IV 7.8.3 (ii), 7.9.5), and consequently the connected components of the U_i^0 and of the U_i correspond bijectively (EGA IV 7.8.3 (vii)). The isomorphism $\mathbf{U}^0 \times_{X^0} X \simeq \mathbf{U}$ therefore gives an isomorphism

$$H^q(\mathbf{U}^0, \boldsymbol{\mu}_n) \simeq H^q(\mathbf{U}, \boldsymbol{\mu}_n).$$

Since k is separably closed, A^0 is a strictly local ring, and we may apply 1.1 to the ring A^0. It follows that $H^q\ (U^0, \boldsymbol{\mu}_n) = 0$. But

$$H^q(U^0, \boldsymbol{\mu}_n) \simeq \varinjlim_{\mathbf{V}^0} H^q(\mathbf{V}^0, \boldsymbol{\mu}_n),$$

where $\mathbf{V}^0$ ranges over the category of hypercoverings of U^0 of order $q+1$, and H^q 595
$(\mathbf{U}^0, \boldsymbol{\mu}_n)$ is a finite group. Consequently, there exists a morphism of hypercoverings $\mathbf{V}^0 \to \mathbf{U}^0$ such that the induced morphism

$$H^q(\mathbf{U}^0, \boldsymbol{\mu}_n) \longrightarrow H^q(\mathbf{V}^0, \boldsymbol{\mu}_n)$$

is the zero morphism. Put $\mathbf{V} = \mathbf{V}^0 \times_{X^0} X$. Then $H^0\ (\mathbf{V}^0, \boldsymbol{\mu}_n) \simeq H^q(\mathbf{V}, \boldsymbol{\mu}_n)$ (by the same argument as for $\mathbf{U}^0$), so

$$H^q(\mathbf{U}, \boldsymbol{\mu}_n) \longrightarrow H^q(\mathbf{V}, \boldsymbol{\mu}_n)$$

is also the zero morphism, C.Q.F.D.

Proof of 1.5. It suffices to find finite A^0-algebras B_j^0 and C_j^0 and A-isomorphisms

$$B_j^0 \otimes_{A^0} A \xrightarrow{\sim} B_j$$
$$C_j^0 \otimes_{A^0} A \xrightarrow{\sim} C_j.$$

Indeed, by ([3] 1.4) the morphism $B_j \to C_j$ is induced by a morphism $B_j^0 \to C_j^0$, so the open subset $W_j \subset \operatorname{Spec} B_j$ is induced by an open subset $W_j^0 \subset \operatorname{Spec} B_j^0$. Moreover, the U-morphisms among the W_j are induced by morphisms among the A-algebras B_j, hence ([3] 1.4) by morphisms among the B_j^0, and therefore by U^0-morphisms among the W_j^0. It follows that the W_j^0 form the set of connected components of a simplicial object $\mathbf{U}^0$ over U^0 that induces $\mathbf{U}$, and one sees immediately that it is a hypercovering of U^0.

596 Now 1.4 (ii) does imply that C_j is induced by an A^0-algebra C_j^0, as is easily verified [124]. For B_j, it suffices to find a monic polynomial $f_j^0(Y) \in A^0[Y]$ such that the two A-algebras

$$A[Y]/(f_j) \quad , \quad A[Y]/(f_j^0) = A \otimes_{A^0} A^0[Y]/(f_j^0)$$

are isomorphic. Indeed, $A^0[Y]/(f_j^0)$ is then reduced, and we take for B_j^0 the normalization (EGA IV 7.8.3 (vi)) of $A[Y]/(f_j^0)$. The ring $B_j^0 \otimes_{A^0} A$ is still normal (EGA IV 7.8.3 (vii)) and birational to B_j, hence equal to that ring.

By 1.4 (i), the ideal $I_j \subset A$ generated by d_j is induced by an ideal of $k[[x_1, \dots, x_{r-1}]][x_r]$ (Weierstrass preparation theorem [5][6]), hence by an ideal I_j^0 of A^0. Moreover, A/I_j is a finite $k[[x_1, \dots, x_{r-1}]]$-algebra, and is therefore isomorphic to A^0/I_j^0. The latter ring is thus complete, and consequently the I_j^0-adic completion of A^0 is complete and isomorphic to A. We can therefore find a monic polynomial $f_j^0 \in A^0\,[Y]$ such that in $A\,[Y]$ we have

$$f_j \equiv f_j^0 (\text{modulo } {d_j}^2 \operatorname{rad} A).$$

By ([3], 1.3), $A[Y]/(f_j) \simeq A[Y]/(f_j^0)$, which proves the lemma.

2. The case of a strictly local ring

The theorem is as follows:

597 Theorem 2.1. [124]. *Let A be an excellent, regular, strictly local ring of equal characteristic, and let x be a "local parameter"*[124] *of A. Let $U = \operatorname{Spec} A[1/x]$. Then, for n prime to the characteristic of A, we have*

$$H^q(U, \boldsymbol{\mu}_n) = \begin{cases} \boldsymbol{\mu}_n(A) & \text{if} \quad q = 0 \\ \mathrm{Z}/n & \text{if} \quad q = 1 \\ 0 & \text{if} \quad q > 1. \end{cases}$$

The proof proceeds by induction on q. We denote by $P(N)$ the assertion of 2.1 for the values $q \leq N$. By 1.3, the assertion $P(1)$ is true.

[124] Indeed, 1.4 (ii) implies that C_j is finite over $k[[X_1, \dots, X_{r-1}]]$, and a fortiori over A^0, and it suffices to take $C_j^0 = C_j$ with the A^0-module structure obtained by restriction of scalars.

[124] Depends on resolution of singularities (cf. Introduction).

[124] By which we mean here: an element belonging to a *regular* system of parameters.

LEMMA 2.2. *Suppose that $P(N)$ is true. Let $A \to A'$ be a local morphism of excellent, regular, strictly local rings of equal characteristic. Let $\{x_1, \dots, x_r\}$ be part of a regular system of parameters of A whose set of images in A is again part of a regular system of parameters. Let*

$$U = \operatorname{Spec} A[1/x_1, \dots, 1/x_r]$$
$$U' = \operatorname{Spec} A'[1/x'_1, \dots, 1/x'_r].$$

Then the canonical morphism

$$H^q(U, \mu_n) \longrightarrow H^q(U', \mu_n)$$

is bijective for every $q \leq N$. 598

Proof.[124] We argue by induction on r: the assertion is trivial for $r = 0$. Let $r > 0$ and suppose the theorem has been proved for $r-1$. Let $B = A/(x)$, $V = \operatorname{Spec} A[1/x_1, \dots, 1/x_{r-1}]$, and $Y = \operatorname{Spec} B[1/x_1, \dots, 1/x_{r-1}]$ (where the image of x_i in B is denoted by the same symbol). There is an open immersion $i : U \to V$, and Y is the complementary closed subset, defined by the element x_r.

Applying $P(N)$ to the strict localizations of V at the geometric points of Y, we find that

$$R^q i_*(\mu_n)_U = \begin{cases} (\mu_n)_V & \text{if} \quad q = 0 \\ (\mathbf{Z}/n)_Y & \text{if} \quad q = 1 \\ 0 & \text{if} \quad 1 < q \leq N, \end{cases} \tag{2.3}$$

because these strict localizations are excellent regular rings of equal characteristic, and the element x_r is a local parameter at every point of Y.

We use a "prime" to denote the analogous objects deduced from A'. We obtain a morphism of spectral sequences

$$\begin{array}{ccccc} E_2^{pq} & = & H^p(V, R^q i_* \mu_n) & \Longrightarrow & H^{p+q}(U, \mu_n) \\ \downarrow{\scriptstyle \varphi_2^{pq}} & & \downarrow{\scriptstyle \varphi} & & \downarrow{\scriptstyle \varphi} \\ E_2'^{pq} & = & H^p(V', R^q i'_* \mu_n) & \Rightarrow & H^{p+q}(U', \mu_n). \end{array}$$

Now it is clear from 1.3 and (2.3) that the base change $V \xleftarrow{g} V'$ induces an isomorphism 599

$$g^* R^q i_* \mu_n \xrightarrow{\sim} R^q i'_* \mu_n$$

for $q \leq N$ (only for $q = 0, 1$ is there anything at all to check). Consequently, the induction hypothesis, applied to the morphism

$$H^q(V, \mu_n) \longrightarrow H^q(V', \mu_n),$$

implies that φ_2^{pq} is an isomorphism for $q = 0$ and $p \leq N$.

[124] A shorter proof than the one that follows, but using the cup-product formalism, would consist in showing that, under the hypothesis $P(N)$, the cohomology $H^*(U, \mathbf{Z}/n\mathbf{Z})$ is, in dimensions $\leq N$, the exterior algebra (over $\mathbf{Z}/n\mathbf{Z}$) on $H^1(U, \mathbf{Z}/n\mathbf{Z}) \simeq (\mathbf{Z}/n\mathbf{Z})^r$, which reduces matters to establishing 2.2 for $q = 1$, a well-known case. The calculation indicated under the hypothesis $P(N)$ is made explicit, for example, in SGA 7.

Similarly, we can apply the induction hypothesis to the morphism of rings $B \to B'$; these are regular, excellent, strictly local rings of equal characteristic, and the elements $x_1, \dots, x_{r-1}$ form part of a system of parameters. Therefore

$$H^q(Y, \mu_n) \xrightarrow{\sim} H^q(Y', \mu_n) \quad \text{if} \quad q \le N.$$

Since $(\mu_n) \simeq (\mathbb{Z}/n)_Y$, it follows that φ_2^{pq} is also bijective for $p \le N$ and for $q = 1$.

Since $R^q i_* \mu_n = R^q i'_* \mu_n = 0$ if $1 < q \le N$, we find $E_2^{pq} = E'^{pq}_2 = 0$ if $1 < q \le N$, and for $1 \le m \le N$ the morphism of spectral sequences therefore reduces to a morphism of exact sequences

$$\begin{tikzcd}
H^{m-2}(Y, \mathbb{Z}/n) \arrow[r] \arrow[d, "a"] & H^m(V, \mu_n) \arrow[r] \arrow[d, "b"] & H^m(U, \mu_n) \arrow[r, "e"] \arrow[d, "c"] & H^{m-1}(Y, \mathbb{Z}/n) \arrow[r] \arrow[d, "d"] & H^{m+1}(V, \mu_n) \arrow[d] \\
H^{m-2}(Y', \mathbb{Z}/n) \arrow[r] & H^m(V', \mu_n) \arrow[r] & H^m(U', \mu_n) \arrow[r, "e"] & H^{m-1}(Y', \mathbb{Z}/n) \arrow[r] & H^{m+1}(V', \mu_n).
\end{tikzcd}$$

600 Now we have already seen that a, b, and d are bijective, and it follows that c is *injective*. To prove that c' is *surjective*, it suffices to prove that the morphism

$$H^m(U, \mu_n) \xrightarrow{e} H^{m-1}(Y, \mathbb{Z}/n)$$

is surjective for $m \le N$.

Let $K \subset A$ be an arbitrary separably closed field, and let $A^0 = k\{x_1, \dots, x_r\}$ be the henselization of $k[x_1, \dots, x_r]$ at the origin. Then the hypotheses of 2.1 are satisfied for the morphism $A^0 \to A$ and the elements x_i. With the evident notation, we have a commutative square

$$\begin{tikzcd}
H^m(U^0, \mu_n) \arrow[r, "e^0"] \arrow[d, "c^0"] & H^{m-1}(Y^0, \mathbb{Z}/n) \arrow[d, "d^0"] \\
H^m(U, \mu_n) \arrow[r, "e"] & H^{m-1}(Y^0, \mathbb{Z}/n)
\end{tikzcd}$$

and we already know that the arrow d^0 is bijective. It therefore suffices to prove that the morphism e^0 is surjective. We are thus reduced (dropping the superscript ^{0}) to verifying surjectivity in the case $A = k\{x_1, \dots, x_r\}$.

For this ring, which is a limit of schemes smooth over k, we proved the purity the-
601 orem in XVI 3.7. Thus $R^q i_* \mu_n = 0$ for every $q > 1$, and consequently there is an exact sequence

$$H^m(U, \mu_n) \xrightarrow{e} H^{m-1}(Y, \mathbb{Z}/n) \to H^{m+1}(V, \mu_n) \xrightarrow{f} H^{m+1}(U, \mu_n)$$

for every $m \ge 1$. The surjectivity of e is therefore equivalent to the injectivity of f.

Let $Z = \operatorname{Spec} k\{x_1, \dots, x_{r-1}\}\,[1/x_1, \dots, 1/x_{r-1}]$. The morphism $\operatorname{Spec} k\{x_1, \dots, x_r\} \to \operatorname{Spec} k\{x_1, \dots, x_{r-1}\}$ given by the inclusion of rings is acyclic. This is the local acyclicity of a smooth morphism (XVI 1.1). Since the open subset V of $\operatorname{Spec} k\{x_1, \dots, x_r\}$ is the inverse image of Z, the morphism

$$H^q(Z, \mu_n) \longrightarrow H^q(V, \mu_n)$$

is bijective for every q (XV 1.3). It therefore suffices to prove that the morphism

$$H^q(Z, \mu_n) \longrightarrow H^q(U, \mu_n)$$

is injective for every q.

Now $k\{x_1, \dots, x_r\}$ is the inductive limit of rings A_α étale over $k\{x_1, \dots, x_{r-1}\}\,[x_r]$, each equipped with a lifting to that ring of the point $x_1 = x_2 = \cdots = x_r = 0$. Consequently, U is the projective limit of the open subsets $U_\alpha = \operatorname{Spec} A_\alpha[1/x_1, \dots, 1/x_r]$, and we have

$$H^q(U, \boldsymbol{\mu}_n) = \varinjlim_\alpha H^q(U_\alpha, \boldsymbol{\mu}_n).$$

But since $A_\alpha[1/x_r]$ is smooth over the henselian ring $k\{x_1, \dots, x_{r-1}\}$, and since the closed fiber of $A[1/x_r]$ is nonempty, we see immediately that the $k\{x_1, \dots, x_{r-1}\}$-algebra $A_\alpha[1/x_r]$ admits sections. Hence U_α admits sections over Z, and it follows that 602

$$H^q(Z, \boldsymbol{\mu}_n) \longrightarrow H^q(U_\alpha, \boldsymbol{\mu}_n)$$

is injective for every q, and therefore that

$$H^q(Z, \boldsymbol{\mu}_n) \longrightarrow H^q(U, \boldsymbol{\mu}_n)$$

is also injective, which proves the lemma.

Lemma 2.4. *Suppose that $P(N)$ is true. Let $A \to A'$ be a regular morphism* (EGA IV 6.8.1) *of excellent, strictly local rings of equal characteristic (not necessarily regular), and let $U \subset \operatorname{Spec} A$ be a regular open subset. Let $U' = U \times_{\operatorname{Spec} A} \operatorname{Spec} A'$. Then, for n prime to the residue characteristic of A, the morphism*

$$H^q(U, \boldsymbol{\mu}_n) \longrightarrow H^q(U', \boldsymbol{\mu}_n)$$

is bijective for $q \leq N$.

Proof. Let $f : X \to \operatorname{Spec} A$ be a resolution of singularities of $\operatorname{Spec} A$ such that the open subset U of $\operatorname{Spec} A$ lifts to an open immersion $i : U \to X$, and such that the closed subset $Y = X - U$ is pure of codimension 1 and has normal crossings. We denote by a prime the base change induced by $A \to A'$. The morphism $g : X' \to X$ is regular (EGA IV 6.8.3), so $f' : X' \to \operatorname{Spec} A'$ is a resolution of $\operatorname{Spec} A$, $i' : U' \to X'$ is an open immersion, and $Y' = X' - U'$ is pure of codimension 1 and has normal crossings. All these schemes are excellent. 603

Let p' be a point of X' and let p be its image in X. Since Y has normal crossings in X, there exist elements $x_1, \dots, x_r \in \mathcal{O}_{X,p}$ such that $\{x_1, \dots, x_r\}$ is part of a system of parameters of $\mathcal{O}_{X,p}$ and such that $\operatorname{Spec} \mathcal{O}_{X,p} \cap Y = V(x_1, \dots, x_r)$. Moreover, it follows immediately from the fact that $g : X' \to X$ is regular that the images x_i' of the x_i in $\mathcal{O}_{X',p'}$ form part of a system of parameters of this ring, and that $\operatorname{Spec} \mathcal{O}_{X',p'} \cap Y = V(x_1', \dots, x_r')$.

Let B' be the strict localization of $\mathcal{O}_{X',p'}$ at a geometric point $\overline{p}$ above p', and let B be the strict localization of $\mathcal{O}_{X,p}$ at the corresponding geometric point of X (these are excellent rings (EGA IV 7.8.3 (ii), 7.9.5)). Let

$$U_{\overline{p}} = \operatorname{Spec} B[1/x_1, \dots, 1/x_r] \quad , \quad U'_{\overline{p}} = \operatorname{Spec} B'[1/x_1', \dots, 1/x_r'].$$

Lemma 2.2 applies, and we deduce that the morphisms

$$(*) \qquad H^q(U_{\overline{p}}, \boldsymbol{\mu}_n) \longrightarrow H^q(U'_{\overline{p}}, \boldsymbol{\mu}_n)$$

are bijective for $q \leq N$.

Let us compute the sheaves $R^q i_* \boldsymbol{\mu}_n$ and $R^q i'_* \boldsymbol{\mu}_n$ for the open immersions $i : U \to X$ and $i' : U' \to X'$: the stalk $(R^q i'_* \boldsymbol{\mu}_n)_{\overline{p}}$ at the geometric point $\overline{p}$ is precisely $H^q(U'_{\overline{p}}, \boldsymbol{\mu}_n)$

(VIII 5). Likewise, $(R^q i_* \mu_n)_{\overline{p}} = H^q(U_{\overline{p}}, \mu_n)$. Thus $(*)$ implies that the base-change morphisms

$$(**) \qquad g^* R^q i_* \mu_n \longrightarrow R^q i'_* \mu_n$$

604 are bijective for $q \leq N$.

By the base-change theorem for a proper morphism (XII 5.5), we have (since X and X' are proper over the spectra of strictly local rings, and the closed fibers of X and X' are equal)

$$H^q(X, F) \xrightarrow{\sim} H^q(X', g^* F)$$

for every torsion sheaf F and every q. Consequently, $(**)$ implies that in the morphism of spectral sequences

$$\begin{array}{ccccc} E_2^{pq} & = & H^p(X, R^q i_* \mu_n) & \Longrightarrow & H^{p+q}(U, \mu_n) \\ \downarrow & & \downarrow{\scriptstyle \varphi_2^{pq}} & & \downarrow{\scriptstyle \varphi} \\ E_2^{\prime pq} & = & H^p(X', R^q i'_* \mu_n) & \Rightarrow & H^{p+q}(U', \mu_n), \end{array}$$

the arrows φ_2^{pq} are bijective for every $q \leq N$. It is easily seen that this implies that φ_∞^{pq} is bijective if $q \leq N$ and $p + q \leq N + 1$. Consequently, the morphism of the abutments

$$H^m(U, \mu_n) \longrightarrow H^m(U', \mu_n)$$

is bijective for $m \leq N$, which proves the lemma.

The following lemma will complete the proof of 2.1:

Lemma 2.5. $P(N) \Rightarrow P(N+1)$.

605 *Proof.* Let A be strictly local, regular, excellent, and of equal characteristic, let $x \in \operatorname{rad} A$ be a local parameter, and let $U = \operatorname{Spec} A[1/x]$. Let $\widehat{A}$ be the completion of A, let $\hat{x}$ be the image of x, let $\widehat{U} = \operatorname{Spec} \widehat{A}[1/x]$, and let $g : \widehat{U} \to U$ be the canonical morphism. It suffices to prove that

$$(2.6) \qquad \begin{aligned} g_*(\mu_n)_{\widehat{U}} &\simeq (\mu_n)_U \quad \text{and} \\ R^q g_* \mu_n &= 0 \quad \text{if} \quad 1 \leq q \leq N. \end{aligned}$$

Indeed, the Leray spectral sequence

$$E_2^{pq} = H^p(U, R^q g_* \mu_n) \Longrightarrow H^{p+q}(U', \mu_n)$$

will give isomorphisms

$$H^q(U, \mu_n) \xrightarrow{\sim} H^q(U', \mu_n)$$

for $q \leq N$, and an exact sequence

$$0 \longrightarrow H^{N+1}(U, \mu_n) \longrightarrow H^{N+1}(U', \mu_n) \longrightarrow H^0(U, R^{N+1} g_* \mu_n).$$

But we already know that $P(1)$ is true (1.3), and it is therefore enough to prove that $H^q(U, /u_n) = 0$ if $1 < q \leq N + 1$. We may thus replace A by $\widehat{A}$, which is isomorphic to a ring of formal power series $\widehat{A} \simeq k[[x_1, \ldots, x_r]]$, with k separably closed and $\overline{x} = x_r$. This case was treated in 1.2.

Let us therefore verify (2.6): the fact that $g_*(\mu_n)_{\widehat{U}} \simeq (\mu_n)_U$ is simply the assertion of (2.3) for $q = 0$. Recall that $R^q g_* \mu_n$ is the sheaf associated with the presheaf $\mathscr{R}^q$, where

$$\mathscr{R}^q(V) = H^q(\widehat{V}, \mu_n)$$

606 for $V \to U$ étale. It suffices to take V separated and of finite type; then V is a disjoint

union of integral schemes, and it suffices to consider such V. Let B be the normalization of A in the field of fractions of $\Gamma(V, O_V)$. Then B is a finite A-algebra (EGA IV 7.8.3 (vi)), and we see immediately that V is a regular open subset of Spec B and that $\widehat{V}$ is the corresponding open subset of Spec $\widehat{B}$, where $\widehat{B} = B \otimes_A \widehat{A}$. It follows that the hypotheses of 2.4 are satisfied for the open subset $V \subset$ Spec B and for the morphism $B \to \widehat{B}$. Hence

$$\mathscr{R}^q(V) \simeq H^q(V, \boldsymbol{\mu}_n) \simeq H^q(\widehat{V}, \boldsymbol{\mu}_n)$$

for $q \leq N$. But the sheaf associated with the presheaf $V \mapsto H^q(V, \boldsymbol{\mu}_n)$ is plainly zero, which proves the lemma.

3. Purity

We shall adopt terminology analogous to that of XVI 3:

Definition 3.1. *A* regular pair (Y, X) of codimension c *is a closed immersion* $i : Y \to X$ *of locally Noetherian regular schemes such that for every* $y \in Y$ *one has* $\operatorname{codim}_y(Y, X) = c$ *(then there exists an open neighborhood* X' *of* y *in* X *and sections* $x_1, \dots, x_c \in \Gamma(X', O_X)$ *that generate the ideal of* Y*, and which form part of a regular system of parameters of* X *at every point of* $Y \cap X'$*). We write* $U = X - Y$*, and* $j : U \to X$ *for the corresponding open immersion.*

A *morphism* $(Y', X') \to (Y, X)$ of regular pairs of codimension c is a morphism 607
$X' \to X$ such that $Y' = Y \times_X X'$. The notion of a regular triple, of a morphism of regular triples, etc., is defined in the analogous evident way.

Theorem 3.2. [124]. (Cohomological purity) *Let* (Y, X) *be a regular pair of codimension* $c > 0$*, where* X *is an excellent prescheme of equal characteristic. Let* F *be a locally constant sheaf on* X *of cyclic groups of order* n *prime to the characteristic of* X*. Then, in the notation of* (V 4, VIII 5),

$$\mathscr{H}^q_Y(X, F) = (R^q i^!)F = 0 \quad \text{if} \quad q \neq 2c,$$
$$(R^q j_*)j^*F = 0 \quad \text{if} \quad q \neq 0, 2c-1, \quad \textit{and}$$
$$\mathscr{H}^{2c}_Y(X, F) = i^*(R^{2c-1} j_*)j^*F$$

is a locally constant sheaf of cyclic groups of order n on Y.

Proof. (cf. XVI 3.7). We see immediately that

$$\mathscr{H}^q_Y(X, F) = 0 \quad \text{if} \quad q = 0, 1.$$

By (V 4.5),

$$\mathscr{H}^q_Y(X, F) = (R^q i^!)F \simeq i^*(R^{q-1} j_*)j^*F \quad \text{if} \quad q > 1,$$

so the assertions for the three sheaves are equivalent. Since the assertions are local on
X for the étale topology, we may assume that $F = \boldsymbol{\mu}_n$. 608

For the case $c = 1$, the calculation of the stalks (VIII 5) of $(R^q j_*)j^*\boldsymbol{\mu}_n$ reduces us to the situation considered in 2.1, whence

$$((R^q j_*)j^*\boldsymbol{\mu}_n)_{\overline{y}} = \begin{cases} \mathbf{Z}/n & \text{if} \quad q = 1 \\ 0 & \text{if} \quad q > 1 \end{cases}$$

for every geometric point $\overline{y}$ of Y. Since these sheaves vanish outside Y in any case, this proves the assertion for $q > 1$. For $q = 1$ $(= 2c - 1)$, we apply the criterion of (IX 2.13 (i)) and the explicit calculation of 1.3.

[124] Depends on resolution of singularities (cf. Introduction).

The case $c > i$ is treated by induction (cf. XVI 3.7): let (Y, X) be a regular pair of codimension $c > i$. By definition, it is clear that, locally on X, we can find a regular triple (Y, Z, X) in which (Z, X) has codimension 1 and (Y, Z) has codimension $c - 1$. Let

$$\begin{tikzcd} Y \arrow[rr, "u"] \arrow[dr, "i"'] & & Z \arrow[dl, "v"] \\ & X & \end{tikzcd}$$

be the closed immersions. We have the spectral sequence

$$E_2^{pq} = (R^p u^!)(R^q v^!)F \Longrightarrow (R^{p+q} i^!)F.$$

By the induction hypothesis, $(R^q v^!)F = 0$ if $q \neq 2$, and is locally constant etc. ...if $q = 2$. Thus, again by the induction hypothesis, applied to the morphism u, $(R^p u^!)(R^1 v^!)F = 0$
609 if $(p, q) \neq (2c - 2, 2)$ and is locally constant etc. ...if $(p, q) = (2c - 2, 2)$, whence the result for the abutment $(R^{p+q} i^!)F$, C.Q.F.D.

3.3. It remains to determine the locally constant sheaves of (3.2) up to *canonical* isomorphism. For each prescheme S we introduce the locally constant sheaves $(\boldsymbol{\mu}_n^{\otimes r})_S =$ the r-fold tensor product of the sheaf $(\boldsymbol{\mu}_n)_S$ over the constant sheaf of rings $\mathbf{Z}_S$. We also set $(\boldsymbol{\mu}_n^{\otimes 0})_S = (\mathbf{Z}/n)_S$. It is immediately seen that if n is invertible on S, then $(\boldsymbol{\mu}_n^{\otimes r})_S$ is a locally constant sheaf of cyclic groups of order n. We have formulas of the form $(\boldsymbol{\mu}_n^{\otimes r}) \otimes (\boldsymbol{\mu}_n^{\otimes s}) = (\boldsymbol{\mu}_n^{\otimes r+s})$, and if $S' \to S$, then $(\boldsymbol{\mu}_n^{\otimes r})_{S'}$ is canonically isomorphic to the inverse image on S' of $(\boldsymbol{\mu}_n^{\otimes r})_S$. We shall often omit the symbol S when no confusion can arise.

Let (Y, X) be a regular pair of codimension 1, and suppose that Y is defined by an equation $x = 0$. We have the exact sequence of sheaves

$$0 \longrightarrow (\mathbf{G}_m)_X \longrightarrow j_*(\mathbf{G}_m)_U \longrightarrow \mathbf{Z}_Y \longrightarrow 0,$$

whence, by applying (1.3), canonical isomorphisms

$$(R^1 j_*)\boldsymbol{\mu}_n) \simeq j_*(\boldsymbol{\mu}\mathbf{G}_m)_U/(j_*(\mathbf{G}_m)_U)^n \simeq (\mathbf{Z}/n)_Y, \tag{3.3.1}$$

which plainly do not depend on the choice of x. Consequently, we have a canonical isomorphism

$$\varphi : \mathbf{Z}/n \xrightarrow{\sim} (R_i^{*!})\boldsymbol{\mu}_n \simeq i^*(R^1 j_*)\boldsymbol{\mu}_n. \tag{3.3.2}$$

610 Theorem 3.4. [124]. *Let C be the family of regular pairs of excellent schemes of equal characteristic. There exists a unique function φ assigning to every regular pair (Y, X) of codimension c an isomorphism (called canonical) of sheaves*

$$\varphi(Y, X) = \varphi : \mathbf{Z}/n \xrightarrow{\sim} \mathscr{H}_Y^{2c}(X, \boldsymbol{\mu}_n^{\otimes c}) = (R^{2c} i^!)\boldsymbol{\mu}_n^{\otimes c}$$

such that φ satisfies the following conditions:

(i) *If $c = 1$, and if Y is defined in X by an equation $x = 0$, then φ is the morphism (3.3′).*

[124] Depends on resolution of singularities (cf. Introduction). This theorem may also be regarded as the conjunction of 3.2 and the theory of variance of Exp. XVIII by means of "local fundamental classes," where homomorphisms $\varphi(Y, X)$ (not necessarily bijective) are studied for Y locally a complete intersection in X.

(ii) *φ is compatible with smooth morphisms of pairs; that is, if $(Y', X') \to (Y, X)$ is a morphism, let $g : Y' \to Y$ be the induced morphism. The diagram of Y'-sheaves*

$$\begin{tikzcd}
\mathbf{Z}/n \arrow[r, "\varphi(Y',X')"] & (R^{2c} i^!)\boldsymbol{\mu}_n^{\boxtimes c} \\
g^*(\mathbf{Z}/n) \arrow[u] \arrow[r, "g^*\varphi(Y,X)"] & g^*(R^{2c} i^!)\boldsymbol{\mu}_n^{\boxtimes c} \arrow[u]
\end{tikzcd}$$

is commutative (which implies in particular that the right-hand vertical arrow is an isomorphism).

(iii) Transitivity: *let (Z, Y, X) be a regular triple, that is, a commutative diagram of closed immersions of regular preschemes etc.,*

$$\begin{tikzcd}
Z \arrow[rr, "v"] \arrow[dr, "w"'] & & Y \arrow[dl, "u"] \\
& X &
\end{tikzcd} .$$

Let a, b, c be the codimensions of u, v, w, respectively (thus $c = a + b$). *We have an isomorphism* **611**

$$\varphi(Y, X) \otimes \boldsymbol{\mu}_n{}^{\otimes b} : \boldsymbol{\mu}_n{}^{\otimes b} \longrightarrow (R^{2a} u^!)_{\boldsymbol{\mu}_n}{}^{\otimes a} \otimes \boldsymbol{\mu}_n{}^{\otimes b} \simeq$$
$$\simeq (R^{2a} u^!)\boldsymbol{\mu}_n{}^{\otimes c}.$$

The spectral sequence $(R^p v^!)(R^q u^!)F \Rightarrow (R^{p+q} w^!)F$ gives an isomorphism

$$\xi : (R^{2b} v^!)(R^{2a} u^!)\boldsymbol{\mu}_n{}^{\otimes c} \xrightarrow{\sim} (R^{2c} w^!)\boldsymbol{\mu}_n{}^{\otimes c}.$$

We thus obtain a diagram of isomorphisms

(3.4.1)
$$\begin{tikzcd}
\mathbf{Z}/n \arrow[r, "\varphi(Z,Y)"] \arrow[d, "\varphi(Z,X)"] & (R^{2b} v^!)\boldsymbol{\mu}_n \boxtimes b \arrow[d, "\eta=(R^{2b}v^!)((Y,X)\boxtimes\boldsymbol{\mu}_n^{\boxtimes b})"] \\
(R^{2c} w^!)\boldsymbol{\mu}_n & (R^{2b} v^!(R^{2a} u^!)\boldsymbol{\mu}_n^{\boxtimes c}) \arrow[l]
\end{tikzcd}$$

and the transitivity assertion is that this diagram is commutative.

Proof. The proof of existence and uniqueness proceeds by induction on the codimension c. For $c = 1$, we define the morphism φ locally by (3.3.1). Since this morphism does not depend on the choice of the parameter x, gluing gives an isomorphism φ for every (Y, X) of codimension 1; and it is clear from the definition of the morphism and from 1.3 that the φ so constructed satisfies (ii). Transitivity does not enter when $c = 1$.

Now suppose that existence and uniqueness have already been proved when the **612**
codimension is $< c$, and that $c > 1$. Applying the induction hypothesis, we immediately find that in the situation of 3.4 (iii), the three arrows of (3.4.1) that are already defined, i.e. $\varphi(Z, Y)$, ξ, η, are compatible with morphisms of regular triples. In this situation set

$$\Psi(Z, Y, X) = \xi\eta\varphi(Z, Y),$$

so Ψ too is compatible with morphisms of triples. Now, if (Z, X) is a regular pair of codimension $c > 1$, locally on X one can find a triple (Z, Y, X) with $0 < \operatorname{codim}(Y, X) < c$. It therefore remains only to prove that the arrow $\Psi(Z, Y, X)$ is independent of Y. Indeed, this will prove the uniqueness of φ in codimension c, and existence will then follow by gluing.

If (Z, Y, X) is arbitrary with $\operatorname{codim}(Y, X) > 1$, locally on X one can find a regular triple (Y, W, X) in which (W, X) has codimension 1. A transitivity diagram chase, which we leave to the reader, therefore reduces us to checking independence of Y when (Y, X) has codimension 1.

Let (Z, Y_0, X) and (Z, Y_1, X) be two such triples. It suffices to check the assertion fiber by fiber, and we are thus reduced to the case in which X is strictly local and Y_i is defined by an equation, say $f_i = 0$ $(i = 0, 1)$. Let $\overline{X} = \operatorname{Spec} O_X[t]$, $\overline{Z} = Z \times_X \overline{X}$, and let $\overline{Y}$ be the closed subscheme of $\overline{X}$ defined by the equation

$$(f_1 - f_0)t + f_0 = 0.$$

613 Then $\overline{Y}$ is plainly regular at the points lying over the sections $\epsilon_0 : \{t = 0\}$ and $\epsilon_1 : \{t = 1\}$ of $\overline{X}/X$. Let C be the closed subset (EGA IV 7.8.3 (iv)) of $\overline{Y}$ consisting of the points at which $\overline{Y}$ is not regular, and replace $\overline{X}$, $\overline{Y}$, $\overline{Z}$ by $\overline{X} - C$, $\overline{Y} - C$, $Z - C \cap \overline{Z}$, respectively. Then $(\overline{Z}, \overline{Y}, \overline{X})$ is a regular triple, and the sections ϵ_i give morphisms of triples

$$(Z, Y_i, X) \longrightarrow (\overline{Z}, \overline{Y}, \overline{X}) \quad (i = 0, 1).$$

Let $F = (R^{2c} w^!)\boldsymbol{\mu}_n{}^{\otimes c}$ be the sheaf on Z. It is a locally constant sheaf isomorphic to $\mathbf{Z}/n$ (by either of the isomorphisms $\Psi(Z, Y_i, X)$), and one sees immediately that the sheaf $\overline{F} = (R^{2c} \overline{w}^!)\boldsymbol{\mu}_n{}^{\otimes c}$ on $\overline{Z}$ is the inverse image on $\overline{Z}$ of F. Since the fibers of $\overline{Z} \to Z$ are connected and nonempty and $\overline{Z} \to Z$ is smooth, hence open, it follows that the induced morphism

$$H^0(Z, F) \longrightarrow H^0(\overline{Z}, \overline{F})$$

is *bijective*; hence each of the morphisms

$$(*) \qquad \begin{tikzcd} & & H^0(Z, F) \\ H^0(Z, F) \arrow[r] & H^0(\overline{Z}, \overline{F}) \arrow[ur, "\epsilon_0^*"] \arrow[dr, "\epsilon_1^*"'] & \\ & & H^0(Z, F) \end{tikzcd}$$

is bijective, and consequently $\epsilon_0^* = \epsilon_1^*$.

To give an isomorphism of a sheaf F with $\mathbf{Z}/n$ is equivalent to giving a global section (the image of 1) that is a generator. Let $\overline{\alpha}$, α_i be the global sections of $\overline{F}$, F given by the
614 isomorphisms $\Psi(\overline{Z}, \overline{Y}, \overline{X})$, $\Psi(Z, Y_i, X)$, respectively. Since $\Psi(.,.,.)$ is compatible with morphisms, the diagram $(*)$ implies that

$$\alpha_0 = \epsilon_0^*(\overline{\alpha}) = \epsilon_1^*(\overline{\alpha}) = \alpha_1,$$

whence $\Psi(Z, Y_0, X) = \Psi(Z, Y_1, X)$, C.Q.F.D.

4. Local acyclicity of a regular morphism

THEOREM 4.1. [124] (local acyclicity). *Let $g : X' \to X$ be a regular morphism* (EGA IV 6.8.1) *of excellent schemes of equal characteristic. Then g is universally locally acyclic for n prime to the characteristic.*

[124] Depends on resolution of singularities (cf. Introduction).

Proof. Since the hypothesis is stable under base change of finite type, it follows at once from XV 1.13 ii) that it suffices to prove that g is locally acyclic for n. Apply the criterion of XV 1.17: let $\overline{x}'$ be a geometric point of X' and $\overline{x}$ the corresponding geometric point of X. It follows immediately from the definition of a regular morphism that the morphism of strict localizations at these points induced by g is again regular. We are thus reduced to the case in which $X = \operatorname{Spec} A$ and $X' = \operatorname{Spec} A'$ are strictly local, g is a local morphism, and by XV 1.17 it suffices to prove that the geometric fibers of g are acyclic 615
for n. Replacing X by an integral closed subscheme and X' by the corresponding closed subscheme reduces us to the case of the geometric generic fiber $X'_{\overline{x}}$. Take $\overline{x} = \operatorname{Spec} \overline{K}$, where $\overline{K}$ is the separable closure of the field of rational functions on X. Then

$$\overline{x} = \varprojlim_{\alpha} U_\alpha$$

where U_α ranges over a filtered category of connected étale X-schemes, which may be assumed *regular* (EGA IV 7.8.3 (v)). Let B_α be the normalization of A in the field of rational functions $R(U_\alpha)$. Set $U'_\alpha = X' \times_X U_\alpha$ and $B'_\alpha = A' \otimes_A B_\alpha$. Then U_α is a regular open subscheme of $\operatorname{Spec} B_\alpha$ and $B_\alpha \to B'_\alpha$ is regular, so by (2.3)

$$H^q(U_\alpha, \boldsymbol{\mu}_n) \xrightarrow{\sim} H^q(U'_\alpha, \boldsymbol{\mu}_n)$$

for every q. Now the fiber $X'_{\overline{x}}$ is the limit of the U'_α, hence

$$\begin{aligned} H^q(X'_{\overline{x}}, \boldsymbol{\mu}_n) &= \varinjlim_{\alpha} H^q(U'_\alpha, \boldsymbol{\mu}_n) \\ &= \varinjlim_{\alpha} H^q(U_\alpha, \boldsymbol{\mu}_n) \\ &= H^q(\overline{x}, \boldsymbol{\mu}_n) = 0 \quad \text{if} \quad a > 0, \qquad \text{C.Q.F.D..} \end{aligned}$$

Corollary 4.2. *(base change by a regular morphism). Let*

$$\begin{tikzcd} X & X' \arrow[l, "g"'] \arrow[d, "f'"] \\ Y \arrow[u, leftarrow, "f"'] & Y' \arrow[l, "g"'] \end{tikzcd}$$

be a Cartesian diagram, where $g : Y' \to Y$ is a regular morphism of excellent preschemes of equal characteristic, and f is quasi-compact and quasi-separated. Then for every abelian 616
torsion sheaf F whose torsion is prime to the characteristic, the base-change morphisms (XII 4.2)

$$g^*(R^q f_*)F \longrightarrow (R^q f'_*)g'^* F$$

are bijective for every $q \geq 0$.

This follows from 4.1 and XVI 1.1.

From 4.1 and XV 1.17 we immediately deduce

Corollary 4.3. *Let A be a strictly local excellent ring of equal characteristic, and let $f : X \to \operatorname{Spec} A$ be a morphism. Let $A \to A'$ be a regular morphism, where A' too is strictly local, excellent, and of equal characteristic. Denote by a $'$ the base change induced by $A \to A'$. For every torsion sheaf F on X, of torsion prime to the characteristic of A, we have*

$$H^q(X, F) \xrightarrow{\sim} H^q(X', F')$$

for every $q \geq 0$.

This corollary applies in particular when A' is the completion $\widehat{A}$ of A. (In this case, the hypothesis "A strictly local" can moreover be replaced by "A henselian local," as is seen immediately.)

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5. Finiteness theorem

Theorem 5.1. [124] (finiteness). *Let $f : X \to S$ be a morphism of finite type between excellent schemes of equal characteristic, and let F be a constructible abelian torsion sheaf on X, of torsion prime to the characteristic. Then $(R^q f_*)F$ is likewise constructible for every q.*

This result generalizes the theorem of XVI 5.1. The proof is nearly the same: one applies the purity result 3.2 instead of XVI 3.7.

Corollary 5.2. *Let A be an excellent strictly local ring of equal characteristic and let $f : X \to \operatorname{Spec} A$ be a morphism of finite type. Let F be a constructible abelian sheaf on X, of torsion prime to the characteristic of A. Then the groups $H^q(X, F)$ are finite for every q.*

6. Cohomological dimension of affine morphisms

Let $f : X \to Y$ be a morphism of finite type between excellent preschemes, and let F be a sheaf on X. Recall that we defined (XIV 2.2) an integer $\delta(F)$ when Y is local. We
618 may therefore regard δ as a function $\delta(F, f)$ on an arbitrary Y, by setting

$$\delta(F, f, y) = \delta(F'), \tag{6.0}$$

where $Y' = \operatorname{Spec} O_{Y,y}$, $X' \to Y'$ is the morphism obtained from f by the base change $Y' \to Y$, and F' is the inverse-image sheaf of F on X'.

We have the following result, which generalizes XVI 3.1:

Theorem 6.1. [124]. *Let $f : X \to Y$ be an affine morphism of finite type, where Y is an excellent scheme of equal characteristic, and let F be an abelian torsion sheaf on X. Then $R^q f_* F$ vanishes at every point $y \in Y$ such that $\delta(F, f, y) < q$.*

The local variant, plainly equivalent to 6.1, is the following:

Theorem 6.1 bis. [124]. *Let A be a strictly local excellent ring of equal characteristic, and let $f : X \to Y = \operatorname{Spec} A$ be an affine morphism of finite type. Let F be a torsion sheaf on X. Then*

$$H^q(X, F) = 0 \quad \text{if} \quad q > \delta(F).$$

Recall (X 5.1) that assertions of this kind are of no interest for F of p-torsion, where
619 p is the characteristic. We may therefore suppose that F is of torsion prime to the characteristic.

We have the following corollary:

Corollary 6.2. *Let A be a henselian, excellent ring of equal characteristic, with residue field k, and let $U \subset \operatorname{Spec} A$ be an affine open subscheme. Let $\ell \in \mathbf{P}$. We have*

$$\operatorname{cd}_\ell U \leq \operatorname{cd}_\ell k + \dim A.$$

Indeed, let $\widetilde{A}$ be the strict localization of A, which is an infinite Galois étale covering with group $G = G(\overline{k}/k)$, and let $\widetilde{U}$ be the affine open subscheme of $\operatorname{Spec} \widetilde{A}$ induced by U. Applying the Hochschild–Serre spectral sequence (VIII 8.4)

$$H^p(G, H^q(U, .)) \Longrightarrow H^{p+q}(U, .)$$

[124] Depends on resolution of singularities (cf. Introduction).

reduces us to treating the case $A = \widetilde{A}$, and this is then a special case of 6.1 bis.

With the notation above, let $R(A)$ be the ring of rational functions on A. We have $\operatorname{Spec} R(A) = \varprojlim_\alpha U_\alpha$, where U_α is an affine open subscheme of $\operatorname{Spec} A$. The theory of passage to the limit therefore gives

Corollary 6.3. *Let A be henselian, excellent, and of equal characteristic, with residue field k. Let $R(A)$ be the field of rational functions on* Spec A. *Then, for $\ell \in \mathbf{P}$, we have*

$$\operatorname{cd}_\ell R(A) \le \operatorname{cd}_\ell k + \dim A$$

with equality if ℓ is invertible in k, and if moreover $\ell \neq 2$ or k is not orderable. **620**

Equality in the latter case follows from X 2.4.

Finally, applying X 4.2, X 4.4, we find

Corollary 6.4. *Let X be a noetherian excellent scheme of equal characteristic, and let $R(X)$ be the ring of rational functions on X. Let $\ell \in \mathbf{P}$ be invertible on X, and suppose that $\ell \neq 2$ or that no residue field of X is orderable. Then*

$$\operatorname{cd}_\ell(X) \le \operatorname{cd}_\ell R(X) + \dim X \le 2 \operatorname{cd}_\ell R(X).$$

Proof of 6.1. We begin with a reduction analogous to that of XIV 3. Induction on δ: Clearly, 6.1 for $\delta \le d$ is equivalent to (6.1 bis) for $\delta \le d$. For $\delta = 0$ this is trivial. We treat the case $\delta \le 1$ in the form 6.1 bis. Since F is the filtered direct limit of its constructible subsheaves (IX 2.9 (iii)), we may suppose that F is constructible; then the support of F is contained in a closed subset $Z \subset X$ with $\delta(z) \le 1$ for every $z \in Z$ (XIV 2.2). Each reduced irreducible component Z_i of Z is then either an affine scheme of finite type and dimension ≤ 1 over the closed point y of Y, or a scheme quasi-finite over a reduced closed subscheme C of Y of dimension 1. In the latter case, $C = \operatorname{Spec} B$, where B is a strictly local integral domain of dimension 1. Such a scheme Z_i is the spectrum of a strictly local integral domain of dimension ≤ 1, or of a field finite over the fraction field of B. Thus in all cases the cohomological dimension of Z_i is at most 1, by X 2.3 **621**
and IX 5.7. We have an exact sequence

$$0 \longrightarrow F \longrightarrow \prod_i F_i \longrightarrow \epsilon \longrightarrow 0$$

where F_i is the restriction of F to Z_i, from which the result for Z follows.

Suppose now that the theorem has been proved for the values $\delta < d$, and let us prove it for $\delta(F) \le d$, assuming $d \ge 2$.

Lemma 6.5. *It suffices to treat the case where $X = \mathbf{E}^1_Y = \operatorname{Spec} O_Y[t]$ is affine space of dimension 1, and where Y is strictly local.*

The proof is that of XIV 4.2: It is clear that one may suppose Y strictly local, and that $\delta(F) \le d$. We must prove that $H^q(X, F) = 0$ if $q > d$. We may embed X in $\mathbf{E}^N_Y$, so it suffices to treat the case $X = \mathbf{E}^N_Y$. Induction on N: Consider the diagram

$$\begin{tikzcd}
\mathbf{E}^N_Y \arrow[rr, "g"] \arrow[dr, "f"'] & & \mathbf{E}^{N-1}_Y \arrow[dl, "h"] \\
& Y &
\end{tikzcd}$$

so that $\mathbf{E}_Y^N$ is affine space of dimension 1 over $\mathbf{E}_Y^{N-1}$. Applying the induction hypothesis and the Leray spectral sequence

$$H^p(\mathbf{E}^{N-1}, R^q g_* F) \Longrightarrow H^{p+q}(\mathbf{E}^N, F)$$

shows that it suffices to prove that

$$\delta(R^q g_* F) \leq d - q.$$

622 This means that if $z \in \mathbf{E}^{N-1}$ is a point such that $\delta(z) \geq d - q$, then $R^q g_* F = 0$ at the point z. But the induction hypothesis on N applies to the morphism g, and hence $R^q g_* F = 0$ at the point z if $\delta(F, g, z) < q$. Thus it suffices to prove the inequality

$$\delta(F, g, z) + \delta(z) \leq \delta(F).$$

We leave this pleasure to the reader.

Lemma 6.6. *To prove 6.1 for the values $\delta \leq d$, it suffices to prove the following: Let B be a complete normal local ring of equal characteristic and dimension d. Let $U \subset \operatorname{Spec} B$ be an affine open subset. Suppose that there exists an element $x \in \operatorname{rad} B$ which is invertible on U. Then*

$$H^q(U, \mathbf{Z}/\ell) = 0 \quad \text{if} \quad q > d.$$

This lemma is moreover a special case of Corollary 6.2.

Proof (cf. XIV 4.4). By 6.5, it suffices to treat the case where Y is strictly local and $X = \mathbf{E}_Y^1$. Consider the diagram

$$\begin{tikzcd} \mathbf{E}_Y^1 \arrow[rr, "i"] \arrow[dr, "f"'] & & \mathbf{P}_Y^1 \arrow[dl, "g"] \\ & Y & \end{tikzcd},$$

where $\mathbf{P}_Y^1$ is the projective line and $\mathbf{E}_Y^1$ is obtained from $\mathbf{P}^1$ by removing the section Y_∞ at infinity. Let F be a torsion sheaf on $\mathbf{E}^1$, with $\delta(F) \leq d$. We have the spectral sequence

$$H^p(\mathbf{P}^1, R^q i_* F) \Longrightarrow H^{p+q}(\mathbf{E}^1, F).$$

623 Now the $R^q i_* F$ for $q > 0$ are concentrated on the section Y_∞, which is isomorphic to Y. Since Y is strictly local, it follows that $H^p(\mathbf{P}^1, R^q i_* F) = 0$ if $p > 0$ and $q > 0$. Furthermore, by XII 5.3, we have $H^p(\mathbf{P}^1, i_* F) = 0$ if $p > 2$. Since we wish to prove that $H^n(\mathbf{E}^1, F) = 0$ if $n > d$, and $d \geq 2$, it suffices to prove that

$$H^0(\mathbf{P}^1, R^q i_* F) = H^0(Y_\infty, R^q i_* F) = 0 \quad \text{if} \quad q > d.$$

This group is isomorphic to the stalk of $R^q i_* F$ at the point y_∞ at infinity of $\mathbf{P}^1$ in the closed fiber.

Now it suffices (VII 3.3 and IX 2.9 (iii)) to treat the case where F is constructible. Then, since $\delta(F) \leq d$, there exists a closed subscheme X of $\mathbf{P}^1$ with $\dim X \leq d$ such that X contains the support of F. Let $\operatorname{Spec} A = \widetilde{X}$ be the strict localization of X at the point y_∞, $\widetilde{U} = \widetilde{X} - \widetilde{X} \times_{\mathbf{P}^1} Y_\infty$, and let F be the sheaf induced by F on $\widetilde{U}$. Note that U is obtained from $\widetilde{X}$ by localizing with respect to an element $x \in \Gamma(X, 0_X)$. We have

$$(R^q i_* F)_{y_\infty} \simeq H^q(\widetilde{U}, \widetilde{F}).$$

Thus it suffices to prove that $H^q(\widetilde{U}, F) = 0$, if $q > d$, for every torsion sheaf $\widetilde{F}$ on $\widetilde{U}$.

We may now apply IX 5.6, with $\varphi = \delta$, and we find that it suffices to prove that

$$H^q(V, \mathbf{Z}/\ell) = 0 \quad \text{if} \quad q > \delta(V)$$

for every V finite and integral over $\widetilde{U}$ and for invertible ℓ. Let $x : \overline{V} \to V$ be the normalization of V. We have the exact sequence 624

$$0 \longrightarrow (\mathbf{Z}/\ell)_V \longrightarrow x_*(\mathbf{Z}/\ell)_{\overline{V}} \longrightarrow C \longrightarrow 0$$

with $\delta(C) < \delta(V)$, and it follows from this (by induction on δ) that it suffices to take V *normal.*

Let B be the normalization of A in the field of rational functions on V. Then B is an excellent strictly local ring of equal characteristic, and V is an affine open subset of $\operatorname{Spec} B$, obtained by localizing with respect to x. Let $\widehat{B}$ be the completion of B, and $\widehat{V}$ the induced open subset (which is obtained from $\operatorname{Spec} \widehat{B}$ by localizing with respect to the image of x in $\widehat{B}$). By 4.3, we have

$$H^q(V, \mathbf{Z}/\ell) \simeq H^q(\widehat{V}, \mathbf{Z}/\ell).$$

We are therefore reduced to proving that

$$H^q(\widehat{V}, \mathbf{Z}/\ell) = 0 \quad \text{if} \quad q > \delta(V) = \dim B = \dim \widehat{B}.$$

By the induction hypothesis, the theorem is true if $\delta(V) < d$, so Theorem 6.1 bis applies in this situation. Consequently, it suffices to treat the case $\dim \widehat{B} = d$, and since $\widehat{B}$ is normal, we are in the situation of the lemma, whence the result.

6.7.0. We shall first prove 6.6 under additional hypotheses, when B has characteristic $p > 0$. Let A be an equidimensional, complete ring of equal characteristic, and let $k \subset A$ be a field such that $A/\operatorname{rad} A$ is a finite extension of k. We say that the k-algebra A is *formally separable* if there exists a k-subalgebra of A of the form $k[[x_1, \dots, x_n]]$ such that the morphism $\operatorname{Spec} A \to \operatorname{Spec} k[[x]]$ is finite and étale at the generic point of $\operatorname{Spec} k[[x]]$ $(x = (x_1, \dots, x_n))$. 625

Lemma 6.7. *Let $k \subset B$ be a field over which $B/\operatorname{rad} B$ is finite. Under the induction hypothesis on d above, the conclusion of 6.6 holds if the hypotheses of 6.6 are satisfied and if, in addition, either of the following conditions is satisfied:*

(i) *k has characteristic 0.*
(ii) *There exists an element $x \in \operatorname{rad} B$ which is invertible on U and such that $\overline{B} = B/(x)$ is a formally separable k-algebra.*

Proof. Choose elements $x_1 = x, x_2, \dots, x_d$ of $\operatorname{rad} B$ such that $\{x_1, \dots, x_d\}$ generates a primary ideal for $\operatorname{rad} B$. Then B is a finite A -algebra, where $A = k[[x_1, \dots, x_d]]$, and $\overline{B}$ is the $[[x_2, \dots, x_d]] = \overline{A}$-algebra $B/x_1 B$. Since in case (ii) $\overline{B}$ is analytically separable, we may in that case choose the elements $x_2, \dots, x_d$ in such a way that $\overline{B}$ becomes a generically étale $k[[x_2, \dots, x_d]]$-algebra. It then follows immediately that B is also a generically étale A-algebra, which is of course also true in case (i).

Let $b \in B$ be an element which generates the field extension $R(A)$ over $k((x))$. Let $\epsilon \in k[[x]]$ be the discriminant of the monic irreducible equation of b over $k[[x]]$. In case (ii), we may choose b so that x_1 does not divide ϵ. Write moreover $\operatorname{Spec} B - U =$ 626
$V(x_1) \cup Y$, where Y contains no irreducible component of $V(x_1)$. Let C be the ring $\Gamma(Y, O_Y)$, where Y is given its induced reduced structure. Changing the $x_2, \dots, x_d$ if necessary, we may suppose that $V(x_d)$ is not contained in the image of Y in $\operatorname{Spec} A$, and that, moreover, in case (ii), $V(x_d)$ is not contained in $V(\epsilon)$.

Lemma 6.8. *Let $A_0 = k[[x_1, \dots, x_{d-1}]]\,\{x_d\}$ be the strict localization of $k[[x_1, \dots, x_{d-1}]]\,[x_d]$ at the point $x_1 = x_2 = \cdots = x_d = 0$. Under the conditions above, there exist an A^0-algebra B^0, an affine open subset $U^0 \subset \operatorname{Spec} B^0$, and an isomorphism $A \otimes_{A^0} B^0 \simeq B$ such that the open subset of $\operatorname{Spec} B$ induced by U^0 is U.*

The proof of this lemma is analogous to that of 1.5: it suffices to find A^0-algebras B^0, C^0 and isomorphisms $A \otimes_{A^0} B^0 \simeq B$, $A \otimes_{A^0} C^0 \simeq C$, because then there will exist ([3] 1.4) a unique (surjective) morphism $B^0 \to C^0$ inducing the morphism $B \to C$. We therefore take $U^0 = \operatorname{Spec} B^0 - \{(\operatorname{Spec} C^0) \cup V(x_1)\}$, and immediately verify that U is induced by U^0. The existence of C^0, and of B^0 in case (ii), is proved as in 1.5. The existence of B^0 in case (i) is a consequence of ([3] 5.1). (It may also be proved directly by applying Abhyankar's lemma to the ramification along $\{x_1 = 0\}$, and descent.)

We can now complete the proof of 6.7. Applying 4.3 once more, we find that

$$H^q(U, \mathbf{Z}/\ell) \simeq H^q(U^0, \mathbf{Z}/\ell),$$

627 from which it follows that it suffices to prove that the latter group is zero for $q > d$. Now $\operatorname{Spec} A^0$ is a limit of schemes affine and étale over $k[[x_1, \dots, x_{d-1}]]\,[x_d]$, so $\operatorname{Spec} B^0$ is a limit of schemes of finite type over $\operatorname{Spec} k[[x_1, \dots, x_{d-1}]]$, each fiber of which has dimension ≤ 1. Since $U^0 \subset \operatorname{Spec} B^0[1/x_1]$, we have

$$U^0 = \varprojlim U^0_\alpha$$

where U^0_α is affine of finite type over the scheme $V^0 = \operatorname{Spec} k[[x_1, \dots, x_{d-1}]]\,[1/x_1]$ and where each fiber of U^0_α/V^0 has dimension ≤ 1.

Let $f^0 : U^0 \to V^0$ be the structural morphism. Since the fibers of f^0_α all have dimension ≤ 1, for $v \in V^0$ we have

$$\delta(\mathbf{Z}/\ell, f^0_\alpha, v) \leq \dim \mathcal{O}_{V^0,v} + 1 \leq d - 1$$

(because $\dim \mathcal{O}_{V^0,v} \leq d-2$). The induction hypothesis on d is therefore applicable to the morphism f^0, and, by applying 6.1, we find

$$R^q f^0_{\alpha *}(\mathbf{Z}/\ell) = 0 \quad \text{at the point} \quad v \quad \text{if} \quad q > \dim \mathcal{O}_{V^0,v} + 1,$$

i.e. if $q > d - \dim \overline{v}$. Thus $R^q f^0_{\alpha *}(\mathbf{Z}/\ell) \neq 0$ at the point v implies that $\dim \overline{v} \geq d - q$. Therefore, for the inclusion $i : V^0 \to \operatorname{Spec} k[[x_1, \dots, x_{d-1}]]$ (cf. (6.0)), we have

$$\delta(R^q f^0_{\alpha *}(\mathbf{Z}/\ell)) \leq d - q,$$

628 and the induction hypothesis, in the form of 6.1 bis, applied to the morphism i, implies that

$$H^p(V^0, R^q f^0_{\alpha *}(\mathbf{Z}/\ell)) = 0 \quad \text{if} \quad p + q > d.$$

The Leray spectral sequence

$$E_2^{pq} = H^p(V^0, R^q f^0_*(\mathbf{Z}/\ell)) \Longrightarrow H^{p+q}(U^0, \mathbf{Z}/\ell)$$

therefore implies that $H^n(U^0_\alpha, \mathbf{Z}/\ell) = 0$ if $n > d$, and since U^0 is the limit of the U^0_α, we likewise find

$$H^n(U^0, \mathbf{Z}/\ell) = 0 \quad \text{if} \quad n > d.$$

This completes the proof of 6.7, and hence of Theorem 6.1 in characteristic 0.

7. Affine morphisms — end of the proof

Recall that to complete the proof of 6.1 in characteristic $p > 0$ (still under the resolution hypothesis), we had reduced by 6.6 to proving the following assertion for $d \geq 2$:

Lemma 7.1. *Let B be a complete normal local ring of equal characteristic, with separably closed residue field, and of dimension d. Let $U \subset \operatorname{Spec} B$ be an affine open subset such that there exists $x \in \operatorname{rad} B$ which is invertible on U. Then*

$$H^q(U, \mathbf{Z}/\ell) = 0 \quad \text{if} \quad q > d.$$

Moreover, by the induction hypothesis, we may suppose that 6.1 is true for the values $\delta < d$, and that 7.1 has already been proved in the particular case 6.7 (ii). 629

First note that the condition that B be normal is not important. Indeed, let B be arbitrary, let $\pi : \overline{B} \to B$ be the normalization of B, and let $\overline{U} \subset \operatorname{Spec} \overline{B}$ be the inverse image of U. We have an exact sequence

$$0 \longrightarrow (\mathbf{Z}/\ell)_U \longrightarrow \pi_*(\mathbf{Z}/\ell)_{\overline{U}} \longrightarrow C \longrightarrow 0,$$

where the dimension of the support of C is $< d$, so $H^q(U, C) = 0$ if $q > d - 1$. Since $H^q(U, pi_* \cdot) \simeq H^q(\overline{U}, \cdot)$ (VIII 5.6), we see that

$$\text{(7.2)} \qquad \begin{aligned} &H^q(U, \mathbf{Z}/\ell) = 0 \quad \text{if} \quad q > d \quad \text{is equivalent to} \\ &H^q(\overline{U}, \mathbf{Z}/\ell) = 0 \quad \text{if} \quad q > d. \end{aligned}$$

7.3.0. We shall briefly recall, without proof, the properties of the notion of formal separability that we need. The reader may consult (EGA IV 18.11.10, 18.11.11): Let A be a complete noetherian local k-algebra whose residue field is a finite extension of k, and let A' denote the completion of the local ring $A \otimes_k k'$ (where $k' = k^{1/p}$). Then the extension $A \to A'$ is *radicial* (possibly infinite), as is easily seen — this is the well-known example of Nagata. It is therefore harmless, for the calculation of étale cohomology, to replace A by A' (VIII 1).

Now the condition that an equidimensional algebra A of dimension d be formally *separable*, that is, be a finite generically étale extension of $k[[x_1, \ldots, x_d]]$, can be expressed in terms of the module $\widehat{\Omega}^1_{A/k}$ (EGA IV 18.11.10), or of the Jacobian criterion (EGA 0_{IV} 22.6, IV 7.1)[124], and one sees that if A is not analytically separable, then the k'-algebra A' is not reduced. For the proof, one may for example reduce to the case where A is given by an equation $f = 0$, $f \in k[[x_1, \ldots, x_{d+1}]]$, and apply the Jacobian criterion and the following lemma, whose pleasant proof we leave to the reader: 630

Lemma 7.3. *Let $f(x_1, \ldots, x_n) \in k[[x_1, \ldots, x_n]]$ be a formal power series. Suppose that for each $i = 1, \ldots, n$ there exists an invertible element $u_i \in k[[x_1, \ldots, x_n]]$ such that $u_i f$ is a series in x_i^p and in the $x_1, \ldots, x_{i-1}, x_{i+1}, \ldots, x_n$. Then there exists an invertible element $v \in k[[x_1, \ldots, x_n]]$ such that*

$$vf \in k[[x_1^p, \ldots, x_n^p]].$$

It follows easily that, by applying a finite number of times the procedure of replacing (A, k) by $(\overline{A}', k')$, where $\overline{A}'$ is the normalization of A', one is reduced to the situation where A is indeed formally separable. Similarly, one can prove that for a given $x \in \operatorname{rad} A$, the ring $(A/(x))_{\text{réd}}$ becomes a formally separable algebra by the same procedure, applied to the algebra A.

[124]The co-editor of this seminar has not succeeded in elucidating the meaning of this reference to a situation which seems quite different to him, nor the justification of the assertion that follows. A. G.

Applying VIII 1.1 and 7.2, we find

Corollary 7.4. *It suffices to prove (7.1) under the additional condition that B contain a field k over which $B/\operatorname{rad} B$ is finite, and such that the k-algebra $(B/(x))_{\text{réd}}$ is formally*
631 *separable.*

The proof of 7.1 in this case consists in a reduction to the particular case treated in 6.7 (ii):

With the notation of 7.1, 7.4, let

$$D(x) = \sum_i r_i X_i$$

be the divisor of x, so that each X_i is a closed irreducible subscheme of Spec B of codimension 1 which is formally separable over k (that is, such that $B/\mathfrak{J}(X_i)$ is a formally separable k-algebra, ∞ which is moreover equivalent to the assertion that $(B/(x))_{\text{réd}}$ is formally separable). Choose $y \in \operatorname{rad} B$ such that we have

$$D(y) = \sum_i (r_i - 1)X_i + \Gamma$$

where Γ contains none of the X_i, and consider the blow-up

$$f : Z \longrightarrow \operatorname{Spec} B$$

of the ideal (x, y). The spectra of the rings $B[y/x]$ and $B[x/y]$ form an affine open covering of Z.

Let B_1 be the strict localization of $B[x/y]$ at the maximal ideal generated by $(\operatorname{rad} B, x/y)$. For each closed point Q of the closed fiber of $\operatorname{Spec} B[y/x] \to \operatorname{Spec} B$, let A_Q denote the strict localization of $B[y/x]$ at Q, and let U_Q be the affine open subset of Spec A_Q that is the inverse image of U.

632 Lemma 7.5. *To prove 7.1 for the ring B, it suffices to prove that for $q > d (\geq 2)$ we have*

(i) $$H^q(\operatorname{Spec} B_1[y/x], \mathbf{Z}/\ell) = 0$$

and

(ii) $$H^q(U_Q, \mathbf{Z}/\ell) = 0 \quad \textit{for each } Q.$$

Proof. We have open immersions

$$U \hookrightarrow \operatorname{Spec} B[1/x] \hookrightarrow \operatorname{Spec} B[y/x],$$

and hence a diagram

$$\begin{tikzcd} U \arrow[rr, "i"] \arrow[dr, "k"'] & & \operatorname{Spec} B[y/x] \arrow[dl, "j"] \\ & Z & \end{tikzcd}.$$

If Q is a closed point of the closed fiber of $\operatorname{Spec} B[y/x] \to \operatorname{Spec} B$, we have $(R^q i_* \mathbf{Z}/\ell)_Q = 0$ if $q > d$, by hypothesis. For each point P of $\operatorname{Spec} B[y/x]$ which is not closed in the closed fiber of $\operatorname{Spec} B[y/x] \to \operatorname{Spec} B$, we have $\delta(F, i, P) < d$, where $F = (\mathbf{Z}/\ell)_U$ (notation of (6.0)). Thus, applying the induction hypothesis and 6.1 to the morphism i, we find that for the morphism $\operatorname{Spec} B[y/x] \to \operatorname{Spec} B$ we have

$$(R^q i_* \mathbf{Z}/\ell) \leq d - q.$$

Consequently, the induction hypothesis in the form 6.1 bis, applied to the latter morphism, implies that

$$H^p(\operatorname{Spec} B[y/x], R^q i_* \mathbf{Z}/\ell) = 0$$

if $q > 0$ and $p > d - q$. With the Leray spectral sequence 633

$$E_2^{pq} = H^p(\operatorname{Spec} B[y/x], R^q i_* \mathbf{Z}/\ell) \Longrightarrow H^{p+q}(U, \mathbf{Z}/\ell).$$

we are reduced, in proving 7.1, to proving that

$$H^p(\operatorname{Spec} B[y/x], i_* \mathbf{Z}/\ell) = 0 \quad \text{if} \quad p > d.$$

We have an exact sequence

$$0 \longrightarrow \mathbf{Z}/\ell \longrightarrow i_*(\mathbf{Z}/\ell)_U \longrightarrow C \longrightarrow 0 \tag{7.6}$$

where $\delta(C) < d$. It therefore suffices (again by induction) to prove that

$$H^p(\operatorname{Spec} B[y/x], \mathbf{Z}/\ell) = 0 \quad \text{if} \quad p > d. \tag{7.7}$$

Now the open immersion $\operatorname{Spec} B[y/x] \xrightarrow{j} Z$ is obtained by removing from Z the closed subset C "at infinity," and in the other open subset $\operatorname{Spec} B[x/y]$ of Z it is identified with the open subset $C = V(x/y)$. The $R^q j_* \mathbf{Z}/\ell$ are concentrated on C, and the morphism $C \to \operatorname{Spec} B$ is a closed immersion. Consequently, we have $H^p(Z, R^q j_* \mathbf{Z}/\ell) = 0$ if p and $q > 0$. Since the closed fiber of $Z \to \operatorname{Spec} B$ has dimension 1 and this morphism is proper, XII 5.3 bis implies that $H^p(Z, j_* \mathbf{Z}/\ell) = 0$ if $p > 2$. The spectral sequence

$$E_2^{pq} = H^p(Z, R^q j_* \mathbf{Z}/\ell) \Longrightarrow H^{p+q}(\operatorname{Spec} B[y/x], \mathbf{Z}\ell)$$

therefore gives isomorphisms

$$H^q(\operatorname{Spec} B[y/x], \mathbf{Z}/\ell) \simeq H^0(Z, R^q j_* \mathbf{Z}/\ell) \simeq (R^q j_* \mathbf{Z}/\ell)_P$$

if $q > 2$, where P is the point "at infinity" of Z. These stalks are none other (VIII 5) than 634

$$(R^q j_* \mathbf{Z}/\ell)_P \simeq H^q(\operatorname{Spec} B_1[y/x], j_* \mathbf{Z}/\ell).$$

By (7.7), it therefore suffices to prove that these latter groups vanish if $q > d$ (≥ 2). We have an exact sequence analogous to (7.6) which, together with the induction hypothesis, shows that it suffices that

$$H^q(\operatorname{Spec} B_1[y/x], \mathbf{Z}/\ell) = 0 \quad \text{if} \quad q > d,$$

whence the lemma.

Proof of 7.5 (i).

Consider the closed subset $C = V(x/y)$ of $\operatorname{Spec} B[x/y]$, with its induced reduced structure. It is immediate that the closed immersion of C into $\operatorname{Spec} B$ induced by the morphism $Z \to \operatorname{Spec} B$ identifies C with the closed subscheme $X = \bigcup X_i$ of $\operatorname{Spec} B$. Since

$$D(x) = \sum r_i X_i$$
$$D(y) = \sum (r_i - 1) X_i + \Gamma,$$

the element x/y generates the ideal $\mathfrak{J}(X_i)$ locally at the generic point of X_i, where the local ring is a discrete valuation ring because B is normal. Consequently, the morphism $\operatorname{Spec} B[x/y] \to \operatorname{Spec} B$ is an isomorphism in a neighborhood of such a point, and the element x/y vanishes to order 1 on each of the irreducible components C_i (corresponding 635
to X_i) of C.

Since the closed subset C lifts to Spec B_1, this remains true if the ring $B[x/y]$ is replaced by B_1, and hence $B_1/(x/y)$ is a *complete* ring which is a formally separable k-algebra.

To prove 7.5 (i), we may first apply 4.3, which allows us to replace B_1 by its completion $\widehat{B}_1$, without affecting the ring $B_1/(x/y)$. By (7.2), we may moreover replace $\widehat{B}_1$ by its normalization $\overline{B}_1$. Since B_1 (and hence $\widehat{B}_1$) is already regular at the generic points of the C_i, the morphism Spec $\overline{B}_1 \to$ Spec $\widehat{B}_1$ is an isomorphism at these points, and so $\overline{B}_1/(x/y)$ is still a formally separable k-algebra. We are thus reduced to the special case 6.7 (ii), which proves the result.

Proof of 7.5 (ii).

Choose the point Q of the closed fiber of Spec $B[y/x] \to$ Spec B. The strict localization A_Q may be written as a limit

$$A_Q = \varinjlim A$$

where A ranges over a filtered category of rings that are étale and of finite type over $B[y/x]$. Let U_A be the affine open subset of Spec A that is the inverse image of U. Since x is invertible on U_A, the morphism $U_A \to$ Spec B is étale, and U_A is normal. Consequently, if C denotes the normalization of B in the field $R(A)$ of rational functions on A, we have an open immersion

$$U_A \longrightarrow \operatorname{Spec} C.$$

636 It suffices (VII 5.7) to prove that for each A, the image of $H^q(U_A, \mathbf{Z}/\ell)$ in $H^q(U_Q, \mathbf{Z}/\ell)$ is zero if $q > d$. Since x is invertible on U_Q, the morphism $U_Q \to$ Spec A factors through Spec $A[1/f]$, if $f \in B$ is any element such that f is divisible by x in $B[y/x]$ and f/x does not vanish at the point Q. It thus suffices to prove that for a suitable such f, we have

$$H^q(U_{A,f}, \mathbf{Z}/\ell) = 0 \quad \text{if} \quad a > d,$$

where $U_{A,f} = U_A - V(f)$ is the affine open subset of Spec $A[1/f]$, which is also an affine open subset of Spec C.

Let $c \in k$ be an element such that $y/x + c$ does not vanish at Q, and let J be the ideal in B of the closed subset $\Gamma \cap X$. Then if

$$f \equiv y + cx (\operatorname{mod} J^N) \quad N > 0, \tag{7.8}$$

the element f satisfies the condition above: Indeed, it is immediate that in Spec $B[y/x]$ we have $V(J) = V(x)$. Thus (7.8) does imply that f is divisible by x, whence

$$f/x \equiv y/x + c (\operatorname{mod} J).$$

Since J vanishes at Q, and $y/x + c$ is nonzero at Q, this proves our assertion.

Now $\Gamma \cap X$ is a closed subset of Spec B of codimension 2. Consequently, we can
637 choose an f satisfying (7.8), call it z, such that $V(z) \cap V(y)$ is likewise of codimension 2. Also choose elements $x_3, \dots, x_d$ such that $y, z, x_3, \dots, x_d$ form a system of parameters of the ring B. Then B, and hence C as well, is a finite $k[[y, z, x_3, \dots, x_d]]$-algebra.

Let C^0 be the normalization of $k[[y, z, x_3, \dots, x_d]]$ in the separable closure of $k((y, z, x))$ in the field of rational functions $R(C)$, so that C is a radicial extension of C^0, and C^0 is a generically étale extension of $k((y, z, x))$ (where $x = x_3, \dots, x_d$).

Choose a linear combination

$$f = az + by \quad (a, b \in k),$$

such that the morphism

$$\operatorname{Spec} C^0 \longrightarrow \operatorname{Spec} k[[z, y, x]]$$

is étale over the generic point of $V(f)$, and such that f/x does not vanish at Q in $\operatorname{Spec} B[y/x]$. It suffices to prove that

$$H^q(U_{A,f}, \mathbf{Z}/\ell) = 0 \quad \text{if} \quad q > d,$$

and hence (VIII 1.1) to prove that

$$H^q(U^0_{A,f}, \mathbf{Z}/\ell) = 0 \quad \text{if} \quad q > d,$$

where $U^0_{A,f}$ is the open subset of $\operatorname{Spec} C^0$ corresponding to the open subset $U_{A,f}$ of $\operatorname{Spec} C$, which is affine, as follows by descent (EGA IV 2.7.1 (xiii)). But by construction, the element f of C^0 is invertible on $U^0_{A,f}$, and $C^0/(f)$ is a formally separable k-algebra. We are therefore in the situation of 6.7 (ii), which completes the proof of 7.5 (ii), and hence that of 6.1.

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